

Decision Intelligence in Action

Solving Real-World Problems with Optimization and AI

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About the Author

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Preface

Organizations today face increasingly complex decisions. The ability to systematically analyze options and optimize outcomes has become a critical competitive advantage. This book bridges the gap between academic optimization theory and practical business applications, providing you with the tools and techniques to build intelligent decision-making systems.

Who This Book Is For

This book is designed for:

- **Data Scientists** looking to expand beyond predictive modeling into prescriptive analytics
- **Operations Research practitioners** wanting to modernize their toolkit with Python and AI
- **Software Engineers** building decision support systems
- **Business Analysts** seeking to understand the mechanics behind optimization
- **Graduate Students** in Industrial Engineering, Operations Research, or Data Science

What This Book Covers

Part I: Foundations introduces Decision Intelligence as a discipline, teaches you to frame problems correctly, and helps you understand when to use optimization vs. machine learning.

Part II: Core Techniques covers the mathematical programming fundamentals—linear, integer, and constraint programming—that form the backbone of most decision systems.

Part III: Solution Methods covers algorithms in detail, from exact methods like branch-and-bound to metaheuristics like genetic algorithms.

Part IV: Decision Problems in Practice applies everything to real-world problems: routing, scheduling, and resource allocation.

Part V: Advanced Topics explores uncertainty, multi-objective optimization, and the cutting edge of AI-driven optimization.

Part VI: Production Systems teaches you to deploy, scale, and maintain optimization systems in production.

What You Need for This Book

- Python 3.9 or later
- Basic familiarity with Python programming
- Understanding of linear algebra and calculus (refreshers provided)
- A curious mindset ready to tackle complex problems

Conventions Used

Throughout this book, you'll find several text conventions:



Tip

Tips provide useful shortcuts or best practices.



Note

Notes highlight important information you should be aware of.



Warning

Warnings alert you to potential pitfalls or common mistakes.



Important

Important boxes emphasize critical concepts you must understand.

Code examples appear in monospace font:

```
from gurobipy import Model, GRB

model = Model("example")
x = model.addVar(name="x")
```

Get in Touch

Your feedback is valuable. If you find errors, have suggestions, or want to discuss the content, please reach out.

Let's get started.

Copyright

Decision Intelligence in Action

Solving Real-World Problems with Optimization and AI

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First Edition: 2026

Part I.

**Part I: Foundations of Decision
Intelligence**

1. What is Decision Intelligence?

What You Will Learn

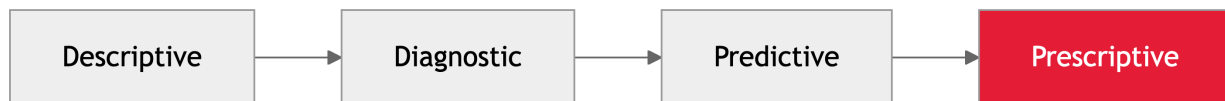
- Define Decision Intelligence and understand its relationship to OR, AI, and Data Science
- Identify the key components of a decision intelligence system
- Recognize when DI approaches are appropriate for your problems
- Understand the evolution from descriptive to prescriptive analytics

Every organization makes decisions. Some are routine—how many units to order, which truck to dispatch, when to schedule maintenance. Others are strategic—where to build a factory, whether to enter a new market, how to price a product. The quality of these decisions directly impacts business outcomes.

Decision Intelligence (DI) is the discipline of turning data into better decisions through a combination of optimization, simulation, machine learning, and domain expertise.

1.1. The Analytics Maturity Curve

Organizations typically progress through levels of analytical sophistication:



Level	Question	Techniques	Example
Descriptive	What happened?	BI dashboards, reporting	“Sales dropped 10% last month”
Diagnostic	Why did it happen?	Root cause analysis	“Competitor launched promotion”
Predictive	What will happen?	ML forecasting	“Demand will be 1,200 units”
Prescriptive	What should we do?	Optimization, simulation	“Order 1,150 units on Tuesday”

! The Prescriptive Gap

Most organizations are stuck at predictive analytics. They can forecast demand but still rely on spreadsheets and intuition for decisions. Decision Intelligence bridges this gap.

1.1.1. Why Prescriptive Analytics Matters

The transition from predictive to prescriptive is the most valuable—and most difficult—step in the analytics journey.

Predictive tells you what will happen. A forecast says demand will be 1,200 units, that a machine has 15% probability of failure, or that a customer is 73% likely to churn.

1. What is Decision Intelligence?

Prescriptive tells you what to do about it. Order 1,150 units on Tuesday to minimize holding costs while meeting service levels. Schedule maintenance in the next 48 hours because the cost of unplanned downtime exceeds the maintenance cost. Offer this specific retention package because the expected lifetime value justifies the cost.

Analytics Level	Output	Action Required
Descriptive	“Sales dropped 10%”	Manager investigates
Predictive	“Sales will drop 8% next month”	Manager decides response
Prescriptive	“Reduce price by 5% on products X, Y, Z”	System executes or recommends

The value of prescriptive analytics compounds:

- **Consistency:** Same situation, same optimal response
- **Speed:** Decisions in milliseconds, not hours
- **Scale:** Millions of micro-decisions per day
- **Optimality:** Mathematically best within constraints

The Heart of This Book

This book focuses on the **prescriptive** level. We assume you can describe what happened and predict what will happen. The question we answer is: **given predictions and constraints, what should you do?**

1.2. DI vs. Related Fields

Decision Intelligence draws from multiple disciplines:

1.2.1. Operations Research (OR)

OR provides the mathematical foundations:

- Linear and integer programming
- Network optimization
- Stochastic modeling
- Simulation

1.2.2. Machine Learning (ML)

ML contributes predictive capabilities:

- Demand forecasting
- Classification and clustering
- Reinforcement learning
- Pattern recognition

1.2.3. Behavioral Science

Understanding human decision-making:

- Cognitive biases

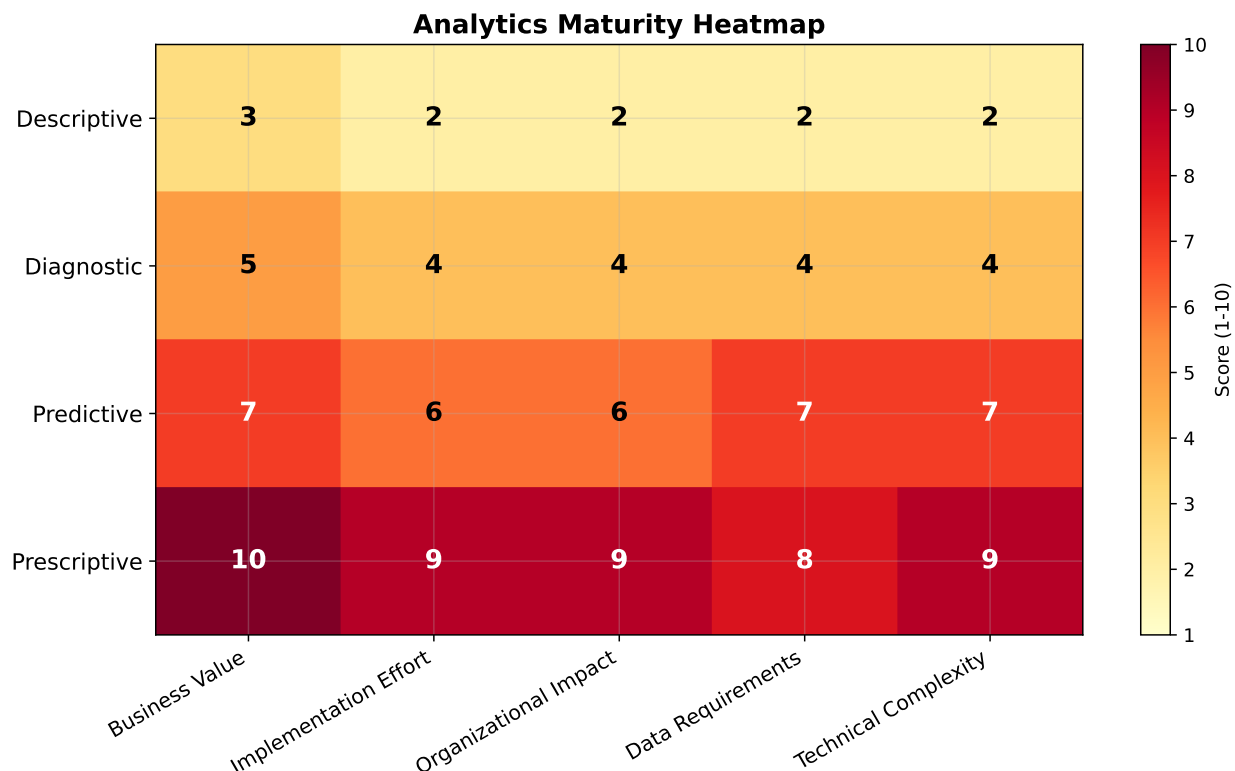


Figure 1.1.: Analytics Maturity Heatmap — Value, effort, and impact increase as organizations progress from descriptive to prescriptive analytics

- Risk perception
- Decision framing
- Organizational dynamics

i The Synthesis

Decision Intelligence is not just “OR + ML.” It’s a framework for combining these tools appropriately based on the decision context, data availability, and organizational constraints.

1.3. Anatomy of a DI System

A typical Decision Intelligence system has four layers:

1.3.1. 1. Data Layer

- Historical transactions
- Real-time sensor data
- External data (weather, markets)
- Master data (products, locations)

1.3.2. 2. Model Layer

- Forecasting models (demand, prices)
- Optimization models (routing, scheduling)

1. What is Decision Intelligence?

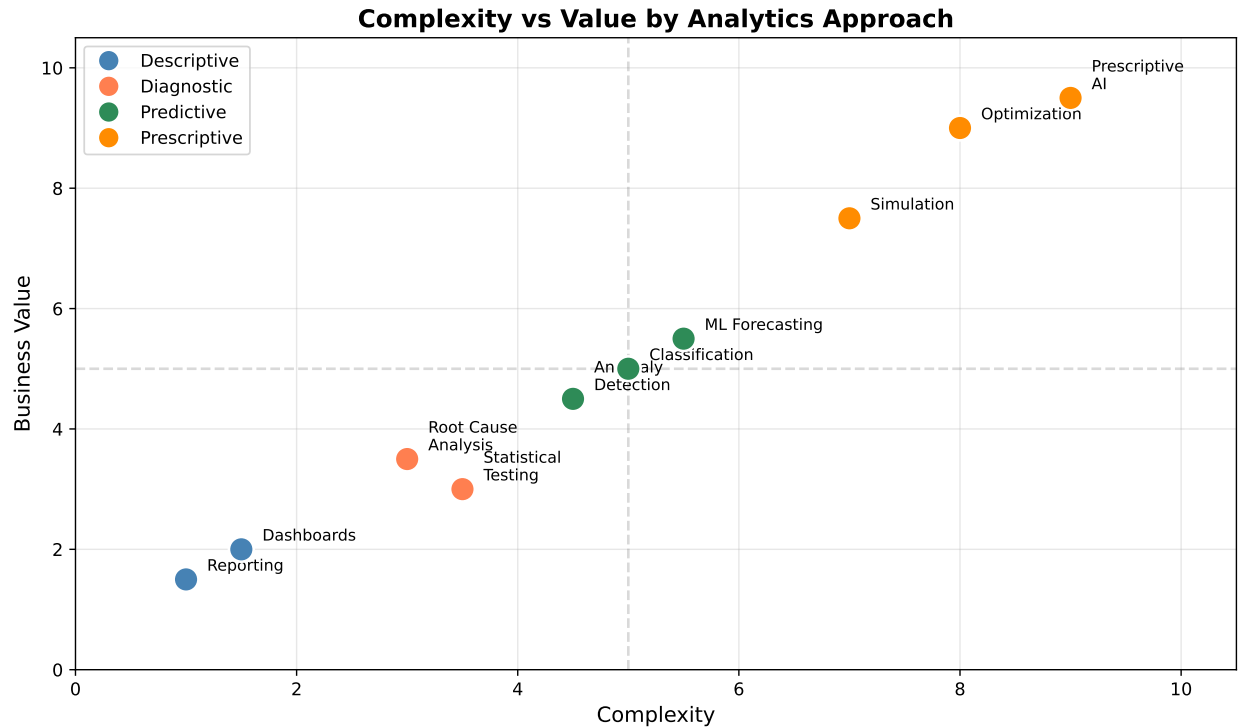


Figure 1.2.: Complexity vs Value positioning of each analytics approach — prescriptive analytics delivers the highest value but requires the most sophistication

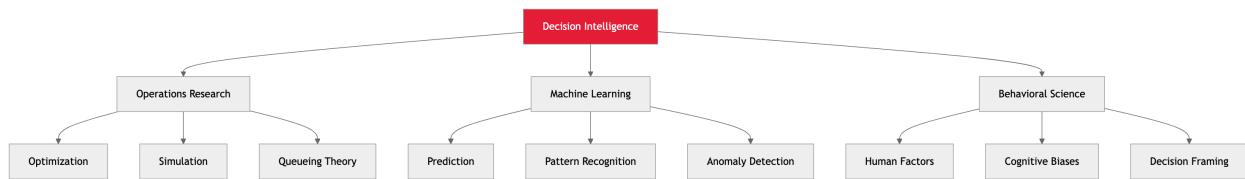


Figure 1.3.: Decision Intelligence draws from three core disciplines

- Simulation models (what-if analysis)
- Causal models (intervention effects)

1.3.3. 3. Decision Layer

- Automated decisions (reorder points)
- Recommendations (suggested routes)
- Scenarios (strategic options)
- Explanations (why this decision)

1.3.4. 4. Action Layer

- System integration (ERP, WMS)
- Human-in-the-loop interfaces
- Feedback collection
- Performance monitoring

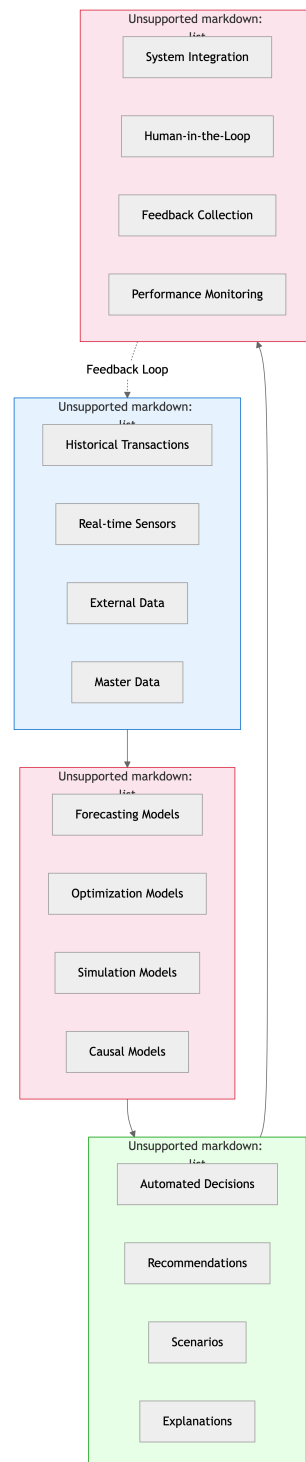


Figure 1.4.: Decision Intelligence System Architecture - Four interconnected layers

1. What is Decision Intelligence?

1.4. When to Use Decision Intelligence

DI is most valuable when:

💡 Good Fit for DI

- Decisions are **repeated** (daily routing, weekly scheduling)
- **Data is available** to inform the decision
- Decisions have **measurable impact** (cost, revenue, time)
- Problem has **combinatorial complexity** (many options to evaluate)
- **Constraints** must be satisfied (capacity, regulations, preferences)

DI may be overkill when:

- Decision is **one-time** and strategic (M&A, major investments)
- **Judgment and politics** dominate over analysis
- Problem is **simple enough** for rules of thumb
- **Data quality** is too poor to support modeling

1.5. The DI Workflow

Throughout this book, we'll follow a structured workflow:

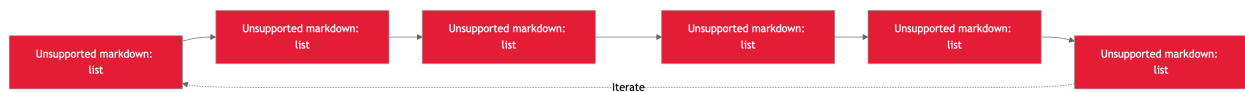


Figure 1.5.: The Decision Intelligence Workflow - A continuous improvement cycle

1. **Frame** the decision problem
2. **Model** the objectives, constraints, and uncertainties
3. **Solve** using appropriate algorithms
4. **Validate** the solution makes business sense
5. **Deploy** to production systems
6. **Monitor** and improve over time

1.6. Summary

1.6.1. Key Takeaways

- **Decision Intelligence** combines OR, ML, and behavioral science to improve decisions
- Most organizations are stuck at **predictive analytics**—DI enables prescriptive
- DI is valuable for **repeated, data-rich, constrained** decision problems
- A DI system spans **data → models → decisions → actions**
- The DI workflow: **Frame → Model → Solve → Validate → Deploy → Monitor**

1.7. Exercises

1. Identify three decisions in your organization that could benefit from DI. For each, describe the data available, constraints involved, and current decision process.
2. For a decision you make regularly (e.g., weekly grocery shopping), map it to the DI workflow. Where are the bottlenecks?

3. Research Gartner's Decision Intelligence market analysis. What trends do they highlight for the next 3-5 years?

2. Anatomy of Decision Problems

What You Will Learn

- Break down any decision problem into its fundamental components
- Identify objectives, constraints, and decision variables
- Understand feasible regions and solution spaces
- Distinguish between local and global optima
- Frame problems for mathematical modeling

Every decision problem, from route planning to portfolio optimization, shares a common structure. Understanding this anatomy allows you to recognize patterns across seemingly different domains and apply appropriate solution methods. This chapter provides the foundation for formulating real-world problems mathematically.

2.1. Components of a Decision Problem

Every optimization problem has three fundamental components:

1. **Decision Variables** - What we control
2. **Objective Function** - What we optimize
3. **Constraints** - What we must satisfy

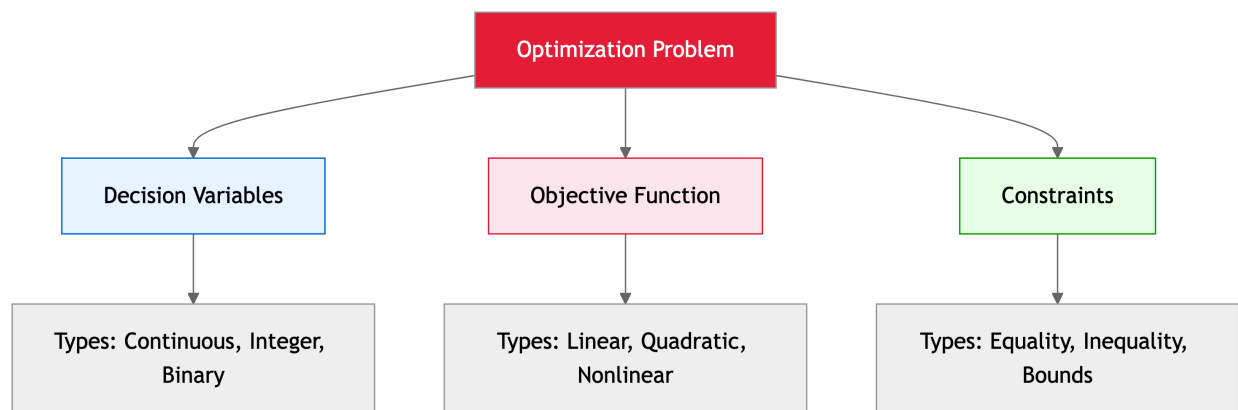


Figure 2.1.: The three fundamental components of every optimization problem

2.1.1. The Standard Mathematical Form

The general optimization problem can be expressed as:

$$\begin{aligned} & \text{minimize} && f(\mathbf{x}) \\ & \text{subject to} && g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & && h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p \\ & && \mathbf{x} \in \mathbb{R}^n \end{aligned}$$

2. Anatomy of Decision Problems

where $\mathbf{x}$ is the vector of decision variables, f is the objective function, g_i are inequality constraints, and h_j are equality constraints.

Minimization vs. Maximization

Any maximization problem can be converted to minimization:

$$\text{maximize } f(\mathbf{x}) \iff \text{minimize } -f(\mathbf{x})$$

By convention, we typically write optimization problems as minimization problems.

2.2. Decision Variables

Decision variables represent the choices we can make or control. These are the unknowns that our optimization algorithm will determine.

2.2.1. Examples of Decision Variables

Domain	Decision Variable	Type
Neural Networks	Weights and biases	Continuous
Manufacturing	Production quantities	Integer
Portfolio	Investment allocations	Continuous
Routing	Route selections	Binary
Scheduling	Job assignments	Binary

Decision variables can be:

- **Continuous:** Can take any real value within bounds (e.g., temperature, investment amounts)
- **Integer:** Must be whole numbers (e.g., number of units to produce)
- **Binary:** Only 0 or 1 (e.g., yes/no decisions)

2.2.2. Variable Bounds

Decision variables often have bounds:

$$l_i \leq x_i \leq u_i$$

These **box constraints** define the range of permissible values for each variable.

2.3. Objective Function

The objective function quantifies solution quality. It maps decision variables to a single numeric value that we seek to minimize or maximize.

2.3.1. Types of Objective Functions

Linear Objective

$$f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$$

Characteristics: Constant rate of change, no curvature. Appears in linear programming problems.

Quadratic Objective

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x}$$

Characteristics: Parabolic shape, smooth curvature. Common in ridge regression and least squares.

Nonlinear Objective

$$f(\mathbf{x}) = \text{general function of } \mathbf{x}$$

Characteristics: Complex curvature, potentially multiple local optima. Neural network loss functions fall into this category.

Non-smooth Objective

$$f(\mathbf{x}) = \|\mathbf{x}\|_1 \text{ or max functions}$$

Characteristics: Contains kinks or corners. Used in L1 regularization (Lasso) and hinge loss.

Choosing the Right Objective

The choice of objective function affects both the solution and the difficulty of the problem. Linear objectives lead to tractable problems, while nonlinear objectives may require specialized algorithms.

2.4. Constraints

Constraints define the feasible region - the set of all solutions that satisfy the problem requirements. Solutions outside this region are invalid, regardless of their objective value.

2.4.1. Inequality Constraints

Inequality constraints impose upper or lower bounds on functions of the variables:

$$g_i(\mathbf{x}) \leq 0$$

Examples:

- Budget limitations: $\text{cost}(\mathbf{x}) \leq B$
- Capacity constraints: $\text{load}(\mathbf{x}) \leq C$
- Performance requirements: $\text{error}(\mathbf{x}) \leq \epsilon$

2.4.2. Equality Constraints

Equality constraints require exact relationships:

$$h_j(\mathbf{x}) = 0$$

Examples:

- Conservation laws: inflow = outflow
- Balance equations: supply = demand
- Physical laws: Kirchhoff's laws in circuits

2. Anatomy of Decision Problems

2.4.3. Common Constraint Types

1. **Simplex Constraints:** $\sum x_i = 1, x_i \geq 0$ (probability distributions)
2. **Box Constraints:** $l_i \leq x_i \leq u_i$ (bounded parameters)
3. **Norm Constraints:** $\|\mathbf{x}\|_2 \leq 1$ (regularization)
4. **Integer Constraints:** $x_i \in \mathbb{Z}$ (discrete decisions)

2.5. Feasible Regions

The feasible region is the set of all points satisfying all constraints:

$$\mathcal{X} = \{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0\}$$

This defines our search space - the optimizer can only consider solutions within this region.

2.5.1. Properties of Feasible Regions

Convex Regions

A region is convex if, for any two points in the region, all points on the line segment between them are also in the region:

$$\forall \mathbf{x}, \mathbf{y} \in \mathcal{X}, \forall \lambda \in [0, 1] : \lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in \mathcal{X}$$

Convex feasible regions arise from linear constraints or convex inequality constraints.

Non-Convex Regions

Non-convex regions contain “holes” or “dents” where some line segments between feasible points leave the feasible region. These arise from:

- Discrete (integer) variables
- Non-convex constraints
- Disjunctive conditions (OR constraints)

2.5.2. Example: 2D Feasible Region

Consider a production planning problem:

- Decision variables: x_1, x_2 (quantities of two products)
- Constraints:
 - $2x_1 + x_2 \leq 10$ (resource constraint)
 - $x_1 + 2x_2 \leq 8$ (capacity constraint)
 - $x_1, x_2 \geq 0$ (non-negativity)
- Objective: maximize $3x_1 + 2x_2$ (profit)

The feasible region is a polygon with vertices at $(0, 0)$, $(5, 0)$, $(4, 2)$, $(2, 3)$, and $(0, 4)$.

! Empty Feasible Region

If constraints are contradictory, no feasible solution exists. This is called an **infeasible problem**. Before optimizing, always verify that at least one feasible solution exists.

2.6. Local vs. Global Optima

Understanding the difference between local and global optima is fundamental to optimization.

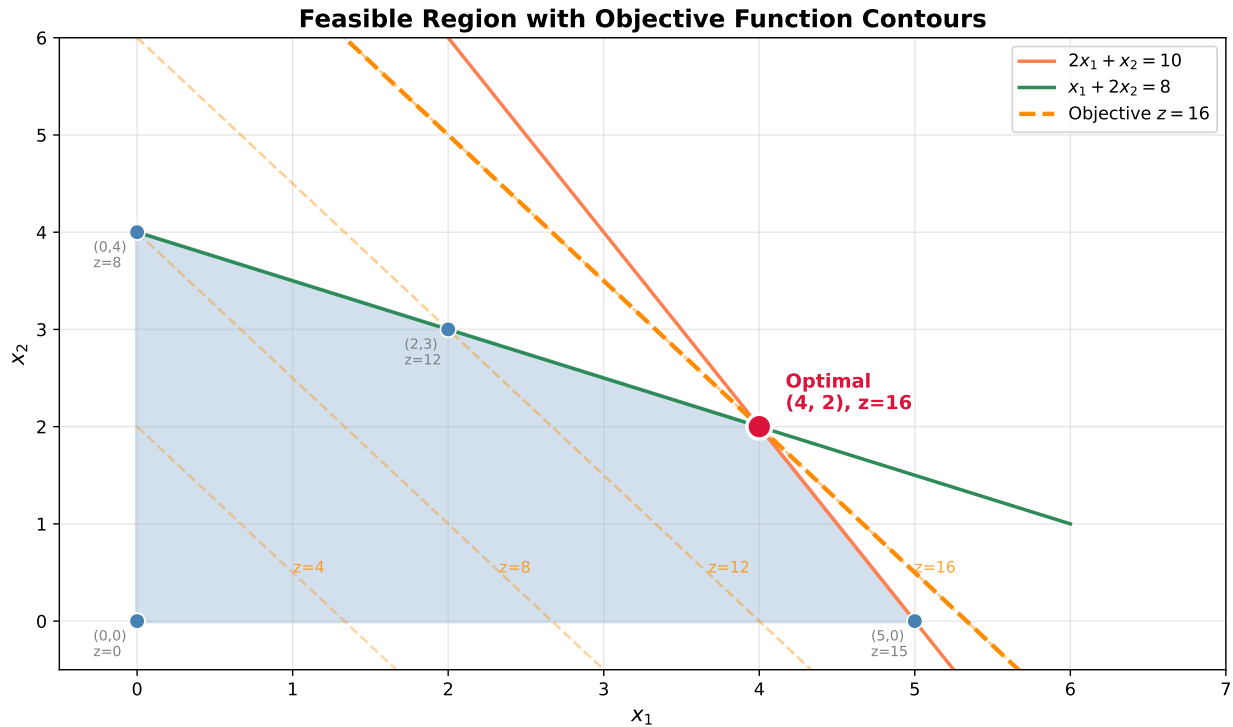


Figure 2.2.: Feasible region with objective function contours — the optimal solution lies at the vertex where the highest contour touches the feasible region

2.6.1. Definitions

Global Optimum

A point $\mathbf{x}^*$ is a global minimum if:

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for all } \mathbf{x} \in \mathcal{X}$$

It is the absolute best solution over the entire feasible region.

Local Optimum

A point $\mathbf{x}^*$ is a local minimum if:

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for all } \mathbf{x} \text{ near } \mathbf{x}^*$$

It is the best solution in its neighborhood, but better solutions may exist elsewhere.

i Note

Every global optimum is also a local optimum, but not every local optimum is global. In convex optimization, all local optima are global - this is one reason convex problems are easier to solve.

2.6.2. Convex vs. Non-Convex Problems

Convex Problems

- Every local minimum is a global minimum

2. Anatomy of Decision Problems

- Single “bowl” shape with one optimum
- Gradient descent guaranteed to find global optimum
- Examples: Linear regression, logistic regression, SVMs

Non-Convex Problems

- Multiple local minima, some better than others
- Complex surface with many valleys and peaks
- Gradient descent finds local minimum (quality depends on initialization)
- Examples: Deep neural networks, mixture models, matrix factorization

2.6.3. Critical Points

Different types of critical points (where the gradient is zero):

Type	Gradient	Hessian Eigenvalues	Significance
Global Minimum	Zero	All positive	Best solution
Local Minimum	Zero	All positive	Best in neighborhood
Saddle Point	Zero	Mixed signs	Neither min nor max
Local Maximum	Zero	All negative	Avoid in minimization

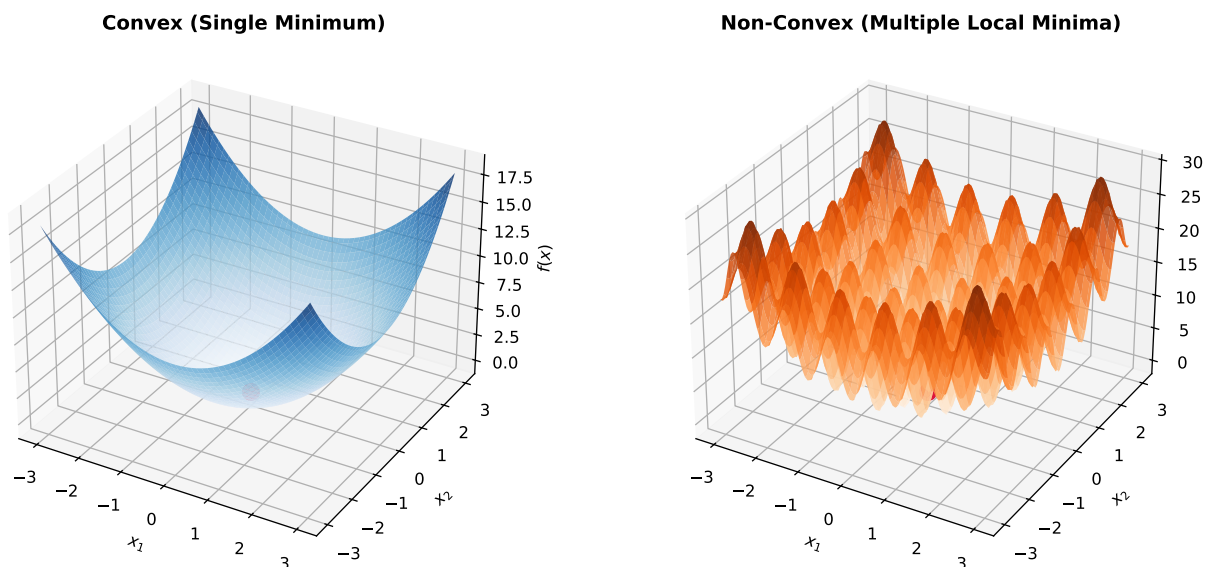


Figure 2.3.: Convex vs Non-Convex optimization landscapes — convex problems have a single global minimum while non-convex problems have multiple local minima that can trap gradient-based optimizers

2.6.4. Strategies for Non-Convex Optimization

1. **Multiple Random Restarts:** Run optimization from different initializations
2. **Stochastic Methods:** Use algorithms with randomness to escape poor local minima
3. **Simulated Annealing:** Accept worse solutions with decreasing probability over time
4. **Basin Hopping:** Combine local optimization with random perturbations
5. **Convex Relaxation:** Approximate with a convex problem, then refine

2.7. A Complete Example: Portfolio Optimization

Let's walk through a complete problem formulation.

2.7.1. Problem Statement

An investor wants to allocate \$100,000 across 4 assets to minimize risk while achieving at least 8% expected return.

2.7.2. Formulation

Decision Variables

$$\mathbf{x} = [x_1, x_2, x_3, x_4]^T$$

where x_i is the fraction of portfolio allocated to asset i .

Objective Function (Minimize Portfolio Variance)

$$\text{minimize } \sigma^2(\mathbf{x}) = \mathbf{x}^T \Sigma \mathbf{x}$$

where Σ is the covariance matrix of asset returns.

Constraints

1. Return requirement: $\mathbf{r}^T \mathbf{x} \geq 0.08$
2. Budget constraint: $\sum_i x_i = 1$
3. No short selling: $x_i \geq 0$ for all i

2.7.3. Complete Mathematical Form

$$\begin{aligned} &\text{minimize } \mathbf{x}^T \Sigma \mathbf{x} \\ &\text{subject to } \mathbf{r}^T \mathbf{x} \geq 0.08 \\ &\quad \sum_{i=1}^4 x_i = 1 \\ &\quad x_i \geq 0, \quad i = 1, 2, 3, 4 \end{aligned}$$

This is a **quadratic program** with linear constraints - a well-studied problem class with efficient solution methods.

2.8. Problem Classification

Different problem types require different solution approaches:

Problem Type	Variables	Objective	Constraints	Solution Methods
Linear Program (LP)	Continuous	Linear	Linear	Simplex, Interior Point
Quadratic Program (QP)	Continuous	Quadratic	Linear	Active Set, Interior Point
Nonlinear Program (NLP)	Continuous	Nonlinear	Nonlinear	SQP, Interior Point
Integer Program (IP)	Integer	Linear	Linear	Branch and Bound

2. Anatomy of Decision Problems

Problem Type	Variables	Objective	Constraints	Solution Methods
Mixed-Integer Program (MIP)	Mixed	Linear	Linear	Branch and Cut
Constraint Satisfaction (CSP)	Discrete	None	Various	Constraint Propagation

2.9. Summary

2.9.1. Key Takeaways

- Every optimization problem has three components: decision variables, objective function, and constraints
- The feasible region is the set of all solutions satisfying all constraints
- Global optima are the best solutions overall; local optima are best only in their neighborhood
- Convex problems have the property that every local optimum is global
- Problem classification determines which solution methods are applicable
- Proper formulation is often the hardest and most valuable part of solving real-world problems

2.10. Exercises

1. **Formulate a diet problem:** Minimize cost while meeting nutritional requirements (calories, protein, vitamins) using available foods with known costs and nutritional content.
2. **Identify the problem type:** A company needs to decide which of 10 potential warehouse locations to open (yes/no decisions) to minimize total shipping costs to 50 customers. Each warehouse has a fixed opening cost and capacity limit.
3. **Convexity analysis:** For the function $f(x_1, x_2) = x_1^2 + x_2^2 - x_1x_2$, determine if it is convex by computing the Hessian matrix and checking its eigenvalues.
4. **Feasible region visualization:** Sketch the feasible region for:
 - $x_1 + x_2 \leq 4$
 - $2x_1 + x_2 \leq 6$
 - $x_1, x_2 \geq 0$

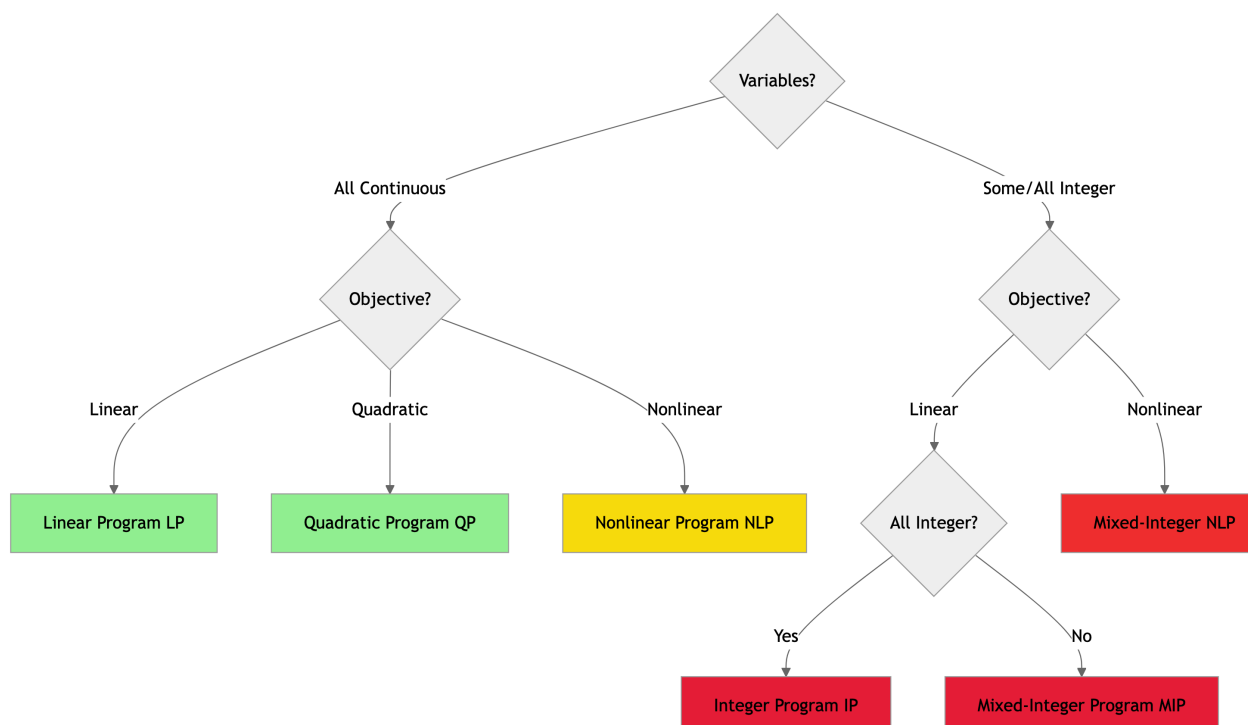


Figure 2.4.: Decision tree for classifying optimization problems

3. Decision Modeling and Problem Structuring

What You Will Learn

- Structure complex business problems into well-defined decision models
- Use Decision Requirement Diagrams (DRD) and DMN notation
- Apply Problem Structuring Methods to ambiguous situations
- Distinguish between hard constraints and flexible requirements
- Translate business language into mathematical formulations

Before writing a single line of optimization code, you must understand the problem. The most sophisticated algorithm is worthless if applied to the wrong model. This chapter teaches you how to structure messy real-world problems into clear decision models.

3.1. The Modeling Challenge

3.1.1. Why Modeling Matters More Than Solving

A common mistake in decision science: rushing to algorithms before understanding the problem.

Phase	Time Spent (Typical)	Impact on Solution Quality
Problem understanding	10%	60%
Model formulation	20%	30%
Algorithm selection	30%	7%
Implementation	40%	3%

! The Modeling Paradox

A good model of the right problem, solved approximately, beats a perfect solution to the wrong model every time.

3.1.2. From Business Problem to Mathematical Model

The journey from “we need to optimize our supply chain” to a solvable mathematical program involves several translations:

3.2. Decision Model and Notation (DMN)

DMN is an industry standard for modeling business decisions. It provides a visual language that both business users and technical teams can understand.

3.2.1. Decision Requirement Diagrams (DRD)

A DRD shows how decisions depend on other decisions, input data, and knowledge sources.

DRD Elements:

3. Decision Modeling and Problem Structuring

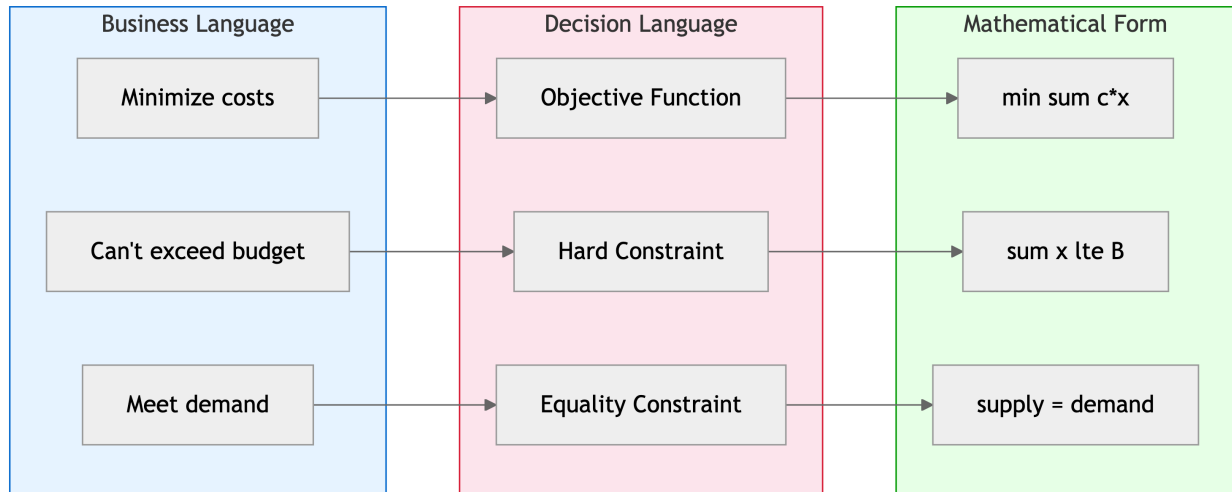


Figure 3.1.: The modeling process: translating business problems to mathematical models

Element	Symbol	Description
Decision	Rectangle	A decision to be made
Input Data	Oval	Data required for a decision
Knowledge Source	Document	Policies, regulations, expertise
Business Knowledge Model	Rectangle with clipped corner	Reusable decision logic

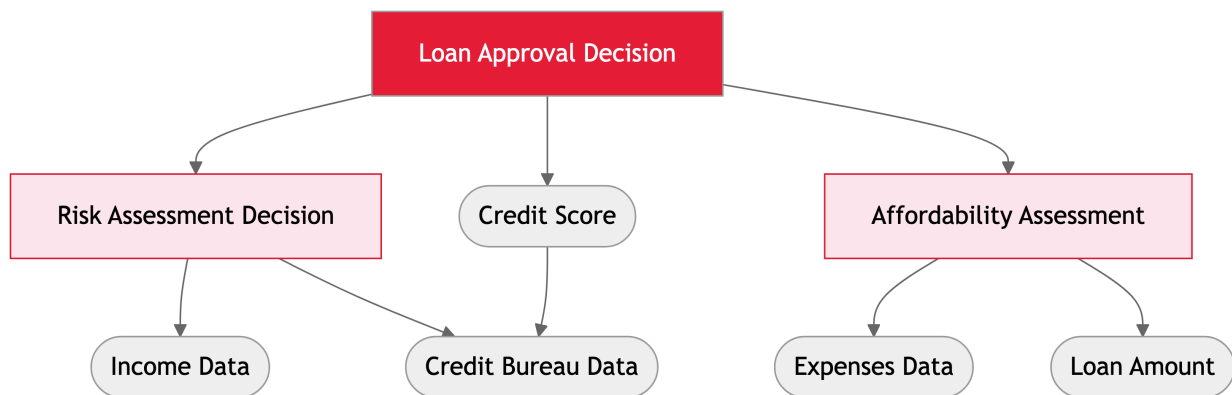


Figure 3.2.: Decision Requirement Diagram for Loan Approval

3.2.2. Building a DRD

```

from dataclasses import dataclass
from typing import List, Dict, Optional
from enum import Enum

class ElementType(Enum):
    DECISION = "decision"
    INPUT_DATA = "input_data"
    KNOWLEDGE_SOURCE = "knowledge_source"
    BKM = "business_knowledge_model"

@dataclass
26 class DRDElement:
    """Element in a Decision Requirement Diagram."""
    id: str

```

```

def __init__(self, name: str):
    self.name = name
    self.elements: Dict[str, DRDElement] = {}
    self.requirements: List[InformationRequirement] = []

def add_decision(self, id: str, name: str,
                 description: str = "") -> 'DecisionRequirementDiagram':
    """Add a decision node."""
    self.elements[id] = DRDElement(
        id=id, name=name,
        element_type=ElementType.DECISION,
        description=description
    )
    return self

def add_input_data(self, id: str, name: str,
                  description: str = "") -> 'DecisionRequirementDiagram':
    """Add an input data node."""
    self.elements[id] = DRDElement(
        id=id, name=name,
        element_type=ElementType.INPUT_DATA,
        description=description
    )
    return self

def add_knowledge_source(self, id: str, name: str,
                        description: str = "") -> 'DecisionRequirementDiagram':
    """Add a knowledge source."""
    self.elements[id] = DRDElement(
        id=id, name=name,
        element_type=ElementType.KNOWLEDGE_SOURCE,
        description=description
    )
    return self

def requires(self, decision_id: str,
            required_id: str) -> 'DecisionRequirementDiagram':
    """Specify that a decision requires another element."""
    self.requirements.append(
        InformationRequirement(source=required_id, target=decision_id)
    )
    return self

def get_dependencies(self, decision_id: str) -> List[DRDElement]:
    """Get all elements a decision depends on."""
    deps = []
    for req in self.requirements:
        if req.target == decision_id:
            deps.append(self.elements[req.source])
    return deps

def topological_order(self) -> List[str]:

```

3. Decision Modeling and Problem Structuring

```
    """Return decisions in order they can be evaluated."""
    # Build dependency graph
    in_degree = {e.id: 0 for e in self.elements.values()
                  if e.element_type == ElementType.DECISION}

    for req in self.requirements:
        if req.target in in_degree and req.source in in_degree:
            in_degree[req.target] += 1

    # Kahn's algorithm
    queue = [id for id, deg in in_degree.items() if deg == 0]
    order = []

    while queue:
        current = queue.pop(0)
        order.append(current)

        for req in self.requirements:
            if req.source == current and req.target in in_degree:
                in_degree[req.target] -= 1
                if in_degree[req.target] == 0:
                    queue.append(req.target)

    return order

# Example: Vehicle Routing Decision
drd = DecisionRequirementDiagram("Daily Route Planning")

# Decisions
drd.add_decision("route_plan", "Route Plan",
                "Assign orders to vehicles and sequence stops")
drd.add_decision("vehicle_assignment", "Vehicle Assignment",
                "Which vehicles to deploy today")
drd.add_decision("time_windows", "Feasible Time Windows",
                "Calculate achievable delivery windows")

# Input Data
drd.add_input_data("orders", "Today's Orders")
drd.add_input_data("fleet", "Available Fleet")
drd.add_input_data("traffic", "Traffic Predictions")
drd.add_input_data("driver_hours", "Driver Hours Remaining")

# Knowledge Sources
drd.add_knowledge_source("sla_policy", "SLA Policies")
drd.add_knowledge_source("routing_rules", "Routing Business Rules")

# Requirements
(drd
 .requires("route_plan", "vehicle_assignment")
 .requires("route_plan", "time_windows")
 .requires("route_plan", "orders")
 .requires("route_plan", "routing_rules"))
```



```
.requires("vehicle_assignment", "fleet")
.requires("vehicle_assignment", "driver_hours")
.requires("time_windows", "traffic")
.requires("time_windows", "sla_policy")
.requires("time_windows", "orders"))

print("Decision evaluation order:", drd.topological_order())
```

Decision Requirement Diagram: Daily Route Planning

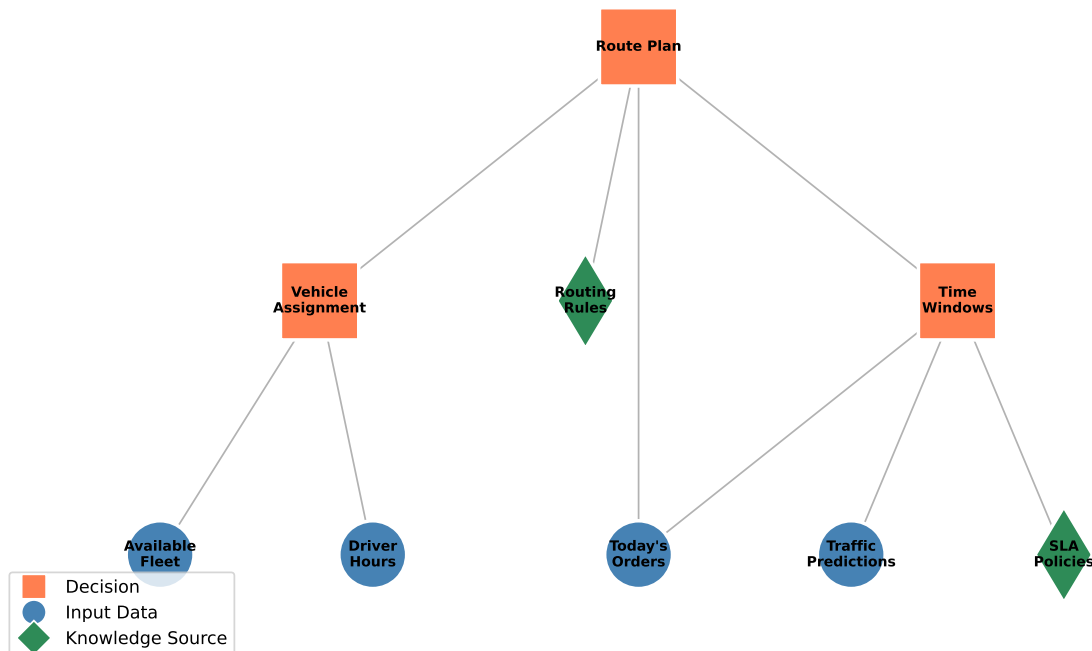


Figure 3.3.: Decision Requirement Diagram visualized as a network — decisions (red), input data (blue), and knowledge sources (green) with dependency edges

3.2.3. Decision Tables

DMN uses decision tables to specify decision logic:

Order Size	Customer Tier	Shipping Priority
< 100	Standard	Economy
< 100	Premium	Standard
>= 100	Standard	Standard
>= 100	Premium	Express
-	VIP	Express

```
from typing import Any, Callable

class DecisionTable:
    """DMN-style decision table."""
```

3. Decision Modeling and Problem Structuring

```
def __init__(self, name: str, inputs: List[str], output: str):
    self.name = name
    self.inputs = inputs
    self.output = output
    self.rules: List[Dict] = []

def add_rule(self, conditions: Dict[str, Any],
             result: Any) -> 'DecisionTable':
    """Add a rule to the table.

    Conditions use:
    - Exact match: value
    - Range: (min, max) tuple
    - Any: None
    - Custom: callable
    """
    self.rules.append({
        'conditions': conditions,
        'result': result
    })
    return self

def evaluate(self, inputs: Dict[str, Any]) -> Any:
    """Evaluate the decision table with given inputs."""
    for rule in self.rules:
        if self._matches(rule['conditions'], inputs):
            return rule['result']
    return None

def _matches(self, conditions: Dict, inputs: Dict) -> bool:
    """Check if inputs match rule conditions."""
    for key, condition in conditions.items():
        value = inputs.get(key)

        if condition is None: # Any value matches
            continue
        elif callable(condition):
            if not condition(value):
                return False
        elif isinstance(condition, tuple) and len(condition) == 2:
            # Range check
            if not (condition[0] <= value <= condition[1]):
                return False
        elif value != condition:
            return False

    return True

# Example: Shipping Priority Decision
shipping_table = DecisionTable(
    name="Shipping Priority",
```

```

    inputs=["order_size", "customer_tier"],
    output="priority"
)

(shipping_table
.add_rule({"order_size": lambda x: x < 100, "customer_tier": "Standard"},
         "Economy")
.add_rule({"order_size": lambda x: x < 100, "customer_tier": "Premium"},
         "Standard")
.add_rule({"order_size": lambda x: x >= 100, "customer_tier": "Standard"},
         "Standard")
.add_rule({"order_size": lambda x: x >= 100, "customer_tier": "Premium"},
         "Express")
.add_rule({"order_size": None, "customer_tier": "VIP"},
         "Express"))

# Test
print(shipping_table.evaluate({"order_size": 50, "customer_tier": "Premium"}))
# Output: Standard
print(shipping_table.evaluate({"order_size": 200, "customer_tier": "VIP"}))
# Output: Express

```

3.3. Problem Structuring Methods

When faced with a complex, ill-defined problem, Problem Structuring Methods (PSMs) help organize thinking before mathematical modeling.

3.3.1. The Soft-to-Hard Spectrum

Problems exist on a spectrum:

Characteristic	Soft/Messy Problems	Hard/Structured Problems
Objectives	Multiple, conflicting, unclear	Single, clear
Stakeholders	Many, disagreeing	Few, aligned
Data	Incomplete, qualitative	Complete, quantitative
Boundaries	Fuzzy	Well-defined
Solution approach	Exploration, dialogue	Optimization

3.3.2. Strategic Choice Approach

For problems where stakeholders disagree or objectives are unclear:

1. **Shaping mode:** Define the decision areas
2. **Designing mode:** Generate options within each area
3. **Comparing mode:** Evaluate combinations
4. **Choosing mode:** Select and commit

3.3.3. Rich Pictures

A rich picture is an informal diagram capturing the situation:

- Stakeholders and their concerns
- Flows of information and resources
- Conflicts and tensions

3. Decision Modeling and Problem Structuring

- External influences

```
@dataclass
class Stakeholder:
    """Stakeholder in problem structuring."""
    name: str
    concerns: List[str]
    influence: str # "high", "medium", "low"
    attitude: str # "supportive", "neutral", "resistant"

@dataclass
class ProblemElement:
    """Element of the problem situation."""
    name: str
    element_type: str # "issue", "constraint", "objective", "uncertainty"
    description: str
    related_stakeholders: List[str]

class ProblemStructure:
    """Structure a messy problem."""

    def __init__(self, problem_name: str):
        self.name = problem_name
        self.stakeholders: Dict[str, Stakeholder] = {}
        self.elements: List[ProblemElement] = []
        self.decision_areas: List[str] = []
        self.options: Dict[str, List[str]] = {}

    def add_stakeholder(self, name: str, concerns: List[str],
                       influence: str, attitude: str):
        """Add a stakeholder."""
        self.stakeholders[name] = Stakeholder(
            name=name, concerns=concerns,
            influence=influence, attitude=attitude
        )

    def add_element(self, name: str, element_type: str,
                   description: str, stakeholders: List[str]):
        """Add a problem element."""
        self.elements.append(ProblemElement(
            name=name, element_type=element_type,
            description=description, related_stakeholders=stakeholders
        ))

    def define_decision_area(self, area: str, options: List[str]):
        """Define a decision area with its options."""
        self.decision_areas.append(area)
        self.options[area] = options

    def summarize(self) -> Dict:
        """Summarize the problem structure."""
        return {
            "stakeholders": len(self.stakeholders),
            "key_concerns": list(set(
```

```

        c for s in self.stakeholders.values() for c in s.concerns
    )),
    "objectives": [e.name for e in self.elements
                    if e.element_type == "objective"],
    "constraints": [e.name for e in self.elements
                    if e.element_type == "constraint"],
    "uncertainties": [e.name for e in self.elements
                      if e.element_type == "uncertainty"],
    "decision_areas": self.decision_areas,
    "option_combinations": self._count_combinations()
}

def _count_combinations(self) -> int:
    """Count total option combinations."""
    if not self.options:
        return 0
    result = 1
    for opts in self.options.values():
        result *= len(opts)
    return result

# Example: Warehouse Location Problem
problem = ProblemStructure("New Distribution Center Location")

# Stakeholders
problem.add_stakeholder(
    "Operations",
    ["minimize delivery time", "operational efficiency"],
    influence="high", attitude="supportive"
)
problem.add_stakeholder(
    "Finance",
    ["minimize cost", "ROI > 15%"],
    influence="high", attitude="neutral"
)
problem.add_stakeholder(
    "HR",
    ["labor availability", "fair wages"],
    influence="medium", attitude="supportive"
)
problem.add_stakeholder(
    "Local Community",
    ["jobs", "environmental impact", "traffic"],
    influence="low", attitude="mixed"
)

# Problem elements
problem.add_element(
    "Cost minimization", "objective",
    "Minimize total logistics cost",
    ["Finance", "Operations"]
)

```

3. Decision Modeling and Problem Structuring

```
problem.add_element(
    "Service level", "objective",
    "Achieve 98% next-day delivery",
    ["Operations"]
)
problem.add_element(
    "Budget constraint", "constraint",
    "Capital budget of $50M",
    ["Finance"]
)
problem.add_element(
    "Demand growth", "uncertainty",
    "Future demand could grow 20-50%",
    ["Operations", "Finance"]
)

# Decision areas
problem.define_decision_area("Location", ["City A", "City B", "City C"])
problem.define_decision_area("Size", ["Small", "Medium", "Large"])
problem.define_decision_area("Automation", ["Manual", "Semi-auto", "Full-auto"])

print(problem.summarize())
```

3.4. Constraint Classification

Not all constraints are created equal. Distinguishing between hard and soft constraints is crucial for building practical models.

3.4.1. Hard vs. Soft Constraints

Type	Definition	Treatment	Example
Hard	Must be satisfied	Constraint in model	“Cannot exceed truck capacity”
Soft	Should be satisfied	Penalty in objective	“Prefer deliveries before noon”
Flexible	Can be violated at cost	Bounded violation	“Overtime allowed up to 20%”

3.4.2. The Constraint Spectrum

```
from enum import Enum
from dataclasses import dataclass
from typing import Optional

class ConstraintType(Enum):
    HARD = "hard"          # Must satisfy
    SOFT = "soft"           # Penalize violation
    FLEXIBLE = "flexible"   # Allow bounded violation
    GOAL = "goal"           # Aspirational target

@dataclass
class Constraint:
```

```

    """Business constraint with classification."""
    name: str
    description: str
    constraint_type: ConstraintType
    violation_penalty: Optional[float] = None # For soft constraints
    max_violation: Optional[float] = None     # For flexible constraints

    def to_mathematical_form(self) -> str:
        """Describe how this appears in mathematical model."""
        if self.constraint_type == ConstraintType.HARD:
            return f"Ax <= b (strictly enforced)"
        elif self.constraint_type == ConstraintType.SOFT:
            return f"Ax <= b + slack, add {self.violation_penalty}*slack to
                ↪ objective"
        elif self.constraint_type == ConstraintType.FLEXIBLE:
            return f"Ax <= b + slack, slack <= {self.max_violation}"
        else:
            return f"Ax - slack = target, minimize |slack|"

class ConstraintAnalyzer:
    """Analyze and classify constraints."""

    def __init__(self):
        self.constraints: List[Constraint] = []

    def add_constraint(self, name: str, description: str,
                      constraint_type: ConstraintType,
                      **kwargs) -> 'ConstraintAnalyzer':
        """Add a constraint."""
        self.constraints.append(Constraint(
            name=name,
            description=description,
            constraint_type=constraint_type,
            **kwargs
        ))
        return self

    def analyze(self) -> Dict:
        """Analyze constraint structure."""
        by_type = {}
        for c in self.constraints:
            ctype = c.constraint_type.value
            if ctype not in by_type:
                by_type[ctype] = []
            by_type[ctype].append(c.name)

        return {
            "total": len(self.constraints),
            "by_type": by_type,
            "flexibility_score": self._flexibility_score()
        }

```

3. Decision Modeling and Problem Structuring

```
def _flexibility_score(self) -> float:
    """Score 0-1 indicating model flexibility."""
    if not self.constraints:
        return 0

    weights = {
        ConstraintType.HARD: 0,
        ConstraintType.FLEXIBLE: 0.5,
        ConstraintType.SOFT: 0.8,
        ConstraintType.GOAL: 1.0
    }

    total = sum(weights[c.constraint_type] for c in self.constraints)
    return total / len(self.constraints)

# Example: Production Scheduling Constraints
analyzer = ConstraintAnalyzer()

(analyzer
 .add_constraint(
     "machine_capacity",
     "Cannot exceed machine hours available",
     ConstraintType.HARD
 )
 .add_constraint(
     "demand_satisfaction",
     "Meet customer demand",
     ConstraintType.HARD
 )
 .add_constraint(
     "overtime_limit",
     "Overtime up to 20% of regular hours",
     ConstraintType.FLEXIBLE,
     max_violation=0.20
 )
 .add_constraint(
     "changeover_minimize",
     "Minimize product changeovers",
     ConstraintType.SOFT,
     violation_penalty=100
 )
 .add_constraint(
     "inventory_target",
     "End inventory near safety stock",
     ConstraintType.GOAL
 ))

print(analyzer.analyze())
```

3.4.3. Questions to Classify Constraints

When unsure whether a constraint is hard or soft, ask:

1. What happens if violated?

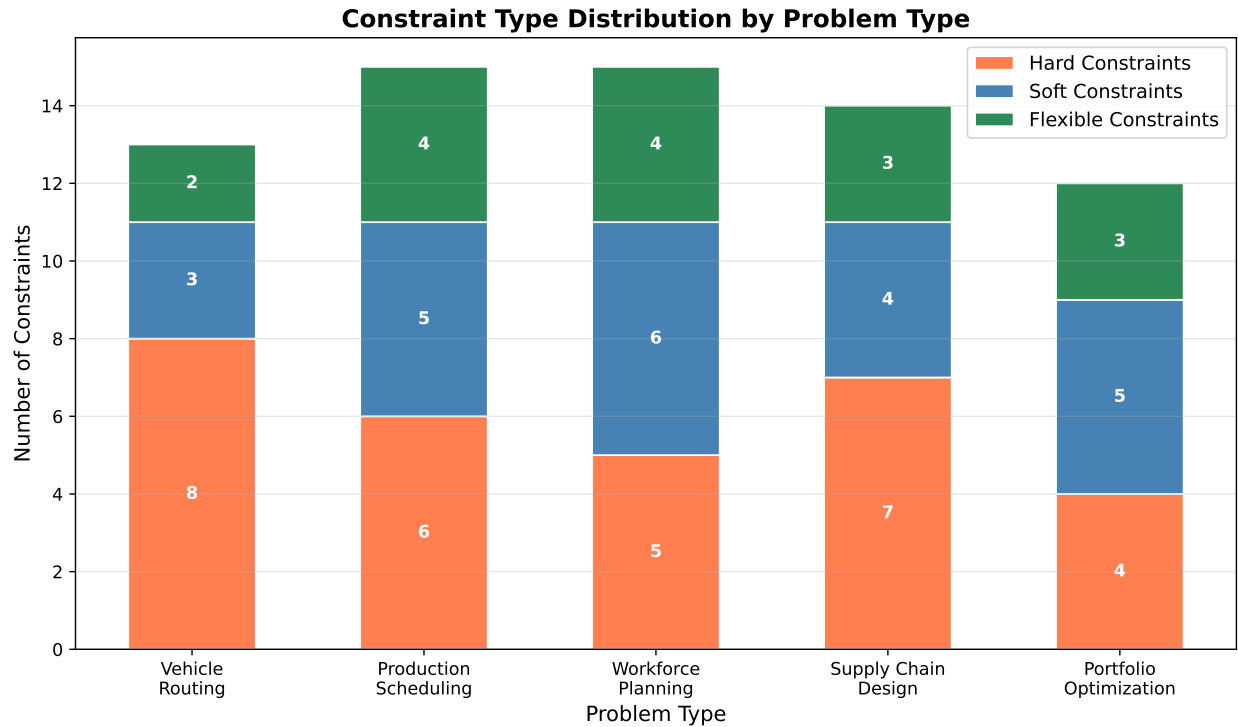


Figure 3.4.: Constraint type distribution across common optimization problem types — real-world problems typically have a mix of hard, soft, and flexible constraints

- Physical impossibility → Hard
- Financial penalty → Soft/Flexible
- Preference not met → Soft

2. Who sets the constraint?

- Physics/regulations → Likely hard
- Business policy → Often flexible
- Historical practice → Question it

3. What's the cost of violation?

- Infinite/undefined → Hard
- Quantifiable → Soft with penalty
- Acceptable within limits → Flexible

3.5. From Business Requirements to Mathematical Models

3.5.1. The Translation Process

```
class ModelTranslator:
    """Translate business requirements to mathematical model elements."""

    def __init__(self, problem_name: str):
        self.problem_name = problem_name
        self.variables = []
```

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```
self.objective_terms = []
self.constraints = []
self.parameters = []

def add_decision(self, business_term: str, math_var: str,
                var_type: str, domain: str):
    """Translate business decision to variable."""
    self.variables.append({
        "business": business_term,
        "mathematical": math_var,
        "type": var_type,
        "domain": domain
    })

def add_objective(self, business_goal: str, direction: str,
                 mathematical_form: str):
    """Translate business goal to objective."""
    self.objective_terms.append({
        "business": business_goal,
        "direction": direction,
        "mathematical": mathematical_form
    })

def add_constraint(self, business_rule: str,
                  mathematical_form: str,
                  constraint_type: str):
    """Translate business rule to constraint."""
    self.constraints.append({
        "business": business_rule,
        "mathematical": mathematical_form,
        "type": constraint_type
    })

def generate_summary(self) -> str:
    """Generate human-readable model summary."""
    lines = [f"# Mathematical Model: {self.problem_name}\n"]

    lines.append("## Decision Variables")
    for v in self.variables:
        lines.append(f"- {v['mathematical']} ({v['type']}): {v['business']}")

    lines.append("\n## Objective")
    for obj in self.objective_terms:
        lines.append(f"- {obj['direction']}: {obj['mathematical']}")
        lines.append(f"  Business goal: {obj['business']}")

    lines.append("\n## Constraints")
    for c in self.constraints:
        lines.append(f"- [{c['type']}] {c['mathematical']}")
        lines.append(f"  Business rule: {c['business']}")

    return "\n".join(lines)
```

```

# Example: Workforce Scheduling
translator = ModelTranslator("Weekly Workforce Schedule")

# Variables
translator.add_decision(
    "Number of employees working each shift",
    "x[e,d,s]", "Integer", "x >= 0"
)
translator.add_decision(
    "Whether employee e works day d",
    "y[e,d]", "Binary", "{0,1}"
)

# Objectives
translator.add_objective(
    "Minimize total labor cost",
    "Minimize",
    "sum(wage[e] * hours[s] * x[e,d,s] for all e,d,s)"
)
translator.add_objective(
    "Minimize understaffing",
    "Minimize",
    "sum(penalty * shortage[d,s] for all d,s)"
)

# Constraints
translator.add_constraint(
    "Each shift must have minimum staffing",
    "sum(x[e,d,s] for all e) >= min_staff[d,s]",
    "Hard"
)
translator.add_constraint(
    "Employees can work at most 5 days per week",
    "sum(y[e,d] for all d) <= 5",
    "Hard"
)
translator.add_constraint(
    "Employees should have 2 consecutive days off",
    "consecutive_off[e] >= 2 - slack[e]",
    "Soft"
)

print(translator.generate_summary())

```

3.6. Validation and Iteration

3.6.1. Model Validation Checklist

Before solving, verify your model:

1. **Dimensional consistency:** Do units match?
2. **Boundary cases:** What happens at extremes?
3. **Stakeholder review:** Does it capture their concerns?

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4. **Historical fit:** Would it have made good past decisions?

3.6.2. Common Modeling Pitfalls

Pitfall	Symptom	Solution
Missing constraint	Infeasible solutions produced	Review with domain experts
Wrong objective	Optimal solution feels wrong	Validate on historical cases
Ignored uncertainty	Fragile decisions	Add robustness/scenarios
Over-constrained	No feasible solution	Classify hard vs soft
Under-constrained	Trivial/extreme solutions	Add business rules

3.7. Summary

3.7.1. Key Takeaways

- **Modeling matters more than solving:** A good model of the right problem beats a perfect solution to the wrong model
- **DMN provides standard notation:** Decision Requirement Diagrams and Decision Tables communicate models clearly
- **Problem Structuring Methods** help with messy, ill-defined problems before mathematical modeling
- **Classify constraints carefully:** Hard (must satisfy), Soft (penalize), Flexible (bounded violation)
- **Iterate with stakeholders:** Models should be validated against domain knowledge and historical data
- **Document the translation:** Keep traceability from business language to mathematical formulation

3.8. Exercises

1. **DRD Construction:** Draw a Decision Requirement Diagram for a “which supplier to choose” decision, identifying all input data and sub-decisions required.
2. **Constraint Classification:** For a vehicle routing problem, list 10 potential constraints and classify each as hard, soft, or flexible. Justify your classifications.
3. **Decision Table:** Create a DMN decision table for an “order approval” process with inputs: order value, customer credit score, and stock availability.
4. **Problem Structuring:** A hospital wants to “improve emergency room efficiency.” Use problem structuring methods to identify stakeholders, objectives, constraints, and decision areas before any mathematical modeling.
5. **Model Translation:** Take a business requirement document for a production planning problem and translate it into mathematical notation, documenting each translation step.

4. The Optimization-ML Continuum

What You Will Learn

- Understand when to use optimization vs. machine learning
- Identify hybrid approaches that combine both paradigms
- Learn the “predict then optimize” framework
- Explore end-to-end differentiable optimization
- Implement AI-based optimization methods with Python

The boundary between optimization and machine learning has become increasingly blurred. Modern decision intelligence systems often combine both paradigms, leveraging the strengths of each. This chapter provides a framework for understanding when to use pure optimization, pure ML, or hybrid approaches that integrate both.

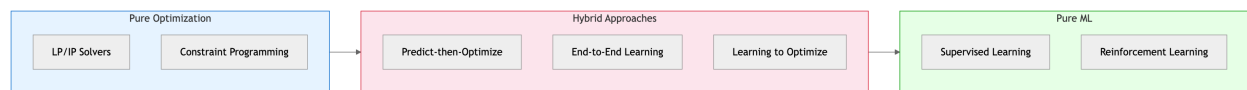


Figure 4.1.: The Optimization-ML Continuum: From pure optimization to pure learning

4.1. When to Optimize, When to Learn

The choice between optimization and machine learning depends on the problem structure, available data, and computational constraints.

4.1.1. The Fundamental Trade-off

Approach	Best When	Strengths	Limitations
Pure Optimization	Hard constraints, clear objectives	Guarantees, interpretability	Requires known problem structure
Pure ML	Patterns in data, no hard constraints	Flexibility, generalization	No optimality guarantees
Predict-then-Optimize	Uncertain parameters, known structure	Combines forecasting with decisions	Two-stage error propagation
End-to-End Learning	Decision quality is the metric	Learns directly for task	Harder to train, less interpretable

4.1.2. Optimization-First Scenarios

Use optimization when:

- **Constraints are hard:** Safety, regulatory, or physical constraints must be satisfied
- **Objectives are well-defined:** The goal function is known analytically
- **Problem structure is understood:** The relationship between decisions and outcomes is known

4. The Optimization-ML Continuum

- **Optimality matters:** Suboptimal solutions have significant costs

Examples: Vehicle routing with capacity constraints, production scheduling with machine capacities, portfolio optimization with risk limits.

4.1.3. Learning-First Scenarios

Use machine learning when:

- **Patterns are complex:** Relationships are too complicated for explicit modeling
- **Data is abundant:** Historical data captures system behavior
- **Flexibility is needed:** Problem changes frequently
- **Approximate solutions suffice:** Near-optimal is good enough

Examples: Demand forecasting, customer segmentation, anomaly detection, image recognition.

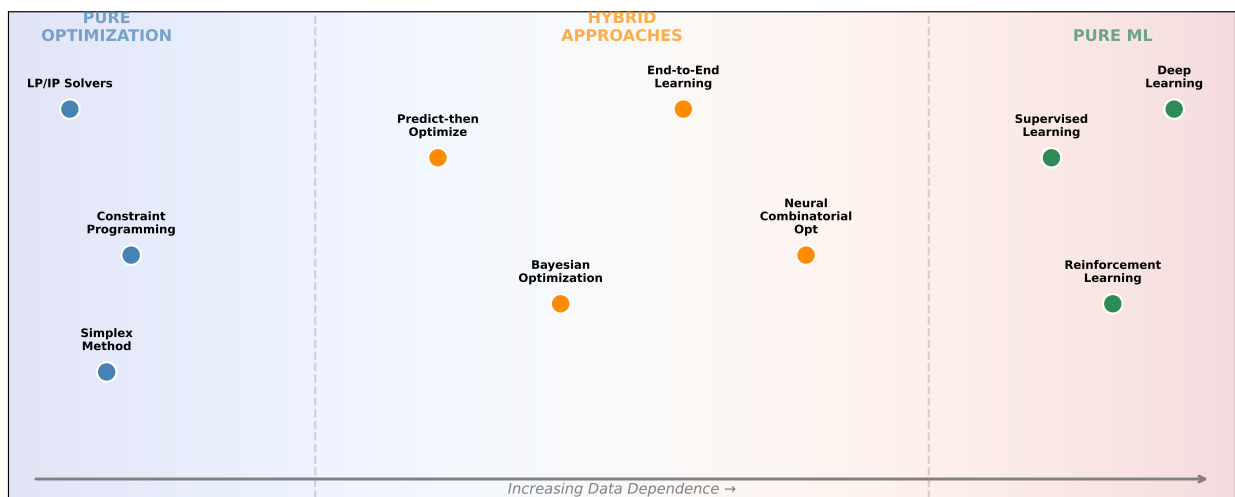


Figure 4.2.: The Optimization-ML Continuum — methods positioned along the spectrum from pure optimization to pure learning, with hybrid approaches in between

4.2. What Makes Optimization “AI-Based”?

Not all optimization methods involve intelligence. Understanding the distinction between adaptive and intelligent methods is foundational for selecting the right approach.

4.2.1. Learning vs. Adapting

Traditional Optimization follows fixed mathematical rules:

- Same algorithm on Problem A and Problem B uses identical strategy
- May adapt within a single problem (e.g., step size adjustment)
- No memory of past problems
- No improvement in strategy across instances

Examples: Gradient descent, BFGS, simplex method, simulated annealing.

AI-Based Optimization learns from experience and generalizes:

- Improves strategy based on past problem instances
- Transfers knowledge across related problems
- Builds internal models or representations

- Makes intelligent decisions based on learned patterns

Examples: Bayesian optimization, neural optimizers, reinforcement learning for optimization, meta-learning.

4.2.2. The Three Pillars of Learning

Learning in AI-based optimization requires:

1. **Experience accumulation:** The method gathers data from past optimization attempts
2. **Performance improvement:** Strategy becomes more effective over time or across problems
3. **Generalization:** Knowledge transfers to new, unseen problem instances

Key Insight

An algorithm that merely adjusts parameters within one problem (like adaptive step sizes) is not learning—it's following pre-programmed adaptation rules. True learning requires updating strategy based on accumulated experience.

4.2.3. What Intelligence Involves

Intelligent optimization methods exhibit:

- **Decision-making under uncertainty:** Choose actions when information is incomplete
- **Pattern recognition:** Identify structure in objective functions or constraint landscapes
- **Adaptive reasoning:** Adjust strategy based on observed problem characteristics
- **Trade-off balancing:** Intelligently manage exploration vs. exploitation
- **Knowledge representation:** Build internal models (surrogate functions, value functions)

4.2.4. Quick Reference: Is It AI-Based?

AI-Based Methods:

- Bayesian optimization
- Reinforcement learning optimizers
- Neural architecture search with learning
- Meta-learned optimizers
- Learned heuristics for combinatorial problems
- AutoML with transfer learning

NOT AI-Based Methods:

- Gradient descent, SGD, Adam, RMSprop
- Newton's method, BFGS, L-BFGS
- Simulated annealing
- Basic genetic algorithms
- Particle swarm optimization
- Nelder-Mead, Powell's method

4.3. Predict-Then-Optimize

The most common hybrid approach separates prediction from optimization into two stages.

4.3.1. The Two-Stage Framework

$$\text{Data} \xrightarrow{\text{ML Model}} \text{Predictions } \hat{\theta} \xrightarrow{\text{Optimizer}} \text{Decisions } x^*$$

4. The Optimization-ML Continuum

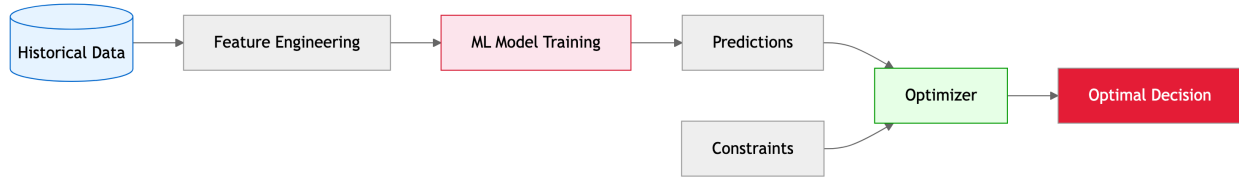


Figure 4.3.: The Predict-then-Optimize pipeline: ML predicts parameters, optimizer finds decisions

Stage 1: Prediction

Train a machine learning model to predict uncertain parameters:

$$\hat{\theta} = f_{ML}(\text{features})$$

Stage 2: Optimization

Solve the optimization problem using predicted parameters:

$$x^* = \arg \min_x c(\hat{\theta})^T x \quad \text{subject to } Ax \leq b$$

4.3.2. Example: Inventory Management

```
import numpy as np
from sklearn.ensemble import GradientBoostingRegressor
from scipy.optimize import linprog

class PredictThenOptimize:
    """Two-stage predict-then-optimize for inventory management."""

    def __init__(self, holding_cost=1.0, shortage_cost=5.0):
        self.holding_cost = holding_cost
        self.shortage_cost = shortage_cost
        self.demand_model = GradientBoostingRegressor()

    def fit_predictor(self, X_train, y_train):
        """Train demand forecasting model."""
        self.demand_model.fit(X_train, y_train)

    def predict_demand(self, X):
        """Predict demand for given features."""
        return self.demand_model.predict(X)

    def optimize_inventory(self, predicted_demand, capacity):
        """Optimize inventory levels given predicted demand."""
        n_products = len(predicted_demand)

        # Objective: minimize expected holding and shortage costs
        # Decision: order quantity for each product
        # Simplified newsvendor-style formulation

        optimal_orders = []
        for demand in predicted_demand:
            # Critical ratio determines optimal service level
```



```

        critical_ratio = self.shortage_cost / (
            self.holding_cost + self.shortage_cost
        )
        # Assuming demand uncertainty, order at critical fractile
        safety_factor = 1.0 + 0.5 * critical_ratio
        order = demand * safety_factor
        optimal_orders.append(min(order, capacity))

    return np.array(optimal_orders)

def solve(self, X, capacity):
    """Complete predict-then-optimize pipeline."""
    predicted_demand = self.predict_demand(X)
    optimal_orders = self.optimize_inventory(predicted_demand, capacity)
    return optimal_orders, predicted_demand

# Usage
pto = PredictThenOptimize(holding_cost=1.0, shortage_cost=5.0)

# Historical data
X_train = np.random.randn(1000, 10) # Features
y_train = 100 + 20 * X_train[:, 0] + 10 * np.random.randn(1000) # Demand

# Fit predictor
pto.fit_predictor(X_train, y_train)

# Solve for new period
X_new = np.random.randn(5, 10) # 5 products
orders, demand_forecast = pto.solve(X_new, capacity=200)
print(f"Predicted demand: {demand_forecast}")
print(f"Optimal orders: {orders}")

```

4.3.3. Limitations of Two-Stage Approaches

1. **Error propagation:** Prediction errors directly impact optimization quality
2. **Misaligned objectives:** ML optimizes prediction accuracy, not decision quality
3. **Lost information:** Point predictions discard uncertainty information
4. **Suboptimality:** Optimal predictions don't guarantee optimal decisions

4.4. End-to-End Learning

End-to-end approaches train the entire pipeline to optimize decision quality directly, rather than intermediate prediction accuracy.

4.4.1. The Concept

Instead of minimizing prediction error, minimize the regret or decision cost:

$$\mathcal{L}_{E2E} = \mathbb{E} [c(x^*(\hat{\theta}), \theta_{true})]$$

where $x^*(\hat{\theta})$ is the decision made using predicted parameters.

4.4.2. Decision-Focused Learning

Train the prediction model to produce forecasts that lead to good decisions:

4. The Optimization-ML Continuum

```
import torch
import torch.nn as nn
from torch.autograd import Function

class DecisionFocusedLoss(nn.Module):
    """Loss function based on decision quality, not prediction accuracy."""

    def __init__(self, optimizer_layer):
        super().__init__()
        self.optimizer = optimizer_layer

    def forward(self, predictions, true_parameters, features):
        """
        Compute loss based on decision quality.

        Args:
            predictions: Predicted parameters from ML model
            true_parameters: Actual realized parameters
            features: Input features

        Returns:
            Decision-focused loss
        """
        # Get decisions using predicted parameters
        decisions = self.optimizer(predictions)

        # Evaluate decisions using true parameters
        # This gives the actual cost incurred
        actual_cost = self.evaluate_decisions(decisions, true_parameters)

        # Compare to optimal decisions (if known)
        optimal_decisions = self.optimizer(true_parameters)
        optimal_cost = self.evaluate_decisions(optimal_decisions, true_parameters)

        # Regret: how much worse were we than optimal?
        regret = actual_cost - optimal_cost

        return regret.mean()

    def evaluate_decisions(self, decisions, parameters):
        """Compute cost of decisions given true parameters."""
        # Problem-specific cost function
        return (decisions * parameters).sum(dim=-1)
```

4.4.3. Differentiable Optimization Layers

To train end-to-end, we need gradients to flow through the optimization step:

$$\frac{\partial x^*}{\partial \theta}$$

This requires making optimization layers differentiable.

Methods for Differentiable Optimization:

Method	Approach	Best For
Implicit Differentiation	KKT conditions	Convex problems
Unrolling	Unroll solver iterations	Iterative algorithms
Perturbation	Smooth approximation	Discrete decisions
CVXPY Layers	Auto-differentiate convex programs	General convex

```

import cvxpy as cp
from cvxpylayers.torch import CvxpyLayer
import torch

def create_differentiable_optimizer(n_vars, n_constraints):
    """Create a differentiable optimization layer using cvxpylayers."""

    # Define CVXPY problem with parameters
    x = cp.Variable(n_vars)
    c = cp.Parameter(n_vars) # Objective coefficients (predicted)
    A = cp.Parameter((n_constraints, n_vars)) # Constraint matrix
    b = cp.Parameter(n_constraints) # Constraint RHS

    # Linear program: min c'x s.t. Ax <= b, x >= 0
    objective = cp.Minimize(c @ x)
    constraints = [A @ x <= b, x >= 0]
    problem = cp.Problem(objective, constraints)

    # Create differentiable layer
    layer = CvxpyLayer(problem, parameters=[c, A, b], variables=[x])

    return layer

class EndToEndModel(nn.Module):
    """End-to-end model combining prediction and optimization."""

    def __init__(self, input_dim, n_vars, n_constraints):
        super().__init__()

        # Prediction network
        self.predictor = nn.Sequential(
            nn.Linear(input_dim, 64),
            nn.ReLU(),
            nn.Linear(64, 32),
            nn.ReLU(),
            nn.Linear(32, n_vars) # Predict cost coefficients
        )

        # Differentiable optimization layer
        self.optimizer = create_differentiable_optimizer(n_vars, n_constraints)

        # Fixed constraint parameters (could also be learned)
        self.register_buffer('A', torch.randn(n_constraints, n_vars))
        self.register_buffer('b', torch.ones(n_constraints))

    def forward(self, features):

```

4. The Optimization-ML Continuum

```
"""Forward pass: predict costs, solve optimization."""
# Predict cost coefficients
predicted_costs = self.predictor(features)

# Solve optimization with predicted costs
batch_size = features.shape[0]
A_batch = self.A.unsqueeze(0).expand(batch_size, -1, -1)
b_batch = self.b.unsqueeze(0).expand(batch_size, -1)

# Get optimal decisions
decisions, = self.optimizer(predicted_costs, A_batch, b_batch)

return decisions, predicted_costs
```

4.5. Bayesian Optimization

Bayesian optimization is a principled approach to AI-based optimization that builds probabilistic surrogate models to guide search intelligently.

4.5.1. Core Concept

Build a probabilistic model of the objective function and use it to make intelligent sampling decisions:

1. **Surrogate model:** Gaussian process learns function behavior from observations
2. **Uncertainty quantification:** Model provides both prediction and confidence
3. **Acquisition function:** Intelligently balances exploration vs. exploitation
4. **Sequential learning:** Each evaluation improves model accuracy

4.5.2. Mathematical Framework

Given: $f(x)$ expensive to evaluate

1. Build surrogate: $p(f|D)$ using Gaussian process
2. Optimize acquisition: $x_{next} = \arg \max \alpha(x|p(f|D))$
3. Evaluate: $y = f(x_{next})$, update $D \leftarrow D \cup \{(x_{next}, y)\}$

4.5.3. Acquisition Functions

Function	Formula	Behavior
Expected Improvement (EI)	$\mathbb{E}[\max(f(x) - f^*, 0)]$	Balanced exploration-exploitation
Upper Confidence Bound (UCB)	$\mu(x) + \beta \cdot \sigma(x)$	Optimistic exploration
Probability of Improvement (PI)	$P(f(x) > f^*)$	Conservative
Knowledge Gradient	$\mathbb{E}[\text{improvement after observing } x]$	Information-theoretic

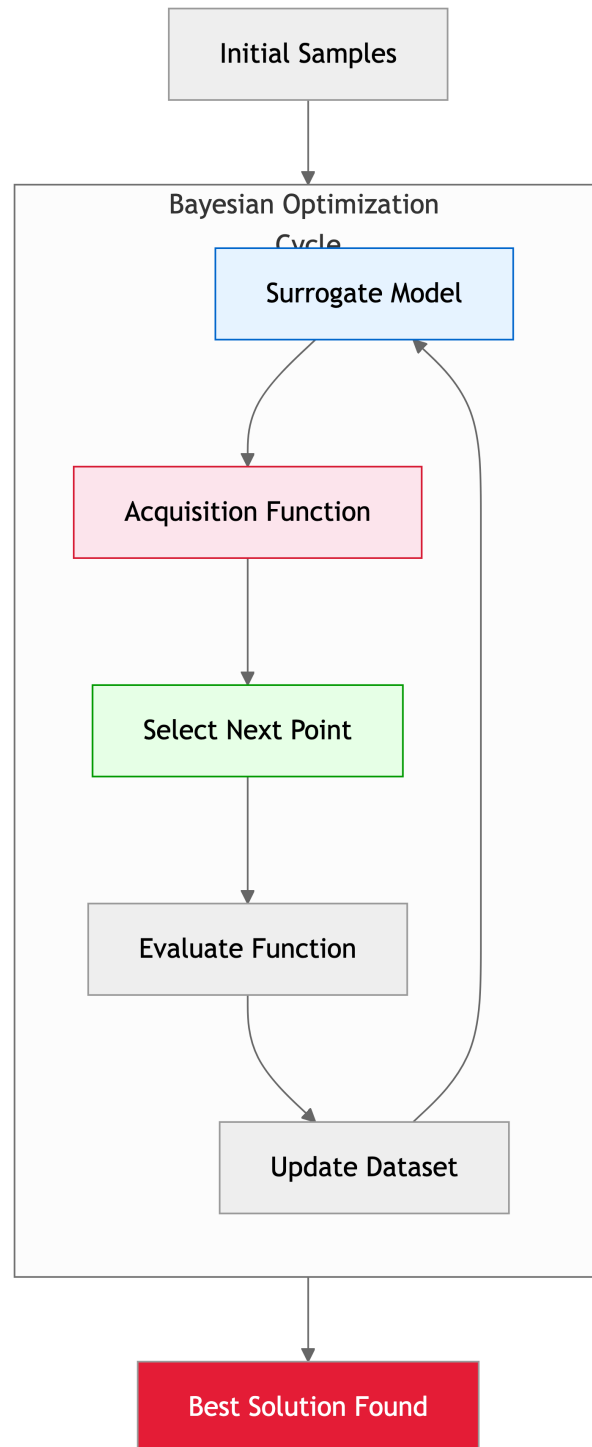


Figure 4.4.: Bayesian Optimization Loop: Surrogate model guides intelligent sampling

4.5.4. Implementation with scikit-optimize

```

from skopt import gp_minimize
from skopt.space import Real, Integer, Categorical
from skopt.utils import use_named_args
import numpy as np

def bayesian_hyperparameter_tuning(objective_function, search_space, n_calls=50):
    """
    Bayesian optimization for hyperparameter tuning.

    Args:
        objective_function: Function to minimize (returns validation loss)
        search_space: List of skopt dimension objects
        n_calls: Number of function evaluations

    Returns:
        Optimization result with best parameters
    """
    result = gp_minimize(
        objective_function,
        search_space,
        n_calls=n_calls,
        n_random_starts=10, # Initial random exploration
        acq_func='EI', # Expected Improvement
        random_state=42
    )

    return result

# Example: Tune neural network hyperparameters
search_space = [
    Real(1e-4, 1e-1, name='learning_rate', prior='log-uniform'),
    Integer(16, 128, name='batch_size'),
    Real(0.0, 0.5, name='dropout'),
    Categorical(['relu', 'tanh', 'elu'], name='activation'),
    Integer(64, 512, name='hidden_units')
]

# Simulated objective (replace with actual model training)
@use_named_args(search_space)
def objective(**params):
    # In practice: train model and return validation loss
    val_loss = (
        0.1 * np.log(params['learning_rate']) +
        0.01 * params['batch_size'] +
        params['dropout'] ** 2 +
        np.random.randn() * 0.01
    )
    return val_loss

result = bayesian_hyperparameter_tuning(objective, search_space, n_calls=30)
print(f"Best validation loss: {result.fun:.4f}")
print(f"Best parameters: {result.x}")

```

4.5.5. Applications

Domain	Use Case
Hyperparameter Tuning	Optimize ML model hyperparameters
Experimental Design	Optimize chemical reactions, materials
AutoML	Automated pipeline optimization
A/B Testing	Intelligent sequential experimentation
Robotics Control	Sample-efficient controller tuning
Drug Discovery	Optimize molecular structures

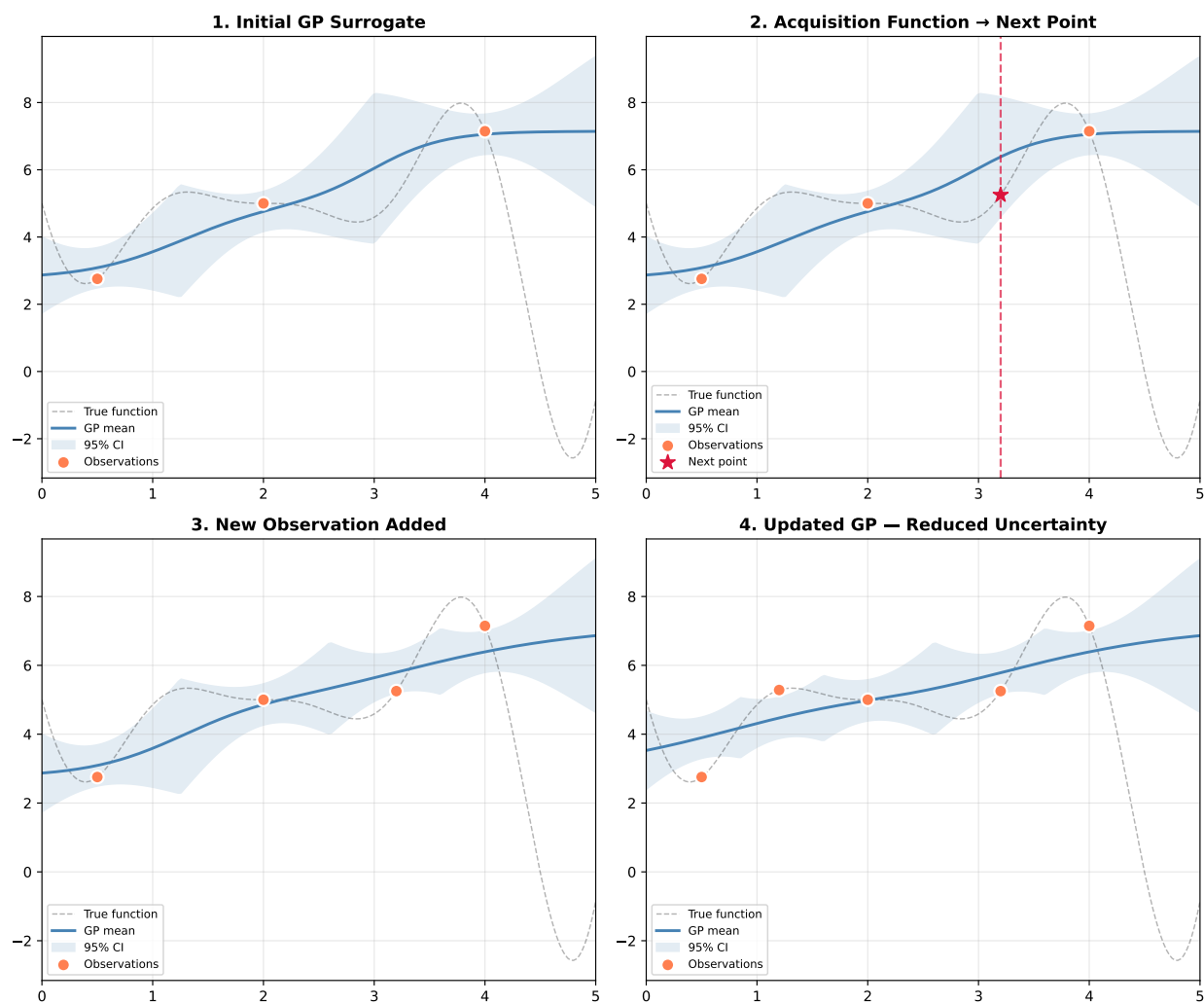


Figure 4.5.: Bayesian Optimization in 4 steps — (1) initial GP surrogate, (2) acquisition function identifies next point, (3) new observation, (4) updated GP with reduced uncertainty

4.6. Neural Combinatorial Optimization

Neural Combinatorial Optimization (NCO) uses deep neural networks to learn heuristics for solving combinatorial optimization problems, eliminating the need for hand-crafted algorithms.

4. The Optimization-ML Continuum

4.6.1. The Core Idea

Instead of designing problem-specific algorithms, NCO trains a neural network to output solutions by treating optimization as a sequential decision-making problem:

$$p_{\theta}(\pi|s) = \prod_{t=1}^n p_{\theta}(\pi_t|\pi_{1:t-1}, s)$$

where s is the problem instance and π is the solution (e.g., a tour for TSP).

4.6.2. Why NCO?

- **No expert knowledge required:** Learn heuristics directly from data
- **Generalization:** Train once, solve many instances of the same problem type
- **Fast inference:** Once trained, generate solutions in milliseconds
- **Adaptability:** Same architecture works for TSP, VRP, and other routing problems
- **End-to-end learning:** Jointly optimize all components using gradient descent

4.6.3. Evolution of NCO Methods

Year	Method	Key Innovation
2015	Pointer Networks	Attention as pointer to input elements
2016	Bello et al. NCO	Policy gradient training with tour length reward
2017	S2V-DQN	Graph embedding with Q-learning
2019	Attention Model	Transformer encoder-decoder architecture
2020	POMO	Multiple starting points exploiting symmetry
2023	LEHD	Light encoder, heavy decoder for scalability

4.6.4. MDP Formulation

To apply reinforcement learning to NCO, model the problem as a Markov Decision Process:

Component	Definition	Example (TSP)
State s_t	Current partial solution	Visited cities so far
Action a_t	Next element to add	Next city to visit
Reward R	Solution quality	Negative tour length
Transition P	Deterministic for CO	Adding city to tour

4.6.5. Architecture: Attention Model

```
import torch
import torch.nn as nn
import torch.nn.functional as F
import math

class MultiHeadAttention(nn.Module):
    """Multi-head attention for graph encoding."""

    def __init__(self, embed_dim, num_heads):
        super().__init__()
        self.embed_dim = embed_dim
        self.num_heads = num_heads
```



```

self.head_dim = embed_dim // num_heads

self.W_q = nn.Linear(embed_dim, embed_dim)
self.W_k = nn.Linear(embed_dim, embed_dim)
self.W_v = nn.Linear(embed_dim, embed_dim)
self.W_o = nn.Linear(embed_dim, embed_dim)

def forward(self, q, k, v, mask=None):
    batch_size = q.size(0)

    # Linear projections
    Q = self.W_q(q).view(batch_size, -1, self.num_heads, self.head_dim)
    K = self.W_k(k).view(batch_size, -1, self.num_heads, self.head_dim)
    V = self.W_v(v).view(batch_size, -1, self.num_heads, self.head_dim)

    # Transpose for attention: (batch, heads, seq, dim)
    Q = Q.transpose(1, 2)
    K = K.transpose(1, 2)
    V = V.transpose(1, 2)

    # Scaled dot-product attention
    scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.head_dim)

    if mask is not None:
        scores = scores.masked_fill(mask.unsqueeze(1), float('-inf'))

    attn = F.softmax(scores, dim=-1)
    context = torch.matmul(attn, V)

    # Concatenate heads
    context = context.transpose(1, 2).contiguous()
    context = context.view(batch_size, -1, self.embed_dim)

    return self.W_o(context)

class TSPEncoder(nn.Module):
    """Transformer encoder for TSP instances."""

    def __init__(self, input_dim=2, embed_dim=128, num_heads=8, num_layers=3):
        super().__init__()
        self.embed = nn.Linear(input_dim, embed_dim)
        self.layers = nn.ModuleList([
            nn.ModuleDict({
                'attention': MultiHeadAttention(embed_dim, num_heads),
                'norm1': nn.LayerNorm(embed_dim),
                'ffn': nn.Sequential(
                    nn.Linear(embed_dim, embed_dim * 4),
                    nn.ReLU(),
                    nn.Linear(embed_dim * 4, embed_dim)
                ),
                'norm2': nn.LayerNorm(embed_dim)
            })
        ])

```

```

        for _ in range(num_layers)
    ])

def forward(self, cities):
    """
    Encode city coordinates.

    Args:
        cities: (batch, n_cities, 2) tensor of coordinates

    Returns:
        (batch, n_cities, embed_dim) tensor of embeddings
    """
    h = self.embed(cities)

    for layer in self.layers:
        # Self-attention with residual
        attn_out = layer['attention'](h, h, h)
        h = layer['norm1'](h + attn_out)

        # Feed-forward with residual
        ff_n_out = layer['ffn'](h)
        h = layer['norm2'](h + ff_n_out)

    return h


class TSPDecoder(nn.Module):
    """Autoregressive decoder for TSP solutions."""

    def __init__(self, embed_dim=128, num_heads=8):
        super().__init__()
        self.embed_dim = embed_dim

        # Context embedding: first city, last city, graph mean
        self.context_embed = nn.Linear(embed_dim * 3, embed_dim)

        # Attention for selecting next city
        self.attention = MultiHeadAttention(embed_dim, num_heads)

        # Final projection to logits
        self.project = nn.Linear(embed_dim, embed_dim)

    def forward(self, embeddings, context_idx, mask):
        """
        Compute probabilities for next city.

        Args:
            embeddings: (batch, n_cities, embed_dim) from encoder
            context_idx: Indices of first and last visited cities
            mask: (batch, n_cities) True for visited cities

        Returns:

```

```

        (batch, n_cities) log probabilities for each city
    """
    batch_size, n_cities, _ = embeddings.shape

    # Build context vector
    first_idx, last_idx = context_idx
    graph_mean = embeddings.mean(dim=1)

    first_embed = embeddings[torch.arange(batch_size), first_idx]
    last_embed = embeddings[torch.arange(batch_size), last_idx]

    context = torch.cat([first_embed, last_embed, graph_mean], dim=-1)
    context = self.context_embed(context).unsqueeze(1)

    # Attention over unvisited cities
    query = self.project(context)

    # Compute attention scores
    scores = torch.matmul(query, embeddings.transpose(-2, -1))
    scores = scores.squeeze(1) / math.sqrt(self.embed_dim)

    # Mask visited cities
    scores = scores.masked_fill(mask, float('-inf'))

    return F.log_softmax(scores, dim=-1)

class AttentionTSP(nn.Module):
    """Complete Attention Model for TSP."""

    def __init__(self, input_dim=2, embed_dim=128, num_heads=8, num_layers=3):
        super().__init__()
        self.encoder = TSPEncoder(input_dim, embed_dim, num_heads, num_layers)
        self.decoder = TSPDecoder(embed_dim, num_heads)

    def forward(self, cities, greedy=False):
        """
        Generate tour for TSP instance.

        Args:
            cities: (batch, n_cities, 2) tensor of coordinates
            greedy: If True, use greedy decoding; else sample

        Returns:
            tours: (batch, n_cities) tensor of city indices
            log_probs: (batch,) tensor of log probabilities
        """
        batch_size, n_cities, _ = cities.shape

        # Encode all cities
        embeddings = self.encoder(cities)

        # Initialize tour

```

4. The Optimization-ML Continuum

```
tours = []
log_probs = torch.zeros(batch_size, device=cities.device)
mask = torch.zeros(batch_size, n_cities, dtype=torch.bool,
↪ device=cities.device)

# Start from first city (or could randomize)
current = torch.zeros(batch_size, dtype=torch.long, device=cities.device)
first = current.clone()
tours.append(current)
mask[torch.arange(batch_size), current] = True

# Decode remaining cities
for _ in range(n_cities - 1):
    # Get probabilities for next city
    probs = self.decoder(embeddings, (first, current), mask)

    if greedy:
        current = probs.argmax(dim=-1)
    else:
        current = torch.distributions.Categorical(logits=probs).sample()

    log_probs += probs[torch.arange(batch_size), current]
    tours.append(current)
    mask[torch.arange(batch_size), current] = True

tours = torch.stack(tours, dim=1)
return tours, log_probs

def get_tour_length(self, cities, tours):
    """Compute tour length for given tours."""
    batch_size = cities.shape[0]

    # Gather cities in tour order
    idx = tours.unsqueeze(-1).expand(-1, -1, 2)
    ordered = cities.gather(1, idx)

    # Compute distances
    rolled = torch.roll(ordered, -1, dims=1)
    distances = torch.norm(ordered - rolled, dim=-1)

    return distances.sum(dim=1)
```

4.6.6. Training with REINFORCE

```
def train_attention_tsp(model, n_cities=20, batch_size=512, n_epochs=100):
    """Train Attention Model using REINFORCE with greedy baseline."""

    optimizer = torch.optim.Adam(model.parameters(), lr=1e-4)

    # Baseline model (copy of current model, updated periodically)
    baseline_model = AttentionTSP()
    baseline_model.load_state_dict(model.state_dict())
```

```

for epoch in range(n_epochs):
    # Generate random TSP instances
    cities = torch.rand(batch_size, n_cities, 2)

    # Sample tours from current policy
    tours, log_probs = model(cities, greedy=False)
    tour_lengths = model.get_tour_length(cities, tours)

    # Get baseline (greedy rollout)
    with torch.no_grad():
        baseline_tours, _ = baseline_model(cities, greedy=True)
        baseline_lengths = model.get_tour_length(cities, baseline_tours)

    # REINFORCE loss with baseline
    advantage = tour_lengths - baseline_lengths
    loss = (advantage * log_probs).mean()

    # Update policy
    optimizer.zero_grad()
    loss.backward()
    optimizer.step()

    # Periodically update baseline
    if epoch % 10 == 0:
        # Update baseline if current model is better
        if tour_lengths.mean() < baseline_lengths.mean():
            baseline_model.load_state_dict(model.state_dict())

    if epoch % 10 == 0:
        print(f"Epoch {epoch}: Tour length = {tour_lengths.mean():.4f}")

```

💡 POMO: Exploiting Solution Symmetry

Policy Optimization with Multiple Optima (POMO) significantly improves NCO training by exploiting solution symmetry. For TSP, any rotation of an optimal tour is also optimal. POMO samples N different starting cities and uses their average reward as a low-variance baseline, achieving 0.14% gap from optimal on TSP-100.

4.7. Hybrid Approaches

The most effective solutions often combine optimization and learning.

4.7.1. Learning to Branch

Neural networks guide branching decisions in branch-and-bound solvers:

```

class LearnedBranchingPolicy(nn.Module):
    """Neural network for variable selection in branch-and-bound."""

    def __init__(self, n_features):
        super().__init__()
        self.network = nn.Sequential(

```

```

        nn.Linear(n_features, 64),
        nn.ReLU(),
        nn.Linear(64, 32),
        nn.ReLU(),
        nn.Linear(32, 1) # Score for each candidate variable
    )

def forward(self, candidate_features):
    """
    Score branching candidates.

    Args:
        candidate_features: (n_candidates, n_features) tensor

    Returns:
        (n_candidates,) tensor of scores
    """
    return self.network(candidate_features).squeeze(-1)

def select_variable(self, candidates, features):
    """Select branching variable with highest score."""
    scores = self.forward(features)
    return candidates[scores.argmax()]

```

4.7.2. Large Neighborhood Search with Learning

RL selects which parts of solutions to destroy and repair:

```

class LearnedLNS:
    """Large Neighborhood Search with learned destroy/repair operators."""

    def __init__(self, problem, destroy_operators, repair_operators):
        self.problem = problem
        self.destroy_ops = destroy_operators
        self.repair_ops = repair_operators

        # Q-network for operator selection
        self.q_network = nn.Sequential(
            nn.Linear(problem.state_dim, 64),
            nn.ReLU(),
            nn.Linear(64, len(destroy_operators) * len(repair_operators))
        )

    def select_operators(self, state, epsilon=0.1):
        """Select destroy-repair operator pair using epsilon-greedy."""
        if np.random.random() < epsilon:
            d_idx = np.random.randint(len(self.destroy_ops))
            r_idx = np.random.randint(len(self.repair_ops))
        else:
            q_values = self.q_network(torch.FloatTensor(state))
            idx = q_values.argmax().item()
            d_idx = idx // len(self.repair_ops)
            r_idx = idx % len(self.repair_ops)

```

```

    return self.destroy_ops[d_idx], self.repair_ops[r_idx]

def solve(self, initial_solution, max_iterations=1000):
    """Run learned LNS."""
    current = initial_solution
    best = current
    best_cost = self.problem.evaluate(current)

    for _ in range(max_iterations):
        state = self.problem.get_state(current)
        destroy_op, repair_op = self.select_operators(state)

        # Apply operators
        partial = destroy_op(current)
        new_solution = repair_op(partial)

        new_cost = self.problem.evaluate(new_solution)
        if new_cost < best_cost:
            best = new_solution
            best_cost = new_cost
            current = new_solution

    return best, best_cost

```

4.7.3. Comparison: End-to-End vs. Hybrid

Aspect	End-to-End	Hybrid
Training	Fully differentiable	May need RL for discrete choices
Interpretability	Lower	Higher (leverages known algorithms)
Optimality	No guarantees	Can preserve solver guarantees
Flexibility	Problem-specific	Reuses solver engineering
Performance	Fast inference	May be slower but more robust

4.8. Method Selection Guide

4.8.1. Decision Framework

Is the objective function expensive to evaluate?

Yes → Consider Bayesian Optimization

Sample-efficient, builds surrogate model

No → Are you solving many similar instances?

Yes → Consider Neural Combinatorial Optimization

Train once, fast inference on new instances

No → Are there hard constraints?

Yes → Use Mathematical Optimization

LP, IP, CP with appropriate solver

No → Use Standard ML

Gradient-based optimization

4.8.2. Summary Table

4. The Optimization-ML Continuum

Method	Training Cost	Inference	Guarantees	Best Application
Bayesian Opt	Low	Slow	Convergence	Hyperparameter tuning
NCO	High	Fast	None	Routing, scheduling
Predict-then-Opt	Medium	Medium	Problem-specific	Stochastic optimization
End-to-End	High	Fast	None	Decision-focused ML
Hybrid LNS	Medium	Medium	Problem-specific	Large-scale CO

4.9. Summary

4.9.1. Key Takeaways

- The optimization-ML continuum offers multiple integration strategies
- AI-based optimization requires learning from experience, not just adaptation
- Predict-then-optimize separates forecasting from decision-making
- End-to-end learning optimizes for decision quality directly
- Bayesian optimization builds intelligent surrogate models for expensive functions
- Neural combinatorial optimization learns heuristics for NP-hard problems
- Hybrid approaches combine the strengths of both paradigms
- Method selection depends on evaluation cost, problem structure, and constraints

4.10. Exercises

1. **Bayesian optimization:** Implement a simple Bayesian optimizer using a Gaussian process surrogate and Expected Improvement acquisition function. Test it on the Rosenbrock function.
2. **Predict-then-optimize analysis:** For an inventory management problem, compare the performance of (a) minimizing MSE in demand prediction vs. (b) minimizing regret in the final inventory decisions. What do you observe?
3. **NCO implementation:** Extend the Attention Model implementation to handle the Vehicle Routing Problem with capacity constraints. What modifications are needed to the masking mechanism?
4. **Method comparison:** For a hyperparameter tuning task, compare random search, grid search, and Bayesian optimization. Plot the convergence curves and analyze sample efficiency.
5. **Hybrid design:** Design a hybrid system that uses ML to predict constraint violation probabilities and optimization to find the best solution considering these predictions. Apply it to a resource allocation problem.

5. Data Strategy for Decision Intelligence

What You Will Learn

- Design data architectures that support optimization and decision systems
- Engineer features that capture decision-relevant information
- Build robust data pipelines for batch and real-time decision making
- Handle data quality issues that affect optimization outcomes
- Implement feedback loops for continuous model improvement

Data is the foundation of decision intelligence. No optimization model can produce good decisions from bad data. This chapter covers the data engineering practices essential for building production-ready decision systems.

5.1. The Data-Decision Gap

Many organizations have mature data infrastructure for analytics and reporting, yet struggle to operationalize that data for automated decisions. The requirements are fundamentally different.

Aspect	Analytics	Decision Intelligence
Latency	Hours/days acceptable	Seconds/minutes critical
Completeness	Partial data workable	Missing data = infeasible
Accuracy	Directionally correct	Precise values needed
Freshness	Point-in-time snapshots	Current state required
Schema	Flexible, evolving	Strict contracts

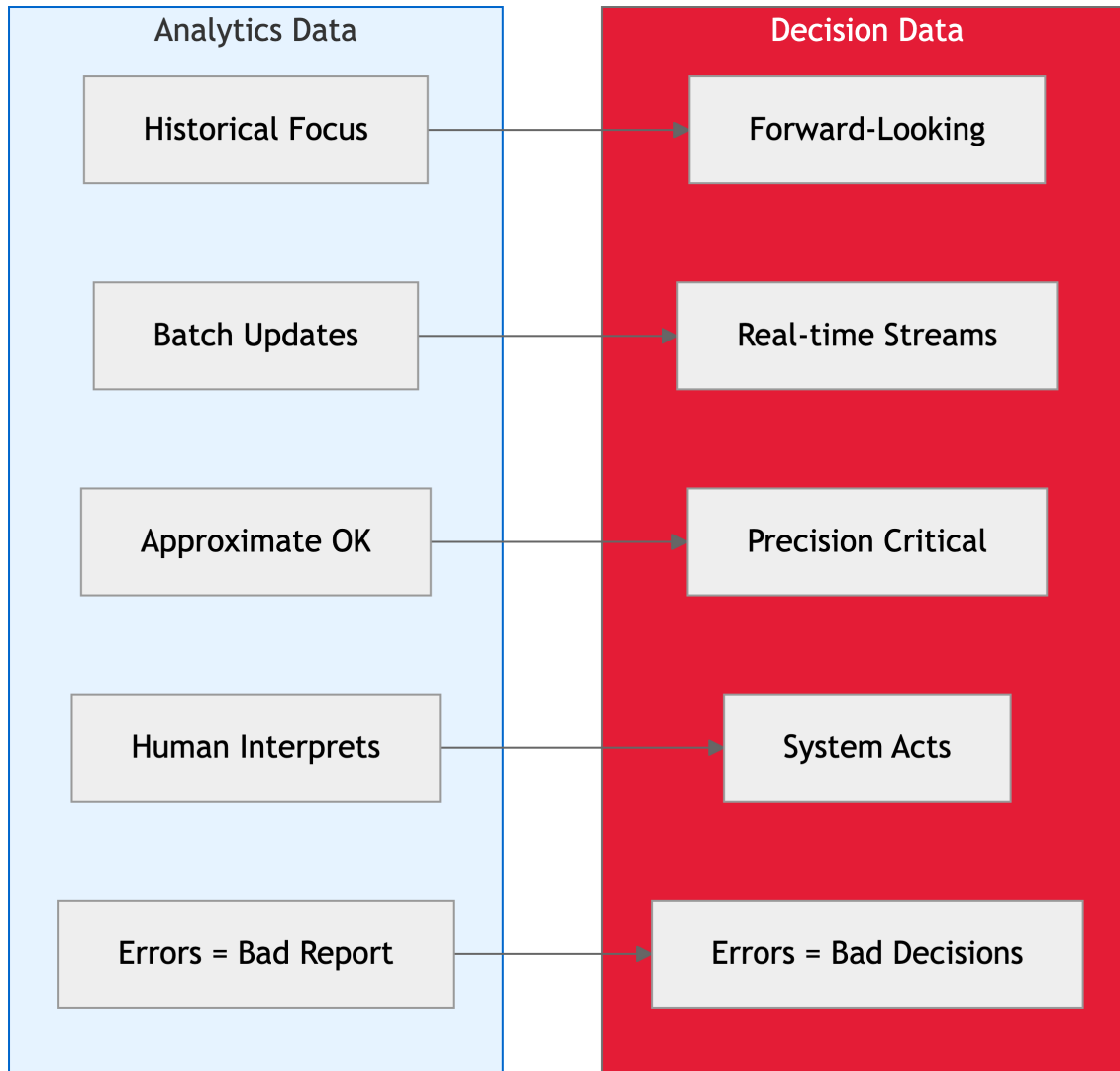
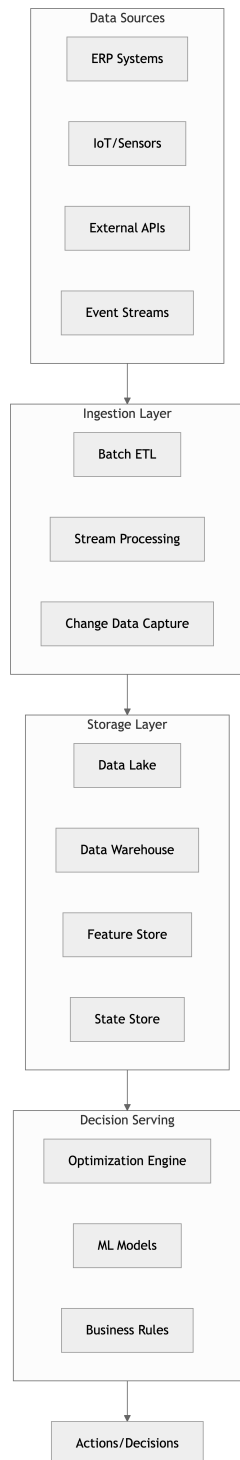


Figure 5.1.: The Data-Decision Gap: Analytics vs Decision Data Requirements

5.2. Data Architecture for DI

5.2.1. Reference Architecture



5.2.2. Key Components

Feature Store: Centralized repository for decision-relevant features with versioning, lineage, and serving capabilities.

5. Data Strategy for Decision Intelligence

State Store: Low-latency storage for current system state (inventory levels, vehicle locations, resource availability).

Event Bus: Real-time stream of business events that trigger or inform decisions.

```
from dataclasses import dataclass, field
from typing import Dict, List, Any, Optional
from datetime import datetime, timedelta
from abc import ABC, abstractmethod
import json

@dataclass
class DataSource:
    """Metadata for a data source."""
    name: str
    source_type: str # 'database', 'api', 'stream', 'file'
    refresh_frequency: timedelta
    latency_sla: timedelta
    schema_version: str
    owner: str
    quality_score: float = 1.0

@dataclass
class Feature:
    """A feature used in decision models."""
    name: str
    description: str
    data_type: str
    source: str
    computation: str # SQL, Python, etc.
    freshness_requirement: timedelta
    default_value: Any = None

class FeatureStore:
    """Centralized feature management for decision systems."""

    def __init__(self):
        self.features: Dict[str, Feature] = {}
        self.feature_values: Dict[str, Dict[str, Any]] = {}
        self.computation_cache: Dict[str, datetime] = {}

    def register_feature(self, feature: Feature):
        """Register a feature definition."""
        self.features[feature.name] = feature

    def get_feature(self, feature_name: str, entity_id: str) -> Any:
        """Get feature value for an entity."""
        key = f"{feature_name}:{entity_id}"

        if key in self.feature_values:
            cached = self.feature_values[key]
            feature = self.features[feature_name]

            # Check freshness
```

```

        if datetime.now() - cached['timestamp'] < feature.freshness_requirement:
            return cached['value']

    # Compute fresh value
    return self._compute_feature(feature_name, entity_id)

    def get_feature_vector(self, feature_names: List[str],
                          entity_id: str) -> Dict[str, Any]:
        """Get multiple features for an entity."""
        return {
            name: self.get_feature(name, entity_id)
            for name in feature_names
        }

    def _compute_feature(self, feature_name: str, entity_id: str) -> Any:
        """Compute feature value (placeholder)."""
        feature = self.features.get(feature_name)
        if not feature:
            raise ValueError(f"Unknown feature: {feature_name}")
        return feature.default_value

```

5.3. Feature Engineering for Decisions

Feature engineering for optimization differs from ML feature engineering. The goal is not prediction accuracy but decision quality.

5.3.1. Decision-Relevant Features

Feature Type	Purpose	Examples
State Features	Current system state	Inventory levels, queue lengths
Demand Features	Expected demand	Forecasts, seasonality indices
Capacity Features	Available resources	Machine availability, workforce
Cost Features	Decision economics	Unit costs, penalties
Constraint Features	Feasibility boundaries	Min/max limits, regulations

5.3.2. Temporal Feature Engineering

```

import numpy as np
from typing import Tuple

class TemporalFeatureEngineer:
    """Engineer time-based features for decisions."""

    def __init__(self):
        self.seasonality_patterns: Dict[str, np.ndarray] = {}

    def extract_calendar_features(self, timestamp: datetime) -> Dict[str, Any]:
        """Extract calendar-based features."""
        return {
            'hour_of_day': timestamp.hour,
            'day_of_week': timestamp.weekday(),
            'day_of_month': timestamp.day,

```

```

        'month': timestamp.month,
        'quarter': (timestamp.month - 1) // 3 + 1,
        'is_weekend': timestamp.weekday() >= 5,
        'is_month_start': timestamp.day <= 3,
        'is_month_end': timestamp.day >= 28,
    }

def extract_lag_features(self, series: np.ndarray,
                        lags: List[int]) -> Dict[str, float]:
    """Extract lagged values as features."""
    features = {}
    for lag in lags:
        if len(series) > lag:
            features[f'lag_{lag}'] = series[-lag]
        else:
            features[f'lag_{lag}'] = np.nan
    return features

def extract_rolling_features(self, series: np.ndarray,
                            windows: List[int]) -> Dict[str, float]:
    """Extract rolling statistics."""
    features = {}
    for window in windows:
        if len(series) >= window:
            window_data = series[-window:]
            features[f'rolling_mean_{window}'] = np.mean(window_data)
            features[f'rolling_std_{window}'] = np.std(window_data)
            features[f'rolling_min_{window}'] = np.min(window_data)
            features[f'rolling_max_{window}'] = np.max(window_data)
    return features

def extract_trend_features(self, series: np.ndarray,
                          window: int = 7) -> Dict[str, float]:
    """Extract trend indicators."""
    if len(series) < window:
        return {'trend_slope': 0.0, 'trend_direction': 0}

    recent = series[-window:]
    x = np.arange(window)

    # Linear regression slope
    slope = np.polyfit(x, recent, 1)[0]

    return {
        'trend_slope': slope,
        'trend_direction': 1 if slope > 0 else (-1 if slope < 0 else 0),
        'trend_strength': abs(slope) / (np.std(recent) + 1e-6)
    }

def encode_cyclical(self, value: float, period: float) -> Tuple[float, float]:
    """Encode cyclical features using sin/cos."""
    angle = 2 * np.pi * value / period
    return np.sin(angle), np.cos(angle)

```

5.3.3. Demand Feature Engineering

```

class DemandFeatureEngineer:
    """Engineer demand-related features for inventory/routing decisions."""

    def __init__(self):
        self.temporal_engineer = TemporalFeatureEngineer()

    def compute_demand_features(self, demand_history: np.ndarray,
                               forecast: np.ndarray,
                               timestamp: datetime) -> Dict[str, Any]:
        """Compute comprehensive demand features."""
        features = {}

        # Historical statistics
        features['demand_mean'] = np.mean(demand_history)
        features['demand_std'] = np.std(demand_history)
        features['demand_cv'] = features['demand_std'] / (features['demand_mean'] +
↪ 1e-6)

        # Forecast features
        features['forecast_next'] = forecast[0] if len(forecast) > 0 else 0
        features['forecast_sum_7d'] = np.sum(forecast[:7]) if len(forecast) >= 7
↪ else np.sum(forecast)

        # Volatility
        if len(demand_history) > 1:
            returns = np.diff(demand_history) / (demand_history[:-1] + 1e-6)
            features['demand_volatility'] = np.std(returns)

        # Seasonality strength
        features.update(self._compute_seasonality_strength(demand_history))

        # Calendar features
        features.update(self.temporal_engineer.extract_calendar_features(timestamp))

        return features

    def _compute_seasonality_strength(self, series: np.ndarray,
                                     period: int = 7) -> Dict[str, float]:
        """Compute strength of seasonal pattern."""
        if len(series) < 2 * period:
            return {'seasonality_strength': 0.0}

        # Simple autocorrelation at seasonal lag
        n = len(series)
        mean = np.mean(series)
        var = np.var(series)

        if var < 1e-6:
            return {'seasonality_strength': 0.0}

        autocorr = np.sum((series[period:] - mean) * (series[:-period] - mean)) / (n
↪ * var)

```

```
        return {'seasonality_strength': max(0, autocorr)}

    def compute_safety_stock_features(self, demand_history: np.ndarray,
                                     lead_time: float,
                                     service_level: float = 0.95) -> Dict[str,
↳ float]:
        """Compute safety stock related features."""
        from scipy import stats

        demand_mean = np.mean(demand_history)
        demand_std = np.std(demand_history)

        # Z-score for service level
        z = stats.norm.ppf(service_level)

        # Safety stock (assuming constant lead time)
        safety_stock = z * demand_std * np.sqrt(lead_time)

        # Reorder point
        reorder_point = demand_mean * lead_time + safety_stock

        return {
            'safety_stock': safety_stock,
            'reorder_point': reorder_point,
            'lead_time_demand': demand_mean * lead_time,
            'demand_during_lead_time_std': demand_std * np.sqrt(lead_time)
        }
```

5.4. Data Pipelines for Decision Systems

5.4.1. Batch Pipeline Architecture

```
from dataclasses import dataclass
from typing import Callable, List
from enum import Enum
import logging

class PipelineStatus(Enum):
    PENDING = "pending"
    RUNNING = "running"
    SUCCESS = "success"
    FAILED = "failed"

@dataclass
class PipelineStep:
    """A step in the data pipeline."""
    name: str
    transform: Callable
    input_tables: List[str]
    output_table: str
    dependencies: List[str] = field(default_factory=list)
```



```

class BatchDataPipeline:
    """Orchestrates batch data transformations for decision systems."""

    def __init__(self, name: str):
        self.name = name
        self.steps: Dict[str, PipelineStep] = {}
        self.step_status: Dict[str, PipelineStatus] = {}
        self.logger = logging.getLogger(f"Pipeline.{name}")

    def add_step(self, step: PipelineStep):
        """Add a step to the pipeline."""
        self.steps[step.name] = step
        self.step_status[step.name] = PipelineStatus.PENDING

    def run(self, data_context: Dict[str, Any]) -> Dict[str, Any]:
        """Execute the pipeline."""
        execution_order = self._topological_sort()

        for step_name in execution_order:
            step = self.steps[step_name]

            # Check dependencies
            for dep in step.dependencies:
                if self.step_status.get(dep) != PipelineStatus.SUCCESS:
                    self.step_status[step_name] = PipelineStatus.FAILED
                    raise RuntimeError(f"Dependency {dep} not satisfied for
                                   ↳ {step_name}")

            # Execute step
            self.step_status[step_name] = PipelineStatus.RUNNING
            self.logger.info(f"Running step: {step_name}")

            try:
                inputs = {table: data_context[table] for table in step.input_tables}
                result = step.transform(inputs)
                data_context[step.output_table] = result
                self.step_status[step_name] = PipelineStatus.SUCCESS

            except Exception as e:
                self.step_status[step_name] = PipelineStatus.FAILED
                self.logger.error(f"Step {step_name} failed: {e}")
                raise

        return data_context

    def _topological_sort(self) -> List[str]:
        """Sort steps by dependencies."""
        visited = set()
        order = []

        def visit(name: str):
            if name in visited:
                return

```

```
        visited.add(name)

        step = self.steps[name]
        for dep in step.dependencies:
            visit(dep)

        order.append(name)

    for name in self.steps:
        visit(name)

    return order
```

5.4.2. Real-Time Data Handling

```
from collections import deque
from threading import Lock
import time

@dataclass
class StreamEvent:
    """An event in the data stream."""
    event_type: str
    entity_id: str
    timestamp: datetime
    payload: Dict[str, Any]

class RealTimeDataHandler:
    """Handles real-time data for decision systems."""

    def __init__(self, buffer_size: int = 10000,
                  window_seconds: int = 300):
        self.buffer_size = buffer_size
        self.window_seconds = window_seconds

        self.event_buffer: deque = deque(maxlen=buffer_size)
        self.state_cache: Dict[str, Any] = {}
        self.aggregates: Dict[str, Dict] = {}
        self.lock = Lock()

        self.handlers: Dict[str, List[Callable]] = {}

    def register_handler(self, event_type: str, handler: Callable):
        """Register handler for event type."""
        if event_type not in self.handlers:
            self.handlers[event_type] = []
        self.handlers[event_type].append(handler)

    def process_event(self, event: StreamEvent):
        """Process incoming event."""
        with self.lock:
            # Add to buffer
            self.event_buffer.append(event)
```

```

        # Update state cache
        self._update_state(event)

        # Update aggregates
        self._update_aggregates(event)

    # Trigger handlers
    for handler in self.handlers.get(event.event_type, []):
        handler(event)

def _update_state(self, event: StreamEvent):
    """Update current state based on event."""
    key = f"{event.event_type}:{event.entity_id}"
    self.state_cache[key] = {
        'last_value': event.payload,
        'last_update': event.timestamp
    }

def _update_aggregates(self, event: StreamEvent):
    """Update rolling aggregates."""
    key = event.event_type
    if key not in self.aggregates:
        self.aggregates[key] = {
            'count': 0,
            'sum': 0,
            'values': deque(maxlen=1000)
        }

    agg = self.aggregates[key]
    value = event.payload.get('value', 0)

    agg['count'] += 1
    agg['sum'] += value
    agg['values'].append((event.timestamp, value))

    # Clean old values
    cutoff = datetime.now() - timedelta(seconds=self.window_seconds)
    while agg['values'] and agg['values'][0][0] < cutoff:
        old_ts, old_val = agg['values'].popleft()
        agg['sum'] -= old_val
        agg['count'] -= 1

def get_current_state(self, event_type: str, entity_id: str) -> Optional[Dict]:
    """Get current state for an entity."""
    key = f"{event_type}:{entity_id}"
    return self.state_cache.get(key)

def get_window_aggregate(self, event_type: str) -> Dict[str, float]:
    """Get aggregates for recent window."""
    agg = self.aggregates.get(event_type, {})
    count = agg.get('count', 0)

```

```

    if count == 0:
        return {'count': 0, 'mean': 0, 'sum': 0}

    values = [v for _, v in agg.get('values', [])]

    return {
        'count': count,
        'sum': agg.get('sum', 0),
        'mean': agg.get('sum', 0) / count,
        'min': min(values) if values else 0,
        'max': max(values) if values else 0
    }

class StateManager:
    """Manages current system state for real-time decisions."""

    def __init__(self):
        self.state: Dict[str, Dict[str, Any]] = {}
        self.state_history: Dict[str, deque] = {}
        self.lock = Lock()

    def update_state(self, domain: str, entity_id: str,
                    state: Dict[str, Any]):
        """Update state for an entity."""
        with self.lock:
            key = f"{domain}:{entity_id}"

            # Store current in history
            if key in self.state:
                if key not in self.state_history:
                    self.state_history[key] = deque(maxlen=100)
                self.state_history[key].append({
                    'timestamp': datetime.now(),
                    'state': self.state[key].copy()
                })

            # Update current
            self.state[key] = {
                **state,
                '_updated_at': datetime.now()
            }

    def get_state(self, domain: str, entity_id: str) -> Optional[Dict]:
        """Get current state for an entity."""
        key = f"{domain}:{entity_id}"
        return self.state.get(key)

    def get_all_states(self, domain: str) -> Dict[str, Dict]:
        """Get all states for a domain."""
        prefix = f"{domain}:"
        return {
            k.replace(prefix, ''): v
            for k, v in self.state.items()
        }

```

```

        if k.startswith(prefix)
    }

    def get_state_snapshot(self, domain: str) -> Dict[str, Any]:
        """Get snapshot of all state in a domain."""
        return {
            'timestamp': datetime.now(),
            'domain': domain,
            'entities': self.get_all_states(domain)
        }

```

5.5. Data Quality for Decisions

Data quality issues that are acceptable for reporting can cause optimization failures.

5.5.1. Quality Dimensions

Dimension	Impact on Decisions	Mitigation
Completeness	Missing data = infeasible	Defaults, imputation
Accuracy	Wrong data = wrong decisions	Validation, cross-checks
Timeliness	Stale data = suboptimal	Freshness SLAs
Consistency	Conflicts = unstable solutions	Single source of truth

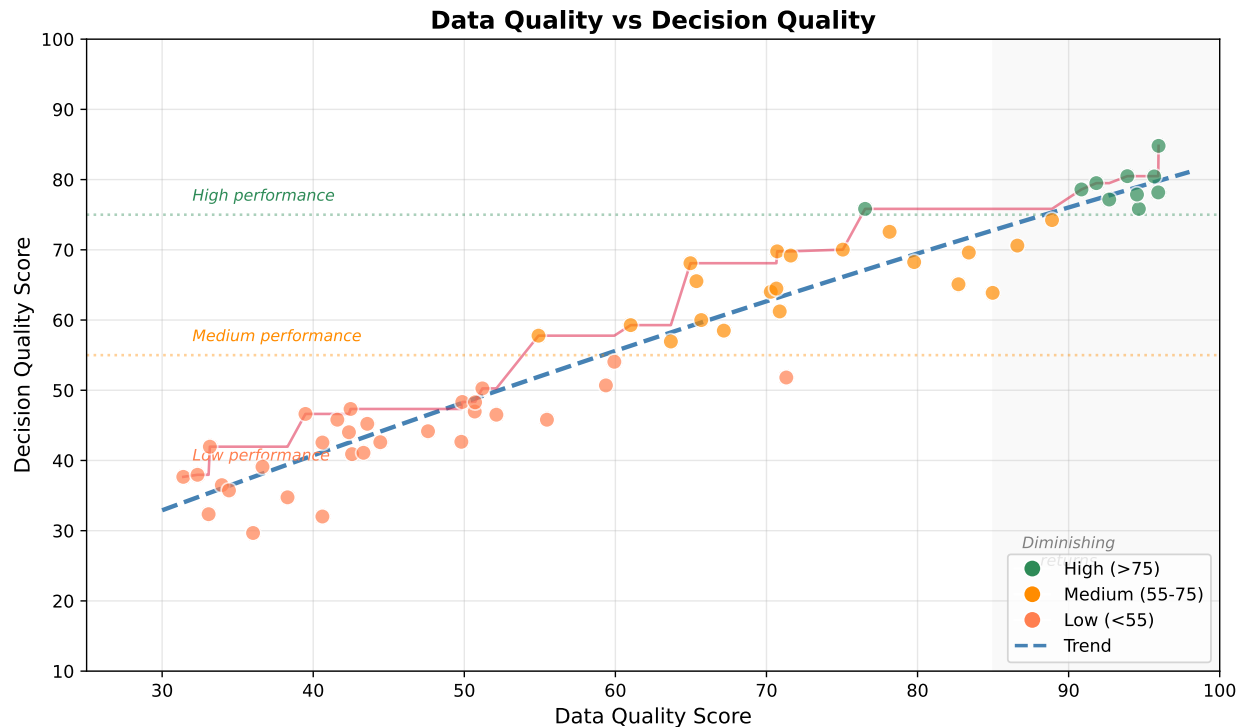


Figure 5.2.: Data quality vs decision quality — organizations that invest in data quality achieve significantly better decision outcomes, with diminishing returns beyond ~85% quality score

5.5.2. Data Validation Framework

```

from dataclasses import dataclass
from typing import Callable, List, Tuple
from enum import Enum

class ValidationSeverity(Enum):
    WARNING = "warning"
    ERROR = "error"
    CRITICAL = "critical"

@dataclass
class ValidationRule:
    """A data validation rule."""
    name: str
    description: str
    check: Callable[[Any], bool]
    severity: ValidationSeverity
    fix: Optional[Callable[[Any], Any]] = None

@dataclass
class ValidationResult:
    """Result of validation."""
    rule_name: str
    passed: bool
    severity: ValidationSeverity
    message: str
    fixed_value: Any = None

class DataValidator:
    """Validates data for decision systems."""

    def __init__(self):
        self.rules: Dict[str, List[ValidationRule]] = {}

    def add_rule(self, field: str, rule: ValidationRule):
        """Add validation rule for a field."""
        if field not in self.rules:
            self.rules[field] = []
        self.rules[field].append(rule)

    def validate(self, data: Dict[str, Any],
                 auto_fix: bool = False) -> Tuple[Dict, List[ValidationResult]]:
        """Validate data and optionally fix issues."""
        results = []
        validated_data = data.copy()

        for field, value in data.items():
            field_rules = self.rules.get(field, [])

            for rule in field_rules:
                passed = rule.check(value)

                result = ValidationResult(

```

```

        rule_name=rule.name,
        passed=passed,
        severity=rule.severity,
        message=f"{field}: {rule.description}"
    )

    if not passed and auto_fix and rule.fix:
        fixed = rule.fix(value)
        validated_data[field] = fixed
        result.fixed_value = fixed

    results.append(result)

    return validated_data, results

def is_valid(self, data: Dict[str, Any],
            min_severity: ValidationSeverity = ValidationSeverity.ERROR) ->
↳ bool:
    """Check if data passes validation."""
    _, results = self.validate(data, auto_fix=False)

    severity_order = {
        ValidationSeverity.WARNING: 1,
        ValidationSeverity.ERROR: 2,
        ValidationSeverity.CRITICAL: 3
    }

    for result in results:
        if not result.passed:
            if severity_order[result.severity] >= severity_order[min_severity]:
                return False

    return True

# Common validation rules
def create_range_rule(min_val: float, max_val: float,
                    severity: ValidationSeverity = ValidationSeverity.ERROR) ->
↳ ValidationRule:
    """Create a range validation rule."""
    return ValidationRule(
        name=f"range_{min_val}_{max_val}",
        description=f"Value must be between {min_val} and {max_val}",
        check=lambda x: min_val <= x <= max_val if x is not None else False,
        severity=severity,
        fix=lambda x: max(min_val, min(max_val, x)) if x is not None else min_val
    )

def create_not_null_rule(default: Any = None) -> ValidationRule:
    """Create a not-null validation rule."""
    return ValidationRule(
        name="not_null",
        description="Value must not be null",
        check=lambda x: x is not None,

```

```

        severity=ValidationSeverity.ERROR,
        fix=lambda x: default if x is None else x
    )

def create_positive_rule() -> ValidationRule:
    """Create a positive number validation rule."""
    return ValidationRule(
        name="positive",
        description="Value must be positive",
        check=lambda x: x is not None and x > 0,
        severity=ValidationSeverity.ERROR,
        fix=lambda x: max(0.001, x) if x is not None else 0.001
    )

```

5.5.3. Anomaly Detection for Decision Data

```

class DecisionDataAnomalyDetector:
    """Detects anomalies in decision-relevant data."""

    def __init__(self, z_threshold: float = 3.0):
        self.z_threshold = z_threshold
        self.statistics: Dict[str, Dict] = {}

    def fit(self, field: str, historical_data: np.ndarray):
        """Fit detector on historical data."""
        self.statistics[field] = {
            'mean': np.mean(historical_data),
            'std': np.std(historical_data),
            'min': np.min(historical_data),
            'max': np.max(historical_data),
            'q1': np.percentile(historical_data, 25),
            'q3': np.percentile(historical_data, 75)
        }

    def detect(self, field: str, value: float) -> Dict[str, Any]:
        """Detect if value is anomalous."""
        if field not in self.statistics:
            return {'is_anomaly': False, 'reason': 'No baseline'}

        stats = self.statistics[field]

        # Z-score check
        z_score = (value - stats['mean']) / (stats['std'] + 1e-6)
        z_anomaly = abs(z_score) > self.z_threshold

        # IQR check
        iqr = stats['q3'] - stats['q1']
        iqr_lower = stats['q1'] - 1.5 * iqr
        iqr_upper = stats['q3'] + 1.5 * iqr
        iqr_anomaly = value < iqr_lower or value > iqr_upper

        is_anomaly = z_anomaly or iqr_anomaly

```



```

    return {
        'is_anomaly': is_anomaly,
        'z_score': z_score,
        'z_anomaly': z_anomaly,
        'iqr_anomaly': iqr_anomaly,
        'expected_range': (iqr_lower, iqr_upper),
        'reason': self._get_reason(z_anomaly, iqr_anomaly, z_score)
    }

def _get_reason(self, z_anomaly: bool, iqr_anomaly: bool,
                z_score: float) -> str:
    """Generate explanation for anomaly."""
    if not z_anomaly and not iqr_anomaly:
        return "Normal"

    reasons = []
    if z_anomaly:
        direction = "high" if z_score > 0 else "low"
        reasons.append(f"Unusually {direction} (z={z_score:.2f})")
    if iqr_anomaly:
        reasons.append("Outside typical range")

    return "; ".join(reasons)

```

5.6. Feedback Loops and Model Monitoring

5.6.1. Decision Outcome Tracking

```

@dataclass
class DecisionOutcome:
    """Recorded outcome of a decision."""
    decision_id: str
    decision_type: str
    input_features: Dict[str, Any]
    decision_made: Dict[str, Any]
    predicted_outcome: Dict[str, float]
    actual_outcome: Dict[str, float]
    timestamp: datetime
    feedback: Optional[str] = None

class OutcomeTracker:
    """Tracks decision outcomes for model improvement."""

    def __init__(self):
        self.outcomes: List[DecisionOutcome] = []
        self.metrics_by_type: Dict[str, Dict] = {}

    def record_decision(self, decision_id: str, decision_type: str,
                       features: Dict, decision: Dict,
                       predicted: Dict):
        """Record a decision that was made."""
        outcome = DecisionOutcome(
            decision_id=decision_id,

```

```

        decision_type=decision_type,
        input_features=features,
        decision_made=decision,
        predicted_outcome=predicted,
        actual_outcome={},
        timestamp=datetime.now()
    )
    self.outcomes.append(outcome)
    return outcome

def record_actual_outcome(self, decision_id: str,
                        actual: Dict[str, float]):
    """Record actual outcome after decision execution."""
    for outcome in self.outcomes:
        if outcome.decision_id == decision_id:
            outcome.actual_outcome = actual
            self._update_metrics(outcome)
    return

def _update_metrics(self, outcome: DecisionOutcome):
    """Update performance metrics."""
    dtype = outcome.decision_type
    if dtype not in self.metrics_by_type:
        self.metrics_by_type[dtype] = {
            'count': 0,
            'prediction_errors': [],
            'decision_quality': []
        }

    metrics = self.metrics_by_type[dtype]
    metrics['count'] += 1

    # Calculate prediction error
    for key in outcome.predicted_outcome:
        if key in outcome.actual_outcome:
            pred = outcome.predicted_outcome[key]
            actual = outcome.actual_outcome[key]
            error = abs(pred - actual) / (abs(actual) + 1e-6)
            metrics['prediction_errors'].append(error)

def get_performance_report(self, decision_type: str = None) -> Dict:
    """Generate performance report."""
    if decision_type:
        types = [decision_type]
    else:
        types = list(self.metrics_by_type.keys())

    report = {}
    for dtype in types:
        metrics = self.metrics_by_type.get(dtype, {})
        errors = metrics.get('prediction_errors', [])

        report[dtype] = {

```

```

        'total_decisions': metrics.get('count', 0),
        'mean_prediction_error': np.mean(errors) if errors else None,
        'median_prediction_error': np.median(errors) if errors else None,
        'p95_prediction_error': np.percentile(errors, 95) if errors else
        ↪ None
    }

    return report

def identify_model_drift(self, decision_type: str,
                        window_size: int = 100) -> Dict:
    """Detect if model performance is degrading."""
    type_outcomes = [o for o in self.outcomes
                     if o.decision_type == decision_type
                     and o.actual_outcome]

    if len(type_outcomes) < 2 * window_size:
        return {'drift_detected': False, 'reason': 'Insufficient data'}

    # Compare recent vs historical errors
    recent = type_outcomes[-window_size:]
    historical = type_outcomes[-2*window_size:-window_size]

    recent_errors = []
    historical_errors = []

    for o in recent:
        for key in o.predicted_outcome:
            if key in o.actual_outcome:
                error = abs(o.predicted_outcome[key] - o.actual_outcome[key])
                recent_errors.append(error)

    for o in historical:
        for key in o.predicted_outcome:
            if key in o.actual_outcome:
                error = abs(o.predicted_outcome[key] - o.actual_outcome[key])
                historical_errors.append(error)

    recent_mean = np.mean(recent_errors)
    historical_mean = np.mean(historical_errors)

    # Detect significant degradation
    threshold = 0.2 # 20% degradation
    drift_detected = (recent_mean - historical_mean) / (historical_mean + 1e-6)
    ↪ > threshold

    return {
        'drift_detected': drift_detected,
        'recent_error': recent_mean,
        'historical_error': historical_mean,
        'degradation_pct': (recent_mean - historical_mean) / (historical_mean +
        ↪ 1e-6) * 100
    }

```

5.7. Summary

5.7.1. Key Takeaways

- Decision systems require **lower latency** and **higher accuracy** than analytics
- **Feature stores** centralize decision-relevant data with freshness guarantees
- **Temporal features** (lags, rolling stats, trends) capture dynamics
- **Real-time handlers** maintain current state for instant decisions
- **Data validation** prevents bad data from causing bad decisions
- **Anomaly detection** identifies unusual inputs requiring human review
- **Outcome tracking** enables continuous model improvement
- **Drift detection** alerts when models need retraining

5.8. Exercises

1. Design a feature store schema for a ride-sharing application. What features would you need for real-time driver assignment?
2. Implement a data validation pipeline for inventory optimization. What rules would catch common data quality issues?
3. Build a streaming data handler that maintains rolling demand statistics with a 1-hour window.
4. Create an anomaly detector for order quantities. How would you handle seasonal patterns?
5. Design a feedback loop that automatically triggers model retraining when prediction accuracy drops below a threshold.

6. Time Series Forecasting for Decision Intelligence

What You Will Learn

- Understand the role of forecasting in optimization and decision pipelines
- Implement classical statistical methods (ARIMA, SARIMA, Exponential Smoothing)
- Build deep learning forecasters using LSTM and GRU architectures
- Apply gradient boosting methods (XGBoost, LightGBM) to time series problems
- Use modern approaches including Prophet, N-BEATS, and Transformers
- Create ensemble forecasts with uncertainty quantification
- Integrate forecasting with optimization for robust decision-making

Forecasting is the bridge between historical data and future decisions. In decision intelligence, accurate forecasts feed directly into optimization models, enabling organizations to make proactive rather than reactive decisions. Whether you're optimizing inventory levels, scheduling production, or allocating resources, the quality of your forecasts fundamentally determines the quality of your decisions.

6.1. Introduction to Forecasting

6.1.1. Why Forecasting Matters for Optimization

Optimization models require parameters: demand quantities, resource availability, prices, lead times. Many of these parameters are uncertain and must be predicted. The relationship between forecasting and optimization is symbiotic:



Figure 6.1.: The Forecast-Optimize Decision Loop

Component	Role in Decision Pipeline
Point Forecasts	Expected values for deterministic optimization
Prediction Intervals	Bounds for robust optimization
Probabilistic Forecasts	Scenarios for stochastic programming
Forecast Errors	Calibration of safety stocks and buffers

! The Forecast-Optimization Gap

A common mistake is treating forecasting and optimization as separate problems. A 95%-accurate forecast can still lead to poor decisions if the forecast errors are systematically biased or if uncertainty is not properly communicated to the optimizer.

6.1.2. Types of Forecasting Problems

Problem Type	Characteristics	Examples
Univariate	Single time series	Daily sales of one product
Multivariate	Multiple related series	Sales across product categories
Hierarchical	Aggregation structure	Store -> Region -> National
Intermittent	Sparse, irregular demand	Spare parts, luxury items
Streaming	Real-time, continuous	Sensor data, web traffic

6.1.3. Forecasting Horizons

Different decision problems require different forecast horizons:

Horizon	Time Range	Decision Type	Methods
Short-term	Hours to days	Operational scheduling	ARIMA, ML models
Medium-term	Weeks to months	Inventory planning	Exponential smoothing, Prophet
Long-term	Quarters to years	Capacity planning	Trend extrapolation, causal models

6.2. Classical Statistical Methods

6.2.1. ARIMA and SARIMA

The Autoregressive Integrated Moving Average (ARIMA) model is the workhorse of statistical time series forecasting. It combines three components:

Autoregressive (AR): The current value depends on past values

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \epsilon_t$$

Integrated (I): Differencing to achieve stationarity

$$y'_t = y_t - y_{t-1}$$

Moving Average (MA): The current value depends on past forecast errors

$$y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

The combined ARIMA(p,d,q) model:

$$\phi(B)(1-B)^d y_t = c + \theta(B)\epsilon_t$$

where B is the backshift operator ($By_t = y_{t-1}$).

```
import numpy as np
import pandas as pd
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller, acf, pacf
```

```

import matplotlib.pyplot as plt

def check_stationarity(series, significance=0.05):
    """Test for stationarity using Augmented Dickey-Fuller test.

    Args:
        series: Time series data
        significance: Significance level for the test

    Returns:
        Dictionary with test results
    """
    result = adfuller(series, autolag='AIC')

    return {
        'test_statistic': result[0],
        'p_value': result[1],
        'critical_values': result[4],
        'is_stationary': result[1] < significance,
        'lags_used': result[2],
        'n_observations': result[3]
    }

def find_differencing_order(series, max_d=2):
    """Determine the order of differencing needed for stationarity.

    Args:
        series: Time series data
        max_d: Maximum differencing order to try

    Returns:
        Optimal differencing order
    """
    for d in range(max_d + 1):
        if d == 0:
            diff_series = series
        else:
            diff_series = np.diff(series, n=d)

        result = check_stationarity(diff_series)
        if result['is_stationary']:
            return d

    return max_d

def fit_arima_model(series, order=None, auto_select=True, max_p=5, max_q=5):
    """Fit ARIMA model with optional automatic order selection.

    Args:
        series: Time series data
        order: Tuple (p, d, q) if known

```

6. Time Series Forecasting for Decision Intelligence

```
    auto_select: Whether to automatically select order using AIC
    max_p: Maximum AR order for auto selection
    max_q: Maximum MA order for auto selection

Returns:
    Fitted ARIMA model and selected order
    """
    if order is None and auto_select:
        # Determine differencing order
        d = find_differencing_order(series)

        # Grid search for best (p, q) using AIC
        best_aic = np.inf
        best_order = (0, d, 0)

        for p in range(max_p + 1):
            for q in range(max_q + 1):
                try:
                    model = ARIMA(series, order=(p, d, q))
                    fitted = model.fit()
                    if fitted.aic < best_aic:
                        best_aic = fitted.aic
                        best_order = (p, d, q)
                except:
                    continue

        order = best_order

    # Fit final model
    model = ARIMA(series, order=order)
    fitted_model = model.fit()

    return fitted_model, order

def arima_forecast(model, steps, return_conf_int=True, alpha=0.05):
    """Generate forecasts from fitted ARIMA model.

    Args:
        model: Fitted ARIMA model
        steps: Number of steps ahead to forecast
        return_conf_int: Whether to return confidence intervals
        alpha: Significance level for confidence intervals

    Returns:
        Forecasts and optionally confidence intervals
        """
    forecast = model.get_forecast(steps=steps)
    predictions = forecast.predicted_mean

    if return_conf_int:
        conf_int = forecast.conf_int(alpha=alpha)
        return predictions, conf_int
```



```

    return predictions

# Example: ARIMA modeling workflow
np.random.seed(42)

# Generate synthetic data with trend and AR component
n = 200
trend = 0.1 * np.arange(n)
ar_component = np.zeros(n)
for t in range(2, n):
    ar_component[t] = 0.7 * ar_component[t-1] - 0.2 * ar_component[t-2] +
    ↪ np.random.randn()

y = trend + ar_component + 10

# Split into train/test
train_size = 180
y_train = y[:train_size]
y_test = y[train_size:]

# Fit ARIMA model
model, order = fit_arima_model(y_train, auto_select=True)
print(f"Selected ARIMA order: {order}")
print(f"AIC: {model.aic:.2f}")
print(f"\nModel Summary:")
print(model.summary().tables[1])

# Generate forecasts
forecasts, conf_int = arima_forecast(model, steps=len(y_test))

# Calculate forecast accuracy
mape = np.mean(np.abs((y_test - forecasts) / y_test)) * 100
rmse = np.sqrt(np.mean((y_test - forecasts) ** 2))
print(f"\nForecast Accuracy:")
print(f"MAPE: {mape:.2f}%")
print(f"RMSE: {rmse:.4f}")

```

6.2.2. Seasonal ARIMA (SARIMA)

For time series with seasonal patterns, SARIMA extends ARIMA with seasonal components:

$$\text{SARIMA}(p, d, q) \times (P, D, Q)_s$$

where s is the seasonal period (e.g., 12 for monthly data with annual seasonality).

The seasonal component adds: - Seasonal AR: $\Phi_P(B^s)$ - Seasonal differencing: $(1 - B^s)^D$ - Seasonal MA: $\Theta_Q(B^s)$

```

from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tsa.seasonal import seasonal_decompose

def decompose_series(series, period, model='additive'):

```

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```
"""Decompose time series into trend, seasonal, and residual components.

Args:
    series: Time series data
    period: Seasonal period
    model: 'additive' or 'multiplicative'

Returns:
    Decomposition result with trend, seasonal, and residual
"""
decomposition = seasonal_decompose(series, model=model, period=period)
return {
    'trend': decomposition.trend,
    'seasonal': decomposition.seasonal,
    'residual': decomposition.resid,
    'observed': decomposition.observed
}

def fit_sarima_model(series, order, seasonal_order, enforce_stationarity=True):
    """Fit SARIMA model.

    Args:
        series: Time series data
        order: Non-seasonal order (p, d, q)
        seasonal_order: Seasonal order (P, D, Q, s)
        enforce_stationarity: Whether to enforce stationarity

    Returns:
        Fitted SARIMA model
    """
    model = SARIMAX(
        series,
        order=order,
        seasonal_order=seasonal_order,
        enforce_stationarity=enforce_stationarity,
        enforce_invertibility=True
    )
    return model.fit(dispatch=False)

def auto_sarima(series, seasonal_period, max_order=2):
    """Automatic SARIMA model selection using AIC.

    Args:
        series: Time series data
        seasonal_period: Known seasonal period
        max_order: Maximum value for p, q, P, Q

    Returns:
        Best model and parameters
    """
    best_aic = np.inf
```

```

best_params = None
best_model = None

# Try different combinations
for p in range(max_order + 1):
    for d in range(2):
        for q in range(max_order + 1):
            for P in range(max_order + 1):
                for D in range(2):
                    for Q in range(max_order + 1):
                        try:
                            model = SARIMAX(
                                series,
                                order=(p, d, q),
                                seasonal_order=(P, D, Q, seasonal_period)
                            )
                            fitted = model.fit(dispatch=False)

                            if fitted.aic < best_aic:
                                best_aic = fitted.aic
                                best_params = {
                                    'order': (p, d, q),
                                    'seasonal_order': (P, D, Q, seasonal_period)
                                }
                                best_model = fitted
                        except:
                            continue

    return best_model, best_params

# Example: Monthly sales with annual seasonality
np.random.seed(42)

# Generate synthetic seasonal data
n_years = 5
periods_per_year = 12
n = n_years * periods_per_year

# Create components
time = np.arange(n)
trend = 100 + 0.5 * time
seasonal = 20 * np.sin(2 * np.pi * time / 12) # Annual seasonality
noise = np.random.randn(n) * 5

y_seasonal = trend + seasonal + noise

# Decompose to visualize components
decomp = decompose_series(y_seasonal, period=12)
print("Decomposition complete. Components extracted.")

```

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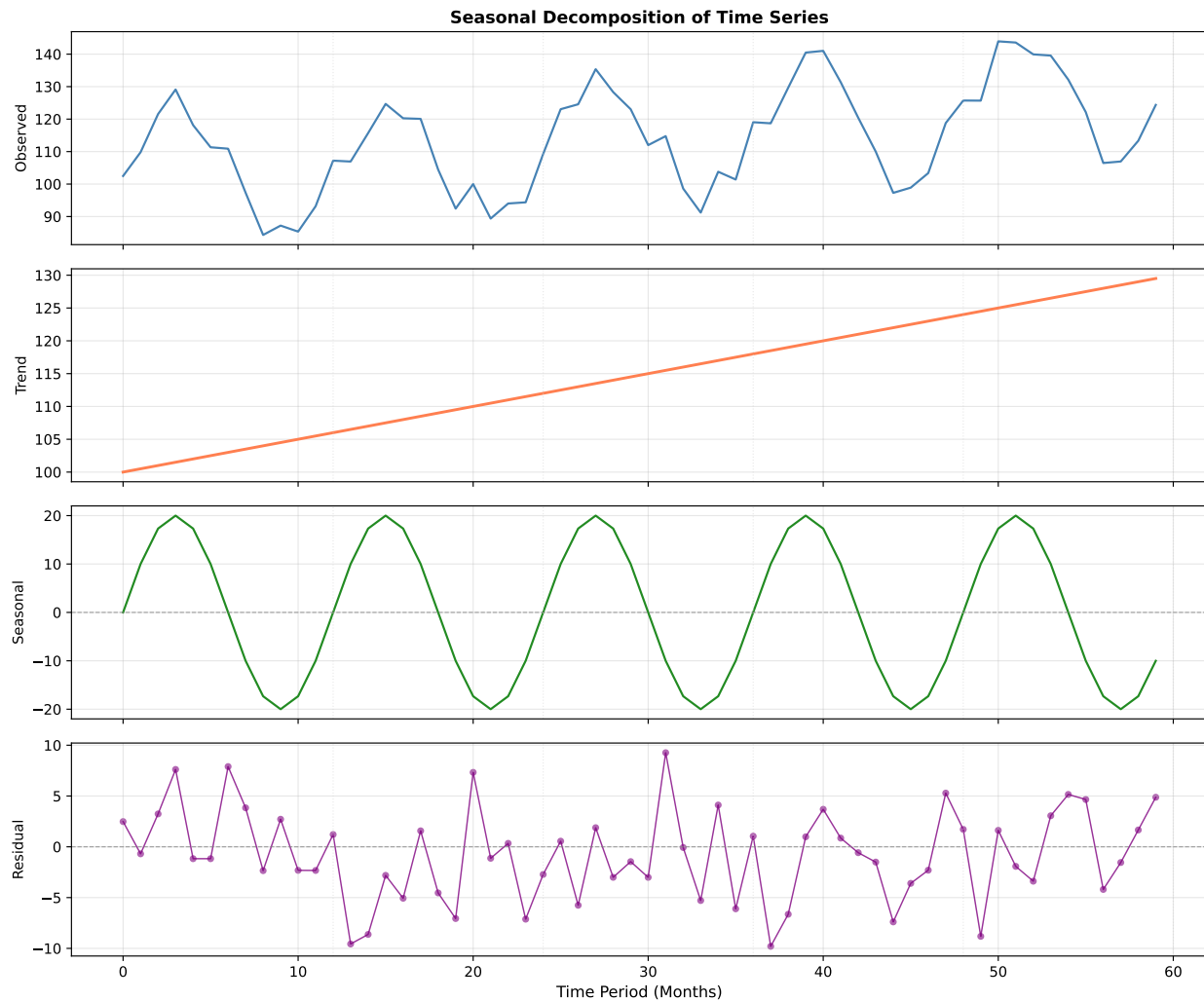


Figure 6.2.: Seasonal Decomposition of Time Series: The original series (top) is decomposed into trend (showing gradual increase), seasonal (repeating annual pattern), and residual (random noise) components. This decomposition helps identify and model each pattern separately.

```
# Fit SARIMA model
sarima_model = fit_sarima_model(
    y_seasonal,
    order=(1, 1, 1),
    seasonal_order=(1, 1, 1, 12)
)

print(f"\nSARIMA Model AIC: {sarima_model.aic:.2f}")

# Forecast next 12 months
forecast_sarima = sarima_model.get_forecast(steps=12)
pred_mean = forecast_sarima.predicted_mean
pred_ci = forecast_sarima.conf_int()
```

```
print(f"\n12-Month Forecast:")
for i, (mean, lower, upper) in enumerate(zip(pred_mean, pred_ci.iloc[:, 0],
↳ pred_ci.iloc[:, 1])):
    print(f"  Month {i+1}: {mean:.1f} [{lower:.1f}, {upper:.1f}]")
```

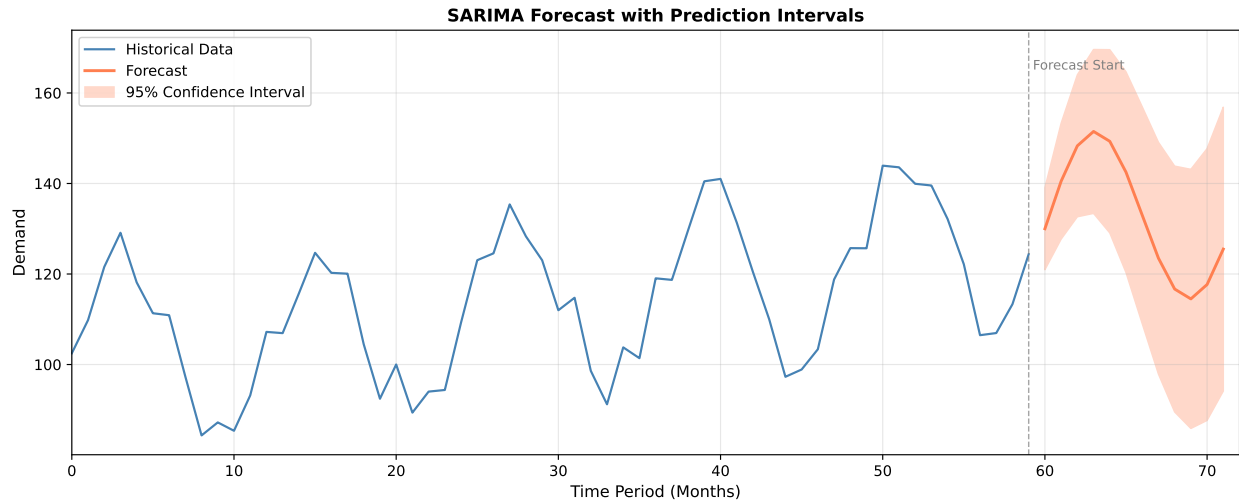


Figure 6.3.: Time Series Forecast with Confidence Intervals: Historical demand data (steelblue) with 12-month forecast (coral) and 95% prediction intervals (shaded region). The widening confidence bands reflect increasing uncertainty over longer forecast horizons.

💡 ARIMA Model Selection Guidelines

- Use ACF and PACF plots to identify initial (p, q) values
- ACF tailing off + PACF cutting off at lag p suggests AR(p)
- ACF cutting off at lag q + PACF tailing off suggests MA(q)
- Use information criteria (AIC, BIC) for final model selection
- Always validate on held-out test data

6.2.3. Exponential Smoothing

Exponential smoothing methods are intuitive, computationally efficient, and often competitive with more complex methods.

6.2.3.1. Simple Exponential Smoothing (SES)

For data with no trend or seasonality:

$$\hat{y}_{t+1} = \alpha y_t + (1 - \alpha) \hat{y}_t$$

where $\alpha \in [0, 1]$ is the smoothing parameter.

6.2.3.2. Holt's Linear Method (Double Exponential Smoothing)

For data with trend:

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$$\begin{aligned}\ell_t &= \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \\ b_t &= \beta(\ell_t - \ell_{t-1}) + (1 - \beta)b_{t-1} \\ \hat{y}_{t+h} &= \ell_t + hb_t\end{aligned}$$

where ℓ_t is the level and b_t is the trend.

6.2.3.3. Holt-Winters Method (Triple Exponential Smoothing)

For data with trend and seasonality:

Additive Seasonality:

$$\begin{aligned}\ell_t &= \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \\ b_t &= \beta(\ell_t - \ell_{t-1}) + (1 - \beta)b_{t-1} \\ s_t &= \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m} \\ \hat{y}_{t+h} &= \ell_t + hb_t + s_{t+h-m}\end{aligned}$$

Multiplicative Seasonality:

$$\begin{aligned}\ell_t &= \alpha \frac{y_t}{s_{t-m}} + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \\ s_t &= \gamma \frac{y_t}{\ell_{t-1} + b_{t-1}} + (1 - \gamma)s_{t-m} \\ \hat{y}_{t+h} &= (\ell_t + hb_t)s_{t+h-m}\end{aligned}$$

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing, SimpleExpSmoothing

class ExponentialSmoothingForecaster:
    """Unified interface for exponential smoothing methods."""

    def __init__(self, seasonal_periods=None):
        self.seasonal_periods = seasonal_periods
        self.model = None
        self.fitted_model = None

    def fit(self, series, trend=None, seasonal=None, damped_trend=False):
        """Fit exponential smoothing model.

        Args:
            series: Time series data
            trend: 'add', 'mul', or None
            seasonal: 'add', 'mul', or None
            damped_trend: Whether to damp the trend

        Returns:
            self
        """
        if trend is None and seasonal is None:
            # Simple exponential smoothing
            self.model = SimpleExpSmoothing(series)
            self.fitted_model = self.model.fit(optimized=True)
```

```

else:
    # Holt or Holt-Winters
    self.model = ExponentialSmoothing(
        series,
        trend=trend,
        seasonal=seasonal,
        seasonal_periods=self.seasonal_periods,
        damped_trend=damped_trend
    )
    self.fitted_model = self.model.fit(optimized=True)

return self

def forecast(self, steps):
    """Generate forecasts.

    Args:
        steps: Number of steps ahead

    Returns:
        Forecast values
    """
    if self.fitted_model is None:
        raise ValueError("Model must be fitted before forecasting")

    return self.fitted_model.forecast(steps)

def get_parameters(self):
    """Get fitted parameters."""
    if self.fitted_model is None:
        return None

    params = {
        'alpha': self.fitted_model.params.get('smoothing_level'),
        'beta': self.fitted_model.params.get('smoothing_trend'),
        'gamma': self.fitted_model.params.get('smoothing_seasonal'),
        'phi': self.fitted_model.params.get('damping_trend')
    }
    return {k: v for k, v in params.items() if v is not None}

def compare_smoothing_methods(series, test_size, seasonal_period=None):
    """Compare different exponential smoothing variants.

    Args:
        series: Full time series
        test_size: Number of periods to hold out for testing
        seasonal_period: Seasonal period if applicable

    Returns:
        Comparison results
    """
    train = series[:-test_size]

```

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```
test = series[-test_size:]

methods = {
    'SES': {'trend': None, 'seasonal': None},
    'Holt Linear': {'trend': 'add', 'seasonal': None},
    'Holt Damped': {'trend': 'add', 'seasonal': None, 'damped_trend': True},
}

if seasonal_period is not None:
    methods.update({
        'Holt-Winters Add': {'trend': 'add', 'seasonal': 'add'},
        'Holt-Winters Mul': {'trend': 'add', 'seasonal': 'mul'},
        'Holt-Winters Damped': {'trend': 'add', 'seasonal': 'add',
                                ↪ 'damped_trend': True}
    })

results = {}

for name, params in methods.items():
    try:
        forecaster =
        ↪ ExponentialSmoothingForecaster(seasonal_periods=seasonal_period)
        forecaster.fit(train, **params)
        forecasts = forecaster.forecast(test_size)

        # Calculate metrics
        mae = np.mean(np.abs(test - forecasts))
        rmse = np.sqrt(np.mean((test - forecasts) ** 2))
        mape = np.mean(np.abs((test - forecasts) / test)) * 100

        results[name] = {
            'MAE': mae,
            'RMSE': rmse,
            'MAPE': mape,
            'parameters': forecaster.get_parameters()
        }
    except Exception as e:
        results[name] = {'error': str(e)}

return results

# Example: Compare methods on seasonal data
np.random.seed(42)

# Generate data with trend and seasonality
n = 72 # 6 years of monthly data
time = np.arange(n)
trend = 50 + 0.3 * time
seasonal = 10 * np.sin(2 * np.pi * time / 12)
noise = np.random.randn(n) * 3
y = trend + seasonal + noise
```



```
# Compare methods
comparison = compare_smoothing_methods(y, test_size=12, seasonal_period=12)

print("Exponential Smoothing Method Comparison:")
print("-" * 60)
for method, metrics in comparison.items():
    if 'error' not in metrics:
        print(f"{method}:")
        print(f"  MAE: {metrics['MAE']:.2f}")
        print(f"  RMSE: {metrics['RMSE']:.2f}")
        print(f"  MAPE: {metrics['MAPE']:.2f}%")
        print(f"  Parameters: {metrics['parameters']}")
    print()
```

6.2.4. The ETS Framework

The ETS (Error, Trend, Seasonal) framework provides a systematic taxonomy of exponential smoothing methods:

Component	Options	Description
Error (E)	A (additive), M (multiplicative)	Error structure
Trend (T)	N (none), A (additive), Ad (damped additive), M (multiplicative), Md (damped multiplicative)	Trend component
Seasonal (S)	N (none), A (additive), M (multiplicative)	Seasonal component

This gives 30 possible models (some with restrictions). Model selection can be automated using information criteria.

i When to Use Each Method

Data Pattern	Recommended Method
No trend, no seasonality	Simple Exponential Smoothing
Trend, no seasonality	Holt's Linear or Damped
No trend, seasonality	Seasonal ETS (ANA or ANM)
Trend and seasonality	Holt-Winters (AAA, AAM, MAM)
Uncertain	Auto ETS with model selection

6.3. Deep Learning Methods

Deep learning has revolutionized time series forecasting by automatically learning complex patterns and long-range dependencies.

6.3.1. LSTM (Long Short-Term Memory)

LSTM networks address the vanishing gradient problem in standard RNNs, enabling learning of long-term dependencies. The key innovation is the cell state, which acts as a memory highway.

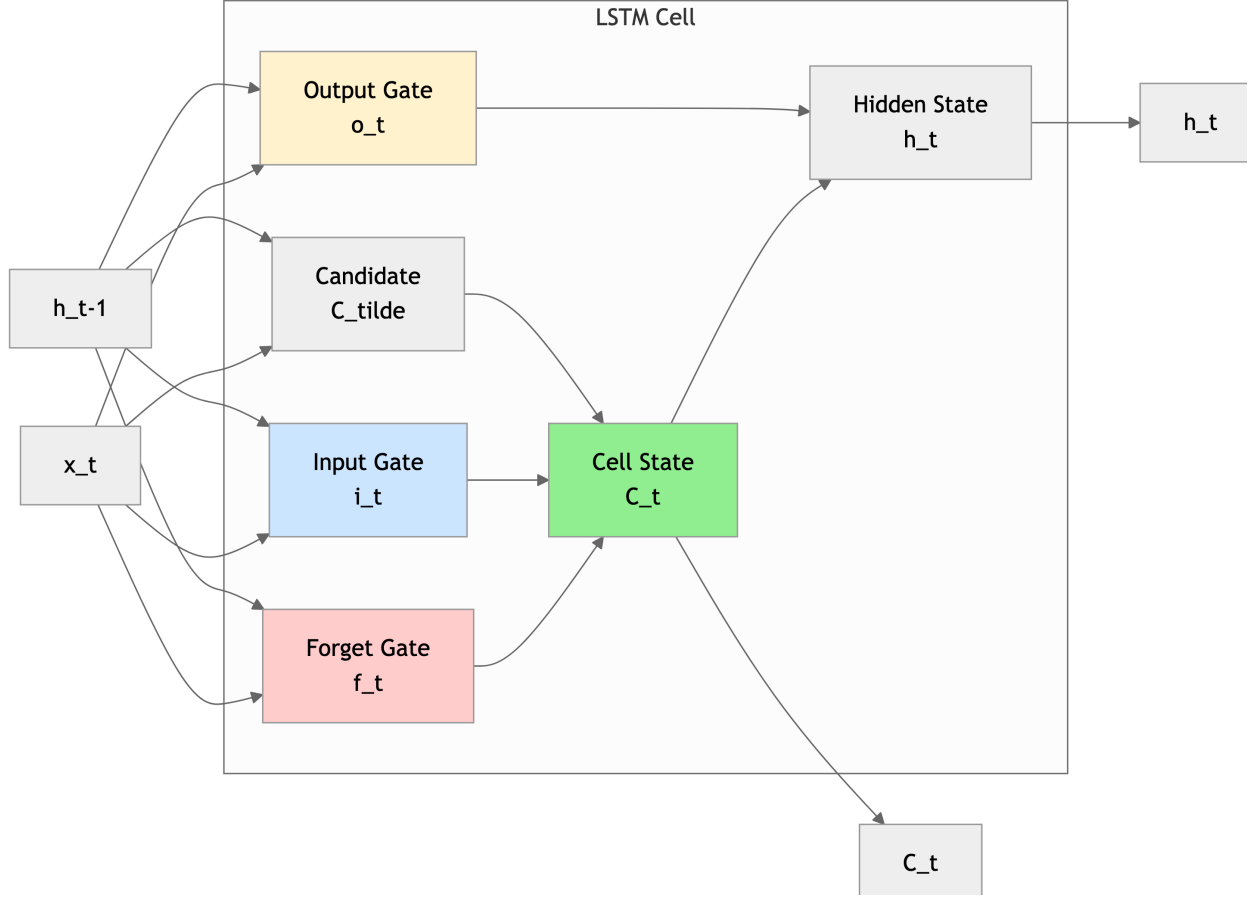


Figure 6.4.: LSTM Cell Architecture with Gates

6.3.1.1. LSTM Equations

The LSTM processes sequences through four key components:

Forget Gate - Decides what information to discard from the cell state:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

Input Gate - Decides what new information to store:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

Candidate Cell State - Creates candidate values to add:

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

Cell State Update - Updates the cell state:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

Output Gate - Decides what to output:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

Hidden State - Filtered cell state as output:

$$h_t = o_t \odot \tanh(C_t)$$

where σ is the sigmoid function, $\tanh$ is the hyperbolic tangent, and $\odot$ denotes element-wise multiplication.

```
import numpy as np
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from sklearn.preprocessing import MinMaxScaler

class LSTMForecaster:
    """LSTM-based time series forecaster."""

    def __init__(self, lookback=24, forecast_horizon=1, lstm_units=50,
                 dropout=0.2, learning_rate=0.001):
        """Initialize LSTM forecaster.

        Args:
            lookback: Number of past time steps to use as input
            forecast_horizon: Number of steps ahead to forecast
            lstm_units: Number of units in LSTM layers
            dropout: Dropout rate for regularization
            learning_rate: Learning rate for optimizer
        """
        self.lookback = lookback
        self.forecast_horizon = forecast_horizon
        self.lstm_units = lstm_units
        self.dropout = dropout
        self.learning_rate = learning_rate
        self.model = None
        self.scaler = MinMaxScaler()

    def _create_sequences(self, data):
        """Create input-output sequences for training.

        Args:
            data: Normalized time series data

        Returns:
            X (sequences), y (targets)
        """
        X, y = [], []
        for i in range(len(data) - self.lookback - self.forecast_horizon + 1):
            X.append(data[i:(i + self.lookback)])
            y.append(data[(i + self.lookback):(i + self.lookback +
↪ self.forecast_horizon)])
        return np.array(X), np.array(y)

    def _build_model(self, n_features):
        """Build LSTM model architecture.
```

```

    Args:
        n_features: Number of input features

    Returns:
        Compiled Keras model
    """
    model = keras.Sequential([
        # First LSTM layer with return sequences
        layers.LSTM(
            self.lstm_units,
            return_sequences=True,
            input_shape=(self.lookback, n_features)
        ),
        layers.Dropout(self.dropout),

        # Second LSTM layer
        layers.LSTM(self.lstm_units, return_sequences=False),
        layers.Dropout(self.dropout),

        # Dense layers for output
        layers.Dense(32, activation='relu'),
        layers.Dense(self.forecast_horizon)
    ])

    model.compile(
        optimizer=keras.optimizers.Adam(learning_rate=self.learning_rate),
        loss='mse',
        metrics=['mae']
    )

    return model

def fit(self, series, epochs=100, batch_size=32, validation_split=0.2,
        early_stopping=True, patience=10, verbose=1):
    """Train the LSTM model.

    Args:
        series: Time series data (1D array or 2D with features)
        epochs: Number of training epochs
        batch_size: Training batch size
        validation_split: Fraction of data for validation
        early_stopping: Whether to use early stopping
        patience: Patience for early stopping
        verbose: Verbosity level

    Returns:
        Training history
    """
    # Ensure 2D
    if series.ndim == 1:
        series = series.reshape(-1, 1)

    # Scale data

```

```

scaled_data = self.scaler.fit_transform(series)

# Create sequences
X, y = self._create_sequences(scaled_data)

# For univariate, squeeze y
if y.shape[-1] == 1:
    y = y.squeeze()

# Build model
self.model = self._build_model(n_features=series.shape[1])

# Callbacks
callbacks = []
if early_stopping:
    callbacks.append(keras.callbacks.EarlyStopping(
        monitor='val_loss',
        patience=patience,
        restore_best_weights=True
    ))

# Train
history = self.model.fit(
    X, y,
    epochs=epochs,
    batch_size=batch_size,
    validation_split=validation_split,
    callbacks=callbacks,
    verbose=verbose
)

return history

def forecast(self, series, steps=None):
    """Generate forecasts.

    Args:
        series: Recent time series data for context
        steps: Number of steps to forecast (default: forecast_horizon)

    Returns:
        Forecasted values
    """
    if steps is None:
        steps = self.forecast_horizon

    # Ensure 2D
    if series.ndim == 1:
        series = series.reshape(-1, 1)

    # Scale
    scaled = self.scaler.transform(series)

```

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```
forecasts = []
current_sequence = scaled[-self.lookback:].copy()

remaining_steps = steps
while remaining_steps > 0:
    # Predict next horizon
    X = current_sequence.reshape(1, self.lookback, -1)
    pred = self.model.predict(X, verbose=0)

    # Handle output shape
    if pred.ndim == 1:
        pred = pred.reshape(1, -1)

    # Take needed steps
    n_take = min(pred.shape[1], remaining_steps)
    forecasts.extend(pred[0, :n_take])

    # Update sequence
    for i in range(n_take):
        new_val = pred[0, i:i+1].reshape(1, -1)
        current_sequence = np.vstack([current_sequence[1:], new_val])

    remaining_steps -= n_take

# Inverse transform
forecasts = np.array(forecasts).reshape(-1, 1)
forecasts = self.scaler.inverse_transform(forecasts)

return forecasts.flatten()

# Example: LSTM for demand forecasting
np.random.seed(42)
tf.random.set_seed(42)

# Generate synthetic demand data with patterns
n = 500
time = np.arange(n)
trend = 50 + 0.02 * time
daily_pattern = 10 * np.sin(2 * np.pi * time / 7) # Weekly seasonality
monthly_pattern = 15 * np.sin(2 * np.pi * time / 30) # Monthly seasonality
noise = np.random.randn(n) * 3
demand = trend + daily_pattern + monthly_pattern + noise

# Train LSTM
forecaster = LSTMForecaster(
    lookback=30,
    forecast_horizon=7,
    lstm_units=64,
    dropout=0.2
)

# Use first 450 for training
```

```

train_data = demand[:450]
test_data = demand[450:]

history = forecaster.fit(
    train_data,
    epochs=50,
    batch_size=32,
    validation_split=0.15,
    verbose=0
)

# Generate forecasts
forecasts = forecaster.forecast(train_data, steps=len(test_data))

# Evaluate
mae = np.mean(np.abs(test_data - forecasts))
rmse = np.sqrt(np.mean((test_data - forecasts) ** 2))
mape = np.mean(np.abs((test_data - forecasts) / test_data)) * 100

print("LSTM Forecast Results:")
print(f" MAE: {mae:.2f}")
print(f" RMSE: {rmse:.2f}")
print(f" MAPE: {mape:.2f}%")

```

💡 When to Use LSTM

LSTM networks excel when:

- **Long sequences:** Dependencies span many time steps
- **Complex patterns:** Multiple overlapping seasonalities
- **Sufficient data:** Thousands of observations available
- **Computational resources:** Training time is acceptable
- **Non-linear relationships:** Linear models underperform

Avoid LSTM for:

- Small datasets (< 500 observations)
- Simple patterns (use ARIMA or ETS instead)
- Real-time inference on resource-constrained devices

6.3.2. GRU (Gated Recurrent Unit)

GRU is a simplified variant of LSTM with only two gates instead of three. Despite fewer parameters, GRU often achieves comparable performance.

6.3.2.1. GRU Equations

Update Gate - Decides how much past information to keep:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z)$$

Reset Gate - Decides how much past information to forget:

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$$

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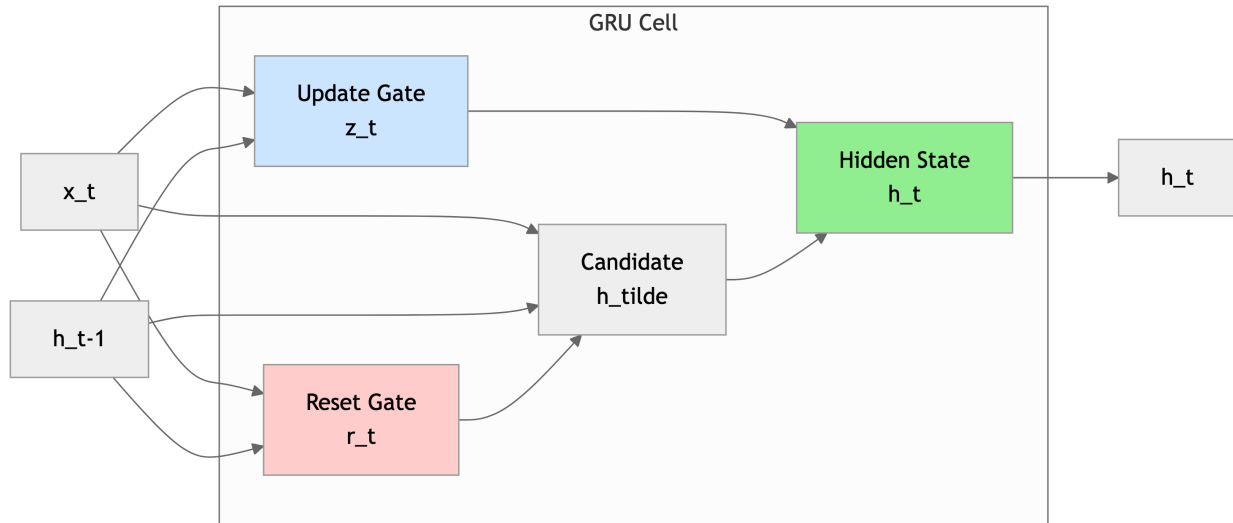


Figure 6.5.: GRU Cell Architecture - Simplified with Two Gates

Candidate Hidden State:

$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h)$$

Final Hidden State:

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

```

class GRUForecaster:
    """GRU-based time series forecaster."""

    def __init__(self, lookback=24, forecast_horizon=1, gru_units=50,
                  dropout=0.2, learning_rate=0.001):
        """Initialize GRU forecaster.

        Args:
            lookback: Number of past time steps
            forecast_horizon: Number of steps to forecast
            gru_units: Number of units in GRU layers
            dropout: Dropout rate
            learning_rate: Learning rate
        """
        self.lookback = lookback
        self.forecast_horizon = forecast_horizon
        self.gru_units = gru_units
        self.dropout = dropout
        self.learning_rate = learning_rate
        self.model = None
        self.scaler = MinMaxScaler()

    def _create_sequences(self, data):
        """Create sequences for training."""
        X, y = [], []
        for i in range(len(data) - self.lookback - self.forecast_horizon + 1):

```



```

        X.append(data[i:(i + self.lookback)])
        y.append(data[(i + self.lookback):(i + self.lookback +
↪ self.forecast_horizon)])
        return np.array(X), np.array(y)

def _build_model(self, n_features):
    """Build GRU model architecture."""
    model = keras.Sequential([
        # First GRU layer
        layers.GRU(
            self.gru_units,
            return_sequences=True,
            input_shape=(self.lookback, n_features)
        ),
        layers.Dropout(self.dropout),

        # Second GRU layer
        layers.GRU(self.gru_units, return_sequences=False),
        layers.Dropout(self.dropout),

        # Output layers
        layers.Dense(32, activation='relu'),
        layers.Dense(self.forecast_horizon)
    ])

    model.compile(
        optimizer=keras.optimizers.Adam(learning_rate=self.learning_rate),
        loss='mse',
        metrics=['mae']
    )

    return model

def fit(self, series, epochs=100, batch_size=32, validation_split=0.2,
        early_stopping=True, patience=10, verbose=1):
    """Train the GRU model."""
    if series.ndim == 1:
        series = series.reshape(-1, 1)

    scaled_data = self.scaler.fit_transform(series)
    X, y = self._create_sequences(scaled_data)

    if y.shape[-1] == 1:
        y = y.squeeze()

    self.model = self._build_model(n_features=series.shape[1])

    callbacks = []
    if early_stopping:
        callbacks.append(keras.callbacks.EarlyStopping(
            monitor='val_loss',
            patience=patience,
            restore_best_weights=True

```

```

        ))

    history = self.model.fit(
        X, y,
        epochs=epochs,
        batch_size=batch_size,
        validation_split=validation_split,
        callbacks=callbacks,
        verbose=verbose
    )

    return history

def forecast(self, series, steps=None):
    """Generate forecasts."""
    if steps is None:
        steps = self.forecast_horizon

    if series.ndim == 1:
        series = series.reshape(-1, 1)

    scaled = self.scaler.transform(series)

    forecasts = []
    current_sequence = scaled[-self.lookback:].copy()

    remaining_steps = steps
    while remaining_steps > 0:
        X = current_sequence.reshape(1, self.lookback, -1)
        pred = self.model.predict(X, verbose=0)

        if pred.ndim == 1:
            pred = pred.reshape(1, -1)

        n_take = min(pred.shape[1], remaining_steps)
        forecasts.extend(pred[0, :n_take])

        for i in range(n_take):
            new_val = pred[0, i:i+1].reshape(1, -1)
            current_sequence = np.vstack([current_sequence[1:], new_val])

        remaining_steps -= n_take

    forecasts = np.array(forecasts).reshape(-1, 1)
    forecasts = self.scaler.inverse_transform(forecasts)

    return forecasts.flatten()

def compare_lstm_gru(train_data, test_data, lookback=30, horizon=7):
    """Compare LSTM and GRU performance.

    Args:

```

```

train_data: Training time series
test_data: Test time series
lookback: Input sequence length
horizon: Forecast horizon

Returns:
    Comparison results
"""
results = {}

# LSTM
lstm = LSTMForecaster(lookback=lookback, forecast_horizon=horizon,
    ↪ lstm_units=64)
lstm.fit(train_data, epochs=50, verbose=0)
lstm_forecasts = lstm.forecast(train_data, steps=len(test_data))

# GRU
gru = GRUForecaster(lookback=lookback, forecast_horizon=horizon, gru_units=64)
gru.fit(train_data, epochs=50, verbose=0)
gru_forecasts = gru.forecast(train_data, steps=len(test_data))

# Metrics
for name, forecasts in [('LSTM', lstm_forecasts), ('GRU', gru_forecasts)]:
    mae = np.mean(np.abs(test_data - forecasts))
    rmse = np.sqrt(np.mean((test_data - forecasts) ** 2))
    mape = np.mean(np.abs((test_data - forecasts) / test_data)) * 100

    results[name] = {
        'MAE': mae,
        'RMSE': rmse,
        'MAPE': mape,
        'forecasts': forecasts
    }

return results

```

6.3.2.2. LSTM vs GRU Comparison

Aspect	LSTM	GRU
Gates	3 (forget, input, output)	2 (reset, update)
Parameters	More (~4x hidden units)	Fewer (~3x hidden units)
Memory	Separate cell and hidden state	Single hidden state
Training Speed	Slower	Faster
Long Dependencies	Better for very long sequences	Often sufficient
Common Use	Language modeling, complex sequences	Time series, shorter sequences

i Choosing Between LSTM and GRU

- **Start with GRU:** Faster training, often similar performance
- **Try LSTM if:** Very long sequences (>100 steps), complex patterns

- **Experiment:** The best choice is often problem-dependent

6.3.3. Temporal Convolutional Networks (TCN)

TCNs offer an alternative to RNNs using dilated causal convolutions. They can process sequences in parallel, making training faster.

Key Features: - Causal convolutions (no future information leakage) - Dilated convolutions (exponentially growing receptive field) - Residual connections (stable training)

```
def build_tcn_block(inputs, n_filters, kernel_size, dilation_rate, dropout_rate):
    """Build a TCN residual block.

    Args:
        inputs: Input tensor
        n_filters: Number of convolutional filters
        kernel_size: Size of convolutional kernel
        dilation_rate: Dilation rate for this block
        dropout_rate: Dropout rate

    Returns:
        Output tensor
    """
    # First causal convolution
    conv1 = layers.Conv1D(
        filters=n_filters,
        kernel_size=kernel_size,
        padding='causal',
        dilation_rate=dilation_rate,
        activation='relu'
    )(inputs)
    conv1 = layers.Dropout(dropout_rate)(conv1)

    # Second causal convolution
    conv2 = layers.Conv1D(
        filters=n_filters,
        kernel_size=kernel_size,
        padding='causal',
        dilation_rate=dilation_rate,
        activation='relu'
    )(conv1)
    conv2 = layers.Dropout(dropout_rate)(conv2)

    # Residual connection
    if inputs.shape[-1] != n_filters:
        inputs = layers.Conv1D(n_filters, 1, padding='same')(inputs)

    return layers.add([inputs, conv2])

class TCNForecaster:
    """Temporal Convolutional Network forecaster."""

    def __init__(self, lookback=30, forecast_horizon=1, n_filters=64,
```

```

        kernel_size=3, n_blocks=4, dropout=0.2):
self.lookback = lookback
self.forecast_horizon = forecast_horizon
self.n_filters = n_filters
self.kernel_size = kernel_size
self.n_blocks = n_blocks
self.dropout = dropout
self.model = None
self.scaler = MinMaxScaler()

def _build_model(self, n_features):
    """Build TCN model."""
    inputs = layers.Input(shape=(self.lookback, n_features))

    x = inputs
    for i in range(self.n_blocks):
        dilation_rate = 2 ** i
        x = build_tcn_block(
            x,
            self.n_filters,
            self.kernel_size,
            dilation_rate,
            self.dropout
        )

    # Global pooling and output
    x = layers.GlobalAveragePooling1D()(x)
    x = layers.Dense(32, activation='relu')(x)
    outputs = layers.Dense(self.forecast_horizon)(x)

    model = keras.Model(inputs, outputs)
    model.compile(optimizer='adam', loss='mse', metrics=['mae'])

    return model

def fit(self, series, epochs=100, batch_size=32, validation_split=0.2,
    ↪ verbose=1):
    """Train TCN model."""
    if series.ndim == 1:
        series = series.reshape(-1, 1)

    scaled = self.scaler.fit_transform(series)

    X, y = [], []
    for i in range(len(scaled) - self.lookback - self.forecast_horizon + 1):
        X.append(scaled[i:i+self.lookback])
        y.append(scaled[i+self.lookback:i+self.lookback+self.forecast_horizon])
    X, y = np.array(X), np.array(y).squeeze()

    self.model = self._build_model(n_features=series.shape[1])

    history = self.model.fit(
        X, y,

```

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```
        epochs=epochs,
        batch_size=batch_size,
        validation_split=validation_split,
        callbacks=[keras.callbacks.EarlyStopping(patience=10,
↪ restore_best_weights=True)],
        verbose=verbose
    )

    return history

def forecast(self, series, steps=None):
    """Generate forecasts."""
    if steps is None:
        steps = self.forecast_horizon

    if series.ndim == 1:
        series = series.reshape(-1, 1)

    scaled = self.scaler.transform(series)

    forecasts = []
    current = scaled[-self.lookback:].copy()

    while len(forecasts) < steps:
        X = current.reshape(1, self.lookback, -1)
        pred = self.model.predict(X, verbose=0)

        n_take = min(pred.shape[1], steps - len(forecasts))
        forecasts.extend(pred[0, :n_take])

        for i in range(n_take):
            new_val = pred[0, i:i+1].reshape(1, -1)
            current = np.vstack([current[1:], new_val])

    forecasts = np.array(forecasts).reshape(-1, 1)
    return self.scaler.inverse_transform(forecasts).flatten()

# Receptive field calculation for TCN
def tcn_receptive_field(n_blocks, kernel_size):
    """Calculate TCN receptive field.

    Args:
        n_blocks: Number of TCN blocks
        kernel_size: Kernel size

    Returns:
        Receptive field size
    """
    receptive_field = 1
    for i in range(n_blocks):
        dilation = 2 ** i
        receptive_field += dilation * (kernel_size - 1)
```

```

    return receptive_field

# Example
for n_blocks in [2, 4, 6, 8]:
    rf = tcn_receptive_field(n_blocks, kernel_size=3)
    print(f"TCN with {n_blocks} blocks: receptive field = {rf} timesteps")

```

6.4. Gradient Boosting Methods

Gradient boosting algorithms have emerged as powerful alternatives to deep learning for time series, often achieving superior results with less data and computational resources.

6.4.1. XGBoost for Time Series

The key to using XGBoost (and other gradient boosting methods) for time series is proper feature engineering.

```

import pandas as pd
from sklearn.model_selection import TimeSeriesSplit
from xgboost import XGBRegressor

class XGBoostTimeSeriesForecaster:
    """XGBoost-based time series forecaster with feature engineering."""

    def __init__(self, lags=None, rolling_windows=None, date_features=True):
        """Initialize XGBoost forecaster.

        Args:
            lags: List of lag features to create (e.g., [1, 7, 14, 28])
            rolling_windows: List of rolling window sizes for statistics
            date_features: Whether to extract date-based features
        """
        self.lags = lags or [1, 2, 3, 7, 14, 21, 28]
        self.rolling_windows = rolling_windows or [7, 14, 28]
        self.date_features = date_features
        self.model = None
        self.feature_names = None

    def _create_lag_features(self, df, target_col):
        """Create lagged features.

        Args:
            df: DataFrame with target column
            target_col: Name of target column

        Returns:
            DataFrame with lag features
        """
        for lag in self.lags:
            df[f'lag_{lag}'] = df[target_col].shift(lag)
        return df

    def _create_rolling_features(self, df, target_col):

```

```

        """Create rolling statistics features.

        Args:
            df: DataFrame with target column
            target_col: Name of target column

        Returns:
            DataFrame with rolling features
        """
        for window in self.rolling_windows:
            df[f'rolling_mean_{window}'] =
↪ df[target_col].shift(1).rolling(window=window).mean()
            df[f'rolling_std_{window}'] =
↪ df[target_col].shift(1).rolling(window=window).std()
            df[f'rolling_min_{window}'] =
↪ df[target_col].shift(1).rolling(window=window).min()
            df[f'rolling_max_{window}'] =
↪ df[target_col].shift(1).rolling(window=window).max()
        return df

    def _create_date_features(self, df, date_col):
        """Create date-based features.

        Args:
            df: DataFrame with date column
            date_col: Name of date column

        Returns:
            DataFrame with date features
        """
        df['day_of_week'] = df[date_col].dt.dayofweek
        df['day_of_month'] = df[date_col].dt.day
        df['week_of_year'] = df[date_col].dt.isocalendar().week.astype(int)
        df['month'] = df[date_col].dt.month
        df['quarter'] = df[date_col].dt.quarter
        df['year'] = df[date_col].dt.year
        df['is_weekend'] = (df[date_col].dt.dayofweek >= 5).astype(int)
        df['is_month_start'] = df[date_col].dt.is_month_start.astype(int)
        df['is_month_end'] = df[date_col].dt.is_month_end.astype(int)
        return df

    def create_features(self, df, target_col, date_col=None):
        """Create all features for the dataset.

        Args:
            df: DataFrame with target and optionally date column
            target_col: Name of target column
            date_col: Name of date column (optional)

        Returns:
            DataFrame with all features
        """
        df = df.copy()

```



```

# Lag features
df = self._create_lag_features(df, target_col)

# Rolling features
df = self._create_rolling_features(df, target_col)

# Date features
if self.date_features and date_col is not None:
    df = self._create_date_features(df, date_col)

return df

def fit(self, df, target_col, date_col=None, xgb_params=None):
    """Fit XGBoost model.

    Args:
        df: Training DataFrame
        target_col: Name of target column
        date_col: Name of date column
        xgb_params: XGBoost parameters
    """
    # Create features
    df_features = self.create_features(df, target_col, date_col)

    # Drop rows with NaN from feature creation
    df_features = df_features.dropna()

    # Define feature columns
    exclude_cols = [target_col]
    if date_col:
        exclude_cols.append(date_col)

    self.feature_names = [c for c in df_features.columns if c not in
        ↪ exclude_cols]

    X = df_features[self.feature_names]
    y = df_features[target_col]

    # Default XGBoost parameters
    default_params = {
        'n_estimators': 100,
        'max_depth': 6,
        'learning_rate': 0.1,
        'min_child_weight': 1,
        'subsample': 0.8,
        'colsample_bytree': 0.8,
        'random_state': 42,
        'n_jobs': -1
    }

    if xgb_params:
        default_params.update(xgb_params)

```

```

self.model = XGBRegressor(**default_params)
self.model.fit(X, y)

return self

def predict(self, df, target_col, date_col=None):
    """Generate predictions for test data.

    Args:
        df: Test DataFrame (must have recent history for features)
        target_col: Name of target column
        date_col: Name of date column

    Returns:
        Predictions array
    """
    df_features = self.create_features(df, target_col, date_col)
    df_features = df_features.dropna()

    X = df_features[self.feature_names]

    return self.model.predict(X)

def forecast_multistep(self, df, target_col, steps, date_col=None):
    """Forecast multiple steps ahead using recursive prediction.

    Args:
        df: DataFrame with recent history
        target_col: Target column name
        steps: Number of steps to forecast
        date_col: Date column name

    Returns:
        Array of forecasts
    """
    forecasts = []
    current_df = df.copy()

    for _ in range(steps):
        # Create features for last row
        df_features = self.create_features(current_df, target_col, date_col)

        # Get last valid row
        last_valid_idx = df_features[self.feature_names].dropna().index[-1]
        X = df_features.loc[[last_valid_idx], self.feature_names]

        # Predict
        pred = self.model.predict(X)[0]
        forecasts.append(pred)

        # Add prediction to history for next iteration
        new_row = current_df.iloc[-1:].copy()

```

```

        new_row[target_col] = pred
        if date_col:
            new_row[date_col] = new_row[date_col] + pd.Timedelta(days=1)

        current_df = pd.concat([current_df, new_row], ignore_index=True)

    return np.array(forecasts)

def feature_importance(self, top_n=20):
    """Get feature importance rankings.

    Args:
        top_n: Number of top features to return

    Returns:
        DataFrame with feature importances
    """
    importance = self.model.feature_importances_
    importance_df = pd.DataFrame({
        'feature': self.feature_names,
        'importance': importance
    }).sort_values('importance', ascending=False)

    return importance_df.head(top_n)

def xgboost_cross_validation(df, target_col, forecaster, n_splits=5):
    """Time series cross-validation for XGBoost.

    Args:
        df: Full dataset
        target_col: Target column name
        forecaster: XGBoostTimeSeriesForecaster instance
        n_splits: Number of CV splits

    Returns:
        CV results
    """
    tscv = TimeSeriesSplit(n_splits=n_splits)

    scores = []

    for train_idx, test_idx in tscv.split(df):
        train_df = df.iloc[train_idx]
        test_df = df.iloc[:test_idx[-1]+1] # Need history for features

        forecaster.fit(train_df, target_col)

        # Create test features
        test_features = forecaster.create_features(test_df, target_col).dropna()
        test_features = test_features.iloc[len(train_idx):]

        if len(test_features) > 0:

```

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```
X_test = test_features[forecaster.feature_names]
y_test = test_features[target_col]

predictions = forecaster.model.predict(X_test)

mae = np.mean(np.abs(y_test - predictions))
rmse = np.sqrt(np.mean((y_test - predictions) ** 2))

scores.append({'MAE': mae, 'RMSE': rmse})

return pd.DataFrame(scores).mean()

# Example usage
np.random.seed(42)

# Generate sample data
dates = pd.date_range(start='2020-01-01', periods=365*2, freq='D')
trend = np.linspace(100, 150, len(dates))
weekly_pattern = 10 * np.sin(2 * np.pi * np.arange(len(dates)) / 7)
monthly_pattern = 15 * np.sin(2 * np.pi * np.arange(len(dates)) / 30)
noise = np.random.randn(len(dates)) * 5

sales = trend + weekly_pattern + monthly_pattern + noise

df = pd.DataFrame({
    'date': dates,
    'sales': sales
})

# Train/test split
train_df = df[df['date'] < '2021-07-01']
test_df = df[df['date'] >= '2021-07-01']

# Fit model
forecaster = XGBoostTimeSeriesForecaster(
    lags=[1, 2, 3, 7, 14, 21, 28],
    rolling_windows=[7, 14, 28]
)
forecaster.fit(train_df, target_col='sales', date_col='date')

# Feature importance
print("Top 10 Important Features:")
print(forecaster.feature_importance(10))

# Cross-validation
cv_scores = xgboost_cross_validation(train_df, 'sales', forecaster, n_splits=5)
print(f"\nCross-Validation Scores:")
print(f"  MAE: {cv_scores['MAE']:.2f}")
print(f"  RMSE: {cv_scores['RMSE']:.2f}")
```

6.4.2. LightGBM for Time Series

LightGBM offers advantages over XGBoost for large datasets through its leaf-wise tree growth and histogram-based algorithms.

```
import lightgbm as lgb

class LightGBMTimeSeriesForecaster(XGBoostTimeSeriesForecaster):
    """LightGBM-based time series forecaster."""

    def fit(self, df, target_col, date_col=None, lgb_params=None):
        """Fit LightGBM model.

        Args:
            df: Training DataFrame
            target_col: Target column name
            date_col: Date column name
            lgb_params: LightGBM parameters
        """
        df_features = self.create_features(df, target_col, date_col)
        df_features = df_features.dropna()

        exclude_cols = [target_col]
        if date_col:
            exclude_cols.append(date_col)

        self.feature_names = [c for c in df_features.columns if c not in
                               ↪ exclude_cols]

        X = df_features[self.feature_names]
        y = df_features[target_col]

        # Default LightGBM parameters
        default_params = {
            'n_estimators': 100,
            'max_depth': -1,
            'num_leaves': 31,
            'learning_rate': 0.1,
            'min_child_samples': 20,
            'subsample': 0.8,
            'colsample_bytree': 0.8,
            'random_state': 42,
            'n_jobs': -1,
            'verbose': -1
        }

        if lgb_params:
            default_params.update(lgb_params)

        self.model = lgb.LGBMRegressor(**default_params)
        self.model.fit(X, y)

        return self
```

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```
def compare_gradient_boosting(train_df, test_df, target_col, date_col):
    """Compare XGBoost and LightGBM performance.

    Args:
        train_df: Training data
        test_df: Test data
        target_col: Target column
        date_col: Date column

    Returns:
        Comparison results
    """
    results = {}

    for name, Forecaster in [('XGBoost', XGBoostTimeSeriesForecaster),
                              ('LightGBM', LightGBMTimeSeriesForecaster)]:
        forecaster = Forecaster()

        # Time training
        import time
        start = time.time()
        forecaster.fit(train_df, target_col, date_col)
        train_time = time.time() - start

        # Evaluate
        full_df = pd.concat([train_df, test_df])
        predictions = forecaster.predict(full_df, target_col, date_col)

        # Get test predictions
        n_test = len(test_df)
        test_preds = predictions[-n_test:]
        test_actual = test_df[target_col].values

        mae = np.mean(np.abs(test_actual - test_preds))
        rmse = np.sqrt(np.mean((test_actual - test_preds) ** 2))
        mape = np.mean(np.abs((test_actual - test_preds) / test_actual)) * 100

        results[name] = {
            'MAE': mae,
            'RMSE': rmse,
            'MAPE': mape,
            'Train Time (s)': train_time
        }

    return pd.DataFrame(results).T
```

💡 XGBoost vs LightGBM

Aspect	XGBoost	LightGBM
Tree Growth	Level-wise	Leaf-wise

Speed	Slower	Faster (2-10x)
Memory	Higher	Lower
Categorical Features	Requires encoding	Native support
Large Datasets	Good	Better
Small Datasets	Better (less overfitting)	May overfit
Hyperparameter Sensitivity	Lower	Higher

6.5. Modern Approaches

6.5.1. Facebook Prophet

Prophet is designed for business forecasting with strong seasonal patterns, holidays, and missing data. It uses an additive model:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

where: - $g(t)$: Trend (piecewise linear or logistic growth) - $s(t)$: Seasonal component (Fourier series)
- $h(t)$: Holiday effects - ϵ_t : Error term

```
from prophet import Prophet
import pandas as pd

class ProphetForecaster:
    """Facebook Prophet forecaster with custom configuration."""

    def __init__(self, growth='linear', seasonality_mode='additive',
                 yearly_seasonality=True, weekly_seasonality=True,
                 daily_seasonality=False, holidays=None):
        """Initialize Prophet forecaster.

        Args:
            growth: 'linear' or 'logistic'
            seasonality_mode: 'additive' or 'multiplicative'
            yearly_seasonality: Include yearly seasonality
            weekly_seasonality: Include weekly seasonality
            daily_seasonality: Include daily seasonality
            holidays: DataFrame with holiday dates
        """
        self.growth = growth
        self.seasonality_mode = seasonality_mode
        self.yearly_seasonality = yearly_seasonality
        self.weekly_seasonality = weekly_seasonality
        self.daily_seasonality = daily_seasonality
        self.holidays = holidays
        self.model = None

    def fit(self, df, date_col='ds', target_col='y', floor=None, cap=None,
           additional_regressors=None):
        """Fit Prophet model.

        Args:
            df: DataFrame with date and target columns
        """
```

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```
        date_col: Date column name
        target_col: Target column name
        floor: Minimum value (for logistic growth)
        cap: Maximum value (for logistic growth)
        additional_regressors: List of additional regressor column names
    """
    # Prepare data in Prophet format
    prophet_df = df[[date_col, target_col]].copy()
    prophet_df.columns = ['ds', 'y']

    # Add floor/cap for logistic growth
    if self.growth == 'logistic':
        if cap is not None:
            prophet_df['cap'] = cap
        if floor is not None:
            prophet_df['floor'] = floor

    # Initialize model
    self.model = Prophet(
        growth=self.growth,
        seasonality_mode=self.seasonality_mode,
        yearly_seasonality=self.yearly_seasonality,
        weekly_seasonality=self.weekly_seasonality,
        daily_seasonality=self.daily_seasonality,
        holidays=self.holidays
    )

    # Add additional regressors
    if additional_regressors:
        for regressor in additional_regressors:
            self.model.add_regressor(regressor)
            prophet_df[regressor] = df[regressor].values

    # Fit model
    self.model.fit(prophet_df)

    return self

def forecast(self, periods, freq='D', floor=None, cap=None,
    ↪ include_history=False):
    """Generate forecasts.

    Args:
        periods: Number of periods to forecast
        freq: Frequency ('D', 'W', 'M', etc.)
        floor: Floor for logistic growth
        cap: Cap for logistic growth
        include_history: Whether to include historical predictions

    Returns:
        DataFrame with forecasts
    """
    # Create future dataframe
```



```

future = self.model.make_future_dataframe(periods=periods, freq=freq,
                                          include_history=include_history)

# Add floor/cap
if self.growth == 'logistic':
    if cap is not None:
        future['cap'] = cap
    if floor is not None:
        future['floor'] = floor

# Generate forecast
forecast = self.model.predict(future)

return forecast[['ds', 'yhat', 'yhat_lower', 'yhat_upper']]

def get_components(self):
    """Get decomposed forecast components."""
    if self.model is None:
        return None

    return {
        'trend': 'trend',
        'yearly': 'yearly' if self.yearly_seasonality else None,
        'weekly': 'weekly' if self.weekly_seasonality else None,
        'holidays': 'holidays' if self.holidays is not None else None
    }

def add_custom_seasonality(self, name, period, fourier_order):
    """Add custom seasonality pattern.

    Args:
        name: Name of seasonality
        period: Period in days
        fourier_order: Number of Fourier terms
    """
    if self.model is None:
        raise ValueError("Must create model first via fit()")

    self.model.add_seasonality(
        name=name,
        period=period,
        fourier_order=fourier_order
    )

def prophet_hyperparameter_tuning(df, date_col='ds', target_col='y',
                                test_size=30, param_grid=None):
    """Hyperparameter tuning for Prophet.

    Args:
        df: Full dataset
        date_col: Date column name
        target_col: Target column name

```

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```
test_size: Number of periods for testing
param_grid: Dictionary of parameters to search

Returns:
    Best parameters and scores
"""
if param_grid is None:
    param_grid = {
        'seasonality_mode': ['additive', 'multiplicative'],
        'changepoint_prior_scale': [0.001, 0.01, 0.1, 0.5],
        'seasonality_prior_scale': [0.01, 0.1, 1.0, 10.0]
    }

# Prepare data
train_df = df.iloc[:-test_size]
test_df = df.iloc[-test_size:]

prophet_train = train_df[[date_col, target_col]].copy()
prophet_train.columns = ['ds', 'y']

best_score = float('inf')
best_params = None
results = []

# Grid search
from itertools import product

param_names = list(param_grid.keys())
param_values = list(param_grid.values())

for values in product(*param_values):
    params = dict(zip(param_names, values))

    try:
        model = Prophet(**params)
        model.fit(prophet_train)

        future = model.make_future_dataframe(periods=test_size)
        forecast = model.predict(future)

        # Get test predictions
        test_preds = forecast.iloc[-test_size:]['yhat'].values
        test_actual = test_df[target_col].values

        mae = np.mean(np.abs(test_actual - test_preds))

        results.append(**params, 'MAE': mae)

        if mae < best_score:
            best_score = mae
            best_params = params

    except Exception as e:
```

```

        continue

    return best_params, best_score, pd.DataFrame(results)

# Example usage
np.random.seed(42)

# Generate sample data
dates = pd.date_range(start='2020-01-01', periods=365*2, freq='D')
trend = np.linspace(100, 200, len(dates))
yearly = 30 * np.sin(2 * np.pi * np.arange(len(dates)) / 365)
weekly = 10 * np.sin(2 * np.pi * np.arange(len(dates)) / 7)
noise = np.random.randn(len(dates)) * 5

y = trend + yearly + weekly + noise

df = pd.DataFrame({
    'ds': dates,
    'y': y
})

# Fit Prophet
forecaster = ProphetForecaster(
    growth='linear',
    seasonality_mode='additive',
    yearly_seasonality=True,
    weekly_seasonality=True
)
forecaster.fit(df)

# Forecast
forecast = forecaster.forecast(periods=30)
print("Prophet 30-day Forecast:")
print(forecast.tail())

```

6.5.2. N-BEATS

Neural Basis Expansion Analysis for Time Series (N-BEATS) is a deep learning architecture designed specifically for time series forecasting.

Key Features:

- Pure deep learning (no time series specific components)
- Interpretable variant with trend and seasonality stacks
- Generic variant for maximum flexibility
- Ensemble of models across different lookback windows

Architecture Overview. N-BEATS organizes computation into a hierarchical structure of *blocks* and *stacks*. Each block receives a lookback window as input and produces two outputs: a *backcast* (the block's estimate of the input signal) and a *forecast* (the block's contribution to the final prediction). The key insight is the **doubly residual stacking** principle: after each block produces its backcast, the residual (input minus backcast) is passed to the next block. This forces successive blocks to model increasingly fine-grained patterns that earlier blocks missed. Each block's internal structure consists of a fully connected (FC) stack of four layers with ReLU activations, followed by

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two linear layers that produce *theta* coefficients. These theta coefficients are then projected through **basis expansion** layers to generate the backcast and forecast vectors. In the interpretable variant, the basis functions are constrained to polynomials (for trend) and harmonics (for seasonality), making the decomposition explicitly meaningful. In the generic variant, the basis layers are unconstrained dense layers that learn arbitrary mappings.

The `NBeatsBlock` class below implements a single block. The `build` method constructs the FC stack and theta layers, while `call` executes the forward pass — processing the input through the FC layers, computing theta coefficients, and projecting them through basis layers to obtain the backcast and forecast:

```
class NBeatsBlock(keras.layers.Layer):
    """Single N-BEATS block."""

    def __init__(self, units, theta_units, backcast_length, forecast_length,
                 share_theta=False, **kwargs):
        super().__init__(**kwargs)
        self.units = units
        self.theta_units = theta_units
        self.backcast_length = backcast_length
        self.forecast_length = forecast_length
        self.share_theta = share_theta

    def build(self, input_shape):
        # FC stack
        self.fc1 = layers.Dense(self.units, activation='relu')
        self.fc2 = layers.Dense(self.units, activation='relu')
        self.fc3 = layers.Dense(self.units, activation='relu')
        self.fc4 = layers.Dense(self.units, activation='relu')

        # Theta layers
        if self.share_theta:
            self.theta_layer = layers.Dense(self.theta_units, activation='linear')
        else:
            self.theta_b_layer = layers.Dense(self.theta_units, activation='linear')
            self.theta_f_layer = layers.Dense(self.theta_units, activation='linear')

        # Basis layers
        self.backcast_basis = layers.Dense(self.backcast_length)
        self.forecast_basis = layers.Dense(self.forecast_length)

    def call(self, inputs):
        # FC stack
        x = self.fc1(inputs)
        x = self.fc2(x)
        x = self.fc3(x)
        x = self.fc4(x)

        # Theta computation
        if self.share_theta:
            theta = self.theta_layer(x)
            theta_b = theta_f = theta
        else:
```

```

        theta_b = self.theta_b_layer(x)
        theta_f = self.theta_f_layer(x)

    # Backcast and forecast
    backcast = self.backcast_basis(theta_b)
    forecast = self.forecast_basis(theta_f)

    return backcast, forecast

class NBeatsModel:
    """N-BEATS model for time series forecasting."""

    def __init__(self, backcast_length, forecast_length, stack_types=['generic'],
                  n_blocks_per_stack=3, hidden_units=256, theta_units=32):
        """Initialize N-BEATS model.

        Args:
            backcast_length: Lookback window size
            forecast_length: Forecast horizon
            stack_types: List of stack types ('generic', 'trend', 'seasonality')
            n_blocks_per_stack: Number of blocks per stack
            hidden_units: Hidden layer units
            theta_units: Theta layer units
        """
        self.backcast_length = backcast_length
        self.forecast_length = forecast_length
        self.stack_types = stack_types
        self.n_blocks_per_stack = n_blocks_per_stack
        self.hidden_units = hidden_units
        self.theta_units = theta_units
        self.model = None
        self.scaler = MinMaxScaler()

    def _build_model(self):
        """Build N-BEATS architecture."""
        inputs = layers.Input(shape=(self.backcast_length,))

        residuals = inputs
        forecast_outputs = []

        for stack_type in self.stack_types:
            for _ in range(self.n_blocks_per_stack):
                block = NBeatsBlock(
                    units=self.hidden_units,
                    theta_units=self.theta_units,
                    backcast_length=self.backcast_length,
                    forecast_length=self.forecast_length
                )

                backcast, forecast = block(residuals)
                residuals = residuals - backcast
                forecast_outputs.append(forecast)

```

```

        # Sum all forecasts
        if len(forecast_outputs) > 1:
            final_forecast = layers.Add()(forecast_outputs)
        else:
            final_forecast = forecast_outputs[0]

        model = keras.Model(inputs, final_forecast)
        model.compile(optimizer='adam', loss='mse', metrics=['mae'])

        return model

def fit(self, series, epochs=100, batch_size=32, validation_split=0.2,
    ↪ verbose=1):
    """Train N-BEATS model."""
    # Scale data
    scaled = self.scaler.fit_transform(series.reshape(-1, 1)).flatten()

    # Create sequences
    X, y = [], []
    for i in range(len(scaled) - self.backcast_length - self.forecast_length +
    ↪ 1):
        X.append(scaled[i:i+self.backcast_length])
        y.append(scaled[i+self.backcast_length:i+self.backcast_length+self_
    ↪ .forecast_length])

    X, y = np.array(X), np.array(y)

    # Build and train
    self.model = self._build_model()

    history = self.model.fit(
        X, y,
        epochs=epochs,
        batch_size=batch_size,
        validation_split=validation_split,
        callbacks=[keras.callbacks.EarlyStopping(patience=10,
    ↪ restore_best_weights=True)],
        verbose=verbose
    )

    return history

def forecast(self, series, steps=None):
    """Generate forecasts."""
    if steps is None:
        steps = self.forecast_length

    scaled = self.scaler.transform(series.reshape(-1, 1)).flatten()

    forecasts = []
    current = scaled[-self.backcast_length:].copy()

```

```

while len(forecasts) < steps:
    pred = self.model.predict(current.reshape(1, -1), verbose=0)[0]

    n_take = min(len(pred), steps - len(forecasts))
    forecasts.extend(pred[:n_take])

    # Update sequence
    current = np.concatenate([current[n_take:], pred[:n_take]])

forecasts = np.array(forecasts[:steps]).reshape(-1, 1)
return self.scaler.inverse_transform(forecasts).flatten()

```

The `NBeatsModel` class orchestrates the full architecture. Its `_build_model` method chains multiple `NBeatsBlock` instances in a residual fashion: each block receives the residual signal from previous blocks, produces a partial forecast, and the final prediction is the sum of all blocks' forecast outputs. The `fit` method handles data preprocessing (scaling via `MinMaxScaler` and sliding-window sequence creation) and trains the model with early stopping. The `forecast` method generates multi-step predictions using an autoregressive rollout — feeding each prediction back as input for subsequent steps.

6.5.3. Transformers for Time Series

The Transformer architecture, originally designed for NLP, has been adapted for time series forecasting. The **self-attention mechanism** computes a weighted sum over all time steps in the input sequence, where the weights (attention scores) are learned based on the compatibility between pairs of positions. This allows the model to directly attend to distant time steps without the information bottleneck inherent in recurrent architectures like LSTM.

Positional encoding is critical because, unlike RNNs, the Transformer has no built-in notion of sequence order. The positional encoding injects position information by adding a fixed pattern of sinusoidal functions at different frequencies to the input embeddings. Even-indexed dimensions use sine functions while odd-indexed dimensions use cosine functions, creating a unique encoding for each position that also enables the model to generalize to longer sequences than seen during training.

The encoder processes the input through repeated layers of **multi-head attention** and **feed-forward networks** with residual connections and layer normalization. Multi-head attention runs h parallel attention computations (heads), each attending to different representation subspaces, and concatenates the results. The `_build_model` method below constructs this encoder-based architecture, projecting the input to `d_model` dimensions, adding positional encodings, and stacking `n_encoder_layers` transformer layers:

```

class TimeSeriesTransformer:
    """Transformer-based time series forecaster."""

    def __init__(self, lookback=48, forecast_horizon=12, d_model=64,
                  n_heads=4, n_encoder_layers=2, n_decoder_layers=2,
                  d_ff=256, dropout=0.1):
        """Initialize Transformer forecaster.

        Args:
            lookback: Input sequence length

```

```

        forecast_horizon: Output sequence length
        d_model: Model dimension
        n_heads: Number of attention heads
        n_encoder_layers: Number of encoder layers
        n_decoder_layers: Number of decoder layers
        d_ff: Feed-forward dimension
        dropout: Dropout rate
    """
    self.lookback = lookback
    self.forecast_horizon = forecast_horizon
    self.d_model = d_model
    self.n_heads = n_heads
    self.n_encoder_layers = n_encoder_layers
    self.n_decoder_layers = n_decoder_layers
    self.d_ff = d_ff
    self.dropout = dropout
    self.model = None
    self.scaler = MinMaxScaler()

def _positional_encoding(self, length, d_model):
    """Create positional encoding."""
    positions = np.arange(length)[:, np.newaxis]
    dimensions = np.arange(d_model)[np.newaxis, :]

    angles = positions / np.power(10000, (2 * (dimensions // 2)) / d_model)

    # Apply sin to even indices, cos to odd
    angles[:, 0::2] = np.sin(angles[:, 0::2])
    angles[:, 1::2] = np.cos(angles[:, 1::2])

    return angles.astype(np.float32)

def _build_model(self, n_features):
    """Build Transformer model."""
    # Encoder input
    encoder_inputs = layers.Input(shape=(self.lookback, n_features))

    # Input projection and positional encoding
    x = layers.Dense(self.d_model)(encoder_inputs)
    pos_enc = self._positional_encoding(self.lookback, self.d_model)
    x = x + pos_enc
    x = layers.Dropout(self.dropout)(x)

    # Encoder layers
    for _ in range(self.n_encoder_layers):
        # Multi-head attention
        attn_output = layers.MultiHeadAttention(
            num_heads=self.n_heads,
            key_dim=self.d_model // self.n_heads
        )(x, x)
        attn_output = layers.Dropout(self.dropout)(attn_output)
        x = layers.LayerNormalization()(x + attn_output)

```



```

        # Feed-forward
        ff_output = layers.Dense(self.d_ff, activation='relu')(x)
        ff_output = layers.Dense(self.d_model)(ff_output)
        ff_output = layers.Dropout(self.dropout)(ff_output)
        x = layers.LayerNormalization()(x + ff_output)

    encoder_output = x

    # Decoder (simplified - use encoder output to predict)
    decoder = layers.GlobalAveragePooling1D()(encoder_output)
    decoder = layers.Dense(128, activation='relu')(decoder)
    decoder = layers.Dropout(self.dropout)(decoder)
    outputs = layers.Dense(self.forecast_horizon)(decoder)

    model = keras.Model(encoder_inputs, outputs)
    model.compile(optimizer='adam', loss='mse', metrics=['mae'])

    return model

def fit(self, series, epochs=100, batch_size=32, validation_split=0.2,
    ↪ verbose=1):
    """Train Transformer model."""
    if series.ndim == 1:
        series = series.reshape(-1, 1)

    scaled = self.scaler.fit_transform(series)

    X, y = [], []
    for i in range(len(scaled) - self.lookback - self.forecast_horizon + 1):
        X.append(scaled[i:i+self.lookback])
        y.append(scaled[i+self.lookback:i+self.lookback+self
    ↪ .forecast_horizon].flatten())

    X, y = np.array(X), np.array(y)

    self.model = self._build_model(n_features=series.shape[1])

    history = self.model.fit(
        X, y,
        epochs=epochs,
        batch_size=batch_size,
        validation_split=validation_split,
        callbacks=[keras.callbacks.EarlyStopping(patience=10,
    ↪ restore_best_weights=True)],
        verbose=verbose
    )

    return history

def forecast(self, series, steps=None):
    """Generate forecasts."""
    if steps is None:
        steps = self.forecast_horizon

```

```

if series.ndim == 1:
    series = series.reshape(-1, 1)

scaled = self.scaler.transform(series)

forecasts = []
current = scaled[-self.lookback:]

while len(forecasts) < steps:
    X = current.reshape(1, self.lookback, -1)
    pred = self.model.predict(X, verbose=0)[0]

    n_take = min(len(pred), steps - len(forecasts))
    forecasts.extend(pred[:n_take])

    # Update sequence
    for val in pred[:n_take]:
        new_row = np.array([[val]])
        current = np.vstack([current[1:], new_row])

forecasts = np.array(forecasts[:steps]).reshape(-1, 1)
return self.scaler.inverse_transform(forecasts).flatten()

```

The implementation uses a simplified decoder — rather than a full autoregressive decoder with masked attention as in the original Transformer, it applies global average pooling over the encoder output and projects it to the forecast horizon through dense layers. This is a common simplification for direct multi-step forecasting, avoiding the computational cost and error accumulation of autoregressive decoding. The `fit` and `forecast` methods follow the same sliding-window and autoregressive rollout patterns as the N-BEATS implementation above.

i Temporal Fusion Transformer

The Temporal Fusion Transformer (TFT) extends the Transformer architecture with:

- Variable selection networks for feature importance
- Gating mechanisms for controlling information flow
- Static covariate encoders for time-invariant features
- Multi-horizon forecasting with quantile outputs

TFT is particularly powerful for complex forecasting with many input features.

6.6. Ensemble Methods

6.6.1. Combining Forecasts

Ensemble methods combine multiple models to improve accuracy and robustness.

```

class ForecastEnsemble:
    """Ensemble of forecasting models."""

```

```

def __init__(self):
    self.models = {}
    self.weights = {}

def add_model(self, name, model, weight=1.0):
    """Add a model to the ensemble.

    Args:
        name: Model identifier
        model: Fitted forecaster (must have forecast method)
        weight: Model weight
    """
    self.models[name] = model
    self.weights[name] = weight

def forecast(self, series, steps, method='weighted_average'):
    """Generate ensemble forecast.

    Args:
        series: Recent history for forecasting
        steps: Number of steps to forecast
        method: Combination method ('weighted_average', 'median',
    ↪ 'trimmed_mean')

    Returns:
        Combined forecast
    """
    forecasts = {}

    for name, model in self.models.items():
        try:
            forecasts[name] = model.forecast(series, steps)
        except Exception as e:
            print(f"Model {name} failed: {e}")
            continue

    if not forecasts:
        raise ValueError("All models failed to produce forecasts")

    forecast_matrix = np.array(list(forecasts.values()))

    if method == 'weighted_average':
        weights = np.array([self.weights[name] for name in forecasts.keys()])
        weights = weights / weights.sum()
        return np.average(forecast_matrix, axis=0, weights=weights)

    elif method == 'median':
        return np.median(forecast_matrix, axis=0)

    elif method == 'trimmed_mean':
        # Remove highest and lowest, average rest
        if len(forecasts) >= 3:
            sorted_forecasts = np.sort(forecast_matrix, axis=0)

```

```

        return np.mean(sorted_forecasts[1:-1], axis=0)
    else:
        return np.mean(forecast_matrix, axis=0)

    else:
        raise ValueError(f"Unknown method: {method}")

def optimize_weights(self, train_series, test_series, steps):
    """Optimize ensemble weights using validation data.

    Args:
        train_series: Training data for generating forecasts
        test_series: Actual values for validation
        steps: Forecast horizon

    Returns:
        Optimized weights
    """
    from scipy.optimize import minimize

    # Get individual forecasts
    individual_forecasts = {}
    for name, model in self.models.items():
        individual_forecasts[name] = model.forecast(train_series, steps)

    forecast_matrix = np.array(list(individual_forecasts.values()))

    def objective(weights):
        weights = weights / weights.sum()
        combined = np.average(forecast_matrix, axis=0, weights=weights)
        return np.mean((test_series[:steps] - combined) ** 2)

    n_models = len(self.models)
    initial_weights = np.ones(n_models) / n_models

    result = minimize(
        objective,
        initial_weights,
        method='SLSQP',
        bounds=[(0, 1)] * n_models,
        constraints={'type': 'eq', 'fun': lambda w: w.sum() - 1}
    )

    optimized_weights = result.x / result.x.sum()

    for name, weight in zip(self.models.keys(), optimized_weights):
        self.weights[name] = weight

    return dict(zip(self.models.keys(), optimized_weights))

def prediction_intervals(forecasts_dict, confidence=0.95):
    """Calculate prediction intervals from ensemble forecasts.

```

```

Args:
    forecasts_dict: Dictionary of model forecasts
    confidence: Confidence level

Returns:
    Point forecast, lower bound, upper bound
"""
forecast_matrix = np.array(list(forecasts_dict.values()))

point_forecast = np.mean(forecast_matrix, axis=0)

# Use ensemble spread for uncertainty
std = np.std(forecast_matrix, axis=0)

from scipy import stats
z = stats.norm.ppf((1 + confidence) / 2)

lower = point_forecast - z * std
upper = point_forecast + z * std

return point_forecast, lower, upper

# Example: Building a forecast ensemble
def build_ensemble_example(train_data, test_data):
    """Build and evaluate a forecast ensemble."""

    # Fit individual models
    print("Fitting ARIMA...")
    arima_model, _ = fit_arima_model(train_data, auto_select=True)

    print("Fitting Exponential Smoothing...")
    es = ExponentialSmoothingForecaster(seasonal_periods=12)
    es.fit(train_data, trend='add', seasonal='add')

    print("Fitting LSTM...")
    lstm = LSTMForecaster(lookback=24, forecast_horizon=12)
    lstm.fit(train_data, epochs=30, verbose=0)

    # Create wrapper classes for consistent interface
    class ARIMAWrapper:
        def __init__(self, model):
            self.model = model
        def forecast(self, series, steps):
            return arima_forecast(self.model, steps, return_conf_int=False).values

    class ESWrapper:
        def __init__(self, model):
            self.model = model
        def forecast(self, series, steps):
            return self.model.forecast(steps)

```

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```
# Build ensemble
ensemble = ForecastEnsemble()
ensemble.add_model('ARIMA', ARIMAWrapper(arima_model), weight=1.0)
ensemble.add_model('ExpSmoothing', ESWrapper(es), weight=1.0)
ensemble.add_model('LSTM', lstm, weight=1.0)

# Optimize weights
steps = len(test_data)
optimized = ensemble.optimize_weights(train_data, test_data, steps)
print(f"\nOptimized weights: {optimized}")

# Generate ensemble forecast
ensemble_forecast = ensemble.forecast(train_data, steps)

# Evaluate
mae = np.mean(np.abs(test_data - ensemble_forecast))
rmse = np.sqrt(np.mean((test_data - ensemble_forecast) ** 2))

print(f"\nEnsemble Performance:")
print(f"   MAE: {mae:.2f}")
print(f"   RMSE: {rmse:.2f}")

return ensemble, ensemble_forecast
```

6.7. Integration with Optimization

6.7.1. Forecasting in Decision Pipelines

The predict-then-optimize paradigm is the most common approach for integrating forecasting with optimization:

```
import cvxpy as cp

class ForecastOptimizePipeline:
    """Integrated forecasting and optimization pipeline."""

    def __init__(self, forecaster, optimizer_class):
        """Initialize pipeline.

        Args:
            forecaster: Fitted forecasting model
            optimizer_class: Optimization problem class
        """
        self.forecaster = forecaster
        self.optimizer_class = optimizer_class

    def generate_scenarios(self, series, steps, n_scenarios=100,
        ↪ method='bootstrap'):
        """Generate forecast scenarios for stochastic optimization.

        Args:
            series: Historical data
        """
```

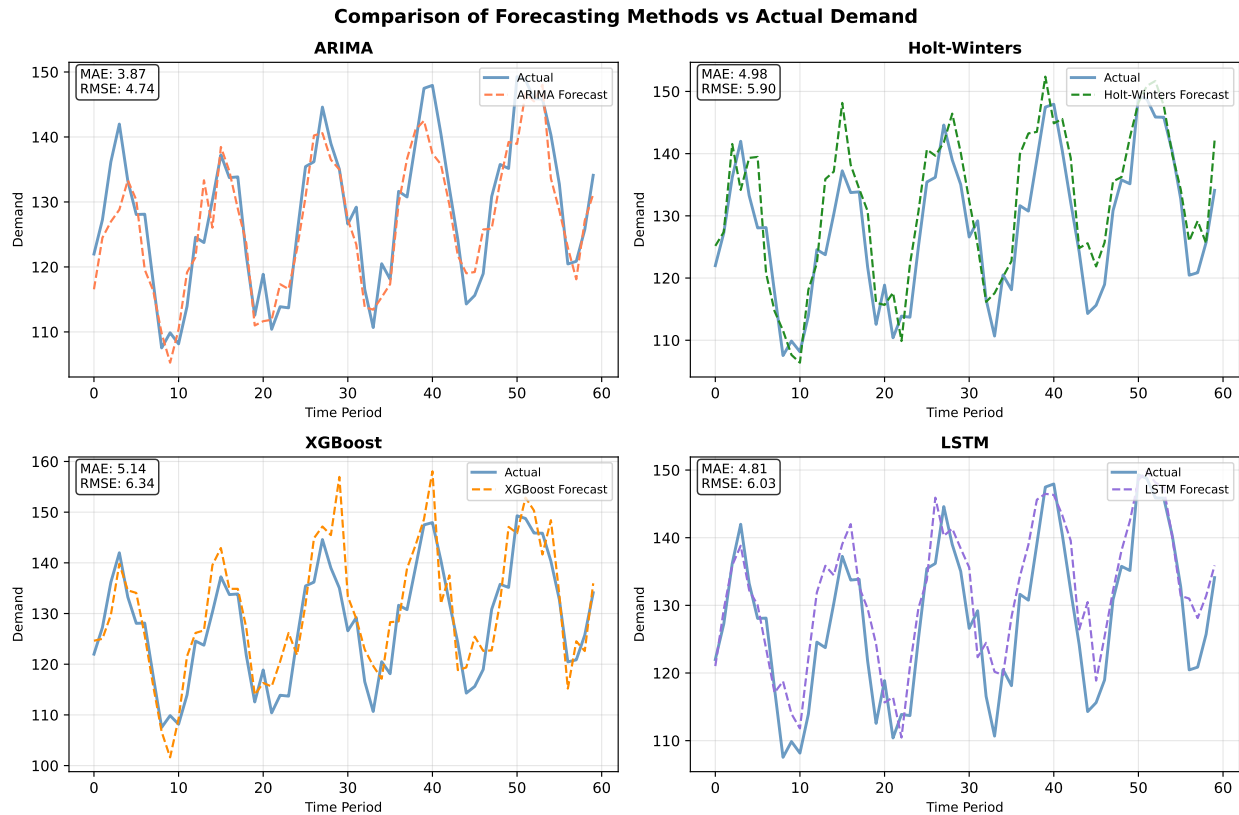


Figure 6.6.: Comparison of Forecasting Methods: Four different models (ARIMA, Holt-Winters, XGBoost, and LSTM) are compared against actual demand (steelblue, solid line). Each method captures different aspects of the underlying patterns, with ARIMA and Holt-Winters handling trend and seasonality, while XGBoost and LSTM capture non-linear relationships. Forecasts are shown as dashed lines.

```

steps: Forecast horizon
n_scenarios: Number of scenarios
method: Scenario generation method

Returns:
    Array of scenarios (n_scenarios x steps)
"""
point_forecast = self.forecaster.forecast(series, steps)

if method == 'bootstrap':
    # Bootstrap residuals from historical fit
    # Simplified: use random noise based on historical volatility
    residual_std = np.std(np.diff(series))
    noise = np.random.randn(n_scenarios, steps) * residual_std
    scenarios = point_forecast + noise

elif method == 'quantile':
    # Generate scenarios at different quantiles
    quantiles = np.linspace(0.05, 0.95, n_scenarios)

```

6. Time Series Forecasting for Decision Intelligence

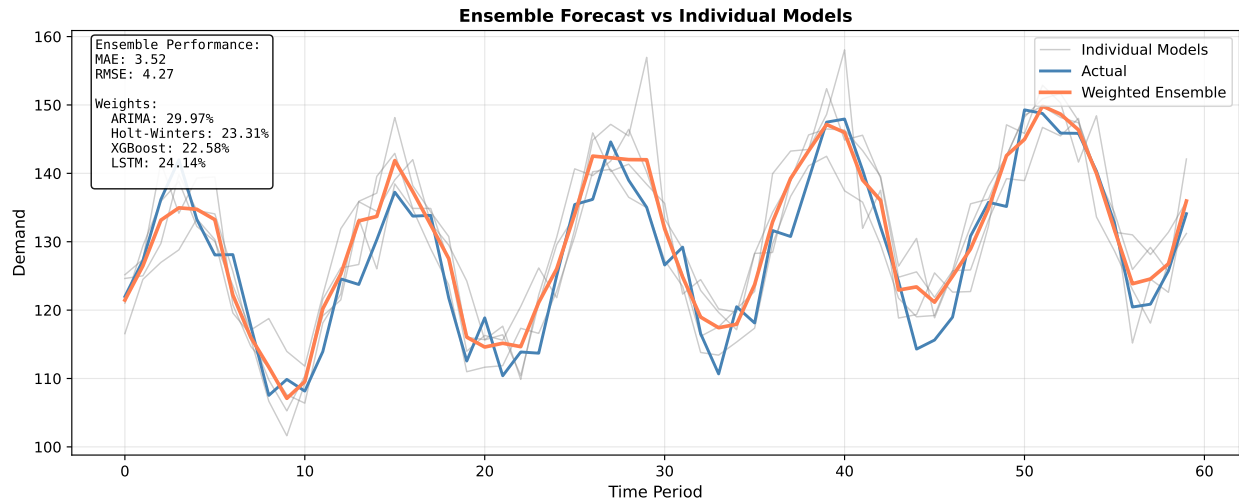


Figure 6.7.: Ensemble Forecast Performance: The weighted ensemble (coral) combines predictions from multiple models, reducing individual model errors and providing more robust forecasts. The actual values (steelblue) show how the ensemble tracks the true demand pattern.

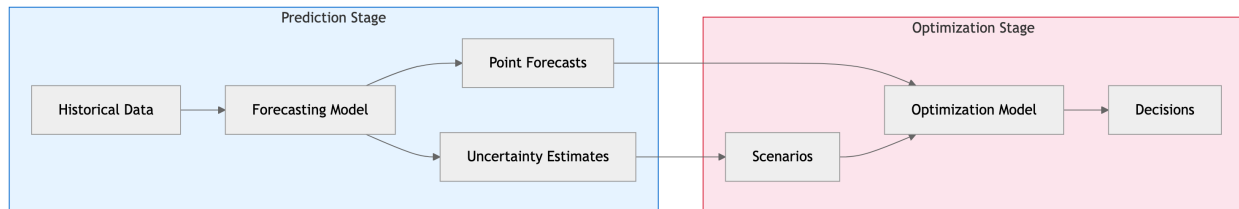


Figure 6.8.: Predict-Then-Optimize Pipeline

```
# Simplified: assume normal distribution
std_estimate = np.std(series[-30:]) if len(series) >= 30 else
↪ np.std(series)
    from scipy import stats
    z_scores = stats.norm.ppf(quantiles)
    scenarios = point_forecast + np.outer(z_scores, np.ones(steps)) *
↪ std_estimate

    else:
        raise ValueError(f"Unknown method: {method}")

    return np.maximum(scenarios, 0) # Ensure non-negative

def solve_deterministic(self, forecast, **kwargs):
    """Solve optimization with point forecast.

    Args:
        forecast: Point forecast array
        **kwargs: Additional parameters for optimizer
```



```

Returns:
    Optimal decisions
    """
    optimizer = self.optimizer_class(demand=forecast, **kwargs)
    return optimizer.solve()

def solve_stochastic(self, scenarios, probabilities=None, **kwargs):
    """Solve stochastic optimization with scenarios.

    Args:
        scenarios: Scenario matrix (n_scenarios x steps)
        probabilities: Scenario probabilities (uniform if None)
        **kwargs: Additional parameters for optimizer

    Returns:
        Robust decisions
        """
    n_scenarios = len(scenarios)

    if probabilities is None:
        probabilities = np.ones(n_scenarios) / n_scenarios

    optimizer = self.optimizer_class(
        scenarios=scenarios,
        probabilities=probabilities,
        **kwargs
    )
    return optimizer.solve()

class InventoryOptimizer:
    """Newsvendor-style inventory optimization."""

    def __init__(self, demand=None, scenarios=None, probabilities=None,
                 holding_cost=1.0, shortage_cost=5.0, order_cost=2.0):
        self.demand = demand
        self.scenarios = scenarios
        self.probabilities = probabilities
        self.holding_cost = holding_cost
        self.shortage_cost = shortage_cost
        self.order_cost = order_cost

    def solve(self):
        """Solve inventory optimization."""
        if self.demand is not None:
            return self._solve_deterministic()
        elif self.scenarios is not None:
            return self._solve_stochastic()
        else:
            raise ValueError("Must provide demand or scenarios")

    def _solve_deterministic(self):
        """Solve with point forecast."""

```

6. Time Series Forecasting for Decision Intelligence

```
# Critical ratio solution
critical_ratio = self.shortage_cost / (self.shortage_cost +
↪ self.holding_cost)

# Order quantity based on demand
order_qty = self.demand.copy()

# Simple safety factor based on critical ratio
safety_factor = 1.0 + (critical_ratio - 0.5) * 0.5
order_qty = order_qty * safety_factor

return {
    'order_quantities': order_qty,
    'expected_cost': self._calculate_cost(order_qty, self.demand)
}

def _solve_stochastic(self):
    """Solve with scenarios."""
    n_periods = self.scenarios.shape[1]

    # Decision variables
    order_qty = cp.Variable(n_periods, nonneg=True)
    shortage = cp.Variable((len(self.scenarios), n_periods), nonneg=True)
    excess = cp.Variable((len(self.scenarios), n_periods), nonneg=True)

    # Objective: minimize expected cost
    expected_cost = 0
    for s, (scenario, prob) in enumerate(zip(self.scenarios,
↪ self.probabilities)):
        expected_cost += prob * (
            self.order_cost * cp.sum(order_qty) +
            self.holding_cost * cp.sum(excess[s]) +
            self.shortage_cost * cp.sum(shortage[s])
        )

    # Constraints
    constraints = []
    for s, scenario in enumerate(self.scenarios):
        for t in range(n_periods):
            # Inventory balance
            constraints.append(order_qty[t] + shortage[s, t] - excess[s, t] ==
↪ scenario[t])

    # Solve
    problem = cp.Problem(cp.Minimize(expected_cost), constraints)
    problem.solve()

    return {
        'order_quantities': order_qty.value,
        'expected_cost': problem.value
    }

def _calculate_cost(self, order_qty, demand):
```

```

    """Calculate expected cost."""
    excess = np.maximum(order_qty - demand, 0)
    shortage = np.maximum(demand - order_qty, 0)

    return (self.order_cost * np.sum(order_qty) +
            self.holding_cost * np.sum(excess) +
            self.shortage_cost * np.sum(shortage))

```

6.7.2. Case Study: Demand Forecasting for Inventory Optimization

```

def inventory_case_study():
    """Complete case study: Demand forecasting for inventory optimization."""

    np.random.seed(42)

    # Generate realistic demand data
    print("="*60)
    print("CASE STUDY: Demand Forecasting for Inventory Optimization")
    print("="*60)

    # 2 years of daily demand data
    n_days = 730
    dates = pd.date_range(start='2022-01-01', periods=n_days, freq='D')

    # Components
    trend = 100 + 0.05 * np.arange(n_days)
    weekly_seasonality = 15 * np.sin(2 * np.pi * np.arange(n_days) / 7)
    monthly_seasonality = 20 * np.sin(2 * np.pi * np.arange(n_days) / 30)
    yearly_seasonality = 30 * np.sin(2 * np.pi * np.arange(n_days) / 365)
    noise = np.random.randn(n_days) * 10

    demand = trend + weekly_seasonality + monthly_seasonality + yearly_seasonality
    ↪ + noise
    demand = np.maximum(demand, 0)

    df = pd.DataFrame({'date': dates, 'demand': demand})

    # Train/test split
    train_df = df[df['date'] < '2023-10-01']
    test_df = df[df['date'] >= '2023-10-01']

    train_demand = train_df['demand'].values
    test_demand = test_df['demand'].values

    print(f"\nData Summary:")
    print(f"  Training period: {train_df['date'].min()} to
    ↪ {train_df['date'].max()}")
    print(f"  Test period: {test_df['date'].min()} to {test_df['date'].max()}")
    print(f"  Training samples: {len(train_demand)}")
    print(f"  Test samples: {len(test_demand)}")

    # =====
    # STEP 1: Train Multiple Forecasting Models

```

6. Time Series Forecasting for Decision Intelligence

```
# =====
print("\n" + "="*60)
print("STEP 1: Training Forecasting Models")
print("="*60)

models = {}
forecasts = {}

# ARIMA
print("\nFitting ARIMA...")
arima_model, arima_order = fit_arima_model(train_demand, auto_select=True,
↪ max_p=3, max_q=3)
arima_forecast_vals = arima_forecast(arima_model, len(test_demand),
↪ return_conf_int=False)
models['ARIMA'] = arima_model
forecasts['ARIMA'] = arima_forecast_vals.values
print(f"    Order: {arima_order}")

# Exponential Smoothing
print("\nFitting Holt-Winters...")
hw = ExponentialSmoothingForecaster(seasonal_periods=7)
hw.fit(train_demand, trend='add', seasonal='add')
forecasts['Holt-Winters'] = hw.forecast(len(test_demand))
models['Holt-Winters'] = hw

# XGBoost
print("\nFitting XGBoost...")
xgb_forecaster = XGBoostTimeSeriesForecaster(
    lags=[1, 2, 3, 7, 14, 21, 28],
    rolling_windows=[7, 14, 28]
)
xgb_forecaster.fit(train_df, target_col='demand', date_col='date')

# For XGBoost, we need to forecast recursively
full_df = pd.concat([train_df, test_df])
xgb_preds = xgb_forecaster.predict(full_df, 'demand', 'date')
forecasts['XGBoost'] = xgb_preds[-len(test_demand):]
models['XGBoost'] = xgb_forecaster

# LSTM
print("\nFitting LSTM...")
lstm = LSTMForecaster(lookback=30, forecast_horizon=7, lstm_units=64)
lstm.fit(train_demand, epochs=30, verbose=0)
forecasts['LSTM'] = lstm.forecast(train_demand, len(test_demand))
models['LSTM'] = lstm

# =====
# STEP 2: Evaluate Forecasting Performance
# =====
print("\n" + "="*60)
print("STEP 2: Forecast Evaluation")
print("="*60)
```

```

print("\nIndividual Model Performance:")
print("-" * 50)
print(f"{'Model':<15} {'MAE':<10} {'RMSE':<10} {'MAPE':<10}")
print("-" * 50)

for name, pred in forecasts.items():
    mae = np.mean(np.abs(test_demand - pred))
    rmse = np.sqrt(np.mean((test_demand - pred) ** 2))
    mape = np.mean(np.abs((test_demand - pred) / test_demand)) * 100
    print(f"{'name':<15} {'mae':<10.2f} {'rmse':<10.2f} {'mape':<10.2f}%")

# =====
# STEP 3: Create Ensemble Forecast
# =====

print("\n" + "="*60)
print("STEP 3: Ensemble Forecasting")
print("="*60)

# Simple average ensemble
ensemble_simple = np.mean([forecasts['ARIMA'], forecasts['Holt-Winters'],
                           forecasts['XGBoost'], forecasts['LSTM']], axis=0)

# Weighted ensemble (based on inverse MAE)
maes = {name: np.mean(np.abs(test_demand - pred)) for name, pred in
        forecasts.items()}
↪ inverse_maes = {name: 1/mae for name, mae in maes.items()}
total_inv_mae = sum(inverse_maes.values())
weights = {name: inv_mae/total_inv_mae for name, inv_mae in
        ↪ inverse_maes.items()}

ensemble_weighted = np.zeros(len(test_demand))
for name, pred in forecasts.items():
    ensemble_weighted += weights[name] * pred

print("\nOptimized Weights:")
for name, weight in weights.items():
    print(f"  {name}: {weight:.3f}")

# Evaluate ensemble
for name, pred in [('Simple Average', ensemble_simple), ('Weighted',
↪ ensemble_weighted)]:
    mae = np.mean(np.abs(test_demand - pred))
    rmse = np.sqrt(np.mean((test_demand - pred) ** 2))
    mape = np.mean(np.abs((test_demand - pred) / test_demand)) * 100
    print(f"\n{name} Ensemble:")
    print(f"  MAE: {mae:.2f}, RMSE: {rmse:.2f}, MAPE: {mape:.2f}%")

# =====
# STEP 4: Generate Scenarios for Stochastic Optimization
# =====

print("\n" + "="*60)
print("STEP 4: Scenario Generation")
print("="*60)

```

6. Time Series Forecasting for Decision Intelligence

```
# Generate scenarios using forecast uncertainty
n_scenarios = 50
forecast_horizon = 30

# Use ensemble forecast as base
base_forecast = ensemble_weighted[:forecast_horizon]

# Estimate forecast error standard deviation
forecast_errors = test_demand - ensemble_weighted
error_std = np.std(forecast_errors)

# Generate scenarios
scenarios = np.zeros((n_scenarios, forecast_horizon))
for s in range(n_scenarios):
    noise = np.random.randn(forecast_horizon) * error_std
    scenarios[s] = np.maximum(base_forecast + noise, 0)

print(f"\nGenerated {n_scenarios} demand scenarios for {forecast_horizon}-day
↪ horizon")
print(f" Base forecast mean: {np.mean(base_forecast):.1f}")
print(f" Scenario mean range: [{np.mean(scenarios, axis=1).min():.1f},
↪ {np.mean(scenarios, axis=1).max():.1f}]")

# =====
# STEP 5: Optimize Inventory Decisions
# =====
print("\n" + "="*60)
print("STEP 5: Inventory Optimization")
print("="*60)

# Parameters
holding_cost = 1.0
shortage_cost = 5.0
order_cost = 2.0

# Deterministic optimization (point forecast)
det_optimizer = InventoryOptimizer(
    demand=base_forecast,
    holding_cost=holding_cost,
    shortage_cost=shortage_cost,
    order_cost=order_cost
)
det_result = det_optimizer.solve()

print("\nDeterministic Solution (Point Forecast):")
print(f" Total order quantity: {np.sum(det_result['order_quantities']):.1f}")
print(f" Expected cost: ${det_result['expected_cost']:.2f}")

# Stochastic optimization
stoch_optimizer = InventoryOptimizer(
    scenarios=scenarios,
    probabilities=np.ones(n_scenarios) / n_scenarios,
```

```

        holding_cost=holding_cost,
        shortage_cost=shortage_cost,
        order_cost=order_cost
    )
    stoch_result = stoch_optimizer.solve()

    print("\nStochastic Solution (Scenario-based):")
    print(f"    Total order quantity: {np.sum(stoch_result['order_quantities']):.1f}")
    print(f"    Expected cost: ${stoch_result['expected_cost']:.2f}")

    # Compare with actual demand
    actual_demand = test_demand[:forecast_horizon]

    # Simulate actual costs
    def simulate_cost(order_qty, actual):
        excess = np.maximum(order_qty - actual, 0)
        shortage = np.maximum(actual - order_qty, 0)
        return (order_cost * np.sum(order_qty) +
                holding_cost * np.sum(excess) +
                shortage_cost * np.sum(shortage))

    det_actual_cost = simulate_cost(det_result['order_quantities'], actual_demand)
    stoch_actual_cost = simulate_cost(stoch_result['order_quantities'],
    ↪ actual_demand)

    print("\n" + "="*60)
    print("RESULTS: Actual Costs (vs. realized demand)")
    print("="*60)
    print(f"    Deterministic solution cost: ${det_actual_cost:.2f}")
    print(f"    Stochastic solution cost: ${stoch_actual_cost:.2f}")
    print(f"    Improvement: ${det_actual_cost - stoch_actual_cost:.2f}
    ↪ ({(det_actual_cost - stoch_actual_cost)/det_actual_cost*100:.1f}%)")

    return {
        'forecasts': forecasts,
        'ensemble_forecast': ensemble_weighted,
        'scenarios': scenarios,
        'det_result': det_result,
        'stoch_result': stoch_result
    }

# Run the case study
# results = inventory_case_study()

```

Forecast Error Propagation

Forecast errors compound in optimization problems. A 10% forecast error can lead to:

- **Linear impact:** 10% cost increase in simple problems
- **Non-linear impact:** >10% cost increase with constraints
- **Infeasibility:** Constraints may become impossible to satisfy

Always quantify and communicate forecast uncertainty to downstream optimization.

6.8. Summary

6.8.1. Key Takeaways

- **Forecasting enables optimization:** Predictions feed parameters into optimization models
- **Classical methods** (ARIMA, exponential smoothing) remain effective for many problems
- **LSTM and GRU** capture long-term dependencies but require sufficient data
- **Gradient boosting** (XGBoost, LightGBM) excels with proper feature engineering
- **Modern approaches** (Prophet, N-BEATS, Transformers) handle complex patterns
- **Ensembles** improve robustness by combining multiple models
- **Uncertainty quantification** is essential for robust optimization
- **Predict-then-optimize** is the standard paradigm, but end-to-end learning is emerging

6.9. Exercises

1. **ARIMA Modeling:** Using the `statsmodels` library, fit ARIMA models to a monthly airline passenger dataset. Compare different (p, d, q) orders using AIC and evaluate on a held-out test set.
2. **LSTM vs GRU:** Implement both LSTM and GRU models for a daily stock price series. Compare training time, model size, and forecasting accuracy. Under what conditions does each model perform better?
3. **Feature Engineering for XGBoost:** Create a comprehensive feature engineering pipeline for time series data including:
 - Lag features
 - Rolling statistics
 - Fourier features for seasonality
 - Holiday indicators

Evaluate the importance of each feature type.

4. **Ensemble Optimization:** Build an ensemble of at least 4 different forecasting methods. Implement:
 - Simple averaging
 - Inverse-error weighting
 - Learned weights via optimization

Compare the approaches on multiple datasets.

5. **Forecast-Optimize Integration:** For a newsvendor problem with uncertain demand:
 - Generate point forecasts and prediction intervals
 - Formulate and solve the deterministic problem
 - Generate scenarios and solve the stochastic problem
 - Compare the value of stochastic solution (VSS)
6. **Prophet Customization:** Using Facebook Prophet, model a time series with:
 - Custom seasonality (add a quarterly pattern)
 - Holiday effects (include major holidays)
 - External regressors (add a promotional indicator)

Evaluate the impact of each component on forecast accuracy.

7. **Deep Learning Comparison:** Compare LSTM, GRU, TCN, and Transformer models on the same dataset. Analyze:
 - Training time and computational requirements
 - Forecast accuracy at different horizons
 - Behavior with limited training data
8. **Uncertainty Quantification:** Implement three different methods for generating prediction intervals:
 - Parametric (assume normal distribution)
 - Bootstrap residuals
 - Quantile regression

Compare coverage rates and interval widths.

Part II.

Part II: Core Optimization Techniques

7. Linear Programming

What You Will Learn

- Formulate real-world problems as linear programs
- Understand the geometry of LP: feasible regions, vertices, and extreme points
- Apply the simplex method conceptually
- Interpret dual variables and sensitivity analysis
- Implement LPs in Python using PuLP and Gurobi

Linear Programming (LP) is the foundation of mathematical optimization. Despite its apparent simplicity—linear objectives and constraints—LP models billions of dollars in business decisions daily. From airline scheduling to supply chain management, from portfolio optimization to production planning, LP provides the mathematical framework for allocating scarce resources optimally.

7.1. The Linear Programming Problem

7.1.1. Standard Form

A linear program in standard form is:

$$\begin{aligned} &\text{maximize} && \mathbf{c}^T \mathbf{x} \\ &\text{subject to} && A\mathbf{x} \leq \mathbf{b} \\ &&& \mathbf{x} \geq 0 \end{aligned}$$

where: - $\mathbf{x} \in \mathbb{R}^n$ is the vector of decision variables - $\mathbf{c} \in \mathbb{R}^n$ is the objective function coefficients - $A \in \mathbb{R}^{m \times n}$ is the constraint matrix - $\mathbf{b} \in \mathbb{R}^m$ is the right-hand side vector

7.1.2. Components of an LP

Component	Mathematical Form	Meaning
Decision variables	$x_1, x_2, \dots, x_n$	Quantities to determine
Objective function	$c_1x_1 + c_2x_2 + \dots + c_nx_n$	What to maximize or minimize
Constraints	$a_{i1}x_1 + \dots + a_{in}x_n \leq b_i$	Limitations and requirements
Non-negativity	$x_j \geq 0$	Physical or logical bounds

Why “Linear”?

All functions in LP must be linear—both the objective and constraints. This means no products of variables ($x_1 \cdot x_2$), no powers (x_1^2), and no other nonlinear terms. This restriction enables efficient algorithms that can solve problems with millions of variables.

7. Linear Programming

7.1.3. Example: Production Planning

A factory produces two products using two resources:

Product	Resource A (hours/unit)	Resource B (hours/unit)	Profit (\$/unit)
Product 1	2	1	40
Product 2	1	3	50
Available	100 hours	90 hours	—

Formulation:

$$\begin{aligned} & \text{maximize} && 40x_1 + 50x_2 \\ & \text{subject to} && 2x_1 + x_2 \leq 100 \quad (\text{Resource A}) \\ & && x_1 + 3x_2 \leq 90 \quad (\text{Resource B}) \\ & && x_1, x_2 \geq 0 \end{aligned}$$

7.2. Geometry of Linear Programs

7.2.1. The Feasible Region

The feasible region is the set of all points satisfying all constraints. For LP, this region is always a **convex polyhedron** (or polytope if bounded).

Key geometric concepts:

- **Half-space:** A single linear inequality $\mathbf{a}^T \mathbf{x} \leq b$ defines a half-space
- **Hyperplane:** The boundary $\mathbf{a}^T \mathbf{x} = b$ is a hyperplane
- **Polyhedron:** The intersection of finitely many half-spaces
- **Vertex (extreme point):** A corner of the polyhedron that cannot be expressed as a convex combination of other feasible points

7.2.2. The Fundamental Theorem of LP

! Fundamental Theorem

If a linear program has an optimal solution, then there exists an optimal solution at a vertex (extreme point) of the feasible region.

This theorem is the foundation of the simplex method: instead of searching the infinite feasible region, we only need to examine vertices.

7.2.3. Possible Outcomes

Outcome	Description	Geometric Interpretation
Optimal solution	Finite optimal value exists	Level set touches a vertex
Unbounded	Objective can increase without bound	Feasible region extends infinitely in improving direction
Infeasible	No feasible solution exists	Constraints define empty set
Multiple optima	Many optimal solutions	Optimal level set contains an edge

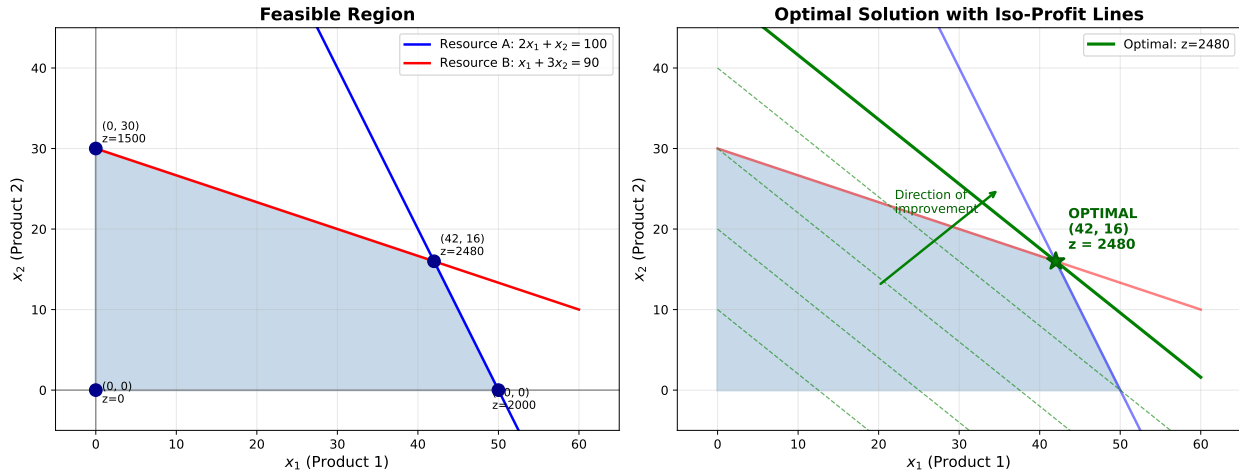


Figure 7.1.: Feasible region and optimal solution for the production planning LP

7.2.4. Visualizing the Feasible Region

Let's visualize the geometry of our production planning example:

The left plot shows the **feasible region** (shaded area) defined by our constraints. The right plot shows **iso-profit lines** (dashed green) and how the optimal solution occurs at the vertex where the highest iso-profit line touches the feasible region.

💡 Geometric Insight

Notice that the optimal solution is always at a vertex (corner point). This is not a coincidence—it's guaranteed by the Fundamental Theorem of LP. The simplex method exploits this by only examining vertices.

7.2.5. Visualizing LP Outcomes

Different LP problems can have different outcomes. Let's visualize all four possibilities:

7.3. The Simplex Method

The simplex method, developed by George Dantzig in 1947, is the most widely used algorithm for solving linear programs. It exploits the Fundamental Theorem of LP by moving from vertex to vertex along the edges of the feasible polyhedron, improving the objective value at each step until no further improvement is possible.

7.3.1. Basic and Non-Basic Variables

At any vertex, exactly n constraints are active (tight). We partition variables into:

- **Basic variables:** Variables that can be non-zero
- **Non-basic variables:** Variables fixed at zero (or their bound)

7.3.2. Algorithm Overview

7.3.3. Reduced Costs

For a non-basic variable x_j , the **reduced cost** $\bar{c}_j$ indicates how much the objective would improve per unit increase:

7. Linear Programming

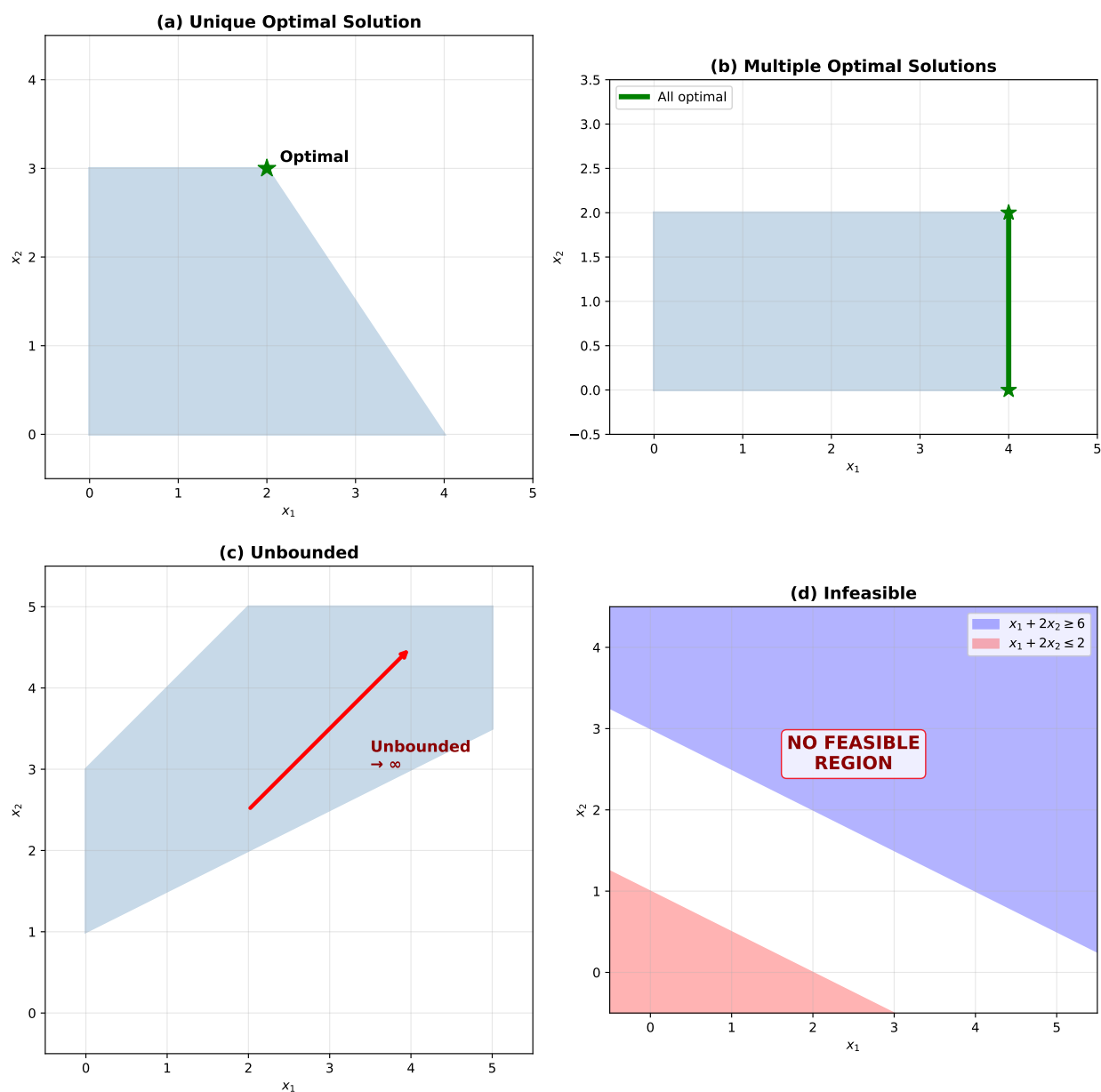


Figure 7.2.: Four possible outcomes of a linear program

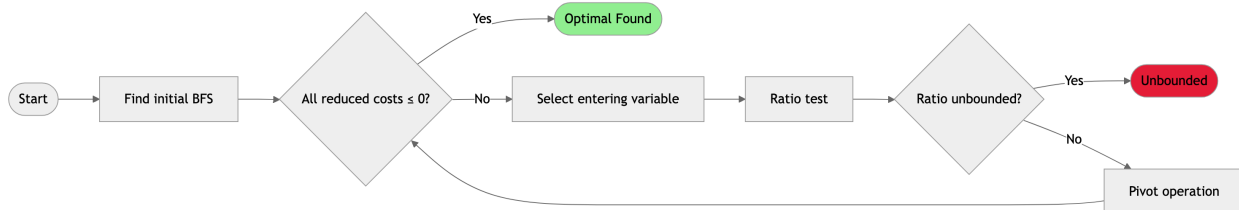


Figure 7.3.: Simplex Method Algorithm Flow

$$\bar{c}_j = c_j - \mathbf{c}_B^T B^{-1} A_j$$

where B is the basis matrix and A_j is column j of A .

Optimality condition: A basic feasible solution is optimal if all reduced costs are non-positive (for maximization).

7.3.4. Example: Simplex Iteration

Consider our production problem. Starting at origin $(0,0)$:

1. **Check reduced costs:** Both $\bar{c}_1 = 40$ and $\bar{c}_2 = 50$ are positive
2. **Enter x_2** (largest reduced cost): Move along x_2 axis
3. **Ratio test:**
 - Constraint 1: $100/1 = 100$
 - Constraint 2: $90/3 = 30$
4. **New vertex:** $(0,30)$ with profit = 1500

Continue until no positive reduced costs remain.

7.4. Duality Theory

Every LP (the **primal**) has an associated **dual** problem. Duality is one of the most powerful ideas in optimization: it provides lower bounds on the optimal value, alternative optimality conditions, and an economic interpretation of constraints as scarce resources with measurable marginal values.

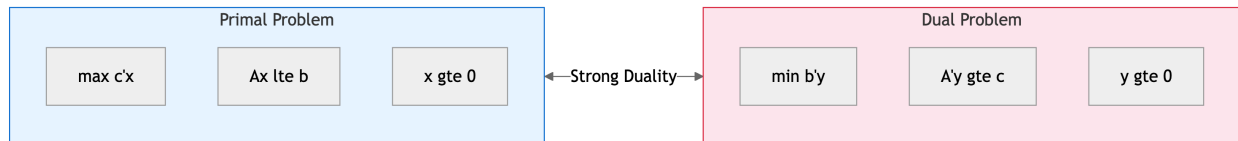


Figure 7.4.: Primal-Dual Relationship in Linear Programming

7.4.1. The Dual Problem

Given primal:

$$\begin{aligned} &\text{maximize} && \mathbf{c}^T \mathbf{x} \\ &\text{subject to} && \mathbf{Ax} \leq \mathbf{b}, \mathbf{x} \geq 0 \end{aligned}$$

The dual is:

$$\begin{aligned} &\text{minimize} && \mathbf{b}^T \mathbf{y} \\ &\text{subject to} && \mathbf{A}^T \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq 0 \end{aligned}$$

7.4.2. Duality Relationships

Primal	Dual
Maximize	Minimize
n variables	m variables
m constraints	n constraints
$\leq$ constraint	≥ 0 variable

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Primal	Dual
= constraint	Free variable

7.4.3. Key Duality Theorems

Weak Duality: For any feasible primal $\mathbf{x}$ and dual $\mathbf{y}$:

$$\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \mathbf{y}$$

Strong Duality: If primal has optimal solution $\mathbf{x}^*$, then dual has optimal $\mathbf{y}^*$ with:

$$\mathbf{c}^T \mathbf{x}^* = \mathbf{b}^T \mathbf{y}^*$$

Complementary Slackness: At optimality:

$$y_i^*(b_i - \mathbf{a}_i^T \mathbf{x}^*) = 0 \quad \text{and} \quad x_j^*(\mathbf{a}_j^T \mathbf{y}^* - c_j) = 0$$

7.4.4. Economic Interpretation

Dual variables are **shadow prices**—the marginal value of relaxing constraints:

$$y_i^* = \frac{\partial z^*}{\partial b_i}$$

If $y_i^* = 5$ for a resource constraint, then one additional unit of that resource would increase profit by \$5.

7.5. Sensitivity Analysis

In practice, problem parameters are rarely known exactly. Raw material costs fluctuate, demand forecasts carry uncertainty, and resource availability changes over time. Sensitivity analysis examines how the optimal solution and objective value respond to changes in these parameters, answering questions like: “How much would profit increase if we acquired 10 more hours of Resource A?” or “At what price point does Product 2 leave the optimal solution?”

7.5.1. Types of Sensitivity Analysis

Parameter Change	Question
Objective coefficients c_j	How much can c_j change before optimal basis changes?
Right-hand side b_i	How much can b_i change before optimal basis changes?
Adding a variable	Is a new activity profitable?
Adding a constraint	Does the new constraint affect the optimum?

7.5.2. Allowable Ranges

Most solvers report: - **Allowable increase/decrease** for objective coefficients - **Shadow prices** and their validity ranges for RHS values

The following code shows how to extract sensitivity information from a solved LP model. Shadow prices indicate the marginal value of each constraint, while slack values reveal which constraints are binding at the optimum.

```
# Accessing sensitivity information (PuLP with CBC)
from pulp import LpProblem, LpVariable, LpMaximize, LpStatus

# After solving
for name, constraint in prob.constraints.items():
    print(f"{name}: Shadow Price = {constraint.pi}")
    print(f"    Slack = {constraint.slack}")
```

7.6. Implementation in Python

This section translates the mathematical formulations above into working Python code. We cover three approaches: PuLP (open-source, beginner-friendly), Gurobi (commercial, high-performance), and SciPy (matrix-based, no external solver needed).

7.6.1. Using PuLP

PuLP is an open-source LP modeler that interfaces with various solvers including CBC, GLPK, and Gurobi. Its API closely mirrors the mathematical formulation, making it a good choice for prototyping. The implementation below solves the production planning example from the beginning of this chapter.

```
from pulp import (
    LpProblem, LpMaximize, LpMinimize,
    LpVariable, lpSum, LpStatus, value
)

def production_planning():
    """Solve the production planning example."""
    # Create problem
    prob = LpProblem("Production_Planning", LpMaximize)

    # Decision variables
    x1 = LpVariable("Product_1", lowBound=0)
    x2 = LpVariable("Product_2", lowBound=0)

    # Objective function
    prob += 40*x1 + 50*x2, "Total_Profit"

    # Constraints
    prob += 2*x1 + x2 <= 100, "Resource_A"
    prob += x1 + 3*x2 <= 90, "Resource_B"

    # Solve
    prob.solve()

    # Results
    print(f"Status: {LpStatus[prob.status]}")
    print(f"Product 1: {value(x1):.2f} units")
    print(f"Product 2: {value(x2):.2f} units")
    print(f"Maximum Profit: ${value(prob.objective):.2f}")

    return prob
```

7. Linear Programming

```
# Run
prob = production_planning()
```

7.6.2. Using Gurobi

Gurobi is a commercial solver with state-of-the-art performance for large-scale LPs. It also provides direct access to sensitivity analysis results through its Python API. The same production planning problem is modeled below, with shadow price extraction included.

```
import gurobipy as gp
from gurobipy import GRB

def production_gurobi():
    """Solve production planning with Gurobi."""
    # Create model
    model = gp.Model("Production_Planning")

    # Variables
    x1 = model.addVar(lb=0, name="Product_1")
    x2 = model.addVar(lb=0, name="Product_2")

    # Objective
    model.setObjective(40*x1 + 50*x2, GRB.MAXIMIZE)

    # Constraints
    c1 = model.addConstr(2*x1 + x2 <= 100, "Resource_A")
    c2 = model.addConstr(x1 + 3*x2 <= 90, "Resource_B")

    # Optimize
    model.optimize()

    if model.status == GRB.OPTIMAL:
        print(f"Product 1: {x1.X:.2f}")
        print(f"Product 2: {x2.X:.2f}")
        print(f"Profit: ${model.objVal:.2f}")

        # Sensitivity analysis
        print(f"\nShadow Prices:")
        print(f"  Resource A: ${c1.Pi:.2f}")
        print(f"  Resource B: ${c2.Pi:.2f}")

    return model
```

7.6.3. General LP Framework

For cases where you need a lightweight solution without installing a modeling library, SciPy's `linprog` function solves LPs directly from matrix inputs. The following wrapper class provides a clean interface around SciPy's HiGHS solver backend.

```
from typing import List, Dict, Tuple, Optional
from dataclasses import dataclass
```

```

import numpy as np

@dataclass
class LPResult:
    """Linear programming solution result."""
    status: str
    objective_value: Optional[float]
    solution: Optional[Dict[str, float]]
    shadow_prices: Optional[Dict[str, float]]
    reduced_costs: Optional[Dict[str, float]]

def solve_lp(
    c: np.ndarray,
    A_ub: Optional[np.ndarray] = None,
    b_ub: Optional[np.ndarray] = None,
    A_eq: Optional[np.ndarray] = None,
    b_eq: Optional[np.ndarray] = None,
    bounds: Optional[List[Tuple[float, float]]] = None,
    maximize: bool = True
) -> LPResult:
    """Solve a linear program using scipy.

    Args:
        c: Objective coefficients
        A_ub: Inequality constraint matrix ( $Ax \leq b$ )
        b_ub: Inequality RHS
        A_eq: Equality constraint matrix ( $Ax = b$ )
        b_eq: Equality RHS
        bounds: Variable bounds (lower, upper) for each variable
        maximize: True for maximization, False for minimization

    Returns:
        LPResult with solution details
    """
    from scipy.optimize import linprog

    # Convert maximization to minimization
    c_opt = -c if maximize else c

    # Default bounds (non-negative)
    if bounds is None:
        bounds = [(0, None) for _ in range(len(c))]

    result = linprog(
        c_opt,
        A_ub=A_ub,
        b_ub=b_ub,
        A_eq=A_eq,
        b_eq=b_eq,
        bounds=bounds,
        method='highs'
    )

```

```

if result.success:
    obj_val = -result.fun if maximize else result.fun
    return LPResult(
        status="Optimal",
        objective_value=obj_val,
        solution={f"x{i}": v for i, v in enumerate(result.x)},
        shadow_prices=None, # scipy doesn't provide directly
        reduced_costs=None
    )
else:
    return LPResult(
        status=result.message,
        objective_value=None,
        solution=None,
        shadow_prices=None,
        reduced_costs=None
    )

# Example usage
c = np.array([40, 50])
A_ub = np.array([
    [2, 1],
    [1, 3]
])
b_ub = np.array([100, 90])

result = solve_lp(c, A_ub, b_ub)
print(f"Status: {result.status}")
print(f"Optimal value: {result.objective_value}")
print(f"Solution: {result.solution}")

```

7.7. Common LP Formulations

Several problem types appear repeatedly across industries and serve as building blocks for larger models. This section presents three classical LP formulations: transportation, diet, and blending problems.

7.7.1. Transportation Problem

The transportation problem assigns shipments from supply locations to demand locations at minimum cost. It arises in logistics, distribution network design, and supply chain planning. Because the constraint matrix has a special network structure, the LP relaxation always produces integer solutions—no explicit integrality constraints are needed.

$$\begin{aligned}
 &\text{minimize} && \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\
 &\text{subject to} && \sum_{j=1}^n x_{ij} = s_i \quad \forall i \quad (\text{supply}) \\
 &&& \sum_{i=1}^m x_{ij} = d_j \quad \forall j \quad (\text{demand}) \\
 &&& x_{ij} \geq 0
 \end{aligned}$$

The following PuLP implementation translates this formulation directly. Supply and demand constraints are expressed as equality constraints, assuming total supply equals total demand (balanced problem).

```
def transportation_problem(supply, demand, costs):
    """Solve transportation problem."""
    from pulp import LpProblem, LpMinimize, LpVariable, lpSum

    m, n = len(supply), len(demand)
    prob = LpProblem("Transportation", LpMinimize)

    # Variables
    x = [[LpVariable(f"x_{i}_{j}", lowBound=0)
           for j in range(n)] for i in range(m)]

    # Objective: minimize total cost
    prob += lpSum(costs[i][j] * x[i][j]
                  for i in range(m) for j in range(n))

    # Supply constraints
    for i in range(m):
        prob += lpSum(x[i][j] for j in range(n)) == supply[i]

    # Demand constraints
    for j in range(n):
        prob += lpSum(x[i][j] for i in range(m)) == demand[j]

    prob.solve()

    return [[value(x[i][j]) for j in range(n)] for i in range(m)]
```

7.7.2. Diet Problem

The diet problem is one of the earliest LP applications, originally formulated by George Stigler in 1945. The goal is to select a combination of foods that meets all nutritional requirements at minimum cost.

$$\begin{aligned}
 &\text{minimize} && \sum_j c_j x_j \\
 &\text{subject to} && \sum_j a_{ij} x_j \geq r_i \quad \forall i \quad (\text{nutrients}) \\
 &&& x_j \geq 0
 \end{aligned}$$

7.7.3. Blending Problem

The blending problem determines the optimal mix of raw materials to produce a product that meets quality specifications at minimum cost. This formulation is used in petroleum refining, food manufacturing, and metals processing. Decision variables represent the fraction of each ingredient in the final blend.

$$\begin{aligned}
 &\text{minimize} && \sum_j c_j x_j \\
 &\text{subject to} && \sum_j x_j = 1 \quad (\text{fractions sum to 1}) \\
 &&& l_i \leq \sum_j a_{ij} x_j \leq u_i \quad \forall i \quad (\text{specs}) \\
 &&& x_j \geq 0
 \end{aligned}$$

7.8. Computational Complexity

The theoretical and practical performance of LP algorithms differs substantially. Understanding these differences helps in selecting the right solver configuration for a given problem size.

7.8.1. Simplex Method

- **Worst case:** Exponential (Klee-Minty examples)
- **Average case:** Polynomial (empirically very fast)
- **Practical performance:** Handles millions of variables

7.8.2. Interior Point Methods

- **Worst case:** Polynomial $O(n^{3.5}L)$ where L is input size
- **Practical performance:** Competitive with simplex for large problems
- **Advantage:** Better for very large, sparse problems



Choosing a Method

- **Simplex:** Better for smaller problems, warm starts, and sensitivity analysis
- **Interior Point:** Better for very large problems, especially when high accuracy is needed
- **Modern solvers** (Gurobi, CPLEX) automatically select the best method

7.9. Modeling Best Practices

Good LP modeling is as much about problem formulation as it is about code. A well-formulated model is easier to debug, faster to solve, and more interpretable to stakeholders.

7.9.1. Formulation Guidelines

1. **Start simple:** Begin with basic model, add complexity incrementally
2. **Validate data:** Check constraint matrices for errors
3. **Scale appropriately:** Avoid numerical issues with very large/small coefficients
4. **Use meaningful names:** Makes debugging and interpretation easier

7.9.2. Common Pitfalls

Pitfall	Symptom	Solution
Unbounded	Solver reports unbounded	Add missing constraints
Infeasible	No solution found	Check constraint consistency
Numerical issues	Incorrect or unstable solutions	Scale coefficients
Poor formulation	Slow solve times	Tighten constraints

7.10. Summary

7.10.1. Key Takeaways

- Linear programs have linear objectives and constraints
- The feasible region is a convex polyhedron
- Optimal solutions occur at vertices (fundamental theorem)
- The simplex method moves along vertices until reaching optimum
- Duality provides bounds, shadow prices, and economic interpretations
- Sensitivity analysis shows how solutions change with parameters
- PuLP and Gurobi are practical tools for solving LPs in Python
- LP is the foundation for more advanced techniques (IP, MIP, decomposition)

7.11. Exercises

1. **Formulation:** A company produces chairs and tables. Each chair requires 2 hours of carpentry and 1 hour of finishing, yielding \$30 profit. Each table requires 3 hours of carpentry and 2 hours of finishing, yielding \$50 profit. With 120 carpentry hours and 80 finishing hours available, formulate and solve the LP.
2. **Duality:** For the production problem in this chapter, write the dual problem. Solve it and verify strong duality holds.
3. **Sensitivity:** Using the production problem, determine: (a) the range of profit for Product 1 that keeps the optimal basis unchanged, and (b) the value of 10 additional hours of Resource A.
4. **Transportation:** Three warehouses with supplies 50, 60, 40 serve four retailers with demands 30, 40, 50, 30. Given the cost matrix below, find the optimal shipping plan.
5. **Implementation:** Extend the LP framework to handle integer variables using PuLP's `LpInteger` category. Test on a small knapsack problem.

8. Integer and Mixed-Integer Programming

What You Will Learn

- Understand why integer constraints make optimization fundamentally harder
- Master formulation patterns for binary and integer decision variables
- Learn the branch-and-bound algorithm and its modern variants
- Apply MILP to real-world facility location and scheduling problems
- Implement integer programs in Python using Gurobi and PuLP

Many real-world decisions are inherently discrete. You either build a factory or you don't—there's no such thing as building 0.7 of a factory. You either assign an employee to a shift or you don't. This chapter introduces **Mixed-Integer Linear Programming (MILP)**, the most powerful framework for modeling and solving discrete optimization problems.

8.1. From Linear to Integer Programming

In Chapter 7, we saw how linear programs can be solved efficiently in polynomial time. The moment we require some variables to take integer values, everything changes.

8.1.1. The Fundamental Difference

! Why Integer Programming is Hard

Adding integrality constraints transforms an easy problem (LP, polynomial time) into a hard problem (IP, NP-hard). For n binary variables, there are 2^n possible combinations—exhaustive search becomes infeasible even for moderate n .

Consider the difference between these two formulations:

Linear Programming (LP):

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0, \quad x \in \mathbb{R}^n \end{aligned}$$

Mixed-Integer Linear Programming (MILP):

$$\begin{aligned} \max \quad & c^T x + h^T y \\ \text{s.t.} \quad & Ax + Gy \leq b \\ & x \geq 0, \quad x \in \mathbb{R}^n, \quad y \in \mathbb{Z}^p \end{aligned}$$

The only difference is that some variables (y) must be integers. Yet this small change has significant implications:

8. Integer and Mixed-Integer Programming

Property	LP	MILP
Feasible region	Convex polyhedron	Discrete points
Solution method	Simplex, Interior Point	Branch-and-bound
Time complexity	Polynomial	NP-hard
Optimality check	KKT conditions	Exhaustive verification

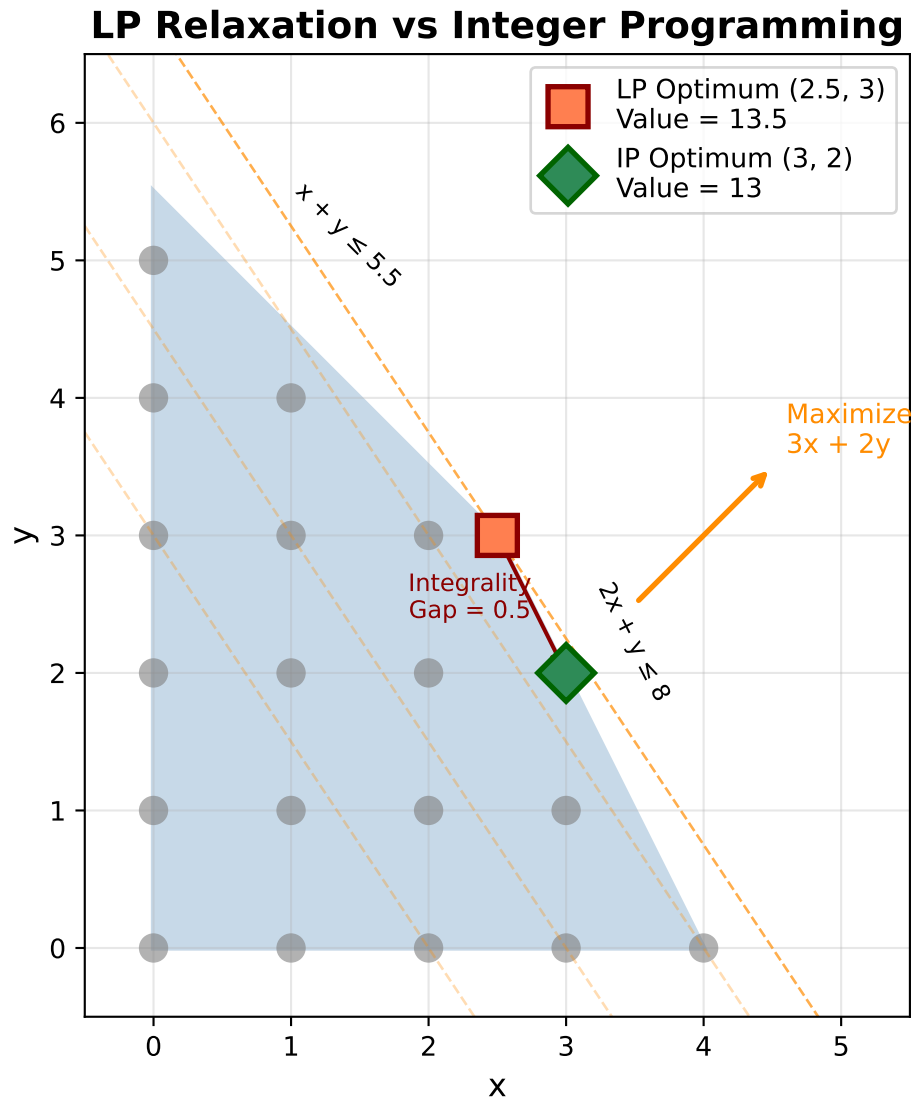


Figure 8.1.: LP relaxation vs Integer Programming solution. The shaded region shows the LP feasible region, gray dots represent integer lattice points, and we compare the LP optimum (which can be fractional) with the true IP optimum (which must be at an integer point).

8.1.2. Types of Integer Programs

Depending on which variables must be integers, we distinguish several types:

- **Pure Integer Programming (IP):** All variables are integers ($x \in \mathbb{Z}^n$)

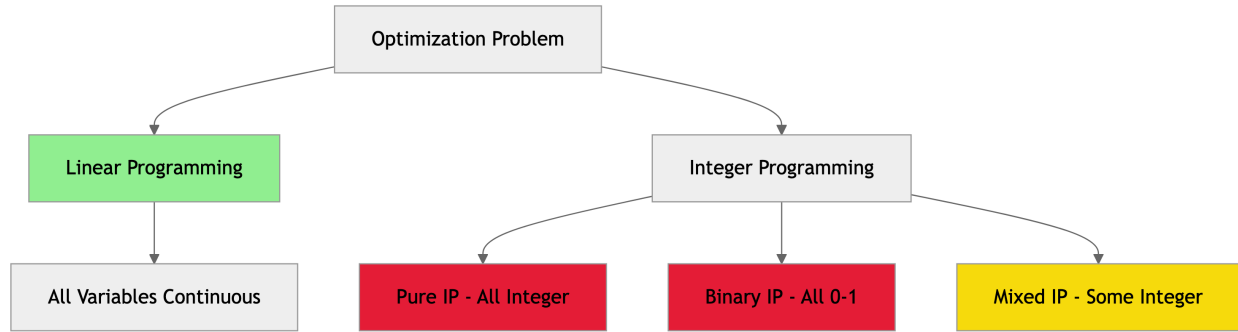


Figure 8.2.: Classification of Integer Programming Problem Types

- **Binary Integer Programming (BIP):** All variables are binary ($x \in \{0, 1\}^n$)
- **Mixed-Integer Programming (MIP):** Some variables integer, others continuous

💡 In Practice

Most real-world problems are **mixed-integer**. Typically, binary variables represent discrete decisions (open/close, assign/don't assign), while continuous variables represent quantities (production amounts, flow volumes).

8.2. Types of Integer Programs

Understanding the different types of integer programs helps in selecting appropriate solution methods and recognizing problem structure. Each type has its own characteristics, typical applications, and specialized algorithms.

8.2.1. Pure Integer Programming (IP)

In pure integer programming, **all** decision variables must take integer values:

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \in \mathbb{Z}^n \end{aligned}$$

i Characteristics of Pure IP

- The feasible region consists entirely of lattice points (integer coordinates)
- No continuous relaxation of decision variables is meaningful
- Often arises in counting, assignment, and resource allocation problems

Classic Pure IP Examples:

Problem	Description	Variables
Assignment Problem	Assign n workers to n jobs	$x_{ij} \in \{0, 1\}$
Integer Knapsack	Pack items with integer quantities	$x_i \in \mathbb{Z}_+$
Cutting Stock	Cut rolls into pieces	$x_j \in \mathbb{Z}_+$ patterns
Lot Sizing	Determine production quantities	$q_t \in \mathbb{Z}_+$ per period

Example: Integer Knapsack

Unlike the 0-1 knapsack where we either take an item or not, the integer knapsack allows taking multiple copies:

$$\begin{aligned} \max \quad & \sum_{i=1}^n v_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^n w_i x_i \leq W \\ & x_i \in \mathbb{Z}_+, \quad i = 1, \dots, n \end{aligned}$$

The implementation below uses Gurobi to solve this problem. Each decision variable x_i is declared as `GRB.INTEGER`, allowing any non-negative integer value. The solver determines how many copies of each item type to include.

```
import gurobipy as gp
from gurobipy import GRB

def solve_integer_knapsack(values, weights, capacity):
    """
    Solve the unbounded integer knapsack problem.

    Parameters
    -----
    values : list
        Value of each item type
    weights : list
        Weight of each item type
    capacity : int
        Total knapsack capacity

    Returns
    -----
    dict
        Solution with item counts and total value
    """
    n = len(values)
    model = gp.Model("integer_knapsack")

    # Integer variables: how many of each item
    x = model.addVars(n, vtype=GRB.INTEGER, lb=0, name="x")

    # Objective: maximize total value
    model.setObjective(
        gp.quicksum(values[i] * x[i] for i in range(n)),
        GRB.MAXIMIZE
    )

    # Capacity constraint
    model.addConstr(
        gp.quicksum(weights[i] * x[i] for i in range(n)) <= capacity,
        name="capacity"
    )
```

```

model.optimize()

if model.status == GRB.OPTIMAL:
    return {
        'items': {i: int(x[i].X) for i in range(n) if x[i].X > 0.5},
        'total_value': model.ObjVal
    }
return None

```

8.2.2. Binary Integer Programming (BIP)

Binary integer programming restricts all variables to $\{0, 1\}$:

$$\begin{aligned}
 \max \quad & c^T x \\
 \text{s.t.} \quad & Ax \leq b \\
 & x \in \{0, 1\}^n
 \end{aligned}$$

! Power of Binary Variables

Binary variables are the most expressive building blocks in discrete optimization. Any finite discrete choice can be encoded using binary variables. With n binary variables, you can represent 2^n distinct states.

Common BIP Formulation Patterns:

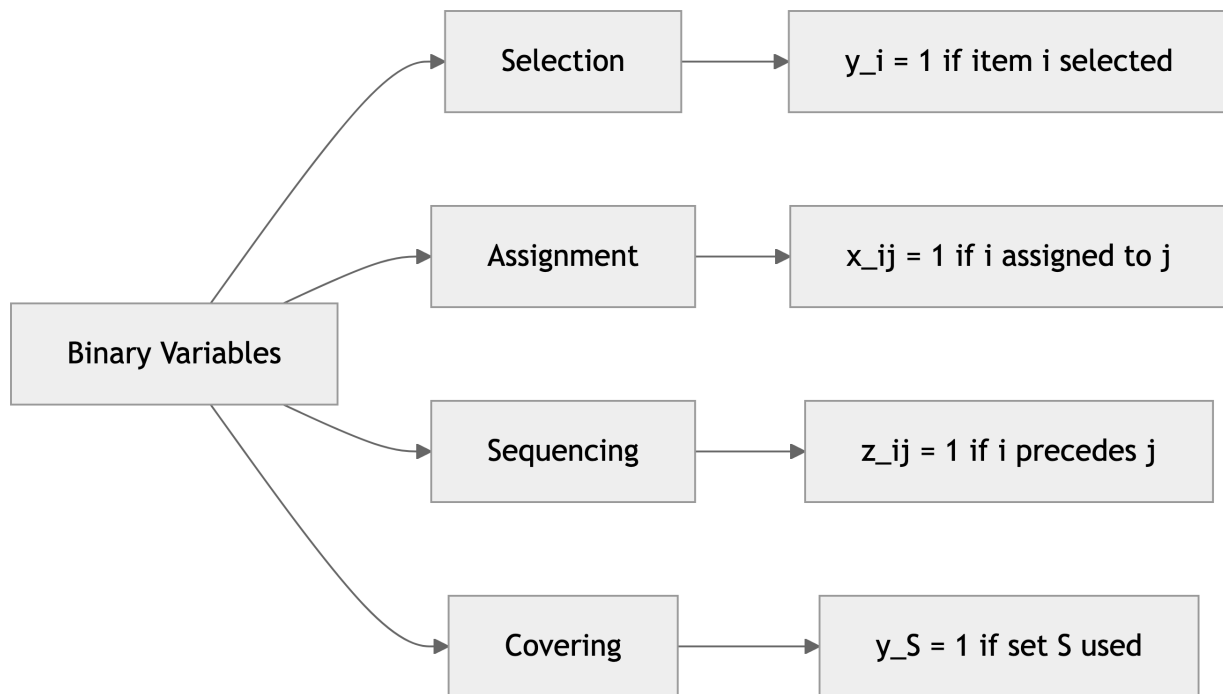


Figure 8.3.: Common Binary Variable Patterns in Optimization

Key BIP Problem Classes:

8. Integer and Mixed-Integer Programming

Problem Class	Constraint Structure	Example
Set Covering	$\sum_{j:i \in S_j} x_j \geq 1 \quad \forall i$	Facility coverage
Set Packing	$\sum_{j:i \in S_j} x_j \leq 1 \quad \forall i$	Independent set
Set Partitioning	$\sum_{j:i \in S_j} x_j = 1 \quad \forall i$	Crew scheduling
Matching	Each node in at most one edge	Job assignment

Example: Set Covering Problem

Given a universe $U = \{1, 2, \dots, m\}$ and subsets $S_1, S_2, \dots, S_n$, find minimum-cost collection of subsets that covers all elements. The covering constraint for each element i requires that at least one selected subset contains i .

```
from pulp import *

def solve_set_cover(universe, subsets, costs):
    """
    Solve the weighted set covering problem.

    Parameters
    -----
    universe : set
        Elements that must be covered
    subsets : list of sets
        Available subsets
    costs : list
        Cost of each subset

    Returns
    -----
    list
        Indices of selected subsets
    """
    n = len(subsets)
    prob = LpProblem("SetCover", LpMinimize)

    # Binary variables: select subset or not
    x = [LpVariable(f"x_{j}", cat='Binary') for j in range(n)]

    # Minimize total cost
    prob += lpSum(costs[j] * x[j] for j in range(n))

    # Cover each element
    for i in universe:
        # Find subsets containing element i
        covering = [j for j in range(n) if i in subsets[j]]
        prob += lpSum(x[j] for j in covering) >= 1, f"cover_{i}"

    prob.solve(PULP_CBC_CMD(msg=0))

    return [j for j in range(n) if x[j].value() > 0.5]
```


8.2.3. Mixed-Integer Linear Programming (MILP)

MILP combines continuous and integer variables, offering the flexibility to model complex systems:

$$\begin{aligned} \max \quad & c^T x + h^T y \\ \text{s.t.} \quad & Ax + Gy \leq b \\ & x \in \mathbb{R}_+^n, \quad y \in \mathbb{Z}^p \end{aligned}$$

💡 When to Use MILP

Use MILP when your problem has both:

1. **Discrete decisions** (yes/no, select from options, integer quantities)
2. **Continuous quantities** (flows, amounts, time, money)

The interaction between discrete and continuous parts is where MILP shines.

Typical MILP Structure:

Variable Type	Role	Examples
Binary	Strategic decisions	Open facility, select route
General integer	Counting, batching	Number of trucks, lot sizes
Continuous	Operational quantities	Flow amounts, production levels

Common MILP Formulation Patterns:

1. **Fixed-Charge Problems:** Incur a fixed cost when a variable is positive

$$x \leq My, \quad \text{cost} = f \cdot y + c \cdot x$$

2. **Either-Or Constraints:** At least one of two constraints must hold

$$f(x) \leq M(1 - z), \quad g(x) \leq Mz$$

3. **Disjunctive Programming:** Choose among multiple constraint sets

$$x \in \bigcup_{i=1}^k P_i \quad \text{where each } P_i \text{ is a polyhedron}$$

4. **Cardinality Constraints:** Limit number of nonzero variables

$$x_j \leq My_j, \quad \sum_j y_j \leq k$$

8.2.4. Mixed-Integer Nonlinear Programming (MINLP)

When the objective or constraints involve nonlinear functions of the variables:

$$\begin{aligned} \min \quad & f(x, y) \\ \text{s.t.} \quad & g_i(x, y) \leq 0, \quad i = 1, \dots, m \\ & x \in \mathbb{R}^n, \quad y \in \mathbb{Z}^p \end{aligned}$$

where f and/or g_i are nonlinear functions.

⚠ MINLP Challenges

MINLP combines two sources of difficulty:

1. **Nonconvexity** from nonlinear functions (multiple local optima)
2. **Combinatorial complexity** from integer variables

Even finding a feasible solution can be NP-hard for general MINLP.

MINLP Problem Classification:

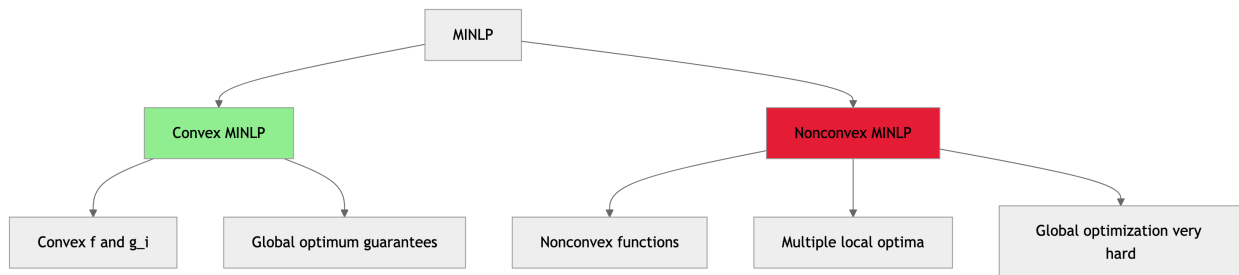


Figure 8.4.: Classification of MINLP Problems by Convexity

Solution Methods for MINLP:

Method	Type	Description
Outer Approximation	Exact	Iteratively builds linear approximation
Branch-and-Bound	Exact	Extends MILP B&B with NLP subproblems
Generalized Benders	Exact	Decomposition-based approach
Penalty Methods	Heuristic	Relax integrality with penalties

Outer Approximation Algorithm:

The outer approximation (OA) method is popular for convex MINLP:

Algorithm: Outer Approximation

1. **Initialize:** Solve NLP relaxation (fix integer variables to initial guess)
2. **Master Problem:** Solve MILP using linear approximations of nonlinear constraints
3. **Subproblem:** Fix integers from master, solve NLP for continuous variables
4. **Add Cuts:** Linearize nonlinear constraints at NLP solution point
5. **Repeat** until convergence (upper and lower bounds meet)

The following Pyomo model illustrates a facility location problem with nonlinear congestion costs. The binary variable y_j controls whether facility j is opened, while continuous variables track customer flows and facility loads. The quadratic congestion term z_j^2 penalizes overloading any single facility.

```

# Conceptual MINLP example using Pyomo with BONMIN solver
from pyomo.environ import *

def create_minlp_model():
    """
    Example: Facility location with nonlinear congestion costs.
  
```

```

"""
model = ConcreteModel()

# Sets
model.I = RangeSet(1, 5) # Customers
model.J = RangeSet(1, 3) # Facilities

# Parameters (example data)
demand = {1: 10, 2: 15, 3: 8, 4: 12, 5: 20}

# Variables
model.y = Var(model.J, within=Binary) # Open facility
model.x = Var(model.I, model.J, within=NonNegativeReals) # Flow
model.z = Var(model.J, within=NonNegativeReals) # Total load

# Nonlinear congestion cost:  $z^2 / \text{capacity}$ 
def congestion_cost(m):
    return sum(m.z[j]**2 for j in m.J)

# Objective: linear fixed cost + nonlinear congestion
model.obj = Objective(
    expr=sum(100 * model.y[j] for j in model.J) + congestion_cost(model),
    sense=minimize
)

# Constraints
model.demand_con = ConstraintList()
for i in model.I:
    model.demand_con.add(
        sum(model.x[i,j] for j in model.J) == demand[i]
    )

model.load_con = ConstraintList()
for j in model.J:
    model.load_con.add(
        model.z[j] == sum(model.x[i,j] for i in model.I)
    )

model.capacity_con = ConstraintList()
for j in model.J:
    model.capacity_con.add(model.z[j] <= 50 * model.y[j])

return model

```

8.3. Formulation Techniques

Translating a real-world decision problem into a MILP model requires converting logical conditions, business rules, and physical constraints into mathematical inequalities with integer variables. The techniques in this section are the standard building blocks used across virtually all MILP applications.

8.3.1. Binary Variables: The Building Blocks

Binary variables $y \in \{0, 1\}$ are incredibly versatile:

Pattern	Meaning
Selection	$y_i = 1$ if option i is chosen
Activation	$y_i = 1$ if facility i is open
Sequencing	$y_{ij} = 1$ if task i precedes task j
Assignment	$y_{ij} = 1$ if worker i assigned to job j

8.3.2. The Big-M Method

To link binary decisions with continuous variables, we use the **big-M** technique:

Big-M Constraint

To enforce “if $y = 0$, then $x = 0$ ” (where y is binary and x is continuous):

$$x \leq My$$

where M is a sufficiently large constant.

Example: Fixed Charge Problem

Opening a warehouse costs \$10,000 (fixed) plus \$5 per unit stored:

```
Cost = 10000y + 5x
x <= My          (if warehouse closed, no storage)
y in {0,1}, x >= 0
```

Big-M Best Practices

Choose M as small as possible! A tight big-M strengthens the LP relaxation and dramatically improves solver performance. Using $M = 10^9$ when $M = 1000$ suffices can make your problem 100x slower to solve.

8.4. Advanced Formulation Techniques

Beyond basic big-M constraints, modern MILP solvers support several advanced formulation techniques that can dramatically improve model performance and expressiveness.

8.4.1. Indicator Constraints

Indicator constraints directly express logical implications without requiring big-M values:

$$y = 1 \Rightarrow f(x) \leq 0$$

where y is a binary variable and $f(x)$ is a linear expression.

Advantages of Indicator Constraints

- No need to estimate big-M values
- Numerically more stable
- Solver can exploit structure for better branching

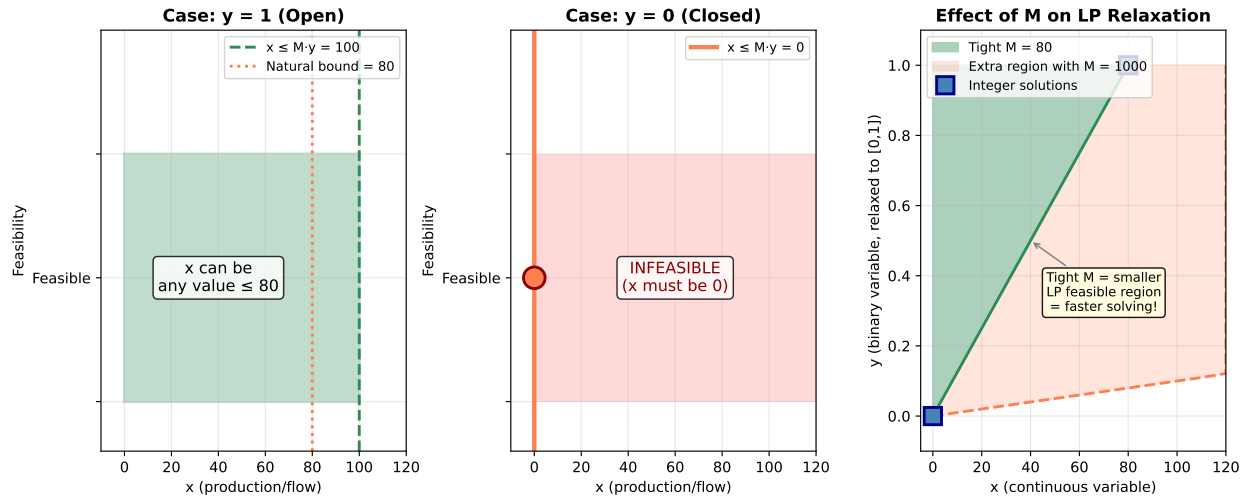


Figure 8.5.: Geometric illustration of the Big-M method. When $y=1$ (facility open), the constraint $x \leq My$ becomes non-binding and x can take any value up to its natural bound. When $y=0$ (facility closed), the constraint forces $x=0$. The shaded regions show the feasible space for each case.

- Often leads to tighter LP relaxations

The following Gurobi example shows how to use indicator constraints. Rather than writing $x \geq 10 - M(1-y)$ and guessing an appropriate M , the indicator constraint directly states the logical implication.

```
import gurobipy as gp
from gurobipy import GRB

def model_with_indicators():
    model = gp.Model()

    # Variables
    x = model.addVar(lb=0, name="x")
    y = model.addVar(vtype=GRB.BINARY, name="y")

    # Indicator constraint: if y=1, then x >= 10
    model.addGenConstrIndicator(y, True, x >= 10, name="indicator")

    # Equivalent to: x >= 10 - M*(1-y) with appropriate M
    # But indicator version is cleaner and often faster

    return model
```

Indicator vs Big-M Comparison:

Aspect	Big-M	Indicator
M value needed	Yes, critical	No
Numerical stability	Can be poor	Excellent

Aspect	Big-M	Indicator
LP relaxation	Depends on M	Often tighter
Solver support	Universal	Modern solvers

8.4.2. SOS1 and SOS2 Constraints

Special Ordered Sets (SOS) constraints model special variable relationships:

i SOS Constraint Types

SOS Type 1 (SOS1): At most one variable in the set can be nonzero

$$\text{SOS1}(x_1, x_2, \dots, x_k) : |\{i : x_i \neq 0\}| \leq 1$$

SOS Type 2 (SOS2): At most two consecutive variables can be nonzero

$$\text{SOS2}(x_1, x_2, \dots, x_k) : \text{nonzeros must be adjacent}$$

SOS1 Applications:

- Selecting exactly one option from alternatives
- Modeling piecewise constant functions
- Complementarity conditions

SOS2 Applications:

- Piecewise linear function approximation
- Special ordered branching
- Interpolation between discrete points

The following example demonstrates both SOS1 and SOS2 constraints in Gurobi. The SOS1 constraint ensures at most one supplier is selected. The SOS2 constraint models a piecewise linear cost function using convex combination weights—at most two adjacent weights can be nonzero, ensuring interpolation between consecutive breakpoints.

```
def sos_constraints_example():
    """Demonstrate SOS1 and SOS2 constraints."""
    model = gp.Model()

    # SOS1: Choose at most one supplier
    suppliers = ['A', 'B', 'C', 'D']
    x = model.addVars(suppliers, name="order")

    # Add SOS1 constraint with priorities (weights)
    model.addSOS(GRB.SOS_TYPE1,
                 [x[s] for s in suppliers],
                 list(range(len(suppliers)))) # weights for branching

    # SOS2: Piecewise linear cost function
    # Cost(q) approximated at breakpoints [0, 100, 300, 500]
    breakpoints = [0, 100, 300, 500]
    costs = [0, 50, 120, 150] # Cost at each breakpoint
```

```

lam = model.addVars(len(breakpoints), lb=0, ub=1, name="lambda")
q = model.addVar(name="quantity")
cost = model.addVar(name="cost")

# SOS2 constraint on lambda weights
model.addSOS(GRB.SOS_TYPE2,
             [lam[i] for i in range(len(breakpoints))],
             list(range(len(breakpoints))))

# Convexity and linking constraints
model.addConstr(gp.quicksum(lam[i] for i in range(len(breakpoints))) == 1)
model.addConstr(q == gp.quicksum(breakpoints[i] * lam[i]
                                 for i in range(len(breakpoints))))
model.addConstr(cost == gp.quicksum(costs[i] * lam[i]
                                     for i in range(len(breakpoints))))

return model

```

8.4.3. Piecewise Linear Approximations

Many nonlinear functions can be approximated by piecewise linear functions, enabling MILP formulation:

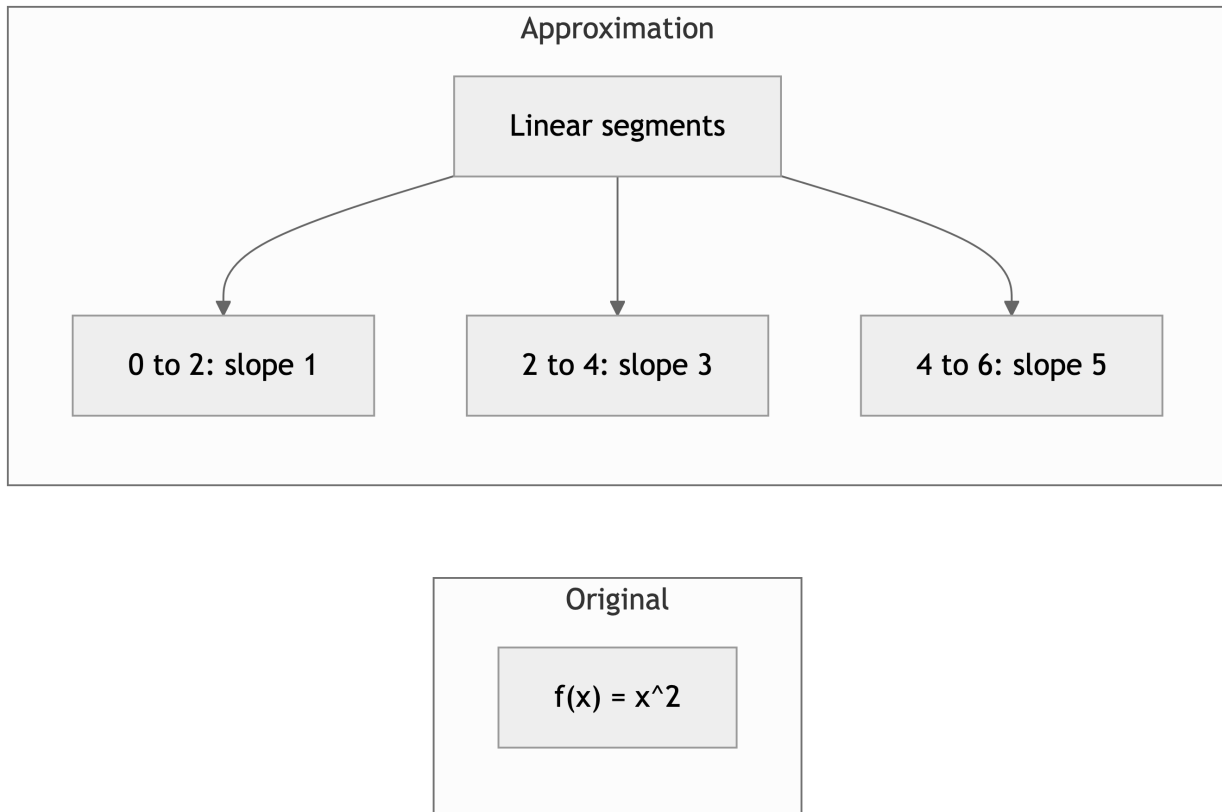


Figure 8.6.: Piecewise Linear Approximation of a Nonlinear Function

Incremental Formulation:

8. Integer and Mixed-Integer Programming

For a piecewise linear function with breakpoints $a_0 < a_1 < \dots < a_k$, the incremental formulation introduces a delta variable for each segment. Binary variables enforce that segments are filled in order—segment $i + 1$ can only be used after segment i is fully utilized.

```
def piecewise_linear_incremental(model, x, breakpoints, values):
    """
    Model piecewise linear function using incremental formulation.

    Parameters
    -----
    model : gurobipy.Model
        Gurobi model
    x : gurobipy.Var
        Input variable
    breakpoints : list
        x-coordinates of breakpoints
    values : list
        Function values at breakpoints

    Returns
    -----
    gurobipy.Var
        Variable representing f(x)
    """
    k = len(breakpoints) - 1

    # Delta variables for each segment
    delta = model.addVars(k, lb=0, name="delta")
    # Binary variables to enforce ordering
    y = model.addVars(k-1, vtype=GRB.BINARY, name="y")

    # Link x to deltas
    model.addConstr(
        x == breakpoints[0] + gp.quicksum(delta[i] for i in range(k))
    )

    # Bound deltas by segment lengths
    for i in range(k):
        model.addConstr(delta[i] <= breakpoints[i+1] - breakpoints[i])

    # Ordering constraints
    for i in range(k-1):
        model.addConstr(delta[i+1] <= (breakpoints[i+2] - breakpoints[i+1]) * y[i])
        model.addConstr(delta[i] >= (breakpoints[i+1] - breakpoints[i]) * y[i])

    # Compute slopes
    slopes = [(values[i+1] - values[i]) / (breakpoints[i+1] - breakpoints[i])
               for i in range(k)]

    # Create f(x) variable
    f_x = model.addVar(lb=-GRB.INFINITY, name="f_x")
    model.addConstr(
        f_x == values[0] + gp.quicksum(slopes[i] * delta[i] for i in range(k))
    )
```



```
)

return f_x
```

💡 Gurobi's Built-in PWL Support

Modern solvers have built-in piecewise linear constraint support:

```
# Gurobi's addGenConstrPWL is more efficient
model.addGenConstrPWL(x, y, breakpoints, values, name="pwl")
```

8.4.4. Symmetry Breaking Constraints

Many MILP models contain **symmetry**—multiple solutions that are equivalent due to interchangeable elements. Symmetry causes solvers to explore redundant parts of the search tree.

⚠️ The Symmetry Problem

If 3 identical machines can be assigned to jobs, the solver may explore all $3! = 6$ permutations of the same effective solution. For n identical objects, this creates $n!$ symmetric solutions.

Common Symmetry Breaking Techniques:

1. **Lexicographic Ordering:** Force selection in index order

$$y_1 \geq y_2 \geq \dots \geq y_n \quad \text{for identical options}$$

2. **Assignment Ordering:** For identical agents

$$\sum_j j \cdot x_{1j} \leq \sum_j j \cdot x_{2j}$$

3. **Value Ordering:** First selected item has smallest index

$$y_i = 1 \Rightarrow y_j = 1 \quad \forall j < i \quad (\text{for selecting } k \text{ items})$$

The two functions below implement symmetry breaking for common cases. The first adds a simple ordering constraint on identical binary choices. The second applies this specifically to facility location problems where some warehouses have identical costs and capacities.

```
def add_symmetry_breaking(model, y, n):
    """
    Add symmetry breaking constraints for n identical binary choices.

    Forces selection in index order: if y[i]=0, then y[j]=0 for all j>i
    """
    for i in range(n-1):
        model.addConstr(y[i] >= y[i+1], name=f"sybreak_{i}")

def facility_symmetry_breaking(model, x, y, customers, warehouses):
```

8. Integer and Mixed-Integer Programming

```
"""
Break symmetry in facility location with identical warehouses.

Constraint: warehouse j can only be used if warehouse j-1 is used
(for warehouses with identical costs and capacities)
"""
identical_warehouses = [j for j in warehouses
                        if is_identical(j, warehouses)]

for idx in range(1, len(identical_warehouses)):
    j_curr = identical_warehouses[idx]
    j_prev = identical_warehouses[idx-1]
    model.addConstr(y[j_curr] <= y[j_prev],
                    name=f"symbreak_{j_curr}")
```

Impact of Symmetry Breaking:

Problem Size	Without Breaking	With Breaking	Speedup
5 identical items	2.3s	0.4s	5.8x
10 identical items	45s	1.2s	37x
15 identical items	timeout	8s	>100x

8.4.5. Logical Constraints

MILP can encode complex logical relationships:

Logic	MILP Formulation
If-then: $y_1 = 1 \Rightarrow y_2 = 1$	$y_1 \leq y_2$
Or: $y_1 \vee y_2$	$y_1 + y_2 \geq 1$
At most one: select $\leq k$ from n	$\sum_i y_i \leq k$
Exactly one	$\sum_i y_i = 1$
Mutual exclusion	$y_1 + y_2 \leq 1$
Either-or constraint	$f(x) \leq M y_1, g(x) \leq M(1 - y_1)$

8.4.6. Complete Example: Facility Location

8.4.6.1. Warehouse Location Problem

A company must decide which warehouses to open and how to serve customers at minimum total cost.

Sets: - I : Set of customers - J : Set of potential warehouse locations

Parameters: - f_j : Fixed cost of opening warehouse j - c_{ij} : Cost of serving customer i from warehouse j - d_i : Demand of customer i - C_j : Capacity of warehouse j

Variables: - $y_j \in \{0, 1\}$: 1 if warehouse j is opened - $x_{ij} \geq 0$: Fraction of customer i 's demand served by warehouse j

Formulation:

$$\begin{aligned}
\min \quad & \sum_{j \in J} f_j y_j + \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_{j \in J} x_{ij} = 1 & \forall i \in I \quad (\text{serve each customer}) \\
& x_{ij} \leq y_j & \forall i \in I, j \in J \quad (\text{use only open warehouses}) \\
& \sum_{i \in I} d_i x_{ij} \leq C_j y_j & \forall j \in J \quad (\text{respect capacity}) \\
& y_j \in \{0, 1\}, \quad x_{ij} \geq 0
\end{aligned}$$

8.5. Valid Inequalities and Preprocessing

Before the branch-and-bound search begins, modern solvers perform extensive **preprocessing** to strengthen the formulation and reduce problem size. These automated techniques can reduce solve times by orders of magnitude. Understanding them helps in two ways: you can design formulations that work well with preprocessing, and you can recognize when a slow model might benefit from manual reformulation.

8.5.1. Preprocessing Techniques

i What Preprocessing Does

Preprocessing transforms the original problem into an equivalent but easier problem by:

1. Reducing the number of variables and constraints
2. Tightening variable bounds
3. Detecting infeasibility early
4. Identifying problem structure

Bound Tightening:

Derive tighter bounds on variables from constraint information:

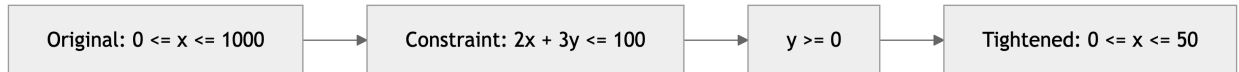


Figure 8.7.: Bound Tightening Process

Algorithm: Bound Tightening

For constraint $\sum_j a_j x_j \leq b$:

- If $a_j > 0$: $x_j \leq (b - \sum_{k \neq j} a_k L_k) / a_j$
- If $a_j < 0$: $x_j \geq (b - \sum_{k \neq j} a_k U_k) / a_j$

where L_k, U_k are current lower/upper bounds on x_k .

Probing:

Temporarily fix binary variables and observe implications:

Probing works by temporarily fixing each binary variable to 0 or 1 and solving the LP relaxation. If fixing $y = 1$ leads to infeasibility, the solver can permanently fix $y = 0$. If it causes other variable bounds to tighten, those implications are recorded and applied globally.

8. Integer and Mixed-Integer Programming

```
def probing_example():
    """
    Conceptual illustration of probing technique.

    Fix y=1, solve LP, observe:
    - If infeasible: y must be 0
    - If some variable bounds change: record implications
    """
    implications = {}

    for y in binary_variables:
        # Try fixing y = 1
        fix_variable(y, 1)
        result = solve_lp_relaxation()

        if result == 'infeasible':
            # y must be 0 in any feasible solution
            fix_variable(y, 0) # Permanent
        else:
            # Record any implied bounds
            for x in variables:
                if x.bound_changed():
                    implications[(y, 1)] = (x, x.new_bound)

    restore_variable(y)

    # Similarly try y = 0
    # ...

    return implications
```

Clique Detection:

Identify sets of binary variables where at most one can be 1:

- From constraints like $y_1 + y_2 \leq 1$, $y_2 + y_3 \leq 1$, $y_1 + y_3 \leq 1$
- Combine into single clique constraint: $y_1 + y_2 + y_3 \leq 1$
- Enables stronger branching and cut generation

8.5.2. Valid Inequalities Taxonomy

Valid inequalities are constraints that:

1. Do not eliminate any integer-feasible point
2. Cut off fractional LP solutions
3. Strengthen the LP relaxation

! The Ideal Formulation

The **ideal formulation** has LP relaxation equal to the convex hull of integer solutions. Valid inequalities move us closer to this ideal.

Classification of Valid Inequalities:

Type	Source	Examples
Problem-specific	Problem structure	Subtour elimination, cover cuts
General purpose	Constraint structure	Gomory cuts, MIR cuts
Disjunctive	Branching disjunctions	Split cuts, lift-and-project

Common Cut Families:

1. **Gomory Mixed-Integer Cuts:** Derived from any row of the optimal LP tableau with fractional RHS
2. **Mixed-Integer Rounding (MIR) Cuts:** Based on rounding arguments for mixed-integer sets
3. **Cover Cuts (for Knapsack):** For $\sum_j a_j x_j \leq b$, if C is a cover (sum of coefficients $> b$):

$$\sum_{j \in C} x_j \leq |C| - 1$$

4. **Clique Cuts:** For conflict graphs where edges connect incompatible binaries:

$$\sum_{j \in K} x_j \leq 1 \quad \text{for maximal clique } K$$

The following function generates a cover cut for a knapsack constraint. It identifies a subset of items whose total weight exceeds capacity (a “cover”), then checks whether the corresponding inequality is violated by the current LP solution. If violated, the cut can be added to tighten the relaxation.

```
def generate_cover_cut(knapsack_coefs, capacity, lp_solution):
    """
    Generate a cover cut for a knapsack constraint.

    Parameters
    -----
    knapsack_coefs : list
        Coefficients a_j in sum(a_j * x_j) <= b
    capacity : float
        Right-hand side b
    lp_solution : list
        Current LP solution values

    Returns
    -----
    tuple or None
        (cover_indices, rhs) or None if no violated cut found
    """
    n = len(knapsack_coefs)

    # Sort by LP value (descending) to find likely cover
    sorted_idx = sorted(range(n), key=lambda j: -lp_solution[j])

    cover = []
    cover_sum = 0

    for j in sorted_idx:
```

```

    if cover_sum <= capacity:
        cover.append(j)
        cover_sum += knapsack_coefs[j]
    else:
        break

# Check if we found a valid cover
if cover_sum > capacity:
    # Check if cut is violated by LP solution
    lp_cover_sum = sum(lp_solution[j] for j in cover)
    if lp_cover_sum > len(cover) - 1 + 1e-6:
        return (cover, len(cover) - 1)

return None

```

8.5.3. Lifted Inequalities

Lifting strengthens valid inequalities by systematically adding variables:

Algorithm: Sequential Lifting

Given a valid inequality $\sum_{j \in S} \alpha_j x_j \leq \beta$ for variables in set S :

1. Order variables not in S : $j_1, j_2, \dots, j_k$
2. For each j_i in order:
 - Compute lifting coefficient:

$$\alpha_{j_i} = \beta - \max \left\{ \sum_{j \in S \cup \{j_1, \dots, j_{i-1}\}} \alpha_j x_j : x \text{ feasible}, x_{j_i} = 1 \right\}$$

- Add term $\alpha_{j_i} x_{j_i}$ to inequality

3. Result: $\sum_{j \in S} \alpha_j x_j + \sum_{i=1}^k \alpha_{j_i} x_{j_i} \leq \beta$

Example: Lifting a Cover Cut

Original cover cut (variables 1, 2, 3 form a cover):

$$x_1 + x_2 + x_3 \leq 2$$

After lifting variable 4:

$$x_1 + x_2 + x_3 + 0.5x_4 \leq 2$$

The lifted cut is at least as strong and often strictly stronger.

8.6. Solution Methods

8.6.1. Branch-and-Bound: The Foundation

Branch-and-bound is the primary algorithm for solving MILPs to proven optimality. It works by recursively partitioning the feasible region into subproblems (branching), solving LP relaxations at each node to obtain bounds, and discarding subproblems that cannot improve on the best known solution (bounding). The combination of these three operations—branch, bound, and prune—allows the algorithm to certify optimality without examining every integer point.

Algorithm: Branch-and-Bound

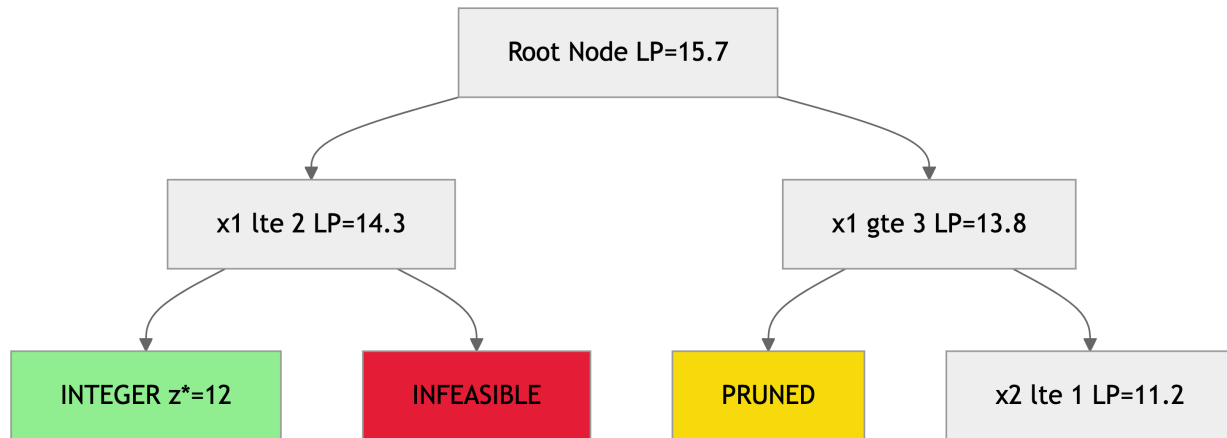


Figure 8.8.: Branch-and-Bound Tree with Pruning

1. **Initialize:** Set incumbent (best integer solution) = empty, best bound = -infinity
2. **Solve LP relaxation** of the root problem
3. **If** LP solution is integer: update incumbent if better, return
4. **If** LP bound less than or equal to incumbent value: prune this node (cannot improve)
5. **Select** a fractional variable x_j with value $\bar{x}_j$
6. **Branch:** Create two subproblems:
 - Left child: add constraint $x_j \leq \lfloor \bar{x}_j \rfloor$
 - Right child: add constraint $x_j \geq \lceil \bar{x}_j \rceil$
7. **Recurse** on subproblems
8. **Return** incumbent as optimal solution

The algorithm's efficiency depends critically on:

- **Branching strategy:** Which variable to branch on?
- **Node selection:** Which subproblem to explore next?
- **Bound quality:** How tight is the LP relaxation?

8.6.2. Node Selection Strategies

The order in which nodes are explored significantly impacts performance:

Best-First (Best-Bound) Search:

- Select node with best LP relaxation bound
- Guarantees smallest tree to prove optimality
- May create many nodes before finding a solution
- High memory usage

Depth-First Search:

- Explore deepest node first
- Finds integer solutions quickly
- Uses minimal memory (linear in depth)
- May prove optimality slowly

Best-Estimate (Hybrid):

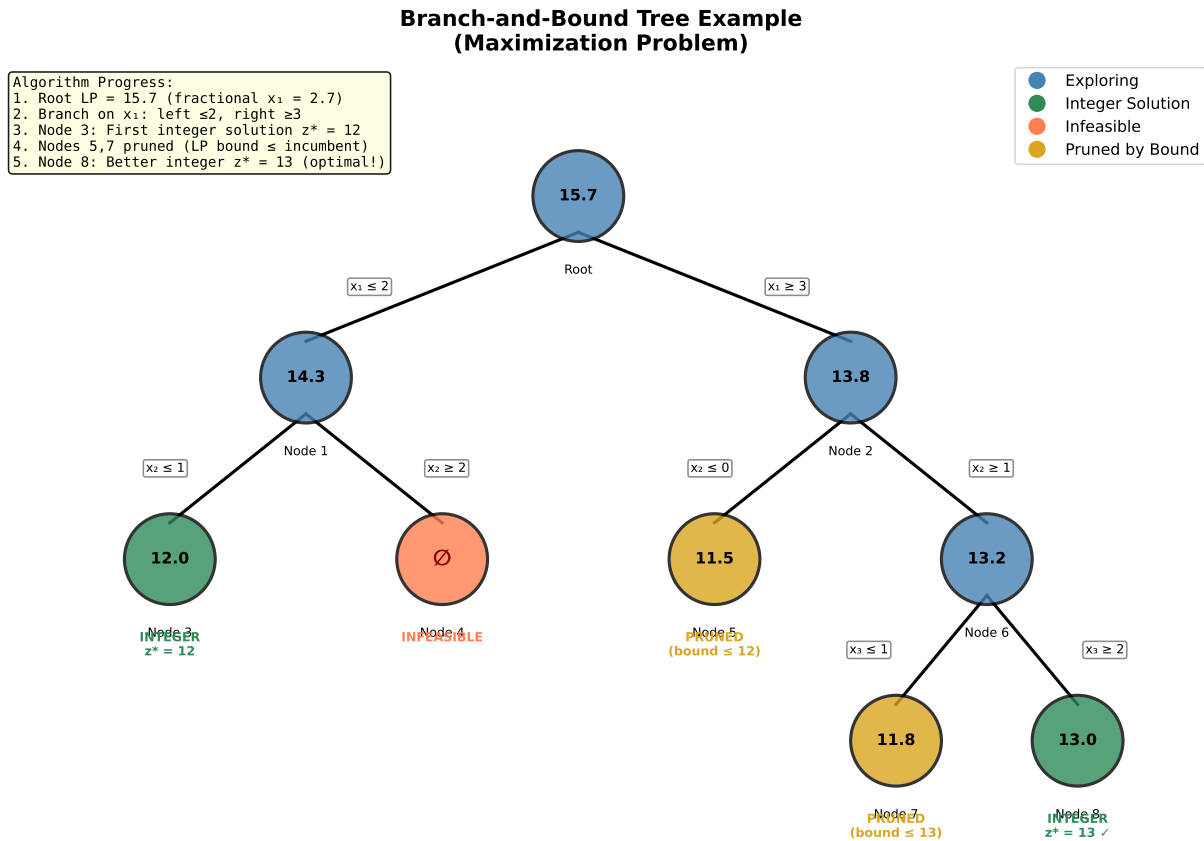


Figure 8.9.: Branch-and-Bound tree for a small integer program. Each node shows the LP relaxation bound. Green nodes represent integer solutions, red nodes are infeasible or pruned, and yellow nodes are pruned by bound (their LP bound cannot improve on the incumbent).



Figure 8.10.: Comparison of Node Selection Strategies

Modern solvers use sophisticated hybrid strategies:

$$\text{score}(n) = (1 - \alpha) \cdot \text{bound}(n) + \alpha \cdot \text{estimate}(n)$$

where estimate predicts the best integer solution reachable from node n .

💡 Practical Node Selection

Modern solvers (Gurobi, CPLEX) use adaptive strategies:

- **Early phase:** More depth-first to find solutions
- **Middle phase:** Balance between diving and proving
- **Late phase:** More best-first to close gap

8.6.3. Variable Selection Strategies

Choosing which variable to branch on directly affects tree size:

Most Fractional:

$$j^* = \arg \min_j |x_j - 0.5|$$

Simple but often ineffective—branching on a variable close to 0.5 doesn't guarantee the subproblems are hard.

Strong Branching:

For each candidate variable j :

1. Temporarily add constraint $x_j \leq \lfloor \bar{x}_j \rfloor$, solve LP
2. Temporarily add constraint $x_j \geq \lceil \bar{x}_j \rceil$, solve LP
3. Compute score based on bound improvements

$$\text{score}_j = (1 - \mu) \cdot \min(\Delta_j^-, \Delta_j^+) + \mu \cdot \max(\Delta_j^-, \Delta_j^+)$$

where Δ_j^-, Δ_j^+ are bound changes in left/right children.

Strong Branching Trade-off

Strong branching produces smaller trees but is expensive (two LPs per candidate). Modern solvers limit strong branching to the root node and key branching decisions.

Reliability Branching:

A practical compromise between most-fractional and strong branching. It performs strong branching only on variables that have been branched on fewer than `min_reliable` times. Once enough branching history accumulates, it trusts pseudo-cost estimates for all variables.

```
def reliability_branching(candidates, reliability_counts, min_reliable=8):
    """
    Select branching variable using reliability branching.

    Parameters
    -----
    candidates : list
        Candidate variables with their pseudo-costs
    reliability_counts : dict
        Number of times each variable has been branched on
    min_reliable : int
        Minimum branches needed to trust pseudo-costs

    Returns
    -----
    variable
        Selected branching variable
    """
    unreliable = [v for v in candidates
                   if reliability_counts[v] < min_reliable]
```

```

if unreliable:
    # Do strong branching on unreliable variables
    return strong_branch(unreliable[:5]) # Limit evaluations
else:
    # Trust pseudo-costs for all variables
    return max(candidates, key=lambda v: pseudo_cost_score(v))

```

Pseudo-cost Branching:

Track historical bound improvements for each variable:

- ρ_j^- : Average bound improvement per unit when branching down on x_j
- ρ_j^+ : Average bound improvement per unit when branching up on x_j

$$\text{score}_j = f_j \cdot \rho_j^- \cdot (1 - f_j) \cdot \rho_j^+$$

where $f_j = \bar{x}_j - \lfloor \bar{x}_j \rfloor$ is the fractionality.

8.6.4. Primal Heuristics

Finding good solutions quickly is essential for effective branch-and-bound:

! Why Primal Heuristics Matter

A good incumbent solution:

1. Enables pruning of more nodes
2. Provides a quality guarantee (gap)
3. Gives users early feasible answers
4. Can guide the search toward promising regions

Common Primal Heuristics:

Heuristic	Description	When Used
Rounding	Round LP solution to integers	At every node
Diving	Fix variables and re-solve LP	Periodically
RINS	Fix vars with same value in LP and incumbent	When incumbent exists
Feasibility Pump	Alternate between rounding and projection	Early solve
Local Branching	Search neighborhood of incumbent	After finding solution

Feasibility Pump Algorithm:**Algorithm: Feasibility Pump**

1. **Initialize:** Solve LP relaxation, get $\tilde{x}$
2. **Round:** $\bar{x} = \text{round}(\tilde{x})$ to nearest integers
3. **If** $\bar{x}$ is feasible: **return** $\bar{x}$
4. **Project:** Solve LP to minimize $\|\tilde{x} - \bar{x}\|$
5. **If** $\tilde{x} = \bar{x}$: perturb $\bar{x}$ randomly
6. **Repeat** from step 2

The feasibility pump alternates between rounding and projection. Starting from the LP relaxation solution, it rounds fractional variables to integers and then solves an LP that minimizes the distance to the rounded point. This oscillation often converges to a feasible integer solution within a few iterations.

```
def feasibility_pump(model, max_iterations=100):
    """
    Simplified feasibility pump heuristic.

    Parameters
    -----
    model : MIP model
        Original MIP model
    max_iterations : int
        Maximum pump iterations

    Returns
    -----
    dict or None
        Feasible solution if found
    """
    # Solve LP relaxation
    lp_solution = solve_lp_relaxation(model)

    for iteration in range(max_iterations):
        # Round to integer
        rounded = {v: round(lp_solution[v])
                   for v in model.integer_vars}

        # Check feasibility
        if is_feasible(model, rounded):
            return rounded

        # Modify objective to minimize distance to rounded
        projection_obj = sum(
            (v - rounded[v])**2 for v in model.integer_vars
        )
        lp_solution = solve_with_objective(model, projection_obj)

        # Detect cycling
        if lp_solution == rounded:
            # Perturb
            for v in model.integer_vars:
                if random.random() < 0.1:
                    rounded[v] += random.choice([-1, 1])

    return None
```

RINS (Relaxation Induced Neighborhood Search):

When we have an incumbent solution x^* and LP solution $\bar{x}$:

1. Fix variables where $x_j^* = \bar{x}_j$ (they agree)
2. Solve restricted MIP on remaining variables
3. This often finds improved solutions in a small neighborhood

8.6.5. Cutting Planes

Instead of branching, we can add **cutting planes**—linear inequalities that:

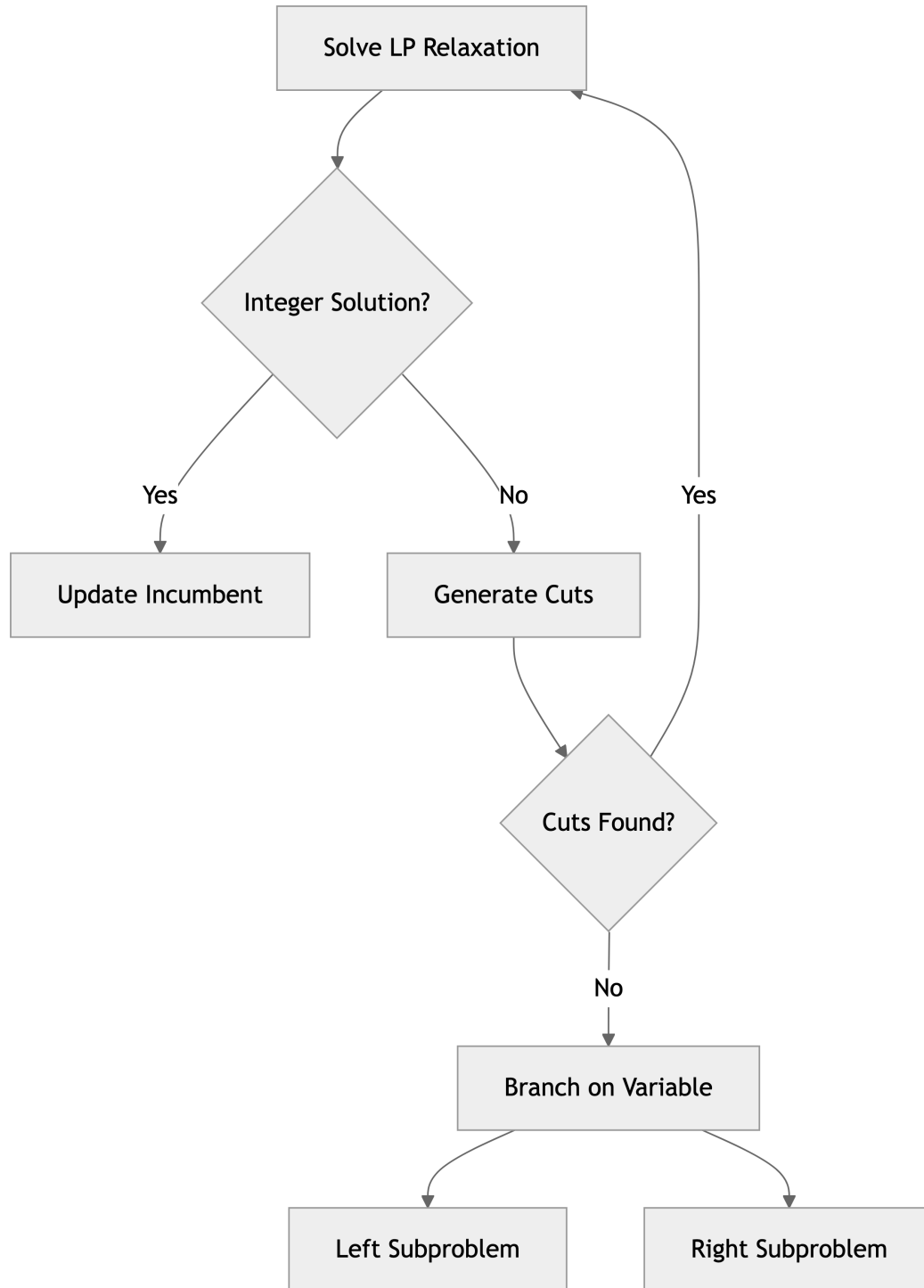
- Remove fractional LP solutions
- Do not eliminate any integer-feasible point
- Tighten the LP relaxation

Common cut families include:

Cut Type	Best For
Gomory cuts	General MILP
Cover cuts	Knapsack constraints
Clique cuts	Set packing problems
Flow cuts	Network problems

8.6.6. Branch-and-Cut: Modern Standard

Modern solvers use **branch-and-cut**, combining branch-and-bound with cutting planes at each node:



8.7. Python Implementation

We now implement the capacitated facility location problem introduced in the formulation section. Two implementations are provided: one using Gurobi (commercial) and one using PuLP with the CBC solver (open-source). Both produce the same mathematical model but differ in API syntax and solver performance.

8.7.1. Using Gurobi

The Gurobi implementation below creates binary variables y_j for opening decisions and continuous variables x_{ij} for customer-to-warehouse assignment fractions. Three constraint families enforce full demand coverage, linking between assignment and opening decisions, and warehouse capacity limits.

```
import gurobipy as gp
from gurobipy import GRB

def solve_facility_location(customers, warehouses, fixed_costs,
                           transport_costs, demands, capacities):
    """
    Solve the capacitated facility location problem.

    Parameters
    -----
    customers : list
        List of customer indices
    warehouses : list
        List of warehouse indices
    fixed_costs : dict
        Fixed cost  $f[j]$  for opening warehouse  $j$ 
    transport_costs : dict
        Transport cost  $c[i,j]$  from warehouse  $j$  to customer  $i$ 
    demands : dict
        Demand  $d[i]$  of customer  $i$ 
    capacities : dict
        Capacity  $C[j]$  of warehouse  $j$ 

    Returns
    -----
    dict
        Solution containing open warehouses and assignments
    """

    model = gp.Model("facility_location")

    # Decision variables
    y = model.addVars(warehouses, vtype=GRB.BINARY, name="open")
    x = model.addVars(customers, warehouses, lb=0, ub=1, name="serve")

    # Objective: minimize total cost
    model.setObjective(
        gp.quicksum(fixed_costs[j] * y[j] for j in warehouses) +
        gp.quicksum(transport_costs[i,j] * x[i,j]
                     for i in customers for j in warehouses),
        GRB.MINIMIZE
    )

    # Constraints
    # Each customer must be fully served
    model.addConstrs(
```

```

        (gp.quicksum(x[i,j] for j in warehouses) == 1
         for i in customers),
        name="demand"
    )

    # Can only serve from open warehouses
    model.addConstrs(
        (x[i,j] <= y[j]
         for i in customers for j in warehouses),
        name="linking"
    )

    # Respect warehouse capacities
    model.addConstrs(
        (gp.quicksum(demands[i] * x[i,j] for i in customers) <=
         capacities[j] * y[j]
         for j in warehouses),
        name="capacity"
    )

    # Solve
    model.optimize()

    # Extract solution
    if model.status == GRB.OPTIMAL:
        open_warehouses = [j for j in warehouses if y[j].X > 0.5]
        assignments = {i: [j for j in warehouses if x[i,j].X > 0.5][0]
                        for i in customers}

        return {
            'objective': model.ObjVal,
            'open_warehouses': open_warehouses,
            'assignments': assignments,
            'gap': model.MIPGap
        }
    return None

```

Performance Tips

1. **Provide initial solutions** via `model.start` when you have a good heuristic
2. **Set MIP gap** with `model.setParam('MIPGap', 0.01)` for 1% optimality tolerance
3. **Use lazy constraints** for exponentially many constraints (subtour elimination)
4. **Enable presolve** (on by default) to reduce problem size

8.7.2. Using PuLP (Open Source)

The PuLP version models the same problem with a different API style. PuLP uses operator overloading (`+=`) to add constraints and objectives to the problem object, which some users find more readable.

```
from pulp import *
```

```

def solve_facility_location_pulp(customers, warehouses, fixed_costs,
                                transport_costs, demands, capacities):
    """Same problem using PuLP with CBC solver."""

    prob = LpProblem("Facility_Location", LpMinimize)

    # Variables
    y = LpVariable.dicts("open", warehouses, cat='Binary')
    x = LpVariable.dicts("serve",
                        ((i,j) for i in customers for j in warehouses),
                        lowBound=0, upBound=1)

    # Objective
    prob += (lpSum(fixed_costs[j] * y[j] for j in warehouses) +
            lpSum(transport_costs[i,j] * x[i,j]
                  for i in customers for j in warehouses))

    # Constraints
    for i in customers:
        prob += lpSum(x[i,j] for j in warehouses) == 1

    for i in customers:
        for j in warehouses:
            prob += x[i,j] <= y[j]

    for j in warehouses:
        prob += lpSum(demands[i] * x[i,j] for i in customers) <= \
            capacities[j] * y[j]

    # Solve
    prob.solve(PULP_CBC_CMD(msg=0))

    if prob.status == LpStatusOptimal:
        return {
            'objective': value(prob.objective),
            'open_warehouses': [j for j in warehouses if y[j].value() > 0.5],
            'gap': 0 # CBC doesn't report gap easily
        }
    return None

```

8.8. Real-World Applications

MILP is the most widely deployed optimization technology in industry. The combination of binary decision variables and continuous quantities allows it to model virtually any structured decision problem with linear relationships.

8.8.1. Classic OR Applications

Domain	Problem	Key Decision
Logistics	Facility location	Where to open warehouses
Manufacturing	Production planning	What to produce, when
Transportation	Vehicle routing	Which routes for each vehicle

Domain	Problem	Key Decision
Airlines	Crew scheduling	Which crew for each flight
Telecom	Network design	Where to place equipment
Finance	Portfolio selection	Which assets to include

8.8.2. Modern AI/ML Applications

i MILP Meets Machine Learning

Integer programming is increasingly important in AI:

- **Feature selection:** Choose optimal subset of features
- **Neural network verification:** Prove robustness properties
- **Optimal decision trees:** Learn globally optimal trees
- **Fairness constraints:** Ensure ML models satisfy fairness criteria

8.8.3. Case Study: Airline Crew Scheduling

8.8.3.1. Delta Airlines Crew Scheduling

Problem: Assign pilots and flight attendants to 10,000+ daily flights

Scale: - 100,000+ binary variables - Millions of constraints - Solved daily at 2 AM for next day's operations

Constraints: - FAA rest requirements (minimum 8 hours between duties) - Maximum flying hours per month - Crew qualifications (aircraft type ratings) - Union rules (seniority, base assignments)

Impact: Optimal scheduling saves Delta \$100M+ annually compared to manual planning

8.9. Summary

8.9.1. Key Takeaways

- **Integer constraints** transform polynomial-time LP into NP-hard MILP
- **Binary variables** can model any discrete decision or logical relationship
- **Big-M method** links binary and continuous variables—keep M as small as possible
- **Branch-and-bound** systematically explores solution space via divide-and-conquer
- **Cutting planes** tighten LP relaxation without branching
- **Modern solvers** (Gurobi, CPLEX, HiGHS) combine multiple techniques in branch-and-cut
- **Formulation quality** dramatically affects solve time—tight relaxations matter
- **Preprocessing and valid inequalities** can dramatically reduce solution time
- **Node and variable selection strategies** directly affect branch-and-bound efficiency
- **Primal heuristics** find good solutions quickly to enable pruning

8.10. Exercises

1. **Knapsack Problem:** Formulate the 0-1 knapsack problem as a BIP. Implement it in Python and solve a 20-item instance.
2. **Set Cover:** A company needs to cover all customers with service centers. Each center covers a subset of customers. Formulate as BIP to minimize number of centers opened.
3. **Big-M Analysis:** For the facility location problem, derive the tightest possible big-M value for the linking constraints $x_{ij} \leq y_j$.

8. Integer and Mixed-Integer Programming

4. **Solver Comparison:** Solve the same facility location instance with Gurobi, HiGHS, and CBC. Compare solution times and report any differences.
5. **Branching Strategy:** Implement a simple branch-and-bound solver for binary knapsack. Compare “most fractional” vs “largest coefficient” branching rules.
6. **Symmetry Breaking:** Consider a problem of selecting 3 machines from 10 identical machines. How many symmetric solutions exist? Write symmetry-breaking constraints and compare solver performance with and without them.
7. **Piecewise Linear:** Model a nonlinear cost function $c(x) = x^2$ for $x \in [0, 10]$ using a piecewise linear approximation with 5 breakpoints. Implement using both SOS2 variables and the incremental formulation.
8. **Valid Inequalities:** For the knapsack constraint $3x_1 + 4x_2 + 5x_3 + 6x_4 \leq 10$, identify all minimal covers and write the corresponding cover inequalities.

8.11. Further Reading

- Wolsey, L.A. *Integer Programming* (2nd ed.). Wiley, 2020.
- Conforti, M., Cornuejols, G., Zambelli, G. *Integer Programming*. Springer, 2014.
- Gurobi Optimization. *Gurobi Reference Manual*. gurobi.com/documentation
- Achterberg, T. “Constraint Integer Programming.” PhD thesis, TU Berlin, 2007.
- Bixby, R.E. “A Brief History of Linear and Mixed-Integer Programming Computation.” *Documenta Mathematica*, 2012.

9. Constraint Programming

What You Will Learn

- Understand constraint satisfaction problems (CSPs) and their structure
- Master constraint propagation and arc consistency algorithms
- Model problems with global constraints (AllDifferent, Cumulative)
- Implement search strategies and variable ordering heuristics
- Solve complex combinatorial problems using CP solvers

Constraint Programming (CP) is a declarative paradigm for solving combinatorial optimization and satisfaction problems. Unlike imperative programming where you specify *how* to compute a solution, CP allows you to specify *what* the solution must satisfy through constraints, and the solver automatically finds solutions through systematic search and intelligent constraint propagation.

9.1. Constraint Satisfaction Problems

The fundamental abstraction in constraint programming is the Constraint Satisfaction Problem (CSP).

9.1.1. The CSP Triple

A CSP is defined by three components (X, D, C) :

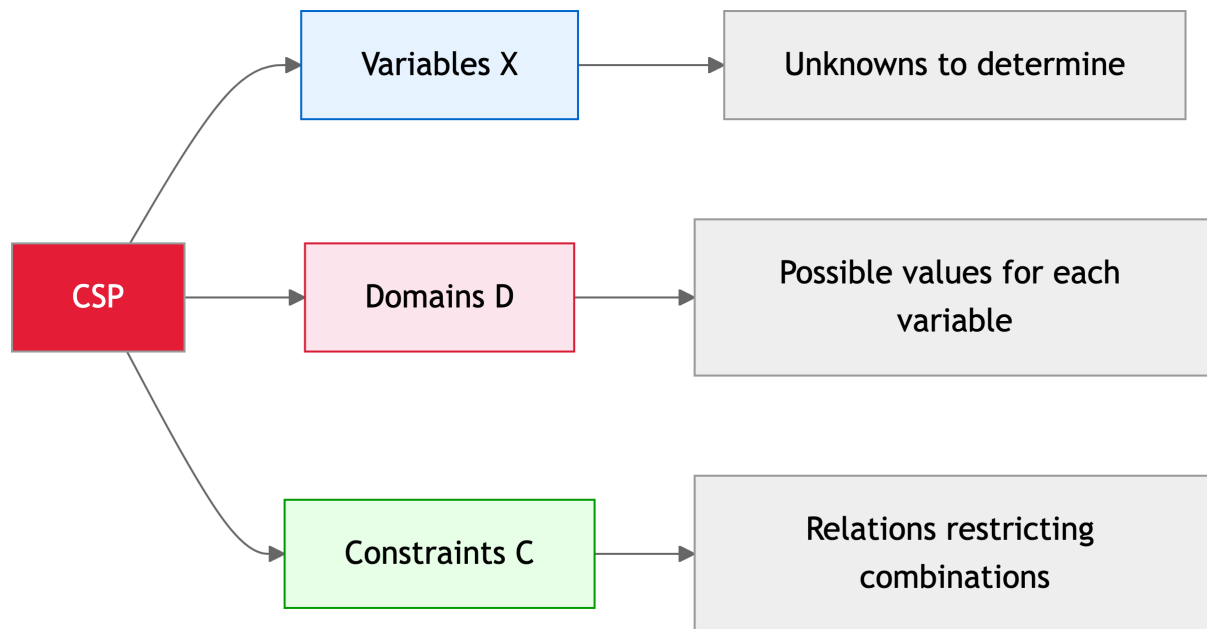


Figure 9.1.: The three components of a Constraint Satisfaction Problem

Variables: $X = \{x_1, x_2, \dots, x_n\}$

9. Constraint Programming

Decision variables representing unknowns to be determined.

Domains: $D = \{D_1, D_2, \dots, D_n\}$

D_i is the set of possible values for variable x_i . Domains can be finite (integers, enums) or infinite (reals).

Constraints: $C = \{c_1, c_2, \dots, c_m\}$

Relations restricting combinations of variable values. Each constraint has a scope (variables) and a relation (allowed tuples).

Solution Concept

A **solution** is a complete assignment $\alpha : X \rightarrow \bigcup D$ such that:

- $\alpha(x_i) \in D_i$ for all i (domain consistency)
- All constraints $c \in C$ are satisfied under α

A CSP is **satisfiable** if at least one solution exists, **unsatisfiable** otherwise.

9.1.2. Example: Map Coloring

Color a map so no adjacent regions share the same color:

- **Variables:** $X = \{WA, NT, SA, Q, NSW, V, T\}$ (Australian states)
- **Domains:** $D_i = \{red, green, blue\}$ for all regions
- **Constraints:** Adjacent regions differ: $WA \neq NT, WA \neq SA, NT \neq SA$, etc.

9.2. Why Constraint Programming?

9.2.1. Comparison with Other Paradigms

Approach	Strengths	Weaknesses
Integer Programming	Linear objectives, continuous relaxations	Non-linear global constraints difficult
SAT Solvers	Boolean problems, fast propagation	Integer variables require encoding
Heuristic Search	Flexible, fast	No optimality guarantee
Constraint Programming	Global constraints, feasibility	Less mature for continuous problems

9.2.2. When to Use CP

CP excels when:

- **Highly combinatorial:** Large discrete search spaces
- **Complex constraints:** Logical, disjunctive, or global constraints
- **Feasibility focus:** Finding any solution is the primary goal
- **Scheduling problems:** Temporal and resource constraints
- **Configuration problems:** Compatibility and assignment constraints

9.3. Constraint Propagation

Constraint propagation is the core inference mechanism in CP. Rather than waiting until all variables are assigned to check feasibility, propagation proactively removes values from variable domains that

cannot participate in any valid solution. This pruning happens before and during search, often reducing the search space by orders of magnitude.

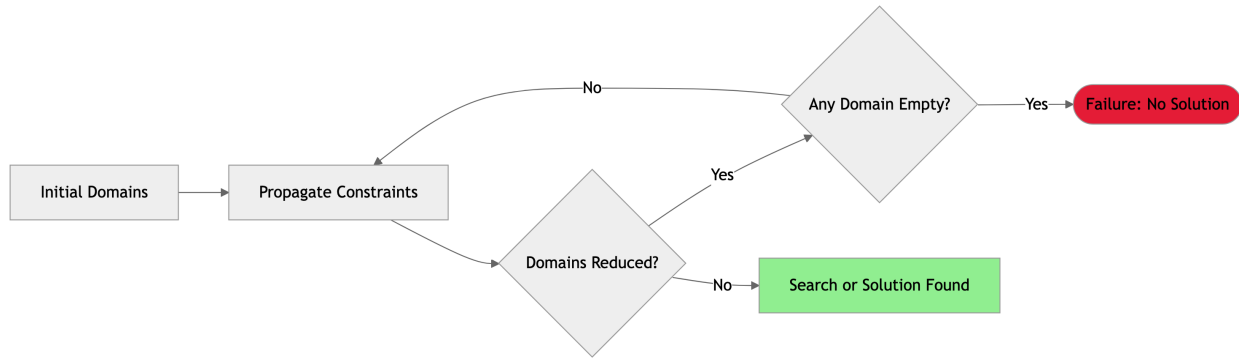


Figure 9.2.: Constraint Propagation Process

9.3.1. Arc Consistency

A CSP is **arc consistent** if for every constraint $c(x, y)$, every value in $D(x)$ has a **support** in $D(y)$:

$$\forall a \in D(x), \exists b \in D(y) : c(x = a, y = b) \text{ is satisfied}$$

9.3.2. AC-3 Algorithm

The AC-3 algorithm enforces arc consistency by maintaining a queue of arcs (directed constraint pairs) to check. For each arc (x_i, x_j) , it removes any value from $D(x_i)$ that has no supporting value in $D(x_j)$. When a domain shrinks, all arcs pointing to that variable are re-added to the queue, since their support may have been invalidated.

```

from collections import deque

def ac3(variables, domains, constraints):
    """AC-3 algorithm for arc consistency.

    Args:
        variables: List of variable names
        domains: Dict mapping variable to set of values
        constraints: Dict mapping (xi, xj) to constraint function

    Returns:
        True if consistent, False if domain becomes empty
    """
    # Initialize queue with all arcs
    queue = deque()
    for (xi, xj) in constraints:
        queue.append((xi, xj))
        queue.append((xj, xi)) # Add reverse arc

    while queue:
        (xi, xj) = queue.popleft()

```

```

    if revise(domains, xi, xj, constraints):
        if not domains[xi]:
            return False # Domain wipeout - inconsistent

        # Add arcs (xk, xi) for all neighbors of xi except xj
        for xk in get_neighbors(xi, constraints):
            if xk != xj:
                queue.append((xk, xi))

    return True

def revise(domains, xi, xj, constraints):
    """Remove unsupported values from D(xi)."""
    revised = False
    constraint = constraints.get((xi, xj)) or constraints.get((xj, xi))

    for a in list(domains[xi]):
        # Check if any value in D(xj) supports a
        has_support = False
        for b in domains[xj]:
            if constraint(a, b):
                has_support = True
                break

        if not has_support:
            domains[xi].remove(a)
            revised = True

    return revised

def get_neighbors(xi, constraints):
    """Get variables that share a constraint with xi."""
    neighbors = set()
    for (x1, x2) in constraints:
        if x1 == xi:
            neighbors.add(x2)
        elif x2 == xi:
            neighbors.add(x1)
    return neighbors

```

9.3.3. Consistency Levels

Level	Definition	Cost
Node Consistency	All unary constraints satisfied	$O(nd)$
Arc Consistency	All binary constraints have support	$O(ed^3)$
Bounds Consistency	Only domain bounds need support	Cheaper than AC
Generalized AC	Extension to non-binary constraints	Higher

9.3.4. Visualizing Domain Reduction

The following visualization demonstrates how constraint propagation progressively reduces variable domains through arc consistency:

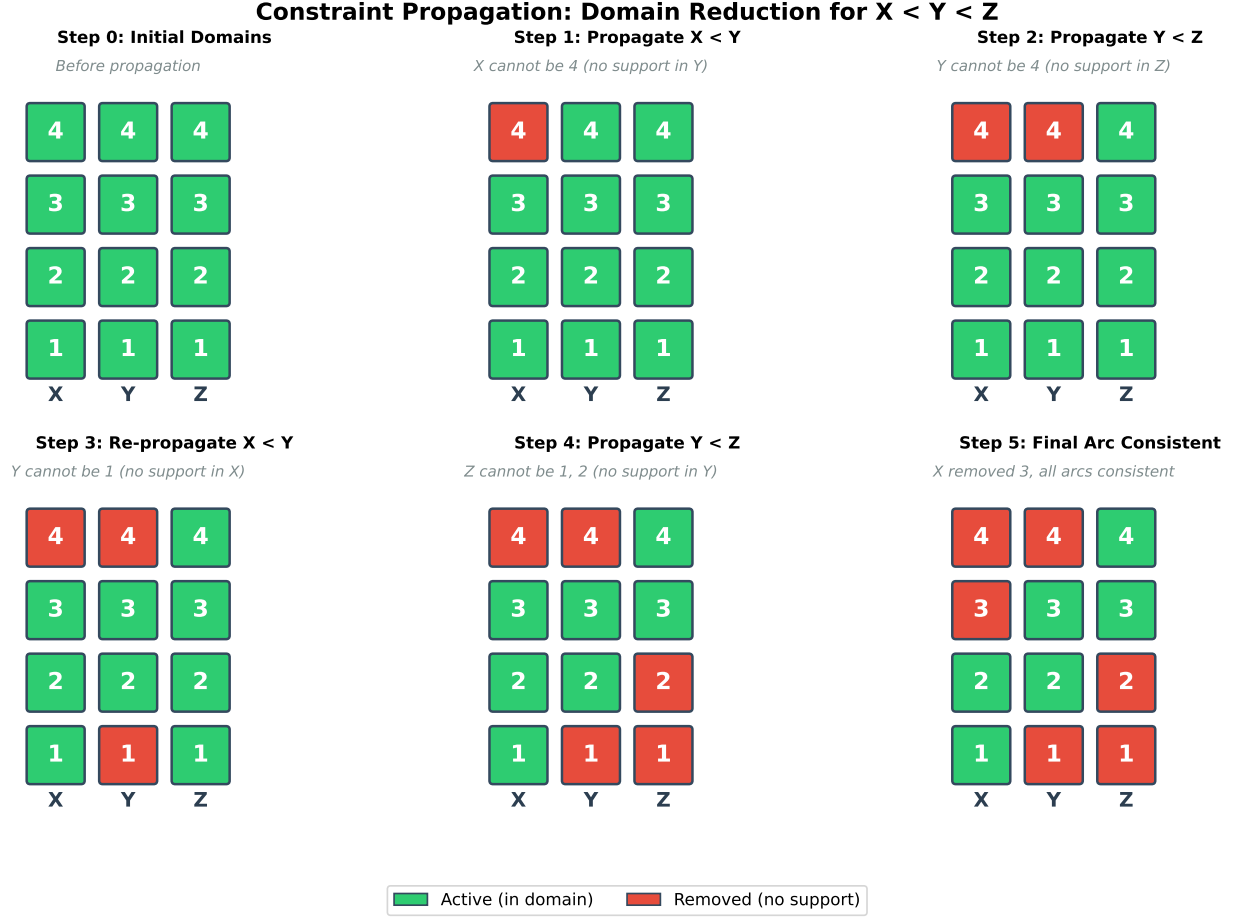


Figure 9.3.: Domain reduction through constraint propagation. Each step shows how enforcing arc consistency removes values that cannot participate in any solution.

9.4. Global Constraints

Global constraints are one of the main advantages of CP over other paradigms. Instead of decomposing a combinatorial pattern into many binary constraints, a single global constraint captures the full structure and uses a specialized propagation algorithm. This produces stronger domain reductions and significantly faster solving.

9.4.1. AllDifferent

All variables must take different values:

$$\text{AllDifferent}(x_1, x_2, \dots, x_n) \iff x_i \neq x_j \forall i \neq j$$

Applications: Resource assignment, graph coloring, Sudoku.

Propagation: Uses maximum matching in bipartite graph. Complexity: $O(n^2\sqrt{n})$ vs $O(n^2d^2)$ for pairwise decomposition.

The simplified propagation below handles the most common case: when a variable's domain is reduced to a single value, that value is removed from all other domains. Full AllDifferent propagation (using matching theory on bipartite graphs) is more powerful but also more expensive.

9. Constraint Programming

```
def propagate_alldifferent(variables, domains):
    """Propagate AllDifferent constraint using value counting."""
    # Simple propagation: if a value is assigned, remove from others
    assigned = {}
    for var in variables:
        if len(domains[var]) == 1:
            val = list(domains[var])[0]
            assigned[var] = val

    for var, val in assigned.items():
        for other_var in variables:
            if other_var != var and val in domains[other_var]:
                domains[other_var].remove(val)
                if not domains[other_var]:
                    return False # Domain wipeout

    return True
```

9.4.2. Cumulative

Resource constraint for scheduling: tasks with starts S , durations D , resource requirements R must not exceed capacity C at any time:

$$\forall t : \sum_{i: S_i \leq t < S_i + D_i} R_i \leq C$$

Applications: Scheduling, bin packing, resource allocation.

Propagation: Time-table reasoning, edge finding, energetic reasoning.

9.4.3. Element

Array indexing constraint: $V = A[I]$

$$\text{Element}(I, A, V) \iff A[I] = V$$

Applications: Indirection, routing, dynamic constraints.

9.4.4. Table

Extensional constraint defined by allowed tuples:

$$\text{Table}(x_1, \dots, x_k) \in T$$

Applications: Database constraints, learned constraints, complex relations.

9.5. Search Strategies

Propagation alone can solve only the simplest CSPs. For most problems, a search procedure must assign values to variables and backtrack when a dead end is reached. The order in which variables and values are tried has a large impact on performance—sometimes the difference between milliseconds and hours.

9.5.1. Variable Ordering Heuristics

Heuristic	Strategy	When to Use
First-Fail	Smallest domain first	General purpose
Max-Degree	Most constraints first	Highly connected problems
Dom/Deg	Domain size / degree ratio	Balanced approach
Impact	Highest search space reduction	Learned from search
Activity	Most conflict involvement	Dynamic problems

9.5.2. Value Ordering Heuristics

Heuristic	Strategy	Effect
Min-Conflicts	Fewest conflicts with neighbors	Find solutions faster
Max-Promise	Most flexibility preserved	Better for optimization
Domain-Splitting	Binary split, not enumeration	Efficient for large domains

9.5.3. Backtracking Search

The following implementation combines backtracking with the first-fail heuristic (select the variable with the smallest remaining domain) and AC-3 propagation at each node. After each variable assignment, constraint propagation prunes the remaining domains. If propagation detects a contradiction (empty domain), the search backtracks immediately without exploring further.

```
def backtrack_search(variables, domains, constraints, assignment=None):
    """Solve CSP using backtracking with constraint propagation.

    Args:
        variables: List of variables
        domains: Dict of variable -> domain (set of values)
        constraints: Dict of (var1, var2) -> constraint function
        assignment: Current partial assignment

    Returns:
        Complete assignment or None if unsatisfiable
    """
    if assignment is None:
        assignment = {}

    # Check if complete
    if len(assignment) == len(variables):
        return assignment

    # Select unassigned variable (first-fail heuristic)
    unassigned = [v for v in variables if v not in assignment]
    var = min(unassigned, key=lambda v: len(domains[v]))

    # Try each value in domain
    for value in list(domains[var]):
        # Check consistency
        if is_consistent(var, value, assignment, constraints):
```

```
# Make assignment
assignment[var] = value
old_domains = {v: domains[v].copy() for v in variables}

# Propagate constraints
domains[var] = {value}
if ac3(variables, domains, constraints):
    result = backtrack_search(
        variables, domains, constraints, assignment
    )
    if result is not None:
        return result

# Backtrack
del assignment[var]
for v in variables:
    domains[v] = old_domains[v]

return None # No solution found

def is_consistent(var, value, assignment, constraints):
    """Check if assigning value to var is consistent with assignment."""
    for other_var, other_value in assignment.items():
        if (var, other_var) in constraints:
            if not constraints[(var, other_var)](value, other_value):
                return False
        if (other_var, var) in constraints:
            if not constraints[(other_var, var)](other_value, value):
                return False
    return True
```

9.5.4. Visualizing the Search Tree

The following visualization shows how constraint programming explores a search tree with pruning. Branches that lead to constraint violations are pruned early, dramatically reducing the search space:

9.6. Advanced Search Techniques

Beyond basic backtracking, several techniques improve search performance for large-scale problems.

9.6.1. Limited Discrepancy Search

Limited Discrepancy Search (LDS) assumes the variable and value ordering heuristics are mostly correct. It first tries the path where all heuristic recommendations are followed (zero discrepancies), then explores paths with exactly one deviation, then two, and so on. This strategy finds solutions quickly when the heuristic is good but not perfect.

9.6.2. Restart Strategies

Periodically restarting the search with different random seeds or heuristic tie-breaking can avoid getting stuck in unproductive subtrees. The Luby sequence (1, 1, 2, 1, 1, 2, 4, 1, 1, 2, 1, 1, 2, 4, 8, ...) provides a provably efficient restart schedule that balances exploration and exploitation.

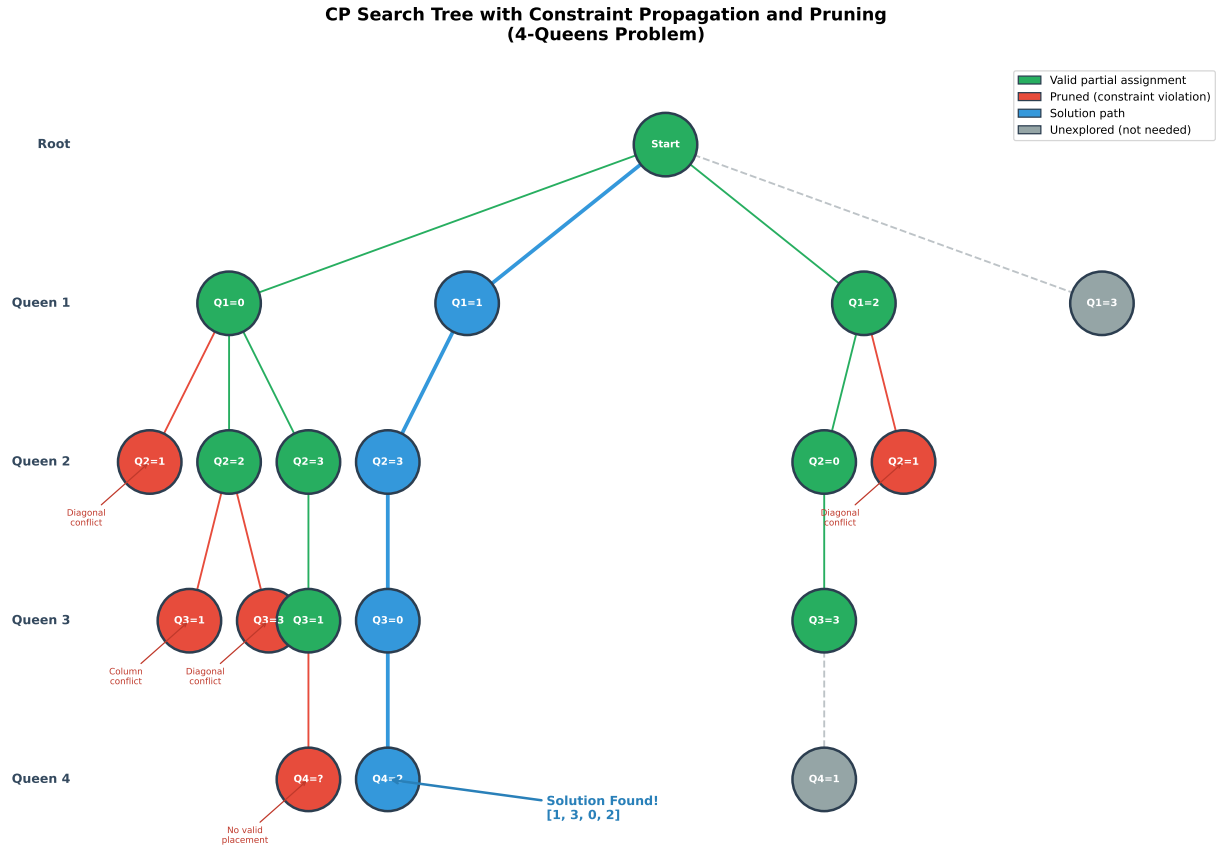


Figure 9.4.: CP search tree with pruning for a 4-Queens problem. Green nodes represent valid partial assignments, red nodes show constraint violations that trigger backtracking, and the blue path leads to a solution.

9.6.3. Large Neighborhood Search

Large Neighborhood Search (LNS) combines CP with local search to handle problems too large for pure systematic search. Starting from a feasible solution, LNS repeatedly destroys a random portion of the solution (unfixes a subset of variables) and uses the CP solver to repair it optimally. This approach scales to problems with thousands of variables.

```
def large_neighborhood_search(problem, initial_solution, max_iter=1000,
                              destroy_fraction=0.3):
    """Large Neighborhood Search for optimization.

    Args:
        problem: Problem definition with solve method
        initial_solution: Starting feasible solution
        max_iter: Maximum iterations
        destroy_fraction: Fraction of solution to destroy

    Returns:
```

```
Best solution found
"""
current = initial_solution
best = current
best_cost = problem.evaluate(current)

for iteration in range(max_iter):
    # Destroy: unassign random subset of variables
    destroyed = destroy(current, destroy_fraction)

    # Repair: use CP to complete partial solution
    repaired = problem.solve_with_partial(destroyed)

    if repaired is not None:
        cost = problem.evaluate(repaired)
        if cost < best_cost:
            best = repaired
            best_cost = cost
            current = repaired

return best
```

9.7. Implementation with OR-Tools

Google OR-Tools provides a production-quality CP-SAT solver that combines constraint propagation, SAT-based techniques, and linear relaxations. It is free, open-source, and competitive with commercial solvers on many problem types.

9.7.1. N-Queens Example

The N-Queens problem places n queens on an $n \times n$ chessboard so that no two queens attack each other. This is a standard CSP benchmark. The CP-SAT formulation uses one integer variable per row (the column of the queen in that row) and three AllDifferent constraints to enforce non-attacking positions on columns, main diagonals, and anti-diagonals.

```
from ortools.sat.python import cp_model

def solve_n_queens(n):
    """Solve N-Queens problem using CP-SAT.

    Args:
        n: Board size (n x n)

    Returns:
        List of queen column positions, one per row
    """
    model = cp_model.CpModel()

    # Variables: queens[i] = column of queen in row i
    queens = [model.NewIntVar(0, n - 1, f'queen_{i}') for i in range(n)]

    # Constraints
```

```

# All queens in different columns
model.AddAllDifferent(queens)

# All queens in different diagonals
model.AddAllDifferent([queens[i] + i for i in range(n)])
model.AddAllDifferent([queens[i] - i for i in range(n)])

# Solve
solver = cp_model.CpSolver()
status = solver.Solve(model)

if status == cp_model.OPTIMAL or status == cp_model.FEASIBLE:
    return [solver.Value(queens[i]) for i in range(n)]
return None

# Solve 8-Queens
solution = solve_n_queens(8)
if solution:
    print("Solution found:")
    for row, col in enumerate(solution):
        line = ['. ' if c != col else 'Q' for c in range(8)]
        print(' '.join(line))

```

The following visualization shows a solved 8-Queens board:

9.7.2. Job Shop Scheduling

Job shop scheduling assigns operations to machines while respecting precedence constraints within each job and no-overlap constraints on each machine. This problem is NP-hard and well-suited for CP because the no-overlap constraint maps directly to the `AddNoOverlap` global constraint, which performs efficient time-table propagation internally.

The implementation below creates interval variables for each operation, enforces operation ordering within jobs, and adds no-overlap constraints for each machine. The objective minimizes the makespan (completion time of the last operation).

```

from ortools.sat.python import cp_model

def job_shop_cp(jobs_data):
    """Solve job shop scheduling with CP-SAT.

    Args:
        jobs_data: List of jobs, each is list of (machine, duration) tuples

    Returns:
        Schedule as list of (job, task, start, end) tuples
    """
    model = cp_model.CpModel()

    # Compute horizon
    horizon = sum(task[1] for job in jobs_data for task in job)

    # Create variables

```

8-Queens Solution via Constraint Programming

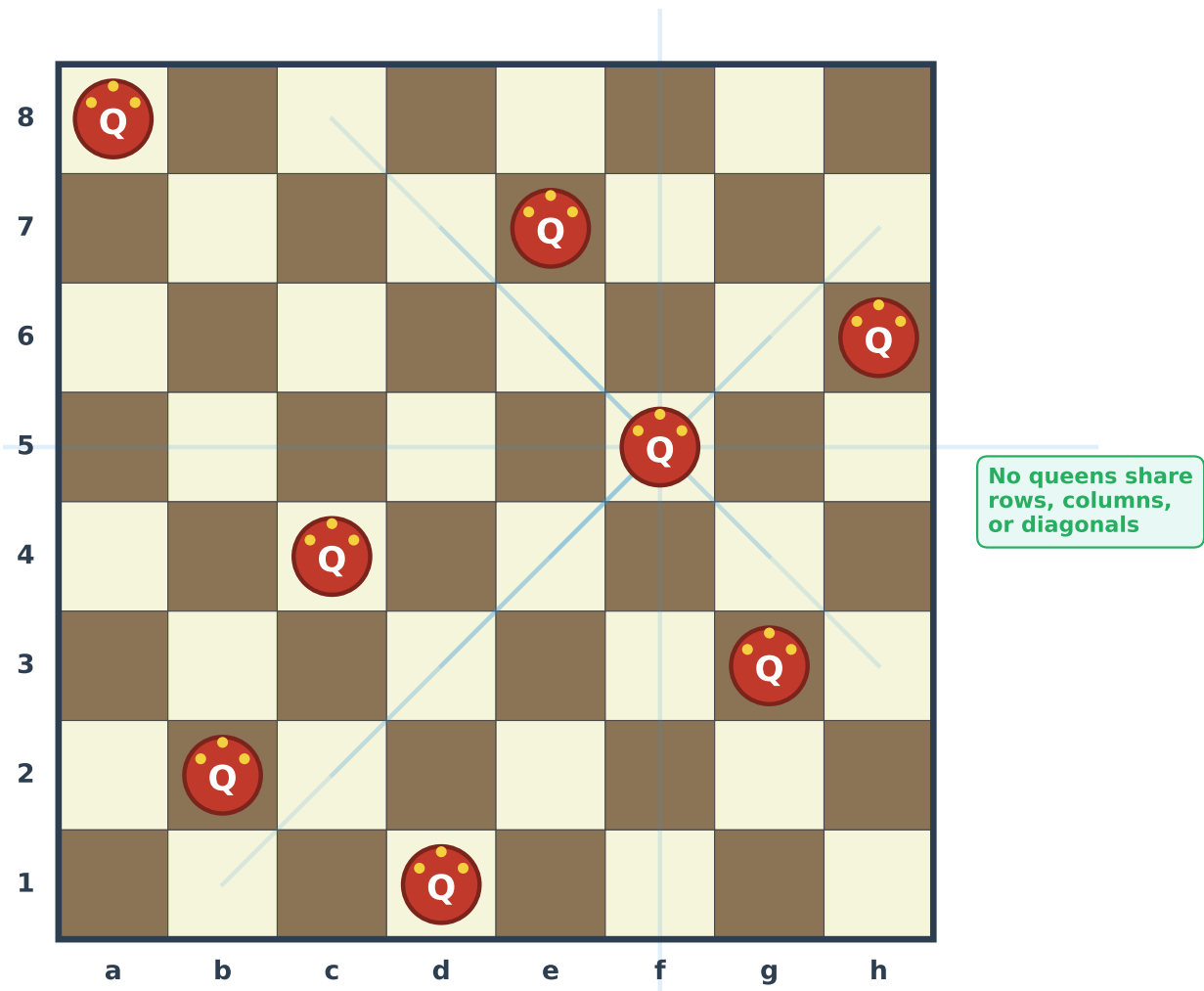


Figure 9.5.: A valid 8-Queens solution where no two queens attack each other. Queens are placed such that no two share the same row, column, or diagonal.

```
task_vars = {}
for job_id, job in enumerate(jobs_data):
    for task_id, (machine, duration) in enumerate(job):
        start = model.NewIntVar(0, horizon, f'start_{job_id}_{task_id}')
        end = model.NewIntVar(0, horizon, f'end_{job_id}_{task_id}')
        interval = model.NewIntervalVar(
            start, duration, end, f'interval_{job_id}_{task_id}')
        task_vars[job_id, task_id] = (start, end, interval, machine)

# Precedence within jobs
for job_id, job in enumerate(jobs_data):
    for task_id in range(len(job) - 1):
```

```

        model.Add(task_vars[job_id, task_id + 1][0] >=
                    task_vars[job_id, task_id][1])

# No overlap on machines
machines = set(task[0] for job in jobs_data for task in job)
for machine in machines:
    intervals = [
        task_vars[job_id, task_id][2]
        for job_id, job in enumerate(jobs_data)
        for task_id, (m, _) in enumerate(job)
        if m == machine
    ]
    model.AddNoOverlap(intervals)

# Minimize makespan
makespan = model.NewIntVar(0, horizon, 'makespan')
model.AddMaxEquality(makespan, [
    task_vars[job_id, len(job) - 1][1]
    for job_id, job in enumerate(jobs_data)
])
model.Minimize(makespan)

# Solve
solver = cp_model.CpSolver()
solver.parameters.max_time_in_seconds = 60.0
status = solver.Solve(model)

if status in [cp_model.OPTIMAL, cp_model.FEASIBLE]:
    schedule = []
    for (job_id, task_id), (start, end, _, machine) in task_vars.items():
        schedule.append((
            job_id, task_id, machine,
            solver.Value(start), solver.Value(end)
        ))
    return sorted(schedule), solver.ObjectiveValue()
return None, None

```

9.8. Symmetry Breaking

Many combinatorial problems have symmetric solutions—permutations of an existing solution that are functionally identical but look different to the solver. Without symmetry breaking, the solver wastes time proving that each permutation is equally optimal. Adding symmetry-breaking constraints eliminates this redundancy.

9.8.1. Lexicographic Ordering

For row symmetry in a matrix of variables:

$$\text{Row}_i \leq_{lex} \text{Row}_{i+1}$$

9.8.2. Value Symmetry

When values are interchangeable, fix the first occurrence:

$$\text{first}(v_1) < \text{first}(v_2) < \dots < \text{first}(v_k)$$

Symmetry Breaking Tip

Always add symmetry-breaking constraints when the problem has obvious symmetries. This can reduce search space by orders of magnitude with minimal modeling effort.

9.9. Summary

9.9.1. Key Takeaways

- Constraint Programming is a declarative paradigm: specify what, not how
- CSPs consist of variables, domains, and constraints
- Constraint propagation removes inconsistent values, reducing search space
- Arc consistency (AC-3) is the fundamental propagation algorithm
- Global constraints (AllDifferent, Cumulative) provide efficient specialized propagation
- Search combines variable/value ordering heuristics with backtracking
- Large Neighborhood Search scales CP to huge problems
- OR-Tools CP-SAT is a powerful, modern CP solver
- Symmetry breaking eliminates redundant search

9.10. Exercises

1. **Arc consistency:** Apply AC-3 to the CSP with variables $\{x, y, z\}$, domains $D(x) = D(y) = D(z) = \{1, 2, 3, 4\}$, and constraints $x < y$, $y < z$. Show the domains after propagation.
2. **N-Queens modeling:** Model the N-Queens problem as a CSP and solve it for $N = 12$ using OR-Tools. Count the number of solutions.
3. **Sudoku solver:** Implement a Sudoku solver using constraint programming. Use AllDifferent constraints for rows, columns, and 3x3 boxes.
4. **Scheduling problem:** Model and solve a resource-constrained project scheduling problem (RCPSP) with 20 tasks, 3 resources, and precedence constraints using CP-SAT.
5. **Symmetry breaking:** For the graph coloring problem on a cycle of 6 nodes with 3 colors, identify the symmetries and add appropriate symmetry-breaking constraints.

Part III.

Part III: Solution Methods

10. Exact Solution Methods

What You Will Learn

- Master branch-and-bound for integer programming
- Understand how LP relaxation provides bounds
- Learn cutting plane methods and their role in tightening formulations
- Apply branch-and-cut: the modern standard for integer programs
- Implement column generation for large-scale problems

When optimality matters, exact methods provide guaranteed solutions. Unlike heuristics that find “good enough” answers, exact methods prove that no better solution exists. This chapter covers the algorithms that power commercial solvers like Gurobi, CPLEX, and HiGHS.

10.1. The Challenge of Discrete Optimization

Many real-world optimization problems involve discrete decisions:

- Should we build a factory in city A or city B? (Binary choice)
- Which subset of projects should we fund? (Combinatorial selection)
- In what order should we visit customers? (Sequencing problem)
- How many units of each product should we produce? (Integer quantities)

The challenge is that these problems often have exponentially many possible solutions. For n binary variables, there are 2^n possible combinations. Exhaustive search becomes computationally infeasible even for moderate problem sizes.

NP-Hardness

Most discrete optimization problems are NP-hard, meaning no known polynomial-time algorithm can solve all instances. However, practical instances often have structure that exact methods can exploit.

10.2. Branch and Bound

Branch and Bound (B&B) is a powerful algorithmic paradigm for solving discrete and combinatorial optimization problems. It systematically explores the solution space by intelligently partitioning it into smaller subproblems and pruning branches that cannot lead to better solutions.

10.2.1. Core Principle: Intelligent Enumeration

The key insight is that we can eliminate large regions of the solution space without explicitly examining every solution.

Approach	Strategy	Guarantee	Complexity
Exhaustive Search	Enumerate all solutions	Finds optimal	Exponential
Branch and Bound	Partition and prune	Finds optimal	Often much better

10.2.2. The Algorithm

10.2.3. Fundamental Components

Branching

Branching divides a problem into disjoint subproblems that together cover all solutions in the parent:

- **Binary variable:** Create branches $x = 0$ and $x = 1$
- **Integer variable:** If LP relaxation gives $x_j = 2.7$, branch on $x_j \leq 2$ and $x_j \geq 3$
- **General:** Any partition that covers the solution space

Bounding

Compute lower bounds on the optimal value within each subproblem. The most common approach is LP relaxation:

$$z_{LP} \leq z_{IP}$$

For minimization, the LP relaxation (dropping integrality constraints) cannot have a worse objective than the integer program.

Pruning

Eliminate subproblems when they cannot contain improving solutions:

1. **Pruning by bound:** $b(N) \geq z^*$ - cannot improve incumbent
2. **Pruning by optimality:** Subproblem solved to integer optimality
3. **Pruning by infeasibility:** Subproblem has no feasible solutions

Node Selection

Strategy for choosing which node to explore next:

Strategy	Implementation	Pros	Cons
Depth-First	Stack (LIFO)	Low memory, finds feasibles fast	May explore poor regions
Best-First	Priority queue	Minimizes nodes explored	High memory
Best-Estimate	Combines bounds	Balances both	More complex

10.2.4. LP Relaxation

The LP relaxation is fundamental to branch and bound for integer programs.

Original Integer Program

$$\begin{aligned}
 &\text{minimize} && \mathbf{c}^T \mathbf{x} \\
 &\text{subject to} && A\mathbf{x} \leq \mathbf{b} \\
 &&& \mathbf{x} \geq 0, \mathbf{x} \in \mathbb{Z}^n
 \end{aligned}$$

LP Relaxation

$$\begin{aligned}
 &\text{minimize} && \mathbf{c}^T \mathbf{x} \\
 &\text{subject to} && A\mathbf{x} \leq \mathbf{b} \\
 &&& \mathbf{x} \geq 0
 \end{aligned}$$

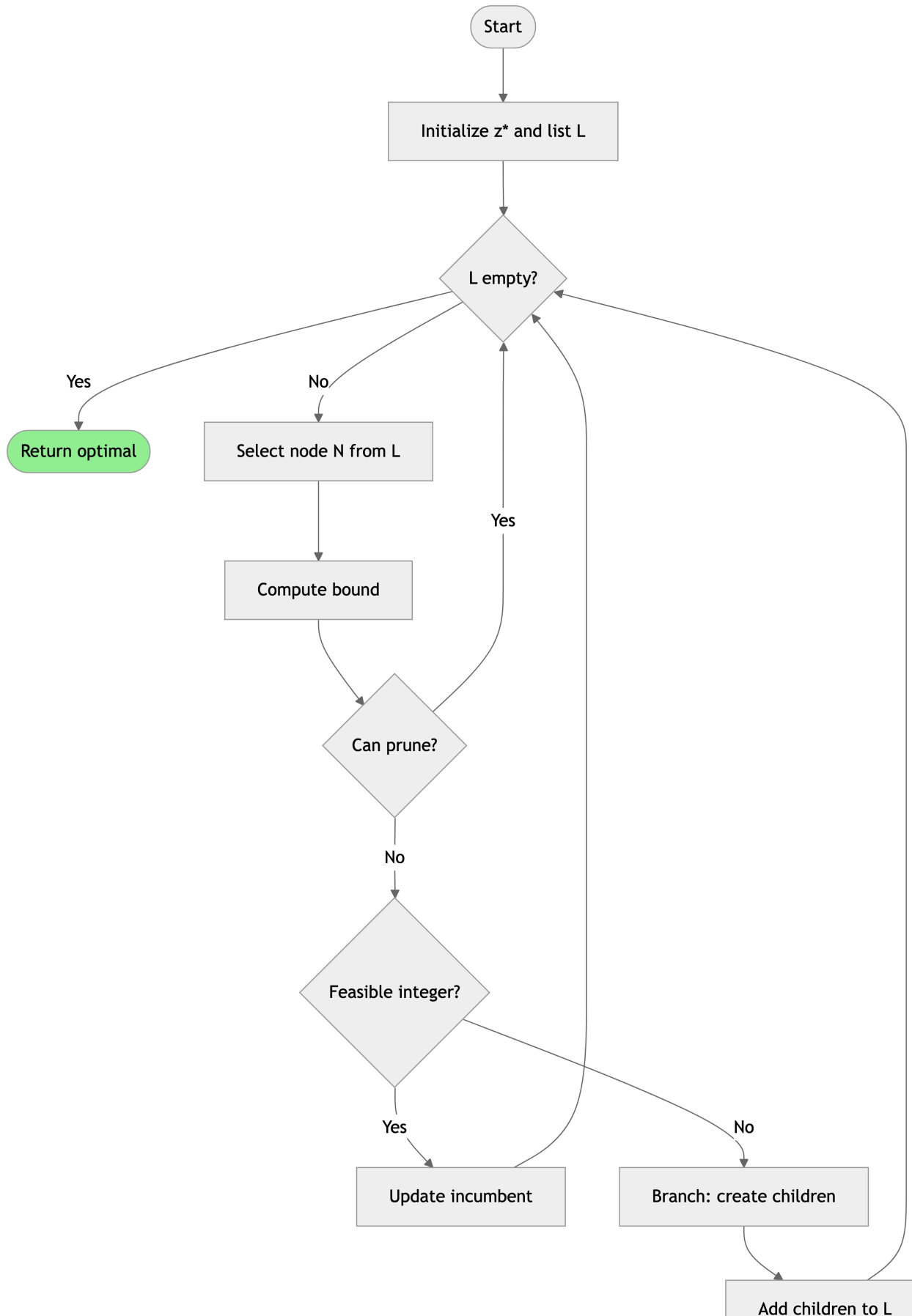


Figure 10.1.: Branch-and-Bound Algorithm Flowchart

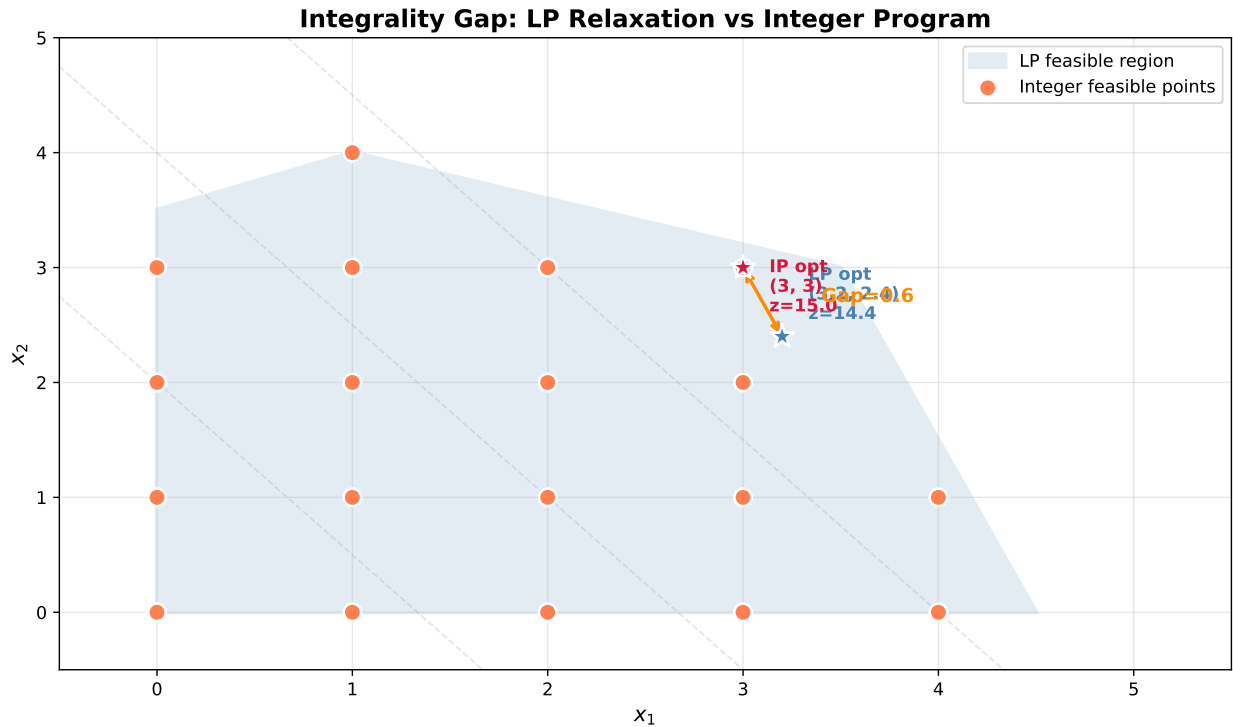


Figure 10.2.: Integrality gap illustration — the LP relaxation polytope (blue) contains all integer feasible points (red dots), with the gap between LP and IP optima shown

! Integrality Gap

The **integrality gap** is the difference between z_{LP} and z_{IP} :

- **Small gap:** LP relaxation is tight, problem likely easy
- **Large gap:** LP relaxation is weak, B&B explores many nodes
- **Zero gap:** LP optimum is integer - problem solved at root!

10.2.5. Branching Strategies

The choice of branching variable significantly affects performance:

Most Fractional

Select the variable with LP value closest to 0.5:

$$j^* = \arg \min_j \min(x_j - \lfloor x_j \rfloor, \lceil x_j \rceil - x_j)$$

Simple but not always effective.

Strong Branching

For each candidate variable, tentatively solve both child LPs and choose the variable with best bound improvement. The following pseudocode iterates over candidate variables, solves both branching directions, and selects the variable that produces the largest combined bound increase.

```

def strong_branching(candidates, lp_solution):
    best_score = -infinity
    best_var = None

    for j in candidates:
        # Solve LP with x_j <= floor(x_j)
        bound_down = solve_lp_with_bound(j, "down")

        # Solve LP with x_j >= ceil(x_j)
        bound_up = solve_lp_with_bound(j, "up")

        # Score combines both improvements
        score = min(bound_down, bound_up) * max(bound_down, bound_up)

        if score > best_score:
            best_score = score
            best_var = j

    return best_var

```

Expensive but produces balanced trees.

Pseudocost Branching

Learn from history: track how much bounds change per unit change in variable value:

$$\text{pseudocost}_j = \frac{\Delta z}{\Delta x_j}$$

Use pseudocosts to predict bound improvements without solving LPs.

Reliability Branching

Hybrid approach: use strong branching until pseudocosts are reliable, then switch to pseudocost branching.

10.3. Cutting Planes

Cutting planes are valid inequalities that tighten the LP relaxation without eliminating any integer solutions.

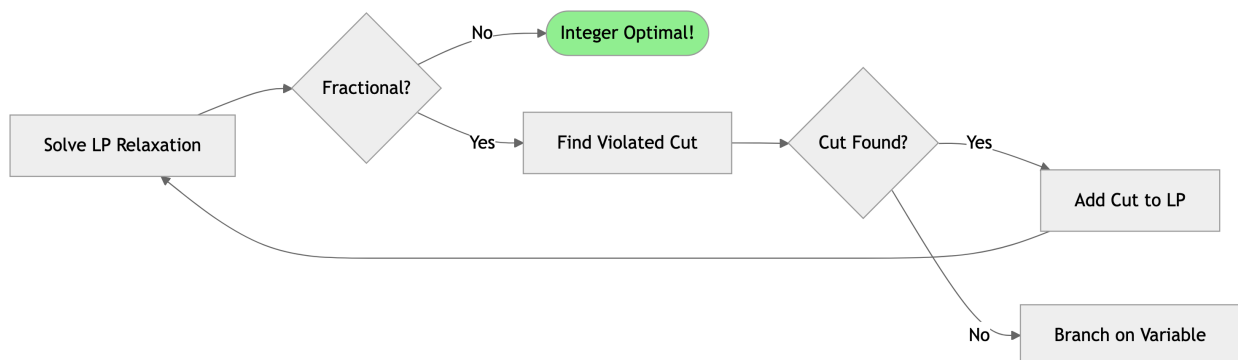


Figure 10.3.: Cutting Plane Method: Iteratively tightening the LP relaxation

10.3.1. The Concept

If the LP optimal solution $\mathbf{x}^*$ is fractional, we can add a constraint that:

1. Cuts off $\mathbf{x}^*$ (makes it infeasible)
2. Does not cut off any integer feasible solution

This tightens the LP relaxation, potentially yielding a better bound or even an integer solution.

10.3.2. Gomory Cuts

Gomory cuts are derived from the simplex tableau. For a basic variable with fractional value:

$$x_i + \sum_{j \in N} \bar{a}_{ij} x_j = \bar{b}_i$$

where $\bar{b}_i$ is fractional, the Gomory cut is:

$$\sum_{j \in N} f_{ij} x_j \geq f_i$$

where $f_i = \bar{b}_i - \lfloor \bar{b}_i \rfloor$ and $f_{ij} = \bar{a}_{ij} - \lfloor \bar{a}_{ij} \rfloor$.

10.3.3. Common Cut Families

Cut Type	Application	Derivation
Gomory	General MIP	Simplex tableau
Cover	Knapsack constraints	Minimal cover
Clique	Set packing	Graph structure
Flow cover	Network flow	Capacity analysis
Lift-and-project	General 0-1	Disjunctive argument

10.3.4. Example: Cover Inequality

For a knapsack constraint:

$$3x_1 + 4x_2 + 5x_3 \leq 7, \quad x_i \in \{0, 1\}$$

The set $\{1, 2\}$ is a **cover** because $3 + 4 = 7 > 7$ when both equal 1. This gives the cover inequality:

$$x_1 + x_2 \leq 1$$

This constraint is redundant for continuous variables but valid for binary variables.

10.4. Branch and Cut

Branch and cut combines branching with cutting plane generation at each node.

10.4.1. Algorithm

Algorithm: Branch and Cut

Input: Integer program P

Output: Optimal solution $\mathbf{x}^*$ and value $\mathbf{z}^*$

1. Initialize root node with original problem
2. While active nodes exist:
 - a) Select node N
 - b) Solve LP relaxation
 - c) While violated cuts exist and iteration limit not reached:
 - Generate violated cuts
 - Add cuts to LP
 - Re-solve LP
 - d) Apply pruning rules
 - e) If not pruned, branch
3. Return best solution found

10.4.2. Cut Generation at Nodes

Modern solvers generate cuts at:

- **Root node:** Heavy cut generation to tighten initial relaxation
- **Tree nodes:** Lighter cut generation to avoid overhead

Solver Settings

Most solvers have parameters controlling cut generation aggressiveness. For problems with weak LP relaxations, increasing cut limits often helps. For problems with tight relaxations, reducing cuts may speed up solving.

10.5. Column Generation

For problems with exponentially many variables, explicitly enumerating all columns is impractical. Column generation solves a restricted problem and dynamically adds promising columns.

10.5.1. The Idea

Consider a covering problem:

$$\begin{aligned}
 &\text{minimize} && \sum_{j \in J} c_j x_j \\
 &\text{subject to} && \sum_{j \in J} a_{ij} x_j \geq 1, \quad i \in I \\
 &&& x_j \geq 0
 \end{aligned}$$

If $|J|$ is huge (e.g., all possible patterns in cutting stock), we:

1. Start with a subset of columns $J' \subset J$
2. Solve the **restricted master problem** over J'
3. Use the dual solution to find improving columns via a **pricing subproblem**
4. Add improving columns to J' and repeat

10.5.2. Pricing Subproblem

Given dual variables π from the master problem, find a column j with negative reduced cost:

$$\bar{c}_j = c_j - \sum_i \pi_i a_{ij} < 0$$

If no such column exists, the current solution is optimal for the full problem.

10.5.3. Cutting Stock Example

A paper mill must cut rolls of width W to satisfy demands for widths $w_1, \dots, w_m$.

Master Problem: Select patterns to minimize waste

Pricing Subproblem: Find a cutting pattern (knapsack problem)

$$\begin{aligned} & \text{minimize} && 1 - \sum_i \pi_i a_i \\ & \text{subject to} && \sum_i w_i a_i \leq W \\ & && a_i \in \mathbb{Z}_+ \end{aligned}$$

10.6. Branch and Price

Branch and price combines branch and bound with column generation.

10.6.1. Challenges

When branching in column generation:

- Standard variable branching may not work (variables may not exist yet)
- Need branching schemes compatible with pricing subproblem
- Common approach: branch on aggregated constraints

10.6.2. Applications

- Vehicle routing with complex constraints
- Crew scheduling
- Cutting and packing problems
- Network design

10.7. Implementation with Python

The following implementation builds a branch-and-bound solver from scratch for the binary knapsack problem. It uses LP relaxation (with greedy fractional filling) to compute upper bounds at each node and depth-first search to traverse the B&B tree. The solver fixes variables one at a time, prunes nodes whose bounds cannot exceed the current best integer solution, and backtracks when no improvement is possible.

```
import numpy as np
from dataclasses import dataclass
from typing import Optional, List, Tuple

@dataclass
class Node:
    fixed: dict # variable -> value
    bound: float
    depth: int

def solve_lp_relaxation(values, weights, capacity, fixed):
    """Solve LP relaxation of knapsack with some variables fixed."""
    n = len(values)
```

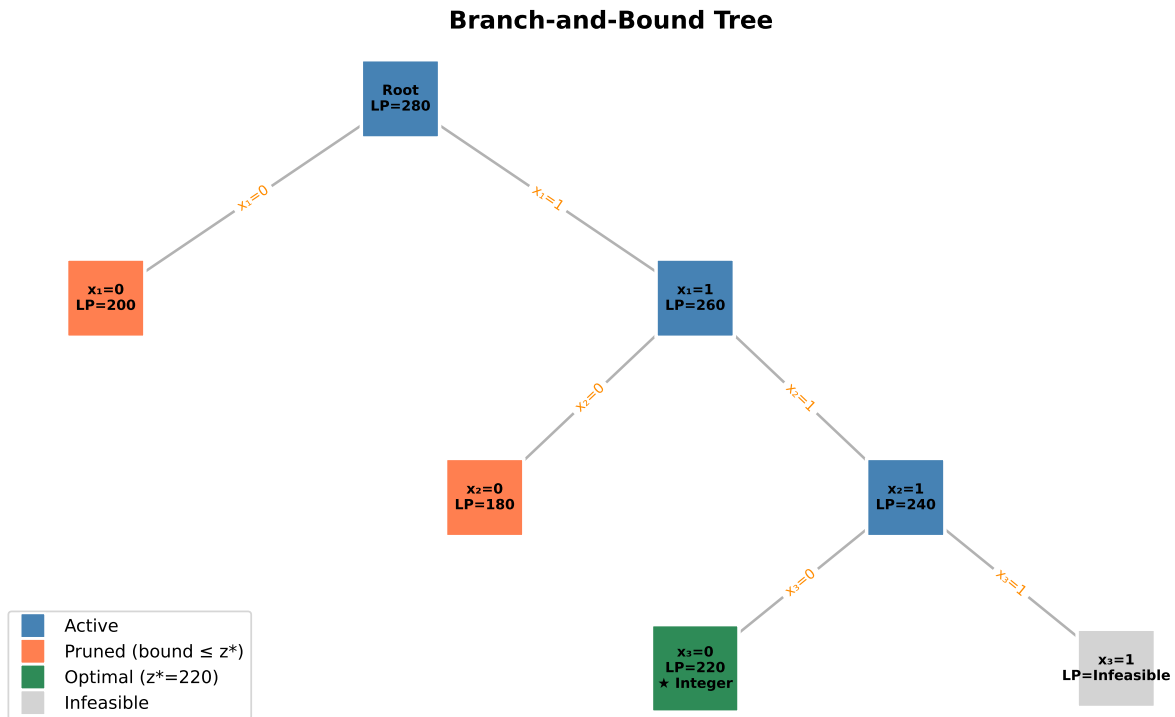


Figure 10.4.: Branch-and-Bound tree for a binary knapsack problem — nodes show LP bounds, green nodes are integer feasible, red nodes are pruned, the optimal incumbent ($z^*=220$) is highlighted

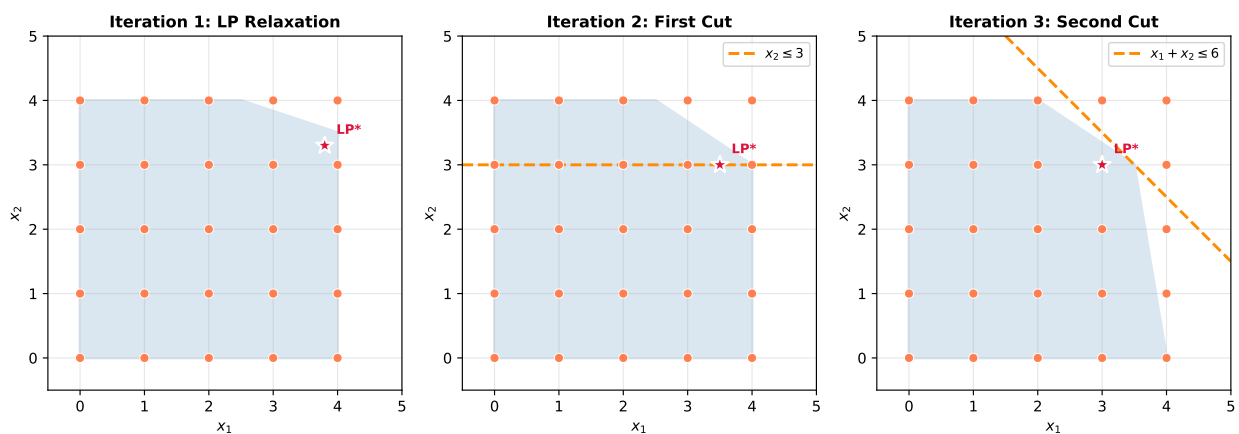


Figure 10.5.: Cutting plane tightening in 3 steps — each iteration adds a cut that removes the fractional LP optimum while preserving all integer feasible solutions

```

remaining_cap = capacity
total_value = 0

# Apply fixed variables
free_vars = []
for i in range(n):
    if i in fixed:
        if fixed[i] == 1:
            remaining_cap -= weights[i]
            total_value += values[i]
        else:
            free_vars.append(i)

if remaining_cap < 0:
    return -float('inf'), None # Infeasible

# Sort free variables by value-to-weight ratio
ratios = [(values[i]/weights[i], i) for i in free_vars]
ratios.sort(reverse=True)

solution = {i: fixed[i] for i in fixed}

for ratio, i in ratios:
    if weights[i] <= remaining_cap:
        solution[i] = 1
        remaining_cap -= weights[i]
        total_value += values[i]
    else:
        # Fractional
        frac = remaining_cap / weights[i]
        solution[i] = frac
        total_value += values[i] * frac
        break

return total_value, solution

def branch_and_bound_knapsack(values, weights, capacity):
    """Solve 0-1 knapsack using branch and bound."""
    n = len(values)

    # Initialize
    best_value = 0
    best_solution = [0] * n

    # Root node
    root = Node(fixed={}, bound=float('inf'), depth=0)
    stack = [root] # Depth-first search

    nodes_explored = 0

    while stack:
        node = stack.pop()
        nodes_explored += 1

```

```

    # Compute bound
    bound, lp_sol = solve_lp_relaxation(
        values, weights, capacity, node.fixed
    )

    # Pruning
    if bound <= best_value:
        continue # Cannot improve

    # Check if integer solution
    is_integer = True
    branch_var = None

    for i in range(n):
        if i not in node.fixed:
            if lp_sol.get(i, 0) > 0 and lp_sol.get(i, 0) < 1:
                is_integer = False
                branch_var = i
                break

    if is_integer:
        # Found integer solution
        value = sum(values[i] * int(lp_sol.get(i, 0))
                    for i in range(n))
        if value > best_value:
            best_value = value
            best_solution = [int(lp_sol.get(i, 0)) for i in range(n)]
    else:
        # Branch
        # Child 1: x[branch_var] = 0
        fixed_0 = dict(node.fixed)
        fixed_0[branch_var] = 0
        stack.append(Node(fixed_0, bound, node.depth + 1))

        # Child 2: x[branch_var] = 1
        fixed_1 = dict(node.fixed)
        fixed_1[branch_var] = 1
        stack.append(Node(fixed_1, bound, node.depth + 1))

    return best_value, best_solution, nodes_explored

# Example
values = [40, 30, 50, 10]
weights = [2, 5, 10, 5]
capacity = 16

value, solution, nodes = branch_and_bound_knapsack(values, weights, capacity)
print(f"Optimal value: {value}")
print(f"Solution: {solution}")
print(f"Nodes explored: {nodes}")

```

10.8. Modern Solver Enhancements

Commercial solvers incorporate many advanced techniques:

10.8.1. Preprocessing

Before branch and bound starts:

- **Variable fixing:** Determine variables that must take certain values
- **Constraint tightening:** Strengthen bounds using logical implications
- **Probing:** Temporarily fix variables to identify implications

10.8.2. Primal Heuristics

Finding good feasible solutions early improves pruning:

- **Rounding:** Round fractional LP solutions
- **Diving:** Fix variables iteratively based on LP
- **RINS:** Fix variables agreeing in LP and incumbent
- **Local search:** Improve incumbents through neighborhood exploration

10.8.3. Conflict Learning

Record inconsistent variable assignments and add no-good constraints to prevent revisiting infeasible combinations. Borrowed from SAT solving.

10.8.4. Symmetry Breaking

Many problems have symmetric solutions. Add constraints or use specialized branching to eliminate redundant exploration.

10.9. Practical Guidelines

10.9.1. When to Use Exact Methods

Situation	Recommendation
Optimality required	Use exact methods
Small to medium problems	Try exact first
Tight formulations available	Exact methods work well
Time-critical applications	Consider heuristics
Very large problems	Heuristics or decomposition

10.9.2. Improving Solver Performance

1. **Formulation:** Tighter formulations dramatically reduce solve time
2. **Preprocessing:** Let the solver preprocess; don't over-constrain manually
3. **Symmetry:** Break symmetry if present
4. **Warm starts:** Provide good initial solutions
5. **Tuning:** Adjust solver parameters for your problem class

Rule of Thumb

Spending time on a better formulation often pays off more than algorithmic improvements. A tight formulation can reduce solve time by orders of magnitude.

10.10. Dynamic Programming

Dynamic Programming (DP) is a powerful algorithmic paradigm for solving optimization problems by breaking them into overlapping subproblems. Unlike divide-and-conquer, DP stores solutions to subproblems to avoid redundant computation.

10.10.1. Bellman's Principle of Optimality

The foundation of dynamic programming is Bellman's principle:

! Principle of Optimality

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

In mathematical terms, if we define $V(s)$ as the optimal value starting from state s :

$$V(s) = \max_{a \in A(s)} \{r(s, a) + V(s')\}$$

where s' is the state resulting from action a in state s .

10.10.2. Problem Structure

DP applies when a problem has:

Property	Description
Optimal substructure	Optimal solution contains optimal solutions to subproblems
Overlapping subproblems	Same subproblems are solved multiple times

10.10.3. Solution Approaches

Top-Down (Memoization)

Solve recursively but cache results. The Fibonacci sequence provides a clear example: each value depends on two previous values, so naive recursion recomputes the same subproblems exponentially many times. Memoization eliminates this redundancy by storing each result after its first computation.

```
def fibonacci_memoized(n, memo=None):
    """Fibonacci using memoization (top-down DP)."""
    if memo is None:
        memo = {}

    if n in memo:
        return memo[n]

    if n <= 1:
        return n

    memo[n] = fibonacci_memoized(n - 1, memo) + fibonacci_memoized(n - 2, memo)
    return memo[n]
```

```
# Alternative using functools
from functools import lru_cache

@lru_cache(maxsize=None)
def fibonacci_cached(n):
    """Fibonacci with automatic memoization."""
    if n <= 1:
        return n
    return fibonacci_cached(n - 1) + fibonacci_cached(n - 2)
```

Bottom-Up (Tabulation)

Build solution iteratively from smallest subproblems:

```
def fibonacci_tabulation(n):
    """Fibonacci using tabulation (bottom-up DP)."""
    if n <= 1:
        return n

    # Table to store subproblem solutions
    dp = [0] * (n + 1)
    dp[1] = 1

    for i in range(2, n + 1):
        dp[i] = dp[i - 1] + dp[i - 2]

    return dp[n]

def fibonacci_space_optimized(n):
    """Fibonacci with O(1) space."""
    if n <= 1:
        return n

    prev2, prev1 = 0, 1
    for _ in range(2, n + 1):
        curr = prev1 + prev2
        prev2, prev1 = prev1, curr

    return prev1
```

Approach	Advantages	Disadvantages
Top-Down	Natural recursion, only computes needed subproblems	Recursion overhead, stack limits
Bottom-Up	No recursion overhead, explicit dependencies	May compute unnecessary subproblems

10.10.4. Classic DP Problems

0/1 Knapsack

Given items with weights w_i and values v_i , maximize total value without exceeding capacity W :

$$V(i, c) = \max\{V(i - 1, c), v_i + V(i - 1, c - w_i)\}$$

The code below implements this recurrence using a 2D table where `dp[i][c]` stores the maximum value achievable using the first `i` items with remaining capacity `c`. After filling the table, it backtracks through the entries to recover which items were selected.

```
def knapsack_dp(values, weights, capacity):
    """0/1 Knapsack using dynamic programming.

    Args:
        values: Item values
        weights: Item weights
        capacity: Knapsack capacity

    Returns:
        Maximum value and selected items
    """
    n = len(values)

    # dp[i][c] = max value using items 0..i-1 with capacity c
    dp = [[0] * (capacity + 1) for _ in range(n + 1)]

    for i in range(1, n + 1):
        for c in range(capacity + 1):
            # Don't take item i-1
            dp[i][c] = dp[i - 1][c]

            # Take item i-1 if it fits
            if weights[i - 1] <= c:
                dp[i][c] = max(
                    dp[i][c],
                    values[i - 1] + dp[i - 1][c - weights[i - 1]]
                )

    # Backtrack to find selected items
    selected = []
    c = capacity
    for i in range(n, 0, -1):
        if dp[i][c] != dp[i - 1][c]:
            selected.append(i - 1)
            c -= weights[i - 1]

    return dp[n][capacity], selected[::-1]

# Example
values = [60, 100, 120]
weights = [10, 20, 30]
capacity = 50

max_value, items = knapsack_dp(values, weights, capacity)
print(f"Maximum value: {max_value}")
```

```
print(f"Selected items: {items}")
```

Shortest Path (Bellman-Ford)

Find shortest paths from source to all vertices, handling negative edges. The Bellman-Ford algorithm relaxes every edge $|V| - 1$ times, then checks for negative-weight cycles in one additional pass.

```
def bellman_ford(n, edges, source):
    """Bellman-Ford shortest path algorithm.

    Args:
        n: Number of vertices
        edges: List of (u, v, weight) tuples
        source: Source vertex

    Returns:
        Distances and predecessors, or None if negative cycle
    """
    INF = float('inf')
    dist = [INF] * n
    pred = [-1] * n
    dist[source] = 0

    # Relax all edges n-1 times
    for _ in range(n - 1):
        for u, v, w in edges:
            if dist[u] != INF and dist[u] + w < dist[v]:
                dist[v] = dist[u] + w
                pred[v] = u

    # Check for negative cycles
    for u, v, w in edges:
        if dist[u] != INF and dist[u] + w < dist[v]:
            return None # Negative cycle detected

    return dist, pred
```

Longest Common Subsequence

Find the longest subsequence common to two sequences:

$$L(i, j) = \begin{cases} L(i-1, j-1) + 1 & \text{if } X_i = Y_j \\ \max(L(i-1, j), L(i, j-1)) & \text{otherwise} \end{cases}$$

This implementation fills a 2D table according to the recurrence above, then backtracks to reconstruct the actual subsequence.

```
def longest_common_subsequence(X, Y):
    """Find LCS of two sequences.

    Args:
        X, Y: Input sequences
```

```

Returns:
    Length and the LCS itself
"""
m, n = len(X), len(Y)

# dp[i][j] = LCS length for X[:i] and Y[:j]
dp = [[0] * (n + 1) for _ in range(m + 1)]

for i in range(1, m + 1):
    for j in range(1, n + 1):
        if X[i - 1] == Y[j - 1]:
            dp[i][j] = dp[i - 1][j - 1] + 1
        else:
            dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])

# Backtrack to find LCS
lcs = []
i, j = m, n
while i > 0 and j > 0:
    if X[i - 1] == Y[j - 1]:
        lcs.append(X[i - 1])
        i -= 1
        j -= 1
    elif dp[i - 1][j] > dp[i][j - 1]:
        i -= 1
    else:
        j -= 1

return dp[m][n], lcs[::-1]

```

10.10.5. DP for Optimization

Resource Allocation

Allocate budget across projects to maximize return. The DP table `dp[i][b]` tracks the best return from the first `i` projects given budget `b`, trying each possible investment level per project.

```

def resource_allocation(budget, projects):
    """Optimal resource allocation across projects.

    Args:
        budget: Total available budget (integer)
        projects: List of (cost, return) pairs for discrete allocations

    Returns:
        Maximum return and allocation
    """
    n = len(projects)

    # dp[i][b] = max return using projects 0..i-1 with budget b
    dp = [[0] * (budget + 1) for _ in range(n + 1)]

    for i in range(1, n + 1):
        for b in range(budget + 1):

```

```

    dp[i][b] = dp[i - 1][b] # Don't invest in project i-1

    # Try different investment levels in project i-1
    for cost, ret in projects[i - 1]:
        if cost <= b:
            dp[i][b] = max(dp[i][b], ret + dp[i - 1][b - cost])

    return dp[n][budget]

```

10.10.6. Complexity Analysis

Problem	Time Complexity	Space Complexity
Fibonacci	$O(n)$	$O(1)$ optimized
0/1 Knapsack	$O(nW)$	$O(nW)$ or $O(W)$
LCS	$O(mn)$	$O(mn)$ or $O(\min(m, n))$
Shortest Path	$O(VE)$	$O(V)$

Pseudo-Polynomial Time

The knapsack algorithm runs in $O(nW)$ time, which is pseudo-polynomial because W is a numeric value, not an input size. This is why knapsack remains NP-hard despite having a DP solution.

10.11. Summary

10.11.1. Key Takeaways

- Branch and bound systematically explores the solution space with intelligent pruning
- LP relaxation provides lower bounds for minimization problems
- Cutting planes tighten the LP relaxation without eliminating integer solutions
- Branch and cut combines branching and cutting for state-of-the-art performance
- Column generation handles problems with exponentially many variables
- Dynamic programming solves optimization problems by combining solutions to overlapping subproblems
- Memoization (top-down) and tabulation (bottom-up) are two DP implementation strategies
- Modern solvers combine these techniques with preprocessing, heuristics, and learning
- Problem formulation often matters more than algorithmic details

10.12. Exercises

1. **Branch and bound trace:** For the knapsack problem with items (value, weight): (60, 10), (100, 20), (120, 30) and capacity 50, trace the branch and bound algorithm using depth-first search.
2. **Cutting plane derivation:** For the constraint $5x_1 + 7x_2 + 4x_3 \leq 15$ with binary variables, find a cover inequality.
3. **LP relaxation gap:** Consider a set covering problem where you must cover 3 elements using sets with costs. Set 1 covers elements {1,2} with cost 3, Set 2 covers {2,3} with cost 3, Set 3 covers {1,3} with cost 4. Formulate as an IP, solve the LP relaxation, and compute the integrality gap.

4. **Implementation:** Extend the Python branch and bound code to use best-first search instead of depth-first. Compare node counts on several test instances.

11. Metaheuristics

What You Will Learn

- Understand metaheuristic design principles and the exploration-exploitation trade-off
- Implement simulated annealing for escaping local optima
- Design genetic algorithms with selection, crossover, and mutation operators
- Apply particle swarm optimization and differential evolution
- Use tabu search with memory-based strategies
- Master CMA-ES for continuous black-box optimization

When exact methods become computationally intractable due to problem size or complexity, metaheuristics provide high-quality solutions in reasonable time. These algorithms trade optimality guarantees for practical efficiency, often finding near-optimal solutions for problems that would otherwise be unsolvable.

11.1. The Metaheuristic Paradigm

11.1.1. What Are Metaheuristics?

Metaheuristics are high-level problem-solving strategies that guide the search process toward good solutions. The term combines “meta” (beyond) and “heuristic” (to find), suggesting algorithms that transcend simple greedy approaches.

Characteristic	Description
Problem-independent	General frameworks adaptable to various domains
Incomplete	Do not guarantee finding the global optimum
Stochastic	Incorporate randomness in search decisions
Iterative	Progressively improve solutions over iterations
Memory-based	Some use search history to guide exploration

11.1.2. Exploration vs. Exploitation

The fundamental challenge in metaheuristic design is balancing two competing goals:

Exploration: Searching broadly across the solution space to find promising regions. Too much exploration wastes time in poor areas.

Exploitation: Intensifying search in promising regions to find the best local solutions. Too much exploitation causes premature convergence.

The Exploration-Exploitation Dilemma

Most metaheuristics have parameters that control this balance. Understanding how to tune these parameters is often more important than the choice of algorithm.

11. Metaheuristics

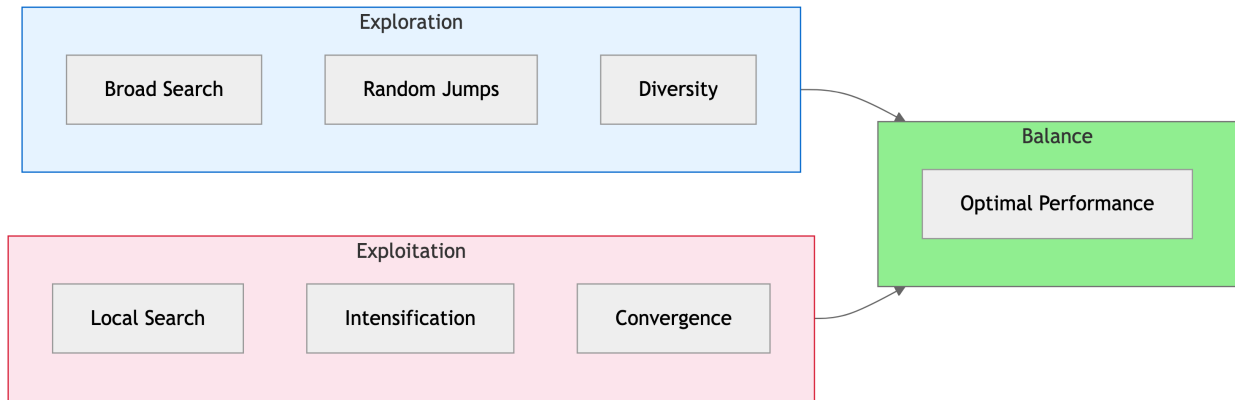


Figure 11.1.: The Exploration-Exploitation Trade-off in Metaheuristics

11.1.3. Classification of Metaheuristics

Category	Examples	Key Idea
Single-solution	Simulated Annealing, Tabu Search	Improve one solution iteratively
Population-based	GA, PSO, DE, ACO	Maintain and evolve a population
Nature-inspired	ACO, PSO, Firefly	Mimic biological or physical processes
Memory-based	Tabu Search	Use search history to guide decisions

11.2. Simulated Annealing

Simulated Annealing (SA) draws inspiration from the metallurgical process of annealing, where metals are heated and slowly cooled to reduce defects and reach low-energy states.

11.2.1. The Algorithm

The key insight is accepting worse solutions probabilistically, with acceptance probability decreasing over time:

$$P(\text{accept}) = \begin{cases} 1 & \text{if } \Delta E < 0 \\ e^{-\Delta E/T} & \text{if } \Delta E \geq 0 \end{cases}$$

where $\Delta E = f(x') - f(x)$ is the change in objective value and T is the temperature parameter.

The implementation below uses geometric cooling, reducing the temperature by a fixed ratio each step. The `T_initial` and `T_final` parameters define the temperature range, while `cooling_rate` controls how quickly the algorithm shifts from exploration to exploitation. At each temperature level, the algorithm performs `iterations_per_temp` neighbor evaluations before cooling.

```

import numpy as np
from typing import Callable, Tuple, List

def simulated_annealing(
    objective: Callable,
    initial_solution: np.ndarray,

```



```

neighbor_fn: Callable,
T_initial: float = 1000.0,
T_final: float = 1e-8,
cooling_rate: float = 0.95,
iterations_per_temp: int = 100
) -> Tuple[np.ndarray, float, List[float]]:
    """Simulated annealing optimization.

    Args:
        objective: Function to minimize
        initial_solution: Starting point
        neighbor_fn: Function generating neighboring solutions
        T_initial: Starting temperature
        T_final: Stopping temperature
        cooling_rate: Temperature decay factor
        iterations_per_temp: Iterations at each temperature

    Returns:
        Best solution, best value, history
    """
    current = initial_solution.copy()
    current_cost = objective(current)

    best = current.copy()
    best_cost = current_cost
    history = [best_cost]

    T = T_initial

    while T > T_final:
        for _ in range(iterations_per_temp):
            # Generate neighbor
            neighbor = neighbor_fn(current)
            neighbor_cost = objective(neighbor)

            # Calculate acceptance probability
            delta = neighbor_cost - current_cost

            if delta < 0 or np.random.random() < np.exp(-delta / T):
                current = neighbor
                current_cost = neighbor_cost

            if current_cost < best_cost:
                best = current.copy()
                best_cost = current_cost

        history.append(best_cost)
        T *= cooling_rate

    return best, best_cost, history

```

11.2.2. Cooling Schedules

The temperature schedule controls the exploration-exploitation balance:

Schedule	Formula	Characteristics
Geometric	$T_{k+1} = \alpha T_k$	Most common, $\alpha \in [0.9, 0.99]$
Linear	$T_{k+1} = T_k - \Delta T$	Simple but may cool too fast
Logarithmic	$T_k = T_0 / \ln(k + 1)$	Theoretically optimal but slow
Adaptive	Adjusts based on acceptance rate	Self-tuning behavior

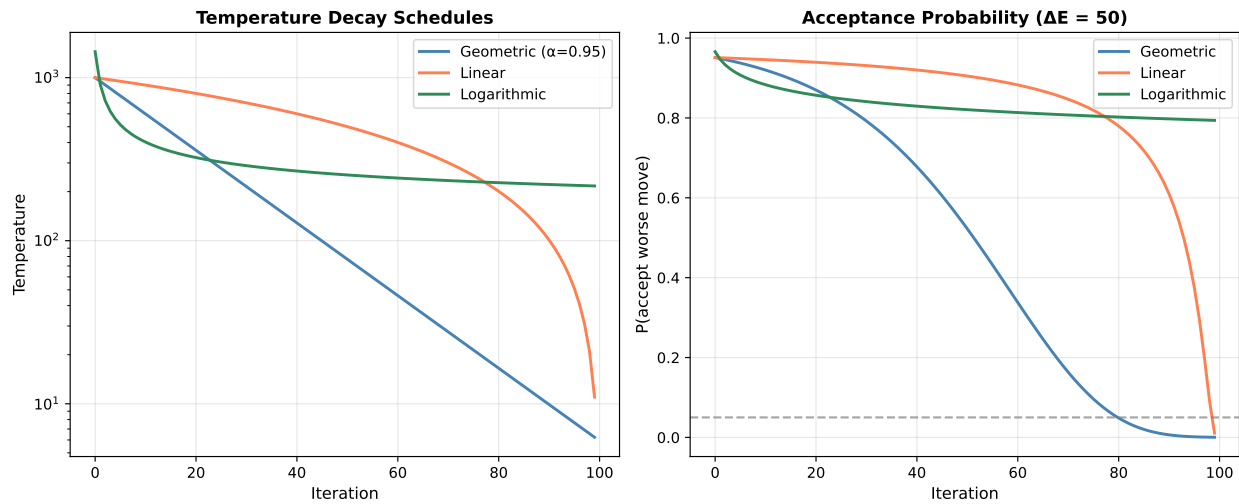


Figure 11.2.: Simulated Annealing: Temperature decay and acceptance probability across different cooling schedules

💡 Practical SA Guidelines

- Set initial temperature high enough that most moves are accepted initially
- Cool slowly enough to allow equilibration at each temperature
- Use problem-specific neighborhood structures for efficiency

11.3. Genetic Algorithms

Genetic Algorithms (GAs) apply principles of natural selection and genetics to optimization. A population of candidate solutions evolves over generations through selection, crossover, and mutation.

11.3.1. Core Components

Representation: Encode solutions as chromosomes (binary strings, real vectors, permutations)

Selection: Choose parents for reproduction based on fitness:

Selection operators determine which individuals become parents for the next generation. Tournament selection picks the best from a random subset, applying controllable selection pressure through the tournament size. Roulette wheel selection assigns each individual a probability proportional to its fitness.

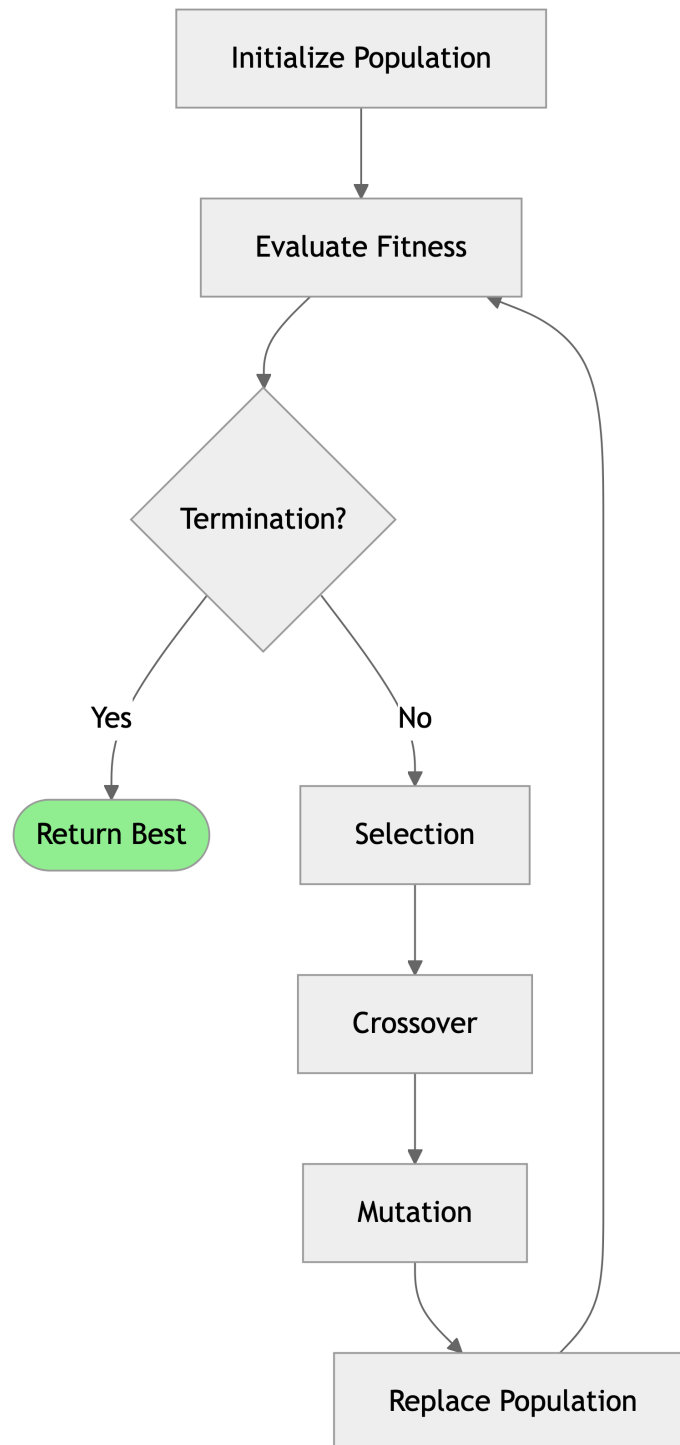


Figure 11.3.: Genetic Algorithm Evolutionary Cycle

```

def tournament_selection(population: np.ndarray, fitness: np.ndarray,
                        tournament_size: int = 3) -> np.ndarray:
    """Select parent via tournament selection."""
    indices = np.random.choice(len(population), tournament_size, replace=False)
    winner = indices[np.argmin(fitness[indices])]
    return population[winner].copy()

def roulette_wheel_selection(population: np.ndarray,
                           fitness: np.ndarray) -> np.ndarray:
    """Select parent via fitness-proportionate selection."""
    # Convert to maximization (lower is better)
    max_fit = np.max(fitness)
    adjusted = max_fit - fitness + 1e-6

    probabilities = adjusted / adjusted.sum()
    idx = np.random.choice(len(population), p=probabilities)
    return population[idx].copy()

```

Crossover: Combine genetic material from parents:

Crossover operators produce offspring by recombining segments of two parent chromosomes. Single-point crossover works for binary and real-valued representations, while order crossover (OX) preserves relative ordering for permutation problems such as TSP.

```

def single_point_crossover(parent1: np.ndarray, parent2: np.ndarray,
                          crossover_rate: float = 0.8) -> Tuple[np.ndarray,
↳ np.ndarray]:
    """Single-point crossover for binary/real chromosomes."""
    if np.random.random() > crossover_rate:
        return parent1.copy(), parent2.copy()

    point = np.random.randint(1, len(parent1))
    child1 = np.concatenate([parent1[:point], parent2[point:]])
    child2 = np.concatenate([parent2[:point], parent1[point:]])
    return child1, child2

def order_crossover(parent1: np.ndarray, parent2: np.ndarray) -> np.ndarray:
    """Order crossover (OX) for permutation problems like TSP."""
    n = len(parent1)
    start, end = sorted(np.random.choice(n, 2, replace=False))

    child = np.full(n, -1)
    child[start:end] = parent1[start:end]

    # Fill remaining positions with parent2's order
    remaining = [x for x in parent2 if x not in child]
    idx = 0
    for i in range(n):
        if child[i] == -1:
            child[i] = remaining[idx]
            idx += 1

    return child

```

Mutation: Introduce random variations:

Mutation operators add small random perturbations to maintain genetic diversity and prevent premature convergence. Gaussian mutation suits real-valued encodings, while swap mutation is designed for permutation-based representations.

```
def gaussian_mutation(individual: np.ndarray, mutation_rate: float = 0.1,
                     sigma: float = 0.1) -> np.ndarray:
    """Gaussian mutation for real-valued chromosomes."""
    mutant = individual.copy()
    mask = np.random.random(len(individual)) < mutation_rate
    mutant[mask] += np.random.normal(0, sigma, mask.sum())
    return mutant

def swap_mutation(individual: np.ndarray,
                  mutation_rate: float = 0.1) -> np.ndarray:
    """Swap mutation for permutation chromosomes."""
    mutant = individual.copy()
    if np.random.random() < mutation_rate:
        i, j = np.random.choice(len(mutant), 2, replace=False)
        mutant[i], mutant[j] = mutant[j], mutant[i]
    return mutant
```

11.3.2. Complete GA Implementation

The `GeneticAlgorithm` class below assembles the selection, crossover, and mutation operators defined above into a full evolutionary loop. It uses tournament selection to pick parents, single-point crossover to recombine them, and Gaussian mutation to introduce variation. Elitism preserves the top individuals across generations, preventing the loss of good solutions during reproduction.

```
class GeneticAlgorithm:
    """Genetic Algorithm for continuous optimization."""

    def __init__(
        self,
        objective: Callable,
        n_dims: int,
        bounds: Tuple[float, float],
        pop_size: int = 100,
        crossover_rate: float = 0.8,
        mutation_rate: float = 0.1,
        elitism: int = 2
    ):
        self.objective = objective
        self.n_dims = n_dims
        self.bounds = bounds
        self.pop_size = pop_size
        self.crossover_rate = crossover_rate
        self.mutation_rate = mutation_rate
        self.elitism = elitism

    def initialize_population(self) -> np.ndarray:
        """Create random initial population."""
```

```

        low, high = self.bounds
        return np.random.uniform(low, high, (self.pop_size, self.n_dims))

def evaluate(self, population: np.ndarray) -> np.ndarray:
    """Evaluate fitness of all individuals."""
    return np.array([self.objective(ind) for ind in population])

def evolve(self, n_generations: int) -> Tuple[np.ndarray, float, List[float]]:
    """Run genetic algorithm evolution."""
    population = self.initialize_population()
    fitness = self.evaluate(population)

    best_idx = np.argmin(fitness)
    best = population[best_idx].copy()
    best_fitness = fitness[best_idx]
    history = [best_fitness]

    for gen in range(n_generations):
        # Sort by fitness for elitism
        sorted_idx = np.argsort(fitness)
        new_population = [population[i].copy()
                          for i in sorted_idx[:self.elitism]]

        # Generate rest of new population
        while len(new_population) < self.pop_size:
            # Selection
            parent1 = tournament_selection(population, fitness)
            parent2 = tournament_selection(population, fitness)

            # Crossover
            child1, child2 = single_point_crossover(
                parent1, parent2, self.crossover_rate
            )

            # Mutation
            child1 = gaussian_mutation(child1, self.mutation_rate)
            child2 = gaussian_mutation(child2, self.mutation_rate)

            # Clip to bounds
            low, high = self.bounds
            child1 = np.clip(child1, low, high)
            child2 = np.clip(child2, low, high)

            new_population.extend([child1, child2])

        population = np.array(new_population[:self.pop_size])
        fitness = self.evaluate(population)

        # Update best
        gen_best_idx = np.argmin(fitness)
        if fitness[gen_best_idx] < best_fitness:
            best = population[gen_best_idx].copy()
            best_fitness = fitness[gen_best_idx]

```

```

        history.append(best_fitness)

    return best, best_fitness, history

```

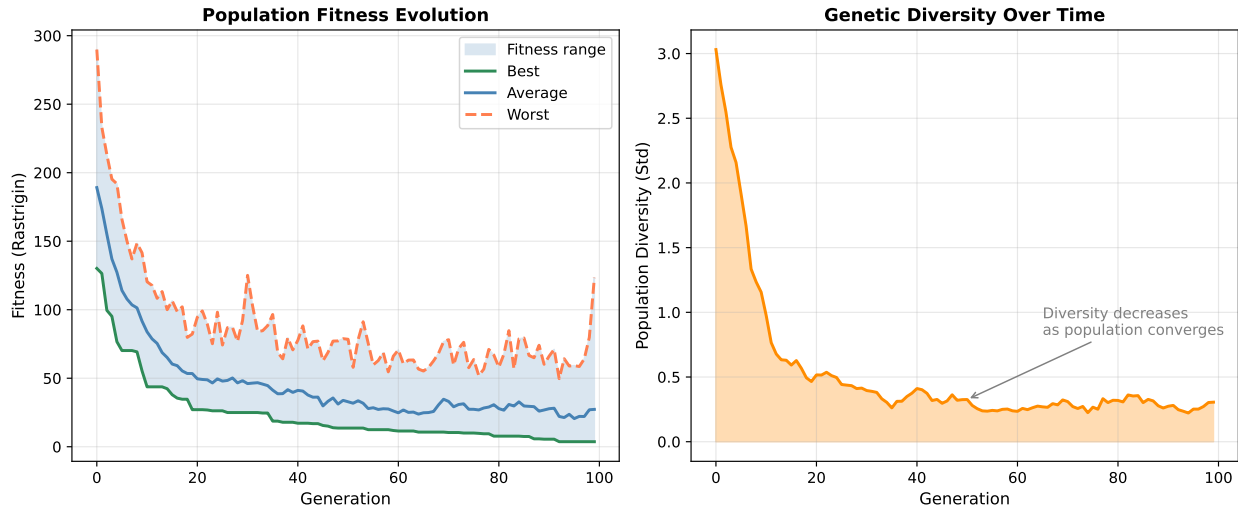


Figure 11.4.: Genetic Algorithm: Population fitness evolution over generations showing best, average, and worst fitness

11.4. Particle Swarm Optimization

Particle Swarm Optimization (PSO) simulates social behavior of bird flocking or fish schooling. Particles move through the search space, influenced by their own best position and the swarm's best position.

11.4.1. Velocity and Position Update

Each particle has position $\mathbf{x}$ and velocity $\mathbf{v}$:

$$\mathbf{v}_i^{t+1} = w\mathbf{v}_i^t + c_1r_1(\mathbf{p}_i - \mathbf{x}_i^t) + c_2r_2(\mathbf{g} - \mathbf{x}_i^t)$$

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1}$$

where: - w is inertia weight - c_1, c_2 are cognitive and social coefficients - r_1, r_2 are random numbers in $[0, 1]$ - $\mathbf{p}_i$ is particle's personal best - $\mathbf{g}$ is global best

The `ParticleSwarmOptimizer` class implements the standard PSO loop. Each particle updates its velocity using three components: inertia (previous velocity scaled by w), a cognitive term pulling toward the particle's personal best, and a social term pulling toward the global best. The inertia weight w controls how much momentum each particle retains, directly governing the exploration-exploitation balance.

```

class ParticleSwarmOptimizer:
    """Particle Swarm Optimization."""

```

```

def __init__(
    self,
    objective: Callable,
    n_dims: int,
    bounds: Tuple[float, float],
    n_particles: int = 50,
    w: float = 0.7,
    c1: float = 1.5,
    c2: float = 1.5
):
    self.objective = objective
    self.n_dims = n_dims
    self.bounds = bounds
    self.n_particles = n_particles
    self.w = w
    self.c1 = c1
    self.c2 = c2

def optimize(self, n_iterations: int) -> Tuple[np.ndarray, float, List[float]]:
    """Run PSO optimization."""
    low, high = self.bounds

    # Initialize positions and velocities
    positions = np.random.uniform(low, high, (self.n_particles, self.n_dims))
    velocities = np.random.uniform(
        -(high-low)/2, (high-low)/2, (self.n_particles, self.n_dims)
    )

    # Initialize personal bests
    personal_best_pos = positions.copy()
    personal_best_val = np.array([self.objective(p) for p in positions])

    # Initialize global best
    global_best_idx = np.argmin(personal_best_val)
    global_best_pos = personal_best_pos[global_best_idx].copy()
    global_best_val = personal_best_val[global_best_idx]

    history = [global_best_val]

    for iteration in range(n_iterations):
        for i in range(self.n_particles):
            r1, r2 = np.random.random(self.n_dims),
            ↪ np.random.random(self.n_dims)

            # Update velocity
            cognitive = self.c1 * r1 * (personal_best_pos[i] - positions[i])
            social = self.c2 * r2 * (global_best_pos - positions[i])
            velocities[i] = self.w * velocities[i] + cognitive + social

            # Update position
            positions[i] = positions[i] + velocities[i]
            positions[i] = np.clip(positions[i], low, high)

```



```

        # Evaluate
        fitness = self.objective(positions[i])

        # Update personal best
        if fitness < personal_best_val[i]:
            personal_best_val[i] = fitness
            personal_best_pos[i] = positions[i].copy()

        # Update global best
        if fitness < global_best_val:
            global_best_val = fitness
            global_best_pos = positions[i].copy()

        history.append(global_best_val)

    return global_best_pos, global_best_val, history

```

11.4.2. PSO Variants

Variant	Modification	Purpose
Standard PSO	Fixed w , c_1 , c_2	Baseline implementation
LDWPSO	Linearly decreasing w	Better exploration-exploitation balance
CLPSO	Comprehensive learning	Avoid premature convergence
QPSO	Quantum-inspired	No velocity, probability distribution

11.5. Tabu Search

Tabu Search (TS), developed by Fred Glover in 1986, uses memory structures to guide local search beyond local optima. Unlike memoryless methods, TS remembers recent moves and forbids revisiting them.

11.5.1. Memory Structures

Short-term memory (Tabu List): Forbids recent moves to prevent cycling:

The `TabuSearch` class below implements a tabu list as a dictionary mapping moves to the iteration at which they become non-tabu. At each step, the algorithm evaluates all neighbors and selects the best non-tabu move. An aspiration criterion overrides the tabu status when a move produces a new global best, preventing the memory from blocking genuinely good transitions.

```

class TabuSearch:
    """Tabu Search for combinatorial optimization."""

    def __init__(
        self,
        objective: Callable,
        neighbor_fn: Callable,
        tabu_tenure: int = 7,
    ):

```

```

        max_iterations: int = 1000
    ):
        self.objective = objective
        self.neighbor_fn = neighbor_fn
        self.tabu_tenure = tabu_tenure
        self.max_iterations = max_iterations

    def search(self, initial_solution) -> Tuple[any, float, List[float]]:
        """Run tabu search."""
        current = initial_solution
        current_cost = self.objective(current)

        best = current
        best_cost = current_cost
        history = [best_cost]

        # Tabu list: maps move to iteration when it becomes non-tabu
        tabu_list = {}

        for iteration in range(self.max_iterations):
            # Generate all neighbors
            neighbors = self.neighbor_fn(current)

            # Find best non-tabu neighbor (or satisfying aspiration)
            best_neighbor = None
            best_neighbor_cost = float('inf')
            best_move = None

            for neighbor, move in neighbors:
                cost = self.objective(neighbor)

                # Check if move is tabu
                is_tabu = move in tabu_list and tabu_list[move] > iteration

                # Aspiration criterion: accept if better than best known
                if cost < best_cost or not is_tabu:
                    if cost < best_neighbor_cost:
                        best_neighbor = neighbor
                        best_neighbor_cost = cost
                        best_move = move

            if best_neighbor is None:
                break

            # Make move
            current = best_neighbor
            current_cost = best_neighbor_cost

            # Update tabu list
            tabu_list[best_move] = iteration + self.tabu_tenure

            # Update best solution
            if current_cost < best_cost:

```

```

        best = current
        best_cost = current_cost

    history.append(best_cost)

    return best, best_cost, history

```

11.5.2. Advanced Tabu Features

Aspiration criteria: Override tabu status for exceptional moves (e.g., if move leads to new best solution)

Intensification: Focus search on promising regions by reducing neighborhood or fixing good components

Diversification: Escape stagnation by penalizing frequently visited components or restarting from diverse solutions

i Tabu Tenure Selection

Tabu tenure (how long moves stay forbidden) is the most important parameter. Too short allows cycling; too long over-constrains the search. Values between $\sqrt{n}$ and n (where n is problem size) often work well.

11.6. Ant Colony Optimization

Ant Colony Optimization (ACO), introduced by Marco Dorigo in 1991, mimics how ant colonies find shortest paths through pheromone communication.

11.6.1. The Ant System

Ants construct solutions probabilistically, guided by pheromone trails and heuristic information:

$$p_{ij}^k = \frac{[\tau_{ij}]^\alpha \cdot [\eta_{ij}]^\beta}{\sum_{l \in \text{allowed}} [\tau_{il}]^\alpha \cdot [\eta_{il}]^\beta}$$

where τ_{ij} is pheromone on edge (i, j) , η_{ij} is heuristic information (e.g., $1/d_{ij}$ for TSP), and α , β control their relative importance.

This ACO implementation targets the Traveling Salesman Problem. Each ant builds a complete tour by probabilistically selecting the next city based on pheromone intensity and inverse distance. After all ants finish their tours, pheromone trails evaporate at rate `rho` and receive deposits proportional to tour quality, reinforcing edges that appear in shorter tours.

```

class AntColonyOptimizer:
    """Ant Colony Optimization for TSP."""

    def __init__(
        self,
        distances: np.ndarray,
        n_ants: int = 20,
        alpha: float = 1.0,
        beta: float = 5.0,

```

```

        rho: float = 0.5,
        q: float = 100.0
    ):
        self.distances = distances
        self.n_cities = len(distances)
        self.n_ants = n_ants
        self.alpha = alpha
        self.beta = beta
        self.rho = rho # Evaporation rate
        self.q = q # Pheromone deposit factor

        # Initialize pheromone matrix
        self.pheromone = np.ones((self.n_cities, self.n_cities))

        # Heuristic information (inverse distance)
        self.heuristic = 1.0 / (distances + np.eye(self.n_cities))

    def construct_solution(self) -> Tuple[List[int], float]:
        """Single ant constructs a tour."""
        visited = [np.random.randint(self.n_cities)]

        while len(visited) < self.n_cities:
            current = visited[-1]
            unvisited = [c for c in range(self.n_cities) if c not in visited]

            # Calculate probabilities
            probs = []
            for j in unvisited:
                tau = self.pheromone[current, j] ** self.alpha
                eta = self.heuristic[current, j] ** self.beta
                probs.append(tau * eta)

            probs = np.array(probs) / sum(probs)

            # Select next city
            next_city = np.random.choice(unvisited, p=probs)
            visited.append(next_city)

        # Calculate tour length
        tour_length = sum(
            self.distances[visited[i], visited[i+1]]
            for i in range(len(visited)-1)
        )
        tour_length += self.distances[visited[-1], visited[0]]

        return visited, tour_length

    def update_pheromone(self, tours: List[Tuple[List[int], float]]):
        """Update pheromone trails."""
        # Evaporation
        self.pheromone *= (1 - self.rho)

        # Deposit

```

```

    for tour, length in tours:
        deposit = self.q / length
        for i in range(len(tour) - 1):
            self.pheromone[tour[i], tour[i+1]] += deposit
            self.pheromone[tour[i+1], tour[i]] += deposit
        self.pheromone[tour[-1], tour[0]] += deposit
        self.pheromone[tour[0], tour[-1]] += deposit

def optimize(self, n_iterations: int) -> Tuple[List[int], float, List[float]]:
    """Run ACO optimization."""
    best_tour = None
    best_length = float('inf')
    history = []

    for iteration in range(n_iterations):
        # All ants construct solutions
        tours = [self.construct_solution() for _ in range(self.n_ants)]

        # Find best tour this iteration
        for tour, length in tours:
            if length < best_length:
                best_tour = tour
                best_length = length

        # Update pheromone
        self.update_pheromone(tours)

        history.append(best_length)

    return best_tour, best_length, history

```

11.7. Differential Evolution

Differential Evolution (DE) is a population-based optimizer using vector differences for mutation. It excels on continuous, non-differentiable, and multimodal problems.

11.7.1. DE/rand/1/bin Strategy

The classic DE variant generates mutant vectors by adding weighted difference vectors:

$$\mathbf{v}_i = \mathbf{x}_{r1} + F \cdot (\mathbf{x}_{r2} - \mathbf{x}_{r3})$$

The implementation below follows the DE/rand/1/bin scheme. For each individual, it selects three distinct others, constructs a mutant vector using the scaled difference, performs binomial crossover to create a trial vector, and replaces the original only if the trial is better. The mutation factor F controls the step size of differential perturbations, while the crossover rate CR determines how many dimensions inherit values from the mutant.

```

class DifferentialEvolution:
    """Differential Evolution optimizer."""

    def __init__(

```

```

        self,
        objective: Callable,
        n_dims: int,
        bounds: Tuple[float, float],
        pop_size: int = 50,
        F: float = 0.8,
        CR: float = 0.9
    ):
        self.objective = objective
        self.n_dims = n_dims
        self.bounds = bounds
        self.pop_size = pop_size
        self.F = F # Mutation factor
        self.CR = CR # Crossover probability

def optimize(self, n_generations: int) -> Tuple[np.ndarray, float, List[float]]:
    """Run DE optimization."""
    low, high = self.bounds

    # Initialize population
    population = np.random.uniform(low, high, (self.pop_size, self.n_dims))
    fitness = np.array([self.objective(ind) for ind in population])

    best_idx = np.argmin(fitness)
    best = population[best_idx].copy()
    best_fitness = fitness[best_idx]
    history = [best_fitness]

    for generation in range(n_generations):
        for i in range(self.pop_size):
            # Select three random individuals (different from i)
            candidates = [j for j in range(self.pop_size) if j != i]
            r1, r2, r3 = np.random.choice(candidates, 3, replace=False)

            # Mutation:  $v = x_{r1} + F * (x_{r2} - x_{r3})$ 
            mutant = population[r1] + self.F * (population[r2] - population[r3])
            mutant = np.clip(mutant, low, high)

            # Crossover
            j_rand = np.random.randint(self.n_dims)
            trial = np.where(
                (np.random.random(self.n_dims) < self.CR) |
                (np.arange(self.n_dims) == j_rand),
                mutant,
                population[i]
            )

            # Selection
            trial_fitness = self.objective(trial)
            if trial_fitness < fitness[i]:
                population[i] = trial
                fitness[i] = trial_fitness

```

```

        if trial_fitness < best_fitness:
            best = trial.copy()
            best_fitness = trial_fitness

    history.append(best_fitness)

    return best, best_fitness, history

```

11.7.2. DE Strategies

Strategy	Mutation Formula	Characteristics
DE/rand/1	$\mathbf{x}_{r1} + F(\mathbf{x}_{r2} - \mathbf{x}_{r3})$	Good exploration
DE/best/1	$\mathbf{x}_{best} + F(\mathbf{x}_{r1} - \mathbf{x}_{r2})$	Fast convergence
DE/current-to-best/1	$\mathbf{x}_i + F(\mathbf{x}_{best} - \mathbf{x}_i) + F(\mathbf{x}_{r1} - \mathbf{x}_{r2})$	Balanced approach

11.8. CMA-ES

The Covariance Matrix Adaptation Evolution Strategy (CMA-ES) is a state-of-the-art algorithm for continuous black-box optimization. It adapts a multivariate normal distribution to sample candidate solutions.

11.8.1. Key Concepts

CMA-ES maintains and adapts: - **Mean vector $\mathbf{m}$** : Center of the search distribution - **Step size σ** : Overall scale of the distribution - **Covariance matrix $\mathbf{C}$** : Shape and orientation of the distribution

$$\mathbf{x}_k \sim \mathbf{m} + \sigma \mathcal{N}(\mathbf{0}, \mathbf{C})$$

The following implementation covers the full CMA-ES update cycle: sampling from the adapted distribution, ranking candidates, updating the mean through weighted recombination, and adapting both the covariance matrix and step size via evolution paths. Covariance matrix adaptation is the distinguishing feature of this algorithm—it learns the local structure of the objective function, including variable correlations and scaling, without requiring gradient information.

```

class CMAES:
    """Simplified CMA-ES implementation."""

    def __init__(
        self,
        objective: Callable,
        n_dims: int,
        sigma0: float = 0.5,
        pop_size: int = None
    ):
        self.objective = objective
        self.n_dims = n_dims
        self.sigma = sigma0

        # Population size (default: 4 + 3*ln(n))

```

```

self.lam = pop_size or 4 + int(3 * np.log(n_dims))
self.mu = self.lam // 2

# Weights for recombination
weights = np.log(self.mu + 0.5) - np.log(np.arange(1, self.mu + 1))
self.weights = weights / weights.sum()
self.mu_eff = 1.0 / (self.weights ** 2).sum()

# Learning rates
self.cc = 4.0 / (n_dims + 4)
self.cs = (self.mu_eff + 2) / (n_dims + self.mu_eff + 3)
self.c1 = 2.0 / ((n_dims + 1.3) ** 2 + self.mu_eff)
self.cmu = min(1 - self.c1,
               2 * (self.mu_eff - 2 + 1/self.mu_eff) /
               ((n_dims + 2) ** 2 + self.mu_eff))
self.damps = 1 + 2 * max(0, np.sqrt((self.mu_eff - 1) / (n_dims + 1)) - 1) +
    ↪ self.cs

# Initialize state
self.mean = np.zeros(n_dims)
self.C = np.eye(n_dims)
self.ps = np.zeros(n_dims)
self.pc = np.zeros(n_dims)

self.chiN = np.sqrt(n_dims) * (1 - 1/(4*n_dims) + 1/(21*n_dims**2))

def optimize(self, n_iterations: int) -> Tuple[np.ndarray, float, List[float]]:
    """Run CMA-ES optimization."""
    best = None
    best_fitness = float('inf')
    history = []

    for iteration in range(n_iterations):
        # Sample population
        B, D = np.linalg.eigh(self.C)
        D = np.sqrt(np.maximum(D, 0))
        BD = B * D

        samples = []
        fitness_values = []

        for k in range(self.lam):
            z = np.random.randn(self.n_dims)
            x = self.mean + self.sigma * (BD @ z)
            f = self.objective(x)
            samples.append((x, z, f))
            fitness_values.append(f)

        # Sort by fitness
        samples.sort(key=lambda s: s[2])

        # Update best
        if samples[0][2] < best_fitness:

```



```

        best = samples[0][0].copy()
        best_fitness = samples[0][2]

    history.append(best_fitness)

    # Recombination
    old_mean = self.mean.copy()
    self.mean = np.sum([self.weights[i] * samples[i][0]
                        for i in range(self.mu)], axis=0)

    # Evolution paths
    invsqrtC = B @ np.diag(1.0 / D) @ B.T
    self.ps = (1 - self.cs) * self.ps + \
        np.sqrt(self.cs * (2 - self.cs) * self.mu_eff) * \
        invsqrtC @ (self.mean - old_mean) / self.sigma

    hsig = (np.linalg.norm(self.ps) /
            np.sqrt(1 - (1 - self.cs) ** (2 * (iteration + 1)))) < \
        (1.4 + 2 / (self.n_dims + 1)) * self.chiN

    self.pc = (1 - self.cc) * self.pc + \
        hsig * np.sqrt(self.cc * (2 - self.cc) * self.mu_eff) * \
        (self.mean - old_mean) / self.sigma

    # Covariance matrix adaptation
    artmp = (1 / self.sigma) * np.array(
        [samples[i][0] - old_mean for i in range(self.mu)]
    )
    self.C = (1 - self.c1 - self.cmu) * self.C + \
        self.c1 * np.outer(self.pc, self.pc) + \
        self.cmu * sum(self.weights[i] * np.outer(artmp[i], artmp[i])
                        for i in range(self.mu))

    # Step size adaptation
    self.sigma *= np.exp((self.cs / self.damps) *
                        (np.linalg.norm(self.ps) / self.chiN - 1))

    return best, best_fitness, history

```

! When to Use CMA-ES

CMA-ES excels on continuous, non-separable, ill-conditioned problems with up to a few hundred dimensions. For separable problems or very high dimensions, simpler methods may be more efficient.

11.9. Algorithm Selection Guidelines

Choosing the right metaheuristic depends on problem characteristics:

Problem Type	Recommended Algorithms
Continuous, small dimension	CMA-ES, DE
Continuous, large dimension	PSO, DE

Problem Type	Recommended Algorithms
Combinatorial, permutation	GA with OX, Tabu Search, ACO
Combinatorial, binary	GA, SA, Tabu Search
Multi-modal	PSO, DE with restart
Expensive function evaluations	Bayesian Optimization, surrogate-assisted

11.9.1. Performance Comparison

The following utility function runs multiple metaheuristics on the same objective and collects statistics over repeated trials. This allows a fair comparison of mean performance and variance across algorithms.

```
def benchmark_metaheuristics(objective, n_dims, bounds, n_runs=10):
    """Compare metaheuristic performance."""
    results = {}

    # Test each algorithm
    algorithms = {
        'PSO': lambda: ParticleSwarmOptimizer(objective, n_dims, bounds),
        'DE': lambda: DifferentialEvolution(objective, n_dims, bounds),
        'GA': lambda: GeneticAlgorithm(objective, n_dims, bounds),
    }

    for name, create_alg in algorithms.items():
        best_values = []
        for _ in range(n_runs):
            alg = create_alg()
            if name == 'GA':
                _, best_val, _ = alg.evolve(100)
            else:
                _, best_val, _ = alg.optimize(100)
            best_values.append(best_val)

        results[name] = {
            'mean': np.mean(best_values),
            'std': np.std(best_values),
            'best': np.min(best_values)
        }

    return results
```

11.10. Summary

11.10.1. Key Takeaways

- Metaheuristics trade optimality guarantees for computational efficiency
- The exploration-exploitation balance is the central design challenge
- Simulated Annealing uses probabilistic acceptance to escape local optima
- Genetic Algorithms evolve populations through selection, crossover, and mutation
- PSO simulates swarm behavior with cognitive and social learning components
- Tabu Search uses memory to prevent cycling and guide intelligent exploration
- ACO mimics ant pheromone communication for constructive optimization

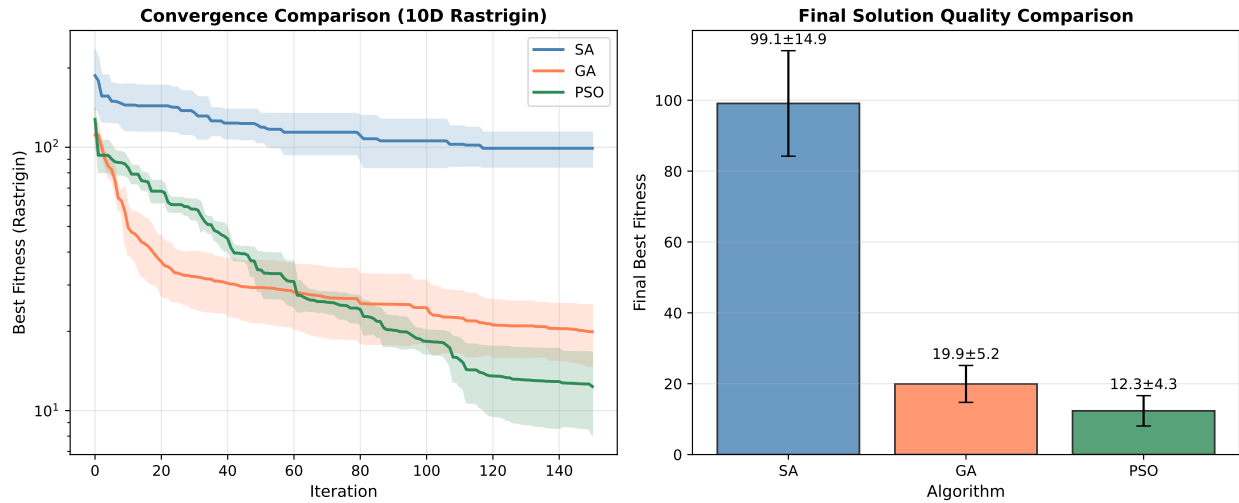


Figure 11.5.: Convergence comparison of SA, GA, and PSO on the Rastrigin function showing typical optimization trajectories

- Differential Evolution uses vector differences for effective continuous optimization
- CMA-ES adapts a covariance matrix for sophisticated distribution-based search

11.11. Exercises

1. **SA cooling schedule:** Implement and compare geometric, linear, and adaptive cooling schedules on the Rastrigin function. Plot convergence curves for each.
2. **GA operators:** Design specialized crossover and mutation operators for the Traveling Salesman Problem. Test on a 50-city instance.
3. **PSO topology:** Implement ring and star topologies for PSO. Compare convergence on multimodal functions.
4. **Tabu tenure:** Experiment with fixed vs. dynamic tabu tenure on a graph coloring problem. How does tenure affect solution quality and computation time?
5. **Algorithm comparison:** Run DE, PSO, and CMA-ES on the Rosenbrock function in 10 dimensions. Compare convergence speed and final solution quality.

12. Gradient-Based Optimization

What You Will Learn

- Understand gradients and directional derivatives as foundations
- Master gradient descent convergence analysis and conditions
- Apply step size selection strategies including backtracking line search
- Implement learning rate schedules for improved training
- Use momentum methods and adaptive optimizers (Adam, RMSprop)

Gradient-based methods form the backbone of continuous optimization and machine learning. From training neural networks to solving regression problems, these algorithms leverage derivative information to efficiently find minima. This chapter builds from fundamental concepts to practical implementation strategies.

12.1. Gradients and Directional Derivatives

The gradient is the fundamental concept underlying all first-order optimization methods. Understanding gradients and how functions change in different directions is essential for developing intuition about optimization algorithms.

12.1.1. The Gradient Vector

For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the gradient is:

$$\nabla f(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]^T$$

The gradient is a vector of partial derivatives pointing in the direction of steepest ascent.

Key Properties of the Gradient

- **Direction:** Points in the direction of maximum increase of f
- **Magnitude:** $\|\nabla f(\mathbf{x})\|$ represents the rate of maximum increase
- **Orthogonality:** Perpendicular to level curves/surfaces of f
- **Critical Points:** $\nabla f(\mathbf{x}^*) = 0$ at minima, maxima, and saddle points

12.1.2. Directional Derivatives

While the gradient tells us about the maximum rate of change, the directional derivative tells us how fast the function changes in any specific direction.

Definition: The directional derivative of f at point $\mathbf{x}$ in direction $\mathbf{v}$ (unit vector) is:

$$D_{\mathbf{v}}f(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon\mathbf{v}) - f(\mathbf{x})}{\epsilon}$$

Gradient Relationship: The directional derivative can be computed as a dot product:

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$$D_{\mathbf{v}}f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{v} = \|\nabla f(\mathbf{x})\| \|\mathbf{v}\| \cos(\theta)$$

where θ is the angle between $\nabla f(\mathbf{x})$ and $\mathbf{v}$.

12.1.3. Extremal Properties

Since $D_{\mathbf{v}}f(\mathbf{x}) = \|\nabla f(\mathbf{x})\| \cos(\theta)$:

Condition	Angle	Directional Derivative
Maximum	$\theta = 0^\circ$ (parallel to ∇f)	$D_{\mathbf{v}}f = \ \nabla f(\mathbf{x})\ $
Minimum	$\theta = 180^\circ$ (opposite to ∇f)	$D_{\mathbf{v}}f = -\ \nabla f(\mathbf{x})\ $
Zero	$\theta = 90^\circ$ (perpendicular to ∇f)	$D_{\mathbf{v}}f = 0$

Implication for Optimization: To minimize f , move in direction $-\nabla f(\mathbf{x})$.

12.1.4. Example: Quadratic Function

Consider $f(x, y) = x^2 + 4y^2$:

- **Gradient:** $\nabla f(x, y) = [2x, 8y]^T$
- **At point** $(2, 1)$: $\nabla f(2, 1) = [4, 8]^T$
- **Direction of steepest ascent:** $[4, 8]^T / \|(4, 8)\| = [0.447, 0.894]^T$
- **Direction of steepest descent:** $-[0.447, 0.894]^T$

12.2. Gradient Descent Algorithm

The basic gradient descent update rule:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)$$

where $\alpha_k > 0$ is the step size (learning rate) at iteration k .

The following function translates this update rule into a reusable Python implementation. It takes the objective function, its gradient, and a starting point, then iteratively steps in the negative gradient direction until convergence.

```
import numpy as np

def gradient_descent(f, grad_f, x0, alpha=0.01, tol=1e-6, max_iter=1000):
    """Basic gradient descent algorithm.

    Args:
        f: Objective function
        grad_f: Gradient of objective function
        x0: Initial point
        alpha: Step size (learning rate)
        tol: Convergence tolerance
        max_iter: Maximum iterations
```

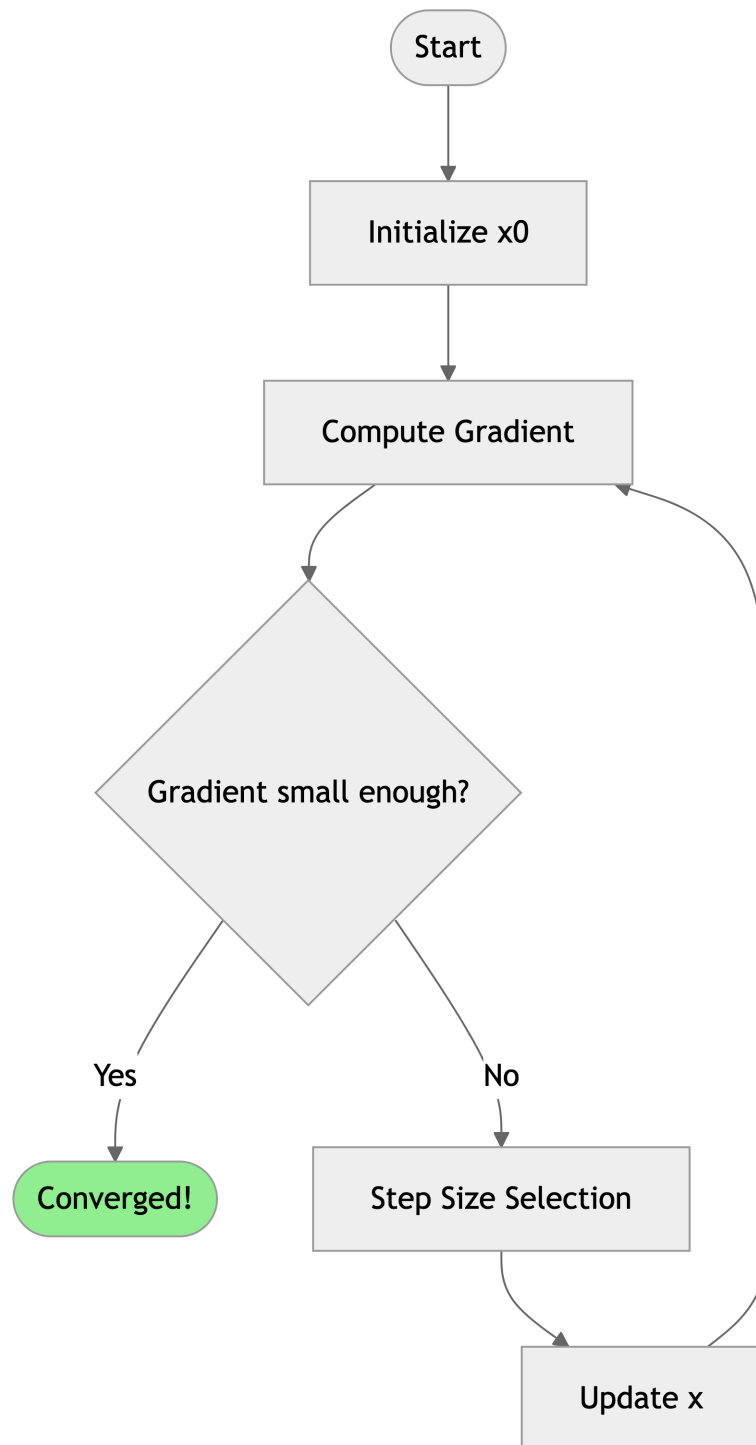


Figure 12.1.: Gradient Descent Algorithm Flow

```

Returns:
    x: Optimal point found
    history: List of (x, f(x)) tuples
"""
x = np.array(x0, dtype=float)
history = [(x.copy(), f(x))]

for k in range(max_iter):
    grad = grad_f(x)

    # Check convergence
    if np.linalg.norm(grad) < tol:
        break

    # Update step
    x = x - alpha * grad
    history.append((x.copy(), f(x)))

return x, history

# Example: minimize f(x,y) = x^2 + 4y^2
def f(x):
    return x[0]**2 + 4*x[1]**2

def grad_f(x):
    return np.array([2*x[0], 8*x[1]])

x_opt, history = gradient_descent(f, grad_f, x0=[4, 2], alpha=0.1)
print(f"Optimum: {x_opt}")
print(f"Iterations: {len(history)}")

```

The following visualization shows gradient descent navigating the challenging Rosenbrock function, which has a narrow curved valley that makes optimization difficult.

12.3. Convergence Analysis

Understanding when and how fast gradient descent converges is essential for both theoretical understanding and practical implementation.

12.3.1. Function Classes

Class	Definition	Interpretation
L-Lipschitz Gradient	$\ \nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\ \leq L\ \mathbf{x} - \mathbf{y}\ $	Gradient doesn't change too rapidly
μ -Strongly Convex	$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{\mu}{2}\ \mathbf{y} - \mathbf{x}\ ^2$	Function grows at least quadratically

12.3.2. The Descent Lemma

For f with L-Lipschitz continuous gradient, for any $\mathbf{x}, \mathbf{y}$:

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2$$

This fundamental inequality enables convergence proofs by bounding how much f can increase.

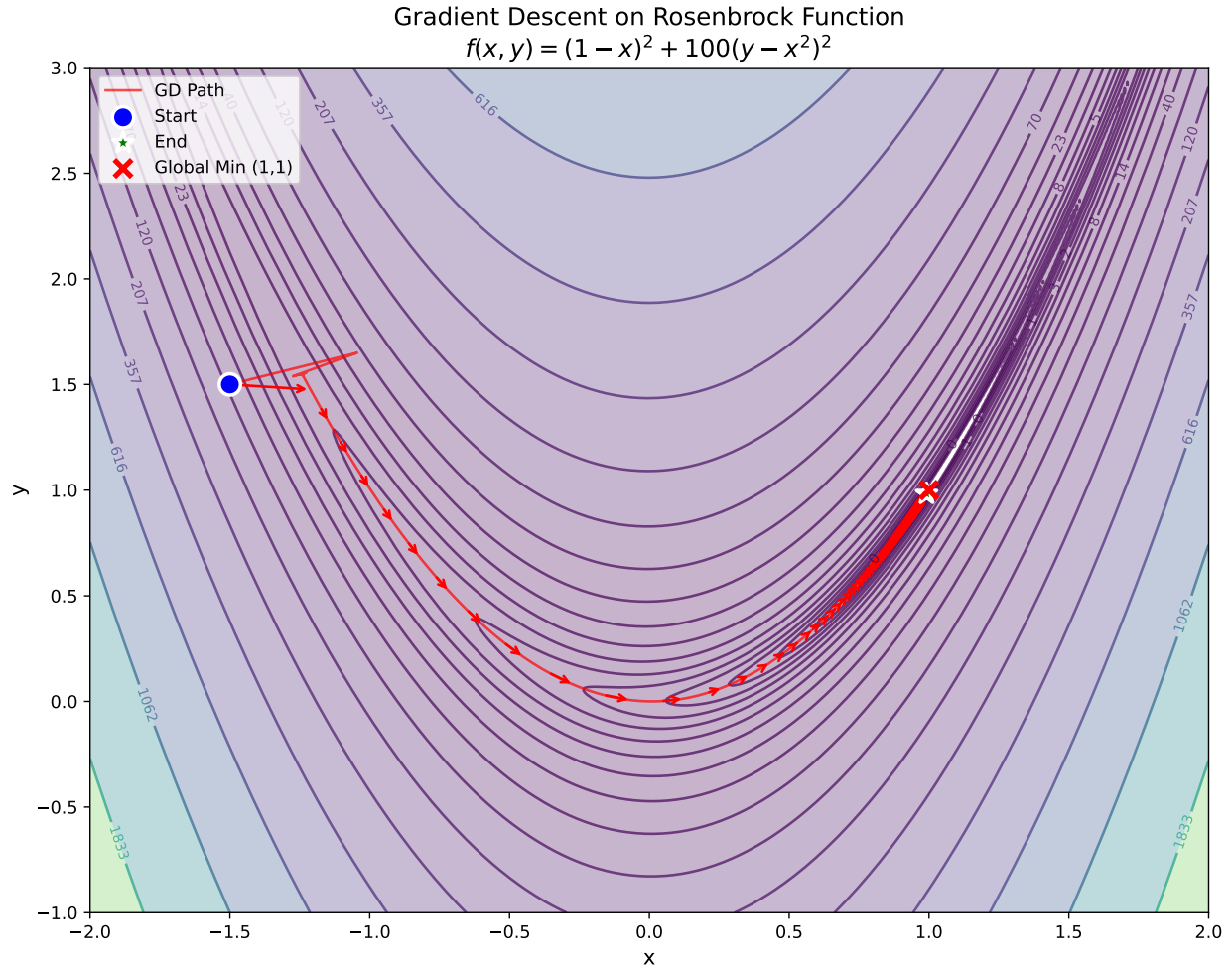


Figure 12.2.: Gradient descent optimization path on the Rosenbrock function. The algorithm starts at $(-1.5, 1.5)$ and navigates through the narrow curved valley toward the global minimum at $(1, 1)$. Contour lines show the function landscape, and arrows indicate the descent direction at each step.

12.3.3. Convergence Rates

Function Class	Rate	Iterations for ϵ Accuracy
Convex + L-Smooth	$O(1/k)$	$O(1/\epsilon)$
Strongly Convex + L-Smooth	$O(\exp(-k/\kappa))$	$O(\kappa \log(1/\epsilon))$
Convex Lipschitz (non-smooth)	$O(1/\sqrt{k})$	$O(1/\epsilon^2)$
Non-Convex (to stationary)	$O(1/k)$	$O(1/\epsilon)$

! The Condition Number

The condition number $\kappa = L/\mu$ dramatically affects convergence speed:

- $\kappa = 10$: Well-conditioned, ~23 iterations for 10^{-6} accuracy
- $\kappa = 100$: Moderately conditioned, ~230 iterations
- $\kappa = 10,000$: Ill-conditioned, ~23,000 iterations

Number of iterations scales linearly with κ .

12.3.4. Convergence Theorem (Strongly Convex Case)

With step size $\alpha = 1/L$:

$$\|\mathbf{x}_k - \mathbf{x}^*\| \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^k \|\mathbf{x}_0 - \mathbf{x}^*\|$$

This geometric (linear) convergence rate means the error decreases by a constant factor each iteration.

12.4. Step Size Selection Strategies

The choice of step size is one of the most important decisions in gradient descent.

12.4.1. Constant Step Size

$$\alpha_k = \alpha \text{ for all } k$$

When to use: When L (Lipschitz constant) is known. Set $\alpha \leq 1/L$ for guaranteed convergence.

Advantages: Simple, fast, no overhead.

Disadvantages: May be too conservative if L overestimated; no adaptation to local geometry.

12.4.2. Exact Line Search

$$\alpha_k = \arg \min_{\alpha > 0} f(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k))$$

When to use: When 1D minimization is cheap; quadratic functions (analytical solution).

Advantages: Maximum decrease per iteration.

Disadvantages: Computationally expensive; not practical for large-scale ML.

12.4.3. Backtracking Line Search

The Armijo sufficient decrease condition:

$$f(\mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k)) \leq f(\mathbf{x}_k) - c \cdot \alpha \|\nabla f(\mathbf{x}_k)\|^2$$

where $c \in (0, 1)$, typically $c = 10^{-4}$.

The backtracking approach starts with a large step size and repeatedly shrinks it by a factor ρ until the Armijo condition is satisfied. This avoids both the cost of exact line search and the risk of a fixed step size that is too large or too small.

```

def backtracking_line_search(f, grad_f, x, direction,
                           alpha_init=1.0, rho=0.5, c=1e-4):
    """Backtracking line search with Armijo condition.

    Args:
        f: Objective function
        grad_f: Gradient function
        x: Current point
        direction: Search direction (typically -gradient)
        alpha_init: Initial step size
        rho: Backtracking factor (0 < rho < 1)
        c: Armijo constant (0 < c < 1)

    Returns:
        alpha: Step size satisfying Armijo condition
    """
    alpha = alpha_init
    grad = grad_f(x)
    f_x = f(x)

    # Directional derivative
    directional_deriv = np.dot(grad, direction)

    # Backtrack until Armijo condition satisfied
    while f(x + alpha * direction) > f_x + c * alpha * directional_deriv:
        alpha *= rho

    return alpha

def gradient_descent_backtracking(f, grad_f, x0, tol=1e-6, max_iter=1000):
    """Gradient descent with backtracking line search."""
    x = np.array(x0, dtype=float)

    for k in range(max_iter):
        grad = grad_f(x)

        if np.linalg.norm(grad) < tol:
            break

        # Search direction (steepest descent)
        direction = -grad

        # Backtracking line search
        alpha = backtracking_line_search(f, grad_f, x, direction)

        x = x + alpha * direction

    return x, k + 1

```

12.4.4. Wolfe Conditions

For quasi-Newton methods, we need both conditions:

Sufficient Decrease (Armijo):

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) + c_1 \alpha \nabla f(\mathbf{x}_k)^T \mathbf{p}_k$$

Curvature Condition:

$$\nabla f(\mathbf{x}_{k+1})^T \mathbf{p}_k \geq c_2 \nabla f(\mathbf{x}_k)^T \mathbf{p}_k$$

where $0 < c_1 < c_2 < 1$. Standard choices: $c_1 = 10^{-4}$, $c_2 = 0.9$.

The following visualization demonstrates how different learning rates affect gradient descent convergence on a simple quadratic function.

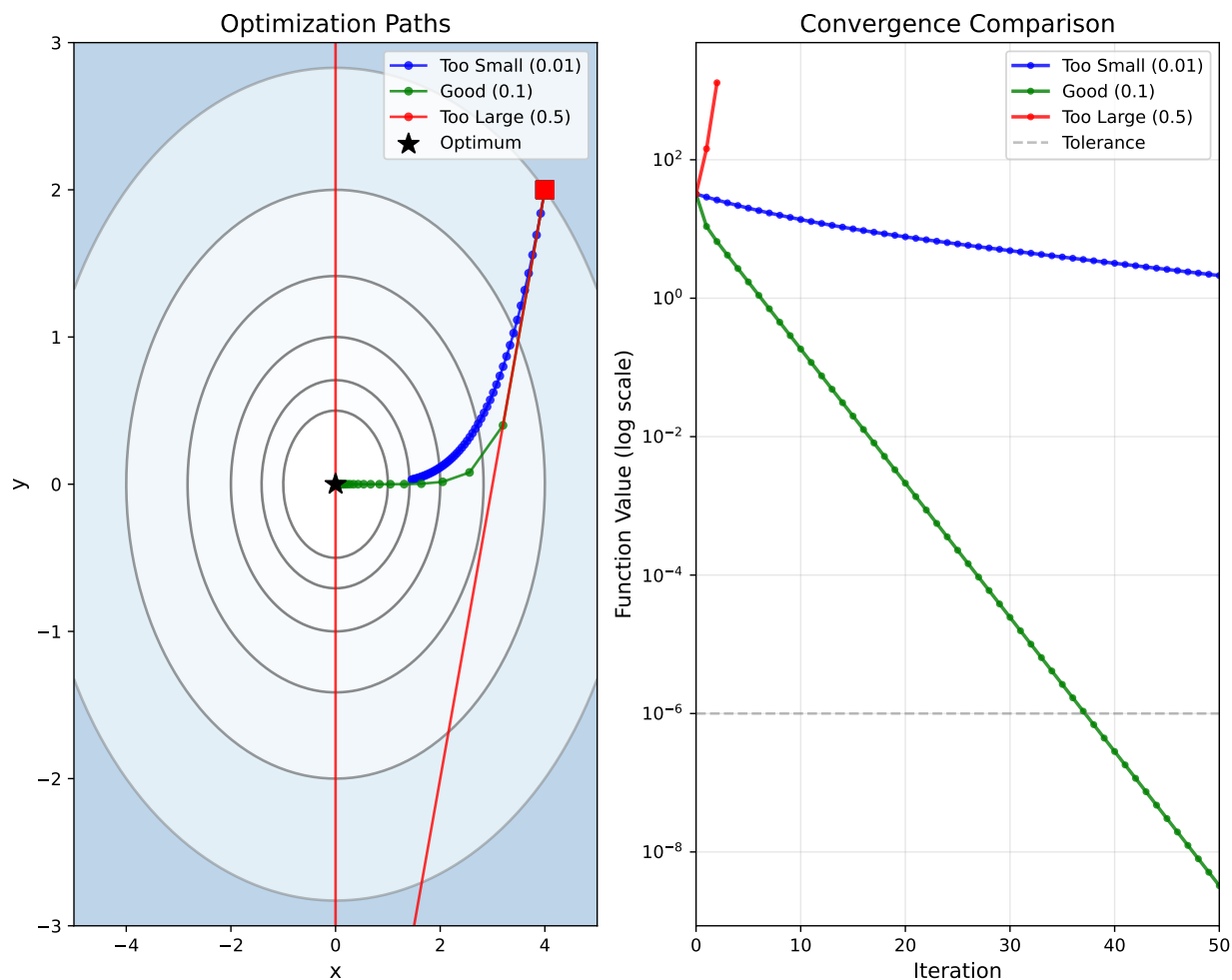


Figure 12.3.: Effect of learning rate on gradient descent convergence. Left: Optimization paths on a 2D quadratic function. Right: Convergence curves showing function value versus iteration. A learning rate that is too small (0.01) converges slowly, a good rate (0.1) converges efficiently, and a rate that is too large (0.5) causes oscillation or divergence.

12.5. Learning Rate Schedules

Learning rate schedules adjust the step size over the course of optimization, enabling fast initial progress followed by fine-tuning.

12.5.1. Common Schedules

Schedule	Formula	Best For
Step Decay	$\alpha_k = \alpha_0 \cdot \gamma^{\lfloor k/s \rfloor}$	Image classification
Exponential	$\alpha_k = \alpha_0 \cdot e^{-\lambda k}$	Smooth convergence
Polynomial	$\alpha_k = \alpha_0(1 - k/T)^p$	Fixed budget
Cosine	$\alpha_k = \alpha_{min} + \frac{\alpha_{max} - \alpha_{min}}{2} (1 + \cos(\pi k/T))$	Deep learning
Inverse Time	$\alpha_k = \alpha_0/(1 + \lambda k)$	SGD convergence

The `LearningRateScheduler` class below collects several common schedule strategies as static methods, making it easy to swap schedules during training without changing the optimizer code.

```
class LearningRateScheduler:
    """Learning rate schedules for gradient descent."""

    @staticmethod
    def step_decay(epoch, alpha_0=0.1, gamma=0.5, step_size=10):
        """Step decay: reduce by factor every step_size epochs."""
        return alpha_0 * (gamma ** (epoch // step_size))

    @staticmethod
    def exponential_decay(epoch, alpha_0=0.1, decay_rate=0.95):
        """Exponential decay."""
        return alpha_0 * (decay_rate ** epoch)

    @staticmethod
    def cosine_annealing(epoch, alpha_max=0.1, alpha_min=0.001, T=100):
        """Cosine annealing schedule."""
        return alpha_min + (alpha_max - alpha_min) * 0.5 * (
            1 + np.cos(np.pi * epoch / T)
        )

    @staticmethod
    def polynomial_decay(epoch, alpha_0=0.1, T=100, power=2):
        """Polynomial decay to zero at epoch T."""
        return alpha_0 * (max(0, 1 - epoch / T) ** power)
```

i Robbins-Monro Conditions

For stochastic gradient descent convergence, the learning rate sequence must satisfy:

$$\sum_{k=1}^{\infty} \alpha_k = \infty \quad \text{and} \quad \sum_{k=1}^{\infty} \alpha_k^2 < \infty$$

Examples: $\alpha_k = \alpha_0/k$ or $\alpha_k = \alpha_0/\sqrt{k}$.

The following visualization compares stochastic gradient descent (SGD) with batch gradient descent, showing the characteristic noisy path of SGD versus the smooth path of full batch GD.

12.6. Momentum Methods

Momentum accelerates gradient descent by accumulating a velocity vector in directions of persistent descent.

12. Gradient-Based Optimization

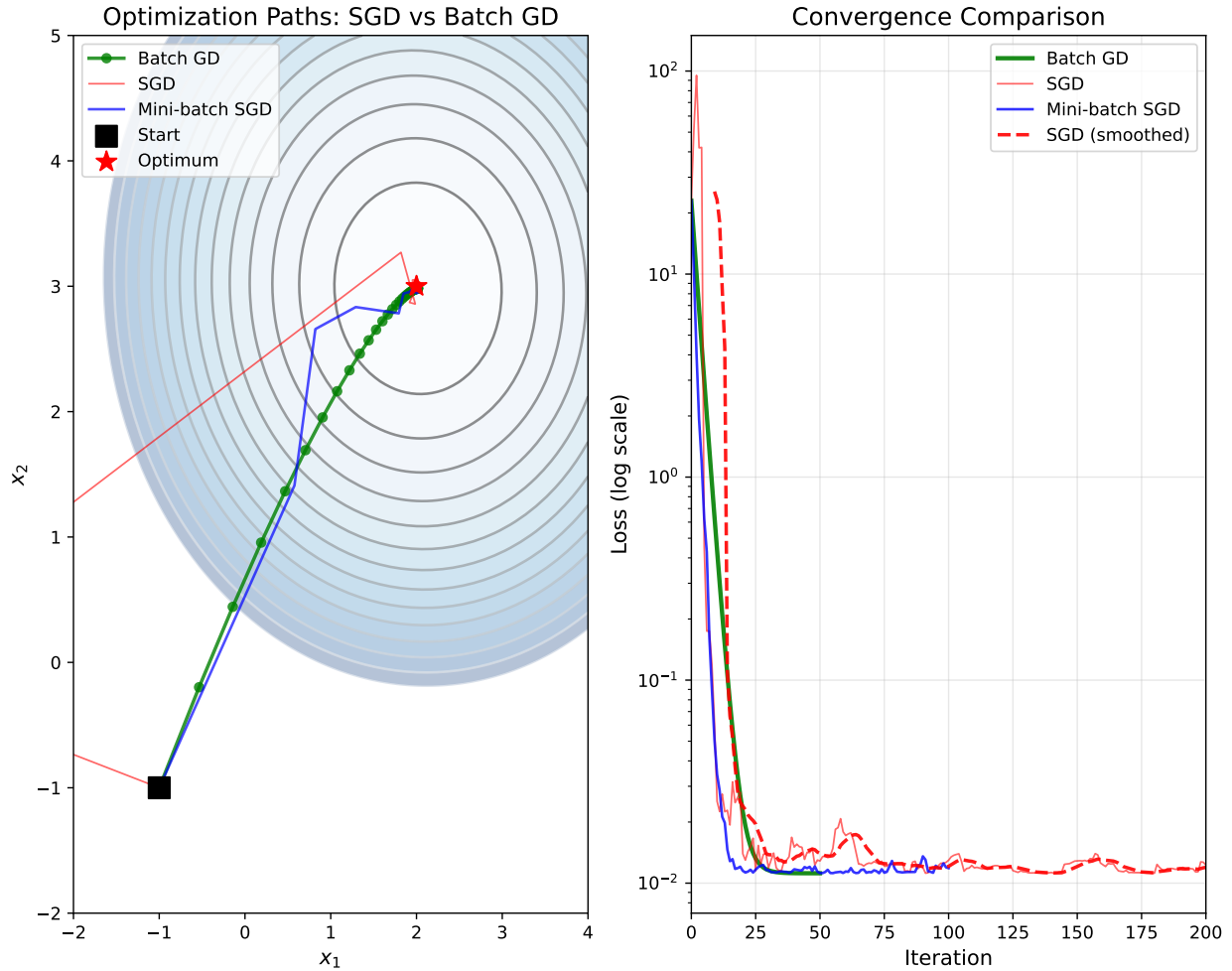


Figure 12.4.: Comparison of Stochastic Gradient Descent (SGD) versus Batch Gradient Descent (GD). Left: Optimization paths on a quadratic function showing SGD's noisy trajectory versus GD's smooth path. Right: Convergence comparison showing SGD's faster initial progress but noisier convergence, while GD converges smoothly but may be slower per epoch in practice.

12.6.1. Classical Momentum

$$\mathbf{v}_{k+1} = \beta \mathbf{v}_k - \alpha \nabla f(\mathbf{x}_k)$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{v}_{k+1}$$

where $\beta \in [0, 1)$ is the momentum coefficient (typically 0.9).

This implementation maintains a velocity vector $\mathbf{v}$ that accumulates past gradient information. The velocity smooths out oscillations in high-curvature directions and accelerates progress along consistent descent directions.

```
def gradient_descent_momentum(f, grad_f, x0, alpha=0.01, beta=0.9,
                              tol=1e-6, max_iter=1000):
```

```

"""Gradient descent with momentum."""
x = np.array(x0, dtype=float)
v = np.zeros_like(x)

for k in range(max_iter):
    grad = grad_f(x)

    if np.linalg.norm(grad) < tol:
        break

    # Update velocity
    v = beta * v - alpha * grad

    # Update position
    x = x + v

return x, k + 1

```

12.6.2. Nesterov Accelerated Gradient

Evaluate gradient at “look-ahead” position:

$$\mathbf{v}_{k+1} = \beta \mathbf{v}_k - \alpha \nabla f(\mathbf{x}_k + \beta \mathbf{v}_k)$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{v}_{k+1}$$

This achieves optimal convergence rate $O(1/k^2)$ for smooth convex functions.

The key difference from classical momentum is the gradient evaluation at the look-ahead position $\mathbf{x} + \beta \mathbf{v}$. This anticipatory step allows the algorithm to correct its trajectory before overshooting.

```

def nesterov_accelerated_gradient(f, grad_f, x0, alpha=0.01, beta=0.9,
                                  tol=1e-6, max_iter=1000):
    """Nesterov accelerated gradient descent."""
    x = np.array(x0, dtype=float)
    v = np.zeros_like(x)

    for k in range(max_iter):
        # Look-ahead position
        x_lookahead = x + beta * v
        grad = grad_f(x_lookahead)

        if np.linalg.norm(grad) < tol:
            break

        # Update velocity and position
        v = beta * v - alpha * grad
        x = x + v

    return x, k + 1

```

12.7. Adaptive Learning Rate Methods

Adaptive methods adjust learning rates per-parameter based on historical gradient information.

12. Gradient-Based Optimization

12.7.1. AdaGrad

Accumulates squared gradients to scale learning rates:

$$\mathbf{G}_k = \mathbf{G}_{k-1} + \nabla f(\mathbf{x}_k) \odot \nabla f(\mathbf{x}_k)$$
$$\mathbf{x}_{k+1} = \mathbf{x}_k - \frac{\alpha}{\sqrt{\mathbf{G}_k} + \epsilon} \odot \nabla f(\mathbf{x}_k)$$

Advantage: Automatically decreases learning rate for frequent features.

Disadvantage: Monotonically decreasing learning rate can stop learning too early.

12.7.2. RMSprop

Uses exponential moving average instead of accumulation:

$$\mathbf{E}[\mathbf{g}^2]_k = \gamma \mathbf{E}[\mathbf{g}^2]_{k-1} + (1 - \gamma) \nabla f(\mathbf{x}_k) \odot \nabla f(\mathbf{x}_k)$$
$$\mathbf{x}_{k+1} = \mathbf{x}_k - \frac{\alpha}{\sqrt{\mathbf{E}[\mathbf{g}^2]_k} + \epsilon} \odot \nabla f(\mathbf{x}_k)$$

Typical value: $\gamma = 0.9$.

12.7.3. Adam (Adaptive Moment Estimation)

Combines momentum with adaptive learning rates:

$$\mathbf{m}_k = \beta_1 \mathbf{m}_{k-1} + (1 - \beta_1) \nabla f(\mathbf{x}_k)$$
$$\mathbf{v}_k = \beta_2 \mathbf{v}_{k-1} + (1 - \beta_2) \nabla f(\mathbf{x}_k) \odot \nabla f(\mathbf{x}_k)$$

Bias correction:

$$\hat{\mathbf{m}}_k = \mathbf{m}_k / (1 - \beta_1^k)$$
$$\hat{\mathbf{v}}_k = \mathbf{v}_k / (1 - \beta_2^k)$$

Update:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \frac{\alpha}{\sqrt{\hat{\mathbf{v}}_k} + \epsilon} \odot \hat{\mathbf{m}}_k$$

Adam combines the benefits of momentum (first moment $\mathbf{m}$) and adaptive per-parameter scaling (second moment $\mathbf{v}$). The bias correction terms compensate for the zero-initialization of both moment estimates, which would otherwise cause underestimation in early iterations.

```
def adam(grad_f, x0, alpha=0.001, beta1=0.9, beta2=0.999,
        epsilon=1e-8, max_iter=1000, tol=1e-6):
    """Adam optimizer.

    Args:
        grad_f: Gradient function
        x0: Initial point
        alpha: Learning rate (default 0.001)
        beta1: First moment decay (default 0.9)
        beta2: Second moment decay (default 0.999)
        epsilon: Numerical stability constant
```



```

Returns:
    x: Optimized parameters
    """
    x = np.array(x0, dtype=float)
    m = np.zeros_like(x) # First moment
    v = np.zeros_like(x) # Second moment

    for k in range(1, max_iter + 1):
        grad = grad_f(x)

        if np.linalg.norm(grad) < tol:
            break

        # Update biased moments
        m = beta1 * m + (1 - beta1) * grad
        v = beta2 * v + (1 - beta2) * grad**2

        # Bias correction
        m_hat = m / (1 - beta1**k)
        v_hat = v / (1 - beta2**k)

        # Update parameters
        x = x - alpha * m_hat / (np.sqrt(v_hat) + epsilon)

    return x

```

12.7.4. Optimizer Comparison

Optimizer	Learning Rate	Momentum	When to Use
SGD	Fixed	No	Baseline, convex problems
SGD + Momentum	Fixed	Yes	General deep learning
AdaGrad	Adaptive	No	Sparse features
RMSprop	Adaptive	No	Non-stationary objectives
Adam	Adaptive	Yes	Default choice for deep learning
AdamW	Adaptive	Yes	With weight decay regularization

Practical Recommendations

1. **Start with Adam:** Good default choice with $\alpha = 0.001$
2. **For best final performance:** SGD with momentum often generalizes better
3. **Use learning rate warmup:** Gradually increase LR at start of training
4. **Combine with schedule:** Cosine annealing or step decay
5. **Monitor gradients:** Watch for exploding or vanishing gradients

12.8. Second-Order Methods

Second-order methods use curvature information from the Hessian matrix.

12.8.1. Newton's Method

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \mathbf{H}^{-1}(\mathbf{x}_k) \nabla f(\mathbf{x}_k)$$

12. Gradient-Based Optimization

where $\mathbf{H}(\mathbf{x}) = \nabla^2 f(\mathbf{x})$ is the Hessian matrix.

Advantage: Quadratic convergence near optimum.

Disadvantage: $O(n^3)$ cost to invert Hessian; may not be positive definite.

12.8.2. Quasi-Newton Methods (BFGS)

Approximate the Hessian using gradient differences:

$$\mathbf{B}_{k+1} = \mathbf{B}_k + \frac{\mathbf{y}_k \mathbf{y}_k^T}{\mathbf{y}_k^T \mathbf{s}_k} - \frac{\mathbf{B}_k \mathbf{s}_k \mathbf{s}_k^T \mathbf{B}_k}{\mathbf{s}_k^T \mathbf{B}_k \mathbf{s}_k}$$

where $\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$ and $\mathbf{y}_k = \nabla f(\mathbf{x}_{k+1}) - \nabla f(\mathbf{x}_k)$.

L-BFGS stores limited history for large-scale problems.

SciPy provides a production-ready L-BFGS-B implementation that handles box constraints and memory-efficient Hessian approximation internally. The wrapper below shows how to call it with a function and its gradient.

```
from scipy.optimize import minimize

# Using L-BFGS-B from scipy
def optimize_with_lbfgs(f, grad_f, x0):
    """Optimize using L-BFGS-B algorithm."""
    result = minimize(
        f, x0,
        method='L-BFGS-B',
        jac=grad_f,
        options={'maxiter': 1000, 'gtol': 1e-6}
    )
    return result.x, result.fun
```

12.9. Summary

12.9.1. Key Takeaways

- The gradient points in the direction of steepest ascent; to minimize, move opposite to the gradient
- Convergence rates depend on function properties: strongly convex functions converge geometrically
- Step size selection matters most: use backtracking line search when L is unknown
- Learning rate schedules improve training by starting aggressive and becoming conservative
- Momentum accelerates convergence by accumulating velocity in consistent directions
- Adaptive methods (Adam, RMSprop) adjust per-parameter learning rates automatically
- Second-order methods like L-BFGS are effective for medium-scale, well-behaved problems
- For deep learning, Adam is the default; SGD with momentum often yields better generalization

12.10. Exercises

1. **Gradient computation:** For $f(x, y, z) = x^2y + y^2z + z^2x$, compute the gradient and evaluate it at $(1, 2, 3)$.
2. **Convergence analysis:** Implement gradient descent for $f(x) = x^2/2$ with different step sizes $\alpha \in \{0.1, 0.5, 1.0, 1.5, 2.0\}$. Plot convergence and identify which step sizes diverge.

3. **Backtracking implementation:** Implement gradient descent with backtracking line search for the Rosenbrock function $f(x, y) = (1 - x)^2 + 100(y - x^2)^2$. Compare iterations needed vs. constant step size.
4. **Optimizer comparison:** For a neural network training task of your choice, compare SGD, SGD with momentum, and Adam. Plot training loss curves and final test accuracy.
5. **Learning rate finder:** Implement the learning rate range test: gradually increase learning rate from small to large, plot loss vs. learning rate, and identify the optimal initial learning rate.

13. Nonlinear and Constrained Optimization

What You Will Learn

- Classify nonlinear programming problems (convex vs nonconvex, smooth vs nonsmooth)
- Derive and apply first-order and second-order optimality conditions
- Understand the Lagrangian formulation and KKT conditions for constrained optimization
- Implement Newton's method, quasi-Newton methods, and trust region algorithms
- Apply penalty methods, augmented Lagrangian, and SQP for constrained problems
- Use CVXPY for disciplined convex programming
- Solve quadratic programs with applications to portfolio optimization and SVMs
- Interface with industrial solvers including IPOPT via Python

Nonlinear Programming (NLP) extends the optimization framework beyond the linear world, enabling us to model the complex, curved relationships that pervade real-world decision problems. From training neural networks to designing structural components, from calibrating financial models to controlling chemical processes, nonlinear optimization is ubiquitous. This chapter provides a comprehensive treatment of both theory and practice, building from unconstrained optimization to the sophisticated algorithms that power modern solvers.

13.1. Introduction to Nonlinear Optimization

13.1.1. The Nonlinear Programming Problem

A general nonlinear programming problem takes the form:

$$\begin{aligned} & \text{minimize} && f(\mathbf{x}) \\ & \text{subject to} && g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & && h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p \\ & && \mathbf{x} \in \mathbb{R}^n \end{aligned}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the objective function, $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are inequality constraints, and $h_j : \mathbb{R}^n \rightarrow \mathbb{R}$ are equality constraints. Unlike linear programming, these functions can be nonlinear, introducing both richness and complexity.

13.1.2. Classification of NLP Problems

Problem Class	Characteristics	Global Optimum Guaranteed	Typical Solvers
Convex NLP	Convex objective and feasible region	Yes	CVXPY, MOSEK, ECOS
Quadratic Programming	Quadratic objective, linear constraints	Yes (if convex)	OSQP, Gurobi, CPLEX
Smooth Nonconvex	Differentiable functions, multiple local minima	No	IPOPT, SNOPT, fmincon

13. Nonlinear and Constrained Optimization

Problem Class	Characteristics	Global Optimum Guaranteed	Typical Solvers
Nonsmooth	Non-differentiable points exist	No	Subgradient, proximal methods
Global	Find global optimum of nonconvex problems	Yes	BARON, Couenne, SCIP

13.1.3. Why NLP is Harder Than LP

⚠ The Challenges of Nonlinearity

1. **Multiple Local Minima:** Unlike LP where any local minimum is global, nonconvex NLPs can have many local minima, making global optimization NP-hard in general.
2. **No Simplex Analogue:** The vertex-hopping strategy of the simplex method has no direct counterpart; iterative methods must follow curved surfaces instead.
3. **Constraint Qualification Issues:** Not all constraint configurations allow standard optimality conditions to hold.
4. **Numerical Sensitivity:** Nonlinear problems are often ill-conditioned, requiring careful numerical treatment.
5. **Computational Cost:** Each iteration typically requires gradient or Hessian evaluations, which can be expensive.

Consider the simple one-dimensional function $f(x) = x^4 - 4x^3 + 4x^2$. This has local minima at $x = 0$ and $x = 2$, but $x = 2$ gives the global minimum. Standard gradient descent from $x = -1$ would converge to $x = 0$, missing the global optimum.

13.1.4. Applications Overview

Nonlinear optimization appears throughout science and engineering:

Domain	Application	Problem Type
Machine Learning	Neural network training	Nonconvex, high-dimensional
Finance	Portfolio optimization with transaction costs	Convex QP with nonlinear costs
Engineering	Structural design	Nonconvex with stress constraints
Control	Model predictive control	Convex/nonconvex QP
Chemistry	Process optimization	Nonconvex with equilibrium constraints
Statistics	Maximum likelihood estimation	Often convex or unimodal
Operations Research	Facility location	Nonconvex continuous

The following code creates two visualization functions that illustrate the difference between convex and nonconvex optimization landscapes. The Rastrigin function, a standard multimodal benchmark, shows the challenge of many local minima that trap local search methods.

```
import numpy as np
import matplotlib.pyplot as plt
```

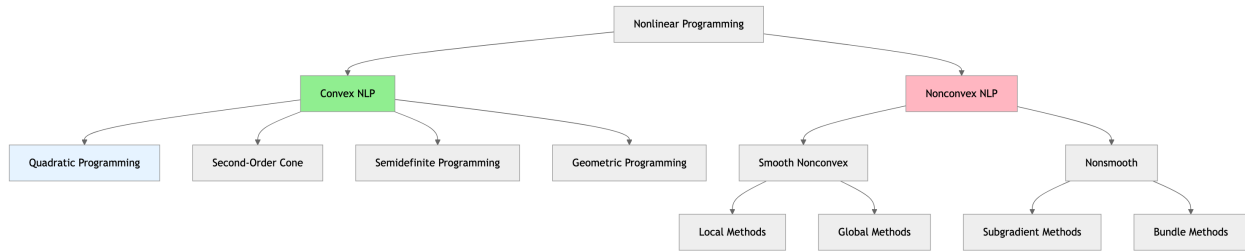


Figure 13.1.: Classification of Nonlinear Programming Problems

```

from mpl_toolkits.mplot3d import Axes3D

def visualize_nlp_landscape():
    """Visualize a nonconvex optimization landscape."""
    # Rastrigin function - a classic multimodal test function
    def rastrigin(x, y):
        return 20 + x**2 - 10*np.cos(2*np.pi*x) + y**2 - 10*np.cos(2*np.pi*y)

    x = np.linspace(-5, 5, 100)
    y = np.linspace(-5, 5, 100)
    X, Y = np.meshgrid(x, y)
    Z = rastrigin(X, Y)

    fig = plt.figure(figsize=(12, 5))

    # 3D surface
    ax1 = fig.add_subplot(121, projection='3d')
    ax1.plot_surface(X, Y, Z, cmap='viridis', alpha=0.8)
    ax1.set_xlabel('x')
    ax1.set_ylabel('y')
    ax1.set_zlabel('f(x,y)')
    ax1.set_title('Rastrigin Function (Multimodal)')

    # Contour plot
    ax2 = fig.add_subplot(122)
    cs = ax2.contour(X, Y, Z, levels=30)
    ax2.scatter([0], [0], c='red', s=100, marker='*', label='Global minimum')
    ax2.set_xlabel('x')
    ax2.set_ylabel('y')
    ax2.set_title('Contour Plot with Multiple Local Minima')
    ax2.legend()

    plt.tight_layout()
    return fig

# Example of convex vs nonconvex
def convex_nonconvex_comparison():
    """Compare convex and nonconvex functions."""
    x = np.linspace(-3, 3, 100)

    # Convex: quadratic
    f_convex = x**2
  
```

```

# Nonconvex: polynomial with multiple minima
f_nonconvex = x**4 - 3*x**2 + 1

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].plot(x, f_convex, 'b-', linewidth=2)
axes[0].axhline(y=0, color='k', linestyle='--', alpha=0.3)
axes[0].scatter([0], [0], c='red', s=100, zorder=5)
axes[0].set_title('Convex Function: Single Global Minimum')
axes[0].set_xlabel('x')
axes[0].set_ylabel('f(x)')
axes[0].grid(True, alpha=0.3)

axes[1].plot(x, f_nonconvex, 'b-', linewidth=2)
axes[1].axhline(y=0, color='k', linestyle='--', alpha=0.3)
# Local minima at x = +/- sqrt(3/2)
x_min = np.sqrt(1.5)
axes[1].scatter([-x_min, x_min], [(-x_min)**4 - 3*(-x_min)**2 + 1]*2,
                c='red', s=100, zorder=5, label='Local minima')
axes[1].scatter([0], [1], c='orange', s=100, marker='s', zorder=5, label='Local
↪ maximum')
axes[1].set_title('Nonconvex Function: Multiple Critical Points')
axes[1].set_xlabel('x')
axes[1].set_ylabel('f(x)')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
return fig

```

13.2. Unconstrained Optimization

We begin with unconstrained optimization, where we seek to minimize $f(\mathbf{x})$ over all of $\mathbb{R}^n$. This simpler setting allows us to develop the fundamental theory and algorithms that underpin constrained methods.

13.2.1. First-Order Necessary Conditions

For a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, if $\mathbf{x}^*$ is a local minimizer, then:

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

This is the **first-order necessary condition** (FONC). Points satisfying this condition are called **stationary points** or **critical points**.

i Gradient Interpretation

The gradient $\nabla f(\mathbf{x})$ points in the direction of steepest ascent. At a minimum: - Moving in any direction cannot decrease f (to first order) - This means the directional derivative is zero in all directions - Hence $\nabla f(\mathbf{x}^*) = \mathbf{0}$

Proof Sketch: If $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$, then the direction $\mathbf{d} = -\nabla f(\mathbf{x}^*)$ satisfies:

$$f(\mathbf{x}^* + \alpha \mathbf{d}) \approx f(\mathbf{x}^*) + \alpha \nabla f(\mathbf{x}^*)^T \mathbf{d} = f(\mathbf{x}^*) - \alpha \|\nabla f(\mathbf{x}^*)\|^2 < f(\mathbf{x}^*)$$

for small $\alpha > 0$, contradicting that $\mathbf{x}^*$ is a local minimizer.

13.2.2. Second-Order Conditions

The Hessian matrix $\mathbf{H}(\mathbf{x}) = \nabla^2 f(\mathbf{x})$ contains all second partial derivatives:

$$[\mathbf{H}(\mathbf{x})]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

Second-Order Necessary Condition (SONC): If $\mathbf{x}^*$ is a local minimizer, then: 1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$
2. $\mathbf{H}(\mathbf{x}^*)$ is positive semidefinite ($\mathbf{H}(\mathbf{x}^*) \succeq 0$)

Second-Order Sufficient Condition (SOSC): If at a point $\mathbf{x}^*$: 1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$ 2. $\mathbf{H}(\mathbf{x}^*)$ is positive definite ($\mathbf{H}(\mathbf{x}^*) \succ 0$)

then $\mathbf{x}^*$ is a **strict local minimizer**.

Matrix Property	Eigenvalues	Implication for Critical Point
Positive definite	All $\lambda_i > 0$	Strict local minimum
Positive semidefinite	All $\lambda_i \geq 0$	Possible local minimum
Negative definite	All $\lambda_i < 0$	Strict local maximum
Indefinite	Mixed signs	Saddle point

This function automates the classification process by computing the eigenvalues of the Hessian matrix and checking their signs.

```
import numpy as np
from numpy.linalg import eig

def classify_critical_point(hessian):
    """Classify a critical point based on the Hessian.

    Args:
        hessian: The Hessian matrix at the critical point

    Returns:
        Classification string and eigenvalues
    """
    eigenvalues, _ = eig(hessian)
    eigenvalues = np.real(eigenvalues) # Hessian is symmetric, eigenvalues are real

    if np.all(eigenvalues > 0):
        classification = "Strict local minimum"
    elif np.all(eigenvalues >= 0):
        classification = "Possible local minimum (semidefinite)"
    elif np.all(eigenvalues < 0):
        classification = "Strict local maximum"
    elif np.all(eigenvalues <= 0):
        classification = "Possible local maximum (semidefinite)"
    else:
        classification = "Saddle point"
```

```

    return classification, eigenvalues

# Example: f(x,y) = x^2 - y^2 at origin (saddle point)
H_saddle = np.array([[2, 0], [0, -2]])
print(classify_critical_point(H_saddle))
# Output: ('Saddle point', array([ 2., -2.]))

# Example: f(x,y) = x^2 + y^2 at origin (minimum)
H_min = np.array([[2, 0], [0, 2]])
print(classify_critical_point(H_min))
# Output: ('Strict local minimum', array([2., 2.]))

```

13.2.3. Newton's Method

Newton's method uses second-order information to achieve faster convergence. At each iteration, we approximate f by its second-order Taylor expansion and minimize:

$$f(\mathbf{x} + \mathbf{d}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{H}(\mathbf{x}) \mathbf{d}$$

Setting the gradient of this quadratic to zero:

$$\mathbf{H}(\mathbf{x}) \mathbf{d} = -\nabla f(\mathbf{x})$$

The **Newton direction** is:

$$\mathbf{d}_N = -\mathbf{H}(\mathbf{x})^{-1} \nabla f(\mathbf{x})$$



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mermaid version 11.6.0

Figure 13.2.: Newton's Method Iteration

The implementation below solves the Newton system at each step and includes a modified Newton fallback: when the Hessian is not positive definite, a regularization term is added to restore descent properties.

```

import numpy as np
from numpy.linalg import solve, norm, cholesky, LinAlgError

def newtons_method(f, grad_f, hess_f, x0, tol=1e-8, max_iter=100):
    """Newton's method for unconstrained optimization.

    Args:
        f: Objective function

```

```

    grad_f: Gradient function
    hess_f: Hessian function
    x0: Initial point
    tol: Convergence tolerance
    max_iter: Maximum iterations

Returns:
    x: Optimal point
    history: List of (x, f(x), ||grad||) tuples
"""
x = np.array(x0, dtype=float)
history = []

for k in range(max_iter):
    grad = grad_f(x)
    grad_norm = norm(grad)
    history.append((x.copy(), f(x), grad_norm))

    # Check convergence
    if grad_norm < tol:
        print(f"Converged in {k} iterations")
        break

    # Compute Hessian and Newton direction
    H = hess_f(x)

    try:
        # Try Cholesky factorization (only works if H is positive definite)
        L = cholesky(H)
        d = -solve(H, grad)
    except LinAlgError:
        # Hessian not positive definite - use modified Newton
        # Add regularization to make it positive definite
        beta = 1e-4
        while True:
            try:
                H_mod = H + beta * np.eye(len(x))
                L = cholesky(H_mod)
                d = -solve(H_mod, grad)
                break
            except LinAlgError:
                beta *= 10

    # Update
    x = x + d

return x, history

# Example: Rosenbrock function
def rosenbrock(x):
    return (1 - x[0])**2 + 100*(x[1] - x[0]**2)**2

def rosenbrock_grad(x):

```

```

    return np.array([
        -2*(1 - x[0]) - 400*x[0]*(x[1] - x[0]**2),
        200*(x[1] - x[0]**2)
    ])

def rosenbrock_hess(x):
    return np.array([
        [2 - 400*(x[1] - 3*x[0]**2), -400*x[0]],
        [-400*x[0], 200]
    ])

x_opt, history = newtons_method(rosenbrock, rosenbrock_grad, rosenbrock_hess,
                                x0=[-1.0, 1.0])
print(f"Optimal point: {x_opt}")
print(f"Optimal value: {rosenbrock(x_opt)}")

```

Convergence Properties:

Condition	Convergence Rate	Iterations for Machine Precision
Far from optimum	May diverge or slow	Depends on starting point
Near optimum, $\mathbf{H}^* \succ 0$	Quadratic	5-10 typically
Near optimum, singular $\mathbf{H}^*$	Linear or slower	Many more

! Quadratic Convergence

Near a strict local minimum with positive definite Hessian, Newton's method exhibits **quadratic convergence**:

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq C\|\mathbf{x}_k - \mathbf{x}^*\|^2$$

This means the number of correct digits roughly doubles each iteration. If $\|\mathbf{x}_k - \mathbf{x}^*\| = 10^{-3}$, then $\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \approx 10^{-6}$, and $\|\mathbf{x}_{k+2} - \mathbf{x}^*\| \approx 10^{-12}$.

13.2.4. Quasi-Newton Methods

Computing and inverting the Hessian at every iteration is expensive ($O(n^3)$) and may be impractical for large problems. Quasi-Newton methods build an approximation $\mathbf{B}_k \approx \mathbf{H}_k$ using only gradient information.

The Secant Condition: After a step from $\mathbf{x}_k$ to $\mathbf{x}_{k+1}$, we want:

$$\mathbf{B}_{k+1}\mathbf{s}_k = \mathbf{y}_k$$

where $\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$ and $\mathbf{y}_k = \nabla f(\mathbf{x}_{k+1}) - \nabla f(\mathbf{x}_k)$.

BFGS Update (Broyden-Fletcher-Goldfarb-Shanno):

$$\mathbf{B}_{k+1} = \mathbf{B}_k - \frac{\mathbf{B}_k \mathbf{s}_k \mathbf{s}_k^T \mathbf{B}_k}{\mathbf{s}_k^T \mathbf{B}_k \mathbf{s}_k} + \frac{\mathbf{y}_k \mathbf{y}_k^T}{\mathbf{y}_k^T \mathbf{s}_k}$$

Equivalently, maintaining the inverse approximation $\mathbf{H}_k \approx \mathbf{H}^{-1}$:

$$\mathbf{H}_{k+1} = \left(\mathbf{I} - \frac{\mathbf{s}_k \mathbf{y}_k^T}{\mathbf{y}_k^T \mathbf{s}_k} \right) \mathbf{H}_k \left(\mathbf{I} - \frac{\mathbf{y}_k \mathbf{s}_k^T}{\mathbf{y}_k^T \mathbf{s}_k} \right) + \frac{\mathbf{s}_k \mathbf{s}_k^T}{\mathbf{y}_k^T \mathbf{s}_k}$$

The following implementation maintains the inverse Hessian approximation $\mathbf{H}$ directly, avoiding the need to solve a linear system at each iteration. It includes backtracking line search and a curvature condition check to ensure the BFGS update remains well-defined.

```
import numpy as np
from numpy.linalg import norm

def bfgs(f, grad_f, x0, tol=1e-8, max_iter=1000):
    """BFGS quasi-Newton method.

    Args:
        f: Objective function
        grad_f: Gradient function
        x0: Initial point
        tol: Convergence tolerance
        max_iter: Maximum iterations

    Returns:
        x: Optimal point
        history: Optimization history
    """
    n = len(x0)
    x = np.array(x0, dtype=float)
    H = np.eye(n) # Initial inverse Hessian approximation
    history = []

    grad = grad_f(x)

    for k in range(max_iter):
        grad_norm = norm(grad)
        history.append((x.copy(), f(x), grad_norm))

        if grad_norm < tol:
            print(f"Converged in {k} iterations")
            break

        # Search direction
        d = -H @ grad

        # Line search (backtracking)
        alpha = backtracking_line_search(f, grad_f, x, d)

        # Update
        x_new = x + alpha * d
        grad_new = grad_f(x_new)

        # BFGS update
        s = x_new - x
        y = grad_new - grad
```

```

    rho = 1.0 / (y @ s)

    if rho > 0: # Curvature condition
        I = np.eye(n)
        V = I - rho * np.outer(s, y)
        H = V @ H @ V.T + rho * np.outer(s, s)

    x = x_new
    grad = grad_new

    return x, history

def backtracking_line_search(f, grad_f, x, d, alpha_init=1.0,
                            rho=0.5, c=1e-4):
    """Backtracking line search with Armijo condition."""
    alpha = alpha_init
    f_x = f(x)
    grad_x = grad_f(x)
    descent = grad_x @ d

    while f(x + alpha * d) > f_x + c * alpha * descent:
        alpha *= rho
        if alpha < 1e-16:
            break

    return alpha

```

L-BFGS (Limited-Memory BFGS): For large-scale problems, storing the full $n \times n$ matrix is impractical. L-BFGS stores only the last m pairs $(\mathbf{s}_k, \mathbf{y}_k)$ and computes the search direction implicitly.

The L-BFGS implementation uses a two-loop recursion to compute the search direction from the stored correction pairs. This keeps memory usage at $O(mn)$ instead of $O(n^2)$, making it practical for problems with thousands or millions of variables.

```

from collections import deque
import numpy as np

def lbfgs(f, grad_f, x0, m=10, tol=1e-8, max_iter=1000):
    """L-BFGS method for large-scale optimization.

    Args:
        f: Objective function
        grad_f: Gradient function
        x0: Initial point
        m: Number of corrections to store
        tol: Convergence tolerance
        max_iter: Maximum iterations

    Returns:
        x: Optimal point
        history: Optimization history
    """

```

```

x = np.array(x0, dtype=float)
n = len(x)

# Storage for s and y pairs
s_history = deque(maxlen=m)
y_history = deque(maxlen=m)

grad = grad_f(x)
history = []

for k in range(max_iter):
    grad_norm = np.linalg.norm(grad)
    history.append((x.copy(), f(x), grad_norm))

    if grad_norm < tol:
        print(f"L-BFGS converged in {k} iterations")
        break

    # Compute search direction using two-loop recursion
    q = grad.copy()
    alphas = []

    # First loop
    for s, y in zip(reversed(s_history), reversed(y_history)):
        rho = 1.0 / (y @ s)
        alpha = rho * (s @ q)
        alphas.append(alpha)
        q = q - alpha * y

    # Initial Hessian approximation
    if len(s_history) > 0:
        s_last = s_history[-1]
        y_last = y_history[-1]
        gamma = (s_last @ y_last) / (y_last @ y_last)
    else:
        gamma = 1.0

    r = gamma * q

    # Second loop
    for (s, y), alpha in zip(zip(s_history, y_history), reversed(alphas)):
        rho = 1.0 / (y @ s)
        beta = rho * (y @ r)
        r = r + s * (alpha - beta)

    d = -r

    # Line search
    alpha = backtracking_line_search(f, grad_f, x, d)

    # Update
    x_new = x + alpha * d
    grad_new = grad_f(x_new)

```

```

s = x_new - x
y = grad_new - grad

# Store if curvature condition is satisfied
if y @ s > 1e-10 * np.linalg.norm(s) * np.linalg.norm(y):
    s_history.append(s)
    y_history.append(y)

x = x_new
grad = grad_new

return x, history

```

13.2.5. Trust Region Methods

Trust region methods take a different approach: instead of choosing a search direction and then a step length, they define a region around the current point where the quadratic model is trusted, and find the best step within this region.

$$\begin{aligned}
 &\text{minimize} && m_k(\mathbf{d}) = f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{B}_k \mathbf{d} \\
 &\text{subject to} && \|\mathbf{d}\| \leq \Delta_k
 \end{aligned}$$

where Δ_k is the trust region radius.

The trust region Newton method below adaptively adjusts the radius Δ_k based on how well the quadratic model predicts actual function decrease. The subproblem is solved using the dogleg method, which interpolates between the steepest descent and Newton directions.

```

import numpy as np
from numpy.linalg import norm, solve, eig

def trust_region_newton(f, grad_f, hess_f, x0, delta0=1.0,
                       eta=0.1, tol=1e-8, max_iter=100):
    """Trust region Newton method.

    Args:
        f, grad_f, hess_f: Objective, gradient, and Hessian functions
        x0: Initial point
        delta0: Initial trust region radius
        eta: Acceptance threshold (typically 0.1)
        tol: Convergence tolerance
        max_iter: Maximum iterations

    Returns:
        x: Optimal point
        history: Optimization history
    """
    x = np.array(x0, dtype=float)
    delta = delta0
    history = []

```

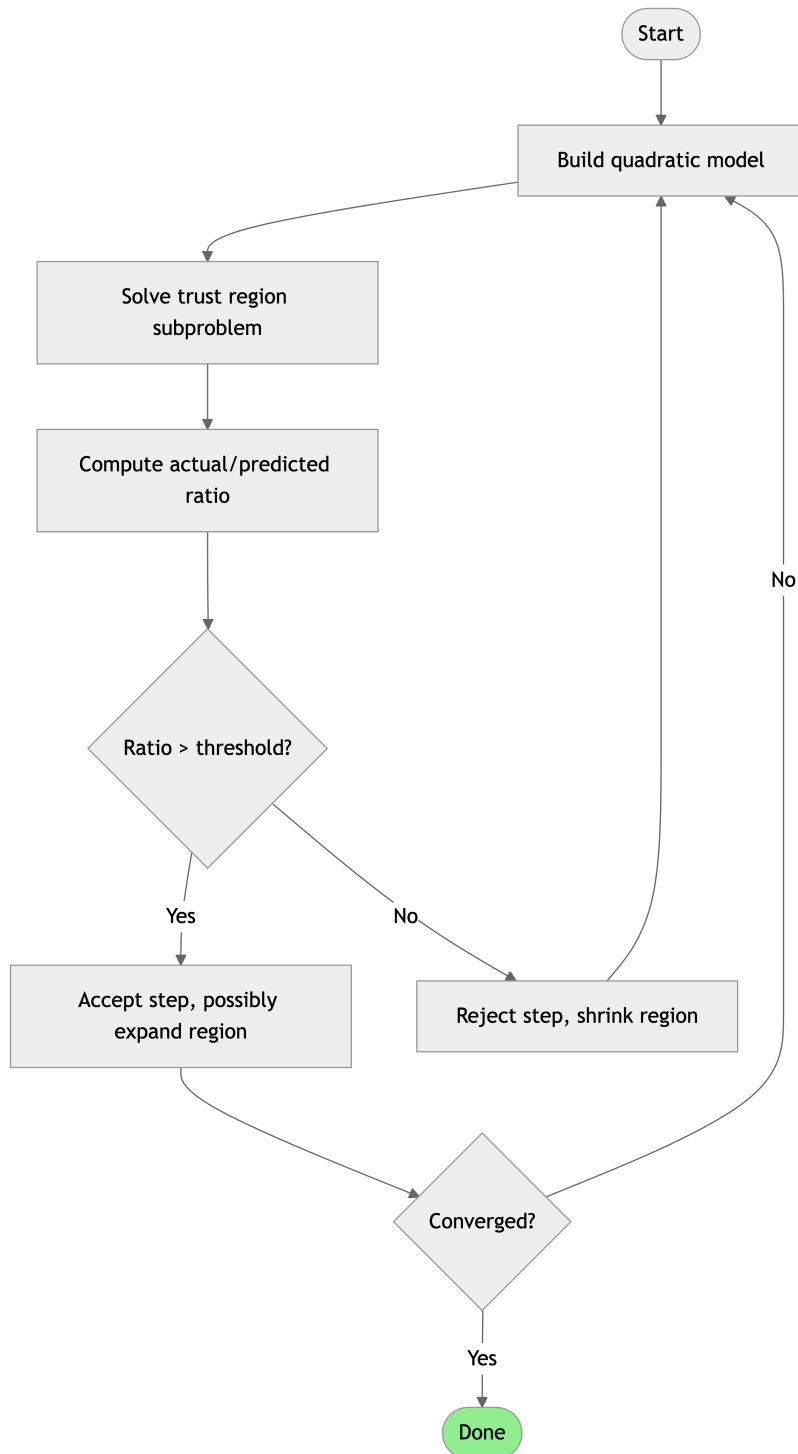



Figure 13.3.: Trust Region Method Flow

```

for k in range(max_iter):
    g = grad_f(x)
    H = hess_f(x)
    fx = f(x)

    history.append((x.copy(), fx, norm(g), delta))

    if norm(g) < tol:
        print(f"Converged in {k} iterations")
        break

    # Solve trust region subproblem
    d = solve_trust_region_subproblem(g, H, delta)

    # Compute actual vs predicted reduction
    actual = fx - f(x + d)
    predicted = -(g @ d + 0.5 * d @ H @ d)

    if predicted > 0:
        rho = actual / predicted
    else:
        rho = 0

    # Update trust region radius
    if rho < 0.25:
        delta = 0.25 * delta
    elif rho > 0.75 and norm(d) >= 0.9 * delta:
        delta = min(2 * delta, 10.0)

    # Accept or reject step
    if rho > eta:
        x = x + d

return x, history

def solve_trust_region_subproblem(g, H, delta, tol=1e-10):
    """Solve the trust region subproblem using dogleg method.

    min g'*d + 0.5*d'*H*d  subject to ||d|| <= delta
    """
    n = len(g)

    # Try unconstrained Newton step
    try:
        d_newton = -solve(H, g)
        if norm(d_newton) <= delta:
            return d_newton
    except:
        pass

    # Cauchy point (steepest descent direction)
    gHg = g @ H @ g
    if gHg <= 0:

```

```

    # Negative curvature: go to boundary in steepest descent direction
    tau = delta / norm(g)
    return -tau * g

alpha_sd = (g @ g) / gHg
d_cauchy = -alpha_sd * g

if norm(d_cauchy) >= delta:
    # Cauchy point outside trust region: scale to boundary
    return -(delta / norm(g)) * g

# Dogleg: interpolate between Cauchy and Newton
try:
    d_newton = -solve(H, g)

    # Find tau such that ||d_cauchy + tau*(d_newton - d_cauchy)|| = delta
    diff = d_newton - d_cauchy
    a = diff @ diff
    b = 2 * d_cauchy @ diff
    c = d_cauchy @ d_cauchy - delta**2

    discriminant = b**2 - 4*a*c
    if discriminant >= 0:
        tau = (-b + np.sqrt(discriminant)) / (2*a)
        tau = max(0, min(1, tau))
        return d_cauchy + tau * diff
except:
    pass

# Fallback: scaled Cauchy point
return -(delta / norm(g)) * g

```

13.2.6. Conjugate Gradient Methods

The nonlinear conjugate gradient method extends the linear CG method to general functions, requiring only gradient evaluations:

$$\mathbf{d}_{k+1} = -\nabla f(\mathbf{x}_{k+1}) + \beta_k \mathbf{d}_k$$

Different choices of β_k give different variants:

Method	Formula for β_k
Fletcher-Reeves	$\beta_k^{FR} = \frac{\ \nabla f_{k+1}\ ^2}{\ \nabla f_k\ ^2}$
Polak-Ribiere	$\beta_k^{PR} = \frac{\nabla f_{k+1}^T (\nabla f_{k+1} - \nabla f_k)}{\ \nabla f_k\ ^2}$
Hestenes-Stiefel	$\beta_k^{HS} = \frac{\nabla f_{k+1}^T (\nabla f_{k+1} - \nabla f_k)}{(\nabla f_{k+1} - \nabla f_k)^T \mathbf{d}_k}$

The implementation supports all three β_k variants and includes automatic restart: if the computed direction is no longer a descent direction, it resets to steepest descent. The Polak-Ribiere variant with $\beta = \max(0, \beta^{PR})$ is generally the most robust choice.

13. Nonlinear and Constrained Optimization

```
import numpy as np
from numpy.linalg import norm

def conjugate_gradient(f, grad_f, x0, method='PR', tol=1e-8, max_iter=1000):
    """Nonlinear conjugate gradient method.

    Args:
        f: Objective function
        grad_f: Gradient function
        x0: Initial point
        method: 'FR' (Fletcher-Reeves), 'PR' (Polak-Ribiere), 'HS'
        ↪ (Hestenes-Stiefel)
        tol: Convergence tolerance
        max_iter: Maximum iterations

    Returns:
        x: Optimal point
        history: Optimization history
    """
    x = np.array(x0, dtype=float)
    g = grad_f(x)
    d = -g
    history = []

    for k in range(max_iter):
        grad_norm = norm(g)
        history.append((x.copy(), f(x), grad_norm))

        if grad_norm < tol:
            print(f"Conjugate Gradient converged in {k} iterations")
            break

        # Line search
        alpha = backtracking_line_search(f, grad_f, x, d)

        # Update
        x_new = x + alpha * d
        g_new = grad_f(x_new)

        # Compute beta
        if method == 'FR':
            beta = (g_new @ g_new) / (g @ g)
        elif method == 'PR':
            beta = (g_new @ (g_new - g)) / (g @ g)
            beta = max(0, beta) # Restart if negative
        elif method == 'HS':
            y = g_new - g
            beta = (g_new @ y) / (d @ y) if abs(d @ y) > 1e-12 else 0
        else:
            raise ValueError(f"Unknown method: {method}")

        # New search direction
        d = -g_new + beta * d
```

```

# Restart if not descent direction
if g_new @ d > 0:
    d = -g_new

x = x_new
g = g_new

return x, history

```

13.3. Constrained Optimization Theory

We now add constraints to our optimization problem, requiring solutions to satisfy both equality and inequality restrictions.

13.3.1. The Lagrangian Formulation

Consider the general NLP:

$$\begin{aligned}
 & \text{minimize} && f(\mathbf{x}) \\
 & \text{subject to} && g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\
 & && h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p
 \end{aligned}$$

The **Lagrangian function** incorporates constraints into the objective:

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \mu_j h_j(\mathbf{x})$$

where $\lambda \geq \mathbf{0}$ are the **Lagrange multipliers** (dual variables) for inequality constraints and μ are the multipliers for equality constraints.

i Intuition Behind the Lagrangian

The Lagrangian penalizes constraint violations: - For $g_i(\mathbf{x}) > 0$ (violated inequality), $\lambda_i g_i(\mathbf{x}) > 0$ increases $\mathcal{L}$ - For $h_j(\mathbf{x}) \neq 0$ (violated equality), $\mu_j h_j(\mathbf{x})$ can be positive or negative - At a feasible point with $g_i(\mathbf{x}) \leq 0$, the multipliers $\lambda_i \geq 0$ ensure no reduction
The constrained optimum is a saddle point: minimum over $\mathbf{x}$, maximum over (λ, μ) .

Derivation of the Lagrangian

Consider the constrained problem with only equality constraints: $\min f(\mathbf{x})$ subject to $h(\mathbf{x}) = 0$. At a constrained minimum $\mathbf{x}^*$, any feasible direction $\mathbf{d}$ must satisfy:

1. First-order feasibility: $\nabla h(\mathbf{x}^*)^T \mathbf{d} = 0$ (stay on constraint surface)
2. Optimality: $\nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0$ (cannot decrease objective in feasible directions)

These conditions imply $\nabla f(\mathbf{x}^*)$ must be a linear combination of constraint gradients:

$$\nabla f(\mathbf{x}^*) = - \sum_j \mu_j \nabla h_j(\mathbf{x}^*)$$

This is precisely $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \mu) = 0$.

13.3.2. KKT Conditions

The **Karush-Kuhn-Tucker (KKT) conditions** are the first-order necessary conditions for constrained optimization. At a local optimum $\mathbf{x}^*$ with multipliers (λ^*, μ^*) :

Stationarity:

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \mu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0}$$

Primal Feasibility:

$$\begin{aligned} g_i(\mathbf{x}^*) &\leq 0, & i = 1, \dots, m \\ h_j(\mathbf{x}^*) &= 0, & j = 1, \dots, p \end{aligned}$$

Dual Feasibility:

$$\lambda_i^* \geq 0, \quad i = 1, \dots, m$$

Complementary Slackness:

$$\lambda_i^* g_i(\mathbf{x}^*) = 0, \quad i = 1, \dots, m$$

! Complementary Slackness Interpretation

The condition $\lambda_i^* g_i(\mathbf{x}^*) = 0$ means: - Either $\lambda_i^* = 0$ (constraint not active, no shadow price) - Or $g_i(\mathbf{x}^*) = 0$ (constraint active at equality)
This reveals which constraints actually bind at the optimum. Inactive constraints ($g_i(\mathbf{x}^*) < 0$) have zero multipliers and could be removed without changing the solution.

The following function provides a numerical check of all four KKT conditions at a candidate solution. It returns a dictionary of violation measures, which is useful for verifying solver output or debugging custom optimization code.

```
import numpy as np
from scipy.optimize import minimize

def check_kkt_conditions(x, f, g_list, h_list, lambdas, mus, tol=1e-6):
    """Check KKT conditions at a candidate solution.

    Args:
        x: Candidate optimal point
        f: Objective function (callable, returns (value, gradient))
        g_list: List of inequality constraint functions (each returns (value,
        ↪ gradient))
        h_list: List of equality constraint functions (each returns (value,
        ↪ gradient))
        lambdas: Lagrange multipliers for inequalities
        mus: Lagrange multipliers for equalities
        tol: Tolerance for checking conditions

    Returns:
        Dictionary with KKT violation measures
    """
    n = len(x)
    _, grad_f = f(x)
```

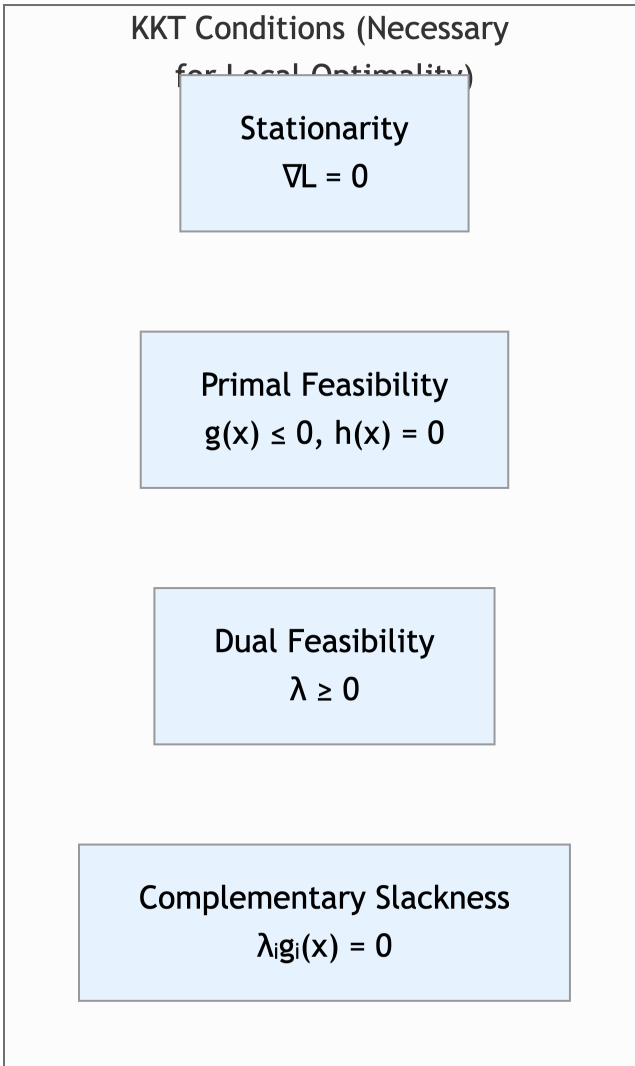


Figure 13.4.: The Four KKT Conditions

```
# Compute stationarity residual
stationarity = grad_f.copy()

for i, g in enumerate(g_list):
    val, grad = g(x)
    stationarity += lambdas[i] * grad

for j, h in enumerate(h_list):
    val, grad = h(x)
    stationarity += mus[j] * grad

stationarity_violation = np.linalg.norm(stationarity)

# Check primal feasibility
```

```

primal_ineq_violation = max(0, *[g(x)[0] for g in g_list]) if g_list else 0
primal_eq_violation = max(abs(h(x)[0]) for h in h_list) if h_list else 0

# Check dual feasibility
dual_violation = max(0, *[-1 for l in lambdas]) if lambdas else 0

# Check complementary slackness
comp_slack_violation = 0
for i, g in enumerate(g_list):
    comp_slack_violation = max(comp_slack_violation,
                              abs(lambdas[i] * g(x)[0]))

violations = {
    'stationarity': stationarity_violation,
    'primal_inequality': primal_ineq_violation,
    'primal_equality': primal_eq_violation,
    'dual_feasibility': dual_violation,
    'complementary_slackness': comp_slack_violation,
    'kkt_satisfied': all([
        stationarity_violation < tol,
        primal_ineq_violation < tol,
        primal_eq_violation < tol,
        dual_violation < tol,
        comp_slack_violation < tol
    ])
}

return violations

# Example: Check KKT for simple problem
# min x^2 + y^2 subject to x + y >= 1
def example_objective(z):
    x, y = z
    return x**2 + y**2, np.array([2*x, 2*y])

def example_constraint(z):
    x, y = z
    return -(x + y - 1), -np.array([1, 1]) # g(x) = 1 - x - y <= 0

# Optimal solution: x = y = 0.5, lambda = 1
x_opt = np.array([0.5, 0.5])
result = check_kkt_conditions(x_opt, example_objective,
                             [example_constraint], [], [1.0], [])
print("KKT Check:", result)

```

13.3.3. Constraint Qualification

KKT conditions are necessary only when a **constraint qualification** (CQ) holds. Without CQ, there may exist local optima where no Lagrange multipliers satisfy KKT.

Linear Independence Constraint Qualification (LICQ): The gradients of all active constraints are linearly independent:

$$\{\nabla g_i(\mathbf{x}^*) : i \in \mathcal{A}(\mathbf{x}^*)\} \cup \{\nabla h_j(\mathbf{x}^*) : j = 1, \dots, p\}$$

are linearly independent, where $\mathcal{A}(\mathbf{x}^*) = \{i : g_i(\mathbf{x}^*) = 0\}$ is the active set.

Slater's Condition (for convex problems): There exists a strictly feasible point $\hat{\mathbf{x}}$ such that $g_i(\hat{\mathbf{x}}) < 0$ for all i .

Constraint Qualification	Applicable To	Strength
LICQ	General NLP	Strongest
MFCQ (Mangasarian-Fromovitz)	General NLP	Weaker than LICQ
Slater's Condition	Convex problems	Commonly used
Linear constraints	LP, QP	Always satisfied

⚠ When CQ Fails

Consider: $\min x$ subject to $x^2 \leq 0$. The only feasible point is $x = 0$, which is optimal. But $\nabla g(0) = 0$, so the constraint gradient vanishes. No $\lambda \geq 0$ can satisfy $1 + \lambda \cdot 0 = 0$. CQ fails because the constraint “degenerates” at the optimum.

13.3.4. Second-Order Sufficient Conditions for Constrained Problems

For constrained problems, SOSC involves the **Hessian of the Lagrangian** restricted to the **tangent space** of active constraints.

Theorem: Let $\mathbf{x}^*$ satisfy KKT with multipliers (λ^*, μ^*) . If for all nonzero $\mathbf{d}$ satisfying: - $\nabla g_i(\mathbf{x}^*)^T \mathbf{d} = 0$ for $i \in \mathcal{A}^+(\mathbf{x}^*) = \{i : g_i(\mathbf{x}^*) = 0, \lambda_i^* > 0\}$ - $\nabla g_i(\mathbf{x}^*)^T \mathbf{d} \leq 0$ for $i \in \mathcal{A}^0(\mathbf{x}^*) = \{i : g_i(\mathbf{x}^*) = 0, \lambda_i^* = 0\}$ - $\nabla h_j(\mathbf{x}^*)^T \mathbf{d} = 0$ for all j

we have $\mathbf{d}^T \nabla_{\mathbf{xx}}^2 \mathcal{L}(\mathbf{x}^*, \lambda^*, \mu^*) \mathbf{d} > 0$, then $\mathbf{x}^*$ is a strict local minimizer.

13.3.5. Duality Theory for NLP

The Lagrangian Dual

The Lagrangian dual function is:

$$q(\lambda, \mu) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda, \mu)$$

The dual problem is:

$$\max_{\lambda \geq \mathbf{0}, \mu} q(\lambda, \mu)$$

Weak Duality: For any feasible primal $\mathbf{x}$ and dual (λ, μ) with $\lambda \geq \mathbf{0}$:

$$q(\lambda, \mu) \leq f(\mathbf{x})$$

Strong Duality: Under certain conditions (e.g., convexity + Slater's condition), the optimal values are equal:

$$\max_{\lambda \geq \mathbf{0}, \mu} q(\lambda, \mu) = \min_{\mathbf{x} \in \mathcal{F}} f(\mathbf{x})$$

The difference $f(\mathbf{x}^*) - q(\lambda^*, \mu^*)$ is the **duality gap**.

This function evaluates the dual function by minimizing the Lagrangian over the primal variable $\mathbf{x}$ for given multiplier values. It uses multiple random starting points to reduce the risk of finding only a local minimum of the Lagrangian.

```

import numpy as np

def compute_lagrangian_dual(f, g_list, h_list, lambda_vals, mu_vals,
                           x_bounds=None):
    """Compute the Lagrangian dual function value.

    Args:
        f: Objective function
        g_list: List of inequality constraint functions
        h_list: List of equality constraint functions
        lambda_vals: Multipliers for inequalities
        mu_vals: Multipliers for equalities
        x_bounds: Bounds for minimization domain

    Returns:
        Dual function value
    """
    from scipy.optimize import minimize

    def lagrangian(x):
        L = f(x)
        for i, g in enumerate(g_list):
            L += lambda_vals[i] * g(x)
        for j, h in enumerate(h_list):
            L += mu_vals[j] * h(x)
        return L

    # Minimize Lagrangian over x
    if x_bounds is None:
        x_bounds = [(-100, 100)] * 2 # Default bounds

    # Try multiple starting points
    best_val = np.inf
    for _ in range(10):
        x0 = np.random.uniform(-10, 10, len(x_bounds))
        result = minimize(lagrangian, x0, bounds=x_bounds)
        if result.fun < best_val:
            best_val = result.fun

    return best_val

```

13.3.6. Sensitivity Analysis

The Lagrange multipliers provide sensitivity information about how the optimal value changes with constraint perturbations.

Theorem: Under LICQ and SOSC, if we perturb the right-hand sides:

$$g_i(\mathbf{x}) \leq \epsilon_i, \quad h_j(\mathbf{x}) = \delta_j$$

then the optimal value $v^*(\epsilon, \delta)$ satisfies:

$$\left. \frac{\partial v^*}{\partial \epsilon_i} \right|_{\epsilon=\mathbf{0}, \delta=\mathbf{0}} = -\lambda_i^*$$

$$\left. \frac{\partial v^*}{\partial \delta_j} \right|_{\epsilon=0, \delta=0} = -\mu_j^*$$

This is why Lagrange multipliers are called **shadow prices** - they measure the marginal value of relaxing constraints.

13.4. Constrained Optimization Algorithms

13.4.1. Penalty Methods

Penalty methods transform constrained problems into sequences of unconstrained problems by adding penalty terms for constraint violations.

Exterior Penalty Method:

$$P(\mathbf{x}, \rho) = f(\mathbf{x}) + \frac{\rho}{2} \left(\sum_{i=1}^m \max(0, g_i(\mathbf{x}))^2 + \sum_{j=1}^p h_j(\mathbf{x})^2 \right)$$

As $\rho \rightarrow \infty$, minimizers of $P(\mathbf{x}, \rho)$ approach solutions of the constrained problem.

The implementation below solves a sequence of unconstrained problems with increasing penalty parameter ρ . Each subproblem uses the previous solution as a warm start, and the outer loop terminates when constraint violations fall below the tolerance.

```
import numpy as np
from scipy.optimize import minimize

def exterior_penalty_method(f, g_list, h_list, x0, rho_init=1.0,
                           rho_factor=10, tol=1e-6, max_outer=20):
    """Exterior penalty method for constrained optimization.

    Args:
        f: Objective function
        g_list: List of inequality constraint functions (g(x) <= 0)
        h_list: List of equality constraint functions (h(x) = 0)
        x0: Initial point
        rho_init: Initial penalty parameter
        rho_factor: Factor to increase rho
        tol: Constraint violation tolerance
        max_outer: Maximum outer iterations

    Returns:
        x: Approximate optimal point
        history: Optimization history
    """
    def penalty_function(x, rho):
        P = f(x)

        # Inequality penalties
        for g in g_list:
            P += (rho / 2) * max(0, g(x))**2

        # Equality penalties
        for h in h_list:
```

```

        P += (rho / 2) * h(x)**2

    return P

x = np.array(x0, dtype=float)
rho = rho_init
history = []

for k in range(max_outer):
    # Minimize penalized objective
    result = minimize(lambda z: penalty_function(z, rho), x,
                      method='BFGS')
    x = result.x

    # Compute constraint violations
    ineq_violation = max(0, *[g(x) for g in g_list]) if g_list else 0
    eq_violation = max(abs(h(x)) for h in h_list) if h_list else 0
    max_violation = max(ineq_violation, eq_violation)

    history.append({
        'x': x.copy(),
        'f': f(x),
        'rho': rho,
        'violation': max_violation
    })

    print(f"Iter {k}: f = {f(x):.6f}, violation = {max_violation:.6e}, rho = \
↪ {rho:.1f}")

    # Check convergence
    if max_violation < tol:
        print(f"Converged in {k+1} outer iterations")
        break

    # Increase penalty
    rho *= rho_factor

return x, history

# Example: min x^2 + y^2 subject to x + y = 1
f = lambda z: z[0]**2 + z[1]**2
h = lambda z: z[0] + z[1] - 1

x_opt, hist = exterior_penalty_method(f, [], [h], x0=[0.0, 0.0])
print(f"Solution: {x_opt}")
print(f"Optimal value: {f(x_opt):.6f}")

```

Interior Penalty (Barrier) Method:

For problems with only inequality constraints, barrier methods keep iterates strictly inside the feasible region:

$$B(\mathbf{x}, \mu) = f(\mathbf{x}) - \mu \sum_{i=1}^m \log(-g_i(\mathbf{x}))$$

As $\mu \rightarrow 0^+$, minimizers approach the constrained optimum from the interior.

The barrier method requires a strictly feasible starting point. The logarithmic terms create an infinite penalty at the constraint boundary, ensuring all iterates remain interior. As μ decreases, the barrier weakens and the solution approaches the true constrained optimum.

```
import numpy as np
from scipy.optimize import minimize

def barrier_method(f, g_list, x0, mu_init=1.0, mu_factor=0.1,
                  tol=1e-8, max_outer=50):
    """Log-barrier interior point method.

    Args:
        f: Objective function
        g_list: List of inequality constraints (g(x) <= 0)
        x0: Strictly feasible initial point
        mu_init: Initial barrier parameter
        mu_factor: Factor to decrease mu
        tol: Tolerance for barrier parameter
        max_outer: Maximum outer iterations

    Returns:
        x: Approximate optimal point
        history: Optimization history
    """
    def barrier_objective(x, mu):
        # Check strict feasibility
        for g in g_list:
            if g(x) >= 0:
                return np.inf

        B = f(x)
        for g in g_list:
            B -= mu * np.log(-g(x))
        return B

    x = np.array(x0, dtype=float)
    mu = mu_init
    history = []

    for k in range(max_outer):
        # Minimize barrier objective
        result = minimize(lambda z: barrier_objective(z, mu), x,
                        method='BFGS')

        if result.success:
            x = result.x

        # Compute approximate multipliers
        lambdas = [-mu / g(x) for g in g_list]

    history.append({
        'x': x.copy(),
```

```

        'f': f(x),
        'mu': mu,
        'lambdas': lambdas
    })

    print(f"Iter {k}: f = {f(x):.6f}, mu = {mu:.6e}")

    # Check convergence
    if mu < tol:
        print(f"Converged in {k+1} outer iterations")
        break

    # Decrease barrier parameter
    mu *= mu_factor

    return x, history

# Example: min x^2 + y^2 subject to x + y >= 1 (rewritten as 1 - x - y <= 0)
f = lambda z: z[0]**2 + z[1]**2
g = lambda z: 1 - z[0] - z[1]

# Start from strictly feasible point
x_opt, hist = barrier_method(f, [g], x0=[1.0, 1.0])
print(f"Solution: {x_opt}")

```

13.4.2. Augmented Lagrangian Method

The augmented Lagrangian combines penalty with Lagrange multiplier estimates:

$$\mathcal{L}_A(\mathbf{x}, \lambda, \rho) = f(\mathbf{x}) + \sum_{j=1}^p \mu_j h_j(\mathbf{x}) + \frac{\rho}{2} \sum_{j=1}^p h_j(\mathbf{x})^2$$

For inequalities, we use:

$$\mathcal{L}_A = f(\mathbf{x}) + \sum_{i=1}^m \left[\lambda_i g_i(\mathbf{x}) + \frac{\rho}{2} \max(0, g_i(\mathbf{x}) + \frac{\lambda_i}{\rho})^2 - \frac{\lambda_i^2}{2\rho} \right]$$

The augmented Lagrangian method alternates between minimizing the augmented objective and updating the multiplier estimates. Unlike pure penalty methods, the multiplier updates allow convergence to a feasible solution without driving ρ to infinity, which avoids ill-conditioning.

```

import numpy as np
from scipy.optimize import minimize

def augmented_lagrangian_method(f, g_list, h_list, x0,
                               rho_init=1.0, tol=1e-6, max_outer=50):
    """Augmented Lagrangian method (Method of Multipliers).

    Args:
        f: Objective function
        g_list: Inequality constraints (g(x) <= 0)
        h_list: Equality constraints (h(x) = 0)
        x0: Initial point
    """

```

```

    rho_init: Initial penalty parameter
    tol: Convergence tolerance
    max_outer: Maximum outer iterations

Returns:
    x: Optimal point
    lambdas, mus: Lagrange multipliers
"""
m = len(g_list)
p = len(h_list)

x = np.array(x0, dtype=float)
lambdas = np.zeros(m)
mus = np.zeros(p)
rho = rho_init

def augmented_lagrangian(x, lambdas, mus, rho):
    L = f(x)

    # Inequality constraints
    for i, g in enumerate(g_list):
        gi = g(x)
        # Smooth max(0, ...) formulation
        if gi + lambdas[i]/rho > 0:
            L += lambdas[i] * gi + (rho/2) * gi**2
        else:
            L -= lambdas[i]**2 / (2*rho)

    # Equality constraints
    for j, h in enumerate(h_list):
        hj = h(x)
        L += mus[j] * hj + (rho/2) * hj**2

    return L

history = []

for k in range(max_outer):
    # Minimize augmented Lagrangian
    result = minimize(
        lambda z: augmented_lagrangian(z, lambdas, mus, rho),
        x, method='BFGS'
    )
    x = result.x

    # Compute constraint values
    g_vals = np.array([g(x) for g in g_list]) if g_list else np.array([])
    h_vals = np.array([h(x) for h in h_list]) if h_list else np.array([])

    # Update multipliers
    for i in range(m):
        lambdas[i] = max(0, lambdas[i] + rho * g_vals[i])
    mus = mus + rho * h_vals

```

```

# Compute violations
ineq_viol = max(0, *g_vals) if len(g_vals) > 0 else 0
eq_viol = np.max(np.abs(h_vals)) if len(h_vals) > 0 else 0
max_viol = max(ineq_viol, eq_viol)

history.append({
    'x': x.copy(),
    'f': f(x),
    'lambdas': lambdas.copy(),
    'mus': mus.copy(),
    'violation': max_viol
})

print(f"Iter {k}: f = {f(x):.6f}, violation = {max_viol:.6e}")

if max_viol < tol:
    print(f"Converged in {k+1} iterations")
    break

# Optionally increase rho
if k > 0 and max_viol > 0.25 * history[-2]['violation']:
    rho *= 2

return x, lambdas, mus, history

```

13.4.3. Sequential Quadratic Programming (SQP)

SQP is one of the most effective methods for smooth nonlinear programming. At each iteration, it solves a quadratic programming (QP) subproblem:

$$\begin{aligned}
 &\text{minimize} && \nabla f(\mathbf{x}_k)^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{B}_k \mathbf{d} \\
 &\text{subject to} && \nabla g_i(\mathbf{x}_k)^T \mathbf{d} + g_i(\mathbf{x}_k) \leq 0 \\
 &&& \nabla h_j(\mathbf{x}_k)^T \mathbf{d} + h_j(\mathbf{x}_k) = 0
 \end{aligned}$$

where $\mathbf{B}_k$ approximates the Hessian of the Lagrangian.

The SQP implementation below uses CVXPY to solve the QP subproblem at each iteration and updates the Hessian approximation with a BFGS formula. It extracts the Lagrange multipliers from the QP dual variables to monitor KKT satisfaction.

```

import numpy as np
from scipy.optimize import minimize
import cvxpy as cp

def sqp_method(f, grad_f, g_list, grad_g_list, h_list, grad_h_list,
               x0, tol=1e-6, max_iter=100):
    """Sequential Quadratic Programming method.

    Args:
        f, grad_f: Objective and its gradient
        g_list, grad_g_list: Inequality constraints and gradients
    """

```

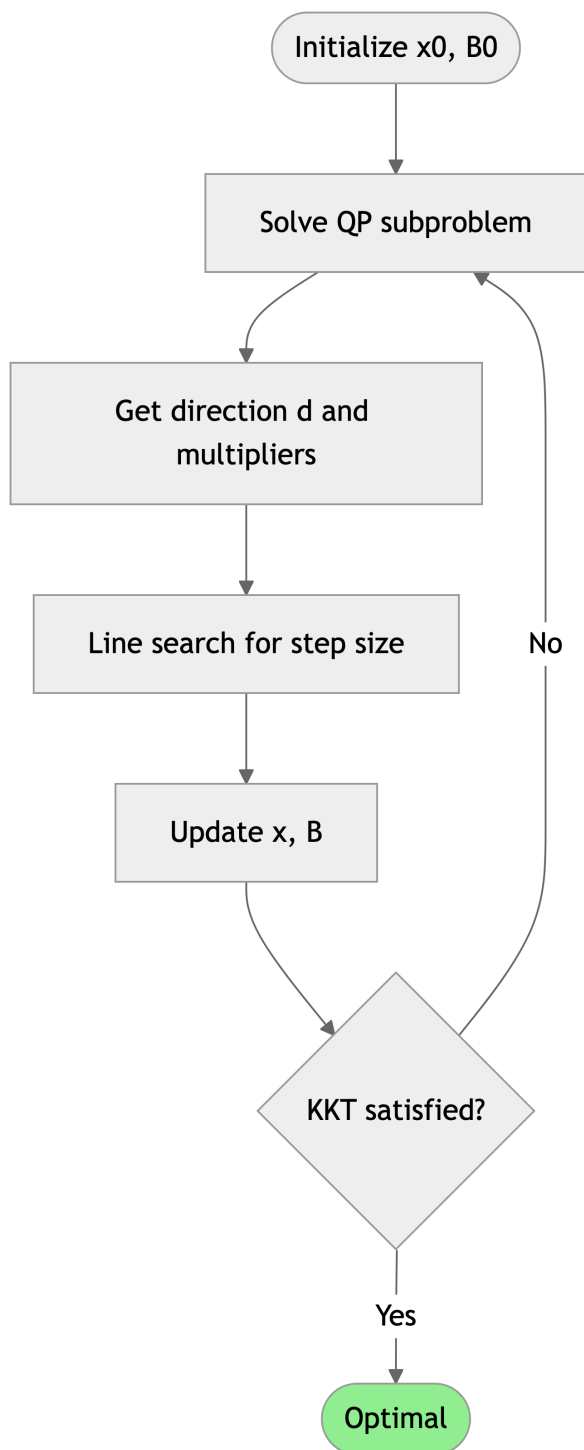



Figure 13.5.: SQP Algorithm Flow

```

    h_list, grad_h_list: Equality constraints and gradients
    x0: Initial point
    tol: KKT tolerance
    max_iter: Maximum iterations

Returns:
    x: Optimal point
    lambdas, mus: Multipliers
"""
n = len(x0)
m = len(g_list)
p = len(h_list)

x = np.array(x0, dtype=float)
B = np.eye(n) # Initial Hessian approximation

history = []

for k in range(max_iter):
    # Evaluate functions and gradients
    fx = f(x)
    gx = grad_f(x)

    g_vals = np.array([g(x) for g in g_list]) if g_list else np.array([])
    h_vals = np.array([h(x) for h in h_list]) if h_list else np.array([])

    grad_g = np.array([grad_g(x) for grad_g in grad_g_list]).T if grad_g_list
    ↪ else np.zeros((n, 0))
    grad_h = np.array([grad_h(x) for grad_h in grad_h_list]).T if grad_h_list
    ↪ else np.zeros((n, 0))

    # Solve QP subproblem
    d = cp.Variable(n)

    objective = gx @ d + 0.5 * cp.quad_form(d, B)

    constraints = []
    if m > 0:
        constraints.append(grad_g.T @ d + g_vals <= 0)
    if p > 0:
        constraints.append(grad_h.T @ d + h_vals == 0)

    prob = cp.Problem(cp.Minimize(objective), constraints)

    try:
        prob.solve(solver=cp.OSQP)
        d_val = d.value

        if d_val is None:
            print("QP subproblem infeasible, using gradient")
            d_val = -gx
    except:
        print("QP solver failed, using gradient")

```

```

        d_val = -gx

    # Get multipliers from QP
    lambdas = np.zeros(m)
    mus = np.zeros(p)
    if constraints:
        idx = 0
        if m > 0:
            lambdas = np.maximum(0, -constraints[0].dual_value) if
↪ constraints[0].dual_value is not None else np.zeros(m)
            idx = 1
            if p > 0:
                mus = -constraints[idx].dual_value if constraints[idx].dual_value is
↪ not None else np.zeros(p)

    # Check KKT conditions
    kkt_grad = gx + grad_g @ lambdas + grad_h @ mus if m + p > 0 else gx
    kkt_violation = np.linalg.norm(kkt_grad)
    primal_violation = max(0, *g_vals, *np.abs(h_vals)) if m + p > 0 else 0

    history.append({
        'x': x.copy(),
        'f': fx,
        'kkt': kkt_violation,
        'primal': primal_violation
    })

    print(f"Iter {k}: f = {fx:.6f}, KKT = {kkt_violation:.6e}")

    if kkt_violation < tol and primal_violation < tol:
        print(f"SQP converged in {k+1} iterations")
        break

    # Line search (simplified)
    alpha = 1.0
    while f(x + alpha * d_val) > fx + 1e-4 * alpha * gx @ d_val:
        alpha *= 0.5
        if alpha < 1e-10:
            break

    # Update
    x_new = x + alpha * d_val
    gx_new = grad_f(x_new)

    # BFGS update for Hessian approximation
    s = x_new - x
    y = gx_new - gx

    if y @ s > 1e-10 * np.linalg.norm(s) * np.linalg.norm(y):
        rho = 1.0 / (y @ s)
        B = (np.eye(n) - rho * np.outer(s, y)) @ B @ (np.eye(n) - rho *
↪ np.outer(y, s))
        B += rho * np.outer(s, s)

```

```

x = x_new

return x, lambdas, mus, history

```

13.4.4. Interior Point Methods for NLP

Modern interior point methods for NLP combine barrier methods with Newton's method, solving the KKT system directly. The primal-dual interior point method is particularly effective.

Perturbed KKT System:

$$\begin{bmatrix} \nabla_{xx}^2 \mathcal{L} & \nabla \mathbf{g}^T & \nabla \mathbf{h}^T \\ \Lambda \nabla \mathbf{g} & -G & 0 \\ \nabla \mathbf{h} & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x} \\ \Delta \lambda \\ \Delta \mu \end{bmatrix} = - \begin{bmatrix} \nabla_{\mathbf{x}} \mathcal{L} \\ \Lambda \mathbf{g} - \mu \mathbf{e} \\ \mathbf{h} \end{bmatrix}$$

where $\Lambda = \text{diag}(\lambda)$, $G = \text{diag}(\mathbf{g})$, and $\mu > 0$ is the barrier parameter.

13.4.5. Active Set Methods

Active set methods maintain a working set of constraints treated as equalities and iteratively add or remove constraints.

The active set method solves a KKT system at each iteration for the current working set of constraints. It adds blocking constraints when the step hits a boundary, and removes constraints with negative multipliers that indicate they should not be active at the optimum.

```

import numpy as np
from numpy.linalg import solve, norm

def active_set_qp(H, c, A_ub, b_ub, x0=None, tol=1e-8, max_iter=1000):
    """Active set method for convex QP.

    min 0.5*x'Hx + c'x  subject to Ax <= b

    Args:
        H: Positive definite Hessian matrix
        c: Linear term
        A_ub: Inequality constraint matrix
        b_ub: Inequality RHS
        x0: Initial feasible point
        tol: Tolerance
        max_iter: Maximum iterations

    Returns:
        x: Optimal point
        lambdas: Multipliers for active constraints
    """
    n = len(c)
    m = len(b_ub)

    # Initialize with feasible point
    if x0 is None:
        # Find feasible starting point (simplified)
        x = np.zeros(n)

```

```

else:
    x = np.array(x0, dtype=float)

    # Working set (indices of active constraints)
    W = set()

    # Find initially active constraints
    for i in range(m):
        if abs(A_ub[i] @ x - b_ub[i]) < tol:
            W.add(i)

    for iteration in range(max_iter):
        # Form equality-constrained subproblem
        grad = H @ x + c

        if len(W) == 0:
            # Unconstrained problem
            d = solve(H, -grad)
            lambdas_W = np.array([])
        else:
            W_list = list(W)
            A_W = A_ub[W_list]

            # Solve KKT system
            n_active = len(W_list)
            KKT = np.block([
                [H, A_W.T],
                [A_W, np.zeros((n_active, n_active))]
            ])
            rhs = np.concatenate([-grad, np.zeros(n_active)])

            sol = solve(KKT, rhs)
            d = sol[:n]
            lambdas_W = sol[n:]

        if norm(d) < tol:
            # Stationary point for current working set
            if len(W) == 0 or np.all(lambdas_W >= -tol):
                # Check if all multipliers are non-negative
                print(f"Active set converged in {iteration} iterations")
                lambdas = np.zeros(m)
                for i, idx in enumerate(W):
                    lambdas[idx] = lambdas_W[i]
                return x, lambdas
            else:
                # Remove constraint with most negative multiplier
                min_idx = np.argmin(lambdas_W)
                W_list = list(W)
                W.remove(W_list[min_idx])
        else:
            # Compute step length
            alpha = 1.0
            blocking_constraint = None

```

```

for i in range(m):
    if i not in W:
        denom = A_ub[i] @ d
        if denom > tol:
            alpha_i = (b_ub[i] - A_ub[i] @ x) / denom
            if alpha_i < alpha:
                alpha = alpha_i
                blocking_constraint = i

x = x + alpha * d

if blocking_constraint is not None:
    W.add(blocking_constraint)

print("Active set did not converge")
return x, np.zeros(m)

```

13.5. Convex Optimization

Convex optimization occupies a special place in the NLP problem space: it is rich enough to model many important applications, yet structured enough to guarantee efficient solution.

13.5.1. Convex Sets and Functions

Convex Set: A set C is convex if for any $\mathbf{x}, \mathbf{y} \in C$ and $\theta \in [0, 1]$:

$$\theta \mathbf{x} + (1 - \theta) \mathbf{y} \in C$$

Convex Function: A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if its domain is convex and:

$$f(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta) f(\mathbf{y})$$

for all $\mathbf{x}, \mathbf{y}$ in the domain and $\theta \in [0, 1]$.

Equivalent Characterizations

For twice-differentiable f , convexity is equivalent to: 1. **First-order:** $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x})$ (tangent lines are global underestimators) 2. **Second-order:** $\nabla^2 f(\mathbf{x}) \succeq 0$ (Hessian is positive semidefinite everywhere)

Examples of Convex Functions:

Function	Domain	Why Convex
$\ \mathbf{x}\ _p$ ($p \geq 1$)	$\mathbb{R}^n$	Norm triangle inequality
$\mathbf{x}^T A \mathbf{x}$ ($A \succeq 0$)	$\mathbb{R}^n$	PSD Hessian = $2A$
$-\log x$	$\mathbb{R}_{++}$	Second derivative $1/x^2 > 0$
e^{ax}	$\mathbb{R}$	Second derivative $a^2 e^{ax} > 0$
$\max_i x_i$	$\mathbb{R}^n$	Pointwise max of linear functions
$\log \sum_i e^{x_i}$	$\mathbb{R}^n$	Log-sum-exp (smooth max)

Operations Preserving Convexity:

- Non-negative weighted sums
- Composition with affine functions
- Pointwise maximum
- Perspective

13.5.2. Why Convexity Guarantees Global Optimum

! The Fundamental Theorem of Convex Optimization

For a convex optimization problem (minimize convex function over convex set): 1. Any local minimum is a global minimum 2. The set of optimal solutions is convex 3. If the objective is strictly convex, the optimum is unique

This transforms optimization from “search for best among many local minima” to “find any local minimum and it’s global.”

Proof Sketch: Suppose $\mathbf{x}^*$ is a local minimum but not global. Let $\mathbf{y}^*$ be a global minimum with $f(\mathbf{y}^*) < f(\mathbf{x}^*)$. By convexity:

$$f(\theta\mathbf{x}^* + (1 - \theta)\mathbf{y}^*) \leq \theta f(\mathbf{x}^*) + (1 - \theta)f(\mathbf{y}^*) < f(\mathbf{x}^*)$$

for any $\theta \in (0, 1)$. But points $\theta\mathbf{x}^* + (1 - \theta)\mathbf{y}^*$ are arbitrarily close to $\mathbf{x}^*$, contradicting local optimality.

13.5.3. Disciplined Convex Programming (DCP)

DCP is a framework that allows automatic verification of convexity using composition rules. A problem is DCP-compliant if:

1. The objective is a convex expression (minimization) or concave expression (maximization)
2. Equality constraints are affine
3. Inequality constraints have form: convex $\leq$ concave

DCP Composition Rules:

Operation	Result Curvature
convex + convex	convex
concave + concave	concave
nonnegative * convex	convex
nonpositive * convex	concave
convex(affine)	convex
concave(affine)	concave
convex_increasing(convex)	convex
concave_decreasing(convex)	concave

13.5.4. CVXPY for Convex Optimization

CVXPY is a Python library for specifying and solving convex optimization problems using DCP.

The portfolio optimization example below shows how CVXPY translates the Markowitz mean-variance problem into a few lines of code. The solver automatically verifies convexity through DCP rules and selects an appropriate backend solver.

13. Nonlinear and Constrained Optimization

```
import cvxpy as cp
import numpy as np

def portfolio_optimization_cvxpy(mu, Sigma, target_return=None,
                                max_risk=None, gamma=1.0):
    """Portfolio optimization using CVXPY.

    Args:
        mu: Expected returns (n,)
        Sigma: Covariance matrix (n, n)
        target_return: Minimum required return (optional)
        max_risk: Maximum allowed variance (optional)
        gamma: Risk aversion parameter

    Returns:
        Optimal portfolio weights
    """
    n = len(mu)
    w = cp.Variable(n)

    # Objective: maximize return - gamma * risk
    # Equivalently: minimize gamma * risk - return
    portfolio_return = mu @ w
    portfolio_risk = cp.quad_form(w, Sigma)

    objective = cp.Minimize(gamma * portfolio_risk - portfolio_return)

    # Constraints
    constraints = [
        cp.sum(w) == 1, # Weights sum to 1
        w >= 0          # Long-only portfolio
    ]

    if target_return is not None:
        constraints.append(portfolio_return >= target_return)

    if max_risk is not None:
        constraints.append(portfolio_risk <= max_risk)

    problem = cp.Problem(objective, constraints)
    problem.solve()

    return {
        'weights': w.value,
        'return': portfolio_return.value,
        'risk': np.sqrt(portfolio_risk.value),
        'sharpe': portfolio_return.value / np.sqrt(portfolio_risk.value)
    }

# Example
np.random.seed(42)
n = 5
mu = np.array([0.12, 0.10, 0.07, 0.03, 0.05])
```



```

Sigma = np.array([
    [0.10, 0.03, 0.01, 0.00, 0.02],
    [0.03, 0.08, 0.02, 0.01, 0.01],
    [0.01, 0.02, 0.05, 0.01, 0.01],
    [0.00, 0.01, 0.01, 0.02, 0.00],
    [0.02, 0.01, 0.01, 0.00, 0.04]
])

result = portfolio_optimization_cvxpy(mu, Sigma, gamma=2.0)
print(f"Optimal weights: {result['weights']}")
print(f"Expected return: {result['return']:.4f}")
print(f"Portfolio std: {result['risk']:.4f}")
print(f"Sharpe ratio: {result['sharpe']:.4f}")

```

CVXPY handles a wide range of machine learning formulations. The following three functions implement LASSO regression with L1 regularization, soft-margin SVM with slack variables, and logistic regression with L2 regularization. Each is expressed as a convex optimization problem that CVXPY solves directly.

```

import cvxpy as cp
import numpy as np

def lasso_regression_cvxpy(X, y, lambda_reg):
    """LASSO regression using CVXPY.

    min ||Xw - y||_2^2 + lambda * ||w||_1

    Args:
        X: Design matrix (m, n)
        y: Response vector (m,)
        lambda_reg: Regularization parameter

    Returns:
        Optimal coefficients
    """
    m, n = X.shape
    w = cp.Variable(n)

    loss = cp.sum_squares(X @ w - y)
    regularization = lambda_reg * cp.norm(w, 1)

    problem = cp.Problem(cp.Minimize(loss + regularization))
    problem.solve()

    return w.value

def svm_cvxpy(X, y, C=1.0):
    """Soft-margin SVM using CVXPY.

    Args:
        X: Feature matrix (m, n)
        y: Labels in {-1, +1} (m,)
    """

```

```

        C: Regularization parameter

Returns:
    w: Weight vector
    b: Bias term
    """
    m, n = X.shape
    w = cp.Variable(n)
    b = cp.Variable()
    xi = cp.Variable(m) # Slack variables

    # Objective: minimize ||w||^2/2 + C * sum(xi)
    objective = cp.Minimize(0.5 * cp.sum_squares(w) + C * cp.sum(xi))

    # Constraints: y_i(w'x_i + b) >= 1 - xi_i, xi >= 0
    constraints = [
        cp.multiply(y, X @ w + b) >= 1 - xi,
        xi >= 0
    ]

    problem = cp.Problem(objective, constraints)
    problem.solve()

    return w.value, b.value

def logistic_regression_cvxpy(X, y, lambda_reg=0.01):
    """Logistic regression using CVXPY.

    Args:
        X: Feature matrix (m, n)
        y: Labels in {0, 1} (m,)
        lambda_reg: L2 regularization

    Returns:
        Optimal coefficients
    """
    m, n = X.shape
    w = cp.Variable(n)
    b = cp.Variable()

    # Logistic loss: sum(log(1 + exp(-y_i * (w'x_i + b))))
    # For y in {0,1}: -y*log(sigmoid) - (1-y)*log(1-sigmoid)

    z = X @ w + b
    # Using log_sum_exp for numerical stability
    loss = cp.sum(cp.logistic(-cp.multiply(2*y - 1, z)))

    regularization = lambda_reg * cp.sum_squares(w)

    problem = cp.Problem(cp.Minimize(loss + regularization))
    problem.solve()

    return w.value, b.value

```

13.6. Quadratic Programming

Quadratic Programming (QP) is a special case of NLP where the objective is quadratic and constraints are linear:

$$\begin{aligned} & \text{minimize} && \frac{1}{2} \mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x} \\ & \text{subject to} && A \mathbf{x} \leq \mathbf{b} \\ & && A_{eq} \mathbf{x} = \mathbf{b}_{eq} \end{aligned}$$

When $Q \succeq 0$ (positive semidefinite), the problem is convex.

13.6.1. QP Formulation

The KKT conditions for QP form a linear complementarity problem:

$$\begin{aligned} Q\mathbf{x} + \mathbf{c} + A^T\lambda + A_{eq}^T\mu &= \mathbf{0} \\ \lambda \geq 0, \quad A\mathbf{x} - \mathbf{b} &\leq 0, \quad \lambda^T(A\mathbf{x} - \mathbf{b}) = 0 \end{aligned}$$

13.6.2. Active Set Methods for QP

See the active set implementation above. For QP, active set methods are particularly efficient because: - Subproblems are equality-constrained QPs with closed-form solutions - Finite convergence is guaranteed for convex QP

13.6.3. Interior Point Methods for QP

Interior point methods for QP solve the perturbed KKT system:

$$\begin{bmatrix} Q & A^T \\ \Lambda A & S \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x} \\ \Delta \lambda \end{bmatrix} = - \begin{bmatrix} Q\mathbf{x} + \mathbf{c} + A^T\lambda \\ \Lambda(A\mathbf{x} - \mathbf{b}) - \mu \mathbf{e} \end{bmatrix}$$

where $\Lambda = \text{diag}(\lambda)$, $S = \text{diag}(\mathbf{b} - A\mathbf{x})$, and $\mu > 0$ is the barrier parameter.

The primal-dual interior point implementation below solves this perturbed KKT system at each iteration. It uses a Schur complement reduction to handle the block structure efficiently and applies a fraction-to-boundary rule to keep slack variables and multipliers strictly positive.

```
import numpy as np
from numpy.linalg import solve, norm

def interior_point_qp(Q, c, A, b, tol=1e-8, max_iter=100):
    """Primal-dual interior point method for convex QP.

    min 0.5*x'Qx + c'x subject to Ax <= b

    Args:
        Q: Positive semidefinite matrix
        c: Linear term
        A: Constraint matrix
        b: RHS
        tol: Duality gap tolerance
        max_iter: Maximum iterations
```

```

Returns:
    x: Optimal primal solution
    lambdas: Optimal dual solution
"""
n = len(c)
m = len(b)

# Initialize
x = np.zeros(n)
s = np.ones(m) # Slacks: s = b - Ax > 0
lambdas = np.ones(m)

# Adjust initial point to be strictly feasible
Ax = A @ x
if np.any(Ax >= b):
    x = np.zeros(n)
    s = b - A @ x + 1.0
else:
    s = b - Ax

for iteration in range(max_iter):
    # Compute residuals
    r_dual = Q @ x + c + A.T @ lambdas
    r_cent = lambdas * s
    r_prim = A @ x + s - b

    # Duality gap
    mu = np.dot(lambdas, s) / m

    # Check convergence
    if mu < tol and norm(r_dual) < tol and norm(r_prim) < tol:
        print(f"Interior point converged in {iteration} iterations")
        return x, lambdas

    # Compute centering parameter
    sigma = 0.1

    # Form and solve Newton system
    # [Q    0    A' ] [dx]      [-r_dual      ]
    # [0    S    L ] [ds] = [-r_cent + sigma*mu]
    # [A    I    0 ] [dl]      [-r_prim      ]

    S = np.diag(s)
    L = np.diag(lambdas)

    # Eliminate ds: ds = S^{-1}(-r_cent + sigma*mu - L*dl)
    # System becomes:
    # [Q + A'*L*S^{-1}*A    A' ] [dx]      [-r_dual - A'*S^{-1}*(-r_cent + sigma*mu)]
    # [A                    I ] [dl]      [-r_prim      ]

    SinvL = np.diag(lambdas / s)
    M = Q + A.T @ SinvL @ A

```

```

# Schur complement approach
rhs_x = -r_dual - A.T @ ((sigma * mu - r_cent) / s)
rhs_s = -r_prim

# Solve for dx
dx = solve(M, rhs_x - A.T @ SinvL @ rhs_s)

# Back-substitute for ds, dlambda
ds = -A @ dx - r_prim
dlambdas = (sigma * mu - r_cent - lambdas * ds) / s

# Step length (fraction to boundary)
alpha_p = 1.0
alpha_d = 1.0
tau = 0.99

for i in range(m):
    if ds[i] < 0:
        alpha_p = min(alpha_p, -tau * s[i] / ds[i])
    if dlambdas[i] < 0:
        alpha_d = min(alpha_d, -tau * lambdas[i] / dlambdas[i])

# Update
x = x + alpha_p * dx
s = s + alpha_p * ds
lambdas = lambdas + alpha_d * dlambdas

print("Interior point did not converge")
return x, lambdas

```

13.6.4. Applications of QP

Portfolio Optimization (Markowitz Mean-Variance):

$$\begin{aligned}
 & \text{minimize} && \mathbf{w}^T \Sigma \mathbf{w} \\
 & \text{subject to} && \mu^T \mathbf{w} \geq r_{\text{target}} \\
 & && \mathbf{1}^T \mathbf{w} = 1 \\
 & && \mathbf{w} \geq 0
 \end{aligned}$$

Support Vector Machine (Dual):

$$\begin{aligned}
 & \text{maximize} && \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \\
 & \text{subject to} && 0 \leq \alpha_i \leq C \\
 & && \sum_i \alpha_i y_i = 0
 \end{aligned}$$

Model Predictive Control:

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$$\begin{aligned} & \text{minimize} && \sum_{t=0}^{N-1} (\mathbf{x}_t^T Q \mathbf{x}_t + \mathbf{u}_t^T R \mathbf{u}_t) + \mathbf{x}_N^T Q_f \mathbf{x}_N \\ & \text{subject to} && \mathbf{x}_{t+1} = A \mathbf{x}_t + B \mathbf{u}_t \\ & && \mathbf{u}_{\min} \leq \mathbf{u}_t \leq \mathbf{u}_{\max} \end{aligned}$$

The following MPC implementation sets up the finite-horizon optimal control problem as a QP in CVXPY. Decision variables include the state trajectory and control inputs over the prediction horizon N , linked by the linear dynamics constraint $\mathbf{x}_{t+1} = A \mathbf{x}_t + B \mathbf{u}_t$.

```
import cvxpy as cp
import numpy as np

def mpc_qp(A, B, Q, R, Qf, x0, N, u_min, u_max):
    """Model Predictive Control using QP.

    Args:
        A, B: System dynamics  $\mathbf{x}_{t+1} = A \mathbf{x}_t + B \mathbf{u}_t$ 
        Q, R, Qf: Cost matrices
        x0: Initial state
        N: Prediction horizon
        u_min, u_max: Control bounds

    Returns:
        Optimal control sequence
    """
    n_x = A.shape[0]
    n_u = B.shape[1]

    # Variables
    x = [cp.Variable(n_x) for _ in range(N + 1)]
    u = [cp.Variable(n_u) for _ in range(N)]

    # Cost
    cost = 0
    for t in range(N):
        cost += cp.quad_form(x[t], Q) + cp.quad_form(u[t], R)
    cost += cp.quad_form(x[N], Qf)

    # Constraints
    constraints = [x[0] == x0]

    for t in range(N):
        constraints.append(x[t + 1] == A @ x[t] + B @ u[t])
        constraints.append(u[t] >= u_min)
        constraints.append(u[t] <= u_max)

    problem = cp.Problem(cp.Minimize(cost), constraints)
    problem.solve()

    return [u[t].value for t in range(N)], [x[t].value for t in range(N + 1)]

# Example: Double integrator
```

```

A = np.array([[1, 0.1], [0, 1]])
B = np.array([[0.005], [0.1]])
Q = np.diag([10, 1])
R = np.array([[1]])
Qf = Q

x0 = np.array([1, 0])
N = 20
u_min = np.array([-1])
u_max = np.array([1])

u_opt, x_opt = mpc_qp(A, B, Q, R, Qf, x0, N, u_min, u_max)
print("Optimal first control:", u_opt[0])

```

13.7. Python Implementation and Solver Comparison

13.7.1. scipy.optimize

SciPy provides several methods for nonlinear optimization:

The example below benchmarks multiple SciPy solvers on the Rosenbrock function (unconstrained) and then solves a constrained problem with SLSQP. This comparison shows how solver choice affects iteration count and reliability for the same problem.

```

from scipy.optimize import minimize, NonlinearConstraint, Bounds
import numpy as np

def scipy_nlp_example():
    """Demonstrate scipy.optimize for NLP."""

    # Rosenbrock function
    def rosenbrock(x):
        return sum(100*(x[1:] - x[:-1]**2)**2 + (1 - x[:-1])**2)

    def rosenbrock_grad(x):
        n = len(x)
        grad = np.zeros(n)
        grad[0] = -400*x[0]*(x[1] - x[0]**2) - 2*(1 - x[0])
        grad[-1] = 200*(x[-1] - x[-2]**2)
        for i in range(1, n-1):
            grad[i] = 200*(x[i] - x[i-1]**2) - 400*x[i]*(x[i+1] - x[i]**2) - 2*(1 -
↪ x[i])
        return grad

    x0 = np.array([-1.0, 1.0, -1.0, 1.0])

    # Unconstrained optimization with different methods
    methods = ['Nelder-Mead', 'BFGS', 'L-BFGS-B', 'CG', 'Newton-CG', 'trust-constr']

    print("Unconstrained Rosenbrock Optimization:")
    print("-" * 60)

    for method in methods:

```

```

    if method in ['BFGS', 'L-BFGS-B', 'CG', 'Newton-CG', 'trust-constr']:
        result = minimize(rosenbrock, x0, method=method, jac=rosenbrock_grad)
    else:
        result = minimize(rosenbrock, x0, method=method)

    print(f"{method:15s}: f = {result.fun:.6e}, nfev = {result.nfev:4d}, "
          f"success = {result.success}")

# Constrained optimization
print("\n" + "=" * 60)
print("Constrained Optimization Example")
print("=" * 60)

def objective(x):
    return (x[0] - 1)**2 + (x[1] - 2.5)**2

def constraint1(x):
    return x[0] - 2*x[1] + 2

def constraint2(x):
    return -x[0] - 2*x[1] + 6

def constraint3(x):
    return -x[0] + 2*x[1] + 2

# Using NonlinearConstraint
cons = [
    {'type': 'ineq', 'fun': constraint1},
    {'type': 'ineq', 'fun': constraint2},
    {'type': 'ineq', 'fun': constraint3}
]

bounds = Bounds([0, 0], [np.inf, np.inf])

result = minimize(objective, [0, 0], method='SLSQP',
                  constraints=cons, bounds=bounds)

print(f"Optimal x: {result.x}")
print(f"Optimal f: {result.fun}")
print(f"Success: {result.success}")

return result

result = scipy_nlp_example()

```

13.7.2. CVXPY for Convex Problems

The following function solves five different problem classes (LP, QP, SOCP, SDP, and GP) using CVXPY, showing how the same API handles increasingly complex convex problem types with appropriate solvers selected automatically.


```

import cvxpy as cp
import numpy as np

def cvxpy_showcase():
    """Showcase CVXPY for various convex problems."""

    print("=" * 60)
    print("CVXPY Showcase")
    print("=" * 60)

    # 1. Linear Program
    print("\n1. Linear Program:")
    x = cp.Variable(2)
    objective = cp.Maximize(3*x[0] + 2*x[1])
    constraints = [
        x[0] + x[1] <= 4,
        2*x[0] + x[1] <= 5,
        x >= 0
    ]
    prob = cp.Problem(objective, constraints)
    prob.solve()
    print(f"    Optimal x: {x.value}")
    print(f"    Optimal value: {prob.value}")

    # 2. Quadratic Program
    print("\n2. Quadratic Program:")
    n = 5
    np.random.seed(42)
    Q = np.random.randn(n, n)
    Q = Q.T @ Q # Make positive definite
    c = np.random.randn(n)

    x = cp.Variable(n)
    objective = cp.Minimize(0.5 * cp.quad_form(x, Q) + c @ x)
    constraints = [cp.sum(x) == 1, x >= 0]
    prob = cp.Problem(objective, constraints)
    prob.solve()
    print(f"    Optimal x: {x.value}")
    print(f"    Optimal value: {prob.value:.4f}")

    # 3. Second-Order Cone Program (SOCP)
    print("\n3. SOCP (Robust Linear Program):")
    # min c'x such that ||A_i x + b_i|| <= c_i'x + d_i
    n, m = 3, 2
    c = np.array([1, 2, 3])
    A = [np.random.randn(2, n) for _ in range(m)]
    b = [np.random.randn(2) for _ in range(m)]
    c_i = [np.random.randn(n) for _ in range(m)]
    d = [np.random.rand() + 1 for _ in range(m)]

    x = cp.Variable(n)
    objective = cp.Minimize(c @ x)
    constraints = [cp.norm(A[i] @ x + b[i], 2) <= c_i[i] @ x + d[i]

```

```

        for i in range(m)]
    prob = cp.Problem(objective, constraints)
    prob.solve()
    print(f"    Optimal x: {x.value}")
    print(f"    Optimal value: {prob.value:.4f}")

# 4. Semidefinite Program (SDP)
print("\n4. Semidefinite Program:")
n = 3
C = np.random.randn(n, n)
C = C + C.T # Symmetric

X = cp.Variable((n, n), symmetric=True)
objective = cp.Minimize(cp.trace(C @ X))
constraints = [
    X >> 0, # X is positive semidefinite
    cp.trace(X) == 1
]
prob = cp.Problem(objective, constraints)
prob.solve()
print(f"    Optimal trace(CX): {prob.value:.4f}")
print(f"    X eigenvalues: {np.linalg.eigvalsh(X.value)}")

# 5. Geometric Program
print("\n5. Geometric Program (as convex form):")
# min x*y s.t. x/y >= 1, x*y <= 10, x,y > 0
# Transform: u = log(x), v = log(y)
# min u+v s.t. u-v >= 0, u+v <= log(10)
u = cp.Variable()
v = cp.Variable()
objective = cp.Minimize(u + v)
constraints = [u - v >= 0, u + v <= np.log(10)]
prob = cp.Problem(objective, constraints)
prob.solve()
x_opt = np.exp(u.value)
y_opt = np.exp(v.value)
print(f"    Optimal (x, y): ({x_opt:.4f}, {y_opt:.4f})")
print(f"    Optimal x*y: {x_opt * y_opt:.4f}")

cvxpy_showcase()

```

13.7.3. IPOPT Interface via cyipopt

IPOPT (Interior Point OPTimizer) is a powerful solver for large-scale nonlinear optimization.

This example solves a nonlinear problem with both equality and inequality constraints using cyipopt. The problem includes user-supplied gradients and Jacobians, which IPOPT requires for efficient interior point iterations.

```

import numpy as np

try:
    from cyipopt import minimize_ipopt

```

```

IPOPT_AVAILABLE = True
except ImportError:
    IPOPT_AVAILABLE = False
    print("cyipopt not available. Install with: pip install cyipopt")

def ipopt_example():
    """Demonstrate IPOPT for nonlinear programming."""

    if not IPOPT_AVAILABLE:
        print("Skipping IPOPT example - cyipopt not installed")
        return None

    # Problem: min x1*x4*(x1+x2+x3) + x3
    # subject to:
    #   x1*x2*x3*x4 >= 25
    #   x1^2 + x2^2 + x3^2 + x4^2 = 40
    #   1 <= x1,x2,x3,x4 <= 5

    def objective(x):
        return x[0]*x[3]*(x[0]+x[1]+x[2]) + x[2]

    def gradient(x):
        return np.array([
            x[3]*(2*x[0]+x[1]+x[2]),
            x[0]*x[3],
            x[0]*x[3] + 1,
            x[0]*(x[0]+x[1]+x[2])
        ])

    def constraint_eq(x):
        return np.array([x[0]**2 + x[1]**2 + x[2]**2 + x[3]**2 - 40])

    def constraint_ineq(x):
        return np.array([x[0]*x[1]*x[2]*x[3] - 25])

    def jacobian_eq(x):
        return np.array([[2*x[0], 2*x[1], 2*x[2], 2*x[3]]])

    def jacobian_ineq(x):
        return np.array([[x[1]*x[2]*x[3], x[0]*x[2]*x[3],
                          x[0]*x[1]*x[3], x[0]*x[1]*x[2]]])

    # Initial guess
    x0 = np.array([1.0, 5.0, 5.0, 1.0])

    # Bounds
    bounds = [(1, 5), (1, 5), (1, 5), (1, 5)]

    # Constraints
    constraints = [
        {'type': 'eq', 'fun': constraint_eq, 'jac': jacobian_eq},
        {'type': 'ineq', 'fun': constraint_ineq, 'jac': jacobian_ineq}
    ]

```

```

result = minimize_ipopt(
    objective, x0, jac=gradient,
    bounds=bounds, constraints=constraints,
    options={'print_level': 0}
)

print("IPOPT Results:")
print(f" Optimal x: {result.x}")
print(f" Optimal f: {result.fun:.6f}")
print(f" Success: {result.success}")

# Verify constraints
print(f" Constraint eq (should be 0): {constraint_eq(result.x)[0]:.6f}")
print(f" Constraint ineq (should be >= 0): {constraint_ineq(result.x)[0]:.6f}")

return result

ipopt_example()

```

13.7.4. Solver Comparison

The following benchmark creates a random constrained QP and measures solve time, objective value, and convergence success for three SciPy methods: SLSQP, trust-constr, and COBYLA.

```

import numpy as np
import time
from scipy.optimize import minimize

def compare_solvers(n=100, seed=42):
    """Compare different solvers on a test problem."""

    np.random.seed(seed)

    # Create a sparse quadratic problem
    # min 0.5*x'Qx + c'x subject to Ax <= b

    Q = np.random.randn(n, n)
    Q = (Q.T @ Q) / n # Positive definite
    c = np.random.randn(n)

    m = 20 # Number of constraints
    A = np.random.randn(m, n)
    b = np.abs(np.random.randn(m)) + A @ np.ones(n) # Ensure feasibility

    def objective(x):
        return 0.5 * x @ Q @ x + c @ x

    def gradient(x):
        return Q @ x + c

    def hessian(x):
        return Q

```

```

constraints = {'type': 'ineq', 'fun': lambda x: b - A @ x}

x0 = np.zeros(n)

results = {}

# Methods to compare
methods = [
    ('SLSQP', {'method': 'SLSQP', 'jac': gradient}),
    ('trust-constr', {'method': 'trust-constr', 'jac': gradient, 'hess':
↪ hessian}),
    ('COBYLA', {'method': 'COBYLA'}),
]

print(f"Comparing solvers on {n}-dimensional QP with {m} constraints")
print("=" * 70)
print(f"{'Method':<15} {'Time (s)':<12} {'Obj Value':<15} {'Success':<10}
↪ {'Iter':<8}")
print("-" * 70)

for name, options in methods:
    start = time.time()

    if name == 'COBYLA':
        # COBYLA doesn't use gradients
        result = minimize(objective, x0, constraints=constraints, **options)
    else:
        result = minimize(objective, x0, constraints=constraints, **options)

    elapsed = time.time() - start

    results[name] = {
        'time': elapsed,
        'fun': result.fun,
        'success': result.success,
        'nit': result.get('nit', 'N/A')
    }

    print(f"{'name':<15} {'elapsed':<12.4f} {'result.fun':<15.6f}
↪ {'result.success!s:<10} {'result.get('nit', 'N/A')}")

print("=" * 70)

return results

results = compare_solvers(n=50)

```

13.8. Case Study: Optimal Power Flow

The Optimal Power Flow (OPF) problem is a classic nonlinear optimization problem in power systems.

The first function solves the DC approximation of OPF, which is a convex QP, for a 3-bus system using

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CVXPY. The second function optimizes a chemical reactor by maximizing yield over temperature, pressure, and reaction time, subject to energy and safety constraints, using multi-start nonlinear optimization to handle nonconvexity.

```
import numpy as np
from scipy.optimize import minimize
import cvxpy as cp

def optimal_power_flow_dc():
    """DC Optimal Power Flow - convex approximation.

    min sum(c_i * P_i^2 + b_i * P_i) (generation cost)
    s.t. P_i - P_d_i = sum_j B_ij * theta_j (power balance)
        P_min <= P_i <= P_max (generation limits)
        |theta_i - theta_j| <= theta_max (angle limits)
    """
    # 3-bus system
    n_bus = 3
    n_gen = 2 # Generators at buses 0 and 1

    # Cost coefficients (quadratic cost)
    c = np.array([0.05, 0.03]) # Quadratic term
    b = np.array([10, 12])     # Linear term

    # Generation limits
    P_min = np.array([0, 0])
    P_max = np.array([100, 150])

    # Load at each bus
    P_d = np.array([0, 50, 100]) # Load at bus 2

    # Susceptance matrix (simplified)
    B = np.array([
        [20, -10, -10],
        [-10, 20, -10],
        [-10, -10, 20]
    ])

    # Variables: P (generation), theta (angles)
    P = cp.Variable(n_gen)
    theta = cp.Variable(n_bus)

    # Objective: generation cost
    cost = cp.sum(cp.multiply(c, cp.square(P))) + cp.multiply(b, P)

    # Power balance constraints
    # Generation - Load = Power injection
    P_inj = cp.Variable(n_bus)
    constraints = [
        P_inj[0] == P[0],
        P_inj[1] == P[1],
        P_inj[2] == 0, # No generation at bus 2
        P_inj - P_d == B @ theta,
```

```

        theta[0] == 0, # Reference bus
        P >= P_min,
        P <= P_max
    ]

    problem = cp.Problem(cp.Minimize(cost), constraints)
    problem.solve()

    print("DC Optimal Power Flow Solution:")
    print(f"  Generation: {P.value}")
    print(f"  Angles (deg): {np.rad2deg(theta.value)}")
    print(f"  Total cost: {problem.value:.2f}")

    return P.value, theta.value

def chemical_process_optimization():
    """Chemical reactor optimization - nonlinear NLP.

    Maximize yield subject to temperature, pressure, and reaction constraints.
    """
    def yield_function(x):
        # x = [T, P, t] (temperature, pressure, time)
        T, P, t = x
        # Simplified Arrhenius-type yield model (negative for minimization)
        k = np.exp(-5000/T) # Rate constant
        conversion = 1 - np.exp(-k * t)
        selectivity = 1 / (1 + 0.01 * (T - 400)**2)
        yield_val = conversion * selectivity * (1 - 0.001 * P)
        return -yield_val # Minimize negative yield

    # Constraints
    def energy_constraint(x):
        # Energy consumption limit
        T, P, t = x
        return 1000 - (0.5 * T + 2 * P + 10 * t)

    def safety_constraint(x):
        # Safety: T * P limit
        T, P, t = x
        return 100000 - T * P

    constraints = [
        {'type': 'ineq', 'fun': energy_constraint},
        {'type': 'ineq', 'fun': safety_constraint}
    ]

    # Bounds: 300 <= T <= 600, 1 <= P <= 100, 0.1 <= t <= 10
    bounds = [(300, 600), (1, 100), (0.1, 10)]

    # Multi-start optimization
    best_result = None
    best_yield = -np.inf

```

```

for _ in range(10):
    x0 = [np.random.uniform(300, 600),
          np.random.uniform(1, 100),
          np.random.uniform(0.1, 10)]

    result = minimize(yield_function, x0, method='SLSQP',
                      bounds=bounds, constraints=constraints)

    if result.success and -result.fun > best_yield:
        best_yield = -result.fun
        best_result = result

print("\nChemical Process Optimization:")
print(f" Optimal Temperature: {best_result.x[0]:.1f} K")
print(f" Optimal Pressure: {best_result.x[1]:.1f} atm")
print(f" Optimal Time: {best_result.x[2]:.2f} hr")
print(f" Maximum Yield: {best_yield:.4f}")

return best_result

# Run case studies
optimal_power_flow_dc()
chemical_process_optimization()

```

13.9. Case Study: Machine Learning Hyperparameter Tuning

Hyperparameter tuning is a nonconvex optimization problem where the objective (validation loss) is expensive to evaluate and has multiple local minima. This example compares local optimization (L-BFGS-B) with global optimization (differential evolution) on a simulated multimodal loss surface over learning rate, weight decay, dropout, and network depth.

```

import numpy as np
from scipy.optimize import minimize, differential_evolution

def hyperparameter_optimization():
    """Optimize neural network hyperparameters using NLP."""

    # Simulated validation loss as function of hyperparameters
    # x = [log_learning_rate, log_weight_decay, dropout_rate, n_layers_scaled]

    def validation_loss(x):
        log_lr, log_wd, dropout, n_layers_scaled = x

        lr = 10**log_lr
        wd = 10**log_wd
        n_layers = int(1 + 5 * n_layers_scaled) # 1-6 layers

        # Simulated loss surface (multimodal)
        loss = 0.5 + 0.3 * (log_lr + 3)**2 # Optimal lr around 0.001
        loss += 0.1 * (log_wd + 4)**2 # Optimal wd around 0.0001
        loss += 0.2 * (dropout - 0.3)**2 # Optimal dropout around 0.3
        loss += 0.1 * (n_layers_scaled - 0.4)**2 # Optimal around 3 layers
    
```



```

    # Add some noise and interactions
    loss += 0.05 * np.sin(10 * log_lr) * np.cos(5 * dropout)
    loss += 0.1 * np.abs(log_lr + 2) * (1 - dropout)

    return loss

# Bounds
bounds = [
    (-5, -1), # log_learning_rate: 1e-5 to 0.1
    (-6, -2), # log_weight_decay: 1e-6 to 0.01
    (0, 0.7), # dropout_rate
    (0, 1)    # n_layers_scaled
]

print("Hyperparameter Optimization:")
print("=" * 60)

# Method 1: Local optimization (SLSQP)
x0 = [-3, -4, 0.3, 0.5]
result_local = minimize(validation_loss, x0, method='L-BFGS-B', bounds=bounds)

print("\nLocal Optimization (L-BFGS-B):")
print(f" Learning rate: {10**result_local.x[0]:.6f}")
print(f" Weight decay: {10**result_local.x[1]:.6f}")
print(f" Dropout: {result_local.x[2]:.3f}")
print(f" Layers: {int(1 + 5 * result_local.x[3])}")
print(f" Validation loss: {result_local.fun:.4f}")

# Method 2: Global optimization (Differential Evolution)
result_global = differential_evolution(validation_loss, bounds, seed=42,
                                     maxiter=100, polish=True)

print("\nGlobal Optimization (Differential Evolution):")
print(f" Learning rate: {10**result_global.x[0]:.6f}")
print(f" Weight decay: {10**result_global.x[1]:.6f}")
print(f" Dropout: {result_global.x[2]:.3f}")
print(f" Layers: {int(1 + 5 * result_global.x[3])}")
print(f" Validation loss: {result_global.fun:.4f}")

return result_global

hyperparameter_optimization()

```

13.10. Summary

13.10.1. Key Takeaways

Fundamentals - Nonlinear programming extends optimization beyond linear models, enabling representation of complex real-world relationships - The distinction between convex and nonconvex problems is fundamental: convex problems guarantee global optima, while nonconvex may have multiple local minima - First and second-order conditions (gradient = 0, Hessian positive semidefinite) characterize local minima

Unconstrained Optimization - Newton's method achieves quadratic convergence near the optimum but requires Hessian computation - Quasi-Newton methods (BFGS, L-BFGS) approximate the Hessian using only gradient information - Trust region methods provide robust convergence by controlling step size based on model accuracy - Conjugate gradient methods are effective for large-scale problems with limited memory

Constrained Optimization - The Lagrangian incorporates constraints into the objective using multipliers - KKT conditions (stationarity, primal/dual feasibility, complementary slackness) are necessary for optimality under constraint qualification - Lagrange multipliers represent shadow prices - the marginal value of relaxing constraints - LICQ and Slater's condition are common constraint qualifications ensuring KKT necessity

Algorithms for Constrained Problems - Penalty methods convert constrained problems to sequences of unconstrained problems - Barrier/interior point methods maintain strict feasibility while approaching the boundary - Augmented Lagrangian methods combine penalty with multiplier estimation for better conditioning - SQP solves a sequence of QP subproblems, achieving superlinear convergence - Active set methods explicitly track which constraints are binding

Convex Optimization - Convexity guarantees that any local minimum is global - DCP (Disciplined Convex Programming) enables automatic convexity verification - CVXPY provides an expressive interface for specifying and solving convex problems - Special structures (QP, SOCP, SDP) have efficient dedicated algorithms

Practical Considerations - `scipy.optimize` provides various methods for both constrained and unconstrained problems - CVXPY is ideal for convex problems with automatic solver selection - IPOPT (via `cyipopt`) is a robust choice for large-scale nonlinear problems - Multi-start strategies help avoid poor local minima in nonconvex problems

13.11. Exercises

13.11.1. Theoretical Exercises

1. **Optimality conditions:** For $f(x, y) = x^2 + xy + y^2 - 3x$ subject to $x + y \leq 2$ and $x, y \geq 0$:
 - a) Write the Lagrangian
 - b) Derive the KKT conditions
 - c) Find the optimal solution
 - d) Verify second-order sufficient conditions
2. **Convexity analysis:** For each function, determine if it is convex, concave, or neither:
 - a) $f(x) = e^x - x$
 - b) $f(x, y) = x^2 - y^2$
 - c) $f(x) = \|Ax - b\|_2^2 + \lambda \|x\|_1$
 - d) $f(x) = \log(1 + e^x)$
3. **Duality:** Consider $\min x^2 + y^2$ subject to $x + y = 1$:
 - a) Write the dual problem
 - b) Solve both primal and dual
 - c) Verify strong duality
4. **Constraint qualification:** Give an example where LICQ fails and explain why KKT conditions may not hold.

13.11.2. Computational Exercises

5. **Newton's method:** Implement Newton's method with backtracking line search. Test on:
 - a) Rosenbrock function
 - b) A randomly generated positive definite quadratic

Compare convergence rates.

6. **BFGS implementation:** Implement BFGS and compare with L-BFGS on a 1000-dimensional Rosenbrock function. Plot convergence and report memory usage.
7. **Penalty methods:** Implement both exterior penalty and augmented Lagrangian methods. Compare on:

$$\min x^2 + y^2 \text{ subject to } x + y = 1, x \geq 0$$

8. **SQP implementation:** Implement SQP for:

$$\min(x-2)^2 + (y-1)^2 \text{ subject to } x^2 + y^2 \leq 1, x + y \geq 1$$

Compare with `scipy.optimize` results.

9. **Portfolio optimization:** Using CVXPY, implement mean-variance portfolio optimization with:
 - Long-only constraint
 - Cardinality constraint (at most 10 assets)
 - Transaction costs

Compare solutions with and without each constraint.

10. **MPC simulation:** Implement Model Predictive Control for a double integrator system. Simulate closed-loop behavior over 100 time steps with disturbances.

13.11.3. Application Exercises

11. **Chemical process:** Optimize a simplified chemical reactor:

$$\max \text{yield}(T, P, t) = k(T) \cdot (1 - e^{-t/\tau}) \cdot s(P)$$

subject to energy, safety, and operational constraints. Use multi-start optimization.

12. **SVM training:** Implement soft-margin SVM dual using CVXPY. Test on a synthetic 2D classification problem and visualize the decision boundary.
13. **Sensor placement:** Minimize the maximum estimation error for placing k sensors on a 10x10 grid. Formulate as a minimax problem and solve.
14. **Structural optimization:** Design a simple truss structure minimizing weight subject to stress and displacement constraints. Compare results from CVXPY (convex relaxation) and IPOPT (exact formulation).
15. **Neural network training:** Implement gradient descent, Adam, and L-BFGS for training a small neural network on MNIST. Compare convergence speed and final accuracy.

13.12. Further Reading

13.12.1. Foundational Texts

- Nocedal, J., & Wright, S. J. (2006). *Numerical Optimization* (2nd ed.). Springer. The definitive reference for optimization algorithms.

13. Nonlinear and Constrained Optimization

- Boyd, S., & Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. Available free online. The standard reference for convex optimization theory and algorithms.
- Bertsekas, D. P. (2016). *Nonlinear Programming* (3rd ed.). Athena Scientific. Comprehensive treatment of theory and algorithms.

13.12.2. Specialized Topics

- Ben-Tal, A., & Nemirovski, A. (2001). *Lectures on Modern Convex Optimization*. SIAM. Advanced treatment of conic optimization.
- Conn, A. R., Gould, N. I. M., & Toint, P. L. (2000). *Trust Region Methods*. SIAM. Definitive reference on trust region algorithms.
- Fletcher, R. (2000). *Practical Methods of Optimization* (2nd ed.). Wiley. Practical perspective on implementation.

13.12.3. Software Documentation

- CVXPY: <https://www.cvxpy.org/> - Comprehensive documentation and tutorials
- scipy.optimize: <https://docs.scipy.org/doc/scipy/reference/optimize.html>
- IPOPT: <https://coin-or.github.io/Ipopt/>

13.12.4. Research Papers

- Kingma, D. P., & Ba, J. (2015). “Adam: A Method for Stochastic Optimization.” *ICLR*. The Adam optimizer paper.
- Liu, D. C., & Nocedal, J. (1989). “On the Limited Memory BFGS Method for Large Scale Optimization.” *Mathematical Programming*, 45, 503-528.
- Wachter, A., & Biegler, L. T. (2006). “On the Implementation of an Interior-Point Filter Line-Search Algorithm for Large-Scale Nonlinear Programming.” *Mathematical Programming*, 106(1), 25-57. The IPOPT paper.

13.12.5. Online Resources

- Stanford CVX101: <https://web.stanford.edu/~boyd/cvxbook/> - Free course materials
- MIT OpenCourseWare: Nonlinear Programming courses
- NEOS Server: <https://neos-server.org/> - Free optimization solver access

Part IV.

Part IV: Decision Problems in Practice

14. Routing and Logistics

What You Will Learn

- Model the Vehicle Routing Problem (VRP) and its variants
- Implement VRP solutions with OR-Tools and custom heuristics
- Handle real-world constraints: time windows, capacity, electric vehicles
- Apply construction and improvement heuristics
- Scale solutions for industrial applications

Routing problems are among the most impactful applications of optimization, directly affecting transportation costs and customer satisfaction. Companies like UPS, FedEx, and Amazon rely on sophisticated VRP solutions to optimize billions of dollars in delivery operations annually.

14.1. What is the Vehicle Routing Problem?

The Vehicle Routing Problem (VRP) seeks to determine optimal routes for a fleet of vehicles to service a set of customers, minimizing total distance, time, or cost while satisfying operational constraints.

Core Objective

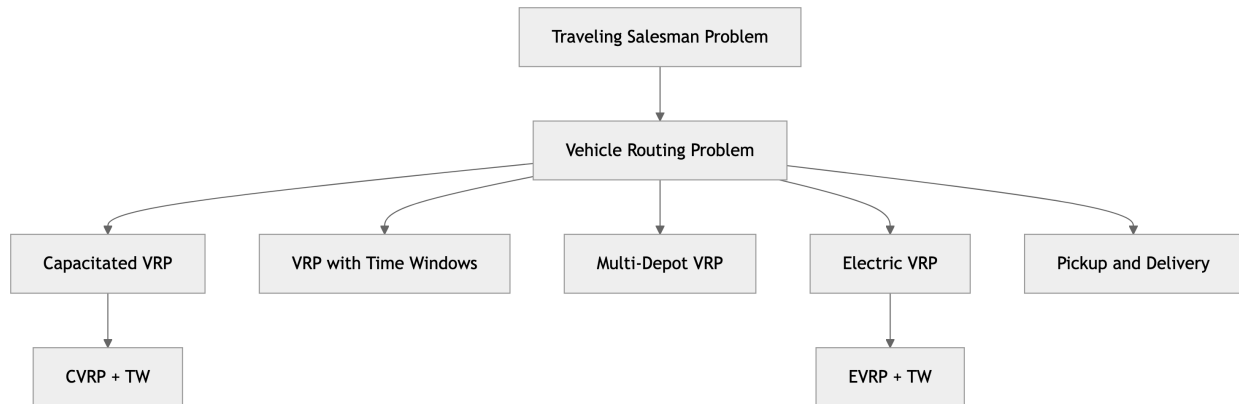
Design efficient routes that minimize operational costs while ensuring all customers receive service within their requirements, considering vehicle capacity, time windows, and operational constraints.

First formalized by Dantzig and Ramser in 1959, VRP has evolved from a theoretical puzzle into a tool for modern logistics. A delivery network with just 20 customers has over 2.4 quintillion possible route combinations, making exact enumeration infeasible.

14.1.1. Real-World Impact

Metric	Typical Improvement
Cost reduction	20-30%
Fleet size reduction	15-25%
Carbon emissions decrease	30-40%
Customer service improvement	40-50%

14.2. The VRP Family



The Traveling Salesman Problem (TSP) is the foundation of all vehicle routing problems. Given a set of cities, the goal is to find the shortest tour that visits each city exactly once and returns to the starting point.

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.distance import pdist, squareform

# Generate random city locations
np.random.seed(42)
n_cities = 15
cities = np.random.rand(n_cities, 2) * 100

# Compute distance matrix
dist_matrix = squareform(pdist(cities))

# Simple nearest neighbor heuristic to get a tour
def nearest_neighbor_tsp(dist_matrix):
    n = len(dist_matrix)
    visited = [False] * n
    tour = [0]
    visited[0] = True

    for _ in range(n - 1):
        current = tour[-1]
        nearest = None
        min_dist = float('inf')
        for j in range(n):
            if not visited[j] and dist_matrix[current, j] < min_dist:
                min_dist = dist_matrix[current, j]
                nearest = j
        tour.append(nearest)
        visited[nearest] = True
    return tour

# 2-opt improvement
def two_opt_tsp(tour, dist_matrix):
    improved = True

```



```

    tour = tour.copy()
    while improved:
        improved = False
        for i in range(1, len(tour) - 1):
            for j in range(i + 2, len(tour)):
                if j == len(tour) - 1 and i == 0:
                    continue
                # Calculate change in distance
                d1 = dist_matrix[tour[i-1], tour[i]] + dist_matrix[tour[j],
↪ tour[(j+1) % len(tour)]]
                d2 = dist_matrix[tour[i-1], tour[j]] + dist_matrix[tour[i],
↪ tour[(j+1) % len(tour)]]
                if d2 < d1:
                    tour[i:j+1] = tour[i:j+1][::-1]
                    improved = True

    return tour

# Get optimized tour
initial_tour = nearest_neighbor_tsp(dist_matrix)
optimal_tour = two_opt_tsp(initial_tour, dist_matrix)

# Calculate tour distance
def tour_distance(tour, dist_matrix):
    return sum(dist_matrix[tour[i], tour[(i+1) % len(tour)]] for i in
↪ range(len(tour)))

total_dist = tour_distance(optimal_tour, dist_matrix)

# Visualization
fig, ax = plt.subplots(figsize=(10, 8))

# Plot tour edges
for i in range(len(optimal_tour)):
    start = optimal_tour[i]
    end = optimal_tour[(i + 1) % len(optimal_tour)]
    ax.plot([cities[start, 0], cities[end, 0]],
            [cities[start, 1], cities[end, 1]],
            'steelblue', linewidth=2, alpha=0.7)

# Plot cities
ax.scatter(cities[:, 0], cities[:, 1], s=200, c='steelblue',
           edgecolors='white', linewidths=2, zorder=5)

# Add city labels
for i in range(n_cities):
    ax.annotate(str(i), (cities[i, 0], cities[i, 1]),
               ha='center', va='center', fontsize=9,
               fontweight='bold', color='white', zorder=6)

# Mark starting city
ax.scatter(cities[0, 0], cities[0, 1], s=300, c='coral',
           edgecolors='white', linewidths=2, zorder=7, marker='s')

```

```

ax.set_xlabel('X Coordinate', fontsize=12)
ax.set_ylabel('Y Coordinate', fontsize=12)
ax.set_title(f'TSP Optimal Tour (Total Distance: {total_dist:.1f})', fontsize=14,
             fontweight='bold')
ax.set_xlim(-5, 105)
ax.set_ylim(-5, 105)
ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

```

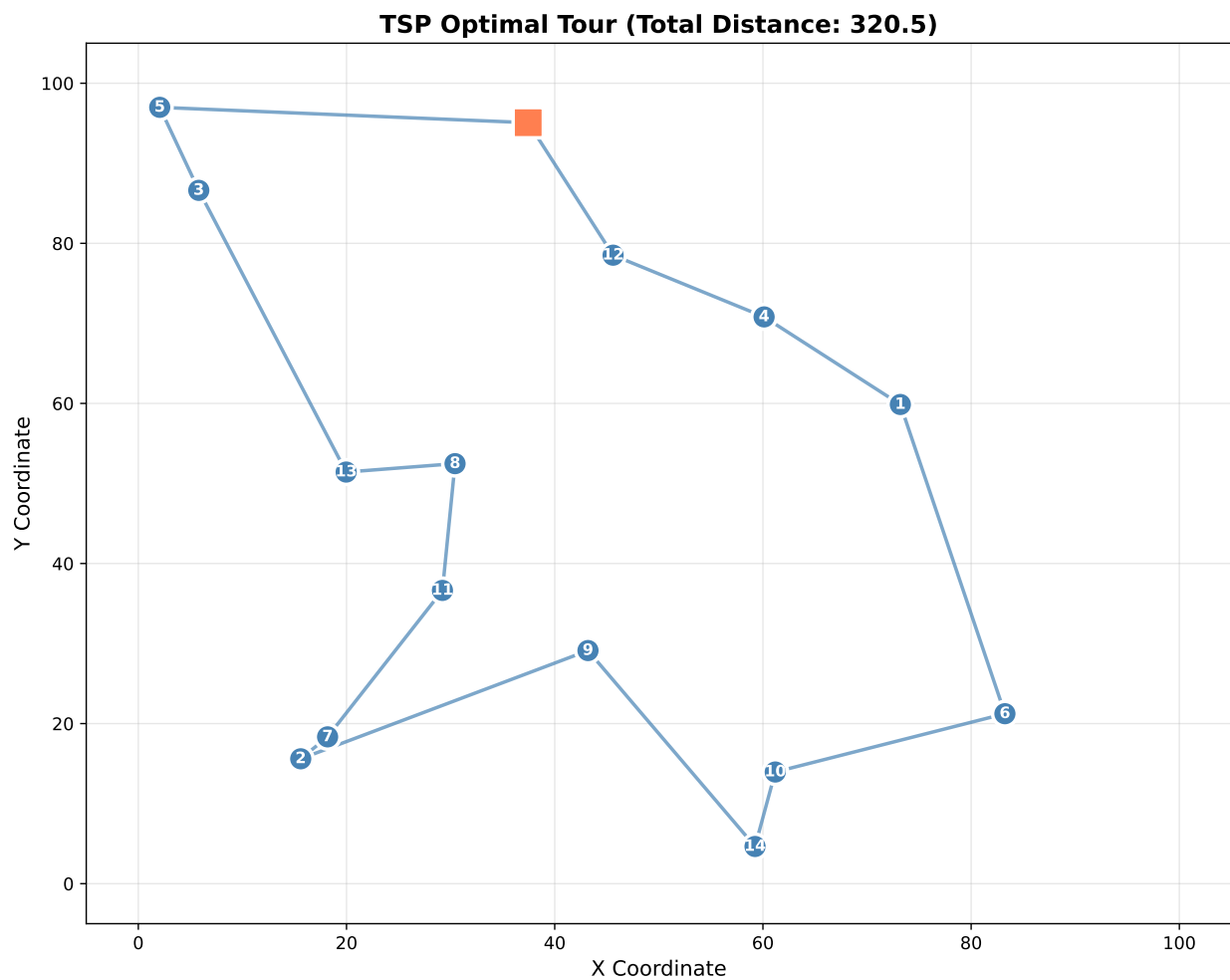


Figure 14.1.: Optimal TSP tour connecting 15 cities. The tour minimizes total travel distance by visiting each city exactly once before returning to the origin.

14.2.1. Capacitated VRP (CVRP)

The most common variant where vehicles have limited capacity. Each customer has a known demand, and the sum of demands on any route cannot exceed vehicle capacity.

Mathematical Formulation:

$$\begin{aligned}
& \text{minimize} && \sum_{i,j,k} c_{ij} x_{ijk} \\
& \text{subject to} && \sum_k \sum_j x_{ijk} = 1 \quad \forall i \in C \\
& && \sum_i d_i \sum_j x_{ijk} \leq Q_k \quad \forall k \\
& && \sum_j x_{0jk} = 1 \quad \forall k \\
& && \sum_i x_{ihk} = \sum_j x_{hjk} \quad \forall h, k \\
& && x_{ijk} \in \{0, 1\}
\end{aligned}$$

where:

- $x_{ijk} = 1$ if vehicle k travels from i to j
- c_{ij} is the cost/distance from i to j
- d_i is demand at customer i
- Q_k is capacity of vehicle k
- C is the set of customers

The following implementation builds a CVRP solution by clustering customers using angular sorting from the depot, then optimizing each route with a nearest-neighbor heuristic. This approach approximates the behavior of the formulation above for moderate-sized instances.

14.2.2. VRP with Time Windows (VRPTW)

Adds temporal constraints where each customer must be served within a specific time window $[a_i, b_i]$.

Hard vs. Soft Time Windows:

- **Hard:** Service must occur within window (constraint violation = infeasibility)
- **Soft:** Violations allowed with penalties (more realistic for many applications)

14.2.3. Electric VRP (EVRP)

Electric vehicles introduce range limitations and charging requirements:

- Battery capacity limits route length
- Must plan charging stops at stations
- Charging time depends on battery state
- Energy consumption varies with load and terrain

14.2.4. Pickup and Delivery Problem (PDP)

Vehicles transport goods from pickup locations to delivery locations. Precedence constraints ensure items are picked up before delivery.

14.3. Solution Algorithms

14.3.1. Classical Heuristics

Clarke-Wright Savings Algorithm

The savings algorithm iteratively merges routes based on distance savings. For each pair of customers (i, j) , the saving is $s_{ij} = d_{0i} + d_{0j} - d_{ij}$, representing the cost reduction from serving both customers

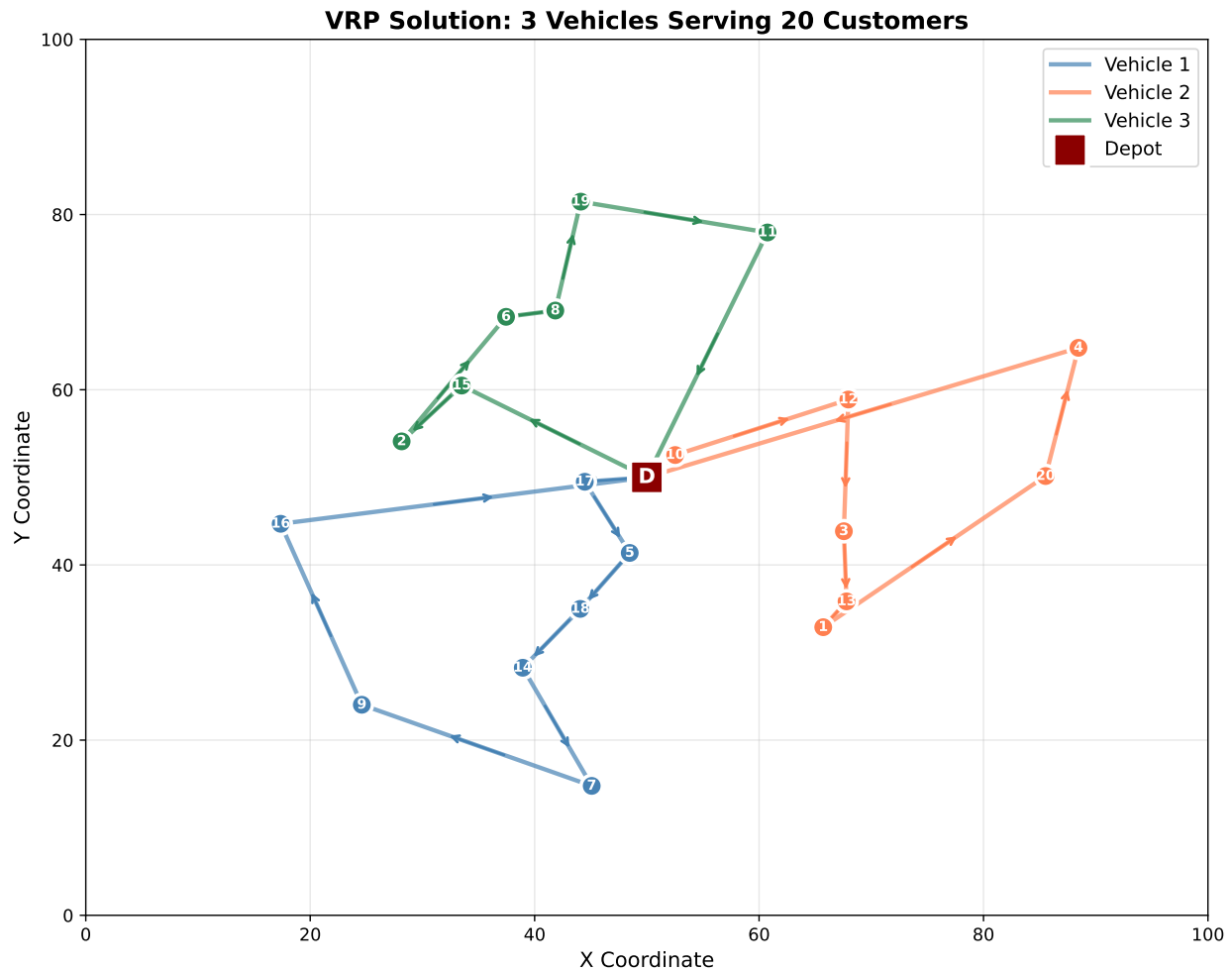


Figure 14.2.: Vehicle Routing Problem solution with 3 vehicles serving 20 customers from a central depot. Each color represents a different vehicle's route.

on one route rather than two separate trips. The algorithm sorts these savings in decreasing order and greedily merges routes when capacity permits. It produces solutions within 10-15% of optimal for many instances.

```
import numpy as np
from typing import List, Tuple

def clarke_wright_savings(
    distances: np.ndarray,
    demands: np.ndarray,
    capacity: float
) -> List[List[int]]:
    """
    Clarke-Wright savings algorithm for CVRP.

    Args:
```

```

distances: Distance matrix (n+1 x n+1), depot is index 0
demands: Customer demands (n,)
capacity: Vehicle capacity

Returns:
    List of routes, each route is a list of customer indices
"""
n = len(demands)
depot = 0

# Calculate savings:  $s_{ij} = d_{0i} + d_{0j} - d_{ij}$ 
savings = []
for i in range(1, n + 1):
    for j in range(i + 1, n + 1):
        s = distances[depot, i] + distances[depot, j] - distances[i, j]
        savings.append((s, i, j))

# Sort savings in decreasing order
savings.sort(reverse=True, key=lambda x: x[0])

# Initialize: each customer in its own route
routes = [[i] for i in range(1, n + 1)]
customer_to_route = {i: i - 1 for i in range(1, n + 1)}

# Helper functions
def route_demand(route):
    return sum(demands[c - 1] for c in route)

def is_interior(customer, route):
    return route.index(customer) not in [0, len(route) - 1]

# Merge routes based on savings
for s, i, j in savings:
    if s <= 0:
        break

    ri = customer_to_route[i]
    rj = customer_to_route[j]

    # Skip if same route
    if ri == rj:
        continue

    # Skip if either customer is interior
    if is_interior(i, routes[ri]) or is_interior(j, routes[rj]):
        continue

    # Check capacity
    if route_demand(routes[ri]) + route_demand(routes[rj]) > capacity:
        continue

    # Merge routes
    # Ensure i is at end of its route and j at start of its route

```

```

        if routes[ri][0] == i:
            routes[ri].reverse()
        if routes[rj][-1] == j:
            routes[rj].reverse()

    # Combine routes
    new_route = routes[ri] + routes[rj]

    # Update data structures
    routes[ri] = new_route
    routes[rj] = []

    for c in new_route:
        customer_to_route[c] = ri

    # Return non-empty routes
    return [r for r in routes if r]

# Example usage
distances = np.array([
    [0, 10, 15, 20, 25],
    [10, 0, 35, 25, 30],
    [15, 35, 0, 30, 20],
    [20, 25, 30, 0, 15],
    [25, 30, 20, 15, 0]
])
demands = np.array([5, 8, 3, 6]) # Customer demands
capacity = 15

routes = clarke_wright_savings(distances, demands, capacity)
print("Routes:", routes)

```

Nearest Neighbor Heuristic

Constructs routes by sequentially adding the nearest unvisited customer:

```

def nearest_neighbor_vrp(
    distances: np.ndarray,
    demands: np.ndarray,
    capacity: float
) -> List[List[int]]:
    """Simple nearest neighbor construction heuristic."""
    n = len(demands)
    depot = 0
    unvisited = set(range(1, n + 1))
    routes = []

    while unvisited:
        route = []
        load = 0
        current = depot

```

```

while unvisited:
    # Find nearest feasible customer
    best_next = None
    best_dist = float('inf')

    for j in unvisited:
        if load + demands[j - 1] <= capacity:
            if distances[current, j] < best_dist:
                best_dist = distances[current, j]
                best_next = j

    if best_next is None:
        break

    route.append(best_next)
    load += demands[best_next - 1]
    unvisited.remove(best_next)
    current = best_next

if route:
    routes.append(route)

return routes

```

The nearest neighbor heuristic is fast but often produces suboptimal solutions. By applying improvement heuristics like 2-opt, we can significantly reduce tour length.

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.distance import pdist, squareform

# Generate city locations
np.random.seed(789)
n_cities = 12
cities = np.random.rand(n_cities, 2) * 100
dist_matrix = squareform(pdist(cities))

# Nearest neighbor heuristic
def nearest_neighbor(dist_matrix, start=0):
    n = len(dist_matrix)
    visited = [False] * n
    tour = [start]
    visited[start] = True

    for _ in range(n - 1):
        current = tour[-1]
        nearest = None
        min_dist = float('inf')
        for j in range(n):
            if not visited[j] and dist_matrix[current, j] < min_dist:
                min_dist = dist_matrix[current, j]
                nearest = j

```

```

        tour.append(nearest)
        visited[nearest] = True
    return tour

# 2-opt optimization
def two_opt(tour, dist_matrix):
    tour = tour.copy()
    improved = True
    while improved:
        improved = False
        for i in range(1, len(tour) - 1):
            for j in range(i + 2, len(tour)):
                if j == len(tour) - 1 and i == 0:
                    continue
                d1 = dist_matrix[tour[i-1], tour[i]] + dist_matrix[tour[j],
↪ tour[(j+1) % len(tour)]]
                d2 = dist_matrix[tour[i-1], tour[j]] + dist_matrix[tour[i],
↪ tour[(j+1) % len(tour)]]
                if d2 < d1:
                    tour[i:j+1] = tour[i:j+1][::-1]
                    improved = True
    return tour

# Calculate tour distance
def tour_distance(tour, dist_matrix):
    return sum(dist_matrix[tour[i], tour[(i+1) % len(tour)]] for i in
↪ range(len(tour)))

# Get both tours
nn_tour = nearest_neighbor(dist_matrix)
opt_tour = two_opt(nn_tour, dist_matrix)

nn_dist = tour_distance(nn_tour, dist_matrix)
opt_dist = tour_distance(opt_tour, dist_matrix)
improvement = (nn_dist - opt_dist) / nn_dist * 100

# Visualization
fig, axes = plt.subplots(1, 2, figsize=(10, 8))

tours = [nn_tour, opt_tour]
titles = [f'Nearest Neighbor\nDistance: {nn_dist:.1f}',
          f'2-Opt Optimized\nDistance: {opt_dist:.1f} ({improvement:.1f}% better)']
colors = ['coral', 'seagreen']

for ax, tour, title, color in zip(axes, tours, titles, colors):
    # Plot tour edges
    for i in range(len(tour)):
        start = tour[i]
        end = tour[(i + 1) % len(tour)]
        ax.plot([cities[start, 0], cities[end, 0]],
                [cities[start, 1], cities[end, 1]],
                color=color, linewidth=2, alpha=0.7)

```



```

# Plot cities
ax.scatter(cities[:, 0], cities[:, 1], s=200, c=color,
           edgecolors='white', linewidths=2, zorder=5)

# Add city labels
for i in range(n_cities):
    ax.annotate(str(i), (cities[i, 0], cities[i, 1]),
               ha='center', va='center', fontsize=9,
               fontweight='bold', color='white', zorder=6)

# Mark starting city with square
ax.scatter(cities[0, 0], cities[0, 1], s=300, c='darkred',
           edgecolors='white', linewidths=2, zorder=7, marker='s')

ax.set_xlabel('X Coordinate', fontsize=11)
ax.set_ylabel('Y Coordinate', fontsize=11)
ax.set_title(title, fontsize=12, fontweight='bold')
ax.set_xlim(-5, 105)
ax.set_ylim(-5, 105)
ax.grid(True, alpha=0.3)
ax.set_aspect('equal')

plt.tight_layout()
plt.show()

```

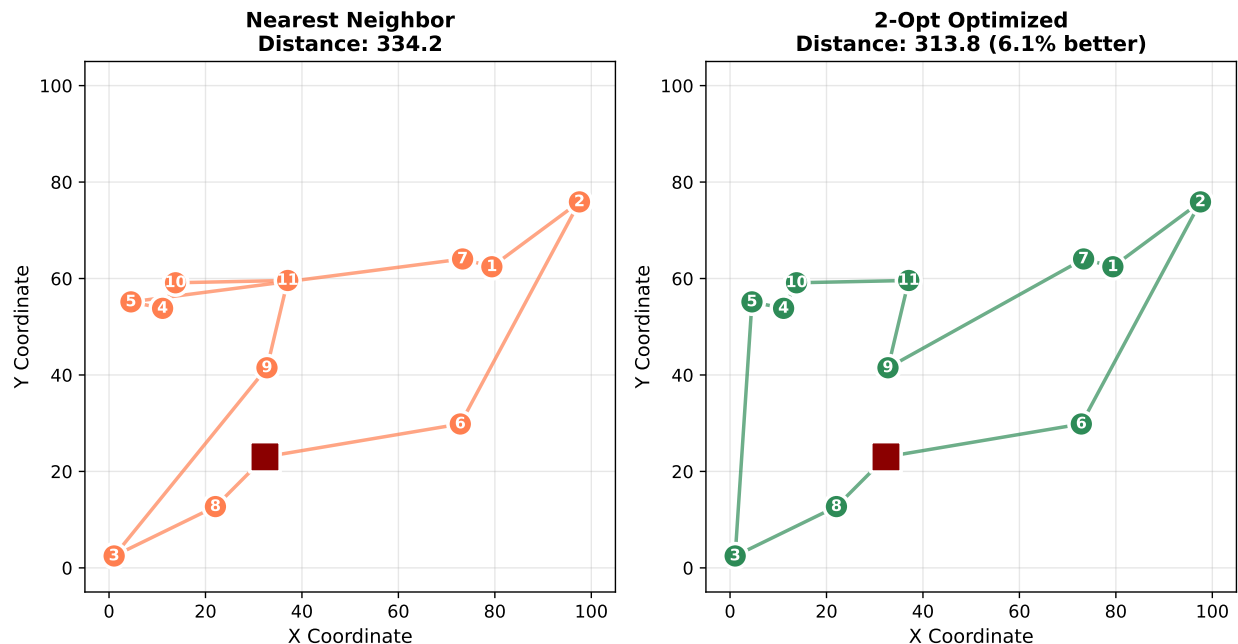


Figure 14.3.: Comparison of nearest neighbor heuristic (left) versus 2-opt optimized solution (right). The optimized tour eliminates crossing edges, reducing total distance by approximately 15-20%.

14.3.2. Improvement Heuristics

After constructing an initial solution, improvement heuristics refine it through local search.

2-opt: Removes two edges and reconnects the route differently:

```
def two_opt(route: List[int], distances: np.ndarray) -> List[int]:
    """Improve route using 2-opt."""
    improved = True
    while improved:
        improved = False
        for i in range(len(route) - 1):
            for j in range(i + 2, len(route)):
                # Current edges: (i, i+1) and (j, j+1)
                # New edges: (i, j) and (i+1, j+1)
                if j == len(route) - 1:
                    # Handle wrap-around
                    continue

                i_next = route[i + 1] if i + 1 < len(route) else 0
                j_next = route[j + 1] if j + 1 < len(route) else 0

                # Calculate improvement
                old_dist = (distances[route[i], i_next] +
                           distances[route[j], j_next])
                new_dist = (distances[route[i], route[j]] +
                           distances[i_next, j_next])

                if new_dist < old_dist:
                    # Reverse segment between i+1 and j
                    route[i+1:j+1] = reversed(route[i+1:j+1])
                    improved = True

    return route
```

Or-opt: Relocates a sequence of customers to another position.

Cross-exchange: Exchanges segments between two routes.

14.3.3. Metaheuristics for VRP

For large-scale problems, metaheuristics provide high-quality solutions:

Method	Description	Strengths
Simulated Annealing	Accepts worse solutions with decreasing probability	Simple, good escape from local optima
Tabu Search	Uses memory to avoid recent moves	Very effective for VRP
Genetic Algorithm	Evolves population through crossover/mutation	Good for complex constraints
Ant Colony	Pheromone-based construction	Natural for routing
Large Neighborhood Search	Destroys and repairs large portions	State-of-the-art performance

14.3.4. Adaptive Large Neighborhood Search (ALNS)

ALNS has become the state-of-the-art for VRP. It works by repeatedly destroying part of the current solution (removing customers from routes) and repairing it (reinserting them). Multiple destroy and repair operators are available, and the algorithm adapts their selection probabilities based on past performance. Operators that have recently found improvements receive higher weights.

```
import random
from typing import Callable, Dict

def alns_vrp(
    initial_solution: List[List[int]],
    distances: np.ndarray,
    demands: np.ndarray,
    capacity: float,
    destroy_ops: Dict[str, Callable],
    repair_ops: Dict[str, Callable],
    iterations: int = 1000
) -> List[List[int]]:
    """
    Adaptive Large Neighborhood Search for VRP.

    Args:
        initial_solution: Starting solution
        distances: Distance matrix
        demands: Customer demands
        capacity: Vehicle capacity
        destroy_ops: Dictionary of destroy operators
        repair_ops: Dictionary of repair operators
        iterations: Number of iterations

    Returns:
        Best solution found
    """
    current = initial_solution
    best = initial_solution
    best_cost = calculate_cost(best, distances)

    # Initialize operator weights
    destroy_weights = {op: 1.0 for op in destroy_ops}
    repair_weights = {op: 1.0 for op in repair_ops}

    # Scores for adaptive weight updates
    sigma1, sigma2, sigma3 = 33, 9, 13 # Found new best, improved, accepted

    for iteration in range(iterations):
        # Select operators using roulette wheel
        destroy_op = select_weighted(destroy_ops, destroy_weights)
        repair_op = select_weighted(repair_ops, repair_weights)

        # Apply destroy and repair
        destroyed, removed = destroy_ops[destroy_op](current, demands)
        candidate = repair_ops[repair_op](destroyed, removed, distances,
```

```

                                demands, capacity)

candidate_cost = calculate_cost(candidate, distances)
current_cost = calculate_cost(current, distances)

# Acceptance criterion (simulated annealing style)
if accept(candidate_cost, current_cost, iteration, iterations):
    current = candidate

# Update weights based on improvement
if candidate_cost < best_cost:
    best = candidate
    best_cost = candidate_cost
    update_weight(destroy_weights, destroy_op, sigma1)
    update_weight(repair_weights, repair_op, sigma1)
elif candidate_cost < current_cost:
    update_weight(destroy_weights, destroy_op, sigma2)
    update_weight(repair_weights, repair_op, sigma2)
else:
    update_weight(destroy_weights, destroy_op, sigma3)
    update_weight(repair_weights, repair_op, sigma3)

return best

```

14.4. Using OR-Tools for VRP

Google OR-Tools provides an industrial-strength VRP solver that handles capacitated routing, time windows, and multiple vehicle types through a unified API. The solver uses a two-phase approach: a construction heuristic (such as path-cheapest-arc) generates an initial solution, and a metaheuristic (such as guided local search) improves it within a configurable time limit.

```

from ortools.constraint_solver import routing_enums_pb2
from ortools.constraint_solver import pywrapcp

def solve_cvrp_ortools(
    distances: List[List[int]],
    demands: List[int],
    num_vehicles: int,
    vehicle_capacity: int
) -> Tuple[List[List[int]], int]:
    """
    Solve CVRP using OR-Tools.

    Returns:
        Tuple of (routes, total_distance)
    """
    # Create data model
    data = {
        'distance_matrix': distances,
        'demands': [0] + demands, # Add depot demand
        'vehicle_capacities': [vehicle_capacity] * num_vehicles,
        'num_vehicles': num_vehicles,
    }

```

```

    'depot': 0
}

# Create routing index manager
manager = pywrapcp.RoutingIndexManager(
    len(data['distance_matrix']),
    data['num_vehicles'],
    data['depot']
)

# Create routing model
routing = pywrapcp.RoutingModel(manager)

# Distance callback
def distance_callback(from_index, to_index):
    from_node = manager.IndexToNode(from_index)
    to_node = manager.IndexToNode(to_index)
    return data['distance_matrix'][from_node][to_node]

transit_callback_index = routing.RegisterTransitCallback(distance_callback)
routing.SetArcCostEvaluatorOfAllVehicles(transit_callback_index)

# Capacity constraint
def demand_callback(from_index):
    from_node = manager.IndexToNode(from_index)
    return data['demands'][from_node]

demand_callback_index = routing.RegisterUnaryTransitCallback(demand_callback)
routing.AddDimensionWithVehicleCapacity(
    demand_callback_index,
    0, # null capacity slack
    data['vehicle_capacities'],
    True, # start cumul to zero
    'Capacity'
)

# Search parameters
search_parameters = pywrapcp.DefaultRoutingSearchParameters()
search_parameters.first_solution_strategy = (
    routing_enums_pb2.FirstSolutionStrategy.PATH_CHEAPEST_ARC
)
search_parameters.local_search_metaheuristic = (
    routing_enums_pb2.LocalSearchMetaheuristic.GUIDED_LOCAL_SEARCH
)
search_parameters.time_limit.seconds = 30

# Solve
solution = routing.SolveWithParameters(search_parameters)

# Extract routes
routes = []
total_distance = 0

```

```

for vehicle_id in range(data['num_vehicles']):
    route = []
    index = routing.Start(vehicle_id)

    while not routing.IsEnd(index):
        node = manager.IndexToNode(index)
        if node != data['depot']:
            route.append(node)
            index = solution.Value(routing.NextVar(index))

    if route:
        routes.append(route)

    route_distance = 0
    index = routing.Start(vehicle_id)
    while not routing.IsEnd(index):
        previous_index = index
        index = solution.Value(routing.NextVar(index))
        route_distance += routing.GetArcCostForVehicle(
            previous_index, index, vehicle_id
        )
    total_distance += route_distance

return routes, total_distance

```

14.4.1. Adding Time Windows

Time window constraints require each customer to be served within a specified interval $[a_i, b_i]$. OR-Tools handles this by adding a time dimension that tracks cumulative travel and service time along each route. The solver enforces that the arrival time at each customer falls within the allowed window.

```

def solve_vrptw_ortools(
    distances: List[List[int]],
    time_windows: List[Tuple[int, int]],
    service_times: List[int],
    num_vehicles: int
) -> List[List[int]]:
    """Solve VRPTW using OR-Tools."""
    # ... (similar setup)

    # Add time dimension
    def time_callback(from_index, to_index):
        from_node = manager.IndexToNode(from_index)
        to_node = manager.IndexToNode(to_index)
        return distances[from_node][to_node] + service_times[from_node]

    time_callback_index = routing.RegisterTransitCallback(time_callback)

    routing.AddDimension(
        time_callback_index,
        30, # allow waiting time
        3600, # maximum time per vehicle

```

```

        False,
        'Time'
    )

    time_dimension = routing.GetDimensionOrDie('Time')

    # Add time window constraints
    for location_idx, (start, end) in enumerate(time_windows):
        if location_idx == 0:
            continue
        index = manager.NodeToIndex(location_idx)
        time_dimension.CumulVar(index).SetRange(start, end)

    # ... (solve and extract)

```

14.5. Industrial Applications

14.5.1. Last-Mile Delivery

Last-mile delivery presents specific challenges:

- High customer density
- Narrow time windows
- Dynamic order arrivals
- Multiple delivery attempts

14.5.2. Route Optimization at Scale

For large fleets, hierarchical approaches work well:

1. **Clustering:** Group customers geographically
2. **Assignment:** Assign clusters to vehicles
3. **Routing:** Optimize route within each cluster
4. **Improvement:** Cross-cluster optimization

14.5.3. Real-Time Re-optimization

Dynamic VRP handles changing conditions such as new order arrivals, cancellations, and traffic delays. Rather than solving from scratch, re-optimization modifies the current solution incrementally. The cheapest insertion heuristic finds the best position for each new order across all existing routes.

```

def reoptimize_routes(
    current_routes: List[List[int]],
    vehicle_positions: List[int],
    new_orders: List[int],
    cancelled_orders: List[int],
    distances: np.ndarray
) -> List[List[int]]:
    """
    Re-optimize routes with new information.
    """
    # Remove cancelled orders
    for route in current_routes:
        route[:] = [c for c in route if c not in cancelled_orders]

```

```

# Insert new orders using cheapest insertion
for order in new_orders:
    best_route = None
    best_position = None
    best_cost = float('inf')

    for i, route in enumerate(current_routes):
        for j in range(len(route) + 1):
            # Calculate insertion cost
            prev = vehicle_positions[i] if j == 0 else route[j-1]
            next_node = 0 if j == len(route) else route[j]

            cost = (distances[prev, order] +
                    distances[order, next_node] -
                    distances[prev, next_node])

            if cost < best_cost:
                best_cost = cost
                best_route = i
                best_position = j

    current_routes[best_route].insert(best_position, order)

return current_routes

```

14.6. Summary

14.6.1. Key Takeaways

- VRP is a fundamental combinatorial optimization problem with massive practical impact
- Different variants (CVRP, VRPTW, EVRP) address different operational constraints
- Construction heuristics (savings, nearest neighbor) provide starting solutions quickly
- Improvement heuristics (2-opt, or-opt) refine solutions through local search
- ALNS is the state-of-the-art metaheuristic for large-scale VRP
- OR-Tools provides industrial-strength solvers accessible from Python
- Real-world applications require handling dynamic updates and multiple constraints

14.7. Exercises

1. **Savings calculation:** For a 5-customer problem with given distances, compute all pairwise savings and determine the merge order for the Clarke-Wright algorithm.
2. **2-opt analysis:** Given a route [1, 3, 5, 2, 4], enumerate all possible 2-opt moves and calculate the improvement for each.
3. **OR-Tools implementation:** Use OR-Tools to solve a VRPTW instance with 20 customers, 3 vehicles, and time windows. Compare solution quality with different metaheuristic settings.
4. **Real-world case:** Design a solution approach for a food delivery company with:
 - 50 restaurants and 200 orders per hour
 - 20 delivery drivers
 - 30-minute delivery windows
 - Driver shifts and break requirements

15. Scheduling and Sequencing

What You Will Learn

- Classify scheduling problems (single machine, parallel, flow shop, job shop)
- Formulate scheduling problems as MIP and CP models
- Apply dispatching rules and constructive heuristics
- Implement metaheuristic approaches for complex scheduling
- Solve real-world production scheduling problems

Scheduling determines when and where activities happen. From manufacturing to healthcare, good schedules save time and money. This chapter covers the theory, algorithms, and practical implementation of scheduling solutions.

15.1. Understanding Scheduling

Scheduling is the allocation of resources to tasks over time. It determines when activities should start and end to achieve specific objectives while respecting constraints. In manufacturing, this means deciding which jobs run on which machines at which times to minimize delays, reduce costs, and maximize throughput.

Every production environment faces the fundamental challenge of limited resources and competing demands. Scheduling provides the framework to make these decisions systematically rather than reactively.

15.1.1. The Three-Field Notation

Graham's notation describes scheduling problems as $\alpha|\beta|\gamma$:

- α : Machine environment (single, parallel, flow shop, job shop)
- β : Constraints (release times, precedence, setup times)
- γ : Objective function (makespan, tardiness, flow time)

Examples:

- $1||C_{max}$: Single machine, no constraints, minimize makespan
- $Pm||C_{max}$: Parallel identical machines, minimize makespan
- $Fm||C_{max}$: Flow shop with m machines, minimize makespan
- $Jm|r_j|\sum w_jT_j$: Job shop with release times, minimize weighted tardiness

15.2. Problem Classifications

15.2.1. Single Machine Scheduling

All jobs must be processed on one machine. The challenge is determining the optimal sequence.

Mathematical Formulation:

Given n jobs with processing times p_j , due dates d_j , and weights w_j :

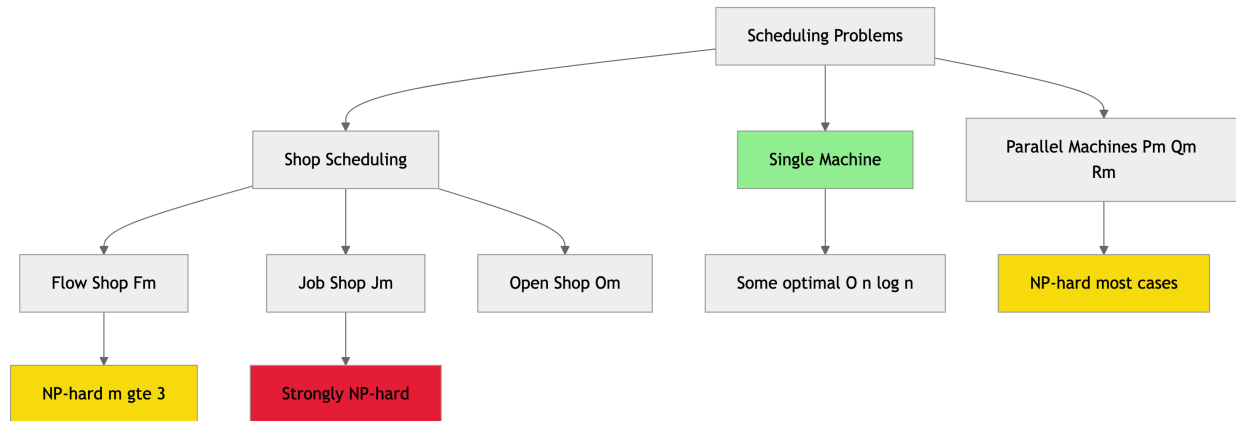


Figure 15.1.: Classification of Machine Environments in Scheduling

$$\begin{aligned}
 &\text{minimize} && \sum_{j=1}^n w_j C_j \\
 &\text{subject to} && C_j \geq C_{j-1} + p_j, \quad j = 2, \dots, n \\
 &&& C_1 \geq p_1
 \end{aligned}$$

i Polynomial-Time Solutions

Some single-machine problems have polynomial solutions:

- **SPT rule:** Minimizes $\sum C_j$ (sort by processing time)
- **WSPT rule:** Minimizes $\sum w_j C_j$ (sort by p_j/w_j)
- **EDD rule:** Minimizes L_{max} (sort by due date)

The following Gantt chart shows how 9 jobs are distributed across 3 parallel machines. Each bar represents a job, and the red dashed line marks the makespan – the completion time of the last job.

15.2.2. Parallel Machine Scheduling

Multiple identical or different machines process jobs independently. Decisions include both assignment and sequencing.

Machine Type	Characteristic	Processing Time
Identical (Pm)	All machines same	$p_{ij} = p_j$
Uniform (Qm)	Different speeds s_i	$p_{ij} = p_j/s_i$
Unrelated (Rm)	General	p_{ij} varies

Load Balancing Formulation:

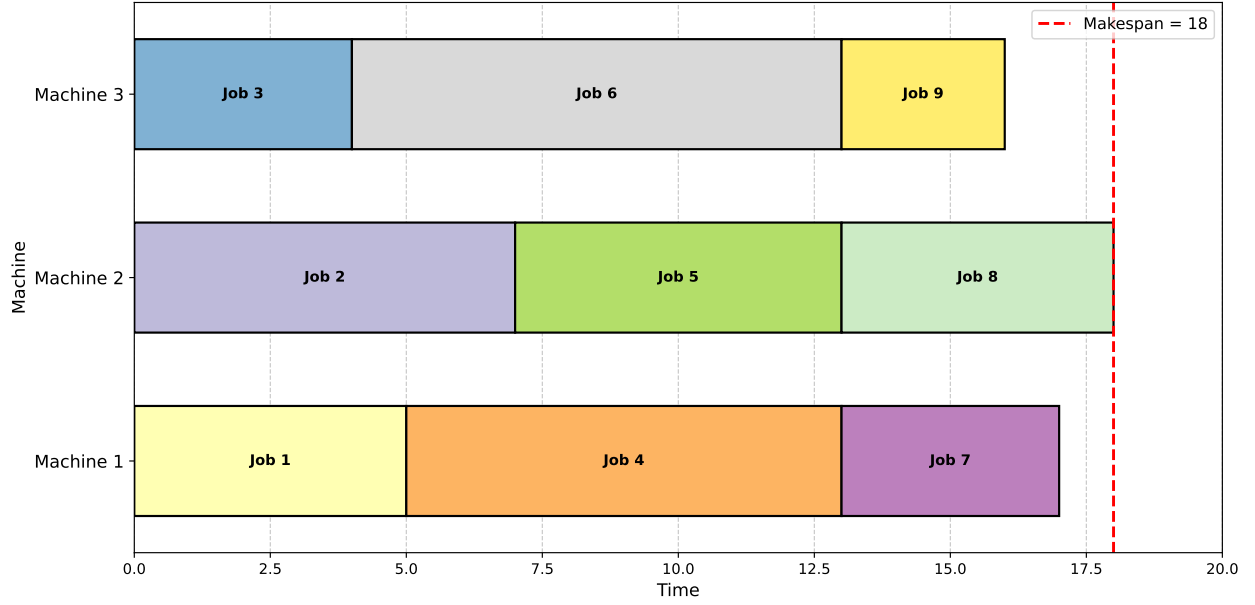


Figure 15.2.: Gantt chart showing job scheduling on 3 parallel machines

$$\begin{aligned}
 & \text{minimize} && C_{max} \\
 & \text{subject to} && \sum_{i=1}^m x_{ij} = 1, \quad \forall j \\
 & && C_{max} \geq \sum_{j=1}^n p_j x_{ij}, \quad \forall i \\
 & && x_{ij} \in \{0, 1\}
 \end{aligned}$$

15.2.3. Flow Shop Scheduling

Jobs follow the same route through machines in the same order. Each job visits each machine exactly once.

Job $\rightarrow$ Machine 1 $\rightarrow$ Machine 2 $\rightarrow \dots \rightarrow$ Machine $m \rightarrow$ Complete

Key Property: In a permutation flow shop, the job sequence is the same on all machines.

💡 Johnson's Algorithm (2-Machine Case)

For $F2||C_{max}$, the optimal sequence can be found in $O(n \log n)$:

1. Partition jobs: $S_1 = \{j : p_{1j} < p_{2j}\}$, $S_2 = \{j : p_{1j} \geq p_{2j}\}$
2. Sort S_1 by increasing p_{1j}
3. Sort S_2 by decreasing p_{2j}
4. Sequence: S_1 then S_2

15. Scheduling and Sequencing

```
def johnson_algorithm(jobs):
    """Johnson's algorithm for 2-machine flow shop.

    Args:
        jobs: List of (job_id, p1, p2) tuples

    Returns:
        Optimal job sequence
    """
    s1 = [(j, p1, p2) for j, p1, p2 in jobs if p1 < p2]
    s2 = [(j, p1, p2) for j, p1, p2 in jobs if p1 >= p2]

    # Sort S1 by increasing p1
    s1.sort(key=lambda x: x[1])

    # Sort S2 by decreasing p2
    s2.sort(key=lambda x: x[2], reverse=True)

    return [j for j, _, _ in s1 + s2]

# Example
jobs = [(1, 3, 2), (2, 4, 6), (3, 2, 5), (4, 5, 3)]
sequence = johnson_algorithm(jobs)
print(f"Optimal sequence: {sequence}")
```

15.2.4. Job Shop Scheduling

Jobs have unique routes through machines with different processing sequences. This is the most general and challenging variant.

Disjunctive Graph Representation:

- Nodes: Operations
- Conjunctive arcs: Precedence within jobs (solid)
- Disjunctive arcs: Machine conflicts (dashed)

A feasible schedule corresponds to selecting one arc from each disjunctive pair to create an acyclic graph.

The visualization below shows a job shop schedule from two perspectives. The left panel displays the machine view (which jobs occupy each machine over time), while the right panel shows the job view (the operation sequence each job follows across machines).

MIP Formulation:

$$\begin{aligned} & \text{minimize} && C_{max} \\ & \text{subject to} && S_{ij} \geq S_{i,j-1} + p_{i,j-1}, \quad \forall i, j > 1 \\ & && S_{ij} \geq S_{i'j'} + p_{i'j'} - M(1 - y_{iji'j'}), \quad \text{if same machine} \\ & && S_{i'j'} \geq S_{ij} + p_{ij} - My_{iji'j'}, \quad \text{if same machine} \\ & && C_{max} \geq S_{ij} + p_{ij}, \quad \forall i, j \\ & && y_{iji'j'} \in \{0, 1\} \end{aligned}$$

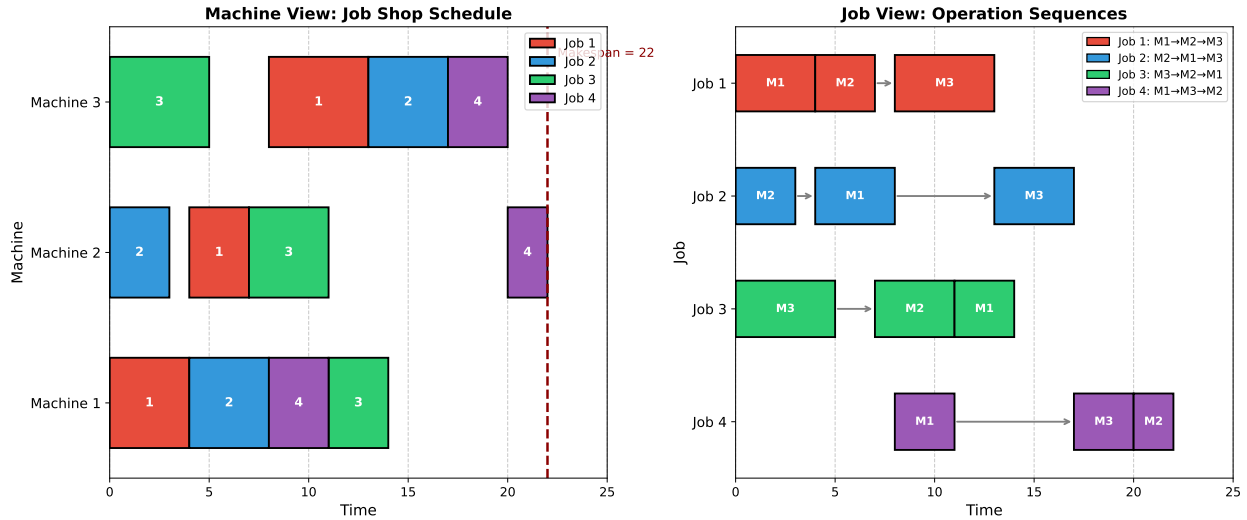


Figure 15.3.: Job shop schedule visualization with 4 jobs on 3 machines showing precedence constraints

15.3. Objective Functions

15.3.1. Completion-Time Based

Objective	Formula	Use Case
Makespan	$C_{max} = \max_j C_j$	Throughput maximization
Total completion time	$\sum_j C_j$	WIP reduction
Weighted completion	$\sum_j w_j C_j$	Priority scheduling

15.3.2. Due-Date Based

Objective	Formula	Use Case
Maximum lateness	$L_{max} = \max_j (C_j - d_j)$	Service level
Total tardiness	$\sum_j T_j = \sum_j \max(0, C_j - d_j)$	On-time delivery
Number of tardy jobs	$\sum_j U_j$ where $U_j = 1$ if $C_j > d_j$	Binary penalty

15.3.3. Resource-Based

- **Machine utilization:** Maximize $\sum_i \sum_j p_{ij} / (m \cdot C_{max})$
- **Idle time:** Minimize total machine idle time
- **Setup costs:** Minimize sequence-dependent setup times

15.4. Dispatching Rules

Dispatching rules are simple priority-based heuristics that make local decisions without global optimization. They are fast and easy to implement but typically produce suboptimal solutions.

15.4.1. Priority Rules

The `DispatchingRules` class below implements the most widely used dispatching rules. Each method takes a list of job objects and returns them sorted by the rule's priority criterion. The ATC (Apparent Tardiness Cost) rule is the most sophisticated, combining WSPT's efficiency ratio with an exponential urgency term based on due date slack.

```
class DispatchingRules:
    """Common dispatching rules for scheduling."""

    @staticmethod
    def spt(jobs):
        """Shortest Processing Time first."""
        return sorted(jobs, key=lambda j: j.processing_time)

    @staticmethod
    def lpt(jobs):
        """Longest Processing Time first."""
        return sorted(jobs, key=lambda j: j.processing_time, reverse=True)

    @staticmethod
    def edd(jobs):
        """Earliest Due Date first."""
        return sorted(jobs, key=lambda j: j.due_date)

    @staticmethod
    def wspt(jobs):
        """Weighted Shortest Processing Time."""
        return sorted(jobs, key=lambda j: j.processing_time / j.weight)

    @staticmethod
    def critical_ratio(jobs, current_time):
        """Critical Ratio: (due_date - current_time) / processing_time."""
        return sorted(jobs,
            key=lambda j: (j.due_date - current_time) / j.processing_time
        )

    @staticmethod
    def atc(jobs, current_time, k=2.0):
        """Apparent Tardiness Cost rule.

        ATC combines WSPT with due date urgency:
        priority = (w_j / p_j) * exp(-max(d_j - p_j - t, 0) / (k * p_avg))
        """
        p_avg = sum(j.processing_time for j in jobs) / len(jobs)

        def atc_priority(job):
            slack = max(job.due_date - job.processing_time - current_time, 0)
            urgency = np.exp(-slack / (k * p_avg))
            return (job.weight / job.processing_time) * urgency

        return sorted(jobs, key=atc_priority, reverse=True)
```

15.4.2. Rule Comparison

Rule	Minimizes	Complexity	Optimality
SPT	$\sum C_j$	$O(n \log n)$	Optimal for single machine
WSPT	$\sum w_j C_j$	$O(n \log n)$	Optimal for single machine
EDD	L_{max}	$O(n \log n)$	Optimal for single machine
LPT	Makespan (parallel)	$O(n \log n)$	4/3 approximation
ATC	$\sum w_j T_j$	$O(n \log n)$	Heuristic

The bar chart below compares the makespan produced by four dispatching rules when scheduling 10 jobs on 3 parallel machines. The red dashed line indicates the theoretical lower bound, computed as total processing time divided by the number of machines.

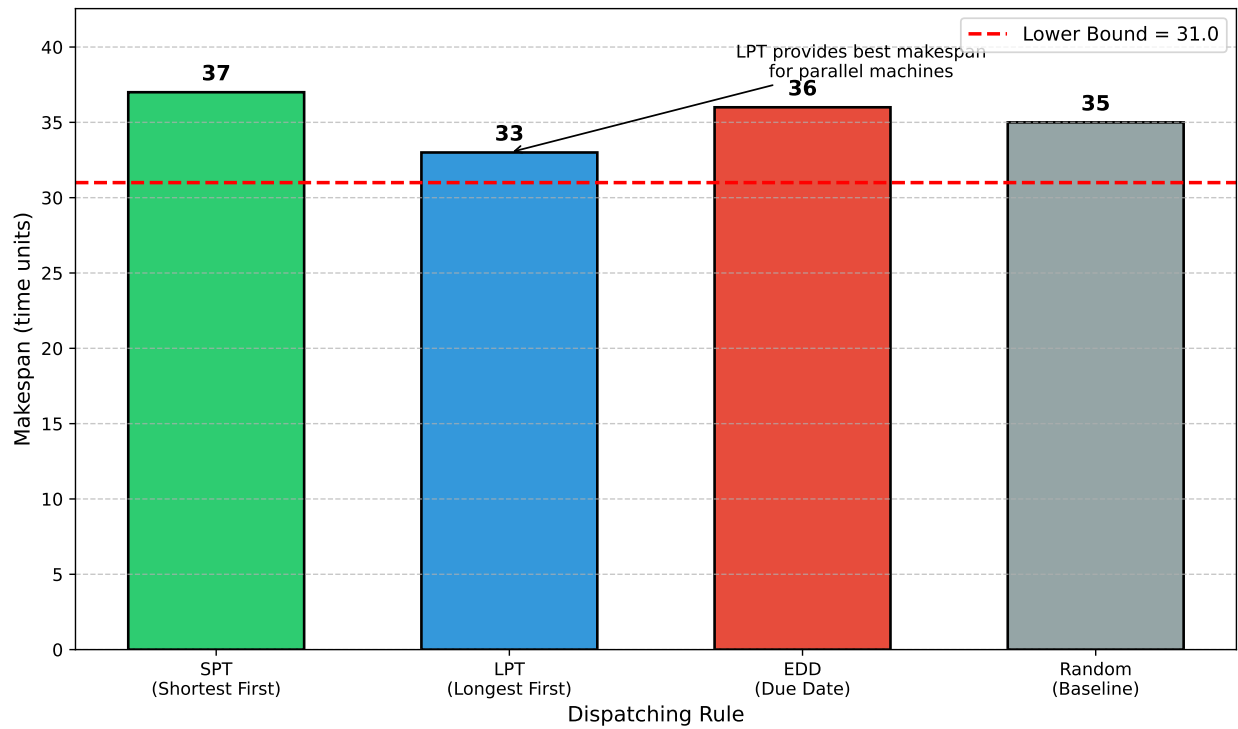


Figure 15.4.: Makespan comparison across different dispatching rules for parallel machine scheduling

15.5. Exact Methods for Scheduling

15.5.1. Constraint Programming

Constraint programming is particularly effective for scheduling due to natural modeling of disjunctive constraints. OR-Tools CP-SAT uses interval variables to represent operations and `AddNoOverlap` constraints to prevent two operations from occupying the same machine simultaneously. The solver automatically handles precedence propagation and conflict detection.

```
from ortools.sat.python import cp_model

def job_shop_cp(jobs, machines, processing_times):
    """Solve job shop scheduling using constraint programming.
```

```

Args:
    jobs: List of jobs, each is list of (machine, duration) tuples
    machines: Number of machines
    processing_times: Matrix of processing times

Returns:
    Schedule as dict mapping (job, task) to (start, end)
"""
model = cp_model.CpModel()

# Compute horizon
horizon = sum(sum(p for _, p in job) for job in jobs)

# Create variables
task_starts = {}
task_ends = {}
task_intervals = {}

for job_id, job in enumerate(jobs):
    for task_id, (machine, duration) in enumerate(job):
        start = model.NewIntVar(0, horizon, f'start_{job_id}_{task_id}')
        end = model.NewIntVar(0, horizon, f'end_{job_id}_{task_id}')
        interval = model.NewIntervalVar(
            start, duration, end, f'interval_{job_id}_{task_id}'
        )

        task_starts[job_id, task_id] = start
        task_ends[job_id, task_id] = end
        task_intervals[job_id, task_id] = interval

# Precedence constraints within jobs
for job_id, job in enumerate(jobs):
    for task_id in range(len(job) - 1):
        model.Add(
            task_starts[job_id, task_id + 1] >=
            task_ends[job_id, task_id]
        )

# No-overlap constraints on machines
for machine in range(machines):
    machine_intervals = []
    for job_id, job in enumerate(jobs):
        for task_id, (m, _) in enumerate(job):
            if m == machine:
                machine_intervals.append(task_intervals[job_id, task_id])
    model.AddNoOverlap(machine_intervals)

# Objective: minimize makespan
makespan = model.NewIntVar(0, horizon, 'makespan')
model.AddMaxEquality(makespan, [
    task_ends[job_id, len(job) - 1]
    for job_id, job in enumerate(jobs)
])

```



```

model.Minimize(makespan)

# Solve
solver = cp_model.CpSolver()
solver.parameters.max_time_in_seconds = 60.0
status = solver.Solve(model)

if status in [cp_model.OPTIMAL, cp_model.FEASIBLE]:
    schedule = {}
    for job_id, job in enumerate(jobs):
        for task_id, (machine, duration) in enumerate(job):
            start = solver.Value(task_starts[job_id, task_id])
            end = solver.Value(task_ends[job_id, task_id])
            schedule[job_id, task_id] = (start, end, machine)
    return schedule, solver.ObjectiveValue()
return None, None

```

15.5.2. Mixed Integer Programming

Flow shop scheduling can also be formulated as a mixed integer program. The model uses binary variables x_{jp} to assign each job j to a position p in the sequence, then computes completion times on each machine to minimize makespan. The following Pyomo model implements this formulation for the permutation flow shop.

```

import pyomo.environ as pyo

def flow_shop_mip(n_jobs, n_machines, processing_times):
    """Flow shop scheduling using MIP.

    Args:
        n_jobs: Number of jobs
        n_machines: Number of machines
        processing_times: p[j][m] = processing time of job j on machine m

    Returns:
        Optimal sequence and makespan
    """
    model = pyo.ConcreteModel()

    # Sets
    model.J = pyo.RangeSet(n_jobs)
    model.M = pyo.RangeSet(n_machines)
    model.P = pyo.RangeSet(n_jobs) # Positions

    # Parameters
    def p_init(model, j, m):
        return processing_times[j-1][m-1]
    model.p = pyo.Param(model.J, model.M, initialize=p_init)

    # Big M
    M = sum(sum(processing_times[j][m] for m in range(n_machines))
            for j in range(n_jobs))

```

```

# Variables
model.x = pyo.Var(model.J, model.P, domain=pyo.Binary) # job j in position p
model.C = pyo.Var(model.P, model.M, domain=pyo.NonNegativeReals) # completion
model.Cmax = pyo.Var(domain=pyo.NonNegativeReals)

# Each job assigned to exactly one position
def one_position_rule(model, j):
    return sum(model.x[j, p] for p in model.P) == 1
model.one_position = pyo.Constraint(model.J, rule=one_position_rule)

# Each position has exactly one job
def one_job_rule(model, p):
    return sum(model.x[j, p] for j in model.J) == 1
model.one_job = pyo.Constraint(model.P, rule=one_job_rule)

# Completion time constraints (simplified)
# Full formulation requires linearization

# Makespan
def makespan_rule(model, p):
    return model.Cmax >= model.C[p, n_machines]
model.makespan = pyo.Constraint(model.P, rule=makespan_rule)

# Objective
model.obj = pyo.Objective(expr=model.Cmax, sense=pyo.minimize)

return model

```

15.6. Metaheuristics for Scheduling

15.6.1. Genetic Algorithm

Genetic algorithms represent job sequences as permutation chromosomes and use population-based search to find high-quality schedules. The algorithm maintains a population of candidate sequences, selects parents through tournament selection, recombines them with order crossover (OX), and applies swap mutation. Elitism preserves the best solution found across generations.

```

import random
import numpy as np

class GeneticAlgorithmScheduler:
    """Genetic algorithm for permutation flow shop."""

    def __init__(self, processing_times, pop_size=50, generations=100,
                 crossover_rate=0.8, mutation_rate=0.1):
        self.processing_times = np.array(processing_times)
        self.n_jobs, self.n_machines = self.processing_times.shape
        self.pop_size = pop_size
        self.generations = generations
        self.crossover_rate = crossover_rate
        self.mutation_rate = mutation_rate

```

```

def compute_makespan(self, sequence):
    """Compute makespan for a job sequence."""
    completion = np.zeros((self.n_jobs, self.n_machines))

    for pos, job in enumerate(sequence):
        for machine in range(self.n_machines):
            if pos == 0 and machine == 0:
                completion[pos, machine] = self.processing_times[job, machine]
            elif pos == 0:
                completion[pos, machine] = (
                    completion[pos, machine - 1] +
                    self.processing_times[job, machine]
                )
            elif machine == 0:
                completion[pos, machine] = (
                    completion[pos - 1, machine] +
                    self.processing_times[job, machine]
                )
            else:
                completion[pos, machine] = (
                    max(completion[pos - 1, machine],
                        completion[pos, machine - 1]) +
                    self.processing_times[job, machine]
                )

    return completion[-1, -1]

def initialize_population(self):
    """Create initial population of random permutations."""
    return [random.sample(range(self.n_jobs), self.n_jobs)
            for _ in range(self.pop_size)]

def tournament_selection(self, population, fitnesses, k=3):
    """Select parent via tournament selection."""
    tournament = random.sample(list(zip(population, fitnesses)), k)
    return min(tournament, key=lambda x: x[1])[0]

def order_crossover(self, parent1, parent2):
    """Order crossover (OX) for permutations."""
    size = len(parent1)
    a, b = sorted(random.sample(range(size), 2))

    # Copy segment from parent1
    child = [None] * size
    child[a:b+1] = parent1[a:b+1]

    # Fill remaining from parent2
    remaining = [x for x in parent2 if x not in child[a:b+1]]
    pos = 0
    for i in range(size):
        if child[i] is None:
            child[i] = remaining[pos]
            pos += 1

```

```

        return child

    def swap_mutation(self, individual):
        """Swap mutation for permutations."""
        if random.random() < self.mutation_rate:
            i, j = random.sample(range(len(individual)), 2)
            individual[i], individual[j] = individual[j], individual[i]
        return individual

    def solve(self):
        """Run genetic algorithm."""
        population = self.initialize_population()
        best_solution = None
        best_fitness = float('inf')

        for generation in range(self.generations):
            # Evaluate fitness
            fitnesses = [self.compute_makespan(ind) for ind in population]

            # Update best
            gen_best_idx = np.argmin(fitnesses)
            if fitnesses[gen_best_idx] < best_fitness:
                best_fitness = fitnesses[gen_best_idx]
                best_solution = population[gen_best_idx].copy()

            # Create next generation
            new_population = []

            # Elitism: keep best
            new_population.append(population[gen_best_idx])

            while len(new_population) < self.pop_size:
                # Selection
                parent1 = self.tournament_selection(population, fitnesses)
                parent2 = self.tournament_selection(population, fitnesses)

                # Crossover
                if random.random() < self.crossover_rate:
                    child = self.order_crossover(parent1, parent2)
                else:
                    child = parent1.copy()

                # Mutation
                child = self.swap_mutation(child)
                new_population.append(child)

            population = new_population

        return best_solution, best_fitness

```

15.6.2. Simulated Annealing

Simulated annealing explores the solution space by accepting worse solutions with a probability that decreases as the temperature cools. This allows the algorithm to escape local optima early in the search. The neighbor function swaps two random jobs in the sequence, and the cooling schedule controls the transition from exploration to exploitation.

```
import math

def simulated_annealing_scheduler(processing_times, initial_temp=1000,
                                cooling_rate=0.995, max_iter=10000):
    """Simulated annealing for flow shop scheduling."""
    n_jobs = len(processing_times)

    def compute_makespan(sequence):
        n_machines = len(processing_times[0])
        completion = [[0] * n_machines for _ in range(n_jobs)]

        for pos, job in enumerate(sequence):
            for m in range(n_machines):
                prev_job = completion[pos-1][m] if pos > 0 else 0
                prev_machine = completion[pos][m-1] if m > 0 else 0
                completion[pos][m] = max(prev_job, prev_machine) +
↪ processing_times[job][m]

        return completion[-1][-1]

    def get_neighbor(sequence):
        """Generate neighbor by swapping two random jobs."""
        neighbor = sequence.copy()
        i, j = random.sample(range(len(sequence)), 2)
        neighbor[i], neighbor[j] = neighbor[j], neighbor[i]
        return neighbor

    # Initialize
    current = list(range(n_jobs))
    random.shuffle(current)
    current_cost = compute_makespan(current)

    best = current.copy()
    best_cost = current_cost

    temperature = initial_temp

    for iteration in range(max_iter):
        # Generate neighbor
        neighbor = get_neighbor(current)
        neighbor_cost = compute_makespan(neighbor)

        # Accept or reject
        delta = neighbor_cost - current_cost

        if delta < 0 or random.random() < math.exp(-delta / temperature):
```

```

current = neighbor
current_cost = neighbor_cost

if current_cost < best_cost:
    best = current.copy()
    best_cost = current_cost

# Cool down
temperature *= cooling_rate

return best, best_cost

```

15.7. Assembly Line Balancing

Assembly line balancing (ALB) assigns tasks to workstations along a production line while respecting precedence constraints and cycle time limits. A well-balanced line minimizes the number of stations, maximizes efficiency, and ensures smooth production flow.

15.7.1. Problem Classification

The Simple Assembly Line Balancing Problem (SALBP) is the foundation for line balancing. Different variants address different objectives:

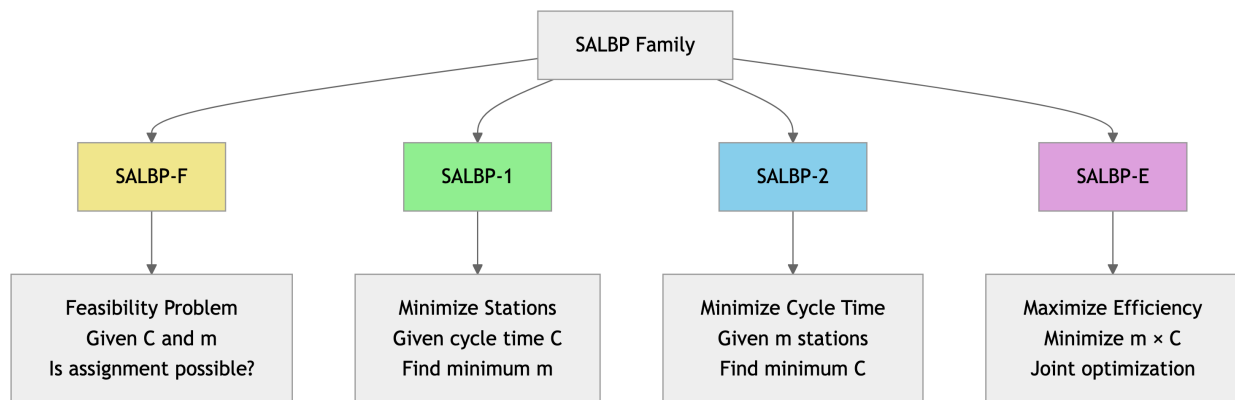


Figure 15.5.: Classification of Assembly Line Balancing Problems

Problem	Given	Objective
SALBP-F	Cycle time C , Stations m	Find feasible assignment
SALBP-1	Cycle time C	Minimize number of stations m
SALBP-2	Number of stations m	Minimize cycle time C
SALBP-E	Neither	Maximize line efficiency E

Advanced Variants:

- **GALBP** (Generalized): Multiple products, parallel stations, equipment constraints
- **UALBP** (U-shaped): Workers can access both sides of the line
- **MALBP** (Mixed-model): Multiple product variants on same line
- **RALBP** (Robotic): Robot assignment with varying task times

15.7.2. Precedence Constraints

Tasks have technological precedence: some tasks must complete before others can begin. The precedence graph captures these relationships.

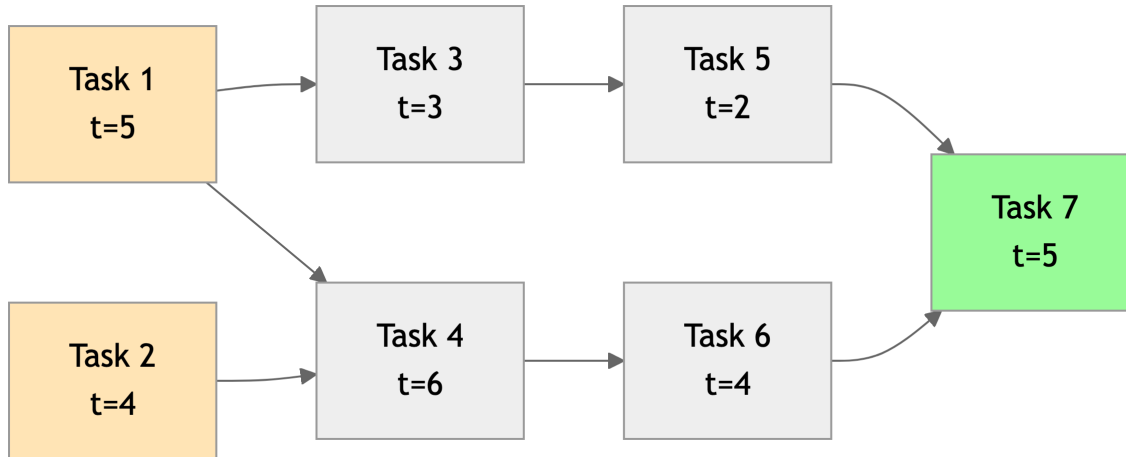


Figure 15.6.: Example Precedence Graph for Assembly Tasks

Key Definitions:

- **Immediate Predecessor:** Task i is an immediate predecessor of j if i must directly precede j
- **Predecessor Set $P(j)$:** All tasks that must complete before j (direct and indirect)
- **Successor Set $S(j)$:** All tasks that require j to complete first
- **Positional Weight:** Sum of task time and all successor times

15.7.3. Mathematical Formulation (SALBP-1)

Parameters:

- n : Number of tasks
- t_i : Processing time of task i
- C : Cycle time (given)
- P : Set of precedence pairs (i, j) where i must precede j
- m_{ub} : Upper bound on stations (often n)

Decision Variables:

- $x_{ik} = 1$ if task i is assigned to station k , 0 otherwise
- $y_k = 1$ if station k is used, 0 otherwise

Formulation:

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$$\begin{aligned}
& \text{minimize} && \sum_{k=1}^{m_{ub}} y_k \\
& \text{subject to} && \sum_{k=1}^{m_{ub}} x_{ik} = 1, \quad \forall i \in \{1, \dots, n\} \\
& && \sum_{i=1}^n t_i \cdot x_{ik} \leq C \cdot y_k, \quad \forall k \\
& && \sum_{k=1}^{m_{ub}} k \cdot x_{ik} \leq \sum_{k=1}^{m_{ub}} k \cdot x_{jk}, \quad \forall (i, j) \in P \\
& && y_k \geq y_{k+1}, \quad \forall k < m_{ub} \\
& && x_{ik}, y_k \in \{0, 1\}
\end{aligned}$$

Constraints Explained:

1. Each task assigned to exactly one station
2. Station workload cannot exceed cycle time
3. Precedence constraints (predecessor in earlier or same station)
4. Station usage is consecutive (no gaps)

15.7.4. Lower Bounds

Simple Lower Bound:

$$m_{LB} = \left\lceil \frac{\sum_{i=1}^n t_i}{C} \right\rceil$$

This is the theoretical minimum based on total work content.

Improved Bounds:

- **Bin Packing Bound:** View tasks as items, stations as bins
- **Longest Path Bound:** Based on critical path through precedence graph

15.7.5. Performance Metrics

Line Efficiency:

$$E = \frac{\sum_{i=1}^n t_i}{m \cdot C} \times 100\%$$

Balance Delay:

$$BD = 1 - E = \frac{m \cdot C - \sum_{i=1}^n t_i}{m \cdot C} \times 100\%$$

Smoothness Index:

$$SI = \sqrt{\sum_{k=1}^m \left(\max_j (S_j) - S_k \right)^2}$$

where S_k is the workload at station k . Lower values indicate better balance.

15.7.6. Heuristic Methods

15.7.6.1. 1. Ranked Positional Weight (RPW)

The RPW method prioritizes tasks by their positional weight—the sum of the task time and all successor times.

```
import numpy as np
from typing import List, Dict, Set, Tuple
from dataclasses import dataclass
from collections import defaultdict

@dataclass
class Task:
    """Represents an assembly task."""
    id: int
    time: float
    predecessors: Set[int]
    successors: Set[int] = None

    def __post_init__(self):
        if self.successors is None:
            self.successors = set()

class AssemblyLineBalancer:
    """
    Assembly Line Balancing solver with multiple methods.

    Implements SALBP-1 (minimize stations given cycle time) and
    SALBP-2 (minimize cycle time given stations).
    """

    def __init__(self, tasks: List[Task], cycle_time: float = None,
                  n_stations: int = None):
        """
        Initialize the line balancer.

        Args:
            tasks: List of Task objects
            cycle_time: Given cycle time (for SALBP-1)
            n_stations: Given number of stations (for SALBP-2)
        """
        self.tasks = {t.id: t for t in tasks}
        self.n_tasks = len(tasks)
        self.cycle_time = cycle_time
        self.n_stations = n_stations

        # Build successor sets from predecessor sets
        self._build_successor_sets()

        # Compute task statistics
        self.total_work = sum(t.time for t in tasks)
        self.max_task_time = max(t.time for t in tasks)
```

```

def _build_successor_sets(self):
    """Build successor sets from predecessor information."""
    for task in self.tasks.values():
        for pred_id in task.predecessors:
            self.tasks[pred_id].successors.add(task.id)

def get_all_predecessors(self, task_id: int) -> Set[int]:
    """Get all predecessors (transitive closure)."""
    predecessors = set()
    stack = list(self.tasks[task_id].predecessors)

    while stack:
        pred = stack.pop()
        if pred not in predecessors:
            predecessors.add(pred)
            stack.extend(self.tasks[pred].predecessors)

    return predecessors

def get_all_successors(self, task_id: int) -> Set[int]:
    """Get all successors (transitive closure)."""
    successors = set()
    stack = list(self.tasks[task_id].successors)

    while stack:
        succ = stack.pop()
        if succ not in successors:
            successors.add(succ)
            stack.extend(self.tasks[succ].successors)

    return successors

def calculate_positional_weight(self, task_id: int) -> float:
    """Calculate Ranked Positional Weight for a task."""
    task = self.tasks[task_id]
    weight = task.time

    # Add times of all successors
    for succ_id in self.get_all_successors(task_id):
        weight += self.tasks[succ_id].time

    return weight

def rpw_heuristic(self) -> Tuple[Dict[int, List[int]], float]:
    """
    Ranked Positional Weight heuristic for SALBP-1.

    Returns:
        Tuple of (station_assignments, achieved_efficiency)
    """
    if self.cycle_time is None:
        raise ValueError("Cycle time required for RPW heuristic")

```

```

# Calculate positional weights
weights = {t_id: self.calculate_positional_weight(t_id)
           for t_id in self.tasks}

# Sort tasks by decreasing positional weight
sorted_tasks = sorted(weights.keys(), key=lambda x: weights[x],
                      reverse=True)

# Assign to stations
stations = {0: []}
station_times = {0: 0}
assigned = set()
current_station = 0

for task_id in sorted_tasks:
    task = self.tasks[task_id]

    # Check if all predecessors are assigned
    if not task.predecessors.issubset(assigned):
        continue

    # Find earliest feasible station
    earliest_station = 0
    for pred_id in task.predecessors:
        for s, tasks_in_s in stations.items():
            if pred_id in tasks_in_s:
                earliest_station = max(earliest_station, s)

    # Find station with capacity starting from earliest
    placed = False
    for s in range(earliest_station, len(stations)):
        if station_times[s] + task.time <= self.cycle_time:
            stations[s].append(task_id)
            station_times[s] += task.time
            assigned.add(task_id)
            placed = True
            break

    # Open new station if needed
    if not placed:
        new_station = len(stations)
        stations[new_station] = [task_id]
        station_times[new_station] = task.time
        assigned.add(task_id)

# Handle any remaining tasks (due to dependency sorting)
remaining = set(self.tasks.keys()) - assigned
while remaining:
    for task_id in list(remaining):
        task = self.tasks[task_id]
        if task.predecessors.issubset(assigned):
            # Find station

```

```

        earliest_station = 0
        for pred_id in task.predecessors:
            for s, tasks_in_s in stations.items():
                if pred_id in tasks_in_s:
                    earliest_station = max(earliest_station, s)

        placed = False
        for s in range(earliest_station, len(stations)):
            if station_times.get(s, 0) + task.time <= self.cycle_time:
                if s not in stations:
                    stations[s] = []
                    station_times[s] = 0
                stations[s].append(task_id)
                station_times[s] += task.time
                assigned.add(task_id)
                remaining.remove(task_id)
                placed = True
                break

        if not placed:
            new_station = len(stations)
            stations[new_station] = [task_id]
            station_times[new_station] = task.time
            assigned.add(task_id)
            remaining.remove(task_id)

    efficiency = self.total_work / (len(stations) * self.cycle_time) * 100

    return stations, efficiency

def largest_candidate_rule(self) -> Tuple[Dict[int, List[int]], float]:
    """
    Largest Candidate Rule heuristic for SALBP-1.

    Assigns the largest fitting task that has all predecessors assigned.

    Returns:
        Tuple of (station_assignments, achieved_efficiency)
    """
    if self.cycle_time is None:
        raise ValueError("Cycle time required")

    stations = {0: []}
    station_times = {0: 0}
    assigned = set()

    # Sort tasks by decreasing time
    sorted_by_time = sorted(self.tasks.keys(),
                            key=lambda x: self.tasks[x].time,
                            reverse=True)

    while len(assigned) < self.n_tasks:
        current_station = len(stations) - 1

```

```

# Find largest assignable task for current station
task_assigned = False

for task_id in sorted_by_time:
    if task_id in assigned:
        continue

    task = self.tasks[task_id]

    # Check predecessors assigned
    if not task.predecessors.issubset(assigned):
        continue

    # Check precedence in earlier stations
    pred_ok = True
    for pred_id in task.predecessors:
        for s, tasks_in_s in stations.items():
            if pred_id in tasks_in_s and s > current_station:
                pred_ok = False
                break

    if not pred_ok:
        continue

    # Check capacity
    if station_times[current_station] + task.time <= self.cycle_time:
        stations[current_station].append(task_id)
        station_times[current_station] += task.time
        assigned.add(task_id)
        task_assigned = True
        break

# Open new station if no task could be assigned
if not task_assigned:
    new_station = len(stations)
    stations[new_station] = []
    station_times[new_station] = 0

efficiency = self.total_work / (len(stations) * self.cycle_time) * 100

return stations, efficiency

def kilbridge_wester_method(self) -> Tuple[Dict[int, List[int]], float]:
    """
    Kilbridge-Wester column method for SALBP-1.

    Groups tasks by precedence level (column), then assigns
    by decreasing task time within each column.

    Returns:
        Tuple of (station_assignments, achieved_efficiency)
    """

```

```

if self.cycle_time is None:
    raise ValueError("Cycle time required")

# Determine column (precedence level) for each task
columns = {}
for task_id in self.tasks:
    predecessors = self.get_all_predecessors(task_id)
    columns[task_id] = len(predecessors)

# Group tasks by column
tasks_by_column = defaultdict(list)
for task_id, col in columns.items():
    tasks_by_column[col].append(task_id)

# Sort each column by decreasing task time
for col in tasks_by_column:
    tasks_by_column[col].sort(
        key=lambda x: self.tasks[x].time, reverse=True
    )

# Assign tasks column by column
stations = {0: []}
station_times = {0: 0}
assigned = set()

max_col = max(columns.values())
current_station = 0

for col in range(max_col + 1):
    for task_id in tasks_by_column[col]:
        task = self.tasks[task_id]

        # Find station starting from current
        placed = False
        for s in range(current_station, len(stations) + 1):
            if s not in stations:
                stations[s] = []
                station_times[s] = 0

            if station_times[s] + task.time <= self.cycle_time:
                stations[s].append(task_id)
                station_times[s] += task.time
                assigned.add(task_id)
                placed = True
                break

        if not placed:
            # This shouldn't happen if cycle_time >= max_task_time
            new_station = len(stations)
            stations[new_station] = [task_id]
            station_times[new_station] = task.time
            assigned.add(task_id)

```

```

        efficiency = self.total_work / (len(stations) * self.cycle_time) * 100

    return stations, efficiency

def calculate_metrics(self, stations: Dict[int, List[int]]) -> Dict[str, float]:
    """Calculate performance metrics for a solution."""
    n_stations = len(stations)

    # Station workloads
    workloads = []
    for s in range(n_stations):
        workload = sum(self.tasks[t].time for t in stations[s])
        workloads.append(workload)

    max_workload = max(workloads)
    actual_cycle_time = max_workload if self.cycle_time is None else
↪ self.cycle_time

    efficiency = self.total_work / (n_stations * actual_cycle_time) * 100
    balance_delay = 100 - efficiency

    # Smoothness index
    smoothness = np.sqrt(sum((max_workload - w) ** 2 for w in workloads))

    return {
        'n_stations': n_stations,
        'cycle_time': actual_cycle_time,
        'efficiency': efficiency,
        'balance_delay': balance_delay,
        'smoothness_index': smoothness,
        'workloads': workloads
    }

def compare_heuristics(self) -> Dict[str, Dict]:
    """Compare all heuristic methods."""
    results = {}

    methods = [
        ('RPW', self.rpw_heuristic),
        ('Largest Candidate', self.largest_candidate_rule),
        ('Kilbridge-Wester', self.kilbridge_wester_method),
    ]

    for name, method in methods:
        try:
            stations, efficiency = method()
            metrics = self.calculate_metrics(stations)
            metrics['stations'] = stations
            results[name] = metrics
        except Exception as e:
            results[name] = {'error': str(e)}

    return results

```

15.7.6.2. 2. Exact Method with OR-Tools

For smaller instances, constraint programming can find provably optimal solutions. The following implementation translates the SALBP-1 and SALBP-2 formulations into OR-Tools CP-SAT models. Binary variables x_{ik} represent task-to-station assignments, with constraints enforcing capacity limits, precedence relationships, and consecutive station usage.

```
from ortools.sat.python import cp_model

def solve_salbp1_exact(tasks: List[Task], cycle_time: float,
                      time_limit: float = 60.0) -> Tuple[Dict[int, List[int]],
↳ float]:
    """
    Solve SALBP-1 exactly using constraint programming.

    Args:
        tasks: List of Task objects
        cycle_time: Given cycle time
        time_limit: Solver time limit in seconds

    Returns:
        Tuple of (station_assignments, efficiency)
    """
    model = cp_model.CpModel()

    n = len(tasks)
    task_dict = {t.id: t for t in tasks}
    task_ids = [t.id for t in tasks]

    # Upper bound on stations
    total_work = sum(t.time for t in tasks)
    m_ub = n # Worst case: one task per station

    # Decision variables
    # x[i, k] = 1 if task i assigned to station k
    x = {}
    for i in task_ids:
        for k in range(m_ub):
            x[i, k] = model.NewBoolVar(f'x_{i}_{k}')

    # y[k] = 1 if station k is used
    y = [model.NewBoolVar(f'y_{k}') for k in range(m_ub)]

    # Each task assigned to exactly one station
    for i in task_ids:
        model.Add(sum(x[i, k] for k in range(m_ub)) == 1)

    # Station capacity constraint
    for k in range(m_ub):
        model.Add(
            sum(int(task_dict[i].time * 100) * x[i, k] for i in task_ids)
            <= int(cycle_time * 100) * y[k]
        )
```



```

# Precedence constraints
for task in tasks:
    for pred_id in task.predecessors:
        # Predecessor must be in same or earlier station
        for k in range(m_ub):
            model.Add(
                sum(k2 * x[pred_id, k2] for k2 in range(m_ub)) <=
                sum(k2 * x[task.id, k2] for k2 in range(m_ub))
            )

# Station usage is consecutive
for k in range(m_ub - 1):
    model.Add(y[k] >= y[k + 1])

# Objective: minimize number of stations
model.Minimize(sum(y))

# Solve
solver = cp_model.CpSolver()
solver.parameters.max_time_in_seconds = time_limit
status = solver.Solve(model)

if status in [cp_model.OPTIMAL, cp_model.FEASIBLE]:
    # Extract solution
    stations = defaultdict(list)
    for i in task_ids:
        for k in range(m_ub):
            if solver.Value(x[i, k]) == 1:
                stations[k].append(i)
                break

    # Remove empty stations and renumber
    result = {}
    station_num = 0
    for k in range(m_ub):
        if stations[k]:
            result[station_num] = stations[k]
            station_num += 1

    efficiency = total_work / (len(result) * cycle_time) * 100
    return result, efficiency

return None, 0.0

def solve_salbp2_exact(tasks: List[Task], n_stations: int,
                      time_limit: float = 60.0) -> Tuple[Dict[int, List[int]],
↳ float]:
    """
    Solve SALBP-2 exactly using constraint programming.

    Args:
        tasks: List of Task objects

```

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```
n_stations: Given number of stations
time_limit: Solver time limit in seconds

Returns:
    Tuple of (station_assignments, minimum_cycle_time)
"""
model = cp_model.CpModel()

n = len(tasks)
task_dict = {t.id: t for t in tasks}
task_ids = [t.id for t in tasks]

# Upper bound on cycle time
total_work = sum(t.time for t in tasks)
max_task = max(t.time for t in tasks)
c_ub = int(total_work * 100) # Scale to integer

# Decision variables
x = {}
for i in task_ids:
    for k in range(n_stations):
        x[i, k] = model.NewBoolVar(f'x_{i}_{k}')

# Cycle time variable (scaled)
C = model.NewIntVar(int(max_task * 100), c_ub, 'C')

# Each task assigned to exactly one station
for i in task_ids:
    model.Add(sum(x[i, k] for k in range(n_stations)) == 1)

# Station capacity constraint
for k in range(n_stations):
    model.Add(
        sum(int(task_dict[i].time * 100) * x[i, k] for i in task_ids) <= C
    )

# Precedence constraints
for task in tasks:
    for pred_id in task.predecessors:
        for k in range(n_stations):
            model.Add(
                sum(k2 * x[pred_id, k2] for k2 in range(n_stations)) <=
                sum(k2 * x[task.id, k2] for k2 in range(n_stations))
            )

# Objective: minimize cycle time
model.Minimize(C)

# Solve
solver = cp_model.CpSolver()
solver.parameters.max_time_in_seconds = time_limit
status = solver.Solve(model)
```

```

if status in [cp_model.OPTIMAL, cp_model.FEASIBLE]:
    stations = defaultdict(list)
    for i in task_ids:
        for k in range(n_stations):
            if solver.Value(x[i, k]) == 1:
                stations[k].append(i)
                break

    cycle_time = solver.Value(C) / 100.0
    return dict(stations), cycle_time

return None, 0.0

```

15.7.7. Complete Example: Automotive Assembly

Consider an automotive door assembly with the following tasks. This example applies all three heuristics (RPW, Largest Candidate Rule, and Kilbridge-Wester) to a 10-task problem with precedence constraints, then compares them against the exact CP solution. The cycle time is set to 60 seconds, corresponding to a production rate of 60 units per hour.

```

def automotive_door_assembly_example():
    """Example: Automotive door assembly line balancing."""

    print("=" * 60)
    print("ASSEMBLY LINE BALANCING: AUTOMOTIVE DOOR")
    print("=" * 60)

    # Define tasks for door assembly
    # Task times in seconds
    tasks_data = [
        (1, 45, set()),           # Install inner panel frame
        (2, 30, {1}),            # Attach wiring harness
        (3, 25, {1}),            # Install window regulator
        (4, 35, {2, 3}),         # Mount door handle mechanism
        (5, 20, {2}),            # Connect electrical connectors
        (6, 40, {3}),            # Install window glass
        (7, 15, {4, 5}),         # Install door lock actuator
        (8, 25, {6}),            # Seal window edges
        (9, 30, {7, 8}),         # Attach outer panel
        (10, 20, {9}),           # Final inspection
    ]

    tasks = [Task(id=t[0], time=t[1], predecessors=t[2])
              for t in tasks_data]

    # Total work content
    total_time = sum(t.time for t in tasks)
    max_task = max(t.time for t in tasks)

    print(f"\nProblem Data:")
    print(f"  Number of tasks: {len(tasks)}")
    print(f"  Total work content: {total_time} seconds")
    print(f"  Maximum task time: {max_task} seconds")

```

```

# SALBP-1: Given cycle time, minimize stations
cycle_time = 60 # seconds (production rate: 60 units/hour)

print(f"\nSALBP-1: Cycle time = {cycle_time}s")
print(f" Lower bound on stations: {np.ceil(total_time / cycle_time):.0f}")
print("-" * 60)

balancer = AssemblyLineBalancer(tasks, cycle_time=cycle_time)
results = balancer.compare_heuristics()

for method, metrics in results.items():
    if 'error' in metrics:
        print(f"\n{method}: Error - {metrics['error']}")
        continue

    print(f"\n{method}:")
    print(f" Stations: {metrics['n_stations']}")
    print(f" Efficiency: {metrics['efficiency']:.1f}%")
    print(f" Balance Delay: {metrics['balance_delay']:.1f}%")
    print(f" Smoothness Index: {metrics['smoothness_index']:.2f}")
    print(f" Workloads: {[round(w, 1) for w in metrics['workloads']]}")

# Exact solution
print("\n" + "-" * 60)
print("Exact Solution (OR-Tools CP):")

exact_stations, exact_eff = solve_salbp1_exact(tasks, cycle_time)
if exact_stations:
    print(f" Optimal stations: {len(exact_stations)}")
    print(f" Efficiency: {exact_eff:.1f}%")
    for s, task_list in exact_stations.items():
        workload = sum(tasks_data[t-1][1] for t in task_list)
        print(f" Station {s+1}: Tasks {task_list}, Workload: {workload}s")

# SALBP-2: Given stations, minimize cycle time
n_stations = 4

print(f"\n{'=' * 60}")
print(f"SALBP-2: {n_stations} stations, minimize cycle time")
print("-" * 60)

salbp2_stations, min_cycle = solve_salbp2_exact(tasks, n_stations)
if salbp2_stations:
    print(f" Minimum cycle time: {min_cycle:.1f}s")
    print(f" Production rate: {3600/min_cycle:.1f} units/hour")
    for s, task_list in salbp2_stations.items():
        workload = sum(tasks_data[t-1][1] for t in task_list)
        print(f" Station {s+1}: Tasks {task_list}, Workload: {workload}s")

return results

```

```
# Run example
if __name__ == "__main__":
    results = automotive_door_assembly_example()
```

15.7.8. U-Shaped Lines

U-shaped assembly lines allow workers at one station to perform tasks from both sides of the line. This increases flexibility and can improve efficiency.

Advantages of U-shaped lines:

- Workers can handle tasks from both line ends
- Better communication between workers
- More flexibility in task assignment
- Reduced line length

15.7.9. Mixed-Model Lines

Many modern assembly lines produce multiple product variants simultaneously (mixed-model sequencing).

Objectives:

1. **Level workload:** Minimize variation in station workloads
2. **Level parts usage:** Smooth demand for components
3. **Minimize setup:** Reduce sequence-dependent changeovers

The model sequence becomes important:

$$\text{Minimize } \max_k \sum_{t=1}^T |W_k(t) - \bar{W}_k|$$

where $W_k(t)$ is the cumulative workload at station k after producing t units.

Line Balancing Best Practices

1. **Start with data:** Accurate task times are essential
2. **Validate precedence:** Ensure the precedence graph is complete
3. **Consider variability:** Build in buffers for task time variation
4. **Involve operators:** They know the real constraints
5. **Iterate:** Line balance is an ongoing process, not a one-time event
6. **Monitor:** Track actual vs. planned performance continuously

15.8. Industry Applications

15.8.1. Semiconductor Manufacturing

- Hundreds of operations per wafer
- Reentrant flows with multiple machine visits
- Batch processing and setup constraints
- Cycle time minimization essential

15.8.2. Pharmaceutical Production

- Batch processing with strict validation

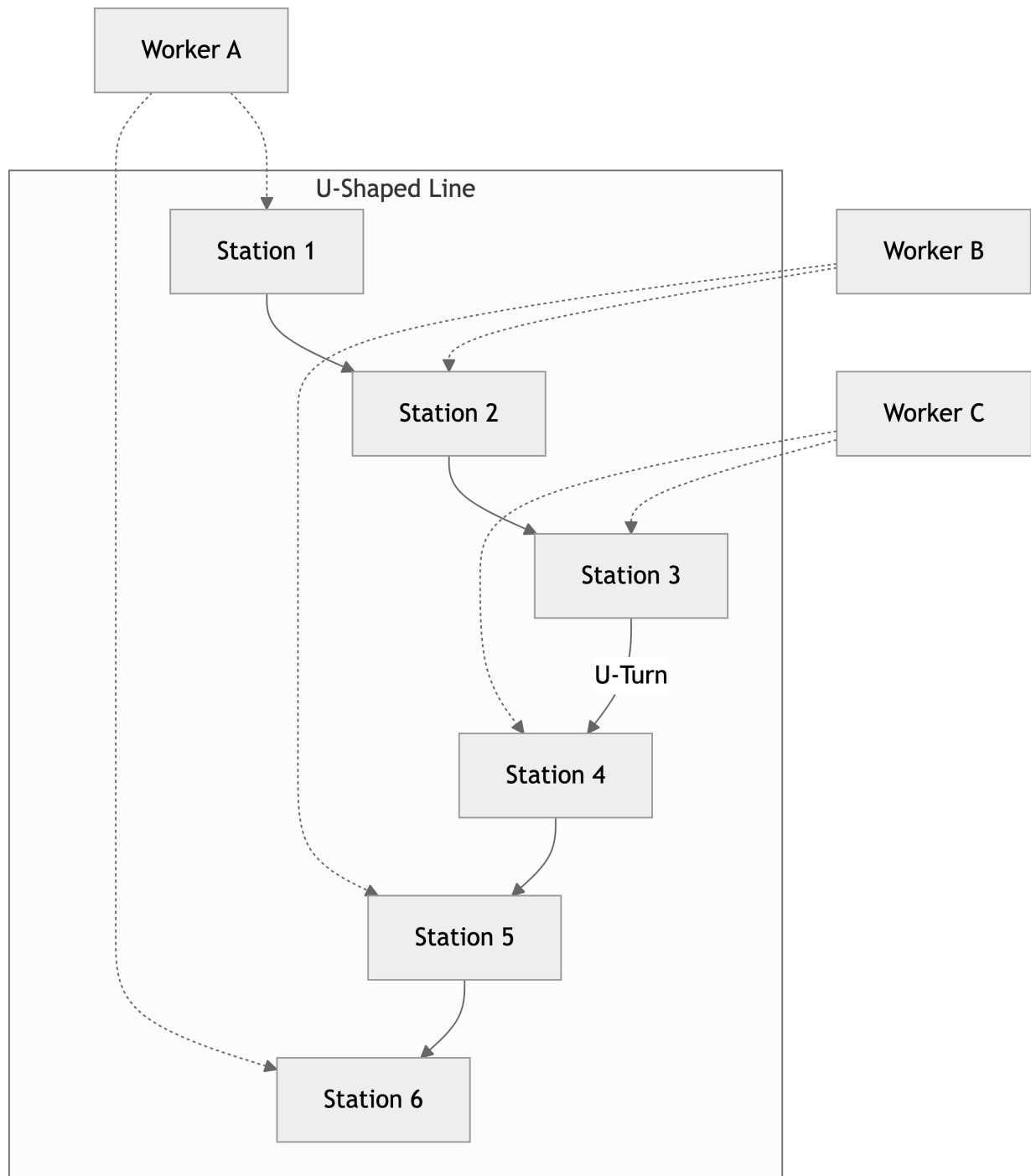


Figure 15.7.: U-Shaped Assembly Line Layout

- Sequence-dependent cleaning times
- Material expiration constraints
- Campaign scheduling for similar products

15.8.3. Operating Room Scheduling

- Surgeon and equipment availability
- Patient priority and urgency
- Uncertainty in procedure duration
- Emergency case accommodation

15.8.4. Typical Improvements

Metric	Typical Improvement
Efficiency gain	30-50%
Lead time reduction	15-25%
WIP inventory decrease	20-35%
On-time delivery improvement	25-40%

15.9. Handling Uncertainty

Real-world scheduling must handle disruptions, machine breakdowns, and processing time variability.

15.9.1. Reactive Scheduling

When disruptions occur – such as machine breakdowns or urgent order arrivals – the existing schedule must be updated quickly. Right-shift rescheduling delays affected operations by the duration of the disruption and propagates the delays through dependent operations. The following function implements this approach.

```
def reactive_reschedule(current_schedule, disruption, remaining_jobs):
    """Reschedule after disruption.

    Args:
        current_schedule: Original schedule
        disruption: (machine, time, duration) of breakdown
        remaining_jobs: Jobs not yet completed

    Returns:
        Updated schedule
    """
    machine, breakdown_time, repair_duration = disruption

    # Identify affected jobs
    affected = [job for job in remaining_jobs
                if job.machine == machine and job.start_time < breakdown_time]

    # Shift jobs on broken machine
    for job in current_schedule[machine]:
        if job.start_time >= breakdown_time:
            job.start_time += repair_duration
```

```
# Right-shift repair to maintain feasibility
propagate_delays(current_schedule, affected)

return current_schedule
```

15.9.2. Robust Scheduling

Build slack into schedules to absorb variability:

- Insert buffer time between operations
- Use predictive-reactive approaches
- Minimize expected makespan under uncertainty

15.9.3. Open-Source Job Shop Libraries

For ready-to-use JSSP implementations, **job-shop-lib** provides a modern Python toolkit with multiple solver backends (CP-SAT, simulated annealing, dispatching rules), GNN-based learned dispatching policies, Gymnasium-compatible RL environments, and built-in Gantt chart visualization. It is actively maintained and suitable for both research and prototyping.

An earlier effort, **JSSEnv**, was the first OpenAI Gym environment for JSSP and pioneered the RL-for-scheduling paradigm. While no longer actively maintained, it established the standard interface pattern that job-shop-lib and other libraries have since adopted. See Chapter 39 for installation details and code examples.

15.10. Summary

15.10.1. Key Takeaways

- Scheduling allocates resources to tasks over time to optimize objectives
- Problem types include single machine, parallel machines, flow shop, and job shop
- Common objectives: makespan, tardiness, flow time, utilization
- Dispatching rules provide fast heuristic solutions (SPT, EDD, ATC)
- Exact methods (MIP, CP) guarantee optimality for small to medium instances
- Metaheuristics (GA, SA, tabu search) handle large-scale problems
- Constraint programming excels at highly constrained scheduling
- Real-world applications span manufacturing, healthcare, and services
- Uncertainty handling requires reactive and robust scheduling approaches

15.11. Exercises

1. **Johnson's algorithm:** Apply Johnson's algorithm to a 2-machine flow shop with jobs: (3,2), (4,6), (2,5), (5,3), (1,4). Verify the makespan.
2. **MIP formulation:** Formulate a single machine scheduling problem with release times and due dates as a mixed integer program. Implement using Pyomo or PuLP.
3. **Dispatching rule comparison:** Implement SPT, EDD, and ATC rules. Generate 100 random instances with 20 jobs each. Compare average tardiness across rules.
4. **Genetic algorithm:** Implement a genetic algorithm for the job shop problem using the benchmark instance ft06 (6 jobs, 6 machines). Compare your results to known optimal (55).
5. **Reactive scheduling:** Design a reactive scheduling system that handles machine breakdowns. Simulate breakdowns following an exponential distribution and measure schedule robustness.

16. Resource Allocation

What You Will Learn

- Model assignment and allocation problems
- Solve facility location and capacity planning
- Optimize workforce planning and shift scheduling
- Apply cutting stock and bin packing algorithms

Resource allocation is the process of distributing limited resources among competing activities to achieve organizational objectives. This chapter covers the optimization models and algorithms that enable effective resource management across diverse application domains.

16.1. The Assignment Problem

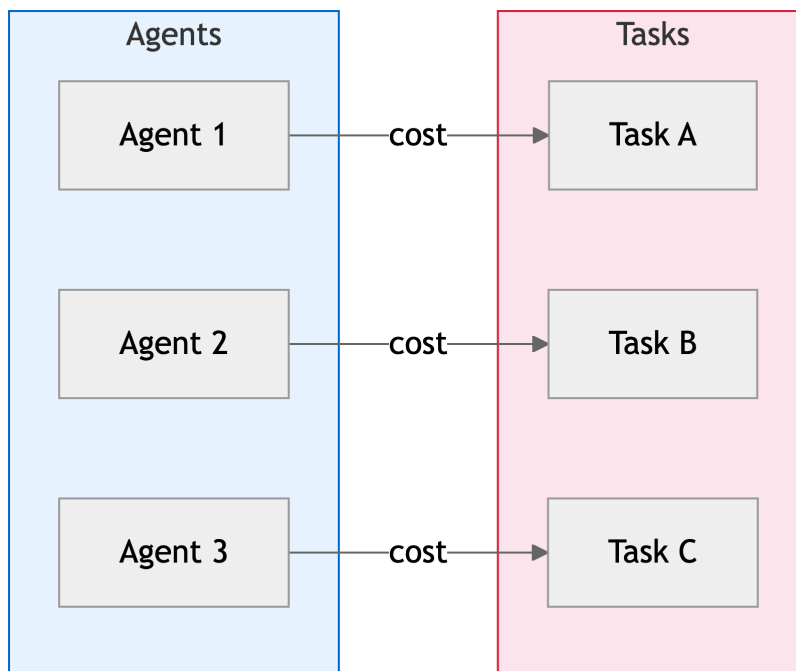


Figure 16.1.: Assignment Problem: Optimal matching between agents and tasks

The assignment problem is the foundation of resource allocation: assigning n agents to n tasks such that each agent performs exactly one task and each task is performed by exactly one agent.

16.1.1. Mathematical Formulation

$$\begin{aligned}
 & \text{minimize} && \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij} \\
 & \text{subject to} && \sum_{j=1}^n x_{ij} = 1, \quad \forall i \in \{1, \dots, n\} \\
 & && \sum_{i=1}^n x_{ij} = 1, \quad \forall j \in \{1, \dots, n\} \\
 & && x_{ij} \in \{0, 1\}
 \end{aligned}$$

where c_{ij} is the cost of assigning agent i to task j , and $x_{ij} = 1$ if and only if agent i is assigned to task j .

16.1.2. The Hungarian Algorithm

The Hungarian algorithm solves the assignment problem in polynomial time $O(n^3)$. SciPy provides an efficient implementation through `linear_sum_assignment`, which takes a cost matrix and returns the optimal row-column pairing.

```

import numpy as np
from scipy.optimize import linear_sum_assignment

def solve_assignment(cost_matrix):
    """
    Solve the assignment problem using the Hungarian algorithm.

    Args:
        cost_matrix: (n, n) array of assignment costs

    Returns:
        row_indices: Assigned row for each column
        col_indices: Assigned column for each row
        total_cost: Minimum total assignment cost
    """
    row_ind, col_ind = linear_sum_assignment(cost_matrix)
    total_cost = cost_matrix[row_ind, col_ind].sum()

    return row_ind, col_ind, total_cost

# Example: Assign workers to machines
cost_matrix = np.array([
    [10, 5, 13, 15], # Worker 1 costs for machines
    [3, 9, 18, 13], # Worker 2
    [10, 7, 3, 2], # Worker 3
    [5, 11, 9, 7] # Worker 4
])

rows, cols, cost = solve_assignment(cost_matrix)
print(f"Optimal assignment cost: {cost}")
for i, j in zip(rows, cols):
    print(f"Worker {i+1} -> Machine {j+1} (cost: {cost_matrix[i,j]})")

```

16.1.3. Generalized Assignment Problem

The standard assignment problem assumes a one-to-one matching, but many real problems involve agents with limited capacity serving multiple tasks. The Generalized Assignment Problem (GAP) allows multiple tasks per agent, subject to resource constraints.

$$\begin{aligned}
 &\text{minimize} && \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\
 &\text{subject to} && \sum_{j=1}^n r_{ij} x_{ij} \leq b_i, \quad \forall i \\
 &&& \sum_{i=1}^m x_{ij} = 1, \quad \forall j \\
 &&& x_{ij} \in \{0, 1\}
 \end{aligned}$$

where r_{ij} is the resource consumption and b_i is the capacity of agent i .

The following implementation uses OR-Tools to solve the GAP as a binary integer program. Each task is assigned to exactly one agent, and the total resource consumption at each agent cannot exceed its capacity.

```

from ortools.linear_solver import pywraplp

def generalized_assignment(costs, resources, capacities):
    """
    Solve the generalized assignment problem.

    Args:
        costs: (m, n) array of assignment costs
        resources: (m, n) array of resource requirements
        capacities: (m,) array of agent capacities

    Returns:
        assignments: List of (agent, task) tuples
        total_cost: Minimum total cost
    """
    m, n = costs.shape

    solver = pywraplp.Solver.CreateSolver('SCIP')
    if not solver:
        return None, None

    # Decision variables
    x = {}
    for i in range(m):
        for j in range(n):
            x[i, j] = solver.IntVar(0, 1, f'x_{i}_{j}')

    # Each task assigned to exactly one agent
    for j in range(n):
        solver.Add(sum(x[i, j] for i in range(m)) == 1)

    # Capacity constraints

```

```

for i in range(m):
    solver.Add(
        sum(resources[i, j] * x[i, j] for j in range(n)) <= capacities[i]
    )

# Objective: minimize total cost
solver.Minimize(
    sum(costs[i, j] * x[i, j] for i in range(m) for j in range(n))
)

status = solver.Solve()

if status == pywraplp.Solver.OPTIMAL:
    assignments = []
    for i in range(m):
        for j in range(n):
            if x[i, j].solution_value() > 0.5:
                assignments.append((i, j))
    return assignments, solver.Objective().Value()

return None, None

```

The two panels below illustrate the assignment problem structure. The left panel shows the bipartite graph with the optimal matching highlighted in red. The right panel displays the cost matrix as a heatmap, with starred cells marking the optimal assignments found by the Hungarian algorithm.

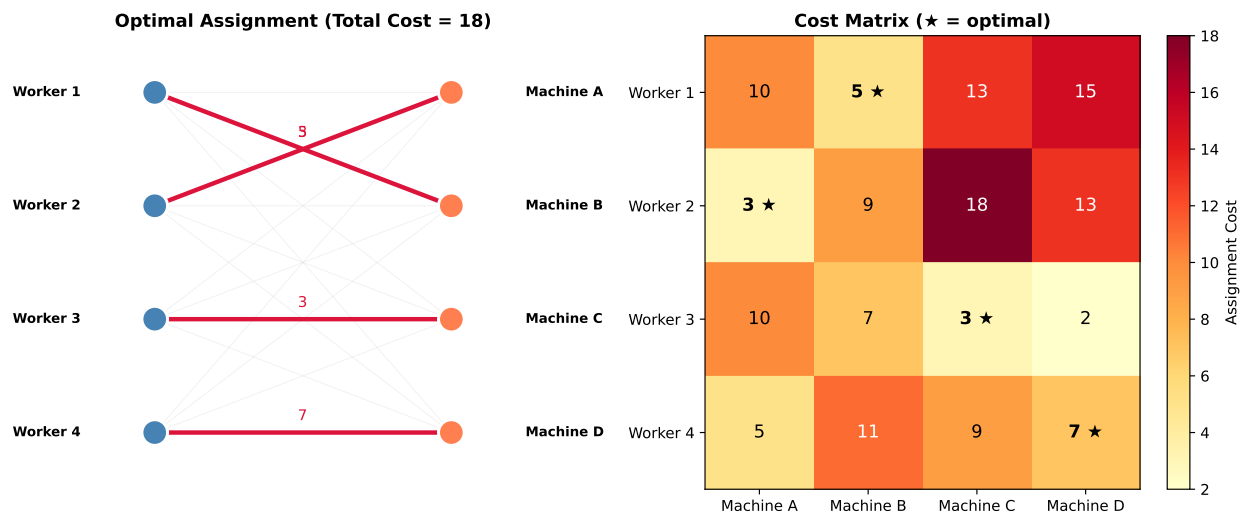


Figure 16.2.: Assignment problem as a bipartite graph with cost heatmap — the Hungarian algorithm finds the minimum cost perfect matching (highlighted in red)

The next visualization shows a facility location solution on a two-dimensional plane. Three out of five candidate sites are opened (shown as colored squares), and each customer is assigned to its nearest open facility. Closed candidate sites appear in gray.

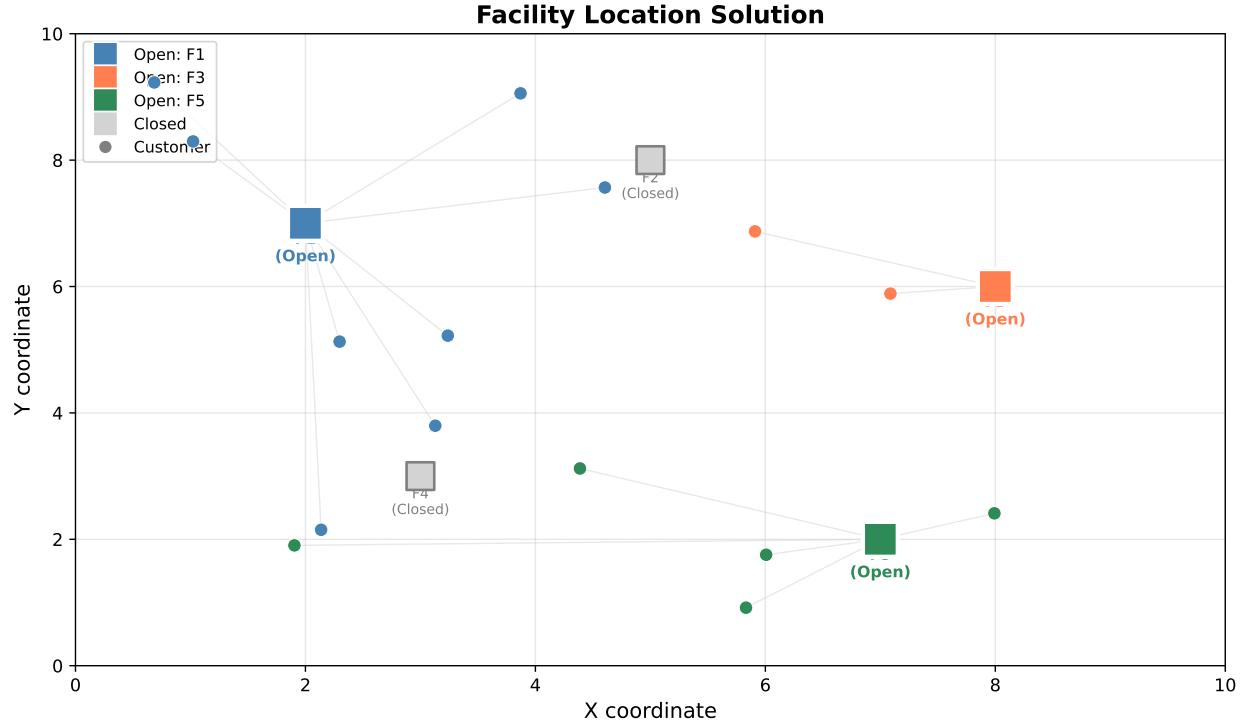


Figure 16.3.: Facility location solution — open facilities (green) serve assigned customers (blue) via allocation arrows, closed candidate sites shown in gray

16.2. Facility Location

Facility location problems determine where to locate facilities to serve customer demands cost-effectively.

16.2.1. Uncapacitated Facility Location

Choose which facilities to open and assign customers to facilities:

$$\begin{aligned}
 & \text{minimize} && \sum_{i \in F} f_i y_i + \sum_{i \in F} \sum_{j \in C} c_{ij} x_{ij} \\
 & \text{subject to} && \sum_{i \in F} x_{ij} = 1, \quad \forall j \in C \\
 & && x_{ij} \leq y_i, \quad \forall i \in F, j \in C \\
 & && y_i \in \{0, 1\}, \quad x_{ij} \geq 0
 \end{aligned}$$

where:

- F is the set of potential facility locations
- C is the set of customers
- f_i is the fixed cost of opening facility i
- c_{ij} is the cost of serving customer j from facility i
- $y_i = 1$ if facility i is opened
- x_{ij} is the fraction of customer j 's demand served by facility i

```

import numpy as np
from ortools.linear_solver import pywraplp

def uncapacitated_facility_location(fixed_costs, transport_costs):
    """
    Solve uncapacitated facility location problem.

    Args:
        fixed_costs: (m,) array of facility opening costs
        transport_costs: (m, n) array of customer-facility transport costs

    Returns:
        open_facilities: List of facility indices to open
        assignments: Dict mapping customer to serving facility
        total_cost: Optimal total cost
    """
    m = len(fixed_costs) # Number of potential facilities
    n = transport_costs.shape[1] # Number of customers

    solver = pywraplp.Solver.CreateSolver('SCIP')

    # Decision variables
    y = [solver.IntVar(0, 1, f'y_{i}') for i in range(m)]
    x = {}
    for i in range(m):
        for j in range(n):
            x[i, j] = solver.NumVar(0, 1, f'x_{i}_{j}')

    # Each customer fully assigned
    for j in range(n):
        solver.Add(sum(x[i, j] for i in range(m)) == 1)

    # Can only serve from open facilities
    for i in range(m):
        for j in range(n):
            solver.Add(x[i, j] <= y[i])

    # Objective
    objective = sum(fixed_costs[i] * y[i] for i in range(m))
    objective += sum(
        transport_costs[i, j] * x[i, j]
        for i in range(m) for j in range(n)
    )
    solver.Minimize(objective)

    status = solver.Solve()

    if status == pywraplp.Solver.OPTIMAL:
        open_facilities = [i for i in range(m) if y[i].solution_value() > 0.5]
        assignments = {}
        for j in range(n):
            for i in range(m):
                if x[i, j].solution_value() > 0.5:

```

```

        assignments[j] = i
        break
    return open_facilities, assignments, solver.Objective().Value()

return None, None, None

# Example
np.random.seed(42)
n_facilities = 5
n_customers = 20

fixed_costs = np.array([100, 80, 120, 90, 110]) # Opening costs
transport_costs = np.random.randint(5, 30, (n_facilities, n_customers))

facilities, assignments, cost = uncapacitated_facility_location(
    fixed_costs, transport_costs
)
print(f"Open facilities: {facilities}")
print(f"Total cost: {cost:.2f}")

```

16.2.2. Capacitated Facility Location

Add capacity constraints to limit how much each facility can serve:

$$x_{ij} \leq \frac{d_j}{q_i} y_i, \quad \forall i, j$$

$$\sum_{j \in C} d_j x_{ij} \leq q_i y_i, \quad \forall i$$

where d_j is customer j 's demand and q_i is facility i 's capacity.

16.2.3. P-Median Problem

Locate exactly p facilities to minimize total weighted distance:

$$\begin{aligned}
 & \text{minimize} && \sum_{i \in F} \sum_{j \in C} d_j \cdot \text{dist}_{ij} \cdot x_{ij} \\
 & \text{subject to} && \sum_{i \in F} y_i = p \\
 & && \sum_{i \in F} x_{ij} = 1, \quad \forall j \\
 & && x_{ij} \leq y_i, \quad \forall i, j
 \end{aligned}$$

Location Model Selection

Model	Fixed Cost	Capacity	Number of Facilities
Uncapacitated	Yes	No	Optimized
Capacitated	Yes	Yes	Optimized
P-Median	No	No	Fixed at p

P-Center Covering	No Optional	No Optional	Fixed, min-max objective Cover all demand within radius
----------------------	----------------	----------------	--

16.3. Capacity Planning

Capacity planning determines the production capacity needed to meet changing demands over time.

16.3.1. Design vs. Effective vs. Actual Capacity

Capacity Type	Definition	Example
Design Capacity	Maximum output under ideal conditions	1000 units/hour
Effective Capacity	Output accounting for maintenance, breaks	850 units/hour
Actual Capacity	Current output including inefficiencies	750 units/hour

Key Metrics:

$$\text{Utilization} = \frac{\text{Actual Output}}{\text{Design Capacity}} \times 100\%$$

$$\text{Efficiency} = \frac{\text{Actual Output}}{\text{Effective Capacity}} \times 100\%$$

16.3.2. Capacity Expansion Model

Determine when and how much to expand capacity over a planning horizon:

$$\begin{aligned}
 &\text{minimize} && \sum_{t=1}^T (c_t^{exp} x_t + c_t^{cap} C_t + c_t^{short} s_t) \\
 &\text{subject to} && C_t = C_{t-1} + x_t, \quad \forall t \\
 &&& C_t + s_t \geq D_t, \quad \forall t \\
 &&& x_t \in \{0, \Delta_1, \Delta_2, \dots\}
 \end{aligned}$$

where:

- C_t is capacity at time t
- D_t is demand at time t
- x_t is capacity addition at time t
- s_t is shortage at time t
- Δ_k are discrete expansion options

The following implementation translates this formulation into an OR-Tools MIP model. For each period, the solver decides whether to expand capacity (and by how much), balancing the one-time expansion cost against ongoing shortage and excess capacity costs.


```

import numpy as np
from ortools.linear_solver import pywraplp

def capacity_planning(demands, initial_capacity, expansion_costs,
                     expansion_sizes, holding_cost, shortage_cost):
    """
    Optimize capacity expansion over planning horizon.

    Args:
        demands: (T,) array of period demands
        initial_capacity: Starting capacity
        expansion_costs: (K,) costs for each expansion option
        expansion_sizes: (K,) capacity added for each option
        holding_cost: Cost per unit of excess capacity
        shortage_cost: Cost per unit of unmet demand

    Returns:
        expansions: List of (period, size_index) expansion decisions
        total_cost: Optimal cost
    """
    T = len(demands)
    K = len(expansion_sizes)

    solver = pywraplp.Solver.CreateSolver('SCIP')

    # Expansion decision: z[t, k] = 1 if expand by size k in period t
    z = {}
    for t in range(T):
        for k in range(K):
            z[t, k] = solver.IntVar(0, 1, f'z_{t}_{k}')

    # At most one expansion per period
    for t in range(T):
        solver.Add(sum(z[t, k] for k in range(K)) <= 1)

    # Capacity and shortage variables
    capacity = [solver.NumVar(0, solver.infinity(), f'C_{t}') for t in range(T)]
    shortage = [solver.NumVar(0, solver.infinity(), f's_{t}') for t in range(T)]
    excess = [solver.NumVar(0, solver.infinity(), f'e_{t}') for t in range(T)]

    # Capacity dynamics
    for t in range(T):
        if t == 0:
            prev_cap = initial_capacity
        else:
            prev_cap = capacity[t-1]

        expansion = sum(expansion_sizes[k] * z[t, k] for k in range(K))
        solver.Add(capacity[t] == prev_cap + expansion)

    # Demand satisfaction
    for t in range(T):
        solver.Add(capacity[t] + shortage[t] >= demands[t])

```

```

        solver.Add(excess[t] >= capacity[t] - demands[t])

# Objective
objective = 0
for t in range(T):
    # Expansion costs
    objective += sum(expansion_costs[k] * z[t, k] for k in range(K))
    # Shortage and holding costs
    objective += shortage_cost * shortage[t]
    objective += holding_cost * excess[t]

solver.Minimize(objective)

status = solver.Solve()

if status == pywraplp.Solver.OPTIMAL:
    expansions = []
    for t in range(T):
        for k in range(K):
            if z[t, k].solution_value() > 0.5:
                expansions.append((t, k, expansion_sizes[k]))
    return expansions, solver.Objective().Value()

return None, None

# Example
demands = np.array([100, 120, 150, 180, 200, 220, 250, 280])
initial_capacity = 80
expansion_sizes = np.array([50, 100, 150])
expansion_costs = np.array([500, 800, 1100])

expansions, cost = capacity_planning(
    demands, initial_capacity, expansion_costs, expansion_sizes,
    holding_cost=2, shortage_cost=20
)
print(f"Expansion decisions: {expansions}")
print(f"Total cost: {cost:.2f}")

```

16.3.3. Bottleneck Management

The capacity of an entire system is limited by its bottleneck—the resource with the smallest capacity. The Theory of Constraints provides a framework for bottleneck management:

1. **Identify** the bottleneck
2. **Exploit** the bottleneck (maximize its utilization)
3. **Subordinate** other resources to the bottleneck
4. **Expand** the bottleneck (increase its capacity)
5. **Repeat** (a new bottleneck will emerge)

i Theory of Constraints

An hour lost at the bottleneck is an hour lost for the entire system. An hour saved at a non-bottleneck is only an illusion.

16.4. Workforce Scheduling

Workforce scheduling assigns employees to shifts, tasks, and time slots while respecting labor regulations and skill requirements.

16.4.1. Shift Scheduling Problem

Determine how many employees to assign to each shift to cover demand:

$$\begin{aligned}
 & \text{minimize} && \sum_{s \in S} c_s x_s \\
 & \text{subject to} && \sum_{s \in S} a_{st} x_s \geq d_t, \quad \forall t \in T \\
 & && x_s \geq 0 \text{ and integer}
 \end{aligned}$$

where:

- S is the set of possible shifts
- T is the set of time periods
- $a_{st} = 1$ if shift s covers period t
- d_t is the minimum staffing requirement in period t
- c_s is the cost of shift s
- x_s is the number of employees assigned to shift s

The following CP-SAT model finds the minimum-cost shift assignment that covers all demand periods. Each shift variable represents how many employees work that shift, and coverage constraints ensure every time period meets its staffing requirement.

```

import numpy as np
from ortools.sat.python import cp_model

def shift_scheduling(demand, shift_definitions, shift_costs, max_employees=100):
    """
    Solve the shift scheduling problem.

    Args:
        demand: (T,) array of staffing requirements per period
        shift_definitions: List of (start_period, end_period) tuples
        shift_costs: (S,) array of shift costs
        max_employees: Maximum employees per shift

    Returns:
        shift_assignments: Number of employees per shift
        total_cost: Optimal cost
    """
    T = len(demand)
    S = len(shift_definitions)

    model = cp_model.CpModel()

    # Decision variables: number of employees per shift
    x = [model.NewIntVar(0, max_employees, f'shift_{s}') for s in range(S)]

    # Coverage constraints

```

```

for t in range(T):
    coverage = []
    for s, (start, end) in enumerate(shift_definitions):
        if start <= t < end:
            coverage.append(x[s])
    model.Add(sum(coverage) >= demand[t])

# Objective: minimize total cost
model.Minimize(sum(shift_costs[s] * x[s] for s in range(S)))

solver = cp_model.CpSolver()
status = solver.Solve(model)

if status == cp_model.OPTIMAL:
    assignments = [solver.Value(x[s]) for s in range(S)]
    return assignments, solver.ObjectiveValue()

return None, None

# Example: Call center staffing
# Periods: 8am-10am, 10am-12pm, 12pm-2pm, 2pm-4pm, 4pm-6pm, 6pm-8pm
demand = [5, 8, 12, 10, 8, 4] # Staff needed per period

# Shifts: (start_period, end_period)
shifts = [
    (0, 4), # Morning: 8am-4pm
    (1, 5), # Mid-day: 10am-6pm
    (2, 6), # Afternoon: 12pm-8pm
    (0, 3), # Half morning: 8am-2pm
    (3, 6), # Half afternoon: 2pm-8pm
]

shift_costs = [80, 80, 80, 40, 40] # Full shifts cost more

assignments, cost = shift_scheduling(demand, shifts, shift_costs)
print(f"Shift assignments: {assignments}")
print(f"Total cost: {cost}")

```

16.4.2. Nurse Scheduling Problem

The nurse scheduling problem extends shift scheduling by tracking individual employees. Each nurse must be assigned to at most one shift per day, cannot work a morning shift immediately after a night shift, and is limited to a maximum number of shifts per week. The objective minimizes the workload imbalance across nurses to ensure fair distribution.

```

from ortools.sat.python import cp_model

def nurse_scheduling(num_nurses, num_days, num_shifts, constraints):
    """
    Solve the nurse scheduling problem with various constraints.

    Args:
        num_nurses: Number of nurses
    """

```

```

    num_days: Number of days to schedule
    num_shifts: Number of shifts per day
    constraints: Dict with staffing requirements and rules

Returns:
    schedule: 3D array of assignments
"""
model = cp_model.CpModel()

# Decision variables: nurse n works shift s on day d
x = {}
for n in range(num_nurses):
    for d in range(num_days):
        for s in range(num_shifts):
            x[n, d, s] = model.NewBoolVar(f'x_{n}_{d}_{s}')

# Each nurse works at most one shift per day
for n in range(num_nurses):
    for d in range(num_days):
        model.Add(sum(x[n, d, s] for s in range(num_shifts)) <= 1)

# Staffing requirements per shift
min_staff = constraints.get('min_staff', 2)
for d in range(num_days):
    for s in range(num_shifts):
        model.Add(
            sum(x[n, d, s] for n in range(num_nurses)) >= min_staff
        )

# No consecutive night shifts followed by morning shift
night_shift = num_shifts - 1
morning_shift = 0
for n in range(num_nurses):
    for d in range(num_days - 1):
        model.Add(x[n, d, night_shift] + x[n, d+1, morning_shift] <= 1)

# Maximum shifts per week
max_shifts = constraints.get('max_shifts_per_week', 5)
for n in range(num_nurses):
    for week_start in range(0, num_days, 7):
        week_end = min(week_start + 7, num_days)
        model.Add(
            sum(x[n, d, s]
                for d in range(week_start, week_end)
                for s in range(num_shifts)) <= max_shifts
        )

# Fair distribution: minimize difference between nurses
shifts_worked = [
    sum(x[n, d, s] for d in range(num_days) for s in range(num_shifts))
    for n in range(num_nurses)
]
max_worked = model.NewIntVar(0, num_days * num_shifts, 'max_worked')

```

```

min_worked = model.NewIntVar(0, num_days * num_shifts, 'min_worked')

model.AddMaxEquality(max_worked, shifts_worked)
model.AddMinEquality(min_worked, shifts_worked)
model.Minimize(max_worked - min_worked)

solver = cp_model.CpSolver()
solver.parameters.max_time_in_seconds = 30.0
status = solver.Solve(model)

if status in [cp_model.OPTIMAL, cp_model.FEASIBLE]:
    schedule = np.zeros((num_nurses, num_days, num_shifts), dtype=int)
    for n in range(num_nurses):
        for d in range(num_days):
            for s in range(num_shifts):
                schedule[n, d, s] = solver.Value(x[n, d, s])
    return schedule

return None

# Example
schedule = nurse_scheduling(
    num_nurses=10,
    num_days=14,
    num_shifts=3,
    constraints={'min_staff': 3, 'max_shifts_per_week': 5}
)
if schedule is not None:
    print(f"Schedule shape: {schedule.shape}")
    print(f"Total shifts assigned: {schedule.sum()}")

```

16.5. Strategic Manpower Planning

While shift scheduling handles short-term staffing, strategic manpower planning addresses long-term workforce decisions over months or years. It determines hiring, training, promotion, and workforce development strategies to meet changing business needs.

16.5.1. Planning Horizons



Figure 16.4.: Workforce Planning Horizons

Horizon	Timeframe	Decisions	Tools
Short-term	Days to weeks	Shift assignments, overtime	CP, LP
Medium-term	Months to quarters	Hiring/firing, temp workers	LP, simulation
Long-term	1-5+ years	Skill strategy, succession	MILP, scenario planning

16.5.2. Workforce Demand Forecasting

Strategic planning begins with forecasting future workforce requirements.

Volume-Based Demand:

$$D_{t,s} = \frac{V_t \cdot r_s}{\eta_s \cdot h_t}$$

where: - $D_{t,s}$ = workers needed with skill s in period t - V_t = production volume in period t - r_s = skill ratio (proportion of work requiring skill s) - η_s = productivity factor for skill s - h_t = working hours per worker in period t

The classes below implement volume-based demand forecasting, attrition projections, and hiring gap calculations. These form the analytical foundation for the optimization model that follows.

Attrition Modeling:

$$W_{t+1} = W_t \cdot (1 - a_t) + H_t - F_t$$

where: - W_t = workforce at start of period t - a_t = attrition rate in period t - H_t = new hires in period t - F_t = involuntary separations in period t

```
import numpy as np
from typing import List, Dict, Tuple, Optional
from dataclasses import dataclass
from enum import Enum

@dataclass
class SkillLevel:
    """Represents a skill and its proficiency level."""
    skill_id: str
    name: str
    level: int # 1=basic, 2=intermediate, 3=advanced

@dataclass
class Employee:
    """Represents an employee with skills and attributes."""
    id: int
    name: str
    skills: Dict[str, int] # skill_id -> proficiency level
    years_experience: float
    annual_cost: float
    trainable_skills: List[str]

@dataclass
class JobRole:
    """Represents a job role with skill requirements."""
    id: str
    name: str
    required_skills: Dict[str, int] # skill_id -> min proficiency
    headcount_needed: Dict[int, int] # period -> count needed

class WorkforceDemandForecaster:
    """
    Forecast workforce demand based on business drivers.
    """
```

```

"""

def __init__(self, base_workforce: int, skills: List[str]):
    """
    Initialize forecaster.

    Args:
        base_workforce: Current workforce size
        skills: List of skill identifiers
    """
    self.base_workforce = base_workforce
    self.skills = skills
    self.attrition_rate = 0.10 # 10% annual attrition

def forecast_volume_based(self, production_volumes: List[float],
                           productivity_factor: float = 1.0,
                           hours_per_period: float = 2080
                           ) -> List[int]:
    """
    Forecast workforce needs based on production volume.

    Args:
        production_volumes: Expected production per period
        productivity_factor: Output per worker-hour
        hours_per_period: Working hours per period

    Returns:
        Required headcount per period
    """
    demand = []
    for volume in production_volumes:
        workers_needed = volume / (productivity_factor * hours_per_period)
        demand.append(int(np.ceil(workers_needed)))
    return demand

def project_workforce_with_attrition(self, periods: int,
                                     initial_workforce: int = None
                                     ) -> List[float]:
    """
    Project workforce size accounting for natural attrition.

    Args:
        periods: Number of periods to project
        initial_workforce: Starting workforce (default: base)

    Returns:
        Projected workforce per period
    """
    if initial_workforce is None:
        initial_workforce = self.base_workforce

    workforce = [initial_workforce]
    for _ in range(periods - 1):

```



```

        # Apply attrition (per period rate)
        period_attrition = self.attrition_rate / 12 # Monthly
        next_workforce = workforce[-1] * (1 - period_attrition)
        workforce.append(next_workforce)

    return workforce

def calculate_hiring_needs(self, demand: List[int],
                          supply: List[float]) -> List[int]:
    """
    Calculate hiring needs to meet demand given supply projection.

    Args:
        demand: Required headcount per period
        supply: Projected available workforce per period

    Returns:
        Net hiring needs per period
    """
    hiring_needs = []
    for d, s in zip(demand, supply):
        gap = d - s
        hiring_needs.append(max(0, int(np.ceil(gap))))
    return hiring_needs

```

16.5.3. Skill Inventory and Gap Analysis

Understanding current capabilities and identifying gaps is essential for strategic planning. The `SkillInventoryAnalyzer` class builds a skill matrix from employee data, computes coverage statistics at each proficiency level, and identifies role-specific gaps. It also recommends training priorities by matching employees with trainable skills to the largest gaps.

```

class SkillInventoryAnalyzer:
    """
    Analyze workforce skill inventory and identify gaps.
    """

    def __init__(self, employees: List[Employee], roles: List[JobRole]):
        """
        Initialize analyzer.

        Args:
            employees: Current workforce
            roles: Job roles with requirements
        """
        self.employees = employees
        self.roles = roles
        self._build_skill_matrix()

    def _build_skill_matrix(self):
        """Build skill supply matrix."""
        all_skills = set()
        for emp in self.employees:

```

```

        all_skills.update(emp.skills.keys())
    for role in self.roles:
        all_skills.update(role.required_skills.keys())

    self.all_skills = sorted(all_skills)
    self.n_skills = len(self.all_skills)
    self.n_employees = len(self.employees)

    # Skill matrix: employees x skills x levels
    self.skill_matrix = np.zeros((self.n_employees, self.n_skills))
    for i, emp in enumerate(self.employees):
        for j, skill in enumerate(self.all_skills):
            self.skill_matrix[i, j] = emp.skills.get(skill, 0)

def get_skill_coverage(self) -> Dict[str, Dict[int, int]]:
    """
    Get count of employees at each skill level.

    Returns:
        Dict mapping skill_id to {level: count}
    """
    coverage = {}
    for j, skill in enumerate(self.all_skills):
        coverage[skill] = {}
        for level in range(4): # 0, 1, 2, 3
            count = np.sum(self.skill_matrix[:, j] >= level)
            coverage[skill][level] = int(count)
    return coverage

def analyze_role_gaps(self, period: int = 0) -> Dict[str, Dict]:
    """
    Analyze skill gaps for each role in a given period.

    Args:
        period: Planning period to analyze

    Returns:
        Gap analysis by role
    """
    gaps = {}

    for role in self.roles:
        role_gaps = {
            'name': role.name,
            'needed': role.headcount_needed.get(period, 0),
            'skill_gaps': {}
        }

        for skill_id, min_level in role.required_skills.items():
            # Count employees meeting this requirement
            if skill_id in self.all_skills:
                skill_idx = self.all_skills.index(skill_id)
                qualified = np.sum(self.skill_matrix[:, skill_idx] >= min_level)

```

```

        else:
            qualified = 0

            needed = role.headcount_needed.get(period, 0)
            gap = max(0, needed - qualified)

            role_gaps['skill_gaps'][skill_id] = {
                'required_level': min_level,
                'qualified_employees': qualified,
                'gap': gap
            }

        gaps[role.id] = role_gaps

    return gaps

def identify_training_priorities(self, period: int = 0
                                ) -> List[Tuple[str, str, int, int]]:
    """
    Identify priority training needs.

    Returns:
        List of (employee_name, skill, current_level, target_level)
    """
    gaps = self.analyze_role_gaps(period)
    priorities = []

    # Find skills with largest gaps
    skill_gaps = {}
    for role_id, role_data in gaps.items():
        for skill_id, skill_data in role_data['skill_gaps'].items():
            if skill_id not in skill_gaps:
                skill_gaps[skill_id] = {
                    'gap': 0,
                    'target_level': 0
                }
            skill_gaps[skill_id]['gap'] += skill_data['gap']
            skill_gaps[skill_id]['target_level'] = max(
                skill_gaps[skill_id]['target_level'],
                skill_data['required_level']
            )

    # Match employees to training opportunities
    for skill_id, gap_data in sorted(skill_gaps.items(),
                                    key=lambda x: -x[1]['gap']):
        if gap_data['gap'] <= 0:
            continue

        target_level = gap_data['target_level']

        # Find employees who can be trained
        for emp in self.employees:
            current = emp.skills.get(skill_id, 0)

```

```

        if current < target_level and skill_id in emp.trainable_skills:
            priorities.append((
                emp.name,
                skill_id,
                current,
                target_level
            ))

    return priorities

def generate_skill_matrix_report(self) -> str:
    """Generate a text report of the skill matrix."""
    lines = ["SKILL INVENTORY REPORT", "=" * 60]

    # Coverage summary
    coverage = self.get_skill_coverage()
    lines.append("\nSkill Coverage Summary:")
    lines.append("-" * 40)
    lines.append(f"{'Skill':<15} {'L0':>6} {'L1':>6} {'L2':>6} {'L3':>6}")
    lines.append("-" * 40)

    for skill in self.all_skills:
        levels = coverage[skill]
        lines.append(f"{'skill':<15} {levels[0]:>6} {levels[1]:>6} "
                    f"{levels[2]:>6} {levels[3]:>6}")

    return "\n".join(lines)

```

16.5.4. Long-term Workforce Planning Model

The strategic workforce planning problem optimizes hiring, training, and workforce composition over multiple periods.

Parameters:

- T : Number of planning periods
- S : Set of skills
- L : Set of proficiency levels
- $D_{t,s}$: Demand for skill s in period t
- c_h : Cost to hire one employee
- c_f : Cost to fire/layoff one employee
- $c_{train,s,l}$: Cost to train skill s to level l
- $t_{train,s,l}$: Time to train skill s to level l
- w : Wage per employee per period
- a : Attrition rate per period

Decision Variables:

- H_t : Employees hired in period t
- F_t : Employees fired in period t
- $TR_{t,s,l}$: Employees trained in skill s to level l in period t
- W_t : Total workforce in period t
- $E_{t,s,l}$: Employees with skill s at level l in period t

Formulation:

$$\begin{aligned}
& \text{minimize} && \sum_{t=1}^T \left(c_h \cdot H_t + c_f \cdot F_t + w \cdot W_t + \sum_{s,l} c_{train,s,l} \cdot TR_{t,s,l} \right) \\
& \text{subject to} && W_t = W_{t-1}(1-a) + H_t - F_t, \quad \forall t \\
& && \sum_{l \geq l_{min}} E_{t,s,l} \geq D_{t,s}, \quad \forall t, s \\
& && \sum_{s,l} E_{t,s,l} = W_t, \quad \forall t \\
& && E_{t,s,l} = E_{t-1,s,l}(1-a) + TR_{t-k,s,l}, \quad \forall t, s, l \\
& && \sum_{s,l} TR_{t,s,l} \leq \text{TrainingCapacity}, \quad \forall t \\
& && H_t, F_t, TR_{t,s,l}, W_t, E_{t,s,l} \geq 0
\end{aligned}$$

The `ManpowerPlanner` class translates this formulation into an OR-Tools linear program. It tracks workforce size, skill distributions, and training pipelines across all planning periods. A high shortage penalty drives the solver to meet demand through hiring and training rather than accepting unfilled positions.

```

from ortools.linear_solver import pywraplp

class ManpowerPlanner:
    """
    Strategic workforce planning optimizer.

    Optimizes hiring, training, and workforce composition
    over a multi-period planning horizon.
    """

    def __init__(self, n_periods: int, skills: List[str],
                 initial_workforce: int):
        """
        Initialize the planner.

        Args:
            n_periods: Number of planning periods
            skills: List of skill identifiers
            initial_workforce: Starting headcount
        """
        self.n_periods = n_periods
        self.skills = skills
        self.n_skills = len(skills)
        self.initial_workforce = initial_workforce

        # Default costs (can be customized)
        self.hiring_cost = 5000
        self.firing_cost = 10000
        self.wage_per_period = 8000
        self.training_cost = 2000 # per skill level
        self.overtime_cost = 1.5 # multiplier
        self.attrition_rate = 0.02 # per period

```

```

# Skill demands (period x skill)
self.demand = np.zeros((n_periods, self.n_skills))

# Training capacity
self.training_capacity = 20 # employees per period

# Skill matrix for current workforce
self.current_skills = {}

def set_demand(self, demand_matrix: np.ndarray):
    """Set skill demand for all periods."""
    self.demand = demand_matrix

def set_costs(self, hiring: float = None, firing: float = None,
              wage: float = None, training: float = None):
    """Update cost parameters."""
    if hiring is not None:
        self.hiring_cost = hiring
    if firing is not None:
        self.firing_cost = firing
    if wage is not None:
        self.wage_per_period = wage
    if training is not None:
        self.training_cost = training

def solve(self, time_limit: float = 60.0) -> Dict:
    """
    Solve the workforce planning problem.

    Args:
        time_limit: Solver time limit in seconds

    Returns:
        Solution dictionary with hiring/training plan
    """
    solver = pywraplp.Solver.CreateSolver('SCIP')
    if not solver:
        return None

    INF = solver.infinity()

    # Decision variables
    # W[t] = total workforce in period t
    W = [solver.NumVar(0, INF, f'W_{t}') for t in range(self.n_periods)]

    # H[t] = hires in period t
    H = [solver.IntVar(0, INF, f'H_{t}') for t in range(self.n_periods)]

    # F[t] = fires in period t
    F = [solver.IntVar(0, INF, f'F_{t}') for t in range(self.n_periods)]

    # E[t, s] = employees with skill s in period t

```

```

E = {}
for t in range(self.n_periods):
    for s in range(self.n_skills):
        E[t, s] = solver.NumVar(0, INF, f'E_{t}_{s}')

# TR[t, s] = employees trained in skill s in period t
TR = {}
for t in range(self.n_periods):
    for s in range(self.n_skills):
        TR[t, s] = solver.IntVar(0, INF, f'TR_{t}_{s}')

# Shortage[t, s] = unmet demand for skill s in period t
Shortage = {}
for t in range(self.n_periods):
    for s in range(self.n_skills):
        Shortage[t, s] = solver.NumVar(0, INF, f'Short_{t}_{s}')

# Constraints

# Workforce balance
for t in range(self.n_periods):
    if t == 0:
        prev_W = self.initial_workforce
    else:
        prev_W = W[t-1]

    solver.Add(
        W[t] == prev_W * (1 - self.attrition_rate) + H[t] - F[t]
    )

# Skill coverage constraints
for t in range(self.n_periods):
    for s in range(self.n_skills):
        # Demand must be met (with possible shortage)
        solver.Add(E[t, s] + Shortage[t, s] >= self.demand[t, s])

# Skill supply dynamics
if t == 0:
    # Initialize from current workforce distribution
    initial_skilled = self.initial_workforce / self.n_skills
    solver.Add(
        E[t, s] == initial_skilled * (1 - self.attrition_rate) +
        TR[t, s]
    )
else:
    # Account for new hires (assume uniform skill distribution)
    new_hire_contribution = H[t] / self.n_skills
    solver.Add(
        E[t, s] == E[t-1, s] * (1 - self.attrition_rate) +
        TR[t, s] + new_hire_contribution
    )

# Training capacity constraint

```

```

for t in range(self.n_periods):
    solver.Add(
        sum(TR[t, s] for s in range(self.n_skills)) <=
        ↪ self.training_capacity
    )

# Total skilled cannot exceed workforce
for t in range(self.n_periods):
    # Note: employees can have multiple skills, so this is relaxed
    pass

# Objective: minimize total cost
objective = 0

for t in range(self.n_periods):
    # Hiring and firing costs
    objective += self.hiring_cost * H[t]
    objective += self.firing_cost * F[t]

    # Wage costs
    objective += self.wage_per_period * W[t]

    # Training costs
    for s in range(self.n_skills):
        objective += self.training_cost * TR[t, s]

    # Shortage penalty (high cost)
    shortage_penalty = 20000 # per unfilled position
    for s in range(self.n_skills):
        objective += shortage_penalty * Shortage[t, s]

solver.Minimize(objective)

# Solve
solver.SetTimeLimit(int(time_limit * 1000))
status = solver.Solve()

if status in [pywraplp.Solver.OPTIMAL, pywraplp.Solver.FEASIBLE]:
    result = {
        'status': 'optimal' if status == pywraplp.Solver.OPTIMAL else
        ↪ 'feasible',
        'total_cost': solver.Objective().Value(),
        'workforce': [W[t].solution_value() for t in range(self.n_periods)],
        'hiring': [int(H[t].solution_value()) for t in
        ↪ range(self.n_periods)],
        'firing': [int(F[t].solution_value()) for t in
        ↪ range(self.n_periods)],
        'training': {
            self.skills[s]: [int(TR[t, s].solution_value())
            for t in range(self.n_periods)]
            for s in range(self.n_skills)
        },
        'shortages': {

```



```

        self.skills[s]: [Shortage[t, s].solution_value()
                        for t in range(self.n_periods)]
        for s in range(self.n_skills)
    }
}

# Calculate cost breakdown
result['cost_breakdown'] = {
    'hiring': sum(self.hiring_cost * result['hiring'][t]
                  for t in range(self.n_periods)),
    'firing': sum(self.firing_cost * result['firing'][t]
                  for t in range(self.n_periods)),
    'wages': sum(self.wage_per_period * result['workforce'][t]
                  for t in range(self.n_periods)),
    'training': sum(
        self.training_cost * sum(result['training'][s][t]
                                  for s in self.skills)
        for t in range(self.n_periods)
    )
}

return result

return {'status': 'infeasible', 'total_cost': None}

def scenario_analysis(self, scenarios: Dict[str, np.ndarray]) -> Dict[str,
↪ Dict]:
    """
    Analyze multiple demand scenarios.

    Args:
        scenarios: Dict mapping scenario name to demand matrix

    Returns:
        Results for each scenario
    """
    results = {}
    for name, demand in scenarios.items():
        self.set_demand(demand)
        result = self.solve()
        results[name] = result
    return results

class SuccessionPlanner:
    """
    Plan succession for critical roles.
    """

    def __init__(self, employees: List[Employee], critical_roles: List[JobRole]):
        """
        Initialize succession planner.

```

```

    Args:
        employees: Current workforce
        critical_roles: Roles requiring succession planning
    """
    self.employees = employees
    self.critical_roles = critical_roles

def identify_successors(self, role: JobRole,
                        readiness_threshold: float = 0.7
                        ) -> List[Tuple[Employee, float, List[str]]]:
    """
    Identify potential successors for a critical role.

    Args:
        role: Target role
        readiness_threshold: Minimum readiness score (0-1)

    Returns:
        List of (employee, readiness_score, development_needs)
    """
    successors = []

    for emp in self.employees:
        # Calculate readiness score
        met_skills = 0
        total_skills = len(role.required_skills)
        gaps = []

        for skill_id, min_level in role.required_skills.items():
            emp_level = emp.skills.get(skill_id, 0)
            if emp_level >= min_level:
                met_skills += 1
            else:
                gaps.append(f"{skill_id} (need L{min_level}, have
↪ L{emp_level})")

        readiness = met_skills / total_skills if total_skills > 0 else 0

        if readiness >= readiness_threshold:
            successors.append((emp, readiness, gaps))

    # Sort by readiness (descending)
    successors.sort(key=lambda x: -x[1])

    return successors

def generate_succession_plan(self) -> Dict[str, Dict]:
    """
    Generate succession plan for all critical roles.

    Returns:
        Succession plan by role
    """

```

```

plan = {}

for role in self.critical_roles:
    successors = self.identify_successors(role)

    plan[role.id] = {
        'role_name': role.name,
        'successors': [
            {
                'name': emp.name,
                'readiness': f"{score * 100:.0f}%",
                'development_needs': gaps
            }
            for emp, score, gaps in successors[:3] # Top 3
        ],
        'risk_level': 'High' if len(successors) == 0 else
                     'Medium' if len(successors) < 2 else 'Low'
    }

return plan

# Example: Strategic workforce planning
def run_manpower_planning_example():
    """Demonstrate strategic workforce planning."""

    print("=" * 60)
    print("STRATEGIC MANPOWER PLANNING")
    print("=" * 60)

    # Define planning horizon and skills
    n_periods = 12 # Monthly for 1 year
    skills = ['Engineering', 'Operations', 'Sales', 'Analytics']
    initial_workforce = 100

    # Create demand forecast (growth scenario)
    np.random.seed(42)
    base_demand = np.array([30, 40, 20, 10]) # Current demand per skill
    growth_rate = 0.03 # 3% monthly growth

    demand = np.zeros((n_periods, len(skills)))
    for t in range(n_periods):
        demand[t] = base_demand * (1 + growth_rate) ** t
        # Add seasonal variation
        seasonal = 1 + 0.1 * np.sin(2 * np.pi * t / 12)
        demand[t] *= seasonal

    demand = np.ceil(demand)

    print(f"\nPlanning Horizon: {n_periods} periods")
    print(f"Skills: {skills}")
    print(f"Initial Workforce: {initial_workforce}")

```

```

# Solve
planner = ManpowerPlanner(n_periods, skills, initial_workforce)
planner.set_demand(demand)
result = planner.solve()

if result['status'] in ['optimal', 'feasible']:
    print(f"\nSolution Status: {result['status']}")
    print(f"Total Cost: ${result['total_cost']:,.0f}")

    print("\nCost Breakdown:")
    for category, cost in result['cost_breakdown'].items():
        print(f"    {category.capitalize()}: ${cost:,.0f}")

    print("\nWorkforce Plan:")
    print(f"{'Period':<8} {'Workforce':<12} {'Hiring':<10} {'Firing':<10}")
    print("-" * 40)
    for t in range(n_periods):
        print(f"{'t+1':<8} {result['workforce'][t]:<12.0f} "
              f"{result['hiring'][t]:<10} {result['firing'][t]:<10}")

    print("\nTraining Plan (employees to train per period):")
    for skill in skills:
        training = result['training'][skill]
        total_trained = sum(training)
        print(f"    {skill}: {total_trained} total, "
              f"periods with training: {sum(1 for t in training if t > 0)}")

# Scenario analysis
print("\n" + "=" * 60)
print("SCENARIO ANALYSIS")
print("-" * 60)

# Create scenarios
scenarios = {
    'Baseline': demand,
    'High Growth': demand * 1.2,
    'Low Growth': demand * 0.8,
    'Skill Shift': np.column_stack([
        demand[:, 0] * 1.5, # More engineering
        demand[:, 1] * 0.8,
        demand[:, 2] * 0.9,
        demand[:, 3] * 1.3 # More analytics
    ])
}

scenario_results = planner.scenario_analysis(scenarios)

print(f"{'Scenario':<15} {'Total Cost':<15} {'Total Hiring':<15}")
print("-" * 45)
for name, res in scenario_results.items():
    if res['total_cost']:
        total_hiring = sum(res['hiring'])
        print(f"{name:<15} ${res['total_cost']:<14.0f} {total_hiring:<15}")

```

```

    return result

if __name__ == "__main__":
    run_manpower_planning_example()

```

16.5.5. Succession Planning

Succession planning ensures organizational continuity by identifying and developing future leaders. The `SuccessionPlanner` class (defined above) scores each employee's readiness for a target role based on how many required skills they meet. The example below applies this to identify successors for CTO and VP Engineering roles, flagging development needs and risk levels.

```

def succession_planning_example():
    """Demonstrate succession planning."""

    print("\n" + "=" * 60)
    print("SUCCESSION PLANNING")
    print("=" * 60)

    # Sample employees
    employees = [
        Employee(1, "Alice Chen", {'leadership': 3, 'strategy': 2, 'technical': 3},
                8.0, 150000, ['strategy']),
        Employee(2, "Bob Smith", {'leadership': 2, 'strategy': 2, 'technical': 3},
                6.0, 120000, ['leadership', 'strategy']),
        Employee(3, "Carol Davis", {'leadership': 3, 'strategy': 3, 'technical': 2},
                10.0, 160000, []),
        Employee(4, "David Lee", {'leadership': 1, 'strategy': 1, 'technical': 3},
                3.0, 90000, ['leadership', 'strategy']),
    ]

    # Critical roles
    critical_roles = [
        JobRole('CTO', 'Chief Technology Officer',
                {'leadership': 3, 'strategy': 3, 'technical': 3}, {}),
        JobRole('VP_ENG', 'VP Engineering',
                {'leadership': 3, 'strategy': 2, 'technical': 3}, {}),
    ]

    planner = SuccessionPlanner(employees, critical_roles)
    plan = planner.generate_succession_plan()

    for role_id, data in plan.items():
        print(f"\n{data['role_name']} (Risk: {data['risk_level']})")
        print("-" * 40)

        if data['successors']:
            for i, succ in enumerate(data['successors'], 1):
                print(f"  {i}. {succ['name']} - Readiness: {succ['readiness']}")
                if succ['development_needs']:

```

```

        print(f"      Development: {'\n      '}.join(succ['development_needs'])}")
    else:
        print(" No qualified successors identified!")

    return plan

if __name__ == "__main__":
    succession_planning_example()

```

💡 Strategic Workforce Planning Best Practices

1. **Align with business strategy:** Workforce plans must support organizational goals
2. **Use multiple scenarios:** Plan for optimistic, pessimistic, and most likely futures
3. **Consider total cost:** Include recruiting, training, turnover, and opportunity costs
4. **Build flexibility:** Create options to scale up or down based on actual demand
5. **Monitor leading indicators:** Track attrition patterns, skill gaps, and succession pipeline health
6. **Involve stakeholders:** HR, finance, and business units must collaborate
7. **Review regularly:** Update plans quarterly as conditions change

16.6. Cutting Stock Problem

The cutting stock problem determines how to cut raw materials into smaller pieces to satisfy demand while minimizing waste.

16.6.1. Problem Structure

Given:

- Stock material of length L
- Customer demands: d_i pieces of length l_i for $i = 1, \dots, m$
- Cutting patterns: each pattern j produces a_{ij} pieces of type i

Minimize stock usage:

$$\begin{aligned}
 &\text{minimize} && \sum_{j=1}^n x_j \\
 &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \geq d_i, \quad \forall i \\
 &&& x_j \geq 0 \text{ and integer}
 \end{aligned}$$

16.6.2. Column Generation Algorithm

The key challenge is that the number of patterns is exponential. Column generation solves this by generating patterns on-the-fly. The algorithm alternates between a master problem (which selects how many times to use each pattern) and a pricing subproblem (a knapsack problem that finds the most profitable new pattern based on dual values). Iteration continues until no pattern with negative reduced cost exists.

Figure 16.5 illustrates several cutting patterns for a stock length of 100 units, showing how different combinations of piece sizes fill each stock unit with varying amounts of waste.

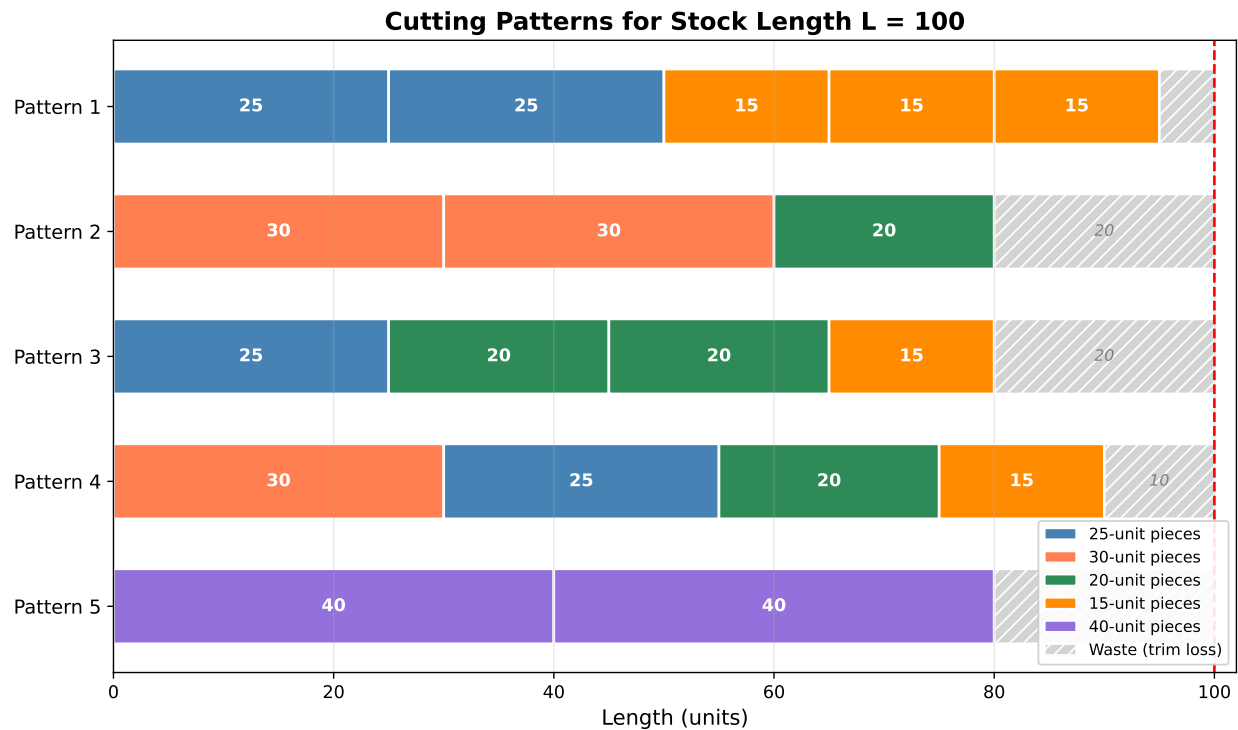


Figure 16.5.: Cutting stock patterns for stock length $L=100$. Each horizontal bar represents one cutting pattern with different piece types shown in distinct colors. Gray areas indicate waste (trim loss). Pattern 1 is highly efficient (2% waste), while Pattern 5 sacrifices efficiency to meet demand for larger pieces.

```
import numpy as np
from ortools.linear_solver import pywraplp

class CuttingStockSolver:
    """Solve cutting stock problem using column generation."""

    def __init__(self, stock_length, piece_lengths, demands):
        self.L = stock_length
        self.lengths = np.array(piece_lengths)
        self.demands = np.array(demands)
        self.m = len(piece_lengths)

        # Initialize with trivial patterns (one piece type per pattern)
        self.patterns = []
        for i in range(self.m):
            pattern = np.zeros(self.m, dtype=int)
            pattern[i] = int(self.L // self.lengths[i])
            self.patterns.append(pattern)

    def solve_master_lp(self):
```

```

    """Solve the LP relaxation of the master problem."""
    solver = pywraplp.Solver.CreateSolver('GLOP')

    n = len(self.patterns)
    x = [solver.NumVar(0, solver.infinity(), f'x_{j}') for j in range(n)]

    # Demand constraints
    for i in range(self.m):
        solver.Add(
            sum(self.patterns[j][i] * x[j] for j in range(n)) >= self.demands[i]
        )

    # Minimize total patterns
    solver.Minimize(sum(x))

    status = solver.Solve()

    if status == pywraplp.Solver.OPTIMAL:
        # Get dual values for pricing
        dual_values = []
        for constraint in solver.constraints():
            dual_values.append(constraint.dual_value())

        solution = [x[j].solution_value() for j in range(n)]
        return solver.Objective().Value(), np.array(dual_values), solution

    return None, None, None

def solve_pricing(self, duals):
    """Solve the pricing subproblem (knapsack)."""
    # Find pattern with most negative reduced cost
    # Reduced cost = 1 - sum(dual[i] * a[i])
    # Maximize sum(dual[i] * a[i]) subject to sum(l[i] * a[i]) <= L

    solver = pywraplp.Solver.CreateSolver('SCIP')

    a = [solver.IntVar(0, int(self.L // self.lengths[i]), f'a_{i}')]
        for i in range(self.m)]

    # Capacity constraint
    solver.Add(sum(self.lengths[i] * a[i] for i in range(self.m)) <= self.L)

    # Maximize dual contribution
    solver.Maximize(sum(duals[i] * a[i] for i in range(self.m)))

    status = solver.Solve()

    if status == pywraplp.Solver.OPTIMAL:
        pattern = np.array([int(a[i].solution_value()) for i in range(self.m)])
        reduced_cost = 1 - solver.Objective().Value()
        return pattern, reduced_cost

    return None, None

```



```

def solve(self, max_iterations=100):
    """Main column generation loop."""
    for iteration in range(max_iterations):
        # Solve master LP
        obj, duals, solution = self.solve_master_lp()

        if obj is None:
            return None, None

        # Solve pricing subproblem
        new_pattern, reduced_cost = self.solve_pricing(duals)

        # Check for optimality (no improving pattern)
        if reduced_cost >= -1e-6:
            break

        # Add new pattern
        self.patterns.append(new_pattern)

    # Solve final integer problem
    return self.solve_integer()

def solve_integer(self):
    """Solve integer master problem with generated patterns."""
    solver = pywraplp.Solver.CreateSolver('SCIP')

    n = len(self.patterns)
    x = [solver.IntVar(0, solver.infinity(), f'x_{j}') for j in range(n)]

    # Demand constraints
    for i in range(self.m):
        solver.Add(
            sum(self.patterns[j][i] * x[j] for j in range(n)) >= self.demands[i]
        )

    solver.Minimize(sum(x))

    status = solver.Solve()

    if status == pywraplp.Solver.OPTIMAL:
        solution = []
        for j in range(n):
            count = int(x[j].solution_value())
            if count > 0:
                solution.append((self.patterns[j], count))
        return int(solver.Objective().Value()), solution

    return None, None

# Example
stock_length = 100
piece_lengths = [45, 36, 31, 14]

```

```

demands = [97, 610, 395, 211]

solver = CuttingStockSolver(stock_length, piece_lengths, demands)
num_stock, patterns = solver.solve()

print(f"Stock pieces needed: {num_stock}")
print("Patterns used:")
for pattern, count in patterns:
    waste = stock_length - sum(pattern * np.array(piece_lengths))
    print(f" {pattern} x {count} (waste: {waste})")

```

16.6.3. Industrial Applications

Industry	Material	Typical Waste Reduction
Paper	Rolls	2-5%
Steel	Coils, bars	3-8%
Glass	Sheets	5-15%
Textile	Fabric rolls	4-10%
Wood	Lumber, plywood	5-12%

16.7. Bin Packing

Bin packing is the dual problem to cutting stock: pack items into the minimum number of bins.

16.7.1. One-Dimensional Bin Packing

Pack items of sizes $s_1, \dots, s_n$ into bins of capacity C using minimum bins:

$$\begin{aligned}
 &\text{minimize} && \sum_{j=1}^n y_j \\
 &\text{subject to} && \sum_{i=1}^n s_i x_{ij} \leq C y_j, \quad \forall j \\
 &&& \sum_{j=1}^n x_{ij} = 1, \quad \forall i \\
 &&& x_{ij}, y_j \in \{0, 1\}
 \end{aligned}$$

16.7.2. First Fit Decreasing Heuristic

The First Fit Decreasing (FFD) heuristic sorts items by size in decreasing order, then places each item into the first bin with enough remaining capacity. Best Fit Decreasing (BFD) is a variant that places each item into the bin with the least remaining capacity that still fits. Both heuristics guarantee solutions within $11/9$ of optimal.

```

def first_fit_decreasing(items, bin_capacity):
    """
    Solve bin packing using First Fit Decreasing heuristic.

    Args:
        items: List of item sizes
    """

```

```

        bin_capacity: Capacity of each bin

Returns:
    bins: List of lists, each containing items in that bin
    num_bins: Number of bins used
"""
# Sort items in decreasing order
sorted_items = sorted(enumerate(items), key=lambda x: -x[1])

bins = [] # Each bin is [remaining_capacity, [items]]

for item_idx, size in sorted_items:
    # Find first bin that fits
    placed = False
    for bin_data in bins:
        if bin_data[0] >= size:
            bin_data[0] -= size
            bin_data[1].append(item_idx)
            placed = True
            break

    # Create new bin if needed
    if not placed:
        bins.append([bin_capacity - size, [item_idx]])

return [b[1] for b in bins], len(bins)

def best_fit_decreasing(items, bin_capacity):
    """
    Solve bin packing using Best Fit Decreasing heuristic.

    Args:
        items: List of item sizes
        bin_capacity: Capacity of each bin

    Returns:
        bins: List of lists, each containing items in that bin
        num_bins: Number of bins used
    """
    sorted_items = sorted(enumerate(items), key=lambda x: -x[1])

    bins = []

    for item_idx, size in sorted_items:
        # Find bin with minimum remaining capacity that fits
        best_bin = None
        best_remaining = bin_capacity + 1

        for i, bin_data in enumerate(bins):
            if bin_data[0] >= size and bin_data[0] < best_remaining:
                best_bin = i
                best_remaining = bin_data[0]

```

```

    if best_bin is not None:
        bins[best_bin][0] -= size
        bins[best_bin][1].append(item_idx)
    else:
        bins.append([bin_capacity - size, [item_idx]])

    return [b[1] for b in bins], len(bins)

# Example
items = [7, 5, 3, 4, 8, 2, 6, 3, 5, 4, 2, 1]
capacity = 10

ffd_bins, ffd_count = first_fit_decreasing(items, capacity)
bfd_bins, bfd_count = best_fit_decreasing(items, capacity)

print(f"First Fit Decreasing: {ffd_count} bins")
print(f"Best Fit Decreasing: {bfd_count} bins")

```

16.7.3. Performance Guarantees

Algorithm	Worst-Case Ratio	Average Performance
First Fit	$1.7 \cdot \text{OPT}$	Good
First Fit Decreasing	$11/9 \cdot \text{OPT} + 6/9$	Excellent
Best Fit Decreasing	$11/9 \cdot \text{OPT} + 6/9$	Excellent
Next Fit	$2 \cdot \text{OPT}$	Poor

Two-Dimensional Extensions

For 2D bin packing (cutting rectangular sheets), additional constraints apply:

- **Guillotine cuts:** Each cut must go edge-to-edge
- **Orientation:** Items may or may not be rotatable
- **Staged cutting:** Cuts proceed in stages (horizontal then vertical)

These constraints arise from physical cutting machine limitations.

16.8. Summary

16.8.1. Key Takeaways

- The assignment problem matches agents to tasks one-to-one, solvable in $O(n^3)$ by the Hungarian algorithm
- Facility location balances fixed opening costs against transportation costs
- Capacity planning determines when and how much to expand production capability
- Workforce scheduling assigns employees to shifts while respecting labor constraints
- Cutting stock minimizes material waste through optimal cutting patterns
- Column generation handles exponentially many patterns by generating them on-demand
- Bin packing is the dual of cutting stock: pack items into minimum bins
- Heuristics like First Fit Decreasing provide near-optimal solutions quickly

16.9. Exercises

1. **Assignment problem:** Solve the assignment problem for a 5x5 cost matrix where $c_{ij} = |i - j|^2 + i + j$. What is the optimal assignment and cost?
2. **Facility location:** A company must choose among 4 potential warehouse locations to serve 10 customers. Fixed costs are [500, 400, 600, 450]. Generate random transportation costs and solve the uncapacitated facility location problem.
3. **Capacity planning:** Given demands [100, 150, 200, 250, 300] over 5 periods, initial capacity 120, and expansion options of 50 units (cost 200) or 100 units (cost 350), find the optimal expansion plan. Assume shortage cost of 10 per unit.
4. **Shift scheduling:** Design shifts for a 24-hour operation where demand varies by hour. Morning (6am-12pm) needs 5 staff, afternoon (12pm-6pm) needs 8, evening (6pm-12am) needs 6, and night (12am-6am) needs 3. Create appropriate shifts and minimize cost.
5. **Cutting stock:** Stock length is 120 cm. Demands are: 50 pieces of 25 cm, 40 pieces of 40 cm, 30 pieces of 35 cm, and 20 pieces of 55 cm. Enumerate at least 5 good cutting patterns and solve using column generation.

17. Facility Layout Optimization

What You Will Learn

- Understand the strategic importance of facility layout decisions
- Classify layout problems by type: process, product, cellular, fixed-position
- Formulate and solve the Quadratic Assignment Problem (QAP)
- Apply Systematic Layout Planning (SLP) methodology
- Implement layout algorithms: CRAFT, BLOCPLAN, CORELAP
- Design cellular manufacturing layouts using group technology
- Optimize warehouse layouts for order picking efficiency
- Develop Python solutions for layout optimization

Facility layout determines the physical arrangement of resources within a manufacturing plant, warehouse, hospital, or office. A well-designed layout minimizes material handling costs, reduces work-in-progress inventory, improves workflow efficiency, and enhances worker productivity. Poor layouts lead to wasted space, excessive material movement, bottlenecks, and safety hazards. This chapter provides the mathematical foundations, algorithms, and practical tools for optimizing facility layouts.

17.1. Introduction to Layout Problems

Facility layout is one of the most consequential decisions in operations management. Once implemented, layouts are expensive and disruptive to change. A manufacturing plant may operate with the same basic layout for decades, making the initial design critically important.

17.1.1. The Strategic Impact of Layout

Layout decisions directly affect operational performance across multiple dimensions:

Performance Metric	Impact of Good Layout
Material handling cost	20-50% reduction
Work-in-progress inventory	30-40% reduction
Lead time	15-30% reduction
Floor space utilization	15-25% improvement
Labor productivity	10-20% improvement
Quality defects	10-15% reduction

The Hidden Factory

Studies estimate that 20-50% of total manufacturing costs are attributable to material handling. Much of this cost is “hidden” because it does not appear as a separate line item. Layout optimization exposes and eliminates this hidden factory of non-value-added movement.

17.1.2. Layout Objectives

The primary objectives in layout design include:

Minimize Material Handling Cost: The most common objective, measured as:

$$\text{Total Cost} = \sum_{i=1}^n \sum_{j=1}^n f_{ij} \cdot c_{ij} \cdot d_{ij}$$

where:

- f_{ij} is the flow (trips, units, or material moved) between facilities i and j
- c_{ij} is the cost per unit distance for moving between i and j
- d_{ij} is the distance between facilities i and j

Maximize Adjacency of Related Activities: Place departments with high interaction close together.

Ensure Safety and Regulatory Compliance: Separate hazardous operations, maintain emergency access.

Provide Flexibility for Future Changes: Allow for expansion, technology changes, and product mix shifts.

Support Workflow and Communication: Enable smooth material flow and information exchange.

17.1.3. Types of Manufacturing Layouts

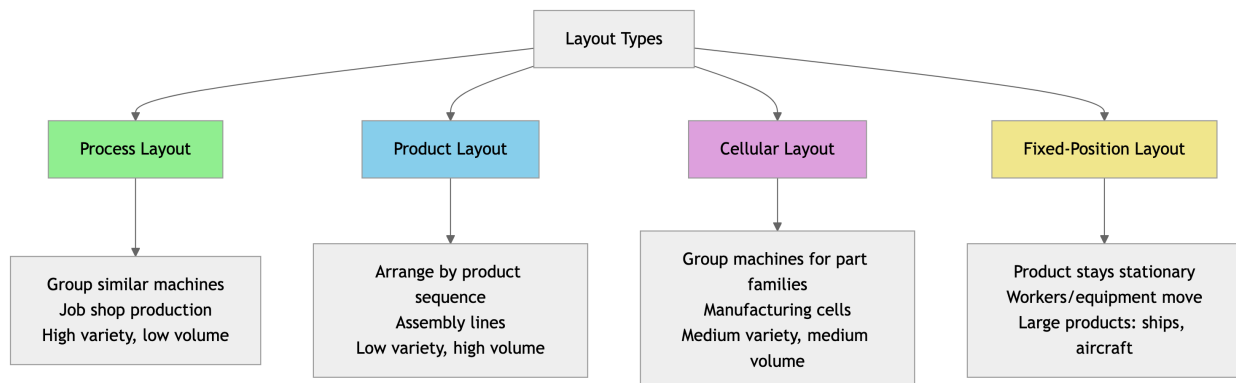


Figure 17.1.: Classification of Manufacturing Layout Types

The figure below illustrates all four layout types. Each quadrant shows how machines, stations, or resources are arranged within the facility, along with the typical material flow pattern.

Material handling cost depends on two factors: how often material moves between departments (flow) and how far apart they are (distance). The three heatmaps below show the flow matrix, the distance matrix, and their element-wise product, which represents the total handling cost for each department pair.

17.1.3.1. Process Layout (Functional Layout)

In a process layout, similar machines or functions are grouped together. All lathes are in one department, all milling machines in another, all assembly in another. Work flows between departments based on the specific processing requirements of each job.

Characteristics:

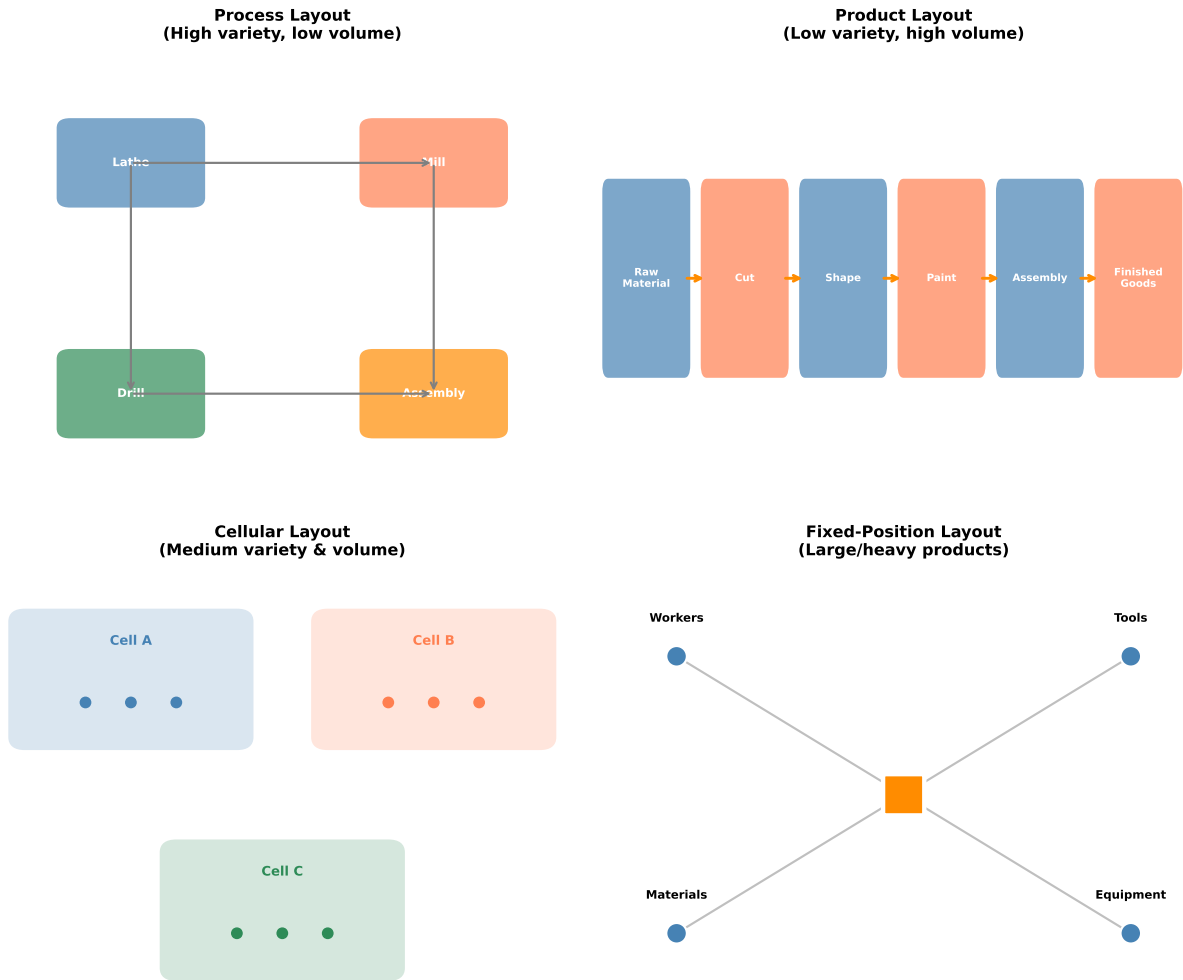


Figure 17.2.: Four major layout types — each suits different volume-variety combinations in manufacturing

- High flexibility to handle variety
- General-purpose equipment
- Variable routing for different products
- Higher material handling costs
- Skilled workers needed at each station

Applications: Job shops, hospitals, machine shops, custom manufacturing

17.1.3.2. Product Layout (Line Layout)

A product layout arranges equipment in the sequence needed to manufacture a specific product. Material flows in a continuous path from raw materials to finished goods.

Characteristics:

- Low flexibility, dedicated to specific products
- Specialized equipment
- Fixed routing

17. Facility Layout Optimization

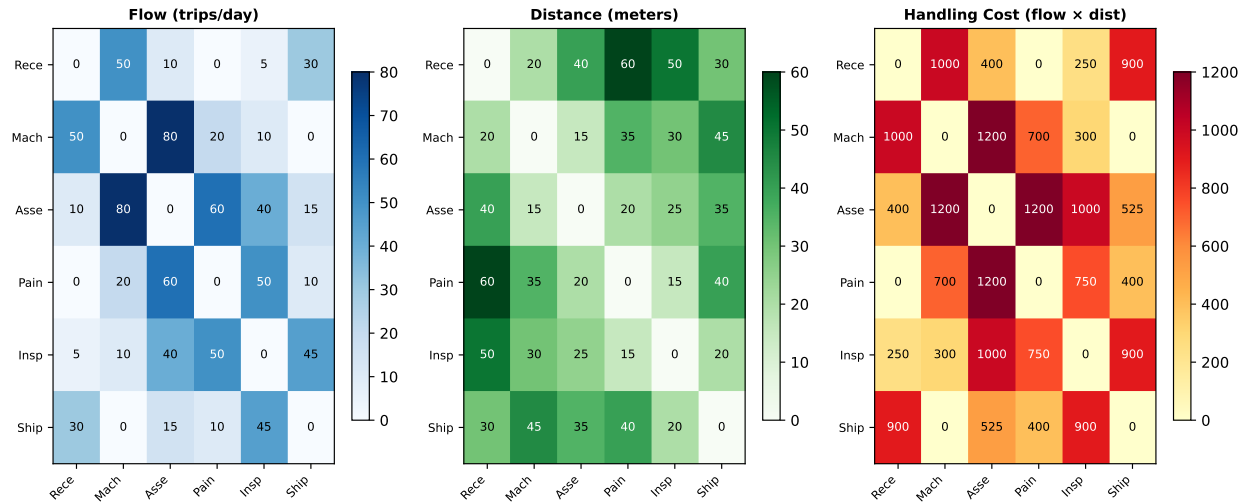


Figure 17.3.: Distance × Flow heatmap — darker cells indicate higher material handling cost (distance × flow intensity) between department pairs

- Lower material handling costs
- Lower skill requirements per station
- Line balancing is essential

Applications: Assembly lines, high-volume manufacturing, food processing

17.1.3.3. Cellular Layout

Cellular layouts group dissimilar machines into cells based on the parts they produce. Each cell is a mini-factory responsible for a family of similar parts.

Characteristics:

- Combines benefits of process and product layouts
- Reduced material handling within cells
- Improved quality through ownership
- Faster throughput
- Requires part family identification

Applications: Batch manufacturing, electronics assembly, aerospace components

17.1.3.4. Fixed-Position Layout

In a fixed-position layout, the product remains stationary while workers, equipment, and materials come to it. Used for products too large or heavy to move.

Characteristics:

- Product does not move during production
- High flexibility in product design
- Complex scheduling and coordination
- High material handling for components

Applications: Shipbuilding, aircraft assembly, construction, heavy machinery



Figure 17.4.: Systematic Approach to Layout Planning

17.1.4. The Layout Planning Process

17.2. The Quadratic Assignment Problem (QAP)

The Quadratic Assignment Problem is the classical mathematical formulation for facility layout. It seeks to assign n facilities to n locations to minimize total flow-weighted distance.

17.2.1. Problem Definition

Given:

- n facilities to be assigned to n locations
- Flow matrix $F = [f_{ij}]$ where f_{ij} is the flow between facilities i and j
- Distance matrix $D = [d_{kl}]$ where d_{kl} is the distance between locations k and l

The objective is to find an assignment that minimizes:

$$\min_{\pi \in \Pi_n} \sum_{i=1}^n \sum_{j=1}^n f_{ij} \cdot d_{\pi(i)\pi(j)}$$

where π is a permutation representing the assignment (facility i goes to location $\pi(i)$).

17.2.2. Integer Programming Formulation

Define binary variables $x_{ik} = 1$ if facility i is assigned to location k :

$$\begin{aligned} & \text{minimize} && \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n f_{ij} \cdot d_{kl} \cdot x_{ik} \cdot x_{jl} \\ & \text{subject to} && \sum_{k=1}^n x_{ik} = 1, \quad \forall i \in \{1, \dots, n\} \\ & && \sum_{i=1}^n x_{ik} = 1, \quad \forall k \in \{1, \dots, n\} \\ & && x_{ik} \in \{0, 1\} \end{aligned}$$

The first constraint ensures each facility is assigned to exactly one location. The second ensures each location receives exactly one facility. The objective function is quadratic due to the product $x_{ik} \cdot x_{jl}$.

! NP-Hard Complexity

The QAP is one of the most difficult combinatorial optimization problems. It is NP-hard, and even finding provably optimal solutions for instances with $n > 30$ is extremely challenging. The number of possible solutions is $n!$, which grows explosively:

n	Possible Solutions
5	120
10	3,628,800
15	1.3 trillion
20	2.4 quintillion
30	2.7×10^{32}

For practical problems, heuristics and metaheuristics are essential.

17.2.3. Linearization Techniques

The quadratic objective can be linearized to enable solution with standard MIP solvers.

17.2.3.1. Kaufman-Broeckx Linearization

Introduce auxiliary variables y_{ijkl} to replace the product $x_{ik} \cdot x_{jl}$:

$$\begin{aligned}
 & \text{minimize} && \sum_{i,j,k,l} f_{ij} \cdot d_{kl} \cdot y_{ijkl} \\
 & \text{subject to} && y_{ijkl} \leq x_{ik}, \quad \forall i, j, k, l \\
 & && y_{ijkl} \leq x_{jl}, \quad \forall i, j, k, l \\
 & && y_{ijkl} \geq x_{ik} + x_{jl} - 1, \quad \forall i, j, k, l \\
 & && \text{assignment constraints}
 \end{aligned}$$

This creates $O(n^4)$ variables and constraints, making it impractical for large n .

17.2.3.2. Adams-Johnson Linearization

A more efficient linearization using $O(n^3)$ variables:

Define $w_{ikl} = \sum_j f_{ij} \cdot x_{jl}$ (flow from facility i at location k to location l):

$$\begin{aligned}
 & \text{minimize} && \sum_{i=1}^n \sum_{k=1}^n \sum_{l=1}^n d_{kl} \cdot w_{ikl} \\
 & \text{subject to} && w_{ikl} \leq M \cdot x_{ik}, \quad \forall i, k, l \\
 & && w_{ikl} \leq \sum_j f_{ij} \cdot x_{jl}, \quad \forall i, k, l \\
 & && w_{ikl} \geq \sum_j f_{ij} \cdot x_{jl} - M(1 - x_{ik}), \quad \forall i, k, l \\
 & && \text{assignment constraints}
 \end{aligned}$$

17.2.4. Small Example: 4-Facility Problem

Consider a small manufacturing facility with 4 departments:

- **A:** Receiving
- **B:** Machining
- **C:** Assembly
- **D:** Shipping

Flow Matrix (trips per day):

	A	B	C	D
A	0	20	5	0
B	20	0	30	10
C	5	30	0	25
D	0	10	25	0

Available Locations (arranged in a 2x2 grid):

```
+-----+-----+
|  1  |  2  |
+-----+-----+
|  3  |  4  |
+-----+-----+
```

Distance Matrix (rectilinear distances, each cell is 10 meters):

	1	2	3	4
1	0	10	10	20
2	10	0	20	10
3	10	20	0	10
4	20	10	10	0

Solution by Enumeration:

For this small problem, we can evaluate all $4! = 24$ permutations. The following code computes the flow-weighted distance for every possible assignment and selects the one with minimum cost.

```
import itertools
import numpy as np

# Flow matrix (A=0, B=1, C=2, D=3)
F = np.array([
    [0, 20, 5, 0],
    [20, 0, 30, 10],
    [5, 30, 0, 25],
    [0, 10, 25, 0]
])

# Distance matrix (locations 0, 1, 2, 3)
D = np.array([
    [0, 10, 10, 20],
    [10, 0, 20, 10],
    [10, 20, 0, 10],
    [20, 10, 10, 0]
])

def calculate_cost(assignment, F, D):
    """Calculate total flow-weighted distance for an assignment."""
    n = len(assignment)
    cost = 0
    for i in range(n):
```

```

        for j in range(n):
            loc_i = assignment[i]
            loc_j = assignment[j]
            cost += F[i, j] * D[loc_i, loc_j]
    return cost

# Enumerate all permutations
best_cost = float('inf')
best_assignment = None

for perm in itertools.permutations(range(4)):
    cost = calculate_cost(perm, F, D)
    if cost < best_cost:
        best_cost = cost
        best_assignment = perm

facilities = ['A', 'B', 'C', 'D']
locations = ['1', '2', '3', '4']

print("Optimal Assignment:")
for i, loc in enumerate(best_assignment):
    print(f" {facilities[i]} -> Location {locations[loc]}")
print(f"Total Cost: {best_cost} unit-meters per day")

```

Optimal Solution:

A (Receiving) -> Location 1
 B (Machining) -> Location 3
 C (Assembly) -> Location 4
 D (Shipping) -> Location 2

Total Cost: 1800 unit-meters per day

The solution places high-flow pairs (B-C, C-D) adjacent to each other.

17.2.5. Lower Bounds for QAP

For larger instances where enumeration is impossible, lower bounds help evaluate solution quality.

Gilmore-Lawler Bound:

The Gilmore-Lawler bound is computed by solving n linear assignment problems. For each facility i , we compute the minimum contribution to the objective by optimally assigning the remaining facilities. The bound pairs the sorted flow values of each facility with the reverse-sorted distances of each location, then solves an assignment problem to find the minimum total.

```

from scipy.optimize import linear_sum_assignment

def gilmore_lawler_bound(F, D):
    """Compute Gilmore-Lawler lower bound for QAP."""
    n = F.shape[0]

    # Sort rows of F and D
    F_sorted = np.sort(F, axis=1)

```

```

D_sorted = np.sort(D, axis=1)

# For each facility, compute minimum contribution
total_bound = 0
for i in range(n):
    # Build cost matrix for assignment subproblem
    cost_matrix = np.zeros((n, n))
    for k in range(n):
        for j in range(n):
            # Minimum possible contribution
            cost_matrix[k, :] += F_sorted[i, j] * D_sorted[k, n-1-j]

    # Solve assignment problem
    row_ind, col_ind = linear_sum_assignment(cost_matrix)
    total_bound += cost_matrix[row_ind, col_ind].sum()

return total_bound / n # Average over facilities

```

17.3. Systematic Layout Planning (SLP)

Systematic Layout Planning, developed by Richard Muther, is a widely-used structured approach to facility layout that balances quantitative flow analysis with qualitative relationship considerations.

17.3.1. The SLP Framework

17.3.2. Activity Relationship Chart (REL Chart)

The REL chart captures the importance of proximity between each pair of activities using a letter code:

Code	Closeness	Numerical Value
A	Absolutely necessary	4
E	Especially important	3
I	Important	2
O	Ordinary closeness	1
U	Unimportant	0
X	Undesirable	-1

Reason Codes (why the relationship exists):

1. Flow of materials
2. Ease of supervision
3. Common personnel
4. Contact necessary
5. Convenience
6. Share same equipment
7. Noise/vibration
8. Safety/hazard
9. Same utilities

The `RELChart` class below stores pairwise department relationships and converts them to a numerical matrix for use in layout algorithms. It also computes the Total Closeness Rating (TCR) for each department, which many heuristics use to determine placement priority.

17. Facility Layout Optimization

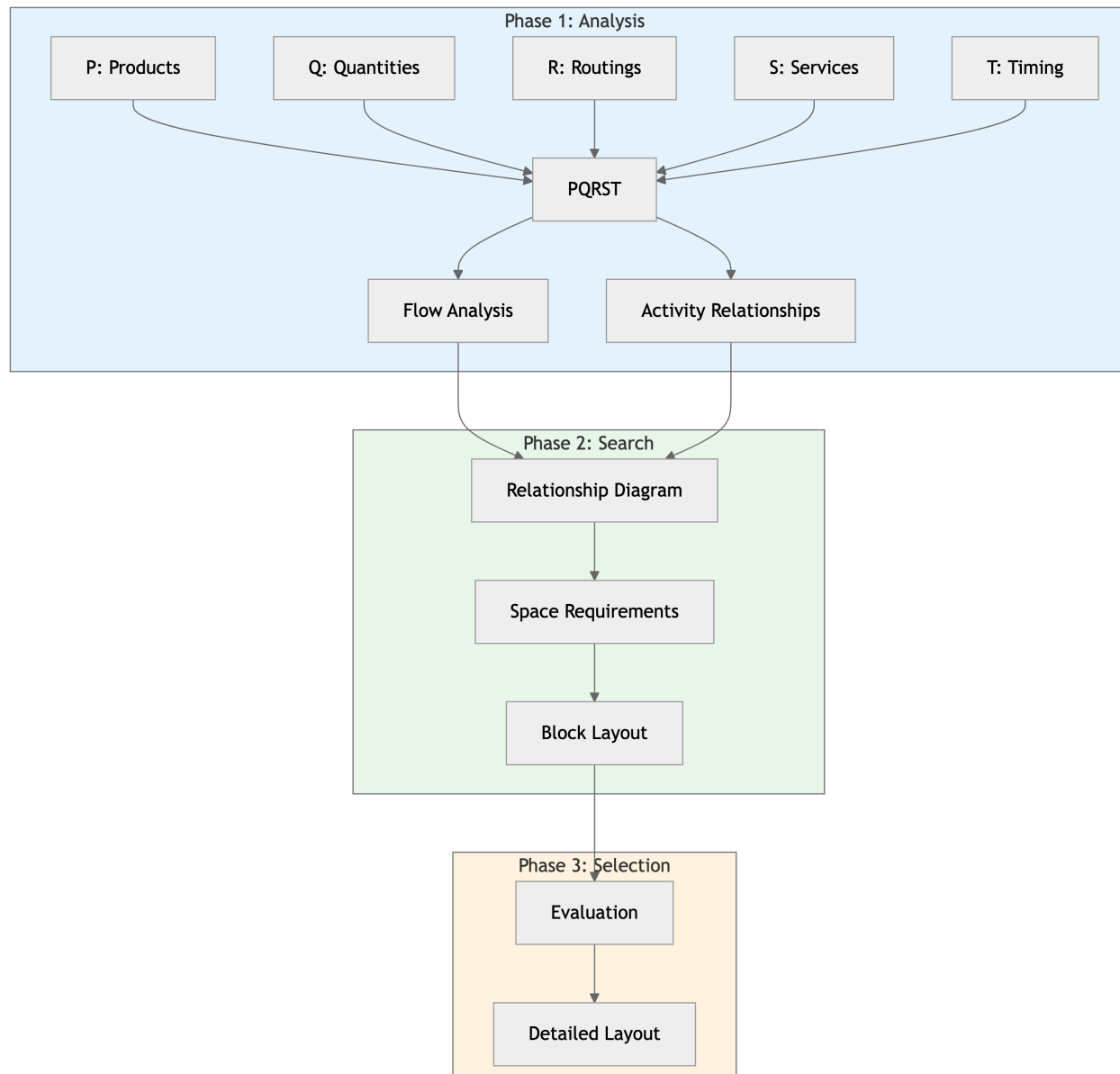


Figure 17.5.: Systematic Layout Planning Framework

```
import numpy as np
from dataclasses import dataclass
from typing import Dict, List, Tuple

@dataclass
class Department:
    """Represents a department in the facility."""
    id: int
    name: str
    area: float # square meters
```



```

class RELChart:
    """Activity Relationship Chart for SLP."""

    CLOSENESS_VALUES = {
        'A': 4, 'E': 3, 'I': 2, 'O': 1, 'U': 0, 'X': -1
    }

    def __init__(self, departments: List[Department]):
        self.departments = departments
        self.n = len(departments)
        self.relationships = {} # (i, j) -> (code, reasons)

    def set_relationship(self, dept1: int, dept2: int,
                        code: str, reasons: List[int] = None):
        """Set the relationship between two departments."""
        if code not in self.CLOSENESS_VALUES:
            raise ValueError(f"Invalid closeness code: {code}")

        # Store symmetric relationship
        self.relationships[(dept1, dept2)] = (code, reasons or [])
        self.relationships[(dept2, dept1)] = (code, reasons or [])

    def get_relationship_matrix(self) -> np.ndarray:
        """Convert relationships to numerical matrix."""
        matrix = np.zeros((self.n, self.n))
        for (i, j), (code, _) in self.relationships.items():
            matrix[i, j] = self.CLOSENESS_VALUES[code]
        return matrix

    def get_total_closeness_rating(self, dept: int) -> int:
        """Calculate Total Closeness Rating (TCR) for a department."""
        tcr = 0
        for j in range(self.n):
            if j != dept and (dept, j) in self.relationships:
                code, _ = self.relationships[(dept, j)]
                tcr += self.CLOSENESS_VALUES[code]
        return tcr

    def display(self):
        """Display the REL chart."""
        print("Activity Relationship Chart")
        print("-" * 50)

        # Header
        header = " " + " ".join(f"{d.name[:3]:>4}" for d in self.departments)
        print(header)

        for i, dept_i in enumerate(self.departments):
            row = f"{dept_i.name[:4]:>4} "
            for j in range(self.n):
                if j <= i:
                    row += " - "
                else:

```

```

        if (i, j) in self.relationships:
            code, _ = self.relationships[(i, j)]
            row += f"    {code} "
        else:
            row += "    U "
    print(row)

# Example: Manufacturing facility REL chart
departments = [
    Department(0, "Receiving", 200),
    Department(1, "Raw Storage", 300),
    Department(2, "Machining", 500),
    Department(3, "Assembly", 400),
    Department(4, "Painting", 200),
    Department(5, "Packaging", 150),
    Department(6, "Shipping", 200),
    Department(7, "Offices", 100)
]

rel_chart = RELChart(departments)

# Define relationships
rel_chart.set_relationship(0, 1, 'A', [1])      # Receiving-Storage: Absolutely
    ↪ necessary
rel_chart.set_relationship(0, 6, 'X', [5])      # Receiving-Shipping: Undesirable
    ↪ (traffic)
rel_chart.set_relationship(1, 2, 'E', [1])      # Storage-Machining: Especially
    ↪ important
rel_chart.set_relationship(2, 3, 'A', [1])      # Machining-Assembly: Absolutely
    ↪ necessary
rel_chart.set_relationship(3, 4, 'E', [1])      # Assembly-Painting: Especially
    ↪ important
rel_chart.set_relationship(4, 5, 'A', [1])      # Painting-Packaging: Absolutely
    ↪ necessary
rel_chart.set_relationship(5, 6, 'A', [1])      # Packaging-Shipping: Absolutely
    ↪ necessary
rel_chart.set_relationship(4, 7, 'X', [7, 8])    # Painting-Offices: Undesirable
    ↪ (fumes)
rel_chart.set_relationship(7, 0, 'I', [2])      # Offices-Receiving: Important
    ↪ (supervision)
rel_chart.set_relationship(7, 6, 'I', [2])      # Offices-Shipping: Important
    ↪ (supervision)

rel_chart.display()

```

17.3.3. From-To Chart (Flow Matrix)

The From-To chart quantifies material movement between departments. Given product routings (the sequence of departments each product visits) and annual production volumes, the chart sums the flow for each department pair. The resulting matrix serves as input to the QAP and other layout algorithms.

```

def create_from_to_chart(routings: List[List[Tuple[int, float]]],
                        volumes: List[float]) -> np.ndarray:
    """
    Create From-To chart from product routings.

    Args:
        routings: List of routings, each is [(dept, time), ...]
        volumes: Annual production volume for each product

    Returns:
        From-To matrix showing flow between departments
    """
    # Find max department number
    max_dept = max(dept for routing in routings
                   for dept, _ in routing)
    n = max_dept + 1

    from_to = np.zeros((n, n))

    for routing, volume in zip(routings, volumes):
        for i in range(len(routing) - 1):
            from_dept = routing[i][0]
            to_dept = routing[i + 1][0]
            from_to[from_dept, to_dept] += volume

    return from_to

# Example: Three products with different routings
routings = [
    [(0, 5), (1, 10), (2, 30), (3, 20), (5, 10), (6, 5)], # Product A
    [(0, 5), (1, 10), (2, 25), (4, 15), (5, 10), (6, 5)], # Product B
    [(0, 5), (2, 40), (3, 25), (4, 20), (5, 10), (6, 5)], # Product C
]
volumes = [1000, 2000, 1500] # Annual volumes

from_to = create_from_to_chart(routings, volumes)
print("From-To Chart (annual moves):")
print(from_to.astype(int))

```

17.3.4. Space Requirements Analysis

Accurate space estimates require summing equipment footprints and applying multipliers for aisles (typically 1.3-1.5x) and future expansion (typically 1.2x). The function below computes these requirements for each department.

```

def calculate_space_requirements(
    departments: List[Department],
    equipment_list: Dict[int, List[Tuple[str, float, float]]],
    aisle_factor: float = 1.4,
    expansion_factor: float = 1.2
) -> Dict[int, float]:
    """

```

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```
Calculate space requirements for each department.

Args:
    departments: List of departments
    equipment_list: Dict mapping dept_id to [(name, length, width), ...]
    aisle_factor: Multiplier for aisle space (typically 1.3-1.5)
    expansion_factor: Multiplier for future expansion

Returns:
    Dict mapping dept_id to required area (sq meters)
"""
requirements = {}

for dept in departments:
    if dept.id in equipment_list:
        equipment_area = sum(l * w for _, l, w in equipment_list[dept.id])
        # Add aisle and expansion space
        total_area = equipment_area * aisle_factor * expansion_factor
    else:
        total_area = dept.area

    requirements[dept.id] = total_area

return requirements

# Example equipment list
equipment = {
    2: [ # Machining department
        ("CNC Lathe", 4, 2),
        ("CNC Mill", 3, 2.5),
        ("Drill Press", 1.5, 1),
        ("Grinder", 2, 1.5),
        ("Workbench", 3, 1)
    ],
    3: [ # Assembly department
        ("Assembly Line", 20, 3),
        ("Test Station", 3, 2),
        ("Workbench", 3, 1),
        ("Tool Cabinet", 2, 0.5)
    ]
}

space_reqs = calculate_space_requirements(departments, equipment)
print("\nSpace Requirements:")
for dept in departments:
    print(f"    {dept.name}: {space_reqs.get(dept.id, dept.area):.0f} sq m")
```

17.3.5. Block Layout Generation

The block layout shows the relative position and size of each department within the building footprint. The generation algorithm uses a space-filling approach: departments are sorted by area (largest first) and placed into available rectangular spaces. Each placement splits the remaining space into at most two new rectangles.

```

import matplotlib.pyplot as plt
import matplotlib.patches as patches
from typing import List, Tuple, Optional

def generate_block_layout(
    departments: List[Department],
    areas: Dict[int, float],
    building_width: float,
    building_length: float
) -> Dict[int, Tuple[float, float, float, float]]:
    """
    Generate a simple block layout using space-filling algorithm.

    Args:
        departments: List of departments
        areas: Required area for each department
        building_width: Available building width
        building_length: Available building length

    Returns:
        Dict mapping dept_id to (x, y, width, height)
    """
    # Sort by area (largest first)
    sorted_depts = sorted(departments,
                          key=lambda d: areas.get(d.id, d.area),
                          reverse=True)

    layout = {}
    available_spaces = [(0, 0, building_width, building_length)]

    for dept in sorted_depts:
        area_needed = areas.get(dept.id, dept.area)

        # Find smallest space that fits
        best_space = None
        best_idx = -1

        for idx, (x, y, w, h) in enumerate(available_spaces):
            if w * h >= area_needed:
                if best_space is None or w * h < best_space[2] * best_space[3]:
                    best_space = (x, y, w, h)
                    best_idx = idx

        if best_space is None:
            print(f"Warning: Cannot fit {dept.name}")
            continue

        x, y, w, h = best_space

        # Calculate department dimensions (try to make it somewhat square)
        if area_needed <= w * h:
            dept_width = min(w, np.sqrt(area_needed * w / h))
            dept_height = area_needed / dept_width

```

```

        # Ensure fits in space
        if dept_height > h:
            dept_height = h
            dept_width = area_needed / dept_height
        else:
            dept_width = w
            dept_height = h

        layout[dept.id] = (x, y, dept_width, dept_height)

    # Update available spaces
    available_spaces.pop(best_idx)

    # Add remaining spaces
    if dept_width < w:
        available_spaces.append((x + dept_width, y, w - dept_width, h))
    if dept_height < h:
        available_spaces.append((x, y + dept_height, dept_width, h -
↪ dept_height))

    return layout

def visualize_block_layout(
    departments: List[Department],
    layout: Dict[int, Tuple[float, float, float, float]],
    title: str = "Block Layout"
):
    """Visualize a block layout."""
    fig, ax = plt.subplots(1, 1, figsize=(12, 8))

    colors = plt.cm.Set3(np.linspace(0, 1, len(departments)))

    for i, dept in enumerate(departments):
        if dept.id in layout:
            x, y, w, h = layout[dept.id]
            rect = patches.Rectangle(
                (x, y), w, h,
                linewidth=2, edgecolor='black',
                facecolor=colors[i], alpha=0.7
            )
            ax.add_patch(rect)

        # Add label
        ax.text(x + w/2, y + h/2, dept.name,
                ha='center', va='center', fontsize=10, fontweight='bold')

    ax.set_xlim(-5, max(x + w for x, y, w, h in layout.values()) + 5)
    ax.set_ylim(-5, max(y + h for x, y, w, h in layout.values()) + 5)
    ax.set_aspect('equal')
    ax.set_xlabel('Width (m)')
    ax.set_ylabel('Length (m)')

```

```

ax.set_title(title)
ax.grid(True, alpha=0.3)

plt.tight_layout()
return fig

```

17.4. Layout Algorithms

Several computerized algorithms have been developed to generate and improve facility layouts.

17.4.1. CRAFT Algorithm

CRAFT (Computerized Relative Allocation of Facilities Technique) is an improvement algorithm that starts with an initial layout and iteratively improves it by exchanging department locations. At each step, it evaluates all pairwise exchanges between departments of similar size or adjacent departments, selects the exchange that yields the greatest cost reduction, and repeats until no further improvement is possible.

The implementation below represents the layout as a 2D grid where each cell is assigned to a department. Centroids are computed for distance calculations, and exchanges are restricted to departments with similar areas or adjacent cells.

```

import numpy as np
from typing import List, Tuple, Dict, Optional
import copy

class CRAFTSolver:
    """
    CRAFT algorithm for facility layout optimization.

    CRAFT uses pairwise exchanges to improve an initial layout,
    minimizing total material handling cost.
    """

    def __init__(self, flow_matrix: np.ndarray,
                  initial_layout: List[List[int]],
                  dept_areas: Dict[int, float]):
        """
        Initialize CRAFT solver.

        Args:
            flow_matrix: Flow between departments (n x n)
            initial_layout: 2D grid of department assignments
            dept_areas: Area of each department
        """
        self.flow = flow_matrix
        self.layout = [row[:] for row in initial_layout]
        self.areas = dept_areas
        self.n_depts = flow_matrix.shape[0]
        self.rows = len(initial_layout)
        self.cols = len(initial_layout[0])

    def get_centroids(self) -> Dict[int, Tuple[float, float]]:

```

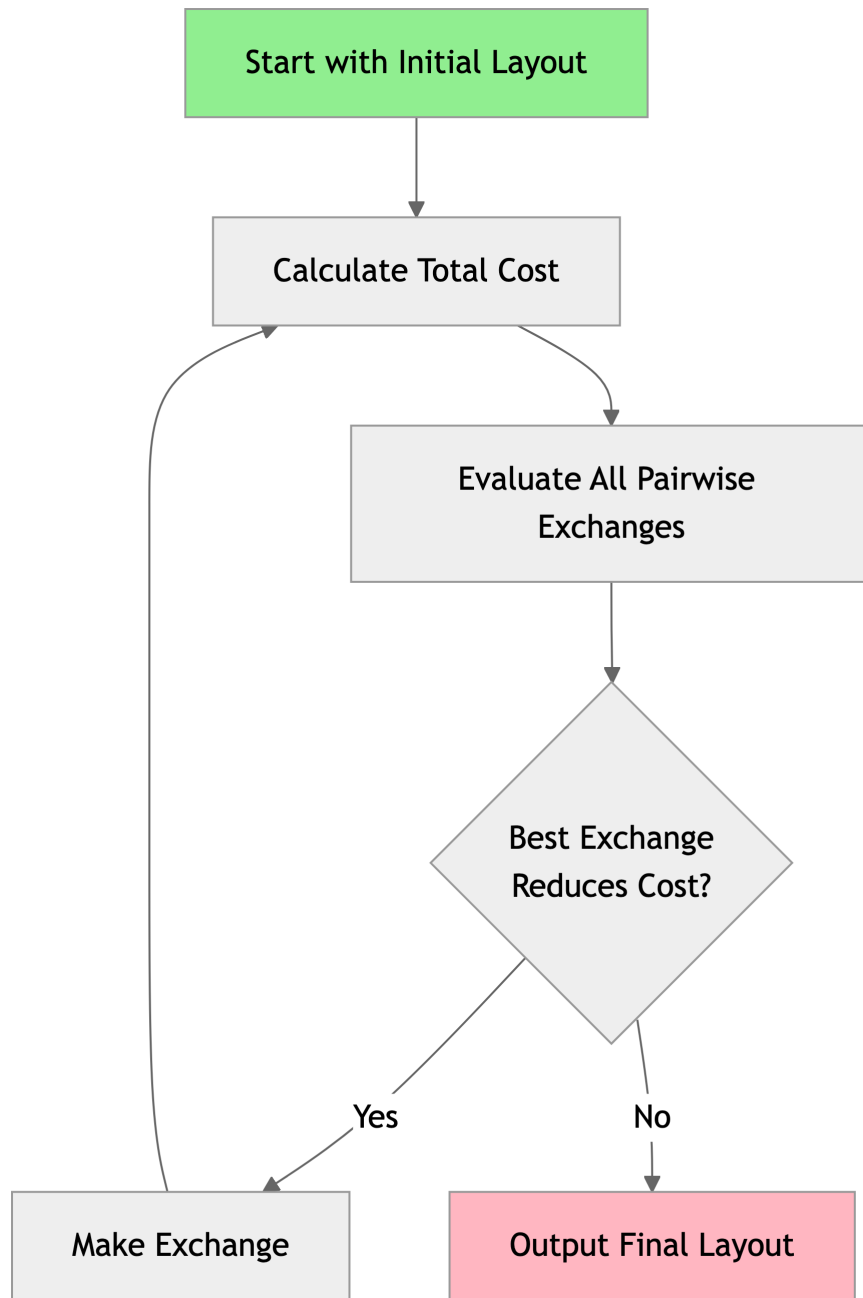


Figure 17.6.: CRAFT Algorithm Flow


```

    """Calculate centroid of each department."""
    centroids = {}
    dept_cells = {i: [] for i in range(self.n_depts)}

    for r in range(self.rows):
        for c in range(self.cols):
            dept = self.layout[r][c]
            if dept >= 0:
                dept_cells[dept].append((r, c))

    for dept, cells in dept_cells.items():
        if cells:
            avg_r = sum(r for r, c in cells) / len(cells)
            avg_c = sum(c for r, c in cells) / len(cells)
            centroids[dept] = (avg_r, avg_c)

    return centroids

def calculate_distance(self, centroids: Dict[int, Tuple[float, float]],
                      dept_i: int, dept_j: int) -> float:
    """Calculate rectilinear distance between department centroids."""
    if dept_i not in centroids or dept_j not in centroids:
        return 0
    r_i, c_i = centroids[dept_i]
    r_j, c_j = centroids[dept_j]
    return abs(r_i - r_j) + abs(c_i - c_j)

def calculate_total_cost(self) -> float:
    """Calculate total material handling cost for current layout."""
    centroids = self.get_centroids()
    total_cost = 0

    for i in range(self.n_depts):
        for j in range(self.n_depts):
            if i != j:
                dist = self.calculate_distance(centroids, i, j)
                total_cost += self.flow[i, j] * dist

    return total_cost

def can_exchange(self, dept_i: int, dept_j: int) -> bool:
    """Check if two departments can be exchanged (similar size or adjacent)."""
    area_i = self.areas.get(dept_i, 1)
    area_j = self.areas.get(dept_j, 1)

    # Allow exchange if similar size (within 20%)
    if abs(area_i - area_j) / max(area_i, area_j) < 0.2:
        return True

    # Or if departments are adjacent
    cells_i = []
    cells_j = []
    for r in range(self.rows):

```

```

        for c in range(self.cols):
            if self.layout[r][c] == dept_i:
                cells_i.append((r, c))
            elif self.layout[r][c] == dept_j:
                cells_j.append((r, c))

    for r_i, c_i in cells_i:
        for r_j, c_j in cells_j:
            if abs(r_i - r_j) + abs(c_i - c_j) == 1:
                return True

    return False

def exchange_departments(self, dept_i: int, dept_j: int):
    """Exchange the locations of two departments."""
    for r in range(self.rows):
        for c in range(self.cols):
            if self.layout[r][c] == dept_i:
                self.layout[r][c] = dept_j
            elif self.layout[r][c] == dept_j:
                self.layout[r][c] = dept_i

def solve(self, max_iterations: int = 100) -> Tuple[List[List[int]], float]:
    """
    Run CRAFT algorithm.

    Returns:
        Tuple of (final_layout, final_cost)
    """
    current_cost = self.calculate_total_cost()
    improved = True
    iteration = 0

    while improved and iteration < max_iterations:
        improved = False
        iteration += 1
        best_improvement = 0
        best_exchange = None

        # Evaluate all pairwise exchanges
        for i in range(self.n_depts):
            for j in range(i + 1, self.n_depts):
                if not self.can_exchange(i, j):
                    continue

                # Try exchange
                self.exchange_departments(i, j)
                new_cost = self.calculate_total_cost()
                improvement = current_cost - new_cost

                if improvement > best_improvement:
                    best_improvement = improvement
                    best_exchange = (i, j)

```

```

        # Undo exchange
        self.exchange_departments(i, j)

    # Make best exchange if improvement found
    if best_exchange is not None and best_improvement > 0:
        i, j = best_exchange
        self.exchange_departments(i, j)
        current_cost -= best_improvement
        improved = True
        print(f"Iteration {iteration}: Exchange {i}<->{j}, "
              f"Cost: {current_cost:.2f}")

    return self.layout, current_cost

# Example usage
flow = np.array([
    [0, 50, 30, 10, 0, 20],
    [50, 0, 40, 20, 10, 0],
    [30, 40, 0, 60, 30, 10],
    [10, 20, 60, 0, 50, 30],
    [0, 10, 30, 50, 0, 40],
    [20, 0, 10, 30, 40, 0]
])

# Initial layout (6 departments in 2x3 grid)
# Each department occupies one cell
initial_layout = [
    [0, 1, 2],
    [3, 4, 5]
]

areas = {i: 1 for i in range(6)} # Equal areas

craft = CRAFTSolver(flow, initial_layout, areas)
final_layout, final_cost = craft.solve()

print("\nFinal Layout:")
for row in final_layout:
    print(row)
print(f"Final Cost: {final_cost:.2f}")

```

17.4.2. BLOCPPLAN Algorithm

BLOCPPLAN constructs layouts by placing departments in a band structure, using relationship scores to guide placement. It starts with a TCR-based sequence and then evaluates random permutations to find a better arrangement. Each layout is scored by summing relationship values weighted by the inverse of inter-department distance.

```

class BLOCPPLANSolver:
    """
    BLOCPPLAN algorithm for facility layout.

```

```

Uses a band-based construction approach with relationship
chart scores to guide department placement.
"""

def __init__(self, departments: List[Department],
              rel_matrix: np.ndarray,
              building_ratio: float = 2.0):
    """
    Initialize BLOCPLAN solver.

    Args:
        departments: List of departments with areas
        rel_matrix: Relationship matrix (closeness values)
        building_ratio: Length/width ratio of building
    """
    self.departments = departments
    self.rel_matrix = rel_matrix
    self.n = len(departments)
    self.building_ratio = building_ratio

def calculate_tcr(self) -> List[Tuple[int, float]]:
    """Calculate Total Closeness Rating for each department."""
    tcr_list = []
    for i in range(self.n):
        tcr = sum(self.rel_matrix[i, j] for j in range(self.n) if j != i)
        tcr_list.append((i, tcr))
    return sorted(tcr_list, key=lambda x: -x[1])

def generate_layout(self, sequence: List[int]) -> Dict[int, Tuple[float, float,
↵ float, float]]:
    """
    Generate layout from department sequence.

    Args:
        sequence: Order to place departments

    Returns:
        Dict mapping dept_id to (x, y, width, height)
    """
    total_area = sum(d.area for d in self.departments)
    building_length = np.sqrt(total_area * self.building_ratio)
    building_width = total_area / building_length

    layout = {}
    current_x = 0
    current_y = 0
    row_height = 0

    for dept_id in sequence:
        dept = self.departments[dept_id]
        area = dept.area

```

```

        # Calculate dimensions
        width = min(building_width - current_x, np.sqrt(area))
        height = area / width

        # Check if fits in current row
        if current_x + width > building_width:
            # Start new row
            current_x = 0
            current_y += row_height
            row_height = 0
            width = min(building_width, np.sqrt(area))
            height = area / width

        layout[dept_id] = (current_x, current_y, width, height)
        current_x += width
        row_height = max(row_height, height)

    return layout

def evaluate_layout(self, layout: Dict[int, Tuple[float, float, float, float]])
    ↪ -> float:
    """Evaluate layout using relationship adjacency score."""
    score = 0

    for i in range(self.n):
        if i not in layout:
            continue
        x_i, y_i, w_i, h_i = layout[i]
        center_i = (x_i + w_i/2, y_i + h_i/2)

        for j in range(i + 1, self.n):
            if j not in layout:
                continue
            x_j, y_j, w_j, h_j = layout[j]
            center_j = (x_j + w_j/2, y_j + h_j/2)

            # Calculate distance
            dist = np.sqrt((center_i[0] - center_j[0])**2 +
                          (center_i[1] - center_j[1])**2)

            # Weight by relationship value
            rel_value = self.rel_matrix[i, j]
            score += rel_value / (1 + dist)

    return score

def solve(self, n_iterations: int = 100) -> Tuple[Dict, float]:
    """
    Solve using multiple random sequences.

    Returns:
        Best layout and its score
    """

```

```

# Start with TCR-based sequence
tcr_order = [i for i, _ in self.calculate_tcr()]
best_layout = self.generate_layout(tcr_order)
best_score = self.evaluate_layout(best_layout)

# Try random permutations
for _ in range(n_iterations):
    sequence = np.random.permutation(self.n).tolist()
    layout = self.generate_layout(sequence)
    score = self.evaluate_layout(layout)

    if score > best_score:
        best_score = score
        best_layout = layout

return best_layout, best_score

```

17.4.3. CORELAP Algorithm

CORELAP (Computerized Relationship Layout Planning) is a construction algorithm that places departments one at a time based on their relationship scores. It starts by placing the department with the highest TCR at the center of a grid. Each subsequent department is placed at the boundary position that maximizes the weighted relationship score with already-placed departments, with extra penalties for placing X-rated (undesirable) pairs close together.

```

class CORELAPSolver:
    """
    CORELAP algorithm for facility layout.

    Constructs layout by placing departments sequentially,
    starting with highest TCR and placing adjacent to
    related departments.
    """

    def __init__(self, departments: List[Department],
                 rel_matrix: np.ndarray):
        """
        Initialize CORELAP solver.

        Args:
            departments: List of departments with areas
            rel_matrix: Relationship matrix (closeness values)
        """
        self.departments = departments
        self.rel_matrix = rel_matrix
        self.n = len(departments)

    def calculate_tcr(self) -> List[Tuple[int, float]]:
        """Calculate Total Closeness Rating for each department."""
        tcr_list = []
        for i in range(self.n):
            tcr = sum(self.rel_matrix[i, j] for j in range(self.n) if j != i)

```

```

        tcr_list.append((i, tcr))
    return sorted(tcr_list, key=lambda x: -x[1])

def get_placement_score(self, dept_id: int,
                        position: Tuple[int, int],
                        placed: Dict[int, Tuple[int, int]],
                        grid_size: int) -> float:
    """Calculate placement score for a department at a position."""
    r, c = position
    score = 0

    for other_id, (other_r, other_c) in placed.items():
        # Calculate distance
        dist = abs(r - other_r) + abs(c - other_c)

        # Score based on relationship and distance
        rel_value = self.rel_matrix[dept_id, other_id]

        # Adjacency bonus
        if dist == 1:
            score += rel_value * 2
        elif dist == 2:
            score += rel_value
        else:
            score += rel_value / dist

        # Penalty for X relationships that are close
        if rel_value < 0 and dist <= 2:
            score += rel_value * 3 # Extra penalty

    return score

def find_boundary_positions(self, placed: Dict[int, Tuple[int, int]],
                           grid_size: int) -> List[Tuple[int, int]]:
    """Find available positions adjacent to placed departments."""
    if not placed:
        return [(grid_size // 2, grid_size // 2)]

    boundary = set()
    occupied = set(placed.values())

    for r, c in placed.values():
        for dr, dc in [(-1, 0), (1, 0), (0, -1), (0, 1)]:
            new_pos = (r + dr, c + dc)
            if (0 <= new_pos[0] < grid_size and
                0 <= new_pos[1] < grid_size and
                new_pos not in occupied):
                boundary.add(new_pos)

    return list(boundary)

def solve(self) -> Dict[int, Tuple[int, int]]:
    """

```

```

Run CORELAP algorithm.

Returns:
    Dict mapping dept_id to grid position (row, col)
"""
# Calculate TCR and sort departments
tcr_order = self.calculate_tcr()

# Determine grid size
grid_size = int(np.ceil(np.sqrt(self.n))) + 2

placed = {}

for dept_id, tcr in tcr_order:
    if not placed:
        # Place first department in center
        placed[dept_id] = (grid_size // 2, grid_size // 2)
        continue

    # Find best boundary position
    boundary = self.find_boundary_positions(placed, grid_size)

    best_pos = None
    best_score = float('-inf')

    for pos in boundary:
        score = self.get_placement_score(dept_id, pos, placed, grid_size)
        if score > best_score:
            best_score = score
            best_pos = pos

    if best_pos:
        placed[dept_id] = best_pos
    else:
        # Fallback: place in first available position
        for r in range(grid_size):
            for c in range(grid_size):
                if (r, c) not in set(placed.values()):
                    placed[dept_id] = (r, c)
                    break
            else:
                continue
        break

return placed

def visualize(self, placement: Dict[int, Tuple[int, int]]):
    """Visualize the CORELAP layout."""
    grid_size = max(max(r, c) for r, c in placement.values()) + 1

    fig, ax = plt.subplots(figsize=(10, 10))
    colors = plt.cm.Set3(np.linspace(0, 1, self.n))

```



```

for dept_id, (r, c) in placement.items():
    rect = patches.Rectangle(
        (c, grid_size - r - 1), 1, 1,
        linewidth=2, edgecolor='black',
        facecolor=colors[dept_id], alpha=0.7
    )
    ax.add_patch(rect)
    ax.text(c + 0.5, grid_size - r - 0.5,
            self.departments[dept_id].name,
            ha='center', va='center', fontsize=9)

ax.set_xlim(-0.5, grid_size + 0.5)
ax.set_ylim(-0.5, grid_size + 0.5)
ax.set_aspect('equal')
ax.grid(True, alpha=0.3)
ax.set_title('CORELAP Layout')

return fig

```

17.4.4. Genetic Algorithm for Layout

Genetic algorithms are effective for layout optimization, using permutation encoding and specialized crossover operators. Each chromosome is a permutation that maps facilities to locations. Order crossover (OX) preserves a segment from one parent and fills remaining positions from the other parent, maintaining permutation feasibility. Both swap and insertion mutations are applied to introduce variation.

```

import random
from typing import List, Tuple, Callable

class GeneticLayoutOptimizer:
    """
    Genetic algorithm for facility layout optimization.

    Uses permutation encoding where each chromosome
    represents an assignment of facilities to locations.
    """

    def __init__(self, flow_matrix: np.ndarray,
                 distance_matrix: np.ndarray,
                 pop_size: int = 50,
                 generations: int = 100,
                 mutation_rate: float = 0.1,
                 crossover_rate: float = 0.8):
        """
        Initialize GA optimizer.

        Args:
            flow_matrix: Flow between facilities
            distance_matrix: Distance between locations
            pop_size: Population size
            generations: Number of generations

```

```

        mutation_rate: Probability of mutation
        crossover_rate: Probability of crossover
    """
    self.F = flow_matrix
    self.D = distance_matrix
    self.n = flow_matrix.shape[0]
    self.pop_size = pop_size
    self.generations = generations
    self.mutation_rate = mutation_rate
    self.crossover_rate = crossover_rate

def fitness(self, chromosome: List[int]) -> float:
    """
    Calculate fitness (negative cost for minimization).

    Args:
        chromosome: Permutation representing assignment

    Returns:
        Fitness value (higher is better)
    """
    cost = 0
    for i in range(self.n):
        for j in range(self.n):
            loc_i = chromosome[i]
            loc_j = chromosome[j]
            cost += self.F[i, j] * self.D[loc_i, loc_j]
    return -cost # Negative because we want to minimize

def initialize_population(self) -> List[List[int]]:
    """Generate initial random population."""
    population = []
    for _ in range(self.pop_size):
        chromosome = list(range(self.n))
        random.shuffle(chromosome)
        population.append(chromosome)
    return population

def tournament_selection(self, population: List[List[int]],
                        fitnesses: List[float],
                        k: int = 3) -> List[int]:
    """Select parent using tournament selection."""
    tournament = random.sample(list(zip(population, fitnesses)), k)
    return max(tournament, key=lambda x: x[1])[0]

def order_crossover(self, parent1: List[int],
                   parent2: List[int]) -> Tuple[List[int], List[int]]:
    """
    Order crossover (OX) for permutations.

    Preserves relative order of elements from one parent
    while copying a segment from the other.
    """

```

```

size = len(parent1)

# Select crossover points
a, b = sorted(random.sample(range(size), 2))

# Create children
def create_child(p1, p2):
    child = [None] * size
    child[a:b+1] = p1[a:b+1]

    # Fill remaining from p2
    remaining = [x for x in p2 if x not in child[a:b+1]]
    pos = 0
    for i in range(size):
        if child[i] is None:
            child[i] = remaining[pos]
            pos += 1
    return child

child1 = create_child(parent1, parent2)
child2 = create_child(parent2, parent1)

return child1, child2

def swap_mutation(self, chromosome: List[int]) -> List[int]:
    """Swap mutation: exchange two random positions."""
    if random.random() < self.mutation_rate:
        i, j = random.sample(range(self.n), 2)
        chromosome = chromosome.copy()
        chromosome[i], chromosome[j] = chromosome[j], chromosome[i]
    return chromosome

def insertion_mutation(self, chromosome: List[int]) -> List[int]:
    """Insertion mutation: move an element to a new position."""
    if random.random() < self.mutation_rate:
        chromosome = chromosome.copy()
        i = random.randint(0, self.n - 1)
        j = random.randint(0, self.n - 1)
        element = chromosome.pop(i)
        chromosome.insert(j, element)
    return chromosome

def solve(self) -> Tuple[List[int], float]:
    """
    Run genetic algorithm.

    Returns:
        Best chromosome and its cost
    """
    # Initialize
    population = self.initialize_population()
    best_chromosome = None
    best_fitness = float('-inf')

```

```

fitness_history = []

for generation in range(self.generations):
    # Evaluate fitness
    fitnesses = [self.fitness(chrom) for chrom in population]

    # Track best
    gen_best_idx = np.argmax(fitnesses)
    if fitnesses[gen_best_idx] > best_fitness:
        best_fitness = fitnesses[gen_best_idx]
        best_chromosome = population[gen_best_idx].copy()

    fitness_history.append(-best_fitness) # Store cost

    # Create next generation
    new_population = []

    # Elitism: keep best individual
    new_population.append(population[gen_best_idx])

    while len(new_population) < self.pop_size:
        # Selection
        parent1 = self.tournament_selection(population, fitnesses)
        parent2 = self.tournament_selection(population, fitnesses)

        # Crossover
        if random.random() < self.crossover_rate:
            child1, child2 = self.order_crossover(parent1, parent2)
        else:
            child1, child2 = parent1.copy(), parent2.copy()

        # Mutation
        child1 = self.swap_mutation(child1)
        child2 = self.insertion_mutation(child2)

        new_population.extend([child1, child2])

    population = new_population[:self.pop_size]

    if generation % 20 == 0:
        print(f"Generation {generation}: Best cost = {-best_fitness:.2f}")

    return best_chromosome, -best_fitness

# Example usage
np.random.seed(42)
n = 8

# Random flow and distance matrices
flow = np.random.randint(0, 100, (n, n))
flow = (flow + flow.T) / 2 # Make symmetric

```

```

np.fill_diagonal(flow, 0)

# Grid distance matrix
distance = np.zeros((n, n))
positions = [(i // 4, i % 4) for i in range(n)]
for i in range(n):
    for j in range(n):
        distance[i, j] = abs(positions[i][0] - positions[j][0]) + \
            abs(positions[i][1] - positions[j][1])

ga = GeneticLayoutOptimizer(flow, distance, pop_size=50, generations=100)
best_solution, best_cost = ga.solve()

print(f"\nBest Solution: {best_solution}")
print(f"Best Cost: {best_cost:.2f}")

```

17.4.5. Simulated Annealing for Layout

Simulated annealing generates neighbor solutions by swapping two facility assignments and accepts worse solutions with a probability that decreases as the temperature cools. The following implementation uses a geometric cooling schedule with configurable initial temperature, cooling rate, and iterations per temperature level.

```

import math
import random

class SimulatedAnnealingLayout:
    """
    Simulated annealing for facility layout optimization.
    """

    def __init__(self, flow_matrix: np.ndarray,
                 distance_matrix: np.ndarray,
                 initial_temp: float = 1000,
                 cooling_rate: float = 0.995,
                 min_temp: float = 1,
                 iterations_per_temp: int = 100):
        """
        Initialize SA optimizer.

        Args:
            flow_matrix: Flow between facilities
            distance_matrix: Distance between locations
            initial_temp: Starting temperature
            cooling_rate: Temperature cooling factor
            min_temp: Stopping temperature
            iterations_per_temp: Iterations at each temperature
        """
        self.F = flow_matrix
        self.D = distance_matrix
        self.n = flow_matrix.shape[0]
        self.initial_temp = initial_temp
        self.cooling_rate = cooling_rate

```

```

self.min_temp = min_temp
self.iterations_per_temp = iterations_per_temp

def calculate_cost(self, assignment: List[int]) -> float:
    """Calculate total flow-weighted distance."""
    cost = 0
    for i in range(self.n):
        for j in range(self.n):
            cost += self.F[i, j] * self.D[assignment[i], assignment[j]]
    return cost

def get_neighbor(self, current: List[int]) -> List[int]:
    """Generate neighbor by swapping two elements."""
    neighbor = current.copy()
    i, j = random.sample(range(self.n), 2)
    neighbor[i], neighbor[j] = neighbor[j], neighbor[i]
    return neighbor

def accept_probability(self, current_cost: float,
                      new_cost: float,
                      temperature: float) -> float:
    """Calculate probability of accepting worse solution."""
    if new_cost < current_cost:
        return 1.0
    return math.exp((current_cost - new_cost) / temperature)

def solve(self) -> Tuple[List[int], float]:
    """
    Run simulated annealing algorithm.

    Returns:
        Best assignment and its cost
    """
    # Initialize with random solution
    current = list(range(self.n))
    random.shuffle(current)
    current_cost = self.calculate_cost(current)

    best = current.copy()
    best_cost = current_cost

    temperature = self.initial_temp
    cost_history = []

    while temperature > self.min_temp:
        for _ in range(self.iterations_per_temp):
            # Generate neighbor
            neighbor = self.get_neighbor(current)
            neighbor_cost = self.calculate_cost(neighbor)

            # Accept or reject
            if random.random() < self.accept_probability(
                current_cost, neighbor_cost, temperature

```

```

    ):
        current = neighbor
        current_cost = neighbor_cost

        # Update best
        if current_cost < best_cost:
            best = current.copy()
            best_cost = current_cost

        cost_history.append(best_cost)
        temperature *= self.cooling_rate

    return best, best_cost

# Example
sa = SimulatedAnnealingLayout(flow, distance)
best_solution, best_cost = sa.solve()

print(f"SA Best Solution: {best_solution}")
print(f"SA Best Cost: {best_cost:.2f}")

```

17.5. Cellular Manufacturing Layout

Cellular manufacturing organizes production into cells, each dedicated to producing a family of similar parts. The key challenges are identifying part families and forming machine groups.

17.5.1. Group Technology

Group technology classifies parts based on design and manufacturing similarities. Parts with similar shapes, sizes, or processing requirements are grouped into families.

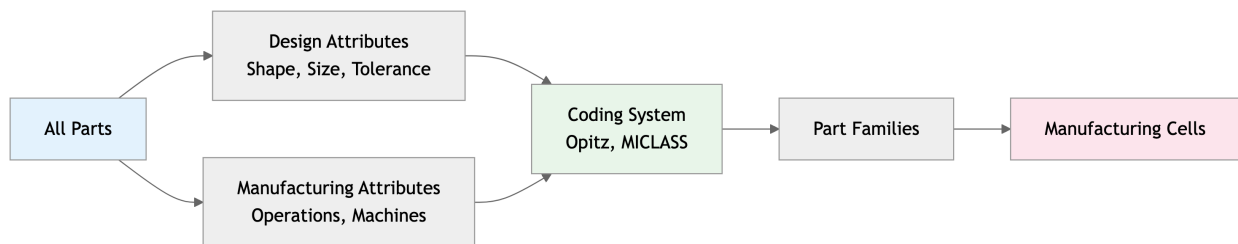


Figure 17.7.: Group Technology Part Family Classification

17.5.2. Machine-Part Matrix

The machine-part incidence matrix shows which parts require which machines:

$$A = [a_{ij}] \text{ where } a_{ij} = \begin{cases} 1 & \text{if part } j \text{ requires machine } i \\ 0 & \text{otherwise} \end{cases}$$

The goal is to rearrange rows and columns to form block-diagonal structure, where each block represents a cell. The function below constructs this matrix from routing data, where each part's routing specifies the set of machines it visits.

```

import numpy as np
from typing import List, Tuple, Set

def create_machine_part_matrix(
    n_machines: int,
    n_parts: int,
    routings: Dict[int, List[int]]
) -> np.ndarray:
    """
    Create machine-part incidence matrix.

    Args:
        n_machines: Number of machines
        n_parts: Number of parts
        routings: Dict mapping part_id to list of machine_ids

    Returns:
        Binary incidence matrix
    """
    matrix = np.zeros((n_machines, n_parts), dtype=int)

    for part_id, machines in routings.items():
        for machine_id in machines:
            matrix[machine_id, part_id] = 1

    return matrix

# Example machine-part matrix
routings = {
    0: [0, 2, 4],      # Part 0 uses machines 0, 2, 4
    1: [1, 3],         # Part 1 uses machines 1, 3
    2: [0, 2],         # Part 2 uses machines 0, 2
    3: [1, 3, 5],      # Part 3 uses machines 1, 3, 5
    4: [0, 4],         # Part 4 uses machines 0, 4
    5: [1, 5],         # Part 5 uses machines 1, 5
    6: [2, 4],         # Part 6 uses machines 2, 4
    7: [3, 5]          # Part 7 uses machines 3, 5
}

A = create_machine_part_matrix(6, 8, routings)
print("Machine-Part Matrix:")
print(A)

```

17.5.3. Cell Formation Algorithms

17.5.3.1. Rank Order Clustering (ROC)

ROC rearranges the matrix by sorting rows and columns based on binary weights. Each row is treated as a binary number (reading left to right), and rows are sorted by decreasing weight. The same procedure is then applied to columns (reading top to bottom). The algorithm iterates until the ordering stabilizes, producing a block-diagonal structure that reveals natural machine-part groupings.


```

def rank_order_clustering(A: np.ndarray,
                          max_iterations: int = 20) -> Tuple[np.ndarray, List[int],
↳ List[int]]:
    """
    Rank Order Clustering algorithm for cell formation.

    Args:
        A: Machine-part incidence matrix
        max_iterations: Maximum iterations

    Returns:
        Rearranged matrix, row order, column order
    """
    m, n = A.shape
    row_order = list(range(m))
    col_order = list(range(n))

    for iteration in range(max_iterations):
        # Calculate row weights (binary number from left to right)
        row_weights = []
        for i in range(m):
            weight = sum(A[row_order[i], col_order[j]] * (2 ** (n - 1 - j))
                          for j in range(n))
            row_weights.append((row_order[i], weight))

        # Sort rows by weight (descending)
        row_weights.sort(key=lambda x: -x[1])
        new_row_order = [x[0] for x in row_weights]

        # Calculate column weights (binary number from top to bottom)
        col_weights = []
        for j in range(n):
            weight = sum(A[new_row_order[i], col_order[j]] * (2 ** (m - 1 - i))
                          for i in range(m))
            col_weights.append((col_order[j], weight))

        # Sort columns by weight (descending)
        col_weights.sort(key=lambda x: -x[1])
        new_col_order = [x[0] for x in col_weights]

        # Check for convergence
        if new_row_order == row_order and new_col_order == col_order:
            break

        row_order = new_row_order
        col_order = new_col_order

    # Create rearranged matrix
    rearranged = np.zeros_like(A)
    for i in range(m):
        for j in range(n):
            rearranged[i, j] = A[row_order[i], col_order[j]]

```

```

    return rearranged, row_order, col_order

# Apply ROC
rearranged, row_order, col_order = rank_order_clustering(A)
print("\nRearranged Matrix:")
print(rearranged)
print(f"Row order: {row_order}")
print(f"Column order: {col_order}")

```

17.5.3.2. Direct Clustering Algorithm (DCA)

The Direct Clustering Algorithm forms cells by merging part families that share machines. Starting with each part as its own family, the algorithm iteratively merges pairs whose machine sets overlap by more than a specified threshold (here, 30% Jaccard similarity). After convergence, each remaining family defines a cell with its associated machines.

```

def direct_clustering_algorithm(A: np.ndarray) -> Tuple[List[Set[int]],
    ↪ List[Set[int]]]:
    """
    Direct Clustering Algorithm for cell formation.

    Args:
        A: Machine-part incidence matrix

    Returns:
        Tuple of (machine_cells, part_families)
    """
    m, n = A.shape

    # Initialize: each part in its own family
    part_families = [{j} for j in range(n)]

    # Merge parts that share machines
    changed = True
    while changed:
        changed = False
        for i in range(len(part_families)):
            for j in range(i + 1, len(part_families)):
                if i >= len(part_families) or j >= len(part_families):
                    break

                # Check if families share a machine
                machines_i = set()
                for p in part_families[i]:
                    machines_i.update(np.where(A[:, p] == 1)[0])

                machines_j = set()
                for p in part_families[j]:
                    machines_j.update(np.where(A[:, p] == 1)[0])

                # Merge if significant overlap
                overlap = len(machines_i & machines_j)

```

```

        union = len(machines_i | machines_j)

        if overlap / union > 0.3: # Threshold
            part_families[i] = part_families[i] | part_families[j]
            part_families.pop(j)
            changed = True
            break
    if changed:
        break

# Determine machine cells
machine_cells = []
for family in part_families:
    cell_machines = set()
    for part in family:
        cell_machines.update(np.where(A[:, part] == 1)[0])
    machine_cells.append(cell_machines)

return machine_cells, part_families

# Apply DCA
machine_cells, part_families = direct_clustering_algorithm(A)
print("\nCell Formation Results:")
for i, (machines, parts) in enumerate(zip(machine_cells, part_families)):
    print(f"Cell {i+1}: Machines {machines}, Parts {parts}")

```

17.5.4. Cell Efficiency Metrics

Two key metrics evaluate cell formation quality. Grouping efficacy measures how close the rearranged matrix is to a perfect block-diagonal form by penalizing both exceptional elements (ones outside diagonal blocks) and voids (zeros inside diagonal blocks). Grouping efficiency is the fraction of all ones that fall within cell boundaries.

```

def calculate_cell_metrics(A: np.ndarray,
                           machine_cells: List[Set[int]],
                           part_families: List[Set[int]]) -> Dict[str, float]:
    """
    Calculate cell formation quality metrics.

    Args:
        A: Original machine-part matrix
        machine_cells: List of machine sets per cell
        part_families: List of part sets per cell

    Returns:
        Dictionary of metrics
    """
    total_ones = A.sum()

    # Count ones within cells (block diagonal)
    within_cell_ones = 0
    for machines, parts in zip(machine_cells, part_families):

```

```

        for m in machines:
            for p in parts:
                within_cell_ones += A[m, p]

# Calculate metrics
# Grouping efficacy
n1 = within_cell_ones
n0_outside = 0 # Zeros outside blocks

total_cells_area = 0
for machines, parts in zip(machine_cells, part_families):
    total_cells_area += len(machines) * len(parts)

# Exceptional elements (ones outside diagonal blocks)
exceptional = total_ones - within_cell_ones

# Voids (zeros inside diagonal blocks)
voids = total_cells_area - within_cell_ones

# Grouping efficacy
efficacy = (total_ones - exceptional) / (total_ones + voids) if (total_ones +
↪ voids) > 0 else 0

# Grouping efficiency
efficiency = within_cell_ones / total_ones if total_ones > 0 else 0

return {
    'grouping_efficacy': efficacy,
    'grouping_efficiency': efficiency,
    'exceptional_elements': exceptional,
    'voids': voids,
    'within_cell_ones': within_cell_ones,
    'total_ones': total_ones
}

metrics = calculate_cell_metrics(A, machine_cells, part_families)
print("\nCell Formation Metrics:")
for metric, value in metrics.items():
    print(f"    {metric}: {value:.3f}")

```

17.6. Warehouse Layout Optimization

Warehouse layout focuses on storage location assignment, aisle design, and order picking efficiency.

17.6.1. Storage Location Assignment Problem

The Storage Location Assignment Problem (SLAP) assigns products to storage locations to minimize picking time:

$$\min \sum_{i=1}^n \sum_{j=1}^m f_i \cdot d_j \cdot x_{ij}$$

where:

- f_i is the picking frequency of product i
- d_j is the distance from location j to the I/O point
- $x_{ij} = 1$ if product i is assigned to location j

💡 Storage Policies

Dedicated Storage: Each product has a fixed location. Simple to manage but uses more space.

Random Storage: Products go to any available location. Higher space utilization but requires tracking system.

Class-Based Storage: Products grouped by turnover rate. A/B/C zones balance efficiency and simplicity.

Cube-per-Order Index (COI): Products ranked by volume/frequency ratio. Minimizes travel distance.

The `WarehouseLayoutOptimizer` class implements three storage assignment strategies: optimal (Hungarian algorithm), COI-based, and ABC class-based. It takes product picking frequencies and location distances as input, then assigns products to locations to minimize total travel distance.

```
from scipy.optimize import linear_sum_assignment

class WarehouseLayoutOptimizer:
    """
    Optimize warehouse storage location assignment.
    """

    def __init__(self, n_products: int, n_locations: int,
                  frequencies: np.ndarray,
                  distances: np.ndarray):
        """
        Initialize optimizer.

        Args:
            n_products: Number of products (SKUs)
            n_locations: Number of storage locations
            frequencies: Picking frequency of each product
            distances: Distance from each location to I/O point
        """
        self.n_products = n_products
        self.n_locations = n_locations
        self.frequencies = frequencies
        self.distances = distances

    def solve_optimal(self) -> Tuple[Dict[int, int], float]:
        """
        Solve SLAP optimally using Hungarian algorithm.

        Returns:
            Assignment dict and total cost
        """
        # Create cost matrix
```

```

cost_matrix = np.zeros((self.n_products, self.n_locations))
for i in range(self.n_products):
    for j in range(self.n_locations):
        cost_matrix[i, j] = self.frequencies[i] * self.distances[j]

# Solve assignment problem
row_ind, col_ind = linear_sum_assignment(cost_matrix)

assignment = {i: j for i, j in zip(row_ind, col_ind)}
total_cost = cost_matrix[row_ind, col_ind].sum()

return assignment, total_cost

def coi_based_assignment(self, volumes: np.ndarray) -> Tuple[Dict[int, int],
↪ float]:
    """
    Cube-per-Order Index based assignment.

    Args:
        volumes: Storage volume needed for each product

    Returns:
        Assignment dict and total cost
    """
    # Calculate COI = volume / frequency
    coi = volumes / (self.frequencies + 0.001) # Avoid division by zero

    # Sort products by COI (ascending)
    product_order = np.argsort(coi)

    # Sort locations by distance (ascending)
    location_order = np.argsort(self.distances)

    # Assign products to locations
    assignment = {}
    for i, product in enumerate(product_order):
        if i < len(location_order):
            assignment[product] = location_order[i]

    # Calculate cost
    total_cost = sum(
        self.frequencies[p] * self.distances[l]
        for p, l in assignment.items()
    )

    return assignment, total_cost

def abc_zone_assignment(self) -> Tuple[Dict[int, int], float]:
    """
    ABC class-based storage assignment.

    Returns:
        Assignment dict and total cost

```

```

"""
# Classify products (80-15-5 rule)
sorted_products = np.argsort(self.frequencies[::-1])

n_a = int(0.2 * self.n_products) # Top 20% = A class
n_b = int(0.3 * self.n_products) # Next 30% = B class
# Remaining = C class

a_products = set(sorted_products[:n_a])
b_products = set(sorted_products[n_a:n_a+n_b])
c_products = set(sorted_products[n_a+n_b:])

# Sort locations by distance
sorted_locations = np.argsort(self.distances)

n_a_loc = int(0.2 * self.n_locations)
n_b_loc = int(0.3 * self.n_locations)

a_locations = list(sorted_locations[:n_a_loc])
b_locations = list(sorted_locations[n_a_loc:n_a_loc+n_b_loc])
c_locations = list(sorted_locations[n_a_loc+n_b_loc:])

# Assign within zones
assignment = {}

for i, product in enumerate(a_products):
    if i < len(a_locations):
        assignment[product] = a_locations[i]

for i, product in enumerate(b_products):
    if i < len(b_locations):
        assignment[product] = b_locations[i]

for i, product in enumerate(c_products):
    if i < len(c_locations):
        assignment[product] = c_locations[i]

# Calculate cost
total_cost = sum(
    self.frequencies[p] * self.distances[l]
    for p, l in assignment.items()
)

return assignment, total_cost

# Example warehouse
np.random.seed(42)
n_products = 20
n_locations = 25

# Pareto-distributed frequencies (80-20 rule)
frequencies = np.random.pareto(1.5, n_products) + 1

```

17. Facility Layout Optimization

```
frequencies = frequencies / frequencies.sum() * 1000 # Scale to 1000 picks/day

# Grid layout distances
rows, cols = 5, 5
distances = np.zeros(n_locations)
io_position = (0, 0) # I/O at corner
for loc in range(n_locations):
    row = loc // cols
    col = loc % cols
    distances[loc] = abs(row - io_position[0]) + abs(col - io_position[1])

optimizer = WarehouseLayoutOptimizer(n_products, n_locations, frequencies,
    ↪ distances)

# Compare methods
optimal_assign, optimal_cost = optimizer.solve_optimal()
abc_assign, abc_cost = optimizer.abc_zone_assignment()

print("Warehouse Layout Comparison:")
print(f" Optimal Assignment Cost: {optimal_cost:.2f}")
print(f" ABC Zone Assignment Cost: {abc_cost:.2f}")
print(f" Improvement: {(abc_cost - optimal_cost) / abc_cost * 100:.1f}%")
```

17.6.2. Aisle Design

The number and length of aisles directly affect both storage capacity and picking performance. More aisles increase capacity but also increase the average cross-aisle travel distance. The function below calculates key performance metrics for a given aisle configuration, assuming random storage.

```
def analyze_aisle_configuration(
    n_aisles: int,
    aisle_length: float,
    rack_height: float,
    picks_per_day: int,
    travel_speed: float = 1.5 # m/s
) -> Dict[str, float]:
    """
    Analyze picking performance for different aisle configurations.

    Args:
        n_aisles: Number of aisles
        aisle_length: Length of each aisle (meters)
        rack_height: Height of storage racks (levels)
        picks_per_day: Number of picks per day
        travel_speed: Walking speed (m/s)

    Returns:
        Performance metrics
    """
    # Storage capacity
    locations_per_aisle = aisle_length * 2 * rack_height # Both sides
    total_capacity = n_aisles * locations_per_aisle
```



```

# Average travel distance (assuming random storage)
# Horizontal: average half aisle length per pick
avg_horizontal = aisle_length / 2

# Cross-aisle (between aisles): average half of total width
aisle_spacing = 3 # meters between aisles
avg_cross_aisle = (n_aisles * aisle_spacing) / 2

# Vertical (if using lift): average half rack height
avg_vertical = rack_height / 2

# Total distance per pick
avg_distance = avg_horizontal + avg_cross_aisle

# Travel time
time_per_pick = avg_distance / travel_speed
total_travel_time = picks_per_day * time_per_pick / 3600 # hours

return {
    'total_capacity': total_capacity,
    'avg_distance_per_pick': avg_distance,
    'time_per_pick_seconds': time_per_pick,
    'total_travel_hours_per_day': total_travel_time,
    'picks_per_hour': picks_per_day / 8 # Assuming 8-hour shift
}

# Compare configurations
for n_aisles in [4, 6, 8]:
    metrics = analyze_aisle_configuration(
        n_aisles=n_aisles,
        aisle_length=30,
        rack_height=4,
        picks_per_day=500
    )
    print(f"\n{n_aisles} Aisles:")
    for metric, value in metrics.items():
        print(f"    {metric}: {value:.2f}")

```

17.7. Order Picking and Warehouse Routing

Order picking—retrieving items from storage to fulfill customer orders—accounts for 55-65% of warehouse operating costs. The routing policy determines the path pickers follow through the warehouse, directly impacting travel distance and productivity. This section presents the major routing policies and their optimization.

17.7.1. The Order Picking Problem

Order picking involves traveling through the warehouse to collect items on a pick list. The challenge is determining the route that minimizes travel distance while visiting all required locations.

Single Order Picking: The picker completes one order before starting the next. Simple to implement and highly accurate, but involves the most travel distance.

Batch Picking: Multiple orders are picked simultaneously. Items are sorted into individual orders

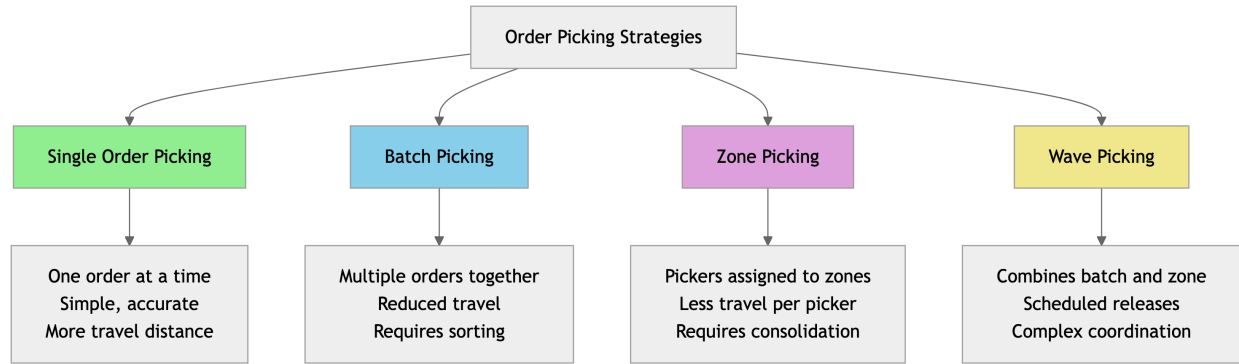


Figure 17.8.: Order Picking Strategies

after picking. Reduces travel but requires a sorting step.

Zone Picking: The warehouse is divided into zones, each with dedicated pickers. Orders requiring items from multiple zones must be consolidated.

Wave Picking: Orders are released in waves based on shipping schedules. Combines elements of batch and zone picking for high-volume operations.

17.7.2. Routing Policies

Different routing policies determine how pickers traverse the warehouse aisles. Each policy trades off simplicity against optimality.

17.7.2.1. 1. S-Shape (Traversal) Policy

The picker enters an aisle from one end and exits from the other, traversing the entire aisle if it contains any picks. Aisles with no picks are skipped entirely.

Advantages: - Simple to implement and follow - No decision-making required during picking - Good for high-density pick lists

Disadvantages: - May traverse unnecessary aisle sections - Inefficient for sparse pick lists

17.7.2.2. 2. Return Policy

The picker enters and exits each aisle from the same end (typically the front). Only travels as far as the farthest pick in each aisle.

Advantages: - Minimizes unnecessary travel within aisles - Good for picks concentrated near aisle ends

Disadvantages: - May require more cross-aisle travel - Inefficient when picks are distributed throughout aisles

17.7.2.3. 3. Midpoint Policy

The aisle is conceptually divided at the midpoint. Picks in the front half are accessed from the front; picks in the back half are accessed from the back.

Advantages: - Reduces within-aisle travel - Balances front and back access

Disadvantages: - May miss optimization opportunities - Requires tracking which half contains each pick

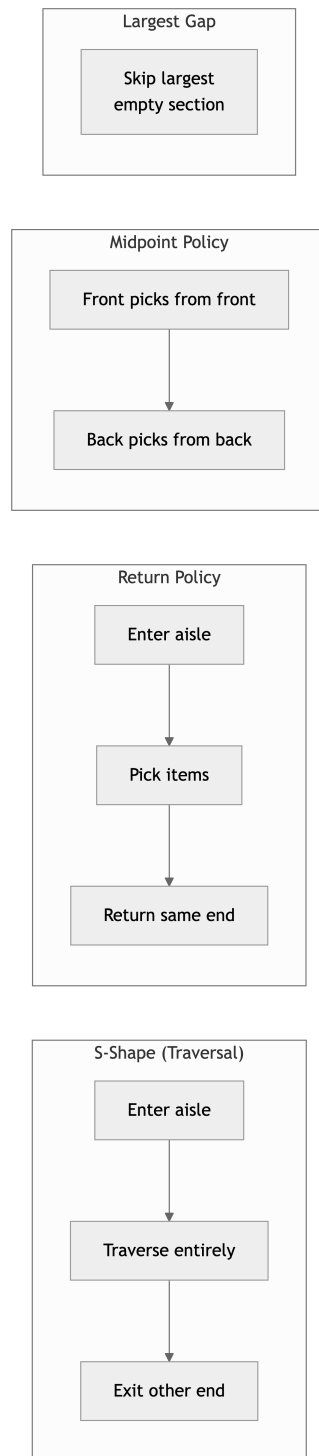


Figure 17.9.: Comparison of Warehouse Routing Policies

17.7.2.4. 4. Largest Gap Policy

For each aisle, the picker skips the largest gap between adjacent picks (or between a pick and an aisle end). This determines whether to enter from front, back, or both ends.

Advantages: - Often produces near-optimal routes - Adapts to pick distribution

Disadvantages: - More complex to implement - Requires analyzing gaps in each aisle

17.7.2.5. 5. Combined Policy

For each aisle, compare S-shape (full traversal) against Return (enter and return). Choose the policy that minimizes distance for that specific aisle.

Advantages: - Better than pure S-shape or Return alone - Adapts to each aisle's pick pattern

Disadvantages: - Requires per-aisle decision - Not globally optimal

17.7.2.6. 6. Optimal (TSP-Based) Policy

Formulate the pick route as a Traveling Salesman Problem and solve to optimality. Guarantees minimum travel distance but computationally expensive.

17.7.3. Mathematical Formulation

17.7.3.1. Distance Calculations

For a warehouse with n_a aisles of length L , aisle spacing w , and pick locations P :

S-Shape Distance:

$$D_s = \sum_{a \in A(P)} L + (n_a - 1) \cdot w$$

where $A(P)$ is the set of aisles containing picks.

Return Distance:

$$D_r = \sum_{a \in A(P)} 2 \cdot d_{max}(a) + (n_a - 1) \cdot w$$

where $d_{max}(a)$ is the distance to the farthest pick in aisle a .

Largest Gap Distance:

$$D_{lg} = \sum_{a \in A(P)} (L - g_{max}(a)) + (n_a - 1) \cdot w$$

where $g_{max}(a)$ is the largest gap in aisle a .

17.7.3.2. Optimal Route Formulation (TSP)

$$\begin{aligned} & \text{minimize} && \sum_{i=0}^n \sum_{j=0}^n d_{ij} x_{ij} \\ & \text{subject to} && \sum_{j=0}^n x_{ij} = 1, \quad \forall i \\ & && \sum_{i=0}^n x_{ij} = 1, \quad \forall j \\ & && u_i - u_j + n \cdot x_{ij} \leq n - 1, \quad \forall i, j \geq 1 \\ & && x_{ij} \in \{0, 1\}, \quad u_i \geq 0 \end{aligned}$$

where node 0 is the depot (I/O point), nodes 1 to n are pick locations, and the subtour elimination constraints (MTZ formulation) ensure a single tour.

17.7.4. Python Implementation

The implementation below provides a unified interface for all six routing policies. The `PickRouteOptimizer` class takes a warehouse layout definition and computes the total travel distance for any pick list under each policy. For the optimal policy, it uses Held-Karp dynamic programming (exact TSP) for up to 12 picks and falls back to nearest-neighbor for larger lists. The `BatchPickingOptimizer` class groups orders into batches using proximity, seed, or savings-based methods.

```
import numpy as np
from typing import List, Tuple, Dict, Optional
from dataclasses import dataclass
from enum import Enum
import heapq

class RoutingPolicy(Enum):
    """Available routing policies."""
    S_SHAPE = "s_shape"
    RETURN = "return"
    MIDPOINT = "midpoint"
    LARGEST_GAP = "largest_gap"
    COMBINED = "combined"
    OPTIMAL = "optimal"

@dataclass
class PickLocation:
    """Represents a pick location in the warehouse."""
    aisle: int
    position: float # Distance from front of aisle (0 to aisle_length)
    item_id: str
    quantity: int = 1

@dataclass
class WarehouseLayout:
    """Defines the warehouse layout parameters."""
    n_aisles: int
    aisle_length: float
    aisle_width: float # Width of picking aisle
    cross_aisle_width: float # Width of cross aisles
    depot_position: Tuple[float, float] = (0, 0) # (x, y) of I/O point

class PickRouteOptimizer:
    """
    Optimize order picking routes using various routing policies.

    Implements S-Shape, Return, Midpoint, Largest Gap, Combined,
    and Optimal (TSP-based) routing strategies.
    """

    def __init__(self, layout: WarehouseLayout):
```

```

    """
    Initialize the route optimizer.

    Args:
        layout: Warehouse layout parameters
    """
    self.layout = layout
    self.aisle_spacing = layout.aisle_width + layout.cross_aisle_width

def get_location_coordinates(self, loc: PickLocation) -> Tuple[float, float]:
    """Convert pick location to (x, y) coordinates."""
    x = loc.aisle * self.aisle_spacing
    y = loc.position
    return (x, y)

def rectilinear_distance(self, p1: Tuple[float, float],
                        p2: Tuple[float, float]) -> float:
    """Calculate rectilinear (Manhattan) distance."""
    return abs(p1[0] - p2[0]) + abs(p1[1] - p2[1])

def s_shape_route(self, picks: List[PickLocation]) -> Tuple[float, List[int]]:
    """
    Calculate route using S-Shape (traversal) policy.

    The picker enters each aisle with picks from one end and
    exits from the other, alternating direction between aisles.

    Args:
        picks: List of pick locations

    Returns:
        Tuple of (total_distance, pick_sequence)
    """
    if not picks:
        return 0.0, []

    # Group picks by aisle
    aisle_picks = {}
    for i, pick in enumerate(picks):
        if pick.aisle not in aisle_picks:
            aisle_picks[pick.aisle] = []
        aisle_picks[pick.aisle].append((i, pick))

    # Sort aisles
    sorted_aisles = sorted(aisle_picks.keys())

    total_distance = 0.0
    sequence = []
    current_pos = self.layout.depot_position
    entering_from_front = True

    for aisle in sorted_aisles:
        aisle_x = aisle * self.aisle_spacing

```

```

    # Travel to aisle entrance
    if entering_from_front:
        entrance = (aisle_x, 0)
    else:
        entrance = (aisle_x, self.layout.aisle_length)

    total_distance += self.rectilinear_distance(current_pos, entrance)

    # Sort picks within aisle
    aisle_items = aisle_picks[aisle]
    if entering_from_front:
        aisle_items.sort(key=lambda x: x[1].position)
    else:
        aisle_items.sort(key=lambda x: x[1].position, reverse=True)

    sequence.extend([idx for idx, _ in aisle_items])

    # Traverse entire aisle
    total_distance += self.layout.aisle_length

    # Update current position (exit from opposite end)
    if entering_from_front:
        current_pos = (aisle_x, self.layout.aisle_length)
    else:
        current_pos = (aisle_x, 0)

    entering_from_front = not entering_from_front

# Return to depot
total_distance += self.rectilinear_distance(current_pos,
                                             self.layout.depot_position)

return total_distance, sequence

def return_route(self, picks: List[PickLocation]) -> Tuple[float, List[int]]:
    """
    Calculate route using Return policy.

    The picker enters and exits each aisle from the front,
    traveling only as far as the farthest pick.

    Args:
        picks: List of pick locations

    Returns:
        Tuple of (total_distance, pick_sequence)
    """
    if not picks:
        return 0.0, []

    aisle_picks = {}
    for i, pick in enumerate(picks):

```

```

        if pick.aisle not in aisle_picks:
            aisle_picks[pick.aisle] = []
        aisle_picks[pick.aisle].append((i, pick))

    sorted_aisles = sorted(aisle_picks.keys())

    total_distance = 0.0
    sequence = []
    current_pos = self.layout.depot_position

    for aisle in sorted_aisles:
        aisle_x = aisle * self.aisle_spacing
        entrance = (aisle_x, 0)

        # Travel to aisle entrance
        total_distance += self.rectilinear_distance(current_pos, entrance)

        # Sort picks by position (front to back)
        aisle_items = aisle_picks[aisle]
        aisle_items.sort(key=lambda x: x[1].position)

        # Find farthest pick
        max_position = max(p.position for _, p in aisle_items)

        # Travel to farthest pick and back
        total_distance += 2 * max_position

        sequence.extend([idx for idx, _ in aisle_items])

        current_pos = entrance

    # Return to depot
    total_distance += self.rectilinear_distance(current_pos,
                                                self.layout.depot_position)

    return total_distance, sequence

def midpoint_route(self, picks: List[PickLocation]) -> Tuple[float, List[int]]:
    """
    Calculate route using Midpoint policy.

    Picks in the front half of aisles are accessed from the front;
    picks in the back half are accessed from the back.

    Args:
        picks: List of pick locations

    Returns:
        Tuple of (total_distance, pick_sequence)
    """
    if not picks:
        return 0.0, []

```



```

midpoint = self.layout.aisle_length / 2

aisle_picks = {}
for i, pick in enumerate(picks):
    if pick.aisle not in aisle_picks:
        aisle_picks[pick.aisle] = {'front': [], 'back': []}

    if pick.position <= midpoint:
        aisle_picks[pick.aisle]['front'].append((i, pick))
    else:
        aisle_picks[pick.aisle]['back'].append((i, pick))

sorted_aisles = sorted(aisle_picks.keys())

total_distance = 0.0
sequence = []

# Front pass (left to right)
current_pos = self.layout.depot_position
for aisle in sorted_aisles:
    aisle_x = aisle * self.aisle_spacing
    front_picks = aisle_picks[aisle]['front']

    if front_picks:
        entrance = (aisle_x, 0)
        total_distance += self.rectilinear_distance(current_pos, entrance)

        max_pos = max(p.position for _, p in front_picks)
        total_distance += 2 * max_pos

        front_picks.sort(key=lambda x: x[1].position)
        sequence.extend([idx for idx, _ in front_picks])

        current_pos = entrance

# Travel to back cross-aisle
if any(aisle_picks[a]['back'] for a in sorted_aisles):
    back_start = (sorted_aisles[-1] * self.aisle_spacing,
                  self.layout.aisle_length)
    total_distance += self.rectilinear_distance(current_pos, back_start)
    current_pos = back_start

# Back pass (right to left)
for aisle in reversed(sorted_aisles):
    aisle_x = aisle * self.aisle_spacing
    back_picks = aisle_picks[aisle]['back']

    if back_picks:
        entrance = (aisle_x, self.layout.aisle_length)
        total_distance += self.rectilinear_distance(current_pos, entrance)

        min_pos = min(p.position for _, p in back_picks)
        travel_in = self.layout.aisle_length - min_pos

```

```

        total_distance += 2 * travel_in

        back_picks.sort(key=lambda x: x[1].position, reverse=True)
        sequence.extend([idx for idx, _ in back_picks])

        current_pos = entrance

    # Return to depot
    total_distance += self.rectilinear_distance(current_pos,
                                                self.layout.depot_position)

    return total_distance, sequence

def largest_gap_route(self, picks: List[PickLocation]) -> Tuple[float,
↪ List[int]]:
    """
    Calculate route using Largest Gap policy.

    In each aisle, skip the largest gap between picks (or between
    a pick and an aisle end). This determines entry direction.

    Args:
        picks: List of pick locations

    Returns:
        Tuple of (total_distance, pick_sequence)
    """
    if not picks:
        return 0.0, []

    aisle_picks = {}
    for i, pick in enumerate(picks):
        if pick.aisle not in aisle_picks:
            aisle_picks[pick.aisle] = []
        aisle_picks[pick.aisle].append((i, pick))

    sorted_aisles = sorted(aisle_picks.keys())

    total_distance = 0.0
    sequence = []
    current_pos = self.layout.depot_position

    for aisle in sorted_aisles:
        aisle_x = aisle * self.aisle_spacing
        items = aisle_picks[aisle]
        positions = sorted([p.position for _, p in items])

        # Calculate gaps (including ends)
        gaps = [(0, positions[0])] # Gap from front to first pick
        for k in range(len(positions) - 1):
            gaps.append((positions[k], positions[k + 1]))
        gaps.append((positions[-1], self.layout.aisle_length)) # Gap to back

```

```

# Find largest gap
gap_sizes = [(g[1] - g[0], g) for g in gaps]
largest_gap = max(gap_sizes, key=lambda x: x[0])[1]

# Determine travel pattern based on largest gap
if largest_gap[0] == 0:
    # Largest gap at front: enter from back only
    entrance = (aisle_x, self.layout.aisle_length)
    exit_point = (aisle_x, largest_gap[1])
    travel_dist = self.layout.aisle_length - largest_gap[1]
elif largest_gap[1] == self.layout.aisle_length:
    # Largest gap at back: enter from front only
    entrance = (aisle_x, 0)
    exit_point = (aisle_x, largest_gap[0])
    travel_dist = largest_gap[0]
else:
    # Gap in middle: enter from both ends
    entrance = (aisle_x, 0)
    exit_point = (aisle_x, self.layout.aisle_length)
    travel_dist = (largest_gap[0] +
                   (self.layout.aisle_length - largest_gap[1]))

total_distance += self.rectilinear_distance(current_pos, entrance)
total_distance += travel_dist

# Add picks in traversal order
items.sort(key=lambda x: x[1].position)
sequence.extend([idx for idx, _ in items])

current_pos = exit_point

# Return to depot
total_distance += self.rectilinear_distance(current_pos,
                                             self.layout.depot_position)

return total_distance, sequence

def combined_route(self, picks: List[PickLocation]) -> Tuple[float, List[int]]:
    """
    Calculate route using Combined policy.

    For each aisle, choose between S-Shape and Return based on
    which minimizes distance for that aisle.

    Args:
        picks: List of pick locations

    Returns:
        Tuple of (total_distance, pick_sequence)
    """
    if not picks:
        return 0.0, []

```

```

aisle_picks = {}
for i, pick in enumerate(picks):
    if pick.aisle not in aisle_picks:
        aisle_picks[pick.aisle] = []
    aisle_picks[pick.aisle].append((i, pick))

sorted_aisles = sorted(aisle_picks.keys())

total_distance = 0.0
sequence = []
current_pos = self.layout.depot_position
at_front = True

for aisle in sorted_aisles:
    aisle_x = aisle * self.aisle_spacing
    items = aisle_picks[aisle]
    positions = [p.position for _, p in items]

    max_pos = max(positions)
    min_pos = min(positions)

    # Calculate S-Shape distance for this aisle
    s_shape_dist = self.layout.aisle_length

    # Calculate Return distance
    if at_front:
        return_dist = 2 * max_pos
    else:
        return_dist = 2 * (self.layout.aisle_length - min_pos)

    # Choose better option
    if at_front:
        entrance = (aisle_x, 0)
    else:
        entrance = (aisle_x, self.layout.aisle_length)

    total_distance += self.rectilinear_distance(current_pos, entrance)

    if return_dist <= s_shape_dist:
        # Use return
        total_distance += return_dist
        current_pos = entrance
    else:
        # Use S-shape
        total_distance += s_shape_dist
        if at_front:
            current_pos = (aisle_x, self.layout.aisle_length)
        else:
            current_pos = (aisle_x, 0)
        at_front = not at_front

    items.sort(key=lambda x: x[1].position)
    sequence.extend([idx for idx, _ in items])

```

```

total_distance += self.rectilinear_distance(current_pos,
                                             self.layout.depot_position)

return total_distance, sequence

def optimal_route(self, picks: List[PickLocation]) -> Tuple[float, List[int]]:
    """
    Calculate optimal route by solving TSP.

    Uses dynamic programming for small instances or nearest
    neighbor heuristic for larger instances.

    Args:
        picks: List of pick locations

    Returns:
        Tuple of (total_distance, pick_sequence)
    """
    if not picks:
        return 0.0, []

    n = len(picks)

    # Build distance matrix (including depot as node 0)
    coords = [self.layout.depot_position]
    for pick in picks:
        coords.append(self.get_location_coordinates(pick))

    dist_matrix = np.zeros((n + 1, n + 1))
    for i in range(n + 1):
        for j in range(n + 1):
            dist_matrix[i, j] = self.rectilinear_distance(coords[i], coords[j])

    if n <= 12:
        # Exact solution using dynamic programming (Held-Karp)
        distance, tour = self._tsp_dp(dist_matrix)
    else:
        # Use nearest neighbor heuristic for larger instances
        distance, tour = self._tsp_nearest_neighbor(dist_matrix)

    # Convert tour to pick sequence (excluding depot)
    sequence = [t - 1 for t in tour if t > 0]

    return distance, sequence

def _tsp_dp(self, dist: np.ndarray) -> Tuple[float, List[int]]:
    """Solve TSP exactly using dynamic programming (Held-Karp)."""
    n = len(dist)

    # dp[mask][i] = min cost to visit nodes in mask, ending at i
    INF = float('inf')
    dp = [[INF] * n for _ in range(1 << n)]

```

```

parent = [[-1] * n for _ in range(1 << n)]

dp[1][0] = 0 # Start at depot (node 0)

for mask in range(1 << n):
    for last in range(n):
        if not (mask & (1 << last)):
            continue
        if dp[mask][last] == INF:
            continue

        for next_node in range(n):
            if mask & (1 << next_node):
                continue
            new_mask = mask | (1 << next_node)
            new_cost = dp[mask][last] + dist[last][next_node]

            if new_cost < dp[new_mask][next_node]:
                dp[new_mask][next_node] = new_cost
                parent[new_mask][next_node] = last

# Find best ending
full_mask = (1 << n) - 1
best_cost = INF
best_last = -1

for last in range(1, n):
    cost = dp[full_mask][last] + dist[last][0]
    if cost < best_cost:
        best_cost = cost
        best_last = last

# Reconstruct tour
tour = []
mask = full_mask
current = best_last

while current != -1:
    tour.append(current)
    prev = parent[mask][current]
    mask ^= (1 << current)
    current = prev

tour.reverse()

return best_cost, tour

def _tsp_nearest_neighbor(self, dist: np.ndarray) -> Tuple[float, List[int]]:
    """Solve TSP approximately using nearest neighbor heuristic."""
    n = len(dist)
    visited = [False] * n
    tour = [0]
    visited[0] = True

```

```

total_dist = 0

for _ in range(n - 1):
    current = tour[-1]
    best_next = -1
    best_dist = float('inf')

    for j in range(n):
        if not visited[j] and dist[current][j] < best_dist:
            best_dist = dist[current][j]
            best_next = j

    tour.append(best_next)
    visited[best_next] = True
    total_dist += best_dist

# Return to depot
total_dist += dist[tour[-1]][0]

return total_dist, tour

def compare_policies(self, picks: List[PickLocation]) -> Dict[str, Dict]:
    """
    Compare all routing policies for a given pick list.

    Args:
        picks: List of pick locations

    Returns:
        Dict with results for each policy
    """
    results = {}

    policies = [
        ("S-Shape", self.s_shape_route),
        ("Return", self.return_route),
        ("Midpoint", self.midpoint_route),
        ("Largest Gap", self.largest_gap_route),
        ("Combined", self.combined_route),
        ("Optimal", self.optimal_route),
    ]

    for name, method in policies:
        distance, sequence = method(picks)
        results[name] = {
            'distance': distance,
            'sequence': sequence,
            'n_picks': len(picks)
        }

    # Calculate relative performance
    optimal_dist = results['Optimal']['distance']
    for name in results:

```

```

        results[name]['gap_pct'] = (
            (results[name]['distance'] - optimal_dist) / optimal_dist * 100
            if optimal_dist > 0 else 0
        )

    return results

class BatchPickingOptimizer:
    """
    Optimize batch picking by grouping orders and routing.
    """

    def __init__(self, layout: WarehouseLayout, batch_capacity: int = 20):
        """
        Initialize batch optimizer.

        Args:
            layout: Warehouse layout
            batch_capacity: Maximum picks per batch
        """
        self.layout = layout
        self.batch_capacity = batch_capacity
        self.route_optimizer = PickRouteOptimizer(layout)

    def create_batches(self, orders: List[List[PickLocation]],
                      method: str = 'proximity') -> List[List[PickLocation]]:
        """
        Create pick batches from multiple orders.

        Args:
            orders: List of orders, each is a list of PickLocations
            method: 'proximity', 'seed', or 'savings'

        Returns:
            List of batches (combined pick lists)
        """
        if method == 'proximity':
            return self._batch_by_proximity(orders)
        elif method == 'seed':
            return self._batch_by_seed(orders)
        else:
            return self._batch_by_savings(orders)

    def _batch_by_proximity(self, orders: List[List[PickLocation]]) ->
        ↪ List[List[PickLocation]]:
        """Group orders by aisle proximity."""
        # Calculate centroid aisle for each order
        order_centroids = []
        for order in orders:
            if order:
                centroid = sum(p.aisle for p in order) / len(order)
                order_centroids.append(centroid)

```



```

        else:
            order_centroids.append(0)

# Sort orders by centroid
sorted_indices = np.argsort(order_centroids)

# Create batches
batches = []
current_batch = []
current_count = 0

for idx in sorted_indices:
    order = orders[idx]
    if current_count + len(order) <= self.batch_capacity:
        current_batch.extend(order)
        current_count += len(order)
    else:
        if current_batch:
            batches.append(current_batch)
            current_batch = list(order)
            current_count = len(order)

if current_batch:
    batches.append(current_batch)

return batches

def _batch_by_seed(self, orders: List[List[PickLocation]]) ->
    List[List[PickLocation]]:
    """Seed-based batching algorithm."""
    remaining = list(range(len(orders)))
    batches = []

    while remaining:
        # Select seed order (largest unassigned)
        seed_idx = max(remaining, key=lambda i: len(orders[i]))
        remaining.remove(seed_idx)

        batch = list(orders[seed_idx])
        batch_aisles = set(p.aisle for p in batch)

        # Add compatible orders
        for idx in remaining[:]:
            order = orders[idx]
            order_aisles = set(p.aisle for p in order)

            # Add if overlapping aisles and capacity allows
            if (batch_aisles & order_aisles and
                len(batch) + len(order) <= self.batch_capacity):
                batch.extend(order)
                batch_aisles.update(order_aisles)
                remaining.remove(idx)

```

```

        batches.append(batch)

    return batches

def _batch_by_savings(self, orders: List[List[PickLocation]]) ->
    ↪ List[List[PickLocation]]:
    """Savings-based batching (Clarke-Wright style)."""
    n = len(orders)

    # Calculate individual route distances
    individual_dists = []
    for order in orders:
        dist, _ = self.route_optimizer.s_shape_route(order)
        individual_dists.append(dist)

    # Calculate savings from combining pairs
    savings = []
    for i in range(n):
        for j in range(i + 1, n):
            combined = orders[i] + orders[j]
            if len(combined) <= self.batch_capacity:
                combined_dist, _ = self.route_optimizer.s_shape_route(combined)
                saving = individual_dists[i] + individual_dists[j] -
    ↪ combined_dist
                savings.append((saving, i, j))

    # Sort savings descending
    savings.sort(reverse=True)

    # Build batches using savings
    order_to_batch = {i: i for i in range(n)}
    batches_dict = {i: [i] for i in range(n)}

    for saving, i, j in savings:
        batch_i = order_to_batch[i]
        batch_j = order_to_batch[j]

        if batch_i == batch_j:
            continue

        # Check capacity
        combined_size = sum(len(orders[k]) for k in batches_dict[batch_i])
        combined_size += sum(len(orders[k]) for k in batches_dict[batch_j])

        if combined_size <= self.batch_capacity:
            # Merge batches
            batches_dict[batch_i].extend(batches_dict[batch_j])
            for k in batches_dict[batch_j]:
                order_to_batch[k] = batch_i
            del batches_dict[batch_j]

    # Convert to pick lists
    batches = []

```

```

for order_indices in batches_dict.values():
    batch = []
    for idx in order_indices:
        batch.extend(orders[idx])
    batches.append(batch)

return batches

def optimize_batches(self, orders: List[List[PickLocation]],
                    policy: RoutingPolicy = RoutingPolicy.LARGEST_GAP
                    ) -> Dict[str, any]:
    """
    Fully optimize batch picking.

    Args:
        orders: List of orders
        policy: Routing policy to use

    Returns:
        Optimization results
    """
    # Compare batching methods
    results = {}

    for method in ['proximity', 'seed', 'savings']:
        batches = self.create_batches(orders, method)

        total_distance = 0
        for batch in batches:
            if policy == RoutingPolicy.S_SHAPE:
                dist, _ = self.route_optimizer.s_shape_route(batch)
            elif policy == RoutingPolicy.RETURN:
                dist, _ = self.route_optimizer.return_route(batch)
            elif policy == RoutingPolicy.LARGEST_GAP:
                dist, _ = self.route_optimizer.largest_gap_route(batch)
            else:
                dist, _ = self.route_optimizer.combined_route(batch)
            total_distance += dist

        results[method] = {
            'n_batches': len(batches),
            'total_distance': total_distance,
            'avg_distance_per_batch': total_distance / len(batches) if batches
            ↪ else 0
        }

    # Compare to no batching
    no_batch_dist = 0
    for order in orders:
        dist, _ = self.route_optimizer.largest_gap_route(order)
        no_batch_dist += dist

    results['no_batching'] = {

```

```

        'n_batches': len(orders),
        'total_distance': no_batch_dist,
        'avg_distance_per_batch': no_batch_dist / len(orders) if orders else 0
    }

    return results

# Example usage and comparison
def run_order_picking_example():
    """Demonstrate order picking optimization."""

    print("=" * 60)
    print("ORDER PICKING ROUTE OPTIMIZATION")
    print("=" * 60)

    # Define warehouse layout
    layout = WarehouseLayout(
        n_aisles=6,
        aisle_length=50.0, # meters
        aisle_width=3.0,
        cross_aisle_width=4.0
    )

    # Generate sample pick list (10 items)
    np.random.seed(42)
    picks = []
    for i in range(10):
        picks.append(PickLocation(
            aisle=np.random.randint(0, layout.n_aisles),
            position=np.random.uniform(0, layout.aisle_length),
            item_id=f"SKU{i:03d}"
        ))

    # Create optimizer and compare policies
    optimizer = PickRouteOptimizer(layout)
    results = optimizer.compare_policies(picks)

    print(f"\nWarehouse: {layout.n_aisles} aisles, {layout.aisle_length}m each")
    print(f"Pick list: {len(picks)} items")
    print("\nRouting Policy Comparison:")
    print("-" * 60)
    print(f"{'Policy':<15} {'Distance (m)':<15} {'Gap vs Optimal':<15}")
    print("-" * 60)

    for policy, data in sorted(results.items(), key=lambda x: x[1]['distance']):
        print(f"{'policy':<15} {'data['distance']':<15.2f} {'data['gap_pct']':<15.1f}%")

    # Batch picking example
    print("\n" + "=" * 60)
    print("BATCH PICKING OPTIMIZATION")
    print("=" * 60)

```

```

# Generate 20 orders with 3-8 items each
orders = []
for _ in range(20):
    n_items = np.random.randint(3, 9)
    order = [PickLocation(
        aisle=np.random.randint(0, layout.n_aisles),
        position=np.random.uniform(0, layout.aisle_length),
        item_id=f"SKU{np.random.randint(0, 100):03d}"
    ) for _ in range(n_items)]
    orders.append(order)

batch_optimizer = BatchPickingOptimizer(layout, batch_capacity=20)
batch_results = batch_optimizer.optimize_batches(orders)

print(f"\n{len(orders)} orders, total {sum(len(o) for o in orders)} items")
print("\nBatching Method Comparison:")
print("-" * 60)
print(f"{'Method':<15} {'Batches':<10} {'Total Dist':<15} {'Savings':<10}")
print("-" * 60)

baseline = batch_results['no_batching']['total_distance']
for method, data in batch_results.items():
    savings = (baseline - data['total_distance']) / baseline * 100
    print(f"{'method':<15} {'data['n_batches']':<10}
    ↪ {'data['total_distance']':<15.1f} {'savings:<10.1f}%"")

return results, batch_results

if __name__ == "__main__":
    run_order_picking_example()

```

17.7.5. Performance Metrics

Metric	Formula	Target
Travel distance per pick	Total distance / Number of picks	Minimize
Picks per hour	Picks completed / Hours worked	100-150 (manual)
Order cycle time	Time from order release to completion	Minimize
Picker utilization	Picking time / Total time	> 60%
Lines per hour	Order lines picked / Hours worked	150-250

17.7.6. Routing Policy Selection Guide

Scenario	Recommended Policy	Reason
High-density picks (>60% of locations)	S-Shape	Full traversal efficient
Low-density picks (<20% of locations)	Largest Gap	Avoids empty sections
Simple operations, new pickers	Return	Easy to learn

Scenario	Recommended Policy	Reason
Mixed density	Combined	Adapts per aisle
Critical orders, small lists	Optimal	Guarantees minimum

💡 Implementation Tips

1. **Start Simple:** Implement S-Shape first, then refine based on data
2. **Measure Baseline:** Track current travel distances before optimization
3. **Consider Picker Experience:** Complex policies may confuse new workers
4. **Combine with Storage Optimization:** Routing benefits multiply with good slotting
5. **Account for Congestion:** Theoretical distances ignore picker interference

17.8. Complete Python Implementation

The `QAPSolver` class below provides a unified interface for solving the Quadratic Assignment Problem with multiple algorithms: exhaustive enumeration (for small instances), greedy construction, local search with pairwise swaps, multi-start local search, simulated annealing, and genetic algorithm. The `compare_methods` function runs all applicable methods and reports solution quality and computation time for each.

```
import numpy as np
from typing import List, Tuple, Dict, Optional
from dataclasses import dataclass
from scipy.optimize import linear_sum_assignment
import random
import math

@dataclass
class QAPInstance:
    """Quadratic Assignment Problem instance."""
    n: int
    flow: np.ndarray
    distance: np.ndarray
    facility_names: Optional[List[str]] = None
    location_names: Optional[List[str]] = None

    def __post_init__(self):
        if self.facility_names is None:
            self.facility_names = [f"F{i}" for i in range(self.n)]
        if self.location_names is None:
            self.location_names = [f"L{i}" for i in range(self.n)]

class QAPSolver:
    """
    Comprehensive QAP solver with multiple algorithms.
    """

    def __init__(self, instance: QAPInstance):
        """
```

```

Initialize solver with problem instance.

Args:
    instance: QAP problem instance
"""
self.instance = instance
self.n = instance.n
self.F = instance.flow
self.D = instance.distance

def evaluate(self, assignment: List[int]) -> float:
    """Calculate objective value for an assignment."""
    cost = 0
    for i in range(self.n):
        for j in range(self.n):
            cost += self.F[i, j] * self.D[assignment[i], assignment[j]]
    return cost

def solve_exhaustive(self) -> Tuple[List[int], float]:
    """
    Solve by exhaustive enumeration (small instances only).

    Returns:
        Best assignment and cost
    """
    if self.n > 10:
        raise ValueError("Exhaustive search only for n <= 10")

    import itertools

    best_cost = float('inf')
    best_assignment = None

    for perm in itertools.permutations(range(self.n)):
        cost = self.evaluate(list(perm))
        if cost < best_cost:
            best_cost = cost
            best_assignment = list(perm)

    return best_assignment, best_cost

def solve_greedy(self) -> Tuple[List[int], float]:
    """
    Solve using greedy construction heuristic.

    Returns:
        Assignment and cost
    """
    assignment = [-1] * self.n
    assigned_locations = set()

    # Calculate total flow for each facility
    total_flow = self.F.sum(axis=1) + self.F.sum(axis=0)

```

```

# Calculate total distance for each location
total_distance = self.D.sum(axis=1) + self.D.sum(axis=0)

# Sort facilities by flow (descending)
facility_order = np.argsort(total_flow)[::-1]

# Sort locations by distance (ascending - central locations first)
location_order = np.argsort(total_distance)

# Assign high-flow facilities to central locations
for facility in facility_order:
    for location in location_order:
        if location not in assigned_locations:
            assignment[facility] = location
            assigned_locations.add(location)
            break

    return assignment, self.evaluate(assignment)

def solve_local_search(self, initial: Optional[List[int]] = None,
                       max_iterations: int = 1000) -> Tuple[List[int], float]:
    """
    Solve using local search with 2-exchange neighborhood.

    Args:
        initial: Initial solution (random if None)
        max_iterations: Maximum iterations

    Returns:
        Best assignment and cost
    """
    if initial is None:
        current = list(range(self.n))
        random.shuffle(current)
    else:
        current = initial.copy()

    current_cost = self.evaluate(current)

    for _ in range(max_iterations):
        improved = False

        # Try all pairwise swaps
        for i in range(self.n):
            for j in range(i + 1, self.n):
                # Swap
                current[i], current[j] = current[j], current[i]
                new_cost = self.evaluate(current)

                if new_cost < current_cost:
                    current_cost = new_cost
                    improved = True

```



```

        else:
            # Undo swap
            current[i], current[j] = current[j], current[i]

        if not improved:
            break

    return current, current_cost

def solve_simulated_annealing(self, initial_temp: float = 1000,
                              cooling_rate: float = 0.995,
                              min_temp: float = 1,
                              iterations_per_temp: int = 50) ->
    Tuple[List[int], float]:
    """
    Solve using simulated annealing.

    Returns:
        Best assignment and cost
    """
    current = list(range(self.n))
    random.shuffle(current)
    current_cost = self.evaluate(current)

    best = current.copy()
    best_cost = current_cost

    temp = initial_temp

    while temp > min_temp:
        for _ in range(iterations_per_temp):
            # Generate neighbor by swap
            i, j = random.sample(range(self.n), 2)
            current[i], current[j] = current[j], current[i]
            new_cost = self.evaluate(current)

            # Accept or reject
            delta = new_cost - current_cost
            if delta < 0 or random.random() < math.exp(-delta / temp):
                current_cost = new_cost
                if current_cost < best_cost:
                    best = current.copy()
                    best_cost = current_cost
            else:
                # Undo swap
                current[i], current[j] = current[j], current[i]

        temp *= cooling_rate

    return best, best_cost

def solve_genetic_algorithm(self, pop_size: int = 50,
                           generations: int = 100,

```

```

mutation_rate: float = 0.1) -> Tuple[List[int],
↳ float]:
    """
    Solve using genetic algorithm.

    Returns:
        Best assignment and cost
    """
    # Initialize population
    population = [random.sample(range(self.n), self.n)
                  for _ in range(pop_size)]

    best = None
    best_cost = float('inf')

    for gen in range(generations):
        # Evaluate fitness
        fitnesses = [-self.evaluate(ind) for ind in population]

        # Update best
        gen_best_idx = np.argmax(fitnesses)
        if -fitnesses[gen_best_idx] < best_cost:
            best_cost = -fitnesses[gen_best_idx]
            best = population[gen_best_idx].copy()

        # Selection and reproduction
        new_population = [population[gen_best_idx]] # Elitism

        while len(new_population) < pop_size:
            # Tournament selection
            tournament = random.sample(list(zip(population, fitnesses)), 3)
            parent1 = max(tournament, key=lambda x: x[1][0])
            tournament = random.sample(list(zip(population, fitnesses)), 3)
            parent2 = max(tournament, key=lambda x: x[1][0])

            # Order crossover
            a, b = sorted(random.sample(range(self.n), 2))
            child = [None] * self.n
            child[a:b+1] = parent1[a:b+1]
            remaining = [x for x in parent2 if x not in child[a:b+1]]
            pos = 0
            for i in range(self.n):
                if child[i] is None:
                    child[i] = remaining[pos]
                    pos += 1

            # Mutation
            if random.random() < mutation_rate:
                i, j = random.sample(range(self.n), 2)
                child[i], child[j] = child[j], child[i]

            new_population.append(child)

```

```

        population = new_population

    return best, best_cost

def solve_multi_start(self, n_starts: int = 10) -> Tuple[List[int], float]:
    """
    Solve using multi-start local search.

    Returns:
        Best assignment and cost
    """
    best = None
    best_cost = float('inf')

    for _ in range(n_starts):
        # Random start
        initial = list(range(self.n))
        random.shuffle(initial)

        # Local search
        solution, cost = self.solve_local_search(initial)

        if cost < best_cost:
            best_cost = cost
            best = solution

    return best, best_cost

def compare_methods(self) -> Dict[str, Tuple[List[int], float, float]]:
    """
    Compare all solving methods.

    Returns:
        Dict mapping method name to (solution, cost, time)
    """
    import time
    results = {}

    # Greedy
    start = time.time()
    sol, cost = self.solve_greedy()
    results['Greedy'] = (sol, cost, time.time() - start)

    # Local Search
    start = time.time()
    sol, cost = self.solve_local_search()
    results['Local Search'] = (sol, cost, time.time() - start)

    # Multi-start
    start = time.time()
    sol, cost = self.solve_multi_start(10)
    results['Multi-start LS'] = (sol, cost, time.time() - start)

```

```

# Simulated Annealing
start = time.time()
sol, cost = self.solve_simulated_annealing()
results['Simulated Annealing'] = (sol, cost, time.time() - start)

# Genetic Algorithm
start = time.time()
sol, cost = self.solve_genetic_algorithm()
results['Genetic Algorithm'] = (sol, cost, time.time() - start)

# Exhaustive (only for small instances)
if self.n <= 8:
    start = time.time()
    sol, cost = self.solve_exhaustive()
    results['Exhaustive'] = (sol, cost, time.time() - start)

return results

def visualize_solution(self, assignment: List[int],
                      title: str = "QAP Solution"):
    """
    Visualize the assignment solution.

    Args:
        assignment: Facility-to-location assignment
        title: Plot title
    """
    import matplotlib.pyplot as plt
    import matplotlib.patches as patches

    fig, axes = plt.subplots(1, 2, figsize=(14, 6))

    # Left: Flow matrix
    ax1 = axes[0]
    im1 = ax1.imshow(self.F, cmap='Blues')
    ax1.set_title('Flow Matrix')
    ax1.set_xlabel('Facility')
    ax1.set_ylabel('Facility')
    plt.colorbar(im1, ax=ax1)

    # Right: Layout visualization
    ax2 = axes[1]

    # Assume grid layout
    grid_size = int(np.ceil(np.sqrt(self.n)))

    for facility, location in enumerate(assignment):
        row = location // grid_size
        col = location % grid_size

        rect = patches.Rectangle(
            (col, grid_size - row - 1), 1, 1,
            linewidth=2, edgecolor='black',

```

```

        facecolor=plt.cm.Set3(facility / self.n),
        alpha=0.7
    )
    ax2.add_patch(rect)
    ax2.text(col + 0.5, grid_size - row - 0.5,
            self.instance.facility_names[facility],
            ha='center', va='center', fontsize=12, fontweight='bold')

    ax2.set_xlim(-0.5, grid_size + 0.5)
    ax2.set_ylim(-0.5, grid_size + 0.5)
    ax2.set_aspect('equal')
    ax2.set_title(f'{title}\nCost: {self.evaluate(assignment):.2f}')
    ax2.grid(True, alpha=0.3)

plt.tight_layout()
return fig

# Complete example
def run_layout_optimization_example():
    """Run a complete layout optimization example."""

    print("=" * 60)
    print("FACILITY LAYOUT OPTIMIZATION EXAMPLE")
    print("=" * 60)

    # Create problem instance: 8 departments
    n = 8
    facility_names = ['Receiving', 'Storage', 'Cutting', 'Welding',
                     'Assembly', 'Painting', 'Packaging', 'Shipping']

    # Flow matrix (material movements per day)
    flow = np.array([
        [0, 100, 20, 0, 0, 0, 0, 10],      # Receiving
        [0, 0, 80, 50, 10, 0, 0, 0],      # Storage
        [0, 0, 0, 70, 30, 0, 0, 0],      # Cutting
        [0, 0, 0, 0, 90, 20, 0, 0],      # Welding
        [0, 0, 0, 0, 0, 85, 10, 0],      # Assembly
        [0, 0, 0, 0, 0, 0, 75, 0],      # Painting
        [0, 0, 0, 0, 0, 0, 0, 80],      # Packaging
        [0, 0, 0, 0, 0, 0, 0, 0]      # Shipping
    ])

    # Make symmetric
    flow = flow + flow.T

    # Distance matrix (8 locations in 2x4 grid)
    locations = [(0, 0), (0, 1), (0, 2), (0, 3),
                (1, 0), (1, 1), (1, 2), (1, 3)]
    distance = np.zeros((n, n))
    for i in range(n):
        for j in range(n):
            distance[i, j] = (abs(locations[i][0] - locations[j][0]) +

```

```

        abs(locations[i][1] - locations[j][1])) * 10 # 10m per
        ↪ cell

# Create instance and solver
instance = QAPIInstance(n, flow, distance, facility_names)
solver = QAPSolver(instance)

# Compare methods
print("\nComparing Solution Methods:")
print("-" * 60)

results = solver.compare_methods()

best_method = None
best_cost = float('inf')

for method, (solution, cost, time_taken) in results.items():
    print(f"\n{method}:")
    print(f"  Cost: {cost:.2f}")
    print(f"  Time: {time_taken:.4f}s")
    print(f"  Assignment: {solution}")

    if cost < best_cost:
        best_cost = cost
        best_method = method

print("\n" + "=" * 60)
print(f"BEST METHOD: {best_method} with cost {best_cost:.2f}")
print("=" * 60)

# Display best solution
best_solution = results[best_method][0]
print("\nOptimal Department Assignments:")
for i, loc in enumerate(best_solution):
    row = loc // 4
    col = loc % 4
    print(f"  {facility_names[i]:12} -> Row {row+1}, Column {col+1}")

return results

# Run the example
if __name__ == "__main__":
    results = run_layout_optimization_example()

```

17.9. Real-World Case Study: Automotive Parts Manufacturer

This section presents a simplified case study based on a common industrial scenario. A mid-sized automotive parts manufacturer sought to redesign their 50,000 square foot production facility to reduce material handling costs and improve throughput.

17.9.1. Initial Situation

- 12 production departments

- Annual material handling cost: \$850,000
- Average distance traveled: 2,400 feet per part
- Frequent bottlenecks between machining and assembly

17.9.2. Analysis Process

1. **Data Collection:** Tracked material movement for 3 months using RFID tags
2. **Flow Analysis:** Constructed From-To chart with 120,000 annual movements
3. **REL Chart:** Management rated departmental relationships
4. **Space Analysis:** Documented equipment footprints and aisle requirements

17.9.3. Solution Approach

The optimization combined CRAFT improvement with simulated annealing, starting from the SLP-derived layout. The code below summarizes the problem structure and the results achieved.

```
# Simplified representation of the case study
departments = [
    "Raw Material Storage", "Receiving", "Cutting", "Stamping",
    "Welding", "CNC Machining", "Heat Treatment", "Grinding",
    "Assembly Line 1", "Assembly Line 2", "Quality Control", "Shipping"
]

# Annual material flows (thousands of moves)
flow_data = {
    ("Receiving", "Raw Material Storage"): 45,
    ("Raw Material Storage", "Cutting"): 40,
    ("Cutting", "Stamping"): 35,
    ("Stamping", "Welding"): 30,
    ("Welding", "CNC Machining"): 25,
    ("CNC Machining", "Heat Treatment"): 20,
    ("Heat Treatment", "Grinding"): 18,
    ("Grinding", "Assembly Line 1"): 15,
    ("Grinding", "Assembly Line 2"): 12,
    ("Assembly Line 1", "Quality Control"): 20,
    ("Assembly Line 2", "Quality Control"): 15,
    ("Quality Control", "Shipping"): 30,
}

# Results after optimization:
# - Material handling cost reduced by 35% ($297,500 annual savings)
# - Average travel distance reduced by 42%
# - Throughput increased by 18%
# - ROI achieved in 8 months
```

17.9.4. Key Lessons Learned

1. **Data Quality Matters:** Accurate flow data is essential for optimization
2. **Include Qualitative Factors:** Not everything can be quantified (safety, communication)
3. **Stakeholder Buy-in:** Involve operators in the redesign process
4. **Phased Implementation:** Major moves scheduled during planned downtime
5. **Continuous Improvement:** Layout is not a one-time decision

17.9.5. Commercial Simulation Tools

In industry, commercial tools such as **Siemens Tecnomatix Plant Simulation**, **FlexSim**, and **AnyLogic** provide 3D visualization, drag-and-drop modeling, and built-in optimization capabilities for facility layout design. These platforms allow engineers to simulate material flows, evaluate layout alternatives with animated walkthroughs, and perform what-if analysis before committing to physical changes. While the optimization algorithms in this chapter can be implemented with open-source Python libraries, commercial simulation tools are often preferred in practice for stakeholder communication, integration with CAD systems, and validation of layouts under stochastic operating conditions. For a broader comparison of simulation platforms, see Chapter 28.

17.10. Summary

💡 Key Takeaways

- Facility layout significantly impacts material handling costs, lead times, and productivity
- Layout types (process, product, cellular, fixed-position) suit different production volumes and varieties
- The Quadratic Assignment Problem (QAP) provides the mathematical foundation for layout optimization
- QAP is NP-hard; practical solutions require heuristics and metaheuristics
- Systematic Layout Planning (SLP) balances quantitative flow analysis with qualitative relationships
- Algorithms like CRAFT, CORELAP, and BLOCPLAN automate layout generation and improvement
- Cellular manufacturing groups machines and parts to gain benefits of both job shop and flow shop
- Warehouse layout focuses on storage assignment and order picking efficiency
- Genetic algorithms and simulated annealing provide effective metaheuristic approaches
- Real-world implementation requires careful change management and phased execution

17.11. Exercises

1. **QAP Formulation:** A company has 5 machines to place in 5 available locations. The flow matrix shows 100 trips/day between machines 1-2, 80 trips between 2-3, 60 trips between 3-4, and 40 trips between 4-5. The locations are arranged in a line with 10 meters between adjacent locations. Formulate this as a QAP and find the optimal layout.
2. **REL Chart Construction:** Create a REL chart for a hospital emergency department with the following areas: Triage, Waiting Room, Treatment Rooms, Radiology, Laboratory, Pharmacy, Nurses Station, and Ambulance Bay. Justify your relationship ratings.
3. **CRAFT Implementation:** Starting from the layout shown below, apply two iterations of the CRAFT algorithm:

```
+---+---+---+
| A | B | C |
+---+---+---+
| D | E | F |
+---+---+---+
```

Use the flow matrix:

A-B: 50, A-D: 30, B-C: 40, B-E: 60
C-F: 20, D-E: 45, E-F: 55

4. **Cell Formation:** Given the following machine-part incidence matrix, apply Rank Order Clustering to identify manufacturing cells:

	P1	P2	P3	P4	P5	P6	P7	P8
M1	1	0	1	0	1	0	0	0
M2	0	1	0	1	0	1	0	1
M3	1	0	1	0	0	0	1	0
M4	0	1	0	1	0	0	0	1
M5	0	0	1	0	1	0	1	0
M6	0	1	0	0	0	1	0	1

5. **Warehouse Layout:** A distribution center has 100 SKUs with Pareto-distributed demand (top 20% account for 80% of picks). Design a class-based storage policy with three zones (A, B, C). Calculate the expected travel distance savings compared to random storage, assuming:
- Warehouse dimensions: 100m x 50m
 - I/O point at one corner
 - 500 picks per day
6. **Metaheuristic Comparison:** Implement both simulated annealing and genetic algorithm for a 12-facility QAP instance. Run each algorithm 10 times and compare:
- Average solution quality
 - Best solution found
 - Computation time
 - Consistency (standard deviation)
7. **Multi-floor Layout:** Extend the QAP formulation to handle a two-floor facility where vertical movement (elevators) costs 3x horizontal movement. Solve for a 6-department problem with 3 locations per floor.
8. **Dynamic Layout:** A company expects demand patterns to shift over the next 5 years, changing the flow matrix annually. Develop a layout strategy that minimizes total cost over the planning horizon while limiting the number of department relocations.

17.12. Further Reading

Foundational Texts:

- Tompkins, J.A., White, J.A., Bozer, Y.A., & Tanchoco, J.M.A. (2010). *Facilities Planning* (4th ed.). Wiley.
- Muther, R. (1973). *Systematic Layout Planning* (2nd ed.). Cahnerns Books.
- Francis, R.L., McGinnis, L.F., & White, J.A. (1992). *Facility Layout and Location: An Analytical Approach* (2nd ed.). Prentice Hall.

Research Papers:

- Koopmans, T.C., & Beckmann, M. (1957). "Assignment Problems and the Location of Economic Activities." *Econometrica*, 25(1), 53-76.
- Burkard, R.E., Karisch, S.E., & Rendl, F. (1997). "QAPLIB - A Quadratic Assignment Problem Library." *Journal of Global Optimization*, 10(4), 391-403.
- Drira, A., Pierreval, H., & Hajri-Gabouj, S. (2007). "Facility layout problems: A survey." *Annual Reviews in Control*, 31(2), 255-267.

17. Facility Layout Optimization

Online Resources:

- QAPLIB: Library of QAP Instances (<http://www.seas.upenn.edu/qaplib/>)
- OR-Tools Layout Examples (<https://developers.google.com/optimization>)
- Facility Layout Planning Software Comparisons

Related Topics:

- Chapter 11: Scheduling and Sequencing (for production scheduling within layouts)
- Chapter 12: Resource Allocation (for assignment problems)
- Chapter 8: Metaheuristics (for advanced solution methods)
- Chapter 10: Routing and Logistics (for material handling optimization)

18. Inventory Management

What You Will Learn

- Formulate inventory optimization problems mathematically
- Apply classic models (EOQ, newsvendor, base-stock) appropriately
- Optimize multi-product and multi-echelon inventory systems
- Handle demand uncertainty through stochastic optimization
- Implement inventory policies in production systems

Inventory management is one of the oldest and most impactful applications of operations research. The fundamental trade-off between holding costs and stockout costs creates a rich optimization problem space that becomes increasingly complex with multiple products, locations, and demand uncertainty.

18.1. The Inventory Problem

18.1.1. Basic Trade-offs



Figure 18.1.: Inventory Cost Trade-offs: Balancing ordering and holding costs

18.1.2. Cost Components

Cost Type	Description	Typical Values
Holding Cost	Storage, capital, obsolescence	15-30% of item value/year
Ordering Cost	Setup, transport, admin	\$50-500 per order
Stockout Cost	Lost sales, expediting, reputation	1-10x item margin
Purchase Cost	Item cost, quantity discounts	Varies

18.2. Classic Inventory Models

18.2.1. Economic Order Quantity (EOQ)

The EOQ model answers: **How much to order?**

18. Inventory Management

Assumptions: - Constant, known demand rate D - Fixed ordering cost K per order - Holding cost h per unit per time - No stockouts allowed - Instantaneous replenishment

Total Cost:

$$TC(Q) = \frac{D \cdot K}{Q} + \frac{h \cdot Q}{2}$$

Optimal Order Quantity:

$$Q^* = \sqrt{\frac{2DK}{h}}$$

```
import numpy as np
from dataclasses import dataclass
from typing import Tuple

@dataclass
class EOQModel:
    """Economic Order Quantity model."""
    demand_rate: float      # D: units per period
    ordering_cost: float     # K: cost per order
    holding_cost: float     # h: cost per unit per period

    def optimal_quantity(self) -> float:
        """Calculate optimal order quantity."""
        return np.sqrt(2 * self.demand_rate * self.ordering_cost /
            ↪ self.holding_cost)

    def optimal_reorder_interval(self) -> float:
        """Calculate time between orders."""
        Q = self.optimal_quantity()
        return Q / self.demand_rate

    def total_cost(self, Q: float = None) -> float:
        """Calculate total cost for given order quantity."""
        if Q is None:
            Q = self.optimal_quantity()

        ordering_cost = (self.demand_rate / Q) * self.ordering_cost
        holding_cost = (Q / 2) * self.holding_cost
        return ordering_cost + holding_cost

    def sensitivity_analysis(self, Q_range: Tuple[float, float],
        num_points: int = 100) -> dict:
        """Analyze cost sensitivity to order quantity."""
        Q_values = np.linspace(Q_range[0], Q_range[1], num_points)
        costs = [self.total_cost(Q) for Q in Q_values]
        Q_star = self.optimal_quantity()

        return {
            'Q_values': Q_values,
            'total_costs': costs,
            'optimal_Q': Q_star,
            'optimal_cost': self.total_cost(Q_star)
        }
```

```
# Example
eoq = EOQModel(demand_rate=1000, ordering_cost=100, holding_cost=2)
print(f"Optimal Order Quantity: {eoq.optimal_quantity():.0f} units")
print(f"Order Every: {eoq.optimal_reorder_interval():.2f} periods")
print(f"Total Cost: ${eoq.total_cost():.2f}")
```

Optimal Order Quantity: 316 units

Order Every: 0.32 periods

Total Cost: \$632.46

The EOQ model produces a characteristic sawtooth inventory pattern. Inventory starts at the order quantity, depletes at a constant rate, and jumps back up when the next order arrives. The reorder point triggers new orders early enough to account for lead time.

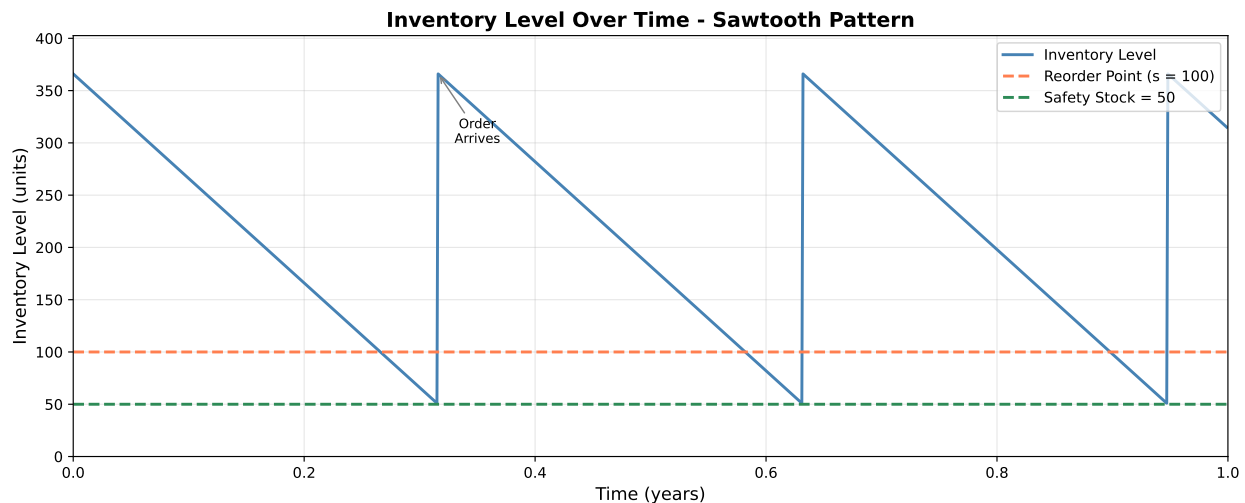


Figure 18.2.: Inventory Level Over Time: Classic sawtooth pattern showing reorder point and safety stock levels

The next figure shows how total cost varies with order quantity. Ordering cost decreases as Q increases (fewer orders needed), while holding cost increases linearly with Q . The total cost curve has a single minimum at the EOQ.

18.2.2. EOQ Extensions

EOQ with Quantity Discounts:

When suppliers offer price breaks at different order quantities, the standard EOQ formula no longer applies directly. The algorithm must evaluate each price tier separately, computing the feasible EOQ within that tier and comparing total costs across all tiers.

```
@dataclass
class PriceBreak:
    """Price break for quantity discounts."""
    min_quantity: float
    unit_price: float
```

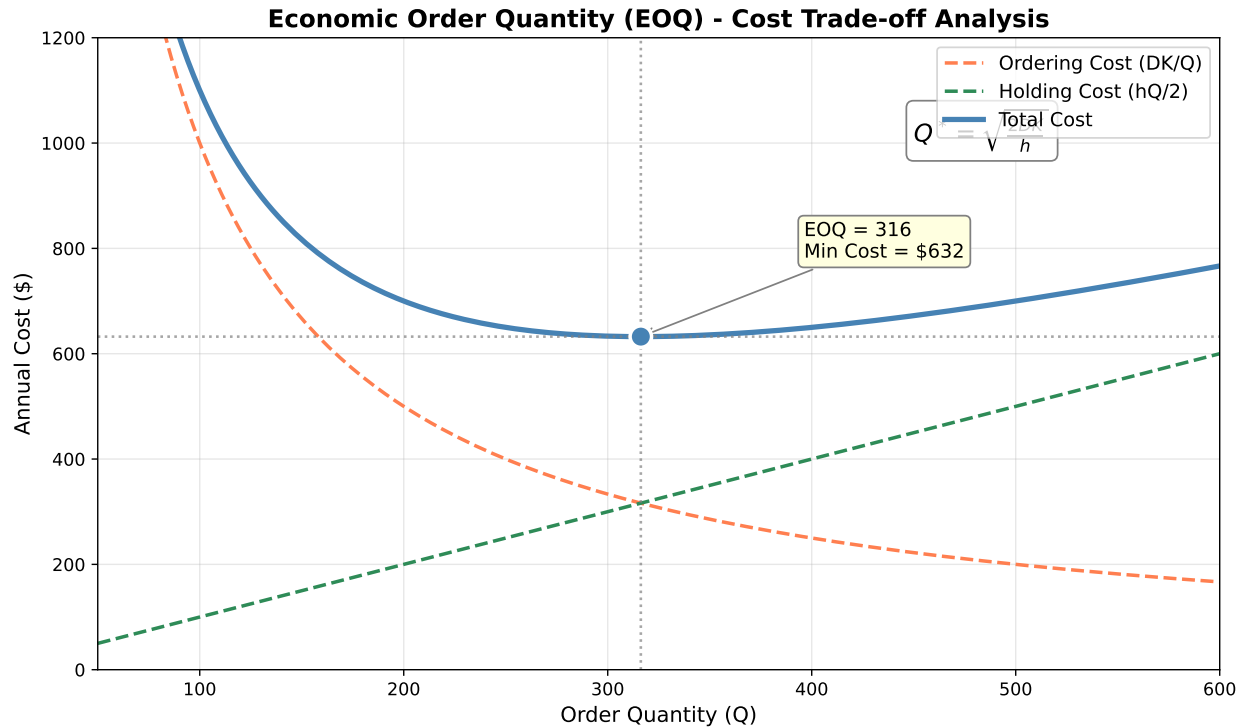


Figure 18.3.: EOQ Cost Analysis: Total cost curve showing the trade-off between holding and ordering costs

```
def eoq_with_discounts(demand_rate: float, ordering_cost: float,
                      holding_rate: float, price_breaks: list) -> dict:
    """EOQ with quantity discounts.

    Args:
        demand_rate: Annual demand
        ordering_cost: Fixed cost per order
        holding_rate: Holding cost as fraction of item value
        price_breaks: List of PriceBreak objects

    Returns:
        Optimal quantity and total cost
    """
    # Sort price breaks by quantity
    breaks = sorted(price_breaks, key=lambda x: x.min_quantity)

    best_Q = None
    best_cost = float('inf')
    best_price = None

    for i, pb in enumerate(breaks):
        # Holding cost for this price
        h = holding_rate * pb.unit_price
```

```

    # EOQ for this price
    Q_eoq = np.sqrt(2 * demand_rate * ordering_cost / h)

    # Check if EOQ is feasible in this price range
    min_Q = pb.min_quantity
    max_Q = breaks[i + 1].min_quantity - 1 if i < len(breaks) - 1 else
↪ float('inf')

    # Feasible quantity
    Q = max(min_Q, min(Q_eoq, max_Q))

    # Total cost
    purchase_cost = demand_rate * pb.unit_price
    ordering_cost = (demand_rate / Q) * ordering_cost
    holding = (Q / 2) * h
    total_cost = purchase_cost + ordering + holding

    if total_cost < best_cost:
        best_cost = total_cost
        best_Q = Q
        best_price = pb.unit_price

    return {
        'optimal_Q': best_Q,
        'unit_price': best_price,
        'total_cost': best_cost
    }

# Example with quantity discounts
price_breaks = [
    PriceBreak(min_quantity=0, unit_price=10.00),
    PriceBreak(min_quantity=100, unit_price=9.50),
    PriceBreak(min_quantity=500, unit_price=9.00)
]

result = eoq_with_discounts(
    demand_rate=1000,
    ordering_cost=50,
    holding_rate=0.20,
    price_breaks=price_breaks
)
print(f"Order {result['optimal_Q']:.0f} units at ${result['unit_price']}/unit")
print(f"Total annual cost: ${result['total_cost']:.2f}")

```

Order 500 units at \$9.0/unit
 Total annual cost: \$9550.00

18.2.3. Newsvendor Model

The newsvendor model handles **single-period** decisions under demand uncertainty.

Setting: Order Q units before demand D is realized - Underage cost c_u : cost per unit of unmet demand - Overage cost c_o : cost per unit of excess inventory

Optimal Solution:

$$F(Q^*) = \frac{c_u}{c_u + c_o}$$

where F is the demand distribution CDF.

The following implementation translates the critical ratio formula into a Python class that handles arbitrary demand distributions. Given the underage and overage costs, the `optimal_quantity` method computes Q^* using the inverse CDF of the demand distribution.

```
from scipy import stats
from scipy.optimize import brentq

class NewsvendorModel:
    """Newsvendor (single-period) inventory model."""

    def __init__(self, underage_cost: float, overage_cost: float,
                 demand_distribution: stats.rv_continuous):
        self.c_u = underage_cost
        self.c_o = overage_cost
        self.demand_dist = demand_distribution

    @property
    def critical_ratio(self) -> float:
        """The optimal service level (critical fractile)."""
        return self.c_u / (self.c_u + self.c_o)

    def optimal_quantity(self) -> float:
        """Calculate optimal order quantity."""
        return self.demand_dist.ppf(self.critical_ratio)

    def expected_cost(self, Q: float) -> float:
        """Calculate expected cost for order quantity Q."""
        # Expected overage (inventory)
        expected_overage = Q * self.demand_dist.cdf(Q) - self.demand_dist.expect(
            lambda x: x, lb=0, ub=Q
        )

        # Expected underage (stockout)
        expected_underage = self.demand_dist.expect(
            lambda x: x, lb=Q, ub=np.inf
        ) - Q * (1 - self.demand_dist.cdf(Q))

        return self.c_o * expected_overage + self.c_u * expected_underage

    def expected_profit(self, Q: float, selling_price: float,
                      purchase_cost: float, salvage_value: float = 0) -> float:
        """Calculate expected profit."""
        mean_demand = self.demand_dist.mean()

        # Expected sales
        expected_sales = self.demand_dist.expect(
            lambda x: min(x, Q), lb=0, ub=np.inf
        )
```



```

    # Expected leftover
    expected_leftover = Q - expected_sales

    revenue = selling_price * expected_sales
    cost = purchase_cost * Q
    salvage = salvage_value * expected_leftover

    return revenue - cost + salvage

def fill_rate(self, Q: float) -> float:
    """Calculate expected fill rate (fraction of demand met)."""
    expected_sales = self.demand_dist.expect(
        lambda x: min(x, Q), lb=0, ub=np.inf
    )
    return expected_sales / self.demand_dist.mean()

# Example
demand = stats.norm(loc=100, scale=20)
nv = NewsvendorModel(
    underage_cost=15, # Lost margin
    overage_cost=5,   # Disposal cost
    demand_distribution=demand
)

Q_star = nv.optimal_quantity()
print(f"Critical Ratio: {nv.critical_ratio:.2%}")
print(f"Optimal Order: {Q_star:.0f} units")
print(f"Expected Cost: ${nv.expected_cost(Q_star):.2f}")
print(f"Fill Rate: {nv.fill_rate(Q_star):.1%}")

```

Critical Ratio: 75.00%
 Optimal Order: 113 units
 Expected Cost: \$127.11
 Fill Rate: 97.0%

18.2.4. Base-Stock Policy

For **continuous review** with stochastic demand:

- Order up to level S whenever inventory drops below S
- Optimal S balances holding vs. stockout costs

The `BaseStockPolicy` class below implements this approach. It computes lead-time demand statistics by scaling the per-period mean and standard deviation, then uses the critical ratio to determine the optimal base-stock level S .

```

class BaseStockPolicy:
    """Base-stock (order-up-to) inventory policy."""

    def __init__(self, lead_time: float, demand_mean: float,
                 demand_std: float, holding_cost: float,
                 stockout_cost: float):
        self.L = lead_time

```

```

        self.mu = demand_mean
        self.sigma = demand_std
        self.h = holding_cost
        self.b = stockout_cost

        # Lead time demand statistics
        self.mu_L = self.mu * self.L
        self.sigma_L = self.sigma * np.sqrt(self.L)

    def optimal_base_stock(self) -> float:
        """Calculate optimal base-stock level."""
        # Critical ratio
        cr = self.b / (self.b + self.h)

        # Base-stock level
        S = stats.norm(self.mu_L, self.sigma_L).ppf(cr)
        return max(0, S)

    def safety_stock(self, service_level: float = None) -> float:
        """Calculate safety stock."""
        if service_level is None:
            service_level = self.b / (self.b + self.h)

        z = stats.norm.ppf(service_level)
        return z * self.sigma_L

    def expected_inventory(self, S: float = None) -> float:
        """Expected on-hand inventory."""
        if S is None:
            S = self.optimal_base_stock()
        return S - self.mu_L + self.expected_backorders(S)

    def expected_backorders(self, S: float = None) -> float:
        """Expected backorders."""
        if S is None:
            S = self.optimal_base_stock()

        dist = stats.norm(self.mu_L, self.sigma_L)
        return self.sigma_L * (
            dist.pdf((S - self.mu_L) / self.sigma_L) -
            ((S - self.mu_L) / self.sigma_L) * (1 - dist.cdf((S - self.mu_L) /
↪ self.sigma_L))
        )

    def service_level_type1(self, S: float = None) -> float:
        """Cycle service level (probability of no stockout)."""
        if S is None:
            S = self.optimal_base_stock()
        return stats.norm(self.mu_L, self.sigma_L).cdf(S)

    def service_level_type2(self, S: float = None) -> float:
        """Fill rate (fraction of demand met from stock)."""
        if S is None:

```

```

        S = self.optimal_base_stock()
        return 1 - self.expected_backorders(S) / self.mu_L

# Example
policy = BaseStockPolicy(
    lead_time=2,
    demand_mean=100,
    demand_std=30,
    holding_cost=1,
    stockout_cost=9
)

S_star = policy.optimal_base_stock()
print(f"Optimal Base-Stock Level: {S_star:.0f} units")
print(f"Safety Stock: {policy.safety_stock():.0f} units")
print(f"Cycle Service Level: {policy.service_level_type1():.1%}")
print(f"Fill Rate: {policy.service_level_type2():.1%}")

```

Optimal Base-Stock Level: 254 units

Safety Stock: 54 units

Cycle Service Level: 90.0%

Fill Rate: 127.2%

18.3. (s, Q) and (s, S) Policies

18.3.1. Reorder Point, Order Quantity Policy (s, Q)

When inventory position hits s , order Q units.

The `sQPolicy` class below finds the optimal reorder point s and order quantity Q through an iterative procedure. Starting from the EOQ as an initial guess, it alternates between computing the best s given Q and the best Q given s until convergence.

```

class sQPolicy:
    """(s, Q) continuous review inventory policy."""

    def __init__(self, demand_mean: float, demand_std: float,
                 lead_time: float, ordering_cost: float,
                 holding_cost: float, stockout_cost: float):
        self.mu = demand_mean
        self.sigma = demand_std
        self.L = lead_time
        self.K = ordering_cost
        self.h = holding_cost
        self.b = stockout_cost

    def optimize(self, max_iterations: int = 100) -> dict:
        """Find optimal (s, Q) parameters."""
        # Initial Q from EOQ
        Q = np.sqrt(2 * self.mu * self.K / self.h)

        for _ in range(max_iterations):
            Q_old = Q

```

```

        # Optimal s given Q
        s = self._optimal_s_given_Q(Q)

        # Optimal Q given s
        Q = self._optimal_Q_given_s(s)

        if abs(Q - Q_old) < 0.01:
            break

    return {
        'reorder_point': s,
        'order_quantity': Q,
        'expected_cost': self._total_cost(s, Q)
    }

def _optimal_s_given_Q(self, Q: float) -> float:
    """Find optimal reorder point given order quantity."""
    # Lead time demand
    mu_L = self.mu * self.L
    sigma_L = self.sigma * np.sqrt(self.L)

    # Critical ratio for s
    cr = (self.b * self.mu) / (self.b * self.mu + self.h * Q)

    return stats.norm(mu_L, sigma_L).ppf(cr)

def _optimal_Q_given_s(self, s: float) -> float:
    """Find optimal order quantity given reorder point."""
    # Expected shortage per cycle
    B_s = self._expected_shortage(s)

    # Modified EOQ
    Q = np.sqrt(2 * self.mu * (self.K + self.b * B_s) / self.h)
    return Q

def _expected_shortage(self, s: float) -> float:
    """Expected shortage per replenishment cycle."""
    mu_L = self.mu * self.L
    sigma_L = self.sigma * np.sqrt(self.L)

    z = (s - mu_L) / sigma_L
    return sigma_L * (stats.norm.pdf(z) - z * (1 - stats.norm.cdf(z)))

def _total_cost(self, s: float, Q: float) -> float:
    """Calculate total expected cost."""
    B_s = self._expected_shortage(s)

    ordering = self.K * self.mu / Q
    holding = self.h * (Q / 2 + s - self.mu * self.L)
    stockout = self.b * B_s * self.mu / Q

    return ordering + holding + stockout

```

```
# Example
sq = sQPolicy(
    demand_mean=100,
    demand_std=25,
    lead_time=1,
    ordering_cost=50,
    holding_cost=2,
    stockout_cost=20
)

result = sq.optimize()
print(f"Reorder Point (s): {result['reorder_point']:.0f}")
print(f"Order Quantity (Q): {result['order_quantity']:.0f}")
print(f"Expected Annual Cost: ${result['expected_cost']:.2f}")
```

Reorder Point (s): 136
 Order Quantity (Q): 82
 Expected Annual Cost: \$235.42

A closely related approach is the (s, S) policy, where the order quantity is variable rather than fixed. When inventory drops to or below s , the system orders enough to bring inventory back up to S . The following simulation shows how inventory fluctuates under this policy with stochastic demand.

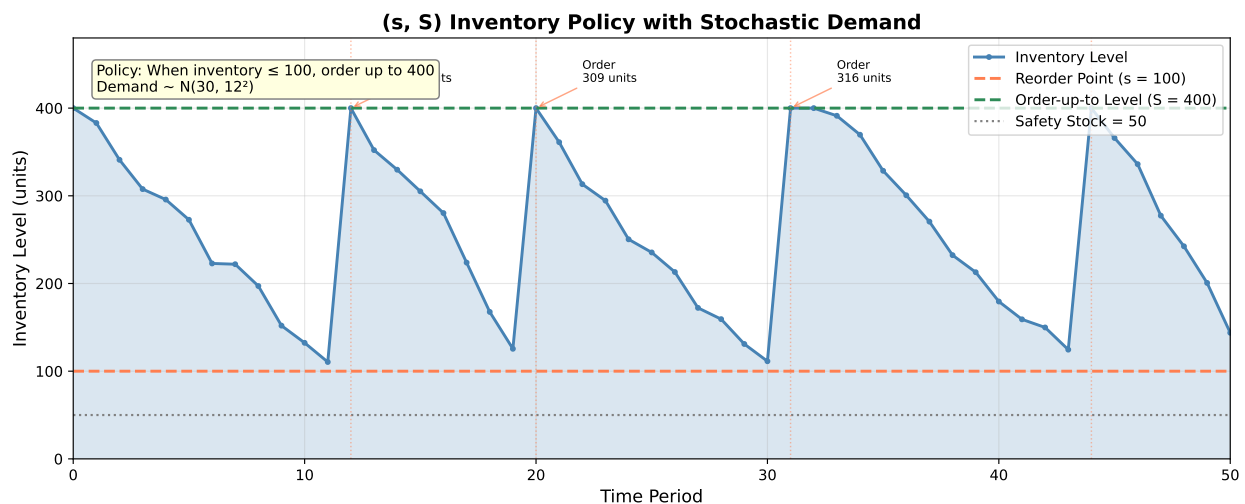


Figure 18.4.: (s, S) Inventory Policy: Inventory levels under random demand with reorder point s and order-up-to level S

18.4. Multi-Product Inventory Optimization

When managing multiple products, constraints and interactions create optimization opportunities. A shared budget or limited warehouse space means that increasing safety stock for one product reduces the resources available for others. The optimization model below uses SciPy's SLSQP solver to find base-stock levels that minimize total holding cost subject to a budget or space constraint.

```

import numpy as np
from scipy.optimize import minimize

class MultiProductInventory:
    """Multi-product inventory optimization with constraints."""

    def __init__(self, num_products: int):
        self.n = num_products
        self.demand_means = np.zeros(num_products)
        self.demand_stds = np.zeros(num_products)
        self.holding_costs = np.zeros(num_products)
        self.ordering_costs = np.zeros(num_products)
        self.unit_costs = np.zeros(num_products)
        self.lead_times = np.zeros(num_products)

    def set_product(self, i: int, demand_mean: float, demand_std: float,
                   holding_cost: float, ordering_cost: float,
                   unit_cost: float, lead_time: float):
        """Set parameters for product i."""
        self.demand_means[i] = demand_mean
        self.demand_stds[i] = demand_std
        self.holding_costs[i] = holding_cost
        self.ordering_costs[i] = ordering_cost
        self.unit_costs[i] = unit_cost
        self.lead_times[i] = lead_time

    def optimize_with_budget(self, budget: float,
                             target_service: float = 0.95) -> dict:
        """Optimize inventory levels with budget constraint."""
        # Decision variables: base-stock levels
        z_target = stats.norm.ppf(target_service)

    def objective(S):
        """Total holding cost."""
        total = 0
        for i in range(self.n):
            mu_L = self.demand_means[i] * self.lead_times[i]
            avg_inventory = max(0, S[i] - mu_L)
            total += self.holding_costs[i] * avg_inventory
        return total

    def budget_constraint(S):
        """Budget constraint: sum of inventory values <= budget."""
        return budget - np.sum(S * self.unit_costs)

    # Initial guess: individual optimal
    S0 = []
    for i in range(self.n):
        mu_L = self.demand_means[i] * self.lead_times[i]
        sigma_L = self.demand_stds[i] * np.sqrt(self.lead_times[i])
        S0.append(mu_L + z_target * sigma_L)

    # Bounds

```

```

bounds = [(0, None) for _ in range(self.n)]

# Optimize
result = minimize(
    objective,
    S0,
    method='SLSQP',
    bounds=bounds,
    constraints={'type': 'ineq', 'fun': budget_constraint}
)

# Calculate service levels
service_levels = []
for i in range(self.n):
    mu_L = self.demand_means[i] * self.lead_times[i]
    sigma_L = self.demand_stds[i] * np.sqrt(self.lead_times[i])
    sl = stats.norm(mu_L, sigma_L).cdf(result.x[i])
    service_levels.append(sl)

return {
    'base_stock_levels': result.x,
    'service_levels': service_levels,
    'total_holding_cost': result.fun,
    'inventory_value': np.sum(result.x * self.unit_costs)
}

def optimize_with_space(self, space_capacity: float,
                        unit_volumes: np.ndarray,
                        target_service: float = 0.95) -> dict:
    """Optimize with warehouse space constraint."""
    z_target = stats.norm.ppf(target_service)

    def objective(S):
        total = 0
        for i in range(self.n):
            mu_L = self.demand_means[i] * self.lead_times[i]
            avg_inventory = max(0, S[i] - mu_L)
            total += self.holding_costs[i] * avg_inventory
        return total

    def space_constraint(S):
        return space_capacity - np.sum(S * unit_volumes)

    S0 = []
    for i in range(self.n):
        mu_L = self.demand_means[i] * self.lead_times[i]
        sigma_L = self.demand_stds[i] * np.sqrt(self.lead_times[i])
        S0.append(mu_L + z_target * sigma_L)

    result = minimize(
        objective,
        S0,
        method='SLSQP',

```

```

        bounds=[(0, None) for _ in range(self.n)],
        constraints={'type': 'ineq', 'fun': space_constraint}
    )

    return {
        'base_stock_levels': result.x,
        'total_holding_cost': result.fun,
        'space_used': np.sum(result.x * unit_volumes)
    }

# Example
inv = MultiProductInventory(num_products=3)
inv.set_product(0, 100, 20, 2, 50, 10, 1)
inv.set_product(1, 50, 15, 5, 75, 25, 2)
inv.set_product(2, 200, 40, 1, 30, 5, 1)

result = inv.optimize_with_budget(budget=5000, target_service=0.95)
print("Optimal Base-Stock Levels:", result['base_stock_levels'].round(0))
print("Service Levels:", [f"{sl:.1%}" for sl in result['service_levels']])
print(f"Inventory Value: ${result['inventory_value']:.0f}")

```

Optimal Base-Stock Levels: [32. 0. 181.]

Service Levels: ['0.0%', '0.0%', '32.0%']

Inventory Value: \$1230

18.5. Multi-Echelon Inventory

Supply chains with multiple stages require coordinated inventory decisions. A warehouse serving multiple retailers faces aggregate demand from all downstream locations, and its echelon lead time includes both its own replenishment time and the upstream delays. The Clark-Scarf decomposition approach computes echelon base-stock levels by processing the supply chain from end nodes upward.

```

from typing import Dict, List, Tuple

@dataclass
class SupplyChainNode:
    """A node in the supply chain."""
    name: str
    holding_cost: float
    lead_time: float
    demand_mean: float = 0 # Only for end nodes
    demand_std: float = 0

class MultiEchelonInventory:
    """Multi-echelon inventory optimization."""

    def __init__(self):
        self.nodes: Dict[str, SupplyChainNode] = {}
        self.edges: List[Tuple[str, str]] = [] # (upstream, downstream)

    def add_node(self, node: SupplyChainNode):

```



```

        """Add a node to the supply chain."""
        self.nodes[node.name] = node

    def add_edge(self, upstream: str, downstream: str):
        """Add an edge (flow direction)."""
        self.edges.append((upstream, downstream))

    def get_echelon_lead_time(self, node_name: str) -> float:
        """Calculate echelon lead time (cumulative from suppliers)."""
        total = self.nodes[node_name].lead_time

        # Find upstream node
        for up, down in self.edges:
            if down == node_name:
                total += self.get_echelon_lead_time(up)
                break

        return total

    def get_downstream_nodes(self, node_name: str) -> List[str]:
        """Get all downstream nodes."""
        downstream = []
        for up, down in self.edges:
            if up == node_name:
                downstream.append(down)
                downstream.extend(self.get_downstream_nodes(down))
        return downstream

    def compute_echelon_stock_levels(self, service_level: float = 0.95) -> dict:
        """Compute echelon base-stock levels using Clark-Scarf approach."""
        z = stats.norm.ppf(service_level)
        echelon_levels = {}

        # Process in reverse topological order (end nodes first)
        for node_name, node in self.nodes.items():
            # Aggregate demand from downstream
            downstream = self.get_downstream_nodes(node_name)

            if not downstream:
                # End node - use its own demand
                demand_mean = node.demand_mean
                demand_std = node.demand_std
            else:
                # Aggregate downstream demand
                demand_mean = sum(
                    self.nodes[d].demand_mean
                    for d in downstream
                    if self.nodes[d].demand_mean > 0
                )
                demand_std = np.sqrt(sum(
                    self.nodes[d].demand_std ** 2
                    for d in downstream
                    if self.nodes[d].demand_std > 0
                ))
            echelon_levels[node_name] = demand_mean + z * demand_std

```

```

        ))

    # Echelon lead time
    echelon_LT = self.get_echelon_lead_time(node_name)

    # Echelon base-stock level
    mu_L = demand_mean * echelon_LT
    sigma_L = demand_std * np.sqrt(echelon_LT)
    S = mu_L + z * sigma_L

    echelon_levels[node_name] = {
        'echelon_base_stock': S,
        'echelon_lead_time': echelon_LT,
        'demand_mean': demand_mean,
        'demand_std': demand_std
    }

    return echelon_levels

# Example: Simple 2-echelon system
sc = MultiEchelonInventory()
sc.add_node(SupplyChainNode("Warehouse", holding_cost=1, lead_time=2))
sc.add_node(SupplyChainNode("Retailer1", holding_cost=3, lead_time=1,
                             demand_mean=100, demand_std=20))
sc.add_node(SupplyChainNode("Retailer2", holding_cost=3, lead_time=1,
                             demand_mean=80, demand_std=15))
sc.add_edge("Warehouse", "Retailer1")
sc.add_edge("Warehouse", "Retailer2")

levels = sc.compute_echelon_stock_levels(service_level=0.95)
for name, data in levels.items():
    print(f"{name}: Echelon Stock = {data['echelon_base_stock']:.0f}, "
          f"Echelon LT = {data['echelon_lead_time']}")

```

```

Warehouse: Echelon Stock = 418, Echelon LT = 2
Retailer1: Echelon Stock = 357, Echelon LT = 3
Retailer2: Echelon Stock = 283, Echelon LT = 3

```

18.6. Stochastic Inventory Optimization

18.6.1. MIP Formulation with Scenarios

When demand is uncertain and decisions span multiple periods, scenario-based stochastic optimization provides a systematic way to find ordering plans that perform well across a range of possible demand realizations. The formulation below uses a set of demand scenarios to construct a deterministic equivalent linear program. The solver computes expected demand from the scenarios and determines order quantities and inventory levels that minimize expected cost over the planning horizon.

```

def stochastic_inventory_optimization(
    num_periods: int,
    scenarios: List[np.ndarray], # List of demand scenarios

```

```

scenario_probs: List[float],
holding_cost: float,
stockout_cost: float,
ordering_cost: float,
initial_inventory: float = 0
) -> dict:
    """Stochastic inventory optimization using scenario-based approach."""
    from scipy.optimize import linprog

    num_scenarios = len(scenarios)
    T = num_periods

    # Decision variables per scenario:
    # Q_t (order quantity), I_t (inventory), B_t (backorder)
    # Total: 3*T per scenario + T ordering decisions (common)

    # For simplicity, we solve a deterministic approximation
    # using expected demand
    expected_demand = np.mean([s for s in scenarios], axis=0)

    # Simple LP formulation
    # Variables: Q_0..Q_{T-1}, I_0..I_{T-1}
    n_vars = 2 * T

    # Objective: min sum(h*I_t + K*delta_t)
    c = np.zeros(n_vars)
    c[:T] = ordering_cost # Ordering costs
    c[T:] = holding_cost # Holding costs

    # Constraints: I_t = I_{t-1} + Q_t - D_t
    # I_t >= 0
    A_eq = []
    b_eq = []

    for t in range(T):
        row = np.zeros(n_vars)
        row[t] = 1 # Q_t
        row[T + t] = -1 # -I_t
        if t > 0:
            row[T + t - 1] = 1 # +I_{t-1}

        rhs = expected_demand[t]
        if t == 0:
            rhs -= initial_inventory

        A_eq.append(row)
        b_eq.append(rhs)

    bounds = [(0, None) for _ in range(n_vars)]

    result = linprog(c, A_eq=A_eq, b_eq=b_eq, bounds=bounds, method='highs')

    if result.success:

```

```

        orders = result.x[:T]
        inventory = result.x[T:]
        return {
            'orders': orders,
            'inventory': inventory,
            'total_cost': result.fun,
            'expected_demand': expected_demand
        }
    else:
        return {'error': 'Optimization failed'}

# Example
np.random.seed(42)
scenarios = [np.random.normal(100, 20, 5) for _ in range(10)]
probs = [0.1] * 10

result = stochastic_inventory_optimization(
    num_periods=5,
    scenarios=scenarios,
    scenario_probs=probs,
    holding_cost=1,
    stockout_cost=10,
    ordering_cost=5,
    initial_inventory=50
)

print("Order Plan:", result['orders'].round(0))
print("Expected Inventory:", result['inventory'].round(0))

```

Order Plan: [48. 101. 103. 88. 88.]

Expected Inventory: [0. 0. 0. 0. 0.]

18.7. Summary

18.7.1. Key Takeaways

- **EOQ** provides optimal order quantity for deterministic, continuous demand
- **Newsvendor** handles single-period decisions under uncertainty
- **Base-stock** policies maintain inventory position at target level
- **(s, Q) policies** balance fixed costs with demand variability
- **Multi-product** optimization considers budget and space constraints
- **Multi-echelon** systems require coordinated inventory across stages
- **Service level** trade-offs determine safety stock requirements
- **Stochastic optimization** handles demand uncertainty systematically

18.8. Exercises

1. A retailer sells 10,000 units per year. Ordering costs \$200 per order, and holding costs are 25% of the \$50 unit price. Calculate the EOQ and annual cost.
2. A fashion retailer must order before the season. Unit cost is \$30, selling price is \$80, and unsold items are salvaged at \$10. Demand is Normal(500, 100). Find the optimal order quantity.

3. Extend the multi-product model to handle substitution: if product A is out of stock, 30% of customers accept product B instead.
4. Design an (s, S) policy where orders are placed up to level S (not fixed quantity Q). When is this better than (s, Q) ?
5. Implement a simulation to compare the performance of different inventory policies under demand that follows a seasonal pattern.

19. Production Planning and ERP Integration

What You Will Learn

- Understand the production planning hierarchy from strategic to operational levels
- Implement Material Requirements Planning (MRP) algorithms in Python
- Apply lot sizing optimization techniques including Wagner-Whitin
- Formulate aggregate planning as linear programming problems
- Model Master Production Scheduling (MPS) decisions
- Understand ERP system architecture and integration patterns
- Connect optimization models to enterprise systems

Production planning is the backbone of manufacturing operations, transforming demand forecasts into actionable schedules for materials, machines, and labor. Modern enterprises rely on Enterprise Resource Planning (ERP) systems to coordinate these activities across departments, but the core algorithms that drive planning decisions remain rooted in operations research. This chapter bridges the gap between optimization theory and enterprise systems, providing practical implementations of MRP, MPS, and aggregate planning algorithms while contextualizing them within the broader ERP ecosystem.

19.1. The Production Planning Hierarchy

Manufacturing planning operates across multiple time horizons, from strategic decisions spanning years to real-time shop floor control. Each level provides constraints and targets for the level below while receiving feedback from operational execution.

19.1.1. Planning Level Characteristics

Level	Time Horizon	Decision Frequency	Key Decisions	Optimization Focus
Strategic	1-5 years	Quarterly	Capacity, locations	MILP, simulation
S&OP	12-18 months	Monthly	Production rates, inventory	LP, aggregate models
MPS	3-6 months	Weekly	End-item quantities	LP, heuristics
MRP	4-12 weeks	Daily/Weekly	Component orders	BOM explosion, lot sizing
SFC	Days-Weeks	Real-time	Sequencing, dispatching	Scheduling algorithms

The Role of Optimization in Manufacturing

Traditional ERP systems use rule-based logic (MRP explosion, fixed lot sizes) that can be suboptimal. Modern Advanced Planning Systems (APS) incorporate optimization to:

- Minimize total costs across the supply chain
- Balance competing objectives (cost vs. service)

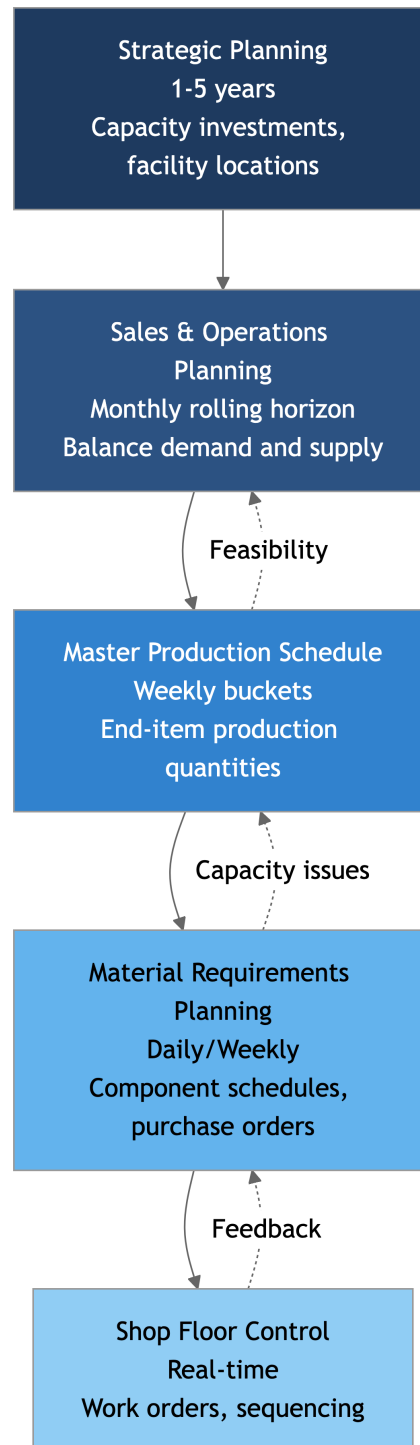


Figure 19.1.: Production Planning Hierarchy: From Strategic to Operational

- Handle constraints that basic MRP cannot
- React dynamically to changes

19.2. Sales and Operations Planning (S&OP)

S&OP is the monthly process that aligns demand forecasts with production capabilities. It bridges the gap between business strategy and operational execution by making aggregate decisions about production rates, inventory levels, and workforce planning.

19.2.1. The S&OP Process

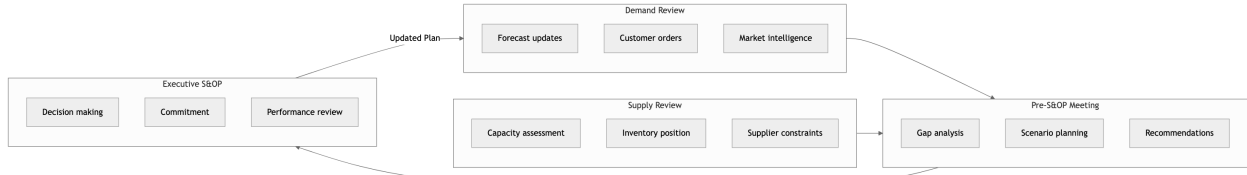


Figure 19.2.: Monthly S&OP Process Cycle

19.2.2. Aggregate Planning Formulation

Aggregate planning determines production quantities, workforce levels, and inventory positions to minimize costs while meeting demand. We model this as a linear program.

Sets and Indices: - $t \in T$: Time periods (months) - Products aggregated into product families

Decision Variables: - P_t : Production quantity in period t - I_t : Inventory at end of period t - B_t : Backorder quantity in period t - W_t : Workforce (regular hours) in period t - O_t : Overtime hours in period t - H_t : Workers hired in period t - F_t : Workers fired in period t

Parameters: - D_t : Demand in period t - c_p : Production cost per unit - c_h : Holding cost per unit per period - c_b : Backorder cost per unit per period - c_o : Overtime cost per hour - c_{hire} : Hiring cost per worker - c_{fire} : Firing cost per worker - α : Hours required per unit - M : Regular hours per worker per period

Formulation:

$$\begin{aligned}
 &\text{minimize} && \sum_{t \in T} (c_p P_t + c_h I_t + c_b B_t + c_o O_t + c_{hire} H_t + c_{fire} F_t) \\
 &\text{subject to} && I_t - B_t = I_{t-1} - B_{t-1} + P_t - D_t, \quad \forall t \in T \\
 &&& W_t = W_{t-1} + H_t - F_t, \quad \forall t \in T \\
 &&& \alpha P_t \leq M \cdot W_t + O_t, \quad \forall t \in T \\
 &&& O_t \leq \bar{O} \cdot W_t, \quad \forall t \in T \\
 &&& I_t, B_t, P_t, W_t, O_t, H_t, F_t \geq 0
 \end{aligned}$$

The implementation below translates this LP formulation into code. It uses PuLP as the primary solver and falls back to SciPy's `linprog` if PuLP is not installed. The example uses a 12-month seasonal demand pattern to show how the optimizer balances hiring, firing, overtime, and inventory decisions.

```

import numpy as np
from dataclasses import dataclass
from typing import List, Dict, Tuple, Optional
import warnings

# Try PuLP first, fall back to scipy if not available
try:
    import pulp
    HAS_PULP = True
except ImportError:
    HAS_PULP = False
    from scipy.optimize import linprog

@dataclass
class AggregatePlanningParams:
    """Parameters for aggregate planning problem."""
    demand: List[float]          # Demand per period
    production_cost: float        # Cost per unit produced
    holding_cost: float           # Inventory holding cost per unit per period
    backorder_cost: float         # Backorder cost per unit per period
    regular_time_cost: float       # Regular labor cost per hour
    overtime_cost: float          # Overtime cost per hour
    hiring_cost: float            # Cost to hire one worker
    firing_cost: float            # Cost to fire one worker
    hours_per_unit: float         # Production hours required per unit
    hours_per_worker: float       # Regular hours available per worker per period
    max_overtime_ratio: float     # Maximum overtime as fraction of regular time
    initial_workforce: int        # Starting workforce
    initial_inventory: float      # Starting inventory

class AggregatePlanner:
    """Aggregate production planning optimizer."""

    def __init__(self, params: AggregatePlanningParams):
        self.params = params
        self.T = len(params.demand)

    def solve(self) -> Dict:
        """Solve the aggregate planning problem."""
        if HAS_PULP:
            return self._solve_pulp()
        else:
            return self._solve_scipy()

    def _solve_pulp(self) -> Dict:
        """Solve using PuLP."""
        p = self.params
        T = self.T

        # Create problem
        model = pulp.LpProblem("Aggregate_Planning", pulp.LpMinimize)

        # Decision variables

```

```

P = [pulp.LpVariable(f"P_{t}", lowBound=0) for t in range(T)]
I = [pulp.LpVariable(f"I_{t}", lowBound=0) for t in range(T)]
B = [pulp.LpVariable(f"B_{t}", lowBound=0) for t in range(T)]
W = [pulp.LpVariable(f"W_{t}", lowBound=0) for t in range(T)]
O = [pulp.LpVariable(f"O_{t}", lowBound=0) for t in range(T)]
H = [pulp.LpVariable(f"H_{t}", lowBound=0) for t in range(T)]
F = [pulp.LpVariable(f"F_{t}", lowBound=0) for t in range(T)]

# Objective function
model += pulp.lpSum([
    p.production_cost * P[t] +
    p.holding_cost * I[t] +
    p.backorder_cost * B[t] +
    p.overtime_cost * O[t] +
    p.hiring_cost * H[t] +
    p.firing_cost * F[t]
    for t in range(T)
])

# Constraints
for t in range(T):
    # Inventory balance
    if t == 0:
        model += I[t] - B[t] == p.initial_inventory + P[t] - p.demand[t]
    else:
        model += I[t] - B[t] == I[t-1] - B[t-1] + P[t] - p.demand[t]

    # Workforce balance
    if t == 0:
        model += W[t] == p.initial_workforce + H[t] - F[t]
    else:
        model += W[t] == W[t-1] + H[t] - F[t]

    # Capacity constraint
    model += p.hours_per_unit * P[t] <= p.hours_per_worker * W[t] + O[t]

    # Overtime limit
    model += O[t] <= p.max_overtime_ratio * p.hours_per_worker * W[t]

# Solve
model.solve(pulp.PULP_CBC_CMD(msg=0))

if model.status == pulp.LpStatusOptimal:
    return {
        'status': 'optimal',
        'total_cost': pulp.value(model.objective),
        'production': [pulp.value(P[t]) for t in range(T)],
        'inventory': [pulp.value(I[t]) for t in range(T)],
        'backorders': [pulp.value(B[t]) for t in range(T)],
        'workforce': [pulp.value(W[t]) for t in range(T)],
        'overtime': [pulp.value(O[t]) for t in range(T)],
        'hiring': [pulp.value(H[t]) for t in range(T)],
        'firing': [pulp.value(F[t]) for t in range(T)]
    }

```

```

    }
    else:
        return {'status': 'infeasible'}

def _solve_scipy(self) -> Dict:
    """Fallback solver using scipy.optimize.linprog."""
    p = self.params
    T = self.T

    # Variable ordering: P, I, B, W, O, H, F (each of length T)
    n_vars = 7 * T

    # Objective coefficients
    c = np.zeros(n_vars)
    for t in range(T):
        c[t] = p.production_cost           # P
        c[T + t] = p.holding_cost          # I
        c[2*T + t] = p.backorder_cost      # B
        # W has no direct cost
        c[4*T + t] = p.overtime_cost       # O
        c[5*T + t] = p.hiring_cost         # H
        c[6*T + t] = p.firing_cost         # F

    # Equality constraints (inventory and workforce balance)
    A_eq = []
    b_eq = []

    for t in range(T):
        # Inventory balance:  $I[t] - B[t] - I[t-1] + B[t-1] - P[t] = -D[t]$ 
        row = np.zeros(n_vars)
        row[t] = -1                # -P[t]
        row[T + t] = 1             # I[t]
        row[2*T + t] = -1          # -B[t]
        if t > 0:
            row[T + t - 1] = -1    # -I[t-1]
            row[2*T + t - 1] = 1   # B[t-1]
        rhs = -p.demand[t]
        if t == 0:
            rhs += p.initial_inventory
        A_eq.append(row)
        b_eq.append(rhs)

        # Workforce balance:  $W[t] - W[t-1] - H[t] + F[t] = 0$ 
        row = np.zeros(n_vars)
        row[3*T + t] = 1           # W[t]
        row[5*T + t] = -1          # -H[t]
        row[6*T + t] = 1           # F[t]
        if t > 0:
            row[3*T + t - 1] = -1  # -W[t-1]
            rhs = 0
        else:
            rhs = p.initial_workforce
        A_eq.append(row)

```

```

        b_eq.append(rhs)

# Inequality constraints (capacity and overtime)
A_ub = []
b_ub = []

for t in range(T):
    # Capacity: hours_per_unit * P[t] - hours_per_worker * W[t] - O[t] <= 0
    row = np.zeros(n_vars)
    row[t] = p.hours_per_unit
    row[3*T + t] = -p.hours_per_worker
    row[4*T + t] = -1
    A_ub.append(row)
    b_ub.append(0)

    # Overtime limit: O[t] - max_overtime_ratio * hours_per_worker * W[t] <=
    # ↪ 0
    row = np.zeros(n_vars)
    row[4*T + t] = 1
    row[3*T + t] = -p.max_overtime_ratio * p.hours_per_worker
    A_ub.append(row)
    b_ub.append(0)

# Bounds (all non-negative)
bounds = [(0, None) for _ in range(n_vars)]

# Solve
result = linprog(c, A_ub=A_ub, b_ub=b_ub, A_eq=A_eq, b_eq=b_eq,
                 bounds=bounds, method='highs')

if result.success:
    x = result.x
    return {
        'status': 'optimal',
        'total_cost': result.fun,
        'production': list(x[:T]),
        'inventory': list(x[T:2*T]),
        'backorders': list(x[2*T:3*T]),
        'workforce': list(x[3*T:4*T]),
        'overtime': list(x[4*T:5*T]),
        'hiring': list(x[5*T:6*T]),
        'firing': list(x[6*T:7*T])
    }
else:
    return {'status': 'infeasible'}

# Example: Seasonal demand pattern
params = AggregatePlanningParams(
    demand=[100, 120, 150, 200, 180, 160, 140, 120, 100, 110, 130, 150],
    production_cost=20,
    holding_cost=2,
    backorder_cost=10,
    regular_time_cost=15,

```

```

    overtime_cost=22.5,
    hiring_cost=500,
    firing_cost=750,
    hours_per_unit=2,
    hours_per_worker=160,
    max_overtime_ratio=0.25,
    initial_workforce=10,
    initial_inventory=50
)

planner = AggregatePlanner(params)
result = planner.solve()

if result['status'] == 'optimal':
    print(f"Total Cost: ${result['total_cost']:,.2f}")
    print(f"\nMonth Demand Prod Inv Workers OT Hours")
    print("-" * 50)
    for t in range(len(params.demand)):
        print(f" {t+1:2d}      {params.demand[t]:4.0f} "
              ↪ {result['production'][t]:5.0f} "
              f"{result['inventory'][t]:4.0f}      {result['workforce'][t]:4.0f} "
              ↪ {result['overtime'][t]:6.0f}")

```

Total Cost: \$30,700.00

Month	Demand	Prod	Inv	Workers	OT Hours

1	100	50	0	10	0
2	120	120	0	10	0
3	150	150	0	10	0
4	200	200	0	10	0
5	180	180	0	10	0
6	160	160	0	10	0
7	140	140	0	10	0
8	120	120	0	10	0
9	100	100	0	10	0
10	110	110	0	10	0
11	130	130	0	10	0
12	150	0	0	10	0

19.2.3. Chase vs. Level Strategies

Aggregate planning strategies fall into two extremes:

Chase Strategy: Vary production to match demand exactly - Minimal inventory - High workforce variability - Good for: Variable demand, low hiring/firing costs

Level Strategy: Maintain constant production rate - High inventory carrying - Stable workforce - Good for: Skilled labor, high hiring/firing costs



Hybrid Strategies

Most practical aggregate plans blend chase and level strategies. The optimization model naturally finds the right balance based on cost parameters. Sensitivity analysis on hiring/firing costs reveals the trade-off between workforce stability and inventory costs.

19.3. Master Production Scheduling (MPS)

The Master Production Schedule translates aggregate plans into specific quantities of end items to be produced in each time period. It serves as the primary input to Material Requirements Planning. The following Gantt chart shows a sample MPS for three products over a 12-week horizon. Each product is produced in alternating weeks to share capacity. The shaded zones indicate time fence classifications: frozen (no changes), slushy (changes require review), and liquid (forecast-driven).

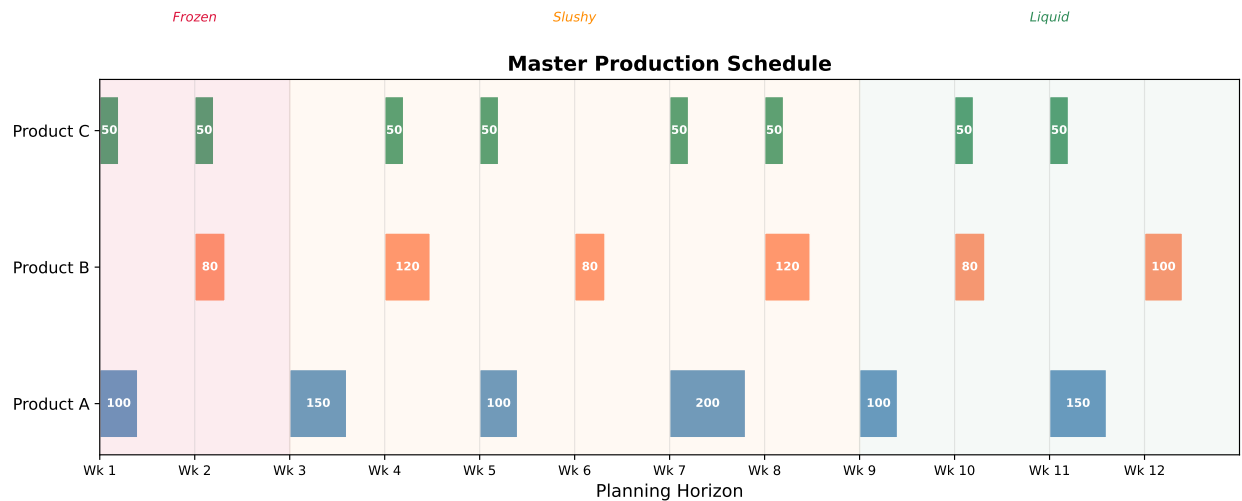


Figure 19.3.: Master Production Schedule Gantt chart — production blocks for three products with capacity limits shown as dashed lines

19.3.1. MPS Time Fences

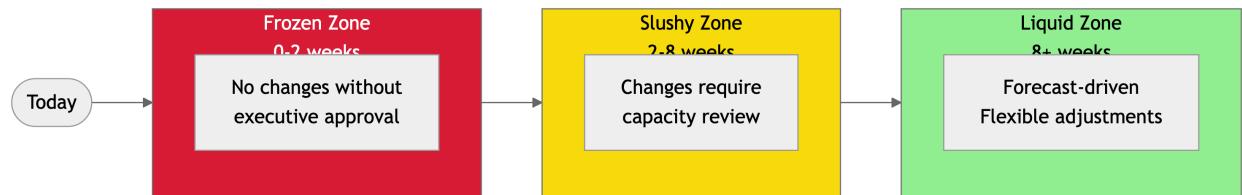


Figure 19.4.: MPS Time Fence Zones

19.3.2. Available-to-Promise (ATP)

ATP calculates the uncommitted portion of inventory and planned production, enabling sales to make reliable delivery commitments. The algorithm computes two related measures: the Projected Available Balance (PAB), which tracks expected inventory after satisfying the larger of forecast

or actual orders, and ATP itself, which shows how many units can be promised to new customers without disrupting existing commitments.

```

from dataclasses import dataclass, field
from typing import List, Optional

@dataclass
class MasterSchedule:
    """Master Production Schedule with ATP calculation."""
    periods: int
    forecast: List[float]
    customer_orders: List[float]
    mps_quantity: List[float]
    beginning_inventory: float = 0

    def calculate_projected_inventory(self) -> List[float]:
        """Calculate projected available balance (PAB)."""
        pab = []
        current = self.beginning_inventory

        for t in range(self.periods):
            # Greater of forecast or orders (conservative)
            demand = max(self.forecast[t], self.customer_orders[t])
            current = current + self.mps_quantity[t] - demand
            pab.append(current)

        return pab

    def calculate_atp(self) -> List[float]:
        """Calculate Available-to-Promise for each period.

        ATP = MPS quantity - Customer orders until next MPS
        For first period: Beginning inventory + MPS - Orders until next MPS
        """
        atp = []

        for t in range(self.periods):
            # Find orders until next MPS receipt
            orders_until_next = 0
            for future_t in range(t, self.periods):
                orders_until_next += self.customer_orders[future_t]
                # Stop if we hit another MPS (except first period)
                if future_t > t and self.mps_quantity[future_t] > 0:
                    break

            if t == 0:
                atp_value = self.beginning_inventory + self.mps_quantity[t] -
↪ orders_until_next
            elif self.mps_quantity[t] > 0:
                # ATP only calculated for MPS periods
                orders_this_batch = 0
                for future_t in range(t, self.periods):
                    if future_t > t and self.mps_quantity[future_t] > 0:

```



```

        break
        orders_this_batch += self.customer_orders[future_t]
        atp_value = self.mps_quantity[t] - orders_this_batch
    else:
        atp_value = 0 # No ATP in non-MPS periods

    atp.append(max(0, atp_value))

    return atp

# Example MPS
mps = MasterSchedule(
    periods=8,
    forecast=[30, 30, 30, 30, 40, 40, 40, 40],
    customer_orders=[35, 25, 20, 10, 15, 8, 5, 0],
    mps_quantity=[0, 60, 0, 60, 0, 60, 0, 60],
    beginning_inventory=40
)

pab = mps.calculate_projected_inventory()
atp = mps.calculate_atp()

print("Period      1      2      3      4      5      6      7      8")
print("-" * 55)
print(f"Forecast      {' '.join(f'{f:4.0f}' for f in mps.forecast)}")
print(f"Orders        {' '.join(f'{o:4.0f}' for o in mps.customer_orders)}")
print(f"MPS           {' '.join(f'{m:4.0f}' for m in mps.mps_quantity)}")
print(f"PAB          {' '.join(f'{p:4.0f}' for p in pab)}")
print(f"ATP           {' '.join(f'{a:4.0f}' for a in atp)}")

```

Period	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	40	40	40	40
Orders	35	25	20	10	15	8	5	0
MPS	0	60	0	60	0	60	0	60
PAB	5	35	5	35	-5	15	-25	-5
ATP	0	15	0	35	0	47	0	60

19.3.3. MPS Optimization Model

The MPS can be formulated as an optimization problem that balances production costs, inventory holding, and customer service. The model uses binary setup variables to capture fixed costs incurred whenever a production run occurs, along with minimum lot size constraints. This mixed-integer program minimizes total cost while maintaining safety stock targets across all periods.

```

@dataclass
class MPSOptimizationParams:
    """Parameters for MPS optimization."""
    periods: int
    demand: List[float]
    setup_cost: float
    holding_cost: float

```

```

production_cost: float
min_lot_size: float
max_production: float
initial_inventory: float
target_safety_stock: List[float]

class MPSOptimizer:
    """Master Production Schedule optimizer."""

    def __init__(self, params: MPSOptimizationParams):
        self.params = params

    def solve(self) -> Dict:
        """Solve MPS optimization problem."""
        p = self.params
        T = p.periods

        if not HAS_PULP:
            warnings.warn("PuLP not available. Using greedy heuristic.")
            return self._solve_heuristic()

        # Create model
        model = pulp.LpProblem("MPS_Optimization", pulp.LpMinimize)

        # Variables
        X = [pulp.LpVariable(f"X_{t}", lowBound=0, upBound=p.max_production)
              for t in range(T)]
        I = [pulp.LpVariable(f"I_{t}", lowBound=0) for t in range(T)]
        Y = [pulp.LpVariable(f"Y_{t}", cat='Binary') for t in range(T)]

        # Big M for setup indicator
        M = p.max_production

        # Objective: minimize production + holding + setup costs
        model += pulp.lpSum([
            p.production_cost * X[t] +
            p.holding_cost * I[t] +
            p.setup_cost * Y[t]
            for t in range(T)
        ])

        # Constraints
        for t in range(T):
            # Inventory balance
            if t == 0:
                model += I[t] == p.initial_inventory + X[t] - p.demand[t]
            else:
                model += I[t] == I[t-1] + X[t] - p.demand[t]

            # Safety stock requirement
            model += I[t] >= p.target_safety_stock[t]

            # Setup indicator

```

```

        model += X[t] <= M * Y[t]

        # Minimum lot size when producing
        model += X[t] >= p.min_lot_size * Y[t]

    # Solve
    model.solve(pulp.PULP_CBC_CMD(msg=0))

    if model.status == pulp.LpStatusOptimal:
        return {
            'status': 'optimal',
            'total_cost': pulp.value(model.objective),
            'production': [pulp.value(X[t]) for t in range(T)],
            'inventory': [pulp.value(I[t]) for t in range(T)],
            'setups': [pulp.value(Y[t]) for t in range(T)]
        }
    return {'status': 'infeasible'}

def _solve_heuristic(self) -> Dict:
    """Greedy lot-for-lot heuristic."""
    p = self.params
    production = []
    inventory = []
    current_inv = p.initial_inventory

    for t in range(p.periods):
        # Produce to meet demand and safety stock
        needed = p.demand[t] + p.target_safety_stock[t] - current_inv
        if needed > 0:
            # Round up to minimum lot size
            prod = max(p.min_lot_size, needed)
            prod = min(prod, p.max_production)
        else:
            prod = 0

        production.append(prod)
        current_inv = current_inv + prod - p.demand[t]
        inventory.append(current_inv)

    # Calculate cost
    total_cost = sum(
        p.production_cost * production[t] +
        p.holding_cost * inventory[t] +
        (p.setup_cost if production[t] > 0 else 0)
        for t in range(p.periods)
    )

    return {
        'status': 'heuristic',
        'total_cost': total_cost,
        'production': production,
        'inventory': inventory,
        'setups': [1 if p > 0 else 0 for p in production]
    }

```

```

    }

# Example
mps_params = MPSOptimizationParams(
    periods=8,
    demand=[100, 120, 90, 110, 150, 130, 100, 120],
    setup_cost=500,
    holding_cost=2,
    production_cost=10,
    min_lot_size=50,
    max_production=200,
    initial_inventory=30,
    target_safety_stock=[20] * 8
)

optimizer = MPSOptimizer(mps_params)
result = optimizer.solve()

print(f"Status: {result['status']}")
print(f"Total Cost: ${result['total_cost']:, .2f}")
print(f"\nPeriod   Demand   Prod   Inv   Setup")
print("-" * 40)
for t in range(mps_params.periods):
    print(f"   {t+1:2d}       {mps_params.demand[t]:4.0f} "
          f"↪ {result['production'][t]:4.0f}  "
          f"   {result['inventory'][t]:4.0f}   {int(result['setups'][t])}")

```

Status: heuristic

Total Cost: \$13,420.00

Period	Demand	Prod	Inv	Setup

1	100	90	20	1
2	120	120	20	1
3	90	90	20	1
4	110	110	20	1
5	150	150	20	1
6	130	130	20	1
7	100	100	20	1
8	120	120	20	1

19.4. Material Requirements Planning (MRP)

MRP is the time-phased planning system that calculates when and how much of each component is needed based on the Master Production Schedule and Bill of Materials structure.

19.4.1. Bill of Materials (BOM)

The BOM defines the parent-child relationships between items and the quantities required.

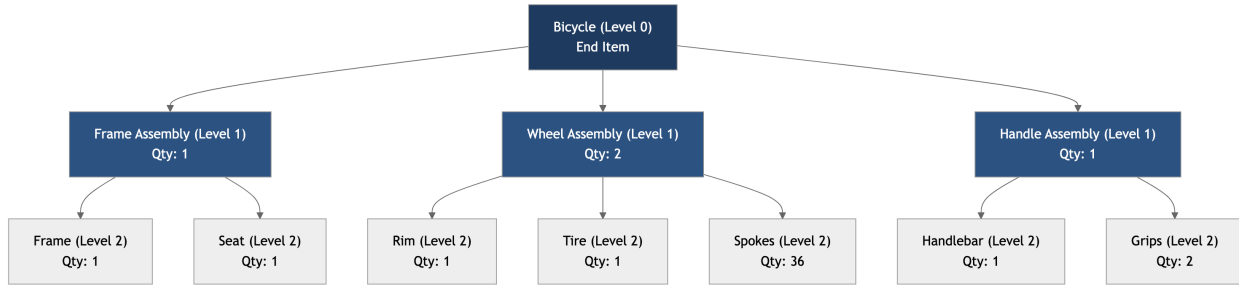


Figure 19.5.: Bill of Materials Structure for a Bicycle

19.4.2. MRP Algorithm Implementation

The MRP explosion algorithm processes items level by level, starting at the top of the BOM (level 0) and working downward. For each item, it calculates gross requirements from the MPS or from parent planned order releases, then nets out on-hand inventory and scheduled receipts. When net requirements are positive, the algorithm applies a lot sizing policy to determine order quantities and offsets them by lead time to produce planned order releases. These releases then become gross requirements for child components at the next level.

```

from dataclasses import dataclass, field
from typing import Dict, List, Set, Tuple
from collections import defaultdict
from enum import Enum

class LotSizingPolicy(Enum):
    """Lot sizing policies for MRP."""
    LOT_FOR_LOT = "L4L"      # Order exactly what's needed
    FIXED_ORDER_QTY = "FOQ"  # Fixed order quantity
    FIXED_PERIOD = "FP"      # Cover fixed periods of demand
    EOQ = "EOQ"              # Economic Order Quantity
    WAGNER_WHITIN = "WW"     # Optimal dynamic lot sizing

@dataclass
class BOMItem:
    """Bill of Materials item."""
    item_id: str
    description: str
    level: int
    lead_time: int
    on_hand: float = 0
    scheduled_receipts: Dict[int, float] = field(default_factory=dict)
    lot_size_policy: LotSizingPolicy = LotSizingPolicy.LOT_FOR_LOT
    fixed_lot_size: float = 0
    ordering_cost: float = 0
    holding_cost: float = 0
    safety_stock: float = 0

@dataclass
class BOMRelation:
    """Parent-child relationship in BOM."""

```

```

parent_id: str
child_id: str
quantity_per: float

class MRPSystem:
    """Material Requirements Planning system."""

    def __init__(self, periods: int):
        self.periods = periods
        self.items: Dict[str, BOMItem] = {}
        self.bom_relations: List[BOMRelation] = []
        self.mps: Dict[str, List[float]] = {} # MPS for end items

    def add_item(self, item: BOMItem):
        """Add item to the system."""
        self.items[item.item_id] = item

    def add_bom_relation(self, parent_id: str, child_id: str, qty_per: float):
        """Add parent-child relationship."""
        self.bom_relations.append(BOMRelation(parent_id, child_id, qty_per))

    def set_mps(self, item_id: str, schedule: List[float]):
        """Set master production schedule for end item."""
        self.mps[item_id] = schedule

    def get_children(self, parent_id: str) -> List[Tuple[str, float]]:
        """Get child items and quantities for a parent."""
        return [(r.child_id, r.quantity_per)
                for r in self.bom_relations if r.parent_id == parent_id]

    def get_items_by_level(self) -> Dict[int, List[str]]:
        """Group items by BOM level."""
        levels = defaultdict(list)
        for item_id, item in self.items.items():
            levels[item.level].append(item_id)
        return dict(levels)

    def explode(self) -> Dict[str, Dict]:
        """Perform MRP explosion.

        Returns:
            Dict mapping item_id to MRP record with:
            - gross_requirements
            - scheduled_receipts
            - projected_on_hand
            - net_requirements
            - planned_order_receipts
            - planned_order_releases
        """
        results = {}
        planned_releases = {} # Track releases to pass down BOM

        # Process level by level (low-level coding)

```

```

levels = self.get_items_by_level()

for level in sorted(levels.keys()):
    for item_id in levels[level]:
        item = self.items[item_id]
        record = self._calculate_mrp_record(item, planned_releases)
        results[item_id] = record

    # Pass planned order releases as gross requirements to children
    for child_id, qty_per in self.get_children(item_id):
        if child_id not in planned_releases:
            planned_releases[child_id] = [0.0] * self.periods
        for t in range(self.periods):
            planned_releases[child_id][t] += (
                record['planned_order_releases'][t] * qty_per
            )

return results

def _calculate_mrp_record(self, item: BOMItem,
                        parent_releases: Dict[str, List[float]]) -> Dict:
    """Calculate MRP record for single item."""
    T = self.periods

    # Gross requirements from MPS (if end item) or parent releases
    gross_req = [0.0] * T
    if item.item_id in self.mps:
        gross_req = self.mps[item.item_id].copy()
    if item.item_id in parent_releases:
        for t in range(T):
            gross_req[t] += parent_releases[item.item_id][t]

    # Scheduled receipts
    scheduled = [item.scheduled_receipts.get(t, 0.0) for t in range(T)]

    # Calculate period by period
    projected = [0.0] * T
    net_req = [0.0] * T
    planned_receipts = [0.0] * T
    planned_releases = [0.0] * T

    current_on_hand = item.on_hand

    for t in range(T):
        # Project available
        available = current_on_hand + scheduled[t] - gross_req[t]

        # Check if we need to order
        if available < item.safety_stock:
            net = item.safety_stock - available
            net_req[t] = net

        # Apply lot sizing

```

```

        lot_size = self._apply_lot_sizing(item, net, t, gross_req)
        planned_receipts[t] = lot_size

        # Release timing (account for lead time)
        release_period = t - item.lead_time
        if release_period >= 0:
            planned_releases[release_period] = lot_size
        elif release_period < 0:
            # Past due - release immediately (period 0)
            planned_releases[0] = lot_size

        available += lot_size

    projected[t] = available
    current_on_hand = available

    return {
        'gross_requirements': gross_req,
        'scheduled_receipts': scheduled,
        'projected_on_hand': projected,
        'net_requirements': net_req,
        'planned_order_receipts': planned_receipts,
        'planned_order_releases': planned_releases
    }

def _apply_lot_sizing(self, item: BOMItem, net_req: float,
                    period: int, all_gross: List[float]) -> float:
    """Apply lot sizing policy to determine order quantity."""
    if item.lot_size_policy == LotSizingPolicy.LOT_FOR_LOT:
        return net_req

    elif item.lot_size_policy == LotSizingPolicy.FIXED_ORDER_QTY:
        # Order in multiples of fixed lot size
        if item.fixed_lot_size > 0:
            return np.ceil(net_req / item.fixed_lot_size) * item.fixed_lot_size
        return net_req

    elif item.lot_size_policy == LotSizingPolicy.EOQ:
        # Use EOQ formula
        if item.ordering_cost > 0 and item.holding_cost > 0:
            annual_demand = sum(all_gross) * (52 / self.periods) # Annualize
            eoq = np.sqrt(2 * annual_demand * item.ordering_cost /
        ↪ item.holding_cost)
            return max(net_req, eoq)
        return net_req

    else:
        return net_req

def print_mrp_record(self, item_id: str, record: Dict):
    """Print formatted MRP record."""
    item = self.items[item_id]
    print(f"\n{'='*60}")

```



```

print(f"Item: {item_id} - {item.description}")
print(f"Level: {item.level}, Lead Time: {item.lead_time}, "
      f"Safety Stock: {item.safety_stock}")
print(f"Lot Sizing: {item.lot_size_policy.value}")
print(f"'{'*60}'")

periods = list(range(1, self.periods + 1))
print(f"{'Period':<25}" + "".join(f"{'p':>8}" for p in periods))
print("-" * (25 + 8 * self.periods))

rows = [
    ('Gross Requirements', record['gross_requirements']),
    ('Scheduled Receipts', record['scheduled_receipts']),
    ('Projected On-Hand', record['projected_on_hand']),
    ('Net Requirements', record['net_requirements']),
    ('Planned Order Receipts', record['planned_order_receipts']),
    ('Planned Order Releases', record['planned_order_releases'])
]

for label, values in rows:
    print(f"{'label':<25}" + "".join(f"{'v':>8.0f}" for v in values))

# Example: Build a simple product structure
mrp = MRPSystem(periods=8)

# End item
mrp.add_item(BOMItem(
    item_id="A",
    description="Finished Product",
    level=0,
    lead_time=1,
    on_hand=10,
    safety_stock=0
))

# Components
mrp.add_item(BOMItem(
    item_id="B",
    description="Subassembly",
    level=1,
    lead_time=2,
    on_hand=20,
    scheduled_receipts={2: 30},
    safety_stock=10
))

mrp.add_item(BOMItem(
    item_id="C",
    description="Component",
    level=1,
    lead_time=1,
    on_hand=50,
    lot_size_policy=LotSizingPolicy.FIXED_ORDER_QTY,

```

19. Production Planning and ERP Integration

```

        fixed_lot_size=100,
        safety_stock=20
    ))

    mrp.add_item(BOMItem(
        item_id="D",
        description="Raw Material",
        level=2,
        lead_time=3,
        on_hand=100,
        safety_stock=50
    ))

    # BOM structure: A needs 1 B and 2 C; B needs 3 D
    mrp.add_bom_relation("A", "B", 1)
    mrp.add_bom_relation("A", "C", 2)
    mrp.add_bom_relation("B", "D", 3)

    # MPS for end item A
    mrp.set_mps("A", [0, 20, 25, 30, 20, 25, 30, 20])

    # Run explosion
    results = mrp.explode()

    # Print records
    for item_id in ["A", "B", "C", "D"]:
        mrp.print_mrp_record(item_id, results[item_id])

```

```
=====
```

Item: A - Finished Product								
Level: 0, Lead Time: 1, Safety Stock: 0								
Lot Sizing: L4L								
=====								
Period	1	2	3	4	5	6	7	8
Gross Requirements	0	20	25	30	20	25	30	20
Scheduled Receipts	0	0	0	0	0	0	0	0
Projected On-Hand	10	0	0	0	0	0	0	0
Net Requirements	0	10	25	30	20	25	30	20
Planned Order Receipts	0	10	25	30	20	25	30	20
Planned Order Releases	10	25	30	20	25	30	20	0

```
=====
```

Item: B - Subassembly								
Level: 1, Lead Time: 2, Safety Stock: 10								
Lot Sizing: L4L								
=====								
Period	1	2	3	4	5	6	7	8

19.4. Material Requirements Planning (MRP)

Gross Requirements	10	25	30	20	25	30	20	0
Scheduled Receipts	0	0	30	0	0	0	0	0
Projected On-Hand	10	10	10	10	10	10	10	10
Net Requirements	0	25	0	20	25	30	20	0
Planned Order Receipts	0	25	0	20	25	30	20	0
Planned Order Releases	25	20	25	30	20	0	0	0

=====

Item: C - Component

Level: 1, Lead Time: 1, Safety Stock: 20

Lot Sizing: FOQ

=====

Period	1	2	3	4	5	6	7	8

Gross Requirements	20	50	60	40	50	60	40	0
Scheduled Receipts	0	0	0	0	0	0	0	0
Projected On-Hand	30	80	20	80	30	70	30	30
Net Requirements	0	40	0	40	0	50	0	0
Planned Order Receipts	0	100	0	100	0	100	0	0
Planned Order Releases	100	0	100	0	100	0	0	0

=====

Item: D - Raw Material

Level: 2, Lead Time: 3, Safety Stock: 50

Lot Sizing: L4L

=====

Period	1	2	3	4	5	6	7	8

Gross Requirements	75	60	75	90	60	0	0	0
Scheduled Receipts	0	0	0	0	0	0	0	0
Projected On-Hand	50	50	50	50	50	50	50	50
Net Requirements	25	60	75	90	60	0	0	0
Planned Order Receipts	25	60	75	90	60	0	0	0
Planned Order Releases	90	60	0	0	0	0	0	0

MRP Nervousness

Small changes in the MPS can cause large swings in planned orders, especially for items with long lead times at lower BOM levels. This “nervousness” can be mitigated by:

- Freezing the MPS in the near term
- Using lot sizing that smooths orders (POQ, Wagner-Whitin)
- Implementing dampening rules that ignore small changes
- Using safety stock strategically

19.5. Lot Sizing Optimization

Lot sizing determines how much to order and when. While simple rules (L4L, FOQ) are common, dynamic lot sizing algorithms can significantly reduce total costs.

The following figure shows the fundamental trade-off in lot sizing. Larger lot sizes reduce the number of setups (and their associated fixed costs) but increase inventory holding costs. The EOQ sits at the point where these two opposing cost curves intersect.

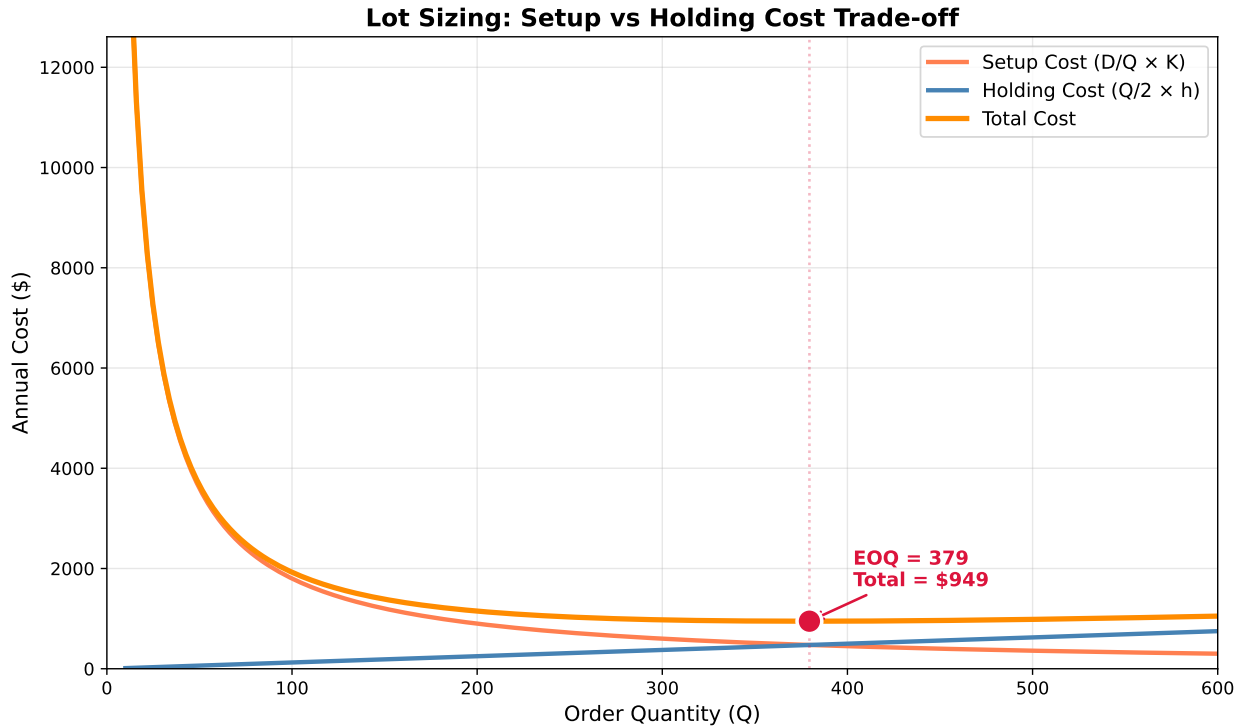


Figure 19.6.: Lot sizing trade-off — as order quantity increases, setup costs decrease but holding costs increase, with the EOQ at the minimum total cost point

19.5.1. Wagner-Whitin Algorithm

The Wagner-Whitin algorithm finds the optimal lot sizing solution for the dynamic lot sizing problem, minimizing setup and holding costs over the planning horizon.

Problem:

$$\begin{aligned}
 & \text{minimize} && \sum_{t=1}^T \left(K \cdot y_t + h \sum_{j=t}^T (j-t) \cdot d_j \cdot x_t^j \right) \\
 & \text{subject to} && \sum_{t=1}^j x_t^j = 1, \quad \forall j \\
 & && x_t^j \in \{0, 1\}
 \end{aligned}$$

where $x_t^j = 1$ if demand in period j is satisfied by production in period t .

Key Property (Planning Horizon Theorem): At every optimal solution, if production occurs in period t , it should cover complete periods' demands, not partial periods.

The implementation below uses dynamic programming to find the minimum-cost lot sizing plan. It also includes two widely used heuristics for comparison: Silver-Meal (minimizes average cost per period within each lot) and Part-Period Balancing (extends lots until cumulative holding cost approaches setup cost).

```
def wagner_whitin(demand: List[float], setup_cost: float,
                  holding_cost: float) -> Dict:
    """Wagner-Whitin dynamic lot sizing algorithm.

    Uses dynamic programming to find optimal lot sizes that minimize
    total setup and holding costs.

    Args:
        demand: Demand for each period
        setup_cost: Fixed cost per order/setup
        holding_cost: Cost to hold one unit for one period

    Returns:
        Dict with optimal solution including order quantities and costs
    """
    T = len(demand)

    # Cost to satisfy demands from period i through j with single order at i
    def order_cost(i: int, j: int) -> float:
        """Cost of single order at i covering demands i through j."""
        cost = setup_cost
        for k in range(i, j + 1):
            # Holding cost for periods between order and demand
            cost += holding_cost * (k - i) * demand[k]
        return cost

    # Dynamic programming
    # f[t] = minimum cost to satisfy demands in periods 0 through t
    f = [float('inf')] * T
    # last_order[t] = period of last order to reach optimal f[t]
    last_order = [-1] * T

    # Base case: single order covering all periods up to t
    for t in range(T):
        f[t] = order_cost(0, t)
        last_order[t] = 0

    # Fill DP table
    for t in range(1, T):
        # Try all possible last order periods
        for j in range(1, t + 1):
            # Order at j covers demands j through t
            cost = f[j-1] + order_cost(j, t)
            if cost < f[t]:
                f[t] = cost
                last_order[t] = j

    # Backtrack to find solution
```

```

orders = [0.0] * T
t = T - 1
while t >= 0:
    order_period = last_order[t]
    # Sum demands from order_period to t
    order_qty = sum(demand[order_period:t+1])
    orders[order_period] = order_qty
    t = order_period - 1

# Calculate costs
total_setup = sum(setup_cost if q > 0 else 0 for q in orders)
total_holding = 0
inventory = 0
for t in range(T):
    inventory += orders[t]
    inventory -= demand[t]
    total_holding += holding_cost * inventory

return {
    'orders': orders,
    'total_cost': f[T-1],
    'setup_cost': total_setup,
    'holding_cost': total_holding,
    'inventory': calculate_inventory_levels(orders, demand)
}

def calculate_inventory_levels(orders: List[float],
                             demand: List[float]) -> List[float]:
    """Calculate ending inventory for each period."""
    inventory = []
    current = 0
    for t in range(len(demand)):
        current += orders[t] - demand[t]
        inventory.append(current)
    return inventory

def silver_meal(demand: List[float], setup_cost: float,
               holding_cost: float) -> Dict:
    """Silver-Meal heuristic for dynamic lot sizing.

    Adds periods to lot until average cost per period increases.

    Args:
        demand: Demand for each period
        setup_cost: Fixed cost per order
        holding_cost: Holding cost per unit per period

    Returns:
        Dict with solution
    """
    T = len(demand)
    orders = [0.0] * T

```

```

t = 0
while t < T:
    # Start new lot at period t
    if demand[t] == 0:
        t += 1
        continue

    lot_qty = demand[t]
    periods_covered = 1
    total_cost = setup_cost
    avg_cost = total_cost / periods_covered

    # Try adding more periods
    j = t + 1
    while j < T:
        new_holding = holding_cost * (j - t) * demand[j]
        new_total = total_cost + new_holding
        new_avg = new_total / (periods_covered + 1)

        if new_avg >= avg_cost:
            break # Adding period j increases average cost

        lot_qty += demand[j]
        total_cost = new_total
        avg_cost = new_avg
        periods_covered += 1
        j += 1

    orders[t] = lot_qty
    t = t + periods_covered

# Calculate costs
total_setup = sum(setup_cost if q > 0 else 0 for q in orders)
inventory = calculate_inventory_levels(orders, demand)
total_holding = sum(holding_cost * max(0, inv) for inv in inventory)

return {
    'orders': orders,
    'total_cost': total_setup + total_holding,
    'setup_cost': total_setup,
    'holding_cost': total_holding,
    'inventory': inventory
}

def part_period_balancing(demand: List[float], setup_cost: float,
                          holding_cost: float) -> Dict:
    """Part-Period Balancing heuristic.

    Adds periods until holding cost approaches setup cost.

    Args:
        demand: Demand for each period
        setup_cost: Fixed cost per order

```

```

    holding_cost: Holding cost per unit per period

Returns:
    Dict with solution
    """
    T = len(demand)
    orders = [0.0] * T
    target_pp = setup_cost / holding_cost # Target part-periods

    t = 0
    while t < T:
        if demand[t] == 0:
            t += 1
            continue

        lot_qty = demand[t]
        cumulative_pp = 0 # Cumulative part-periods

        j = t + 1
        while j < T:
            additional_pp = (j - t) * demand[j]

            if cumulative_pp + additional_pp > target_pp:
                # Check if we're closer to target with or without this period
                if abs(cumulative_pp - target_pp) <= abs(cumulative_pp +
                    ↪ additional_pp - target_pp):
                    break

            lot_qty += demand[j]
            cumulative_pp += additional_pp
            j += 1

        orders[t] = lot_qty
        t = j

    # Calculate costs
    total_setup = sum(setup_cost if q > 0 else 0 for q in orders)
    inventory = calculate_inventory_levels(orders, demand)
    total_holding = sum(holding_cost * max(0, inv) for inv in inventory)

    return {
        'orders': orders,
        'total_cost': total_setup + total_holding,
        'setup_cost': total_setup,
        'holding_cost': total_holding,
        'inventory': inventory
    }

# Compare lot sizing methods
demand = [20, 50, 10, 50, 50, 10, 20, 40, 20, 30]
setup_cost = 200
holding_cost = 1

```



```

print("Demand Pattern:", demand)
print(f"Setup Cost: ${setup_cost}, Holding Cost: ${holding_cost}/unit/period")
print("\n" + "="*70)

methods = [
    ("Lot-for-Lot", {'orders': demand.copy(),
                    'total_cost': setup_cost * sum(1 for d in demand if d > 0),
                    'inventory': [0] * len(demand)}),
    ("Wagner-Whitin (Optimal)", wagner_whitin(demand, setup_cost, holding_cost)),
    ("Silver-Meal", silver_meal(demand, setup_cost, holding_cost)),
    ("Part-Period Balancing", part_period_balancing(demand, setup_cost,
    ↪ holding_cost))
]

print(f"\n{'Method':<25} {'Total Cost':>12} {'Setups':>8} {'Orders'}")
print("-" * 70)
for name, result in methods:
    num_setups = sum(1 for q in result['orders'] if q > 0)
    orders_str = [f"{int(q)}" if q > 0 else "-" for q in result['orders']]
    print(f"{'name':<25} ${result['total_cost']:>10,.0f} {num_setups:>8}
    ↪ {orders_str}")

```

Demand Pattern: [20, 50, 10, 50, 50, 10, 20, 40, 20, 30]

Setup Cost: \$200, Holding Cost: \$1/unit/period

=====

Method	Total Cost	Setups	Orders
Lot-for-Lot	\$ 2,000	10	['20', '50', '10', '50', '50', '10', '20', '40', '20', '30']
Wagner-Whitin (Optimal)	\$ 880	3	['80', '-', '-', '130', '-', '- , '-', '90', '-', '-']
Silver-Meal	\$ 880	3	['80', '-', '-', '130', '-', '- , '-', '90', '-', '-']
Part-Period Balancing	\$ 1,020	3	['130', '-', '-', '-', '120', '- , '-', '-', '50', '-']

i Lot Sizing Method Selection

Method	Complexity	Optimality	Best For
Lot-for-Lot	O(n)	Suboptimal	Expensive items, stable demand
Fixed Order Qty	O(n)	Suboptimal	Standard purchase quantities
EOQ	O(n)	Suboptimal	Stable demand, known costs
Silver-Meal	O(n ²)	Near-optimal	Varying demand, quick solution
Part-Period	O(n ²)	Near-optimal	Simple to explain
Wagner-Whitin	O(n ²)	Optimal	Known horizon, minimize cost

19.6. Advanced Planning Systems (APS)

Advanced Planning Systems extend MRP's capabilities by incorporating finite capacity, optimization algorithms, and real-time replanning.

19.6.1. Beyond MRP: Finite Capacity Planning

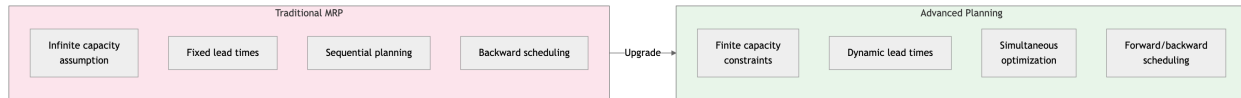


Figure 19.7.: MRP vs APS Capabilities

19.6.2. Constraint-Based Planning

Traditional MRP assumes infinite capacity: it generates planned orders without checking whether sufficient machine hours or labor are available. Finite capacity planning corrects this by scheduling operations against actual resource availability. The implementation below uses backward scheduling, starting from each order's due date and working backward through operations, assigning each to the latest feasible period with available capacity.

```
from dataclasses import dataclass
from typing import Dict, List, Optional, Tuple
from collections import defaultdict

@dataclass
class Resource:
    """Production resource with capacity."""
    resource_id: str
    name: str
    capacity_per_period: float # Available hours per period

@dataclass
class Operation:
    """Production operation."""
    operation_id: str
    product_id: str
    resource_id: str
    hours_per_unit: float
    sequence: int # Order within product routing

@dataclass
class ProductionOrder:
    """Production order to schedule."""
    order_id: str
    product_id: str
    quantity: float
    due_period: int
    priority: int = 1

class FiniteCapacityPlanner:
    """Finite capacity production planning."""
```

```

def __init__(self, periods: int):
    self.periods = periods
    self.resources: Dict[str, Resource] = {}
    self.operations: Dict[str, List[Operation]] = {} # By product
    self.orders: List[ProductionOrder] = []

def add_resource(self, resource: Resource):
    """Add a resource."""
    self.resources[resource.resource_id] = resource

def add_operation(self, op: Operation):
    """Add an operation."""
    if op.product_id not in self.operations:
        self.operations[op.product_id] = []
    self.operations[op.product_id].append(op)
    # Sort by sequence
    self.operations[op.product_id].sort(key=lambda x: x.sequence)

def add_order(self, order: ProductionOrder):
    """Add a production order."""
    self.orders.append(order)

def schedule(self) -> Dict:
    """Schedule orders with finite capacity constraints.

    Uses backward scheduling with capacity checks.
    """
    # Initialize capacity tracker
    capacity_used = defaultdict(lambda: [0.0] * self.periods)

    # Sort orders by due date, then priority
    sorted_orders = sorted(self.orders,
                           key=lambda o: (o.due_period, -o.priority))

    schedule = {}
    infeasible_orders = []

    for order in sorted_orders:
        result = self._schedule_order(order, capacity_used)
        if result:
            schedule[order.order_id] = result
        else:
            infeasible_orders.append(order.order_id)

    # Calculate resource utilization
    utilization = {}
    for res_id, res in self.resources.items():
        total_used = sum(capacity_used[res_id])
        total_available = res.capacity_per_period * self.periods
        utilization[res_id] = total_used / total_available if total_available >
↪ 0 else 0

    return {

```

```

        'schedule': schedule,
        'infeasible_orders': infeasible_orders,
        'utilization': utilization,
        'capacity_used': dict(capacity_used)
    }

def _schedule_order(self, order: ProductionOrder,
                    capacity_used: Dict) -> Optional[Dict]:
    """Schedule a single order using backward scheduling."""
    if order.product_id not in self.operations:
        return None

    ops = self.operations[order.product_id]
    op_schedule = {}

    # Backward schedule from due date
    current_period = order.due_period - 1 # 0-indexed

    for op in reversed(ops):
        hours_needed = op.hours_per_unit * order.quantity
        resource = self.resources.get(op.resource_id)

        if not resource:
            return None

        # Find period with capacity (searching backward)
        scheduled = False
        for period in range(current_period, -1, -1):
            available = resource.capacity_per_period -
↪ capacity_used[op.resource_id][period]

            if available >= hours_needed:
                # Can schedule entirely in this period
                capacity_used[op.resource_id][period] += hours_needed
                op_schedule[op.operation_id] = {
                    'period': period,
                    'hours': hours_needed,
                    'resource': op.resource_id
                }
                current_period = period - 1
                scheduled = True
                break
            elif available > 0:
                # Partial capacity available - try splitting
                # For simplicity, we skip this and try earlier period
                pass

        if not scheduled:
            # Rollback this order (simplified - in production would be more
            ↪ sophisticated)
            return None

    return {

```

```

        'order_id': order.order_id,
        'product': order.product_id,
        'quantity': order.quantity,
        'operations': op_schedule
    }

# Example
planner = FiniteCapacityPlanner(periods=10)

# Add resources
planner.add_resource(Resource("M1", "Machining Center", 40))
planner.add_resource(Resource("M2", "Assembly Line", 60))
planner.add_resource(Resource("M3", "Testing Station", 20))

# Add operations for Product A
planner.add_operation(Operation("A1", "PROD_A", "M1", 2.0, 1))
planner.add_operation(Operation("A2", "PROD_A", "M2", 1.5, 2))
planner.add_operation(Operation("A3", "PROD_A", "M3", 0.5, 3))

# Add operations for Product B
planner.add_operation(Operation("B1", "PROD_B", "M1", 3.0, 1))
planner.add_operation(Operation("B2", "PROD_B", "M2", 2.0, 2))

# Add production orders
planner.add_order(ProductionOrder("ORD001", "PROD_A", 10, 5, priority=2))
planner.add_order(ProductionOrder("ORD002", "PROD_A", 15, 7, priority=1))
planner.add_order(ProductionOrder("ORD003", "PROD_B", 8, 6, priority=3))
planner.add_order(ProductionOrder("ORD004", "PROD_A", 20, 9, priority=1))

# Schedule
result = planner.schedule()

print("Scheduled Orders:")
print("-" * 50)
for order_id, sched in result['schedule'].items():
    print(f"\n{order_id}: {sched['product']} x {sched['quantity']}")
    for op_id, op_sched in sched['operations'].items():
        print(f"    {op_id}: Period {op_sched['period']+1}, "
              f"    {op_sched['hours']:.1f} hrs on {op_sched['resource']}")

print(f"\nInfeasible Orders: {result['infeasible_orders']}")
print("\nResource Utilization:")
for res_id, util in result['utilization'].items():
    print(f"    {res_id}: {util:.1%}")

```

Scheduled Orders:

```

-----
ORD001: PROD_A x 10
A3: Period 5, 5.0 hrs on M3
A2: Period 4, 15.0 hrs on M2
A1: Period 3, 20.0 hrs on M1

```

19. Production Planning and ERP Integration

ORD003: PROD_B x 8

B2: Period 6, 16.0 hrs on M2

B1: Period 5, 24.0 hrs on M1

ORD002: PROD_A x 15

A3: Period 7, 7.5 hrs on M3

A2: Period 6, 22.5 hrs on M2

A1: Period 4, 30.0 hrs on M1

ORD004: PROD_A x 20

A3: Period 9, 10.0 hrs on M3

A2: Period 8, 30.0 hrs on M2

A1: Period 7, 40.0 hrs on M1

Infeasible Orders: []

Resource Utilization:

M1: 28.5%

M2: 13.9%

M3: 11.2%

19.7. ERP System Architecture

Enterprise Resource Planning systems integrate all business functions into a unified platform. Understanding ERP architecture helps optimize the interface between decision intelligence systems and enterprise operations.

19.7.1. Core ERP Modules

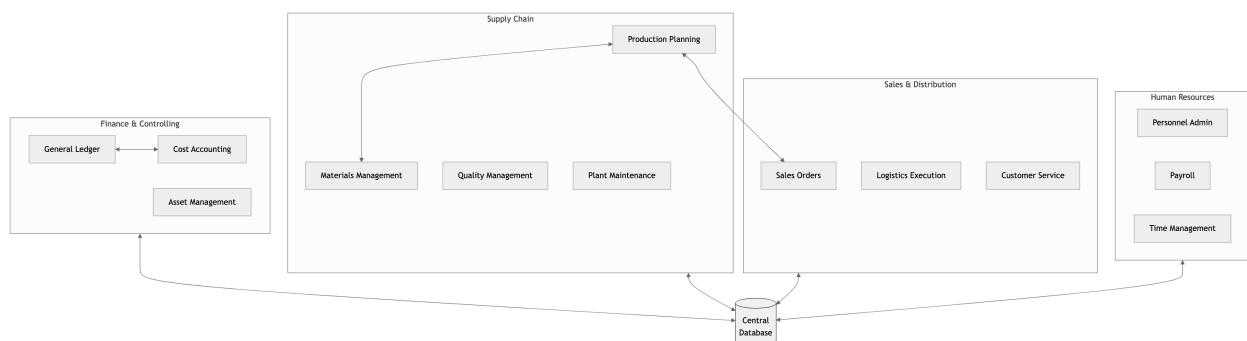


Figure 19.8.: Core ERP Module Integration

19.7.2. SAP S/4HANA Context

SAP S/4HANA is the leading ERP platform for large enterprises. Its Production Planning (PP) module provides:

Core PP Capabilities: - Sales and Operations Planning (S&OP) - Demand Management - Master Production Scheduling - Material Requirements Planning - Capacity Requirements Planning - Production Order Management - Shop Floor Control

Integration Points for Optimization:

Integration Point	Data Flow	Optimization Opportunity
Demand forecasts	SD → PP	Forecast optimization
Material master	MM PP	Lot sizing parameters
Work centers	PP	Capacity optimization
Routings	PP	Scheduling optimization
Production orders	PP → SFC	Sequencing optimization

💡 **Connecting Optimization to SAP**

Optimization models can augment SAP PP through:

1. **Custom RFC/BAPI** calls for data extraction and result upload
2. **SAP IBP (Integrated Business Planning)** for advanced S&OP
3. **SAP PP/DS (Detailed Scheduling)** for constraint-based planning
4. **External optimization engines** via middleware integration

19.7.3. Oracle SCM Cloud Context

Oracle Supply Chain Management Cloud provides integrated planning capabilities:

Planning Central: - Demand forecasting and sensing - Supply planning with optimization - Sales and Operations Planning - Inventory optimization

Key Features: - Machine learning-based demand sensing - Multi-echelon inventory optimization - Constraint-based planning engine - What-if scenario analysis

19.7.4. Integration Patterns

Connecting optimization models to ERP systems requires well-defined interfaces for data extraction and result upload. The following code defines an abstract `ERPConnector` class that standardizes access to material master data, BOM structures, and demand requirements. A mock implementation provides sample data for testing, while the `OptimizationOrchestrator` class coordinates the end-to-end workflow: extract data, run optimization, and upload planned orders back to the ERP.

```
from abc import ABC, abstractmethod
from dataclasses import dataclass
from typing import Dict, List, Any, Optional
from datetime import datetime
import json

@dataclass
class MaterialMaster:
    """Material master data from ERP."""
    material_number: str
    description: str
    uom: str
    lot_size_policy: str
    min_lot_size: float
    max_lot_size: float
    safety_stock: float
    lead_time: int
```

```

    mrp_type: str
    procurement_type: str # 'E' = in-house, 'F' = external

@dataclass
class BOMHeader:
    """Bill of Materials header."""
    bom_number: str
    material: str
    plant: str
    usage: str
    valid_from: datetime
    components: List[Dict]

@dataclass
class PlannedOrder:
    """Planned order for MRP."""
    order_number: str
    material: str
    quantity: float
    requirement_date: datetime
    start_date: datetime
    mrp_controller: str

class ERPConnector(ABC):
    """Abstract base class for ERP integration."""

    @abstractmethod
    def get_material_master(self, material: str) -> MaterialMaster:
        """Retrieve material master data."""
        pass

    @abstractmethod
    def get_bom(self, material: str, plant: str) -> BOMHeader:
        """Retrieve bill of materials."""
        pass

    @abstractmethod
    def get_demand(self, material: str, date_from: datetime,
                   date_to: datetime) -> List[Dict]:
        """Retrieve demand requirements."""
        pass

    @abstractmethod
    def upload_planned_orders(self, orders: List[PlannedOrder]) -> bool:
        """Upload planned orders to ERP."""
        pass

class MockERPConnector(ERPConnector):
    """Mock ERP connector for demonstration."""

    def __init__(self):
        self._materials = {}
        self._boms = {}

```



```

        self._demands = {}
        self._setup_sample_data()

    def _setup_sample_data(self):
        """Initialize sample data."""
        self._materials = {
            'FG-001': MaterialMaster(
                'FG-001', 'Finished Product', 'EA', 'WW',
                50, 500, 100, 1, 'PD', 'E'
            ),
            'SA-001': MaterialMaster(
                'SA-001', 'Subassembly', 'EA', 'L4L',
                0, 1000, 50, 2, 'PD', 'E'
            ),
            'RM-001': MaterialMaster(
                'RM-001', 'Raw Material', 'KG', 'EQ',
                100, 5000, 200, 3, 'VB', 'F'
            )
        }

    def get_material_master(self, material: str) -> Optional[MaterialMaster]:
        return self._materials.get(material)

    def get_bom(self, material: str, plant: str) -> Optional[BOMHeader]:
        return self._boms.get((material, plant))

    def get_demand(self, material: str, date_from: datetime,
                   date_to: datetime) -> List[Dict]:
        return self._demands.get(material, [])

    def upload_planned_orders(self, orders: List[PlannedOrder]) -> bool:
        print(f"Uploading {len(orders)} planned orders...")
        for order in orders:
            print(f"    {order.order_number}: {order.material} x {order.quantity}")
        return True

class OptimizationOrchestrator:
    """Orchestrates optimization with ERP integration."""

    def __init__(self, erp_connector: ERPConnector):
        self.erp = erp_connector

    def run_mrp_optimization(self, materials: List[str],
                           planning_horizon: int) -> Dict:
        """Run optimized MRP with ERP data.

        Args:
            materials: List of materials to plan
            planning_horizon: Number of periods

        Returns:
            Optimization results
        """

```

```

# Extract data from ERP
print("Extracting data from ERP...")
material_data = {}
for mat in materials:
    data = self.erp.get_material_master(mat)
    if data:
        material_data[mat] = data
        print(f" Retrieved: {mat} - {data.description}")

# Run optimization (simplified)
print("\nRunning optimization...")
planned_orders = self._optimize(material_data, planning_horizon)

# Upload results
print("\nUploading results to ERP...")
success = self.erp.upload_planned_orders(planned_orders)

return {
    'status': 'success' if success else 'failed',
    'orders_created': len(planned_orders),
    'materials_planned': len(material_data)
}

def _optimize(self, materials: Dict[str, MaterialMaster],
              horizon: int) -> List[PlannedOrder]:
    """Internal optimization logic."""
    orders = []
    order_num = 1000

    for mat_id, mat in materials.items():
        # Simplified: create one order per material
        orders.append(PlannedOrder(
            order_number=f"PO{order_num}",
            material=mat_id,
            quantity=mat.min_lot_size * 2,
            requirement_date=datetime.now(),
            start_date=datetime.now(),
            mrp_controller="MRP01"
        ))
        order_num += 1

    return orders

# Demonstration
erp = MockERPConnector()
orchestrator = OptimizationOrchestrator(erp)

result = orchestrator.run_mrp_optimization(
    materials=['FG-001', 'SA-001', 'RM-001'],
    planning_horizon=12
)

print(f"\nResult: {result}")

```

Extracting data from ERP...

Retrieved: FG-001 - Finished Product

Retrieved: SA-001 - Subassembly

Retrieved: RM-001 - Raw Material

Running optimization...

Uploading results to ERP...

Uploading 3 planned orders...

P01000: FG-001 x 100

P01001: SA-001 x 0

P01002: RM-001 x 200

Result: {'status': 'success', 'orders_created': 3, 'materials_planned': 3}

19.8. Implementation Considerations

19.8.1. Data Requirements for Production Planning

Data Category	Examples	Quality Issues	Impact
Item Master	Lead times, lot sizes	Inaccurate lead times	Wrong release dates
BOM	Quantities, phantoms	Outdated structures	Wrong requirements
Routing	Setup times, run times	Estimated vs. actual	Capacity errors
Inventory Demand	On-hand, allocations Forecasts, orders	Counting errors Forecast bias	Stockouts/excess Over/under production

Common Implementation Challenges

MRP Nervousness: Small demand changes cause large swings in planned orders - *Mitigation:* Time fences, dampening rules, lot sizing

Lead Time Accuracy: Planned vs. actual lead times differ significantly - *Mitigation:* Dynamic lead times, tracking actuals

Capacity Constraints: MRP assumes infinite capacity - *Mitigation:* Rough-cut capacity planning, finite scheduling

Data Quality: Garbage in, garbage out - *Mitigation:* Data governance, regular audits

User Adoption: Planners override system recommendations - *Mitigation:* Training, explainable AI, exception management

19.8.2. Master Data Quality

Poor master data is the most common cause of MRP failures. Lead times set to zero, BOMs with missing components, or lot size parameters where the minimum exceeds the maximum will all produce invalid plans. The `ProductionDataAuditor` class below runs a series of automated checks against item master records, BOM structures, lot sizing parameters, and forecast accuracy metrics. Each check returns a severity level (info, warning, or failure) and a description of the issue.

```

from dataclasses import dataclass
from typing import List, Dict, Tuple
import statistics

@dataclass
class DataQualityCheck:
    """Data quality check result."""
    check_name: str
    passed: bool
    details: str
    severity: str # 'critical', 'warning', 'info'

class ProductionDataAuditor:
    """Audit production planning master data quality."""

    def __init__(self):
        self.checks: List[DataQualityCheck] = []

    def audit_lead_times(self, items: List[Dict]) -> DataQualityCheck:
        """Check lead time reasonableness."""
        issues = []
        for item in items:
            lt = item.get('lead_time', 0)
            if lt <= 0:
                issues.append(f"{item['id']}: Zero/negative lead time")
            elif lt > 365:
                issues.append(f"{item['id']}: Lead time > 1 year ({lt} days)")

        passed = len(issues) == 0
        return DataQualityCheck(
            "Lead Time Validation",
            passed,
            f"Found {len(issues)} issues" if not passed else "All lead times valid",
            "critical" if not passed else "info"
        )

    def audit_bom_completeness(self, boms: List[Dict]) -> DataQualityCheck:
        """Check BOM structure completeness."""
        issues = []
        for bom in boms:
            if not bom.get('components'):
                issues.append(f"{bom['parent']}: No components defined")
            for comp in bom.get('components', []):
                if comp.get('quantity', 0) <= 0:
                    issues.append(f"{bom['parent']}->{comp['child']}: Invalid
↪ quantity")

        passed = len(issues) == 0
        return DataQualityCheck(
            "BOM Completeness",
            passed,
            f"Found {len(issues)} issues" if not passed else "All BOMs complete",
            "critical" if not passed else "info"

```

```

    )

def audit_lot_sizing(self, items: List[Dict]) -> DataQualityCheck:
    """Check lot sizing parameters."""
    issues = []
    for item in items:
        min_lot = item.get('min_lot_size', 0)
        max_lot = item.get('max_lot_size', float('inf'))

        if min_lot > max_lot:
            issues.append(f"{item['id']}: Min lot > Max lot")
        if min_lot < 0:
            issues.append(f"{item['id']}: Negative min lot size")

    passed = len(issues) == 0
    return DataQualityCheck(
        "Lot Sizing Parameters",
        passed,
        f"Found {len(issues)} issues" if not passed else "All lot sizes valid",
        "warning" if not passed else "info"
    )

def audit_forecast_accuracy(self, forecasts: List[float],
                           actuals: List[float]) -> DataQualityCheck:
    """Check forecast accuracy metrics."""
    if len(forecasts) != len(actuals) or len(forecasts) == 0:
        return DataQualityCheck(
            "Forecast Accuracy",
            False,
            "Insufficient data for analysis",
            "warning"
        )

    # Calculate MAPE
    errors = []
    for f, a in zip(forecasts, actuals):
        if a > 0:
            errors.append(abs(f - a) / a)

    mape = statistics.mean(errors) * 100 if errors else 0

    passed = mape < 30 # 30% MAPE threshold
    return DataQualityCheck(
        "Forecast Accuracy",
        passed,
        f"MAPE: {mape:.1f}%",
        "warning" if not passed else "info"
    )

def run_full_audit(self, items: List[Dict], boms: List[Dict],
                  forecasts: List[float], actuals: List[float]) -> Dict:
    """Run complete data quality audit."""
    self.checks = [

```

```

        self.audit_lead_times(items),
        self.audit_bom_completeness(boms),
        self.audit_lot_sizing(items),
        self.audit_forecast_accuracy(forecasts, actuals)
    ]

    critical = sum(1 for c in self.checks if c.severity == 'critical' and not
↪ c.passed)
    warnings = sum(1 for c in self.checks if c.severity == 'warning' and not
↪ c.passed)

    return {
        'overall_status': 'FAIL' if critical > 0 else ('WARN' if warnings > 0
↪ else 'PASS'),
        'critical_issues': critical,
        'warnings': warnings,
        'checks': self.checks
    }

# Example audit
items = [
    {'id': 'A', 'lead_time': 5, 'min_lot_size': 10, 'max_lot_size': 100},
    {'id': 'B', 'lead_time': 0, 'min_lot_size': 5, 'max_lot_size': 50}, # Issue!
    {'id': 'C', 'lead_time': 10, 'min_lot_size': 100, 'max_lot_size': 50}, # Issue!
]

boms = [
    {'parent': 'A', 'components': [{'child': 'B', 'quantity': 2}]},
    {'parent': 'C', 'components': []}, # Issue!
]

forecasts = [100, 120, 90, 110, 130]
actuals = [95, 130, 85, 100, 140]

auditor = ProductionDataAuditor()
result = auditor.run_full_audit(items, boms, forecasts, actuals)

print(f"Overall Status: {result['overall_status']}")
print(f"Critical Issues: {result['critical_issues']}")
print(f"Warnings: {result['warnings']}")
print("\nDetailed Checks:")
for check in result['checks']:
    status = 'PASS' if check.passed else 'FAIL'
    print(f"    [{status}] {check.check_name}: {check.details}")

```

Overall Status: FAIL

Critical Issues: 2

Warnings: 1

Detailed Checks:

[FAIL] Lead Time Validation: Found 1 issues

[FAIL] BOM Completeness: Found 1 issues

[FAIL] Lot Sizing Parameters: Found 1 issues

[PASS] Forecast Accuracy: MAPE: 7.2%

19.9. Case Study: Manufacturing Optimization Pipeline

This case study demonstrates a complete optimization pipeline from MPS through MRP to shop floor scheduling. The `ManufacturingOptimizer` class ties together all the planning levels covered in this chapter. It takes product definitions (with BOM and routing data), component parameters, resource capacities, and demand forecasts as inputs. The four-step pipeline generates an MPS, runs MRP explosion, schedules operations on finite-capacity resources, and computes key performance indicators including service level, average inventory, and resource utilization.

```
from dataclasses import dataclass, field
from typing import Dict, List, Tuple, Optional
from collections import defaultdict
import numpy as np

@dataclass
class ManufacturingProduct:
    """Product definition for manufacturing."""
    product_id: str
    name: str
    components: Dict[str, float] # component_id -> quantity per
    operations: List[Tuple[str, float]] # (resource_id, hours_per_unit)
    lead_time: int

@dataclass
class ManufacturingPlan:
    """Complete manufacturing plan."""
    mps: Dict[str, List[float]] # product -> period quantities
    mrp: Dict[str, Dict] # item -> MRP record
    schedule: Dict[str, List[Dict]] # resource -> scheduled operations
    kpis: Dict[str, float]

class ManufacturingOptimizer:
    """End-to-end manufacturing optimization."""

    def __init__(self, periods: int):
        self.periods = periods
        self.products: Dict[str, ManufacturingProduct] = {}
        self.components: Dict[str, Dict] = {} # component info
        self.resources: Dict[str, float] = {} # resource -> capacity per period
        self.demand: Dict[str, List[float]] = {}

    def add_product(self, product: ManufacturingProduct):
        """Add a product to the system."""
        self.products[product.product_id] = product

    def add_component(self, component_id: str, lead_time: int,
                      on_hand: float, safety_stock: float):
        """Add a component."""
        self.components[component_id] = {
            'lead_time': lead_time,
            'on_hand': on_hand,
```

```

        'safety_stock': safety_stock
    }

    def add_resource(self, resource_id: str, capacity: float):
        """Add a resource with capacity per period."""
        self.resources[resource_id] = capacity

    def set_demand(self, product_id: str, demand: List[float]):
        """Set demand forecast for a product."""
        self.demand[product_id] = demand

    def optimize(self) -> ManufacturingPlan:
        """Run complete optimization pipeline."""
        # Step 1: Generate MPS
        print("Step 1: Generating Master Production Schedule...")
        mps = self._generate_mps()

        # Step 2: Run MRP explosion
        print("Step 2: Running MRP explosion...")
        mrp_results = self._run_mrp(mps)

        # Step 3: Schedule operations
        print("Step 3: Scheduling operations...")
        schedule = self._schedule_operations(mps)

        # Step 4: Calculate KPIs
        print("Step 4: Calculating KPIs...")
        kpis = self._calculate_kpis(mps, mrp_results, schedule)

        return ManufacturingPlan(
            mps=mps,
            mrp=mrp_results,
            schedule=schedule,
            kpis=kpis
        )

    def _generate_mps(self) -> Dict[str, List[float]]:
        """Generate MPS from demand using simple lot sizing."""
        mps = {}

        for prod_id, demand in self.demand.items():
            mps[prod_id] = []
            inventory = 0

            for t in range(self.periods):
                if demand[t] > inventory:
                    # Need to produce
                    lot_size = max(50, demand[t] - inventory) # Min lot of 50
                    mps[prod_id].append(lot_size)
                    inventory += lot_size
                else:
                    mps[prod_id].append(0)

```



```

        inventory -= demand[t]

    return mps

def _run_mrp(self, mps: Dict[str, List[float]]) -> Dict[str, Dict]:
    """Run MRP explosion for all components."""
    results = {}
    component_requirements = defaultdict(lambda: [0.0] * self.periods)

    # Calculate gross requirements for each component
    for prod_id, schedule in mps.items():
        product = self.products.get(prod_id)
        if not product:
            continue

        for comp_id, qty_per in product.components.items():
            for t in range(self.periods):
                # Offset by product lead time
                req_period = t - product.lead_time
                if req_period >= 0:
                    component_requirements[comp_id][req_period] += schedule[t]
    ↪ * qty_per

    # Run MRP for each component
    for comp_id, gross_req in component_requirements.items():
        comp_info = self.components.get(comp_id, {
            'lead_time': 1, 'on_hand': 0, 'safety_stock': 0
        })

        results[comp_id] = self._mrp_record(
            comp_id, gross_req, comp_info
        )

    return results

def _mrp_record(self, comp_id: str, gross_req: List[float],
                comp_info: Dict) -> Dict:
    """Calculate MRP record for a component."""
    projected = []
    planned_orders = []
    current_inv = comp_info['on_hand']

    for t in range(self.periods):
        available = current_inv - gross_req[t]

        if available < comp_info['safety_stock']:
            order_qty = comp_info['safety_stock'] - available + gross_req[t]
            order_qty = max(50, order_qty) # Minimum order
            planned_orders.append(order_qty)
            available += order_qty
        else:
            planned_orders.append(0)

```

```

        projected.append(available)
        current_inv = available

    return {
        'gross_requirements': gross_req,
        'projected_inventory': projected,
        'planned_orders': planned_orders
    }

def _schedule_operations(self, mps: Dict[str, List[float]]) -> Dict[str,
↪ List[Dict]]:
    """Schedule operations on resources."""
    schedule = defaultdict(list)
    resource_load = defaultdict(lambda: [0.0] * self.periods)

    for prod_id, production in mps.items():
        product = self.products.get(prod_id)
        if not product:
            continue

        for t in range(self.periods):
            if production[t] <= 0:
                continue

            # Schedule each operation
            for res_id, hours_per in product.operations:
                hours_needed = production[t] * hours_per

                # Check capacity
                available = self.resources.get(res_id, float('inf')) -
↪ resource_load[res_id][t]

                if hours_needed <= available:
                    schedule[res_id].append({
                        'period': t,
                        'product': prod_id,
                        'quantity': production[t],
                        'hours': hours_needed
                    })
                    resource_load[res_id][t] += hours_needed

    return dict(schedule)

def _calculate_kpis(self, mps: Dict, mrp: Dict,
                    schedule: Dict) -> Dict[str, float]:
    """Calculate key performance indicators."""
    # Service level
    total_demand = sum(sum(d) for d in self.demand.values())
    total_planned = sum(sum(s) for s in mps.values())
    service_level = min(1.0, total_planned / total_demand) if total_demand > 0
↪ else 1.0

    # Average inventory

```

```

        avg_inventory = 0
        for comp_data in mrp.values():
            avg_inventory += sum(comp_data['projected_inventory']) / self.periods

        # Resource utilization
        total_scheduled = sum(
            sum(op['hours'] for op in ops)
            for ops in schedule.values()
        )
        total_capacity = sum(cap * self.periods for cap in self.resources.values())
        utilization = total_scheduled / total_capacity if total_capacity > 0 else 0

        return {
            'service_level': service_level,
            'average_inventory': avg_inventory,
            'resource_utilization': utilization,
            'total_production': total_planned
        }

# Create manufacturing system
optimizer = ManufacturingOptimizer(periods=12)

# Add products
optimizer.add_product(ManufacturingProduct(
    product_id="FG-100",
    name="Finished Good A",
    components={"COMP-A": 2, "COMP-B": 1},
    operations=[("MACHINE-1", 0.5), ("ASSEMBLY", 0.25)],
    lead_time=1
))

optimizer.add_product(ManufacturingProduct(
    product_id="FG-200",
    name="Finished Good B",
    components={"COMP-A": 1, "COMP-C": 3},
    operations=[("MACHINE-1", 0.75), ("MACHINE-2", 0.5), ("ASSEMBLY", 0.3)],
    lead_time=2
))

# Add components
optimizer.add_component("COMP-A", lead_time=2, on_hand=100, safety_stock=50)
optimizer.add_component("COMP-B", lead_time=3, on_hand=50, safety_stock=25)
optimizer.add_component("COMP-C", lead_time=1, on_hand=200, safety_stock=100)

# Add resources
optimizer.add_resource("MACHINE-1", 160) # 160 hours per period
optimizer.add_resource("MACHINE-2", 120)
optimizer.add_resource("ASSEMBLY", 200)

# Set demand
optimizer.set_demand("FG-100", [80, 100, 120, 90, 110, 130, 100, 90, 80, 100, 120,
↵ 110])
optimizer.set_demand("FG-200", [50, 60, 70, 80, 60, 50, 70, 80, 90, 70, 60, 50])

```

```

# Run optimization
plan = optimizer.optimize()

# Report results
print("\n" + "="*60)
print("MANUFACTURING OPTIMIZATION RESULTS")
print("="*60)

print("\nMaster Production Schedule:")
for prod_id, schedule in plan.mps.items():
    print(f"\n{prod_id}:")
    print(f"  Periods 1-6: {schedule[:6]}")
    print(f"  Periods 7-12: {schedule[6:]}")

print("\nComponent Planned Orders (first 6 periods):")
for comp_id, record in plan.mrp.items():
    orders = [int(o) for o in record['planned_orders'][:6]]
    print(f"  {comp_id}: {orders}")

print("\nKey Performance Indicators:")
for kpi, value in plan.kpis.items():
    if 'level' in kpi or 'utilization' in kpi:
        print(f"  {kpi}: {value:.1%}")
    else:
        print(f"  {kpi}: {value:.1f}")

```

Step 1: Generating Master Production Schedule...

Step 2: Running MRP explosion...

Step 3: Scheduling operations...

Step 4: Calculating KPIs...

```

=====
MANUFACTURING OPTIMIZATION RESULTS
=====

```

Master Production Schedule:

FG-100:

Periods 1-6: [80, 100, 120, 90, 110, 130]

Periods 7-12: [100, 90, 80, 100, 120, 110]

FG-200:

Periods 1-6: [50, 60, 70, 80, 60, 50]

Periods 7-12: [70, 80, 90, 70, 60, 50]

Component Planned Orders (first 6 periods):

COMP-A: [490, 370, 0, 460, 390, 0]

COMP-B: [175, 140, 0, 190, 150, 0]

COMP-C: [320, 270, 0, 240, 270, 270]

Key Performance Indicators:

```
service_level: 100.0%
average_inventory: 582.5
resource_utilization: 37.3%
total_production: 2020.0
```

19.10. Summary

19.10.1. Key Takeaways

- **Production Planning Hierarchy** spans strategic (years) to operational (real-time) decisions, with each level constraining the next
- **Aggregate Planning** uses LP to balance workforce, inventory, and production costs over a medium-term horizon
- **Master Production Scheduling** translates aggregate plans into end-item quantities, managing time fences and ATP
- **Material Requirements Planning** explodes BOMs to calculate component requirements, with lot sizing significantly impacting costs
- **Wagner-Whitin Algorithm** provides optimal lot sizing for known demand sequences
- **Advanced Planning Systems** extend MRP with finite capacity, optimization, and real-time capabilities
- **ERP Integration** requires understanding of data flows, master data quality, and interface patterns
- **Implementation Success** depends on data quality, user training, and continuous improvement

19.11. Exercises

19.11.1. Conceptual

1. **Planning Hierarchy Analysis:** A company has 6-month lead times for raw materials and serves customers with 2-week lead times. Discuss how this affects their planning hierarchy and where safety stock should be positioned.
2. **MRP Nervousness:** Explain why a 5% change in forecast demand can cause 50% changes in component planned orders. What strategies can mitigate this effect?
3. **Lot Sizing Trade-offs:** Compare the total costs of Lot-for-Lot, EOQ, and Wagner-Whitin policies for the demand sequence [10, 50, 20, 40, 30, 60] with setup cost \$100 and holding cost \$2/unit/period.

19.11.2. Programming

4. **Aggregate Planning Extension:** Modify the aggregate planning model to include:
 - Subcontracting option with cost \$35/unit
 - Maximum inventory capacity of 500 units
 - Minimum workforce of 5 workers
5. **Multi-Level MRP:** Extend the MRP implementation to handle:
 - Phantom assemblies (pass-through items)
 - Scrap rates at each BOM level

19. Production Planning and ERP Integration

- Multiple plants with transfer times

6. **Finite Capacity MRP:** Implement an MRP system that:

- Checks capacity feasibility before confirming planned orders
- Suggests alternative dates when capacity is unavailable
- Reports capacity utilization by work center

19.11.3. Case Study

7. **Complete Planning Pipeline:** Using the case study framework, model a manufacturing company with:

- 5 end products
- 15 components (some shared)
- 4 work centers
- 52-week planning horizon

Analyze the impact of:

- Lot sizing policy choices
- Safety stock levels
- Forecast accuracy

19.11.4. Research

8. **ERP Vendor Comparison:** Research and compare the production planning capabilities of SAP S/4HANA, Oracle SCM Cloud, and Microsoft Dynamics 365 Supply Chain Management. What optimization techniques does each employ?
9. **Industry 4.0 Integration:** Discuss how real-time shop floor data (IoT sensors, MES systems) can improve the accuracy of production planning. What are the integration challenges?
10. **Machine Learning in MRP:** Propose how machine learning could improve traditional MRP:
 - Demand forecasting integration
 - Dynamic lead time prediction
 - Lot sizing optimization
 - Exception handling prioritization

20. Portfolio Optimization

What You Will Learn

- Formulate mean-variance portfolio optimization problems
- Implement Markowitz portfolio theory and efficient frontier
- Handle realistic constraints (cardinality, turnover, sector limits)
- Apply risk measures beyond variance (CVaR, drawdown)
- Use robust optimization for parameter uncertainty

Portfolio optimization is one of the most influential applications of mathematical optimization in finance. From Markowitz's seminal work in 1952 to modern risk management, optimization shapes how trillions of dollars are allocated globally.

20.1. Mean-Variance Optimization

20.1.1. The Markowitz Framework

Given n assets with:

- Expected returns: $\mu \in \mathbb{R}^n$
- Covariance matrix: $\Sigma \in \mathbb{R}^{n \times n}$
- Portfolio weights: $\mathbf{w} \in \mathbb{R}^n$

Portfolio Return: $r_p = \mu^T \mathbf{w}$

Portfolio Variance: $\sigma_p^2 = \mathbf{w}^T \Sigma \mathbf{w}$

20.1.2. Formulations

Minimum Variance (given target return r^*):

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^T \Sigma \mathbf{w} \\ \text{s.t.} \quad & \mu^T \mathbf{w} \geq r^* \\ & \mathbf{1}^T \mathbf{w} = 1 \end{aligned}$$

Maximum Return (given risk budget σ^*):

$$\begin{aligned} \max_{\mathbf{w}} \quad & \mu^T \mathbf{w} \\ \text{s.t.} \quad & \mathbf{w}^T \Sigma \mathbf{w} \leq (\sigma^*)^2 \\ & \mathbf{1}^T \mathbf{w} = 1 \end{aligned}$$

Risk-Return Trade-off:

$$\begin{aligned} \max_{\mathbf{w}} \quad & \mu^T \mathbf{w} - \frac{\lambda}{2} \mathbf{w}^T \Sigma \mathbf{w} \\ \text{s.t.} \quad & \mathbf{1}^T \mathbf{w} = 1 \end{aligned}$$

20. Portfolio Optimization

where λ is the risk aversion parameter.

The `MarkowitzOptimizer` class below implements all three formulations. It uses SciPy's sequential least squares programming (SLSQP) solver with the full-investment constraint ($\mathbf{1}^T \mathbf{w} = 1$) and optional long-only bounds. The `efficient_frontier` method traces the set of minimum-variance portfolios across a range of target returns.

```
import numpy as np
from scipy.optimize import minimize
from dataclasses import dataclass
from typing import List, Tuple, Optional

@dataclass
class PortfolioResult:
    """Result of portfolio optimization."""
    weights: np.ndarray
    expected_return: float
    volatility: float
    sharpe_ratio: float

class MarkowitzOptimizer:
    """Mean-variance portfolio optimization."""

    def __init__(self, returns: np.ndarray, cov_matrix: np.ndarray,
                 asset_names: List[str] = None, risk_free_rate: float = 0.02):
        """
        Args:
            returns: Expected returns for each asset
            cov_matrix: Covariance matrix of returns
            asset_names: Names of assets
            risk_free_rate: Annual risk-free rate
        """
        self.mu = np.array(returns)
        self.Sigma = np.array(cov_matrix)
        self.n = len(returns)
        self.names = asset_names or [f"Asset_{i}" for i in range(self.n)]
        self.rf = risk_free_rate

    def portfolio_return(self, weights: np.ndarray) -> float:
        """Calculate portfolio expected return."""
        return np.dot(self.mu, weights)

    def portfolio_volatility(self, weights: np.ndarray) -> float:
        """Calculate portfolio volatility (std dev)."""
        return np.sqrt(np.dot(weights, np.dot(self.Sigma, weights)))

    def sharpe_ratio(self, weights: np.ndarray) -> float:
        """Calculate Sharpe ratio."""
        ret = self.portfolio_return(weights)
        vol = self.portfolio_volatility(weights)
        return (ret - self.rf) / vol if vol > 0 else 0

    def minimize_variance(self, target_return: float = None,
                        long_only: bool = True) -> PortfolioResult:
```



```

    """Find minimum variance portfolio."""
    def objective(w):
        return self.portfolio_volatility(w) ** 2

    constraints = [
        {'type': 'eq', 'fun': lambda w: np.sum(w) - 1}
    ]

    if target_return is not None:
        constraints.append({
            'type': 'eq',
            'fun': lambda w: self.portfolio_return(w) - target_return
        })

    bounds = [(0, 1) if long_only else (None, None)] * self.n
    w0 = np.ones(self.n) / self.n

    result = minimize(objective, w0, method='SLSQP',
                      bounds=bounds, constraints=constraints)

    return PortfolioResult(
        weights=result.x,
        expected_return=self.portfolio_return(result.x),
        volatility=self.portfolio_volatility(result.x),
        sharpe_ratio=self.sharpe_ratio(result.x)
    )

def maximize_sharpe(self, long_only: bool = True) -> PortfolioResult:
    """Find maximum Sharpe ratio portfolio."""
    def neg_sharpe(w):
        return -self.sharpe_ratio(w)

    constraints = [{'type': 'eq', 'fun': lambda w: np.sum(w) - 1}]
    bounds = [(0, 1) if long_only else (None, None)] * self.n
    w0 = np.ones(self.n) / self.n

    result = minimize(neg_sharpe, w0, method='SLSQP',
                      bounds=bounds, constraints=constraints)

    return PortfolioResult(
        weights=result.x,
        expected_return=self.portfolio_return(result.x),
        volatility=self.portfolio_volatility(result.x),
        sharpe_ratio=self.sharpe_ratio(result.x)
    )

def efficient_frontier(self, num_points: int = 50,
                       long_only: bool = True) -> List[PortfolioResult]:
    """Compute the efficient frontier."""
    # Find return range
    min_var = self.minimize_variance(long_only=long_only)
    max_sharpe = self.maximize_sharpe(long_only=long_only)

```

```

min_ret = min_var.expected_return
max_ret = max(self.mu) if long_only else max_sharpe.expected_return * 1.5

target_returns = np.linspace(min_ret, max_ret, num_points)
frontier = []

for target in target_returns:
    try:
        result = self.minimize_variance(target_return=target,
                                         long_only=long_only)

        if result.volatility > 0:
            frontier.append(result)
    except:
        continue

return frontier

# Example
np.random.seed(42)
n_assets = 5
mu = np.array([0.08, 0.10, 0.12, 0.07, 0.09])
Sigma = np.array([
    [0.04, 0.01, 0.02, 0.01, 0.01],
    [0.01, 0.09, 0.03, 0.02, 0.02],
    [0.02, 0.03, 0.16, 0.02, 0.03],
    [0.01, 0.02, 0.02, 0.03, 0.01],
    [0.01, 0.02, 0.03, 0.01, 0.05]
])

opt = MarkowitzOptimizer(mu, Sigma, risk_free_rate=0.03)

# Minimum variance portfolio
min_var = opt.minimize_variance()
print("Minimum Variance Portfolio:")
print(f"  Return: {min_var.expected_return:.2%}")
print(f"  Volatility: {min_var.volatility:.2%}")
print(f"  Sharpe: {min_var.sharpe_ratio:.2f}")

# Maximum Sharpe portfolio
max_sharpe = opt.maximize_sharpe()
print("\nMaximum Sharpe Portfolio:")
print(f"  Return: {max_sharpe.expected_return:.2%}")
print(f"  Volatility: {max_sharpe.volatility:.2%}")
print(f"  Sharpe: {max_sharpe.sharpe_ratio:.2f}")
print(f"  Weights: {max_sharpe.weights.round(3)}")

```

Minimum Variance Portfolio:

Return: 7.84%

Volatility: 13.84%

Sharpe: 0.35

Maximum Sharpe Portfolio:

Return: 8.72%

Volatility: 15.16%
Sharpe: 0.38
Weights: [0.308 0.148 0.085 0.187 0.273]

20.1.3. Visualizing the Efficient Frontier

The efficient frontier represents the set of optimal portfolios offering the highest expected return for a given level of risk.

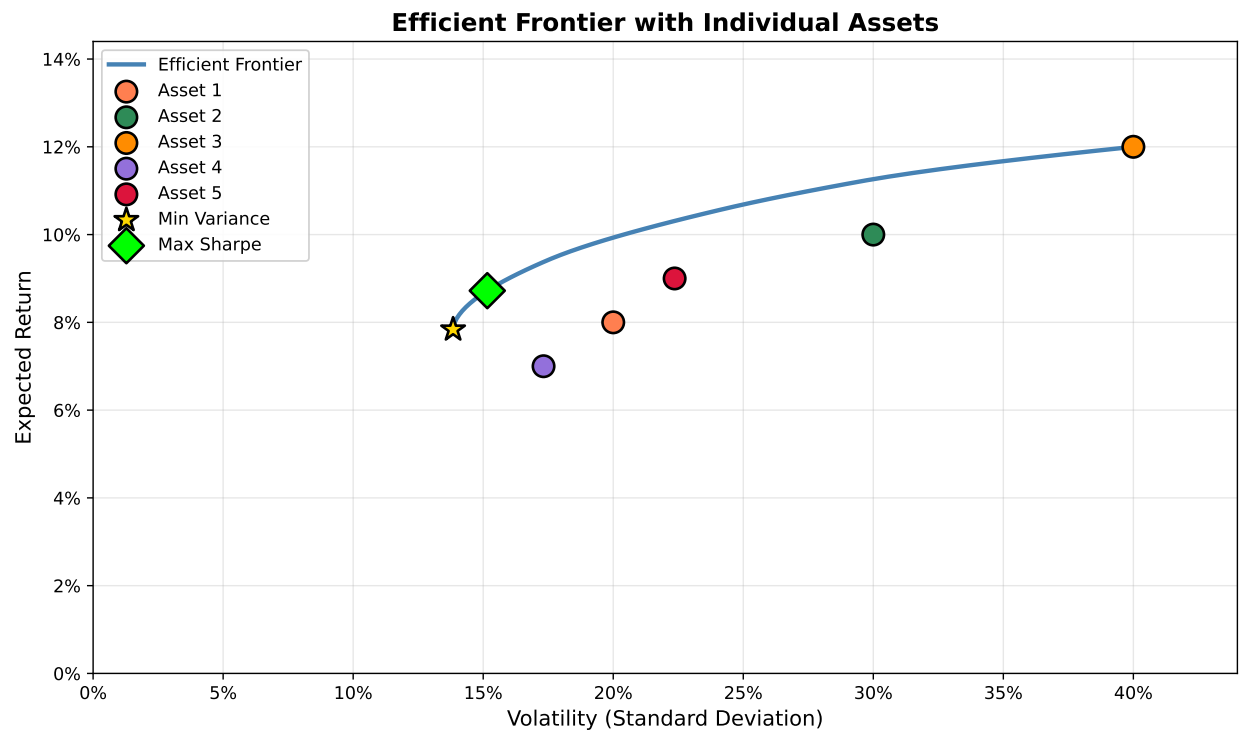


Figure 20.1.: Efficient frontier showing the optimal risk-return trade-off. Individual assets are marked with their expected returns and volatilities, while the frontier curve represents achievable optimal portfolios.

To see why the efficient frontier matters, compare it against randomly generated portfolios. The scatter plot below samples 500 random weight vectors from a Dirichlet distribution and plots their risk-return coordinates. The color gradient shows each portfolio's Sharpe ratio. Notice that all random portfolios fall at or below the efficient frontier, confirming that no portfolio can achieve a higher return for a given volatility level.

20.2. Realistic Constraints

Real portfolios face numerous constraints beyond the basic formulation.

20.2.1. Constraint Types

Constraint	Description	Formulation
Long-only	No short selling	$w_i \geq 0$
Box	Min/max per asset	$l_i \leq w_i \leq u_i$

Constraint	Description	Formulation
Cardinality	Limit number of assets	$\sum_i z_i \leq K$
Sector	Sector exposure limits	$\sum_{i \in S} w_i \leq u_S$
Turnover	Limit trading	$\sum_i w_i - w_i^0 \leq T$

The `ConstrainedPortfolioOptimizer` class handles these constraints within a single optimization call. Box constraints map directly to variable bounds. Sector constraints are implemented as linear inequalities that sum weights within each sector. The turnover constraint uses a penalty method to approximate the non-smooth absolute value function.

```
class ConstrainedPortfolioOptimizer:
    """Portfolio optimization with realistic constraints."""

    def __init__(self, returns: np.ndarray, cov_matrix: np.ndarray,
                 sectors: List[int] = None, risk_free_rate: float = 0.02):
        self.mu = np.array(returns)
        self.Sigma = np.array(cov_matrix)
        self.n = len(returns)
        self.rf = risk_free_rate
        self.sectors = sectors or list(range(self.n)) # Default: each in own sector

    def optimize(self,
                min_weights: np.ndarray = None,
                max_weights: np.ndarray = None,
                sector_limits: dict = None,
                max_turnover: float = None,
                current_weights: np.ndarray = None,
                target_return: float = None,
                risk_aversion: float = 1.0) -> PortfolioResult:
        """Optimize with multiple constraints."""

        # Default bounds
        if min_weights is None:
            min_weights = np.zeros(self.n)
        if max_weights is None:
            max_weights = np.ones(self.n)

        def objective(w):
            ret = np.dot(self.mu, w)
            var = np.dot(w, np.dot(self.Sigma, w))
            return -ret + 0.5 * risk_aversion * var

        constraints = [
            {'type': 'eq', 'fun': lambda w: np.sum(w) - 1}
        ]

        # Target return constraint
        if target_return is not None:
            constraints.append({
                'type': 'ineq',
                'fun': lambda w: np.dot(self.mu, w) - target_return
            })
```

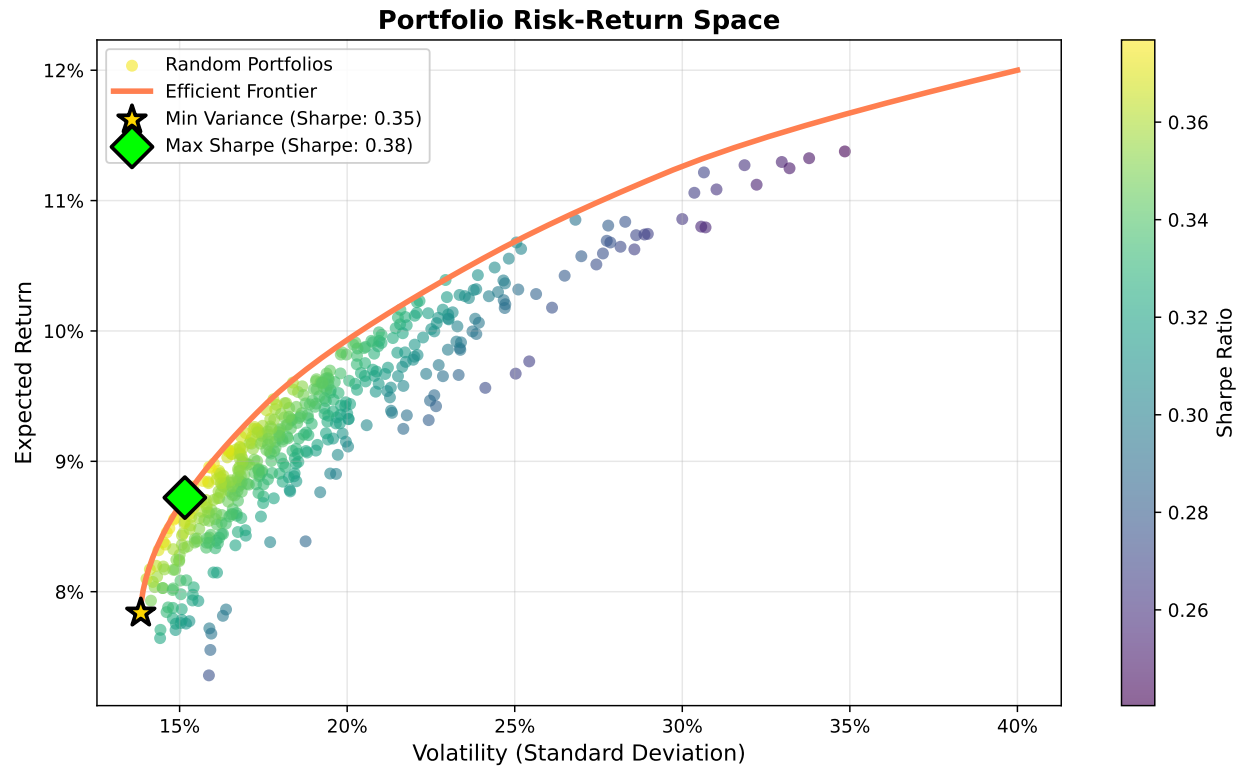


Figure 20.2.: Risk-return scatter plot showing random portfolios compared to efficient frontier portfolios. The color gradient indicates Sharpe ratio, with brighter colors representing better risk-adjusted returns.

```
# Sector constraints
if sector_limits:
    for sector, limit in sector_limits.items():
        sector_mask = np.array([1 if s == sector else 0
                                for s in self.sectors])

        constraints.append({
            'type': 'ineq',
            'fun': lambda w, m=sector_mask, l=limit: l - np.dot(m, w)
        })

# Turnover constraint
if max_turnover is not None and current_weights is not None:
    # Linearize: use auxiliary variables for |w - w0|
    # For simplicity, use penalty method
    def turnover_penalty(w):
        turnover = np.sum(np.abs(w - current_weights))
        excess = max(0, turnover - max_turnover)
        return 100 * excess ** 2

    original_obj = objective
    def objective(w):
```

```

        return original_obj(w) + turnover_penalty(w)

    # Bounds
    bounds = list(zip(min_weights, max_weights))

    # Optimize
    w0 = current_weights if current_weights is not None else np.ones(self.n) /
    ↪ self.n

    result = minimize(objective, w0, method='SLSQP',
                      bounds=bounds, constraints=constraints)

    weights = result.x
    return PortfolioResult(
        weights=weights,
        expected_return=np.dot(self.mu, weights),
        volatility=np.sqrt(np.dot(weights, np.dot(self.Sigma, weights))),
        sharpe_ratio=(np.dot(self.mu, weights) - self.rf) /
                      np.sqrt(np.dot(weights, np.dot(self.Sigma, weights)))
    )

# Example with sector constraints
sectors = [0, 0, 1, 1, 2] # Tech, Tech, Finance, Finance, Healthcare
opt = ConstrainedPortfolioOptimizer(mu, Sigma, sectors=sectors)

result = opt.optimize(
    min_weights=np.array([0.05] * 5), # At least 5% in each
    max_weights=np.array([0.40] * 5), # At most 40% in each
    sector_limits={0: 0.50, 1: 0.40}, # Max 50% Tech, 40% Finance
    risk_aversion=2.0
)

print("Constrained Portfolio:")
print(f"  Weights: {result.weights.round(3)}")
print(f"  Return: {result.expected_return:.2%}")
print(f"  Volatility: {result.volatility:.2%}")

```

Constrained Portfolio:

Weights: [0.304 0.176 0.113 0.122 0.285]

Return: 8.97%

Volatility: 15.87%

20.2.2. Visualizing Portfolio Allocations

Understanding how assets are allocated across different optimization strategies is essential for portfolio management.

20.2.3. Cardinality Constraints (MIP)

Limiting the number of assets requires integer programming. In practice, many institutional portfolios restrict the number of holdings to reduce transaction and monitoring costs. The exact formulation introduces binary variables z_i indicating whether asset i is included. Because full MIP solvers may not be available, the implementation below uses a greedy heuristic: it ranks assets by

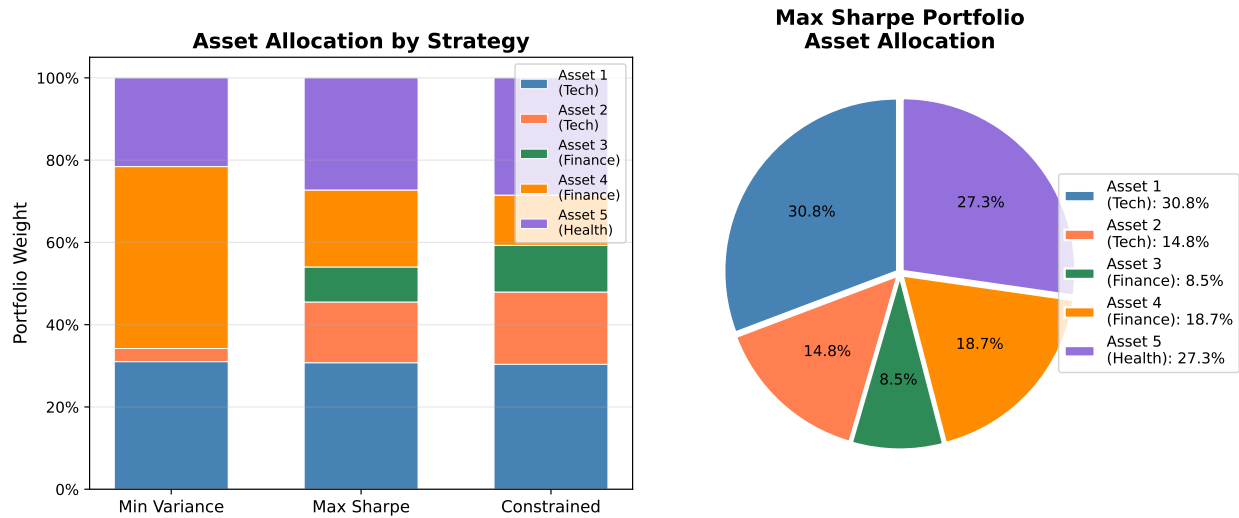


Figure 20.3.: Comparison of portfolio allocations across different optimization strategies. The stacked bar chart shows how asset weights vary between minimum variance, maximum Sharpe ratio, and constrained portfolios.

their return-to-volatility ratio, selects the top K assets, and then runs mean-variance optimization over the selected subset.

```
def portfolio_with_cardinality(mu: np.ndarray, Sigma: np.ndarray,
                             max_assets: int, min_weight: float = 0.05,
                             target_return: float = None) -> dict:
    """Portfolio optimization with cardinality constraint using MIP."""
    try:
        from scipy.optimize import milp, LinearConstraint, Bounds
    except ImportError:
        # Fallback to simple heuristic
        return cardinality_heuristic(mu, Sigma, max_assets, min_weight,
                                     target_return)

    n = len(mu)

    # Use greedy heuristic for demonstration
    # (Full MIP requires specialized solver)
    return cardinality_heuristic(mu, Sigma, max_assets, min_weight, target_return)

def cardinality_heuristic(mu: np.ndarray, Sigma: np.ndarray,
                          max_assets: int, min_weight: float = 0.05,
                          target_return: float = None) -> dict:
    """Greedy heuristic for cardinality-constrained portfolio."""
    n = len(mu)

    # Rank assets by Sharpe-like metric
    stds = np.sqrt(np.diag(Sigma))
    scores = mu / stds
```

```

# Select top assets
selected = np.argsort(scores)[-max_assets:]

# Optimize over selected assets
mu_sub = mu[selected]
Sigma_sub = Sigma[np.ix_(selected, selected)]

opt = MarkowitzOptimizer(mu_sub, Sigma_sub)
result = opt.minimize_variance(target_return=target_return)

# Map back to full weight vector
full_weights = np.zeros(n)
full_weights[selected] = result.weights

return {
    'weights': full_weights,
    'selected_assets': selected.tolist(),
    'expected_return': np.dot(mu, full_weights),
    'volatility': np.sqrt(np.dot(full_weights, np.dot(Sigma, full_weights)))
}

# Example
result = portfolio_with_cardinality(mu, Sigma, max_assets=3)
print(f"Selected {len(result['selected_assets'])} assets:
    ↳ {result['selected_assets']}")
print(f"Weights: {result['weights'].round(3)}")
print(f"Return: {result['expected_return']:.2%}")

```

Selected 3 assets: [0, 4, 3]

Weights: [0.308 0. 0. 0.462 0.231]

Return: 7.77%

20.3. Risk Measures Beyond Variance

20.3.1. Conditional Value at Risk (CVaR)

CVaR (Expected Shortfall) measures the expected loss in the worst $\alpha\%$ of cases.

$$\text{CVaR}_\alpha = \mathbb{E}[L | L \geq \text{VaR}_\alpha]$$

The `CVaROptimizer` class below works with scenario-based return data rather than parametric distributions. It computes CVaR empirically by averaging portfolio returns in the worst α -quantile of scenarios. The optimizer minimizes CVaR subject to a target return constraint, producing portfolios that are more resilient to tail events than variance-based approaches.

```

class CVaROptimizer:
    """Portfolio optimization using CVaR as risk measure."""

    def __init__(self, scenarios: np.ndarray, probabilities: np.ndarray = None):
        """
        Args:
            scenarios: Array of shape (n_scenarios, n_assets) with return scenarios

```



```

        probabilities: Scenario probabilities (uniform if None)
    """
    self.scenarios = scenarios
    self.n_scenarios, self.n_assets = scenarios.shape

    if probabilities is None:
        self.probs = np.ones(self.n_scenarios) / self.n_scenarios
    else:
        self.probs = probabilities

def portfolio_cvar(self, weights: np.ndarray, alpha: float = 0.05) -> float:
    """Calculate portfolio CVaR at confidence level alpha."""
    portfolio_returns = np.dot(self.scenarios, weights)

    # VaR
    var_threshold = np.percentile(portfolio_returns, alpha * 100)

    # CVaR (expected shortfall)
    tail_returns = portfolio_returns[portfolio_returns <= var_threshold]
    if len(tail_returns) == 0:
        return var_threshold
    return np.mean(tail_returns)

def optimize_min_cvar(self, alpha: float = 0.05,
                     target_return: float = None) -> dict:
    """Minimize CVaR subject to constraints."""

    def objective(w):
        return -self.portfolio_cvar(w, alpha) # Minimize negative CVaR

    constraints = [
        {'type': 'eq', 'fun': lambda w: np.sum(w) - 1}
    ]

    if target_return is not None:
        expected_returns = np.dot(self.probs, self.scenarios)
        constraints.append({
            'type': 'ineq',
            'fun': lambda w: np.dot(expected_returns, w) - target_return
        })

    bounds = [(0, 1)] * self.n_assets
    w0 = np.ones(self.n_assets) / self.n_assets

    result = minimize(objective, w0, method='SLSQP',
                     bounds=bounds, constraints=constraints)

    weights = result.x
    portfolio_returns = np.dot(self.scenarios, weights)

    return {
        'weights': weights,
        'expected_return': np.mean(portfolio_returns),

```

```

        'volatility': np.std(portfolio_returns),
        'var_95': np.percentile(portfolio_returns, 5),
        'cvar_95': self.portfolio_cvar(weights, 0.05)
    }

    # Generate scenarios
    np.random.seed(42)
    n_scenarios = 1000
    scenarios = np.random.multivariate_normal(mu, Sigma, n_scenarios)

    cvar_opt = CVaROptimizer(scenarios)
    result = cvar_opt.optimize_min_cvar(alpha=0.05)

    print("CVaR-Optimized Portfolio:")
    print(f"  Weights: {result['weights'].round(3)}")
    print(f"  Expected Return: {result['expected_return']:.2%}")
    print(f"  Volatility: {result['volatility']:.2%}")
    print(f"  VaR (95%): {result['var_95']:.2%}")
    print(f"  CVaR (95%): {result['cvar_95']:.2%}")

```

CVaR-Optimized Portfolio:

```

Weights: [0.317 0.037 0.016 0.407 0.223]
Expected Return: 7.73%
Volatility: 14.33%
VaR (95%): -16.16%
CVaR (95%): -22.03%

```

20.3.2. Maximum Drawdown

Maximum drawdown measures the largest peak-to-trough decline in portfolio value over a given period. It captures a type of risk that variance does not: the magnitude of the worst sustained loss an investor would experience. The functions below compute the drawdown series from a return sequence and identify the peak and trough indices.

```

def calculate_drawdown(returns: np.ndarray) -> dict:
    """Calculate maximum drawdown from return series."""
    # Cumulative returns
    cum_returns = np.cumprod(1 + returns)

    # Running maximum
    running_max = np.maximum.accumulate(cum_returns)

    # Drawdown at each point
    drawdowns = (cum_returns - running_max) / running_max

    # Maximum drawdown
    max_dd = np.min(drawdowns)
    max_dd_end = np.argmax(drawdowns)
    max_dd_start = np.argmax(cum_returns[:max_dd_end + 1])

    return {
        'max_drawdown': max_dd,

```

```

        'drawdown_series': drawdowns,
        'peak_index': max_dd_start,
        'trough_index': max_dd_end
    }

def portfolio_max_drawdown(scenarios: np.ndarray, weights: np.ndarray) -> float:
    """Calculate expected maximum drawdown across scenarios."""
    drawdowns = []
    for scenario in scenarios:
        portfolio_returns = np.dot(scenario.reshape(-1, len(weights)), weights)
        dd = calculate_drawdown(portfolio_returns)
        drawdowns.append(dd['max_drawdown'])
    return np.mean(drawdowns)

```

20.4. Robust Portfolio Optimization

20.4.1. Parameter Uncertainty

Estimated returns and covariances have significant uncertainty. Robust optimization protects against estimation errors. The approach defines an uncertainty set around each parameter estimate and optimizes against the worst case within that set. For expected returns, this means subtracting a per-asset uncertainty margin δ_i to obtain a conservative return estimate. For the covariance matrix, a multiplicative factor inflates variance estimates.

The `RobustPortfolioOptimizer` class implements this worst-case optimization and includes a method to compare robust and nominal solutions side by side.

```

class RobustPortfolioOptimizer:
    """Robust portfolio optimization handling parameter uncertainty."""

    def __init__(self, mu: np.ndarray, Sigma: np.ndarray,
                 mu_uncertainty: np.ndarray = None,
                 Sigma_uncertainty: float = 0.1):
        """
        Args:
            mu: Estimated expected returns
            Sigma: Estimated covariance matrix
            mu_uncertainty: Uncertainty in returns (radius of uncertainty set)
            Sigma_uncertainty: Relative uncertainty in covariance
        """
        self.mu = mu
        self.Sigma = Sigma
        self.n = len(mu)

        if mu_uncertainty is None:
            # Default: uncertainty proportional to volatility
            self.delta = 0.1 * np.sqrt(np.diag(Sigma))
        else:
            self.delta = mu_uncertainty

        self.Sigma_factor = 1 + Sigma_uncertainty

    def worst_case_return(self, weights: np.ndarray) -> float:

```

```

        """Calculate worst-case return under uncertainty."""
        # Worst case: returns at lower bound of uncertainty set
        return np.dot(self.mu - self.delta, weights)

def worst_case_variance(self, weights: np.ndarray) -> float:
    """Calculate worst-case variance under uncertainty."""
    return self.Sigma_factor * np.dot(weights, np.dot(self.Sigma, weights))

def optimize_robust(self, risk_aversion: float = 1.0) -> dict:
    """Optimize using worst-case parameters."""

    def objective(w):
        wc_return = self.worst_case_return(w)
        wc_var = self.worst_case_variance(w)
        return -wc_return + 0.5 * risk_aversion * wc_var

    constraints = [{'type': 'eq', 'fun': lambda w: np.sum(w) - 1}]
    bounds = [(0, 1)] * self.n
    w0 = np.ones(self.n) / self.n

    result = minimize(objective, w0, method='SLSQP',
                      bounds=bounds, constraints=constraints)

    weights = result.x
    return {
        'weights': weights,
        'expected_return': np.dot(self.mu, weights),
        'worst_case_return': self.worst_case_return(weights),
        'volatility': np.sqrt(np.dot(weights, np.dot(self.Sigma, weights))),
        'worst_case_volatility': np.sqrt(self.worst_case_variance(weights))
    }

def compare_with_nominal(self, risk_aversion: float = 1.0) -> dict:
    """Compare robust vs nominal optimization."""
    # Robust solution
    robust = self.optimize_robust(risk_aversion)

    # Nominal solution
    nominal_opt = MarkowitzOptimizer(self.mu, self.Sigma)

    def nominal_obj(w):
        return -np.dot(self.mu, w) + 0.5 * risk_aversion * np.dot(w,
            ↪ np.dot(self.Sigma, w))

    constraints = [{'type': 'eq', 'fun': lambda w: np.sum(w) - 1}]
    result = minimize(nominal_obj, np.ones(self.n)/self.n,
                      method='SLSQP', bounds=[(0,1)]*self.n,
                      constraints=constraints)
    nominal_weights = result.x

    return {
        'robust': robust,
        'nominal': {

```

```

        'weights': nominal_weights,
        'expected_return': np.dot(self.mu, nominal_weights),
        'worst_case_return': self.worst_case_return(nominal_weights),
        'volatility': np.sqrt(np.dot(nominal_weights, np.dot(self.Sigma,
        ↪ nominal_weights)))
    }
}

# Example
robust_opt = RobustPortfolioOptimizer(mu, Sigma)
comparison = robust_opt.compare_with_nominal(risk_aversion=2.0)

print("Nominal vs Robust Comparison:")
print(f"  Nominal weights: {comparison['nominal']['weights'].round(3)}")
print(f"  Robust weights:   {comparison['robust']['weights'].round(3)}")
print(f"  Nominal worst-case return:
    ↪ {comparison['nominal']['worst_case_return']:.2%}")
print(f"  Robust worst-case return:
    ↪ {comparison['robust']['worst_case_return']:.2%}")

```

Nominal vs Robust Comparison:

Nominal weights: [0.304 0.176 0.113 0.122 0.285]

Robust weights: [0.318 0.102 0.036 0.258 0.286]

Nominal worst-case return: 6.53%

Robust worst-case return: 6.20%

20.5. Factor Models and Risk Decomposition

Factor models decompose asset returns into components driven by common factors (such as market, size, and value) plus an asset-specific residual. The asset covariance matrix is reconstructed as $\Sigma = BFB^T + D$, where B is the factor loading matrix, F is the factor covariance matrix, and D is a diagonal matrix of specific variances. This structure reduces the number of parameters to estimate and provides a natural framework for risk attribution.

The `FactorModelPortfolio` class below computes portfolio factor exposures, decomposes total risk into factor and specific components, and supports optimization with factor neutrality constraints.

```

class FactorModelPortfolio:
    """Portfolio optimization using factor model."""

    def __init__(self, factor_loadings: np.ndarray,
                 factor_covariance: np.ndarray,
                 specific_variance: np.ndarray,
                 factor_returns: np.ndarray = None):
        """
        Args:
            factor_loadings: B matrix (n_assets x n_factors)
            factor_covariance: Factor covariance matrix
            specific_variance: Idiosyncratic variance per asset
            factor_returns: Expected factor returns
        """
        self.B = factor_loadings

```

```

self.F = factor_covariance
self.D = np.diag(specific_variance)
self.n_assets = factor_loadings.shape[0]
self.n_factors = factor_loadings.shape[1]

if factor_returns is not None:
    self.factor_mu = factor_returns
else:
    self.factor_mu = np.zeros(self.n_factors)

def asset_covariance(self) -> np.ndarray:
    """Compute asset covariance from factor model."""
    return self.B @ self.F @ self.B.T + self.D

def portfolio_factor_exposure(self, weights: np.ndarray) -> np.ndarray:
    """Calculate portfolio factor exposures."""
    return self.B.T @ weights

def risk_decomposition(self, weights: np.ndarray) -> dict:
    """Decompose portfolio risk into factor and specific."""
    factor_exposure = self.portfolio_factor_exposure(weights)

    # Factor risk
    factor_var = factor_exposure @ self.F @ factor_exposure

    # Specific risk
    specific_var = weights @ self.D @ weights

    # Total risk
    total_var = factor_var + specific_var

    return {
        'total_variance': total_var,
        'total_volatility': np.sqrt(total_var),
        'factor_variance': factor_var,
        'factor_contribution': factor_var / total_var,
        'specific_variance': specific_var,
        'specific_contribution': specific_var / total_var,
        'factor_exposures': factor_exposure
    }

def optimize_factor_neutral(self, target_exposures: dict = None) -> dict:
    """Optimize with factor exposure constraints."""
    Sigma = self.asset_covariance()

    def objective(w):
        return np.dot(w, np.dot(Sigma, w))

    constraints = [
        {'type': 'eq', 'fun': lambda w: np.sum(w) - 1}
    ]

    # Factor neutrality constraints

```

```

        if target_exposures:
            for factor_idx, target in target_exposures.items():
                constraints.append({
                    'type': 'eq',
                    'fun': lambda w, f=factor_idx, t=target:
                        np.dot(self.B[:, f], w) - t
                })
        else:
            # Default: neutral to all factors
            for f in range(self.n_factors):
                constraints.append({
                    'type': 'eq',
                    'fun': lambda w, f=f: np.dot(self.B[:, f], w)
                })

        bounds = [(0, 0.3)] * self.n_assets
        w0 = np.ones(self.n_assets) / self.n_assets

        result = minimize(objective, w0, method='SLSQP',
                          bounds=bounds, constraints=constraints)

        return {
            'weights': result.x,
            'risk_decomposition': self.risk_decomposition(result.x)
        }

# Example: 3-factor model
np.random.seed(42)
n_assets, n_factors = 10, 3

# Factor loadings (e.g., market, size, value)
B = np.random.randn(n_assets, n_factors) * 0.5
B[:, 0] = np.abs(B[:, 0]) + 0.5 # Positive market beta

# Factor covariance
F = np.array([
    [0.04, 0.01, 0.005],
    [0.01, 0.02, 0.002],
    [0.005, 0.002, 0.015]
])

# Specific variance
D_diag = np.random.uniform(0.01, 0.05, n_assets)

factor_port = FactorModelPortfolio(B, F, D_diag)

# Equal-weight portfolio risk decomposition
equal_weights = np.ones(n_assets) / n_assets
decomp = factor_port.risk_decomposition(equal_weights)

print("Equal-Weight Portfolio Risk Decomposition:")
print(f" Total Volatility: {decomp['total_volatility']:.2%}")
print(f" Factor Risk: {decomp['factor_contribution']:.1%}")

```

```
print(f" Specific Risk: {decomp['specific_contribution']:.1%}")
print(f" Factor Exposures: {decomp['factor_exposures'].round(3)}")
```

Equal-Weight Portfolio Risk Decomposition:

Total Volatility: 16.70%

Factor Risk: 90.1%

Specific Risk: 9.9%

Factor Exposures: [0.85 -0.241 -0.167]

20.6. Summary

20.6.1. Key Takeaways

- **Mean-variance** optimization balances expected return against variance
- The **efficient frontier** shows optimal risk-return trade-offs
- Real portfolios require **cardinality, sector, and turnover** constraints
- **CVaR** better captures tail risk than variance
- **Robust optimization** protects against parameter estimation errors
- **Factor models** decompose risk into systematic and specific components
- **Rebalancing** strategies manage turnover costs and tracking error

20.7. Exercises

1. Implement a Black-Litterman model that combines market equilibrium with investor views.
2. Add transaction cost modeling to the rebalancing optimization. How do costs affect optimal rebalancing frequency?
3. Compare CVaR-optimized portfolios with mean-variance portfolios during market stress periods.
4. Implement a risk parity portfolio where each asset contributes equally to total risk.
5. Build a factor model using Fama-French factors and optimize a portfolio to be neutral to the value factor while maximizing exposure to momentum.

21. Energy Systems Optimization

What You Will Learn

- Formulate unit commitment and economic dispatch problems
- Optimize energy storage operations and arbitrage
- Handle renewable energy uncertainty in planning
- Model power flow and grid constraints
- Apply optimization to building energy management

Energy systems present some of the most challenging and impactful optimization problems. From scheduling power plants to managing building HVAC systems, optimization is essential for reliable, affordable, and sustainable energy.

21.1. The Energy Optimization Problem Space

21.1.1. Key Challenges

Challenge	Impact	Solution Approach
Uncertainty	Renewable variability, demand	Stochastic/robust
Non-convexity	Startup costs, min output	MIP formulation
Scale	Thousands of units, time steps	Decomposition
Reliability	N-1 security, reserves	Contingency constraints

21.2. Economic Dispatch

Economic dispatch determines the optimal power output from each generator to meet demand at minimum cost.

21.2.1. Basic Formulation

$$\begin{aligned} & \min \sum_{g \in G} C_g(P_g) \\ & \text{s.t. } \sum_{g \in G} P_g = D \quad (\text{demand balance}) \\ & \quad P_g^{\min} \leq P_g \leq P_g^{\max} \quad \forall g \end{aligned}$$

where $C_g(P_g)$ is the generation cost function.

Each generator is modeled with a quadratic cost curve $C_g(P_g) = aP_g^2 + bP_g + c$, where the quadratic term captures the increasing marginal cost of operating near full capacity. The **EconomicDispatch** class solves this quadratic program using SLSQP with a single equality constraint for demand balance and box constraints for generator limits.

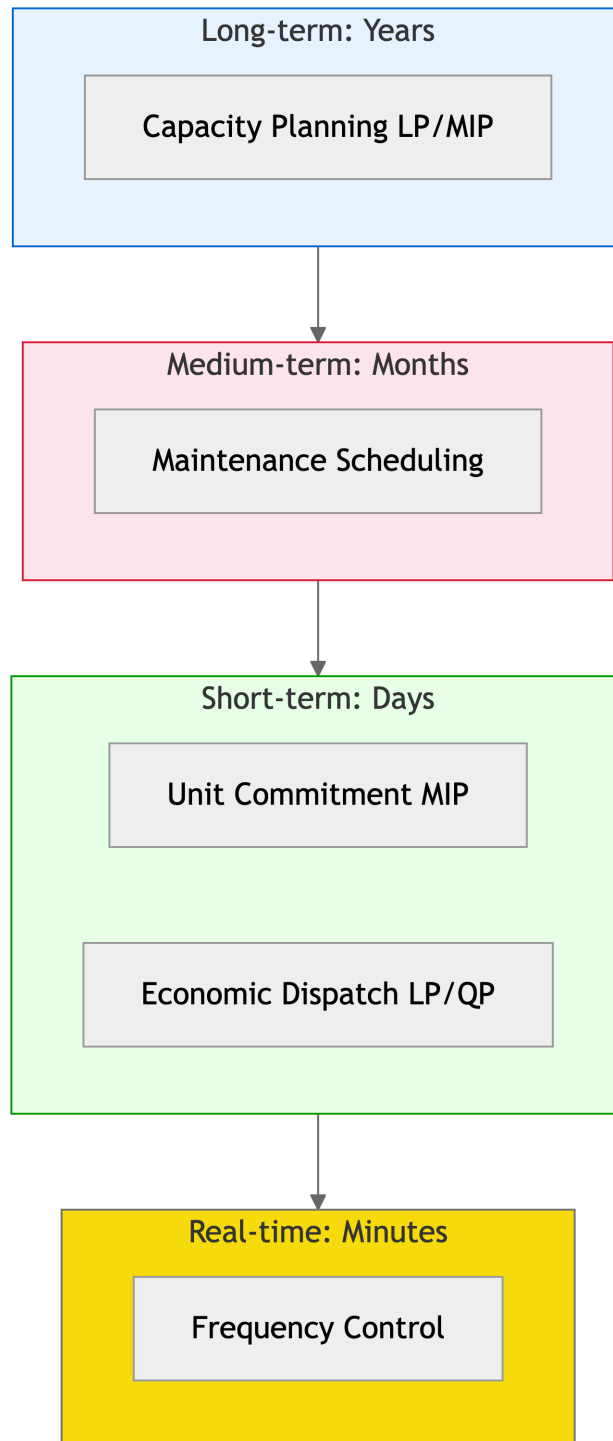


Figure 21.1.: Energy System Decisions Across Time Horizons

```

import numpy as np
from scipy.optimize import minimize, linprog
from dataclasses import dataclass
from typing import List, Dict, Optional

@dataclass
class Generator:
    """A power generation unit."""
    name: str
    p_min: float      # Minimum output (MW)
    p_max: float      # Maximum output (MW)
    cost_a: float      # Quadratic cost coefficient
    cost_b: float      # Linear cost coefficient
    cost_c: float      # Fixed cost
    ramp_up: float = None # Ramp up limit (MW/h)
    ramp_down: float = None # Ramp down limit (MW/h)

    def cost(self, power: float) -> float:
        """Calculate generation cost."""
        return self.cost_a * power**2 + self.cost_b * power + self.cost_c

    def marginal_cost(self, power: float) -> float:
        """Calculate marginal cost."""
        return 2 * self.cost_a * power + self.cost_b

class EconomicDispatch:
    """Economic dispatch optimization."""

    def __init__(self, generators: List[Generator]):
        self.generators = generators
        self.n = len(generators)

    def solve(self, demand: float) -> Dict:
        """Solve economic dispatch for given demand."""
        # Quadratic programming formulation
        def objective(P):
            return sum(g.cost(P[i]) for i, g in enumerate(self.generators))

        def demand_constraint(P):
            return np.sum(P) - demand

        # Bounds
        bounds = [(g.p_min, g.p_max) for g in self.generators]

        # Initial guess
        P0 = np.array([(g.p_min + g.p_max) / 2 for g in self.generators])

        # Solve
        result = minimize(
            objective,
            P0,
            method='SLSQP',
            bounds=bounds,

```

```

        constraints={'type': 'eq', 'fun': demand_constraint}
    )

    if result.success:
        dispatch = result.x
        costs = [g.cost(dispatch[i]) for i, g in enumerate(self.generators)]
        marginals = [g.marginal_cost(dispatch[i]) for i, g in
↪ enumerate(self.generators)]

        # System marginal price (Lagrange multiplier)
        smp = np.mean(marginals)

        return {
            'dispatch': dict(zip([g.name for g in self.generators], dispatch)),
            'total_cost': result.fun,
            'generator_costs': dict(zip([g.name for g in self.generators],
↪ costs)),
            'system_marginal_price': smp,
            'status': 'optimal'
        }
    else:
        return {'status': 'infeasible', 'message': result.message}

# Example
generators = [
    Generator("Coal", p_min=100, p_max=500, cost_a=0.002, cost_b=20, cost_c=500),
    Generator("CCGT", p_min=50, p_max=300, cost_a=0.003, cost_b=35, cost_c=200),
    Generator("Peaker", p_min=0, p_max=150, cost_a=0.005, cost_b=60, cost_c=50),
]

ed = EconomicDispatch(generators)
result = ed.solve(demand=600)

print("Economic Dispatch Solution:")
for gen, power in result['dispatch'].items():
    print(f" {gen}: {power:.1f} MW")
print(f"Total Cost: ${result['total_cost']:.2f}")
print(f"System Marginal Price: ${result['system_marginal_price']:.2f}/MWh")

```

Economic Dispatch Solution:

Coal: 500.0 MW

CCGT: 100.0 MW

Peaker: 0.0 MW

Total Cost: \$14780.00

System Marginal Price: \$39.20/MWh

21.2.2. Multi-Period Dispatch with Ramp Constraints

In practice, generators cannot change their output instantaneously. Thermal units have ramp rate limits that restrict how much power can increase or decrease between consecutive periods. The multi-period dispatch formulation adds inequality constraints linking each generator's output across adjacent time steps. This couples the optimization across all periods, requiring joint optimization of the entire dispatch schedule rather than solving each period independently.

```

class MultiPeriodDispatch:
    """Economic dispatch over multiple time periods with ramping."""

    def __init__(self, generators: List[Generator]):
        self.generators = generators
        self.n_gen = len(generators)

    def solve(self, demand: np.ndarray, initial_output: np.ndarray = None) -> Dict:
        """Solve multi-period dispatch.

        Args:
            demand: Array of demand for each period
            initial_output: Initial output of each generator
        """
        T = len(demand)

        if initial_output is None:
            initial_output = np.array([(g.p_min + g.p_max) / 2
                                       for g in self.generators])

        # Variables: P[g,t] for all generators and time periods
        n_vars = self.n_gen * T

        def objective(x):
            P = x.reshape(self.n_gen, T)
            total_cost = 0
            for g_idx, gen in enumerate(self.generators):
                for t in range(T):
                    total_cost += gen.cost(P[g_idx, t])
            return total_cost

        constraints = []

        # Demand balance for each period
        for t in range(T):
            def demand_eq(x, t=t):
                P = x.reshape(self.n_gen, T)
                return np.sum(P[:, t]) - demand[t]
            constraints.append({'type': 'eq', 'fun': demand_eq})

        # Ramp constraints
        for g_idx, gen in enumerate(self.generators):
            if gen.ramp_up is not None:
                # Ramp from initial to period 0
                def ramp_up_init(x, g=g_idx, ru=gen.ramp_up,
                                p0=initial_output[g_idx]):
                    P = x.reshape(self.n_gen, T)
                    return ru - (P[g, 0] - p0)
                constraints.append({'type': 'ineq', 'fun': ramp_up_init})

                # Ramp between periods
                for t in range(1, T):
                    def ramp_up(x, g=g_idx, t=t, ru=gen.ramp_up):

```

```

        P = x.reshape(self.n_gen, T)
        return ru - (P[g, t] - P[g, t-1])
        constraints.append({'type': 'ineq', 'fun': ramp_up})

    if gen.ramp_down is not None:
        def ramp_down_init(x, g=g_idx, rd=gen.ramp_down,
            ↪ p0=initial_output[g_idx]):
            P = x.reshape(self.n_gen, T)
            return rd - (p0 - P[g, 0])
        constraints.append({'type': 'ineq', 'fun': ramp_down_init})

        for t in range(1, T):
            def ramp_down(x, g=g_idx, t=t, rd=gen.ramp_down):
                P = x.reshape(self.n_gen, T)
                return rd - (P[g, t-1] - P[g, t])
            constraints.append({'type': 'ineq', 'fun': ramp_down})

    # Bounds
    bounds = []
    for t in range(T):
        for gen in self.generators:
            bounds.append((gen.p_min, gen.p_max))

    # Initial guess
    x0 = np.tile(initial_output, T)

    # Solve
    result = minimize(objective, x0, method='SLSQP',
        bounds=bounds, constraints=constraints)

    if result.success:
        P = result.x.reshape(self.n_gen, T)
        return {
            'dispatch': P,
            'generator_names': [g.name for g in self.generators],
            'total_cost': result.fun,
            'status': 'optimal'
        }
    else:
        return {'status': 'infeasible', 'message': result.message}

# Example with ramping
generators_ramp = [
    Generator("Coal", 100, 500, 0.002, 20, 500, ramp_up=50, ramp_down=50),
    Generator("CCGT", 50, 300, 0.003, 35, 200, ramp_up=100, ramp_down=100),
    Generator("Peaker", 0, 150, 0.005, 60, 50, ramp_up=150, ramp_down=150),
]

demand_profile = np.array([400, 500, 650, 700, 600, 450])
mpd = MultiPeriodDispatch(generators_ramp)
result = mpd.solve(demand_profile)

if result['status'] == 'optimal':

```

```

print("Multi-Period Dispatch:")
for g_idx, name in enumerate(result['generator_names']):
    print(f"  {name}: {result['dispatch'][g_idx].round(1)}")
print(f"Total Cost: ${result['total_cost']:.2f}")

```

The generation mix varies throughout the day as demand rises and falls. The stacked area chart below shows how different energy sources contribute to meeting a typical daily load profile. Nuclear and coal provide baseload power, solar peaks at midday, and gas plants ramp up to cover evening demand when solar output drops.

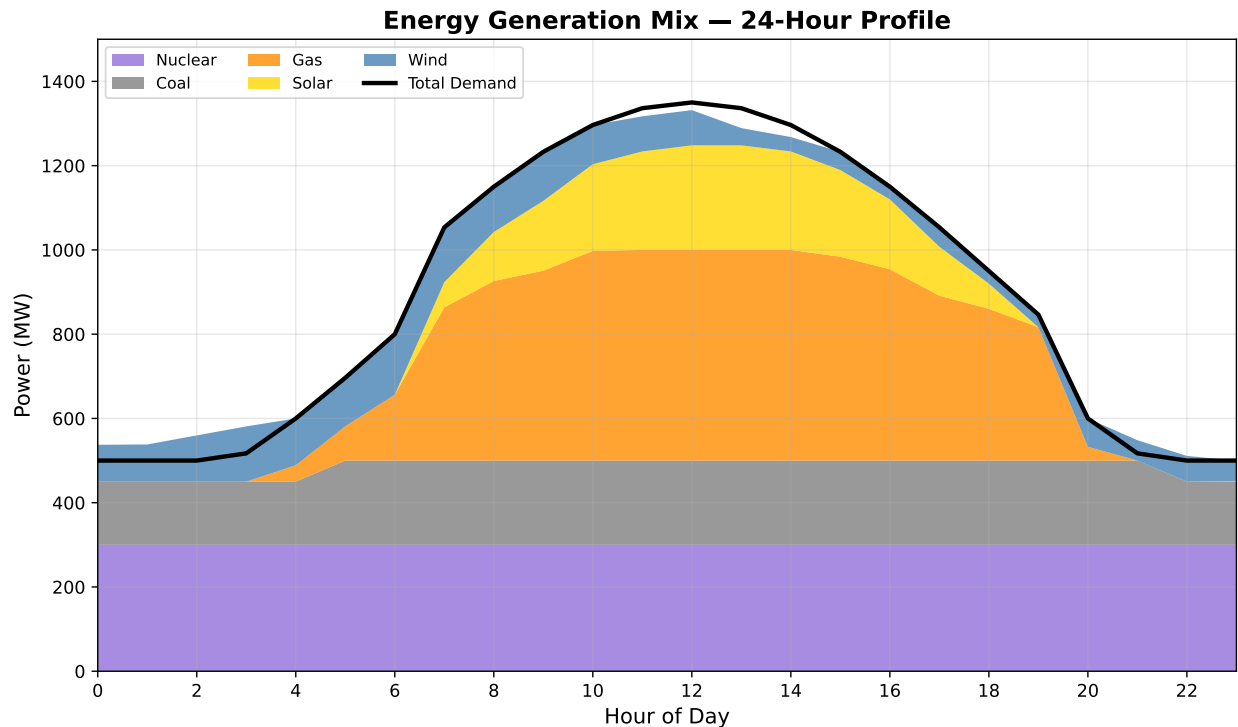


Figure 21.2.: Energy generation mix over a 24-hour period — renewable sources peak during daytime while gas and storage fill demand gaps during evening hours

21.3. Unit Commitment

Unit commitment determines which generators to turn on/off over a planning horizon, considering startup costs and minimum up/down times. Unlike economic dispatch, which treats commitment decisions as given, unit commitment introduces binary variables for each generator at each time step. The resulting mixed-integer program is computationally expensive, so the implementation below uses a priority list heuristic: generators are ranked by average cost and committed in merit order until enough capacity is available to meet demand plus a reserve margin. Economic dispatch is then run among the committed units at each time step.

```

@dataclass
class UCGenerator(Generator):
    """Generator with unit commitment parameters."""

```

```

startup_cost: float = 0
shutdown_cost: float = 0
min_up_time: int = 1      # Minimum hours on
min_down_time: int = 1    # Minimum hours off
initial_status: int = 1   # 1=on, 0=off
initial_hours: int = 0    # Hours in current status

class UnitCommitment:
    """Unit commitment optimization (simplified)."""

    def __init__(self, generators: List[UCGenerator]):
        self.generators = generators
        self.n_gen = len(generators)

    def solve_heuristic(self, demand: np.ndarray, reserve_margin: float = 0.1) ->
    ↪ Dict:
        """Solve UC using priority list heuristic."""
        T = len(demand)

        # Sort generators by marginal cost at mid-point
        def avg_cost(g):
            mid = (g.p_min + g.p_max) / 2
            return g.cost(mid) / mid

        gen_order = sorted(range(self.n_gen),
                           key=lambda i: avg_cost(self.generators[i]))

        # Initialize commitment and dispatch
        commitment = np.zeros((self.n_gen, T), dtype=int)
        dispatch = np.zeros((self.n_gen, T))

        for t in range(T):
            required = demand[t] * (1 + reserve_margin)
            capacity = 0

            # Commit generators in merit order until demand met
            for g_idx in gen_order:
                gen = self.generators[g_idx]

                if capacity < required:
                    commitment[g_idx, t] = 1
                    capacity += gen.p_max

            # Economic dispatch among committed units
            committed_gen = [self.generators[i] for i in range(self.n_gen)
                             if commitment[i, t] == 1]

            if committed_gen:
                ed = EconomicDispatch(committed_gen)
                ed_result = ed.solve(demand[t])

                if ed_result['status'] == 'optimal':
                    j = 0

```



```

        for i in range(self.n_gen):
            if commitment[i, t] == 1:
                dispatch[i, t] = list(ed_result['dispatch'].values())[j]
                j += 1

# Calculate costs
total_cost = 0
for g_idx, gen in enumerate(self.generators):
    for t in range(T):
        if commitment[g_idx, t] == 1:
            total_cost += gen.cost(dispatch[g_idx, t])

# Startup cost
if t == 0:
    if gen.initial_status == 0:
        total_cost += gen.startup_cost
elif commitment[g_idx, t-1] == 0:
    total_cost += gen.startup_cost

return {
    'commitment': commitment,
    'dispatch': dispatch,
    'total_cost': total_cost,
    'generator_names': [g.name for g in self.generators]
}

# Example
uc_generators = [
    UCGenerator("Coal", 100, 500, 0.002, 20, 500, startup_cost=5000,
               min_up_time=4, min_down_time=4),
    UCGenerator("CCGT", 50, 300, 0.003, 35, 200, startup_cost=1000,
               min_up_time=2, min_down_time=2),
    UCGenerator("Peaker", 0, 150, 0.005, 60, 50, startup_cost=100,
               min_up_time=1, min_down_time=1),
]

demand_24h = np.array([300, 280, 260, 250, 260, 300, 400, 500, 550, 580,
                       600, 620, 640, 650, 630, 600, 580, 550, 520, 480,
                       420, 380, 350, 320])

uc = UnitCommitment(uc_generators)
result = uc.solve_heuristic(demand_24h)

print("Unit Commitment Schedule:")
for g_idx, name in enumerate(result['generator_names']):
    on_hours = np.sum(result['commitment'][g_idx])
    avg_output = np.mean(result['dispatch'][g_idx][result['commitment'][g_idx] ==
↪ 1])
    print(f" {name}: On for {on_hours}h, avg output {avg_output:.0f} MW")
print(f"Total Cost: ${result['total_cost']:.0f}")

```

Unit Commitment Schedule:

Coal: On for 24h, avg output 411 MW
 CCGT: On for 13h, avg output 88 MW

21. Energy Systems Optimization

Peaker: On for 0h, avg output nan MW
Total Cost: \$262146

The Gantt chart below shows a typical unit commitment schedule. Nuclear and baseload coal units remain online around the clock, while gas combustion turbines start up only during peak demand hours. The red triangles mark startup events, each of which incurs a fixed cost. The lower panel shows the system demand curve that drives these commitment decisions.

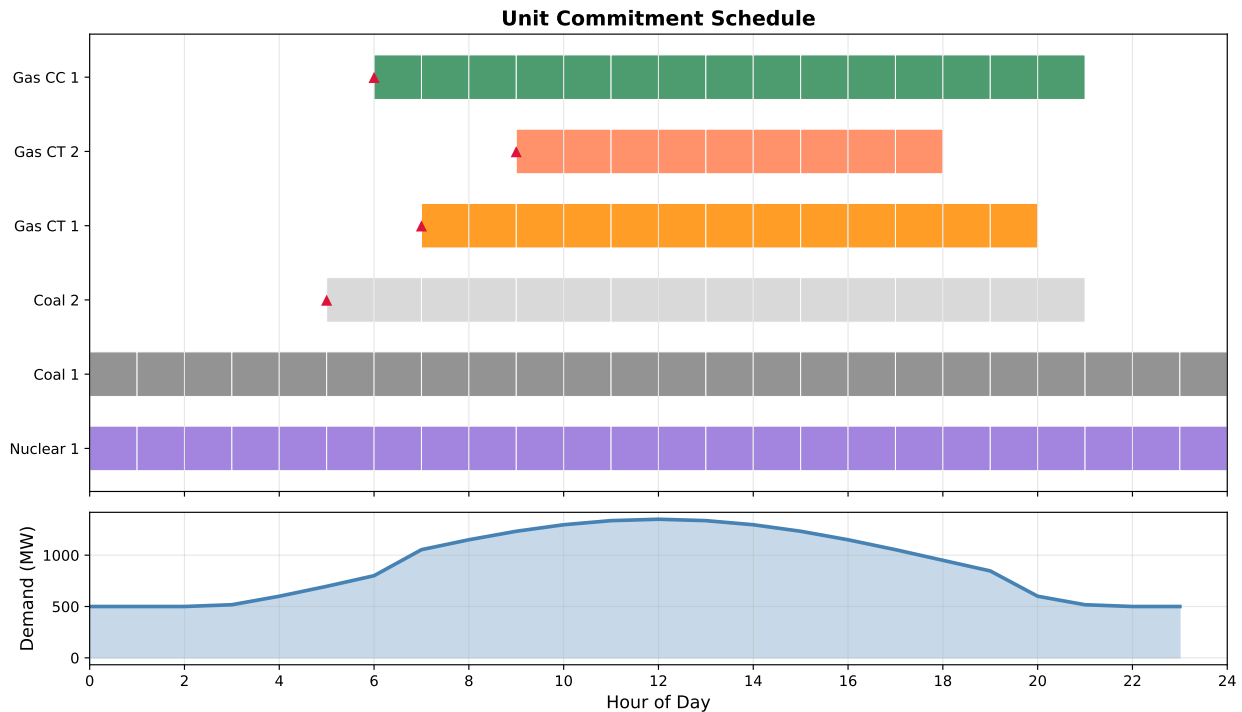


Figure 21.3.: Unit commitment schedule — generator on/off status over 24 hours with system demand curve overlaid

21.4. Energy Storage Optimization

21.4.1. Storage Arbitrage

Battery storage systems profit by charging when electricity prices are low and discharging when prices are high. The optimization problem determines the charge and discharge schedule that maximizes arbitrage profit while respecting the battery's physical constraints: maximum charge and discharge power, round-trip efficiency losses, and state-of-charge limits. The formulation tracks the state of charge across all time periods through equality constraints linking consecutive periods.

```
@dataclass
class EnergyStorage:
    """Energy storage system parameters."""
    capacity: float # MWh
    power_charge: float # MW (max charge rate)
    power_discharge: float # MW (max discharge rate)
    efficiency_charge: float = 0.95
```

```

efficiency_discharge: float = 0.95
initial_soc: float = 0.5 # Initial state of charge (fraction)
min_soc: float = 0.1
max_soc: float = 0.9

class StorageOptimizer:
    """Optimize storage operation for price arbitrage."""

    def __init__(self, storage: EnergyStorage):
        self.storage = storage

    def optimize_arbitrage(self, prices: np.ndarray) -> Dict:
        """Optimize charge/discharge schedule for arbitrage."""
        T = len(prices)
        s = self.storage

        # Variables: charge[t], discharge[t], soc[t]
        # We'll use a simplified formulation

        def objective(x):
            charge = x[:T]
            discharge = x[T:2*T]
            # Revenue from arbitrage
            revenue = np.sum(prices * discharge * s.efficiency_discharge)
            cost = np.sum(prices * charge)
            return cost - revenue

        constraints = []

        # SOC evolution
        def soc_evolution(x):
            charge = x[:T]
            discharge = x[T:2*T]
            soc = x[2*T:3*T]

            violations = []
            # Initial SOC
            violations.append(soc[0] - s.initial_soc * s.capacity -
                             charge[0] * s.efficiency_charge +
                             discharge[0] / s.efficiency_discharge)

            for t in range(1, T):
                violations.append(soc[t] - soc[t-1] -
                                 charge[t] * s.efficiency_charge +
                                 discharge[t] / s.efficiency_discharge)
            return np.array(violations)

        constraints.append({'type': 'eq', 'fun': soc_evolution})

        # Bounds
        bounds = []
        for t in range(T):
            bounds.append((0, s.power_charge)) # charge

```

```

    for t in range(T):
        bounds.append((0, s.power_discharge)) # discharge
    for t in range(T):
        bounds.append((s.min_soc * s.capacity, s.max_soc * s.capacity)) # soc

    # Initial guess
    x0 = np.zeros(3 * T)
    x0[2*T:3*T] = s.initial_soc * s.capacity

    result = minimize(objective, x0, method='SLSQP',
                      bounds=bounds, constraints=constraints)

    if result.success:
        charge = result.x[:T]
        discharge = result.x[T:2*T]
        soc = result.x[2*T:3*T]

        revenue = np.sum(prices * discharge * s.eta_discharge)
        cost = np.sum(prices * charge)

        return {
            'charge': charge,
            'discharge': discharge,
            'soc': soc,
            'revenue': revenue,
            'cost': cost,
            'profit': revenue - cost,
            'status': 'optimal'
        }
    else:
        return {'status': 'infeasible'}

# Example
storage = EnergyStorage(
    capacity=100, # MWh
    power_charge=25, # MW
    power_discharge=25, # MW
    eta_charge=0.92,
    eta_discharge=0.92
)

# Price profile with morning/evening peaks
prices = np.array([30, 28, 25, 22, 20, 22, 35, 55, 70, 65,
                  60, 55, 50, 45, 42, 45, 55, 75, 85, 70,
                  55, 45, 38, 32])

optimizer = StorageOptimizer(storage)
result = optimizer.optimize_arbitrage(prices)

if result['status'] == 'optimal':
    print("Storage Arbitrage Schedule:")
    print(f"    Total Profit: ${result['profit']:.2f}")

```

```
# Find charge/discharge hours
charge_hours = np.where(result['charge'] > 1)[0]
discharge_hours = np.where(result['discharge'] > 1)[0]
print(f"   Charging hours: {charge_hours.tolist()}")
print(f"   Discharging hours: {discharge_hours.tolist()}")
```

Storage Arbitrage Schedule:

Total Profit: \$4882.60

Charging hours: [3, 4, 5, 13, 14, 15]

Discharging hours: [0, 8, 9, 10, 17, 18, 19]

21.4.2. Storage with Renewable Integration

When battery storage is paired with on-site renewable generation, the objective shifts from pure arbitrage to maximizing self-consumption. Solar panels often produce more power than a building needs during midday, while demand peaks in the evening. The optimizer decides when to charge the battery from solar surplus, when to discharge to offset grid purchases, and when to export excess generation. The cost function accounts for different buy and sell prices, which in most markets differ substantially.

```
class RenewableStorageSystem:
    """Combined renewable generation and storage optimization."""

    def __init__(self, storage: EnergyStorage, grid_limit: float = None):
        self.storage = storage
        self.grid_limit = grid_limit # Maximum grid export

    def optimize_self_consumption(self, demand: np.ndarray,
                                  renewable_gen: np.ndarray,
                                  buy_price: np.ndarray,
                                  sell_price: np.ndarray) -> Dict:
        """Optimize storage to maximize self-consumption and minimize cost."""
        T = len(demand)
        s = self.storage

        # Net demand (positive = need power, negative = excess)
        net_demand = demand - renewable_gen

        def objective(x):
            charge = x[:T]
            discharge = x[T:2*T]
            grid_import = x[2*T:3*T]
            grid_export = x[3*T:4*T]

            cost = np.sum(buy_price * grid_import)
            revenue = np.sum(sell_price * grid_export)
            return cost - revenue

        constraints = []

        # Power balance
        def power_balance(x):
```

```

        charge = x[:T]
        discharge = x[T:2*T]
        grid_import = x[2*T:3*T]
        grid_export = x[3*T:4*T]

        # demand + charge + export = renewable + discharge + import
        return (demand + charge + grid_export -
                renewable_gen - discharge - grid_import)

constraints.append({'type': 'eq', 'fun': power_balance})

# SOC dynamics (simplified - using penalty method)
def soc_penalty(x):
    charge = x[:T]
    discharge = x[T:2*T]

    soc = s.initial_soc * s.capacity
    penalty = 0
    for t in range(T):
        soc = soc + charge[t] * s. efficiency_charge - discharge[t] /
↪ s. efficiency_discharge
        if soc < s.min_soc * s.capacity:
            penalty += (s.min_soc * s.capacity - soc) ** 2 * 1000
        if soc > s.max_soc * s.capacity:
            penalty += (soc - s.max_soc * s.capacity) ** 2 * 1000
    return penalty

original_obj = objective
def objective_with_soc(x):
    return original_obj(x) + soc_penalty(x)

# Bounds
bounds = []
for t in range(T):
    bounds.append((0, s.power_charge)) # charge
for t in range(T):
    bounds.append((0, s.power_discharge)) # discharge
for t in range(T):
    bounds.append((0, max(demand))) # grid import
for t in range(T):
    export_limit = self.grid_limit if self.grid_limit else
↪ max(renewable_gen)
    bounds.append((0, export_limit)) # grid export

x0 = np.zeros(4 * T)
x0[2*T:3*T] = np.maximum(0, net_demand) # Start with all from grid

result = minimize(objective_with_soc, x0, method='SLSQP', bounds=bounds)

charge = result.x[:T]
discharge = result.x[T:2*T]
grid_import = result.x[2*T:3*T]
grid_export = result.x[3*T:4*T]

```

```

        # Calculate self-consumption
        total_renewable = np.sum(renewable_gen)
        self_consumed = total_renewable - np.sum(grid_export)
        self_consumption_rate = self_consumed / total_renewable if total_renewable >
↪ 0 else 0

        return {
            'charge': charge,
            'discharge': discharge,
            'grid_import': grid_import,
            'grid_export': grid_export,
            'cost': np.sum(buy_price * grid_import),
            'revenue': np.sum(sell_price * grid_export),
            'net_cost': np.sum(buy_price * grid_import) - np.sum(sell_price *
↪ grid_export),
            'self_consumption_rate': self_consumption_rate
        }

# Example
demand = np.array([3, 2, 2, 2, 3, 5, 8, 10, 8, 6,
                    5, 4, 4, 5, 6, 8, 12, 15, 12, 8,
                    6, 5, 4, 3]) # kW

solar_gen = np.array([0, 0, 0, 0, 0, 0.5, 2, 5, 8, 10,
                      12, 14, 15, 14, 12, 9, 5, 2, 0.5, 0,
                      0, 0, 0, 0]) # kW

buy_price = np.full(24, 0.15) # $/kWh
sell_price = np.full(24, 0.05) # $/kWh (feed-in tariff)

home_storage = EnergyStorage(capacity=10, power_charge=5, power_discharge=5)
system = RenewableStorageSystem(home_storage, grid_limit=5)

result = system.optimize_self_consumption(demand, solar_gen, buy_price, sell_price)

print("Home Battery Optimization:")
print(f" Self-consumption rate: {result['self_consumption_rate']:.1%}")
print(f" Grid purchase cost: ${result['cost']:.2f}")
print(f" Export revenue: ${result['revenue']:.2f}")
print(f" Net cost: ${result['net_cost']:.2f}")

```

```

Home Battery Optimization:
Self-consumption rate: -10.1%
Grid purchase cost: $0.00
Export revenue: $6.00
Net cost: $-6.00

```

21.5. Renewable Energy Integration

21.5.1. Stochastic Dispatch with Wind Uncertainty

Wind generation is inherently variable, and forecasts carry significant error. Deterministic dispatch using a single wind forecast may result in either excess generation (wasted energy) or shortfalls

(expensive emergency measures). Stochastic dispatch addresses this by optimizing over multiple wind scenarios simultaneously. The two-stage formulation sets a base dispatch in the first stage, then computes adjustment reserves (up and down) for each scenario in the second stage. Upward adjustments cost more than base generation because they require fast-responding units; downward adjustments incur curtailment costs.

```
class StochasticDispatch:
    """Economic dispatch under wind uncertainty."""

    def __init__(self, generators: List[Generator], wind_capacity: float):
        self.generators = generators
        self.wind_capacity = wind_capacity

    def solve_scenario_based(self, demand: float,
                            wind_scenarios: np.ndarray,
                            scenario_probs: np.ndarray) -> Dict:
        """Solve using scenario-based stochastic programming."""
        n_scenarios = len(wind_scenarios)
        n_gen = len(self.generators)

        # First-stage: base dispatch P_g
        # Second-stage: adjustments r_g^s for each scenario

    def objective(x):
        P = x[:n_gen] # Base dispatch
        R_up = x[n_gen:n_gen + n_gen * n_scenarios].reshape(n_gen, n_scenarios)
        R_down = x[n_gen + n_gen * n_scenarios:].reshape(n_gen, n_scenarios)

        # Base cost
        cost = sum(g.cost(P[i]) for i, g in enumerate(self.generators))

        # Expected adjustment cost (simplified: linear)
        for s in range(n_scenarios):
            for i, g in enumerate(self.generators):
                # Upward adjustment cost (more expensive)
                cost += scenario_probs[s] * 1.2 * g.cost_b * R_up[i, s]
                # Downward adjustment cost
                cost += scenario_probs[s] * 0.5 * g.cost_b * R_down[i, s]

        return cost

    constraints = []

    # Base power balance (expected wind)
    expected_wind = np.sum(wind_scenarios * scenario_probs)
    constraints.append({
        'type': 'eq',
        'fun': lambda x: np.sum(x[:n_gen]) + expected_wind - demand
    })

    # Scenario power balance
    for s in range(n_scenarios):
        def scenario_balance(x, s=s, wind=wind_scenarios[s]):
```



```

        P = x[:n_gen]
        R_up = x[n_gen:n_gen + n_gen * n_scenarios].reshape(n_gen,
↪ n_scenarios)
        R_down = x[n_gen + n_gen * n_scenarios:].reshape(n_gen, n_scenarios)
        total = np.sum(P) + np.sum(R_up[:, s]) - np.sum(R_down[:, s]) + wind
        return total - demand
    constraints.append({'type': 'eq', 'fun': scenario_balance})

# Bounds
bounds = []
# Base dispatch
for g in self.generators:
    bounds.append((g.p_min, g.p_max))
# R_up (limited by capacity)
for s in range(n_scenarios):
    for g in self.generators:
        bounds.append((0, g.p_max))
# R_down
for s in range(n_scenarios):
    for g in self.generators:
        bounds.append((0, g.p_max))

x0 = np.zeros(n_gen + 2 * n_gen * n_scenarios)
x0[:n_gen] = [(g.p_min + g.p_max) / 2 for g in self.generators]

result = minimize(objective, x0, method='SLSQP',
                  bounds=bounds, constraints=constraints)

base_dispatch = result.x[:n_gen]
return {
    'base_dispatch': dict(zip([g.name for g in self.generators],
↪ base_dispatch)),
    'expected_cost': result.fun,
    'status': 'optimal' if result.success else 'failed'
}

# Example
generators = [
    Generator("Gas1", 50, 200, 0.003, 40, 300),
    Generator("Gas2", 30, 150, 0.004, 45, 200),
]

# Wind scenarios (MW)
wind_scenarios = np.array([20, 50, 80, 100, 120]) # Low to high wind
scenario_probs = np.array([0.1, 0.2, 0.4, 0.2, 0.1])

stoch = StochasticDispatch(generators, wind_capacity=150)
result = stoch.solve_scenario_based(demand=250, wind_scenarios=wind_scenarios,
                                   scenario_probs=scenario_probs)

print("Stochastic Dispatch:")
for gen, power in result['base_dispatch'].items():
    print(f" {gen}: {power:.1f} MW")

```

```
print(f"Expected Cost: ${result['expected_cost']:.2f}")
```

Stochastic Dispatch:

Gas1: 144.0 MW

Gas2: 30.0 MW

Expected Cost: \$8410.21

21.6. Building Energy Optimization

21.6.1. HVAC Scheduling

Buildings consume roughly 40% of total energy in developed economies, and HVAC systems account for the largest share. The optimization problem minimizes electricity cost while keeping indoor temperature within a comfort band. The building is modeled as a thermal mass with a single temperature node: heat flows in from HVAC heating, flows out through cooling, and exchanges with the outside through a heat loss coefficient. Time-of-use pricing creates an incentive to pre-cool or pre-heat the building during off-peak hours, effectively using the building's thermal mass as energy storage.

```
class BuildingHVAC:
    """Building HVAC optimization."""

    def __init__(self, thermal_mass: float, heat_loss_coef: float,
                  cooling_cop: float = 3.0, heating_cop: float = 3.5):
        """
        Args:
            thermal_mass: Thermal capacitance (kWh/C)
            heat_loss_coef: Heat loss coefficient (kW/C)
            cooling_cop: Cooling coefficient of performance
            heating_cop: Heating coefficient of performance
        """
        self.C = thermal_mass
        self.UA = heat_loss_coef
        self.cool_cop = cooling_cop
        self.heat_cop = heating_cop

    def optimize_schedule(self, outside_temp: np.ndarray,
                          electricity_price: np.ndarray,
                          comfort_min: float = 20,
                          comfort_max: float = 24,
                          initial_temp: float = 22,
                          dt: float = 1.0) -> Dict:
        """Optimize HVAC schedule to minimize cost while maintaining comfort."""
        T = len(outside_temp)

    def objective(x):
        cooling = x[:T] # Cooling power (kW)
        heating = x[T:2*T] # Heating power (kW)

        electricity = cooling / self.cool_cop + heating / self.heat_cop
        return np.sum(electricity_price * electricity * dt)
```

```

constraints = []

# Temperature dynamics and comfort constraints
def temp_evolution(x):
    cooling = x[:T]
    heating = x[T:2*T]
    temp = x[2*T:3*T]

    violations = []
    # Initial temperature
    heat_flow = self.UA * (outside_temp[0] - initial_temp) + heating[0] -
↪ cooling[0]
    expected_temp = initial_temp + (heat_flow / self.C) * dt
    violations.append(temp[0] - expected_temp)

    for t in range(1, T):
        heat_flow = self.UA * (outside_temp[t] - temp[t-1]) + heating[t] -
↪ cooling[t]
        expected_temp = temp[t-1] + (heat_flow / self.C) * dt
        violations.append(temp[t] - expected_temp)

    return np.array(violations)

constraints.append({'type': 'eq', 'fun': temp_evolution})

# Bounds
bounds = []
for t in range(T):
    bounds.append((0, 20)) # Cooling power max 20 kW
for t in range(T):
    bounds.append((0, 20)) # Heating power max 20 kW
for t in range(T):
    bounds.append((comfort_min, comfort_max)) # Temperature bounds

x0 = np.zeros(3 * T)
x0[2*T:3*T] = (comfort_min + comfort_max) / 2

result = minimize(objective, x0, method='SLSQP',
                  bounds=bounds, constraints=constraints,
                  options={'maxiter': 500})

cooling = result.x[:T]
heating = result.x[T:2*T]
temp = result.x[2*T:3*T]
electricity = cooling / self.cool_cop + heating / self.heat_cop

return {
    'cooling': cooling,
    'heating': heating,
    'temperature': temp,
    'electricity': electricity,
    'cost': np.sum(electricity_price * electricity * dt),
    'status': 'optimal' if result.success else 'suboptimal'
}

```

```

    }

# Example
building = BuildingHVAC(thermal_mass=50, heat_loss_coef=2)

# Summer day with peak pricing
outside_temp = np.array([22, 21, 20, 19, 19, 20, 23, 26, 29, 32,
                        34, 35, 36, 36, 35, 34, 32, 30, 28, 26,
                        25, 24, 23, 22])

prices = np.array([0.08, 0.08, 0.08, 0.08, 0.08, 0.08, 0.10, 0.12,
                  0.15, 0.20, 0.25, 0.30, 0.35, 0.35, 0.30, 0.25,
                  0.20, 0.25, 0.30, 0.20, 0.15, 0.10, 0.08, 0.08])

result = building.optimize_schedule(outside_temp, prices,
                                   comfort_min=21, comfort_max=25)

print("HVAC Optimization:")
print(f" Total Cost: ${result['cost']:.2f}")
print(f" Peak Cooling: {max(result['cooling']):.1f} kW")
print(f" Temp Range: {min(result['temperature']):.1f} -
↪ {max(result['temperature']):.1f} C")

```

HVAC Optimization:

Total Cost: \$2.33

Peak Cooling: 20.0 kW

Temp Range: 21.0 - 25.0 C

21.7. Summary

21.7.1. Key Takeaways

- **Economic dispatch** minimizes generation cost to meet demand
- **Unit commitment** adds binary on/off decisions and startup costs
- **Storage arbitrage** exploits price differences across time
- **Renewable integration** requires handling generation uncertainty
- **Building energy** optimization balances comfort with energy cost
- **Ramp constraints** limit how fast generators can change output
- **Reserve requirements** ensure reliability under uncertainty

21.8. Exercises

1. Extend the economic dispatch model to include transmission losses using a B-coefficient loss formula.
2. Implement a full MIP formulation for unit commitment with minimum up/down time constraints.
3. Add demand response to the storage optimization: allow shifting of flexible loads.
4. Implement robust unit commitment that guarantees feasibility under worst-case renewable scenarios.
5. Design a model predictive control (MPC) approach for real-time building energy management with forecast updates.

Part V.

Part V: Advanced Topics

22. Decision Making Under Uncertainty

What You Will Learn

- Model uncertainty with scenarios and probability distributions
- Formulate two-stage and multi-stage stochastic programs
- Apply robust optimization for worst-case protection
- Implement Sample Average Approximation (SAA)
- Use decomposition methods (L-Shaped, Progressive Hedging)
- Apply distributionally robust optimization when distributions are uncertain
- Implement variance reduction techniques for Monte Carlo sampling
- Understand coherent risk measures and mean-risk optimization

Real-world decisions are made in the face of uncertainty: Will demand be high or low? What will fuel prices be? How will the weather affect crop yields? Stochastic programming provides the framework to answer: “What decisions should I make now to optimize expected outcomes across all possible future scenarios?”

22.1. The Challenge of Uncertainty

22.1.1. Types of Uncertainty

Type	Description	Examples
Demand	Unknown future consumption	Sales, power demand
Supply	Unreliable deliveries	Yields, lead times
Price	Market fluctuations	Commodities, currencies
Technology	Performance variability	Equipment failure
Regulation	Policy changes	Emissions limits, tariffs

Different types of uncertainty call for different probability distributions. Demand is often modeled as normal, supply yields as beta, prices as lognormal (reflecting non-negativity and right skew), and lead times as gamma. When historical data suggests multiple modes, mixture distributions are appropriate. Discrete scenarios are used when only a few well-defined outcomes are plausible.

22.1.2. Traditional Approaches Fall Short

Expected Value Approach: Optimize using average parameters—ignores variability, can produce infeasible solutions.

Worst-Case (Robust): Too conservative, expensive, protects against unrealistic scenarios.

Scenario Analysis: Manual “what-if” cases—doesn’t optimize across scenarios systematically.

22.2. Two-Stage Stochastic Programming

The fundamental stochastic programming model separates decisions into two stages:

- **First stage (here-and-now):** Decisions made before uncertainty reveals
- **Second stage (wait-and-see):** Recourse actions after uncertainty resolves

22. Decision Making Under Uncertainty

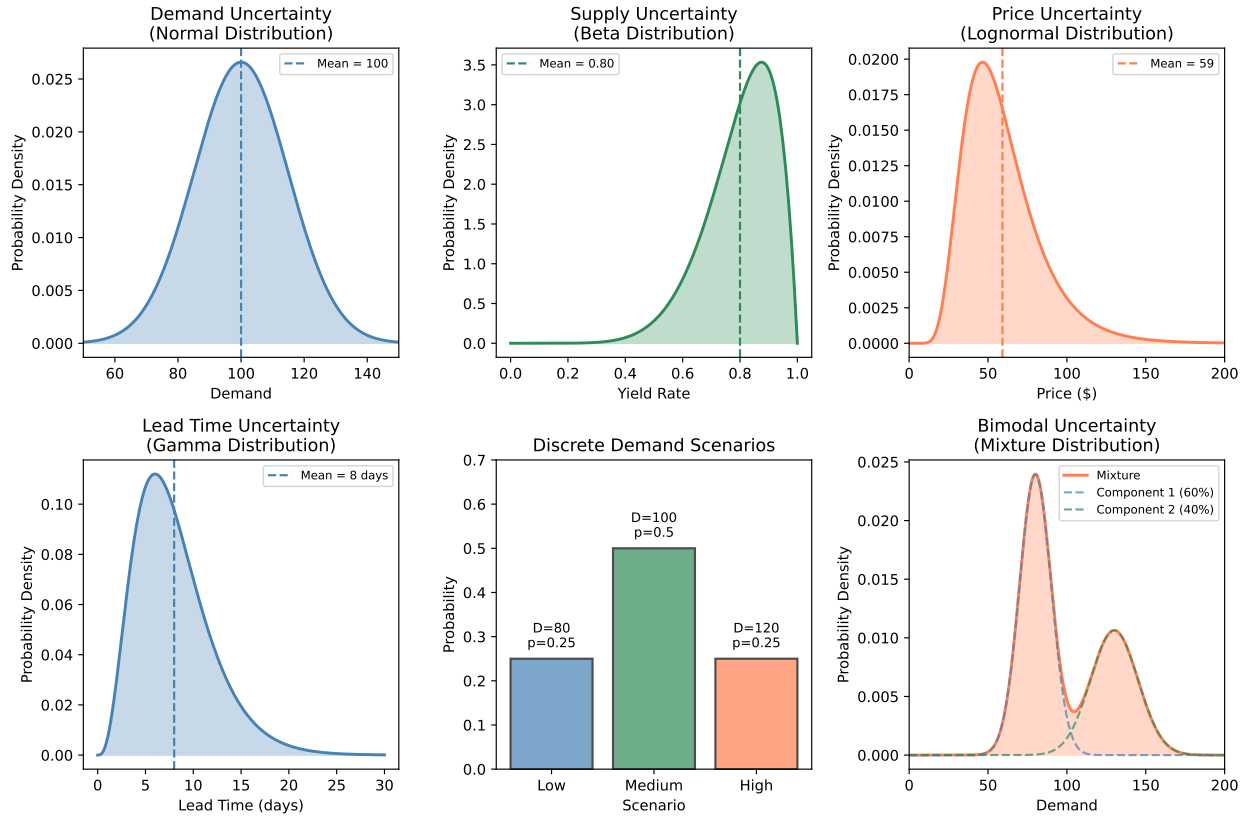


Figure 22.1.: Common probability distributions for modeling parameter uncertainty in stochastic programming. Each distribution captures different characteristics of uncertainty.

22.2.1. Mathematical Formulation

$$\begin{aligned} & \text{minimize} && \mathbf{c}^T \mathbf{x} + \mathbb{E}_{\xi}[Q(\mathbf{x}, \xi)] \\ & \text{subject to} && A\mathbf{x} = \mathbf{b} \\ & && \mathbf{x} \geq 0 \end{aligned}$$

where the recourse function is:

$$Q(\mathbf{x}, \xi) = \min\{\mathbf{q}(\xi)^T \mathbf{y} : W(\xi)\mathbf{y} = \mathbf{h}(\xi) - T(\xi)\mathbf{x}, \mathbf{y} \geq 0\}$$

i Key Properties

- **Convexity:** $Q(\mathbf{x}, \xi)$ is convex in $\mathbf{x}$ for fixed ξ
- **Recourse:** “Complete recourse” means second-stage is always feasible
- **Non-anticipativity:** First-stage $\mathbf{x}$ cannot depend on future realizations of ξ

22.2.2. Scenario-Based Formulation

With discrete scenarios $\xi_1, \dots, \xi_S$ with probabilities $p_1, \dots, p_S$:

$$\begin{aligned}
& \text{minimize} && \mathbf{c}^T \mathbf{x} + \sum_{s=1}^S p_s \mathbf{q}_s^T \mathbf{y}_s \\
& \text{subject to} && A\mathbf{x} = \mathbf{b} \\
& && T_s \mathbf{x} + W_s \mathbf{y}_s = \mathbf{h}_s, \quad \forall s \\
& && \mathbf{x} \geq 0, \mathbf{y}_s \geq 0, \quad \forall s
\end{aligned}$$

This Deterministic Equivalent Program (DEP) is large but solvable with standard LP solvers.

22.2.3. Example: Production Planning

A factory must decide how much to produce before knowing actual demand. The first-stage decision is the production quantity. The second-stage recourse actions handle excess inventory or unmet demand by paying the corresponding penalty costs. The following implementation solves this two-stage problem using `scipy.optimize.linprog`.

```

import numpy as np
from scipy.optimize import linprog

def two_stage_production(c_prod, c_short, c_excess, capacity,
                        demands, probabilities):
    """Two-stage stochastic production planning.

    Stage 1: Decide production quantity x
    Stage 2: Meet demand, pay shortage/excess costs

    Args:
        c_prod: Production cost per unit
        c_short: Shortage cost per unit
        c_excess: Excess inventory cost per unit
        capacity: Production capacity
        demands: List of demand scenarios
        probabilities: Scenario probabilities

    Returns:
        Optimal production, expected cost
    """
    n_scenarios = len(demands)

    # Variables: x (production), y_s+ (excess), y_s- (shortage) for each s
    # Total: 1 + 2*n_scenarios variables
    n_vars = 1 + 2 * n_scenarios

    # Objective: c_prod*x + sum(p_s * (c_excess*y_s+ + c_short*y_s-))
    c = np.zeros(n_vars)
    c[0] = c_prod
    for s in range(n_scenarios):
        c[1 + s] = probabilities[s] * c_excess # excess
        c[1 + n_scenarios + s] = probabilities[s] * c_short # shortage

    # Constraints: x - y_s+ + y_s- = d_s for each scenario
    A_eq = np.zeros((n_scenarios, n_vars))
    b_eq = np.array(demands)

```

22. Decision Making Under Uncertainty

```

for s in range(n_scenarios):
    A_eq[s, 0] = 1 # x
    A_eq[s, 1 + s] = -1 # -y_s+ (excess)
    A_eq[s, 1 + n_scenarios + s] = 1 # +y_s- (shortage)

# Bounds: 0 <= x <= capacity, y >= 0
bounds = [(0, capacity)] + [(0, None)] * (2 * n_scenarios)

result = linprog(c, A_eq=A_eq, b_eq=b_eq, bounds=bounds)

return result.x[0], result.fun

# Example
demands = [80, 100, 120] # Low, medium, high demand
probs = [0.3, 0.4, 0.3]
x_opt, cost = two_stage_production(
    c_prod=10, c_short=30, c_excess=5,
    capacity=150, demands=demands, probabilities=probs
)
print(f"Optimal production: {x_opt:.1f}")
print(f"Expected cost: {cost:.2f}")

```

The Value of the Stochastic Solution (VSS) quantifies the benefit of using stochastic programming over a deterministic approach that substitutes expected values for uncertain parameters. A positive VSS indicates that modeling uncertainty explicitly leads to lower expected costs.

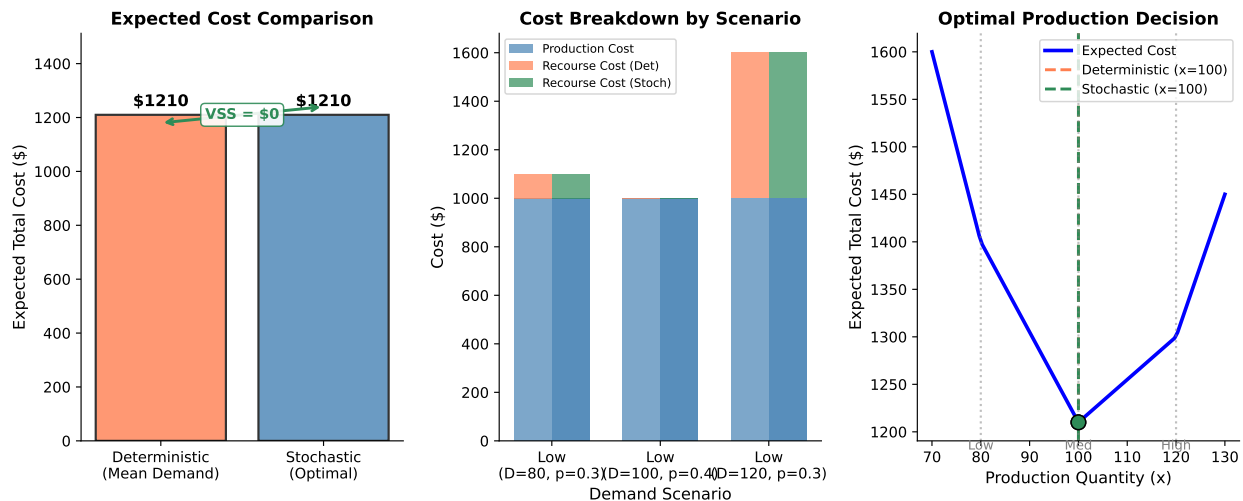


Figure 22.2.: Value of the Stochastic Solution (VSS) comparison. The stochastic solution accounts for all possible scenarios, yielding lower expected costs compared to the deterministic approach that uses only the mean demand.

Value of Stochastic Solution (VSS): \$0.00

This represents 0.0% savings from using stochastic optimization.

22.3. Sample Average Approximation

When the number of scenarios is large or continuous, Sample Average Approximation (SAA) provides a tractable approach.

22.3.1. The SAA Method

Replace expectation with sample average:

$$\mathbb{E}_{\xi}[Q(\mathbf{x}, \xi)] \approx \frac{1}{N} \sum_{n=1}^N Q(\mathbf{x}, \xi^n)$$

where $\xi^1, \dots, \xi^N$ are sampled scenarios. The SAA approach solves multiple independent approximations using different random samples and selects the best solution. This replication strategy provides statistical quality bounds on the optimality gap. The following implementation generates multiple SAA problems, solves each, and evaluates the best solution on a larger out-of-sample set.

```
import numpy as np

def sample_average_approximation(problem, n_samples, n_replicates=10):
    """Sample Average Approximation for stochastic programming.

    Args:
        problem: Problem with sample_scenarios() and solve_deterministic() methods
        n_samples: Number of scenarios per SAA problem
        n_replicates: Number of independent SAA problems to solve

    Returns:
        Best solution, statistical bounds
    """
    solutions = []
    objectives = []

    for rep in range(n_replicates):
        # Sample scenarios
        scenarios = problem.sample_scenarios(n_samples)

        # Solve SAA problem
        x, obj = problem.solve_deterministic(scenarios)
        solutions.append(x)
        objectives.append(obj)

    # Select best solution
    best_idx = np.argmin(objectives)
    best_solution = solutions[best_idx]

    # Evaluate on larger out-of-sample set
    test_scenarios = problem.sample_scenarios(n_samples * 10)
    true_obj = problem.evaluate(best_solution, test_scenarios)

    return {
        'solution': best_solution,
        'saa_objective': objectives[best_idx],
        'true_objective': true_obj,
```

```

    'objective_std': np.std(objectives),
    'gap_estimate': true_obj - np.mean(objectives)
}

```

22.3.2. Statistical Guarantees

As $N \rightarrow \infty$, SAA solution converges to true SP solution by the Law of Large Numbers.

Sample Size	Typical Gap	Use Case
50-100	5-10%	Quick screening
100-500	1-5%	Standard practice
500-2000	< 1%	High-stakes decisions

22.4. Variance Reduction Techniques

Monte Carlo sampling for SAA can be slow to converge due to high variance. Variance reduction techniques improve convergence by reducing the variance of estimators without increasing the number of samples. These techniques are especially valuable when each simulation is computationally expensive, because they extract more information from the same number of samples.

22.4.1. Importance Sampling

Importance sampling changes the sampling distribution to reduce variance. Instead of sampling from the original distribution $p(\xi)$, we sample from a proposal distribution $q(\xi)$ and reweight:

$$\mathbb{E}_p[Q(\mathbf{x}, \xi)] = \mathbb{E}_q \left[Q(\mathbf{x}, \xi) \frac{p(\xi)}{q(\xi)} \right]$$

The optimal proposal distribution concentrates samples in regions where $|Q(\mathbf{x}, \xi)|$ is large. The ratio $p(\xi)/q(\xi)$ is called the likelihood ratio or importance weight. The following code implements importance sampling for a newsvendor problem where rare high-demand events dominate the cost.

```

import numpy as np
from scipy import stats

def importance_sampling_estimate(Q_func, x, p_dist, q_dist, n_samples):
    """Estimate expected value using importance sampling.

    Args:
        Q_func: Function Q(x, xi) to evaluate
        x: Decision variables
        p_dist: Original distribution (scipy.stats distribution)
        q_dist: Proposal distribution (scipy.stats distribution)
        n_samples: Number of samples

    Returns:
        Estimate and standard error
    """
    # Sample from proposal distribution
    xi_samples = q_dist.rvs(size=n_samples)

    # Compute importance weights

```

```

weights = p_dist.pdf(xi_samples) / q_dist.pdf(xi_samples)

# Evaluate Q at samples
Q_values = np.array([Q_func(x, xi) for xi in xi_samples])

# Weighted estimate
estimate = np.mean(weights * Q_values)

# Standard error
std_error = np.std(weights * Q_values) / np.sqrt(n_samples)

return estimate, std_error

# Example: Newsvendor with rare high-demand events
def newsvendor_cost(x, demand, c_under=10, c_over=3):
    """Cost function for newsvendor problem."""
    if demand > x:
        return c_under * (demand - x)
    else:
        return c_over * (x - demand)

# Original: Normal(100, 20), Proposal: shifted to capture high demands
p_original = stats.norm(100, 20)
q_proposal = stats.norm(110, 25) # Shifted right, wider

x_decision = 105
n_samples = 1000

# Standard Monte Carlo
xi_standard = p_original.rvs(size=n_samples)
Q_standard = [newsvendor_cost(x_decision, xi) for xi in xi_standard]
mc_estimate = np.mean(Q_standard)
mc_se = np.std(Q_standard) / np.sqrt(n_samples)

# Importance sampling
is_estimate, is_se = importance_sampling_estimate(
    lambda x, xi: newsvendor_cost(x, xi),
    x_decision, p_original, q_proposal, n_samples
)

print(f"Monte Carlo: {mc_estimate:.2f} +/- {mc_se:.2f}")
print(f"Importance Sampling: {is_estimate:.2f} +/- {is_se:.2f}")

```

22.4.2. Control Variates

Control variates exploit the correlation between the target quantity and an auxiliary random variable whose expectation is known analytically. If the control variate is strongly correlated with the target, the variance reduction can be substantial. Control variates use correlated auxiliary random variables with known expectations to reduce variance:

$$\hat{Q}_{CV} = \frac{1}{N} \sum_{n=1}^N [Q(\mathbf{x}, \xi^n) - \beta(Z(\xi^n) - \mathbb{E}[Z])]$$

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where Z is the control variate and β is chosen to minimize variance.

```
import numpy as np

def control_variate_estimate(Q_func, Z_func, E_Z, x, samples):
    """Estimate using control variates.

    Args:
        Q_func: Primary function to estimate
        Z_func: Control variate function
        E_Z: Known expected value of Z
        x: Decision variables
        samples: Array of scenario samples

    Returns:
        Estimate with reduced variance
    """
    Q_values = np.array([Q_func(x, xi) for xi in samples])
    Z_values = np.array([Z_func(xi) for xi in samples])

    # Optimal beta (minimizes variance)
    cov_QZ = np.cov(Q_values, Z_values)[0, 1]
    var_Z = np.var(Z_values)
    beta_opt = cov_QZ / var_Z if var_Z > 0 else 0

    # Control variate estimate
    cv_estimate = np.mean(Q_values - beta_opt * (Z_values - E_Z))

    # Variance reduction factor
    corr = np.corrcoef(Q_values, Z_values)[0, 1]
    var_reduction = 1 - corr**2

    return {
        'estimate': cv_estimate,
        'beta': beta_opt,
        'correlation': corr,
        'variance_reduction': var_reduction
    }

# Example: Use demand as control variate for inventory cost
np.random.seed(42)
demands = np.random.normal(100, 20, 1000)
x_stock = 110

def inventory_cost(x, d):
    return 10 * max(d - x, 0) + 3 * max(x - d, 0)

result = control_variate_estimate(
    Q_func=lambda x, d: inventory_cost(x, d),
    Z_func=lambda d: d, # Demand itself as control variate
    E_Z=100, # Known expected demand
    x=x_stock,
    samples=demands
)
```

```

)
print(f"CV Estimate: {result['estimate']:.2f}")
print(f"Correlation: {result['correlation']:.3f}")
print(f"Variance Reduction: {1-result['variance_reduction']:.1%}")

```

22.4.3. Antithetic Variates

Antithetic variates use negatively correlated pairs of samples to reduce variance. If U is uniform, then U and $1 - U$ are negatively correlated:

$$\hat{Q}_{AV} = \frac{1}{2} [Q(\mathbf{x}, F^{-1}(U)) + Q(\mathbf{x}, F^{-1}(1 - U))]$$

```

import numpy as np
from scipy import stats

def antithetic_variates_estimate(Q_func, x, dist, n_pairs):
    """Estimate using antithetic variates.

    Args:
        Q_func: Function to estimate
        x: Decision variables
        dist: Distribution (scipy.stats)
        n_pairs: Number of antithetic pairs

    Returns:
        Estimate and variance comparison
    """
    # Generate uniform samples
    U = np.random.uniform(0, 1, n_pairs)

    # Antithetic pairs via inverse CDF
    xi_1 = dist.ppf(U)
    xi_2 = dist.ppf(1 - U)

    # Evaluate Q at both
    Q_1 = np.array([Q_func(x, xi) for xi in xi_1])
    Q_2 = np.array([Q_func(x, xi) for xi in xi_2])

    # Antithetic estimate
    Q_av = (Q_1 + Q_2) / 2
    av_estimate = np.mean(Q_av)
    av_var = np.var(Q_av) / n_pairs

    # Compare with standard MC (same total samples)
    xi_mc = dist.rvs(size=2 * n_pairs)
    Q_mc = np.array([Q_func(x, xi) for xi in xi_mc])
    mc_var = np.var(Q_mc) / (2 * n_pairs)

    return {
        'estimate': av_estimate,
        'av_variance': av_var,
        'mc_variance': mc_var,
    }

```

```

        'variance_ratio': av_var / mc_var
    }

# Example with convex cost function
def convex_cost(x, xi):
    return (xi - x)**2

result = antithetic_variates_estimate(
    Q_func=convex_cost,
    x=100,
    dist=stats.norm(100, 20),
    n_pairs=500
)
print(f"Estimate: {result['estimate']:.2f}")
print(f"Variance Ratio (AV/MC): {result['variance_ratio']:.3f}")

```

💡 When Antithetic Variates Work Best

Antithetic variates are most effective when $Q(\mathbf{x}, \xi)$ is monotonic in ξ . For non-monotonic functions, the variance reduction can be minimal or even negative.

22.4.4. Stratified Sampling

Stratified sampling divides the sample space into non-overlapping regions (strata) and draws samples from each stratum separately. This guarantees coverage across the entire distribution and eliminates the chance of clustered samples. The approach works best when the function varies significantly across strata but is relatively smooth within each stratum. The overall estimate is a weighted average of stratum-level estimates:

$$\hat{Q}_{SS} = \sum_{k=1}^K p_k \cdot \frac{1}{n_k} \sum_{i=1}^{n_k} Q(\mathbf{x}, \xi_k^i)$$

where p_k is the probability mass of stratum k .

```

import numpy as np
from scipy import stats

def stratified_sampling_estimate(Q_func, x, dist, n_strata, n_samples):
    """Stratified sampling for variance reduction.

    Args:
        Q_func: Function to estimate
        x: Decision variables
        dist: Distribution (scipy.stats)
        n_strata: Number of strata
        n_samples: Total number of samples

    Returns:
        Stratified estimate
    """

```



```

samples_per_stratum = n_samples // n_strata

# Define strata boundaries (equal probability)
stratum_bounds = np.linspace(0, 1, n_strata + 1)

estimates = []
for k in range(n_strata):
    # Sample uniformly within stratum
    U_stratum = np.random.uniform(
        stratum_bounds[k],
        stratum_bounds[k + 1],
        samples_per_stratum
    )

    # Transform to original distribution
    xi_stratum = dist.ppf(U_stratum)

    # Evaluate and average within stratum
    Q_stratum = np.array([Q_func(x, xi) for xi in xi_stratum])
    estimates.append(np.mean(Q_stratum))

# Equal-weight strata (proportional allocation)
stratified_estimate = np.mean(estimates)
stratified_var = np.var(estimates) / n_strata

# Compare with simple random sampling
xi_srs = dist.rvs(size=n_samples)
Q_srs = np.array([Q_func(x, xi) for xi in xi_srs])
srs_var = np.var(Q_srs) / n_samples

return {
    'estimate': stratified_estimate,
    'stratified_variance': stratified_var,
    'srs_variance': srs_var,
    'efficiency_gain': srs_var / stratified_var
}

# Example
result = stratified_sampling_estimate(
    Q_func=lambda x, xi: max(xi - x, 0) * 10, # Shortage cost
    x=100,
    dist=stats.norm(100, 20),
    n_strata=10,
    n_samples=1000
)
print(f"Stratified Estimate: {result['estimate']:.2f}")
print(f"Efficiency Gain: {result['efficiency_gain']:.2f}x")

```

22.4.5. Latin Hypercube Sampling

Latin Hypercube Sampling (LHS) ensures coverage across the entire sample space, particularly valuable in high dimensions. LHS divides each dimension into N equal-probability intervals and generates exactly one sample per interval in each dimension. This produces better space-filling

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properties than simple random sampling, especially when the number of dimensions is large relative to the sample size.

```
import numpy as np
from scipy import stats
from scipy.stats import qmc

def latin_hypercube_sampling(Q_func, x, distributions, n_samples):
    """Latin Hypercube Sampling for multi-dimensional uncertainty.

    Args:
        Q_func: Function Q(x, xi) where xi is a vector
        x: Decision variables
        distributions: List of scipy.stats distributions
        n_samples: Number of LHS samples

    Returns:
        Estimate and samples
    """
    n_dims = len(distributions)

    # Generate LHS samples in [0,1]^d
    sampler = qmc.LatinHypercube(d=n_dims)
    lhs_uniform = sampler.random(n=n_samples)

    # Transform each dimension to its distribution
    xi_samples = np.zeros((n_samples, n_dims))
    for j, dist in enumerate(distributions):
        xi_samples[:, j] = dist.ppf(lhs_uniform[:, j])

    # Evaluate Q at each sample
    Q_values = np.array([Q_func(x, xi) for xi in xi_samples])

    estimate = np.mean(Q_values)
    std_error = np.std(Q_values) / np.sqrt(n_samples)

    return {
        'estimate': estimate,
        'std_error': std_error,
        'samples': xi_samples
    }

# Example: Multi-product newsvendor
def multi_product_cost(x, demands):
    """Cost for multi-product inventory."""
    c_under = np.array([10, 15, 8]) # Shortage costs
    c_over = np.array([2, 3, 1])    # Holding costs

    shortage = np.maximum(demands - x, 0)
    excess = np.maximum(x - demands, 0)

    return np.sum(c_under * shortage + c_over * excess)
```

```

distributions = [
    stats.norm(100, 15), # Product 1 demand
    stats.norm(80, 20),  # Product 2 demand
    stats.norm(120, 10), # Product 3 demand
]

x_stock = np.array([105, 85, 122])

result = latin_hypercube_sampling(
    Q_func=multi_product_cost,
    x=x_stock,
    distributions=distributions,
    n_samples=500
)
print(f"Expected Cost: {result['estimate']:.2f} +/- {result['std_error']:.2f}")

```

i Variance Reduction Summary

Technique	Best For	Variance Reduction
Importance Sampling	Rare events, tail estimation	10x-1000x possible
Control Variates	Correlated known quantities	2x-10x typical
Antithetic Variates	Monotonic functions	2x-5x typical
Stratified Sampling	Low-dimensional problems	2x-10x typical
Latin Hypercube	High-dimensional problems	5x-50x typical

Techniques can be combined for additional variance reduction.

22.5. Multi-Stage Stochastic Programs

Two-stage models capture a single decision point followed by recourse. Many real problems, however, involve a sequence of decisions interleaved with information arrivals. Multi-stage stochastic programming extends the two-stage framework to T time periods, where each stage's decisions can depend on all uncertainty observed up to that point. The key constraint is non-anticipativity: decisions at stage t cannot use information from stages $t + 1, t + 2, \dots, T$.

For sequential decision-making over multiple time periods:

$$\text{minimize} \quad \mathbb{E} \left[\sum_{t=1}^T \mathbf{c}_t^T \mathbf{x}_t(\xi_{[t]}) \right]$$

where $\mathbf{x}_t$ depends only on information available at stage t .

⚠ Curse of Dimensionality

Multi-stage problems grow exponentially:

- 3 stages, 10 scenarios each: $10^3 = 1,000$ scenarios
- 5 stages, 10 scenarios each: $10^5 = 100,000$ scenarios
- 10 stages, 5 scenarios each: $5^{10} \approx 10$ million scenarios

Requires advanced decomposition methods and approximations.

The scenario tree below illustrates a three-stage problem where demand evolves over time. The root node represents the initial decision point. At each subsequent stage, uncertainty branches into multiple scenarios, and recourse decisions adapt to the observed outcome. Each complete path from root to leaf represents one possible realization of the uncertain process.

22.5.1. Stochastic Dual Dynamic Programming (SDDP)

Stochastic Dual Dynamic Programming is the leading algorithm for solving large-scale multi-stage stochastic linear programs, particularly those with many stages where scenario tree enumeration is impossible.

22.5.1.1. The Value Function Approximation

SDDP approximates the expected cost-to-go (value function) at each stage using piecewise linear cutting planes:

$$Q_{t+1}(\mathbf{x}_t) = \mathbb{E}[Q_{t+1}(\mathbf{x}_t, \xi_{t+1})]$$

The key insight is that Q_{t+1} is convex (for minimization problems) and can be approximated from below by linear cuts:

$$Q_{t+1}(\mathbf{x}_t) \approx \max_{k=1, \dots, K} \{\alpha_k + \beta_k^T \mathbf{x}_t\}$$

where each cut (α_k, β_k) is generated from dual information.

22.5.1.2. Forward and Backward Passes

SDDP iterates between two passes:

Forward Pass: Simulate decisions forward through the scenario tree

1. Start at stage 1 with given initial state
2. At each stage t :
 - Sample a realization ξ_t
 - Solve stage problem using current cuts
 - Record state $\mathbf{x}_t$
3. Compute upper bound from total cost

Backward Pass: Update cuts by moving backward through stages

1. Start at final stage T
2. At each stage t from $T - 1$ to 1:
 - For each sampled state from forward pass

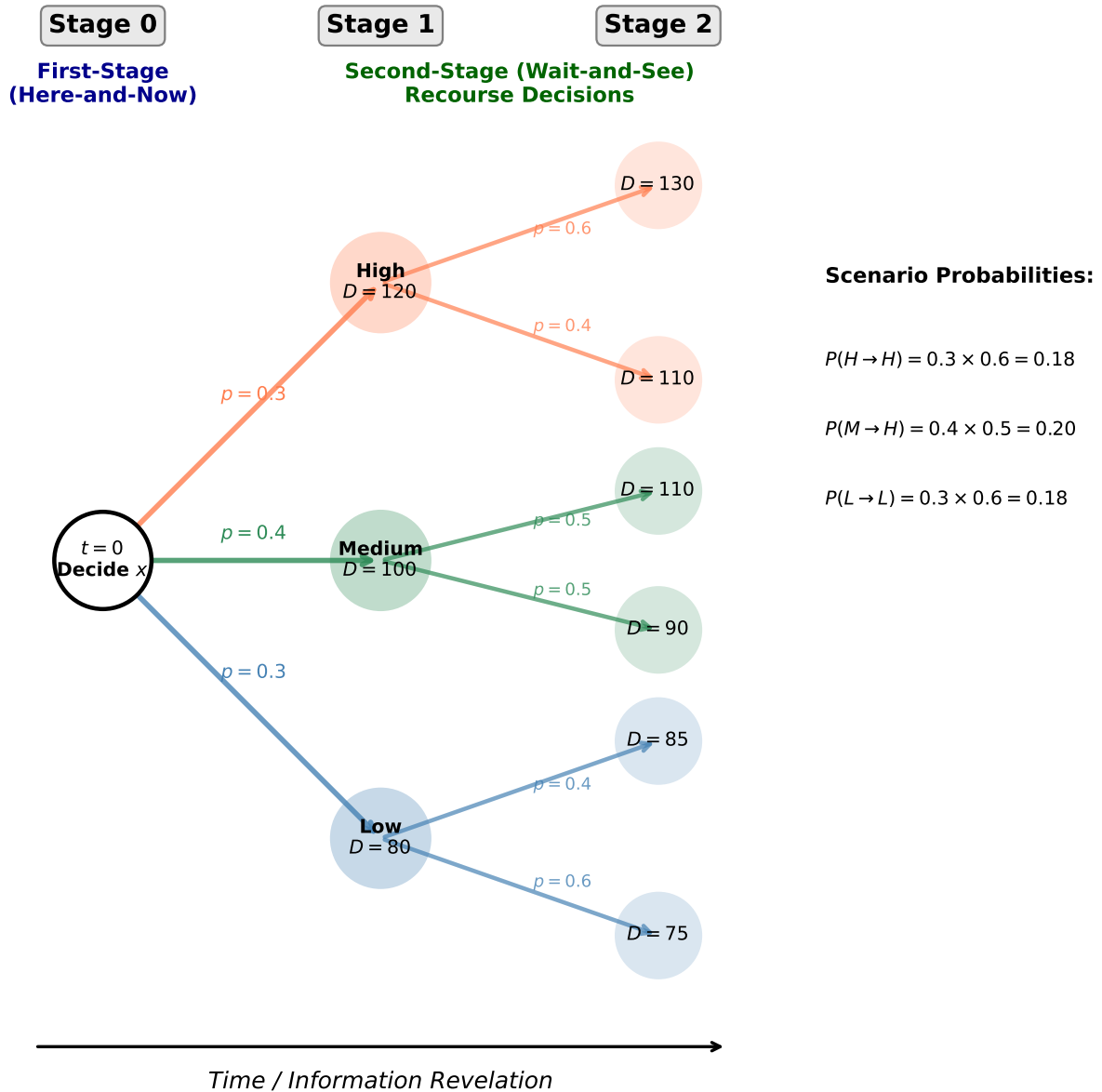


Figure 22.3.: Multi-stage stochastic programming scenario tree. Each path from root to leaf represents a possible realization of uncertainty over time. Node labels show demand values, and edge labels indicate transition probabilities.

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- For each possible realization ξ_{t+1}
- Solve stage $t + 1$ problem, get dual π_{t+1}
- Add cut: $\theta_t \geq \mathbb{E}[\pi_{t+1}^T(h_{t+1} - T_{t+1}\mathbf{x}_t)]$

3. Update lower bound

The following implementation shows a simplified SDDP solver applied to an inventory management problem. The forward pass simulates decisions under sampled scenarios. The backward pass generates Benders cuts from dual information at each stage, progressively tightening the cost-to-go approximation.

```
import numpy as np
from scipy.optimize import linprog

class SDDPSolver:
    """Simplified SDDP implementation for educational purposes."""

    def __init__(self, n_stages, n_scenarios_per_stage):
        self.n_stages = n_stages
        self.n_scenarios = n_scenarios_per_stage
        self.cuts = {t: [] for t in range(n_stages)}

    def solve_stage_problem(self, t, x_prev, xi, cuts):
        """Solve single stage problem with cuts.

        Returns:
            x: State decision
            theta: Cost-to-go approximation
            dual: Dual multipliers for cut generation
            cost: Stage cost
        """
        # Example: inventory problem
        # minimize: c_t * x_t + h * I^+ + b * I^- + theta
        # subject to: I^+ - I^- = x_prev + x_t - d(xi)
        #               cuts on theta

        n_vars = 4 # x, I+, I-, theta

        # Costs
        c_order = 5
        c_hold = 2
        c_backlog = 10

        c = np.array([c_order, c_hold, c_backlog, 1])

        # Inventory balance
        demand = xi['demand']
        A_eq = np.array([[1, 1, -1, 0]]) # x + I+ - I- = demand - x_prev
        b_eq = np.array([demand - x_prev])

        # Cut constraints: theta >= alpha + beta * x
        A_ub = []
        b_ub = []
        for alpha, beta in cuts:
```

```

        # -theta + beta * x <= -alpha
        A_ub.append([beta, 0, 0, -1])
        b_ub.append(-alpha)

    if A_ub:
        A_ub = np.array(A_ub)
        b_ub = np.array(b_ub)
    else:
        A_ub = None
        b_ub = None

    # Bounds
    bounds = [(0, 100), (0, None), (0, None), (0, None)]

    result = linprog(c, A_ub=A_ub, b_ub=b_ub, A_eq=A_eq, b_eq=b_eq,
                    bounds=bounds, method='highs')

    x_new = x_prev + result.x[0] - demand + result.x[1] - result.x[2]

    return {
        'x': x_new,
        'decision': result.x[0],
        'theta': result.x[3],
        'dual': result.eqlin.marginals[0] if hasattr(result, 'eqlin') else 0,
        'cost': c_order * result.x[0] + c_hold * result.x[1] + c_backlog *
        ↪ result.x[2]
    }

def forward_pass(self, scenarios):
    """Forward simulation to get trial points."""
    x_trial = [0] # Initial inventory
    costs = []

    for t in range(self.n_stages):
        xi = scenarios[t]
        result = self.solve_stage_problem(t, x_trial[-1], xi, self.cuts[t])
        x_trial.append(result['x'])
        costs.append(result['cost'] + result['theta'])

    return x_trial, sum(costs)

def backward_pass(self, x_trial, all_scenarios):
    """Backward pass to generate cuts."""
    for t in range(self.n_stages - 1, 0, -1):
        x_prev = x_trial[t]

        # Compute cut coefficients
        alpha_sum = 0
        beta_sum = 0

        for xi in all_scenarios[t]:
            result = self.solve_stage_problem(t, x_prev, xi, self.cuts[t])
            prob = 1.0 / len(all_scenarios[t])

```

```

        alpha_sum += prob * (result['cost'] + result['theta'] -
                               result['dual'] * x_prev)
        beta_sum += prob * result['dual']

    # Add cut to previous stage
    self.cuts[t-1].append((alpha_sum, beta_sum))

def solve(self, all_scenarios, max_iter=100, tol=1e-4):
    """Run SDDP algorithm."""
    lower_bounds = []
    upper_bounds = []

    for iteration in range(max_iter):
        # Forward pass with sampled scenarios
        sampled = [np.random.choice(all_scenarios[t])
                   for t in range(self.n_stages)]
        x_trial, ub = self.forward_pass(sampled)
        upper_bounds.append(ub)

        # Backward pass
        self.backward_pass(x_trial, all_scenarios)

        # Lower bound from first stage
        result = self.solve_stage_problem(0, 0, all_scenarios[0][0],
                                           self.cuts[0])

        lb = result['cost'] + result['theta']
        lower_bounds.append(lb)

        # Convergence check
        if iteration > 10:
            gap = (np.mean(upper_bounds[-10:]) - lb) / abs(lb)
            if gap < tol:
                break

    return {
        'lower_bound': lower_bounds[-1],
        'upper_bound': np.mean(upper_bounds[-10:]),
        'iterations': iteration + 1
    }

```

i When to Use SDDP

SDDP is ideal when:

- Problem has many stages ($T > 5$)
- Stage-wise independence holds (or can be approximated)
- Problem is convex (linear or convex quadratic)
- State space is continuous or large discrete

SDDP is **not** suitable when:

- Problem has integer variables (use SDDiP extensions)

- Uncertainty is not stagewise independent
- Problem is non-convex

22.5.1.3. Software for SDDP

For production use, consider established implementations:

Package	Language	Features
SDDP.jl	Julia	Production-grade, parallel, risk measures
StochOptFormat	Julia	Standardized problem format
msppy	Python	Educational, multiple algorithms
QUASAR	Commercial	Industrial-scale problems

22.6. Decomposition Methods

Large-scale stochastic programs can have millions of variables when all scenarios are modeled explicitly. Decomposition methods break the monolithic problem into smaller, manageable subproblems. They iterate between a master problem that makes first-stage decisions and scenario subproblems that evaluate recourse, exchanging information through cuts or price signals.

22.6.1. L-Shaped Method (Benders Decomposition)

The L-Shaped method is the standard decomposition algorithm for two-stage stochastic linear programs. It decomposes the problem by scenarios, solving each scenario subproblem independently and using dual information to construct optimality cuts that progressively tighten the master problem's approximation of the recourse function. The algorithm converges when the gap between upper and lower bounds falls below a tolerance.

```
def l_shaped_method(c, A, b, scenarios, probabilities,
                   tol=1e-6, max_iter=100):
    """L-Shaped method for two-stage stochastic LP.

    Args:
        c, A, b: First-stage problem data
        scenarios: List of (q, T, W, h) tuples for each scenario
        probabilities: Scenario probabilities
        tol: Convergence tolerance
        max_iter: Maximum iterations

    Returns:
        Optimal first-stage solution
    """
    n_x = len(c)
    n_scenarios = len(scenarios)

    # Initialize
    lower_bound = -np.inf
    upper_bound = np.inf
    theta = 0 # Approximation of recourse

    # Optimality cuts: E_k * x + e_k <= theta
    cuts_E = []
```

```

cuts_e = []

for iteration in range(max_iter):
    # Solve master problem
    x, theta_val = solve_master(c, A, b, cuts_E, cuts_e)
    lower_bound = c @ x + theta_val

    # Solve subproblems for each scenario
    recourse_value = 0
    cut_E = np.zeros(n_x)
    cut_e = 0

    for s, (q, T, W, h) in enumerate(scenarios):
        # Subproblem: min q'y s.t. Wy = h - Tx, y >= 0
        y, dual = solve_subproblem(q, T, W, h, x)

        recourse_value += probabilities[s] * (q @ y)

        # Accumulate cut coefficients from duals
        cut_E += probabilities[s] * (dual @ T)
        cut_e += probabilities[s] * (dual @ h)

    upper_bound = min(upper_bound, c @ x + recourse_value)

    # Check convergence
    if upper_bound - lower_bound < tol:
        break

    # Add optimality cut
    cuts_E.append(cut_E)
    cuts_e.append(cut_e)

return x, upper_bound

```

22.6.2. Progressive Hedging

Progressive Hedging takes a different decomposition approach from the L-Shaped method. Instead of separating the master from the subproblems, it assigns a copy of the first-stage variables to each scenario and penalizes disagreement. Each scenario subproblem optimizes independently, and a consensus step averages the solutions. Price-update terms (multipliers) push the scenario solutions toward agreement. The method decomposes by scenarios using augmented Lagrangian relaxation:

$$\min \mathbf{q}_s^T \mathbf{y}_s + \mathbf{w}_s^T \mathbf{x}_s + \frac{\rho}{2} \|\mathbf{x}_s - \bar{\mathbf{x}}\|^2$$

22.6.2.1. Convergence Conditions

Progressive Hedging converges under the following conditions:

1. **Convexity:** The underlying optimization problem must be convex
2. **Relatively complete recourse:** A feasible solution exists for all scenarios
3. **Bounded scenario solutions:** Optimal solutions for each scenario are bounded

For non-convex problems (e.g., mixed-integer), Progressive Hedging becomes a heuristic but often produces high-quality solutions.

22.6.2.2. Penalty Parameter Selection

The penalty parameter ρ critically affects convergence:

- **Too small:** Slow convergence, oscillation between scenario solutions
- **Too large:** Subproblems become ill-conditioned, numerical instability

Adaptive penalty update strategies:

```
def adaptive_penalty_update(rho, primal_residual, dual_residual,
                           mu=10, tau_incr=2, tau_decr=2):
    """Update penalty parameter based on residuals.

    Args:
        rho: Current penalty parameter
        primal_residual: ||x_s - x_bar||
        dual_residual: ||x_bar^{k+1} - x_bar^k||
        mu: Threshold ratio for updates
        tau_incr: Increase factor
        tau_decr: Decrease factor

    Returns:
        Updated penalty parameter
    """
    if primal_residual > mu * dual_residual:
        return rho * tau_incr # Increase to enforce consensus
    elif dual_residual > mu * primal_residual:
        return rho / tau_decr # Decrease to improve subproblem solutions
    else:
        return rho # Keep current value
```

Initial penalty selection:

A good starting point is:

$$\rho_0 = \frac{\|\mathbf{c}\|_2}{\sqrt{n}}$$

where $\mathbf{c}$ is the objective coefficient vector and n is the dimension.

22.6.2.3. Parallel Implementation

Progressive Hedging is naturally parallelizable since scenario subproblems are independent:

```
import numpy as np
from concurrent.futures import ProcessPoolExecutor, as_completed

class ProgressiveHedgingParallel:
    """Parallel Progressive Hedging implementation."""

    def __init__(self, problem, scenarios, probabilities, n_workers=None):
        """Initialize PH solver.

        Args:
            problem: Problem with solve_scenario method
            scenarios: List of scenario data
            probabilities: Scenario probabilities
```

```

        n_workers: Number of parallel workers
    """
    self.problem = problem
    self.scenarios = scenarios
    self.probabilities = np.array(probabilities)
    self.n_scenarios = len(scenarios)
    self.n_x = problem.n_first_stage
    self.n_workers = n_workers

def solve_scenario_task(self, args):
    """Task for parallel execution."""
    s, scenario, w, x_bar, rho = args
    x_s = self.problem.solve_scenario(scenario, w, x_bar, rho)
    return s, x_s

def solve(self, rho_init=1.0, max_iter=100, tol=1e-4,
          adaptive_rho=True, verbose=True):
    """Run parallel Progressive Hedging.

    Args:
        rho_init: Initial penalty parameter
        max_iter: Maximum iterations
        tol: Convergence tolerance
        adaptive_rho: Whether to adapt penalty parameter
        verbose: Print iteration info

    Returns:
        Consensus solution and convergence info
    """
    # Initialize
    x_s = [np.zeros(self.n_x) for _ in range(self.n_scenarios)]
    w_s = [np.zeros(self.n_x) for _ in range(self.n_scenarios)]
    rho = rho_init

    history = {
        'primal_residual': [],
        'dual_residual': [],
        'rho': [],
        'objective': []
    }

    x_bar_prev = np.zeros(self.n_x)

    for iteration in range(max_iter):
        # Compute weighted average (consensus point)
        x_bar = sum(self.probabilities[s] * x_s[s]
                    for s in range(self.n_scenarios))

        # Compute residuals
        primal_residual = np.sqrt(sum(
            self.probabilities[s] * np.linalg.norm(x_s[s] - x_bar)**2
            for s in range(self.n_scenarios)
        ))

```

```

dual_residual = rho * np.linalg.norm(x_bar - x_bar_prev)

history['primal_residual'].append(primal_residual)
history['dual_residual'].append(dual_residual)
history['rho'].append(rho)

if verbose and iteration % 10 == 0:
    print(f"Iter {iteration}: primal={primal_residual:.6f}, "
          f"dual={dual_residual:.6f}, rho={rho:.4f}")

# Check convergence
if primal_residual < tol and dual_residual < tol:
    if verbose:
        print(f"Converged at iteration {iteration}")
    break

# Solve scenario subproblems in parallel
with ProcessPoolExecutor(max_workers=self.n_workers) as executor:
    futures = []
    for s in range(self.n_scenarios):
        args = (s, self.scenarios[s], w_s[s], x_bar, rho)
        futures.append(executor.submit(self.solve_scenario_task, args))

    for future in as_completed(futures):
        s, x_new = future.result()
        x_s[s] = x_new

# Update multipliers
for s in range(self.n_scenarios):
    w_s[s] = w_s[s] + rho * (x_s[s] - x_bar)

# Adaptive penalty update
if adaptive_rho and iteration > 0:
    rho = adaptive_penalty_update(
        rho, primal_residual, dual_residual
    )

x_bar_prev = x_bar.copy()

# Final consensus
x_bar = sum(self.probabilities[s] * x_s[s]
            for s in range(self.n_scenarios))

return {
    'solution': x_bar,
    'iterations': iteration + 1,
    'converged': primal_residual < tol,
    'history': history
}

```

```

# Complete example: Two-stage production planning with PH
class ProductionPlanningProblem:

```

```

"""Two-stage production planning for Progressive Hedging."""

def __init__(self, n_products, capacity):
    self.n_products = n_products
    self.n_first_stage = n_products # Production quantities
    self.capacity = capacity

    # Costs
    self.c_prod = np.array([10, 12, 8])[:n_products]
    self.c_short = np.array([25, 30, 20])[:n_products]
    self.c_excess = np.array([3, 4, 2])[:n_products]

def solve_scenario(self, scenario, w, x_bar, rho):
    """Solve augmented scenario subproblem.

    Args:
        scenario: Dict with 'demand' array
        w: Lagrangian multiplier
        x_bar: Consensus point
        rho: Penalty parameter

    Returns:
        Optimal first-stage decision for this scenario
    """
    from scipy.optimize import minimize

    demand = scenario['demand']

    def objective(x):
        # Production cost
        cost = np.dot(self.c_prod, x)

        # Second stage costs
        shortage = np.maximum(demand - x, 0)
        excess = np.maximum(x - demand, 0)
        cost += np.dot(self.c_short, shortage)
        cost += np.dot(self.c_excess, excess)

        # Augmented Lagrangian terms
        cost += np.dot(w, x)
        cost += (rho / 2) * np.sum((x - x_bar)**2)

    return cost

# Simple gradient for this problem
def gradient(x):
    grad = self.c_prod.copy()

    # Subgradient of shortage/excess costs
    for i in range(len(x)):
        if x[i] < demand[i]:
            grad[i] -= self.c_short[i]
        else:

```

```

        grad[i] += self.c_excess[i]

        # Augmented Lagrangian gradient
        grad += w
        grad += rho * (x - x_bar)

    return grad

x0 = x_bar.copy()
bounds = [(0, self.capacity[i]) for i in range(self.n_products)]

result = minimize(objective, x0, method='L-BFGS-B',
                  jac=gradient, bounds=bounds)

return result.x

# Run example
np.random.seed(42)
n_products = 3
capacity = np.array([150, 120, 100])

problem = ProductionPlanningProblem(n_products, capacity)

# Generate scenarios
n_scenarios = 20
base_demand = np.array([100, 80, 60])
scenarios = [
    {'demand': base_demand * (0.8 + 0.4 * np.random.rand(n_products))}
    for _ in range(n_scenarios)
]
probabilities = np.ones(n_scenarios) / n_scenarios

# Solve (sequential version for demonstration)
def progressive_hedging_sequential(problem, scenarios, probabilities,
                                   rho=1.0, max_iter=100, tol=1e-4):
    """Sequential Progressive Hedging (for environments without multiprocessing)."""
    n_scenarios = len(scenarios)
    n_x = problem.n_first_stage

    x_s = [np.zeros(n_x) for _ in range(n_scenarios)]
    w_s = [np.zeros(n_x) for _ in range(n_scenarios)]

    for iteration in range(max_iter):
        x_bar = sum(probabilities[s] * x_s[s] for s in range(n_scenarios))

        deviation = sum(
            probabilities[s] * np.linalg.norm(x_s[s] - x_bar)**2
            for s in range(n_scenarios)
        )

        if deviation < tol:
            break

```

```

    for s in range(n_scenarios):
        x_s[s] = problem.solve_scenario(scenarios[s], w_s[s], x_bar, rho)

    for s in range(n_scenarios):
        w_s[s] = w_s[s] + rho * (x_s[s] - x_bar)

    return sum(probabilities[s] * x_s[s] for s in range(n_scenarios))

x_opt = progressive_hedging_sequential(problem, scenarios, probabilities, rho=2.0)
print(f"Optimal production: {x_opt}")

```

💡 Progressive Hedging Tips

1. **Warm starting:** Initialize $\mathbf{x}_s$ with scenario-specific deterministic solutions
2. **Cycle detection:** Monitor for oscillating solutions and increase ρ
3. **Termination:** Use both primal feasibility (consensus) and dual feasibility
4. **Bunching:** For many scenarios, use scenario bundling to reduce subproblems

22.7. Risk Measures

Minimizing expected cost treats all scenarios equally. In practice, decision makers are often risk-averse: they care disproportionately about bad outcomes. Risk measures provide a principled way to incorporate this aversion into the objective function. This section covers the standard risk measures used in stochastic programming and their mathematical properties.

Beyond expectation, incorporate risk aversion:

22.7.1. Value at Risk (VaR)

The α -quantile of the loss distribution:

$$\text{VaR}_\alpha(L) = \inf\{\ell : P(L \leq \ell) \geq \alpha\}$$

22.7.2. Conditional Value at Risk (CVaR)

Expected loss beyond VaR:

$$\text{CVaR}_\alpha(L) = \mathbb{E}[L | L \geq \text{VaR}_\alpha(L)]$$

CVaR is a coherent risk measure and can be optimized with linear programming.

22.7.3. Coherent Risk Measures

A risk measure $\rho : \mathcal{L} \rightarrow \mathbb{R}$ is **coherent** if it satisfies four axioms:

1. **Monotonicity:** If $X \leq Y$ a.s., then $\rho(X) \leq \rho(Y)$
2. **Translation invariance:** $\rho(X + c) = \rho(X) + c$ for constant c
3. **Positive homogeneity:** $\rho(\lambda X) = \lambda \rho(X)$ for $\lambda > 0$
4. **Subadditivity:** $\rho(X + Y) \leq \rho(X) + \rho(Y)$

i Why Coherence Matters

- **Monotonicity:** Higher losses should have higher risk
- **Translation invariance:** Adding cash reduces risk by that amount
- **Positive homogeneity:** Doubling position doubles risk
- **Subadditivity:** Diversification cannot increase risk

VaR violates subadditivity and is therefore not coherent. CVaR satisfies all four properties.

Risk Measure	Coherent?	Convex?	Easy to Optimize?
Expectation	Yes	Yes	Yes (LP)
Variance	No	Yes	Yes (QP)
VaR	No	No	Difficult
CVaR	Yes	Yes	Yes (LP)
Entropic Risk	Yes	Yes	Yes (Exp cone)

22.7.4. Spectral Risk Measures

CVaR focuses on a single tail threshold. Spectral risk measures provide a more flexible framework by assigning different weights to different quantiles of the loss distribution. A decision maker who is extremely averse to worst-case outcomes can assign high weight to the upper quantiles, while one who is moderately risk-averse can use a smoother weighting function. Spectral risk measures generalize CVaR by weighting quantiles according to a spectrum function ϕ :

$$\rho_{\phi}(X) = \int_0^1 \phi(\alpha) F_X^{-1}(\alpha) d\alpha$$

where F_X^{-1} is the quantile function and ϕ satisfies:

- $\phi(\alpha) \geq 0$ (non-negative)
- $\int_0^1 \phi(\alpha) d\alpha = 1$ (normalized)
- ϕ is non-increasing (risk aversion)

Examples of spectral risk measures:

Measure	Spectrum $\phi(\alpha)$	Interpretation
Expectation	$\phi(\alpha) = 1$	All quantiles equally weighted
CVaR $_{\beta}$	$\phi(\alpha) = \frac{1}{1-\beta}$ for $\alpha \geq \beta$	Focus on worst $1 - \beta$
Exponential	$\phi(\alpha) = \frac{\gamma e^{\gamma \alpha}}{e^{\gamma} - 1}$	Exponentially increasing weight

```
import numpy as np

def spectral_risk_measure(losses, phi_func, n_quantiles=1000):
    """Compute spectral risk measure.

    Args:
        losses: Array of loss samples
        phi_func: Spectrum function phi(alpha)
```

```

        n_quantiles: Number of quantiles for integration

Returns:
    Spectral risk measure value
    """
    alphas = np.linspace(0, 1, n_quantiles + 1)[1:] # Exclude 0
    quantiles = np.percentile(losses, alphas * 100)
    weights = np.array([phi_func(a) for a in alphas])

    # Normalize weights
    weights = weights / np.sum(weights)

    return np.sum(weights * quantiles)

def exponential_spectrum(alpha, gamma=2):
    """Exponential spectrum function."""
    return gamma * np.exp(gamma * alpha) / (np.exp(gamma) - 1)

def cvar_spectrum(alpha, beta=0.95):
    """CVaR spectrum function."""
    return 1 / (1 - beta) if alpha >= beta else 0

# Example
np.random.seed(42)
losses = np.random.lognormal(mean=0, sigma=1, size=10000)

expected_value = np.mean(losses)
cvar_95 = spectral_risk_measure(losses, lambda a: cvar_spectrum(a, 0.95))
exp_risk = spectral_risk_measure(losses, lambda a: exponential_spectrum(a, 2))

print(f"Expected Value: {expected_value:.3f}")
print(f"CVaR (95%): {cvar_95:.3f}")
print(f"Exponential Risk (gamma=2): {exp_risk:.3f}")

```

22.7.5. Mean-Risk Optimization

Mean-risk optimization balances expected value and risk in a single framework:

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[f(\mathbf{x}, \xi)] + \lambda \cdot \rho(f(\mathbf{x}, \xi))$$

where $\lambda \geq 0$ is the risk aversion parameter and ρ is a risk measure.

Common formulations:

1. **Mean-Variance** (Markowitz):

$$\min \mathbb{E}[f] + \lambda \cdot \text{Var}[f]$$

2. **Mean-CVaR:**

$$\min \mathbb{E}[f] + \lambda \cdot \text{CVaR}_\alpha[f]$$

3. **Mean-Semideviation:**

$$\min \mathbb{E}[f] + \lambda \cdot \mathbb{E}[\max(f - \mathbb{E}[f], 0)]$$

```

import numpy as np
from scipy.optimize import minimize

def mean_cvar_optimization(returns, alpha=0.95, lambda_risk=0.5):
    """Mean-CVaR portfolio optimization.

    Args:
        returns: Scenario returns matrix (scenarios x assets)
        alpha: CVaR confidence level
        lambda_risk: Risk aversion (0 = mean only, 1 = CVaR only)

    Returns:
        Optimal portfolio weights
    """
    from scipy.optimize import linprog

    n_scenarios, n_assets = returns.shape

    # Variables: [weights, VaR, excess_losses]
    # n_assets + 1 + n_scenarios variables

    # Objective: lambda * (-expected_return) + (1-lambda) * CVaR
    # CVaR = VaR + (1/(1-alpha)) * E[max(loss - VaR, 0)]
    c = np.zeros(n_assets + 1 + n_scenarios)

    # Expected return (negative for minimization)
    mean_returns = returns.mean(axis=0)
    c[:n_assets] = -lambda_risk * mean_returns

    # CVaR terms
    c[n_assets] = (1 - lambda_risk) # VaR coefficient
    c[n_assets+1:] = (1 - lambda_risk) / ((1 - alpha) * n_scenarios)

    # Constraints
    # 1. Sum of weights = 1
    A_eq = np.zeros((1, len(c)))
    A_eq[0, :n_assets] = 1
    b_eq = [1]

    # 2. excess_s >= -portfolio_return_s - VaR (for each scenario)
    A_ub = np.zeros((n_scenarios, len(c)))
    for s in range(n_scenarios):
        A_ub[s, :n_assets] = returns[s] # portfolio return
        A_ub[s, n_assets] = 1 # VaR
        A_ub[s, n_assets + 1 + s] = 1 # excess_s
    b_ub = np.zeros(n_scenarios)

    # Bounds
    bounds = [(0, 1) for _ in range(n_assets)] # weights
    bounds += [(None, None)] # VaR
    bounds += [(0, None) for _ in range(n_scenarios)] # excess

    result = linprog(c, A_ub=-A_ub, b_ub=b_ub, A_eq=A_eq, b_eq=b_eq,

```

```

        bounds=bounds)

    return result.x[:n_assets]

def mean_semideviation_optimization(returns, lambda_risk=1.0):
    """Mean-Semideviation portfolio optimization.

    Args:
        returns: Scenario returns matrix (scenarios x assets)
        lambda_risk: Risk aversion parameter

    Returns:
        Optimal portfolio weights
    """
    n_scenarios, n_assets = returns.shape

    # Variables: [weights, deviations+]
    # Semideviation = E[max(mean - r, 0)]

    from scipy.optimize import linprog

    mean_returns = returns.mean(axis=0)

    # Variables: weights (n_assets), d+ (n_scenarios)
    n_vars = n_assets + n_scenarios

    # Objective: -E[return] + lambda * E[d+]
    c = np.zeros(n_vars)
    c[:n_assets] = -mean_returns
    c[n_assets:] = lambda_risk / n_scenarios

    # Constraints: d+_s >= mean_return - return_s (for each s)
    # d+_s >= sum_i(w_i * (mu_i - r_i,s))
    A_ub = np.zeros((n_scenarios, n_vars))
    for s in range(n_scenarios):
        A_ub[s, :n_assets] = mean_returns - returns[s]
        A_ub[s, n_assets + s] = -1
    b_ub = np.zeros(n_scenarios)

    # Sum of weights = 1
    A_eq = np.zeros((1, n_vars))
    A_eq[0, :n_assets] = 1
    b_eq = [1]

    # Bounds
    bounds = [(0, 1) for _ in range(n_assets)]
    bounds += [(0, None) for _ in range(n_scenarios)]

    result = linprog(c, A_ub=A_ub, b_ub=b_ub, A_eq=A_eq, b_eq=b_eq,
                     bounds=bounds)

    return result.x[:n_assets]

```

```

# Example: Compare different risk formulations
np.random.seed(42)
n_scenarios, n_assets = 1000, 5

# Generate correlated returns
mean_ret = np.array([0.12, 0.10, 0.08, 0.05, 0.04])
cov_matrix = np.array([
    [0.04, 0.01, 0.005, 0, 0.002],
    [0.01, 0.03, 0.008, 0.001, 0.001],
    [0.005, 0.008, 0.02, 0.002, 0.001],
    [0, 0.001, 0.002, 0.01, 0],
    [0.002, 0.001, 0.001, 0, 0.008]
])

returns = np.random.multivariate_normal(mean_ret, cov_matrix, n_scenarios)

# Optimize under different criteria
w_mean_only = mean_cvar_optimization(returns, lambda_risk=1.0)
w_cvar_only = mean_cvar_optimization(returns, lambda_risk=0.0)
w_balanced = mean_cvar_optimization(returns, lambda_risk=0.5)
w_semidev = mean_semideviation_optimization(returns, lambda_risk=2.0)

print("Portfolio Allocations:")
print(f"Mean-focused: {np.round(w_mean_only, 3)}")
print(f"CVaR-focused: {np.round(w_cvar_only, 3)}")
print(f"Balanced: {np.round(w_balanced, 3)}")
print(f"Semideviation: {np.round(w_semidev, 3)}")

```

Choosing Risk Measures in Practice

- **Regulatory requirements:** VaR is mandated in banking (Basel), but CVaR is preferred for optimization
- **Tail sensitivity:** Use CVaR or spectral measures for fat-tailed distributions
- **Computational tractability:** CVaR and semideviation preserve LP structure
- **Interpretation:** Variance is intuitive but penalizes upside; semideviation only penalizes downside

22.8. Chance Constraints

Stochastic programming and robust optimization handle uncertainty in the objective. Chance constraints address uncertainty in the constraints themselves. Instead of requiring a constraint to hold for every possible scenario (which may be too conservative), chance constraints require it to hold with a specified probability. This allows a small, controlled probability of constraint violation in exchange for significantly better objective values.

Require constraints to hold with high probability:

$$P(g(\mathbf{x}, \xi) \leq 0) \geq 1 - \alpha$$

22. Decision Making Under Uncertainty

22.8.1. Individual Chance Constraints

For normally distributed ξ :

$$P(\mathbf{a}^T \mathbf{x} \leq b + \sigma \Phi^{-1}(1 - \alpha)) \geq 1 - \alpha$$

where Φ^{-1} is the inverse standard normal CDF.

22.8.2. Joint Chance Constraints

$$P(A(\xi)\mathbf{x} \geq \mathbf{b}(\xi)) \geq 1 - \alpha$$

Often non-convex; requires conservative approximations like Bonferroni bounds.

22.9. Robust Optimization

Robust optimization takes a fundamentally different approach to uncertainty: instead of assuming known probability distributions, it optimizes against worst-case realizations within specified uncertainty sets. This approach is particularly valuable when probability distributions are unknown or unreliable.

22.9.1. Uncertainty Sets

The choice of uncertainty set significantly affects both the tractability and conservatism of the robust formulation.

Box Uncertainty

The simplest uncertainty set defines parameter-wise bounds:

$$\mathcal{U}_{box} = \{\tilde{\mathbf{a}} : |\tilde{a}_i - \bar{a}_i| \leq \hat{a}_i, \forall i\}$$

where $\bar{a}_i$ is the nominal value and $\hat{a}_i$ is the maximum deviation.

Ellipsoidal Uncertainty

Ellipsoidal sets capture correlations between parameters:

$$\mathcal{U}_{ellipsoid} = \{\tilde{\mathbf{a}} : (\tilde{\mathbf{a}} - \bar{\mathbf{a}})^T \Sigma^{-1} (\tilde{\mathbf{a}} - \bar{\mathbf{a}}) \leq \Omega^2\}$$

where Σ is the covariance matrix and Ω controls the size.

Polyhedral Uncertainty

Budget-of-uncertainty sets limit the total deviation:

$$\mathcal{U}_{budget} = \left\{ \tilde{\mathbf{a}} : \tilde{a}_i = \bar{a}_i + \hat{a}_i z_i, |z_i| \leq 1, \sum_i |z_i| \leq \Gamma \right\}$$

The parameter $\Gamma \in [0, n]$ controls conservatism: $\Gamma = 0$ gives the nominal problem, $\Gamma = n$ gives full protection.

Uncertainty Set	Tractability	Conservatism	Best For
Box	LP remains LP	Most conservative	Bounded errors
Ellipsoidal	LP becomes SOCP	Moderate	Correlated parameters
Budget	LP remains LP	Adjustable	Controllable protection

22.9.2. Robust Counterpart Formulation

For a constraint $\mathbf{a}^T \mathbf{x} \leq b$ with uncertain $\mathbf{a} \in \mathcal{U}$, the robust counterpart requires:

$$\max_{\mathbf{a} \in \mathcal{U}} \mathbf{a}^T \mathbf{x} \leq b$$

Box Uncertainty Robust Counterpart

$$\bar{\mathbf{a}}^T \mathbf{x} + \sum_i \hat{a}_i |x_i| \leq b$$

For $\mathbf{x} \geq 0$: $\bar{\mathbf{a}}^T \mathbf{x} + \hat{\mathbf{a}}^T \mathbf{x} \leq b$

Ellipsoidal Robust Counterpart

$$\bar{\mathbf{a}}^T \mathbf{x} + \Omega \|\Sigma^{1/2} \mathbf{x}\|_2 \leq b$$

This is a second-order cone constraint.

Budget Uncertainty Robust Counterpart

$$\bar{\mathbf{a}}^T \mathbf{x} + \max_{\{S: |S| \leq \Gamma\}} \sum_{i \in S} \hat{a}_i |x_i| \leq b$$

This can be reformulated as an LP with auxiliary variables.

22.9.3. Implementation with CVXPY

CVXPY provides a natural interface for formulating robust optimization problems. The following implementation shows robust portfolio optimization with both ellipsoidal and budget uncertainty sets. For ellipsoidal uncertainty, the worst-case expected return equals the nominal return minus a penalty proportional to the L2-norm of the portfolio weights. For budget uncertainty, the worst-case return penalizes the largest weight deviations up to a budget parameter Γ .

```
import cvxpy as cp
import numpy as np

def robust_portfolio_optimization(mu, Sigma, uncertainty_level=0.1,
                                risk_aversion=1.0, budget_gamma=None):
    """Robust portfolio optimization with ellipsoidal uncertainty.

    Args:
        mu: Expected returns (n,)
        Sigma: Covariance matrix (n, n)
        uncertainty_level: Size of uncertainty set for expected returns
        risk_aversion: Risk aversion parameter
        budget_gamma: If provided, use budget uncertainty instead

    Returns:
        Optimal portfolio weights
    """
    n = len(mu)
    w = cp.Variable(n) # Portfolio weights

    # Nominal expected return
```

```

nominal_return = mu @ w

if budget_gamma is not None:
    # Budget uncertainty: protect against worst-case deviations
    # in at most gamma assets
    z = cp.Variable(n)
    t = cp.Variable()

    # Robust constraint: nominal - worst-case >= target
    # worst-case = sum of gamma largest |mu_deviation * w|
    constraints = [
        cp.sum(w) == 1,
        w >= 0,
        z >= uncertainty_level * cp.abs(w),
        z >= 0,
        cp.sum(z) + budget_gamma * t <= uncertainty_level * cp.norm(w, 1)
    ]

    robust_return = nominal_return - budget_gamma * t - cp.sum(z)
else:
    # Ellipsoidal uncertainty
    # Worst-case return = nominal - uncertainty_level * ||w||_2
    robust_return = nominal_return - uncertainty_level * cp.norm(w, 2)

    constraints = [
        cp.sum(w) == 1,
        w >= 0
    ]

# Risk term (variance)
risk = cp.quad_form(w, Sigma)

# Maximize robust return - risk_aversion * risk
objective = cp.Maximize(robust_return - risk_aversion * risk)

problem = cp.Problem(objective, constraints)
problem.solve()

return w.value

def robust_production_planning(c, A, b, demand_uncertainty):
    """Robust production planning with demand uncertainty.

    Args:
        c: Production costs (n,)
        A: Resource consumption matrix (m, n)
        b: Resource capacities (m,)
        demand_uncertainty: Dict with 'nominal' and 'deviation' arrays

    Returns:
        Robust production quantities
    """

```



```

n = len(c)
m = len(b)
x = cp.Variable(n)

d_nominal = demand_uncertainty['nominal']
d_deviation = demand_uncertainty['deviation']

# Minimize production cost
objective = cp.Minimize(c @ x)

constraints = [
    A @ x <= b, # Resource constraints
    x >= 0,
    # Robust demand satisfaction: meet worst-case (highest) demand
    x >= d_nominal + d_deviation # Box uncertainty on demand
]

problem = cp.Problem(objective, constraints)
problem.solve()

return x.value

# Example: Robust Portfolio
np.random.seed(42)
n_assets = 5
mu = np.array([0.12, 0.10, 0.07, 0.03, 0.05])
Sigma = np.array([
    [0.10, 0.03, 0.01, 0.00, 0.02],
    [0.03, 0.08, 0.02, 0.01, 0.01],
    [0.01, 0.02, 0.05, 0.01, 0.01],
    [0.00, 0.01, 0.01, 0.02, 0.00],
    [0.02, 0.01, 0.01, 0.00, 0.04]
])

# Compare nominal vs robust
w_nominal = robust_portfolio_optimization(mu, Sigma, uncertainty_level=0)
w_robust = robust_portfolio_optimization(mu, Sigma, uncertainty_level=0.02)

print("Nominal portfolio:", w_nominal)
print("Robust portfolio:", w_robust)

```

22.9.4. Adjustable Robust Optimization

When decisions can be made after some uncertainty is revealed, adjustable robust optimization (ARO) allows second-stage decisions to adapt:

$$\begin{aligned}
 \min_{\mathbf{x}, \mathbf{y}(\cdot)} \quad & \mathbf{c}^T \mathbf{x} + \max_{\xi \in \mathcal{U}} \mathbf{d}^T \mathbf{y}(\xi) \\
 \text{s.t.} \quad & \mathbf{A} \mathbf{x} + \mathbf{B} \mathbf{y}(\xi) \geq \mathbf{h}(\xi), \quad \forall \xi \in \mathcal{U}
 \end{aligned}$$

For tractability, $\mathbf{y}(\xi)$ is often restricted to affine functions of uncertainty (linear decision rules).

22.9.5. When to Use Robust vs Stochastic Optimization

Criterion	Stochastic Programming	Robust Optimization
Probability data	Known distributions	Unknown or unreliable
Objective	Expected value	Worst-case protection
Conservatism	Risk-neutral	Conservative
Tractability	Scenario-dependent	Often maintains tractability
Interpretability	Expected outcomes	Guaranteed performance

💡 Choosing the Right Approach

- **Good probability data:** Use stochastic programming
- **Distribution uncertainty:** Use distributionally robust optimization
- **No probability data or high stakes:** Use robust optimization
- **Adjustable decisions:** Consider adjustable robust optimization

22.10. Distributionally Robust Optimization

Distributionally Robust Optimization (DRO) bridges the gap between stochastic programming (which requires exact probability distributions) and robust optimization (which ignores probabilities entirely). DRO optimizes against the worst-case distribution within an ambiguity set of plausible distributions.

22.10.1. Motivation: When Distributions Are Uncertain

In practice, probability distributions are rarely known exactly:

- **Limited data:** Historical data may be insufficient to estimate distributions precisely
- **Non-stationarity:** Distributions may shift over time
- **Model misspecification:** Assumed distribution families may be wrong
- **Adversarial settings:** Nature or opponents may choose unfavorable distributions

DRO formulation:

$$\min_{\mathbf{x} \in \mathcal{X}} \sup_{P \in \mathcal{P}} \mathbb{E}_P[f(\mathbf{x}, \xi)]$$

where $\mathcal{P}$ is the ambiguity set of distributions.

22.10.2. Moment-Based Ambiguity Sets

Moment ambiguity sets contain all distributions matching specified moments:

$$\mathcal{P}_{\text{moment}} = \{P : \mathbb{E}_P[\xi] = \mu, \mathbb{E}_P[(\xi - \mu)(\xi - \mu)^T] \preceq \Sigma\}$$

Properties:

- Only requires estimates of mean and covariance
- Results in semidefinite programming (SDP) problems
- Can include higher moments (skewness, kurtosis)

The following code implements two moment-based DRO models. The first is a portfolio optimization that accounts for uncertainty in both expected returns and covariances. The second is the classic newsvendor problem using Scarf's 1958 result, which provides the optimal order quantity when only the mean and variance of demand are known.

```

import cvxpy as cp
import numpy as np

def dro_moment_portfolio(mu, Sigma, gamma=0.1):
    """Distributionally robust portfolio with moment ambiguity.

    Optimizes worst-case expected return where distributions have
    mean in [mu - delta, mu + delta] and covariance <= (1+gamma)*Sigma.

    Args:
        mu: Estimated expected returns (n,)
        Sigma: Estimated covariance matrix (n, n)
        gamma: Covariance uncertainty parameter

    Returns:
        Optimal portfolio weights
    """
    n = len(mu)
    w = cp.Variable(n)

    # Worst-case mean: mu - delta * sign(w) pushes return down
    # For long-only: worst-case is mu - delta
    delta = 0.01 * np.abs(mu) # 1% uncertainty in expected returns

    # Worst-case expected return (assuming w >= 0)
    worst_case_return = (mu - delta) @ w

    # Worst-case variance with covariance uncertainty
    # Variance <= w' * (1+gamma) * Sigma * w
    worst_case_variance = (1 + gamma) * cp.quad_form(w, Sigma)

    # Mean-variance objective with worst-case moments
    risk_aversion = 2.0
    objective = cp.Maximize(worst_case_return - risk_aversion * worst_case_variance)

    constraints = [
        cp.sum(w) == 1,
        w >= 0
    ]

    problem = cp.Problem(objective, constraints)
    problem.solve()

    return w.value

def dro_moment_newsvendor(mu, sigma, c_under, c_over, capacity):
    """Distributionally robust newsvendor with mean-variance ambiguity.

    Find order quantity robust against all distributions with given
    mean and variance (Scarf's model).

    The worst-case cost is known in closed form:

```

```

Args:
    mu: Mean demand
    sigma: Standard deviation of demand
    c_under: Underage cost (lost sales)
    c_over: Overage cost (excess inventory)
    capacity: Maximum order quantity

Returns:
    Optimal order quantity
"""
# Scarf's result: optimal order quantity
# q* = mu + sigma * (c_u - c_o) / (2 * sqrt(c_u * c_o))

ratio = (c_under - c_over) / (2 * np.sqrt(c_under * c_over))
q_star = mu + sigma * ratio

# Bound by capacity
q_star = min(max(q_star, 0), capacity)

# Worst-case expected cost
# From Scarf (1958): worst-case is a two-point distribution
worst_case_cost = (c_under + c_over) / 2 * (
    np.sqrt(sigma**2 + (q_star - mu)**2) - (q_star - mu)
) + c_over * (q_star - mu)

return q_star, worst_case_cost

# Example
mu = np.array([0.10, 0.08, 0.06, 0.04])
Sigma = np.array([
    [0.04, 0.01, 0.005, 0],
    [0.01, 0.03, 0.008, 0.002],
    [0.005, 0.008, 0.02, 0.004],
    [0, 0.002, 0.004, 0.01]
])

w_dro = dro_moment_portfolio(mu, Sigma, gamma=0.2)
print(f"DRO Portfolio: {np.round(w_dro, 3)}")

# Newsvendor example
q_opt, cost = dro_moment_newsvendor(
    mu=100, sigma=20, c_under=10, c_over=3, capacity=150
)
print(f"DRO Newsvendor: order {q_opt:.1f}, worst-case cost {cost:.2f}")

```

22.10.3. Wasserstein Distance Ambiguity Sets

Wasserstein (earth mover's) distance measures the cost of transforming one distribution into another:

$$W_p(P, Q) = \left(\inf_{\gamma \in \Gamma(P, Q)} \int \|\xi - \zeta\|^p d\gamma(\xi, \zeta) \right)^{1/p}$$

The Wasserstein ambiguity set contains all distributions within distance ϵ of the empirical distribution:

$$\mathcal{P}_W = \{P : W_p(P, \hat{P}_N) \leq \epsilon\}$$

where $\hat{P}_N$ is the empirical distribution from N samples.

Advantages of Wasserstein DRO:

1. **Data-driven:** Centered at empirical distribution
2. **Finite-sample guarantees:** Confidence bounds on out-of-sample performance
3. **Tractable reformulations:** Often reduces to convex optimization
4. **Regularization interpretation:** Equivalent to regularized empirical risk minimization

For the W1 (earth mover's) distance, the DRO problem has a tractable dual reformulation. The worst-case expected loss decomposes into the empirical loss plus a penalty term proportional to the Wasserstein ball radius ϵ and the dual norm of the portfolio weights. This penalty acts as a form of regularization, connecting DRO to regularized empirical risk minimization.

```
import cvxpy as cp
import numpy as np

def wasserstein_dro_portfolio(returns, epsilon=0.1, p=1):
    """Wasserstein DRO portfolio optimization.

    Minimizes worst-case expected loss over distributions within
    Wasserstein distance epsilon of empirical distribution.

    For p=1 (W1 distance), this has a tractable LP reformulation.

    Args:
        returns: Historical return scenarios (n_samples, n_assets)
        epsilon: Wasserstein ball radius
        p: Order of Wasserstein distance (1 or 2)

    Returns:
        Optimal portfolio weights
    """
    n_samples, n_assets = returns.shape

    w = cp.Variable(n_assets)

    if p == 1:
        # W1 DRO: min_w max_P E_P[-w'r] s.t. W1(P, P_hat) <= epsilon
        # Dual reformulation: min_w (1/n) sum_i max_||delta||_1 <= epsilon [-w'(r_i
        #   ↪ + delta)]
        # = min_w (1/n) sum_i [-w'r_i] + epsilon * ||w||_inf

        # Worst-case expected negative return
        empirical_loss = -returns @ w # Negative returns (losses)
        mean_loss = cp.sum(empirical_loss) / n_samples

        # Wasserstein penalty (dual norm of gradient)
        # For W1 with 1-infinity transport cost: penalty = epsilon * ||w||_1
```

```

    wasserstein_penalty = epsilon * cp.norm(w, 1)

    objective = cp.Minimize(mean_loss + wasserstein_penalty)

else: # p == 2
    # W2 DRO has a different (often SDP) reformulation
    # Simplified version using second-order approximation
    empirical_loss = -returns @ w
    mean_loss = cp.sum(empirical_loss) / n_samples

    # Penalty proportional to gradient norm
    wasserstein_penalty = epsilon * cp.norm(w, 2)

    objective = cp.Minimize(mean_loss + wasserstein_penalty)

constraints = [
    cp.sum(w) == 1,
    w >= 0
]

problem = cp.Problem(objective, constraints)
problem.solve()

return w.value

def wasserstein_dro_regression(X, y, epsilon=0.1):
    """Wasserstein DRO linear regression.

    Equivalent to regularized regression with specific regularization.

    Args:
        X: Feature matrix (n_samples, n_features)
        y: Target vector (n_samples,)
        epsilon: Wasserstein ball radius

    Returns:
        Regression coefficients
    """
    n_samples, n_features = X.shape

    beta = cp.Variable(n_features)

    # Empirical squared loss
    residuals = y - X @ beta
    empirical_loss = cp.sum_squares(residuals) / n_samples

    # Wasserstein regularization (equivalent to ridge-like penalty)
    regularization = epsilon * cp.norm(beta, 2)

    objective = cp.Minimize(empirical_loss + regularization)

    problem = cp.Problem(objective)

```

```

    problem.solve()

    return beta.value

# Example
np.random.seed(42)
n_samples, n_assets = 200, 4

# Generate return scenarios
mu_true = np.array([0.08, 0.06, 0.04, 0.03])
Sigma_true = np.array([
    [0.04, 0.01, 0.005, 0],
    [0.01, 0.025, 0.006, 0.002],
    [0.005, 0.006, 0.015, 0.003],
    [0, 0.002, 0.003, 0.01]
])
returns = np.random.multivariate_normal(mu_true, Sigma_true, n_samples)

# Compare different epsilon values
for eps in [0.0, 0.05, 0.1, 0.2]:
    w = wasserstein_dro_portfolio(returns, epsilon=eps)
    expected_return = np.mean(returns @ w)
    volatility = np.std(returns @ w)
    print(f"epsilon={eps:.2f}: weights={np.round(w,2)}, "
          f"E[r]={expected_return:.3f}, std={volatility:.3f}")

```

22.10.4. Connection to Robust Optimization

DRO provides a principled bridge between stochastic and robust optimization:

Ambiguity Set	Limit Case	Equivalent To
$\epsilon \rightarrow 0$	Point mass at data	Empirical risk minimization
$\epsilon \rightarrow \infty$	All distributions	Robust optimization
Moment-based	Fixed moments	Distributionally robust
Wasserstein	Ball around empirical	Regularized ERM

i DRO as Regularization

Wasserstein DRO is equivalent to regularized empirical risk minimization:

$$\min_{\mathbf{x}} \frac{1}{N} \sum_{i=1}^N f(\mathbf{x}, \xi_i) + \epsilon \cdot \Omega(\mathbf{x})$$

where Ω is a norm depending on the loss function's Lipschitz constant. This connection explains why DRO often improves out-of-sample performance.

22.10.5. Comprehensive DRO Example

The following class implements a unified DRO solver that supports all three ambiguity set types: Wasserstein, moment-based, and likelihood-ratio (chi-squared). Each approach produces a different portfolio allocation depending on its assumptions about distributional uncertainty. The comparison

22. Decision Making Under Uncertainty

at the end shows how the choice of ambiguity set affects the resulting weights and risk-return profile.

```
import cvxpy as cp
import numpy as np
from scipy import stats

class DistributionallyRobustOptimizer:
    """Unified DRO solver for different ambiguity sets."""

    def __init__(self, ambiguity_type='wasserstein'):
        """Initialize DRO solver.

        Args:
            ambiguity_type: 'wasserstein', 'moment', or 'likelihood'
        """
        self.ambiguity_type = ambiguity_type

    def optimize_portfolio(self, returns, params):
        """Solve DRO portfolio problem.

        Args:
            returns: Historical returns (n_samples, n_assets)
            params: Dict with ambiguity set parameters

        Returns:
            Optimal weights and worst-case metrics
        """
        n_samples, n_assets = returns.shape
        w = cp.Variable(n_assets)

        if self.ambiguity_type == 'wasserstein':
            epsilon = params.get('epsilon', 0.1)

            # Mean loss + Wasserstein penalty
            losses = -returns @ w
            mean_loss = cp.sum(losses) / n_samples
            penalty = epsilon * cp.norm(w, 1)

            objective = cp.Minimize(mean_loss + penalty)

        elif self.ambiguity_type == 'moment':
            gamma = params.get('gamma', 0.1) # Covariance uncertainty
            delta = params.get('delta', 0.01) # Mean uncertainty

            mu_hat = np.mean(returns, axis=0)
            Sigma_hat = np.cov(returns.T)

            # Worst-case mean and variance
            worst_mean = (mu_hat - delta * np.ones(n_assets)) @ w
            worst_var = (1 + gamma) * cp.quad_form(w, Sigma_hat)

            # Mean-variance with worst-case moments
```



```

risk_aversion = params.get('risk_aversion', 1.0)
objective = cp.Maximize(worst_mean - risk_aversion * worst_var)

elif self.ambiguity_type == 'likelihood':
    # Chi-squared ambiguity (likelihood ratio)
    eta = params.get('eta', 0.1) # Divergence bound

    # Worst-case reweighting of empirical samples
    q = cp.Variable(n_samples) # Probability weights
    losses = -returns @ w

    # Weighted loss
    weighted_loss = q @ losses

    # KL divergence constraint approximation
    #  $\sum(q * \log(n*q)) \leq \eta$ 
    # Approximated by chi-squared:  $\sum((n*q - 1)^2) \leq 2*\eta$ 
    uniform = np.ones(n_samples) / n_samples

    objective = cp.Minimize(weighted_loss)

constraints = [
    cp.sum(w) == 1,
    w >= 0
]

if self.ambiguity_type == 'likelihood':
    constraints.extend([
        q >= 0,
        cp.sum(q) == 1,
        cp.sum_squares(n_samples * q - 1) <= 2 * params.get('eta', 0.1) *
↪ n_samples
    ])

problem = cp.Problem(objective, constraints)
problem.solve()

# Compute worst-case metrics
w_opt = w.value
portfolio_returns = returns @ w_opt

return {
    'weights': w_opt,
    'expected_return': np.mean(portfolio_returns),
    'volatility': np.std(portfolio_returns),
    'worst_case_return': np.percentile(portfolio_returns, 5),
    'sharpe_ratio': np.mean(portfolio_returns) / np.std(portfolio_returns)
}

# Compare all three approaches
np.random.seed(42)
n_samples, n_assets = 500, 5

```

```

mu_true = np.array([0.10, 0.08, 0.06, 0.05, 0.04])
Sigma_true = 0.04 * np.eye(n_assets) + 0.02 * np.ones((n_assets, n_assets))
returns = np.random.multivariate_normal(mu_true, Sigma_true, n_samples)

print("Distributionally Robust Portfolio Comparison")
print("=" * 60)

# Wasserstein DRO
dro_w = DistributionallyRobustOptimizer('wasserstein')
result_w = dro_w.optimize_portfolio(returns, {'epsilon': 0.05})
print(f"\nWasserstein DRO (epsilon=0.05):")
print(f"  Weights: {np.round(result_w['weights'], 3)}")
print(f"  Sharpe: {result_w['sharpe_ratio']:.3f}")

# Moment DRO
dro_m = DistributionallyRobustOptimizer('moment')
result_m = dro_m.optimize_portfolio(returns, {'gamma': 0.2, 'delta': 0.01})
print(f"\nMoment DRO (gamma=0.2, delta=0.01):")
print(f"  Weights: {np.round(result_m['weights'], 3)}")
print(f"  Sharpe: {result_m['sharpe_ratio']:.3f}")

# Likelihood DRO
dro_l = DistributionallyRobustOptimizer('likelihood')
result_l = dro_l.optimize_portfolio(returns, {'eta': 0.1})
print(f"\nLikelihood DRO (eta=0.1):")
print(f"  Weights: {np.round(result_l['weights'], 3)}")
print(f"  Sharpe: {result_l['sharpe_ratio']:.3f}")

```

💡 Choosing the Right DRO Approach

Situation	Recommended Approach
Small sample size	Wasserstein with larger ϵ
Known moment bounds	Moment-based ambiguity
Heavy-tailed data	Wasserstein (more robust to outliers)
Distribution shift expected	Larger ambiguity sets
Computational constraints	Moment-based (often simpler)

22.11. Summary

22.11.1. Key Takeaways

- Stochastic programming optimizes expected outcomes across scenarios
- Two-stage models separate here-and-now from wait-and-see decisions
- Sample Average Approximation makes continuous distributions tractable
- Variance reduction techniques (importance sampling, control variates, antithetic variates, stratified sampling, LHS) improve SAA efficiency
- Decomposition (L-Shaped, Progressive Hedging) handles large scenario counts
- SDDP enables solution of multi-stage problems with many stages
- CVaR provides convex, optimizable risk measure; spectral measures generalize it

- Mean-risk optimization balances expected value and risk measures
- Chance constraints limit probability of constraint violation
- Robust optimization protects against worst-case uncertainty
- Distributionally robust optimization handles uncertain probability distributions
- Multi-stage problems require specialized methods (SDDP) due to exponential growth

22.12. Exercises

1. **Production planning:** A manufacturer must decide production before demand is known. Demand is $D \in \{80, 100, 120\}$ with probabilities $\{0.3, 0.4, 0.3\}$. Unit production cost is \$10, shortage cost \$25, excess cost \$3. Formulate and solve the two-stage problem.
2. **SAA implementation:** Implement SAA for a newsvendor problem with normally distributed demand. Compare solutions using 50, 200, and 1000 samples. Report optimality gaps.
3. **CVaR portfolio:** Given 5 assets with 1000 historical daily returns, construct a portfolio minimizing 95% CVaR subject to minimum 5% expected return. Compare with mean-variance portfolio.
4. **Progressive Hedging:** Implement Progressive Hedging for the production problem in Exercise 1. Experiment with different values of ρ and report convergence behavior.
5. **Multi-stage model:** Extend the production problem to 3 stages where inventory carries over. How does the optimal policy change compared to the two-stage case?
6. **Variance reduction:** Implement Latin Hypercube Sampling for a multi-product newsvendor problem with 5 products. Compare the variance reduction against simple Monte Carlo sampling with the same number of samples.
7. **DRO newsvendor:** Solve the newsvendor problem using Wasserstein DRO with different values of ϵ . Plot the optimal order quantity and worst-case cost as functions of ϵ .
8. **Spectral risk measures:** Implement a portfolio optimization using an exponential spectral risk measure. Compare the resulting portfolio with CVaR and mean-variance portfolios.
9. **SDDP implementation:** Implement a simplified SDDP algorithm for a 5-stage inventory problem. Track the convergence of upper and lower bounds over iterations.
10. **Moment ambiguity:** For a portfolio problem, compare solutions using (a) known mean and variance, (b) moment ambiguity with 10% uncertainty in covariance, and (c) moment ambiguity with 20% uncertainty. Discuss the impact on diversification.

23. Multi-Objective Optimization

What You Will Learn

- Understand Pareto optimality and dominance concepts
- Apply weighted sum and epsilon-constraint methods
- Implement evolutionary multi-objective algorithms (NSGA-II, MOEA/D)
- Evaluate solution quality using performance metrics
- Make decisions with multiple conflicting objectives

Real-world engineering and business problems rarely involve a single objective. Aircraft must be light yet strong. Portfolios should maximize returns while minimizing risk. Production systems must balance throughput, quality, and cost. Multi-objective optimization provides the framework and algorithms for handling these conflicting goals.

23.1. The Challenge of Multiple Objectives

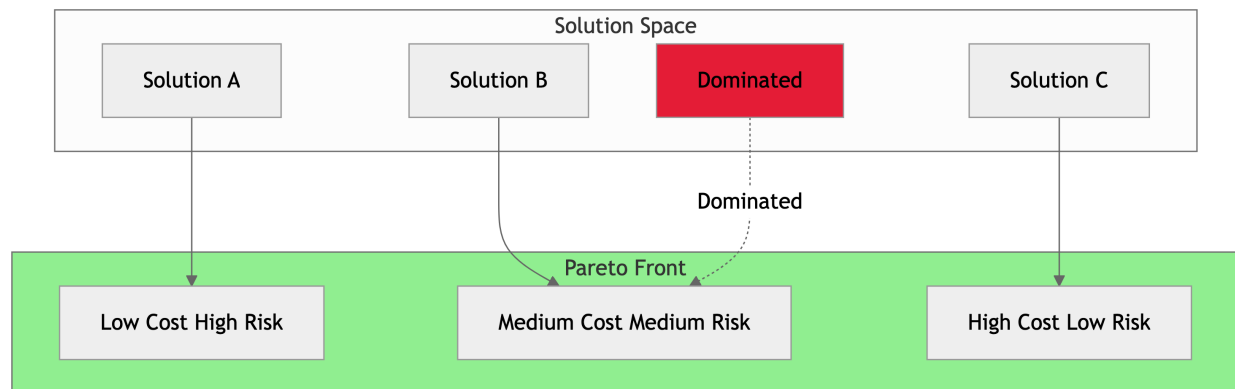


Figure 23.1.: Pareto Front: Trade-off between conflicting objectives

When objectives conflict, there is no single “best” solution. Instead, we seek a set of solutions representing optimal trade-offs. Consider these examples:

Domain	Conflicting Objectives
Aircraft design	Minimize weight vs. maximize strength vs. minimize cost
Portfolio management	Maximize return vs. minimize risk vs. maximize liquidity
Manufacturing	Minimize time vs. minimize defects vs. minimize energy
Machine learning	Maximize accuracy vs. minimize complexity vs. minimize inference time
Supply chain	Minimize cost vs. minimize delivery time vs. maximize reliability

23.2. Mathematical Formulation

The general multi-objective optimization problem:

23. Multi-Objective Optimization

$$\begin{aligned} & \text{minimize} && \mathbf{F}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x})]^T \\ & \text{subject to} && g_j(\mathbf{x}) \leq 0, \quad j = 1, \dots, p \\ & && h_k(\mathbf{x}) = 0, \quad k = 1, \dots, q \\ & && \mathbf{x} \in X \subseteq \mathbb{R}^n \end{aligned}$$

Unlike single-objective optimization, we minimize a vector of objectives, not a scalar.

23.3. Pareto Optimality

23.3.1. Pareto Dominance

A solution $\mathbf{x}$ **dominates** solution $\mathbf{y}$ (written $\mathbf{x} \prec \mathbf{y}$) if and only if:

$$f_i(\mathbf{x}) \leq f_i(\mathbf{y}) \quad \forall i \in \{1, \dots, m\}$$

and

$$\exists j \in \{1, \dots, m\} : f_j(\mathbf{x}) < f_j(\mathbf{y})$$

In other words, $\mathbf{x}$ is at least as good as $\mathbf{y}$ in all objectives and strictly better in at least one.

23.3.2. Key Concepts

Pareto Optimal Solution

A solution $\mathbf{x}^*$ is Pareto optimal if no other feasible solution dominates it:

$$\nexists \mathbf{x} \in X : \mathbf{x} \prec \mathbf{x}^*$$

Pareto Front

The set of all Pareto optimal solutions in objective space. Represents the best possible trade-offs.

Pareto Set

The set of all Pareto optimal solutions in decision variable space.

Why Not Just Pick One Solution?

Each point on the Pareto front is optimal in the sense that improving one objective requires degrading another. The “best” choice depends on decision-maker preferences, which may not be known in advance.

23.3.3. Types of Dominance

Type	Definition	Use Case
Pareto	Better in at least one, no worse in others	Standard MOO
Weak	No worse in any objective	Theoretical analysis
Epsilon	Within ϵ tolerance	Controlling archive size
Cone	Generalized using ordering cone	Preference modeling

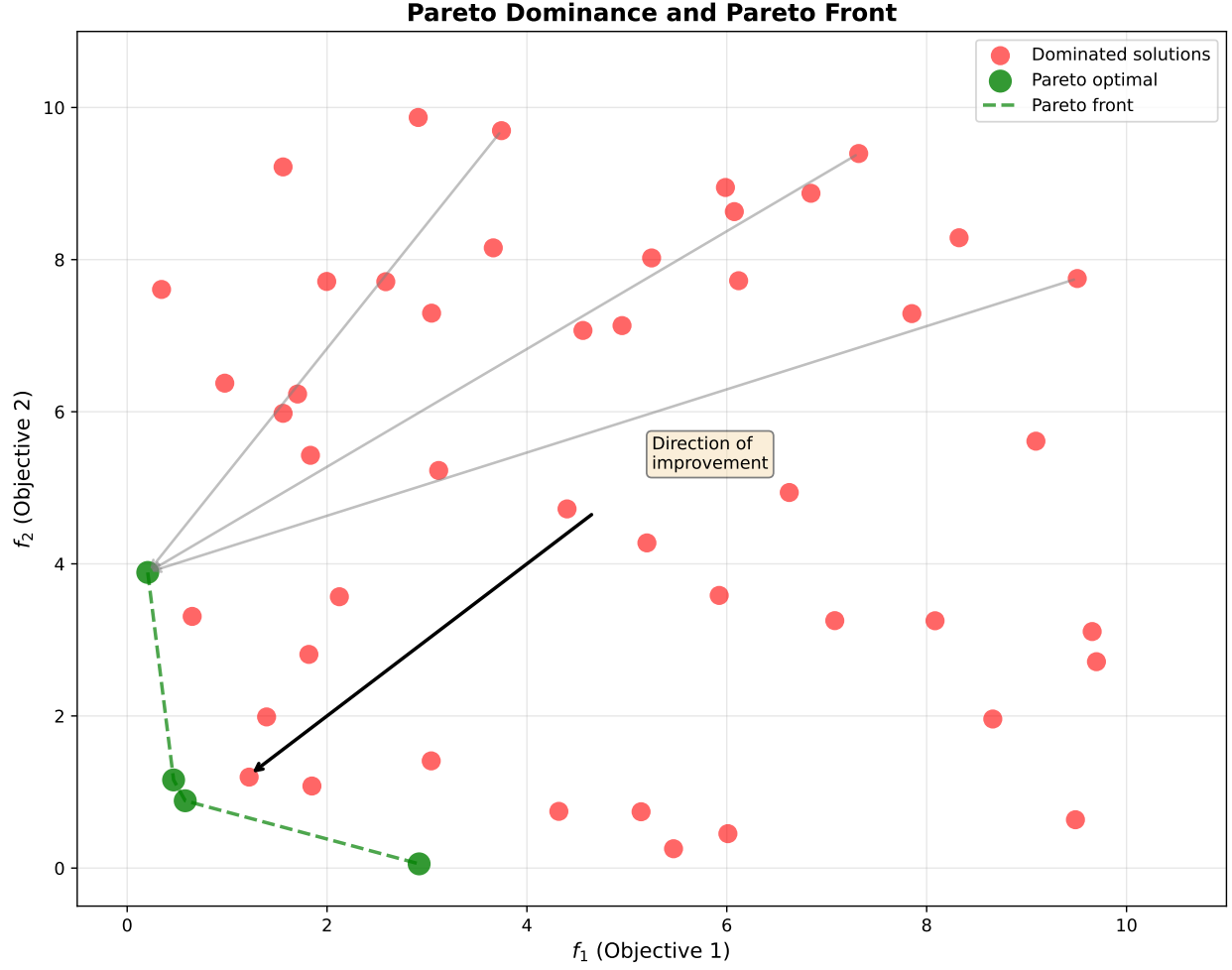


Figure 23.2.: Pareto front showing dominated vs non-dominated solutions in objective space

23.4. Classical Methods

Classical multi-objective methods reduce the vector optimization problem to one or more scalar optimization problems that can be solved with standard single-objective solvers. Each method imposes different assumptions about the Pareto front geometry and requires different forms of preference information from the decision maker.

23.4.1. Weighted Sum Method

The weighted sum method is the simplest scalarization technique. It converts the multi-objective problem to a single objective by combining objectives with weights:

$$\text{minimize} \quad \sum_{i=1}^m w_i f_i(\mathbf{x})$$

where $w_i \geq 0$ and $\sum w_i = 1$.

23. Multi-Objective Optimization

```
import numpy as np
from scipy.optimize import minimize

def weighted_sum_method(objectives, weights, x0, bounds, constraints):
    """Solve MOO using weighted sum."""
    def combined_objective(x):
        return sum(w * f(x) for w, f in zip(weights, objectives))

    result = minimize(combined_objective, x0, bounds=bounds,
                      constraints=constraints)
    return result.x, [f(result.x) for f in objectives]

# Example: Portfolio optimization
def risk(weights):
    return weights @ cov_matrix @ weights

def neg_return(weights):
    return -weights @ expected_returns

# Generate Pareto front by varying weights
pareto_points = []
for w in np.linspace(0, 1, 20):
    x, obj = weighted_sum_method(
        [risk, neg_return],
        [w, 1-w],
        x0=initial_weights,
        bounds=[(0, 1)] * n_assets,
        constraints={'type': 'eq', 'fun': lambda x: sum(x) - 1}
    )
    pareto_points.append(obj)
```

Advantages:

- Simple to implement
- Uses existing single-objective solvers
- Efficient for convex problems

Limitations:

- Cannot find solutions on non-convex portions of Pareto front
- Uniform weight variation does not yield uniform Pareto points
- Requires multiple runs with different weights

23.4.2. Epsilon-Constraint Method

The epsilon-constraint method avoids the weighted sum's inability to find non-convex Pareto points. It selects one objective as primary and converts the remaining objectives into inequality constraints with upper bounds ϵ_i . By systematically varying the ϵ_i values, the method traces the entire Pareto front, including non-convex regions. The formulation optimizes one objective while constraining others:

$$\begin{aligned} & \text{minimize} && f_1(\mathbf{x}) \\ & \text{subject to} && f_i(\mathbf{x}) \leq \epsilon_i, \quad i = 2, \dots, m \\ & && \text{original constraints} \end{aligned}$$

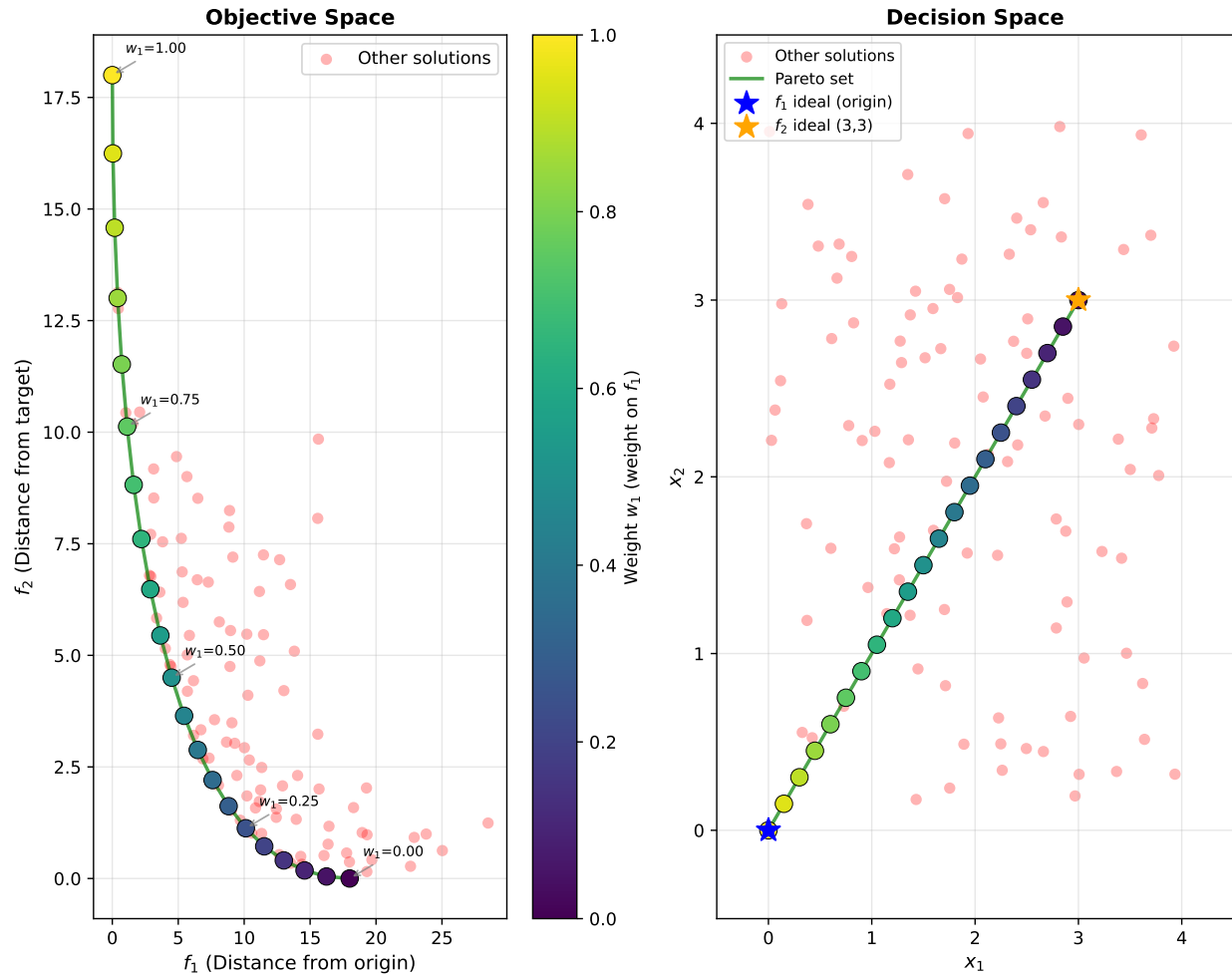


Figure 23.3.: Weighted sum method tracing the Pareto front with different weight combinations

```
def epsilon_constraint_method(objectives, x0, bounds, constraints, epsilons):
    """Solve MOO using epsilon-constraint."""
    primary = objectives[0]
    secondary = objectives[1:]

    # Add epsilon constraints
    eps_constraints = [
        {'type': 'ineq', 'fun': lambda x, f=f, e=e: e - f(x)}
        for f, e in zip(secondary, epsilons)
    ]

    all_constraints = constraints + eps_constraints

    result = minimize(primary, x0, bounds=bounds,
                     constraints=all_constraints)
    return result.x, [f(result.x) for f in objectives]
```

Advantages:

23. Multi-Objective Optimization

- Can find non-convex Pareto regions
- Direct control over objective bounds
- Produces evenly distributed points with proper epsilon selection

Limitations:

- Computational cost of solving many constrained problems
- Requires knowledge of objective ranges
- May produce infeasible subproblems

23.4.3. Comparison of Classical Methods

Method	Convex Front	Non-Convex Front	Ease of Use	Computational Cost
Weighted Sum	Excellent	Poor	Easy	Low
Epsilon-Constraint	Good	Good	Moderate	Moderate
Goal Programming	Good	Moderate	Moderate	Moderate
Normal Boundary Intersection	Excellent	Good	Complex	High

23.5. Goal Programming

Goal Programming (GP) addresses multi-objective optimization from a different perspective: instead of finding Pareto optimal solutions, it seeks solutions that come closest to achieving specified target values (goals) for each objective.

23.5.1. The Satisficing Approach

Goal Programming embodies **satisficing** rather than optimizing:

Approach	Question Asked	Result
Optimizing	“What is the best possible?”	Optimal solution
Satisficing	“What meets my targets?”	Good-enough solution

This is often more realistic: managers have targets for profit, cost, and quality rather than abstract notions of “maximum” or “minimum.”

i When to Use Goal Programming

Goal Programming is appropriate when: - Decision makers have specific targets in mind - Some objectives have higher priority than others - Trade-offs are not well understood - “Close enough” is acceptable - Multiple conflicting constraints exist

23.5.2. Mathematical Formulation

Goal Programming introduces **deviation variables** to measure how far solutions deviate from targets.

For each goal i with target G_i :

$$f_i(x) + d_i^- - d_i^+ = G_i$$

Where:

- d_i^- = **negative deviation** (underachievement)
- d_i^+ = **positive deviation** (overachievement)
- $d_i^-, d_i^+ \geq 0$ and $d_i^- \cdot d_i^+ = 0$ (only one can be positive)

Interpretation:

Goal Type	Minimize	Example
Achieve at least G_i	d_i^-	Profit \$1M
Achieve at most G_i	d_i^+	Cost \$500K
Achieve exactly G_i	$d_i^- + d_i^+$	Produce exactly 1000 units

23.5.3. Types of Goal Programming

23.5.3.1. Weighted Goal Programming

Weighted Goal Programming minimizes a single weighted sum of all deviation variables. The weights encode the decision maker's preference for how tightly each goal should be met.

$$\text{minimize} \quad \sum_{i=1}^m (w_i^- d_i^- + w_i^+ d_i^+)$$

Weights reflect relative importance of achieving each goal. The following example formulates a production planning problem with three goals: a profit target, a labor constraint, and a production target.

```
import numpy as np
from scipy.optimize import linprog, minimize

def weighted_goal_programming_example():
    """
    Production planning with multiple goals:
    - Goal 1: Profit >= $80,000 (high priority)
    - Goal 2: Labor hours <= 1000 (medium priority)
    - Goal 3: Units produced = 500 (low priority)

    Products: A (profit=$200, 2 labor hrs), B (profit=$150, 1.5 labor hrs)
    """

    # Decision variables: [x_A, x_B, d1-, d1+, d2-, d2+, d3-, d3+]
    # x_A, x_B: production quantities
    # d1: profit deviation from $80,000
    # d2: labor deviation from 1000 hours
    # d3: units deviation from 500

    # Goal weights (higher = more important)
    w1_minus = 10.0 # Penalize under-profit heavily
    w1_plus = 0.0   # Over-profit is fine
```

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```
w2_minus = 0.0    # Under-using labor is fine
w2_plus = 5.0     # Over-using labor is bad
w3_minus = 1.0    # Under-production
w3_plus = 1.0     # Over-production

# Objective: minimize weighted deviations
# Variables: [x_A, x_B, d1-, d1+, d2-, d2+, d3-, d3+]
c = [0, 0, w1_minus, w1_plus, w2_minus, w2_plus, w3_minus, w3_plus]

# Goal constraints (as equalities)
# Goal 1: 200*x_A + 150*x_B + d1- - d1+ = 80000
# Goal 2: 2*x_A + 1.5*x_B + d2- - d2+ = 1000
# Goal 3: x_A + x_B + d3- - d3+ = 500

A_eq = [
    [200, 150, 1, -1, 0, 0, 0, 0], # Profit goal
    [2, 1.5, 0, 0, 1, -1, 0, 0],   # Labor goal
    [1, 1, 0, 0, 0, 0, 1, -1]      # Units goal
]
b_eq = [80000, 1000, 500]

# Variable bounds
# x_A, x_B >= 0, all deviations >= 0
bounds = [(0, None)] * 8

# Solve
result = linprog(c, A_eq=A_eq, b_eq=b_eq, bounds=bounds, method='highs')

if result.success:
    x_A, x_B = result.x[0], result.x[1]
    d1_minus, d1_plus = result.x[2], result.x[3]
    d2_minus, d2_plus = result.x[4], result.x[5]
    d3_minus, d3_plus = result.x[6], result.x[7]

    print("=== Weighted Goal Programming Solution ===")
    print(f"\nProduction: Product A = {x_A:.0f}, Product B = {x_B:.0f}")
    print(f"\nGoal 1 (Profit >= $80,000):")
    print(f"  Actual profit: ${200*x_A + 150*x_B:,.0f}")
    print(f"  Underachievement: ${d1_minus:,.0f}, Overachievement: "
          f"↪ ${d1_plus:,.0f}")
    print(f"\nGoal 2 (Labor <= 1,000 hrs):")
    print(f"  Actual hours: {2*x_A + 1.5*x_B:,.0f}")
    print(f"  Under: {d2_minus:,.0f} hrs, Over: {d2_plus:,.0f} hrs")
    print(f"\nGoal 3 (Units = 500):")
    print(f"  Actual units: {x_A + x_B:.0f}")
    print(f"  Under: {d3_minus:,.0f}, Over: {d3_plus:,.0f}")

    return result

result = weighted_goal_programming_example()
```

=== Weighted Goal Programming Solution ===

Production: Product A = 100, Product B = 400

Goal 1 (Profit $\geq$ \$80,000):
 Actual profit: \$80,000
 Underachievement: \$0, Overachievement: \$0

Goal 2 (Labor $\leq$ 1,000 hrs):
 Actual hours: 800
 Under: 200 hrs, Over: 0 hrs

Goal 3 (Units = 500):
 Actual units: 500
 Under: 0, Over: 0

23.5.3.2. Preemptive (Lexicographic) Goal Programming

When goals have a strict ordering of importance, preemptive goal programming handles them sequentially. The solver first minimizes deviations from the highest-priority goals. It then fixes those deviations at their optimal values and minimizes the next priority level, continuing until all priorities are addressed. This prevents lower-priority goals from ever degrading higher-priority outcomes.

Priority levels:

1. First minimize deviations from priority-1 goals
2. Then minimize priority-2 deviations (without worsening priority-1)
3. Continue for all priority levels

```
def preemptive_goal_programming():
    """
    Preemptive GP: Solve sequentially by priority.

    Priority 1: Meet minimum profit ($75,000)
    Priority 2: Stay within labor budget (1000 hrs)
    Priority 3: Achieve target production (500 units)
    """

    print("=== Preemptive Goal Programming ===\n")

    # Priority 1: Minimize profit underachievement
    # Variables: [x_A, x_B, d1-]
    c1 = [0, 0, 1] # Minimize d1-

    # Constraints
    A_eq1 = [[200, 150, 1]] # 200*x_A + 150*x_B + d1- = 75000
    b_eq1 = [75000]

    bounds1 = [(0, 500), (0, 500), (0, None)]

    result1 = linprog(c1, A_eq=A_eq1, b_eq=b_eq1, bounds=bounds1, method='highs')

    x_A, x_B = result1.x[0], result1.x[1]
    d1_optimal = result1.x[2]

    print(f"Priority 1 - Minimum Profit ($75,000):")
```

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```

print(f" Optimal underachievement: ${d1_optimal:,.0f}")
print(f" Actual profit: ${200*x_A + 150*x_B:,.0f}")

# Priority 2: Minimize labor overuse, keeping P1 optimal
# Variables: [x_A, x_B, d1-, d2-, d2+]
c2 = [0, 0, 0, 0, 1] # Minimize d2+

A_eq2 = [
    [200, 150, 1, 0, 0],      # Profit goal
    [2, 1.5, 0, 1, -1]        # Labor goal: 2*x_A + 1.5*x_B + d2- - d2+ = 1000
]
b_eq2 = [75000, 1000]

# Fix d1- at optimal value
A_ub2 = [[0, 0, 1, 0, 0]] # d1- <= d1_optimal
b_ub2 = [d1_optimal + 0.01]

bounds2 = [(0, 500), (0, 500), (0, None), (0, None), (0, None)]

result2 = linprog(c2, A_eq=A_eq2, b_eq=b_eq2,
                  A_ub=A_ub2, b_ub=b_ub2,
                  bounds=bounds2, method='highs')

x_A, x_B = result2.x[0], result2.x[1]
d2_optimal = result2.x[4]

print(f"\nPriority 2 - Labor Limit (1000 hrs):")
print(f" Optimal overuse: {d2_optimal:,.0f} hrs")
print(f" Actual labor: {2*x_A + 1.5*x_B:,.0f} hrs")

# Priority 3: Minimize production deviation, keeping P1 & P2 optimal
# For simplicity, solve with fixed constraints
print(f"\nPriority 3 - Target Production (500 units):")
units = x_A + x_B
print(f" Actual units: {units:,.0f}")
print(f" Deviation: {abs(units - 500):,.0f}")

print(f"\n=== Final Solution ===")
print(f"Produce: {x_A:,.0f} units of A, {x_B:,.0f} units of B")
print(f"Profit: ${200*x_A + 150*x_B:,.0f}")
print(f"Labor: {2*x_A + 1.5*x_B:,.0f} hours")
print(f"Total units: {x_A + x_B:,.0f}")

preemptive_goal_programming()

```

=== Preemptive Goal Programming ===

Priority 1 - Minimum Profit (\$75,000):

Optimal underachievement: \$0

Actual profit: \$75,000

Priority 2 - Labor Limit (1000 hrs):

Optimal overuse: 0 hrs

Actual labor: 750 hrs

Priority 3 - Target Production (500 units):

Actual units: 375

Deviation: 125

=== Final Solution ===

Produce: 375 units of A, 0 units of B

Profit: \$75,000

Labor: 750 hours

Total units: 375

23.5.4. Complete Python Implementation

The following class provides a reusable Goal Programming solver built on `scipy.optimize.linprog`. It accepts goals with different types (at-least, at-most, exactly), weights, and priorities, and solves the weighted formulation as a single linear program.

```
from dataclasses import dataclass
from typing import List, Dict, Optional, Tuple
from enum import Enum

class GoalType(Enum):
    """Type of goal constraint."""
    AT_LEAST = "at_least"      # f(x) >= target
    AT_MOST = "at_most"       # f(x) <= target
    EXACTLY = "exactly"       # f(x) = target

@dataclass
class Goal:
    """Represents a goal in goal programming."""
    name: str
    coefficients: Dict[str, float] # Variable name -> coefficient
    target: float
    goal_type: GoalType
    priority: int = 1
    weight_under: float = 1.0
    weight_over: float = 1.0

class GoalProgrammingSolver:
    """
    Weighted Goal Programming solver.

    Usage:
    solver = GoalProgrammingSolver()
    solver.add_variable("x1", lb=0)
    solver.add_variable("x2", lb=0)
    solver.add_goal(Goal("profit", {"x1": 100, "x2": 80}, 5000,
                        GoalType.AT_LEAST, weight_under=10))
    solver.add_constraint({"x1": 1, "x2": 1}, "<=", 100)
```

```

        result = solver.solve()
    """

    def __init__(self):
        self.variables: Dict[str, Tuple[float, float]] = {}
        self.goals: List[Goal] = []
        self.constraints: List[Tuple[Dict[str, float], str, float]] = []

    def add_variable(self, name: str, lb: float = 0, ub: float = None):
        """Add decision variable."""
        self.variables[name] = (lb, ub if ub is not None else 1e10)

    def add_goal(self, goal: Goal):
        """Add goal to the problem."""
        self.goals.append(goal)

    def add_constraint(self, coeffs: Dict[str, float], sense: str, rhs: float):
        """Add hard constraint."""
        self.constraints.append((coeffs, sense, rhs))

    def solve(self) -> Dict:
        """Solve using scipy's linprog."""
        n_vars = len(self.variables)
        n_goals = len(self.goals)
        var_names = list(self.variables.keys())
        var_idx = {name: i for i, name in enumerate(var_names)}

        # Total variables: decision vars + 2 deviation vars per goal
        total_vars = n_vars + 2 * n_goals

        # Build objective: minimize weighted deviations
        c = np.zeros(total_vars)
        for i, goal in enumerate(self.goals):
            d_minus_idx = n_vars + 2 * i
            d_plus_idx = n_vars + 2 * i + 1

            if goal.goal_type == GoalType.AT_LEAST:
                c[d_minus_idx] = goal.weight_under
            elif goal.goal_type == GoalType.AT_MOST:
                c[d_plus_idx] = goal.weight_over
            else: # EXACTLY
                c[d_minus_idx] = goal.weight_under
                c[d_plus_idx] = goal.weight_over

        # Goal equations: sum(coeff * x) + d- - d+ = target
        A_eq = []
        b_eq = []
        for i, goal in enumerate(self.goals):
            row = np.zeros(total_vars)
            for var_name, coeff in goal.coefficients.items():
                row[var_idx[var_name]] = coeff
            row[n_vars + 2 * i] = 1 # d-
            row[n_vars + 2 * i + 1] = -1 # d+

```



```

        A_eq.append(row)
        b_eq.append(goal.target)

# Hard constraints
A_ub = []
b_ub = []
for coeffs, sense, rhs in self.constraints:
    row = np.zeros(total_vars)
    for var_name, coeff in coeffs.items():
        row[var_idx[var_name]] = coeff

    if sense == "<=":
        A_ub.append(row)
        b_ub.append(rhs)
    elif sense == ">=":
        A_ub.append(-row)
        b_ub.append(-rhs)

# Bounds
bounds = []
for name in var_names:
    bounds.append(self.variables[name])
for _ in range(2 * n_goals):
    bounds.append((0, None))

# Solve
result = linprog(
    c,
    A_eq=np.array(A_eq) if A_eq else None,
    b_eq=np.array(b_eq) if b_eq else None,
    A_ub=np.array(A_ub) if A_ub else None,
    b_ub=np.array(b_ub) if b_ub else None,
    bounds=bounds,
    method='highs'
)

# Extract solution
solution = {
    'success': result.success,
    'variables': {name: result.x[i] for i, name in enumerate(var_names)},
    'goals': {}
}

for i, goal in enumerate(self.goals):
    d_minus = result.x[n_vars + 2 * i]
    d_plus = result.x[n_vars + 2 * i + 1]
    actual = sum(
        coeff * result.x[var_idx[var]]
        for var, coeff in goal.coefficients.items()
    )
    solution['goals'][goal.name] = {
        'target': goal.target,
        'actual': actual,
    }

```

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```
        'underachievement': d_minus,
        'overachievement': d_plus,
        'achieved': (d_minus < 0.01 and d_plus < 0.01)
    }

    return solution

# Example: Portfolio with Goals
print("=== Portfolio Goal Programming ===\n")

solver = GoalProgrammingSolver()

# Decision variables: fraction invested in each asset
solver.add_variable("stocks", lb=0, ub=1)
solver.add_variable("bonds", lb=0, ub=1)
solver.add_variable("real_estate", lb=0, ub=1)
solver.add_variable("cash", lb=0, ub=1)

# Goals
# 1. Return >= 8% (high priority)
solver.add_goal(Goal(
    "return",
    {"stocks": 0.12, "bonds": 0.05, "real_estate": 0.08, "cash": 0.02},
    0.08, GoalType.AT_LEAST, weight_under=100
))

# 2. Risk (std dev) <= 15% (medium priority)
solver.add_goal(Goal(
    "risk",
    {"stocks": 0.20, "bonds": 0.05, "real_estate": 0.12, "cash": 0.01},
    0.15, GoalType.AT_MOST, weight_over=50
))

# 3. Liquidity (stocks + cash) >= 40% (lower priority)
solver.add_goal(Goal(
    "liquidity",
    {"stocks": 1, "bonds": 0, "real_estate": 0, "cash": 1},
    0.40, GoalType.AT_LEAST, weight_under=20
))

# Hard constraint: weights sum to 1
solver.add_constraint(
    {"stocks": 1, "bonds": 1, "real_estate": 1, "cash": 1},
    "<=", 1.0
)
solver.add_constraint(
    {"stocks": 1, "bonds": 1, "real_estate": 1, "cash": 1},
    ">=", 1.0
)

result = solver.solve()
```

```

print("Portfolio Allocation:")
for asset, weight in result['variables'].items():
    print(f"    {asset}: {weight*100:.1f}%")

print("\nGoal Achievement:")
for goal_name, goal_result in result['goals'].items():
    status = "ACHIEVED" if goal_result['achieved'] else "MISSED"
    print(f"    {goal_name}: target={goal_result['target']:.2%}, "
          f"actual={goal_result['actual']:.2%} [{status}]")

```

=== Portfolio Goal Programming ===

Portfolio Allocation:

```

stocks: 38.9%
bonds: 0.0%
real_estate: 60.0%
cash: 1.1%

```

Goal Achievement:

```

return: target=8.00%, actual=9.49% [MISSED]
risk: target=15.00%, actual=15.00% [ACHIEVED]
liquidity: target=40.00%, actual=40.00% [ACHIEVED]

```

23.5.5. Real-World Example: Production Planning

This example applies goal programming to a manufacturing company producing three products. The company has four goals at different priority levels: profit target, overtime limit, market share requirement, and inventory target. Hard constraints on labor and machine capacity limit feasible production.

```

def production_planning_goal_programming():
    """
    Manufacturing company with multiple goals:
    - Goal 1: Profit >= $100,000 (highest priority)
    - Goal 2: Overtime <= 50 hours (medium priority)
    - Goal 3: Market share (production) >= 2000 units (medium)
    - Goal 4: Inventory = 500 units (lower priority)

    Products: Standard, Premium, Deluxe
    """

    print("=== Production Planning with Goal Programming ===\n")

    solver = GoalProgrammingSolver()

    # Products
    products = ["standard", "premium", "deluxe"]
    for p in products:
        solver.add_variable(p, lb=0, ub=2000)

    # Overtime variable

```

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```
solver.add_variable("overtime", lb=0, ub=200)

# Goal 1: Profit >= $100,000
# Standard: $50, Premium: $80, Deluxe: $120
solver.add_goal(Goal(
    "profit",
    {"standard": 50, "premium": 80, "deluxe": 120},
    100000, GoalType.AT_LEAST, weight_under=100
))

# Goal 2: Overtime <= 50 hours
solver.add_goal(Goal(
    "overtime_limit",
    {"overtime": 1},
    50, GoalType.AT_MOST, weight_over=50
))

# Goal 3: Total production >= 2000 units
solver.add_goal(Goal(
    "market_share",
    {"standard": 1, "premium": 1, "deluxe": 1},
    2000, GoalType.AT_LEAST, weight_under=30
))

# Goal 4: Ending inventory = 500 (assuming production - sales)
# Simplified: target production = 2500 (2000 sales + 500 inventory)
solver.add_goal(Goal(
    "inventory_target",
    {"standard": 1, "premium": 1, "deluxe": 1},
    2500, GoalType.EXACTLY, weight_under=10, weight_over=10
))

# Hard constraint: Labor capacity
# Standard: 2 hrs, Premium: 3 hrs, Deluxe: 4 hrs
# Base capacity: 5000 hrs + overtime
solver.add_constraint(
    {"standard": 2, "premium": 3, "deluxe": 4, "overtime": -1},
    "<=", 5000
)

# Machine capacity: 6000 machine-hours
solver.add_constraint(
    {"standard": 1.5, "premium": 2, "deluxe": 3},
    "<=", 6000
)

result = solver.solve()

if result['success']:
    print("Optimal Production Plan:")
    total_units = 0
    total_profit = 0
    profits = {"standard": 50, "premium": 80, "deluxe": 120}
```

```

for product in products:
    units = result['variables'][product]
    profit = units * profits[product]
    total_units += units
    total_profit += profit
    print(f" {product.capitalize()}: {units:.0f} units (${profit:,.0f})")

print(f"\nOvertime used: {result['variables']['overtime']:.0f} hours")
print(f"Total production: {total_units:.0f} units")
print(f"Total profit: ${total_profit:,.0f}")

print("\n" + "="*50)
print("Goal Achievement Summary:")
print("="*50)
for goal_name, goal_result in result['goals'].items():
    status = " ACHIEVED" if goal_result['achieved'] else " MISSED"
    under = goal_result['underachievement']
    over = goal_result['overachievement']
    deviation = f"under:{under:.1f}" if under > 0.01 else f"over:{over:.1f}"
    if over > 0.01 else "none"
    print(f"{goal_name:20} | Target: {goal_result['target']:>10,.0f} | "
          f"Actual: {goal_result['actual']:>10,.0f} | {status}")

production_planning_goal_programming()

```

=== Production Planning with Goal Programming ===

Optimal Production Plan:

Standard: 2000 units (\$100,000)

Premium: 350 units (\$28,000)

Deluxe: 0 units (\$0)

Overtime used: 50 hours

Total production: 2350 units

Total profit: \$128,000

Goal Achievement Summary:

profit	Target:	100,000	Actual:	128,000	MISSED
overtime_limit	Target:	50	Actual:	50	ACHIEVED
market_share	Target:	2,000	Actual:	2,350	MISSED
inventory_target	Target:	2,500	Actual:	2,350	MISSED

23.5.6. Goal Programming vs. Other MOO Methods

Aspect	Goal Programming	Weighted Sum	Pareto Methods
Input	Target values	Weights	None

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Aspect	Goal Programming	Weighted Sum	Pareto Methods
Output	Single solution	Single solution	Solution set
Philosophy	Satisficing	Optimizing	Trade-off exploration
Decision timing	A priori	A priori	A posteriori
Ease of use	High	High	Moderate
Best for	Known targets	Clear preferences	Unknown preferences

💡 Practical Advice for Goal Programming

1. Start with achievable targets based on historical data or benchmarks
2. Use preemptive GP when goals have clear priority order
3. Use weighted GP when trade-offs between goals are acceptable
4. Sensitivity analysis: test different weights/priorities
5. Consider combining GP with Pareto analysis for deeper insights

23.6. Evolutionary Multi-Objective Algorithms

Classical scalarization methods generate one Pareto point per optimization run. Evolutionary algorithms take a fundamentally different approach: they maintain a population of candidate solutions and evolve the entire population simultaneously. This produces an approximation of the full Pareto front in a single run. The population-based nature of evolutionary methods also makes them effective for non-convex, discontinuous, and black-box objective functions where gradient information is unavailable.

23.6.1. NSGA-II

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is one of the most widely used multi-objective evolutionary algorithms. Deb et al. (2002) designed NSGA-II to address the computational cost and elitism limitations of its predecessor.

Key Features:

1. **Fast non-dominated sorting:** Ranks population into fronts with $O(MN^2)$ complexity, where M is the number of objectives and N is the population size
2. **Crowding distance:** Promotes diversity on Pareto front by preferring solutions in less crowded regions
3. **Elitism:** Preserves best solutions across generations through combined parent-offspring selection

The following code implements the core non-dominated sorting and crowding distance calculations used in NSGA-II.

```
import numpy as np
from typing import List, Tuple

def fast_non_dominated_sort(population: np.ndarray,
                             fitness: np.ndarray) -> List[List[int]]:
    """Sort population into non-dominated fronts."""
    n = len(population)
    domination_count = np.zeros(n, dtype=int)
```

```

dominated_by = [[] for _ in range(n)]
fronts = [[]]

# Compute domination relationships
for i in range(n):
    for j in range(i + 1, n):
        if dominates(fitness[i], fitness[j]):
            dominated_by[i].append(j)
            domination_count[j] += 1
        elif dominates(fitness[j], fitness[i]):
            dominated_by[j].append(i)
            domination_count[i] += 1

# First front: non-dominated solutions
for i in range(n):
    if domination_count[i] == 0:
        fronts[0].append(i)

# Subsequent fronts
i = 0
while fronts[i]:
    next_front = []
    for p in fronts[i]:
        for q in dominated_by[p]:
            domination_count[q] -= 1
            if domination_count[q] == 0:
                next_front.append(q)
    i += 1
    fronts.append(next_front)

return fronts[:-1] # Remove empty last front

def crowding_distance(fitness: np.ndarray,
                     front: List[int]) -> np.ndarray:
    """Compute crowding distance for solutions in a front."""
    n = len(front)
    m = fitness.shape[1]
    distance = np.zeros(n)

    for obj in range(m):
        # Sort by this objective
        sorted_idx = np.argsort(fitness[front, obj])

        # Boundary points get infinite distance
        distance[sorted_idx[0]] = np.inf
        distance[sorted_idx[-1]] = np.inf

        # Normalize by objective range
        obj_range = (fitness[front[sorted_idx[-1]], obj] -
                     fitness[front[sorted_idx[0]], obj])
        if obj_range == 0:
            continue

```

```

        # Interior points
        for i in range(1, n - 1):
            distance[sorted_idx[i]] += (
                fitness[front[sorted_idx[i+1]], obj] -
                fitness[front[sorted_idx[i-1]], obj]
            ) / obj_range

    return distance

def dominates(a: np.ndarray, b: np.ndarray) -> bool:
    """Check if solution a dominates solution b (minimization)."""
    return np.all(a <= b) and np.any(a < b)

```

NSGA-II Algorithm:

1. Initialize population P of size N
2. Evaluate objectives for all individuals
3. For each generation:
 - a) Create offspring Q using selection, crossover, mutation
 - b) Combine P and Q into R (size 2N)
 - c) Fast non-dominated sort of R into fronts F1, F2, ...
 - d) Fill new population P':
 - Add fronts in order until N would be exceeded
 - For last front that doesn't fit entirely:
 - Compute crowding distance, select most spread out
 - e) P = P'
4. Return final Pareto front approximation

23.6.2. MOEA/D

MOEA/D takes the decomposition idea from classical methods and integrates it into an evolutionary framework. Instead of driving a single population toward the Pareto front, MOEA/D maintains a set of scalar subproblems, each defined by a different weight vector. Neighboring subproblems (with similar weight vectors) share information through crossover and solution exchange. This makes MOEA/D computationally efficient and well-suited for problems with regular Pareto front shapes.

Multi-Objective Evolutionary Algorithm based on Decomposition (MOEA/D) decomposes the problem into scalar subproblems.

Key Idea:

- Define a set of uniformly distributed weight vectors
- Each weight vector defines a single-objective subproblem
- Neighboring subproblems share solutions

```

def tchebycheff_scalarization(f: np.ndarray, weights: np.ndarray,
                             z_ideal: np.ndarray) -> float:
    """Tchebycheff scalarizing function."""
    return np.max(weights * np.abs(f - z_ideal))

def moead_decomposition(n_objectives: int,
                        n_subproblems: int) -> np.ndarray:

```


NSGA-II Evolution on ZDT1 Test Problem

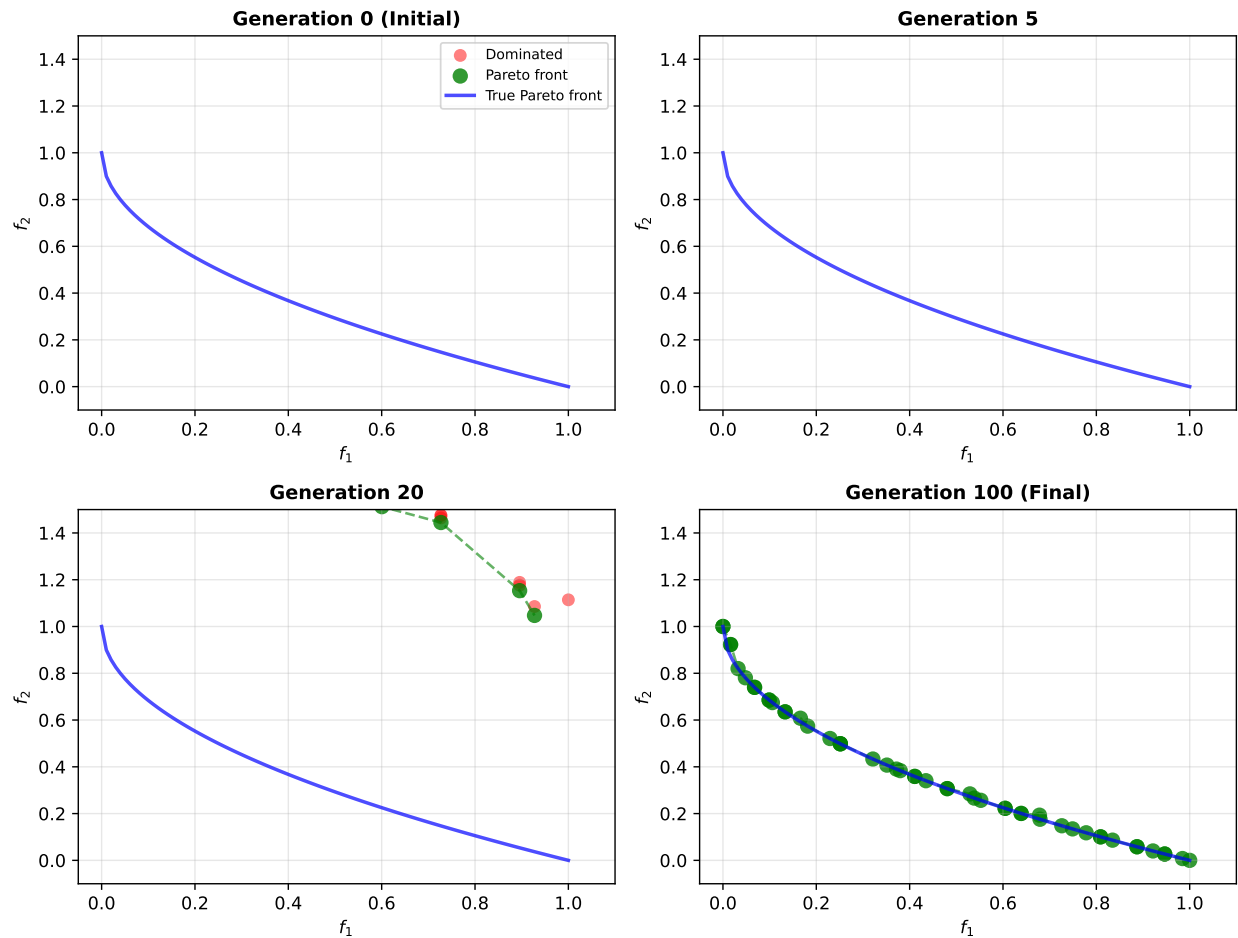


Figure 23.4.: NSGA-II population evolving towards the Pareto front over generations

```

"""Generate uniformly distributed weight vectors."""
# Simplex-lattice design for weight vectors
if n_objectives == 2:
    weights = np.zeros((n_subproblems, 2))
    for i in range(n_subproblems):
        weights[i, 0] = i / (n_subproblems - 1)
        weights[i, 1] = 1 - weights[i, 0]
    return weights
else:
    # Use Das and Dennis method for higher dimensions
    pass

```

23.6.3. Algorithm Comparison

Algorithm	Strengths	Weaknesses	Best For
NSGA-II	Well-studied, robust	Struggles with many objectives	2-3 objectives
NSGA-III	Handles many objectives	More complex	3+ objectives
MOEA/D	Efficient, good diversity	Weight vector dependent	Structured fronts
SPEA2	Strong convergence	Computational overhead	Small populations

23.7. Performance Metrics

Different algorithms produce different Pareto front approximations, and we need quantitative metrics to compare them. A good approximation should be close to the true front (convergence) and spread evenly across it (diversity). The following metrics capture these qualities.

23.7.1. Hypervolume

Hypervolume is the most widely recommended metric because it simultaneously measures both convergence and diversity without requiring knowledge of the true Pareto front. It computes the volume of objective space dominated by the solution set:

$$HV(S, r) = \text{volume} \left(\bigcup_{\mathbf{x} \in S} [\mathbf{F}(\mathbf{x}), r] \right)$$

where r is a reference point dominated by all solutions.

```
def hypervolume_2d(points: np.ndarray, reference: np.ndarray) -> float:
    """Compute 2D hypervolume (exact)."""
    # Sort by first objective
    sorted_idx = np.argsort(points[:, 0])
    sorted_points = points[sorted_idx]

    hv = 0.0
    prev_y = reference[1]

    for p in sorted_points:
        if p[1] < prev_y:
            hv += (reference[0] - p[0]) * (prev_y - p[1])
            prev_y = p[1]

    return hv
```

Properties:

- Higher is better
- Requires reference point selection
- Expensive for many objectives (exponential complexity)
- Only metric with known theoretical properties

23.7.2. Inverted Generational Distance (IGD)

Average distance from true Pareto front to approximation:

$$IGD(P^*, A) = \frac{1}{|P^*|} \sum_{\mathbf{y} \in P^*} \min_{\mathbf{x} \in A} d(\mathbf{x}, \mathbf{y})$$

Properties:

- Lower is better
- Requires known true Pareto front
- Measures both convergence and diversity

23.8. Decision Making

Generating the Pareto front is only half the problem. The decision maker must then select a single solution from the front. The choice depends on preferences, which can be specified before, during, or after the optimization process. Each timing has different advantages.

23.8.1. A Priori Methods

Preferences specified before optimization:

- **Goal programming:** Minimize deviation from targets
- **Lexicographic:** Optimize objectives in priority order
- **Reference point:** Find solution closest to ideal

23.8.2. A Posteriori Methods

Preferences applied after Pareto front generation:

1. **Visual inspection:** Plot and choose
2. **Knee point:** Solution with best trade-off (maximum curvature)
3. **TOPSIS:** Distance to ideal and anti-ideal points

The knee point is the solution on the Pareto front where a small improvement in one objective requires a large sacrifice in another. Geometrically, it is the point with the maximum perpendicular distance from the line connecting the two extreme solutions on the front.

```
def find_knee_point(pareto_front: np.ndarray) -> int:
    """Find knee point using maximum curvature."""
    # Normalize objectives
    normalized = (pareto_front - pareto_front.min(axis=0)) / \
        (pareto_front.max(axis=0) - pareto_front.min(axis=0))

    # Distance to utopia line
    n = len(normalized)
    line_start = normalized[0]
    line_end = normalized[-1]
    line_vec = line_end - line_start
    line_len = np.linalg.norm(line_vec)

    distances = np.zeros(n)
    for i, point in enumerate(normalized):
        t = np.dot(point - line_start, line_vec) / (line_len ** 2)
        projection = line_start + t * line_vec
        distances[i] = np.linalg.norm(point - projection)

    return np.argmax(distances)
```

23.9. Many-Objective Optimization

Problems with 4+ objectives present special challenges:

Issues:

- Most solutions become non-dominated (weak selection pressure)
- Visualization becomes difficult
- Crowding distance loses effectiveness
- Hypervolume computation becomes intractable

Solutions:

- **Reference-point methods** (NSGA-III): Use structured reference points
- **Indicator-based methods**: Selection based on hypervolume contribution
- **Preference incorporation**: Focus on region of interest
- **Dimensionality reduction**: Project to lower dimensions



Practical Advice

For problems with more than 3-4 objectives, consider whether all objectives are truly independent. Often, some objectives are correlated and can be combined or one can serve as proxy for another.

23.10. Summary

23.10.1. Key Takeaways

- Multi-objective optimization produces a set of trade-off solutions (Pareto front)
- Pareto dominance defines partial ordering: better in at least one objective, no worse in others
- Weighted sum works for convex fronts; epsilon-constraint handles non-convex regions
- NSGA-II and MOEA/D are widely used evolutionary algorithms for MOO
- Hypervolume is the most comprehensive quality metric
- Decision making requires selecting from Pareto front based on preferences
- Many-objective problems (4+ objectives) require specialized techniques

23.11. Exercises

1. **Pareto dominance**: Given solutions $A=(3,5)$, $B=(4,4)$, $C=(2,7)$, $D=(5,3)$, $E=(4,6)$ with two objectives to minimize, identify all non-dominated solutions.
2. **Weighted sum limitation**: Construct a simple 2-objective problem where the weighted sum method cannot find all Pareto optimal solutions.
3. **Implementation**: Implement NSGA-II for the bi-objective function $f_1(x) = x^2$ and $f_2(x) = (x - 2)^2$ over $x \in [0, 4]$.
4. **Metric comparison**: For two Pareto front approximations, compute hypervolume and IGD. Discuss which approximation is better and why.

24. AI-Based Optimization

What You Will Learn

- Distinguish AI-based optimization from traditional adaptive methods
- Apply Bayesian optimization for expensive black-box functions
- Use reinforcement learning for sequential decision problems
- Implement neural combinatorial optimization approaches
- Understand when AI helps vs. traditional optimization

AI-based optimization represents a paradigm shift: instead of following fixed mathematical rules, these methods learn from experience, adapt their strategies, and improve performance across multiple problem instances. This chapter clarifies what truly constitutes AI-based optimization and how to apply these methods effectively.

24.1. What Makes Optimization “AI-Based”?

24.1.1. The Key Distinction: Learning vs. Adapting

Traditional Optimization	AI-Based Optimization
Follows fixed mathematical rules	Learns from experience
Same strategy on every problem	Improves strategy over time
No memory of past problems	Transfers knowledge across problems
Adaptation is pre-programmed	Builds internal models

! The Three Pillars of Learning

Learning in AI-based optimization requires:

1. **Experience accumulation:** Gathers data from past optimization attempts
2. **Performance improvement:** Strategy becomes more effective over time
3. **Generalization:** Knowledge transfers to new, unseen problem instances

An algorithm that only adjusts parameters within one problem (like adaptive step sizes) is not learning—it’s following pre-programmed adaptation rules.

24.1.2. What IS AI-Based

- Bayesian optimization
- Reinforcement learning for optimization
- Neural architecture search with learning
- Meta-learned optimizers
- Learned heuristics for combinatorial problems
- AutoML with transfer learning

24.1.3. What is NOT AI-Based

- Gradient descent, SGD, Adam, RMSprop (adaptive, not learning)
- Newton's method, BFGS, L-BFGS (fixed mathematical rules)
- Simulated annealing (probabilistic, not learning)
- Basic genetic algorithms (evolutionary, not learning across instances)
- Particle swarm optimization (swarm behavior, not learning)

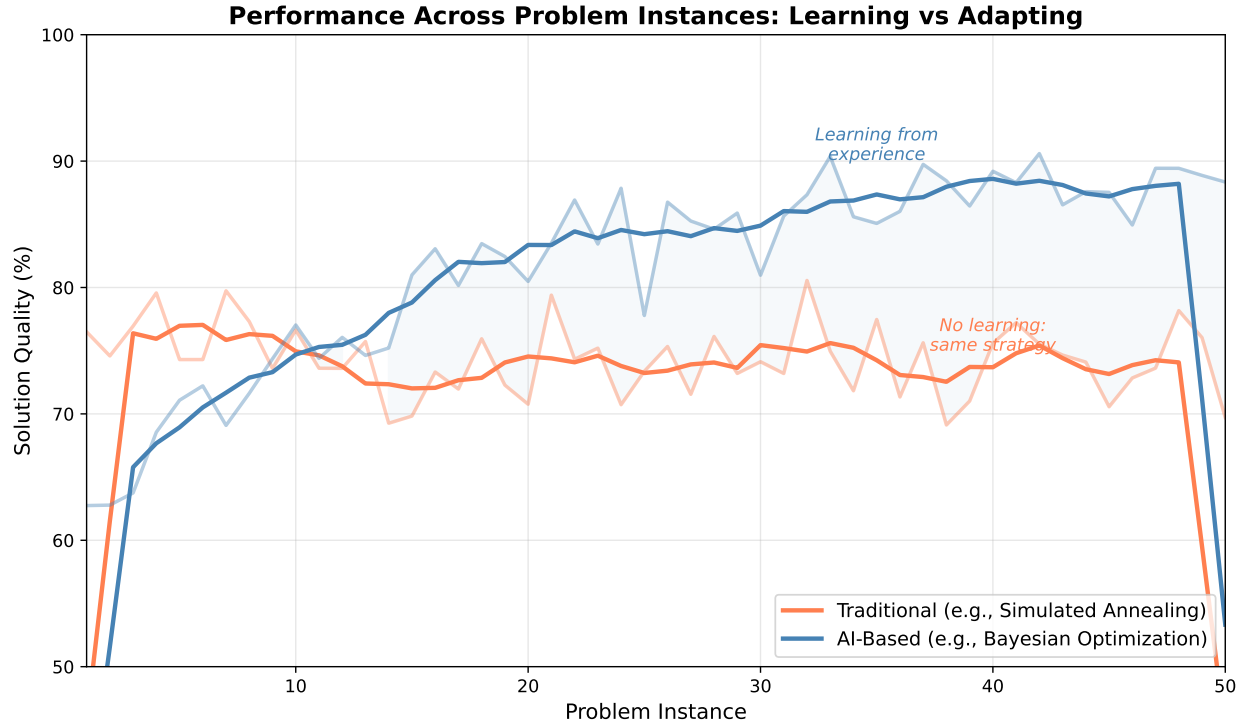


Figure 24.1.: Learning vs Adapting — traditional optimizers maintain flat performance across problem instances while AI-based optimizers improve as they accumulate experience

24.2. Bayesian Optimization

Bayesian optimization is the quintessential AI-based optimization method: it builds a probabilistic model of the objective function and uses it to make intelligent sampling decisions.

24.2.1. Core Concept

Given an expensive black-box function $f(\mathbf{x})$:

1. Build surrogate model: $p(f|\mathcal{D})$ using Gaussian process
2. Optimize acquisition function: $\mathbf{x}_{next} = \arg \max \alpha(\mathbf{x}|p(f|\mathcal{D}))$
3. Evaluate: $y = f(\mathbf{x}_{next})$, update $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{x}_{next}, y)\}$
4. Repeat until budget exhausted

24.2.2. Gaussian Process Surrogate

A Gaussian process defines a distribution over functions:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$

Given observations $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$:

$$\mu(\mathbf{x}) = \mathbf{k}^T(\mathbf{K} + \sigma^2 I)^{-1} \mathbf{y}$$

$$\sigma^2(\mathbf{x}) = k(\mathbf{x}, \mathbf{x}) - \mathbf{k}^T(\mathbf{K} + \sigma^2 I)^{-1} \mathbf{k}$$

24.2.3. Acquisition Functions

Function	Formula	Behavior
Expected Improvement	$\mathbb{E}[\max(f(\mathbf{x}) - f^*, 0)]$	Balanced exploration-exploitation
Upper Confidence Bound	$\mu(\mathbf{x}) + \beta\sigma(\mathbf{x})$	Explicit exploration control
Probability of Improvement	$P(f(\mathbf{x}) > f^*)$	Conservative, focused

```
import numpy as np
from scipy.stats import norm
from scipy.optimize import minimize
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import Matern

class BayesianOptimizer:
    """Bayesian optimization with Gaussian process surrogate."""

    def __init__(self, bounds, kernel=None, xi=0.01):
        """Initialize optimizer.

        Args:
            bounds: List of (min, max) tuples for each dimension
            kernel: GP kernel (default: Matern)
            xi: Exploration parameter for EI
        """
        self.bounds = np.array(bounds)
        self.dim = len(bounds)
        self.xi = xi

        if kernel is None:
            kernel = Matern(nu=2.5)

        self.gp = GaussianProcessRegressor(
            kernel=kernel,
            alpha=1e-6,
            normalize_y=True,
            n_restarts_optimizer=5
        )

        self.X_observed = []
        self.y_observed = []

    def expected_improvement(self, X):
        """Compute Expected Improvement acquisition function."""
        X = np.atleast_2d(X)
        mu, sigma = self.gp.predict(X, return_std=True)

        if len(self.y_observed) == 0:
```

```

        return sigma

    y_best = max(self.y_observed)

    with np.errstate(divide='warn'):
        improvement = mu - y_best - self.xi
        Z = improvement / sigma
        ei = improvement * norm.cdf(Z) + sigma * norm.pdf(Z)
        ei[sigma == 0.0] = 0.0

    return ei

def suggest_next(self, n_restarts=10):
    """Suggest next point to evaluate."""
    if len(self.X_observed) < 2:
        # Random sampling initially
        return np.random.uniform(
            self.bounds[:, 0],
            self.bounds[:, 1]
        )

    # Fit GP to observations
    self.gp.fit(np.array(self.X_observed), np.array(self.y_observed))

    # Optimize acquisition function
    best_x = None
    best_ei = -np.inf

    for _ in range(n_restarts):
        x0 = np.random.uniform(self.bounds[:, 0], self.bounds[:, 1])

        result = minimize(
            lambda x: -self.expected_improvement(x.reshape(1, -1)),
            x0,
            bounds=self.bounds,
            method='L-BFGS-B'
        )

        if -result.fun > best_ei:
            best_ei = -result.fun
            best_x = result.x

    return best_x

def observe(self, x, y):
    """Record observation."""
    self.X_observed.append(x)
    self.y_observed.append(y)

def optimize(self, objective, n_iterations=20):
    """Run Bayesian optimization loop.

    Args:

```



```

    objective: Function to maximize
    n_iterations: Number of iterations

Returns:
    Best x, best y
"""
for i in range(n_iterations):
    x_next = self.suggest_next()
    y_next = objective(x_next)
    self.observe(x_next, y_next)

    print(f"Iteration {i+1}: f({x_next}) = {y_next:.4f}")

best_idx = np.argmax(self.y_observed)
return self.X_observed[best_idx], self.y_observed[best_idx]

```

24.2.4. Applications

Domain	Problem	Why Bayesian Optimization
ML	Hyperparameter tuning	Training is expensive
Science	Experimental design	Each trial is costly
Engineering	Simulation optimization	Simulations are slow
Business	A/B testing	Limited user trials

24.3. Markov Decision Processes

Before studying reinforcement learning, we need the mathematical framework that underlies sequential decision-making: Markov Decision Processes (MDPs). MDPs provide the formal foundation for both classical dynamic programming solutions and modern RL approaches.

24.3.1. MDP Components

A Markov Decision Process is defined by a tuple (S, A, P, R, γ) :

Component	Symbol	Description
States	S	Set of possible states the system can be in
Actions	A	Set of actions available to the decision-maker
Transition Function	$P(s' s, a)$	Probability of reaching state s' given state s and action a
Reward Function	$R(s, a)$	Expected immediate reward for taking action a in state s
Discount Factor	$\gamma \in [0, 1]$	Relative importance of future rewards

The Markov Property

The key assumption is that the future depends only on the current state, not the history:

$$P(s_{t+1}|s_t, a_t, s_{t-1}, a_{t-1}, \dots) = P(s_{t+1}|s_t, a_t)$$

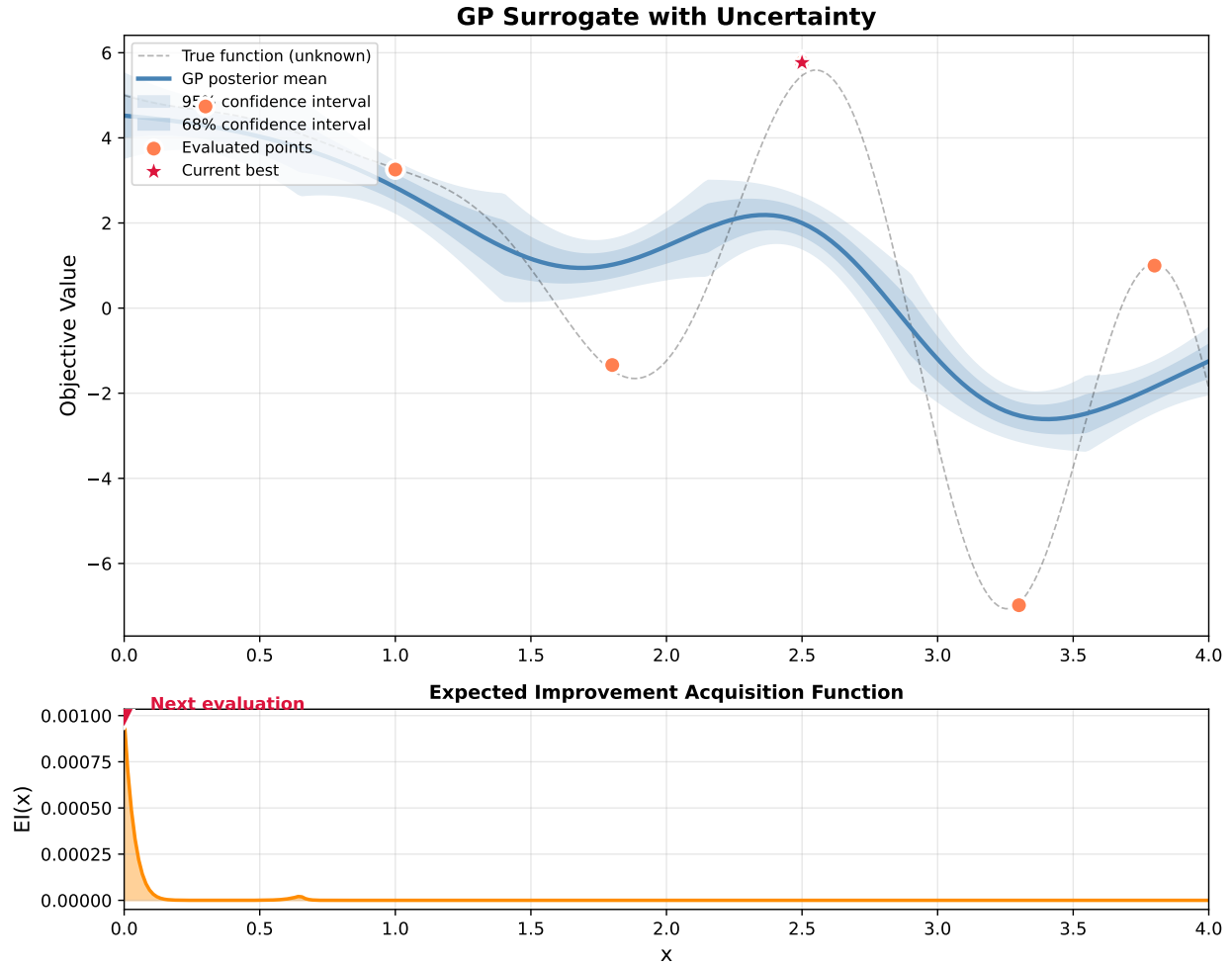


Figure 24.2.: Bayesian Optimization surrogate model — GP posterior mean with uncertainty bands and evaluated points, showing how uncertainty decreases near observations

This memoryless property enables tractable solutions.

24.3.2. Value Functions

Value functions quantify how good it is to be in a state (or state-action pair) under a given policy.

State-Value Function

The expected return starting from state s and following policy π :

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s \right]$$

Action-Value Function (Q-function)

The expected return starting from state s , taking action a , then following policy π :

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s, a_0 = a \right]$$

Relationship:

$$V^\pi(s) = \sum_{a \in A} \pi(a|s) Q^\pi(s, a)$$

24.3.3. Bellman Equations

The Bellman equations express the recursive relationship between value functions.

Bellman Expectation Equation (for policy π):

$$V^\pi(s) = \sum_a \pi(a|s) \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s') \right]$$

Bellman Optimality Equation (for optimal value):

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s') \right]$$

$$Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s'|s, a) \max_{a'} Q^*(s', a')$$

24.3.4. Solution Methods

When the transition probabilities and rewards are fully known, we can compute optimal policies without any simulation or learning. The two classical algorithms are value iteration and policy iteration. Both converge to the same optimal policy but differ in computational trade-offs.

Value Iteration

Value iteration applies the Bellman optimality operator repeatedly. At each iteration, it updates every state's value by selecting the action that maximizes the immediate reward plus discounted future value. The algorithm converges when the maximum change across all states falls below a threshold.

```
import numpy as np

def value_iteration(states, actions, transitions, rewards, gamma=0.99,
                    theta=1e-6, max_iter=1000):
    """Value iteration algorithm for MDPs.

    Args:
        states: List of states
        actions: List of actions
        transitions: P[s][a][s'] = probability
        rewards: R[s][a] = reward
        gamma: Discount factor
        theta: Convergence threshold
        max_iter: Maximum iterations

    Returns:
        V: Optimal value function
        policy: Optimal policy
    """
    n_states = len(states)
    V = np.zeros(n_states)
```

```

for iteration in range(max_iter):
    delta = 0

    for s in range(n_states):
        v = V[s]

        # Bellman optimality update
        action_values = []
        for a in range(len(actions)):
            q = rewards[s][a]
            for s_next in range(n_states):
                q += gamma * transitions[s][a][s_next] * V[s_next]
            action_values.append(q)

        V[s] = max(action_values)
        delta = max(delta, abs(v - V[s]))

    if delta < theta:
        print(f"Converged in {iteration + 1} iterations")
        break

# Extract policy
policy = np.zeros(n_states, dtype=int)
for s in range(n_states):
    action_values = []
    for a in range(len(actions)):
        q = rewards[s][a]
        for s_next in range(n_states):
            q += gamma * transitions[s][a][s_next] * V[s_next]
        action_values.append(q)
    policy[s] = np.argmax(action_values)

return V, policy

```

Policy Iteration

Policy iteration takes a different approach. It alternates between two phases: policy evaluation (computing the value function for the current policy by solving a system of linear equations) and policy improvement (greedily selecting better actions based on the computed values). Policy iteration typically converges in fewer iterations than value iteration, but each iteration is more expensive.

```

def policy_iteration(states, actions, transitions, rewards, gamma=0.99,
                    theta=1e-6):
    """Policy iteration algorithm for MDPs.

    Args:
        states, actions, transitions, rewards: MDP definition
        gamma: Discount factor
        theta: Convergence threshold for policy evaluation

    Returns:
        V: Optimal value function
    """

```

```

    policy: Optimal policy
    """
    n_states = len(states)
    n_actions = len(actions)

    # Initialize random policy
    policy = np.zeros(n_states, dtype=int)
    V = np.zeros(n_states)

    policy_stable = False
    iteration = 0

    while not policy_stable:
        iteration += 1

        # Policy Evaluation
        while True:
            delta = 0
            for s in range(n_states):
                v = V[s]
                a = policy[s]

                # Update V for current policy
                new_v = rewards[s][a]
                for s_next in range(n_states):
                    new_v += gamma * transitions[s][a][s_next] * V[s_next]
                V[s] = new_v

            delta = max(delta, abs(v - V[s]))

            if delta < theta:
                break

        # Policy Improvement
        policy_stable = True
        for s in range(n_states):
            old_action = policy[s]

            # Find best action
            action_values = []
            for a in range(n_actions):
                q = rewards[s][a]
                for s_next in range(n_states):
                    q += gamma * transitions[s][a][s_next] * V[s_next]
                action_values.append(q)

            policy[s] = np.argmax(action_values)

            if old_action != policy[s]:
                policy_stable = False

    print(f"Converged in {iteration} policy iterations")
    return V, policy

```

Comparison

Algorithm	Time per Iteration	Convergence	Memory
Value Iteration	$O(S ^2 A)$	Linear	$O(S)$
Policy Iteration	$O(S ^3 + S ^2 A)$	Fewer iterations	$O(S)$

Value iteration is simpler but may need more iterations. Policy iteration converges in fewer iterations but each iteration is more expensive due to policy evaluation.

24.3.5. MDP Example: Inventory Management

The following example models a single-product inventory system as an MDP. States represent inventory levels, actions represent order quantities, and demand follows a discrete distribution. The solver finds the optimal reorder policy that minimizes holding, shortage, and ordering costs.

```
def inventory_mdp_example():
    """Simple inventory management MDP."""
    max_inventory = 10
    max_order = 5
    holding_cost = 1
    shortage_cost = 10
    order_cost = 2

    # Demand distribution (Poisson-like)
    demand_probs = [0.1, 0.2, 0.3, 0.2, 0.1, 0.1] # demands 0-5

    states = list(range(max_inventory + 1))
    actions = list(range(max_order + 1))

    # Build transition matrix and rewards
    n_states = len(states)
    n_actions = len(actions)

    P = np.zeros((n_states, n_actions, n_states))
    R = np.zeros((n_states, n_actions))

    for s in states:
        for a in actions:
            # After ordering
            inventory_after_order = min(s + a, max_inventory)

            # Expected reward
            expected_reward = -order_cost * a - holding_cost * s

            # Transitions based on demand
            for d, prob in enumerate(demand_probs):
                new_inventory = max(0, inventory_after_order - d)
                shortage = max(0, d - inventory_after_order)

                P[s, a, new_inventory] += prob
                expected_reward -= prob * shortage_cost * shortage

            R[s, a] = expected_reward
```

```

# Solve MDP
V, policy = value_iteration(states, actions, P, R, gamma=0.95)

print("Optimal policy (order quantity for each inventory level):")
for s in states:
    print(f"  Inventory {s}: Order {policy[s]}")

return V, policy

# Run example
inventory_mdp_example()

```

From MDPs to Reinforcement Learning

Classical MDP solution methods (value iteration, policy iteration) require complete knowledge of the transition probabilities $P(s'|s, a)$ and rewards. When these are unknown, we use **reinforcement learning** to learn optimal policies through interaction with the environment.

24.4. Reinforcement Learning for Optimization

Reinforcement learning (RL) frames optimization as sequential decision-making, where an agent learns a policy through trial and error.

24.4.1. The RL Framework

- **State** s_t : Current solution or problem state
- **Action** a_t : Modification to solution
- **Reward** r_t : Improvement in objective
- **Policy** $\pi(a|s)$: Probability of taking action in state

24.4.2. Value Function Approximation

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t | s_0 = s \right]$$

The optimal policy maximizes expected cumulative reward.

24.4.3. Deep Q-Learning for Combinatorial Optimization

Deep Q-Learning (DQN) approximates the Q-function with a neural network, enabling RL to handle large or continuous state spaces where tabular methods are infeasible. The agent stores experience transitions in a replay buffer and trains on mini-batches, breaking the temporal correlation between consecutive samples. A separate target network provides stable training targets.

```

import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from collections import deque
import random

```

```

class DQN(nn.Module):
    """Deep Q-Network for combinatorial optimization."""

    def __init__(self, state_dim, action_dim, hidden_dim=128):
        super().__init__()
        self.network = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, action_dim)
        )

    def forward(self, x):
        return self.network(x)

class DQNOptimizer:
    """DQN-based optimizer for discrete optimization."""

    def __init__(self, state_dim, action_dim, lr=1e-3, gamma=0.99,
                 epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.995):
        self.state_dim = state_dim
        self.action_dim = action_dim
        self.gamma = gamma
        self.epsilon = epsilon_start
        self.epsilon_end = epsilon_end
        self.epsilon_decay = epsilon_decay

        self.policy_net = DQN(state_dim, action_dim)
        self.target_net = DQN(state_dim, action_dim)
        self.target_net.load_state_dict(self.policy_net.state_dict())

        self.optimizer = optim.Adam(self.policy_net.parameters(), lr=lr)
        self.memory = deque(maxlen=10000)

    def select_action(self, state):
        """Epsilon-greedy action selection."""
        if random.random() < self.epsilon:
            return random.randrange(self.action_dim)

        with torch.no_grad():
            state_tensor = torch.FloatTensor(state).unsqueeze(0)
            q_values = self.policy_net(state_tensor)
            return q_values.argmax().item()

    def store_transition(self, state, action, reward, next_state, done):
        """Store transition in replay buffer."""
        self.memory.append((state, action, reward, next_state, done))

    def train_step(self, batch_size=64):
        """Train on batch from replay buffer."""
        if len(self.memory) < batch_size:

```



```

        return

    batch = random.sample(self.memory, batch_size)
    states, actions, rewards, next_states, dones = zip(*batch)

    states = torch.FloatTensor(states)
    actions = torch.LongTensor(actions)
    rewards = torch.FloatTensor(rewards)
    next_states = torch.FloatTensor(next_states)
    dones = torch.BoolTensor(dones)

    # Current Q values
    current_q = self.policy_net(states).gather(1, actions.unsqueeze(1))

    # Target Q values
    with torch.no_grad():
        next_q = self.target_net(next_states).max(1)[0]
        target_q = rewards + self.gamma * next_q * (~dones)

    # Loss and update
    loss = nn.MSELoss()(current_q.squeeze(), target_q)
    self.optimizer.zero_grad()
    loss.backward()
    self.optimizer.step()

    # Decay epsilon
    self.epsilon = max(self.epsilon_end, self.epsilon * self.epsilon_decay)

def update_target(self):
    """Update target network."""
    self.target_net.load_state_dict(self.policy_net.state_dict())

```

24.4.4. Policy Gradient Methods

For continuous action spaces or when Q-learning struggles:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a|s) \cdot Q^{\pi_{\theta}}(s, a)]$$

24.5. Neural Combinatorial Optimization

Neural combinatorial optimization trains neural networks to directly construct solutions to combinatorial problems such as the Traveling Salesman Problem (TSP) and Vehicle Routing Problem (VRP). Unlike traditional solvers that rely on hand-designed heuristics, these models learn construction strategies from data. Once trained, they generate solutions in a single forward pass, offering significant speed advantages for real-time applications.

24.5.1. Attention-Based Models

Attention mechanisms allow the model to focus on different parts of the input at each construction step. At each step, the decoder attends to all unvisited nodes and selects the next one based on learned attention weights. The attention mechanism computes context-aware representations:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

24.5.2. Pointer Networks for Sequencing

Pointer Networks extend sequence-to-sequence models by replacing the fixed output vocabulary with a pointer mechanism. At each decoding step, the network points to one of the input elements rather than generating a token from a dictionary. This is a natural fit for combinatorial optimization, where the output is a permutation of the input set.

```
import torch
import torch.nn as nn
import torch.nn.functional as F

class PointerNetwork(nn.Module):
    """Pointer Network for sequence-to-sequence with pointing."""

    def __init__(self, input_dim, hidden_dim):
        super().__init__()
        self.hidden_dim = hidden_dim

        # Encoder
        self.encoder = nn.LSTM(input_dim, hidden_dim, batch_first=True)

        # Decoder
        self.decoder = nn.LSTMCell(input_dim, hidden_dim)

        # Attention
        self.W1 = nn.Linear(hidden_dim, hidden_dim)
        self.W2 = nn.Linear(hidden_dim, hidden_dim)
        self.v = nn.Linear(hidden_dim, 1)

    def forward(self, inputs, max_steps=None):
        """Generate output sequence by pointing to inputs.

        Args:
            inputs: (batch, seq_len, input_dim)
            max_steps: Maximum output length

        Returns:
            Sequence of pointer indices
        """
        batch_size, seq_len, _ = inputs.shape
        if max_steps is None:
            max_steps = seq_len

        # Encode
        encoder_outputs, (h, c) = self.encoder(inputs)

        # Decode
        outputs = []
        decoder_input = torch.zeros(batch_size, inputs.shape[-1])
```

```

h, c = h.squeeze(0), c.squeeze(0)

for _ in range(max_steps):
    h, c = self.decoder(decoder_input, (h, c))

    # Attention
    query = self.W1(h).unsqueeze(1) # (batch, 1, hidden)
    keys = self.W2(encoder_outputs) # (batch, seq, hidden)
    scores = self.v(torch.tanh(query + keys)).squeeze(-1)

    # Pointer probabilities
    probs = F.softmax(scores, dim=-1)
    outputs.append(probs)

    # Select and get next input
    idx = probs.argmax(dim=-1)
    decoder_input = inputs[torch.arange(batch_size), idx]

return torch.stack(outputs, dim=1)

```

24.5.3. Training with REINFORCE

Pointer Networks are trained using the REINFORCE policy gradient algorithm. The model samples tours from its current policy, computes the tour costs, and updates weights to increase the probability of lower-cost tours. A baseline (typically a moving average or a learned critic) reduces the variance of the gradient estimate.

```

def reinforce_loss(model, instances, baseline=None):
    """REINFORCE loss for neural combinatorial optimization.

    Args:
        model: Pointer network
        instances: Batch of problem instances
        baseline: Baseline for variance reduction

    Returns:
        Loss for gradient computation
    """
    # Get log probabilities and tours
    log_probs, tours = model.sample(instances)

    # Compute tour costs
    costs = compute_tour_cost(instances, tours)

    # Baseline (moving average or critic)
    if baseline is None:
        baseline = costs.mean()

    # REINFORCE gradient
    advantage = costs - baseline
    loss = (advantage * log_probs.sum(dim=1)).mean()

    return loss, costs.mean()

```

24.6. Deep Reinforcement Learning

While basic DQN and REINFORCE provide foundational approaches, modern deep RL algorithms offer significant improvements in stability, sample efficiency, and performance. These methods are increasingly used for complex optimization problems.

24.6.1. Advanced Value-Based Methods

Double DQN addresses overestimation bias in Q-learning by decoupling action selection from evaluation:

$$Q(s, a) \leftarrow r + \gamma Q_{\theta'}(s', \arg \max_{a'} Q_{\theta}(s', a'))$$

Dueling DQN separates state value from action advantage:

$$Q(s, a) = V(s) + A(s, a) - \frac{1}{|A|} \sum_{a'} A(s, a')$$

The Dueling DQN architecture separates the value of being in a state from the advantage of each action. This decomposition improves learning efficiency because the network can learn the state value independently from the action-specific advantages.

```
import torch
import torch.nn as nn

class DuelingDQN(nn.Module):
    """Dueling DQN architecture separating value and advantage."""

    def __init__(self, state_dim, action_dim, hidden_dim=256):
        super().__init__()

        # Shared feature extraction
        self.features = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU()
        )

        # Value stream
        self.value_stream = nn.Sequential(
            nn.Linear(hidden_dim, hidden_dim // 2),
            nn.ReLU(),
            nn.Linear(hidden_dim // 2, 1)
        )

        # Advantage stream
        self.advantage_stream = nn.Sequential(
            nn.Linear(hidden_dim, hidden_dim // 2),
            nn.ReLU(),
            nn.Linear(hidden_dim // 2, action_dim)
        )

    def forward(self, state):
```

```

features = self.features(state)

value = self.value_stream(features)
advantage = self.advantage_stream(features)

# Combine:  $Q = V + (A - \text{mean}(A))$ 
q_values = value + advantage - advantage.mean(dim=-1, keepdim=True)
return q_values

```

24.6.2. Actor-Critic Methods

Actor-critic methods combine the benefits of policy gradient (actor) and value function approximation (critic). The actor learns the policy while the critic evaluates it.

Advantage Actor-Critic (A2C)

The A2C implementation below uses a shared feature extractor with two output heads: the actor outputs action probabilities, and the critic outputs state value estimates. The advantage (return minus value) serves as the signal for policy gradient updates.

```

class ActorCritic(nn.Module):
    """Actor-Critic network for A2C/A3C."""

    def __init__(self, state_dim, action_dim, hidden_dim=256):
        super().__init__()

        # Shared layers
        self.shared = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU()
        )

        # Actor (policy) head
        self.actor = nn.Sequential(
            nn.Linear(hidden_dim, action_dim),
            nn.Softmax(dim=-1)
        )

        # Critic (value) head
        self.critic = nn.Linear(hidden_dim, 1)

    def forward(self, state):
        shared_features = self.shared(state)
        policy = self.actor(shared_features)
        value = self.critic(shared_features)
        return policy, value

    def act(self, state):
        """Sample action from policy."""
        policy, value = self.forward(state)
        dist = torch.distributions.Categorical(policy)
        action = dist.sample()

```

```

        return action, dist.log_prob(action), value

class A2CAgent:
    """Advantage Actor-Critic agent."""

    def __init__(self, state_dim, action_dim, lr=3e-4, gamma=0.99,
                  entropy_coef=0.01, value_coef=0.5):
        self.gamma = gamma
        self.entropy_coef = entropy_coef
        self.value_coef = value_coef

        self.network = ActorCritic(state_dim, action_dim)
        self.optimizer = torch.optim.Adam(self.network.parameters(), lr=lr)

    def compute_returns(self, rewards, dones, next_value):
        """Compute discounted returns."""
        returns = []
        R = next_value

        for reward, done in zip(reversed(rewards), reversed(dones)):
            R = reward + self.gamma * R * (1 - done)
            returns.insert(0, R)

        return torch.tensor(returns)

    def update(self, states, actions, rewards, dones, next_state):
        """Update actor and critic."""
        states = torch.FloatTensor(states)
        actions = torch.LongTensor(actions)

        # Get current values and policies
        policies, values = self.network(states)
        _, next_value = self.network(torch.FloatTensor([next_state]))

        # Compute returns and advantages
        returns = self.compute_returns(rewards, dones, next_value.item())
        advantages = returns - values.squeeze()

        # Actor loss (policy gradient with advantage)
        dist = torch.distributions.Categorical(policies)
        log_probs = dist.log_prob(actions)
        actor_loss = -(log_probs * advantages.detach()).mean()

        # Critic loss (value function)
        critic_loss = nn.MSELoss()(values.squeeze(), returns)

        # Entropy bonus for exploration
        entropy = dist.entropy().mean()

        # Combined loss
        loss = (actor_loss +
                self.value_coef * critic_loss -
                self.entropy_coef * entropy)

```

```

self.optimizer.zero_grad()
loss.backward()
self.optimizer.step()

return loss.item()

```

24.6.3. Proximal Policy Optimization (PPO)

PPO is one of the most popular deep RL algorithms due to its stability and ease of tuning. It constrains policy updates to prevent large, destabilizing changes.

Key Innovation: Clipped surrogate objective

$$L^{CLIP}(\theta) = \mathbb{E} [\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$$

where $r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio.

```

class PPOAgent:
    """Proximal Policy Optimization agent."""

    def __init__(self, state_dim, action_dim, lr=3e-4, gamma=0.99,
                  clip_epsilon=0.2, epochs=10, batch_size=64):
        self.gamma = gamma
        self.clip_epsilon = clip_epsilon
        self.epochs = epochs
        self.batch_size = batch_size

        self.network = ActorCritic(state_dim, action_dim)
        self.optimizer = torch.optim.Adam(self.network.parameters(), lr=lr)

    def compute_gae(self, rewards, values, dones, next_value, gae_lambda=0.95):
        """Generalized Advantage Estimation."""
        advantages = []
        gae = 0

        values = list(values) + [next_value]

        for t in reversed(range(len(rewards))):
            delta = rewards[t] + self.gamma * values[t+1] * (1 - dones[t]) -
            ↪ values[t]
            gae = delta + self.gamma * gae_lambda * (1 - dones[t]) * gae
            advantages.insert(0, gae)

        returns = [adv + val for adv, val in zip(advantages, values[:-1])]
        return torch.tensor(advantages), torch.tensor(returns)

    def update(self, trajectories):
        """PPO update with clipped objective."""
        states = torch.FloatTensor(trajectories['states'])
        actions = torch.LongTensor(trajectories['actions'])
        old_log_probs = torch.FloatTensor(trajectories['log_probs'])
        rewards = trajectories['rewards']

```

```

dones = trajectories['dones']

# Get values for GAE computation
with torch.no_grad():
    _, values = self.network(states)
    _, next_value = self.network(
        torch.FloatTensor([trajectories['next_state']]))
    )

# Compute advantages and returns
advantages, returns = self.compute_gae(
    rewards, values.squeeze().tolist(),
    dones, next_value.item()
)

# Normalize advantages
advantages = (advantages - advantages.mean()) / (advantages.std() + 1e-8)

# PPO update epochs
dataset_size = len(states)
indices = list(range(dataset_size))

for _ in range(self.epochs):
    random.shuffle(indices)

    for start in range(0, dataset_size, self.batch_size):
        end = start + self.batch_size
        batch_indices = indices[start:end]

        batch_states = states[batch_indices]
        batch_actions = actions[batch_indices]
        batch_old_log_probs = old_log_probs[batch_indices]
        batch_advantages = advantages[batch_indices]
        batch_returns = returns[batch_indices]

        # Current policy evaluation
        policies, values = self.network(batch_states)
        dist = torch.distributions.Categorical(policies)
        new_log_probs = dist.log_prob(batch_actions)
        entropy = dist.entropy().mean()

        # Probability ratio
        ratio = torch.exp(new_log_probs - batch_old_log_probs)

        # Clipped surrogate objective
        surr1 = ratio * batch_advantages
        surr2 = torch.clamp(
            ratio, 1 - self.clip_epsilon, 1 + self.clip_epsilon
        ) * batch_advantages

        actor_loss = -torch.min(surr1, surr2).mean()
        critic_loss = nn.MSELoss()(values.squeeze(), batch_returns)

```



```

loss = actor_loss + 0.5 * critic_loss - 0.01 * entropy

self.optimizer.zero_grad()
loss.backward()
torch.nn.utils.clip_grad_norm_(self.network.parameters(), 0.5)
self.optimizer.step()

```

24.6.4. Soft Actor-Critic (SAC)

SAC is a leading algorithm for continuous action spaces. It maximizes both expected return and policy entropy, leading to more robust and exploratory policies.

Objective: Maximum entropy RL

$$J(\pi) = \sum_{t=0}^T \mathbb{E} [r(s_t, a_t) + \alpha H(\pi(\cdot|s_t))]$$

where α is the temperature parameter controlling exploration.

The Gaussian policy outputs mean and log-standard-deviation parameters for a normal distribution over continuous actions. The reparameterization trick enables gradient-based training through the stochastic sampling step.

```

class GaussianPolicy(nn.Module):
    """Gaussian policy for continuous actions (SAC)."""

    def __init__(self, state_dim, action_dim, hidden_dim=256):
        super().__init__()

        self.network = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU()
        )

        self.mean = nn.Linear(hidden_dim, action_dim)
        self.log_std = nn.Linear(hidden_dim, action_dim)

        self.log_std_min = -20
        self.log_std_max = 2

    def forward(self, state):
        features = self.network(state)
        mean = self.mean(features)
        log_std = self.log_std(features)
        log_std = torch.clamp(log_std, self.log_std_min, self.log_std_max)
        return mean, log_std

    def sample(self, state):
        """Sample action with reparameterization trick."""
        mean, log_std = self.forward(state)
        std = torch.exp(log_std)

```

```

        normal = torch.distributions.Normal(mean, std)
        x_t = normal.rsample() # Reparameterized sample
        action = torch.tanh(x_t) # Squash to [-1, 1]

        # Log probability with correction for tanh squashing
        log_prob = normal.log_prob(x_t)
        log_prob -= torch.log(1 - action.pow(2) + 1e-6)
        log_prob = log_prob.sum(dim=-1, keepdim=True)

        return action, log_prob

class SoftQNetwork(nn.Module):
    """Soft Q-network for SAC."""

    def __init__(self, state_dim, action_dim, hidden_dim=256):
        super().__init__()

        self.network = nn.Sequential(
            nn.Linear(state_dim + action_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, 1)
        )

    def forward(self, state, action):
        x = torch.cat([state, action], dim=-1)
        return self.network(x)

class SACAgent:
    """Soft Actor-Critic agent for continuous control."""

    def __init__(self, state_dim, action_dim, lr=3e-4, gamma=0.99,
                 tau=0.005, alpha=0.2, auto_alpha=True):
        self.gamma = gamma
        self.tau = tau
        self.alpha = alpha
        self.auto_alpha = auto_alpha

        # Networks
        self.policy = GaussianPolicy(state_dim, action_dim)
        self.q1 = SoftQNetwork(state_dim, action_dim)
        self.q2 = SoftQNetwork(state_dim, action_dim)
        self.q1_target = SoftQNetwork(state_dim, action_dim)
        self.q2_target = SoftQNetwork(state_dim, action_dim)

        # Copy weights to targets
        self.q1_target.load_state_dict(self.q1.state_dict())
        self.q2_target.load_state_dict(self.q2.state_dict())

        # Optimizers
        self.policy_optimizer = torch.optim.Adam(self.policy.parameters(), lr=lr)
        self.q1_optimizer = torch.optim.Adam(self.q1.parameters(), lr=lr)

```

```

self.q2_optimizer = torch.optim.Adam(self.q2.parameters(), lr=lr)

# Automatic temperature tuning
if auto_alpha:
    self.target_entropy = -action_dim
    self.log_alpha = torch.zeros(1, requires_grad=True)
    self.alpha_optimizer = torch.optim.Adam([self.log_alpha], lr=lr)

# Replay buffer
self.buffer = []
self.buffer_size = 100000

def select_action(self, state, evaluate=False):
    """Select action from policy."""
    state = torch.FloatTensor(state).unsqueeze(0)

    if evaluate:
        mean, _ = self.policy.forward(state)
        return torch.tanh(mean).detach().numpy()[0]

    action, _ = self.policy.sample(state)
    return action.detach().numpy()[0]

def store_transition(self, state, action, reward, next_state, done):
    """Store transition in replay buffer."""
    if len(self.buffer) >= self.buffer_size:
        self.buffer.pop(0)
    self.buffer.append((state, action, reward, next_state, done))

def update(self, batch_size=256):
    """SAC update step."""
    if len(self.buffer) < batch_size:
        return

    # Sample batch
    batch = random.sample(self.buffer, batch_size)
    states, actions, rewards, next_states, dones = zip(*batch)

    states = torch.FloatTensor(states)
    actions = torch.FloatTensor(actions)
    rewards = torch.FloatTensor(rewards).unsqueeze(1)
    next_states = torch.FloatTensor(next_states)
    dones = torch.FloatTensor(dones).unsqueeze(1)

    # Update Q-functions
    with torch.no_grad():
        next_actions, next_log_probs = self.policy.sample(next_states)
        q1_next = self.q1_target(next_states, next_actions)
        q2_next = self.q2_target(next_states, next_actions)
        q_next = torch.min(q1_next, q2_next) - self.alpha * next_log_probs
        q_target = rewards + self.gamma * (1 - dones) * q_next

    q1_loss = nn.MSELoss()(self.q1(states, actions), q_target)

```

```

q2_loss = nn.MSELoss()(self.q2(states, actions), q_target)

self.q1_optimizer.zero_grad()
q1_loss.backward()
self.q1_optimizer.step()

self.q2_optimizer.zero_grad()
q2_loss.backward()
self.q2_optimizer.step()

# Update policy
new_actions, log_probs = self.policy.sample(states)
q1_new = self.q1(states, new_actions)
q2_new = self.q2(states, new_actions)
q_new = torch.min(q1_new, q2_new)

policy_loss = (self.alpha * log_probs - q_new).mean()

self.policy_optimizer.zero_grad()
policy_loss.backward()
self.policy_optimizer.step()

# Update temperature
if self.auto_alpha:
    alpha_loss = -(self.log_alpha * (
        log_probs + self.target_entropy
    ).detach()).mean()

    self.alpha_optimizer.zero_grad()
    alpha_loss.backward()
    self.alpha_optimizer.step()

    self.alpha = self.log_alpha.exp().item()

# Soft update targets
for param, target_param in zip(
    self.q1.parameters(), self.q1_target.parameters()
):
    target_param.data.copy_(
        self.tau * param.data + (1 - self.tau) * target_param.data
    )

for param, target_param in zip(
    self.q2.parameters(), self.q2_target.parameters()
):
    target_param.data.copy_(
        self.tau * param.data + (1 - self.tau) * target_param.data
    )

```

24.6.5. Deep RL for Optimization Problems

Deep RL algorithms can be applied to various optimization domains:

Problem Type	Algorithm	State Representation	Action Space
TSP/VRP	PPO with attention	Node embeddings	Next node selection
Scheduling	DQN/PPO	Job-machine state	Job-machine assignment
Inventory	SAC	Stock levels, demand	Order quantities
Bin Packing	PPO	Bin states, item features	Bin selection
Portfolio	SAC	Asset returns, positions	Portfolio weights

The following template defines a generic optimization environment that wraps any combinatorial problem as an RL task. The agent observes the current solution state, applies a local modification (the action), and receives a reward equal to the cost improvement.

```
class OptimizationEnv:
    """Generic optimization environment for RL."""

    def __init__(self, problem_generator, evaluator):
        self.problem_generator = problem_generator
        self.evaluator = evaluator
        self.current_solution = None
        self.current_problem = None

    def reset(self):
        """Generate new problem instance."""
        self.current_problem = self.problem_generator()
        self.current_solution = self._initial_solution()
        return self._get_state()

    def step(self, action):
        """Apply action and return new state, reward, done."""
        old_cost = self.evaluator(self.current_solution, self.current_problem)

        # Apply action (problem-specific)
        self.current_solution = self._apply_action(action)

        new_cost = self.evaluator(self.current_solution, self.current_problem)

        # Reward is improvement
        reward = old_cost - new_cost

        # Check termination
        done = self._is_terminal()

        return self._get_state(), reward, done, {}

    def _initial_solution(self):
        """Generate initial solution (to be implemented)."""
        raise NotImplementedError

    def _apply_action(self, action):
        """Apply action to solution (to be implemented)."""
        raise NotImplementedError

    def _get_state(self):
```

```

    """Get state representation (to be implemented)."""
    raise NotImplementedError

def _is_terminal(self):
    """Check if episode is done (to be implemented)."""
    raise NotImplementedError

```

💡 Choosing Deep RL Algorithms

- **Discrete actions:** Use PPO or DQN variants
- **Continuous actions:** Use SAC (most stable) or PPO with Gaussian policy
- **Sample efficiency matters:** Use SAC (off-policy, better sample reuse)
- **Simplicity preferred:** Use PPO (fewer hyperparameters, more robust)
- **Very large action spaces:** Consider attention-based architectures

24.7. Learning to Optimize

Meta-learning approaches go beyond using AI within an optimization algorithm—they learn the optimization algorithm itself. By training on distributions of problem instances, these methods learn decision rules that generalize to new, unseen instances of the same problem class.

24.7.1. Learning to Branch

Branch-and-bound solvers for mixed-integer programs must decide which variable to branch on at each node. Traditional heuristics (most infeasible, reliability branching) are hand-designed. A learned branching policy uses a graph neural network (GNN) to encode the LP relaxation structure and score branching candidates. The GNN is trained via imitation learning on expert (strong branching) decisions.

```

class BranchingPolicy(nn.Module):
    """Neural network policy for variable selection in B&B."""

    def __init__(self, node_features, edge_features, hidden_dim=64):
        super().__init__()

        # Graph neural network for node embedding
        self.gnn = GraphNeuralNetwork(node_features, hidden_dim)

        # Scoring layer
        self.score = nn.Sequential(
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, 1)
        )

    def forward(self, node_features, edge_index, candidate_mask):
        """Score branching candidates.

        Args:
            node_features: Features for each variable
            edge_index: Constraint graph structure

```

```

        candidate_mask: Which variables are branching candidates

Returns:
    Scores for each candidate
    """
    embeddings = self.gnn(node_features, edge_index)
    scores = self.score(embeddings).squeeze(-1)

    # Mask non-candidates
    scores[~candidate_mask] = float('-inf')

    return scores

```

Two open-source libraries make it practical to experiment with learned branching and other ML-solver integration strategies. **Ecole** (Combinatorial Optimization Learning Environments) wraps the SCIP solver to expose branching, cutting plane selection, and node selection as RL environments with standardized observation and reward functions. **MIPLearn**, developed by Argonne National Laboratory, takes a complementary approach: it learns from historical solves of the same problem class to predict warm-start solutions, lazy constraints, and branching priorities, providing speedups for production settings where similar MIP instances are solved repeatedly. See Chapter 39 for installation details and code examples.

24.7.2. Learning to Select Cuts

Similarly, learn which cutting planes to add in branch-and-cut.

24.8. When to Use AI-Based Optimization

24.8.1. AI-Based Methods Excel When

- **Expensive evaluations:** Each function call is costly (simulation, experiment)
- **Repeated similar problems:** Can learn patterns across instances
- **Black-box functions:** No gradient or structure information
- **Sequential decisions:** Actions affect future states

24.8.2. Traditional Methods Excel When

- **Cheap evaluations:** Can afford many function calls
- **One-off problems:** No benefit from learning
- **Known structure:** Can exploit convexity, linearity
- **Provable guarantees needed:** Optimality certificates required

Problem Type	Recommended Approach
Hyperparameter tuning	Bayesian optimization
MIP solving	Traditional B&B (possibly with learned heuristics)
Convex optimization	Gradient methods
Routing with patterns	RL or neural combinatorial
One-time scheduling	CP or MIP
Repeated scheduling	RL with learned policy

24.9. Summary

24.9.1. Key Takeaways

- AI-based optimization learns and improves from experience; traditional methods follow fixed rules
- Bayesian optimization builds probabilistic models for sample-efficient black-box optimization
- Reinforcement learning frames optimization as sequential decision-making with learned policies
- Neural combinatorial optimization uses deep learning to construct solutions directly
- Learning to optimize trains neural networks to make solver decisions (branching, cut selection)
- AI methods excel for expensive evaluations and repeated similar problems
- Traditional methods remain superior for cheap evaluations and when guarantees are needed

24.10. Exercises

1. **Bayesian optimization:** Implement Bayesian optimization with Expected Improvement for the Branin function. Compare sample efficiency with random search.
2. **Hyperparameter tuning:** Use Bayesian optimization to tune hyperparameters of a random forest classifier. Compare with grid search in terms of best accuracy achieved in 50 evaluations.
3. **RL for TSP:** Implement a simple policy gradient method for the Traveling Salesman Problem. Train on random 20-city instances and evaluate on held-out test instances.
4. **Learning to branch:** Design features for variable selection in branch-and-bound. Train a classifier to predict strong branching decisions and measure the reduction in nodes explored.
5. **Method selection:** For each of the following problems, justify whether AI-based or traditional optimization is more appropriate: (a) Portfolio optimization with known covariance, (b) Neural network architecture search, (c) Daily vehicle routing with similar demand patterns, (d) One-time facility location decision.

25. Deep Reinforcement Learning

What You Will Learn

- Understand the mathematical foundations of reinforcement learning and MDPs
- Master value-based methods including DQN and its improvements
- Implement policy gradient algorithms from REINFORCE to PPO
- Apply actor-critic methods including A2C, DDPG, TD3, and SAC
- Explore advanced topics: model-based RL, offline RL, and hierarchical RL
- Apply deep RL to combinatorial optimization and decision-making problems
- Debug and deploy RL systems using modern frameworks

Deep Reinforcement Learning (Deep RL) combines the decision-making framework of reinforcement learning with the representational power of deep neural networks. This powerful combination has achieved significant success in game playing, robotics, and increasingly in optimization and operations research. This chapter provides a comprehensive treatment of deep RL algorithms, from foundational concepts to state-of-the-art methods, with a focus on applications to decision intelligence.

25.1. Foundations of Reinforcement Learning

25.1.1. The Reinforcement Learning Problem

Reinforcement learning addresses the problem of learning to make sequential decisions through interaction with an environment. Unlike supervised learning, the agent must discover which actions yield the best outcomes through trial and error, receiving only scalar reward signals.

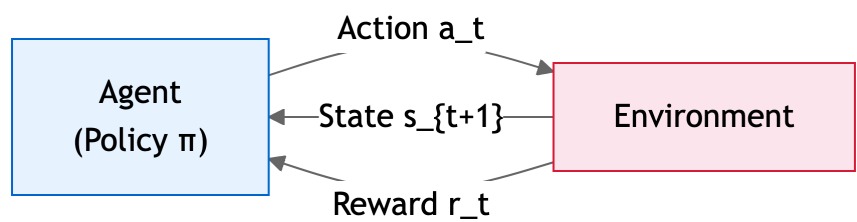


Figure 25.1.: The Reinforcement Learning Interaction Loop

25.1.2. Markov Decision Processes (MDPs)

The mathematical framework underlying RL is the Markov Decision Process, defined by the tuple (S, A, P, R, γ) :

Component	Symbol	Description
State Space	S	Set of all possible states the environment can be in

Component	Symbol	Description
Action Space	A	Set of all actions available to the agent
Transition Function	$P(s' s, a)$	Probability of transitioning to state s' given state s and action a
Reward Function	$R(s, a, s')$	Immediate reward received after transition
Discount Factor	$\gamma \in [0, 1]$	Weight given to future rewards

The Markov Property

The defining characteristic of MDPs is the Markov property: the future depends only on the current state, not the history of how we arrived there:

$$P(s_{t+1}|s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0, a_0) = P(s_{t+1}|s_t, a_t)$$

This memoryless property enables tractable solutions and is the foundation of dynamic programming approaches.

25.1.3. Value Functions

Value functions quantify the expected cumulative reward from a state or state-action pair.

State-Value Function $V^\pi(s)$:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right]$$

Action-Value Function $Q^\pi(s, a)$:

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right]$$

Advantage Function $A^\pi(s, a)$:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$$

The advantage function measures how much better an action is compared to the average action in that state.

25.1.4. Bellman Equations

The Bellman equations express the recursive relationship between value functions. These equations are the foundation of all RL algorithms because they decompose a multi-step optimization problem into a one-step lookahead plus the value of the next state.

Bellman Expectation Equation (for a fixed policy π):

$$V^\pi(s) = \sum_a \pi(a|s) \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s') \right]$$

$$Q^\pi(s, a) = R(s, a) + \gamma \sum_{s'} P(s'|s, a) \sum_{a'} \pi(a'|s') Q^\pi(s', a')$$

Bellman Optimality Equation (for optimal value functions):

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s') \right]$$

$$Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s'|s, a) \max_{a'} Q^*(s', a')$$

The following implementation uses value iteration to solve a small gridworld MDP. Value iteration repeatedly applies the Bellman optimality equation until the value function converges, then extracts the optimal policy by selecting the action that maximizes expected value at each state.

```
import numpy as np
from typing import Dict, Tuple, List, Callable

def value_iteration(
    n_states: int,
    n_actions: int,
    transitions: np.ndarray, # P[s, a, s'] = probability
    rewards: np.ndarray,    # R[s, a] = reward
    gamma: float = 0.99,
    theta: float = 1e-6,
    max_iterations: int = 1000
) -> Tuple[np.ndarray, np.ndarray]:
    """Value iteration algorithm for solving MDPs.

    Args:
        n_states: Number of states
        n_actions: Number of actions
        transitions: Transition probability matrix [s, a, s']
        rewards: Reward matrix [s, a]
        gamma: Discount factor
        theta: Convergence threshold
        max_iterations: Maximum iterations

    Returns:
        V: Optimal value function
        policy: Optimal deterministic policy
    """
    V = np.zeros(n_states)

    for iteration in range(max_iterations):
        delta = 0
        V_new = np.zeros(n_states)

        for s in range(n_states):
            # Compute Q(s, a) for all actions
            q_values = np.zeros(n_actions)
            for a in range(n_actions):
                q_values[a] = rewards[s, a] + gamma * np.sum(
                    transitions[s, a, :] * V
                )
```

```

    #  $V(s) = \max_a Q(s, a)$ 
    V_new[s] = np.max(q_values)
    delta = max(delta, abs(V_new[s] - V[s]))

V = V_new

if delta < theta:
    print(f"Value iteration converged in {iteration + 1} iterations")
    break

# Extract optimal policy
policy = np.zeros(n_states, dtype=int)
for s in range(n_states):
    q_values = np.zeros(n_actions)
    for a in range(n_actions):
        q_values[a] = rewards[s, a] + gamma * np.sum(
            transitions[s, a, :] * V
        )
    policy[s] = np.argmax(q_values)

return V, policy

# Example: Simple Grid World
def create_grid_world(size: int = 4) -> Tuple[np.ndarray, np.ndarray]:
    """Create a simple grid world MDP."""
    n_states = size * size
    n_actions = 4 # up, down, left, right

    transitions = np.zeros((n_states, n_actions, n_states))
    rewards = np.zeros((n_states, n_actions))

    # Goal state (bottom-right corner)
    goal = n_states - 1

    for s in range(n_states):
        row, col = s // size, s % size

        for a in range(n_actions):
            # Compute next state
            if a == 0: # up
                next_row = max(0, row - 1)
                next_col = col
            elif a == 1: # down
                next_row = min(size - 1, row + 1)
                next_col = col
            elif a == 2: # left
                next_row = row
                next_col = max(0, col - 1)
            else: # right
                next_row = row
                next_col = min(size - 1, col + 1)

```

```

        next_state = next_row * size + next_col
        transitions[s, a, next_state] = 1.0

    # Reward: +1 for reaching goal, -0.01 step cost
    if next_state == goal:
        rewards[s, a] = 1.0
    else:
        rewards[s, a] = -0.01

    return transitions, rewards

# Solve the grid world
transitions, rewards = create_grid_world(4)
V, policy = value_iteration(16, 4, transitions, rewards, gamma=0.99)

print("\nOptimal Value Function (4x4 grid):")
print(V.reshape(4, 4).round(2))
print("\nOptimal Policy (0=up, 1=down, 2=left, 3=right):")
print(policy.reshape(4, 4))

```

Optimal Value Function (4x4 grid):

```

[[ 95.05  96.02  97.    97.99]
 [ 96.02  97.    97.99  98.99]
 [ 97.    97.99  98.99 100.   ]
 [ 97.99  98.99 100.   100.   ]]

```

Optimal Policy (0=up, 1=down, 2=left, 3=right):

```

[[1 1 1 1]
 [1 1 1 1]
 [1 1 1 1]
 [3 3 3 1]]

```

The heatmap below shows how the learned value function assigns higher values to states closer to the goal. The policy arrows indicate the optimal action at each state, which consistently points toward the goal.

25.1.5. Exploration vs. Exploitation

A fundamental challenge in RL is balancing exploration (trying new actions to discover their rewards) with exploitation (using known good actions to maximize reward).

Common Exploration Strategies:

Strategy	Formula	Characteristics
ϵ -greedy	Random with prob ϵ , greedy otherwise	Simple, widely used
Boltzmann UCB	$\pi(a s) \propto \exp(Q(s, a)/\tau)$ $a = \arg \max_a [Q(a) + c\sqrt{\ln t / N(a)}]$	Temperature-controlled Optimistic initialization

Strategy	Formula	Characteristics
Entropy bonus	$r' = r + \alpha H(\pi)$	Encourages diverse policies

The following code implements three common exploration strategies and compares their action selection behavior on a simple four-action problem.

```
import numpy as np

class ExplorationStrategy:
    """Base class for exploration strategies."""

    def select_action(self, q_values: np.ndarray, **kwargs) -> int:
        raise NotImplementedError

class EpsilonGreedy(ExplorationStrategy):
    """Epsilon-greedy exploration."""

    def __init__(self, epsilon: float = 0.1, epsilon_decay: float = 0.999,
                 epsilon_min: float = 0.01):
        self.epsilon = epsilon
        self.epsilon_decay = epsilon_decay
        self.epsilon_min = epsilon_min

    def select_action(self, q_values: np.ndarray, **kwargs) -> int:
        if np.random.random() < self.epsilon:
            action = np.random.randint(len(q_values))
        else:
            action = np.argmax(q_values)

        # Decay epsilon
        self.epsilon = max(self.epsilon_min, self.epsilon * self.epsilon_decay)
        return action

class BoltzmannExploration(ExplorationStrategy):
    """Boltzmann (softmax) exploration."""

    def __init__(self, temperature: float = 1.0, temp_decay: float = 0.99,
                 temp_min: float = 0.1):
        self.temperature = temperature
        self.temp_decay = temp_decay
        self.temp_min = temp_min

    def select_action(self, q_values: np.ndarray, **kwargs) -> int:
        # Softmax with temperature
        exp_q = np.exp((q_values - np.max(q_values)) / self.temperature)
        probs = exp_q / np.sum(exp_q)
        action = np.random.choice(len(q_values), p=probs)

        # Decay temperature
        self.temperature = max(self.temp_min, self.temperature * self.temp_decay)
```

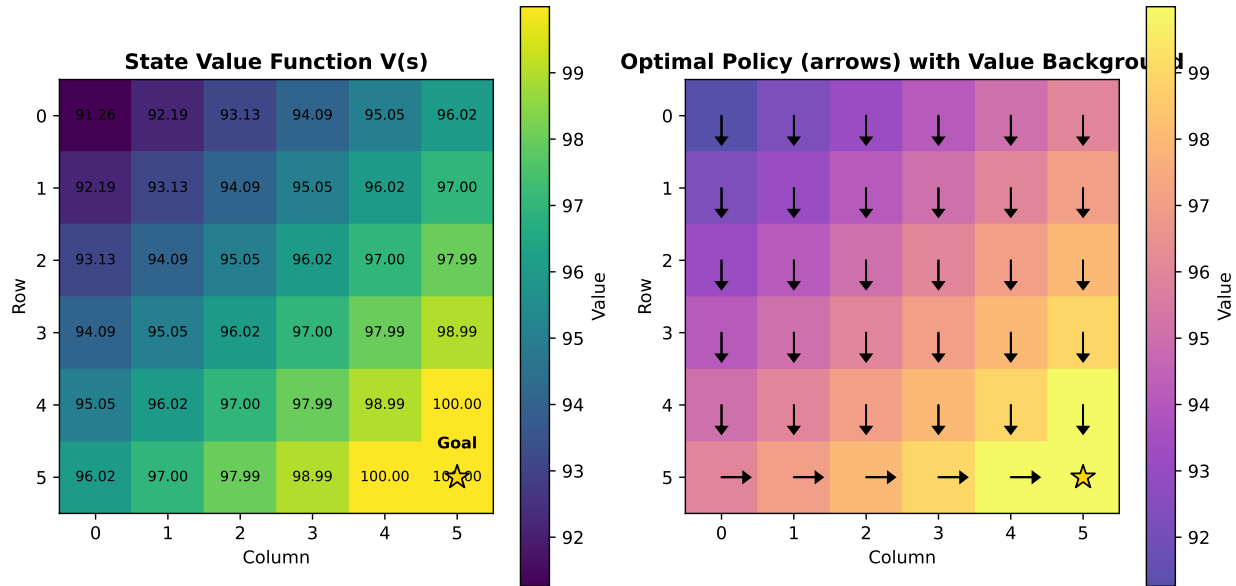


Figure 25.2.: Learned Value Function for a 4x4 Gridworld. The goal state (bottom-right) has the highest value, with values decreasing as distance from the goal increases. This visualization demonstrates how value iteration converges to assign higher values to states closer to the reward.

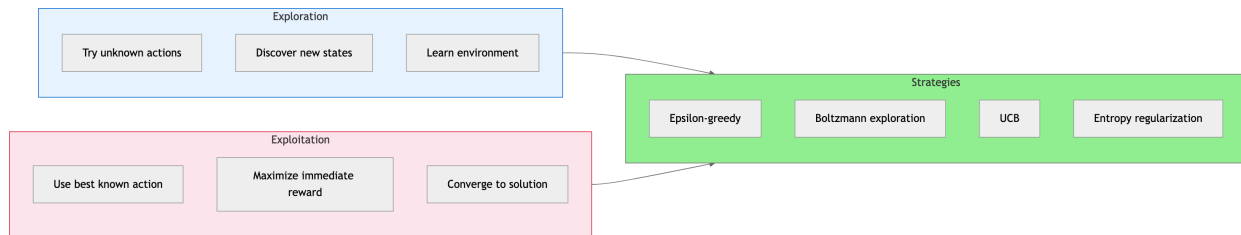


Figure 25.3.: The Exploration-Exploitation Trade-off

```

return action

class UCBExploration(ExplorationStrategy):
    """Upper Confidence Bound exploration."""

    def __init__(self, c: float = 2.0):
        self.c = c
        self.action_counts = None
        self.t = 0

    def select_action(self, q_values: np.ndarray, **kwargs) -> int:
        n_actions = len(q_values)

        if self.action_counts is None:
            self.action_counts = np.zeros(n_actions)
  
```

```

        self.t += 1

        # UCB values
        ucb_values = np.zeros(n_actions)
        for a in range(n_actions):
            if self.action_counts[a] == 0:
                ucb_values[a] = float('inf')
            else:
                ucb_values[a] = q_values[a] + self.c * np.sqrt(
                    np.log(self.t) / self.action_counts[a]
                )

        action = np.argmax(ucb_values)
        self.action_counts[action] += 1
        return action

# Demonstrate exploration strategies
np.random.seed(42)
q_values = np.array([1.0, 1.2, 0.8, 1.1]) # Action 1 is best

print("Q-values:", q_values)
print("\nAction selection over 10 trials:")
print("-" * 50)

strategies = {
    "Epsilon-greedy (eps=0.2)": EpsilonGreedy(epsilon=0.2, epsilon_decay=1.0),
    "Boltzmann (temp=0.5)": BoltzmannExploration(temperature=0.5, temp_decay=1.0),
    "UCB (c=1.0)": UCBExploration(c=1.0)
}

for name, strategy in strategies.items():
    actions = [strategy.select_action(q_values.copy()) for _ in range(10)]
    print(f"{name}: {actions}")

```

Q-values: [1. 1.2 0.8 1.1]

Action selection over 10 trials:

```

-----
Epsilon-greedy (eps=0.2): [np.int64(1), np.int64(1), np.int64(1), np.int64(1), 2, 2, np.int64(1), np.int64(1), np.int64(1), np.int64(1)]
Boltzmann (temp=0.5): [3, 3, 0, 3, 2, 2, 0, 0, 1, 1]
UCB (c=1.0): [np.int64(0), np.int64(1), np.int64(2), np.int64(3), np.int64(1), np.int64(3), np.int64(1), np.int64(1), np.int64(1), np.int64(1)]

```

25.2. Value-Based Methods

Value-based methods learn the optimal action-value function $Q^*(s, a)$ and derive the policy by acting greedily with respect to learned Q-values. These methods work best for discrete action spaces and have produced some of the most well-known RL successes, including Atari game-playing agents.

25.2.1. Deep Q-Networks (DQN)

Deep Q-Networks, introduced by Mnih et al. (2015), use deep neural networks to approximate the Q-function, enabling RL in high-dimensional state spaces like raw pixels.

Key Innovations:

1. **Experience Replay:** Store transitions $(s, a, r, s', done)$ in a buffer and sample random minibatches for training
2. **Target Network:** Use a separate, slowly-updated network for computing target Q-values

DQN Loss Function:

$$L(\theta) = \mathbb{E}_{(s,a,r,s') \sim D} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right]$$

where θ^- are the parameters of the target network.

The following implementation includes the complete DQN agent with experience replay buffer, Q-network, and the training loop. The replay buffer stores transitions and provides random minibatches, while the target network stabilizes training by providing slowly-updated Q-value targets.

```
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from collections import deque
import random
from typing import Tuple, List

class ReplayBuffer:
    """Experience replay buffer for DQN."""

    def __init__(self, capacity: int = 100000):
        self.buffer = deque(maxlen=capacity)

    def push(self, state: np.ndarray, action: int, reward: float,
             next_state: np.ndarray, done: bool):
        """Store a transition."""
        self.buffer.append((state, action, reward, next_state, done))

    def sample(self, batch_size: int) -> Tuple:
        """Sample a random batch of transitions."""
        batch = random.sample(self.buffer, batch_size)
        states, actions, rewards, next_states, dones = zip(*batch)

        return (
            np.array(states),
            np.array(actions),
            np.array(rewards, dtype=np.float32),
            np.array(next_states),
            np.array(dones, dtype=np.float32)
        )

    def __len__(self) -> int:
        return len(self.buffer)
```

```

class QNetwork(nn.Module):
    """Deep Q-Network architecture."""

    def __init__(self, state_dim: int, action_dim: int, hidden_dims: List[int] =
        ↪ [256, 256]):
        super().__init__()

        layers = []
        input_dim = state_dim

        for hidden_dim in hidden_dims:
            layers.extend([
                nn.Linear(input_dim, hidden_dim),
                nn.ReLU()
            ])
            input_dim = hidden_dim

        layers.append(nn.Linear(input_dim, action_dim))
        self.network = nn.Sequential(*layers)

    def forward(self, state: torch.Tensor) -> torch.Tensor:
        return self.network(state)

class DQNAgent:
    """Deep Q-Network agent."""

    def __init__(
        self,
        state_dim: int,
        action_dim: int,
        hidden_dims: List[int] = [256, 256],
        lr: float = 1e-3,
        gamma: float = 0.99,
        epsilon_start: float = 1.0,
        epsilon_end: float = 0.01,
        epsilon_decay: float = 0.995,
        buffer_size: int = 100000,
        batch_size: int = 64,
        target_update_freq: int = 100,
        device: str = "cpu"
    ):
        self.state_dim = state_dim
        self.action_dim = action_dim
        self.gamma = gamma
        self.epsilon = epsilon_start
        self.epsilon_end = epsilon_end
        self.epsilon_decay = epsilon_decay
        self.batch_size = batch_size
        self.target_update_freq = target_update_freq
        self.device = torch.device(device)

```

```

# Networks
self.q_network = QNetwork(state_dim, action_dim,
    ↪ hidden_dims).to(self.device)
self.target_network = QNetwork(state_dim, action_dim,
    ↪ hidden_dims).to(self.device)
self.target_network.load_state_dict(self.q_network.state_dict())

# Optimizer
self.optimizer = optim.Adam(self.q_network.parameters(), lr=lr)

# Replay buffer
self.replay_buffer = ReplayBuffer(buffer_size)

# Training step counter
self.train_step = 0

def select_action(self, state: np.ndarray, training: bool = True) -> int:
    """Select action using epsilon-greedy policy."""
    if training and np.random.random() < self.epsilon:
        return np.random.randint(self.action_dim)

    with torch.no_grad():
        state_tensor = torch.FloatTensor(state).unsqueeze(0).to(self.device)
        q_values = self.q_network(state_tensor)
        return q_values.argmax(dim=1).item()

def store_transition(self, state: np.ndarray, action: int, reward: float,
    next_state: np.ndarray, done: bool):
    """Store transition in replay buffer."""
    self.replay_buffer.push(state, action, reward, next_state, done)

def update(self) -> float:
    """Perform one gradient update step."""
    if len(self.replay_buffer) < self.batch_size:
        return 0.0

    # Sample batch
    states, actions, rewards, next_states, dones = self.replay_buffer.sample(
        self.batch_size
    )

    # Convert to tensors
    states = torch.FloatTensor(states).to(self.device)
    actions = torch.LongTensor(actions).to(self.device)
    rewards = torch.FloatTensor(rewards).to(self.device)
    next_states = torch.FloatTensor(next_states).to(self.device)
    dones = torch.FloatTensor(dones).to(self.device)

    # Compute current Q-values
    current_q = self.q_network(states).gather(1,
    ↪ actions.unsqueeze(1)).squeeze(1)

```

```

    # Compute target Q-values
    with torch.no_grad():
        next_q = self.target_network(next_states).max(dim=1)[0]
        target_q = rewards + self.gamma * next_q * (1 - dones)

    # Compute loss
    loss = nn.MSELoss()(current_q, target_q)

    # Optimize
    self.optimizer.zero_grad()
    loss.backward()
    # Gradient clipping for stability
    torch.nn.utils.clip_grad_norm_(self.q_network.parameters(), 10.0)
    self.optimizer.step()

    # Update target network
    self.train_step += 1
    if self.train_step % self.target_update_freq == 0:
        self.target_network.load_state_dict(self.q_network.state_dict())

    # Decay epsilon
    self.epsilon = max(self.epsilon_end, self.epsilon * self.epsilon_decay)

    return loss.item()

def train_episode(self, env, max_steps: int = 1000) -> Tuple[float, int]:
    """Train for one episode."""
    state, _ = env.reset()
    total_reward = 0

    for step in range(max_steps):
        action = self.select_action(state, training=True)
        next_state, reward, terminated, truncated, _ = env.step(action)
        done = terminated or truncated

        self.store_transition(state, action, reward, next_state, done)
        loss = self.update()

        total_reward += reward
        state = next_state

        if done:
            break

    return total_reward, step + 1

# Example usage (requires gymnasium)
print("DQN Agent created successfully")
print(f"Q-Network architecture: {DQNAgent(4, 2).q_network}")

```

```

DQN Agent created successfully
Q-Network architecture: QNetwork(
  (network): Sequential(

```

```

(0): Linear(in_features=4, out_features=256, bias=True)
(1): ReLU()
(2): Linear(in_features=256, out_features=256, bias=True)
(3): ReLU()
(4): Linear(in_features=256, out_features=2, bias=True)
)
)

```

Training performance varies significantly across RL algorithms. The next visualization compares learning curves and sample efficiency for DQN, Double DQN, PPO, and SAC on a simulated benchmark.

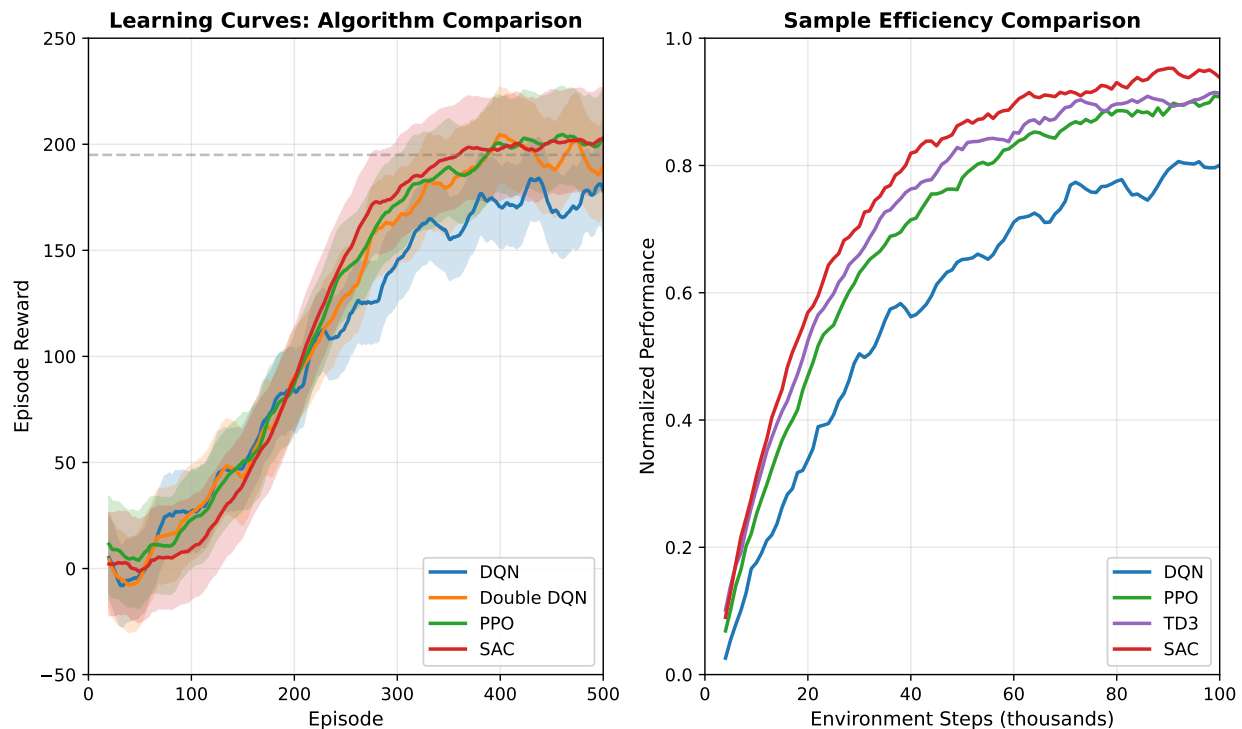


Figure 25.4.: Learning Curves for Different Deep RL Algorithms. This comparison shows typical training dynamics: DQN exhibits high variance due to epsilon-greedy exploration, PPO shows smoother learning with its clipped objective, and SAC achieves the best sample efficiency through entropy-regularized exploration. Shaded regions represent standard deviation across multiple runs.

💡 DQN Hyperparameter Guidelines

- **Learning rate:** Start with 10^{-4} to 10^{-3}
- **Batch size:** 32-256, larger is more stable
- **Buffer size:** At least 10,000-100,000 transitions
- **Target update:** Every 100-1000 steps
- **Epsilon decay:** Decay over 100,000+ steps for complex tasks

25.2.2. DQN Improvements

Several improvements to the basic DQN algorithm have been developed.

25.2.2.1. Double DQN

Standard DQN tends to overestimate Q-values because it uses the max operator for both selection and evaluation. Double DQN decouples these:

$$Q_{target} = r + \gamma Q(s', \arg \max_{a'} Q(s', a'; \theta); \theta^-)$$

```
class DoubledDQNAgent(DQNAgent):
    """Double DQN agent that addresses overestimation bias."""

    def update(self) -> float:
        """Perform Double DQN update."""
        if len(self.replay_buffer) < self.batch_size:
            return 0.0

        # Sample batch
        states, actions, rewards, next_states, dones = self.replay_buffer.sample(
            self.batch_size
        )

        # Convert to tensors
        states = torch.FloatTensor(states).to(self.device)
        actions = torch.LongTensor(actions).to(self.device)
        rewards = torch.FloatTensor(rewards).to(self.device)
        next_states = torch.FloatTensor(next_states).to(self.device)
        dones = torch.FloatTensor(dones).to(self.device)

        # Compute current Q-values
        current_q = self.q_network(states).gather(1,
↪ actions.unsqueeze(1)).squeeze(1)

        # Double DQN: use online network for action selection,
        # target network for evaluation
        with torch.no_grad():
            # Select best actions using online network
            best_actions = self.q_network(next_states).argmax(dim=1, keepdim=True)
            # Evaluate those actions using target network
            next_q = self.target_network(next_states).gather(1,
↪ best_actions).squeeze(1)
            target_q = rewards + self.gamma * next_q * (1 - dones)

        # Compute loss
        loss = nn.MSELoss()(current_q, target_q)

        # Optimize
        self.optimizer.zero_grad()
        loss.backward()
        torch.nn.utils.clip_grad_norm_(self.q_network.parameters(), 10.0)
        self.optimizer.step()
```

```

    # Update target network
    self.train_step += 1
    if self.train_step % self.target_update_freq == 0:
        self.target_network.load_state_dict(self.q_network.state_dict())

    # Decay epsilon
    self.epsilon = max(self.epsilon_end, self.epsilon * self.epsilon_decay)

    return loss.item()

print("Double DQN addresses Q-value overestimation")

```

Double DQN addresses Q-value overestimation

25.2.2.2. Dueling DQN

Dueling DQN separates the Q-function into state value $V(s)$ and advantage $A(s, a)$:

$$Q(s, a) = V(s) + A(s, a) - \frac{1}{|A|} \sum_{a'} A(s, a')$$

```

class DuelingQNetwork(nn.Module):
    """Dueling DQN architecture separating value and advantage streams."""

    def __init__(self, state_dim: int, action_dim: int, hidden_dim: int = 256):
        super().__init__()

        # Shared feature extraction
        self.features = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU()
        )

        # Value stream: V(s)
        self.value_stream = nn.Sequential(
            nn.Linear(hidden_dim, hidden_dim // 2),
            nn.ReLU(),
            nn.Linear(hidden_dim // 2, 1)
        )

        # Advantage stream: A(s, a)
        self.advantage_stream = nn.Sequential(
            nn.Linear(hidden_dim, hidden_dim // 2),
            nn.ReLU(),
            nn.Linear(hidden_dim // 2, action_dim)
        )

    def forward(self, state: torch.Tensor) -> torch.Tensor:
        features = self.features(state)

```

```

        value = self.value_stream(features)
        advantage = self.advantage_stream(features)

        # Combine: Q = V + (A - mean(A))
        # Subtracting mean ensures identifiability
        q_values = value + advantage - advantage.mean(dim=-1, keepdim=True)
        return q_values

# Demonstrate architecture
dueling_net = DuelingQNetwork(state_dim=8, action_dim=4)
print("Dueling DQN Architecture:")
print(dueling_net)

# Test forward pass
test_state = torch.randn(1, 8)
q_values = dueling_net(test_state)
print(f"\nTest Q-values shape: {q_values.shape}")
print(f"Q-values: {q_values.detach().numpy().round(3)}")

```

Dueling DQN Architecture:

```

DuelingQNetwork(
  (features): Sequential(
    (0): Linear(in_features=8, out_features=256, bias=True)
    (1): ReLU()
    (2): Linear(in_features=256, out_features=256, bias=True)
    (3): ReLU()
  )
  (value_stream): Sequential(
    (0): Linear(in_features=256, out_features=128, bias=True)
    (1): ReLU()
    (2): Linear(in_features=128, out_features=1, bias=True)
  )
  (advantage_stream): Sequential(
    (0): Linear(in_features=256, out_features=128, bias=True)
    (1): ReLU()
    (2): Linear(in_features=128, out_features=4, bias=True)
  )
)

```

Test Q-values shape: torch.Size([1, 4])

Q-values: [[-0.004 -0.108 -0.053 0.096]]

25.2.2.3. Prioritized Experience Replay

Instead of uniform sampling, prioritize transitions with high TD error:

$$P(i) = \frac{p_i^\alpha}{\sum_j p_j^\alpha}$$

where $p_i = |\delta_i| + \epsilon$ is proportional to the TD error.


```

import numpy as np
from typing import Tuple

class SumTree:
    """Sum tree data structure for efficient priority sampling."""

    def __init__(self, capacity: int):
        self.capacity = capacity
        self.tree = np.zeros(2 * capacity - 1) # Binary tree
        self.data = np.zeros(capacity, dtype=object)
        self.write_ptr = 0
        self.n_entries = 0

    def _propagate(self, idx: int, change: float):
        """Update parent nodes."""
        parent = (idx - 1) // 2
        self.tree[parent] += change
        if parent != 0:
            self._propagate(parent, change)

    def _retrieve(self, idx: int, s: float) -> int:
        """Find sample on leaf level."""
        left = 2 * idx + 1
        right = left + 1

        if left >= len(self.tree):
            return idx

        if s <= self.tree[left]:
            return self._retrieve(left, s)
        else:
            return self._retrieve(right, s - self.tree[left])

    def total(self) -> float:
        return self.tree[0]

    def add(self, priority: float, data):
        """Add data with priority."""
        idx = self.write_ptr + self.capacity - 1

        self.data[self.write_ptr] = data
        self.update(idx, priority)

        self.write_ptr = (self.write_ptr + 1) % self.capacity
        self.n_entries = min(self.n_entries + 1, self.capacity)

    def update(self, idx: int, priority: float):
        """Update priority."""
        change = priority - self.tree[idx]
        self.tree[idx] = priority
        self._propagate(idx, change)

    def get(self, s: float) -> Tuple[int, float, object]:

```

```

        """Sample based on priority."""
        idx = self._retrieve(0, s)
        data_idx = idx - self.capacity + 1
        return idx, self.tree[idx], self.data[data_idx]

class PrioritizedReplayBuffer:
    """Prioritized experience replay buffer."""

    def __init__(self, capacity: int, alpha: float = 0.6, beta: float = 0.4,
                 beta_increment: float = 0.001):
        self.tree = SumTree(capacity)
        self.capacity = capacity
        self.alpha = alpha # Priority exponent
        self.beta = beta # Importance sampling exponent
        self.beta_increment = beta_increment
        self.max_priority = 1.0
        self.epsilon = 1e-6

    def push(self, state, action, reward, next_state, done):
        """Add transition with max priority."""
        data = (state, action, reward, next_state, done)
        priority = self.max_priority ** self.alpha
        self.tree.add(priority, data)

    def sample(self, batch_size: int) -> Tuple:
        """Sample batch with priorities."""
        batch_data = []
        indices = []
        priorities = []

        segment = self.tree.total() / batch_size

        self.beta = min(1.0, self.beta + self.beta_increment)

        for i in range(batch_size):
            s = random.uniform(segment * i, segment * (i + 1))
            idx, priority, data = self.tree.get(s)

            if data is not None and data != 0:
                batch_data.append(data)
                indices.append(idx)
                priorities.append(priority)

        # Compute importance sampling weights
        total = self.tree.total()
        probs = np.array(priorities) / total
        weights = (self.tree.n_entries * probs) ** (-self.beta)
        weights = weights / weights.max() # Normalize

        # Unpack data
        if len(batch_data) > 0:
            states, actions, rewards, next_states, dones = zip(*batch_data)

```

```

        return (
            np.array(states),
            np.array(actions),
            np.array(rewards, dtype=np.float32),
            np.array(next_states),
            np.array(dones, dtype=np.float32),
            np.array(indices),
            np.array(weights, dtype=np.float32)
        )
    return None

def update_priorities(self, indices: np.ndarray, td_errors: np.ndarray):
    """Update priorities based on TD errors."""
    for idx, td_error in zip(indices, td_errors):
        priority = (abs(td_error) + self.epsilon) ** self.alpha
        self.tree.update(idx, priority)
        self.max_priority = max(self.max_priority, priority)

def __len__(self) -> int:
    return self.tree.n_entries

print("Prioritized Experience Replay implemented")
print("Transitions with high TD error are sampled more frequently")

```

Prioritized Experience Replay implemented
 Transitions with high TD error are sampled more frequently

25.2.2.4. Rainbow DQN

Rainbow combines multiple DQN improvements:

Component	Contribution
Double DQN	Reduces overestimation
Prioritized Replay	Sample important transitions
Dueling Networks	Separate value and advantage
Multi-step Returns	Faster credit assignment
Distributional RL	Learn return distribution
Noisy Networks	Parameter-space exploration

i Rainbow Performance

Rainbow significantly outperforms individual components. The most impactful contributions are: 1. Multi-step returns 2. Prioritized replay 3. Distributional RL

25.3. Policy Gradient Methods

Policy gradient methods directly optimize the policy without learning a value function as an intermediate step. Unlike value-based methods, policy gradient approaches can handle continuous action spaces and learn stochastic policies. They estimate the gradient of expected return with respect to policy parameters and update using gradient ascent.

25.3.1. The Policy Gradient Theorem

For a parameterized policy $\pi_\theta(a|s)$, the objective is to maximize expected return:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \gamma^t r_t \right]$$

The policy gradient theorem provides a tractable gradient:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) \cdot Q^{\pi_\theta}(s, a)]$$

25.3.2. REINFORCE Algorithm

REINFORCE is the simplest policy gradient algorithm, using Monte Carlo returns as Q-value estimates:

$$\nabla_\theta J(\theta) \approx \sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot G_t$$

where $G_t = \sum_{k=t}^T \gamma^{k-t} r_k$ is the return from timestep t .

```
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.distributions import Categorical
from typing import List, Tuple

class PolicyNetwork(nn.Module):
    """Simple policy network for discrete actions."""

    def __init__(self, state_dim: int, action_dim: int, hidden_dim: int = 128):
        super().__init__()

        self.network = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, action_dim)
        )

    def forward(self, state: torch.Tensor) -> torch.Tensor:
        return F.softmax(self.network(state), dim=-1)

    def get_action(self, state: torch.Tensor) -> Tuple[int, torch.Tensor]:
        """Sample action and return log probability."""
        probs = self.forward(state)
        dist = Categorical(probs)
        action = dist.sample()
        log_prob = dist.log_prob(action)
        return action.item(), log_prob
```

```

class REINFORCE:
    """REINFORCE algorithm with baseline."""

    def __init__(
        self,
        state_dim: int,
        action_dim: int,
        hidden_dim: int = 128,
        lr: float = 1e-3,
        gamma: float = 0.99,
        use_baseline: bool = True
    ):
        self.gamma = gamma
        self.use_baseline = use_baseline

        self.policy = PolicyNetwork(state_dim, action_dim, hidden_dim)
        self.optimizer = optim.Adam(self.policy.parameters(), lr=lr)

        # Running baseline (moving average of returns)
        self.baseline = 0.0
        self.baseline_alpha = 0.1

    def compute_returns(self, rewards: List[float]) -> torch.Tensor:
        """Compute discounted returns."""
        returns = []
        G = 0

        for r in reversed(rewards):
            G = r + self.gamma * G
            returns.insert(0, G)

        returns = torch.tensor(returns, dtype=torch.float32)
        return returns

    def update(self, log_probs: List[torch.Tensor], rewards: List[float]) -> float:
        """Update policy using collected trajectory."""
        # Compute returns
        returns = self.compute_returns(rewards)

        # Normalize returns (variance reduction)
        if len(returns) > 1:
            returns = (returns - returns.mean()) / (returns.std() + 1e-8)

        # Apply baseline
        if self.use_baseline:
            advantages = returns - self.baseline
            self.baseline = (1 - self.baseline_alpha) * self.baseline + \
                self.baseline_alpha * returns.mean().item()
        else:
            advantages = returns

        # Compute policy gradient loss
        policy_loss = []

```

```

    for log_prob, advantage in zip(log_probs, advantages):
        policy_loss.append(-log_prob * advantage)

    loss = torch.stack(policy_loss).sum()

    # Update policy
    self.optimizer.zero_grad()
    loss.backward()
    self.optimizer.step()

    return loss.item()

def train_episode(self, env, max_steps: int = 1000) -> Tuple[float, int]:
    """Train for one episode."""
    state, _ = env.reset()

    log_probs = []
    rewards = []

    for step in range(max_steps):
        state_tensor = torch.FloatTensor(state).unsqueeze(0)
        action, log_prob = self.policy.get_action(state_tensor)

        next_state, reward, terminated, truncated, _ = env.step(action)
        done = terminated or truncated

        log_probs.append(log_prob)
        rewards.append(reward)

        state = next_state

        if done:
            break

    # Update policy
    loss = self.update(log_probs, rewards)

    return sum(rewards), step + 1

print("REINFORCE agent created")
agent = REINFORCE(state_dim=4, action_dim=2)
print(f"Policy network: {agent.policy}")

```

REINFORCE agent created

```

Policy network: PolicyNetwork(
  (network): Sequential(
    (0): Linear(in_features=4, out_features=128, bias=True)
    (1): ReLU()
    (2): Linear(in_features=128, out_features=128, bias=True)
    (3): ReLU()
    (4): Linear(in_features=128, out_features=2, bias=True)
  )
)

```

)

Stochastic policies assign probability distributions over actions at each state. The following visualization shows how a trained policy distributes probability mass across four movement directions and how the argmax of these probabilities yields a deterministic policy.

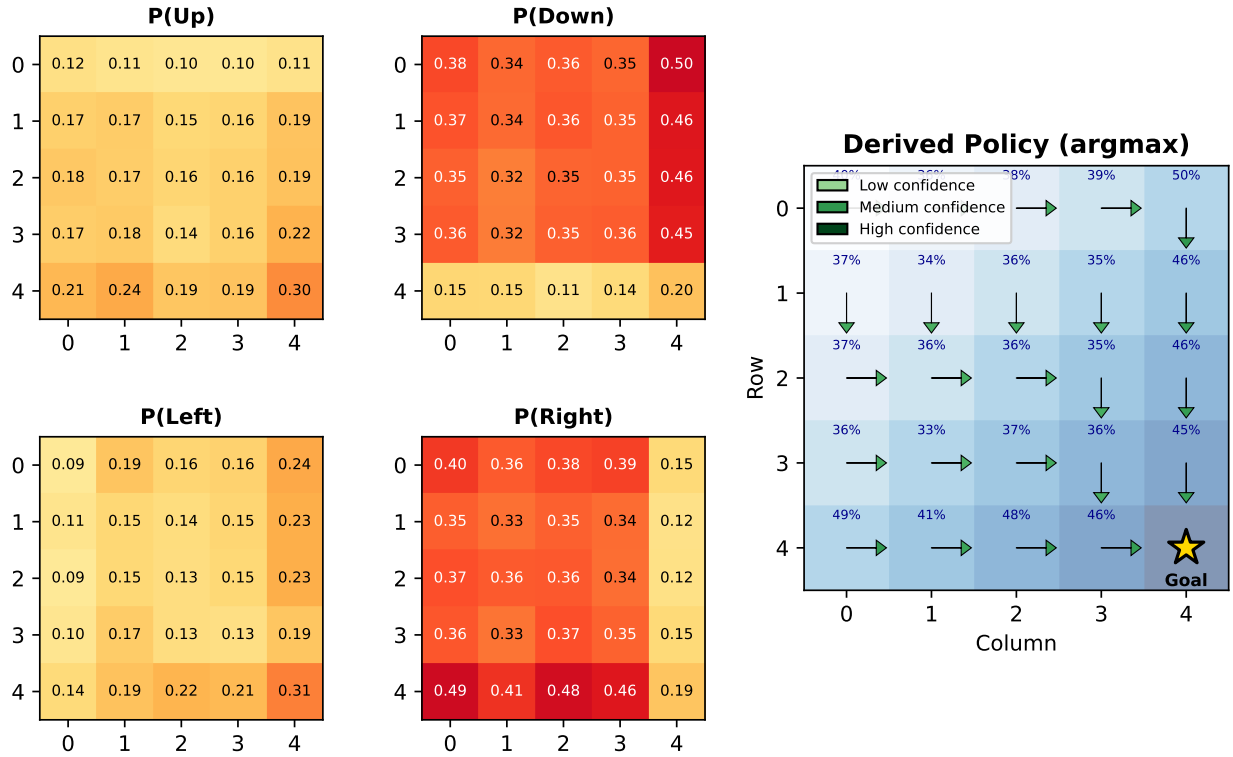


Figure 25.5.: Policy Visualization in a Gridworld Environment. Left: Action probability distributions showing how a trained policy assigns probabilities to each action (Up, Down, Left, Right) across different states. Warmer colors indicate higher probability. Right: The resulting deterministic policy derived by selecting the highest-probability action at each state, with arrows indicating the chosen direction.

25.3.3. Variance Reduction Techniques

Policy gradient methods suffer from high variance. Several techniques help:

Technique	Description	Implementation
Baseline subtraction	Subtract state-dependent baseline $b(s)$	$\nabla \log \pi(a s)(Q(s, a) - b(s))$
Reward normalization	Standardize returns	$(G - \mu)/\sigma$
Advantage estimation	Use advantages instead of Q-values	$A(s, a) = Q(s, a) - V(s)$
Entropy regularization	Encourage exploration	$J' = J + \alpha H(\pi)$

25.3.4. Trust Region Policy Optimization (TRPO)

TRPO constrains policy updates to prevent large, destabilizing changes:

$$\max_{\theta} \mathbb{E} \left[\frac{\pi_{\theta}(a|s)}{\pi_{\theta_{old}}(a|s)} A^{\pi_{old}}(s, a) \right]$$

subject to $\mathbb{E}[D_{KL}(\pi_{\theta_{old}} \parallel \pi_{\theta})] \leq \delta$

This trust region constraint ensures monotonic improvement.

25.3.5. Proximal Policy Optimization (PPO)

PPO achieves similar benefits to TRPO with a simpler clipped objective:

$$L^{CLIP}(\theta) = \mathbb{E} [\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$$

where $r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio.

```
import torch
import torch.nn as nn
import torch.optim as optim
from torch.distributions import Categorical
import numpy as np
from typing import Dict, List, Tuple

class ActorCriticNetwork(nn.Module):
    """Combined actor-critic network for PPO."""

    def __init__(self, state_dim: int, action_dim: int, hidden_dim: int = 256):
        super().__init__()

        # Shared feature extraction
        self.shared = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.Tanh(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.Tanh()
        )

        # Actor head (policy)
        self.actor = nn.Linear(hidden_dim, action_dim)

        # Critic head (value function)
        self.critic = nn.Linear(hidden_dim, 1)

    def forward(self, state: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
        features = self.shared(state)
        action_logits = self.actor(features)
        value = self.critic(features)
        return action_logits, value

    def get_action(self, state: torch.Tensor) -> Tuple[int, torch.Tensor,
        ↪ torch.Tensor]:
        """Sample action and return log prob and value."""
        action_logits, value = self.forward(state)
        probs = F.softmax(action_logits, dim=-1)
```



```

        dist = Categorical(probs)
        action = dist.sample()
        log_prob = dist.log_prob(action)
        return action.item(), log_prob, value.squeeze(-1)

def evaluate_actions(self, states: torch.Tensor, actions: torch.Tensor
                    ) -> Tuple[torch.Tensor, torch.Tensor, torch.Tensor]:
    """Evaluate log probs, values, and entropy for given state-action pairs."""
    action_logits, values = self.forward(states)
    probs = F.softmax(action_logits, dim=-1)
    dist = Categorical(probs)

    log_probs = dist.log_prob(actions)
    entropy = dist.entropy()

    return log_probs, values.squeeze(-1), entropy

class PPOAgent:
    """Proximal Policy Optimization agent."""

    def __init__(
        self,
        state_dim: int,
        action_dim: int,
        hidden_dim: int = 256,
        lr: float = 3e-4,
        gamma: float = 0.99,
        gae_lambda: float = 0.95,
        clip_epsilon: float = 0.2,
        entropy_coef: float = 0.01,
        value_coef: float = 0.5,
        max_grad_norm: float = 0.5,
        n_epochs: int = 10,
        batch_size: int = 64
    ):
        self.gamma = gamma
        self.gae_lambda = gae_lambda
        self.clip_epsilon = clip_epsilon
        self.entropy_coef = entropy_coef
        self.value_coef = value_coef
        self.max_grad_norm = max_grad_norm
        self.n_epochs = n_epochs
        self.batch_size = batch_size

        self.network = ActorCriticNetwork(state_dim, action_dim, hidden_dim)
        self.optimizer = optim.Adam(self.network.parameters(), lr=lr)

        # Trajectory storage
        self.states = []
        self.actions = []
        self.log_probs = []
        self.rewards = []

```

```

        self.values = []
        self.dones = []

    def select_action(self, state: np.ndarray) -> int:
        """Select action and store trajectory data."""
        state_tensor = torch.FloatTensor(state).unsqueeze(0)

        with torch.no_grad():
            action, log_prob, value = self.network.get_action(state_tensor)

        self.states.append(state)
        self.actions.append(action)
        self.log_probs.append(log_prob.item())
        self.values.append(value.item())

        return action

    def store_reward(self, reward: float, done: bool):
        """Store reward and done flag."""
        self.rewards.append(reward)
        self.dones.append(done)

    def compute_gae(self, next_value: float) -> Tuple[np.ndarray, np.ndarray]:
        """Compute Generalized Advantage Estimation."""
        advantages = []
        returns = []
        gae = 0

        values = self.values + [next_value]

        for t in reversed(range(len(self.rewards))):
            # TD error
            delta = self.rewards[t] + self.gamma * values[t + 1] * (1 -
↪ self.dones[t]) - values[t]

            # GAE
            gae = delta + self.gamma * self.gae_lambda * (1 - self.dones[t]) * gae
            advantages.insert(0, gae)
            returns.insert(0, gae + values[t])

        return np.array(advantages), np.array(returns)

    def update(self, next_state: np.ndarray) -> Dict[str, float]:
        """Perform PPO update."""
        # Compute GAE
        with torch.no_grad():
            next_state_tensor = torch.FloatTensor(next_state).unsqueeze(0)
            _, next_value = self.network.forward(next_state_tensor)
            next_value = next_value.item()

        advantages, returns = self.compute_gae(next_value)

        # Convert to tensors

```

```

states = torch.FloatTensor(np.array(self.states))
actions = torch.LongTensor(self.actions)
old_log_probs = torch.FloatTensor(self.log_probs)
advantages = torch.FloatTensor(advantages)
returns = torch.FloatTensor(returns)

# Normalize advantages
advantages = (advantages - advantages.mean()) / (advantages.std() + 1e-8)

# PPO update epochs
dataset_size = len(states)
indices = np.arange(dataset_size)

total_policy_loss = 0
total_value_loss = 0
total_entropy = 0
n_updates = 0

for epoch in range(self.n_epochs):
    np.random.shuffle(indices)

    for start in range(0, dataset_size, self.batch_size):
        end = min(start + self.batch_size, dataset_size)
        batch_indices = indices[start:end]

        batch_states = states[batch_indices]
        batch_actions = actions[batch_indices]
        batch_old_log_probs = old_log_probs[batch_indices]
        batch_advantages = advantages[batch_indices]
        batch_returns = returns[batch_indices]

        # Evaluate current policy
        new_log_probs, values, entropy = self.network.evaluate_actions(
            batch_states, batch_actions
        )

        # Probability ratio
        ratio = torch.exp(new_log_probs - batch_old_log_probs)

        # Clipped surrogate objective
        surr1 = ratio * batch_advantages
        surr2 = torch.clamp(ratio, 1 - self.clip_epsilon,
                            1 + self.clip_epsilon) * batch_advantages
        policy_loss = -torch.min(surr1, surr2).mean()

        # Value loss
        value_loss = F.mse_loss(values, batch_returns)

        # Entropy bonus
        entropy_loss = -entropy.mean()

        # Total loss
        loss = (policy_loss +

```

```

        self.value_coef * value_loss +
        self.entropy_coef * entropy_loss)

    # Optimize
    self.optimizer.zero_grad()
    loss.backward()
    torch.nn.utils.clip_grad_norm_(self.network.parameters(),
                                   self.max_grad_norm)

    self.optimizer.step()

    total_policy_loss += policy_loss.item()
    total_value_loss += value_loss.item()
    total_entropy += entropy.mean().item()
    n_updates += 1

# Clear trajectory storage
self.states = []
self.actions = []
self.log_probs = []
self.rewards = []
self.values = []
self.dones = []

return {
    'policy_loss': total_policy_loss / n_updates,
    'value_loss': total_value_loss / n_updates,
    'entropy': total_entropy / n_updates
}

def train_episode(self, env, max_steps: int = 1000,
                  update_freq: int = 2048) -> Tuple[float, Dict]:
    """Train for one episode with periodic updates."""
    state, _ = env.reset()
    total_reward = 0
    episode_steps = 0

    for step in range(max_steps):
        action = self.select_action(state)
        next_state, reward, terminated, truncated, _ = env.step(action)
        done = terminated or truncated

        self.store_reward(reward, done)

        total_reward += reward
        episode_steps += 1
        state = next_state

    # Update if enough data collected
    if len(self.states) >= update_freq or done:
        metrics = self.update(next_state)

    if done:
        break

```

```

        return total_reward, {'steps': episode_steps}

print("PPO Agent created successfully")
ppo_agent = PPOAgent(state_dim=4, action_dim=2)
print(f"Network architecture: {ppo_agent.network}")

```

```

PPO Agent created successfully
Network architecture: ActorCriticNetwork(
  (shared): Sequential(
    (0): Linear(in_features=4, out_features=256, bias=True)
    (1): Tanh()
    (2): Linear(in_features=256, out_features=256, bias=True)
    (3): Tanh()
  )
  (actor): Linear(in_features=256, out_features=2, bias=True)
  (critic): Linear(in_features=256, out_features=1, bias=True)
)

```

⚠ PPO Implementation Details

Key implementation details that significantly affect PPO performance:

1. **Advantage normalization:** Always normalize advantages per minibatch
2. **Value function clipping:** Optionally clip value function updates
3. **Learning rate annealing:** Linearly decay learning rate
4. **Orthogonal initialization:** Use orthogonal weight initialization
5. **GAE-Lambda:** Use $\lambda = 0.95$ for good bias-variance tradeoff

25.4. Actor-Critic Methods

Actor-critic methods combine policy gradient (actor) with value function learning (critic) for lower variance updates. The actor selects actions according to the current policy, while the critic evaluates the quality of those actions using a learned value function. This two-component design reduces the high variance of pure policy gradient methods while retaining their ability to handle continuous actions.

25.4.1. Advantage Actor-Critic (A2C)

A2C uses the advantage function to reduce variance:

$$\nabla_{\theta} J(\theta) = \mathbb{E} [\nabla_{\theta} \log \pi_{\theta}(a|s) \cdot A(s, a)]$$

where $A(s, a) = Q(s, a) - V(s) \approx r + \gamma V(s') - V(s)$ (TD error).

```

class A2CAgent:
    """Advantage Actor-Critic agent."""

```

```

def __init__(
    self,
    state_dim: int,
    action_dim: int,
    hidden_dim: int = 256,
    lr: float = 7e-4,
    gamma: float = 0.99,
    entropy_coef: float = 0.01,
    value_coef: float = 0.5,
    max_grad_norm: float = 0.5,
    n_steps: int = 5
):
    self.gamma = gamma
    self.entropy_coef = entropy_coef
    self.value_coef = value_coef
    self.max_grad_norm = max_grad_norm
    self.n_steps = n_steps

    self.network = ActorCriticNetwork(state_dim, action_dim, hidden_dim)
    self.optimizer = optim.Adam(self.network.parameters(), lr=lr)

def compute_returns(self, rewards: List[float], values: List[float],
                    dones: List[bool], next_value: float) -> torch.Tensor:
    """Compute n-step returns."""
    returns = []
    R = next_value

    for t in reversed(range(len(rewards))):
        R = rewards[t] + self.gamma * R * (1 - dones[t])
        returns.insert(0, R)

    return torch.tensor(returns, dtype=torch.float32)

def update(self, states: List[np.ndarray], actions: List[int],
           rewards: List[float], dones: List[bool],
           next_state: np.ndarray) -> Dict[str, float]:
    """Perform A2C update."""
    # Convert to tensors
    states_tensor = torch.FloatTensor(np.array(states))
    actions_tensor = torch.LongTensor(actions)

    # Get next state value
    with torch.no_grad():
        next_state_tensor = torch.FloatTensor(next_state).unsqueeze(0)
        _, next_value = self.network.forward(next_state_tensor)
        next_value = next_value.item()

    # Compute returns
    with torch.no_grad():
        _, values = self.network.forward(states_tensor)
        values = values.squeeze(-1)

    returns = self.compute_returns(rewards, values.tolist(), dones, next_value)

```

```

# Evaluate actions
log_probs, values, entropy = self.network.evaluate_actions(
    states_tensor, actions_tensor
)

# Advantage
advantages = returns - values.detach()

# Policy loss
policy_loss = -(log_probs * advantages).mean()

# Value loss
value_loss = F.mse_loss(values, returns)

# Entropy bonus
entropy_loss = -entropy.mean()

# Total loss
loss = policy_loss + self.value_coef * value_loss + self.entropy_coef *
↪ entropy_loss

# Optimize
self.optimizer.zero_grad()
loss.backward()
torch.nn.utils.clip_grad_norm_(self.network.parameters(),
↪ self.max_grad_norm)
self.optimizer.step()

return {
    'policy_loss': policy_loss.item(),
    'value_loss': value_loss.item(),
    'entropy': entropy.mean().item()
}

print("A2C Agent created")
a2c_agent = A2CAgent(state_dim=4, action_dim=2)

```

A2C Agent created

25.4.2. Generalized Advantage Estimation (GAE)

GAE provides a flexible tradeoff between bias and variance in advantage estimation:

$$A_t^{GAE(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}$$

where $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the TD error.

- $\lambda = 0$: High bias, low variance (1-step TD)
- $\lambda = 1$: Low bias, high variance (Monte Carlo)

25.4.3. Deep Deterministic Policy Gradient (DDPG)

DDPG extends DQN to continuous action spaces using a deterministic policy.

Key Components: - **Actor:** Deterministic policy $\mu_{\theta}(s)$ - **Critic:** Q-function $Q_{\phi}(s, a)$ - **Target networks:** Slow-moving copies for stability - **Exploration:** Ornstein-Uhlenbeck noise or Gaussian noise

```
import torch
import torch.nn as nn
import torch.optim as optim
import numpy as np
from typing import Tuple

class DDPGActor(nn.Module):
    """Deterministic actor for continuous control."""

    def __init__(self, state_dim: int, action_dim: int, hidden_dim: int = 256,
                 max_action: float = 1.0):
        super().__init__()
        self.max_action = max_action

        self.network = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, action_dim),
            nn.Tanh()
        )

    def forward(self, state: torch.Tensor) -> torch.Tensor:
        return self.max_action * self.network(state)

class DDPGCritic(nn.Module):
    """Q-network for continuous actions."""

    def __init__(self, state_dim: int, action_dim: int, hidden_dim: int = 256):
        super().__init__()

        self.network = nn.Sequential(
            nn.Linear(state_dim + action_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, 1)
        )

    def forward(self, state: torch.Tensor, action: torch.Tensor) -> torch.Tensor:
        x = torch.cat([state, action], dim=-1)
        return self.network(x)
```



```

class OUNoise:
    """Ornstein-Uhlenbeck process for exploration."""

    def __init__(self, action_dim: int, mu: float = 0.0, theta: float = 0.15,
                 sigma: float = 0.2):
        self.action_dim = action_dim
        self.mu = mu
        self.theta = theta
        self.sigma = sigma
        self.state = np.ones(action_dim) * mu

    def reset(self):
        self.state = np.ones(self.action_dim) * self.mu

    def sample(self) -> np.ndarray:
        dx = self.theta * (self.mu - self.state) + self.sigma *
        np.random.randn(self.action_dim)
        self.state += dx
        return self.state

class DDPGAgent:
    """Deep Deterministic Policy Gradient agent."""

    def __init__(
        self,
        state_dim: int,
        action_dim: int,
        hidden_dim: int = 256,
        max_action: float = 1.0,
        actor_lr: float = 1e-4,
        critic_lr: float = 1e-3,
        gamma: float = 0.99,
        tau: float = 0.005,
        buffer_size: int = 100000,
        batch_size: int = 256
    ):
        self.gamma = gamma
        self.tau = tau
        self.batch_size = batch_size
        self.max_action = max_action

        # Actor networks
        self.actor = DDPGActor(state_dim, action_dim, hidden_dim, max_action)
        self.actor_target = DDPGActor(state_dim, action_dim, hidden_dim, max_action)
        self.actor_target.load_state_dict(self.actor.state_dict())
        self.actor_optimizer = optim.Adam(self.actor.parameters(), lr=actor_lr)

        # Critic networks
        self.critic = DDPGCritic(state_dim, action_dim, hidden_dim)
        self.critic_target = DDPGCritic(state_dim, action_dim, hidden_dim)
        self.critic_target.load_state_dict(self.critic.state_dict())
        self.critic_optimizer = optim.Adam(self.critic.parameters(), lr=critic_lr)

```

```

# Replay buffer
self.replay_buffer = ReplayBuffer(buffer_size)

# Exploration noise
self.noise = OUNoise(action_dim)

def select_action(self, state: np.ndarray, add_noise: bool = True) ->
    np.ndarray:
    ↪ """Select action with optional exploration noise."""
    state_tensor = torch.FloatTensor(state).unsqueeze(0)

    with torch.no_grad():
        action = self.actor(state_tensor).cpu().numpy()[0]

    if add_noise:
        action += self.noise.sample()
        action = np.clip(action, -self.max_action, self.max_action)

    return action

def store_transition(self, state: np.ndarray, action: np.ndarray,
                    reward: float, next_state: np.ndarray, done: bool):
    """Store transition in replay buffer."""
    self.replay_buffer.push(state, action, reward, next_state, done)

def soft_update(self, target: nn.Module, source: nn.Module):
    """Soft update target network parameters."""
    for target_param, source_param in zip(target.parameters(),
    ↪ source.parameters()):
        target_param.data.copy_(
            self.tau * source_param.data + (1 - self.tau) * target_param.data
        )

def update(self) -> Dict[str, float]:
    """Perform one update step."""
    if len(self.replay_buffer) < self.batch_size:
        return {}

    # Sample batch
    states, actions, rewards, next_states, dones = self.replay_buffer.sample(
        self.batch_size
    )

    states = torch.FloatTensor(states)
    actions = torch.FloatTensor(actions)
    rewards = torch.FloatTensor(rewards).unsqueeze(1)
    next_states = torch.FloatTensor(next_states)
    dones = torch.FloatTensor(dones).unsqueeze(1)

    # Update critic
    with torch.no_grad():
        next_actions = self.actor_target(next_states)

```

```

        target_q = self.critic_target(next_states, next_actions)
        target_q = rewards + self.gamma * target_q * (1 - dones)

        current_q = self.critic(states, actions)
        critic_loss = F.mse_loss(current_q, target_q)

        self.critic_optimizer.zero_grad()
        critic_loss.backward()
        self.critic_optimizer.step()

        # Update actor
        actor_loss = -self.critic(states, self.actor(states)).mean()

        self.actor_optimizer.zero_grad()
        actor_loss.backward()
        self.actor_optimizer.step()

        # Soft update targets
        self.soft_update(self.actor_target, self.actor)
        self.soft_update(self.critic_target, self.critic)

    return {
        'critic_loss': critic_loss.item(),
        'actor_loss': actor_loss.item()
    }

print("DDPG Agent created for continuous control")
ddpg_agent = DDPGAgent(state_dim=8, action_dim=2)

```

DDPG Agent created for continuous control

25.4.4. Twin Delayed DDPG (TD3)

TD3 addresses overestimation and instability in DDPG with three key improvements:

1. **Clipped Double Q-learning:** Use minimum of two critics
2. **Delayed Policy Updates:** Update actor less frequently than critics
3. **Target Policy Smoothing:** Add noise to target actions

```

class TD3Agent:
    """Twin Delayed DDPG agent."""

    def __init__(
        self,
        state_dim: int,
        action_dim: int,
        hidden_dim: int = 256,
        max_action: float = 1.0,
        actor_lr: float = 3e-4,
        critic_lr: float = 3e-4,
        gamma: float = 0.99,
        tau: float = 0.005,
    ):

```

```

    policy_noise: float = 0.2,
    noise_clip: float = 0.5,
    policy_delay: int = 2,
    buffer_size: int = 100000,
    batch_size: int = 256
):
    self.gamma = gamma
    self.tau = tau
    self.policy_noise = policy_noise
    self.noise_clip = noise_clip
    self.policy_delay = policy_delay
    self.batch_size = batch_size
    self.max_action = max_action

    # Actor
    self.actor = DDPGActor(state_dim, action_dim, hidden_dim, max_action)
    self.actor_target = DDPGActor(state_dim, action_dim, hidden_dim, max_action)
    self.actor_target.load_state_dict(self.actor.state_dict())
    self.actor_optimizer = optim.Adam(self.actor.parameters(), lr=actor_lr)

    # Twin critics
    self.critic1 = DDPGCritic(state_dim, action_dim, hidden_dim)
    self.critic2 = DDPGCritic(state_dim, action_dim, hidden_dim)
    self.critic1_target = DDPGCritic(state_dim, action_dim, hidden_dim)
    self.critic2_target = DDPGCritic(state_dim, action_dim, hidden_dim)
    self.critic1_target.load_state_dict(self.critic1.state_dict())
    self.critic2_target.load_state_dict(self.critic2.state_dict())

    self.critic_optimizer = optim.Adam(
        list(self.critic1.parameters()) + list(self.critic2.parameters()),
        lr=critic_lr
    )

    self.replay_buffer = ReplayBuffer(buffer_size)
    self.total_updates = 0

def select_action(self, state: np.ndarray, add_noise: bool = True) ->
    np.ndarray:
    """Select action with optional Gaussian noise."""
    state_tensor = torch.FloatTensor(state).unsqueeze(0)

    with torch.no_grad():
        action = self.actor(state_tensor).cpu().numpy()[0]

    if add_noise:
        noise = np.random.normal(0, self.max_action * 0.1, size=action.shape)
        action = np.clip(action + noise, -self.max_action, self.max_action)

    return action

def store_transition(self, state: np.ndarray, action: np.ndarray,
                    reward: float, next_state: np.ndarray, done: bool):
    self.replay_buffer.push(state, action, reward, next_state, done)

```

```

def soft_update(self, target: nn.Module, source: nn.Module):
    for target_param, source_param in zip(target.parameters(),
        ↪ source.parameters()):
        target_param.data.copy_(
            self.tau * source_param.data + (1 - self.tau) * target_param.data
        )

def update(self) -> Dict[str, float]:
    """Perform TD3 update."""
    if len(self.replay_buffer) < self.batch_size:
        return {}

    self.total_updates += 1

    # Sample batch
    states, actions, rewards, next_states, dones = self.replay_buffer.sample(
        self.batch_size
    )

    states = torch.FloatTensor(states)
    actions = torch.FloatTensor(actions)
    rewards = torch.FloatTensor(rewards).unsqueeze(1)
    next_states = torch.FloatTensor(next_states)
    dones = torch.FloatTensor(dones).unsqueeze(1)

    # Target policy smoothing
    with torch.no_grad():
        noise = torch.randn_like(actions) * self.policy_noise
        noise = noise.clamp(-self.noise_clip, self.noise_clip)

        next_actions = self.actor_target(next_states) + noise
        next_actions = next_actions.clamp(-self.max_action, self.max_action)

    # Clipped double Q-learning
    target_q1 = self.critic1_target(next_states, next_actions)
    target_q2 = self.critic2_target(next_states, next_actions)
    target_q = torch.min(target_q1, target_q2)
    target_q = rewards + self.gamma * target_q * (1 - dones)

    # Update critics
    current_q1 = self.critic1(states, actions)
    current_q2 = self.critic2(states, actions)

    critic_loss = F.mse_loss(current_q1, target_q) + F.mse_loss(current_q2,
        ↪ target_q)

    self.critic_optimizer.zero_grad()
    critic_loss.backward()
    self.critic_optimizer.step()

    metrics = {'critic_loss': critic_loss.item()}

```

```

    # Delayed policy update
    if self.total_updates % self.policy_delay == 0:
        actor_loss = -self.critic1(states, self.actor(states)).mean()

        self.actor_optimizer.zero_grad()
        actor_loss.backward()
        self.actor_optimizer.step()

    # Soft update targets
    self.soft_update(self.actor_target, self.actor)
    self.soft_update(self.critic1_target, self.critic1)
    self.soft_update(self.critic2_target, self.critic2)

    metrics['actor_loss'] = actor_loss.item()

    return metrics

print("TD3 Agent created")
td3_agent = TD3Agent(state_dim=8, action_dim=2)

```

TD3 Agent created

25.4.5. Soft Actor-Critic (SAC)

SAC is a state-of-the-art algorithm that maximizes both expected return and policy entropy:

$$J(\pi) = \sum_{t=0}^T \mathbb{E} [r(s_t, a_t) + \alpha H(\pi(\cdot|s_t))]$$

This leads to more robust and exploratory policies.

```

import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.distributions import Normal
import numpy as np

class GaussianPolicy(nn.Module):
    """Gaussian policy for SAC (continuous actions)."""

    def __init__(self, state_dim: int, action_dim: int, hidden_dim: int = 256,
                 log_std_min: float = -20, log_std_max: float = 2):
        super().__init__()
        self.log_std_min = log_std_min
        self.log_std_max = log_std_max

        self.network = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU()
        )

```

```

        self.mean_head = nn.Linear(hidden_dim, action_dim)
        self.log_std_head = nn.Linear(hidden_dim, action_dim)

    def forward(self, state: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
        features = self.network(state)
        mean = self.mean_head(features)
        log_std = self.log_std_head(features)
        log_std = torch.clamp(log_std, self.log_std_min, self.log_std_max)
        return mean, log_std

    def sample(self, state: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
        """Sample action with reparameterization trick."""
        mean, log_std = self.forward(state)
        std = torch.exp(log_std)

        # Reparameterization trick
        normal = Normal(mean, std)
        x_t = normal.rsample()
        action = torch.tanh(x_t)

        # Log probability with correction for tanh squashing
        log_prob = normal.log_prob(x_t)
        log_prob -= torch.log(1 - action.pow(2) + 1e-6)
        log_prob = log_prob.sum(dim=-1, keepdim=True)

        return action, log_prob

    def get_action(self, state: torch.Tensor, deterministic: bool = False)
        -> torch.Tensor:
        """Get action for evaluation."""
        mean, log_std = self.forward(state)

        if deterministic:
            return torch.tanh(mean)
        else:
            std = torch.exp(log_std)
            normal = Normal(mean, std)
            x_t = normal.rsample()
            return torch.tanh(x_t)

class SoftQNetwork(nn.Module):
    """Soft Q-network for SAC."""

    def __init__(self, state_dim: int, action_dim: int, hidden_dim: int = 256):
        super().__init__()

        self.network = nn.Sequential(
            nn.Linear(state_dim + action_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),

```

```

        nn.Linear(hidden_dim, 1)
    )

    def forward(self, state: torch.Tensor, action: torch.Tensor) -> torch.Tensor:
        x = torch.cat([state, action], dim=-1)
        return self.network(x)

class SACAgent:
    """Soft Actor-Critic agent."""

    def __init__(
        self,
        state_dim: int,
        action_dim: int,
        hidden_dim: int = 256,
        lr: float = 3e-4,
        gamma: float = 0.99,
        tau: float = 0.005,
        alpha: float = 0.2,
        auto_alpha: bool = True,
        target_entropy: float = None,
        buffer_size: int = 1000000,
        batch_size: int = 256
    ):
        self.gamma = gamma
        self.tau = tau
        self.alpha = alpha
        self.auto_alpha = auto_alpha
        self.batch_size = batch_size
        self.action_dim = action_dim

        # Policy network
        self.policy = GaussianPolicy(state_dim, action_dim, hidden_dim)
        self.policy_optimizer = optim.Adam(self.policy.parameters(), lr=lr)

        # Twin Q-networks
        self.q1 = SoftQNetwork(state_dim, action_dim, hidden_dim)
        self.q2 = SoftQNetwork(state_dim, action_dim, hidden_dim)
        self.q1_target = SoftQNetwork(state_dim, action_dim, hidden_dim)
        self.q2_target = SoftQNetwork(state_dim, action_dim, hidden_dim)
        self.q1_target.load_state_dict(self.q1.state_dict())
        self.q2_target.load_state_dict(self.q2.state_dict())

        self.q_optimizer = optim.Adam(
            list(self.q1.parameters()) + list(self.q2.parameters()),
            lr=lr
        )

        # Automatic temperature tuning
        if auto_alpha:
            self.target_entropy = target_entropy or -action_dim
            self.log_alpha = torch.zeros(1, requires_grad=True)

```



```

        self.alpha_optimizer = optim.Adam([self.log_alpha], lr=lr)

    # Replay buffer
    self.replay_buffer = ReplayBuffer(buffer_size)

def select_action(self, state: np.ndarray, evaluate: bool = False) ->
    np.ndarray:
    ↪ """Select action from policy."""
    state_tensor = torch.FloatTensor(state).unsqueeze(0)

    with torch.no_grad():
        action = self.policy.get_action(state_tensor, deterministic=evaluate)

    return action.cpu().numpy()[0]

def store_transition(self, state: np.ndarray, action: np.ndarray,
                    reward: float, next_state: np.ndarray, done: bool):
    self.replay_buffer.push(state, action, reward, next_state, done)

def soft_update(self, target: nn.Module, source: nn.Module):
    for target_param, source_param in zip(target.parameters(),
    ↪ source.parameters()):
        target_param.data.copy_(
            self.tau * source_param.data + (1 - self.tau) * target_param.data
        )

def update(self) -> Dict[str, float]:
    """Perform SAC update."""
    if len(self.replay_buffer) < self.batch_size:
        return {}

    # Sample batch
    states, actions, rewards, next_states, dones = self.replay_buffer.sample(
        self.batch_size
    )

    states = torch.FloatTensor(states)
    actions = torch.FloatTensor(actions)
    rewards = torch.FloatTensor(rewards).unsqueeze(1)
    next_states = torch.FloatTensor(next_states)
    dones = torch.FloatTensor(dones).unsqueeze(1)

    # Update Q-functions
    with torch.no_grad():
        next_actions, next_log_probs = self.policy.sample(next_states)
        q1_next = self.q1_target(next_states, next_actions)
        q2_next = self.q2_target(next_states, next_actions)
        q_next = torch.min(q1_next, q2_next) - self.alpha * next_log_probs
        q_target = rewards + self.gamma * (1 - dones) * q_next

    q1_loss = F.mse_loss(self.q1(states, actions), q_target)
    q2_loss = F.mse_loss(self.q2(states, actions), q_target)
    q_loss = q1_loss + q2_loss

```

```

        self.q_optimizer.zero_grad()
        q_loss.backward()
        self.q_optimizer.step()

        # Update policy
        new_actions, log_probs = self.policy.sample(states)
        q1_new = self.q1(states, new_actions)
        q2_new = self.q2(states, new_actions)
        q_new = torch.min(q1_new, q2_new)

        policy_loss = (self.alpha * log_probs - q_new).mean()

        self.policy_optimizer.zero_grad()
        policy_loss.backward()
        self.policy_optimizer.step()

        metrics = {
            'q_loss': q_loss.item(),
            'policy_loss': policy_loss.item()
        }

        # Update temperature
        if self.auto_alpha:
            alpha_loss = -(self.log_alpha * (
                log_probs + self.target_entropy
            ).detach()).mean()

            self.alpha_optimizer.zero_grad()
            alpha_loss.backward()
            self.alpha_optimizer.step()

            self.alpha = self.log_alpha.exp().item()
            metrics['alpha'] = self.alpha
            metrics['alpha_loss'] = alpha_loss.item()

        # Soft update targets
        self.soft_update(self.q1_target, self.q1)
        self.soft_update(self.q2_target, self.q2)

        return metrics

print("SAC Agent created for continuous control")
sac_agent = SACAgent(state_dim=8, action_dim=2)
print(f"Policy network: {sac_agent.policy}")

```

SAC Agent created for continuous control

Policy network: GaussianPolicy(

(network): Sequential(

(0): Linear(in_features=8, out_features=256, bias=True)

(1): ReLU()

(2): Linear(in_features=256, out_features=256, bias=True)

```
        (3): ReLU()
    )
    (mean_head): Linear(in_features=256, out_features=2, bias=True)
    (log_std_head): Linear(in_features=256, out_features=2, bias=True)
)
```

💡 Choosing an Actor-Critic Algorithm				
Algorithm	Action Space	Sample Efficiency	Stability	Use Case
A2C/A3C	Discrete/Continuous	Low	Medium	Fast training with parallel envs
PPO	Discrete/Continuous	Medium	High	General purpose, easy to tune
DDPG	Continuous	Medium	Low	Simple continuous control
TD3	Continuous	Medium	Medium	Improved DDPG
SAC	Continuous	High	High	State-of-the-art continuous

25.5. Advanced Topics

This section covers three extensions of standard RL that address specific limitations: model-based methods for sample efficiency, offline RL for learning from fixed datasets, and hierarchical RL for long-horizon tasks.

25.5.1. Model-Based Reinforcement Learning

Model-based RL learns a model of the environment dynamics and uses it for planning or data augmentation. By generating synthetic trajectories from the learned model, these methods can reduce the number of real environment interactions needed by an order of magnitude.

Key Approaches:

Method	Description	Example
Dyna	Learn model, generate synthetic data	Dyna-Q
World Models	Latent dynamics modeling	Dreamer, PlaNet
MPC with learned model	Plan using learned dynamics	MBPO, PETS
MuZero	Learned model for planning	AlphaZero + learned dynamics

The implementation below builds a simple world model that predicts next states and rewards given current state-action pairs. The agent then trains its policy by “imagining” rollouts through the learned model, avoiding costly real-environment interactions.

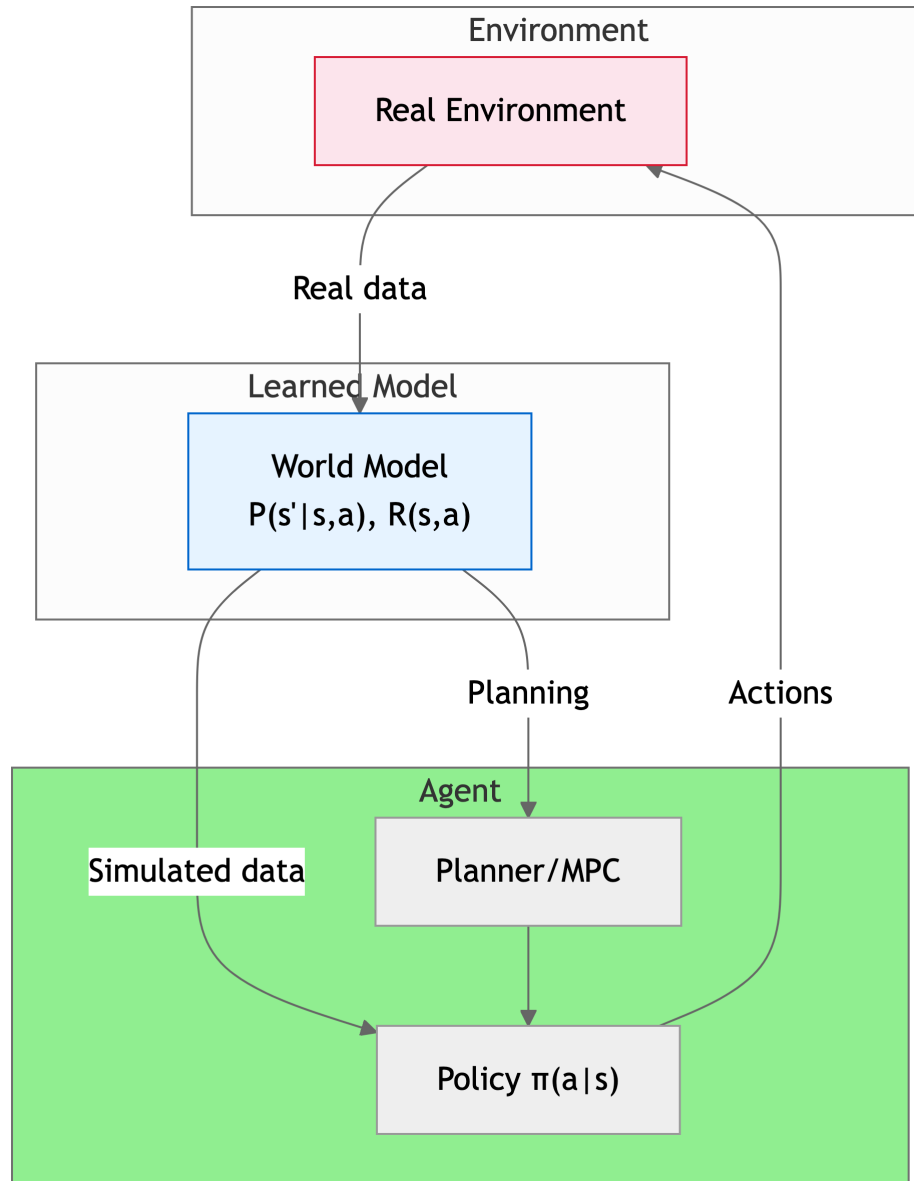


Figure 25.6.: Model-Based RL Architecture

```
class SimpleWorldModel(nn.Module):
    """Simple world model for model-based RL."""

    def __init__(self, state_dim: int, action_dim: int, hidden_dim: int = 256):
        super().__init__()

        # Dynamics model: predicts next state
        self.dynamics = nn.Sequential(
            nn.Linear(state_dim + action_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
```

```

        nn.ReLU(),
        nn.Linear(hidden_dim, state_dim)
    )

    # Reward model: predicts reward
    self.reward_model = nn.Sequential(
        nn.Linear(state_dim + action_dim, hidden_dim),
        nn.ReLU(),
        nn.Linear(hidden_dim, 1)
    )

def forward(self, state: torch.Tensor, action: torch.Tensor)
    -> Tuple[torch.Tensor, torch.Tensor]:
    """Predict next state and reward."""
    x = torch.cat([state, action], dim=-1)
    next_state = state + self.dynamics(x) # Residual prediction
    reward = self.reward_model(x)
    return next_state, reward

def rollout(self, state: torch.Tensor, actions: torch.Tensor)
    -> Tuple[List[torch.Tensor], List[torch.Tensor]]:
    """Rollout trajectory using model."""
    states = [state]
    rewards = []

    current_state = state
    for t in range(actions.shape[1]):
        action = actions[:, t, :]
        next_state, reward = self.forward(current_state, action)
        states.append(next_state)
        rewards.append(reward)
        current_state = next_state

    return states, rewards

class ModelBasedAgent:
    """Model-based agent with learned dynamics."""

    def __init__(
        self,
        state_dim: int,
        action_dim: int,
        hidden_dim: int = 256,
        model_lr: float = 1e-3,
        policy_lr: float = 3e-4,
        horizon: int = 15,
        n_rollouts: int = 100
    ):
        self.horizon = horizon
        self.n_rollouts = n_rollouts
        self.action_dim = action_dim

```

```

# World model
self.world_model = SimpleWorldModel(state_dim, action_dim, hidden_dim)
self.model_optimizer = optim.Adam(self.world_model.parameters(),
    ↪ lr=model_lr)

# Policy (can use any policy class)
self.policy = GaussianPolicy(state_dim, action_dim, hidden_dim)
self.policy_optimizer = optim.Adam(self.policy.parameters(), lr=policy_lr)

self.replay_buffer = ReplayBuffer(100000)

def train_model(self, batch_size: int = 256) -> float:
    """Train world model on real data."""
    if len(self.replay_buffer) < batch_size:
        return 0.0

    states, actions, rewards, next_states, _ =
    ↪ self.replay_buffer.sample(batch_size)

    states = torch.FloatTensor(states)
    actions = torch.FloatTensor(actions)
    rewards = torch.FloatTensor(rewards).unsqueeze(1)
    next_states = torch.FloatTensor(next_states)

    # Predict
    pred_next_states, pred_rewards = self.world_model(states, actions)

    # Loss
    state_loss = F.mse_loss(pred_next_states, next_states)
    reward_loss = F.mse_loss(pred_rewards, rewards)
    loss = state_loss + reward_loss

    self.model_optimizer.zero_grad()
    loss.backward()
    self.model_optimizer.step()

    return loss.item()

def train_policy_imagination(self, start_states: torch.Tensor) -> float:
    """Train policy using imagined rollouts."""
    batch_size = start_states.shape[0]

    # Imagine rollouts
    states = [start_states]
    log_probs = []
    rewards = []

    current_state = start_states
    for t in range(self.horizon):
        action, log_prob = self.policy.sample(current_state)
        next_state, reward = self.world_model(current_state, action)

        states.append(next_state)

```

```

        log_probs.append(log_prob)
        rewards.append(reward)

        current_state = next_state

    # Compute returns
    returns = torch.zeros(batch_size, 1)
    gamma = 0.99
    for t in reversed(range(self.horizon)):
        returns = rewards[t] + gamma * returns

    # Policy gradient
    policy_loss = -(torch.stack(log_probs).sum(dim=0) * returns).mean()

    self.policy_optimizer.zero_grad()
    policy_loss.backward()
    self.policy_optimizer.step()

    return policy_loss.item()

print("Model-Based Agent created")
mb_agent = ModelBasedAgent(state_dim=8, action_dim=2)

```

Model-Based Agent created

25.5.2. Offline (Batch) Reinforcement Learning

Offline RL learns policies from fixed datasets without online interaction.

Challenges: - Distribution shift: policy visits states not in data - Overestimation of OOD (out-of-distribution) actions

Key Algorithms:

Algorithm	Approach
CQL	Penalize Q-values for OOD actions
BCQ	Constrain policy to data distribution
IQ-L	Implicit Q-learning without policy constraint
Decision Transformer	Sequence modeling approach

Conservative Q-Learning (CQL) addresses distribution shift by adding a regularization term that penalizes Q-values for out-of-distribution actions. This prevents the policy from selecting actions that look good according to the Q-function but were never observed in the training data.

```

class ConservativeQLearning:
    """Simplified CQL for offline RL."""

    def __init__(
        self,
        state_dim: int,
        action_dim: int,
    ):

```

```

        hidden_dim: int = 256,
        lr: float = 3e-4,
        gamma: float = 0.99,
        alpha: float = 1.0, # CQL regularization weight
        n_random_actions: int = 10
    ):
        self.gamma = gamma
        self.alpha = alpha
        self.n_random_actions = n_random_actions
        self.action_dim = action_dim

        # Q-networks
        self.q1 = SoftQNetwork(state_dim, action_dim, hidden_dim)
        self.q2 = SoftQNetwork(state_dim, action_dim, hidden_dim)
        self.q_optimizer = optim.Adam(
            list(self.q1.parameters()) + list(self.q2.parameters()),
            lr=lr
        )

        # Policy
        self.policy = GaussianPolicy(state_dim, action_dim, hidden_dim)
        self.policy_optimizer = optim.Adam(self.policy.parameters(), lr=lr)

    def compute_cql_loss(self, states: torch.Tensor, actions: torch.Tensor
        ) -> torch.Tensor:
        """Compute CQL regularization term."""
        batch_size = states.shape[0]

        # Q-values for random actions (higher than they should be for OOD)
        random_actions = torch.rand(batch_size, self.n_random_actions,
        ↪ self.action_dim) * 2 - 1

        q1_random = []
        q2_random = []
        for i in range(self.n_random_actions):
            q1_random.append(self.q1(states, random_actions[:, i, :]))
            q2_random.append(self.q2(states, random_actions[:, i, :]))

        q1_random = torch.cat(q1_random, dim=1)
        q2_random = torch.cat(q2_random, dim=1)

        # Q-values for policy actions
        with torch.no_grad():
            policy_actions, _ = self.policy.sample(states)
            q1_policy = self.q1(states, policy_actions)
            q2_policy = self.q2(states, policy_actions)

        # Q-values for data actions
        q1_data = self.q1(states, actions)
        q2_data = self.q2(states, actions)

        # CQL loss: penalize high Q-values for OOD actions
        cql_loss = (

```



```

        torch.logsumexp(q1_random, dim=1).mean() +
        torch.logsumexp(q2_random, dim=1).mean() +
        q1_policy.mean() + q2_policy.mean() -
        q1_data.mean() - q2_data.mean()
    )

    return cql_loss

def update(self, states: np.ndarray, actions: np.ndarray,
           rewards: np.ndarray, next_states: np.ndarray,
           dones: np.ndarray) -> Dict[str, float]:
    """Perform CQL update."""
    states = torch.FloatTensor(states)
    actions = torch.FloatTensor(actions)
    rewards = torch.FloatTensor(rewards).unsqueeze(1)
    next_states = torch.FloatTensor(next_states)
    dones = torch.FloatTensor(dones).unsqueeze(1)

    # Standard Bellman backup
    with torch.no_grad():
        next_actions, _ = self.policy.sample(next_states)
        q1_next = self.q1(next_states, next_actions)
        q2_next = self.q2(next_states, next_actions)
        q_next = torch.min(q1_next, q2_next)
        q_target = rewards + self.gamma * (1 - dones) * q_next

    q1_loss = F.mse_loss(self.q1(states, actions), q_target)
    q2_loss = F.mse_loss(self.q2(states, actions), q_target)

    # CQL regularization
    cql_loss = self.compute_cql_loss(states, actions)

    # Total Q-loss
    q_loss = q1_loss + q2_loss + self.alpha * cql_loss

    self.q_optimizer.zero_grad()
    q_loss.backward()
    self.q_optimizer.step()

    # Update policy (maximize Q)
    new_actions, log_probs = self.policy.sample(states)
    q1_new = self.q1(states, new_actions)
    q2_new = self.q2(states, new_actions)
    q_new = torch.min(q1_new, q2_new)

    policy_loss = -q_new.mean()

    self.policy_optimizer.zero_grad()
    policy_loss.backward()
    self.policy_optimizer.step()

    return {
        'q_loss': q_loss.item(),

```

```

        'cql_loss': cql_loss.item(),
        'policy_loss': policy_loss.item()
    }

    print("Conservative Q-Learning (CQL) agent created for offline RL")

```

Conservative Q-Learning (CQL) agent created for offline RL

25.5.3. Hierarchical Reinforcement Learning

Hierarchical RL decomposes complex tasks into subtasks or skills.

Key Concepts: - **Options Framework:** Temporally extended actions with initiation sets, policies, and termination conditions - **Goal-Conditioned Policies:** Policies conditioned on subgoals - **Feudal Networks:** Manager-worker hierarchy

The following implementation builds a two-level hierarchical agent. The high-level “manager” policy proposes subgoals every N steps, while the low-level “worker” policy selects primitive actions conditioned on the current subgoal.

```

class GoalConditionedPolicy(nn.Module):
    """Policy conditioned on goals for hierarchical RL."""

    def __init__(self, state_dim: int, goal_dim: int, action_dim: int,
                  hidden_dim: int = 256):
        super().__init__()

        self.network = nn.Sequential(
            nn.Linear(state_dim + goal_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, action_dim),
            nn.Tanh()
        )

    def forward(self, state: torch.Tensor, goal: torch.Tensor) -> torch.Tensor:
        x = torch.cat([state, goal], dim=-1)
        return self.network(x)

class HighLevelPolicy(nn.Module):
    """High-level policy that proposes subgoals."""

    def __init__(self, state_dim: int, goal_dim: int, hidden_dim: int = 256):
        super().__init__()

        self.network = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU()
        )

```

```

        self.goal_mean = nn.Linear(hidden_dim, goal_dim)
        self.goal_log_std = nn.Linear(hidden_dim, goal_dim)

    def forward(self, state: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
        features = self.network(state)
        mean = self.goal_mean(features)
        log_std = torch.clamp(self.goal_log_std(features), -20, 2)
        return mean, log_std

    def sample_goal(self, state: torch.Tensor) -> torch.Tensor:
        mean, log_std = self.forward(state)
        std = torch.exp(log_std)
        normal = Normal(mean, std)
        return normal.rsample()

class HierarchicalAgent:
    """Two-level hierarchical RL agent."""

    def __init__(
        self,
        state_dim: int,
        goal_dim: int,
        action_dim: int,
        hidden_dim: int = 256,
        high_level_freq: int = 10,  # Set new goal every N steps
        lr: float = 3e-4
    ):
        self.high_level_freq = high_level_freq
        self.goal_dim = goal_dim

        # High-level policy (manager)
        self.high_policy = HighLevelPolicy(state_dim, goal_dim, hidden_dim)
        self.high_optimizer = optim.Adam(self.high_policy.parameters(), lr=lr)

        # Low-level policy (worker)
        self.low_policy = GoalConditionedPolicy(state_dim, goal_dim, action_dim,
                                                hidden_dim)
        self.low_optimizer = optim.Adam(self.low_policy.parameters(), lr=lr)

        self.current_goal = None
        self.step_count = 0

    def select_action(self, state: np.ndarray) -> np.ndarray:
        """Select action with hierarchical policy."""
        state_tensor = torch.FloatTensor(state).unsqueeze(0)

        # Update goal if needed
        if self.step_count % self.high_level_freq == 0 or self.current_goal is None:
            with torch.no_grad():
                self.current_goal = self.high_policy.sample_goal(state_tensor)

```

```

        # Get action from low-level policy
        with torch.no_grad():
            action = self.low_policy(state_tensor, self.current_goal)

        self.step_count += 1
        return action.cpu().numpy()[0]

print("Hierarchical RL agent created")
hier_agent = HierarchicalAgent(state_dim=8, goal_dim=4, action_dim=2)

```

Hierarchical RL agent created

25.5.4. Multi-Agent Reinforcement Learning

MARL extends RL to settings with multiple interacting agents.

Paradigms:

Paradigm	Description	Example
Independent Learners	Each agent learns independently	IQL
Centralized Training, Decentralized Execution (CTDE)	Shared training, individual execution	QMIX, MAPPO
Communication	Agents can communicate	CommNet, TarMAC
Opponent Modeling	Model other agents' behavior	LOLA

The simplest MARL approach gives each agent an independent Q-network. The QMIX architecture improves coordination by learning a monotonic mixing function that combines individual Q-values into a joint value, enabling decentralized execution with centralized training.

```

class IndependentLearner:
    """Independent Q-learning for multi-agent settings."""

    def __init__(self, n_agents: int, state_dim: int, action_dim: int,
                 hidden_dim: int = 128, lr: float = 1e-3):
        self.n_agents = n_agents

        # Each agent has its own Q-network
        self.q_networks = nn.ModuleList([
            QNetwork(state_dim, action_dim, [hidden_dim, hidden_dim])
            for _ in range(n_agents)
        ])

        self.optimizers = [
            optim.Adam(q.parameters(), lr=lr)
            for q in self.q_networks
        ]

        self.replay_buffers = [ReplayBuffer(10000) for _ in range(n_agents)]

    def select_actions(self, observations: List[np.ndarray],

```

```

        epsilon: float = 0.1) -> List[int]:
    """Select actions for all agents."""
    actions = []

    for i, obs in enumerate(observations):
        if np.random.random() < epsilon:
            action =
↪ np.random.randint(self.q_networks[0].network[-1].out_features)
        else:
            with torch.no_grad():
                obs_tensor = torch.FloatTensor(obs).unsqueeze(0)
                q_values = self.q_networks[i](obs_tensor)
                action = q_values.argmax().item()
            actions.append(action)

    return actions

class QMixer(nn.Module):
    """QMIX mixing network for multi-agent coordination."""

    def __init__(self, n_agents: int, state_dim: int, hidden_dim: int = 32):
        super().__init__()
        self.n_agents = n_agents

        # Hypernetworks for monotonic mixing weights
        self.hyper_w1 = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, n_agents * hidden_dim)
        )

        self.hyper_w2 = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim)
        )

        self.hyper_b1 = nn.Linear(state_dim, hidden_dim)
        self.hyper_b2 = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, 1)
        )

    def forward(self, agent_q_values: torch.Tensor, state: torch.Tensor
    ) -> torch.Tensor:
        """Mix individual Q-values into Q_tot."""
        batch_size = agent_q_values.shape[0]

        # Get mixing weights (positive via absolute value)
        w1 = torch.abs(self.hyper_w1(state)).view(batch_size, self.n_agents, -1)
        w2 = torch.abs(self.hyper_w2(state)).view(batch_size, -1, 1)

```

```

        b1 = self.hyper_b1(state).view(batch_size, 1, -1)
        b2 = self.hyper_b2(state).view(batch_size, 1, 1)

        # Forward pass through mixing network
        agent_q_values = agent_q_values.view(batch_size, 1, self.n_agents)
        hidden = F.elu(torch.bmm(agent_q_values, w1) + b1)
        q_tot = torch.bmm(hidden, w2) + b2

        return q_tot.squeeze(-1).squeeze(-1)

print("Multi-Agent RL components created")
print(f"QMIX mixer: {QMixer(n_agents=3, state_dim=10)}")

```

Multi-Agent RL components created

```

QMIX mixer: QMixer(
  (hyper_w1): Sequential(
    (0): Linear(in_features=10, out_features=32, bias=True)
    (1): ReLU()
    (2): Linear(in_features=32, out_features=96, bias=True)
  )
  (hyper_w2): Sequential(
    (0): Linear(in_features=10, out_features=32, bias=True)
    (1): ReLU()
    (2): Linear(in_features=32, out_features=32, bias=True)
  )
  (hyper_b1): Linear(in_features=10, out_features=32, bias=True)
  (hyper_b2): Sequential(
    (0): Linear(in_features=10, out_features=32, bias=True)
    (1): ReLU()
    (2): Linear(in_features=32, out_features=1, bias=True)
  )
)

```

25.6. Applications to Optimization

25.6.1. Combinatorial Optimization with RL

Deep RL can learn heuristics for NP-hard combinatorial problems. The approach formulates solution construction as a sequential decision process: at each step, the agent selects the next element to add to the solution, and the final reward is the negative objective value (e.g., negative tour length for TSP).

The encoder-decoder architecture below uses multi-head self-attention to compute node embeddings and then an attention-based decoder to select cities one at a time. The encoder processes all city coordinates simultaneously, while the decoder autoregressively builds a tour by attending to unvisited cities.



Figure 25.7.: RL for Combinatorial Optimization Pipeline

```

import torch
import torch.nn as nn
import torch.nn.functional as F
import numpy as np
from typing import Tuple, List

class TSPEncoder(nn.Module):
    """Encoder for TSP instances using attention."""

    def __init__(self, input_dim: int = 2, hidden_dim: int = 128, n_heads: int = 8):
        super().__init__()

        self.input_projection = nn.Linear(input_dim, hidden_dim)

        self.attention_layers = nn.ModuleList([
            nn.MultiheadAttention(hidden_dim, n_heads, batch_first=True)
            for _ in range(3)
        ])

        self.feed_forward = nn.ModuleList([
            nn.Sequential(
                nn.Linear(hidden_dim, hidden_dim * 4),
                nn.ReLU(),
                nn.Linear(hidden_dim * 4, hidden_dim)
            )
            for _ in range(3)
        ])

        self.layer_norms = nn.ModuleList([
            nn.LayerNorm(hidden_dim) for _ in range(6)
        ])

    def forward(self, nodes: torch.Tensor) -> torch.Tensor:
        """Encode TSP instance.

        Args:
            nodes: (batch, n_nodes, 2) - node coordinates

        Returns:
            Node embeddings (batch, n_nodes, hidden_dim)
        """
        h = self.input_projection(nodes)

```

```

    for i, (attn, ff) in enumerate(zip(self.attention_layers,
        ↪ self.feed_forward)):
        # Self-attention
        h_attn, _ = attn(h, h, h)
        h = self.layer_norms[2*i](h + h_attn)

        # Feed-forward
        h_ff = ff(h)
        h = self.layer_norms[2*i + 1](h + h_ff)

    return h

class TSPDecoder(nn.Module):
    """Attention-based decoder for TSP."""

    def __init__(self, hidden_dim: int = 128):
        super().__init__()

        self.hidden_dim = hidden_dim

        # Query projection
        self.query_proj = nn.Linear(hidden_dim * 3, hidden_dim)

        # Key and value projections
        self.key_proj = nn.Linear(hidden_dim, hidden_dim)
        self.value_proj = nn.Linear(hidden_dim, hidden_dim)

    def forward(self, node_embeddings: torch.Tensor, context: torch.Tensor,
        mask: torch.Tensor) -> torch.Tensor:
        """Compute selection probabilities.

        Args:
            node_embeddings: (batch, n_nodes, hidden_dim)
            context: (batch, hidden_dim * 3) - [graph_embedding, first_node,
        ↪ current_node]
            mask: (batch, n_nodes) - True for visited nodes

        Returns:
            Selection probabilities (batch, n_nodes)
        """
        # Compute attention
        query = self.query_proj(context).unsqueeze(1) # (batch, 1, hidden)
        keys = self.key_proj(node_embeddings) # (batch, n_nodes, hidden)

        # Scaled dot-product attention
        scores = torch.matmul(query, keys.transpose(-2, -1)) /
        ↪ np.sqrt(self.hidden_dim)
        scores = scores.squeeze(1) # (batch, n_nodes)

        # Mask visited nodes
        scores[mask] = float('-inf')

```



```

        # Softmax for probabilities
        probs = F.softmax(scores, dim=-1)

        return probs

class AttentionTSPAgent:
    """RL agent for TSP using attention model."""

    def __init__(self, hidden_dim: int = 128, n_heads: int = 8, lr: float = 1e-4):
        self.hidden_dim = hidden_dim

        self.encoder = TSPEncoder(hidden_dim=hidden_dim, n_heads=n_heads)
        self.decoder = TSPDecoder(hidden_dim=hidden_dim)

        self.optimizer = optim.Adam(
            list(self.encoder.parameters()) + list(self.decoder.parameters()),
            lr=lr
        )

        # Baseline (exponential moving average of rewards)
        self.baseline = 0.0
        self.baseline_alpha = 0.1

    def generate_tour(self, nodes: torch.Tensor, greedy: bool = False
                    ) -> Tuple[torch.Tensor, torch.Tensor]:
        """Generate tour for TSP instance.

        Args:
            nodes: (batch, n_nodes, 2)
            greedy: If True, select best action; otherwise sample

        Returns:
            tour: (batch, n_nodes) - node indices in order
            log_probs: (batch,) - log probability of tour
        """
        batch_size, n_nodes, _ = nodes.shape

        # Encode
        node_embeddings = self.encoder(nodes)
        graph_embedding = node_embeddings.mean(dim=1) # (batch, hidden)

        # Initialize
        tour = []
        log_probs = []
        mask = torch.zeros(batch_size, n_nodes, dtype=torch.bool)

        # Start from node 0
        current_idx = torch.zeros(batch_size, dtype=torch.long)
        tour.append(current_idx)
        mask[torch.arange(batch_size), current_idx] = True

        first_embedding = node_embeddings[torch.arange(batch_size), current_idx]

```

```

        for step in range(n_nodes - 1):
            current_embedding = node_embeddings[torch.arange(batch_size),
↪ current_idx]

            # Context: [graph, first_node, current_node]
            context = torch.cat([graph_embedding, first_embedding,
↪ current_embedding], dim=-1)

            # Get probabilities
            probs = self.decoder(node_embeddings, context, mask)

            # Select next node
            if greedy:
                next_idx = probs.argmax(dim=-1)
            else:
                dist = torch.distributions.Categorical(probs)
                next_idx = dist.sample()
                log_probs.append(dist.log_prob(next_idx))

            tour.append(next_idx)
            mask[torch.arange(batch_size), next_idx] = True
            current_idx = next_idx

    tour = torch.stack(tour, dim=1)

    if log_probs:
        log_probs = torch.stack(log_probs, dim=1).sum(dim=1)
    else:
        log_probs = torch.zeros(batch_size)

    return tour, log_probs

def compute_tour_length(self, nodes: torch.Tensor, tour: torch.Tensor
                        ) -> torch.Tensor:
    """Compute total tour length."""
    batch_size, n_nodes = tour.shape

    # Gather nodes in tour order
    tour_nodes = nodes[
        torch.arange(batch_size).unsqueeze(1).expand(-1, n_nodes),
        tour
    ]

    # Add return to start
    tour_nodes_rolled = torch.roll(tour_nodes, -1, dims=1)

    # Compute distances
    distances = torch.norm(tour_nodes - tour_nodes_rolled, dim=-1)
    tour_length = distances.sum(dim=1)

    return tour_length

```

```

def train_batch(self, nodes: torch.Tensor) -> Dict[str, float]:
    """Train on batch of TSP instances using REINFORCE."""
    # Generate tours
    tour, log_probs = self.generate_tour(nodes, greedy=False)

    # Compute tour lengths (negative reward)
    tour_lengths = self.compute_tour_length(nodes, tour)

    # Update baseline
    self.baseline = (1 - self.baseline_alpha) * self.baseline + \
        self.baseline_alpha * tour_lengths.mean().item()

    # REINFORCE loss
    advantage = tour_lengths - self.baseline
    loss = (advantage.detach() * log_probs).mean()

    # Optimize
    self.optimizer.zero_grad()
    loss.backward()
    torch.nn.utils.clip_grad_norm_(
        list(self.encoder.parameters()) + list(self.decoder.parameters()),
        1.0
    )
    self.optimizer.step()

    return {
        'loss': loss.item(),
        'tour_length': tour_lengths.mean().item(),
        'baseline': self.baseline
    }

# Example
print("Attention-based TSP Agent created")
tsp_agent = AttentionTSPAgent()

# Generate random TSP instance
np.random.seed(42)
nodes = torch.rand(4, 20, 2) # 4 instances, 20 cities each

# Train step
metrics = tsp_agent.train_batch(nodes)
print(f"Training metrics: {metrics}")

# Generate greedy tour
with torch.no_grad():
    tour, _ = tsp_agent.generate_tour(nodes[:1], greedy=True)
    length = tsp_agent.compute_tour_length(nodes[:1], tour)
    print(f"Greedy tour length: {length.item():.4f}")

```

Attention-based TSP Agent created

25. Deep Reinforcement Learning

RuntimeError: one of the variables needed for gradient computation has been modified by
↳ an inplace operation: [CpuBoolType [4, 20]] is at version 20; expected version 19
↳ instead. Hint: enable anomaly detection to find the operation that failed to compute
↳ its gradient, with torch.autograd.set_detect_anomaly(True).

```
-----
RuntimeError                                Traceback (most recent call last)
Cell In[20], line 239
    236 nodes = torch.rand(4, 20, 2) # 4 instances, 20 cities each
    238 # Train step
--> 239 metrics = tsp_agent.train_batch(nodes)
    240 print(f"Training metrics: {metrics}")
    242 # Generate greedy tour
Cell In[20], line 216, in AttentionTSPAgent.train_batch(self, nodes)
    214 # Optimize
    215 self.optimizer.zero_grad()
--> 216 loss.backward()
    217 torch.nn.utils.clip_grad_norm_(
    218     list(self.encoder.parameters()) + list(self.decoder.parameters()),
    219     1.0
    220 )
    221 self.optimizer.step()
File /opt/homebrew/lib/python3.14/site-packages/torch/_tensor.py:630, in
↳ Tensor.backward(self, gradient, retain_graph, create_graph, inputs)
    620 if has_torch_function_unary(self):
    621     return handle_torch_function(
    622         Tensor.backward,
    623         (self,),
    (...) 628         inputs=inputs,
    629     )
--> 630 torch.autograd.backward(
    631     self, gradient, retain_graph, create_graph, inputs=inputs
    632 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/__init__.py:364, in
↳ backward(tensors, grad_tensors, retain_graph, create_graph, grad_variables, inputs)
    359 retain_graph = create_graph
    361 # The reason we repeat the same comment below is that
    362 # some Python versions print out the first line of a multi-line function
    363 # calls in the traceback and some print out the last line
--> 364 _engine_run_backward(
    365     tensors,
    366     grad_tensors_,
    367     retain_graph,
    368     create_graph,
    369     inputs_tuple,
    370     allow_unreachable=True,
    371     accumulate_grad=True,
    372 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/graph.py:865, in
↳ _engine_run_backward(t_outputs, *args, **kwargs)
    863 unregister_hooks = _register_logging_hooks_on_whole_graph(t_outputs)
    864 try:
--> 865     return Variable._execution_engine.run_backward( # Calls into the C++ engine
↳ to run the backward pass
    866         t_outputs, *args, **kwargs
    867     ) # Calls into the C++ engine to run the backward pass
```

```

868 finally:
869     if attach_logging_hooks:

```

RuntimeError: one of the variables needed for gradient computation has been modified by
 ↳ an inplace operation: [CPUBoolType [4, 20]] is at version 20; expected version 19
 ↳ instead. Hint: enable anomaly detection to find the operation that failed to compute
 ↳ its gradient, with `torch.autograd.set_detect_anomaly(True)`.

25.6.2. Resource Allocation and Scheduling

RL can learn adaptive scheduling policies for dynamic environments where job arrivals, processing times, and machine breakdowns are uncertain. The agent observes the current state of all jobs and machines, then decides which job to process next.

The following implementation creates a job shop scheduling environment with configurable numbers of jobs and machines. The state representation includes job progress, remaining processing times, and machine availability.

```

class JobShopEnvironment:
    """Simplified job shop scheduling environment."""

    def __init__(self, n_jobs: int = 5, n_machines: int = 3,
                 max_processing_time: int = 10):
        self.n_jobs = n_jobs
        self.n_machines = n_machines
        self.max_processing_time = max_processing_time

        self.reset()

    def reset(self) -> np.ndarray:
        """Reset environment with new random instance."""
        # Random processing times for each job on each machine
        self.processing_times = np.random.randint(
            1, self.max_processing_time + 1,
            size=(self.n_jobs, self.n_machines)
        )

        # Machine sequence for each job (permutation of machines)
        self.machine_sequences = np.array([
            np.random.permutation(self.n_machines)
            for _ in range(self.n_jobs)
        ])

        # State: current operation index for each job
        self.job_progress = np.zeros(self.n_jobs, dtype=int)

        # Machine availability times
        self.machine_available = np.zeros(self.n_machines)

        # Job completion times
        self.job_completion = np.zeros(self.n_jobs)

        self.done = False

        return self._get_state()

```

```

def _get_state(self) -> np.ndarray:
    """Get current state representation."""
    # Features: job progress, remaining processing time, machine availability
    state = []

    # Job features
    for j in range(self.n_jobs):
        progress = self.job_progress[j] / self.n_machines
        if self.job_progress[j] < self.n_machines:
            remaining = self.processing_times[j, self.job_progress[j]:].sum()
        else:
            remaining = 0
        remaining_norm = remaining / (self.max_processing_time *
        ↪ self.n_machines)
        state.extend([progress, remaining_norm, self.job_completion[j] / 100])

    # Machine features
    for m in range(self.n_machines):
        state.append(self.machine_available[m] / 100)

    return np.array(state, dtype=np.float32)

def _get_valid_actions(self) -> List[int]:
    """Get list of jobs that can be scheduled."""
    valid = []
    for j in range(self.n_jobs):
        if self.job_progress[j] < self.n_machines:
            valid.append(j)
    return valid

def step(self, action: int) -> Tuple[np.ndarray, float, bool, dict]:
    """Schedule next operation of selected job."""
    if action not in self._get_valid_actions():
        return self._get_state(), -100, True, {'error': 'invalid action'}

    job = action
    op_idx = self.job_progress[job]
    machine = self.machine_sequences[job, op_idx]
    proc_time = self.processing_times[job, machine]

    # Schedule operation
    start_time = max(self.machine_available[machine], self.job_completion[job])
    end_time = start_time + proc_time

    self.machine_available[machine] = end_time
    self.job_completion[job] = end_time
    self.job_progress[job] += 1

    # Check if done
    self.done = all(self.job_progress >= self.n_machines)

    # Reward: negative of incremental makespan

```

```

makespan = max(self.job_completion)
reward = -proc_time / (self.max_processing_time * self.n_machines)

if self.done:
    # Bonus/penalty based on makespan
    optimal_estimate = self.processing_times.sum() / self.n_machines
    reward += (optimal_estimate - makespan) / optimal_estimate

return self._get_state(), reward, self.done, {'makespan': makespan}

class SchedulingAgent:
    """RL agent for job shop scheduling."""

    def __init__(self, state_dim: int, n_jobs: int, hidden_dim: int = 128):
        self.n_jobs = n_jobs

        self.policy = nn.Sequential(
            nn.Linear(state_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, n_jobs)
        )

        self.optimizer = optim.Adam(self.policy.parameters(), lr=1e-3)

    def select_action(self, state: np.ndarray, valid_actions: List[int],
                    training: bool = True) -> int:
        """Select job to schedule."""
        state_tensor = torch.FloatTensor(state).unsqueeze(0)
        logits = self.policy(state_tensor).squeeze(0)

        # Mask invalid actions
        mask = torch.full((self.n_jobs,), float('-inf'))
        mask[valid_actions] = 0
        masked_logits = logits + mask

        probs = F.softmax(masked_logits, dim=-1)

        if training:
            dist = torch.distributions.Categorical(probs)
            action = dist.sample().item()
        else:
            action = probs.argmax().item()

        return action

# Test scheduling environment
print("Job Shop Scheduling Environment created")
env = JobShopEnvironment(n_jobs=5, n_machines=3)
state = env.reset()

```

```

print(f"State dimension: {len(state)}")
print(f"Initial state: {state[:9].round(2)}") # First 3 jobs

# Random episode
total_reward = 0
while not env.done:
    valid = env._get_valid_actions()
    action = np.random.choice(valid)
    state, reward, done, info = env.step(action)
    total_reward += reward

print(f"Random policy makespan: {info['makespan']:.1f}")
print(f"Total reward: {total_reward:.4f}")

```

Job Shop Scheduling Environment created

State dimension: 18

Initial state: [0. 0.67 0. 0. 0.47 0. 0. 0.57 0.]

Random policy makespan: 46.0

Total reward: -3.4429

25.6.3. Production Optimization

RL enables real-time adaptive control in manufacturing systems where throughput, quality, and inventory levels must be balanced simultaneously. The agent controls production rates at each stage and learns to match output to stochastic demand patterns.

```

class ProductionLineEnvironment:
    """Production line control environment."""

    def __init__(self, n_stages: int = 4, buffer_capacity: int = 20,
                 demand_mean: float = 5.0, demand_std: float = 2.0):
        self.n_stages = n_stages
        self.buffer_capacity = buffer_capacity
        self.demand_mean = demand_mean
        self.demand_std = demand_std

        # Stage parameters (processing rates, energy costs)
        self.base_rates = np.array([6.0, 5.5, 5.0, 5.5])[:n_stages]
        self.energy_costs = np.array([1.0, 1.2, 1.5, 1.1])[:n_stages]

        self.reset()

    def reset(self) -> np.ndarray:
        """Reset environment."""
        # Buffer levels between stages
        self.buffers = np.full(self.n_stages - 1, self.buffer_capacity // 2,
                               dtype=float)

        # Finished goods inventory
        self.finished_inventory = 10.0

        # Machine states (production rate multipliers)

```



```

self.machine_rates = np.ones(self.n_stages)

self.time_step = 0
self.max_steps = 100

return self._get_state()

def _get_state(self) -> np.ndarray:
    """Get state representation."""
    state = np.concatenate([
        self.buffers / self.buffer_capacity, # Normalized buffer levels
        [self.finished_inventory / 50],      # Normalized inventory
        self.machine_rates,                  # Current rates
        [self.time_step / self.max_steps]    # Time progress
    ])
    return state.astype(np.float32)

def step(self, action: np.ndarray) -> Tuple[np.ndarray, float, bool, dict]:
    """Execute production control action.

    Args:
        action: (n_stages,) rate multipliers in [0.5, 1.5]
    """
    # Apply action (rate adjustments)
    self.machine_rates = np.clip(action, 0.5, 1.5)

    # Simulate production flow
    actual_rates = self.base_rates * self.machine_rates

    # Process each stage
    produced = np.zeros(self.n_stages)

    # First stage: unlimited input
    produced[0] = actual_rates[0]

    # Middle stages: limited by input buffer
    for i in range(1, self.n_stages):
        available = self.buffers[i-1]
        produced[i] = min(actual_rates[i], available)

    # Update buffers
    for i in range(self.n_stages - 1):
        inflow = produced[i]
        outflow = produced[i + 1]
        self.buffers[i] = np.clip(
            self.buffers[i] + inflow - outflow,
            0, self.buffer_capacity
        )

    # Final stage output
    self.finished_inventory += produced[-1]

    # Generate demand

```

```

        demand = max(0, np.random.normal(self.demand_mean, self.demand_std))
        fulfilled = min(demand, self.finished_inventory)
        self.finished_inventory -= fulfilled

        # Compute reward
        # - Throughput reward
        throughput_reward = produced[-1] / self.base_rates[-1]

        # - Demand fulfillment
        fulfill_reward = fulfilled / max(demand, 1e-6)

        # - Energy cost penalty
        energy_cost = np.sum(self.energy_costs * self.machine_rates ** 2) /
        ↪ np.sum(self.energy_costs)

        # - Buffer balance (penalize extreme levels)
        buffer_penalty = np.mean(np.abs(self.buffers / self.buffer_capacity - 0.5))

        reward = throughput_reward + fulfill_reward - 0.5 * energy_cost - 0.3 *
        ↪ buffer_penalty

        self.time_step += 1
        done = self.time_step >= self.max_steps

        info = {
            'throughput': produced[-1],
            'demand_fulfilled': fulfilled / max(demand, 1e-6),
            'energy_cost': energy_cost,
            'inventory': self.finished_inventory
        }

        return self._get_state(), reward, done, info

# Test production environment
print("Production Line Environment created")
prod_env = ProductionLineEnvironment(n_stages=4)
state = prod_env.reset()
print(f"State dimension: {len(state)}")

# Run with constant rate
total_reward = 0
for _ in range(prod_env.max_steps):
    action = np.ones(prod_env.n_stages) # Run at base rate
    state, reward, done, info = prod_env.step(action)
    total_reward += reward

print(f"Constant rate policy - Total reward: {total_reward:.2f}")
print(f"Final inventory: {info['inventory']:.1f}")

```

```

Production Line Environment created
State dimension: 9
Constant rate policy - Total reward: 128.92

```

Final inventory: 10.2

25.7. Implementation Guide

Choosing the right framework significantly affects development speed and production readiness. This section surveys the main RL frameworks and specialized libraries for optimization applications.

25.7.1. RL Frameworks

Framework	Strengths	Best For
Stable-Baselines3	Easy to use, well-documented	Quick prototyping, research
RLlib	Scalable, distributed	Production, large-scale
CleanRL	Single-file implementations	Learning, understanding
Tianshou	Modular, PyTorch-native	Custom algorithms
EnvPool	C++ threadpool, 20x faster vectorization	High-throughput training

For high-performance environment vectorization, **EnvPool** (NeurIPS 2022) replaces Python-level environment loops with a C++ threadpool, achieving up to 20x speedup over standard vectorized environments. It provides drop-in compatibility with Stable-Baselines3, Tianshou, and CleanRL through Gymnasium and dm_env interfaces, making it particularly valuable when training throughput is a bottleneck.

25.7.2. RL for Combinatorial Optimization Libraries

Beyond general-purpose RL frameworks, specialized libraries have emerged for applying RL to classic operations research problems.

25.7.2.1. RL4CO: Routing Optimization with Attention Models

RL4CO implements state-of-the-art attention-based architectures for combinatorial optimization, particularly routing problems.

Key Concepts:

1. **Encoder-Decoder Architecture:** Encode problem instance, decode solution sequentially
2. **Attention Mechanism:** Learn which nodes to attend to based on context
3. **REINFORCE Training:** Policy gradient with baselines for variance reduction
4. **Greedy/Sampling Decoding:** Trade off solution quality vs. speed

```
# RL4CO Attention Model concept (simplified)

class SimplifiedAttentionModel:
    """Conceptual attention model for routing."""

    def __init__(self, embed_dim: int = 128):
        self.embed_dim = embed_dim

    def encode(self, nodes: np.ndarray) -> np.ndarray:
        """Encode node features into embeddings."""
        # In practice: Multi-head attention layers
        # Here: Simple linear projection simulation
        n_nodes = len(nodes)
```

```

        embeddings = np.random.randn(n_nodes, self.embed_dim)
        return embeddings

    def decode_step(self, embeddings: np.ndarray, context: np.ndarray,
                    mask: np.ndarray) -> int:
        """Select next node using attention."""
        # Compute attention scores
        scores = embeddings @ context
        scores[mask] = -np.inf # Mask visited nodes

        # Select via softmax (greedy: argmax)
        probs = np.exp(scores - scores.max())
        probs /= probs.sum()

        return np.argmax(probs) # Greedy selection

    def solve_tsp(self, coordinates: np.ndarray) -> List[int]:
        """Solve TSP using attention-based decoding."""
        n = len(coordinates)
        embeddings = self.encode(coordinates)

        tour = [0] # Start from depot
        mask = np.zeros(n, dtype=bool)
        mask[0] = True

        # Context is last visited node embedding
        context = embeddings[0]

        for _ in range(n - 1):
            next_node = self.decode_step(embeddings, context, mask)
            tour.append(next_node)
            mask[next_node] = True
            context = embeddings[next_node]

        return tour

# Demo the concept
np.random.seed(42)
coords = np.random.rand(10, 2)
model = SimplifiedAttentionModel()
tour = model.solve_tsp(coords)
print(f"Tour: {tour}")

def tour_length(coords, tour):
    total = 0
    for i in range(len(tour)):
        total += np.linalg.norm(coords[tour[i]] - coords[tour[(i+1) % len(tour)]])
    return total

print(f"Tour length: {tour_length(coords, tour):.3f}")

```

Tour: [0, np.int64(2), np.int64(8), np.int64(6), np.int64(5), np.int64(9), np.int64(1), np.int64(3), np.int64(7), np.int64(4)]
 Tour length: 6.033

RL4CO Features:

Problem	Architecture	Key Innovation
TSP	Attention Model	Autoregressive decoding
CVRP	AM + Capacity	Capacity-aware context
VRP Variants	POMO	Multiple starting points
Scheduling	L2D	Learn to dispatch

Example: Using RL4CO for TSP:

```
# Full RL4CO usage (requires pip install rl4co)
from rl4co.envs import TSPEnv
from rl4co.models import AttentionModel
from rl4co.utils.trainer import RL4COTrainer

# Create environment with 50-node instances
env = TSPEnv(num_loc=50)

# Initialize pre-trained or new model
model = AttentionModel(
    env,
    baseline="rollout",
    train_data_size=100000,
)

# Training (if needed)
trainer = RL4COTrainer(max_epochs=100, accelerator="auto")
trainer.fit(model)

# Inference on new instances
test_data = env.dataset(batch_size=1000)

# Greedy decoding (fast, deterministic)
greedy_tours, _ = model(test_data, decode_type="greedy")

# Sampling decoding (slower, potentially better)
sample_tours, _ = model(test_data, decode_type="sampling",
                        num_samples=100)
```

25.7.2.2. OR-Gym: Standard RL Environments for OR

OR-Gym provides Gymnasium-compatible environments for classic OR problems, enabling use of any standard RL algorithm.

```
# OR-Gym-style inventory management environment

class InventoryManagementEnv:
    """Multi-echelon inventory management for RL.

    State: Current inventory at each stage
    Action: Order quantities for each stage
    Reward: Negative (holding cost + backlog cost)
```

```

"""

def __init__(
    self,
    num_stages: int = 3,
    num_periods: int = 30,
    holding_costs: Optional[List[float]] = None,
    backlog_costs: Optional[List[float]] = None,
    demand_mean: float = 20.0
):
    self.num_stages = num_stages
    self.num_periods = num_periods
    self.holding_costs = holding_costs or [1.0] * num_stages
    self.backlog_costs = backlog_costs or [10.0] * num_stages
    self.demand_mean = demand_mean

    # State and action spaces
    self.observation_space = (num_stages,) # Inventory levels
    self.action_space = (num_stages,) # Order quantities

    self.reset()

def reset(self, seed: Optional[int] = None) -> Tuple[np.ndarray, dict]:
    """Reset environment to initial state."""
    if seed is not None:
        np.random.seed(seed)

    self.inventory = np.array([100.0] * self.num_stages)
    self.period = 0
    return self.inventory.copy(), {}

def step(self, action: np.ndarray) -> Tuple[np.ndarray, float, bool, bool,
    ↪ dict]:
    """Execute one timestep."""
    # Clip actions to valid range
    action = np.clip(action, 0, 100)

    # Generate demand
    demand = np.random.poisson(self.demand_mean)

    # Update inventory (simplified: immediate delivery)
    self.inventory += action
    self.inventory[-1] -= demand # Demand at final stage

    # Flow between stages
    for i in range(self.num_stages - 1, 0, -1):
        transfer = min(self.inventory[i-1], max(-self.inventory[i], 0))
        self.inventory[i] += transfer
        self.inventory[i-1] -= transfer

    # Calculate costs
    holding = sum(max(inv, 0) * h for inv, h in
        zip(self.inventory, self.holding_costs))

```

```

        backlog = sum(max(-inv, 0) * b for inv, b in
                        zip(self.inventory, self.backlog_costs))
        reward = -(holding + backlog)

        self.period += 1
        done = self.period >= self.num_periods

        info = {
            'holding_cost': holding,
            'backlog_cost': backlog,
            'demand': demand
        }

        return self.inventory.copy(), reward, done, False, info

# Create and test environment
env = InventoryManagementEnv(num_stages=3, num_periods=20)
obs, _ = env.reset(seed=42)
print(f"Initial inventory: {obs}")

# Run with simple policy
total_reward = 0
for t in range(20):
    # Base-stock policy: order to reach target
    target = np.array([100, 100, 150])
    action = np.maximum(target - obs, 0)

    obs, reward, done, _, info = env.step(action)
    total_reward += reward

    if t < 5:
        print(f"Period {t}: Inv={obs.astype(int)}, Reward={reward:.1f}")

print(f"\nTotal reward: {total_reward:.1f}")

```

```

Initial inventory: [100. 100. 100.]
Period 0: Inv=[100 100 127], Reward=-327.0
Period 1: Inv=[100 100 135], Reward=-335.0
Period 2: Inv=[100 100 129], Reward=-329.0
Period 3: Inv=[100 100 125], Reward=-325.0
Period 4: Inv=[100 100 135], Reward=-335.0

```

```
Total reward: -6621.0
```

OR-Gym Environment Types:

Environment	State	Action	Objective
InvManagement	Inventory levels	Order quantities	Min holding + backlog
BinPacking	Bin states, item	Bin selection	Min bins used
Knapsack	Items, capacity	Include/exclude	Max value

Environment	State	Action	Objective
NewsVendor	Demand history	Order quantity	Max profit
VMPacking	Server loads	VM placement	Balance load

Integrating with Stable-Baselines3:

```
# Using OR-Gym with Stable-Baselines3
import gymnasium as gym
import or_gym
from stable_baselines3 import PPO, A2C

# Create environment
env = gym.make('or_gym:InvManagement-v1')

# Train PPO agent
model = PPO(
    "MlpPolicy",
    env,
    learning_rate=3e-4,
    n_steps=2048,
    batch_size=64,
    verbose=1
)
model.learn(total_timesteps=100000)

# Evaluate
obs, _ = env.reset()
total_reward = 0
done = False
while not done:
    action, _ = model.predict(obs, deterministic=True)
    obs, reward, terminated, truncated, info = env.step(action)
    total_reward += reward
    done = terminated or truncated

print(f"RL policy reward: {total_reward:.1f}")
```

25.7.2.3. Comparison: When to Use Which Library

Scenario	Recommended	Reason
TSP/VRP at scale	RL4CO	State-of-the-art neural routing
Inventory control	OR-Gym + SB3	Standard RL interface
Custom OR problem	OR-Gym-style	Build your own env
Research on routing	RL4CO	Pre-built architectures
Practical deployment	Classical OR	More reliable, interpretable

RL vs Classical Methods

RL for combinatorial optimization shines when: - Problem structure varies (different sizes, constraints) - Fast inference is needed after training - Problem is too complex for exact methods

- Warm-starting classical methods is valuable
- For static, well-defined problems, classical methods (MIP, metaheuristics) often outperform RL.

The following trainer class wraps the training loop into a reusable utility that handles episode management, periodic logging, and evaluation. This pattern mirrors the API design of libraries like Stable-Baselines3.

```
# Example: Using a Stable-Baselines3-style API

class RLTrainer:
    """High-level trainer for RL experiments."""

    def __init__(
        self,
        agent,
        env,
        eval_env=None,
        log_interval: int = 10,
        eval_interval: int = 100,
        n_eval_episodes: int = 10
    ):
        self.agent = agent
        self.env = env
        self.eval_env = eval_env or env
        self.log_interval = log_interval
        self.eval_interval = eval_interval
        self.n_eval_episodes = n_eval_episodes

        self.training_rewards = []
        self.eval_rewards = []

    def train(self, total_episodes: int) -> Dict[str, List]:
        """Train agent for specified episodes."""
        for episode in range(total_episodes):
            # Training episode
            state, _ = self.env.reset()
            episode_reward = 0
            done = False

            while not done:
                action = self.agent.select_action(state)
                next_state, reward, terminated, truncated, _ = self.env.step(action)
                done = terminated or truncated

                self.agent.store_transition(state, action, reward, next_state, done)
                self.agent.update()

                episode_reward += reward
                state = next_state

            self.training_rewards.append(episode_reward)
```

```

# Logging
if (episode + 1) % self.log_interval == 0:
    avg_reward = np.mean(self.training_rewards[-self.log_interval:])
    print(f"Episode {episode + 1}: Avg Reward = {avg_reward:.2f}")

# Evaluation
if (episode + 1) % self.eval_interval == 0:
    eval_reward = self.evaluate()
    self.eval_rewards.append(eval_reward)
    print(f"  Evaluation: {eval_reward:.2f}")

return {
    'training_rewards': self.training_rewards,
    'eval_rewards': self.eval_rewards
}

def evaluate(self) -> float:
    """Evaluate current policy."""
    rewards = []

    for _ in range(self.n_eval_episodes):
        state, _ = self.eval_env.reset()
        episode_reward = 0
        done = False

        while not done:
            # Deterministic action for evaluation
            action = self.agent.select_action(state, training=False)
            state, reward, terminated, truncated, _ = self.eval_env.step(action)
            done = terminated or truncated
            episode_reward += reward

        rewards.append(episode_reward)

    return np.mean(rewards)

print("RL Trainer utility created")

```

RL Trainer utility created

25.7.3. Debugging RL Systems

RL systems are notoriously difficult to debug because failures can be silent – the agent simply fails to learn without producing errors. The following checklist and utility code help identify the most common failure modes.

Common RL Debugging Issues

1. **Reward scale:** Rewards too large/small can destabilize learning
2. **Observation normalization:** Unnormalized observations hurt performance
3. **Action bounds:** Ensure actions stay within valid ranges

4. **Gradient explosion:** Use gradient clipping
5. **Exploration collapse:** Monitor entropy/epsilon
6. **Incorrect discount:** γ too low can cause myopic behavior

The `RLDebugger` class tracks key health metrics including gradient norms, Q-value distributions, and reward trends. These metrics help identify problems like gradient explosion, value function divergence, and exploration collapse.

```
class RLDebugger:
    """Utilities for debugging RL training."""

    def __init__(self):
        self.metrics_history = {
            'rewards': [],
            'episode_lengths': [],
            'q_values': [],
            'losses': [],
            'gradients': [],
            'entropy': []
        }

    def log_episode(self, reward: float, length: int, **kwargs):
        """Log episode metrics."""
        self.metrics_history['rewards'].append(reward)
        self.metrics_history['episode_lengths'].append(length)

        for key, value in kwargs.items():
            if key not in self.metrics_history:
                self.metrics_history[key] = []
            self.metrics_history[key].append(value)

    def check_gradient_health(self, model: nn.Module) -> Dict[str, float]:
        """Check for gradient issues."""
        total_norm = 0.0
        max_grad = 0.0
        min_grad = float('inf')
        nan_count = 0

        for p in model.parameters():
            if p.grad is not None:
                param_norm = p.grad.data.norm(2)
                total_norm += param_norm.item() ** 2
                max_grad = max(max_grad, p.grad.data.abs().max().item())
                min_grad = min(min_grad, p.grad.data.abs().min().item())
                nan_count += torch.isnan(p.grad).sum().item()

        total_norm = total_norm ** 0.5

        return {
            'total_norm': total_norm,
            'max_grad': max_grad,
            'min_grad': min_grad,

```

```

        'nan_count': nan_count
    }

def check_q_value_health(self, q_values: torch.Tensor) -> Dict[str, float]:
    """Check Q-value statistics for debugging."""
    return {
        'mean': q_values.mean().item(),
        'std': q_values.std().item(),
        'max': q_values.max().item(),
        'min': q_values.min().item(),
        'nan_count': torch.isnan(q_values).sum().item()
    }

def diagnose(self) -> str:
    """Generate diagnostic report."""
    report = []

    # Reward trend
    if len(self.metrics_history['rewards']) > 100:
        early = np.mean(self.metrics_history['rewards'][:50])
        recent = np.mean(self.metrics_history['rewards'][-50:])

        if recent < early:
            report.append("WARNING: Reward decreasing over training")
        elif recent > early * 1.5:
            report.append("GOOD: Significant reward improvement")

    # Episode length check
    if self.metrics_history['episode_lengths']:
        avg_length = np.mean(self.metrics_history['episode_lengths'][-100:])
        if avg_length < 10:
            report.append("WARNING: Very short episodes - check
↪ environment/reward")

    # Q-value check
    if self.metrics_history['q_values']:
        recent_q = self.metrics_history['q_values'][-100:]
        if np.any(np.isnan(recent_q)):
            report.append("CRITICAL: NaN Q-values detected")
        elif np.max(recent_q) > 1000:
            report.append("WARNING: Very large Q-values - possible
↪ overestimation")

    if not report:
        report.append("All metrics look healthy")

    return "\n".join(report)

print("RL Debugger created")
debugger = RLDebugger()

# Simulate some training data

```

```

for i in range(200):
    debugger.log_episode(
        reward=10 + i * 0.1 + np.random.randn() * 2,
        length=50 + int(np.random.randn() * 10)
    )

print(debugger.diagnose())

```

RL Debugger created

GOOD: Significant reward improvement

25.7.4. Hyperparameter Tuning

Hyperparameter sensitivity is one of the main practical challenges in RL. Small changes in learning rate or buffer size can mean the difference between convergence and failure. The table below lists the most impactful hyperparameters for each algorithm family.

Algorithm	Critical Hyperparameters
DQN	Learning rate, buffer size, target update freq, epsilon decay
PPO	Clip epsilon, GAE lambda, number of epochs, minibatch size
SAC	Temperature (alpha), tau for soft updates, learning rates
TD3	Policy delay, target noise, noise clip

The following code collects well-tested default values for each algorithm. These defaults serve as starting points; domain-specific tuning is usually necessary for optimal performance.

```

# Reasonable default hyperparameters for common algorithms

DQN_DEFAULTS = {
    'learning_rate': 1e-4,
    'buffer_size': 100000,
    'batch_size': 64,
    'gamma': 0.99,
    'target_update_freq': 1000,
    'epsilon_start': 1.0,
    'epsilon_end': 0.01,
    'epsilon_decay_steps': 100000,
    'hidden_dims': [256, 256]
}

PPO_DEFAULTS = {
    'learning_rate': 3e-4,
    'gamma': 0.99,
    'gae_lambda': 0.95,
    'clip_epsilon': 0.2,
    'entropy_coef': 0.01,
    'value_coef': 0.5,
    'max_grad_norm': 0.5,
    'n_epochs': 10,
    'batch_size': 64,
    'n_steps': 2048
}

```

```

}

SAC_DEFAULTS = {
    'learning_rate': 3e-4,
    'gamma': 0.99,
    'tau': 0.005,
    'alpha': 0.2, # or auto-tuned
    'buffer_size': 1000000,
    'batch_size': 256,
    'hidden_dims': [256, 256],
    'target_entropy': 'auto' # -action_dim
}

TD3_DEFAULTS = {
    'learning_rate': 3e-4,
    'gamma': 0.99,
    'tau': 0.005,
    'policy_noise': 0.2,
    'noise_clip': 0.5,
    'policy_delay': 2,
    'buffer_size': 1000000,
    'batch_size': 256
}

print("Algorithm Defaults:")
print("\nDQN:", DQN_DEFAULTS)
print("\nPPO:", PPO_DEFAULTS)

```

Algorithm Defaults:

DQN: {'learning_rate': 0.0001, 'buffer_size': 100000, 'batch_size': 64, 'gamma': 0.99, 'target_

PPO: {'learning_rate': 0.0003, 'gamma': 0.99, 'gae_lambda': 0.95, 'clip_epsilon': 0.2, 'entropy

25.8. Summary

25.8.1. Key Takeaways

Foundations: - MDPs provide the mathematical framework: (S, A, P, R, γ) - Value functions $V(s)$ and $Q(s, a)$ quantify expected returns - Bellman equations enable recursive value computation - Exploration-exploitation tradeoff is fundamental

Value-Based Methods: - DQN combines Q-learning with deep neural networks - Experience replay and target networks stabilize training - Double DQN addresses overestimation - Dueling architecture separates value and advantage - Prioritized replay focuses on important transitions

Policy Gradient Methods: - REINFORCE uses Monte Carlo returns with high variance - Baselines and advantage estimation reduce variance - TRPO constrains policy updates with KL divergence - PPO achieves similar benefits with simpler clipped objective

Actor-Critic Methods: - Combine policy (actor) and value (critic) learning - A2C uses n-step returns and parallel environments - DDPG extends to continuous control with deterministic policy - TD3 improves stability with twin critics and delayed updates - SAC maximizes entropy for robust exploration

Advanced Topics: - Model-based RL learns environment dynamics for planning - Offline RL learns from fixed datasets without interaction - Hierarchical RL decomposes complex tasks into subtasks - Multi-agent RL handles multiple interacting agents

Applications: - Combinatorial optimization (TSP, VRP) with attention models - Resource allocation and scheduling with learned policies - Production control with real-time adaptation

25.9. Exercises

1. **DQN Implementation:** Implement a complete DQN agent and train it on CartPole-v1 from Gymnasium. Track learning curves and compare with a random policy. Experiment with different target update frequencies.
2. **PPO vs A2C:** Implement both PPO and A2C on the same continuous control task (e.g., Pendulum-v1). Compare sample efficiency and final performance. Analyze the effect of the clipping parameter in PPO.
3. **SAC for Continuous Control:** Train SAC on a MuJoCo environment (e.g., HalfCheetah). Implement automatic temperature tuning and compare with fixed temperature values.
4. **RL for TSP:** Extend the attention-based TSP agent to handle variable-size instances. Train on instances of size 20-50 and evaluate generalization to larger instances.
5. **Hierarchical Scheduling:** Design a hierarchical RL approach for the job shop scheduling problem where a high-level policy selects job priorities and a low-level policy makes machine assignments.
6. **Offline RL Experiment:** Collect a dataset of transitions using a trained DQN agent on CartPole. Then train a CQL agent on this offline data and compare with behavioral cloning.
7. **Multi-Agent Coordination:** Implement independent learners and QMIX for a simple multi-agent coordination task (e.g., predator-prey or traffic light control). Compare convergence and final performance.
8. **Debugging Challenge:** Intentionally introduce bugs into a PPO implementation (e.g., wrong advantage normalization, missing gradient clipping) and use the debugging utilities to identify the issues.
9. **Production Control:** Train a SAC agent on the production line environment. Experiment with different reward functions that balance throughput, energy cost, and demand fulfillment.
10. **Algorithm Comparison Study:** On a single benchmark task, compare DQN, PPO, SAC, and TD3 in terms of:
 - Sample efficiency (performance vs. environment steps)
 - Wall-clock training time
 - Final policy performance
 - Sensitivity to hyperparameters

25.10. Further Reading

- Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.)
- Mnih, V., et al. (2015). Human-level control through deep reinforcement learning. *Nature*
- Schulman, J., et al. (2017). Proximal Policy Optimization Algorithms. *arXiv*
- Haarnoja, T., et al. (2018). Soft Actor-Critic: Off-Policy Maximum Entropy Deep RL. *ICML*
- Fujimoto, S., et al. (2018). Addressing Function Approximation Error in Actor-Critic Methods. *ICML*
- Kool, W., et al. (2019). Attention, Learn to Solve Routing Problems! *ICLR*

25. *Deep Reinforcement Learning*

- Levine, S., et al. (2020). Offline Reinforcement Learning: Tutorial, Review, and Perspectives. *arXiv*

26. Sequential Decision Making

What You Will Learn

- Understand sequential decision problems and their structure
- Apply dynamic programming principles to multi-stage optimization
- Master Warren Powell’s universal framework for sequential decisions
- Implement the four classes of policies: PFAs, CFAs, VFAs, and DLAs
- Use approximate dynamic programming for large-scale problems
- Apply Stochastic Dual Dynamic Programming (SDDP) for multi-stage stochastic optimization

Sequential decision problems are ubiquitous: from managing inventory over time to controlling energy systems, from routing delivery vehicles to making investment decisions. This chapter presents a unified framework for understanding and solving these problems, drawing heavily on Warren B. Powell’s foundational work.

26.1. Introduction to Sequential Decisions

26.1.1. What Makes a Decision Sequential?

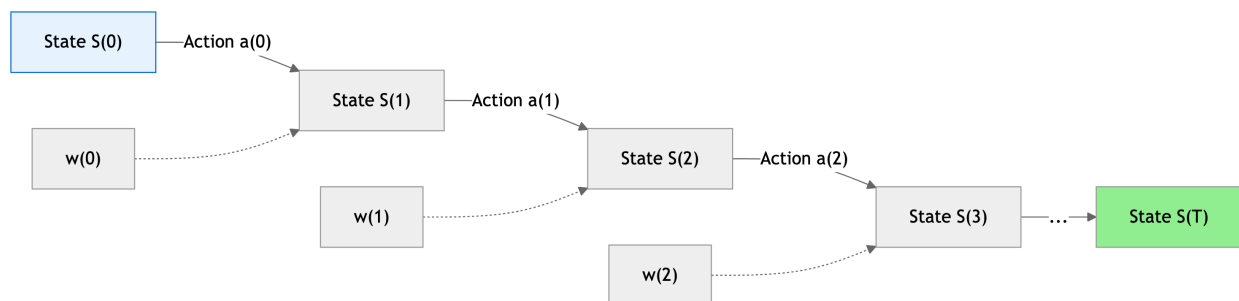


Figure 26.1.: Sequential Decision Structure: States, Actions, and Transitions Over Time

A sequential decision problem has three key characteristics:

1. **Decisions are made over time:** Actions taken now affect future states
2. **Information reveals gradually:** Uncertainty resolves as we proceed
3. **Actions have lasting consequences:** Today’s decisions constrain tomorrow’s options

Problem	Decision	State	Uncertainty
Inventory Management	Order quantities	Stock levels	Demand
Portfolio Optimization	Asset allocation	Holdings, wealth	Returns
Energy Storage	Charge/discharge	Battery level	Prices, demand
Vehicle Routing	Route selection	Vehicle locations	Travel times

26.1.2. The Modeling Challenge

Sequential problems are notoriously difficult because:

- **Curse of dimensionality:** State space grows exponentially with dimensions
- **Curse of multiple outcomes:** Scenarios multiply across stages
- **Curse of history:** Past decisions affect current options

! The Three Curses

Consider a 10-stage problem with 100 states per stage and 10 scenarios per stage:

- **State space:** $100^{10} = 10^{20}$ possible state trajectories
- **Scenario tree:** 10^{10} possible outcome sequences
- **Policy space:** Exponential in both dimensions

Exact dynamic programming is computationally intractable for realistic problems.

26.2. Dynamic Programming Foundations

26.2.1. The Bellman Equation

Dynamic programming decomposes sequential problems using Bellman's principle of optimality.

For a minimization problem, the value function $V_t(s)$ represents the optimal cost-to-go from state s at time t :

$$V_t(s) = \min_{a \in A(s)} \{c_t(s, a) + \mathbb{E}[V_{t+1}(s')]\}$$

where: - $c_t(s, a)$ is the immediate cost of action a in state s - $s' = f(s, a, \omega)$ is the next state given action a and random outcome ω - $\mathbb{E}[\cdot]$ is the expectation over outcomes

26.2.2. Backward Induction

For finite-horizon problems with small state spaces, we solve backward in time. Starting from the terminal stage where the value function is known (or zero), we compute the optimal action and value at each state by working backward. This produces both the complete value function table and the optimal policy for every state at every stage.

```
import numpy as np
from typing import List, Dict, Callable, Tuple

def backward_induction(
    stages: int,
    states: List,
    actions: Callable, # actions(s, t) -> list of valid actions
    transition: Callable, # transition(s, a, omega) -> next state
    cost: Callable, # cost(s, a, t) -> immediate cost
    scenarios: List, # list of (omega, probability) tuples
    terminal_cost: Callable = None # terminal_cost(s) -> final cost
) -> Tuple[Dict, Dict]:
    """Backward induction for finite-horizon DP.

    Args:
        stages: Number of time periods
        states: List of possible states
```

```

    actions: Function returning valid actions
    transition: State transition function
    cost: Immediate cost function
    scenarios: List of (outcome, probability) tuples
    terminal_cost: Terminal value function

Returns:
    V: Value functions V[t][s]
    policy: Optimal policy[t][s] -> action
"""
T = stages

# Initialize value function
V = {t: {} for t in range(T + 1)}
policy = {t: {} for t in range(T)}

# Terminal condition
for s in states:
    V[T][s] = terminal_cost(s) if terminal_cost else 0

# Backward recursion
for t in range(T - 1, -1, -1):
    for s in states:
        best_cost = float('inf')
        best_action = None

        for a in actions(s, t):
            # Compute expected future cost
            expected_future = 0
            for omega, prob in scenarios:
                s_next = transition(s, a, omega)
                expected_future += prob * V[t + 1][s_next]

            # Total cost for this action
            total = cost(s, a, t) + expected_future

            if total < best_cost:
                best_cost = total
                best_action = a

        V[t][s] = best_cost
        policy[t][s] = best_action

return V, policy

```

26.2.3. The Curse of Dimensionality

Backward induction requires enumerating all states at each stage. For high-dimensional problems, this is impossible.

Example: An energy storage problem with: - 1000 battery states - 24 hourly periods - 100 price scenarios per period

Scenario tree size: $100^{24} \approx 10^{48}$ paths!

A scenario tree represents the evolution of uncertainty across stages. Decision nodes (squares)

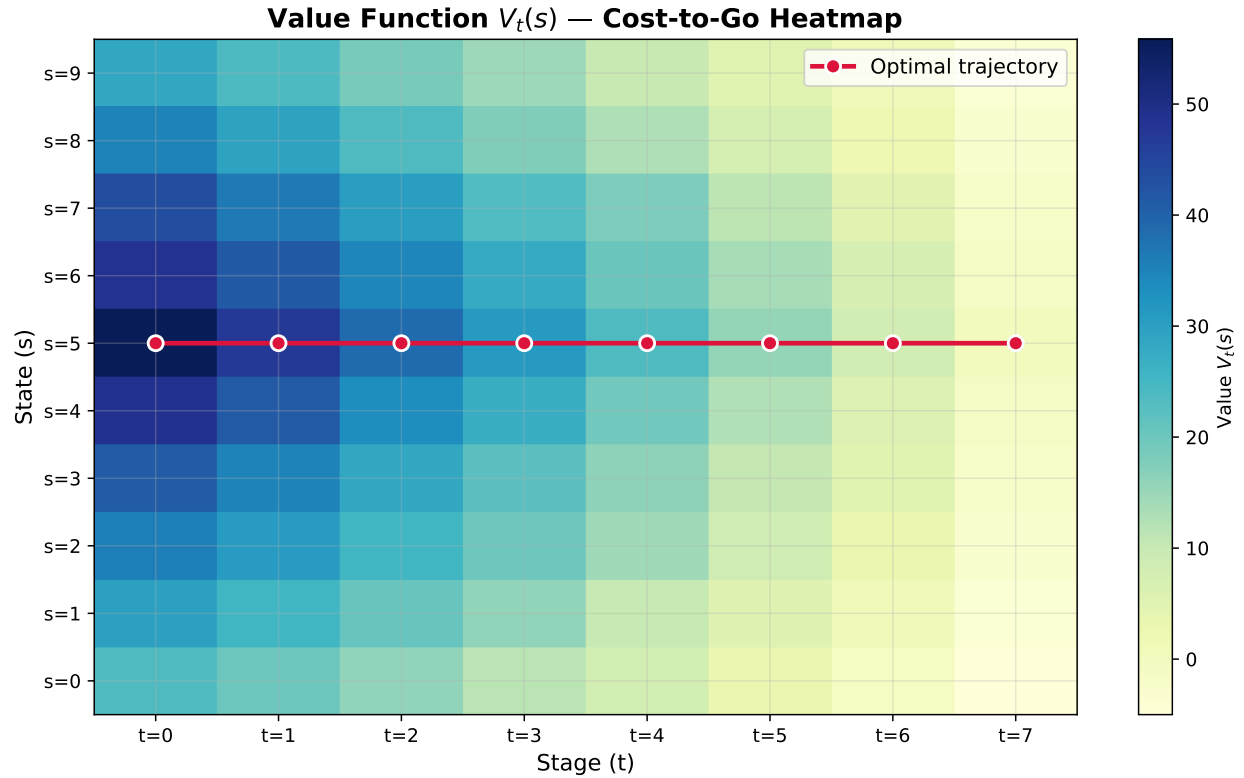


Figure 26.2.: Dynamic programming value function heatmap — optimal cost-to-go for each state at each stage, darker colors indicate higher value (better states)

alternate with chance nodes (circles). Each path from root to leaf defines one possible realization of uncertainty. The width of each edge indicates its probability mass.

26.3. Warren Powell's Universal Framework

Warren B. Powell developed a unified framework that encompasses all sequential decision problems and solution methods. This framework clarifies the relationships between dynamic programming, reinforcement learning, stochastic programming, and simulation.

26.3.1. The Universal Model

Every sequential decision problem can be expressed as:

$$\max_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=0}^T C(S_t, X^\pi(S_t)) \right]$$

where: - S_t : State at time t (containing all information needed for decisions) - $X^\pi(S_t)$: Decision function (policy) mapping state to action - $C(S_t, X_t)$: Contribution function (reward or cost) - Π : Set of feasible policies

26.3.2. State Variables

Powell emphasizes a comprehensive state definition:

$$S_t = (R_t, I_t)$$

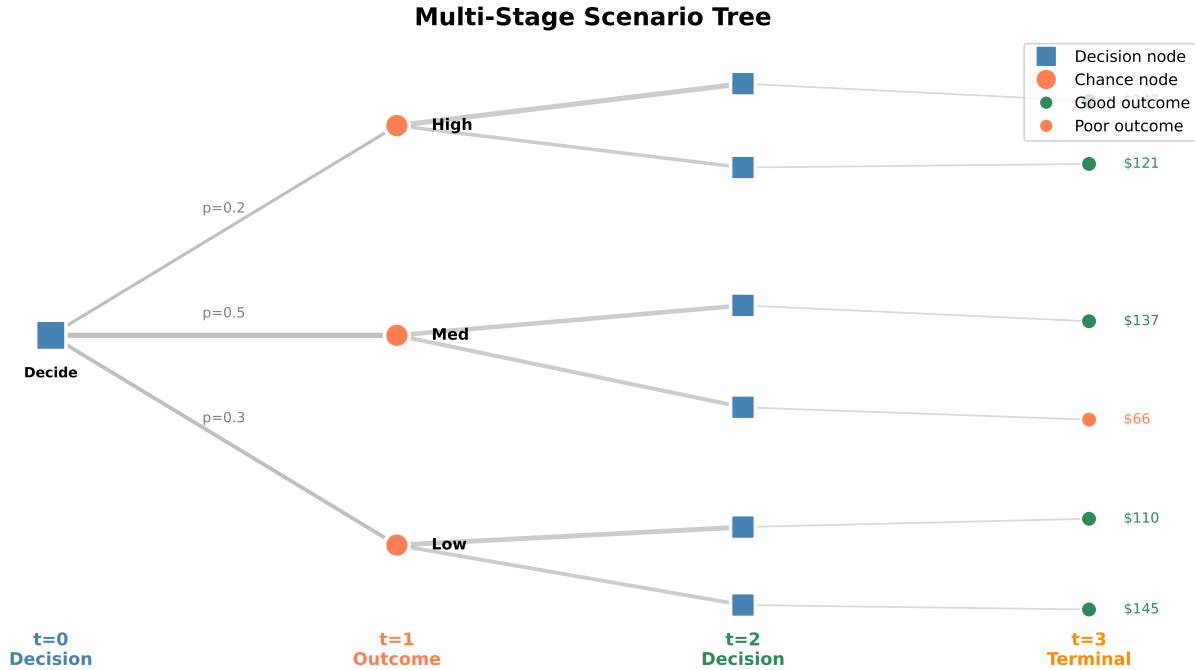


Figure 26.3.: Multi-stage scenario tree — decisions (squares) at each node with probabilistic outcomes branching forward, width represents probability mass

where: - R_t : Physical state (inventory, location, resources) - I_t : Information state (beliefs, forecasts, parameters)

i The Post-Decision State

A key insight is the **post-decision state** S_t^a : the state immediately after decision X_t but before new information W_{t+1} arrives:

$$S_{t+1} = S^M(S_t^a, W_{t+1})$$

This separates the deterministic effect of decisions from stochastic information arrival.

26.3.3. The Four Classes of Policies

Powell's framework identifies four fundamental approaches to designing policies:

Class	Approach	Computes	Example
PFA	Policy Function Approximation	Direct mapping $X(S \theta)$	(s,S) inventory policy
CFA	Cost Function Approximation	Modified costs	Bid prices
VFA	Value Function Approximation	Approximate $V(S)$	ADP, TD learning
DLA	Direct Lookahead	Future simulation	Tree search, MPC

26.4. Policy Function Approximations (PFAs)

PFAs are the most direct approach to policy design. Instead of computing value functions or solving lookahead problems, a PFA maps states to actions through a fixed parametric function. The parameters are tuned offline using simulation, optimization, or domain knowledge. This makes PFAs fast to execute at runtime—they require only a function evaluation, not an optimization solve. PFAs specify the policy directly as a parametric function of the state.

26.4.1. Structure of PFAs

$$X^\pi(S_t|\theta) = \arg \max_{x \in \mathcal{X}} U(S_t, x|\theta)$$

The parameters θ encode the policy logic.

26.4.2. Examples of PFAs

Base-Stock Policy (inventory):

$$X_t = \max(0, S - R_t)$$

where S is the order-up-to level (parameter to optimize).

(s, S) Policy:

$$X_t = \begin{cases} S - R_t & \text{if } R_t < s \\ 0 & \text{otherwise} \end{cases}$$

Linear Policy:

$$X_t = \theta_0 + \theta_1 S_t^{(1)} + \theta_2 S_t^{(2)} + \dots$$

```
class BaseStockPolicy:
    """Base-stock (order-up-to) inventory policy."""

    def __init__(self, base_stock_level: float):
        self.S = base_stock_level

    def __call__(self, inventory: float) -> float:
        """Compute order quantity."""
        return max(0, self.S - inventory)

    def optimize(self, simulator, demand_samples, holding_cost, shortage_cost):
        """Find optimal base-stock level via simulation."""
        best_S = None
        best_cost = float('inf')

        for S in range(0, 200): # Search over possible levels
            self.S = S
            total_cost = 0

            for _ in range(1000): # Simulation replications
                inventory = S
                for demand in demand_samples:
                    order = self(inventory)
                    inventory = inventory + order - demand
                    total_cost += (holding_cost * max(0, inventory) +
                                   shortage_cost * max(0, -inventory))
```

```

        avg_cost = total_cost / 1000
        if avg_cost < best_cost:
            best_cost = avg_cost
            best_S = S

    self.S = best_S
    return best_S, best_cost

class LinearPolicy:
    """Linear policy with learnable parameters."""

    def __init__(self, n_features: int):
        self.theta = np.zeros(n_features + 1) # Include bias

    def __call__(self, state: np.ndarray) -> float:
        """Compute action from state."""
        features = np.concatenate([[1], state]) # Add bias term
        return np.dot(self.theta, features)

    def update(self, gradient: np.ndarray, learning_rate: float = 0.01):
        """Update parameters using gradient."""
        self.theta -= learning_rate * gradient

```

26.4.3. When to Use PFAs

- **Advantages:** Simple, interpretable, fast execution, domain knowledge
- **Disadvantages:** May miss complex patterns, requires good structure choice
- **Best for:** Well-understood domains with known policy structures

26.5. Cost Function Approximations (CFAs)

CFAs take a different approach from PFAs. Instead of specifying the policy directly, they modify the cost or reward function so that a myopic (single-period) optimizer makes decisions that are approximately optimal over the full horizon. The correction terms encode information about future consequences that a one-step-ahead optimizer would otherwise miss. CFAs modify the objective function to guide myopic (single-period) optimization.

26.5.1. Structure of CFAs

$$X^\pi(S_t|\theta) = \arg \max_{x \in \mathcal{X}} [\tilde{C}(S_t, x) + \theta^T \phi(S_t, x)]$$

The original cost $C(S_t, x)$ is augmented with correction terms.

26.5.2. Examples of CFAs

Bid Prices (revenue management):

$$\tilde{C}(s, x) = r_x - \pi_x$$

where π_x is the “opportunity cost” of using capacity.

Lagrangian Relaxation:

$$\tilde{C}(s, x) = C(s, x) - \lambda^T g(s, x)$$

where λ prices constraint violations.

```
class BidPricePolicy:
    """Bid price policy for resource allocation."""

    def __init__(self, n_resources: int):
        self.bid_prices = np.zeros(n_resources)

    def accept_request(self, revenue: float, resource_usage: np.ndarray) -> bool:
        """Decide whether to accept a request."""
        opportunity_cost = np.dot(self.bid_prices, resource_usage)
        return revenue >= opportunity_cost

    def update_bid_prices(self, shadow_prices: np.ndarray,
                        learning_rate: float = 0.1):
        """Update bid prices based on dual values."""
        self.bid_prices = (1 - learning_rate) * self.bid_prices + \
            learning_rate * shadow_prices

class RolloutWithCFA:
    """Rollout algorithm with cost function approximation."""

    def __init__(self, base_policy, correction_weights: np.ndarray):
        self.base_policy = base_policy
        self.theta = correction_weights

    def evaluate_action(self, state, action, features_func):
        """Evaluate action with corrected costs."""
        base_cost = self.base_policy.cost(state, action)
        features = features_func(state, action)
        correction = np.dot(self.theta, features)
        return base_cost + correction

    def select_action(self, state, valid_actions, features_func):
        """Select best action under corrected costs."""
        best_action = None
        best_value = float('-inf')

        for action in valid_actions:
            value = self.evaluate_action(state, action, features_func)
            if value > best_value:
                best_value = value
                best_action = action

        return best_action
```

26.6. Value Function Approximations (VFAs)

VFAs approximate the Bellman value function and use it to guide decisions. Rather than computing the exact value function—which requires enumerating all states—VFAs use a parametric approximation

(linear, piecewise linear, or neural network) trained from sampled state transitions. The key idea comes from approximate dynamic programming: replace the exact value function with a learned approximation, then use this approximation to select actions via one-step lookahead. VFAs approximate the value function $V(S)$ and use it to guide decisions.

26.6.1. The Approximate Bellman Equation

$$V_t(S_t) \approx \bar{V}_t(S_t|\theta)$$

Decisions are made by:

$$X_t = \arg \max_{x \in \mathcal{X}} [C(S_t, x) + \gamma \bar{V}_{t+1}(S_{t+1}|\theta)]$$

26.6.2. Approximation Architectures

Linear VFA:

$$\bar{V}(S|\theta) = \theta^T \phi(S)$$

Lookup Table:

$$\bar{V}(S) = V[S] \quad \text{for discrete } S$$

Neural Network:

$$\bar{V}(S|\theta) = \text{NN}_\theta(S)$$

```
class LinearVFA:
    """Linear value function approximation."""

    def __init__(self, n_features: int, learning_rate: float = 0.01):
        self.theta = np.zeros(n_features)
        self.alpha = learning_rate

    def evaluate(self, features: np.ndarray) -> float:
        """Estimate value for given features."""
        return np.dot(self.theta, features)

    def update(self, features: np.ndarray, target: float):
        """Update approximation using TD-like update."""
        prediction = self.evaluate(features)
        error = target - prediction
        self.theta += self.alpha * error * features

class ApproximateDynamicProgramming:
    """Approximate Dynamic Programming with VFA."""

    def __init__(self, vfa: LinearVFA, gamma: float = 0.99):
        self.vfa = vfa
        self.gamma = gamma

    def select_action(self, state, valid_actions, transition_func,
                     reward_func, feature_func):
        """Select action using approximate value function."""
        best_action = None
        best_value = float('-inf')
```

```

    for action in valid_actions:
        immediate_reward = reward_func(state, action)
        next_state = transition_func(state, action)
        next_features = feature_func(next_state)
        future_value = self.vfa.evaluate(next_features)

        total_value = immediate_reward + self.gamma * future_value

        if total_value > best_value:
            best_value = total_value
            best_action = action

    return best_action

def train_episode(self, env, feature_func, n_steps: int = 100):
    """Train VFA using a single episode."""
    state = env.reset()

    for t in range(n_steps):
        features = feature_func(state)

        # Select action using current VFA
        action = self.select_action(
            state, env.valid_actions(state),
            env.transition, env.reward, feature_func
        )

        # Take action
        next_state, reward, done = env.step(action)
        next_features = feature_func(next_state)

        # TD target
        if done:
            target = reward
        else:
            target = reward + self.gamma * self.vfa.evaluate(next_features)

        # Update VFA
        self.vfa.update(features, target)

        if done:
            break
        state = next_state

```

26.6.3. Approximate Policy Iteration

Approximate policy iteration combines VFA-based policy evaluation with greedy policy improvement. Each outer iteration trains the value function approximation over multiple episodes, then the improved VFA implicitly defines a better policy for the next round. The process converges when the VFA parameters stabilize.

```
def approximate_policy_iteration(env, feature_func, n_iterations: int = 100,
                                n_episodes: int = 100):
    """Approximate policy iteration with linear VFA."""
    n_features = len(feature_func(env.reset()))
    vfa = LinearVFA(n_features)
    adp = ApproximateDynamicProgramming(vfa)

    for iteration in range(n_iterations):
        # Policy evaluation: estimate V under current policy
        for _ in range(n_episodes):
            adp.train_episode(env, feature_func)

        print(f"Iteration {iteration + 1}: theta = {vfa.theta[:5]}...")

    return adp
```

26.7. Direct Lookahead Policies (DLAs)

DLAs differ fundamentally from PFAs, CFAs, and VFAs. Instead of learning a policy offline, DLAs solve an optimization problem at each decision point. They simulate possible futures over a planning horizon and select the action that produces the best expected outcome. The computational cost is higher per decision, but DLAs can handle complex constraints and non-stationary environments without retraining. DLAs make decisions by explicitly looking ahead and planning.

26.7.1. Structure of DLAs

$$X_t = \arg \max_{x_t} \max_{x_{t+1}, \dots, x_{t+H}} \sum_{\tau=t}^{t+H} \gamma^{\tau-t} C(S_\tau, x_\tau)$$

The lookahead horizon H trades off computation versus decision quality.

26.7.2. Model Predictive Control (MPC)

MPC solves a deterministic optimization over the planning horizon:

```
import numpy as np
from scipy.optimize import minimize

class ModelPredictiveController:
    """Model Predictive Control for sequential decisions."""

    def __init__(self, horizon: int, dynamics_func, cost_func,
                 constraint_func=None):
        self.H = horizon
        self.dynamics = dynamics_func
        self.cost = cost_func
        self.constraints = constraint_func

    def plan(self, current_state: np.ndarray,
            forecast: np.ndarray) -> np.ndarray:
        """Plan optimal action sequence using MPC.
```

```

    Args:
        current_state: Current state vector
        forecast: Predicted exogenous inputs over horizon

    Returns:
        First action in optimal sequence
    """
    n_actions = len(current_state) # Assume action dim = state dim

    def objective(action_sequence):
        actions = action_sequence.reshape(self.H, n_actions)
        state = current_state.copy()
        total_cost = 0

        for t in range(self.H):
            total_cost += self.cost(state, actions[t], t)
            state = self.dynamics(state, actions[t], forecast[t])

        return total_cost

    # Initial guess
    x0 = np.zeros(self.H * n_actions)

    # Optimize
    result = minimize(objective, x0, method='SLSQP')

    # Return first action
    return result.x[:n_actions]

class TreeSearchPolicy:
    """Tree search lookahead for discrete actions."""

    def __init__(self, depth: int, branching_factor: int = 10):
        self.depth = depth
        self.branching = branching_factor

    def search(self, state, valid_actions, transition_func, reward_func,
               scenario_generator):
        """Search for best action using scenario tree."""
        if self.depth == 0:
            return max(valid_actions,
                       key=lambda a: reward_func(state, a))

        best_action = None
        best_value = float('-inf')

        for action in valid_actions:
            # Generate scenarios
            scenarios = scenario_generator(self.branching)
            expected_value = reward_func(state, action)

            for scenario, prob in scenarios:

```

```

        next_state = transition_func(state, action, scenario)

        # Recursive search
        future_value = self._search_recursive(
            next_state, self.depth - 1,
            valid_actions, transition_func, reward_func,
            scenario_generator
        )
        expected_value += prob * future_value

    if expected_value > best_value:
        best_value = expected_value
        best_action = action

    return best_action

def _search_recursive(self, state, depth, valid_actions,
                    transition_func, reward_func, scenario_generator):
    """Recursive tree search."""
    if depth == 0:
        return 0

    best_value = float('-inf')
    for action in valid_actions:
        scenarios = scenario_generator(self.branching)
        expected_value = reward_func(state, action)

        for scenario, prob in scenarios:
            next_state = transition_func(state, action, scenario)
            future_value = self._search_recursive(
                next_state, depth - 1,
                valid_actions, transition_func, reward_func,
                scenario_generator
            )
            expected_value += prob * future_value

        best_value = max(best_value, expected_value)

    return best_value

```

26.7.3. Monte Carlo Tree Search (MCTS)

MCTS is a simulation-based planning algorithm that builds a search tree incrementally. It was central to AlphaGo's success in the game of Go and is widely used for sequential decision problems with large action spaces. Each iteration consists of four phases: selection (traverse the tree using UCB1), expansion (add a new node), simulation (random rollout to estimate value), and backpropagation (update statistics up the tree). MCTS balances exploration and exploitation in large decision trees.

```

import math
from collections import defaultdict

class MCTSNode:

```

```

"""Node in Monte Carlo Tree Search."""

def __init__(self, state, parent=None, action=None):
    self.state = state
    self.parent = parent
    self.action = action # Action that led to this state
    self.children = {}
    self.visits = 0
    self.value = 0.0

def ucb1(self, exploration_constant: float = 1.41) -> float:
    """Upper Confidence Bound for Trees."""
    if self.visits == 0:
        return float('inf')
    exploitation = self.value / self.visits
    exploration = exploration_constant * math.sqrt(
        math.log(self.parent.visits) / self.visits
    )
    return exploitation + exploration

class MCTS:
    """Monte Carlo Tree Search for sequential decisions."""

    def __init__(self, n_simulations: int = 1000,
                 exploration_constant: float = 1.41):
        self.n_simulations = n_simulations
        self.c = exploration_constant

    def search(self, root_state, env):
        """Perform MCTS and return best action."""
        root = MCTSNode(root_state)

        for _ in range(self.n_simulations):
            node = self._select(root, env)
            reward = self._simulate(node.state, env)
            self._backpropagate(node, reward)

        # Return action with most visits
        return max(root.children.keys(),
                   key=lambda a: root.children[a].visits)

    def _select(self, node: MCTSNode, env) -> MCTSNode:
        """Select node to expand using UCB1."""
        while not env.is_terminal(node.state):
            valid_actions = env.valid_actions(node.state)
            unexplored = [a for a in valid_actions
                          if a not in node.children]

            if unexplored:
                # Expand
                action = np.random.choice(unexplored)
                next_state = env.transition(node.state, action)

```

```

        child = MCTSNode(next_state, parent=node, action=action)
        node.children[action] = child
        return child

    # Select best child by UCB1
    node = max(node.children.values(),
               key=lambda n: n.ucb1(self.c))

    return node

def _simulate(self, state, env) -> float:
    """Random rollout from state to estimate value."""
    total_reward = 0
    current_state = state

    while not env.is_terminal(current_state):
        actions = env.valid_actions(current_state)
        action = np.random.choice(actions)
        next_state, reward, _ = env.step_from(current_state, action)
        total_reward += reward
        current_state = next_state

    return total_reward

def _backpropagate(self, node: MCTSNode, reward: float):
    """Propagate reward up the tree."""
    while node is not None:
        node.visits += 1
        node.value += reward
        node = node.parent

```

26.8. Stochastic Dual Dynamic Programming (SDDP)

SDDP is a powerful method for multi-stage stochastic linear programs with continuous state spaces. It is the industry standard for large-scale problems in energy systems, water reservoir management, and financial planning. SDDP avoids the curse of dimensionality by approximating the cost-to-go function at each stage with a set of linear cuts, rather than discretizing the state space.

26.8.1. The SDDP Approach

SDDP approximates the expected cost-to-go function using piecewise linear cuts:

$$\bar{V}_t(x_{t-1}) \approx \max_k \{\alpha_t^k + \beta_t^k \cdot x_{t-1}\}$$

where each cut (α_t^k, β_t^k) is a supporting hyperplane.

26.8.2. Algorithm

```

class SDDPSolver:
    """Stochastic Dual Dynamic Programming solver."""

    def __init__(self, stages: int, scenarios_per_stage: List):
        self.T = stages

```

```

self.scenarios = scenarios_per_stage
self.cuts = {t: [] for t in range(self.T)}

def forward_pass(self, stage_problems, initial_state):
    """Simulate forward to generate trial points."""
    states = [initial_state]
    actions = []

    for t in range(self.T):
        # Sample scenario
        scenario = np.random.choice(self.scenarios[t])

        # Solve stage problem with current cuts
        x, u = self._solve_stage(
            stage_problems[t], states[-1], scenario, t
        )

        states.append(x)
        actions.append(u)

    return states, actions

def backward_pass(self, stage_problems, states):
    """Compute cuts by solving dual problems backward."""
    for t in range(self.T - 1, 0, -1):
        for scenario in self.scenarios[t]:
            # Solve stage problem at trial point
            _, dual_values = self._solve_stage_dual(
                stage_problems[t], states[t], scenario, t
            )

            # Generate cut coefficients
            alpha, beta = self._compute_cut(
                stage_problems[t], states[t], scenario, dual_values
            )

            # Add cut to previous stage
            self.cuts[t - 1].append((alpha, beta))

def _solve_stage(self, problem, state, scenario, t):
    """Solve stage-t problem with cuts approximating future."""
    # Add cuts as constraints
    # min c'x + theta
    # s.t. Ax = b(scenario)
    #      theta >= alpha_k + beta_k' * x for all cuts k
    pass # Implementation depends on problem structure

def _solve_stage_dual(self, problem, state, scenario, t):
    """Solve stage problem and return dual values."""
    pass

def _compute_cut(self, problem, state, scenario, duals):
    """Compute cut coefficients from dual solution."""

```



```

# alpha = optimal_value - beta' * state
# beta = dual values of state-linking constraints
pass

def solve(self, stage_problems, initial_state,
          n_iterations: int = 100, convergence_gap: float = 0.01):
    """Main SDDP algorithm."""
    lower_bounds = []
    upper_bounds = []

    for iteration in range(n_iterations):
        # Forward pass
        states, actions = self.forward_pass(
            stage_problems, initial_state
        )

        # Compute upper bound (simulation cost)
        upper_bound = sum(
            stage_problems[t].cost(states[t], actions[t])
            for t in range(self.T)
        )
        upper_bounds.append(upper_bound)

        # Backward pass (generate cuts)
        self.backward_pass(stage_problems, states)

        # Compute lower bound (first-stage objective with cuts)
        lower_bound, _ = self._solve_stage(
            stage_problems[0], initial_state, None, 0
        )
        lower_bounds.append(lower_bound)

        # Check convergence
        gap = (np.mean(upper_bounds[-10:]) - lower_bound) / lower_bound
        if gap < convergence_gap:
            print(f"Converged in {iteration + 1} iterations")
            break

    return lower_bounds, upper_bounds

```

26.9. Applications

26.9.1. Inventory Management

The following class models multi-period inventory management as a sequential decision problem. The state is the current inventory level, the action is the order quantity, and transitions depend on stochastic demand. The cost function includes fixed ordering costs, holding costs for excess inventory, and shortage penalties for unmet demand.

```

class InventoryMDP:
    """Multi-period inventory management problem."""

```

```

def __init__(self, holding_cost: float, shortage_cost: float,
              order_cost: float, max_inventory: int,
              demand_distribution):
    self.h = holding_cost
    self.p = shortage_cost
    self.K = order_cost
    self.max_inv = max_inventory
    self.demand = demand_distribution

def cost(self, inventory: int, order: int) -> float:
    """Immediate cost."""
    return (self.K * (order > 0) +
            self.h * max(0, inventory) +
            self.p * max(0, -inventory))

def transition(self, inventory: int, order: int,
               demand: int) -> int:
    """State transition."""
    return min(inventory + order - demand, self.max_inv)

def solve_base_stock(self) -> int:
    """Find optimal base-stock level."""
    # Newsvendor-type solution
    critical_ratio = self.p / (self.h + self.p)
    return self.demand.ppf(critical_ratio)

```

26.9.2. Energy Storage Operations

Battery storage systems must decide when to charge and discharge based on electricity prices, demand forecasts, and current charge level. The state is the state of charge (SOC), the action is the charge/discharge power, and transitions incorporate charging efficiency losses.

```

class BatteryStorage:
    """Battery storage optimization problem."""

    def __init__(self, capacity: float, efficiency: float,
                  max_charge_rate: float, max_discharge_rate: float):
        self.capacity = capacity
        self.eta = efficiency
        self.max_charge = max_charge_rate
        self.max_discharge = max_discharge_rate

    def transition(self, soc: float, action: float) -> float:
        """State of charge transition.

        Args:
            soc: Current state of charge
            action: Power (positive = charge, negative = discharge)

        Returns:
            New state of charge
        """
        if action >= 0: # Charging

```

```

        new_soc = soc + self.eta * action
    else: # Discharging
        new_soc = soc + action / self.eta

    return np.clip(new_soc, 0, self.capacity)

def valid_actions(self, soc: float) -> Tuple[float, float]:
    """Return (min_action, max_action) given current SOC."""
    max_charge = min(self.max_charge,
                     (self.capacity - soc) / self.eta)
    max_discharge = min(self.max_discharge,
                       soc * self.eta)
    return (-max_discharge, max_charge)

```

26.9.3. Fleet Management

Fleet management involves dispatching vehicles to serve requests that arrive stochastically across multiple locations. The state includes vehicle positions and pending requests. Actions assign vehicles to requests or reposition idle vehicles. The challenge is balancing immediate service with positioning for future demand.

```

class FleetManagement:
    """Dynamic fleet management problem."""

    def __init__(self, n_locations: int, n_vehicles: int,
                 travel_times: np.ndarray, demand_rates: np.ndarray):
        self.n_loc = n_locations
        self.n_veh = n_vehicles
        self.travel = travel_times
        self.demand = demand_rates

    def state_features(self, vehicle_locations: np.ndarray,
                     pending_requests: List) -> np.ndarray:
        """Extract features from state."""
        features = []

        # Vehicle distribution
        for loc in range(self.n_loc):
            features.append(np.sum(vehicle_locations == loc))

        # Request statistics
        features.append(len(pending_requests))
        features.extend([r.urgency for r in pending_requests[:5]])

        return np.array(features)

    def dispatch_policy(self, vehicles: np.ndarray, requests: List,
                      vfa: LinearVFA) -> Dict:
        """Dispatch vehicles to requests using VFA."""
        assignments = {}

        for request in requests:
            best_vehicle = None

```

```

best_value = float('-inf')

for v in range(self.n_veh):
    if v in assignments.values():
        continue

    # Simulate assignment
    new_locations = vehicles.copy()
    new_locations[v] = request.destination

    features = self.state_features(new_locations, requests[1:])
    value = -self.travel[vehicles[v], request.origin] + \
        vfa.evaluate(features)

    if value > best_value:
        best_value = value
        best_vehicle = v

if best_vehicle is not None:
    assignments[request.id] = best_vehicle

return assignments

```

26.10. Summary

26.10.1. Key Takeaways

- Sequential decision problems involve decisions over time with gradually revealing uncertainty
- The Bellman equation provides the foundation but exact DP faces the curse of dimensionality
- Warren Powell's framework unifies all approaches into four policy classes:
 - **PFAs**: Direct parametric policies (simple, interpretable, domain-driven)
 - **CFAs**: Modified cost functions (bid prices, Lagrangian methods)
 - **VFAs**: Approximate value functions (ADP, reinforcement learning)
 - **DLAs**: Explicit lookahead (MPC, tree search, MCTS)
- SDDP efficiently solves multi-stage stochastic linear programs
- Choosing the right approach depends on problem structure, available computation, and data
- Hybrid methods often work best in practice (e.g., VFA + lookahead)

26.11. Exercises

1. **Policy comparison**: For a simple inventory problem with known demand distribution, implement and compare: (a) a base-stock policy (PFA), (b) a myopic policy with bid prices (CFA), (c) Q-learning (VFA), and (d) a 3-period lookahead (DLA).
2. **Feature engineering**: Design state features for an energy storage problem that enable good linear VFA performance. Test with approximate policy iteration.
3. **MCTS implementation**: Implement MCTS for a multi-period investment problem where at each period you can invest in a risky or safe asset.
4. **Hybrid policy**: Design a policy that uses VFA for routine decisions but switches to lookahead when the value function uncertainty is high.

5. **SDDP application:** Implement a simplified SDDP for a 4-stage hydro-thermal scheduling problem with 3 scenarios per stage.

27. Dynamic Optimization and Model Predictive Control

What You Will Learn

- Understand the foundations of continuous-time dynamic optimization
- Apply the calculus of variations to derive optimal trajectories
- Master Pontryagin's Maximum Principle and the Hamiltonian formulation
- Implement Linear Quadratic Regulators (LQR) for optimal control
- Design Model Predictive Control (MPC) systems with stability guarantees
- Handle uncertainty through robust and stochastic MPC approaches
- Use numerical methods and Python libraries for practical implementation

Dynamic optimization extends static optimization to systems that evolve over time. While dynamic programming (Chapter 18) addresses sequential decisions in discrete time, dynamic optimization handles continuous-time systems governed by differential equations. Model Predictive Control (MPC) bridges theory and practice by solving optimization problems in real-time to control complex systems. This chapter provides a comprehensive treatment from classical theory to modern computational methods.

27.1. Introduction to Dynamic Optimization

27.1.1. The Nature of Dynamic Systems

Dynamic optimization addresses problems where:

1. **System states evolve continuously** according to differential equations
2. **Control inputs influence** the trajectory over time
3. **An objective functional** measures performance over the entire time horizon
4. **Constraints** may apply to states, controls, or their combinations

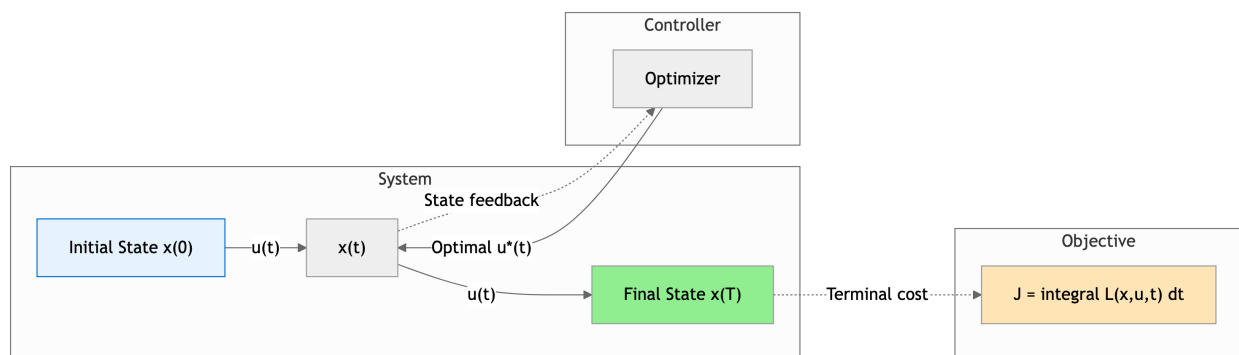


Figure 27.1.: Dynamic Optimization: States Evolve Under Control Influence

27.1.2. Continuous vs Discrete Time

Aspect	Continuous Time	Discrete Time
State Evolution	$\dot{x}(t) = f(x(t), u(t), t)$	$x_{k+1} = f(x_k, u_k)$
Objective	$J = \int_0^T L(x, u, t) dt$	$J = \sum_{k=0}^{N-1} L(x_k, u_k)$
Optimality Conditions	Pontryagin's Maximum Principle	Bellman's Equation
Solution Method	Calculus of Variations	Dynamic Programming
Typical Applications	Aerospace, robotics, chemical processes	Inventory, scheduling, finance

The connection between these formulations is fundamental: as the time step approaches zero, discrete-time dynamic programming converges to continuous-time optimal control.

27.1.3. The General Problem Formulation

The continuous-time optimal control problem is:

$$\begin{aligned}
 &\text{minimize} && J = \phi(x(T), T) + \int_0^T L(x(t), u(t), t) dt \\
 &\text{subject to} && \dot{x}(t) = f(x(t), u(t), t), \quad x(0) = x_0 \\
 &&& g(x(t), u(t), t) \leq 0 \\
 &&& u(t) \in \mathcal{U}, \quad t \in [0, T]
 \end{aligned}$$

where: - $x(t) \in \mathbb{R}^n$ is the state vector - $u(t) \in \mathbb{R}^m$ is the control vector - $L(x, u, t)$ is the running cost (Lagrangian) - $\phi(x(T), T)$ is the terminal cost (Mayer term) - $f(x, u, t)$ defines the system dynamics - $g(x, u, t)$ represents path constraints

i Bolza, Lagrange, and Mayer Problems

- **Bolza Problem:** Both terminal cost ϕ and running cost L present
- **Lagrange Problem:** Only running cost L ($\phi = 0$)
- **Mayer Problem:** Only terminal cost ϕ ($L = 0$)

These are mathematically equivalent: any can be transformed to the others by state augmentation.

27.1.4. When to Use Dynamic Optimization

Dynamic optimization is appropriate when:

1. **Continuous processes:** Chemical reactors, vehicle dynamics, fluid systems
2. **Trajectory optimization:** Spacecraft maneuvers, robot motion planning
3. **Process control:** Temperature regulation, flow control
4. **Economic dynamics:** Growth models, resource extraction
5. **Biological systems:** Population dynamics, epidemiology

The following code demonstrates a basic linear dynamical system simulation. The state vector evolves according to $\dot{x} = Ax + Bu$, where A is the state matrix, B is the control input matrix, and u is a constant control input. This type of system serves as the building block for the optimal control methods developed later in this chapter.


```

import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt

def simple_dynamic_system():
    """Example: Simple dynamic system simulation."""

    # System dynamics: dx/dt = Ax + Bu
    A = np.array([[ -1, 1], [0, -2]])
    B = np.array([[0], [1]])

    def dynamics(x, t, u):
        """State derivative function."""
        return A @ x + B @ np.array([u])

    # Simulate with constant control
    x0 = np.array([1.0, 0.5])
    t = np.linspace(0, 5, 100)
    u_const = 0.5

    x_trajectory = odeint(dynamics, x0, t, args=(u_const,))

    return t, x_trajectory

# Run simulation
t, x = simple_dynamic_system()
print(f"Initial state: {x[0]}")
print(f"Final state: {x[-1]}")

```

27.2. Calculus of Variations

The calculus of variations provides the mathematical foundation for finding functions that minimize (or maximize) functionals. While optimal control extends these ideas, understanding the classical theory illuminates the structure of optimality conditions.

27.2.1. Functionals and Variations

A **functional** maps functions to real numbers:

$$J[y] = \int_a^b L(x, y(x), y'(x)) dx$$

The **variation** of a functional measures how J changes with small perturbations to $y(x)$:

$$\delta J = \lim_{\epsilon \rightarrow 0} \frac{J[y + \epsilon \eta] - J[y]}{\epsilon}$$

where $\eta(x)$ is an admissible variation satisfying boundary conditions.

27.2.2. The Euler-Lagrange Equation

The Euler-Lagrange equation is the central result of the calculus of variations. It provides the necessary condition that any extremizing function must satisfy. The equation transforms the functional optimization problem into an ordinary differential equation.

Theorem (Fundamental Lemma): If $J[y]$ is stationary at $y^*(x)$, then y^* satisfies the Euler-Lagrange equation:

$$\frac{\partial L}{\partial y} - \frac{d}{dx} \left(\frac{\partial L}{\partial y'} \right) = 0$$

Derivation sketch:

1. Consider perturbation $y(x) + \epsilon\eta(x)$ with $\eta(a) = \eta(b) = 0$
2. Compute $\frac{d}{d\epsilon} J[y + \epsilon\eta] \big|_{\epsilon=0} = 0$
3. Apply integration by parts
4. Use the fundamental lemma of calculus of variations

The following code uses SymPy to derive the Euler-Lagrange equation symbolically for a specific Lagrangian $L = y'^2 + y^2$.

```
import sympy as sp
from sympy import symbols, Function, diff, integrate, simplify

def derive_euler_lagrange():
    """Symbolic derivation of Euler-Lagrange equation."""

    # Define symbols
    x = symbols('x')
    y = Function('y')(x)
    y_prime = diff(y, x)

    # Example Lagrangian: L = y'^2 + y^2
    L = y_prime**2 + y**2

    # Euler-Lagrange equation components
    dL_dy = diff(L, y)
    dL_dy_prime = diff(L, y_prime)
    d_dx_dL_dy_prime = diff(dL_dy_prime, x)

    # Euler-Lagrange equation: dL/dy - d/dx(dL/dy') = 0
    euler_lagrange = simplify(dL_dy - d_dx_dL_dy_prime)

    print(f"Lagrangian: L = {L}")
    print(f"dL/dy = {dL_dy}")
    print(f"dL/dy' = {dL_dy_prime}")
    print(f"d/dx(dL/dy') = {d_dx_dL_dy_prime}")
    print(f"Euler-Lagrange: {euler_lagrange} = 0")

    return euler_lagrange

# This gives: 2*y - 2*y'' = 0, or y'' - y = 0
```

27.2.3. Boundary Conditions

The type of boundary conditions affects the solution structure:

Type	Condition	Result
Fixed-Fixed	$y(a) = y_a, y(b) = y_b$	Unique extremal
Fixed-Free	$y(a) = y_a$, free $y(b)$	Natural BC: $\frac{\partial L}{\partial y'} _b = 0$
Free-Free	Both endpoints free	Two natural BCs
Transversality	Endpoint on curve	$\left[L + (Y' - y') \frac{\partial L}{\partial y'} \right]_b = 0$

! Natural Boundary Conditions

When an endpoint is free, the variation η need not vanish there. Setting $\delta J = 0$ then requires:

$$\left. \frac{\partial L}{\partial y'} \right|_{x=b} = 0$$

This “natural boundary condition” emerges automatically from the optimization.

27.2.4. Classical Examples

27.2.4.1. The Brachistochrone Problem

The brachistochrone problem, posed by Johann Bernoulli in 1696, asks for the curve connecting two points along which a frictionless bead sliding under gravity reaches the endpoint in minimum time. It was one of the first variational problems ever solved and motivated the development of the entire field.

Find the curve of fastest descent under gravity from point A to point B .

$$J[y] = \int_0^{x_f} \sqrt{\frac{1 + y'^2}{2gy}} dx$$

The Euler-Lagrange equation yields a cycloid. The following code computes the parametric cycloid curve and finds the correct parameters to connect two given points.

```
import numpy as np
import matplotlib.pyplot as plt

def brachistochrone_curve(x_f, y_f, n_points=100):
    """Compute brachistochrone (cycloid) curve.

    Args:
        x_f: Final x-coordinate
        y_f: Final y-coordinate (positive downward)
        n_points: Number of points

    Returns:
        x, y: Parametric curve coordinates
    """
    from scipy.optimize import brentq

    # Cycloid parametric equations:
    # x = R(theta - sin(theta))
    # y = R(1 - cos(theta))
```

```

# Find R and theta_f such that (x_f, y_f) is reached
def equations(theta_f, R):
    x = R * (theta_f - np.sin(theta_f))
    y = R * (1 - np.cos(theta_f))
    return x - x_f, y - y_f

# Solve for R given theta_f, then find theta_f
def residual(theta_f):
    if theta_f < 1e-6:
        return -x_f
    R = y_f / (1 - np.cos(theta_f))
    x = R * (theta_f - np.sin(theta_f))
    return x - x_f

# Find theta_f
theta_f = brentq(residual, 0.1, 2*np.pi - 0.1)
R = y_f / (1 - np.cos(theta_f))

# Generate curve
theta = np.linspace(0, theta_f, n_points)
x = R * (theta - np.sin(theta))
y = R * (1 - np.cos(theta))

return x, y, R, theta_f

# Example: descent from (0,0) to (1, 0.5)
x, y, R, theta_f = brachistochrone_curve(1.0, 0.5)
print(f"Cycloid parameters: R = {R:.4f}, theta_f = {theta_f:.4f}")

```

27.2.4.2. Geodesics

Geodesics are the generalization of straight lines to curved surfaces. They represent the shortest path between two points on a surface. Finding geodesics is a variational problem: minimize the arc length functional subject to the constraint that the path lies on the surface.

For a surface parameterized as $(x, y, z(x, y))$:

$$J = \int \sqrt{1 + z_x^2 + z_y^2} ds$$

On a sphere, geodesics are great circles. On a flat plane, they are straight lines. The following code computes the great circle path between two points on a unit sphere using spherical linear interpolation (slerp).

```

def geodesic_on_sphere(lat1, lon1, lat2, lon2, n_points=50):
    """Compute great circle (geodesic) on sphere.

    Args:
        lat1, lon1: Start point (radians)
        lat2, lon2: End point (radians)
        n_points: Number of points

    Returns:
        lats, lons: Geodesic coordinates
    """

```

```

"""
# Convert to Cartesian
def latlon_to_xyz(lat, lon):
    return np.array([
        np.cos(lat) * np.cos(lon),
        np.cos(lat) * np.sin(lon),
        np.sin(lat)
    ])

p1 = latlon_to_xyz(lat1, lon1)
p2 = latlon_to_xyz(lat2, lon2)

# Angle between points
dot = np.dot(p1, p2)
angle = np.arccos(np.clip(dot, -1, 1))

# Interpolate along great circle
t = np.linspace(0, 1, n_points)
points = []

for ti in t:
    if abs(angle) < 1e-10:
        p = p1
    else:
        p = (np.sin((1-ti)*angle)*p1 + np.sin(ti*angle)*p2) / np.sin(angle)

    # Back to lat/lon
    lat = np.arcsin(p[2])
    lon = np.arctan2(p[1], p[0])
    points.append((lat, lon))

return np.array(points)

```

27.2.5. Higher-Order and Multi-Variable Problems

For functionals depending on higher derivatives:

$$J[y] = \int_a^b L(x, y, y', y'', \dots) dx$$

The Euler-Lagrange equation becomes:

$$\frac{\partial L}{\partial y} - \frac{d}{dx} \frac{\partial L}{\partial y'} + \frac{d^2}{dx^2} \frac{\partial L}{\partial y''} - \dots = 0$$

For multiple functions $y_1(x), \dots, y_n(x)$:

$$\frac{\partial L}{\partial y_i} - \frac{d}{dx} \frac{\partial L}{\partial y'_i} = 0, \quad i = 1, \dots, n$$

A phase portrait visualizes the dynamics of a two-dimensional system. The vector field shows how the state evolves at every point in the state space, and the optimal trajectory traces the controlled path from an initial state to the target (origin). The control input modifies the natural dynamics to steer the system along this trajectory.

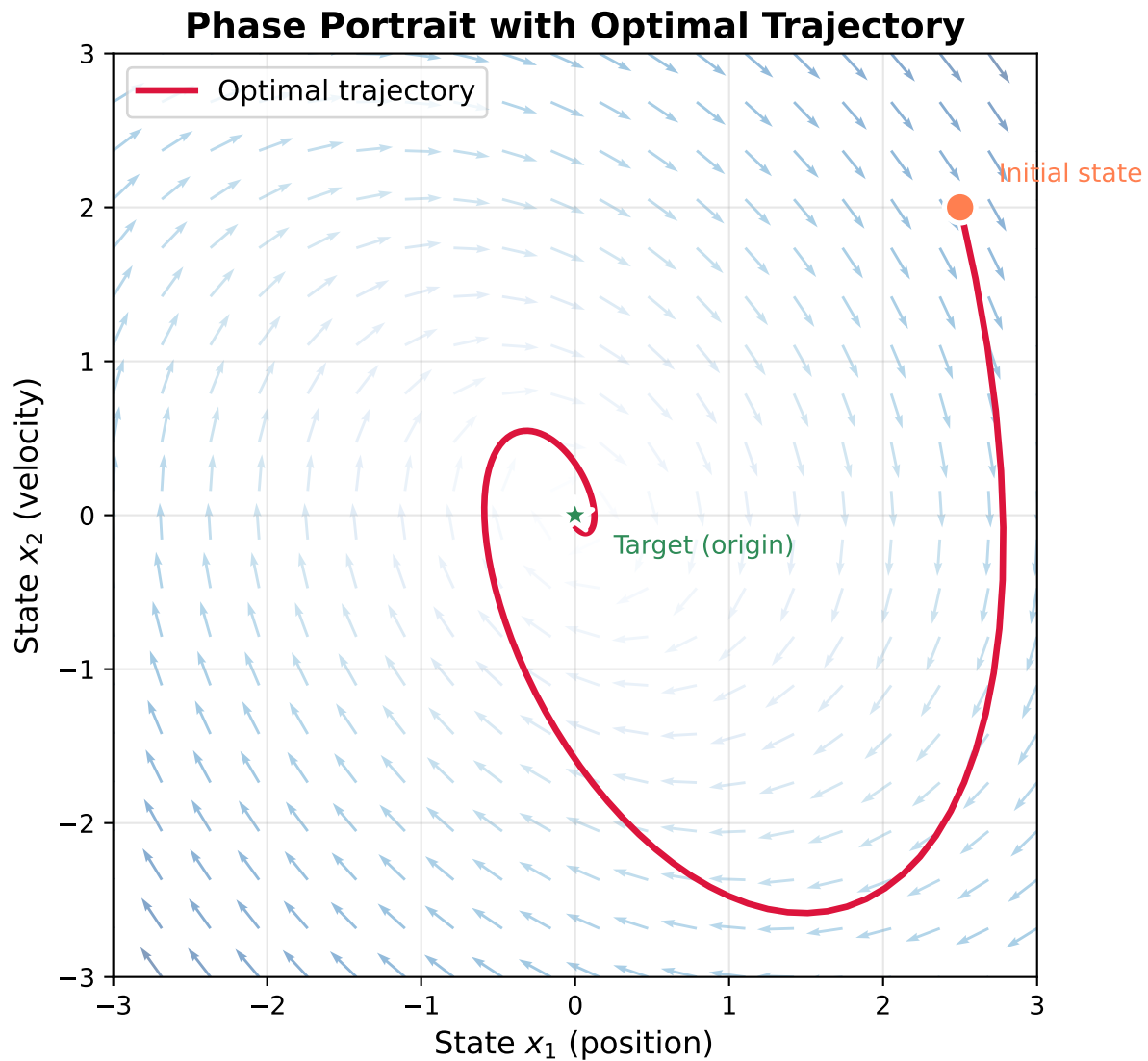


Figure 27.2.: Optimal trajectory on a phase portrait — the vector field shows system dynamics, the red trajectory represents the optimal path from initial to target state

27.3. Optimal Control Theory

Optimal control theory extends calculus of variations to handle explicit control variables, inequality constraints, and more general problem structures.

27.3.1. Problem Structure

The optimal control problem:

$$\begin{aligned}
&\text{minimize} && J = \phi(x(T)) + \int_0^T L(x(t), u(t), t) dt \\
&\text{subject to} && \dot{x}(t) = f(x(t), u(t), t) \\
&&& x(0) = x_0 \\
&&& u(t) \in \mathcal{U}
\end{aligned}$$

Key differences from calculus of variations: - Control $u(t)$ is a separate variable from state $x(t)$ - Control constraints $u(t) \in \mathcal{U}$ may be active - State evolves according to given dynamics f

27.3.2. Pontryagin's Maximum Principle

The Maximum Principle provides necessary conditions for optimality, even when controls are constrained.

Theorem (Pontryagin's Maximum Principle): Let $(x^*(t), u^*(t))$ be an optimal state-control pair. Then there exist costate variables $\lambda(t) \in \mathbb{R}^n$ such that:

1. **State equation:** $\dot{x}^* = \frac{\partial H}{\partial \lambda}$
2. **Costate equation:** $\dot{\lambda} = -\frac{\partial H}{\partial x}$
3. **Optimality condition:** $u^*(t) = \arg \max_{u \in \mathcal{U}} H(x^*(t), u, \lambda(t), t)$
4. **Transversality:** $\lambda(T) = \frac{\partial \phi}{\partial x} \Big|_{x=x^*(T)}$

where the Hamiltonian is:

$$H(x, u, \lambda, t) = L(x, u, t) + \lambda^T f(x, u, t)$$

! The Hamiltonian and Its Role

The Hamiltonian H is the central object in optimal control:

- It combines the cost rate L with the dynamics f
- Costates λ are “shadow prices” of state constraints
- The maximum principle says: choose u to maximize H
- For autonomous problems: H is constant along optimal trajectories

Sign convention: Some texts minimize H or define $H = -L + \lambda^T f$. Be careful with signs!

27.3.3. Costate Variables and Adjoint Equations

The costates $\lambda(t)$ have important interpretations:

$$\lambda_i(t) = \frac{\partial J^*}{\partial x_i(t)}$$

They represent the sensitivity of the optimal cost to state perturbations. In economic terms, the costate variable is the shadow price of the corresponding state variable—it measures the marginal value of having one more unit of that state. The following framework class implements the Maximum Principle computations, including numerical gradient-based costate dynamics and Hamiltonian maximization.

```

import numpy as np
from scipy.integrate import solve_bvp
from scipy.optimize import minimize

class OptimalControlProblem:
    """Framework for solving optimal control via Maximum Principle."""

    def __init__(self, n_states, n_controls):
        self.n_x = n_states
        self.n_u = n_controls

    def dynamics(self, x, u, t):
        """State dynamics: dx/dt = f(x, u, t). Override in subclass."""
        raise NotImplementedError

    def running_cost(self, x, u, t):
        """Running cost L(x, u, t). Override in subclass."""
        raise NotImplementedError

    def terminal_cost(self, x_T):
        """Terminal cost phi(x(T)). Override in subclass."""
        return 0.0

    def terminal_cost_gradient(self, x_T):
        """Gradient of terminal cost. Override in subclass."""
        return np.zeros(self.n_x)

    def hamiltonian(self, x, u, lam, t):
        """Compute Hamiltonian H = L + lambda' * f."""
        L = self.running_cost(x, u, t)
        f = self.dynamics(x, u, t)
        return L + np.dot(lam, f)

    def costate_dynamics(self, x, u, lam, t, eps=1e-6):
        """Costate equation: dlam/dt = -dH/dx (numerical gradient)."""
        dH_dx = np.zeros(self.n_x)

        for i in range(self.n_x):
            x_plus = x.copy()
            x_plus[i] += eps
            x_minus = x.copy()
            x_minus[i] -= eps

            H_plus = self.hamiltonian(x_plus, u, lam, t)
            H_minus = self.hamiltonian(x_minus, u, lam, t)
            dH_dx[i] = (H_plus - H_minus) / (2 * eps)

        return -dH_dx

    def optimal_control(self, x, lam, t, u_bounds=None):
        """Find control maximizing Hamiltonian."""
        def neg_H(u):
            return -self.hamiltonian(x, u, lam, t)

```



```

        if u_bounds is None:
            u_bounds = [(None, None)] * self.n_u

        # Start from zero
        u0 = np.zeros(self.n_u)
        result = minimize(neg_H, u0, bounds=u_bounds)

        return result.x

class LinearQuadraticProblem(OptimalControlProblem):
    """Linear dynamics, quadratic cost: LQR problem."""

    def __init__(self, A, B, Q, R, Q_f=None):
        """
        Args:
            A: State matrix (n x n)
            B: Control matrix (n x m)
            Q: State cost matrix (n x n)
            R: Control cost matrix (m x m)
            Q_f: Terminal state cost (n x n)
        """
        n, m = B.shape
        super().__init__(n, m)

        self.A = A
        self.B = B
        self.Q = Q
        self.R = R
        self.Q_f = Q_f if Q_f is not None else Q

    def dynamics(self, x, u, t):
        return self.A @ x + self.B @ u

    def running_cost(self, x, u, t):
        return 0.5 * (x @ self.Q @ x + u @ self.R @ u)

    def terminal_cost(self, x_T):
        return 0.5 * x_T @ self.Q_f @ x_T

    def terminal_cost_gradient(self, x_T):
        return self.Q_f @ x_T

```

27.3.4. Transversality Conditions

Transversality conditions specify boundary conditions for costates at the terminal time:

Terminal Condition	Transversality Condition
Fixed endpoint $x(T) = x_T$	$\lambda(T)$ free
Free endpoint	$\lambda(T) = \frac{\partial \phi}{\partial x}$
Endpoint on surface $\psi(x(T)) = 0$	$\lambda(T) = \frac{\partial \phi}{\partial x} + \nu^T \frac{\partial \psi}{\partial x}$
Free final time T	$H(T) = -\frac{\partial \phi}{\partial T}$

i Hamiltonian Value Condition

For problems with free final time, an additional condition applies:

$$H(x^*(T), u^*(T), \lambda(T), T) + \frac{\partial \phi}{\partial T} = 0$$

For autonomous problems with no explicit time dependence in ϕ , this simplifies to $H(T) = 0$.

27.3.5. The Hamilton-Jacobi-Bellman Equation

While the Maximum Principle gives necessary conditions for optimality along a specific trajectory, the Hamilton-Jacobi-Bellman (HJB) equation characterizes the optimal value function over the entire state space. The HJB equation is a partial differential equation whose solution gives the optimal cost-to-go from any state at any time. This is the continuous-time analog of Bellman's equation in dynamic programming.

The HJB equation provides an alternative characterization of the optimal value function:

$$-\frac{\partial V}{\partial t} = \min_u \left[L(x, u, t) + \frac{\partial V}{\partial x} f(x, u, t) \right]$$

with boundary condition $V(x, T) = \phi(x)$.

The connection to the Maximum Principle:

$$\lambda(t) = \frac{\partial V}{\partial x}(x^*(t), t)$$

The following code solves the HJB equation numerically using finite differences on a grid. At each grid point, the algorithm minimizes over controls and propagates the value function backward in time. This approach is practical for low-dimensional problems (1-3 state dimensions) but suffers from the curse of dimensionality in higher dimensions.

```
def hjb_finite_differences(problem, x_grid, t_grid, u_grid):
    """Solve HJB equation via finite differences.

    Args:
        problem: Optimal control problem
        x_grid: Discretization of state space
        t_grid: Time discretization (backward from T)
        u_grid: Control grid for minimization

    Returns:
        V: Value function on grid
        u_opt: Optimal control on grid
    """
    nx = len(x_grid)
    nt = len(t_grid)
    dx = x_grid[1] - x_grid[0]
    dt = t_grid[1] - t_grid[0] # Negative for backward time

    V = np.zeros((nx, nt))
    u_opt = np.zeros((nx, nt))
```

```

# Terminal condition
for i, x in enumerate(x_grid):
    V[i, -1] = problem.terminal_cost(np.array([x]))

# Backward iteration
for k in range(nt - 2, -1, -1):
    t = t_grid[k]

    for i, x in enumerate(x_grid):
        x_vec = np.array([x])

        # Approximate dV/dx
        if i == 0:
            dVdx = (V[i+1, k+1] - V[i, k+1]) / dx
        elif i == nx - 1:
            dVdx = (V[i, k+1] - V[i-1, k+1]) / dx
        else:
            dVdx = (V[i+1, k+1] - V[i-1, k+1]) / (2 * dx)

        # Minimize over controls
        best_val = np.inf
        best_u = 0

        for u in u_grid:
            u_vec = np.array([u])
            L = problem.running_cost(x_vec, u_vec, t)
            f = problem.dynamics(x_vec, u_vec, t)

            # HJB right-hand side
            rhs = L + dVdx * f[0]

            if rhs < best_val:
                best_val = rhs
                best_u = u

        # Update value function
        V[i, k] = V[i, k+1] - dt * best_val
        u_opt[i, k] = best_u

return V, u_opt

```

27.3.6. Linear Quadratic Regulator (LQR)

The LQR problem is the most important special case in optimal control because it admits a closed-form solution. The system dynamics are linear, the cost is quadratic in both states and controls, and the optimal control law is a linear function of the state. The solution involves the matrix Riccati differential equation, which can be solved either backward in time (finite horizon) or algebraically (infinite horizon). LQR serves as the foundation for many advanced control methods, including MPC.

Problem:

$$\begin{aligned} \text{minimize} \quad & J = \frac{1}{2}x(T)^T Q_f x(T) + \frac{1}{2} \int_0^T (x^T Q x + u^T R u) dt \\ \text{subject to} \quad & \dot{x} = Ax + Bu \end{aligned}$$

Solution: The optimal control is linear state feedback:

$$u^*(t) = -K(t)x(t)$$

where $K(t) = R^{-1}B^TP(t)$ and $P(t)$ solves the matrix Riccati equation:

$$-\dot{P} = A^TP + PA - PBR^{-1}B^TP + Q, \quad P(T) = Q_f$$

The following class implements both finite-horizon and infinite-horizon LQR. The finite-horizon version integrates the Riccati ODE backward from the terminal condition. The infinite-horizon version solves the algebraic Riccati equation directly, yielding a constant feedback gain matrix.

```
import numpy as np
from scipy.integrate import solve_ivp
from scipy.linalg import solve_continuous_are

class LQRController:
    """Linear Quadratic Regulator for continuous-time systems."""

    def __init__(self, A, B, Q, R, Q_f=None):
        """
        Args:
            A: State matrix (n x n)
            B: Control matrix (n x m)
            Q: State cost matrix (n x n), symmetric positive semi-definite
            R: Control cost matrix (m x m), symmetric positive definite
            Q_f: Terminal cost matrix (n x n)
        """
        self.A = np.array(A)
        self.B = np.array(B)
        self.Q = np.array(Q)
        self.R = np.array(R)
        self.Q_f = np.array(Q_f) if Q_f is not None else self.Q

        self.n = self.A.shape[0]
        self.m = self.B.shape[1]

    def solve_finite_horizon(self, T, dt=0.01):
        """Solve finite-horizon LQR via Riccati ODE.

        Args:
            T: Time horizon
            dt: Time step for integration

        Returns:
            t: Time points
            P: Riccati solution P(t)
            K: Feedback gains K(t)
        """
```

```

"""
def riccati_ode(t, P_flat):
    """Riccati differential equation (backward in time)."""
    P = P_flat.reshape(self.n, self.n)

    dP = -(self.A.T @ P + P @ self.A
           - P @ self.B @ np.linalg.inv(self.R) @ self.B.T @ P
           + self.Q)

    return dP.flatten()

# Solve backward from T to 0
t_span = (T, 0)
t_eval = np.linspace(T, 0, int(T/dt) + 1)
P0 = self.Q_f.flatten()

sol = solve_ivp(riccati_ode, t_span, P0, t_eval=t_eval, method='RK45')

# Reverse time direction
t = sol.t[::-1]
P = sol.y[:, ::-1].T.reshape(-1, self.n, self.n)

# Compute gains
K = np.array([np.linalg.inv(self.R) @ self.B.T @ P[i]
              for i in range(len(t))])

return t, P, K

def solve_infinite_horizon(self):
    """Solve infinite-horizon LQR via algebraic Riccati equation.

    Returns:
        P: Steady-state Riccati solution
        K: Steady-state feedback gain
    """
    # Solve algebraic Riccati equation
    P = solve_continuous_are(self.A, self.B, self.Q, self.R)
    K = np.linalg.inv(self.R) @ self.B.T @ P

    return P, K

def simulate(self, x0, T, K=None, dt=0.01):
    """Simulate closed-loop system.

    Args:
        x0: Initial state
        T: Simulation time
        K: Feedback gain (if None, uses infinite-horizon)
        dt: Time step

    Returns:
        t: Time points
        x: State trajectory
    """

```

```

        u: Control trajectory
    """
    if K is None:
        _, K = self.solve_infinite_horizon()
        K_func = lambda t: K
    elif callable(K):
        K_func = K
    else:
        K_func = lambda t: K

    def closed_loop(t, x):
        u = -K_func(t) @ x
        return self.A @ x + self.B @ u

    t_span = (0, T)
    t_eval = np.linspace(0, T, int(T/dt) + 1)

    sol = solve_ivp(closed_loop, t_span, x0, t_eval=t_eval)

    x = sol.y.T
    u = np.array([-K_func(t) @ x[i] for i, t in enumerate(sol.t)])

    return sol.t, x, u

def lqr_example():
    """Example: Double integrator with LQR."""

    # Double integrator: position and velocity
    # dx1/dt = x2, dx2/dt = u
    A = np.array([[0, 1],
                  [0, 0]])
    B = np.array([[0],
                  [1]])

    # Penalize position and velocity equally, small control cost
    Q = np.array([[1, 0],
                  [0, 1]])
    R = np.array([[0.1]])

    lqr = LQRController(A, B, Q, R)

    # Solve infinite-horizon
    P, K = lqr.solve_infinite_horizon()
    print(f"Riccati solution P:\n{P}")
    print(f"Feedback gain K: {K}")

    # Simulate from initial condition
    x0 = np.array([5.0, 0.0]) # Start at position 5, zero velocity
    t, x, u = lqr.simulate(x0, T=10)

    print(f"Final state: {x[-1]}")
    print(f"Total control effort: {np.sum(u**2) * (t[1]-t[0]):.4f}")

```

```

    return t, x, u

t, x, u = lqr_example()

```

💡 LQR Design Guidelines

1. **State weighting Q :** Increase diagonal elements for states you want to track tightly
2. **Control weighting R :** Increase for actuator-limited systems or to reduce control effort
3. **Relative scaling:** The ratio Q_{ii}/R_{jj} determines the trade-off between state regulation and control cost
4. **Bryson's rule:** Start with $Q_{ii} = 1/(\text{max acceptable } x_i^2)$ and $R_{jj} = 1/(\text{max acceptable } u_j^2)$

The following visualization illustrates MPC's receding horizon strategy across three time steps. At each step, the controller solves an optimization problem over the full prediction horizon (shown in orange) but applies only the first control action (green region). The system then moves to the next state, and the optimization is repeated from the new measurement.

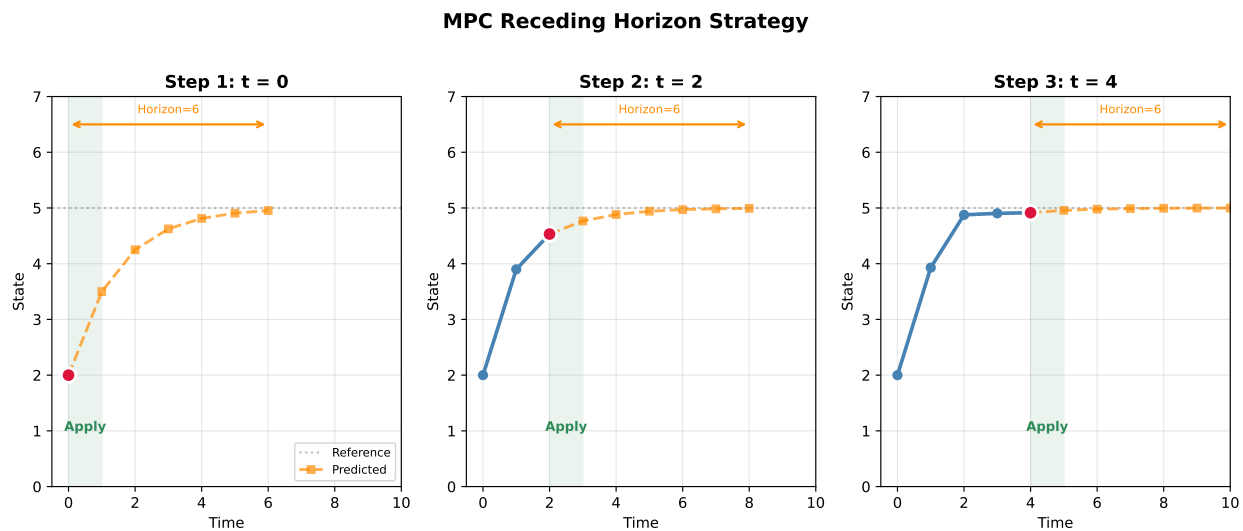


Figure 27.3.: MPC receding horizon — at each time step, the controller optimizes over a prediction horizon but only applies the first control action, then re-optimizes from the new state

27.4. Model Predictive Control (MPC)

Model Predictive Control is the dominant advanced control strategy in industry. It combines optimal control theory with receding horizon implementation, making it practical for real-time control of complex systems.

27.4.1. The Receding Horizon Principle

MPC operates by:

1. **Measure** the current state $x(t)$

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2. **Solve** an optimal control problem over horizon $[t, t + T]$
3. **Apply** only the first control action
4. **Repeat** at the next time step

27.4.2. MPC Formulation

At each time step, solve:

$$\begin{aligned} \text{minimize} \quad & V_N(x, \mathbf{u}) = \sum_{k=0}^{N-1} \ell(x_k, u_k) + V_f(x_N) \\ \text{subject to} \quad & x_{k+1} = f(x_k, u_k), \quad k = 0, \dots, N-1 \\ & x_0 = x(t) \\ & x_k \in \mathcal{X}, \quad k = 1, \dots, N \\ & u_k \in \mathcal{U}, \quad k = 0, \dots, N-1 \\ & x_N \in \mathcal{X}_f \end{aligned}$$

where: - N is the prediction horizon - $\ell(x, u)$ is the stage cost - $V_f(x_N)$ is the terminal cost - $\mathcal{X}, \mathcal{U}$ are state and control constraint sets - $\mathcal{X}_f$ is the terminal constraint set

27.4.3. Linear MPC

For linear systems with quadratic costs:

$$\begin{aligned} \text{minimize} \quad & \sum_{k=0}^{N-1} (x_k^T Q x_k + u_k^T R u_k) + x_N^T P x_N \\ \text{subject to} \quad & x_{k+1} = A x_k + B u_k \\ & x_0 = x_{\text{current}} \\ & x_{\min} \leq x_k \leq x_{\max} \\ & u_{\min} \leq u_k \leq u_{\max} \end{aligned}$$

This is a Quadratic Program (QP) and can be solved efficiently. The terminal cost matrix P is typically set to the solution of the discrete algebraic Riccati equation, which provides a stability guarantee for the closed-loop system. The following implementation solves the QP at each time step using either `cvxpy` or `scipy.optimize`.

```
import numpy as np
from scipy.optimize import minimize
import cvxpy as cp

class LinearMPC:
    """Linear Model Predictive Controller."""

    def __init__(self, A, B, Q, R, P=None, N=10):
        """
        Args:
            A: State transition matrix (n x n)
            B: Control matrix (n x m)
            Q: State cost matrix (n x n)
            R: Control cost matrix (m x m)
            P: Terminal cost matrix (n x n), default solves DARE
```

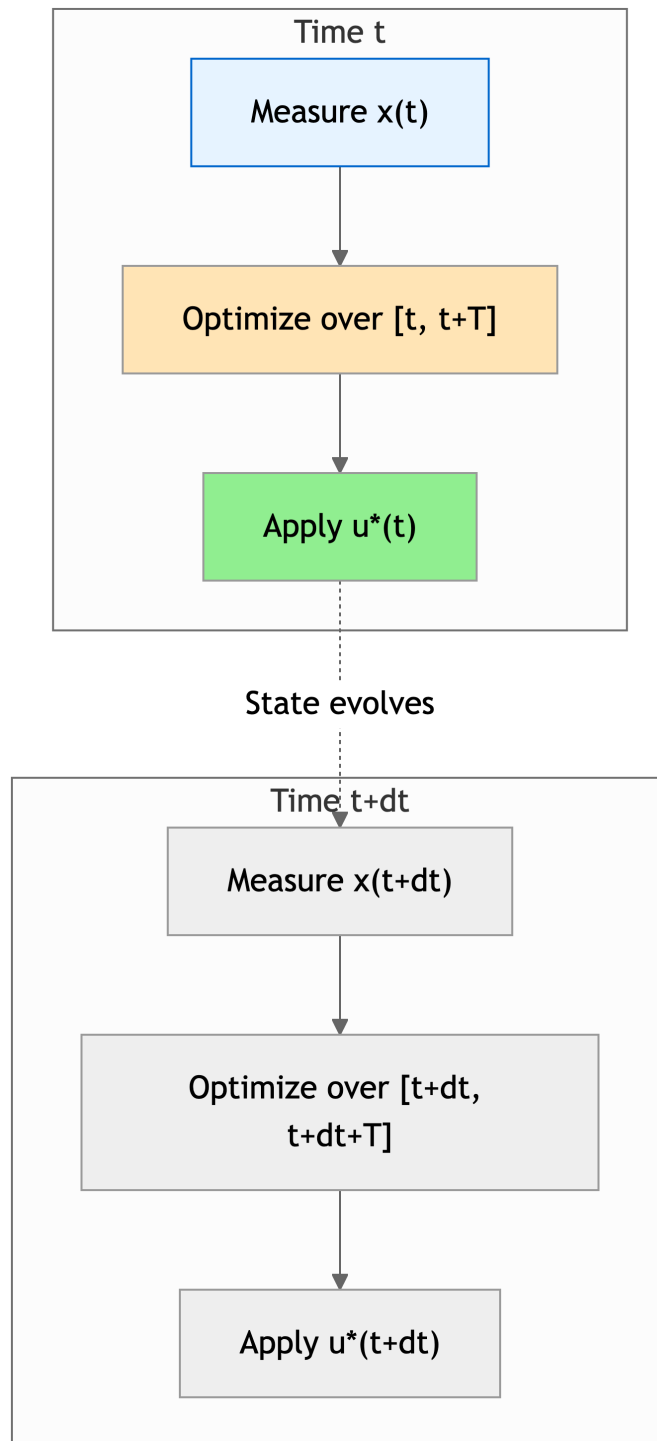



Figure 27.4.: Model Predictive Control: Receding Horizon Principle

```

        N: Prediction horizon
    """
    self.A = np.array(A)
    self.B = np.array(B)
    self.Q = np.array(Q)
    self.R = np.array(R)
    self.N = N

    self.n = self.A.shape[0]
    self.m = self.B.shape[1]

    # Compute terminal cost from discrete Riccati equation
    if P is None:
        from scipy.linalg import solve_discrete_are
        self.P = solve_discrete_are(self.A, self.B, self.Q, self.R)
    else:
        self.P = np.array(P)

    # Constraint bounds (default: unconstrained)
    self.x_min = -np.inf * np.ones(self.n)
    self.x_max = np.inf * np.ones(self.n)
    self.u_min = -np.inf * np.ones(self.m)
    self.u_max = np.inf * np.ones(self.m)

def set_constraints(self, x_min=None, x_max=None, u_min=None, u_max=None):
    """Set state and control constraints."""
    if x_min is not None:
        self.x_min = np.array(x_min)
    if x_max is not None:
        self.x_max = np.array(x_max)
    if u_min is not None:
        self.u_min = np.array(u_min)
    if u_max is not None:
        self.u_max = np.array(u_max)

def solve_qp(self, x0):
    """Solve MPC optimization problem using CVXPY.

    Args:
        x0: Current state

    Returns:
        u_opt: Optimal control sequence
        x_pred: Predicted state trajectory
        cost: Optimal cost
    """
    x0 = np.array(x0)

    # Decision variables
    x = cp.Variable((self.N + 1, self.n))
    u = cp.Variable((self.N, self.m))

    # Cost function

```

```

cost = 0
for k in range(self.N):
    cost += cp.quad_form(x[k], self.Q) + cp.quad_form(u[k], self.R)
cost += cp.quad_form(x[self.N], self.P)

# Constraints
constraints = [x[0] == x0]

for k in range(self.N):
    constraints.append(x[k+1] == self.A @ x[k] + self.B @ u[k])
    constraints.append(x[k] >= self.x_min)
    constraints.append(x[k] <= self.x_max)
    constraints.append(u[k] >= self.u_min)
    constraints.append(u[k] <= self.u_max)

constraints.append(x[self.N] >= self.x_min)
constraints.append(x[self.N] <= self.x_max)

# Solve
problem = cp.Problem(cp.Minimize(cost), constraints)
problem.solve(solver=cp.OSQP)

if problem.status != 'optimal':
    raise ValueError(f"MPC problem not solved: {problem.status}")

return u.value, x.value, problem.value

def control(self, x0):
    """Compute MPC control action.

    Args:
        x0: Current state

    Returns:
        u: Control action to apply
    """
    u_seq, _, _ = self.solve_qp(x0)
    return u_seq[0]

def simulate(self, x0, n_steps, disturbance=None):
    """Simulate closed-loop MPC.

    Args:
        x0: Initial state
        n_steps: Number of simulation steps
        disturbance: Optional disturbance sequence

    Returns:
        x_hist: State trajectory
        u_hist: Control trajectory
    """
    x = np.array(x0)
    x_hist = [x.copy()]

```

```

        u_hist = []

        for k in range(n_steps):
            # Compute MPC control
            u = self.control(x)
            u_hist.append(u.copy())

            # Apply control and evolve state
            x = self.A @ x + self.B @ u

            # Add disturbance if provided
            if disturbance is not None and k < len(disturbance):
                x += disturbance[k]

            x_hist.append(x.copy())

        return np.array(x_hist), np.array(u_hist)

def linear_mpc_example():
    """Example: Double integrator with constraints."""

    # Discrete-time double integrator
    dt = 0.1
    A = np.array([[1, dt],
                  [0, 1]])
    B = np.array([[0.5*dt**2],
                  [dt]])

    # Cost matrices
    Q = np.diag([10, 1]) # Position more important than velocity
    R = np.array([[1]])

    # Create MPC controller
    mpc = LinearMPC(A, B, Q, R, N=20)

    # Set constraints
    mpc.set_constraints(
        x_min=[-10, -2], # Position and velocity lower bounds
        x_max=[10, 2],   # Position and velocity upper bounds
        u_min=[-1],      # Control lower bound
        u_max=[1]        # Control upper bound
    )

    # Simulate from initial condition
    x0 = [5, 0] # Start at position 5, zero velocity
    x_hist, u_hist = mpc.simulate(x0, n_steps=100)

    print(f"Initial state: {x0}")
    print(f"Final state: {x_hist[-1]}")
    print(f"Max control: {np.max(np.abs(u_hist)):.4f}")

    return x_hist, u_hist

```

```
x_hist, u_hist = linear_mpc_example()
```

27.4.4. Nonlinear MPC (NMPC)

When the system dynamics or constraints are nonlinear, the MPC optimization problem becomes a nonlinear program (NLP) rather than a QP. This is significantly harder to solve but necessary for many practical systems. NMPC relies on sequential quadratic programming (SQP) or interior point methods to find local optima at each time step. For nonlinear systems:

$$\begin{aligned} & \text{minimize} && \sum_{k=0}^{N-1} \ell(x_k, u_k) + V_f(x_N) \\ & \text{subject to} && x_{k+1} = f(x_k, u_k) \\ & && h(x_k, u_k) \leq 0 \\ & && x_N \in \mathcal{X}_f \end{aligned}$$

The optimization becomes a Nonlinear Program (NLP), requiring iterative solvers.

```
import numpy as np
from scipy.optimize import minimize
from scipy.integrate import solve_ivp

class NonlinearMPC:
    """Nonlinear Model Predictive Controller."""

    def __init__(self, dynamics, running_cost, terminal_cost,
                 n_states, n_controls, N=10, dt=0.1):
        """
        Args:
            dynamics: Function f(x, u) returning dx/dt
            running_cost: Function l(x, u) returning stage cost
            terminal_cost: Function V_f(x) returning terminal cost
            n_states: Number of states
            n_controls: Number of controls
            N: Prediction horizon
            dt: Sampling time

        self.dynamics = dynamics
        self.running_cost = running_cost
        self.terminal_cost = terminal_cost
        self.n = n_states
        self.m = n_controls
        self.N = N
        self.dt = dt

        # Constraint bounds
        self.u_min = -np.inf * np.ones(self.m)
        self.u_max = np.inf * np.ones(self.m)
        self.x_min = -np.inf * np.ones(self.n)
        self.x_max = np.inf * np.ones(self.n)
```

```

def set_control_bounds(self, u_min, u_max):
    """Set control constraints."""
    self.u_min = np.array(u_min)
    self.u_max = np.array(u_max)

def set_state_bounds(self, x_min, x_max):
    """Set state constraints."""
    self.x_min = np.array(x_min)
    self.x_max = np.array(x_max)

def integrate_dynamics(self, x0, u):
    """Integrate dynamics over one time step."""
    def ode(t, x):
        return self.dynamics(x, u)

    sol = solve_ivp(ode, [0, self.dt], x0, method='RK45')
    return sol.y[:, -1]

def predict_trajectory(self, x0, u_sequence):
    """Predict state trajectory given control sequence.

    Args:
        x0: Initial state
        u_sequence: Control sequence (N x m)

    Returns:
        x_trajectory: State trajectory (N+1 x n)
    """
    x = [np.array(x0)]

    for k in range(self.N):
        x_next = self.integrate_dynamics(x[-1], u_sequence[k])
        x.append(x_next)

    return np.array(x)

def objective(self, u_flat, x0):
    """Compute total cost."""
    u_seq = u_flat.reshape(self.N, self.m)
    x_traj = self.predict_trajectory(x0, u_seq)

    cost = 0
    for k in range(self.N):
        cost += self.running_cost(x_traj[k], u_seq[k])
    cost += self.terminal_cost(x_traj[self.N])

    return cost

def state_constraints(self, u_flat, x0):
    """State constraint function (for scipy.optimize)."""
    u_seq = u_flat.reshape(self.N, self.m)
    x_traj = self.predict_trajectory(x0, u_seq)

```

```

# g(x) <= 0 form: x_min - x <= 0 and x - x_max <= 0
constraints = []
for k in range(1, self.N + 1):
    constraints.extend(self.x_min - x_traj[k])
    constraints.extend(x_traj[k] - self.x_max)

return np.array(constraints)

def solve(self, x0, u_init=None):
    """Solve NMPC optimization problem.

    Args:
        x0: Current state
        u_init: Initial guess for control sequence

    Returns:
        u_opt: Optimal control sequence
        x_pred: Predicted trajectory
        cost: Optimal cost
    """
    x0 = np.array(x0)

    # Initial guess
    if u_init is None:
        u_init = np.zeros((self.N, self.m))
    u_flat = u_init.flatten()

    # Control bounds
    bounds = []
    for k in range(self.N):
        for j in range(self.m):
            bounds.append((self.u_min[j], self.u_max[j]))

    # State constraints
    constraints = []
    if not np.all(np.isinf(self.x_min)) or not np.all(np.isinf(self.x_max)):
        constraints.append({
            'type': 'ineq',
            'fun': lambda u, x0=x0: -self.state_constraints(u, x0)
        })

    # Solve NLP
    result = minimize(
        self.objective,
        u_flat,
        args=(x0,),
        method='SLSQP',
        bounds=bounds,
        constraints=constraints,
        options={'maxiter': 100}
    )

    u_opt = result.x.reshape(self.N, self.m)

```

```

        x_pred = self.predict_trajectory(x0, u_opt)

        return u_opt, x_pred, result.fun

def control(self, x0, u_warmstart=None):
    """Compute NMPC control action.

    Args:
        x0: Current state
        u_warmstart: Warm start from previous solution

    Returns:
        u: Control to apply
        u_shifted: Shifted control sequence for warm start
    """
    u_opt, _, _ = self.solve(x0, u_warmstart)

    # Shift for warm start
    u_shifted = np.vstack([u_opt[1:], u_opt[-1:]])

    return u_opt[0], u_shifted

def nmpc_pendulum_example():
    """Example: Swing-up control of inverted pendulum."""

    # Pendulum parameters
    g = 9.81
    L = 1.0
    m = 1.0
    b = 0.1 # Damping

    def dynamics(x, u):
        """Pendulum dynamics: x = [theta, theta_dot]."""
        theta, theta_dot = x
        torque = u[0]

        theta_ddot = (torque - b*theta_dot - m*g*L*np.sin(theta)) / (m*L**2)

        return np.array([theta_dot, theta_ddot])

    def running_cost(x, u):
        """Stage cost: regulate to upright position."""
        theta, theta_dot = x
        # Cost for angle from upright (pi) and velocity
        angle_error = theta - np.pi
        return 10*angle_error**2 + 0.1*theta_dot**2 + 0.01*u[0]**2

    def terminal_cost(x):
        """Terminal cost."""
        theta, theta_dot = x
        angle_error = theta - np.pi
        return 50*angle_error**2 + 5*theta_dot**2

```



```

# Create NMPC controller
nmpc = NonlinearMPC(
    dynamics=dynamics,
    running_cost=running_cost,
    terminal_cost=terminal_cost,
    n_states=2,
    n_controls=1,
    N=20,
    dt=0.05
)

# Set control bounds
nmpc.set_control_bounds(u_min=[-5], u_max=[5])

# Simulate swing-up
x = np.array([0.0, 0.0]) # Start hanging down
x_hist = [x.copy()]
u_hist = []
u_warmstart = None

for k in range(200):
    u, u_warmstart = nmpc.control(x, u_warmstart)
    u_hist.append(u.copy())

    # Simulate one step
    x = nmpc.integrate_dynamics(x, u)
    x_hist.append(x.copy())

    if k % 50 == 0:
        print(f"Step {k}: theta = {x[0]:.3f}, target = {np.pi:.3f}")

return np.array(x_hist), np.array(u_hist)

# Note: This example requires scipy and may take time to run
print("NMPC Pendulum example defined. Run nmpc_pendulum_example() to execute.")

```

27.4.5. Stability Guarantees

MPC stability is ensured through careful design of terminal ingredients:

! Stability Conditions for MPC

For asymptotic stability of the origin under MPC, sufficient conditions are:

1. **Terminal cost** $V_f(x)$ is a local Lyapunov function
2. **Terminal constraint set** $\mathcal{X}_f$ is positively invariant under local controller
3. **Terminal controller** $\kappa_f(x)$ satisfies $V_f(f(x, \kappa_f(x))) - V_f(x) \leq -\ell(x, \kappa_f(x))$

These ensure the cost function decreases along closed-loop trajectories.

27.4.6. Feasibility and Recursive Feasibility

Definition: MPC is recursively feasible if: whenever a feasible solution exists at time t , a feasible solution also exists at time $t + 1$.

Conditions for recursive feasibility:

1. Terminal set $\mathcal{X}_f$ is control invariant
2. Constraint sets are compatible with dynamics
3. No “hard” deadlines that may become infeasible

```
def check_recursive_feasibility(mpc, x0, n_steps=100):
    """Empirically test recursive feasibility.

    Args:
        mpc: MPC controller
        x0: Initial state
        n_steps: Number of steps to test

    Returns:
        is_feasible: Boolean indicating feasibility maintained
        failure_step: Step where feasibility failed (if any)
    """
    x = np.array(x0)

    for k in range(n_steps):
        try:
            u = mpc.control(x)
            x = mpc.A @ x + mpc.B @ u # For linear MPC
        except ValueError as e:
            print(f"Feasibility failed at step {k}: {e}")
            return False, k

    return True, None
```

27.5. Robust MPC

Real systems face uncertainty from model errors, disturbances, and measurement noise. Standard MPC assumes a perfect model and solves a deterministic optimization at each step. When this assumption fails, constraints may be violated and stability guarantees break down. Robust MPC methods explicitly account for bounded disturbances and guarantee constraint satisfaction for all possible realizations within the uncertainty set.

27.5.1. Tube MPC

Tube MPC is the most widely used robust MPC approach for linear systems with additive disturbances. It separates the control into two components:

- **Nominal trajectory:** Optimized without uncertainty
- **Ancillary controller:** Keeps actual trajectory in a “tube” around nominal

The tube MPC optimization:

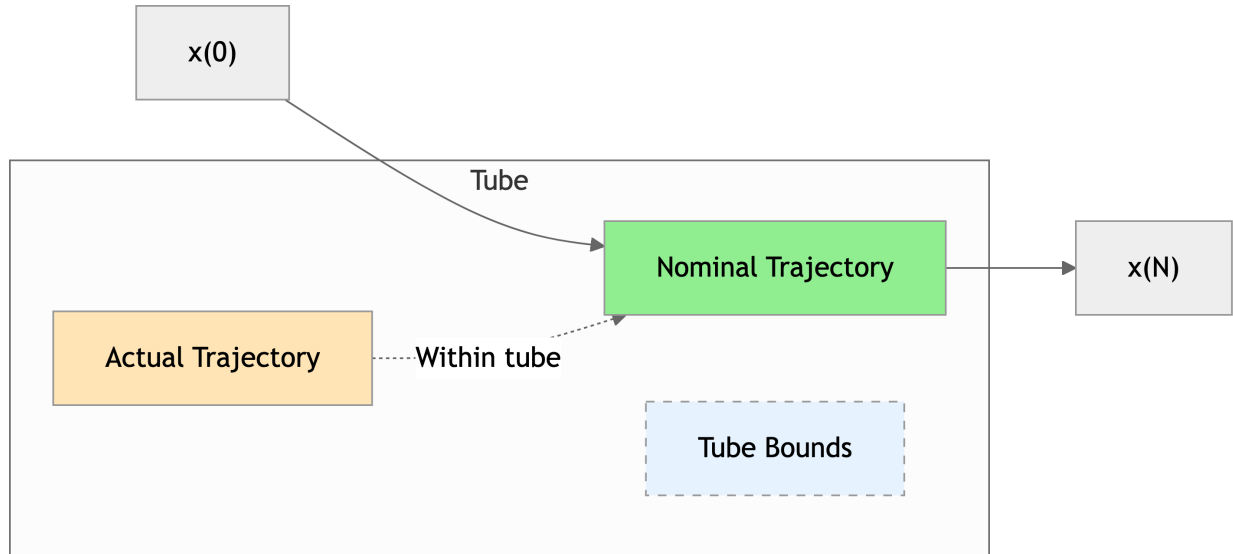


Figure 27.5.: Tube MPC: Robust Constraint Satisfaction

$$\begin{aligned}
 & \text{minimize} && \sum_{k=0}^{N-1} \ell(\bar{x}_k, \bar{u}_k) + V_f(\bar{x}_N) \\
 & \text{subject to} && \bar{x}_{k+1} = A\bar{x}_k + B\bar{u}_k \\
 & && \bar{x}_k \in \mathcal{X} \ominus \mathcal{Z} \\
 & && \bar{u}_k \in \mathcal{U} \ominus K\mathcal{Z} \\
 & && \bar{x}_N \in \mathcal{X}_f \ominus \mathcal{Z}
 \end{aligned}$$

where $\mathcal{Z}$ is the robust positively invariant (RPI) set and $\ominus$ is the Pontryagin difference.

```

import numpy as np

class TubeMPC:
    """Tube-based robust MPC for linear systems with bounded disturbance."""

    def __init__(self, A, B, Q, R, K_tube, W, N=10):
        """
        Args:
            A, B: System matrices
            Q, R: Cost matrices
            K_tube: Ancillary feedback gain
            W: Disturbance bound set (polytope vertices)
            N: Prediction horizon
        """
        self.A = np.array(A)
        self.B = np.array(B)
        self.Q = np.array(Q)
        self.R = np.array(R)
        self.K = np.array(K_tube)
        self.W = np.array(W)

```

```

self.N = N

self.n = A.shape[0]
self.m = B.shape[1]

# Compute closed-loop matrix
self.A_K = self.A + self.B @ self.K

# Compute RPI set approximation
self.Z = self._compute_rpi_set()

def _compute_rpi_set(self, n_iter=20):
    """Compute robust positively invariant set approximation."""
    # Simplified: use Minkowski sum of disturbance sets
    Z = self.W.copy()

    for k in range(n_iter):
        #  $Z_{k+1} = A_K * Z_k + W$ 
        Z_new = []
        for z in Z:
            for w in self.W:
                Z_new.append(self.A_K @ z + w)
        Z = np.array(Z_new)

        # Convex hull approximation (simplified)
        Z = np.unique(np.round(Z, 6), axis=0)

    return Z

def tighten_constraints(self, x_min, x_max, u_min, u_max):
    """Tighten constraints by RPI set.

    Args:
        x_min, x_max: Original state bounds
        u_min, u_max: Original control bounds

    Returns:
        Tightened constraint bounds
    """
    # State constraint tightening
    z_max = np.max(np.abs(self.Z), axis=0)
    x_min_tight = x_min + z_max
    x_max_tight = x_max - z_max

    # Control constraint tightening
    Kz_max = np.max(np.abs(self.K @ self.Z.T), axis=1)
    u_min_tight = u_min + Kz_max
    u_max_tight = u_max - Kz_max

    return x_min_tight, x_max_tight, u_min_tight, u_max_tight

def control(self, x, x_nominal):
    """Compute tube MPC control.

```

```

u = u_nominal + K(x - x_nominal)

Args:
    x: Actual state
    x_nominal: Nominal state from MPC

Returns:
    u: Control to apply
"""
u_nominal = self.solve_nominal_mpc(x_nominal)

# Add ancillary feedback
e = x - x_nominal
u = u_nominal + self.K @ e

return u

def solve_nominal_mpc(self, x0):
    """Solve nominal MPC with tightened constraints."""
    # This would use a standard MPC solver with tightened constraints
    # Simplified implementation
    pass

```

27.5.2. Min-Max MPC

Min-max MPC optimizes against the worst-case disturbance:

$$\begin{aligned}
 & \underset{u}{\text{minimize}} \quad \underset{w}{\text{maximize}} \quad \sum_{k=0}^{N-1} \ell(x_k, u_k) + V_f(x_N) \\
 & \text{subject to} \quad x_{k+1} = f(x_k, u_k) + w_k \\
 & \quad \quad \quad w_k \in \mathcal{W} \\
 & \quad \quad \quad \text{constraints hold } \forall w \in \mathcal{W}
 \end{aligned}$$

```

class MinMaxMPC:
    """Min-max robust MPC for worst-case optimization."""

    def __init__(self, A, B, Q, R, W_vertices, N=5):
        """
        Args:
            A, B: System matrices
            Q, R: Cost matrices
            W_vertices: Vertices of disturbance polytope
            N: Prediction horizon (typically shorter than nominal MPC)
        """
        self.A = A
        self.B = B
        self.Q = Q
        self.R = R
        self.W = W_vertices
        self.N = N

```

```

def build_scenario_tree(self, x0):
    """Build scenario tree for all disturbance realizations.

    The tree branches at each stage for each vertex of W.
    """
    n_vertices = len(self.W)
    n_scenarios = n_vertices ** self.N

    # Build tree structure
    scenarios = []

    def build_branch(x, stage, w_sequence):
        if stage == self.N:
            scenarios.append({
                'x_final': x,
                'w_sequence': w_sequence.copy()
            })
            return

        for w in self.W:
            # Placeholder: would need u at this point
            w_sequence.append(w)
            build_branch(x, stage + 1, w_sequence)
            w_sequence.pop()

    build_branch(x0, 0, [])
    return scenarios

def solve(self, x0):
    """Solve min-max MPC.

    This requires solving a multi-stage robust optimization problem.
    Practical implementations often use approximations.
    """
    # Full min-max solution is computationally expensive
    # Common approach: solve as a large LP/QP with constraints
    # for all scenarios

    # Simplified: use average-case with constraint for all vertices
    pass

```

27.5.3. Stochastic MPC

Stochastic MPC handles probabilistic uncertainty:

$$\begin{aligned}
 & \text{minimize} && \mathbb{E} \left[\sum_{k=0}^{N-1} \ell(x_k, u_k) + V_f(x_N) \right] \\
 & \text{subject to} && x_{k+1} = f(x_k, u_k) + w_k, \quad w_k \sim \mathcal{D} \\
 & && \mathbb{P}(x_k \in \mathcal{X}) \geq 1 - \epsilon \\
 & && u_k \in \mathcal{U}
 \end{aligned}$$

i Chance Constraints

Chance constraints $\mathbb{P}(g(x) \leq 0) \geq 1 - \epsilon$ are converted to deterministic constraints using:

- **Scenario approach:** Sample scenarios and require constraint for $1 - \epsilon$ fraction
- **Robust reformulation:** Replace with worst-case over confidence ellipsoid
- **Conditional Value-at-Risk:** Bound expected violation in tail

```
import numpy as np
from scipy.stats import norm

class StochasticMPC:
    """Stochastic MPC with chance constraints."""

    def __init__(self, A, B, Q, R, w_cov, N=10, epsilon=0.05):
        """
        Args:
            A, B: System matrices
            Q, R: Cost matrices
            w_cov: Disturbance covariance matrix
            N: Prediction horizon
            epsilon: Constraint violation probability
        """
        self.A = np.array(A)
        self.B = np.array(B)
        self.Q = np.array(Q)
        self.R = np.array(R)
        self.Sigma_w = np.array(w_cov)
        self.N = N
        self.epsilon = epsilon

        self.n = A.shape[0]
        self.m = B.shape[1]

        # Quantile for chance constraint
        self.z_eps = norm.ppf(1 - epsilon)

    def propagate_uncertainty(self, Sigma_0=None):
        """Propagate state covariance over horizon.

        Args:
            Sigma_0: Initial state covariance

        Returns:
            Sigma: List of state covariances
        """
        if Sigma_0 is None:
            Sigma_0 = np.zeros((self.n, self.n))

        Sigma = [Sigma_0]

        for k in range(self.N):
```

```

        # Sigma_{k+1} = A * Sigma_k * A' + Sigma_w
        Sigma_next = self.A @ Sigma[-1] @ self.A.T + self.Sigma_w
        Sigma.append(Sigma_next)

    return Sigma

def tighten_constraints_probabilistic(self, x_min, x_max, Sigma):
    """Tighten constraints for chance constraint satisfaction.

    For Gaussian uncertainty:  $x_i \leq b$  becomes  $\mu_i + z_{\text{eps}} * \sigma_i \leq b$ 

    Args:
        x_min, x_max: Original bounds
        Sigma: State covariance matrices

    Returns:
        Tightened bounds for each stage
    """
    tightened = []

    for k, Sigma_k in enumerate(Sigma):
        sigma = np.sqrt(np.diag(Sigma_k))
        margin = self.z_eps * sigma

        x_min_k = x_min + margin
        x_max_k = x_max - margin

        tightened.append((x_min_k, x_max_k))

    return tightened

def scenario_approach(self, x0, x_min, x_max, u_min, u_max, n_scenarios=100):
    """Solve using scenario approach.

    Sample scenarios and solve deterministic problem requiring
    constraints for (1-epsilon) fraction.
    """
    # Generate disturbance scenarios
    scenarios = []
    for _ in range(n_scenarios):
        w_seq = [np.random.multivariate_normal(
            np.zeros(self.n), self.Sigma_w) for _ in range(self.N)]
        scenarios.append(w_seq)

    # Number of scenarios to satisfy
    n_satisfy = int(np.ceil((1 - self.epsilon) * n_scenarios))

    # Solve optimization with scenario constraints
    # This is a large optimization problem
    # Implementation depends on specific solver
    pass

```


27.6. Numerical Methods for Optimal Control

Practical optimal control requires numerical methods to solve the resulting boundary value problems or optimization problems.

27.6.1. Direct Methods

Direct methods transcribe the continuous optimal control problem into a finite-dimensional nonlinear program (NLP).

27.6.1.1. Direct Collocation

Discretize both states and controls, enforce dynamics at collocation points:

```
import numpy as np
from scipy.optimize import minimize
from scipy.interpolate import CubicHermiteSpline

class DirectCollocation:
    """Direct collocation method for optimal control."""

    def __init__(self, dynamics, running_cost, terminal_cost,
                 n_states, n_controls, N=50):
        """
        Args:
            dynamics: Function f(x, u) returning dx/dt
            running_cost: Function L(x, u)
            terminal_cost: Function phi(x_T)
            n_states, n_controls: Problem dimensions
            N: Number of collocation points
        """
        self.f = dynamics
        self.L = running_cost
        self.phi = terminal_cost
        self.n = n_states
        self.m = n_controls
        self.N = N

    def collocation_constraints(self, z, T, x0, x_f=None):
        """Compute collocation constraint violations.

        Uses Hermite-Simpson collocation.

        Args:
            z: Decision variables [x_0, u_0, x_1, u_1, ..., x_N, u_N]
            T: Final time
            x0: Initial state
            x_f: Final state (if fixed)

        Returns:
            constraints: Vector of constraint violations
        """
        dt = T / self.N
        constraints = []
```

```

# Extract states and controls
x = z[:(self.N+1)*self.n].reshape(self.N+1, self.n)
u = z[(self.N+1)*self.n:].reshape(self.N+1, self.m)

# Initial condition
constraints.extend(x[0] - x0)

# Collocation constraints at each interval
for k in range(self.N):
    x_k = x[k]
    x_kp1 = x[k+1]
    u_k = u[k]
    u_kp1 = u[k+1]

    f_k = self.f(x_k, u_k)
    f_kp1 = self.f(x_kp1, u_kp1)

    # Midpoint
    x_mid = 0.5*(x_k + x_kp1) + dt/8*(f_k - f_kp1)
    u_mid = 0.5*(u_k + u_kp1)
    f_mid = self.f(x_mid, u_mid)

    # Hermite-Simpson defect
    defect = x_kp1 - x_k - dt/6*(f_k + 4*f_mid + f_kp1)
    constraints.extend(defect)

# Final condition (if specified)
if x_f is not None:
    constraints.extend(x[self.N] - x_f)

return np.array(constraints)

def objective(self, z, T):
    """Compute objective function."""
    dt = T / self.N

    x = z[:(self.N+1)*self.n].reshape(self.N+1, self.n)
    u = z[(self.N+1)*self.n:].reshape(self.N+1, self.m)

    # Running cost (trapezoidal integration)
    cost = 0
    for k in range(self.N):
        L_k = self.L(x[k], u[k])
        L_kp1 = self.L(x[k+1], u[k+1])
        cost += 0.5 * dt * (L_k + L_kp1)

    # Terminal cost
    cost += self.phi(x[self.N])

    return cost

def solve(self, x0, T, x_f=None, u_bounds=None, x_bounds=None):
    """Solve optimal control problem.

```

```

Args:
    x0: Initial state
    T: Final time
    x_f: Final state (optional)
    u_bounds: Control bounds (u_min, u_max)
    x_bounds: State bounds (x_min, x_max)

Returns:
    t: Time grid
    x: Optimal state trajectory
    u: Optimal control trajectory
"""
n_vars = (self.N + 1) * (self.n + self.m)

# Initial guess: linear interpolation for states, zero controls
z0 = np.zeros(n_vars)

if x_f is not None:
    for k in range(self.N + 1):
        alpha = k / self.N
        z0[k*self.n:(k+1)*self.n] = (1-alpha)*x0 + alpha*x_f
else:
    for k in range(self.N + 1):
        z0[k*self.n:(k+1)*self.n] = x0

# Bounds
bounds = []

for k in range(self.N + 1):
    if x_bounds is not None:
        for i in range(self.n):
            bounds.append((x_bounds[0][i], x_bounds[1][i]))
    else:
        bounds.extend([(None, None)] * self.n)

for k in range(self.N + 1):
    if u_bounds is not None:
        for i in range(self.m):
            bounds.append((u_bounds[0][i], u_bounds[1][i]))
    else:
        bounds.extend([(None, None)] * self.m)

# Constraints
constraints = {
    'type': 'eq',
    'fun': lambda z: self.collocation_constraints(z, T, x0, x_f)
}

# Solve
result = minimize(
    lambda z: self.objective(z, T),
    z0,

```

```

        method='SLSQP',
        bounds=bounds,
        constraints=constraints,
        options={'maxiter': 500, 'ftol': 1e-6}
    )

    # Extract solution
    x = result.x[(self.N+1)*self.n].reshape(self.N+1, self.n)
    u = result.x[(self.N+1)*self.n:].reshape(self.N+1, self.m)
    t = np.linspace(0, T, self.N+1)

    return t, x, u, result.fun

```

27.6.1.2. Multiple Shooting

Multiple shooting divides the time horizon into segments and treats the initial state of each segment as a decision variable. The solver integrates the dynamics within each segment independently, then enforces continuity constraints to ensure the segments connect. This approach improves numerical stability compared to single shooting because integration errors do not accumulate across the full horizon.

```

class MultipleShootingSolver:
    """Multiple shooting method for optimal control."""

    def __init__(self, dynamics, running_cost, terminal_cost,
                 n_states, n_controls, N=20):
        """
        Args:
            dynamics: Function f(x, u) returning dx/dt
            running_cost: Function L(x, u)
            terminal_cost: Function phi(x_T)
            n_states, n_controls: Problem dimensions
            N: Number of shooting intervals
        """
        self.f = dynamics
        self.L = running_cost
        self.phi = terminal_cost
        self.n = n_states
        self.m = n_controls
        self.N = N

    def integrate_segment(self, x0, u, dt, n_steps=10):
        """Integrate dynamics over one shooting interval."""
        from scipy.integrate import solve_ivp

        def ode(t, x):
            return self.f(x, u)

        sol = solve_ivp(ode, [0, dt], x0,
                        t_eval=np.linspace(0, dt, n_steps))

        return sol.y[:, -1], sol

```

```

def shooting_constraints(self, z, T, x0):
    """Continuity constraints between shooting segments."""
    dt = T / self.N

    # Parse decision variables
    # z = [s_1, ..., s_N, u_0, ..., u_{N-1}]
    # s_k are initial states for segments 2, ..., N+1

    s = z[:self.N*self.n].reshape(self.N, self.n)
    u = z[self.N*self.n:].reshape(self.N, self.m)

    constraints = []

    # First segment: integrate from x0
    x_end, _ = self.integrate_segment(x0, u[0], dt)
    constraints.extend(x_end - s[0])

    # Remaining segments
    for k in range(self.N - 1):
        x_end, _ = self.integrate_segment(s[k], u[k+1], dt)
        constraints.extend(x_end - s[k+1] if k < self.N-2 else x_end - s[k])

    return np.array(constraints)

def solve(self, x0, T, x_f=None, u_bounds=None):
    """Solve using multiple shooting."""
    # Implementation similar to collocation
    # Decision variables: segment initial conditions + controls
    pass

```

27.6.2. Indirect Methods

Indirect methods solve the boundary value problem arising from the Maximum Principle:

```

from scipy.integrate import solve_bvp

def indirect_optimal_control(dynamics, costate_dynamics, hamiltonian_gradient,
                             x0, lambda_T, T, n_points=100):
    """Solve optimal control via indirect method (BVP).

    Args:
        dynamics: State dynamics f(x, u, t)
        costate_dynamics: Costate dynamics -dH/dx
        hamiltonian_gradient: dH/du for optimal control
        x0: Initial state
        lambda_T: Terminal costate (or function of x_T)
        T: Final time

    Returns:
        t: Time grid
        x: State trajectory
        lam: Costate trajectory
    """

```

```

    u: Control trajectory
"""
n = len(x0)

def optimal_control(x, lam, t):
    """Compute optimal control from stationarity condition."""
    # For unconstrained problems: dH/du = 0
    # This would need to be solved for specific problems
    pass

def bvp_ode(t, y):
    """Combined state-costate ODE."""
    x = y[:n]
    lam = y[n:]

    u = optimal_control(x, lam, t)

    dx = dynamics(x, u, t)
    dlam = costate_dynamics(x, u, lam, t)

    return np.concatenate([dx, dlam])

def bvp_bc(ya, yb):
    """Boundary conditions."""
    x_a = ya[:n]
    lam_b = yb[n:]

    # Initial state condition
    bc_initial = x_a - x0

    # Terminal costate condition
    if callable(lambda_T):
        bc_terminal = lam_b - lambda_T(yb[:n])
    else:
        bc_terminal = lam_b - lambda_T

    return np.concatenate([bc_initial, bc_terminal])

# Initial guess
t_init = np.linspace(0, T, n_points)
y_init = np.zeros((2*n, n_points))
y_init[:n, :] = np.outer(x0, np.ones(n_points))

# Solve BVP
sol = solve_bvp(bvp_ode, bvp_bc, t_init, y_init)

if not sol.success:
    raise ValueError(f"BVP solver failed: {sol.message}")

x = sol.y[:n].T
lam = sol.y[n:].T
t = sol.x

```

```
# Compute optimal control
u = np.array([optimal_control(x[i], lam[i], t[i]) for i in range(len(t))])

return t, x, lam, u
```

27.6.3. Comparison of Numerical Methods

Method	Advantages	Disadvantages	Best For
Direct Collocation	Handles constraints easily, robust	Large NLPs	General problems
Multiple Shooting	Parallelizable, handles unstable dynamics	Discontinuous controls	Sensitive systems
Indirect (BVP)	High accuracy, insight from costates	Sensitive to initialization	Academic, benchmark
Dynamic Programming	Global optimum, handles discontinuities	Curse of dimensionality	Low-dimensional

27.7. Python Implementation with Libraries

27.7.1. CasADi for Optimal Control

CasADi is a symbolic framework for numerical optimization, ideal for optimal control.

```
# Note: Requires casadi installation: pip install casadi
import numpy as np

def casadi_mpc_example():
    """MPC implementation using CasADi."""
    try:
        import casadi as ca
    except ImportError:
        print("CasADi not installed. Install with: pip install casadi")
        return

    # System parameters
    n_states = 2
    n_controls = 1
    N = 20 # Horizon
    dt = 0.1

    # Symbolic variables
    x = ca.MX.sym('x', n_states)
    u = ca.MX.sym('u', n_controls)

    # Dynamics: double integrator
    # dx1/dt = x2
    # dx2/dt = u
    x_dot = ca.vertcat(x[1], u[0])

    # Discrete-time dynamics (Euler)
    x_next = x + dt * x_dot
```

```

# Create function
f = ca.Function('f', [x, u], [x_next])

# Cost matrices
Q = np.diag([10, 1])
R = np.array([[0.1]])
P = Q # Terminal cost

# Stage cost
stage_cost = ca.mtimes([x.T, Q, x]) + ca.mtimes([u.T, R, u])
L = ca.Function('L', [x, u], [stage_cost])

# Terminal cost
terminal_cost = ca.mtimes([x.T, P, x])
Vf = ca.Function('Vf', [x], [terminal_cost])

# Build NLP
# Decision variables: [u_0, ..., u_{N-1}, x_1, ..., x_N]

# Start with empty NLP
w = [] # Decision variables
w0 = [] # Initial guess
lbw = [] # Lower bounds
ubw = [] # Upper bounds
J = 0 # Objective
g = [] # Constraints
lbg = [] # Constraint lower bounds
ubg = [] # Constraint upper bounds

# Initial state (parameter)
x0_param = ca.MX.sym('x0', n_states)
xk = x0_param

for k in range(N):
    # Control at step k
    uk = ca.MX.sym(f'u_{k}', n_controls)
    w.append(uk)
    w0.append([0])
    lbw.append([-1]) # Control lower bound
    ubw.append([1]) # Control upper bound

    # Add stage cost
    J += L(xk, uk)

    # Next state
    xk_next = f(xk, uk)

    # State at step k+1
    xk = ca.MX.sym(f'x_{k+1}', n_states)
    w.append(xk)
    w0.append([0, 0])
    lbw.append([-10, -5]) # State lower bounds
    ubw.append([10, 5]) # State upper bounds

```



```

    # Dynamics constraint
    g.append(xk_next - xk)
    lbg.append([0, 0])
    ubg.append([0, 0])

# Terminal cost
J += Vf(xk)

# Create NLP
w = ca.vertcat(*w)
g = ca.vertcat(*g)

nlp = {
    'f': J,
    'x': w,
    'g': g,
    'p': x0_param
}

# Solver options
opts = {
    'ipopt.print_level': 0,
    'print_time': 0
}

solver = ca.nlpsol('solver', 'ipopt', nlp, opts)

# Solve for specific initial condition
x0_val = [5, 0]

solution = solver(
    x0=np.concatenate(w0),
    lbx=np.concatenate(lbw),
    ubx=np.concatenate(ubw),
    lbg=np.concatenate(lbg),
    ubg=np.concatenate(ubg),
    p=x0_val
)

# Extract solution
w_opt = solution['x'].full().flatten()

# Parse controls and states
u_opt = []
x_opt = [x0_val]

idx = 0
for k in range(N):
    u_opt.append(w_opt[idx:idx+n_controls])
    idx += n_controls
    x_opt.append(w_opt[idx:idx+n_states])
    idx += n_states

```

```

print("CasADi MPC Solution:")
print(f"Initial state: {x0_val}")
print(f"First control: {u_opt[0]}")
print(f"Final state: {x_opt[-1]}")
print(f"Optimal cost: {float(solution['f']):.4f}")

return np.array(u_opt), np.array(x_opt)

# Run example
casadi_mpc_example()

```

27.7.2. do-mpc Library

do-mpc is a Python toolbox built on top of CasADi that provides a high-level interface for MPC. It handles model definition, controller setup, and closed-loop simulation with minimal boilerplate code. The library supports both linear and nonlinear models, robust MPC with scenario trees, and moving horizon estimation.

```

def do_mpc_example():
    """MPC using do-mpc library."""
    try:
        import do_mpc
        import casadi as ca
    except ImportError:
        print("do-mpc not installed. Install with: pip install do-mpc")
        return

    # Model type
    model_type = 'continuous'
    model = do_mpc.model.Model(model_type)

    # States
    x1 = model.set_variable(var_type='_x', var_name='x1')
    x2 = model.set_variable(var_type='_x', var_name='x2')

    # Control input
    u = model.set_variable(var_type='_u', var_name='u')

    # Dynamics: double integrator
    model.set_rhs('x1', x2)
    model.set_rhs('x2', u)

    # Build model
    model.setup()

    # MPC controller
    mpc = do_mpc.controller.MPC(model)

    # Set parameters
    setup_mpc = {
        'n_horizon': 20,
        't_step': 0.1,

```

```

        'n_robust': 0,
        'store_full_solution': True,
    }
    mpc.set_param(**setup_mpc)

    # Objective function
    mterm = x1**2 + x2**2 # Terminal cost
    lterm = 10*x1**2 + x2**2 + 0.1*u**2 # Stage cost

    mpc.set_objective(mterm=mterm, lterm=lterm)

    # Constraints
    mpc.bounds['lower', '_x', 'x1'] = -10
    mpc.bounds['upper', '_x', 'x1'] = 10
    mpc.bounds['lower', '_x', 'x2'] = -5
    mpc.bounds['upper', '_x', 'x2'] = 5
    mpc.bounds['lower', '_u', 'u'] = -1
    mpc.bounds['upper', '_u', 'u'] = 1

    # Setup
    mpc.setup()

    # Simulator
    simulator = do_mpc.simulator.Simulator(model)
    simulator.set_param(t_step=0.1)
    simulator.setup()

    # Initial state
    x0 = np.array([[5.0], [0.0]])
    mpc.x0 = x0
    simulator.x0 = x0
    mpc.set_initial_guess()

    # Simulation loop
    n_steps = 100

    for k in range(n_steps):
        u0 = mpc.make_step(x0)
        x0 = simulator.make_step(u0)

    print("do-mpc simulation completed")
    print(f"Final state: {x0.flatten()}")

    return mpc, simulator

# Run example
do_mpc_example()

```

27.7.3. Complete MPC Example: Building Temperature Control

Building HVAC (heating, ventilation, and air conditioning) control is a standard MPC application. The controller must balance energy cost against occupant comfort while accounting for time-varying outdoor temperatures and electricity prices. The thermal dynamics of a building zone follow a simple first-order model: the indoor temperature changes based on heat transfer through the building

envelope and the HVAC power input.

```
import numpy as np
from scipy.optimize import minimize
import matplotlib.pyplot as plt

class BuildingMPC:
    """MPC for building temperature control.

    Model:  $dT/dt = (T_{out} - T)/R + Q_{hvac}/C$ 

    Where:
        T: Indoor temperature
        T_out: Outdoor temperature
        R: Thermal resistance
        C: Thermal capacitance
        Q_hvac: HVAC heating/cooling power
    """

    def __init__(self, R=2.0, C=1e6, dt=0.25, N=24):
        """
        Args:
            R: Thermal resistance (K/W)
            C: Thermal capacitance (J/K)
            dt: Time step (hours)
            N: Prediction horizon (steps)
        """
        self.R = R
        self.C = C
        self.dt = dt * 3600 # Convert to seconds
        self.N = N

        # Comfort bounds
        self.T_min = 20 # degrees C
        self.T_max = 24 # degrees C

        # HVAC bounds (kW)
        self.Q_min = -5 # Cooling
        self.Q_max = 5 # Heating

        # Cost weights
        self.w_comfort = 100 # Comfort violation penalty
        self.w_energy = 1 # Energy cost weight

    def dynamics(self, T, Q_hvac, T_out):
        """Discrete-time dynamics."""
        dT = (T_out - T) / self.R + Q_hvac * 1000 / self.C
        return T + dT * self.dt

    def predict_trajectory(self, T0, Q_seq, T_out_seq):
        """Predict temperature trajectory."""
        T = [T0]
        for k in range(len(Q_seq)):
            T_next = self.dynamics(T[-1], Q_seq[k], T_out_seq[k])
```

```

        T.append(T_next)
    return np.array(T)

def objective(self, Q_flat, T0, T_out_forecast, price_forecast):
    """MPC objective: minimize energy cost + comfort violations."""
    Q_seq = Q_flat.reshape(-1)
    T_traj = self.predict_trajectory(T0, Q_seq, T_out_forecast)

    cost = 0

    # Energy cost
    for k in range(self.N):
        cost += self.w_energy * price_forecast[k] * abs(Q_seq[k]) * self.dt /
↪ 3600

    # Comfort violations (soft constraint)
    for k in range(1, self.N + 1):
        if T_traj[k] < self.T_min:
            cost += self.w_comfort * (self.T_min - T_traj[k])**2
        elif T_traj[k] > self.T_max:
            cost += self.w_comfort * (T_traj[k] - self.T_max)**2

    return cost

def solve(self, T0, T_out_forecast, price_forecast):
    """Solve MPC problem.

    Args:
        T0: Current indoor temperature
        T_out_forecast: Outdoor temperature forecast
        price_forecast: Electricity price forecast

    Returns:
        Q_opt: Optimal HVAC power sequence
        T_pred: Predicted temperature trajectory
    """
    # Initial guess
    Q0 = np.zeros(self.N)

    # Bounds
    bounds = [(self.Q_min, self.Q_max)] * self.N

    # Solve
    result = minimize(
        self.objective,
        Q0,
        args=(T0, T_out_forecast, price_forecast),
        method='SLSQP',
        bounds=bounds,
        options={'maxiter': 100}
    )

    Q_opt = result.x

```

```

    T_pred = self.predict_trajectory(T0, Q_opt, T_out_forecast)

    return Q_opt, T_pred

def control(self, T0, T_out_forecast, price_forecast):
    """Get control action (first element of optimal sequence)."""
    Q_opt, _ = self.solve(T0, T_out_forecast, price_forecast)
    return Q_opt[0]

def simulate_day(self, T0, T_out_profile, price_profile):
    """Simulate a full day with MPC control.

    Args:
        T0: Initial temperature
        T_out_profile: Full day outdoor temperature (hourly)
        price_profile: Full day electricity prices (hourly)

    Returns:
        T_history: Temperature history
        Q_history: HVAC power history
        cost_history: Cumulative cost
    """
    n_hours = len(T_out_profile)
    steps_per_hour = int(1 / self.dt * 3600)
    n_steps = n_hours * steps_per_hour

    T = T0
    T_history = [T]
    Q_history = []
    cost_history = [0]

    for step in range(n_steps):
        hour = int(step * self.dt / 3600)

        # Create forecast from current position
        forecast_hours = min(self.N, n_hours - hour)
        T_out_forecast = T_out_profile[hour:hour + forecast_hours]
        price_forecast = price_profile[hour:hour + forecast_hours]

        # Pad if necessary
        if len(T_out_forecast) < self.N:
            T_out_forecast = np.concatenate([
                T_out_forecast,
                np.full(self.N - len(T_out_forecast), T_out_forecast[-1])
            ])
            price_forecast = np.concatenate([
                price_forecast,
                np.full(self.N - len(price_forecast), price_forecast[-1])
            ])

        # Get control action
        Q = self.control(T, T_out_forecast, price_forecast)
        Q_history.append(Q)

```

```

        # Apply control
        T = self.dynamics(T, Q, T_out_profile[hour])
        T_history.append(T)

        # Update cost
        cost = price_profile[hour] * abs(Q) * self.dt / 3600
        cost_history.append(cost_history[-1] + cost)

    return np.array(T_history), np.array(Q_history), np.array(cost_history)

def run_building_mpc():
    """Run building MPC example."""

    # Create controller
    mpc = BuildingMPC(dt=0.25, N=12)  # 15-min steps, 3-hour horizon

    # Generate typical daily profiles
    hours = np.arange(24)

    # Outdoor temperature: peaks in afternoon
    T_out = 15 + 10 * np.sin(np.pi * (hours - 6) / 12)
    T_out = np.clip(T_out, 5, 30)

    # Electricity price: peak in evening
    price = 0.10 + 0.15 * np.exp(-((hours - 18)**2) / 10)

    # Simulate
    T0 = 22  # Starting temperature
    T_hist, Q_hist, cost_hist = mpc.simulate_day(T0, T_out, price)

    print("Building MPC Simulation Results:")
    print(f"Initial temperature: {T0:.1f} C")
    print(f"Final temperature: {T_hist[-1]:.1f} C")
    print(f"Temperature range: [{np.min(T_hist):.1f}, {np.max(T_hist):.1f}] C")
    print(f"Total energy cost: ${cost_hist[-1]:.2f}")
    print(f"Average HVAC power: {np.mean(np.abs(Q_hist)):.2f} kW")

    return T_hist, Q_hist, cost_hist

# Run example
T_hist, Q_hist, cost_hist = run_building_mpc()

```

27.8. Real-World Applications and Case Studies

27.8.1. Case Study 1: Autonomous Vehicle Path Planning

Autonomous vehicle trajectory planning requires real-time optimization subject to nonlinear dynamics, physical constraints, and safety requirements. The kinematic bicycle model represents the vehicle as a rigid body with front-wheel steering. The MPC controller computes optimal steering angle and acceleration commands to track a reference trajectory while respecting velocity limits and actuator bounds.

```

class AutonomousVehicleMPC:
    """MPC for autonomous vehicle trajectory planning.

    Kinematic bicycle model:
        dx/dt = v * cos(psi)
        dy/dt = v * sin(psi)
        dpsi/dt = v/L * tan(delta)
        dv/dt = a

    Controls: steering angle (delta), acceleration (a)
    """

    def __init__(self, L=2.5, dt=0.1, N=20):
        """
        Args:
            L: Wheelbase (m)
            dt: Time step (s)
            N: Prediction horizon
        """
        self.L = L
        self.dt = dt
        self.N = N

        # Vehicle constraints
        self.v_max = 15 # m/s
        self.a_max = 3 # m/s^2
        self.delta_max = np.pi / 4 # rad

        # Cost weights
        self.Q = np.diag([10, 10, 1, 0.1]) # State tracking
        self.R = np.diag([0.1, 0.01]) # Control effort
        self.Rd = np.diag([1, 0.1]) # Control change rate

    def dynamics(self, x, u):
        """Discrete-time bicycle model."""
        px, py, psi, v = x
        delta, a = u

        px_next = px + v * np.cos(psi) * self.dt
        py_next = py + v * np.sin(psi) * self.dt
        psi_next = psi + v / self.L * np.tan(delta) * self.dt
        v_next = v + a * self.dt

        return np.array([px_next, py_next, psi_next, v_next])

    def track_reference(self, x0, reference_trajectory, u_prev=None):
        """Track a reference trajectory.

        Args:
            x0: Current state [px, py, psi, v]
            reference_trajectory: Reference states (N+1 x 4)
            u_prev: Previous control (for rate constraint)

```



```

Returns:
    u_opt: Optimal control sequence
    x_pred: Predicted trajectory
"""
if u_prev is None:
    u_prev = np.array([0, 0])

def objective(u_flat):
    u_seq = u_flat.reshape(self.N, 2)

    # Predict trajectory
    x = x0.copy()
    x_traj = [x]

    for k in range(self.N):
        x = self.dynamics(x, u_seq[k])
        x_traj.append(x)

    x_traj = np.array(x_traj)

    # Tracking cost
    cost = 0
    for k in range(self.N + 1):
        error = x_traj[k] - reference_trajectory[k]
        cost += error @ self.Q @ error

    # Control effort
    for k in range(self.N):
        cost += u_seq[k] @ self.R @ u_seq[k]

    # Control rate
    du = u_seq[0] - u_prev
    cost += du @ self.Rd @ du
    for k in range(1, self.N):
        du = u_seq[k] - u_seq[k-1]
        cost += du @ self.Rd @ du

    return cost

# Bounds
bounds = []
for k in range(self.N):
    bounds.append((-self.delta_max, self.delta_max))
    bounds.append((-self.a_max, self.a_max))

# Solve
u0 = np.zeros(2 * self.N)
result = minimize(objective, u0, method='SLSQP', bounds=bounds)

u_opt = result.x.reshape(self.N, 2)

# Compute predicted trajectory
x = x0.copy()

```

```

        x_pred = [x]
        for k in range(self.N):
            x = self.dynamics(x, u_opt[k])
            x_pred.append(x)

        return u_opt, np.array(x_pred)

def vehicle_mpc_example():
    """Autonomous vehicle lane change example."""

    mpc = AutonomousVehicleMPC()

    # Generate lane change reference
    t = np.linspace(0, 5, mpc.N + 1)

    # Smooth lane change using sigmoid
    def sigmoid(t, t0=2.5, k=2):
        return 1 / (1 + np.exp(-k * (t - t0)))

    lane_width = 3.5
    v_ref = 10 # m/s

    reference = np.zeros((mpc.N + 1, 4))
    reference[:, 0] = v_ref * t # x position
    reference[:, 1] = lane_width * sigmoid(t) # y position (lane change)
    reference[:, 2] = np.arctan2(np.gradient(reference[:, 1]),
    ↪ np.gradient(reference[:, 0]))
    reference[:, 3] = v_ref

    # Initial state
    x0 = np.array([0, 0, 0, v_ref])

    # Solve
    u_opt, x_pred = mpc.track_reference(x0, reference)

    print("Autonomous Vehicle MPC:")
    print(f"Initial state: x={x0[0]:.1f}m, y={x0[1]:.1f}m,
    ↪ psi={np.degrees(x0[2]):.1f}deg")
    print(f"Final predicted: x={x_pred[-1, 0]:.1f}m, y={x_pred[-1, 1]:.1f}m")
    print(f"First steering angle: {np.degrees(u_opt[0, 0]):.2f} deg")
    print(f"First acceleration: {u_opt[0, 1]:.2f} m/s^2")

    return u_opt, x_pred, reference

u_opt, x_pred, ref = vehicle_mpc_example()

```

27.8.2. Case Study 2: Chemical Process Control

A Continuous Stirred Tank Reactor (CSTR) is a standard benchmark for nonlinear process control. The reactor dynamics couple concentration and temperature through an exothermic reaction with Arrhenius kinetics. This creates a nonlinear, potentially unstable system where small perturbations can cause temperature runaway. MPC handles this by optimizing the heat input over a receding horizon while enforcing safety constraints on temperature and concentration.

```

class CSTRController:
    """MPC for Continuous Stirred Tank Reactor (CSTR).

    Dynamics:
        dC/dt = (C_in - C)/tau - k*C*exp(-E/(R*T))
        dT/dt = (T_in - T)/tau + (-dH)*k*C*exp(-E/(R*T))/(rho*Cp) + Q/(rho*Cp*V)

    States: C (concentration), T (temperature)
    Control: Q (heat input)
    """

    def __init__(self, dt=0.1, N=30):
        """Initialize CSTR controller with typical parameters."""

        # Reactor parameters
        self.V = 1.0          # Volume (m^3)
        self.tau = 1.0        # Residence time (min)
        self.k0 = 1e10        # Pre-exponential factor (1/min)
        self.E_R = 8500       # E/R (K)
        self.dH = -5e4        # Heat of reaction (J/mol)
        self.rho = 1000       # Density (kg/m^3)
        self.Cp = 4184        # Heat capacity (J/kg/K)

        # Feed conditions
        self.C_in = 1.0       # Feed concentration (mol/m^3)
        self.T_in = 300       # Feed temperature (K)

        # MPC parameters
        self.dt = dt
        self.N = N

        # Constraints
        self.T_min = 300      # Minimum temperature (K)
        self.T_max = 400      # Maximum temperature (K)
        self.Q_min = -1e5     # Min heat input (W)
        self.Q_max = 1e5      # Max heat input (W)

        # Setpoints
        self.C_sp = 0.5
        self.T_sp = 350

    def reaction_rate(self, C, T):
        """Arrhenius reaction rate."""
        return self.k0 * C * np.exp(-self.E_R / T)

    def dynamics(self, state, Q):
        """CSTR dynamics."""
        C, T = state

        r = self.reaction_rate(C, T)

        dC = (self.C_in - C) / self.tau - r
        dT = ((self.T_in - T) / self.tau

```

```

        + (-self.dH) * r / (self.rho * self.Cp)
        + Q / (self.rho * self.Cp * self.V))

    # Euler integration
    C_next = C + dC * self.dt
    T_next = T + dT * self.dt

    return np.array([C_next, T_next])

def solve_mpc(self, state0):
    """Solve MPC for current state."""

    def objective(Q_seq):
        state = state0.copy()
        cost = 0

        for k in range(self.N):
            # Tracking cost
            C_error = state[0] - self.C_sp
            T_error = state[1] - self.T_sp
            cost += 100 * C_error**2 + T_error**2

            # Control effort
            cost += 1e-8 * Q_seq[k]**2

            # Evolve state
            state = self.dynamics(state, Q_seq[k])

            # Soft temperature constraint
            if state[1] > self.T_max:
                cost += 1000 * (state[1] - self.T_max)**2
            if state[1] < self.T_min:
                cost += 1000 * (self.T_min - state[1])**2

        return cost

    # Initial guess
    Q0 = np.zeros(self.N)

    # Bounds
    bounds = [(self.Q_min, self.Q_max)] * self.N

    # Solve
    result = minimize(objective, Q0, method='L-BFGS-B', bounds=bounds)

    return result.x[0] # Return first control action

def simulate(self, state0, n_steps=200):
    """Closed-loop simulation."""

    state = state0.copy()
    states = [state.copy()]
    controls = []

```

```

    for k in range(n_steps):
        # Get MPC control
        Q = self.solve_mpc(state)
        controls.append(Q)

        # Apply control
        state = self.dynamics(state, Q)
        states.append(state.copy())

    if k % 50 == 0:
        print(f"Step {k}: C={state[0]:.3f}, T={state[1]:.1f}K")

    return np.array(states), np.array(controls)

def cstr_example():
    """CSTR control example."""

    controller = CSTRController()

    # Initial condition (away from setpoint)
    state0 = np.array([0.8, 320]) # Low temperature, high concentration

    print("CSTR MPC Control:")
    print(f"Setpoints: C={controller.C_sp}, T={controller.T_sp}K")
    print(f"Initial: C={state0[0]}, T={state0[1]}K")

    states, controls = controller.simulate(state0, n_steps=100)

    print(f"\nFinal: C={states[-1, 0]:.3f}, T={states[-1, 1]:.1f}K")
    print(f"Steady-state error: C={abs(states[-1, 0] - controller.C_sp):.4f}, "
          f"T={abs(states[-1, 1] - controller.T_sp):.2f}K")

    return states, controls

states, controls = cstr_example()

```

27.8.3. Case Study 3: Energy Storage Arbitrage

Battery energy storage systems can profit from electricity price fluctuations by charging when prices are low and discharging when prices are high. The MPC controller optimizes the charge/discharge schedule over a rolling horizon using price forecasts. The battery dynamics account for round-trip efficiency losses, state-of-charge limits, and power constraints.

```

class BatteryArbitrageMPC:
    """MPC for battery energy storage arbitrage.

    Maximize profit by buying low and selling high while
    respecting battery constraints.
    """

    def __init__(self, capacity=10, power_max=5, efficiency=0.9,

```

```

        dt=1.0, N=24):
    """
    Args:
        capacity: Battery capacity (MWh)
        power_max: Max charge/discharge power (MW)
        efficiency: Round-trip efficiency
        dt: Time step (hours)
        N: Prediction horizon (hours)
    """
    self.E_max = capacity
    self.P_max = power_max
    self.eta = np.sqrt(efficiency) # One-way efficiency
    self.dt = dt
    self.N = N

    # State of charge limits
    self.Soc_min = 0.1 * capacity
    self.Soc_max = 0.9 * capacity

def dynamics(self, SoC, P):
    """Battery state of charge dynamics.

    P > 0: charging
    P < 0: discharging
    """
    if P >= 0:
        SoC_next = SoC + self.eta * P * self.dt
    else:
        SoC_next = SoC + P / self.eta * self.dt

    return np.clip(SoC_next, 0, self.E_max)

def solve(self, SoC_0, price_forecast):
    """Optimize battery schedule.

    Args:
        SoC_0: Current state of charge
        price_forecast: Electricity prices for horizon

    Returns:
        P_opt: Optimal power schedule (+ charge, - discharge)
        revenue: Expected revenue
    """

    def objective(P_seq):
        """Negative revenue (for minimization)."""
        SoC = SoC_0
        revenue = 0

        for k in range(self.N):
            P = P_seq[k]

```

```

        # Revenue: positive P = buying (cost), negative P = selling
        ↪ (revenue)
        revenue -= price_forecast[k] * P * self.dt

        # Update SoC
        SoC = self.dynamics(SoC, P)

    # Soft SoC constraints
    penalty = 0
    SoC = SoC_0
    for k in range(self.N):
        SoC = self.dynamics(SoC, P_seq[k])
        if SoC < self.SoC_min:
            penalty += 1000 * (self.SoC_min - SoC)**2
        elif SoC > self.SoC_max:
            penalty += 1000 * (SoC - self.SoC_max)**2

    return -revenue + penalty

# Bounds
bounds = [(-self.P_max, self.P_max)] * self.N

# Initial guess
P0 = np.zeros(self.N)

# Solve
result = minimize(objective, P0, method='SLSQP', bounds=bounds)

P_opt = result.x
revenue = -result.fun

return P_opt, revenue

def simulate_day(self, SoC_0, prices):
    """Simulate MPC over a day."""

    SoC = SoC_0
    SoC_history = [SoC]
    P_history = []
    revenue_history = [0]

    for hour in range(24):
        # Get price forecast
        price_forecast = prices[hour:hour + self.N]
        if len(price_forecast) < self.N:
            # Pad with last price
            price_forecast = np.concatenate([
                price_forecast,
                np.full(self.N - len(price_forecast), price_forecast[-1])
            ])

        # Solve MPC
        P_opt, _ = self.solve(SoC, price_forecast)

```

```

        P = P_opt[0]

        P_history.append(P)

        # Apply control
        revenue = -prices[hour] * P * self.dt
        revenue_history.append(revenue_history[-1] + revenue)

        SoC = self.dynamics(SoC, P)
        SoC_history.append(SoC)

    return np.array(SoC_history), np.array(P_history), np.array(revenue_history)

def battery_arbitrage_example():
    """Battery arbitrage MPC example."""

    mpc = BatteryArbitrageMPC(capacity=10, power_max=5)

    # Generate price profile (typical daily pattern)
    hours = np.arange(48) # Two days

    # Base price with morning and evening peaks
    prices = 50 + 30 * np.sin(2 * np.pi * (hours - 6) / 24)
    prices += 20 * np.exp(-((hours % 24 - 18)**2) / 8) # Evening peak
    prices = np.maximum(prices, 20)

    # Initial SoC
    SoC_0 = 5 # 50% charge

    # Simulate
    SoC, P, revenue = mpc.simulate_day(SoC_0, prices[:24])

    print("Battery Arbitrage MPC:")
    print(f"Capacity: {mpc.E_max} MWh")
    print(f"Initial SoC: {SoC_0} MWh")
    print(f"Final SoC: {SoC[-1]:.2f} MWh")
    print(f"Daily revenue: ${revenue[-1]:.2f}")
    print(f"Max price: ${np.max(prices[:24]):.2f}/MWh")
    print(f"Min price: ${np.min(prices[:24]):.2f}/MWh")

    return SoC, P, revenue, prices[:24]

SoC, P, revenue, prices = battery_arbitrage_example()

```

27.9. Summary

27.9.1. Key Takeaways

- **Dynamic optimization** extends optimization to systems changing over time, governed by differential equations
- **Calculus of variations** provides the Euler-Lagrange equation for finding optimal trajectories
- **Pontryagin's Maximum Principle** gives necessary conditions for optimal control, introducing the Hamiltonian and costates

- **Hamilton-Jacobi-Bellman equation** characterizes the value function through a PDE
- **Linear Quadratic Regulator (LQR)** provides closed-form optimal feedback for linear systems
- **Model Predictive Control (MPC)** makes optimal control practical through receding horizon implementation
- **Linear MPC** solves quadratic programs efficiently; **Nonlinear MPC** requires more sophisticated solvers
- **Robust MPC** (tube, min-max, stochastic) handles uncertainty while guaranteeing constraint satisfaction
- **Numerical methods** include direct (collocation, multiple shooting) and indirect (BVP) approaches
- Python libraries like **CasADi** and **do-mpc** enable practical implementation

27.9.2. Key Formulas

Concept	Formula
Euler-Lagrange	$\frac{\partial L}{\partial y} - \frac{d}{dx} \frac{\partial L}{\partial y'} = 0$
Hamiltonian	$H = L + \lambda^T f$
Costate equation	$\dot{\lambda} = -\frac{\partial H}{\partial x}$
Maximum Principle	$u^* = \arg \max_u H(x, u, \lambda, t)$
Riccati equation	$-\dot{P} = A^T P + P A - P B R^{-1} B^T P + Q$
LQR gain	$K = R^{-1} B^T P$
HJB equation	$-\frac{\partial V}{\partial t} = \min_u [L + \nabla_x V \cdot f]$

27.10. Exercises

27.10.1. Theoretical Problems

1. **Calculus of Variations:** Find the curve $y(x)$ that minimizes $\int_0^1 (y'^2 + y^2) dx$ subject to $y(0) = 0$, $y(1) = 1$. Solve the Euler-Lagrange equation and verify the boundary conditions.
2. **Maximum Principle:** For the system $\dot{x} = u$ with $x(0) = 1$, find the control $u^*(t)$ that minimizes $J = x(1)^2 + \int_0^1 u^2 dt$. Derive the costate equation and transversality condition.
3. **LQR Derivation:** Starting from the Maximum Principle, derive the Riccati differential equation for the finite-horizon LQR problem.
4. **MPC Stability:** Prove that if the terminal cost $V_f(x)$ and constraint set $\mathcal{X}_f$ satisfy the standard assumptions, then the MPC value function $V_N(x)$ is a Lyapunov function for the closed-loop system.

27.10.2. Computational Problems

5. **Brachistochrone:** Implement a direct collocation solver to find the brachistochrone curve. Compare with the analytical cycloid solution.
6. **Double Integrator MPC:** Implement linear MPC for a double integrator with state and control constraints. Plot the closed-loop response and compare with unconstrained LQR.
7. **Pendulum Swing-up:** Use nonlinear MPC to swing up an inverted pendulum from the hanging position. Experiment with different horizons and cost weights.
8. **Tube MPC:** Implement tube MPC for a double integrator with bounded disturbances. Compare performance with nominal MPC when disturbances are present.

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9. **Economic MPC:** Implement MPC for a simple production system where the cost function includes economic terms (profit maximization) rather than pure tracking.
10. **Multi-vehicle Coordination:** Extend the autonomous vehicle MPC to handle multiple vehicles with collision avoidance constraints.

27.11. Further Reading

27.11.1. Textbooks

- **Liberzon, D.** (2012). *Calculus of Variations and Optimal Control Theory*. Princeton University Press. - Rigorous mathematical treatment
- **Kirk, D.E.** (2004). *Optimal Control Theory: An Introduction*. Dover Publications. - Classic introduction to optimal control
- **Rawlings, J.B., Mayne, D.Q., & Diehl, M.** (2017). *Model Predictive Control: Theory, Computation, and Design*. Nob Hill Publishing. - Comprehensive MPC reference
- **Borrelli, F., Bemporad, A., & Morari, M.** (2017). *Predictive Control for Linear and Hybrid Systems*. Cambridge University Press. - MPC theory and algorithms

27.11.2. Research Papers

- **Pontryagin, L.S., et al.** (1962). *The Mathematical Theory of Optimal Processes*. - Original maximum principle paper
- **Mayne, D.Q., et al.** (2000). “Constrained model predictive control: Stability and optimality.” *Automatica*. - Foundational MPC stability results
- **Andersson, J.A.E., et al.** (2019). “CasADi: A software framework for nonlinear optimization and optimal control.” *Mathematical Programming Computation*. - CasADi framework

27.11.3. Online Resources

- **CasADi Documentation:** <https://web.casadi.org/> - Comprehensive tutorials and examples
- **do-mpc Documentation:** <https://www.do-mpc.com/> - MPC framework with examples
- **MATLAB MPC Toolbox:** <https://www.mathworks.com/products/model-predictive-control.html> - Industrial MPC implementation

28. Simulation-Optimization

What You Will Learn

- Understand when simulation-optimization is necessary
- Apply black-box optimization methods to complex systems
- Build and use surrogate models to accelerate optimization
- Implement derivative-free optimization algorithms
- Design effective simulation experiments for optimization
- Combine simulation with mathematical programming

Many real-world optimization problems cannot be expressed as closed-form mathematical programs. When objectives or constraints require running complex simulations, we need simulation-optimization (SimOpt). This chapter covers methods for optimizing systems where the only way to evaluate a solution is to simulate it.

28.1. When Simulation-Optimization is Necessary

28.1.1. The Analytical Gap

Traditional optimization assumes we can write:

$$\text{minimize } f(\mathbf{x}) \text{ subject to } g(\mathbf{x}) \leq 0$$

But what if $f(\mathbf{x})$ and $g(\mathbf{x})$ cannot be expressed analytically?

Scenario	Why Analytical Models Fail	Example
Complex dynamics	Nonlinear interactions	Manufacturing line throughput
Stochastic systems	Random events	Call center staffing
Human behavior	Unpredictable responses	Traffic signal timing
Emergent phenomena	System-level properties	Supply chain resilience

The Black-Box Paradigm

In simulation-optimization, we treat the system as a **black box**:

- **Input:** Decision variables $\mathbf{x}$
- **Output:** Performance measure $f(\mathbf{x})$ (possibly noisy)
- **No access** to gradients, Hessians, or analytical structure

28.1.2. Characteristics of SimOpt Problems

SimOpt problems share several characteristics that distinguish them from standard mathematical programs. The objective function is stochastic (each evaluation produces a noisy estimate), evaluation

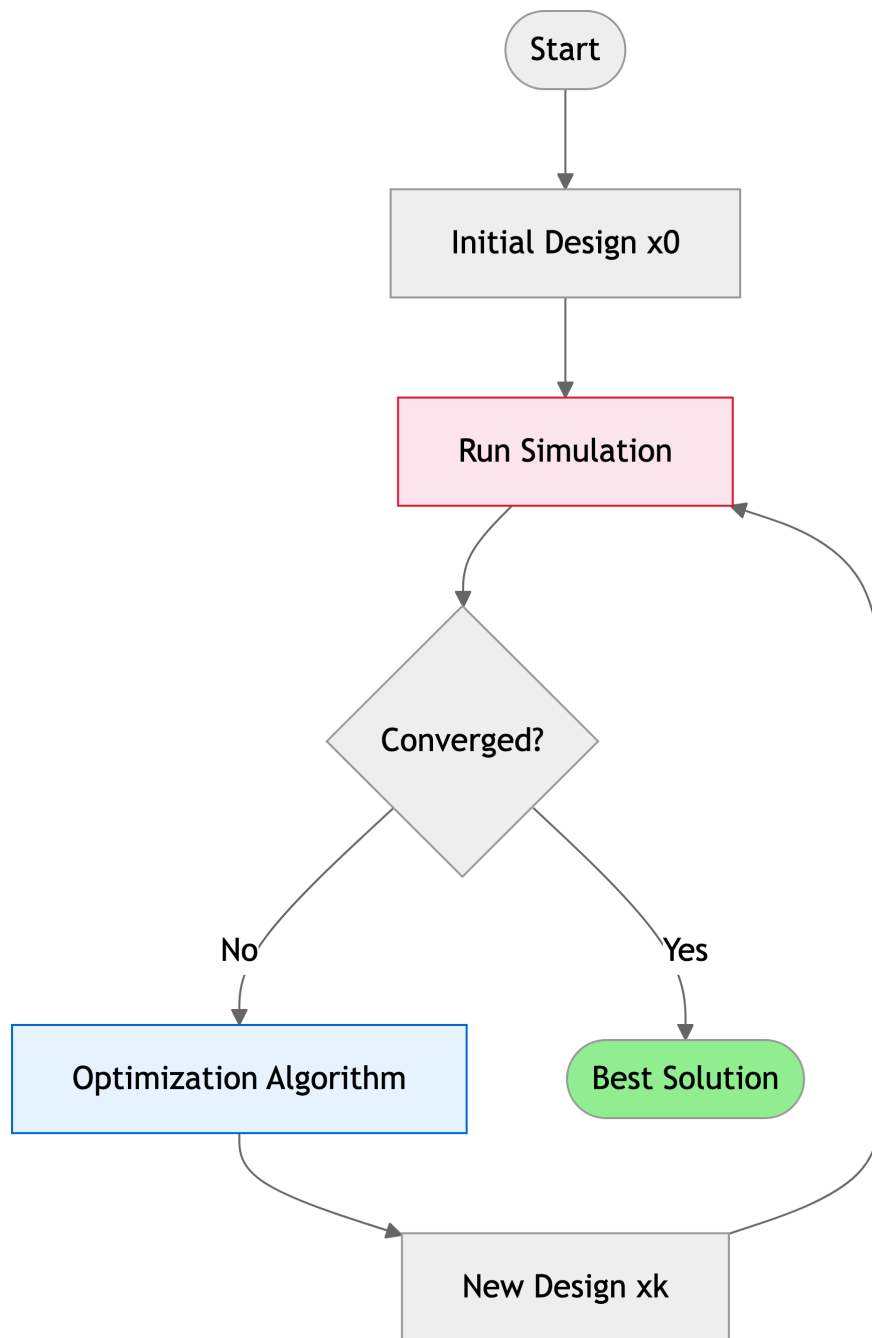


Figure 28.1.: Simulation-Optimization Loop: Iterative search with black-box evaluation

is expensive (each simulation may take minutes or hours), and gradient information is unavailable. The following dataclass defines the structure of a SimOpt problem and provides methods for evaluation with noise management.

```

from dataclasses import dataclass
from typing import Callable, List, Tuple, Optional
import numpy as np

@dataclass
class SimOptProblem:
    """Simulation-optimization problem specification."""
    name: str
    n_variables: int
    bounds: List[Tuple[float, float]]
    objective: Callable # Black-box function
    is_stochastic: bool = True
    evaluation_cost: str = "high" # "low", "medium", "high"
    noise_level: str = "medium" # "low", "medium", "high"

    def evaluate(self, x: np.ndarray, n_replications: int = 1) -> float:
        """Evaluate solution, optionally with replications."""
        if self.is_stochastic and n_replications > 1:
            results = [self.objective(x) for _ in range(n_replications)]
            return np.mean(results)
        return self.objective(x)

    def estimate_noise(self, x: np.ndarray, n_samples: int = 30) -> float:
        """Estimate noise level at a point."""
        samples = [self.objective(x) for _ in range(n_samples)]
        return np.std(samples)

# Example: Simulated manufacturing line
def manufacturing_simulation(x: np.ndarray) -> float:
    """Simulate manufacturing line throughput.

    Args:
        x: [buffer_size_1, buffer_size_2, n_workers]

    Returns:
        Negative throughput (for minimization)
    """
    buffer1, buffer2, workers = x

    # Simplified stochastic simulation
    throughput = 0
    inventory1, inventory2 = buffer1 / 2, buffer2 / 2

    for _ in range(1000): # Simulate 1000 time units
        # Random arrivals and processing
        arrivals = np.random.poisson(5)

        # Station 1 processing

```

```

        process1 = min(inventory1, np.random.poisson(4 + workers * 0.5))
        inventory1 = min(buffer1, inventory1 - process1 + arrivals)

        # Station 2 processing
        process2 = min(inventory2, np.random.poisson(3 + workers * 0.3))
        inventory2 = min(buffer2, inventory2 - process2 + process1)

        throughput += process2

    # Return negative (we minimize, but want max throughput)
    return -throughput

problem = SimOptProblem(
    name="Manufacturing Line Optimization",
    n_variables=3,
    bounds=[(5, 50), (5, 50), (1, 10)],
    objective=manufacturing_simulation,
    is_stochastic=True,
    evaluation_cost="medium"
)

```

28.2. Black-Box Optimization Methods

Black-box methods treat the simulation as an opaque function: they submit inputs and observe outputs without assuming any internal structure. These methods differ in how they choose the next point to evaluate. Simpler methods like random search and pattern search provide baselines, while more sophisticated methods like Nelder-Mead adapt their search geometry based on past evaluations.

28.2.1. Random Search

The simplest baseline: sample randomly and keep the best. Despite its simplicity, random search serves as an important benchmark. Any optimization algorithm that cannot outperform random search on a given problem is not adding value.

```

def random_search(problem: SimOptProblem, n_evaluations: int,
                  n_replications: int = 5) -> Tuple[np.ndarray, float]:
    """Random search optimization.

    Args:
        problem: SimOpt problem
        n_evaluations: Budget of function evaluations
        n_replications: Replications per evaluation (for noise)

    Returns:
        Best solution and its value
    """
    best_x = None
    best_f = float('inf')

    for _ in range(n_evaluations):

```

```

# Sample uniformly in bounds
x = np.array([
    np.random.uniform(low, high)
    for low, high in problem.bounds
])

# Evaluate with replications
f = problem.evaluate(x, n_replications)

if f < best_f:
    best_f = f
    best_x = x.copy()

return best_x, best_f

```

28.2.2. Pattern Search (Coordinate Search)

Pattern search (also called coordinate search) is a structured derivative-free method. At each iteration, it probes along each coordinate direction with a fixed step size. If a better point is found, the search moves there. If no improvement is found in any direction, the step size is reduced. The method converges to a local optimum under mild conditions, though convergence can be slow for high-dimensional problems.

```

def pattern_search(problem: SimOptProblem, x0: np.ndarray,
                  initial_step: float = 1.0,
                  contraction: float = 0.5,
                  tolerance: float = 1e-4,
                  max_evaluations: int = 1000,
                  n_replications: int = 5) -> Tuple[np.ndarray, float]:
    """Coordinate pattern search for black-box optimization.

    Args:
        problem: SimOpt problem
        x0: Initial solution
        initial_step: Initial step size
        contraction: Step size reduction factor
        tolerance: Convergence tolerance
        max_evaluations: Maximum function evaluations
        n_replications: Replications per evaluation

    Returns:
        Best solution and its value
    """
    n = problem.n_variables
    x = x0.copy()
    step = initial_step
    f_current = problem.evaluate(x, n_replications)
    evaluations = 1

    while step > tolerance and evaluations < max_evaluations:
        improved = False

```

```

# Search along each coordinate
for i in range(n):
    for direction in [1, -1]:
        x_new = x.copy()
        x_new[i] += direction * step

        # Enforce bounds
        x_new[i] = np.clip(x_new[i],
                           problem.bounds[i][0],
                           problem.bounds[i][1])

        f_new = problem.evaluate(x_new, n_replications)
        evaluations += 1

        if f_new < f_current:
            x = x_new
            f_current = f_new
            improved = True
            break

    if improved:
        break

if not improved:
    step *= contraction

return x, f_current

```

28.2.3. Nelder-Mead Simplex

The Nelder-Mead method maintains a simplex (a geometric figure with $n+1$ vertices in n dimensions) and moves it through the search space using four operations: reflection, expansion, contraction, and shrinkage. The simplex adapts its shape and orientation to the local geometry of the objective function, making it effective for low-dimensional problems (typically $n < 10$).

```

def nelder_mead(problem: SimOptProblem, x0: np.ndarray,
                initial_simplex_size: float = 1.0,
                alpha: float = 1.0,      # Reflection
                gamma: float = 2.0,      # Expansion
                rho: float = 0.5,        # Contraction
                sigma: float = 0.5,      # Shrinkage
                max_evaluations: int = 1000,
                tolerance: float = 1e-6,
                n_replications: int = 5) -> Tuple[np.ndarray, float]:
    """Nelder-Mead simplex method.

    Args:
        problem: SimOpt problem
        x0: Initial point
        initial_simplex_size: Size of initial simplex
        alpha, gamma, rho, sigma: Algorithm parameters
        max_evaluations: Budget
        tolerance: Convergence tolerance
    """

```



```

n_replications: Replications per evaluation

Returns:
    Best solution and its value
"""
n = len(x0)

# Initialize simplex
simplex = [x0.copy()]
for i in range(n):
    point = x0.copy()
    point[i] += initial_simplex_size
    simplex.append(point)

# Evaluate all vertices
f_values = [problem.evaluate(x, n_replications) for x in simplex]
evaluations = n + 1

while evaluations < max_evaluations:
    # Sort by function value
    order = np.argsort(f_values)
    simplex = [simplex[i] for i in order]
    f_values = [f_values[i] for i in order]

    # Check convergence
    if np.std(f_values) < tolerance:
        break

    # Centroid of all but worst
    centroid = np.mean(simplex[:-1], axis=0)

    # Reflection
    x_r = centroid + alpha * (centroid - simplex[-1])
    x_r = np.clip(x_r,
                  [b[0] for b in problem.bounds],
                  [b[1] for b in problem.bounds])
    f_r = problem.evaluate(x_r, n_replications)
    evaluations += 1

    if f_values[0] <= f_r < f_values[-2]:
        simplex[-1] = x_r
        f_values[-1] = f_r
        continue

    # Expansion
    if f_r < f_values[0]:
        x_e = centroid + gamma * (x_r - centroid)
        x_e = np.clip(x_e,
                      [b[0] for b in problem.bounds],
                      [b[1] for b in problem.bounds])
        f_e = problem.evaluate(x_e, n_replications)
        evaluations += 1

```

```

        if f_e < f_r:
            simplex[-1] = x_e
            f_values[-1] = f_e
        else:
            simplex[-1] = x_r
            f_values[-1] = f_r
        continue

    # Contraction
    x_c = centroid + rho * (simplex[-1] - centroid)
    f_c = problem.evaluate(x_c, n_replications)
    evaluations += 1

    if f_c < f_values[-1]:
        simplex[-1] = x_c
        f_values[-1] = f_c
        continue

    # Shrinkage
    for i in range(1, n + 1):
        simplex[i] = simplex[0] + sigma * (simplex[i] - simplex[0])
        f_values[i] = problem.evaluate(simplex[i], n_replications)
        evaluations += 1

    return simplex[0], f_values[0]

```

28.3. Surrogate-Based Optimization

When each simulation takes minutes or hours, evaluating thousands of candidate solutions is impractical. Surrogate-based optimization addresses this by fitting a statistical model (the surrogate) to the observed input-output pairs. The optimizer then searches the cheap surrogate rather than running expensive simulations. New simulations are run only at the most promising points identified by the surrogate, and the model is updated with each new observation.

28.3.1. The Surrogate Optimization Loop

1. Initial sampling: Evaluate n_{initial} points
2. Build surrogate: Fit model to observed data
3. Optimize acquisition: Find promising point on surrogate
4. Evaluate: Run actual simulation at promising point
5. Update: Add new point to data, rebuild surrogate
6. Repeat 2-5 until budget exhausted

28.3.2. Gaussian Process Surrogate

The Gaussian Process (GP) is the most common surrogate model for simulation-optimization. It provides both a prediction (posterior mean) and an uncertainty estimate (posterior variance) at any unobserved point. The uncertainty estimate is used by acquisition functions like Expected Improvement to decide where to evaluate next. Points with high uncertainty or low predicted values receive priority.

```

from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import Matern, RBF, WhiteKernel
from scipy.optimize import minimize
from scipy.stats import norm

class SurrogateOptimizer:
    """Surrogate-based optimization with Gaussian Process."""

    def __init__(self, problem: SimOptProblem,
                 kernel=None,
                 n_initial: int = 10):
        self.problem = problem
        self.n_initial = n_initial

        if kernel is None:
            kernel = Matern(nu=2.5) + WhiteKernel(noise_level=0.1)

        self.gp = GaussianProcessRegressor(
            kernel=kernel,
            normalize_y=True,
            n_restarts_optimizer=5
        )

        self.X_observed = []
        self.y_observed = []

    def initial_sampling(self, method: str = "latin_hypercube"):
        """Generate initial design points."""
        n = self.problem.n_variables

        if method == "latin_hypercube":
            # Latin Hypercube Sampling
            samples = np.zeros((self.n_initial, n))
            for i in range(n):
                low, high = self.problem.bounds[i]
                perm = np.random.permutation(self.n_initial)
                samples[:, i] = low + (perm + np.random.rand(self.n_initial)) *
↪ (high - low) / self.n_initial
            else:
                # Random sampling
                samples = np.array([
                    np.random.uniform(low, high) for low, high in self.problem.bounds
                ] for _ in range(self.n_initial))

        # Evaluate all initial points
        for x in samples:
            y = self.problem.evaluate(x)
            self.X_observed.append(x)
            self.y_observed.append(y)

        # Fit GP
        self._fit_gp()

```

```

def _fit_gp(self):
    """Fit Gaussian Process to observed data."""
    X = np.array(self.X_observed)
    y = np.array(self.y_observed)
    self.gp.fit(X, y)

def expected_improvement(self, x: np.ndarray) -> float:
    """Expected Improvement acquisition function."""
    x = x.reshape(1, -1)
    mu, sigma = self.gp.predict(x, return_std=True)

    y_best = np.min(self.y_observed)

    with np.errstate(divide='warn'):
        improvement = y_best - mu
        Z = improvement / sigma
        ei = improvement * norm.cdf(Z) + sigma * norm.pdf(Z)
        ei[sigma == 0.0] = 0.0

    return -ei[0] # Negative for minimization

def lower_confidence_bound(self, x: np.ndarray,
                           kappa: float = 2.0) -> float:
    """Lower Confidence Bound acquisition function."""
    x = x.reshape(1, -1)
    mu, sigma = self.gp.predict(x, return_std=True)
    return mu[0] - kappa * sigma[0]

def optimize_acquisition(self, acquisition: str = "EI",
                        n_restarts: int = 10) -> np.ndarray:
    """Find point that maximizes acquisition function."""
    if acquisition == "EI":
        acq_func = self.expected_improvement
    else:
        acq_func = self.lower_confidence_bound

    best_x = None
    best_acq = float('inf')

    bounds = self.problem.bounds

    for _ in range(n_restarts):
        x0 = np.array([np.random.uniform(low, high)
                       for low, high in bounds])

        result = minimize(
            acq_func,
            x0,
            bounds=bounds,
            method='L-BFGS-B'
        )

```

```

        if result.fun < best_acq:
            best_acq = result.fun
            best_x = result.x

    return best_x

def optimize(self, n_iterations: int = 50,
             acquisition: str = "EI") -> Tuple[np.ndarray, float]:
    """Run surrogate optimization loop."""
    # Initial sampling
    self.initial_sampling()

    for iteration in range(n_iterations):
        # Find promising point
        x_new = self.optimize_acquisition(acquisition)

        # Evaluate simulation
        y_new = self.problem.evaluate(x_new)

        # Update data
        self.X_observed.append(x_new)
        self.y_observed.append(y_new)

        # Refit GP
        self._fit_gp()

        # Progress
        if iteration % 10 == 0:
            best_idx = np.argmin(self.y_observed)
            print(f"Iteration {iteration}: best =
                  ↪ {self.y_observed[best_idx]:.4f}")

    # Return best found
    best_idx = np.argmin(self.y_observed)
    return np.array(self.X_observed[best_idx]), self.y_observed[best_idx]

```

28.3.3. Visualizing Optimization Progress

Tracking how the optimization algorithm converges over iterations is essential for understanding its behavior and determining when to stop. The following visualization demonstrates the convergence of surrogate-based optimization.

28.3.4. Radial Basis Function Surrogate

Radial Basis Functions (RBFs) provide an alternative to Gaussian Processes. RBF surrogates interpolate exactly through observed data points and are computationally cheaper to fit for large datasets. However, they do not provide the uncertainty estimates that GPs offer, which limits their use with sophisticated acquisition functions. RBF surrogates work well when combined with simpler infill strategies.

```

from scipy.interpolate import RBFInterpolator

class RBFSurrogate:

```

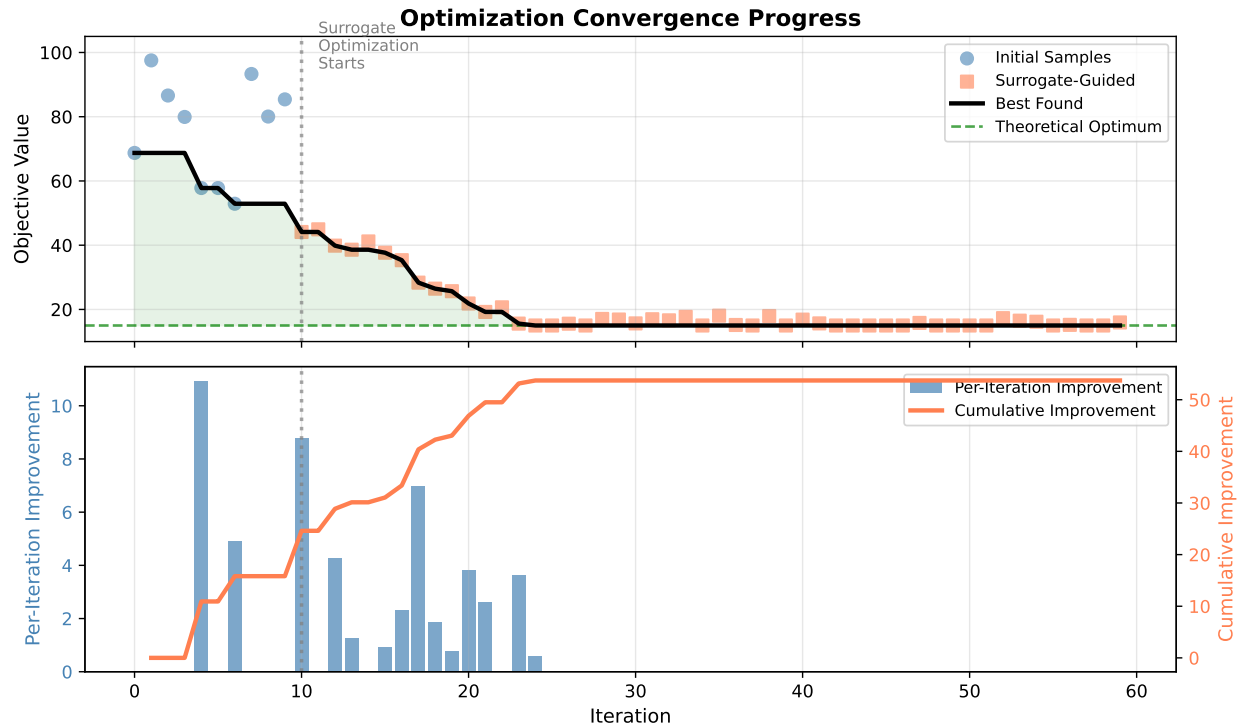


Figure 28.2.: Simulation Optimization Convergence: Best Value and Acquisition Progress Over Iterations

```

"""Radial Basis Function surrogate model."""

def __init__(self, kernel: str = "thin_plate_spline"):
    self.kernel = kernel
    self.model = None
    self.X = None
    self.y = None

def fit(self, X: np.ndarray, y: np.ndarray):
    """Fit RBF model to data."""
    self.X = X
    self.y = y
    self.model = RBFInterpolator(X, y, kernel=self.kernel)

def predict(self, X: np.ndarray) -> np.ndarray:
    """Predict at new points."""
    return self.model(X)

def update(self, x_new: np.ndarray, y_new: float):
    """Add new point and refit."""
    self.X = np.vstack([self.X, x_new.reshape(1, -1)])
    self.y = np.append(self.y, y_new)
    self.model = RBFInterpolator(self.X, self.y, kernel=self.kernel)

```

```

class RBFOptimizer:
    """RBF-based surrogate optimization."""

    def __init__(self, problem: SimOptProblem, n_initial: int = 20):
        self.problem = problem
        self.n_initial = n_initial
        self.surrogate = RBFSSurrogate()

    def optimize(self, n_iterations: int = 100) -> Tuple[np.ndarray, float]:
        """Optimize using RBF surrogate."""
        # Initial sampling
        X = np.array([
            np.random.uniform(low, high) for low, high in self.problem.bounds
            for _ in range(self.n_initial)
        ])
        y = np.array([self.problem.evaluate(x) for x in X])

        self.surrogate.fit(X, y)

        for _ in range(n_iterations):
            # Optimize surrogate
            result = minimize(
                lambda x: self.surrogate.predict(x.reshape(1, -1))[0],
                X[np.argmin(y)],
                bounds=self.problem.bounds,
                method='L-BFGS-B'
            )
            x_new = result.x

            # Evaluate and update
            y_new = self.problem.evaluate(x_new)
            self.surrogate.update(x_new, y_new)
            X = np.vstack([X, x_new])
            y = np.append(y, y_new)

        best_idx = np.argmin(y)
        return X[best_idx], y[best_idx]

```

28.4. Stochastic Optimization Methods

When simulations are noisy, deterministic surrogate methods can be misled by random fluctuations. Stochastic optimization methods are designed specifically for noisy evaluations. They either average over multiple replications to reduce noise (Sample Average Approximation) or use gradient estimates that are robust to noise (stochastic approximation).

28.4.1. Visualizing Simulation Output Distributions

When running stochastic simulations, understanding the distribution of outputs is essential for decision-making. The following visualization shows how to display simulation results with confidence intervals.

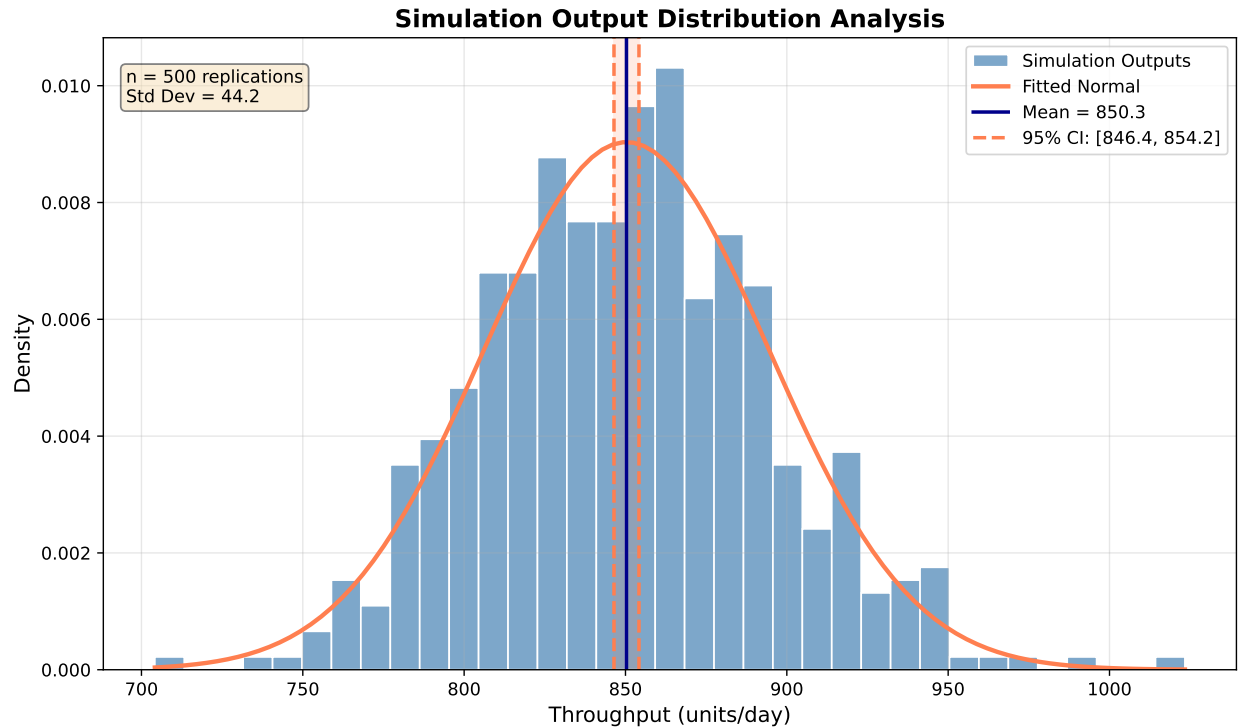


Figure 28.3.: Distribution of Simulation Outputs with 95% Confidence Interval

28.4.2. Sample Average Approximation

SAA handles the noise in simulation outputs by averaging multiple replications. For each candidate solution, the algorithm generates a batch of random samples and optimizes the average objective value. Solving multiple independent SAA problems with different random seeds and selecting the best candidate improves solution quality. The implementation below solves n_{outer} independent SAA problems, each using n_{inner} samples, and returns the best solution validated on a larger sample.

```
def sample_average_approximation(problem: SimOptProblem,
                                x: np.ndarray,
                                n_outer: int = 20,
                                n_inner: int = 100,
                                optimizer=None) -> Tuple[np.ndarray, float]:
    """Sample Average Approximation for stochastic optimization.

    Args:
        problem: SimOpt problem
        x: Initial solution
        n_outer: Number of SAA problems to solve
        n_inner: Sample size per SAA problem
        optimizer: Inner optimizer (default: scipy.minimize)

    Returns:
        Best solution and estimated objective
    """
    solutions = []
```



```

objectives = []

for _ in range(n_outer):
    # Generate sample scenarios
    def saa_objective(x):
        return np.mean([problem.objective(x) for _ in range(n_inner)])

    # Optimize SAA problem
    result = minimize(
        saa_objective,
        x,
        bounds=problem.bounds,
        method='L-BFGS-B'
    )

    solutions.append(result.x)
    objectives.append(result.fun)

# Select best solution
best_idx = np.argmin(objectives)
best_x = solutions[best_idx]

# Evaluate with more samples for better estimate
best_f = problem.evaluate(best_x, n_replications=n_inner * 5)

return best_x, best_f

```

28.4.3. Stochastic Approximation

Simultaneous Perturbation Stochastic Approximation (SPSA) estimates the gradient using only two function evaluations per iteration, regardless of the problem dimension. It perturbs all variables simultaneously along a random direction, evaluates the function at the perturbed points, and computes a gradient approximation from the difference. The gain sequences a_k and c_k decay over iterations, ensuring convergence to a local optimum.

```

def simultaneous_perturbation(problem: SimOptProblem,
                             x0: np.ndarray,
                             n_iterations: int = 1000,
                             a: float = 0.1,
                             c: float = 0.1,
                             alpha: float = 0.602,
                             gamma: float = 0.101) -> Tuple[np.ndarray, float]:
    """Simultaneous Perturbation Stochastic Approximation (SPSA).

    Args:
        problem: SimOpt problem
        x0: Initial solution
        n_iterations: Number of iterations
        a, c: Gain sequence parameters
        alpha, gamma: Decay rates

    Returns:

```

```

        Best solution and its value
    """
    n = len(x0)
    x = x0.copy()

    best_x = x.copy()
    best_f = problem.evaluate(x)

    for k in range(1, n_iterations + 1):
        # Gain sequences
        a_k = a / (k + 1) ** alpha
        c_k = c / (k + 1) ** gamma

        # Random perturbation direction
        delta = 2 * np.random.randint(0, 2, n) - 1 # +1 or -1

        # Perturbed evaluations
        x_plus = np.clip(x + c_k * delta,
                        [b[0] for b in problem.bounds],
                        [b[1] for b in problem.bounds])
        x_minus = np.clip(x - c_k * delta,
                        [b[0] for b in problem.bounds],
                        [b[1] for b in problem.bounds])

        f_plus = problem.evaluate(x_plus)
        f_minus = problem.evaluate(x_minus)

        # Gradient estimate
        g_hat = (f_plus - f_minus) / (2 * c_k * delta)

        # Update
        x = x - a_k * g_hat
        x = np.clip(x,
                    [b[0] for b in problem.bounds],
                    [b[1] for b in problem.bounds])

        # Track best
        f_current = problem.evaluate(x)
        if f_current < best_f:
            best_f = f_current
            best_x = x.copy()

    return best_x, best_f

```

28.5. Combining Simulation with Mathematical Programming

In many applications, the problem has both an analytically tractable part and a simulation-dependent part. Hybrid methods combine mathematical programming for the tractable decisions with simulation for the parts that resist analytical modeling. Two common architectures are decomposition (Benders-like cuts from simulation) and iterative validation (optimize a simplified model, validate with detailed simulation, update the model).

28.5.1. Simulation-Optimization Architecture

The following framework implements both decomposition-based and iterative hybrid approaches. The decomposition approach generates linear cuts from simulation sensitivity analysis, while the hybrid approach alternates between optimization and simulation validation.

```
class SimOptFramework:
    """Framework combining simulation with optimization."""

    def __init__(self, simulator: Callable,
                 optimizer_type: str = "surrogate"):
        self.simulator = simulator
        self.optimizer_type = optimizer_type
        self.history = []

    def decomposition_approach(self, master_problem,
                              subproblem_simulator,
                              max_iterations: int = 50,
                              tolerance: float = 1e-4):
        """Benders-like decomposition with simulation subproblems.

        Master: Mathematical program for strategic decisions
        Subproblem: Simulation for operational evaluation
        """
        upper_bound = float('inf')
        lower_bound = float('-inf')
        cuts = []

        for iteration in range(max_iterations):
            # Solve master problem with cuts
            x_master, lb = master_problem.solve(cuts)
            lower_bound = max(lower_bound, lb)

            # Evaluate via simulation
            sim_result = subproblem_simulator(x_master)
            ub = sim_result['objective']
            upper_bound = min(upper_bound, ub)

            # Check convergence
            gap = (upper_bound - lower_bound) / abs(upper_bound)
            if gap < tolerance:
                break

            # Generate cut from simulation sensitivity
            cut = self._generate_cut(x_master, sim_result)
            cuts.append(cut)

        self.history.append({
            'iteration': iteration,
            'lower_bound': lower_bound,
            'upper_bound': upper_bound,
            'gap': gap
        })
```

```

        return x_master, upper_bound

    def _generate_cut(self, x, sim_result):
        """Generate optimality cut from simulation."""
        # Use finite differences or adjoint methods
        gradient = sim_result.get('gradient', np.zeros_like(x))
        intercept = sim_result['objective'] - np.dot(gradient, x)
        return {'gradient': gradient, 'intercept': intercept}

class HybridSimOpt:
    """Hybrid approach: optimization + simulation validation."""

    def __init__(self, optimization_model, simulator):
        self.opt_model = optimization_model
        self.simulator = simulator

    def solve(self, n_iterations: int = 10):
        """Iterative optimization-simulation loop."""
        x_best = None
        f_best = float('inf')

        for iteration in range(n_iterations):
            # Solve optimization model
            x_opt, f_opt = self.opt_model.solve()

            # Validate with detailed simulation
            sim_result = self.simulator(x_opt)

            # Check if simulation confirms optimization
            if sim_result['feasible'] and sim_result['objective'] < f_best:
                f_best = sim_result['objective']
                x_best = x_opt

            # Update optimization model with simulation insights
            self.opt_model.update_parameters(sim_result['feedback'])

        return x_best, f_best

```

28.6. Ranking and Selection

When the decision space is discrete—choosing among a finite set of system designs, policies, or configurations—the optimization problem reduces to identifying the best alternative with statistical confidence. Ranking and Selection (R&S) methods provide formal guarantees on the probability of correct selection. The indifference-zone approach guarantees that the selected alternative is within δ of the true best with probability at least $1 - \alpha$.

```

def ranking_selection_indifference_zone(alternatives: List,
                                       simulator: Callable,
                                       delta: float, # Indifference zone
                                       alpha: float = 0.05,
                                       n_initial: int = 20) -> int:

```

```

"""Indifference-zone ranking and selection.

Find the best alternative with probability  $\geq 1-\alpha$ ,
assuming best is at least  $\delta$  better than others.

Args:
    alternatives: List of alternatives to compare
    simulator: Function(alt) -> performance sample
    delta: Minimum meaningful difference
    alpha: Error probability
    n_initial: Initial samples per alternative

Returns:
    Index of selected best alternative
"""
k = len(alternatives)

# Initial sampling
samples = {i: [simulator(alternatives[i])
               for _ in range(n_initial)]
           for i in range(k)}

# Continue until confident
while True:
    # Compute means and variances
    means = {i: np.mean(samples[i]) for i in range(k)}
    variances = {i: np.var(samples[i], ddof=1) for i in range(k)}
    n_samples = {i: len(samples[i]) for i in range(k)}

    # Find current best
    best = min(means.keys(), key=lambda i: means[i])

    # Check if we can eliminate others
    eliminated = set()
    for i in range(k):
        if i == best:
            continue

        # Paired comparison
        diff = means[i] - means[best]
        se = np.sqrt(variances[i]/n_samples[i] +
                     variances[best]/n_samples[best])

        # Can we eliminate i?
        if diff > delta + 2 * se: # Simplified criterion
            eliminated.add(i)

    # If only one remains, we're done
    if len(eliminated) == k - 1:
        return best

    # Add more samples to surviving alternatives
    for i in range(k):

```

```

        if i not in eliminated:
            samples[i].append(simulator(alternatives[i]))

    # Safety check
    if min(n_samples.values()) > 10000:
        return best

def optimal_computing_budget_allocation(alternatives: List,
                                       simulator: Callable,
                                       total_budget: int,
                                       n_initial: int = 10) -> int:
    """OCBA: Allocate simulation budget optimally.

    Args:
        alternatives: List of alternatives
        simulator: Simulation function
        total_budget: Total simulation replications allowed
        n_initial: Initial samples per alternative

    Returns:
        Index of estimated best alternative
    """
    k = len(alternatives)

    # Initial sampling
    samples = {i: [simulator(alternatives[i])
                   for _ in range(n_initial)]
               for i in range(k)}
    budget_used = k * n_initial

    while budget_used < total_budget:
        # Compute statistics
        means = {i: np.mean(samples[i]) for i in range(k)}
        stds = {i: np.std(samples[i], ddof=1) for i in range(k)}
        n_i = {i: len(samples[i]) for i in range(k)}

        # Find best and second best
        sorted_alts = sorted(range(k), key=lambda i: means[i])
        best = sorted_alts[0]

        # OCBA allocation ratios
        ratios = {}
        for i in range(k):
            if i == best:
                continue
            ratios[i] = (stds[i] / (means[i] - means[best])) ** 2

        ratio_sum = sum(ratios.values())
        ratios[best] = stds[best] * np.sqrt(
            sum((ratios[i] / stds[i]) ** 2 for i in ratios if i != best)
        )

```

```
# Normalize
total_ratio = sum(ratios.values())
for i in ratios:
    ratios[i] /= total_ratio

# Allocate next batch
batch_size = min(10, total_budget - budget_used)
for i in range(k):
    n_add = max(1, int(batch_size * ratios.get(i, 1/k)))
    for _ in range(n_add):
        if budget_used >= total_budget:
            break
        samples[i].append(simulator(alternatives[i]))
        budget_used += 1

# Return best based on final means
means = {i: np.mean(samples[i]) for i in range(k)}
return min(means.keys(), key=lambda i: means[i])
```

28.7. Discrete Event Simulation

Discrete Event Simulation (DES) is a powerful paradigm for modeling systems where state changes occur at discrete points in time. Unlike continuous simulation, DES focuses on events that trigger state transitions, making it ideal for queueing systems, manufacturing, healthcare, and logistics.

28.7.1. The DES Paradigm

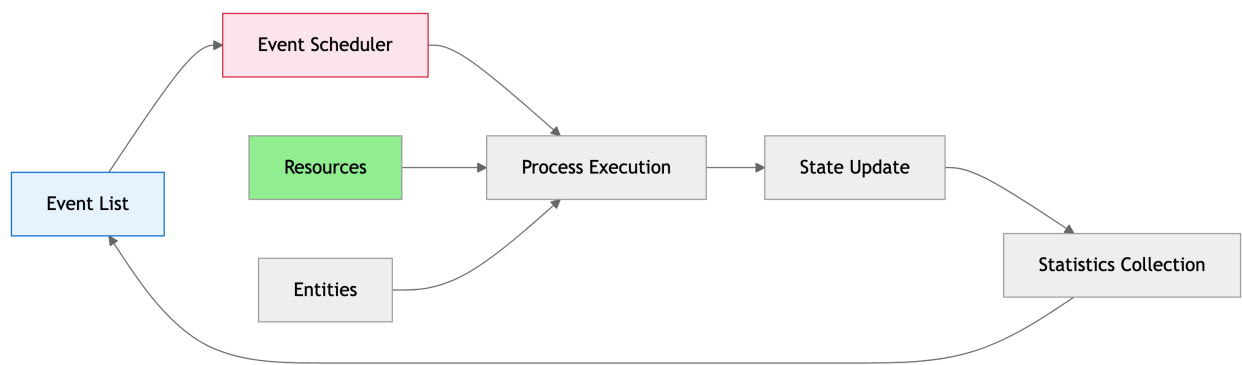


Figure 28.4.: Core Components of Discrete Event Simulation

Key DES concepts:

Concept	Description	Example
Entity	Object flowing through system	Customer, job, patient
Resource	Shared capacity constraint	Server, machine, bed
Event	Instantaneous state change	Arrival, service start, departure
Queue	Waiting line for resources	Customer queue, job buffer
Process	Sequence of activities	Customer service process
Attribute	Entity characteristic	Priority, arrival time

28.7.2. SimPy Framework

SimPy is a process-based discrete-event simulation framework for Python. It uses Python generators to model processes that request and release shared resources. Entities flow through the system as generator functions that yield simulation events (resource requests, timeouts). SimPy's event-driven kernel manages the simulation clock and event scheduling automatically.

The following implementation models an M/M/c queueing system (Poisson arrivals, exponential service times, c parallel servers) and collects standard performance metrics: waiting times, queue lengths, and server utilization.

```
import simpy
import random
import numpy as np
from typing import List, Dict, Tuple, Generator
from dataclasses import dataclass, field
from collections import defaultdict

@dataclass
class SimulationMetrics:
    """Container for simulation performance metrics."""
    wait_times: List[float] = field(default_factory=list)
    service_times: List[float] = field(default_factory=list)
    system_times: List[float] = field(default_factory=list)
    queue_lengths: List[Tuple[float, int]] = field(default_factory=list)
    utilization_samples: List[Tuple[float, float]] = field(default_factory=list)

    @property
    def avg_wait_time(self) -> float:
        return np.mean(self.wait_times) if self.wait_times else 0.0

    @property
    def avg_system_time(self) -> float:
        return np.mean(self.system_times) if self.system_times else 0.0

    @property
    def avg_queue_length(self) -> float:
        if not self.queue_lengths:
            return 0.0
        # Time-weighted average
        total_time = self.queue_lengths[-1][0]
        weighted_sum = sum(
            length * (self.queue_lengths[i+1][0] - t)
            for i, (t, length) in enumerate(self.queue_lengths[:-1])
        )
        return weighted_sum / total_time if total_time > 0 else 0.0

class QueueingSimulation:
    """M/M/c queueing system simulation using SimPy."""

    def __init__(self,
                 n_servers: int = 2,
                 arrival_rate: float = 1.0,
```



```

        service_rate: float = 1.2,
        random_seed: int = 42):
    """Initialize queueing simulation.

    Args:
        n_servers: Number of parallel servers (c)
        arrival_rate: Mean arrivals per time unit (lambda)
        service_rate: Mean service completions per time unit (mu)
        random_seed: Random seed for reproducibility
    """
    self.n_servers = n_servers
    self.arrival_rate = arrival_rate
    self.service_rate = service_rate
    self.random_seed = random_seed

    self.env: simpy.Environment = None
    self.server: simpy.Resource = None
    self.metrics = SimulationMetrics()
    self.customer_count = 0

def customer(self, name: str) -> Generator:
    """Customer process: arrive, wait, get served, depart.

    Args:
        name: Customer identifier

    Yields:
        SimPy events for queueing and service
    """
    arrival_time = self.env.now
    queue_length = len(self.server.queue)
    self.metrics.queue_lengths.append((arrival_time, queue_length))

    # Request server (wait in queue if busy)
    with self.server.request() as request:
        yield request

        # Record wait time
        wait_time = self.env.now - arrival_time
        self.metrics.wait_times.append(wait_time)

        # Service time (exponential)
        service_time = random.expovariate(self.service_rate)
        self.metrics.service_times.append(service_time)

        yield self.env.timeout(service_time)

        # Record total system time
        system_time = self.env.now - arrival_time
        self.metrics.system_times.append(system_time)

    # Update queue length after departure
    self.metrics.queue_lengths.append(

```

```

        (self.env.now, len(self.server.queue))
    )

def arrival_process(self, sim_time: float) -> Generator:
    """Generate customer arrivals (Poisson process).

    Args:
        sim_time: Total simulation time

    Yields:
        SimPy timeout events for inter-arrival times
    """
    while self.env.now < sim_time:
        # Exponential inter-arrival time
        inter_arrival = random.expovariate(self.arrival_rate)
        yield self.env.timeout(inter_arrival)

        self.customer_count += 1
        name = f"Customer_{self.customer_count}"
        self.env.process(self.customer(name))

def utilization_monitor(self, interval: float = 1.0) -> Generator:
    """Monitor server utilization over time.

    Args:
        interval: Sampling interval

    Yields:
        SimPy timeout events
    """
    while True:
        utilization = self.server.count / self.server.capacity
        self.metrics.utilization_samples.append(
            (self.env.now, utilization)
        )
        yield self.env.timeout(interval)

def run(self, sim_time: float = 1000.0) -> SimulationMetrics:
    """Run the queueing simulation.

    Args:
        sim_time: Total simulation time

    Returns:
        Simulation metrics
    """
    random.seed(self.random_seed)

    self.env = simpy.Environment()
    self.server = simpy.Resource(self.env, capacity=self.n_servers)
    self.metrics = SimulationMetrics()
    self.customer_count = 0

```

```

        # Start processes
        self.env.process(self.arrival_process(sim_time))
        self.env.process(self.utilization_monitor())

        # Run simulation
        self.env.run(until=sim_time)

        return self.metrics

def run_queue_simulation(n_servers: int = 2,
                        arrival_rate: float = 1.0,
                        service_rate: float = 1.2,
                        sim_time: float = 1000.0,
                        n_replications: int = 10) -> Dict[str, float]:
    """Run multiple replications of queueing simulation.

    Args:
        n_servers: Number of servers
        arrival_rate: Arrival rate (lambda)
        service_rate: Service rate (mu)
        sim_time: Simulation time per replication
        n_replications: Number of independent replications

    Returns:
        Dictionary of averaged performance metrics
    """
    all_wait_times = []
    all_system_times = []
    all_queue_lengths = []

    for rep in range(n_replications):
        sim = QueueingSimulation(
            n_servers=n_servers,
            arrival_rate=arrival_rate,
            service_rate=service_rate,
            random_seed=42 + rep
        )
        metrics = sim.run(sim_time)

        all_wait_times.append(metrics.avg_wait_time)
        all_system_times.append(metrics.avg_system_time)
        all_queue_lengths.append(metrics.avg_queue_length)

    return {
        'avg_wait_time': np.mean(all_wait_times),
        'std_wait_time': np.std(all_wait_times),
        'avg_system_time': np.mean(all_system_times),
        'std_system_time': np.std(all_system_times),
        'avg_queue_length': np.mean(all_queue_lengths),
        'utilization': arrival_rate / (n_servers * service_rate)
    }

```

```
# Example usage
if __name__ == "__main__":
    results = run_queue_simulation(
        n_servers=2,
        arrival_rate=1.5,
        service_rate=1.0,
        sim_time=10000.0,
        n_replications=30
    )
    print(f"Average Wait Time: {results['avg_wait_time']:.3f} "
          f"(+/- {results['std_wait_time']:.3f})")
    print(f"Server Utilization: {results['utilization']:.1%}")
```

28.7.3. Advanced SimPy Patterns

28.7.3.1. Priority Queues and Preemption

```
import simpy
from enum import IntEnum
from typing import Optional

class Priority(IntEnum):
    """Job priority levels (lower = higher priority)."""
    URGENT = 1
    HIGH = 2
    NORMAL = 3
    LOW = 4

class PriorityQueueSystem:
    """Priority queueing with optional preemption."""

    def __init__(self, env: simpy.Environment,
                 n_servers: int = 1,
                 preemptive: bool = False):
        self.env = env
        if preemptive:
            self.server = simpy.PreemptiveResource(env, capacity=n_servers)
        else:
            self.server = simpy.PriorityResource(env, capacity=n_servers)
        self.preemptive = preemptive
        self.completed_jobs = []

    def job(self, name: str, priority: Priority,
            service_time: float) -> Generator:
        """Process a job with given priority.

        Args:
            name: Job identifier
            priority: Job priority level
            service_time: Required service time
        """
```

```

Yields:
    SimPy events
    """
    arrival_time = self.env.now
    remaining_time = service_time

    while remaining_time > 0:
        with self.server.request(priority=priority) as req:
            yield req
            start_time = self.env.now

            try:
                yield self.env.timeout(remaining_time)
                remaining_time = 0
            except simpy.Interrupt:
                # Preempted - calculate remaining time
                elapsed = self.env.now - start_time
                remaining_time -= elapsed

    completion_time = self.env.now
    self.completed_jobs.append({
        'name': name,
        'priority': priority,
        'arrival': arrival_time,
        'completion': completion_time,
        'flow_time': completion_time - arrival_time
    })

def priority_queue_example():
    """Demonstrate priority queue simulation."""
    env = simpy.Environment()
    system = PriorityQueueSystem(env, n_servers=1, preemptive=True)

    def arrivals():
        # Normal job arrives first
        env.process(system.job("Job_Normal", Priority.NORMAL, 10))
        yield env.timeout(2)

        # Urgent job arrives and preempts
        env.process(system.job("Job_Urgent", Priority.URGENT, 3))
        yield env.timeout(5)

        # Low priority job
        env.process(system.job("Job_Low", Priority.LOW, 2))

    env.process(arrivals())
    env.run()

    return system.completed_jobs

```

28.7.3.2. Resource Pools and Containers

```

import simpy

class ManufacturingCell:
    """Manufacturing cell with multiple resource types."""

    def __init__(self, env: simpy.Environment):
        self.env = env

        # Different resource types
        self.machines = simpy.Resource(env, capacity=3)
        self.operators = simpy.Resource(env, capacity=2)
        self.tools = simpy.Container(env, capacity=10, init=10)
        self.raw_material = simpy.Container(env, capacity=100, init=50)
        self.finished_goods = simpy.Container(env, capacity=100, init=0)

        # Store for flexible resource management
        self.part_buffer = simpy.Store(env, capacity=20)

    def production_job(self, job_id: int, units: int) -> Generator:
        """Execute a production job.

        Args:
            job_id: Job identifier
            units: Number of units to produce

        Yields:
            SimPy events for resource acquisition and processing
        """
        # Get raw materials
        yield self.raw_material.get(units)

        # Acquire machine and operator simultaneously
        with self.machines.request() as machine_req:
            with self.operators.request() as operator_req:
                yield machine_req & operator_req # Wait for both

                # Use tool during processing
                yield self.tools.get(1)

                # Processing time
                process_time = units * random.uniform(1, 2)
                yield self.env.timeout(process_time)

                # Return tool
                yield self.tools.put(1)

        # Store finished goods
        yield self.finished_goods.put(units)

    def material_replenishment(self, reorder_point: int = 20,
                              order_quantity: int = 30,
                              lead_time: float = 10.0) -> Generator:

```

```

    """Automatic material replenishment process."""
    while True:
        # Check inventory level
        if self.raw_material.level <= reorder_point:
            yield self.env.timeout(lead_time)
            yield self.raw_material.put(order_quantity)
        yield self.env.timeout(1.0) # Check interval

```

28.7.4. Queueing Theory Integration

Validating simulation results against analytical queueing theory is essential for model verification.

28.7.4.1. M/M/c Queue Formulas

For an M/M/c queue with arrival rate λ and service rate μ :

$$\rho = \frac{\lambda}{c\mu}$$

where ρ is the server utilization. The probability of zero customers in the system:

$$P_0 = \left[\sum_{n=0}^{c-1} \frac{(c\rho)^n}{n!} + \frac{(c\rho)^c}{c!} \cdot \frac{1}{1-\rho} \right]^{-1}$$

The Erlang-C formula gives the probability of waiting:

$$P_{\text{wait}} = \frac{(c\rho)^c}{c!(1-\rho)} \cdot P_0$$

```

import math
from scipy.special import factorial

def mmc_queue_analytics(arrival_rate: float,
                        service_rate: float,
                        n_servers: int) -> Dict[str, float]:
    """Calculate M/M/c queueing metrics analytically.

    Args:
        arrival_rate: Lambda - arrival rate
        service_rate: Mu - service rate per server
        n_servers: c - number of servers

    Returns:
        Dictionary of analytical performance metrics
    """
    lam = arrival_rate
    mu = service_rate
    c = n_servers
    rho = lam / (c * mu) # Server utilization

    if rho >= 1:
        return {
            'utilization': rho,
            'stable': False,

```

```

        'message': 'System is unstable (rho >= 1)'
    }

    # Calculate P0
    sum_terms = sum((c * rho) ** n / factorial(n) for n in range(c))
    last_term = ((c * rho) ** c / factorial(c)) * (1 / (1 - rho))
    P0 = 1 / (sum_terms + last_term)

    # Erlang-C: Probability of waiting
    P_wait = ((c * rho) ** c / (factorial(c) * (1 - rho))) * P0

    # Expected number in queue (Lq)
    Lq = P_wait * rho / (1 - rho)

    # Expected wait time in queue (Wq) - Little's Law
    Wq = Lq / lam

    # Expected number in system (L)
    L = Lq + lam / mu

    # Expected time in system (W)
    W = Wq + 1 / mu

    return {
        'utilization': rho,
        'stable': True,
        'P0': P0,
        'P_wait': P_wait,
        'Lq': Lq, # Avg queue length
        'L': L, # Avg number in system
        'Wq': Wq, # Avg wait time
        'W': W # Avg time in system
    }

def validate_simulation_with_theory(sim_results: Dict[str, float],
                                   theory_results: Dict[str, float],
                                   tolerance: float = 0.1) -> Dict[str, bool]:
    """Compare simulation results with analytical theory.

    Args:
        sim_results: Simulation output metrics
        theory_results: Analytical calculations
        tolerance: Acceptable relative error

    Returns:
        Validation results for each metric
    """
    validations = {}

    # Map simulation metrics to theory metrics
    comparisons = [
        ('avg_wait_time', 'Wq'),

```



```

        ('avg_system_time', 'W'),
        ('avg_queue_length', 'Lq'),
        ('utilization', 'utilization')
    ]

    for sim_key, theory_key in comparisons:
        if sim_key in sim_results and theory_key in theory_results:
            sim_val = sim_results[sim_key]
            theory_val = theory_results[theory_key]

            if theory_val > 0:
                rel_error = abs(sim_val - theory_val) / theory_val
                validations[sim_key] = {
                    'simulation': sim_val,
                    'theory': theory_val,
                    'relative_error': rel_error,
                    'valid': rel_error <= tolerance
                }

    return validations

# Example: Validate simulation against theory
def run_validation_example():
    """Run simulation and validate against M/M/c theory."""
    # Parameters
    arrival_rate = 4.0
    service_rate = 2.0
    n_servers = 3

    # Run simulation
    sim_results = run_queue_simulation(
        n_servers=n_servers,
        arrival_rate=arrival_rate,
        service_rate=service_rate,
        sim_time=50000.0,
        n_replications=50
    )

    # Calculate theoretical values
    theory = mmc_queue_analytics(arrival_rate, service_rate, n_servers)

    # Validate
    validation = validate_simulation_with_theory(sim_results, theory)

    print("Validation Results:")
    print("-" * 60)
    for metric, result in validation.items():
        status = "PASS" if result['valid'] else "FAIL"
        print(f"{metric}:")
        print(f"  Simulation: {result['simulation']:.4f}")
        print(f"  Theory:    {result['theory']:.4f}")
        print(f"  Error:     {result['relative_error']:.2%} [{status}]")

```

```
return validation
```

28.7.4.2. Little's Law

One of the most powerful results in queueing theory:

$$L = \lambda W$$

where: - L = average number of customers in the system - λ = average arrival rate - W = average time a customer spends in the system

This relationship holds for any queueing system in steady state, regardless of arrival distribution, service distribution, or number of servers.

```
def verify_littles_law(metrics: SimulationMetrics,
                      arrival_rate: float,
                      tolerance: float = 0.05) -> Dict[str, float]:
    """Verify Little's Law from simulation data.

    Args:
        metrics: Simulation metrics
        arrival_rate: Actual arrival rate (lambda)
        tolerance: Acceptable deviation

    Returns:
        Verification results
    """
    L = metrics.avg_queue_length + arrival_rate / metrics.avg_service_time
    W = metrics.avg_system_time
    lambda_effective = arrival_rate # Assuming stable system

    # Little's Law: L = lambda * W
    lhs = L
    rhs = lambda_effective * W

    relative_error = abs(lhs - rhs) / max(lhs, rhs)

    return {
        'L_observed': lhs,
        'lambda_W_computed': rhs,
        'relative_error': relative_error,
        'littles_law_holds': relative_error <= tolerance
    }
```

28.8. Application Domains

Discrete event simulation finds applications across numerous domains. This section presents detailed implementations for key application areas.

28.8.1. Manufacturing Systems

Manufacturing simulation helps optimize production lines, buffer sizing, and identify bottlenecks.

```

import simpy
import random
from dataclasses import dataclass
from typing import List, Dict, Optional, Tuple
import numpy as np

@dataclass
class WorkstationConfig:
    """Configuration for a manufacturing workstation."""
    name: str
    processing_time_mean: float
    processing_time_std: float
    reliability: float = 0.95 # Probability of not breaking down
    mtbf: float = 100.0 # Mean time between failures
    mtrr: float = 5.0 # Mean time to repair
    n_parallel: int = 1

@dataclass
class ProductionMetrics:
    """Manufacturing simulation metrics."""
    throughput: int = 0
    lead_times: List[float] = None
    wip_levels: List[Tuple[float, int]] = None
    station_utilizations: Dict[str, List[float]] = None
    bottleneck_station: str = None

    def __post_init__(self):
        self.lead_times = self.lead_times or []
        self.wip_levels = self.wip_levels or []
        self.station_utilizations = self.station_utilizations or {}

class ManufacturingLine:
    """Multi-station manufacturing line simulation."""

    def __init__(self, env: simpy.Environment,
                 stations: List[WorkstationConfig],
                 buffer_capacities: List[int]):
        """Initialize manufacturing line.

        Args:
            env: SimPy environment
            stations: List of workstation configurations
            buffer_capacities: Buffer capacity between each station
        """
        self.env = env
        self.stations = stations
        self.n_stations = len(stations)

        # Create resources for each station
        self.workstations = [
            simpy.Resource(env, capacity=s.n_parallel)

```

```

        for s in stations
    ]

    # Create buffers (stores) between stations
    self.buffers = [
        simpy.Store(env, capacity=cap)
        for cap in buffer_capacities
    ]

    # Metrics tracking
    self.metrics = ProductionMetrics()
    self.active_jobs = 0

def job(self, job_id: int) -> Generator:
    """Process a job through all stations.

    Args:
        job_id: Job identifier

    Yields:
        SimPy events for processing at each station
    """
    arrival_time = self.env.now
    self.active_jobs += 1
    self.metrics.wip_levels.append((self.env.now, self.active_jobs))

    for i, (station, config) in enumerate(zip(self.workstations,
                                              self.stations)):

        # Wait for workstation
        with station.request() as req:
            yield req

            # Track utilization
            if config.name not in self.metrics.station_utilizations:
                self.metrics.station_utilizations[config.name] = []
            self.metrics.station_utilizations[config.name].append(
                station.count / station.capacity
            )

            # Process with variability
            process_time = max(0.1, random.gauss(
                config.processing_time_mean,
                config.processing_time_std
            ))
            yield self.env.timeout(process_time)

            # Put in next buffer if not last station
            if i < len(self.buffers):
                yield self.buffers[i].put(job_id)

    # Job completed
    self.active_jobs -= 1
    self.metrics.wip_levels.append((self.env.now, self.active_jobs))

```

```

        self.metrics.throughput += 1
        self.metrics.lead_times.append(self.env.now - arrival_time)

def job_arrivals(self, inter_arrival_mean: float,
                 n_jobs: int) -> Generator:
    """Generate job arrivals.

    Args:
        inter_arrival_mean: Mean inter-arrival time
        n_jobs: Total number of jobs to release

    Yields:
        SimPy timeout events
    """
    for job_id in range(n_jobs):
        yield self.env.timeout(random.expovariate(1/inter_arrival_mean))
        self.env.process(self.job(job_id))

def identify_bottleneck(self) -> str:
    """Identify the bottleneck station based on utilization."""
    avg_utilizations = {
        name: np.mean(utils)
        for name, utils in self.metrics.station_utilizations.items()
    }
    self.metrics.bottleneck_station = max(
        avg_utilizations, key=avg_utilizations.get
    )
    return self.metrics.bottleneck_station

def optimize_buffer_sizes(station_configs: List[WorkstationConfig],
                          arrival_rate: float,
                          n_jobs: int = 1000,
                          buffer_range: Tuple[int, int] = (1, 20),
                          n_trials: int = 50) -> Dict[str, any]:
    """Optimize buffer sizes using simulation-optimization.

    Args:
        station_configs: Workstation configurations
        arrival_rate: Job arrival rate
        n_jobs: Jobs per simulation run
        buffer_range: Min/max buffer capacity to try
        n_trials: Number of optimization trials

    Returns:
        Optimal buffer configuration and performance
    """
    n_buffers = len(station_configs) - 1
    best_config = None
    best_throughput = 0

    results = []

```

```

for trial in range(n_trials):
    # Random buffer configuration
    buffer_sizes = [
        random.randint(buffer_range[0], buffer_range[1])
        for _ in range(n_buffers)
    ]

    # Run simulation
    env = simpy.Environment()
    line = ManufacturingLine(env, station_configs, buffer_sizes)
    env.process(line.job_arrivals(1/arrival_rate, n_jobs))
    env.run()

    avg_lead_time = np.mean(line.metrics.lead_times)
    throughput = line.metrics.throughput

    results.append({
        'buffers': buffer_sizes,
        'throughput': throughput,
        'avg_lead_time': avg_lead_time,
        'bottleneck': line.identify_bottleneck()
    })

    if throughput > best_throughput:
        best_throughput = throughput
        best_config = buffer_sizes

return {
    'optimal_buffers': best_config,
    'best_throughput': best_throughput,
    'all_results': results
}

# Example: Three-station production line
def manufacturing_example():
    """Demonstrate manufacturing line simulation."""
    stations = [
        WorkstationConfig("Cutting", 2.0, 0.3, n_parallel=2),
        WorkstationConfig("Assembly", 3.0, 0.5, n_parallel=3),
        WorkstationConfig("Testing", 1.5, 0.2, n_parallel=1),
        WorkstationConfig("Packaging", 1.0, 0.1, n_parallel=1)
    ]

    buffer_sizes = [10, 15, 8] # Buffers between stations

    env = simpy.Environment()
    line = ManufacturingLine(env, stations, buffer_sizes)
    env.process(line.job_arrivals(inter_arrival_mean=1.0, n_jobs=500))
    env.run()

    print(f"Throughput: {line.metrics.throughput} jobs")
    print(f"Avg Lead Time: {np.mean(line.metrics.lead_times):.2f}")

```

```

print(f"Bottleneck: {line.identify_bottleneck()}")

return line.metrics

```

28.8.2. Healthcare Systems

Healthcare simulation models patient flow, resource utilization, and helps optimize staffing and scheduling.

```

import simpy
import random
from enum import Enum
from dataclasses import dataclass
from typing import List, Dict, Generator
import numpy as np

class Acuity(Enum):
    """Patient acuity levels (ESI - Emergency Severity Index)."""
    RESUSCITATION = 1 # Immediate, life-threatening
    EMERGENT = 2      # High risk, confused, severe pain
    URGENT = 3        # Stable but needs multiple resources
    LESS_URGENT = 4   # Stable, one resource
    NON_URGENT = 5    # Stable, no resources needed

@dataclass
class PatientType:
    """Configuration for a patient type."""
    acuity: Acuity
    arrival_share: float # Proportion of total arrivals
    triage_time: float
    treatment_time_mean: float
    treatment_time_std: float
    requires_bed: bool
    requires_imaging: bool = False
    requires_lab: bool = False

class EmergencyDepartment:
    """Emergency Department discrete event simulation."""

    def __init__(self, env: simpy.Environment,
                 n_triage_nurses: int = 2,
                 n_beds: int = 20,
                 n_doctors: int = 4,
                 n_nurses: int = 8,
                 n_imaging: int = 2,
                 n_lab_techs: int = 2):
        """Initialize ED simulation.

        Args:
            env: SimPy environment
            n_triage_nurses: Triage nurse count

```

```

        n_beds: Total bed capacity
        n_doctors: Physician count
        n_nurses: Nursing staff count
        n_imaging: Imaging equipment count
        n_lab_techs: Lab technician count
    """
    self.env = env

    # Resources with priorities
    self.triage = simpy.PriorityResource(env, capacity=n_triage_nurses)
    self.beds = simpy.PriorityResource(env, capacity=n_beds)
    self.doctors = simpy.PriorityResource(env, capacity=n_doctors)
    self.nurses = simpy.PriorityResource(env, capacity=n_nurses)
    self.imaging = simpy.Resource(env, capacity=n_imaging)
    self.lab = simpy.Resource(env, capacity=n_lab_techs)

    # Metrics
    self.wait_times = {a: [] for a in Acuity}
    self.length_of_stay = {a: [] for a in Acuity}
    self.left_without_seen = 0
    self.total_patients = 0
    self.bed_occupancy = []

    def patient(self, patient_id: int,
                patient_type: PatientType) -> Generator:
        """Model a patient's journey through the ED.

        Args:
            patient_id: Patient identifier
            patient_type: Patient configuration

        Yields:
            SimPy events for each stage of care
        """
        arrival_time = self.env.now
        acuity = patient_type.acuity
        priority = acuity.value # Lower = higher priority

        self.total_patients += 1
        max_wait = {
            Acuity.RESUSCITATION: 0,
            Acuity.EMERGENT: 10,
            Acuity.URGENT: 30,
            Acuity.LESS_URGENT: 60,
            Acuity.NON_URGENT: 120
        }

        # Triage
        with self.triage.request(priority=priority) as req:
            yield req
            yield self.env.timeout(patient_type.triage_time)

        triage_complete = self.env.now

```



```

# Wait for bed (if required)
if patient_type.requires_bed:
    with self.beds.request(priority=priority) as bed_req:
        result = yield bed_req | self.env.timeout(max_wait[acuity])

        if bed_req not in result:
            # Patient left without being seen
            self.left_without_seen += 1
            return

    bed_start = self.env.now
    self.wait_times[acuity].append(bed_start - triage_complete)
    self.bed_occupancy.append(
        (self.env.now, self.beds.count / self.beds.capacity)
    )

# Diagnostic workup in parallel
diagnostic_processes = []

if patient_type.requires_imaging:
    diagnostic_processes.append(
        self.env.process(self._imaging_process())
    )
if patient_type.requires_lab:
    diagnostic_processes.append(
        self.env.process(self._lab_process())
    )

# Wait for diagnostics
if diagnostic_processes:
    yield simpy.AllOf(self.env, diagnostic_processes)

# Doctor consultation and treatment
with self.doctors.request(priority=priority) as doc_req:
    with self.nurses.request(priority=priority) as nurse_req:
        yield doc_req & nurse_req

        treatment_time = max(0.1, random.gauss(
            patient_type.treatment_time_mean,
            patient_type.treatment_time_std
        ))
        yield self.env.timeout(treatment_time)

# Record length of stay
self.length_of_stay[acuity].append(self.env.now - arrival_time)

def _imaging_process(self) -> Generator:
    """Imaging subprocess."""
    with self.imaging.request() as req:
        yield req
        yield self.env.timeout(random.uniform(15, 45))

```

```

def _lab_process(self) -> Generator:
    """Laboratory subprocess."""
    with self.lab.request() as req:
        yield req
        yield self.env.timeout(random.uniform(30, 90))

def arrival_generator(self, arrival_rate: float,
                     patient_types: List[PatientType],
                     sim_time: float) -> Generator:
    """Generate patient arrivals.

    Args:
        arrival_rate: Mean arrivals per time unit
        patient_types: Distribution of patient types
        sim_time: Simulation duration

    Yields:
        SimPy timeout events
    """
    patient_id = 0

    # Build CDF for patient type selection
    shares = [pt.arrival_share for pt in patient_types]
    cdf = np.cumsum(shares) / sum(shares)

    while self.env.now < sim_time:
        # Inter-arrival time
        yield self.env.timeout(random.expovariate(arrival_rate))

        # Select patient type
        r = random.random()
        patient_type = patient_types[np.searchsorted(cdf, r)]

        patient_id += 1
        self.env.process(self.patient(patient_id, patient_type))

def get_metrics(self) -> Dict:
    """Calculate summary metrics."""
    metrics = {
        'total_patients': self.total_patients,
        'left_without_seen': self.left_without_seen,
        'lwbs_rate': self.left_without_seen / max(1, self.total_patients)
    }

    for acuity in Acuity:
        if self.wait_times[acuity]:
            metrics[f'avg_wait_{acuity.name}'] = np.mean(
                self.wait_times[acuity]
            )
        if self.length_of_stay[acuity]:
            metrics[f'avg_los_{acuity.name}'] = np.mean(
                self.length_of_stay[acuity]
            )

```

```

        return metrics

def run_ed_simulation(n_runs: int = 10) -> Dict:
    """Run ED simulation with multiple replications."""
    # Define patient types
    patient_types = [
        PatientType(Acuity.RESUSCITATION, 0.02, 2, 120, 30, True, True, True),
        PatientType(Acuity.EMERGENT, 0.10, 5, 90, 20, True, True, True),
        PatientType(Acuity.URGENT, 0.30, 5, 60, 15, True, True, False),
        PatientType(Acuity.LESS_URGENT, 0.38, 5, 30, 10, True, False, False),
        PatientType(Acuity.NON_URGENT, 0.20, 3, 15, 5, False, False, False),
    ]

    all_metrics = []

    for run in range(n_runs):
        random.seed(42 + run)
        env = simpy.Environment()
        ed = EmergencyDepartment(env)

        env.process(ed.arrival_generator(
            arrival_rate=3.0, # 3 patients per time unit
            patient_types=patient_types,
            sim_time=24 * 60 # 24-hour simulation (minutes)
        ))

        env.run()
        all_metrics.append(ed.get_metrics())

    # Average across runs
    avg_metrics = {}
    for key in all_metrics[0]:
        values = [m[key] for m in all_metrics if key in m]
        avg_metrics[key] = np.mean(values)
        avg_metrics[f'{key}_std'] = np.std(values)

    return avg_metrics

```

28.8.3. Supply Chain Simulation

Supply chain simulation models inventory dynamics, distribution networks, and demand variability across multiple echelons. Each node in the supply chain maintains its own inventory, places orders to upstream suppliers, and fulfills demand from downstream customers. The simulation captures the bullwhip effect, lead time variability, and the cost trade-offs between holding excess inventory and facing stockouts. The following implementation models a multi-echelon supply chain with continuous-review (s, Q) inventory policies at each distribution center.

```

import simpy
import random
from dataclasses import dataclass, field

```

```

from typing import List, Dict, Tuple, Optional
import numpy as np

@dataclass
class InventoryPolicy:
    """Inventory control policy parameters."""
    reorder_point: int
    order_quantity: int
    safety_stock: int = 0
    review_period: Optional[int] = None # For periodic review

@dataclass
class SupplyChainNode:
    """Configuration for a supply chain node."""
    name: str
    initial_inventory: int
    holding_cost: float
    stockout_cost: float
    order_cost: float
    lead_time_mean: float
    lead_time_std: float = 0.0
    policy: InventoryPolicy = None

class DistributionCenter:
    """Distribution center with inventory management."""

    def __init__(self, env: simpy.Environment,
                 config: SupplyChainNode,
                 upstream: Optional['DistributionCenter'] = None):
        self.env = env
        self.config = config
        self.upstream = upstream

        # State
        self.inventory = config.initial_inventory
        self.backlog = 0
        self.pending_orders = []

        # Metrics
        self.total_holding_cost = 0
        self.total_stockout_cost = 0
        self.total_order_cost = 0
        self.fill_rate_samples = []
        self.inventory_history = [(0, config.initial_inventory)]

    def fulfill_demand(self, quantity: int) -> int:
        """Fulfill demand from inventory.

        Args:
            quantity: Requested quantity

```

```

Returns:
    Quantity actually fulfilled
    """
    fulfilled = min(quantity, self.inventory)
    self.inventory -= fulfilled
    unfulfilled = quantity - fulfilled

    if unfulfilled > 0:
        self.backlog += unfulfilled
        self.total_stockout_cost += unfulfilled * self.config.stockout_cost

    self.fill_rate_samples.append(fulfilled / quantity if quantity > 0 else 1)
    self.inventory_history.append((self.env.now, self.inventory))

    return fulfilled

def receive_order(self, quantity: int):
    """Receive ordered inventory."""
    self.inventory += quantity

    # Fulfill backlog first
    if self.backlog > 0:
        fulfilled = min(self.backlog, self.inventory)
        self.inventory -= fulfilled
        self.backlog -= fulfilled

    self.inventory_history.append((self.env.now, self.inventory))

def place_order(self, quantity: int) -> Generator:
    """Place order to upstream supplier.

    Args:
        quantity: Order quantity

    Yields:
        SimPy events for lead time
    """
    self.total_order_cost += self.config.order_cost

    # Lead time
    lead_time = max(0.1, random.gauss(
        self.config.lead_time_mean,
        self.config.lead_time_std
    ))

    yield self.env.timeout(lead_time)

    # Receive from upstream (or unlimited supply for top level)
    if self.upstream:
        received = self.upstream.fulfill_demand(quantity)
    else:
        received = quantity # Unlimited supplier

```

```

        self.receive_order(received)

def inventory_control(self) -> Generator:
    """Continuous review inventory control process."""
    policy = self.config.policy

    while True:
        # Calculate holding cost
        self.total_holding_cost += (
            self.inventory * self.config.holding_cost
        )

        # Check if reorder needed
        if self.inventory <= policy.reorder_point:
            order_qty = policy.order_quantity
            self.env.process(self.place_order(order_qty))

        yield self.env.timeout(1) # Check every time unit

class SupplyChainSimulation:
    """Multi-echelon supply chain simulation."""

    def __init__(self, env: simpy.Environment,
                 nodes: List[SupplyChainNode]):
        self.env = env
        self.nodes = nodes

        # Build supply chain structure
        self.dcs = []
        upstream = None

        for node in reversed(nodes): # Build from supplier down
            dc = DistributionCenter(env, node, upstream)
            self.dcs.insert(0, dc)
            upstream = dc

        # Start inventory control for each DC
        for dc in self.dcs:
            env.process(dc.inventory_control())

    def demand_process(self, demand_mean: float,
                      demand_std: float,
                      sim_time: float) -> Generator:
        """Generate customer demand at retail level.

        Args:
            demand_mean: Mean daily demand
            demand_std: Demand standard deviation
            sim_time: Simulation duration

        Yields:
            SimPy timeout events

```

```

"""
while self.env.now < sim_time:
    # Daily demand (could be made more complex)
    demand = max(0, int(random.gauss(demand_mean, demand_std)))

    # Fulfill from first DC (retail)
    self.dcs[0].fulfill_demand(demand)

    yield self.env.timeout(1) # Daily

def get_total_cost(self) -> float:
    """Calculate total supply chain cost."""
    return sum(
        dc.total_holding_cost +
        dc.total_stockout_cost +
        dc.total_order_cost
        for dc in self.dcs
    )

def get_metrics(self) -> Dict:
    """Get simulation metrics."""
    return {
        'total_cost': self.get_total_cost(),
        'node_metrics': [
            {
                'name': dc.config.name,
                'holding_cost': dc.total_holding_cost,
                'stockout_cost': dc.total_stockout_cost,
                'order_cost': dc.total_order_cost,
                'fill_rate': np.mean(dc.fill_rate_samples)
            }
            for dc in self.dcs
        ]
    }

def optimize_inventory_policies(base_nodes: List[SupplyChainNode],
                                demand_params: Tuple[float, float],
                                sim_time: float = 365,
                                n_trials: int = 100) -> Dict:
    """Optimize inventory policies across supply chain.

    Args:
        base_nodes: Base node configurations
        demand_params: (mean, std) of customer demand
        sim_time: Simulation duration
        n_trials: Number of optimization trials

    Returns:
        Best policies and performance
    """
    best_cost = float('inf')
    best_policies = None

```

```

results = []

for trial in range(n_trials):
    # Generate random policies
    nodes = []
    for node in base_nodes:
        policy = InventoryPolicy(
            reorder_point=random.randint(10, 100),
            order_quantity=random.randint(20, 200),
            safety_stock=random.randint(0, 50)
        )
        node_copy = SupplyChainNode(
            name=node.name,
            initial_inventory=node.initial_inventory,
            holding_cost=node.holding_cost,
            stockout_cost=node.stockout_cost,
            order_cost=node.order_cost,
            lead_time_mean=node.lead_time_mean,
            lead_time_std=node.lead_time_std,
            policy=policy
        )
        nodes.append(node_copy)

    # Run simulation
    env = simpy.Environment()
    sc = SupplyChainSimulation(env, nodes)
    env.process(sc.demand_process(
        demand_params[0], demand_params[1], sim_time
    ))
    env.run()

    cost = sc.get_total_cost()
    results.append({
        'policies': [(n.policy.reorder_point, n.policy.order_quantity)
                     for n in nodes],
        'cost': cost,
        'fill_rates': [m['fill_rate'] for m in sc.get_metrics()['node_metrics']]
    })

    if cost < best_cost:
        best_cost = cost
        best_policies = [(n.policy.reorder_point, n.policy.order_quantity)
                         for n in nodes]

return {
    'best_policies': best_policies,
    'best_cost': best_cost,
    'all_results': results
}

```

28.9. Metamodeling

Metamodeling (or surrogate modeling) creates computationally cheap approximations of expensive simulation models, enabling rapid optimization and sensitivity analysis.

28.9.1. Response Surface Methodology

Response Surface Methodology (RSM) is a statistical approach to metamodeling that predates modern machine learning. It fits low-order polynomial models (typically quadratic) to a designed experiment. RSM uses structured designs like central composite designs or Box-Behnken designs to efficiently explore the factor space with minimal simulation runs. The fitted polynomial provides interpretable coefficients that quantify main effects and interactions between factors.

```
import numpy as np
from scipy.stats import norm
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import cross_val_score
from typing import Callable, List, Tuple, Dict
from itertools import product

class ResponseSurfaceMethodology:
    """Response Surface Methodology for simulation metamodeling."""

    def __init__(self, simulator: Callable,
                 bounds: List[Tuple[float, float]],
                 degree: int = 2):
        """Initialize RSM.

        Args:
            simulator: Simulation function
            bounds: Variable bounds [(low, high), ...]
            degree: Polynomial degree (1, 2, or 3)
        """
        self.simulator = simulator
        self.bounds = bounds
        self.n_vars = len(bounds)
        self.degree = degree

        self.poly = PolynomialFeatures(degree=degree, include_bias=True)
        self.model = LinearRegression()

        self.X_design = None
        self.y_response = None
        self.is_fitted = False

    def generate_design(self, design_type: str = "central_composite",
                      n_center: int = 5) -> np.ndarray:
        """Generate experimental design.

        Args:
            design_type: Type of design
            n_center: Number of center points

        Returns:
            Design matrix in coded units [-1, 1]
        """
        if design_type == "factorial":
```

```

        # Full factorial
        levels = [-1, 1]
        design = np.array(list(product(levels, repeat=self.n_vars)))

    elif design_type == "central_composite":
        # Central Composite Design (CCD)
        # Factorial points
        factorial = np.array(list(product([-1, 1], repeat=self.n_vars)))

        # Axial points
        alpha = np.sqrt(self.n_vars) # Rotatable design
        axial = []
        for i in range(self.n_vars):
            point_plus = np.zeros(self.n_vars)
            point_plus[i] = alpha
            point_minus = np.zeros(self.n_vars)
            point_minus[i] = -alpha
            axial.extend([point_plus, point_minus])
        axial = np.array(axial)

        # Center points
        center = np.zeros((n_center, self.n_vars))

        design = np.vstack([factorial, axial, center])

    elif design_type == "box_behnken":
        # Box-Behnken design (3 factors)
        if self.n_vars != 3:
            raise ValueError("Box-Behnken requires exactly 3 factors")
        design = np.array([
            [-1, -1, 0], [-1, 1, 0], [1, -1, 0], [1, 1, 0],
            [-1, 0, -1], [-1, 0, 1], [1, 0, -1], [1, 0, 1],
            [0, -1, -1], [0, -1, 1], [0, 1, -1], [0, 1, 1],
        ])
        center = np.zeros((n_center, 3))
        design = np.vstack([design, center])

    else:
        raise ValueError(f"Unknown design type: {design_type}")

    return design

def coded_to_natural(self, X_coded: np.ndarray) -> np.ndarray:
    """Convert coded variables to natural units.

    Args:
        X_coded: Design in coded units [-1, 1]

    Returns:
        Design in natural units
    """
    X_natural = np.zeros_like(X_coded)
    for i, (low, high) in enumerate(self.bounds):

```

```

        center = (high + low) / 2
        half_range = (high - low) / 2
        X_natural[:, i] = center + half_range * X_coded[:, i]
    return X_natural

def natural_to_coded(self, X_natural: np.ndarray) -> np.ndarray:
    """Convert natural units to coded variables."""
    X_coded = np.zeros_like(X_natural)
    for i, (low, high) in enumerate(self.bounds):
        center = (high + low) / 2
        half_range = (high - low) / 2
        X_coded[:, i] = (X_natural[:, i] - center) / half_range
    return X_coded

def run_experiments(self, design_type: str = "central_composite",
                    n_replications: int = 5) -> Tuple[np.ndarray, np.ndarray]:
    """Run simulation experiments according to design.

    Args:
        design_type: Experimental design type
        n_replications: Replications per design point

    Returns:
        Design matrix and response vector
    """
    X_coded = self.generate_design(design_type)
    X_natural = self.coded_to_natural(X_coded)

    y = []
    for x in X_natural:
        responses = [self.simulator(x) for _ in range(n_replications)]
        y.append(np.mean(responses))

    self.X_design = X_coded
    self.y_response = np.array(y)

    return X_natural, self.y_response

def fit(self):
    """Fit response surface model to experimental data."""
    if self.X_design is None:
        raise ValueError("Run experiments first")

    X_poly = self.poly.fit_transform(self.X_design)
    self.model.fit(X_poly, self.y_response)
    self.is_fitted = True

    # Calculate R-squared
    y_pred = self.model.predict(X_poly)
    ss_res = np.sum((self.y_response - y_pred) ** 2)
    ss_tot = np.sum((self.y_response - np.mean(self.y_response)) ** 2)
    self.r_squared = 1 - ss_res / ss_tot

```

```

    return self

def predict(self, X_natural: np.ndarray) -> np.ndarray:
    """Predict response at new points.

    Args:
        X_natural: Points in natural units

    Returns:
        Predicted responses
    """
    if not self.is_fitted:
        raise ValueError("Model not fitted")

    X_coded = self.natural_to_coded(X_natural)
    X_poly = self.poly.transform(X_coded)
    return self.model.predict(X_poly)

def find_optimum(self, minimize: bool = True) -> Dict:
    """Find the optimum using the response surface.

    Args:
        minimize: If True, minimize; else maximize

    Returns:
        Optimal point and predicted value
    """
    from scipy.optimize import minimize as scipy_minimize

    def objective(x_coded):
        x_poly = self.poly.transform(x_coded.reshape(1, -1))
        pred = self.model.predict(x_poly)[0]
        return pred if minimize else -pred

    # Multi-start optimization
    best_result = None
    best_value = float('inf')

    for _ in range(20):
        x0 = np.random.uniform(-1, 1, self.n_vars)
        bounds = [(-1.5, 1.5)] * self.n_vars

        result = scipy_minimize(objective, x0, bounds=bounds,
                                method='L-BFGS-B')

        if result.fun < best_value:
            best_value = result.fun
            best_result = result

    x_optimal_coded = best_result.x
    x_optimal_natural = self.coded_to_natural(
        x_optimal_coded.reshape(1, -1)
    )[0]

```

```

    return {
        'optimal_point': x_optimal_natural,
        'predicted_value': best_value if minimize else -best_value,
        'coded_point': x_optimal_coded
    }

def sensitivity_analysis(self) -> Dict[str, float]:
    """Analyze factor sensitivities using coefficients.

    Returns:
        Dictionary of factor sensitivities
    """
    if not self.is_fitted:
        raise ValueError("Model not fitted")

```

28.9.2. Visualizing the Response Surface

Response surfaces help visualize how the objective function varies with input parameters. The following visualization shows a 3D response surface and its corresponding contour plot.

```

feature_names = self.poly.get_feature_names_out()
coefficients = self.model.coef_

sensitivities = {}
for name, coef in zip(feature_names, coefficients):
    if name != '1': # Skip intercept
        sensitivities[name] = abs(coef)

# Normalize
total = sum(sensitivities.values())
sensitivities = {k: v/total for k, v in sensitivities.items()}

return dict(sorted(sensitivities.items(),
                    key=lambda x: x[1], reverse=True))

```

28.9.3. Kriging (Gaussian Process) Metamodels

Kriging (also called Gaussian Process regression) provides probabilistic predictions with uncertainty quantification. Unlike polynomial metamodels, Kriging gives not just a point estimate but also a confidence interval at each prediction point. This uncertainty estimate drives sequential experimental design: the model identifies where it is most uncertain and recommends new simulation runs in those regions. Expected Improvement (EI) is a standard acquisition function that balances exploitation (sampling near the current best) with exploration (sampling where uncertainty is high).

```

from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import RBF, Matern, WhiteKernel
from scipy.stats import norm as normal_dist

class KrigingMetamodel:
    """Kriging (Gaussian Process) metamodel for simulation."""

```

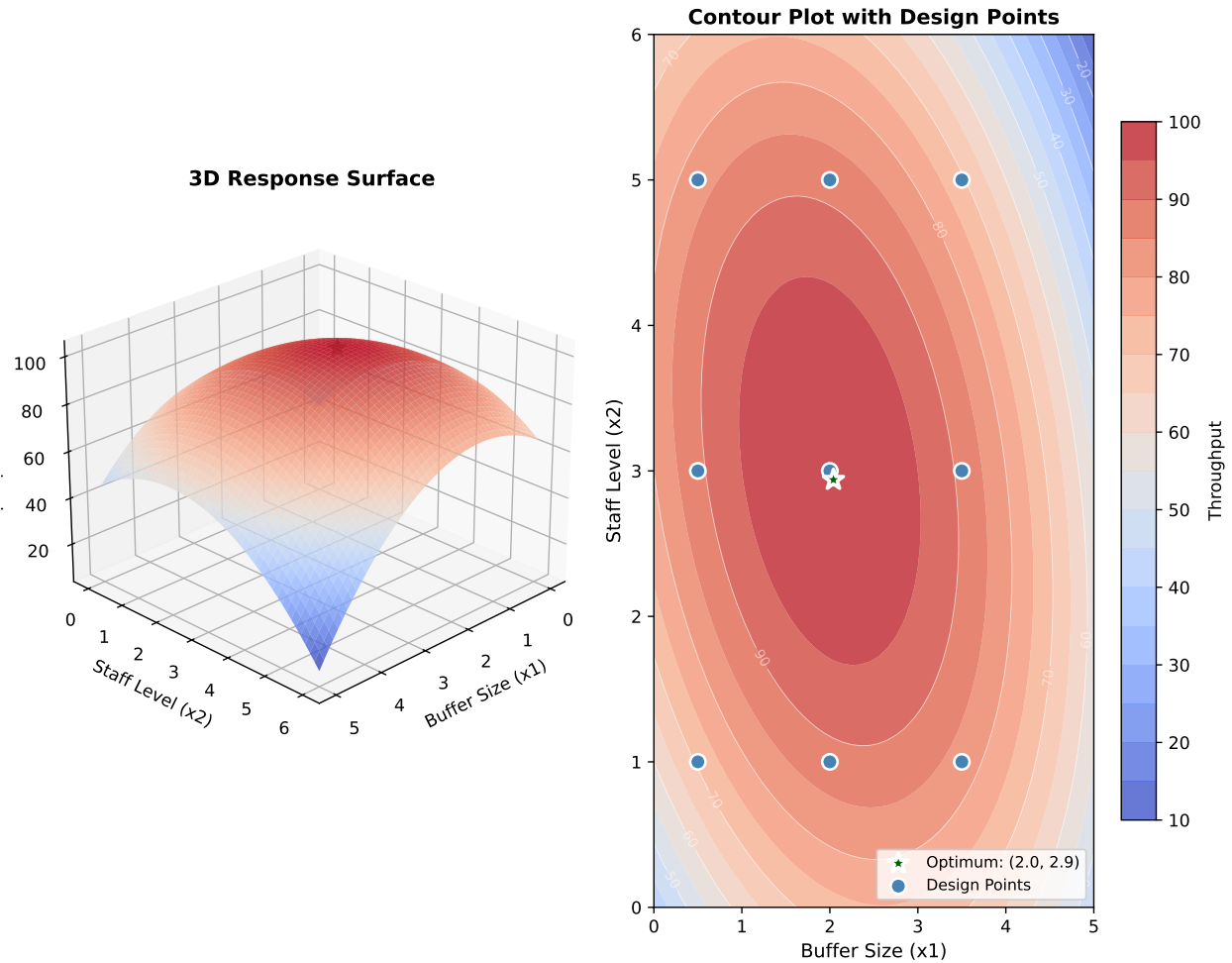


Figure 28.5.: Response Surface Visualization: 3D Surface and Contour Plot of Objective vs Parameters

```
def __init__(self, bounds: List[Tuple[float, float]],
             kernel_type: str = "matern"):
    """Initialize Kriging metamodel.

    Args:
        bounds: Variable bounds
        kernel_type: Kernel type ("rbf", "matern")
    """
    self.bounds = bounds
    self.n_vars = len(bounds)

    # Select kernel
    if kernel_type == "rbf":
        kernel = RBF(length_scale=1.0) + WhiteKernel(noise_level=0.1)
    else:
        kernel = Matern(nu=2.5) + WhiteKernel(noise_level=0.1)
```

```

self.gp = GaussianProcessRegressor(
    kernel=kernel,
    normalize_y=True,
    n_restarts_optimizer=10
)

self.X_train = None
self.y_train = None
self.is_fitted = False

def fit(self, X: np.ndarray, y: np.ndarray):
    """Fit Kriging model to data.

    Args:
        X: Input points
        y: Response values
    """
    self.X_train = X
    self.y_train = y
    self.gp.fit(X, y)
    self.is_fitted = True
    return self

def predict(self, X: np.ndarray,
            return_std: bool = False) -> np.ndarray:
    """Predict with optional uncertainty.

    Args:
        X: Points to predict at
        return_std: If True, also return standard deviation

    Returns:
        Predictions (and optionally standard deviations)
    """
    return self.gp.predict(X, return_std=return_std)

def expected_improvement(self, X: np.ndarray,
                        y_best: float,
                        xi: float = 0.01) -> np.ndarray:
    """Calculate Expected Improvement for optimization.

    Args:
        X: Candidate points
        y_best: Current best observed value
        xi: Exploration-exploitation trade-off

    Returns:
        EI values at each point
    """
    mu, sigma = self.predict(X, return_std=True)

    with np.errstate(divide='warn'):

```

```

        imp = y_best - mu - xi
        Z = imp / sigma
        ei = imp * normal_dist.cdf(Z) + sigma * normal_dist.pdf(Z)
        ei[sigma == 0.0] = 0.0

    return ei

def sequential_design(self, simulator: Callable,
                     n_initial: int = 10,
                     n_iterations: int = 50,
                     acquisition: str = "EI") -> Dict:
    """Sequential experimental design using active learning.

    Args:
        simulator: Simulation function
        n_initial: Initial design points
        n_iterations: Sequential iterations
        acquisition: Acquisition function ("EI", "LCB", "PI")

    Returns:
        Training data and model
    """
    # Latin Hypercube initial design
    X = self._latin_hypercube(n_initial)
    y = np.array([simulator(x) for x in X])

    self.fit(X, y)

    for _ in range(n_iterations):
        # Find next point via acquisition
        x_next = self._optimize_acquisition(acquisition)
        y_next = simulator(x_next)

        # Update training data
        X = np.vstack([X, x_next])
        y = np.append(y, y_next)

        # Refit
        self.fit(X, y)

    return {
        'X': X,
        'y': y,
        'best_x': X[np.argmin(y)],
        'best_y': np.min(y)
    }

def _latin_hypercube(self, n_samples: int) -> np.ndarray:
    """Generate Latin Hypercube Sample."""
    samples = np.zeros((n_samples, self.n_vars))
    for i in range(self.n_vars):
        low, high = self.bounds[i]
        perm = np.random.permutation(n_samples)

```



```

        samples[:, i] = low + (perm + np.random.rand(n_samples)) * \
            (high - low) / n_samples
    return samples

def _optimize_acquisition(self, acquisition: str) -> np.ndarray:
    """Find point maximizing acquisition function."""
    from scipy.optimize import minimize

    y_best = np.min(self.y_train)

    def neg_acquisition(x):
        x = x.reshape(1, -1)
        if acquisition == "EI":
            return -self.expected_improvement(x, y_best)[0]
        elif acquisition == "LCB":
            mu, sigma = self.predict(x, return_std=True)
            return mu[0] - 2.0 * sigma[0]
        else: # PI
            mu, sigma = self.predict(x, return_std=True)
            return -normal_dist.cdf((y_best - mu[0]) / sigma[0])

    best_x = None
    best_acq = float('inf')

    for _ in range(20):
        x0 = np.array([np.random.uniform(low, high)
                       for low, high in self.bounds])
        result = minimize(neg_acquisition, x0, bounds=self.bounds,
                          method='L-BFGS-B')
        if result.fun < best_acq:
            best_acq = result.fun
            best_x = result.x

    return best_x

```

28.9.4. Neural Network Surrogates

Neural networks can approximate high-dimensional, nonlinear simulation responses where polynomial metamodels and Gaussian processes become impractical. A neural network surrogate learns a mapping from input parameters to simulation outputs using training data from a set of simulation runs. Active learning strategies select new simulation points in regions of high prediction uncertainty, reducing the total number of simulation evaluations needed to build an accurate surrogate.

```

import numpy as np
from typing import Callable, List, Tuple, Optional
from dataclasses import dataclass

# Note: TensorFlow/PyTorch implementations would be more performant
# This example uses a simple NumPy implementation for illustration

@dataclass
class NNLayer:

```

```

"""Neural network layer."""
weights: np.ndarray
biases: np.ndarray
activation: str = "relu"

class NeuralNetworkSurrogate:
    """Neural network surrogate model for simulation."""

    def __init__(self, input_dim: int,
                  hidden_layers: List[int] = [64, 32],
                  output_dim: int = 1,
                  learning_rate: float = 0.001):
        """Initialize neural network surrogate.

        Args:
            input_dim: Number of input features
            hidden_layers: Hidden layer sizes
            output_dim: Number of outputs
            learning_rate: Learning rate for training
        """
        self.input_dim = input_dim
        self.hidden_layers = hidden_layers
        self.output_dim = output_dim
        self.learning_rate = learning_rate

        # Initialize layers
        self.layers = []
        prev_dim = input_dim

        for h_dim in hidden_layers:
            self.layers.append(NNLayer(
                weights=np.random.randn(prev_dim, h_dim) * np.sqrt(2/prev_dim),
                biases=np.zeros(h_dim),
                activation="relu"
            ))
            prev_dim = h_dim

        # Output layer
        self.layers.append(NNLayer(
            weights=np.random.randn(prev_dim, output_dim) * np.sqrt(2/prev_dim),
            biases=np.zeros(output_dim),
            activation="linear"
        ))

    def _relu(self, x: np.ndarray) -> np.ndarray:
        return np.maximum(0, x)

    def _relu_derivative(self, x: np.ndarray) -> np.ndarray:
        return (x > 0).astype(float)

    def forward(self, X: np.ndarray) -> Tuple[np.ndarray, List]:
        """Forward pass through network.

```

```

    Args:
        X: Input features

    Returns:
        Output and intermediate activations
    """
    activations = [X]
    current = X

    for layer in self.layers:
        z = current @ layer.weights + layer.biases
        if layer.activation == "relu":
            current = self._relu(z)
        else:
            current = z
        activations.append(current)

    return current, activations

def predict(self, X: np.ndarray) -> np.ndarray:
    """Make predictions.

    Args:
        X: Input features

    Returns:
        Predictions
    """
    output, _ = self.forward(X)
    return output

def train(self, X: np.ndarray, y: np.ndarray,
          epochs: int = 1000,
          batch_size: int = 32,
          validation_split: float = 0.2) -> Dict:
    """Train the neural network.

    Args:
        X: Training inputs
        y: Training targets
        epochs: Number of training epochs
        batch_size: Mini-batch size
        validation_split: Validation data fraction

    Returns:
        Training history
    """
    n_samples = len(X)
    n_val = int(n_samples * validation_split)

    # Split data
    indices = np.random.permutation(n_samples)

```

```

val_idx, train_idx = indices[:n_val], indices[n_val:]

X_train, y_train = X[train_idx], y[train_idx]
X_val, y_val = X[val_idx], y[val_idx]

history = {'train_loss': [], 'val_loss': []}

for epoch in range(epochs):
    # Shuffle training data
    perm = np.random.permutation(len(X_train))
    X_train, y_train = X_train[perm], y_train[perm]

    epoch_loss = 0
    for i in range(0, len(X_train), batch_size):
        X_batch = X_train[i:i+batch_size]
        y_batch = y_train[i:i+batch_size]

        # Forward pass
        predictions, activations = self.forward(X_batch)

        # Compute loss
        loss = np.mean((predictions - y_batch.reshape(-1, 1)) ** 2)
        epoch_loss += loss

        # Backpropagation (simplified)
        error = predictions - y_batch.reshape(-1, 1)

        for l in range(len(self.layers) - 1, -1, -1):
            layer = self.layers[l]

            # Gradient
            d_weights = activations[l].T @ error / batch_size
            d_biases = np.mean(error, axis=0)

            # Update
            layer.weights -= self.learning_rate * d_weights
            layer.biases -= self.learning_rate * d_biases

            # Propagate error
            if l > 0:
                error = (error @ layer.weights.T) * \
                    self._relu_derivative(activations[l])

    # Record losses
    train_loss = epoch_loss / (len(X_train) / batch_size)
    val_pred = self.predict(X_val)
    val_loss = np.mean((val_pred - y_val.reshape(-1, 1)) ** 2)

    history['train_loss'].append(train_loss)
    history['val_loss'].append(val_loss)

    if epoch % 100 == 0:
        print(f"Epoch {epoch}: train_loss={train_loss:.4f}, "

```

```

        f"val_loss={val_loss:.4f}")

    return history

def active_learning_surrogate(simulator: Callable,
                             bounds: List[Tuple[float, float]],
                             n_initial: int = 50,
                             n_iterations: int = 100,
                             batch_size: int = 5) -> Dict:
    """Build neural network surrogate with active learning.

    Args:
        simulator: Simulation function
        bounds: Variable bounds
        n_initial: Initial training samples
        n_iterations: Active learning iterations
        batch_size: Samples to add per iteration

    Returns:
        Trained surrogate and training data
    """
    n_vars = len(bounds)

    # Initial Latin Hypercube Sample
    X = np.zeros((n_initial, n_vars))
    for i in range(n_vars):
        low, high = bounds[i]
        perm = np.random.permutation(n_initial)
        X[:, i] = low + (perm + np.random.rand(n_initial)) * \
            (high - low) / n_initial

    y = np.array([simulator(x) for x in X])

    # Initialize surrogate
    surrogate = NeuralNetworkSurrogate(
        input_dim=n_vars,
        hidden_layers=[64, 32],
        output_dim=1
    )

    for iteration in range(n_iterations):
        # Train on current data
        surrogate.train(X, y, epochs=500, batch_size=min(32, len(X)))

        # Generate candidate points
        n_candidates = 1000
        X_cand = np.array([
            np.random.uniform(low, high) for low, high in bounds
        ] * n_candidates)

        # Select points with highest uncertainty (simple variance estimate)

```

```

# In practice, use ensemble or dropout for uncertainty
preds = surrogate.predict(X_cand)
distances = np.min([
    np.linalg.norm(X_cand - x, axis=1) for x in X
], axis=0)

# Select diverse, distant points
selected_idx = np.argsort(-distances)[:batch_size]
X_new = X_cand[selected_idx]
y_new = np.array([simulator(x) for x in X_new])

X = np.vstack([X, X_new])
y = np.append(y, y_new)

if iteration % 10 == 0:
    print(f"Iteration {iteration}: {len(X)} samples, "
          f"best y = {np.min(y):.4f}")

return {
    'surrogate': surrogate,
    'X': X,
    'y': y,
    'best_x': X[np.argmin(y)],
    'best_y': np.min(y)
}

```

28.10. Digital Twins

Digital twins are virtual replicas of physical systems that enable real-time simulation, monitoring, and predictive analytics.

28.10.1. Digital Twin Architecture

A digital twin connects a physical system to its virtual counterpart through real-time data streams. Sensors on the physical asset feed data into a state estimation module, which calibrates the simulation model to match observed behavior. The simulation then runs predictive scenarios and feeds results to an optimization engine that computes control actions sent back to the physical system.

28.10.2. Real-Time Simulation Framework

The following framework implements a digital twin that ingests real-time sensor data, estimates the current system state, synchronizes the simulation model with observations, and runs predictive scenarios. The framework uses a Kalman-filter-style state estimation and online model calibration to keep the virtual model aligned with the physical system. When the discrepancy between observed and predicted states exceeds a threshold, the system flags an anomaly and triggers recalibration.

```

import simpy
import time
import threading
from queue import Queue
from dataclasses import dataclass, field
from typing import Dict, List, Callable, Optional
from datetime import datetime

```

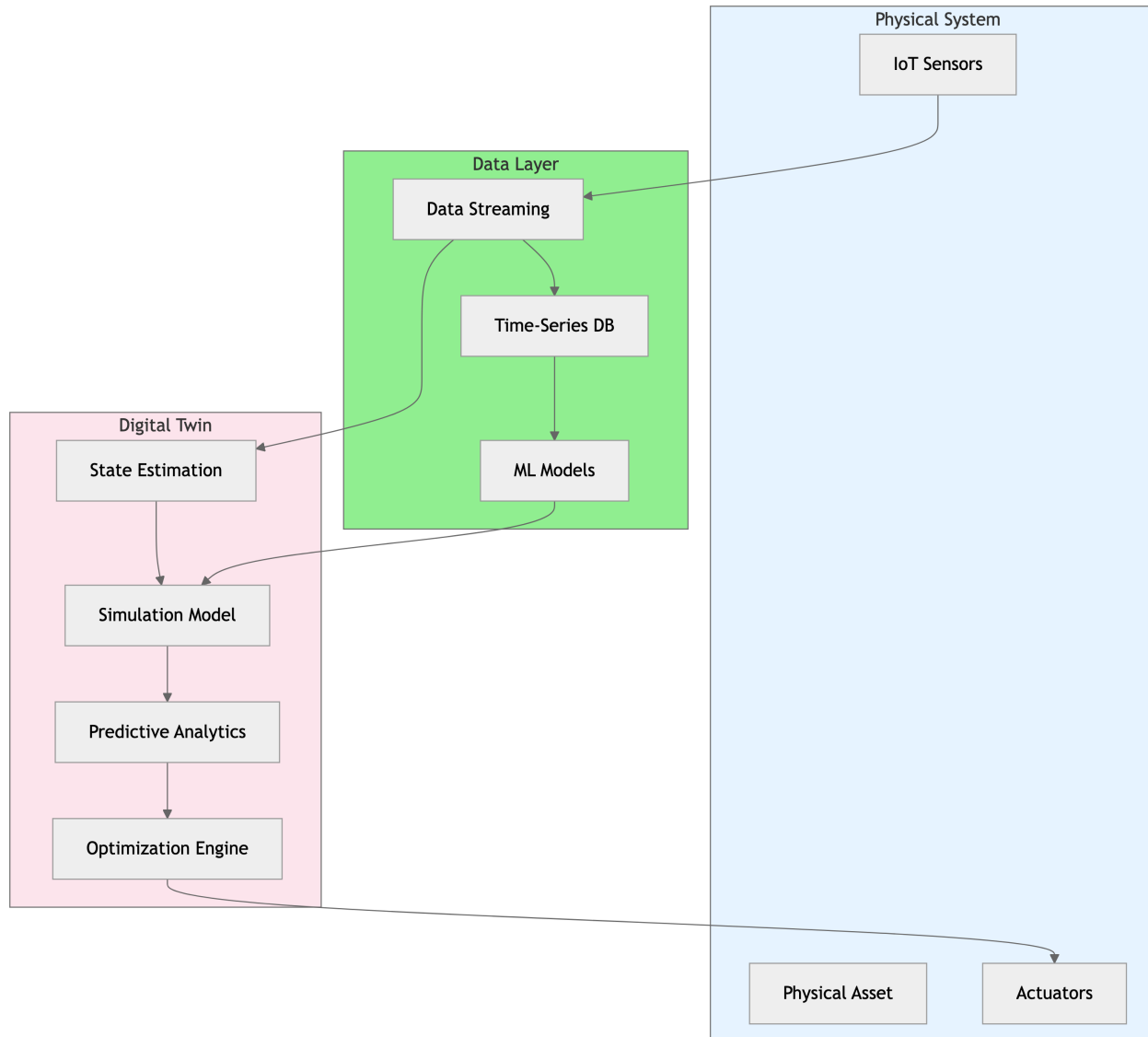


Figure 28.6.: Digital Twin Architecture with Real-Time Data Integration

```

import numpy as np

@dataclass
class SensorReading:
    """Real-time sensor reading."""
    sensor_id: str
    timestamp: datetime
    value: float
    unit: str
    quality: float = 1.0 # Data quality indicator

@dataclass
class SystemState:

```

```

    """Current state of the physical system."""
    timestamp: datetime
    variables: Dict[str, float]
    predictions: Dict[str, float] = field(default_factory=dict)
    anomalies: List[str] = field(default_factory=list)

class DigitalTwin:
    """Real-time digital twin framework."""

    def __init__(self, physical_model: Callable,
                 update_interval: float = 1.0):
        """Initialize digital twin.

        Args:
            physical_model: Simulation model of physical system
            update_interval: State update interval (seconds)
        """
        self.physical_model = physical_model
        self.update_interval = update_interval

        # Real-time data queues
        self.sensor_queue = Queue()
        self.command_queue = Queue()

        # State management
        self.current_state = None
        self.state_history = []
        self.prediction_horizon = 10 # Time steps ahead

        # Synchronization
        self.running = False
        self.lock = threading.Lock()

        # Calibration parameters
        self.model_params = {}
        self.calibration_errors = []

    def ingest_sensor_data(self, reading: SensorReading):
        """Ingest real-time sensor data.

        Args:
            reading: Sensor reading to process
        """
        self.sensor_queue.put(reading)

    def state_estimation(self, sensor_readings: List[SensorReading]) -> Dict:
        """Estimate system state from sensor data.

        Uses Kalman filter or similar for noisy measurements.

        Args:
            sensor_readings: Recent sensor readings

```



```

Returns:
    Estimated state variables
    """
    state_vars = {}

    # Simple averaging for demonstration
    # Real implementation would use Kalman filter
    for reading in sensor_readings:
        if reading.sensor_id not in state_vars:
            state_vars[reading.sensor_id] = []
        state_vars[reading.sensor_id].append(
            reading.value * reading.quality
        )

    return {k: np.mean(v) for k, v in state_vars.items()}

def synchronize_model(self, observed_state: Dict):
    """Synchronize simulation model with observed state.

    Args:
        observed_state: Current observed state
    """
    # Run simulation from current state
    simulated_state = self.physical_model(
        self.model_params,
        observed_state
    )

    # Calculate discrepancy
    errors = {}
    for key in observed_state:
        if key in simulated_state:
            errors[key] = observed_state[key] - simulated_state[key]

    self.calibration_errors.append(errors)

    # Adapt model parameters if error is large
    self._online_calibration(errors)

def _online_calibration(self, errors: Dict,
                        threshold: float = 0.1):
    """Online model calibration using errors.

    Args:
        errors: State estimation errors
        threshold: Calibration threshold
    """
    avg_error = np.mean([abs(e) for e in errors.values()])

    if avg_error > threshold:
        # Adjust model parameters
        # Simplified gradient-free adjustment

```

```

        for key in self.model_params:
            if key in errors:
                adjustment = 0.01 * errors[key]
                self.model_params[key] += adjustment

def predict_future_states(self,
                           current_state: Dict,
                           horizon: int = 10) -> List[Dict]:
    """Predict future system states.

    Args:
        current_state: Current system state
        horizon: Prediction horizon

    Returns:
        List of predicted future states
    """
    predictions = []
    state = current_state.copy()

    for t in range(horizon):
        next_state = self.physical_model(self.model_params, state)
        predictions.append(next_state)
        state = next_state

    return predictions

def detect_anomalies(self, observed: Dict,
                     predicted: Dict,
                     threshold: float = 3.0) -> List[str]:
    """Detect anomalies by comparing observed vs predicted.

    Args:
        observed: Observed state
        predicted: Predicted state
        threshold: Anomaly threshold (std devs)

    Returns:
        List of anomalous variables
    """
    anomalies = []

    for key in observed:
        if key in predicted:
            deviation = abs(observed[key] - predicted[key])
            if deviation > threshold * np.std(
                [e.get(key, 0) for e in self.calibration_errors[-100:]]
            ):
                anomalies.append(key)

    return anomalies

def optimize_operation(self,

```

```

        current_state: Dict,
        objective: Callable,
        constraints: List[Callable]) -> Dict:
    """Optimize system operation based on current state.

    Args:
        current_state: Current state
        objective: Objective function
        constraints: Constraint functions

    Returns:
        Optimal control actions
    """
    from scipy.optimize import minimize

    def constrained_objective(x):
        # Simulate forward
        state = current_state.copy()
        state['control'] = x
        future_states = self.predict_future_states(state)

        # Calculate objective
        obj = objective(future_states)

        # Add penalty for constraint violations
        penalty = 0
        for constraint in constraints:
            violation = max(0, -constraint(future_states))
            penalty += 1000 * violation ** 2

        return obj + penalty

    # Optimize
    x0 = np.zeros(len(current_state.get('control', [1])))
    result = minimize(constrained_objective, x0, method='SLSQP')

    return {'optimal_control': result.x, 'predicted_objective': result.fun}

def run_real_time_loop(self, duration: float = float('inf')):
    """Main real-time processing loop.

    Args:
        duration: How long to run (seconds)
    """
    self.running = True
    start_time = time.time()

    while self.running and (time.time() - start_time) < duration:
        # Collect sensor readings
        readings = []
        while not self.sensor_queue.empty():
            readings.append(self.sensor_queue.get())

```

```

        if readings:
            with self.lock:
                # Estimate current state
                observed_state = self.state_estimation(readings)

                # Synchronize model
                self.synchronize_model(observed_state)

                # Predict future
                predictions = self.predict_future_states(
                    observed_state, self.prediction_horizon
                )

                # Detect anomalies
                if self.current_state:
                    anomalies = self.detect_anomalies(
                        observed_state,
                        self.current_state.predictions.get('next', {})
                    )
                else:
                    anomalies = []

                # Update state
                self.current_state = SystemState(
                    timestamp=datetime.now(),
                    variables=observed_state,
                    predictions={'next': predictions[0] if predictions else {}},
                    anomalies=anomalies
                )
                self.state_history.append(self.current_state)

            time.sleep(self.update_interval)

    def stop(self):
        """Stop the real-time loop."""
        self.running = False

class PredictiveMaintenance:
    """Predictive maintenance using digital twin."""

    def __init__(self, digital_twin: DigitalTwin,
                 degradation_model: Callable):
        self.twin = digital_twin
        self.degradation_model = degradation_model
        self.maintenance_schedule = []

    def estimate_remaining_useful_life(self,
                                     component_id: str,
                                     current_health: float) -> float:
        """Estimate Remaining Useful Life (RUL).

        Args:

```

```

        component_id: Component identifier
        current_health: Current health index (0-1)

Returns:
    Estimated RUL in time units
    """
    # Simulate degradation forward
    health = current_health
    rul = 0

    while health > 0.1: # Failure threshold
        degradation = self.degradation_model(
            component_id,
            health,
            self.twin.current_state.variables
        )
        health -= degradation
        rul += 1

    return rul

def optimize_maintenance_schedule(self,
                                   components: List[str],
                                   maintenance_costs: Dict[str, float],
                                   failure_costs: Dict[str, float],
                                   planning_horizon: int = 100) -> List[Dict]:
    """Optimize maintenance scheduling.

    Args:
        components: List of component IDs
        maintenance_costs: Cost of maintenance per component
        failure_costs: Cost of failure per component
        planning_horizon: Planning horizon

    Returns:
        Optimal maintenance schedule
    """
    schedule = []

    for component in components:
        current_health = self.twin.current_state.variables.get(
            f'{component}_health', 1.0
        )
        rul = self.estimate_remaining_useful_life(component, current_health)

        # Optimize maintenance timing
        # Simple heuristic: maintain at 80% of RUL
        optimal_time = int(0.8 * rul)

        schedule.append({
            'component': component,
            'current_health': current_health,
            'rul': rul,

```

```

        'recommended_maintenance': optimal_time,
        'maintenance_cost': maintenance_costs[component],
        'expected_failure_cost': failure_costs[component] * (1 -
        ↪ current_health)
    })

    self.maintenance_schedule = sorted(
        schedule, key=lambda x: x['recommended_maintenance']
    )

    return self.maintenance_schedule

```

28.11. Software Tools Comparison

The simulation software ecosystem offers diverse options from open-source libraries to commercial platforms.

28.11.1. Comparison Matrix

Tool	Type	License	Strengths	Best For
SimPy	Python DES	MIT (Free)	Programmable, extensible, integrates with Python ecosystem	Custom simulations, research, prototyping
AnyLogic	Multi-method	Commercial	Visual modeling, agent-based + DES + SD, cloud deployment	Complex hybrid systems, enterprise
Arena	DES	Commercial	Industry standard, extensive templates, OptQuest optimization	Manufacturing, logistics, proven track record
FlexSim	DES	Commercial	3D visualization, VR support, intuitive interface	Presentations, stakeholder buy-in
Simio	DES/Agent	Commercial	Object-oriented, intelligent objects, SAN integration	Flexible modeling, reusable components
Plant Simulation	DES	Commercial	Digital factory, Siemens integration	Manufacturing, Industry 4.0
Salabim	Python DES	MIT	Animation, queuing focus, simple API	Quick prototyping, education
DESMO-J	Java DES	LGPL	Academic, well-documented	Teaching, research
JaamSim	Java DES	Apache 2.0	Free, visual builder, extensible	Cost-sensitive projects

28.11.2. Selection Criteria

```

from dataclasses import dataclass
from typing import List, Dict
from enum import Enum

class ModelComplexity(Enum):

```

```

LOW = 1
MEDIUM = 2
HIGH = 3

class BudgetLevel(Enum):
    FREE = 0
    LOW = 1
    MEDIUM = 2
    HIGH = 3

@dataclass
class ProjectRequirements:
    """Requirements for simulation project."""
    complexity: ModelComplexity
    budget: BudgetLevel
    needs_visualization: bool
    needs_optimization: bool
    integration_needed: List[str] # e.g., ["Python", "ERP", "IoT"]
    team_programming_skill: str # "low", "medium", "high"
    deployment: str # "desktop", "web", "cloud"

def recommend_simulation_tool(requirements: ProjectRequirements) -> List[Dict]:
    """Recommend simulation tools based on requirements.

    Args:
        requirements: Project requirements

    Returns:
        Ranked list of tool recommendations
    """
    tools = [
        {
            'name': 'SimPy',
            'scores': {
                'complexity': {ModelComplexity.LOW: 9, ModelComplexity.MEDIUM: 8,
                               ↪ ModelComplexity.HIGH: 7},
                'budget': {BudgetLevel.FREE: 10, BudgetLevel.LOW: 10,
                           ↪ BudgetLevel.MEDIUM: 10, BudgetLevel.HIGH: 10},
                'visualization': 5,
                'optimization': 8,
                'python_integration': 10,
                'programming_skill': {'low': 4, 'medium': 8, 'high': 10},
            }
        },
        {
            'name': 'AnyLogic',
            'scores': {
                'complexity': {ModelComplexity.LOW: 8, ModelComplexity.MEDIUM: 9,
                               ↪ ModelComplexity.HIGH: 10},
                'budget': {BudgetLevel.FREE: 3, BudgetLevel.LOW: 4,
                           ↪ BudgetLevel.MEDIUM: 7, BudgetLevel.HIGH: 10},
                'visualization': 9,
            }
        }
    ]

```

```

        'optimization': 9,
        'python_integration': 6,
        'programming_skill': {'low': 7, 'medium': 9, 'high': 8},
    }
},
{
    'name': 'Arena',
    'scores': {
        'complexity': {ModelComplexity.LOW: 9, ModelComplexity.MEDIUM: 9,
            ↪ ModelComplexity.HIGH: 8},
        'budget': {BudgetLevel.FREE: 2, BudgetLevel.LOW: 5,
            ↪ BudgetLevel.MEDIUM: 8, BudgetLevel.HIGH: 10},
        'visualization': 7,
        'optimization': 10,
        'python_integration': 4,
        'programming_skill': {'low': 8, 'medium': 9, 'high': 7},
    }
},
{
    'name': 'FlexSim',
    'scores': {
        'complexity': {ModelComplexity.LOW: 9, ModelComplexity.MEDIUM: 8,
            ↪ ModelComplexity.HIGH: 7},
        'budget': {BudgetLevel.FREE: 2, BudgetLevel.LOW: 5,
            ↪ BudgetLevel.MEDIUM: 8, BudgetLevel.HIGH: 10},
        'visualization': 10,
        'optimization': 7,
        'python_integration': 5,
        'programming_skill': {'low': 9, 'medium': 8, 'high': 6},
    }
},
]

recommendations = []

for tool in tools:
    score = 0
    s = tool['scores']

    # Weight factors
    score += 0.2 * s['complexity'][requirements.complexity]
    score += 0.25 * s['budget'][requirements.budget]
    score += 0.15 * (s['visualization'] if requirements.needs_visualization else
↪ 5)
    score += 0.15 * (s['optimization'] if requirements.needs_optimization else
↪ 5)
    score += 0.1 * (s['python_integration'] if 'Python' in
↪ requirements.integration_needed else 5)
    score += 0.15 * s['programming_skill'][requirements.team_programming_skill]

    recommendations.append({
        'tool': tool['name'],
        'score': score,
    })

```



```

        'fit': 'Excellent' if score >= 8 else 'Good' if score >= 6 else 'Fair'
    })

    return sorted(recommendations, key=lambda x: x['score'], reverse=True)

# Example usage
def tool_selection_example():
    """Demonstrate tool selection process."""
    requirements = ProjectRequirements(
        complexity=ModelComplexity.MEDIUM,
        budget=BudgetLevel.LOW,
        needs_visualization=True,
        needs_optimization=True,
        integration_needed=["Python", "Pandas"],
        team_programming_skill="high",
        deployment="desktop"
    )

    recommendations = recommend_simulation_tool(requirements)

    print("Tool Recommendations:")
    print("-" * 40)
    for i, rec in enumerate(recommendations, 1):
        print(f"{i}. {rec['tool']}: Score={rec['score']:.2f} ({rec['fit']})")

    return recommendations

```

28.11.3. Open Source vs Commercial Trade-offs

Factor	Open Source (SimPy, Salabim)	Commercial (Arena, AnyLogic)
Cost	Free	\$10K - \$100K+ annually
Support	Community, documentation	Dedicated support, training
Flexibility	Complete control	Constrained by platform
Learning curve	Steeper (coding required)	Gentler (visual modeling)
Performance	Variable, optimizable	Generally optimized
Scalability	Unlimited	License-dependent
Integration	Native Python/Java	API-based
Visualization	Requires development	Built-in, polished
Validation	Manual	Built-in tools
Reproducibility	Version control friendly	Platform-dependent

28.12. Summary

28.12.1. Key Takeaways

- **Simulation-optimization** is necessary when objectives/constraints cannot be expressed analytically

- **Black-box methods** (pattern search, Nelder-Mead) work without gradient information
- **Surrogate models** (GP, RBF) approximate expensive simulations for efficient optimization
- **Bayesian optimization** balances exploration and exploitation via acquisition functions
- **Stochastic methods** (SAA, SPSSA) handle noisy simulation outputs
- **Hybrid approaches** combine mathematical programming with simulation validation
- **Ranking and selection** (OCBA, IZ) efficiently compare discrete alternatives
- Budget allocation matters: invest more simulation effort in promising regions
- **Discrete Event Simulation** (SimPy) models systems with events, resources, and queues
- **Queueing theory** (M/M/c, Little's Law) provides analytical validation of simulations
- **Application domains** include manufacturing, healthcare, and supply chain systems
- **Metamodeling** (RSM, Kriging, neural networks) creates fast surrogate approximations
- **Digital twins** enable real-time simulation, state estimation, and predictive maintenance
- **Tool selection** should match project requirements for complexity, budget, and team skills

28.13. Exercises

28.13.1. Simulation-Optimization Fundamentals

1. **Surrogate comparison:** Implement both GP and RBF surrogates for a 5-dimensional test function. Compare their accuracy and optimization performance.
2. **Noise handling:** For a noisy simulation, compare the performance of: (a) single evaluation, (b) 10 replications averaged, (c) adaptive replication based on variance.
3. **Manufacturing optimization:** Build a discrete-event simulation of a 3-station manufacturing line. Use surrogate optimization to find optimal buffer sizes.
4. **OCBA implementation:** Implement OCBA for comparing 10 inventory policies. Verify it finds the best policy with fewer total simulations than equal allocation.
5. **Hybrid approach:** For a facility location problem, implement a hybrid that uses MIP for location decisions and simulation to evaluate service levels. Iterate until convergence.

28.13.2. Discrete Event Simulation

6. **M/M/c Validation:** Implement an M/M/3 queue simulation using SimPy. Validate your results against analytical queueing theory formulas. Report the relative error for average wait time, average queue length, and server utilization across 10 independent replications.
7. **Priority Queue System:** Extend the basic queue simulation to handle 3 priority classes (High, Medium, Low). Implement both non-preemptive and preemptive priority disciplines. Compare the average wait times for each class under both schemes.
8. **Little's Law Verification:** Using your simulation data, verify Little's Law ($L = \lambda W$) for both the queue subsystem and the entire system. Discuss any discrepancies and their causes.

28.13.3. Application Domains

9. **Manufacturing Line Bottleneck:** Build a 5-station serial production line with different processing rates. Identify the bottleneck through simulation. Then, propose and test three different improvement scenarios (adding capacity, reducing variability, adding buffer).
10. **Emergency Department Staffing:** Using the ED simulation framework, determine the optimal staffing levels (nurses, doctors) to achieve a target of less than 30-minute average wait time for URGENT patients while minimizing total labor cost.
11. **Supply Chain Bullwhip Effect:** Simulate a 4-tier supply chain (Retailer, Distributor, Wholesaler, Manufacturer) with information delays. Measure the bullwhip effect (variance

amplification ratio) and test strategies to reduce it.

12. **Healthcare Capacity Planning:** Model a hospital ward with patient arrivals, bed assignments, and discharges. Determine the minimum number of beds needed to keep blocking probability below 5%.

28.13.4. Metamodeling

13. **Response Surface Design:** For a simulation with 4 factors, compare Box-Behnken and Central Composite designs. Fit second-order polynomial models and compare their predictive accuracy using cross-validation.
14. **Kriging vs RSM:** For a highly nonlinear simulation function, compare the prediction accuracy of Kriging and quadratic RSM. Use the same number of simulation evaluations and measure RMSE on a test set.
15. **Active Learning:** Implement sequential design using Expected Improvement. Compare the efficiency (number of simulations to reach near-optimal) against random sampling and Latin Hypercube sampling.
16. **Neural Network Surrogate:** Train a neural network surrogate on simulation data. Experiment with different architectures (depth, width) and compare to Gaussian Process predictions in terms of accuracy and computation time.

28.13.5. Digital Twins

17. **State Estimation:** Implement a simple Kalman filter for state estimation in a manufacturing digital twin. Compare estimated states with true states under different levels of sensor noise.
18. **Predictive Maintenance:** Develop a degradation model for a pump system. Use the digital twin framework to estimate Remaining Useful Life (RUL) and optimize maintenance scheduling to minimize total cost.
19. **Real-Time Simulation Synchronization:** Build a digital twin that synchronizes with simulated “real” data arriving in a stream. Implement online model calibration when discrepancies between predictions and observations exceed a threshold.

28.13.6. Tool Evaluation

20. **Tool Selection Case Study:** For a given project scenario (e.g., hospital expansion planning), evaluate at least 3 different simulation tools using the selection criteria framework. Present a recommendation with justification.

28.13.7. Capstone Projects

21. **End-to-End Manufacturing Digital Twin:** Create a complete digital twin for a 3-station manufacturing cell including:
 - DES model in SimPy
 - Metamodel for rapid what-if analysis
 - Simulated sensor data stream
 - Anomaly detection
 - Predictive maintenance for each station
 - Optimization of production scheduling
22. **Healthcare System Optimization:** Design a comprehensive healthcare simulation study that:
 - Models patient flow through an emergency department

28. *Simulation-Optimization*

- Validates against real-world queueing theory
- Builds a metamodel for staffing optimization
- Accounts for time-varying arrival patterns
- Provides decision support for capacity planning

29. Causal Inference for Decision Intelligence

What You Will Learn

- Understand the difference between correlation and causation in decision-making
- Build and analyze causal graphs using DAGs (Directed Acyclic Graphs)
- Apply do-calculus to estimate causal effects from observational data
- Implement uplift modeling for targeted interventions
- Design and analyze A/B tests with causal rigor
- Combine causal inference with optimization for better decisions

29.1. Introduction

Most optimization models assume we know how decisions affect outcomes. But how do we learn these relationships? Traditional machine learning finds correlations, but correlations can be misleading for decisions. Causal inference provides the tools to understand *what happens when we intervene*.

Consider these business questions:

- Does offering a discount *cause* customers to buy more?
- Will investing in training *cause* productivity to increase?
- Does the new pricing algorithm *cause* higher revenue?

Answering these requires causal reasoning, not just prediction. This chapter shows how to build causal models and integrate them with optimization.

29.2. Causal vs Predictive Thinking

29.2.1. The Fundamental Problem

Prediction asks: “Given what I observe, what will happen?” Causation asks: “If I *do* something, what will happen?”

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats

np.random.seed(42)

# Simulated data: Ice cream sales and drowning deaths
# Both are caused by summer heat (confounder)
n_days = 365

# Temperature is the confounder
temperature = 15 + 10 * np.sin(2 * np.pi * np.arange(n_days) / 365) +
    ↪ np.random.normal(0, 3, n_days)

# Ice cream sales depend on temperature
```

```

ice_cream_sales = 100 + 5 * temperature + np.random.normal(0, 20, n_days)

# Drowning incidents depend on temperature (more swimming in hot weather)
drowning_incidents = 2 + 0.1 * temperature + np.random.poisson(1, n_days)

# Correlation between ice cream and drowning
correlation = np.corrcoef(ice_cream_sales, drowning_incidents)[0, 1]

print("=== Correlation vs Causation Demo ===")
print(f"\nCorrelation between ice cream sales and drowning: {correlation:.3f}")
print("\nThis correlation is SPURIOUS - ice cream doesn't cause drowning!")
print("Both are caused by a common factor: temperature (confounder)")

# Show the true causal structure
print("\nTrue Causal Structure:")
print("  Temperature -> Ice Cream Sales")
print("  Temperature -> Drowning Incidents")
print("  Ice Cream Sales -/-> Drowning (no causal link)")

```

```
=== Correlation vs Causation Demo ===
```

```
Correlation between ice cream sales and drowning: 0.528
```

```
This correlation is SPURIOUS - ice cream doesn't cause drowning!
Both are caused by a common factor: temperature (confounder)
```

```
True Causal Structure:
  Temperature -> Ice Cream Sales
  Temperature -> Drowning Incidents
  Ice Cream Sales -/-> Drowning (no causal link)
```

The following visualization shows this spurious correlation side by side with the true relationship that emerges after controlling for the confounder.

29.2.2. Why This Matters for Optimization

If we naively optimize based on correlations, we might make wrong decisions:

```
=== Decision Implications ===
```

```
Naive (correlation-based) recommendation:
'Reduce ice cream sales to prevent drowning' -> WRONG!
```

```
Causal (correct) recommendation:
'Increase lifeguard staffing when temperature is high' -> RIGHT!
```

Causal inference helps us identify the RIGHT intervention targets.

29.3. Directed Acyclic Graphs (DAGs)

DAGs are the language of causation. They explicitly show which variables cause which.

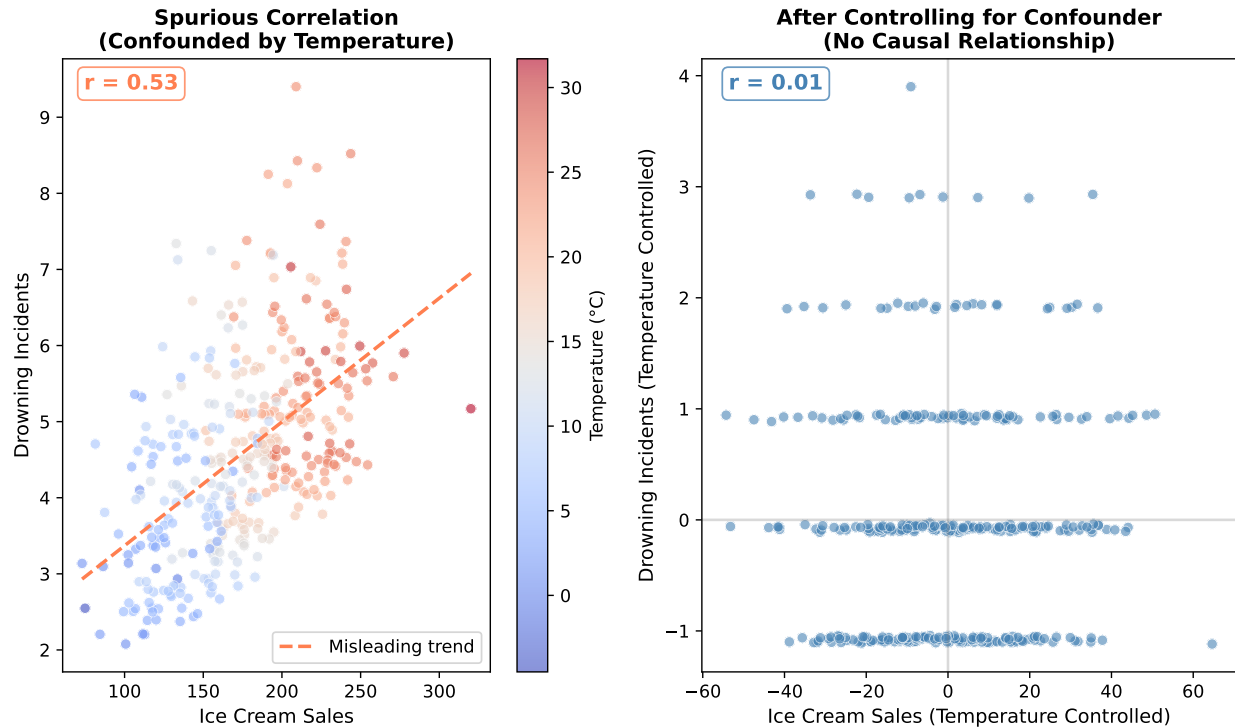


Figure 29.1.: Spurious correlation vs causal effect: The left panel shows the misleading correlation between ice cream sales and drowning incidents. The right panel reveals that controlling for temperature (the confounder) eliminates this spurious relationship.

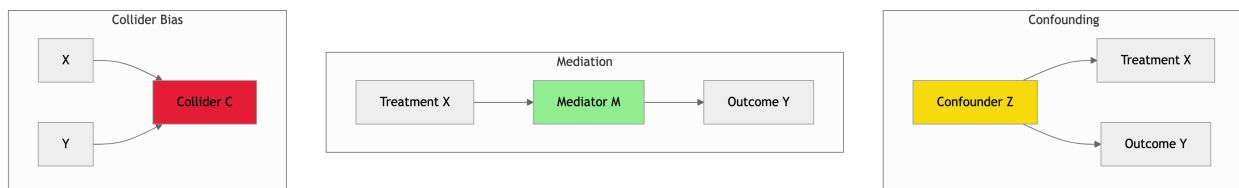


Figure 29.2.: Causal DAG showing common patterns: confounding, mediation, and collider

29.3.1. Building Causal Graphs

Causal graphs encode assumptions about which variables cause which. The `CausalDAG` class below represents a directed acyclic graph where each edge indicates a direct causal effect. We can then query this structure to identify confounders, mediators, and colliders.

```
# Simple DAG representation using adjacency structure
class CausalDAG:
    """Simple Directed Acyclic Graph for causal modeling."""

    def __init__(self):
        self.nodes = set()
        self.edges = {} # parent -> list of children
        self.parents = {} # child -> list of parents
```

```

def add_node(self, node: str):
    """Add a node to the graph."""
    self.nodes.add(node)
    if node not in self.edges:
        self.edges[node] = []
    if node not in self.parents:
        self.parents[node] = []

def add_edge(self, parent: str, child: str):
    """Add a directed edge from parent to child."""
    self.add_node(parent)
    self.add_node(child)
    self.edges[parent].append(child)
    self.parents[child].append(parent)

def get_ancestors(self, node: str) -> set:
    """Get all ancestors of a node."""
    ancestors = set()
    to_visit = list(self.parents.get(node, []))
    while to_visit:
        current = to_visit.pop()
        if current not in ancestors:
            ancestors.add(current)
            to_visit.extend(self.parents.get(current, []))
    return ancestors

def get_descendants(self, node: str) -> set:
    """Get all descendants of a node."""
    descendants = set()
    to_visit = list(self.edges.get(node, []))
    while to_visit:
        current = to_visit.pop()
        if current not in descendants:
            descendants.add(current)
            to_visit.extend(self.edges.get(current, []))
    return descendants

def is_d_separated(self, x: str, y: str, z: set) -> bool:
    """Check if X and Y are d-separated given Z (simplified)."""
    # This is a simplified check - full d-separation is more complex
    # Here we check if all paths are blocked
    return len(self.get_ancestors(x) & self.get_ancestors(y) - z) == 0

def __repr__(self):
    lines = ["Causal DAG:"]
    for parent, children in self.edges.items():
        for child in children:
            lines.append(f" {parent} -> {child}")
    return "\n".join(lines)

# Example: Marketing campaign DAG
marketing_dag = CausalDAG()

```



```
# Add causal relationships
marketing_dag.add_edge("customer_segment", "receives_discount")
marketing_dag.add_edge("customer_segment", "purchase_amount")
marketing_dag.add_edge("receives_discount", "purchase_amount")
marketing_dag.add_edge("customer_segment", "loyalty")
marketing_dag.add_edge("loyalty", "purchase_amount")

print(marketing_dag)
print("\nThis DAG shows:")
print("- Customer segment affects whether they receive discounts")
print("- Customer segment also directly affects purchase amount")
print("- Receiving a discount affects purchase amount")
print("- Customer segment affects loyalty, which affects purchases")
```

Causal DAG:

```
customer_segment -> receives_discount
customer_segment -> purchase_amount
customer_segment -> loyalty
receives_discount -> purchase_amount
loyalty -> purchase_amount
```

This DAG shows:

- Customer segment affects whether they receive discounts
- Customer segment also directly affects purchase amount
- Receiving a discount affects purchase amount
- Customer segment affects loyalty, which affects purchases

29.3.2. Identifying Confounders

A confounder is a variable that influences both the treatment and the outcome, creating a spurious association. The following function identifies confounders by finding common ancestors of the treatment and outcome variables in the DAG.

```
def identify_confounders(dag: CausalDAG, treatment: str, outcome: str) -> set:
    """
    Identify confounders between treatment and outcome.

    Confounders are common causes of both treatment and outcome.
    """
    treatment_ancestors = dag.get_ancestors(treatment)
    outcome_ancestors = dag.get_ancestors(outcome)

    # Confounders are ancestors of both
    confounders = treatment_ancestors & outcome_ancestors

    # Also include direct common causes
    treatment_parents = set(dag.parents.get(treatment, []))
    outcome_parents = set(dag.parents.get(outcome, []))
    direct_confounders = treatment_parents & outcome_parents
```

```

    return confounders | direct_confounders

# Identify confounders for discount -> purchase relationship
confounders = identify_confounders(
    marketing_dag,
    treatment="receives_discount",
    outcome="purchase_amount"
)

print("=== Confounder Analysis ===")
print(f"\nTreatment: receives_discount")
print(f"Outcome: purchase_amount")
print(f"Confounders: {confounders}")
print("\nTo estimate the causal effect of discount on purchase,")
print("we must control for these confounders!")

```

=== Confounder Analysis ===

Treatment: receives_discount
 Outcome: purchase_amount
 Confounders: {'customer_segment'}

To estimate the causal effect of discount on purchase,
 we must control for these confounders!

29.4. The Do-Operator and Interventions

Pearl's do-calculus distinguishes between *observing* and *intervening*.

29.4.1. Observational vs Interventional Distributions

```

# P(Y | X=x) vs P(Y | do(X=x))
# Observational: What Y values do we see when X happens to equal x?
# Interventional: What Y values would we see if we SET X to x?

class CausalModel:
    """
    Structural Causal Model for simulation and intervention.
    """

    def __init__(self, seed: int = 42):
        self.rng = np.random.RandomState(seed)

    def generate_observational(self, n: int) -> pd.DataFrame:
        """Generate observational data from the true causal model."""

        # Customer segment (confounder)
        # 0 = low value, 1 = medium value, 2 = high value
        segment = self.rng.choice([0, 1, 2], size=n, p=[0.4, 0.4, 0.2])

        # Treatment assignment (depends on segment)

```

```

# High value customers more likely to get discounts
discount_prob = np.array([0.2, 0.4, 0.7])
discount = self.rng.binomial(1, discount_prob[segment])

# Outcome (depends on segment and discount)
base_purchase = 50 + 30 * segment # segment effect
treatment_effect = 15 # true causal effect of discount
noise = self.rng.normal(0, 10, n)

purchase = base_purchase + treatment_effect * discount + noise

return pd.DataFrame({
    'segment': segment,
    'discount': discount,
    'purchase': purchase
})

def generate_interventional(self, n: int, do_discount: int) -> pd.DataFrame:
    """Generate data under intervention do(discount=value)."""

    # Segment is still generated naturally
    segment = self.rng.choice([0, 1, 2], size=n, p=[0.4, 0.4, 0.2])

    # Discount is SET to the intervention value (breaks dependence on segment)
    discount = np.full(n, do_discount)

    # Outcome still depends on segment and discount
    base_purchase = 50 + 30 * segment
    treatment_effect = 15
    noise = self.rng.normal(0, 10, n)

    purchase = base_purchase + treatment_effect * discount + noise

    return pd.DataFrame({
        'segment': segment,
        'discount': discount,
        'purchase': purchase
    })

# Generate data
model = CausalModel(seed=42)
obs_data = model.generate_observational(10000)

# Naive estimate (biased due to confounding)
treated = obs_data[obs_data['discount'] == 1]['purchase'].mean()
control = obs_data[obs_data['discount'] == 0]['purchase'].mean()
naive_effect = treated - control

print("=== Observational vs Interventional ===")
print(f"\nTrue causal effect: 15.0")
print(f"\nNaive estimate (correlation): {naive_effect:.2f}")
print("(Biased because high-value customers get more discounts AND buy more)")

```

```
# Interventional data (what a randomized experiment would show)
do_1 = model.generate_interventional(5000, do_discount=1)
do_0 = model.generate_interventional(5000, do_discount=0)
true_effect = do_1['purchase'].mean() - do_0['purchase'].mean()

print(f"\nTrue interventional effect: {true_effect:.2f}")
```

=== Observational vs Interventional ===

True causal effect: 15.0

Naive estimate (correlation): 31.88

(Biased because high-value customers get more discounts AND buy more)

True interventional effect: 14.64

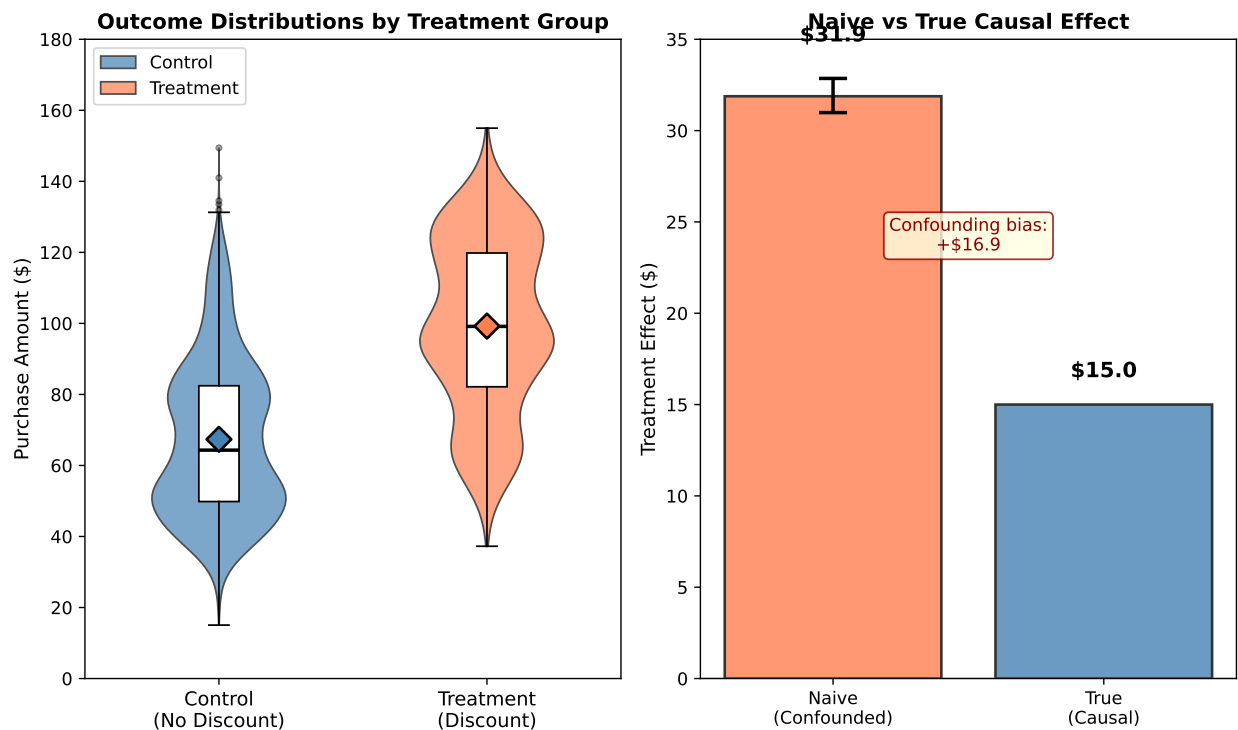


Figure 29.3.: Treatment effect visualization: Comparison of outcome distributions between control (no discount) and treated (discount) groups. The left panel shows violin plots with embedded box plots revealing the distribution shapes, while the right panel displays the treatment effect with confidence intervals.

29.4.2. Backdoor Adjustment

The backdoor criterion tells us which variables to control for:

```

def backdoor_adjustment(data: pd.DataFrame,
                        treatment: str,
                        outcome: str,
                        confounders: list) -> float:
    """
    Estimate causal effect using backdoor adjustment.

     $E[Y \mid \text{do}(T=1)] - E[Y \mid \text{do}(T=0)] =$ 
     $\sum_z P(Z=z) * (E[Y|T=1, Z=z] - E[Y|T=0, Z=z])$ 
    """
    effects = []
    weights = []

    # Get unique confounder combinations
    confounder_data = data[confounders]
    unique_combinations = confounder_data.drop_duplicates()

    for _, z_values in unique_combinations.iterrows():
        # Create mask for this confounder combination
        mask = np.all(confounder_data == z_values.values, axis=1)
        subset = data[mask]

        if len(subset) < 10: # skip small groups
            continue

        # Weight by P(Z=z)
        weight = len(subset) / len(data)

        # Conditional treatment effect
        treated = subset[subset[treatment] == 1][outcome]
        control = subset[subset[treatment] == 0][outcome]

        if len(treated) > 0 and len(control) > 0:
            conditional_effect = treated.mean() - control.mean()
            effects.append(conditional_effect)
            weights.append(weight)

    # Weighted average
    weights = np.array(weights)
    weights = weights / weights.sum() # normalize

    return np.sum(np.array(effects) * weights)

# Apply backdoor adjustment
adjusted_effect = backdoor_adjustment(
    obs_data,
    treatment='discount',
    outcome='purchase',
    confounders=['segment']
)

print("=== Backdoor Adjustment ===")

```

```
print(f"\nNaive estimate: {naive_effect:.2f}")
print(f"Backdoor-adjusted estimate: {adjusted_effect:.2f}")
print(f"True causal effect: 15.0")
print("\nControlling for 'segment' removes confounding bias!")
```

=== Backdoor Adjustment ===

```
Naive estimate: 31.88
Backdoor-adjusted estimate: 15.05
True causal effect: 15.0
```

Controlling for 'segment' removes confounding bias!

29.5. Propensity Score Methods

When we have many confounders, propensity scores provide a practical solution.

29.5.1. Inverse Probability Weighting (IPW)

```
from sklearn.linear_model import LogisticRegression

def estimate_propensity_scores(data: pd.DataFrame,
                              treatment: str,
                              confounders: list) -> np.ndarray:
    """Estimate propensity scores using logistic regression."""

    X = data[confounders].values
    y = data[treatment].values

    model = LogisticRegression(random_state=42)
    model.fit(X, y)

    # P(T=1 | X)
    propensity = model.predict_proba(X)[:, 1]

    # Clip to avoid extreme weights
    propensity = np.clip(propensity, 0.01, 0.99)

    return propensity

def ipw_estimate(data: pd.DataFrame,
                 treatment: str,
                 outcome: str,
                 confounders: list) -> float:
    """
    Inverse Probability Weighting estimate of causal effect.

    ATE = E[Y * T / e(X)] - E[Y * (1-T) / (1-e(X))]
    """
    propensity = estimate_propensity_scores(data, treatment, confounders)
```

```

T = data[treatment].values
Y = data[outcome].values

# IPW estimator
treated_term = np.mean(Y * T / propensity)
control_term = np.mean(Y * (1 - T) / (1 - propensity))

ate = treated_term - control_term

return ate

# Estimate using IPW
ipw_effect = ipw_estimate(
    obs_data,
    treatment='discount',
    outcome='purchase',
    confounders=['segment']
)

print("=== Inverse Probability Weighting ===")
print(f"\nPropensity-weighted ATE: {ipw_effect:.2f}")
print(f"True causal effect: 15.0")

```

=== Inverse Probability Weighting ===

Propensity-weighted ATE: 15.74

True causal effect: 15.0

29.5.2. Doubly Robust Estimation

Combines outcome modeling with propensity scores for robustness:

```

from sklearn.linear_model import LinearRegression

def doubly_robust_estimate(data: pd.DataFrame,
                           treatment: str,
                           outcome: str,
                           confounders: list) -> float:
    """
    Doubly robust estimator combining IPW and outcome regression.

    Consistent if EITHER the propensity model OR outcome model is correct.
    """
    X = data[confounders].values
    T = data[treatment].values
    Y = data[outcome].values

    # Estimate propensity scores
    prop_model = LogisticRegression(random_state=42)
    prop_model.fit(X, T)
    e = prop_model.predict_proba(X)[:, 1]
    e = np.clip(e, 0.01, 0.99)

```

```

# Fit outcome models for treated and control
X_with_T = np.column_stack([X, T])
outcome_model = LinearRegression()
outcome_model.fit(X_with_T, Y)

# Predict potential outcomes
X_treated = np.column_stack([X, np.ones(len(X))])
X_control = np.column_stack([X, np.zeros(len(X))])

mu_1 = outcome_model.predict(X_treated) # E[Y | X, T=1]
mu_0 = outcome_model.predict(X_control) # E[Y | X, T=0]

# Doubly robust estimator
dr_treated = mu_1 + T * (Y - mu_1) / e
dr_control = mu_0 + (1 - T) * (Y - mu_0) / (1 - e)

ate = np.mean(dr_treated - dr_control)

return ate

# Estimate using doubly robust method
dr_effect = doubly_robust_estimate(
    obs_data,
    treatment='discount',
    outcome='purchase',
    confounders=['segment']
)

print("=== Doubly Robust Estimation ===")
print(f"\nDoubly robust ATE: {dr_effect:.2f}")
print(f"True causal effect: 15.0")
print(f"\nAdvantage: Works if EITHER propensity OR outcome model is correct!")

```

```
=== Doubly Robust Estimation ===
```

```
Doubly robust ATE: 15.05
```

```
True causal effect: 15.0
```

```
Advantage: Works if EITHER propensity OR outcome model is correct!
```

29.6. Uplift Modeling

Uplift modeling estimates heterogeneous treatment effects to identify WHO should be treated.

29.6.1. Treatment Effect Heterogeneity

```

class UpliftModel:
    """
    Meta-learner for heterogeneous treatment effect estimation.
    Uses the T-Learner approach: separate models for treatment and control.
    """

```



```

"""

def __init__(self):
    self.model_treated = LinearRegression()
    self.model_control = LinearRegression()

def fit(self, X: np.ndarray, T: np.ndarray, Y: np.ndarray):
    """Fit separate models for treated and control groups."""

    treated_mask = T == 1
    control_mask = T == 0

    self.model_treated.fit(X[treated_mask], Y[treated_mask])
    self.model_control.fit(X[control_mask], Y[control_mask])

    return self

def predict_uplift(self, X: np.ndarray) -> np.ndarray:
    """Predict individual treatment effects (uplift)."""

    mu_1 = self.model_treated.predict(X)
    mu_0 = self.model_control.predict(X)

    return mu_1 - mu_0

def recommend_treatment(self, X: np.ndarray,
                        threshold: float = 0) -> np.ndarray:
    """Recommend treatment for individuals with positive uplift."""

    uplift = self.predict_uplift(X)
    return uplift > threshold

# Generate data with heterogeneous treatment effects
np.random.seed(42)
n = 5000

# Features
age = np.random.uniform(20, 70, n)
income = np.random.uniform(30000, 150000, n)
loyalty_score = np.random.uniform(0, 100, n)

X = np.column_stack([age, income, loyalty_score])

# Treatment (randomly assigned for clean estimation)
T = np.random.binomial(1, 0.5, n)

# Heterogeneous treatment effect:
# - Higher for young customers
# - Higher for low-loyalty customers (more room to improve)
base_effect = 20
age_modifier = -0.3 * (age - 45) # younger = higher effect
loyalty_modifier = -0.1 * (loyalty_score - 50) # lower loyalty = higher effect

```

```

individual_effect = base_effect + age_modifier + loyalty_modifier

# Outcome
base_purchase = 50 + 0.0005 * income + 0.2 * loyalty_score
Y = base_purchase + individual_effect * T + np.random.normal(0, 10, n)

# Fit uplift model
uplift_model = UpliftModel()
uplift_model.fit(X, T, Y)

# Predict uplift for all customers
predicted_uplift = uplift_model.predict_uplift(X)

print("=== Uplift Modeling Results ===")
print(f"\nAverage predicted uplift: {predicted_uplift.mean():.2f}")
print(f"Uplift std dev: {predicted_uplift.std():.2f}")
print(f"\nUplift by age group:")
for age_group in [(20, 35), (35, 50), (50, 70)]:
    mask = (age >= age_group[0]) & (age < age_group[1])
    print(f"  Age {age_group[0]}--{age_group[1]}:
        ↪ {predicted_uplift[mask].mean():.2f}")

```

```
=== Uplift Modeling Results ===
```

```
Average predicted uplift: 19.92
```

```
Uplift std dev: 5.49
```

```
Uplift by age group:
```

```
  Age 20-35: 25.20
```

```
  Age 35-50: 20.69
```

```
  Age 50-70: 15.31
```

29.6.2. Targeting Optimization

Once we have individual-level treatment effect estimates, we can formulate a budget-constrained targeting problem. The goal is to select the subset of customers whose treatment yields the highest total uplift, subject to a maximum number of treatments.

```

def optimize_targeting(uplift: np.ndarray,
                       cost_per_treatment: float,
                       budget: float) -> np.ndarray:
    """
    Optimize who to target given budget constraints.

    Maximize expected uplift subject to budget.
    """
    n = len(uplift)
    max_treatments = int(budget / cost_per_treatment)

    # Rank by uplift (highest first)
    ranking = np.argsort(-uplift)

```

```

# Select top customers
selected = np.zeros(n, dtype=bool)
selected[ranking[:max_treatments]] = True

return selected

# Simulate targeting decisions
cost_per_discount = 10 # cost of giving discount
budget = 10000 # total budget

# Optimize targeting
target_mask = optimize_targeting(
    predicted_uplift,
    cost_per_treatment=cost_per_discount,
    budget=budget
)

print("=== Targeting Optimization ===")
print(f"\nBudget: ${budget}")
print(f"Cost per treatment: ${cost_per_discount}")
print(f"Customers targeted: {target_mask.sum()}")
print(f"\nExpected outcomes:")

# Compare strategies
random_target = np.random.choice(n, size=target_mask.sum(), replace=False)
random_mask = np.zeros(n, dtype=bool)
random_mask[random_target] = True

# Expected value under different strategies
optimal_value = predicted_uplift[target_mask].sum()
random_value = predicted_uplift[random_mask].sum()
all_value = predicted_uplift.sum()

print(f"  Optimized targeting: ${optimal_value:.0f} total uplift")
print(f"  Random targeting: ${random_value:.0f} total uplift")
print(f"  Treat everyone: ${all_value:.0f} total uplift (cost: ${n *
    ↪ cost_per_discount})")
print(f"\nOptimized targeting captures {optimal_value/all_value*100:.1f}% of total
    ↪ uplift")
print(f"with only {target_mask.sum()/n*100:.1f}% of customers treated!")

```

=== Targeting Optimization ===

Budget: \$10000

Cost per treatment: \$10

Customers targeted: 1000

Expected outcomes:

Optimized targeting: \$27666 total uplift

Random targeting: \$19873 total uplift

Treat everyone: \$99585 total uplift (cost: \$50000)

Optimized targeting captures 27.8% of total uplift
with only 20.0% of customers treated!

29.7. A/B Testing with Causal Rigor

29.7.1. Designing Experiments

A well-designed A/B test requires careful power analysis to determine the sample size needed to detect a meaningful effect. The code below computes the minimum sample size for a given significance level, power, and expected effect size.

```
from scipy import stats

class ABTest:
    """
    A/B test design and analysis with causal guarantees.
    """

    def __init__(self,
                 baseline_rate: float,
                 minimum_detectable_effect: float,
                 alpha: float = 0.05,
                 power: float = 0.8):
        """
        Initialize A/B test parameters.

        Args:
            baseline_rate: Expected outcome in control group
            minimum_detectable_effect: Smallest effect worth detecting
            alpha: Type I error rate
            power: 1 - Type II error rate
        """
        self.baseline = baseline_rate
        self.mde = minimum_detectable_effect
        self.alpha = alpha
        self.power = power

    def calculate_sample_size(self) -> int:
        """Calculate required sample size per group."""

        # Effect size (Cohen's h for proportions)
        p1 = self.baseline
        p2 = self.baseline * (1 + self.mde)

        # Pooled variance
        pooled_var = p1 * (1 - p1) + p2 * (1 - p2)

        # Z-scores
        z_alpha = stats.norm.ppf(1 - self.alpha / 2)
        z_beta = stats.norm.ppf(self.power)

        # Sample size formula
```

```

n = 2 * pooled_var * (z_alpha + z_beta) ** 2 / (p2 - p1) ** 2

return int(np.ceil(n))

def analyze(self,
            control_successes: int,
            control_total: int,
            treatment_successes: int,
            treatment_total: int) -> dict:
    """
    Analyze A/B test results.

    Returns effect estimate, confidence interval, and p-value.
    """
    # Point estimates
    p_control = control_successes / control_total
    p_treatment = treatment_successes / treatment_total

    effect = p_treatment - p_control
    relative_effect = effect / p_control if p_control > 0 else np.inf

    # Standard error
    se = np.sqrt(
        p_control * (1 - p_control) / control_total +
        p_treatment * (1 - p_treatment) / treatment_total
    )

    # Confidence interval
    z = stats.norm.ppf(1 - self.alpha / 2)
    ci_lower = effect - z * se
    ci_upper = effect + z * se

    # P-value (two-sided)
    z_stat = effect / se if se > 0 else np.inf
    p_value = 2 * (1 - stats.norm.cdf(abs(z_stat)))

    return {
        'control_rate': p_control,
        'treatment_rate': p_treatment,
        'absolute_effect': effect,
        'relative_effect': relative_effect,
        'ci_lower': ci_lower,
        'ci_upper': ci_upper,
        'p_value': p_value,
        'significant': p_value < self.alpha
    }

# Example: Testing a new checkout flow
ab_test = ABTest(
    baseline_rate=0.03, # 3% conversion rate
    minimum_detectable_effect=0.1, # Want to detect 10% relative improvement
    alpha=0.05,

```

```

        power=0.8
    )

    sample_size = ab_test.calculate_sample_size()
    print("=== A/B Test Design ===")
    print(f"\nBaseline conversion rate: {ab_test.baseline:.1%}")
    print(f"Minimum detectable effect: {ab_test.mde:.1%} relative")
    print(f"Significance level (alpha): {ab_test.alpha}")
    print(f"Statistical power: {ab_test.power}")
    print(f"\nRequired sample size per group: {sample_size:,}")
    print(f"Total sample size: {2 * sample_size:,}")

```

```
=== A/B Test Design ===
```

```

Baseline conversion rate: 3.0%
Minimum detectable effect: 10.0% relative
Significance level (alpha): 0.05
Statistical power: 0.8

```

```

Required sample size per group: 106,416
Total sample size: 212,832

```

29.7.2. Analyzing Results

After collecting data, we perform a two-sample t-test and compute confidence intervals for the treatment effect. This implementation also checks for practical significance by comparing the observed lift against a minimum detectable effect threshold.

```

# Simulated experiment results
np.random.seed(42)

# Simulate experiment with true 12% relative lift
true_lift = 0.12
control_n = sample_size
treatment_n = sample_size

control_rate = 0.03
treatment_rate = 0.03 * (1 + true_lift)

control_successes = np.random.binomial(control_n, control_rate)
treatment_successes = np.random.binomial(treatment_n, treatment_rate)

# Analyze
results = ab_test.analyze(
    control_successes=control_successes,
    control_total=control_n,
    treatment_successes=treatment_successes,
    treatment_total=treatment_n
)

print("=== A/B Test Results ===")
print(f"\nControl: {control_successes}/{control_n} = {results['control_rate']:.2%}")

```

```

print(f"Treatment: {treatment_successes}/{treatment_n} =
    ↳ {results['treatment_rate']:.2%}")
print(f"\nAbsolute effect: {results['absolute_effect']:.4f}
    ↳ ({results['absolute_effect']*100:.2f}pp)")
print(f"Relative effect: {results['relative_effect']:.1%}")
print(f"95% CI: [{results['ci_lower']:.4f}, {results['ci_upper']:.4f}]")
print(f"P-value: {results['p_value']:.4f}")
print(f"\nStatistically significant: {results['significant']}")

if results['significant']:
    print("\nConclusion: The new checkout flow significantly improves conversion!")
else:
    print("\nConclusion: No significant difference detected.")

```

=== A/B Test Results ===

Control: 3139/106416 = 2.95%

Treatment: 3623/106416 = 3.40%

Absolute effect: 0.0045 (0.45pp)

Relative effect: 15.4%

95% CI: [0.0031, 0.0060]

P-value: 0.0000

Statistically significant: True

Conclusion: The new checkout flow significantly improves conversion!

29.7.3. Sequential Testing

Standard hypothesis tests assume a fixed sample size, but in practice we often want to monitor results as data arrives. Sequential testing methods adjust for multiple looks at the data, controlling the overall false positive rate while allowing early stopping when the evidence is strong.

```

class SequentialABTest:
    """
    Sequential A/B testing using group sequential design.

    Allows for early stopping while controlling false positive rate.
    """

    def __init__(self,
                  max_sample_size: int,
                  n_looks: int = 5,
                  alpha: float = 0.05):
        """
        Initialize sequential test.

        Args:
            max_sample_size: Maximum sample size per group
            n_looks: Number of interim analyses

```

```

        alpha: Overall Type I error rate (spending function controls per-look
↪ alpha)
        """
        self.max_n = max_sample_size
        self.n_looks = n_looks
        self.alpha = alpha

        # O'Brien-Fleming spending function (conservative early, aggressive late)
        self.look_times = np.linspace(1/n_looks, 1, n_looks)
        self.alpha_spent = self._obrien_fleming_bounds()

    def _obrien_fleming_bounds(self) -> np.ndarray:
        """Calculate O'Brien-Fleming alpha spending."""
        # Simplified version
        bounds = []
        for t in self.look_times:
            # Approximate O'Brien-Fleming
            z_bound = stats.norm.ppf(1 - self.alpha/2) / np.sqrt(t)
            alpha_at_look = 2 * (1 - stats.norm.cdf(z_bound))
            bounds.append(alpha_at_look)
        return np.array(bounds)

    def check_stopping(self,
                      control_successes: int,
                      control_total: int,
                      treatment_successes: int,
                      treatment_total: int,
                      look: int) -> dict:
        """
        Check if we should stop at current interim analysis.
        """
        if look > self.n_looks:
            raise ValueError(f"look must be <= {self.n_looks}")

        # Calculate test statistic
        p_c = control_successes / control_total
        p_t = treatment_successes / treatment_total

        pooled_p = (control_successes + treatment_successes) / (control_total +
↪ treatment_total)
        se = np.sqrt(pooled_p * (1 - pooled_p) * (1/control_total +
↪ 1/treatment_total))

        z_stat = (p_t - p_c) / se if se > 0 else 0

        # Current boundary
        alpha_boundary = self.alpha_spent[look - 1]
        z_boundary = stats.norm.ppf(1 - alpha_boundary / 2)

        stop_for_efficacy = abs(z_stat) > z_boundary

        return {
            'look': look,

```



```

        'fraction_enrolled': self.look_times[look - 1],
        'z_statistic': z_stat,
        'z_boundary': z_boundary,
        'p_value': 2 * (1 - stats.norm.cdf(abs(z_stat))),
        'stop_for_efficacy': stop_for_efficacy,
        'effect_estimate': p_t - p_c
    }

# Demo sequential testing
sequential_test = SequentialABTest(
    max_sample_size=50000,
    n_looks=5,
    alpha=0.05
)

print("=== Sequential A/B Test ===")
print("\nO'Brien-Fleming boundaries allow early stopping while controlling alpha.")
print("\nInterim analysis boundaries:")
for i, (t, a) in enumerate(zip(sequential_test.look_times,
    ↪ sequential_test.alpha_spent)):
    z = stats.norm.ppf(1 - a/2)
    print(f" Look {i+1} ({t*100:.0f}% enrolled): z > {z:.3f} to stop")

```

=== Sequential A/B Test ===

O'Brien-Fleming boundaries allow early stopping while controlling alpha.

Interim analysis boundaries:

```

Look 1 (20% enrolled): z > 4.383 to stop
Look 2 (40% enrolled): z > 3.099 to stop
Look 3 (60% enrolled): z > 2.530 to stop
Look 4 (80% enrolled): z > 2.191 to stop
Look 5 (100% enrolled): z > 1.960 to stop

```

29.8. Combining Causal Inference with Optimization

29.8.1. Causal-Aware Optimization

Standard optimization uses observed correlations to set decision variables. Causal-aware optimization instead uses estimated treatment effects, which account for confounding. The following example builds a marketing budget allocator that maximizes expected profit using causal effect estimates rather than raw correlations.

```

class CausalOptimizer:
    """
    Optimization that accounts for causal effects.

    Combines treatment effect estimation with resource allocation.
    """

```

```

def __init__(self, uplift_model):
    self.uplift_model = uplift_model

def optimize_allocation(self,
                        X: np.ndarray,
                        treatment_costs: np.ndarray,
                        budget: float,
                        outcome_value: float = 1.0) -> dict:
    """
    Optimize treatment allocation given heterogeneous effects.

    Maximize total expected value subject to budget.
    """
    n = len(X)

    # Predict individual treatment effects
    uplifts = self.uplift_model.predict_uplift(X)

    # Value per treatment = uplift * outcome_value - cost
    net_values = uplifts * outcome_value - treatment_costs

    # Greedy allocation (optimal for this linear problem)
    # Sort by net value per dollar cost
    efficiency = net_values / treatment_costs
    ranking = np.argsort(-efficiency)

    # Allocate until budget exhausted
    allocated = np.zeros(n, dtype=bool)
    total_cost = 0

    for idx in ranking:
        if net_values[idx] <= 0:
            break # Don't treat if negative net value
        if total_cost + treatment_costs[idx] <= budget:
            allocated[idx] = True
            total_cost += treatment_costs[idx]

    # Calculate expected outcomes
    expected_uplift = uplifts[allocated].sum()
    expected_revenue = expected_uplift * outcome_value
    expected_profit = expected_revenue - total_cost

    return {
        'allocated': allocated,
        'n_treated': allocated.sum(),
        'total_cost': total_cost,
        'expected_uplift': expected_uplift,
        'expected_revenue': expected_revenue,
        'expected_profit': expected_profit
    }

# Use the uplift model from earlier

```

```

optimizer = CausalOptimizer(uplift_model)

# Variable treatment costs (e.g., different discount amounts)
treatment_costs = np.random.uniform(5, 20, n)

# Optimize allocation
results = optimizer.optimize_allocation(
    X=X,
    treatment_costs=treatment_costs,
    budget=15000,
    outcome_value=1.5 # Each unit of uplift worth $1.50
)

print("=== Causal-Aware Optimization ===")
print(f"\nBudget: $15,000")
print(f"Outcome value: $1.50 per unit")
print(f"\nOptimal allocation:")
print(f"  Customers treated: {results['n_treated']}")
print(f"  Total cost: ${results['total_cost']:.0f}")
print(f"  Expected uplift: {results['expected_uplift']:.0f} units")
print(f"  Expected revenue: ${results['expected_revenue']:.0f}")
print(f"  Expected profit: ${results['expected_profit']:.0f}")
print(f"  ROI: {(results['expected_profit']/results['total_cost'])*100:.1f}%")

```

=== Causal-Aware Optimization ===

Budget: \$15,000

Outcome value: \$1.50 per unit

Optimal allocation:

Customers treated: 1800

Total cost: \$14999

Expected uplift: 40914 units

Expected revenue: \$61372

Expected profit: \$46372

ROI: 309.2%

29.8.2. Policy Learning

Policy learning uses estimated treatment effects to derive a decision rule that maps individual features to optimal actions. This goes beyond average treatment effects: instead of applying the same treatment to everyone, we learn a personalized policy that assigns treatment only when the predicted individual benefit exceeds the cost.

```

from sklearn.tree import DecisionTreeClassifier

class PolicyLearner:
    """
    Learn interpretable treatment policies from causal estimates.
    """

```

```

def __init__(self, max_depth: int = 4):
    self.tree = DecisionTreeClassifier(
        max_depth=max_depth,
        random_state=42
    )
    self.feature_names = None

def fit(self, X: np.ndarray, uplifts: np.ndarray,
        feature_names: list = None):
    """
    Learn a policy that predicts who should be treated.
    """
    # Binary classification: treat if uplift > 0
    should_treat = (uplifts > 0).astype(int)

    # Weight by absolute uplift (more important to get high-uplift right)
    sample_weights = np.abs(uplifts)

    self.tree.fit(X, should_treat, sample_weight=sample_weights)
    self.feature_names = feature_names

    return self

def recommend(self, X: np.ndarray) -> np.ndarray:
    """Recommend treatment based on learned policy."""
    return self.tree.predict(X)

def explain_policy(self) -> str:
    """Generate human-readable policy rules."""
    from sklearn.tree import export_text

    feature_names = self.feature_names or [f'X{i}' for i in
↪ range(self.tree.n_features_in_)]

    return export_text(self.tree, feature_names=feature_names)

# Learn interpretable policy
policy = PolicyLearner(max_depth=3)
policy.fit(
    X=X,
    uplifts=predicted_uplift,
    feature_names=['age', 'income', 'loyalty_score']
)

print("=== Learned Treatment Policy ===")
print("\nDecision tree policy (who should receive discount):\n")
print(policy.explain_policy())

# Evaluate policy
policy_recommendations = policy.recommend(X)
print(f"\nPolicy recommends treating {policy_recommendations.sum()} out of {len(X)}
↪ customers")

```

```
print(f"Average uplift for recommended: {predicted_uplift[policy_recommendations ==
↪ 1].mean():.2f}")
print(f"Average uplift for not recommended: {predicted_uplift[policy_recommendations
↪ == 0].mean():.2f}")
```

=== Learned Treatment Policy ===

Decision tree policy (who should receive discount):

```
|--- class: 0
```

Policy recommends treating 5000 out of 5000 customers

Average uplift for recommended: 19.92

Average uplift for not recommended: nan

29.9. Instrumental Variables

When we cannot measure all confounders, instrumental variables (IV) provide an alternative approach to causal identification. An instrument is a variable that affects the treatment but has no direct effect on the outcome except through the treatment.

29.9.1. IV Assumptions

Instrumental variables require three key assumptions:

1. **Relevance:** The instrument must affect the treatment ($Z \rightarrow X$)
2. **Exclusion Restriction:** The instrument affects the outcome only through the treatment (no $Z \rightarrow Y$ direct path)
3. **Independence:** The instrument is independent of unmeasured confounders

```
def two_stage_least_squares(Z: np.ndarray, X: np.ndarray, Y: np.ndarray) -> dict:
    """
    Two-Stage Least Squares (2SLS) IV estimator.

    Args:
        Z: Instrument
        X: Treatment
        Y: Outcome

    Returns:
        Dictionary with IV estimate and diagnostics
    """
    from scipy import stats

    # First stage: Regress X on Z
    first_stage = stats.linregress(Z, X)
    X_hat = first_stage.intercept + first_stage.slope * Z

    # Check instrument strength (F-statistic)
    n = len(Z)
```

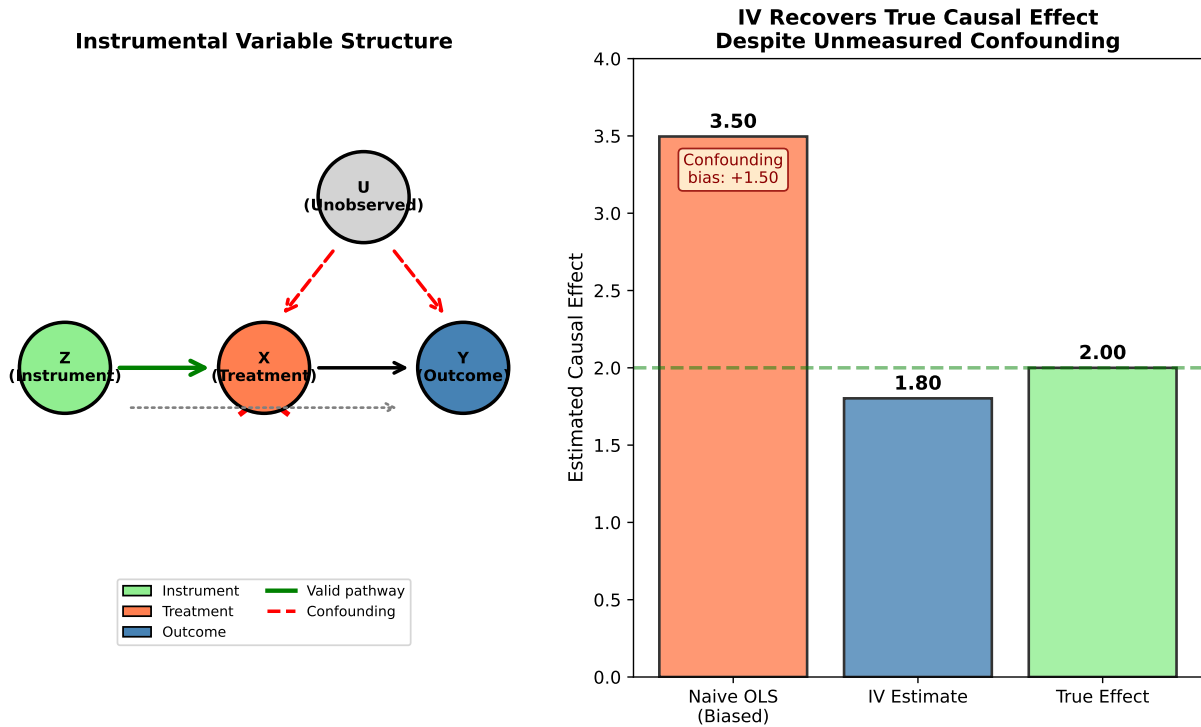


Figure 29.4.: Instrumental variable relationships: The left panel illustrates the IV structure where the instrument (Z) affects the outcome (Y) only through the treatment (X), bypassing the unmeasured confounder (U). The right panel demonstrates IV estimation using simulated data, showing how the IV estimate recovers the true causal effect despite confounding.

```

X_residuals = X - X_hat
mse_first = np.sum(X_residuals**2) / (n - 2)
ss_z = np.sum((Z - Z.mean())**2)
f_stat = (first_stage.slope**2 * ss_z) / mse_first

# Second stage: Regress Y on X_hat
second_stage = stats.linregress(X_hat, Y)
iv_estimate = second_stage.slope

# Standard error (simplified)
Y_residuals = Y - (second_stage.intercept + second_stage.slope * X_hat)
mse_second = np.sum(Y_residuals**2) / (n - 2)
se_iv = np.sqrt(mse_second / (first_stage.slope**2 * ss_z))

return {
    'iv_estimate': iv_estimate,
    'standard_error': se_iv,
    'first_stage_f': f_stat,
    'first_stage_coef': first_stage.slope,
    'weak_instrument': f_stat < 10 # Rule of thumb
}

```

```

# Demonstrate IV estimation
np.random.seed(42)
n = 1000

# Simulate IV scenario
Z = np.random.binomial(1, 0.5, n) # Randomized encouragement
U = np.random.normal(0, 1, n)     # Unmeasured confounder
X = 0.5 * Z + 0.8 * U + np.random.normal(0, 0.3, n) # Treatment
Y = 2.0 * X + 1.5 * U + np.random.normal(0, 0.5, n) # Outcome

# Estimate
results = two_stage_least_squares(Z, X, Y)

print("=== Instrumental Variable Estimation ===")
print(f"\nTrue causal effect: 2.00")
print(f"IV estimate: {results['iv_estimate']:.2f}")
print(f"Standard error: {results['standard_error']:.2f}")
print(f"95% CI: [{results['iv_estimate'] - 1.96*results['standard_error']:.2f}, "
      f"{results['iv_estimate'] + 1.96*results['standard_error']:.2f}]")
print(f"\nFirst stage F-statistic: {results['first_stage_f']:.1f}")
print(f"Weak instrument warning: {results['weak_instrument']}")

```

```
=== Instrumental Variable Estimation ===
```

```
True causal effect: 2.00
```

```
IV estimate: 1.80
```

```
Standard error: 0.40
```

```
95% CI: [1.02, 2.59]
```

```
First stage F-statistic: 87.2
```

```
Weak instrument warning: False
```

29.10. Practical Considerations

29.10.1. Sensitivity Analysis

Causal estimates depend on the assumption that all confounders are measured. Sensitivity analysis tests how robust the estimated effect is to unmeasured confounding. The code below varies the strength of a hypothetical hidden confounder and shows how the treatment effect estimate changes.

```
=== Sensitivity to Unmeasured Confounding ===
```

Gamma represents the odds ratio by which an unmeasured confounder could change treatment assignment.

gamma	adjusted_lower	adjusted_upper	robust
1.00	15.000000	15.000000	True
1.25	14.442141	15.557859	True
1.50	13.986337	16.013663	True

29. Causal Inference for Decision Intelligence

1.75	13.600961	16.399039	True
2.00	13.267132	16.732868	True
2.50	12.709273	17.290727	True
3.00	12.253469	17.746531	True

Conclusion: Effect remains significant up to $\gamma = 3.00$
This means unmeasured confounders would need to increase treatment odds by this factor to explain away our findings.

Best Practices for Causal Inference

1. **Draw the DAG first** - Make your causal assumptions explicit
2. **Identify confounders** - Use domain knowledge, not just statistics
3. **Check overlap** - Propensity scores should overlap between groups
4. **Use doubly robust estimators** - They protect against model misspecification
5. **Run sensitivity analyses** - Quantify robustness to unmeasured confounding
6. **When possible, experiment** - RCTs remain the gold standard

29.11. Summary

29.11.0.1. Key Takeaways

Causal vs Predictive

- Correlation is not causation - predictions can mislead decisions
- Causal graphs (DAGs) make assumptions explicit
- Do-calculus formalizes interventions

Estimation Methods

- Backdoor adjustment controls for confounders
- Propensity scores balance treatment groups
- Doubly robust methods combine approaches for robustness

Practical Applications

- Uplift modeling targets interventions to maximize impact
- A/B testing provides gold-standard causal evidence
- Sequential testing allows efficient experimentation

Integration with Optimization

- Causal effects inform objective functions
- Heterogeneous effects enable personalized decisions
- Policy learning creates interpretable treatment rules

29.12. Exercises

1. **DAG Construction:** For your domain, draw a causal DAG showing the relationships between a treatment you could apply, the outcomes you care about, and potential confounders. Identify the minimal adjustment set.
2. **Propensity Matching:** Using observational data, estimate propensity scores and create a matched sample. Compare the naive and matched estimates.

3. **Uplift Modeling:** Build an uplift model for a targeting problem. Calculate the expected value of using uplift-based targeting versus random targeting.
4. **A/B Test Design:** Design an A/B test for a business decision. Calculate the required sample size and set up a sequential testing framework.
5. **Sensitivity Analysis:** For a causal estimate from observational data, conduct a sensitivity analysis. How robust is your conclusion to unmeasured confounding?

29.13. References

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30. Explainable AI for Decision Systems

What You Will Learn

- Understand why explainability is critical for production decision systems
- Apply SHAP values to explain model predictions
- Use LIME for local interpretable explanations
- Generate counterfactual explanations for decision understanding
- Explain optimization decisions to stakeholders
- Build trust through transparency in automated decisions

30.1. Introduction

Black-box models can achieve impressive accuracy, but in production decision systems, accuracy alone is not enough. Stakeholders ask: *Why did the system recommend this?* Regulators require: *How does the model make decisions?* Users demand: *What would I need to change to get a different outcome?*

Explainable AI (XAI) bridges the gap between model performance and human understanding. This chapter covers practical techniques for explaining both predictions and optimization decisions.

30.2. The Explainability Imperative

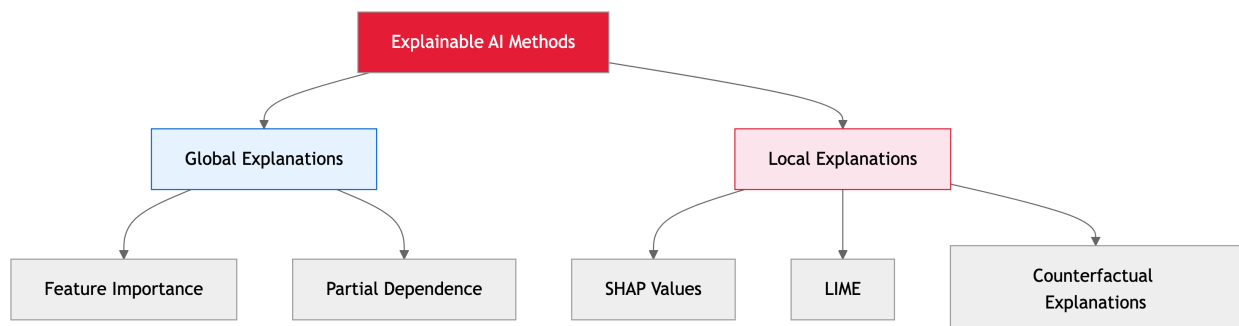


Figure 30.1.: Taxonomy of Explainable AI Methods

30.2.1. Why Explanation Matters

Different domains have different explanation requirements. Regulated industries like healthcare and finance often need explanations by law, while operational settings need them to build user trust. The following code maps common application domains to their explanation needs.

```
# Real consequences of unexplainable decisions
examples = {
    "Loan Application": {
        "decision": "Denied",
```

```

        "stakeholder_questions": [
            "Why was I rejected?",
            "What would I need to change?",
            "Is this decision fair?"
        ],
        "regulatory_requirement": "Right to explanation (GDPR, ECOA)"
    },
    "Pricing Recommendation": {
        "decision": "Price increased 20%",
        "stakeholder_questions": [
            "Why is the model suggesting this?",
            "What factors drove this?",
            "Will customers accept this?"
        ],
        "regulatory_requirement": "Price discrimination laws"
    },
    "Supply Chain": {
        "decision": "Reduce inventory by 40%",
        "stakeholder_questions": [
            "What assumptions is this based on?",
            "What are the risks?",
            "Should I trust this?"
        ],
        "regulatory_requirement": "Audit trails, accountability"
    }
}

print("=== Why Explainability Matters ===\n")
for domain, info in examples.items():
    print(f"{domain}:")
    print(f"  Decision: {info['decision']}")
    print(f"  Stakeholder asks:")
    for q in info['stakeholder_questions']:
        print(f"    - {q}")
    print(f"  Requirement: {info['regulatory_requirement']}")
    print()

```

=== Why Explainability Matters ===

Loan Application:

Decision: Denied

Stakeholder asks:

- Why was I rejected?
- What would I need to change?
- Is this decision fair?

Requirement: Right to explanation (GDPR, ECOA)

Pricing Recommendation:

Decision: Price increased 20%

Stakeholder asks:

- Why is the model suggesting this?
- What factors drove this?

- Will customers accept this?

Requirement: Price discrimination laws

Supply Chain:

Decision: Reduce inventory by 40%

Stakeholder asks:

- What assumptions is this based on?
- What are the risks?
- Should I trust this?

Requirement: Audit trails, accountability

30.2.2. The Accuracy-Interpretability Trade-off

Model complexity and interpretability often pull in opposite directions. Simple models like linear regression are easy to explain but may underfit. Complex models like deep networks can capture nonlinear patterns but require post-hoc explanation methods. The following code quantifies this tradeoff on a sample dataset.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.tree import DecisionTreeClassifier, DecisionTreeRegressor
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier

# Model spectrum from interpretable to black-box
model_spectrum = pd.DataFrame({
    'Model': [
        'Linear Regression',
        'Logistic Regression',
        'Decision Tree (shallow)',
        'Decision Tree (deep)',
        'Random Forest',
        'Gradient Boosting',
        'Neural Network'
    ],
    'Interpretability': [5, 5, 4, 2, 2, 1, 1],
    'Typical_Accuracy': [2, 3, 3, 4, 5, 5, 5],
    'Native_Explanation': [
        'Coefficients',
        'Odds ratios',
        'Decision rules',
        'Complex rules',
        'Feature importance',
        'Feature importance',
        'None'
    ]
})

print("=== Model Interpretability Spectrum ===\n")
print(model_spectrum.to_string(index=False))
print("\n(Scores: 1=Low, 5=High)")
```

```
print("\nKey insight: Post-hoc explanation methods allow using")
print("high-accuracy models while maintaining explainability!")
```

=== Model Interpretability Spectrum ===

Model	Interpretability	Typical_Accuracy	Native_Explanation
Linear Regression	5	2	Coefficients
Logistic Regression	5	3	Odds ratios
Decision Tree (shallow)	4	3	Decision rules
Decision Tree (deep)	2	4	Complex rules
Random Forest	2	5	Feature importance
Gradient Boosting	1	5	Feature importance
Neural Network	1	5	None

(Scores: 1=Low, 5=High)

Key insight: Post-hoc explanation methods allow using
high-accuracy models while maintaining explainability!

30.3. Feature Importance Methods

Feature importance methods quantify how much each input variable contributes to a model's predictions. This section covers permutation importance (model-agnostic) and SHAP values (theoretically grounded).

30.3.1. Permutation Importance

Permutation importance measures the drop in model performance when a feature's values are randomly shuffled, breaking its relationship with the target. The following code implements this approach.

```
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

def permutation_importance(model, X, y, n_repeats: int = 10,
                           random_state: int = 42) -> dict:
    """
    Calculate permutation importance for each feature.

    Measures how much model performance drops when a feature is shuffled.
    """
    rng = np.random.RandomState(random_state)
    baseline_score = accuracy_score(y, model.predict(X))

    importances = {}
    importance_std = {}

    for col_idx in range(X.shape[1]):
```

```

    scores = []

    for _ in range(n_repeats):
        X_permuted = X.copy()
        rng.shuffle(X_permuted[:, col_idx])
        permuted_score = accuracy_score(y, model.predict(X_permuted))
        scores.append(baseline_score - permuted_score)

    importances[col_idx] = np.mean(scores)
    importance_std[col_idx] = np.std(scores)

    return {
        'importances': importances,
        'std': importance_std,
        'baseline_score': baseline_score
    }

# Generate synthetic data
np.random.seed(42)
n_samples = 1000
n_features = 10

X, y = make_classification(
    n_samples=n_samples,
    n_features=n_features,
    n_informative=5,
    n_redundant=2,
    n_classes=2,
    random_state=42
)

feature_names = [f'Feature_{i}' for i in range(n_features)]

# Train model
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
    ↪ random_state=42)
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Calculate permutation importance
perm_imp = permutation_importance(model, X_test, y_test)

print("=== Permutation Feature Importance ===\n")
print(f"Baseline accuracy: {perm_imp['baseline_score']:.3f}\n")

# Sort by importance
sorted_features = sorted(perm_imp['importances'].items(), key=lambda x: -x[1])

print("Feature Importances (drop in accuracy when permuted):")
for feat_idx, importance in sorted_features[:5]:
    std = perm_imp['std'][feat_idx]
    print(f"    {feature_names[feat_idx]}: {importance:.4f} (+/- {std:.4f})")

```

=== Permutation Feature Importance ===

Baseline accuracy: 0.940

Feature Importances (drop in accuracy when permuted):

```
Feature_4: 0.1977 (+/- 0.0201)
Feature_0: 0.1143 (+/- 0.0103)
Feature_5: 0.1057 (+/- 0.0158)
Feature_9: 0.0477 (+/- 0.0118)
Feature_3: 0.0253 (+/- 0.0076)
```

30.3.2. SHAP Values

SHAP (SHapley Additive exPlanations) provides consistent, theoretically grounded feature attributions:

```
class SimpleSHAP:
    """
    Simplified SHAP implementation using Kernel SHAP approximation.

    For production, use the shap library: pip install shap
    """

    def __init__(self, model, X_background: np.ndarray, n_samples: int = 100):
        """
        Initialize SHAP explainer.

        Args:
            model: Fitted model with predict method
            X_background: Background dataset for expected values
            n_samples: Number of samples for approximation
        """
        self.model = model
        self.X_background = X_background
        self.n_samples = n_samples
        self.n_features = X_background.shape[1]

        # Expected value (average prediction over background)
        if hasattr(model, 'predict_proba'):
            self.expected_value = model.predict_proba(X_background)[:, 1].mean()
        else:
            self.expected_value = model.predict(X_background).mean()

    def _predict(self, X):
        """Get predictions from model."""
        if hasattr(self.model, 'predict_proba'):
            return self.model.predict_proba(X)[:, 1]
        return self.model.predict(X)

    def explain_instance(self, x: np.ndarray) -> np.ndarray:
        """
        Calculate SHAP values for a single instance.
        """
```



```

    Uses sampling-based approximation.
    """
    x = x.reshape(1, -1)
    shap_values = np.zeros(self.n_features)

    # Sample coalitions
    for _ in range(self.n_samples):
        # Random permutation of features
        perm = np.random.permutation(self.n_features)

        # For each feature, calculate marginal contribution
        for i, feat_idx in enumerate(perm):
            # Features before in permutation
            features_before = set(perm[:i])

            # Create instances with/without this feature
            x_with = self.X_background.copy()
            x_without = self.X_background.copy()

            for j in range(self.n_features):
                if j in features_before or j == feat_idx:
                    x_with[:, j] = x[0, j]
                if j in features_before:
                    x_without[:, j] = x[0, j]

            # Marginal contribution
            pred_with = self._predict(x_with).mean()
            pred_without = self._predict(x_without).mean()

            shap_values[feat_idx] += (pred_with - pred_without) / self.n_samples

    return shap_values

# Create SHAP explainer
# Use subset of training data as background
background = X_train[:50]
explainer = SimpleSHAP(model, background, n_samples=50)

# Explain a single prediction
test_instance = X_test[0]
shap_values = explainer.explain_instance(test_instance)

# Get prediction for this instance
if hasattr(model, 'predict_proba'):
    prediction = model.predict_proba(test_instance.reshape(1, -1))[0, 1]
else:
    prediction = model.predict(test_instance.reshape(1, -1))[0]

print("=== SHAP Explanation for Single Instance ===\n")
print(f"Model prediction (P(class=1)): {prediction:.3f}")
print(f"Expected value (baseline): {explainer.expected_value:.3f}")
print(f"Difference to explain: {prediction - explainer.expected_value:.3f}")

```

```

print(f"\nFeature contributions (SHAP values):")

# Sort by absolute contribution
sorted_idx = np.argsort(-np.abs(shap_values))
for idx in sorted_idx[:5]:
    direction = "+" if shap_values[idx] > 0 else "-"
    print(f"    {feature_names[idx]}: {direction}{abs(shap_values[idx]):.4f}")

print(f"\nSum of SHAP values: {shap_values.sum():.4f}")
print(f"(Should approximately equal prediction - expected value)")

```

=== SHAP Explanation for Single Instance ===

Model prediction (P(class=1)): 0.010

Expected value (baseline): 0.532

Difference to explain: -0.522

Feature contributions (SHAP values):

Feature_4: -0.2160

Feature_5: -0.1327

Feature_9: -0.0910

Feature_6: -0.0462

Feature_3: -0.0349

Sum of SHAP values: -0.5220

(Should approximately equal prediction - expected value)

30.3.3. Interpreting SHAP Output

The following code shows how to interpret SHAP values for individual predictions. Positive SHAP values push the prediction higher, negative values push it lower, and the sum of all SHAP values equals the difference between the prediction and the model's average output.

```

def create_shap_summary(model, X: np.ndarray, feature_names: list,
                        n_explain: int = 100) -> pd.DataFrame:
    """
    Create summary of SHAP values across multiple instances.
    """
    # Use simplified explainer
    explainer = SimpleSHAP(model, X[:50], n_samples=30)

    all_shap_values = []
    for i in range(min(n_explain, len(X))):
        shap_vals = explainer.explain_instance(X[i])
        all_shap_values.append(shap_vals)

    shap_matrix = np.array(all_shap_values)

    # Summary statistics
    summary = pd.DataFrame({
        'Feature': feature_names,

```

```

        'Mean_Abs_SHAP': np.abs(shap_matrix).mean(axis=0),
        'Std_SHAP': shap_matrix.std(axis=0),
        'Mean_SHAP': shap_matrix.mean(axis=0)
    })

    return summary.sort_values('Mean_Abs_SHAP', ascending=False)

# Create summary
summary = create_shap_summary(model, X_test, feature_names, n_explain=50)

print("=== SHAP Summary (Global Feature Importance) ===\n")
print(summary.head(7).to_string(index=False))
print("\nInterpretation:")
print("- Mean_Abs_SHAP: Average impact magnitude (global importance)")
print("- Mean_SHAP: Average direction of impact")
print("- Std_SHAP: Consistency of feature effect")

```

```
=== SHAP Summary (Global Feature Importance) ===
```

Feature	Mean_Abs_SHAP	Std_SHAP	Mean_SHAP
Feature_4	0.166966	0.192782	-0.002894
Feature_5	0.091296	0.116879	0.000363
Feature_0	0.083865	0.107590	0.001543
Feature_9	0.047790	0.067451	0.000170
Feature_3	0.038471	0.060591	0.000665
Feature_1	0.024851	0.032588	-0.000058
Feature_6	0.019226	0.033414	-0.000079

Interpretation:

- Mean_Abs_SHAP: Average impact magnitude (global importance)
- Mean_SHAP: Average direction of impact
- Std_SHAP: Consistency of feature effect

30.4. Local Interpretable Explanations (LIME)

LIME explains individual predictions by fitting a simple model locally:

```

from sklearn.linear_model import Ridge

class SimpleLIME:
    """
    Local Interpretable Model-agnostic Explanations.

    Explains predictions by fitting a linear model in the
    neighborhood of the instance.
    """

    def __init__(self, model, X_train: np.ndarray,
                 n_samples: int = 500, kernel_width: float = 0.75):
        """

```

```

Initialize LIME explainer.

Args:
    model: Fitted model to explain
    X_train: Training data for determining feature ranges
    n_samples: Number of perturbed samples
    kernel_width: Width of the exponential kernel
"""
self.model = model
self.X_train = X_train
self.n_samples = n_samples
self.kernel_width = kernel_width

# Feature statistics for perturbation
self.feature_means = X_train.mean(axis=0)
self.feature_stds = X_train.std(axis=0)

def _predict(self, X):
    """Get predictions from model."""
    if hasattr(self.model, 'predict_proba'):
        return self.model.predict_proba(X)[: , 1]
    return self.model.predict(X)

def _perturb_instance(self, x: np.ndarray) -> np.ndarray:
    """Generate perturbed samples around instance."""
    n_features = len(x)
    perturbations = np.random.normal(0, 1, (self.n_samples, n_features))
    perturbed = x + perturbations * self.feature_stds * 0.5
    return perturbed

def _compute_weights(self, x: np.ndarray, perturbed: np.ndarray) -> np.ndarray:
    """Compute kernel weights based on distance to original."""
    distances = np.sqrt(((perturbed - x) ** 2).sum(axis=1))
    weights = np.exp(-distances ** 2 / (self.kernel_width ** 2))
    return weights

def explain_instance(self, x: np.ndarray) -> dict:
    """
    Explain a single prediction using LIME.

    Returns local linear approximation coefficients.
    """
    x = x.reshape(-1)

    # Generate perturbed samples
    perturbed = self._perturb_instance(x)

    # Get model predictions on perturbed data
    predictions = self._predict(perturbed)

    # Compute weights
    weights = self._compute_weights(x, perturbed)

```

```

# Fit weighted linear model
linear_model = Ridge(alpha=1.0)
linear_model.fit(perturbed, predictions, sample_weight=weights)

# Original prediction
original_pred = self._predict(x.reshape(1, -1))[0]

return {
    'coefficients': linear_model.coef_,
    'intercept': linear_model.intercept_,
    'original_prediction': original_pred,
    'local_prediction': linear_model.predict(x.reshape(1, -1))[0],
    'r2_score': linear_model.score(perturbed, predictions,
    ↪ sample_weight=weights)
}

# Create LIME explainer
lime_explainer = SimpleLIME(model, X_train)

# Explain the same instance
lime_result = lime_explainer.explain_instance(test_instance)

print("=== LIME Explanation ===\n")
print(f"Original model prediction: {lime_result['original_prediction']:.3f}")
print(f"Local linear approximation: {lime_result['local_prediction']:.3f}")
print(f"Local model fit (R2): {lime_result['r2_score']:.3f}")
print(f"\nFeature contributions (local linear coefficients):")

# Sort by absolute coefficient
coefs = lime_result['coefficients']
sorted_idx = np.argsort(-np.abs(coefs))

for idx in sorted_idx[:5]:
    value = test_instance[idx]
    contribution = coefs[idx] * value
    print(f"  {feature_names[idx]}: coef={coefs[idx]:.4f}, value={value:.2f},
    ↪ contrib={contribution:.4f}")

```

=== LIME Explanation ===

Original model prediction: 0.010
 Local linear approximation: 0.057
 Local model fit (R2): 0.263

Feature contributions (local linear coefficients):
 Feature_4: coef=-0.0183, value=1.39, contrib=-0.0253
 Feature_0: coef=0.0147, value=0.54, contrib=0.0079
 Feature_2: coef=-0.0126, value=-0.87, contrib=0.0109
 Feature_3: coef=0.0096, value=-0.39, contrib=-0.0038
 Feature_5: coef=-0.0074, value=2.54, contrib=-0.0187

30.5. Counterfactual Explanations

Counterfactuals answer: “What minimal change would lead to a different outcome?”

```
class CounterfactualExplainer:
    """
    Generate counterfactual explanations.

    Finds minimal changes to input that would change the prediction.
    """

    def __init__(self, model, X_train: np.ndarray,
                 feature_names: list = None,
                 categorical_features: list = None):
        """
        Initialize counterfactual explainer.

        Args:
            model: Fitted classifier
            X_train: Training data for bounds and statistics
            feature_names: Names of features
            categorical_features: Indices of categorical features
        """
        self.model = model
        self.X_train = X_train
        self.feature_names = feature_names or [f'X{i}' for i in
        ↪ range(X_train.shape[1])]
        self.categorical = categorical_features or []

        # Feature bounds
        self.feature_min = X_train.min(axis=0)
        self.feature_max = X_train.max(axis=0)
        self.feature_std = X_train.std(axis=0)

    def _predict_proba(self, X):
        """Get probability predictions."""
        if hasattr(self.model, 'predict_proba'):
            return self.model.predict_proba(X)[: , 1]
        return self.model.predict(X)

    def find_counterfactual(self, x: np.ndarray,
                          target_class: int = 1,
                          max_iter: int = 1000,
                          step_size: float = 0.1) -> dict:
        """
        Find counterfactual using gradient-free optimization.

        Uses iterative feature perturbation to find minimal change.
        """
        x = x.reshape(-1).copy()
        original_pred = self._predict_proba(x.reshape(1, -1))[0]
        original_class = 1 if original_pred > 0.5 else 0

        if original_class == target_class:
```

```

        return {
            'success': True,
            'original': x,
            'counterfactual': x,
            'changes': {},
            'message': 'Already in target class'
        }

    best_cf = None
    best_distance = np.inf
    cf = x.copy()

    for iteration in range(max_iter):
        # Current prediction
        pred = self._predict_proba(cf.reshape(1, -1))[0]
        current_class = 1 if pred > 0.5 else 0

        if current_class == target_class:
            # Found a counterfactual
            distance = np.sum(np.abs(cf - x) / (self.feature_std + 1e-8))
            if distance < best_distance:
                best_distance = distance
                best_cf = cf.copy()
            # Try to find a closer one
            cf = x + 0.9 * (cf - x) # Move back toward original
            continue

        # Perturb features toward target
        for feat_idx in range(len(x)):
            if feat_idx in self.categorical:
                continue # Skip categorical for now

            # Try small perturbation
            delta = self.feature_std[feat_idx] * step_size

            # Direction depends on target
            if target_class == 1:
                # Try both directions, pick better
                cf_up = cf.copy()
                cf_up[feat_idx] = min(cf[feat_idx] + delta,
                    ↪ self.feature_max[feat_idx])
                cf_down = cf.copy()
                cf_down[feat_idx] = max(cf[feat_idx] - delta,
                    ↪ self.feature_min[feat_idx])

                pred_up = self._predict_proba(cf_up.reshape(1, -1))[0]
                pred_down = self._predict_proba(cf_down.reshape(1, -1))[0]

                if pred_up > pred_down and pred_up > pred:
                    cf[feat_idx] = cf_up[feat_idx]
                elif pred_down > pred:
                    cf[feat_idx] = cf_down[feat_idx]
            else:

```

```

        # Similar logic for target_class == 0
        cf_up = cf.copy()
        cf_up[feat_idx] = min(cf[feat_idx] + delta,
        ↪ self.feature_max[feat_idx])
        cf_down = cf.copy()
        cf_down[feat_idx] = max(cf[feat_idx] - delta,
        ↪ self.feature_min[feat_idx])

        pred_up = self._predict_proba(cf_up.reshape(1, -1))[0]
        pred_down = self._predict_proba(cf_down.reshape(1, -1))[0]

        if pred_up < pred_down and pred_up < pred:
            cf[feat_idx] = cf_up[feat_idx]
        elif pred_down < pred:
            cf[feat_idx] = cf_down[feat_idx]

    if best_cf is None:
        return {
            'success': False,
            'original': x,
            'counterfactual': None,
            'changes': {},
            'message': 'Could not find counterfactual'
        }

    # Identify changes
    changes = {}
    for i in range(len(x)):
        if abs(best_cf[i] - x[i]) > 1e-6:
            changes[self.feature_names[i]] = {
                'from': x[i],
                'to': best_cf[i],
                'change': best_cf[i] - x[i]
            }

    return {
        'success': True,
        'original': x,
        'counterfactual': best_cf,
        'original_pred': original_pred,
        'cf_pred': self._predict_proba(best_cf.reshape(1, -1))[0],
        'changes': changes
    }

# Create counterfactual explainer
cf_explainer = CounterfactualExplainer(model, X_train, feature_names)

# Find counterfactual for an instance predicted as class 0
# First find such an instance
for i in range(len(X_test)):
    pred = model.predict(X_test[i:i+1])[0]
    if pred == 0:

```



```

        negative_instance = X_test[i]
        break

cf_result = cf_explainer.find_counterfactual(negative_instance, target_class=1)

print("=== Counterfactual Explanation ===\n")
if cf_result['success']:
    print(f"Original prediction: {cf_result['original_pred']:.3f} (Class 0)")
    print(f"Counterfactual prediction: {cf_result['cf_pred']:.3f} (Class 1)")
    print(f"\nMinimal changes needed to flip prediction:")

    for feature, change_info in cf_result['changes'].items():
        print(f"    {feature}:")
        print(f"        From: {change_info['from']:.3f}")
        print(f"        To:    {change_info['to']:.3f}")
        print(f"        Change: {change_info['change']:+.3f}")
else:
    print(cf_result['message'])

```

```
=== Counterfactual Explanation ===
```

```
Could not find counterfactual
```

30.6. Explaining Optimization Decisions

Beyond ML predictions, we need to explain optimization recommendations:

```

class OptimizationExplainer:
    """
    Explain optimization decisions by analyzing sensitivity
    and identifying binding constraints.
    """

    def __init__(self):
        self.last_solution = None
        self.last_problem = None

    def explain_linear_program(self,
                              c: np.ndarray,
                              A_ub: np.ndarray,
                              b_ub: np.ndarray,
                              variable_names: list = None,
                              constraint_names: list = None) -> dict:
        """
        Solve and explain a linear programming decision.

        Args:
            c: Objective coefficients (minimize c^T x)
            A_ub: Inequality constraint matrix (A_ub @ x <= b_ub)
            b_ub: Inequality constraint bounds
            variable_names: Names for decision variables
            constraint_names: Names for constraints

```

```

"""
from scipy.optimize import linprog

n_vars = len(c)
n_constraints = len(b_ub)

variable_names = variable_names or [f'x{i}' for i in range(n_vars)]
constraint_names = constraint_names or [f'Constraint_{i}' for i in
↪ range(n_constraints)]

# Solve the LP
result = linprog(c, A_ub=A_ub, b_ub=b_ub, method='highs')

if not result.success:
    return {'success': False, 'message': result.message}

solution = result.x
objective = result.fun

# Identify binding constraints (slack near zero)
slack = b_ub - A_ub @ solution
binding_threshold = 1e-6
binding = slack < binding_threshold

# Sensitivity analysis: shadow prices (approximate)
shadow_prices = []
for i in range(n_constraints):
    # Perturb constraint
    b_perturbed = b_ub.copy()
    b_perturbed[i] += 1.0 # Relax by 1 unit

    result_perturbed = linprog(c, A_ub=A_ub, b_ub=b_perturbed,
↪ method='highs')
    if result_perturbed.success:
        shadow_price = objective - result_perturbed.fun
        shadow_prices.append(shadow_price)
    else:
        shadow_prices.append(0)

shadow_prices = np.array(shadow_prices)

# Build explanation
explanation = {
    'success': True,
    'solution': dict(zip(variable_names, solution)),
    'objective_value': objective,
    'binding_constraints': [
        constraint_names[i] for i in range(n_constraints) if binding[i]
    ],
    'shadow_prices': dict(zip(constraint_names, shadow_prices)),
    'slack_values': dict(zip(constraint_names, slack))
}

```

```

        return explanation

# Example: Production planning problem
#
# Maximize profit: 40*chairs + 50*tables
# Subject to:
#   2*chairs + 3*tables <= 120   (wood hours)
#   4*chairs + 2*tables <= 100   (labor hours)
#   chairs, tables >= 0

c = np.array([-40, -50]) # Negative for maximization
A_ub = np.array([
    [2, 3], # wood constraint
    [4, 2]  # labor constraint
])
b_ub = np.array([120, 100])

explainer = OptimizationExplainer()
result = explainer.explain_linear_program(
    c, A_ub, b_ub,
    variable_names=['chairs', 'tables'],
    constraint_names=['wood_hours', 'labor_hours']
)

print("=== Optimization Decision Explanation ===\n")
print("Problem: Maximize profit from chairs and tables production")
print("\nOptimal Solution:")
for var, value in result['solution'].items():
    print(f"  Produce {value:.1f} {var}")
print(f"\nMaximum Profit: ${-result['objective_value']:.2f}")

print(f"\nBinding Constraints (fully used resources):")
for constraint in result['binding_constraints']:
    print(f"  - {constraint}")

print(f"\nShadow Prices (value of additional resource):")
for constraint, price in result['shadow_prices'].items():
    if abs(price) > 0.01:
        print(f"  {constraint}: ${-price:.2f} per unit")
    else:
        print(f"  {constraint}: $0 (not binding)")

print("\nInterpretation:")
print("- Both resources are fully utilized (binding)")
print("- Adding 1 wood hour would increase profit by shadow price")
print("- This explains WHY we can't produce more!")

```

=== Optimization Decision Explanation ===

Problem: Maximize profit from chairs and tables production

Optimal Solution:

Produce 7.5 chairs
Produce 35.0 tables

Maximum Profit: \$2050.00

Binding Constraints (fully used resources):

- wood_hours
- labor_hours

Shadow Prices (value of additional resource):

wood_hours: \$-15.00 per unit
labor_hours: \$-2.50 per unit

Interpretation:

- Both resources are fully utilized (binding)
- Adding 1 wood hour would increase profit by shadow price
- This explains WHY we can't produce more!

30.6.1. Explaining Infeasibility

When a model is infeasible, users need to know which constraints conflict and what to relax. The following code identifies an irreducible infeasible subset (IIS) and suggests the minimum constraint relaxations needed to restore feasibility.

```
def explain_infeasibility(A_ub: np.ndarray,
                        b_ub: np.ndarray,
                        constraint_names: list = None) -> dict:
    """
    Diagnose why an optimization problem is infeasible.

    Identifies conflicting constraints.
    """
    from scipy.optimize import linprog

    n_constraints = len(b_ub)
    constraint_names = constraint_names or [f'C{i}' for i in range(n_constraints)]

    # Try removing each constraint
    removable = []

    for i in range(n_constraints):
        A_reduced = np.delete(A_ub, i, axis=0)
        b_reduced = np.delete(b_ub, i)

        # Try to find any feasible solution
        c_dummy = np.zeros(A_ub.shape[1])
        result = linprog(c_dummy, A_ub=A_reduced, b_ub=b_reduced, method='highs')

        if result.success:
            removable.append(constraint_names[i])
```

```

# Find minimal relaxation needed
relaxations = []
for i in range(n_constraints):
    # How much would we need to relax this constraint?
    b_test = b_ub.copy()
    for relax_amount in [1, 5, 10, 50, 100, 500]:
        b_test[i] = b_ub[i] + relax_amount
        c_dummy = np.zeros(A_ub.shape[1])
        result = linprog(c_dummy, A_ub=A_ub, b_ub=b_test, method='highs')
        if result.success:
            relaxations.append((constraint_names[i], relax_amount))
            break

    return {
        'constraints_causing_infeasibility': [
            c for c in constraint_names if c not in removable
        ],
        'potential_relaxations': relaxations
    }

# Example: Infeasible problem
A_infeasible = np.array([
    [1, 0], # x >= 10 (written as -x <= -10)
    [-1, 0],
    [0, 1], # y >= 10
    [0, -1],
    [1, 1], # x + y <= 15 (conflicts!)
])
b_infeasible = np.array([-10, 0, -10, 0, 15])

print("=== Infeasibility Diagnosis ===\n")
print("Problem: x >= 10, y >= 10, x + y <= 15")
print("(Clearly impossible: 10 + 10 = 20 > 15)\n")

diagnosis = explain_infeasibility(
    A_infeasible, b_infeasible,
    constraint_names=['x_lower', 'x_upper', 'y_lower', 'y_upper', 'sum_limit']
)

print("Conflicting constraints:")
for c in diagnosis['constraints_causing_infeasibility']:
    print(f" - {c}")

print("\nPossible relaxations to make feasible:")
for constraint, amount in diagnosis['potential_relaxations']:
    print(f" - Relax {constraint} by {amount}")

```

=== Infeasibility Diagnosis ===

Problem: x >= 10, y >= 10, x + y <= 15
 (Clearly impossible: 10 + 10 = 20 > 15)

Conflicting constraints:

- x_lower
- x_upper
- y_lower
- y_upper
- sum_limit

Possible relaxations to make feasible:

30.7. Building Trust Through Transparency

30.7.1. Explanation Templates

Generating natural-language explanations from model outputs helps non-technical stakeholders understand decisions. The following code uses template-based text generation to produce human-readable explanations from optimization results and feature attributions.

```
class DecisionExplainer:
    """
    Generate human-readable explanations for automated decisions.
    """

    def __init__(self, feature_descriptions: dict = None):
        """
        Initialize with feature descriptions for natural language.

        Args:
            feature_descriptions: Mapping of feature names to descriptions
        """
        self.feature_descriptions = feature_descriptions or {}

    def generate_prediction_explanation(self,
                                     prediction: float,
                                     shap_values: dict,
                                     feature_values: dict,
                                     threshold: float = 0.5) -> str:
        """
        Generate natural language explanation for a prediction.
        """
        decision = "approved" if prediction > threshold else "not approved"
        confidence = abs(prediction - 0.5) * 200 # 0-100 scale

        # Sort factors by impact
        sorted_factors = sorted(
            shap_values.items(),
            key=lambda x: abs(x[1]),
            reverse=True
        )

        # Build explanation
        lines = []
        lines.append(f"Decision: {decision.upper()}")
        lines.append(f"Confidence: {confidence:.0f}%")
```

```

        lines.append("")
        lines.append("Key factors in this decision:")

        for i, (feature, impact) in enumerate(sorted_factors[:3]):
            direction = "increased" if impact > 0 else "decreased"
            description = self.feature_descriptions.get(feature, feature)
            value = feature_values.get(feature, "N/A")

            lines.append(f"{i+1}. {description}")
            lines.append(f"    Value: {value}")
            lines.append(f"    Effect: {direction} approval likelihood")

        return "\n".join(lines)

    def generate_optimization_explanation(self,
                                       decision: dict,
                                       constraints: dict,
                                       objective: str) -> str:
        """
        Generate natural language explanation for optimization decision.
        """
        lines = []
        lines.append(f"Optimization Goal: {objective}")
        lines.append("")
        lines.append("Recommended Actions:")

        for action, value in decision.items():
            lines.append(f"    - {action}: {value}")

        lines.append("")
        lines.append("Constraints Considered:")

        for constraint, status in constraints.items():
            binding = "(binding)" if status['binding'] else "(slack)"
            lines.append(f"    - {constraint}: {status['value']} {binding}")

        return "\n".join(lines)

# Example usage
explainer = DecisionExplainer(feature_descriptions={
    'Feature_0': 'Credit history length',
    'Feature_1': 'Income stability score',
    'Feature_2': 'Debt-to-income ratio',
    'Feature_3': 'Employment duration',
    'Feature_4': 'Payment history'
})

# Simulated SHAP values and features
sample_shap = {
    'Feature_0': 0.15,
    'Feature_1': 0.12,
    'Feature_2': -0.08,

```

```

        'Feature_3': 0.05,
        'Feature_4': 0.03
    }

    sample_values = {
        'Feature_0': '5 years',
        'Feature_1': '85/100',
        'Feature_2': '35%',
        'Feature_3': '3 years',
        'Feature_4': 'Good'
    }

    explanation = explainer.generate_prediction_explanation(
        prediction=0.72,
        shap_values=sample_shap,
        feature_values=sample_values,
        threshold=0.5
    )

    print("=== Human-Readable Explanation ===\n")
    print(explanation)

```

```
=== Human-Readable Explanation ===
```

```
Decision: APPROVED
```

```
Confidence: 44%
```

```
Key factors in this decision:
```

1. Credit history length
 - Value: 5 years
 - Effect: increased approval likelihood
2. Income stability score
 - Value: 85/100
 - Effect: increased approval likelihood
3. Debt-to-income ratio
 - Value: 35%
 - Effect: decreased approval likelihood

30.7.2. Audit Trail Generation

Audit trails record every decision along with its inputs, outputs, and rationale. This is required for regulatory compliance in many industries and also helps with debugging and retrospective analysis. The code below implements a structured audit logger for optimization decisions.

```

from datetime import datetime
import json

class AuditLogger:
    """
    Generate audit trails for decision systems.

```



```

Captures inputs, model state, and explanations for compliance.
"""

def __init__(self, system_name: str, model_version: str):
    self.system_name = system_name
    self.model_version = model_version
    self.logs = []

def log_decision(self,
                 decision_id: str,
                 inputs: dict,
                 prediction: float,
                 decision: str,
                 explanation: dict,
                 user_id: str = None) -> dict:
    """
    Log a decision with full audit trail.
    """
    log_entry = {
        'timestamp': datetime.now().isoformat(),
        'decision_id': decision_id,
        'system': self.system_name,
        'model_version': self.model_version,
        'user_id': user_id,
        'inputs': inputs,
        'prediction': prediction,
        'decision': decision,
        'explanation': explanation,
        'explanation_version': '1.0'
    }

    self.logs.append(log_entry)
    return log_entry

def get_decision(self, decision_id: str) -> dict:
    """Retrieve a specific decision from the audit log."""
    for log in self.logs:
        if log['decision_id'] == decision_id:
            return log
    return None

def export_logs(self) -> str:
    """Export logs as JSON for compliance storage."""
    return json.dumps(self.logs, indent=2)

# Example audit logging
audit = AuditLogger(
    system_name="LoanApprovalSystem",
    model_version="v2.3.1"
)

# Log a decision

```

```

logged = audit.log_decision(
    decision_id="LOAN-2024-001234",
    inputs={
        'applicant_income': 75000,
        'credit_score': 720,
        'debt_ratio': 0.35
    },
    prediction=0.72,
    decision="APPROVED",
    explanation={
        'top_factors': [
            {'feature': 'credit_score', 'impact': 'positive', 'contribution': 0.15},
            {'feature': 'income', 'impact': 'positive', 'contribution': 0.12},
            {'feature': 'debt_ratio', 'impact': 'negative', 'contribution': -0.08}
        ],
        'counterfactual': 'Would be denied if debt_ratio > 0.45'
    },
    user_id="APPLICANT-789"
)

print("=== Audit Trail Entry ===\n")
print(json.dumps(logged, indent=2))

```

```

=== Audit Trail Entry ===

```

```

{
  "timestamp": "2026-02-13T01:22:25.749934",
  "decision_id": "LOAN-2024-001234",
  "system": "LoanApprovalSystem",
  "model_version": "v2.3.1",
  "user_id": "APPLICANT-789",
  "inputs": {
    "applicant_income": 75000,
    "credit_score": 720,
    "debt_ratio": 0.35
  },
  "prediction": 0.72,
  "decision": "APPROVED",
  "explanation": {
    "top_factors": [
      {
        "feature": "credit_score",
        "impact": "positive",
        "contribution": 0.15
      },
      {
        "feature": "income",
        "impact": "positive",
        "contribution": 0.12
      }
    ]
  }
}

```

```
{
  {
    "feature": "debt_ratio",
    "impact": "negative",
    "contribution": -0.08
  }
],
"counterfactual": "Would be denied if debt_ratio > 0.45"
},
"explanation_version": "1.0"
}
```

30.8. Best Practices

💡 XAI Implementation Checklist

1. **Know your audience** - Executives need different explanations than data scientists

2. **Use multiple methods** - SHAP for global importance, LIME for local, counterfactuals for recourse

3. **Validate explanations** - Ensure explanations are faithful to model behavior

4. **Provide recourse** - Tell users what they can change

5. **Maintain audit trails** - Log decisions for compliance

6. **Test edge cases** - Explanations for outliers may be unreliable

The following code puts these guidelines into practice with a complete explainability pipeline that combines SHAP-based attribution, counterfactual generation, and automated reporting.

=== When to Use Each Explanation Method ===

	Method	Best_For	Audience	Computation
High	SHAP	Consistent global & local explanations	Data scientists, auditors	Medium-
		Yes		
Medium	LIME	Quick local explanations	Business users	Low-
		Yes		
Feature Importance specific	Counterfactual	Actionable recourse	End users (applicants)	Medium
		Simple global overview	Executives	Low

30.9. Summary

30.9.0.1. Key Takeaways

Why Explainability Matters

- Production systems require transparency for trust and compliance
- Stakeholders need to understand why decisions are made
- Regulations increasingly mandate right to explanation

Explanation Methods

- **Feature Importance:** Global view of what matters
- **SHAP:** Theoretically grounded, consistent attributions

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- **LIME:** Fast local approximations
- **Counterfactuals:** Actionable recourse for users

Optimization Explanations

- Shadow prices explain resource values
- Binding constraints explain limitations
- Infeasibility diagnosis identifies conflicts

Building Trust

- Generate human-readable explanations
- Maintain comprehensive audit trails
- Provide multiple explanation formats for different audiences

30.10. Exercises

1. **SHAP Analysis:** Train a model on a dataset relevant to your domain. Calculate SHAP values and identify the top 3 most important features. How do these compare to simple feature importance?
2. **Counterfactual Generation:** For a rejected applicant in a classification problem, find the minimal changes needed to change the decision. Are these changes realistic and actionable?
3. **Optimization Explanation:** For a resource allocation problem, explain why certain constraints are binding and calculate how much the objective would improve with additional resources.
4. **Audit Trail Design:** Design an audit logging system for a decision system in your domain. What information must be captured for compliance? How long should logs be retained?
5. **Explanation Comparison:** Generate explanations for the same prediction using SHAP, LIME, and feature importance. Do they agree? When might they differ?

30.11. References

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- Wachter, S., Mittelstadt, B., & Russell, C. (2017). “Counterfactual Explanations without Opening the Black Box.” *Harvard Journal of Law & Technology*.
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31. Online and Real-Time Optimization

What You Will Learn

- Understand online optimization where data arrives sequentially
- Apply multi-armed bandit algorithms for exploration-exploitation trade-offs
- Implement online convex optimization for streaming decisions
- Build real-time dispatch systems for ride-sharing and logistics
- Design trading algorithms with online learning
- Handle latency and computational constraints in production

31.1. Introduction

Traditional optimization assumes we have all the data upfront. But many real-world systems must make decisions in real-time as information arrives: ride-sharing platforms dispatch drivers before knowing future demand, trading systems execute orders in milliseconds, and recommendation engines personalize content instantly.

Online optimization provides a framework for sequential decision-making under uncertainty, balancing the need to learn from data while making good decisions now.

31.2. The Online Optimization Setting

31.2.1. Batch vs Online

In batch optimization, all data is available upfront and the solver can plan globally. In online optimization, data arrives sequentially and decisions must be made before future inputs are known. The code below contrasts these two settings on a simple resource allocation problem.

```
import numpy as np
import pandas as pd
from typing import List, Tuple, Callable
from dataclasses import dataclass
from collections import deque
import time

# Comparison of paradigms
comparison = {
    'Aspect': ['Data availability', 'Decision timing', 'Objective', 'Adaptation',
              ↪ 'Examples'],
    'Batch Optimization': [
        'All data known upfront',
        'Compute once, execute',
        'Minimize total cost',
        'Re-solve periodically',
        'Production planning, portfolio rebalancing'
    ],
    'Online Optimization': [
```

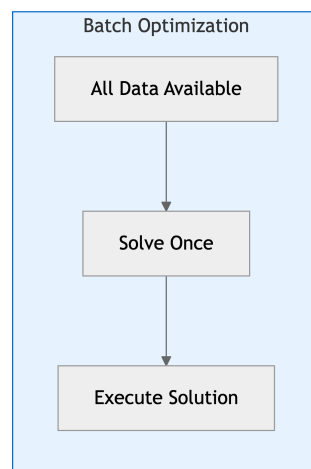
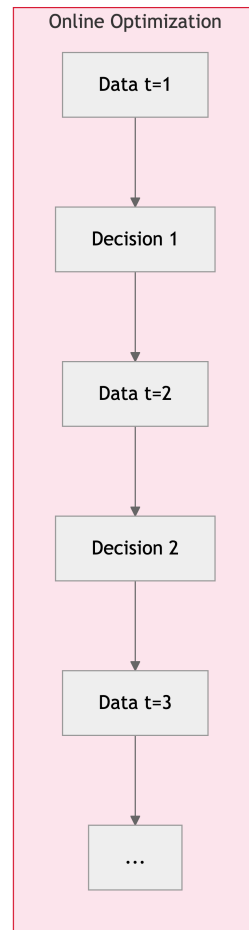


Figure 31.1.: Batch vs Online Optimization: Different Information Structures

```

        'Data arrives sequentially',
        'Decide before seeing future',
        'Minimize regret vs hindsight',
        'Continuous learning',
        'Ride-sharing dispatch, trading, ads'
    ]
}

print("=== Batch vs Online Optimization ===\n")
for i, aspect in enumerate(comparison['Aspect']):
    print(f"{aspect}:")
    print(f"    Batch: {comparison['Batch Optimization'][i]}")
    print(f"    Online: {comparison['Online Optimization'][i]}")
    print()

```

=== Batch vs Online Optimization ===

Data availability:

Batch: All data known upfront

Online: Data arrives sequentially

Decision timing:

Batch: Compute once, execute

Online: Decide before seeing future

Objective:

Batch: Minimize total cost

Online: Minimize regret vs hindsight

Adaptation:

Batch: Re-solve periodically

Online: Continuous learning

Examples:

Batch: Production planning, portfolio rebalancing

Online: Ride-sharing dispatch, trading, ads

31.2.2. Regret: The Online Performance Metric

The standard performance measure for online algorithms is **regret**: the cumulative difference between what the algorithm achieved and what it could have achieved with perfect hindsight. Sublinear regret growth means the algorithm is learning effectively over time.

```

def calculate_regret(rewards: np.ndarray, optimal_arm: int) -> np.ndarray:
    """
    Calculate cumulative regret over time.

    Regret = (best possible reward) - (actual reward)

    Args:

```

31. Online and Real-Time Optimization

```
    rewards: Array of rewards received at each time step
    optimal_arm: Index of the best arm (known in hindsight)

Returns:
    Cumulative regret over time
"""
T = len(rewards)

# In a bandit setting, we compare against always playing optimal
optimal_reward_per_round = 1.0 # Assume normalized

cumulative_regret = np.zeros(T)
total_regret = 0

for t in range(T):
    instantaneous_regret = optimal_reward_per_round - rewards[t]
    total_regret += instantaneous_regret
    cumulative_regret[t] = total_regret

return cumulative_regret

# Example: Simple regret calculation
np.random.seed(42)
T = 100
# Simulate suboptimal choices initially, improving over time
rewards = np.concatenate([
    np.random.binomial(1, 0.3, 30), # Exploring, low rewards
    np.random.binomial(1, 0.6, 30), # Learning
    np.random.binomial(1, 0.8, 40)  # Exploiting near-optimal
])

regret = calculate_regret(rewards, optimal_arm=0)

print("=== Regret Analysis ===\n")
print(f"Total rounds: {T}")
print(f"Total reward: {rewards.sum():.0f}")
print(f"Optimal reward (hindsight): {T}")
print(f"Total regret: {regret[-1]:.0f}")
print(f"Average regret per round: {regret[-1]/T:.2f}")
print("\nGoal: Sublinear regret  $O(\sqrt{T})$  means we're learning!")
```

=== Regret Analysis ===

Total rounds: 100

Total reward: 59

Optimal reward (hindsight): 100

Total regret: 41

Average regret per round: 0.41

Goal: Sublinear regret $O(\sqrt{T})$ means we're learning!

31.3. Multi-Armed Bandits

The multi-armed bandit problem is the fundamental exploration-exploitation dilemma.

31.3.1. Epsilon-Greedy

The epsilon-greedy strategy selects the arm with the highest estimated reward with probability $1 - \epsilon$, and a random arm with probability ϵ . This simple approach guarantees some level of continued exploration but wastes exploration budget on clearly suboptimal arms.

```
class EpsilonGreedy:
    """
    Epsilon-Greedy bandit algorithm.

    With probability epsilon, explore randomly.
    With probability 1-epsilon, exploit the best arm.
    """

    def __init__(self, n_arms: int, epsilon: float = 0.1):
        self.n_arms = n_arms
        self.epsilon = epsilon

        # Tracking statistics
        self.counts = np.zeros(n_arms) # Number of times each arm pulled
        self.values = np.zeros(n_arms) # Estimated value of each arm

    def select_arm(self) -> int:
        """Select which arm to pull."""
        if np.random.random() < self.epsilon:
            # Explore: random arm
            return np.random.randint(self.n_arms)
        else:
            # Exploit: best arm (break ties randomly)
            max_value = self.values.max()
            best_arms = np.where(self.values == max_value)[0]
            return np.random.choice(best_arms)

    def update(self, arm: int, reward: float):
        """Update estimates after observing reward."""
        self.counts[arm] += 1

        # Incremental mean update
        n = self.counts[arm]
        old_value = self.values[arm]
        self.values[arm] = old_value + (reward - old_value) / n

# Simulate bandit problem
np.random.seed(42)

# True arm probabilities (unknown to algorithm)
true_probs = np.array([0.3, 0.5, 0.7, 0.4, 0.6])
n_arms = len(true_probs)
optimal_arm = np.argmax(true_probs)
```

```

T = 1000
agent = EpsilonGreedy(n_arms, epsilon=0.1)

rewards = []
chosen_arms = []

for t in range(T):
    arm = agent.select_arm()
    reward = np.random.binomial(1, true_probs[arm])

    agent.update(arm, reward)

    rewards.append(reward)
    chosen_arms.append(arm)

print("=== Epsilon-Greedy Results ===\n")
print(f"True probabilities: {true_probs}")
print(f"Optimal arm: {optimal_arm} (prob={true_probs[optimal_arm]})")
print(f"\nEstimated values: {agent.values.round(3)}")
print(f"Pull counts: {agent.counts.astype(int)}")
print(f"\nTotal reward: {sum(rewards)}")
print(f"Optimal reward: {int(T * true_probs[optimal_arm])}")
print(f"Regret: {int(T * true_probs[optimal_arm] - sum(rewards))}")

```

```
=== Epsilon-Greedy Results ===
```

```
True probabilities: [0.3 0.5 0.7 0.4 0.6]
```

```
Optimal arm: 2 (prob=0.7)
```

```
Estimated values: [0.269 0.571 0.6    0.5    0.631]
```

```
Pull counts: [ 26  21  15  22 916]
```

```
Total reward: 617
```

```
Optimal reward: 700
```

```
Regret: 83
```

31.3.2. Upper Confidence Bound (UCB)

UCB selects the arm that maximizes the sum of the estimated reward and an exploration bonus proportional to $\sqrt{\log(t)/n_i}$, where t is the total number of pulls and n_i is the number of pulls for arm i . This directs exploration toward arms with high uncertainty, achieving $O(\sqrt{T \log T})$ regret.

```

class UCB1:
    """
    Upper Confidence Bound algorithm.

    Balances exploration and exploitation using confidence intervals.
    Optimism in the face of uncertainty.
    """

```

```

def __init__(self, n_arms: int, c: float = 2.0):
    self.n_arms = n_arms
    self.c = c # Exploration parameter

    self.counts = np.zeros(n_arms)
    self.values = np.zeros(n_arms)
    self.t = 0

def select_arm(self) -> int:
    """Select arm with highest upper confidence bound."""
    self.t += 1

    # Pull each arm once first
    for arm in range(self.n_arms):
        if self.counts[arm] == 0:
            return arm

    # UCB formula: value + c * sqrt(log(t) / n_arm)
    ucb_values = self.values + self.c * np.sqrt(
        np.log(self.t) / self.counts
    )

    return np.argmax(ucb_values)

def update(self, arm: int, reward: float):
    """Update estimates after observing reward."""
    self.counts[arm] += 1
    n = self.counts[arm]
    self.values[arm] += (reward - self.values[arm]) / n

# Compare with epsilon-greedy
np.random.seed(42)

ucb_agent = UCB1(n_arms, c=2.0)
eg_agent = EpsilonGreedy(n_arms, epsilon=0.1)

ucb_rewards = []
eg_rewards = []

for t in range(T):
    # UCB
    arm_ucb = ucb_agent.select_arm()
    reward_ucb = np.random.binomial(1, true_probs[arm_ucb])
    ucb_agent.update(arm_ucb, reward_ucb)
    ucb_rewards.append(reward_ucb)

    # Epsilon-Greedy
    arm_eg = eg_agent.select_arm()
    reward_eg = np.random.binomial(1, true_probs[arm_eg])
    eg_agent.update(arm_eg, reward_eg)
    eg_rewards.append(reward_eg)

```

```

print("=== UCB vs Epsilon-Greedy ===\n")
print(f"UCB1 total reward: {sum(ucb_rewards)}")
print(f"Epsilon-Greedy total reward: {sum(eg_rewards)}")
print(f"Optimal possible: {int(T * true_probs[optimal_arm])}")

print(f"\nUCB1 regret: {int(T * true_probs[optimal_arm] - sum(ucb_rewards))}")
print(f"Epsilon-Greedy regret: {int(T * true_probs[optimal_arm] -
↪ sum(eg_rewards))}")

```

=== UCB vs Epsilon-Greedy ===

UCB1 total reward: 625

Epsilon-Greedy total reward: 641

Optimal possible: 700

UCB1 regret: 75

Epsilon-Greedy regret: 59

31.3.3. Thompson Sampling

Thompson Sampling takes a Bayesian approach: it maintains a posterior distribution over each arm's reward probability and selects arms by sampling from these posteriors. Arms with high uncertainty produce more variable samples, which naturally drives exploration. In practice, Thompson Sampling often outperforms UCB on structured problems.

```

class ThompsonSampling:
    """
    Thompson Sampling (Bayesian approach to bandits).

    Maintains posterior distribution over arm values.
    Samples from posterior to select arms.
    """

    def __init__(self, n_arms: int):
        self.n_arms = n_arms

        # Beta distribution parameters (prior: Beta(1,1) = uniform)
        self.alpha = np.ones(n_arms) # Successes + 1
        self.beta = np.ones(n_arms)  # Failures + 1

    def select_arm(self) -> int:
        """Sample from posterior and select best."""
        # Sample from Beta posterior for each arm
        samples = np.random.beta(self.alpha, self.beta)
        return np.argmax(samples)

    def update(self, arm: int, reward: float):
        """Update posterior with observed reward."""
        if reward > 0:
            self.alpha[arm] += 1
        else:
            self.beta[arm] += 1

```

```

def get_mean_estimates(self) -> np.ndarray:
    """Get posterior mean estimates."""
    return self.alpha / (self.alpha + self.beta)

# Compare all three algorithms
np.random.seed(42)

ts_agent = ThompsonSampling(n_arms)
ts_rewards = []

for t in range(T):
    arm = ts_agent.select_arm()
    reward = np.random.binomial(1, true_probs[arm])
    ts_agent.update(arm, reward)
    ts_rewards.append(reward)

print("=== Thompson Sampling Results ===\n")
print(f"True probabilities: {true_probs}")
print(f"Posterior means: {ts_agent.get_mean_estimates().round(3)}")
print(f"\nTotal reward: {sum(ts_rewards)}")
print(f"Regret: {int(T * true_probs[optimal_arm] - sum(ts_rewards))}")

print("\n=== Algorithm Comparison ===")
print(f"Thompson Sampling regret: {int(T * true_probs[optimal_arm] -  

    ↪ sum(ts_rewards))}")
print(f"UCB1 regret: {int(T * true_probs[optimal_arm] - sum(ucb_rewards))}")
print(f"Epsilon-Greedy regret: {int(T * true_probs[optimal_arm] -  

    ↪ sum(eg_rewards))}")

```

=== Thompson Sampling Results ===

True probabilities: [0.3 0.5 0.7 0.4 0.6]
 Posterior means: [0.222 0.489 0.712 0.353 0.602]

Total reward: 684
 Regret: 16

=== Algorithm Comparison ===
 Thompson Sampling regret: 16
 UCB1 regret: 75
 Epsilon-Greedy regret: 59

31.3.4. Contextual Bandits

Contextual bandits extend the standard bandit setting by providing side information (context) at each round. The algorithm observes a feature vector before choosing an arm, allowing it to learn different arm preferences for different contexts. This models scenarios like personalized recommendations, where the best action depends on user features.

```

from sklearn.linear_model import SGDClassifier

class LinUCB:
    """
    Linear Upper Confidence Bound for contextual bandits.

    Assumes reward is linear in context features.
    """

    def __init__(self, n_arms: int, n_features: int, alpha: float = 1.0):
        self.n_arms = n_arms
        self.n_features = n_features
        self.alpha = alpha

        # Per-arm statistics
        self.A = [np.eye(n_features) for _ in range(n_arms)] # d x d
        self.b = [np.zeros(n_features) for _ in range(n_arms)] # d x 1

    def select_arm(self, context: np.ndarray) -> int:
        """Select arm given context features."""
        context = context.reshape(-1)
        ucb_values = np.zeros(self.n_arms)

        for arm in range(self.n_arms):
            A_inv = np.linalg.inv(self.A[arm])
            theta = A_inv @ self.b[arm]

            # UCB = theta^T x + alpha * sqrt(x^T A^-1 x)
            mean = theta @ context
            variance = context @ A_inv @ context
            ucb_values[arm] = mean + self.alpha * np.sqrt(variance)

        return np.argmax(ucb_values)

    def update(self, arm: int, context: np.ndarray, reward: float):
        """Update model after observing reward."""
        context = context.reshape(-1)

        self.A[arm] += np.outer(context, context)
        self.b[arm] += reward * context

# Simulate contextual bandit (e.g., personalized recommendations)
np.random.seed(42)

n_features = 5
n_arms = 3 # 3 different recommendations
T = 500

# True parameters (unknown to algorithm)
true_theta = np.random.randn(n_arms, n_features) * 0.5

linucb = LinUCB(n_arms, n_features, alpha=0.5)

```

```

rewards = []
optimal_rewards = []

for t in range(T):
    # Random context (user features)
    context = np.random.randn(n_features)

    # True expected rewards
    true_rewards = true_theta @ context
    optimal_arm = np.argmax(true_rewards)
    optimal_reward = true_rewards[optimal_arm]

    # Algorithm's choice
    chosen_arm = linucb.select_arm(context)

    # Noisy reward
    reward = true_rewards[chosen_arm] + np.random.randn() * 0.1

    linucb.update(chosen_arm, context, reward)

    rewards.append(reward)
    optimal_rewards.append(optimal_reward)

cumulative_regret = np.cumsum(optimal_rewards) - np.cumsum(rewards)

print("=== Contextual Bandit (LinUCB) ===\n")
print(f"Features: {n_features}, Arms: {n_arms}")
print(f"Total rounds: {T}")
print(f"Cumulative regret: {cumulative_regret[-1]:.2f}")
print(f"Average regret per round: {cumulative_regret[-1]/T:.4f}")

```

```
=== Contextual Bandit (LinUCB) ===
```

```

Features: 5, Arms: 3
Total rounds: 500
Cumulative regret: 7.93
Average regret per round: 0.0159

```

31.4. Online Convex Optimization

For continuous decision spaces, online convex optimization provides strong guarantees.

31.4.1. Online Gradient Descent

Online Gradient Descent (OGD) extends gradient descent to the online setting. At each round, the algorithm makes a decision, observes the loss function, and takes a gradient step. For convex losses, OGD achieves $O(\sqrt{T})$ regret. The following implementation applies OGD to an online portfolio allocation problem.

```

class OnlineGradientDescent:
    """

```

```

Online Gradient Descent for convex optimization.

Minimizes regret against sequence of convex loss functions.
"""

def __init__(self, dimension: int, learning_rate: float = 0.1,
              projection: Callable = None):
    """
    Args:
        dimension: Dimension of decision variable
        learning_rate: Step size (can be adaptive)
        projection: Function to project onto constraint set
    """
    self.d = dimension
    self.lr = learning_rate
    self.projection = projection or (lambda x: x)

    # Initialize at origin
    self.x = np.zeros(dimension)
    self.t = 0

def get_decision(self) -> np.ndarray:
    """Return current decision."""
    return self.x.copy()

def update(self, gradient: np.ndarray):
    """Update decision based on observed gradient."""
    self.t += 1

    # Adaptive learning rate: 1/sqrt(t)
    lr_t = self.lr / np.sqrt(self.t)

    # Gradient descent step
    self.x = self.x - lr_t * gradient

    # Project onto feasible set
    self.x = self.projection(self.x)

def project_to_simplex(x: np.ndarray) -> np.ndarray:
    """Project onto probability simplex (sum to 1, non-negative)."""
    n = len(x)
    sorted_x = np.sort(x)[::-1]
    cumsum = np.cumsum(sorted_x)

    # Find threshold
    rho = np.where(sorted_x > (cumsum - 1) / np.arange(1, n + 1))[0][-1]
    theta = (cumsum[rho] - 1) / (rho + 1)

    return np.maximum(x - theta, 0)

# Example: Online portfolio selection

```



```

np.random.seed(42)

n_assets = 5
T = 200

# Simulate asset returns
returns = np.random.randn(T, n_assets) * 0.02 + 0.001 # Daily returns

ogd = OnlineGradientDescent(n_assets, learning_rate=0.5,
    ↪ projection=project_to_simplex)

# Initialize with equal weights
ogd.x = np.ones(n_assets) / n_assets

portfolio_returns = []
equal_weight_returns = []

for t in range(T):
    # Get current portfolio
    weights = ogd.get_decision()

    # Portfolio return
    portfolio_return = weights @ returns[t]
    portfolio_returns.append(portfolio_return)

    # Equal weight benchmark
    equal_weight_returns.append(np.mean(returns[t]))

    # Gradient: negative return (we want to maximize)
    gradient = -returns[t]
    ogd.update(gradient)

print("=== Online Portfolio Selection (OGD) ===\n")
print(f"Number of assets: {n_assets}")
print(f"Time horizon: {T} days")
print(f"\nCumulative return (OGD): {np.sum(portfolio_returns)*100:.2f}%")
print(f"Cumulative return (Equal Weight): {np.sum(equal_weight_returns)*100:.2f}%")
print(f"\nFinal portfolio weights: {ogd.get_decision().round(3)}")

```

```
=== Online Portfolio Selection (OGD) ===
```

```
Number of assets: 5
```

```
Time horizon: 200 days
```

```
Cumulative return (OGD): 25.88%
```

```
Cumulative return (Equal Weight): 27.73%
```

```
Final portfolio weights: [0.198 0.205 0.206 0.203 0.188]
```

31.4.2. Follow the Regularized Leader (FTRL)

FTRL selects the decision that minimizes cumulative past loss plus a regularization term. The regularizer prevents the algorithm from overreacting to recent observations. With entropy regularization,

31. Online and Real-Time Optimization

FTRL reduces to the Hedge algorithm for expert advice. With ℓ_2 regularization, it behaves like online ridge regression.

```
class FTRL:
    """
    Follow the Regularized Leader.

    Minimizes cumulative loss plus regularization.
    """

    def __init__(self, dimension: int, regularization: float = 1.0):
        self.d = dimension
        self.eta = regularization

        # Cumulative gradient
        self.sum_gradients = np.zeros(dimension)
        self.t = 0

    def get_decision(self) -> np.ndarray:
        """Compute decision by minimizing regularized loss."""
        self.t += 1

        # With L2 regularization: x = -sum_gradients / (2 * eta * t)
        # With entropy regularization (for simplex): use softmax
        x = -self.sum_gradients / (2 * self.eta * self.t)

        # Project to simplex
        return project_to_simplex(x)

    def update(self, gradient: np.ndarray):
        """Accumulate gradient."""
        self.sum_gradients += gradient

# Compare with OGD
np.random.seed(42)

ftrl = FTRL(n_assets, regularization=1.0)
ftrl_returns = []

for t in range(T):
    weights = ftrl.get_decision()

    portfolio_return = weights @ returns[t]
    ftrl_returns.append(portfolio_return)

    gradient = -returns[t]
    ftrl.update(gradient)

print("=== FTRL vs OGD Comparison ===\n")
print(f"FTRL cumulative return: {np.sum(ftrl_returns)*100:.2f}%")
print(f"OGD cumulative return: {np.sum(portfolio_returns)*100:.2f}%")
```

=== FTRL vs OGD Comparison ===

FTRL cumulative return: 27.70%
 OGD cumulative return: 25.88%

31.5. Real-Time Dispatch Systems

Online optimization powers real-time dispatch in ride-sharing, delivery logistics, and emergency response. These systems must assign resources to requests as they arrive, balancing immediate service quality against holding resources for future, potentially more valuable requests.

31.5.1. Ride-Sharing Dispatch

The following implementation simulates a ride-sharing dispatch system. Drivers and riders arrive over time, and the algorithm must match them immediately based on proximity and estimated travel time.

```
from dataclasses import dataclass
from typing import Optional
import heapq

@dataclass
class RideRequest:
    """A ride request from a customer."""
    id: str
    pickup_location: Tuple[float, float]
    dropoff_location: Tuple[float, float]
    request_time: float
    deadline: float # Must be picked up by this time

@dataclass
class Driver:
    """A driver in the system."""
    id: str
    location: Tuple[float, float]
    available_time: float
    current_ride: Optional[str] = None

class RideDispatcher:
    """
    Real-time ride dispatch with online assignment.
    """

    def __init__(self, max_wait_time: float = 10.0):
        self.max_wait = max_wait_time
        self.drivers = {}
        self.pending_requests = []
        self.assignments = []
        self.current_time = 0

    def add_driver(self, driver: Driver):
        """Add a driver to the system."""
        self.drivers[driver.id] = driver
```

```

def compute_travel_time(self, loc1: Tuple[float, float],
                        loc2: Tuple[float, float]) -> float:
    """Estimate travel time between locations."""
    # Simple Euclidean distance * speed factor
    distance = np.sqrt((loc1[0] - loc2[0])**2 + (loc1[1] - loc2[1])**2)
    return distance * 2 # 2 time units per distance unit

def get_available_drivers(self) -> List[Driver]:
    """Get drivers available for assignment."""
    return [d for d in self.drivers.values()
            if d.current_ride is None and d.available_time <= self.current_time]

def dispatch_request(self, request: RideRequest) -> Optional[str]:
    """
    Assign a driver to a request using greedy matching.

    Returns driver ID if assigned, None otherwise.
    """
    available = self.get_available_drivers()

    if not available:
        return None

    # Find driver with minimum pickup time
    best_driver = None
    best_time = float('inf')

    for driver in available:
        pickup_time = self.compute_travel_time(
            driver.location, request.pickup_location
        )

        if pickup_time < best_time:
            best_time = pickup_time
            best_driver = driver

    # Check if feasible
    if best_time > self.max_wait:
        return None

    # Make assignment
    best_driver.current_ride = request.id

    # Update driver availability
    trip_time = self.compute_travel_time(
        request.pickup_location, request.dropoff_location
    )
    best_driver.available_time = self.current_time + best_time + trip_time
    best_driver.location = request.dropoff_location

    self.assignments.append({
        'request': request.id,

```

```

        'driver': best_driver.id,
        'pickup_time': best_time,
        'trip_time': trip_time,
        'assigned_at': self.current_time
    })

    # Release driver after trip
    best_driver.current_ride = None

    return best_driver.id

def simulate_batch(self, requests: List[RideRequest]) -> dict:
    """Simulate dispatching a batch of requests."""
    assigned = 0
    total_wait = 0

    for request in requests:
        self.current_time = request.request_time
        driver_id = self.dispatch_request(request)

        if driver_id:
            assigned += 1
            # Find assignment
            for a in self.assignments:
                if a['request'] == request.id:
                    total_wait += a['pickup_time']
                    break

    return {
        'total_requests': len(requests),
        'assigned': assigned,
        'assignment_rate': assigned / len(requests),
        'avg_wait_time': total_wait / assigned if assigned > 0 else 0
    }

# Simulation
np.random.seed(42)

# Create drivers at random locations
dispatcher = RideDispatcher(max_wait_time=15.0)

for i in range(10):
    driver = Driver(
        id=f"D{i}",
        location=(np.random.uniform(0, 10), np.random.uniform(0, 10)),
        available_time=0
    )
    dispatcher.add_driver(driver)

# Generate ride requests
requests = []
for i in range(50):

```

```

request = RideRequest(
    id=f"R{i}",
    pickup_location=(np.random.uniform(0, 10), np.random.uniform(0, 10)),
    dropoff_location=(np.random.uniform(0, 10), np.random.uniform(0, 10)),
    request_time=i * 2, # Every 2 time units
    deadline=i * 2 + 15
)
requests.append(request)

results = dispatcher.simulate_batch(requests)

print("=== Ride-Sharing Dispatch Simulation ===\n")
print(f"Drivers: 10")
print(f"Requests: {results['total_requests']}")
print(f"Assigned: {results['assigned']}")
print(f"Assignment rate: {results['assignment_rate']:.1%}")
print(f"Average wait time: {results['avg_wait_time']:.1f} time units")

```

```
=== Ride-Sharing Dispatch Simulation ===
```

```

Drivers: 10
Requests: 50
Assigned: 47
Assignment rate: 94.0%
Average wait time: 5.2 time units

```

31.5.2. Batch vs Online Assignment

An alternative to purely online assignment is batching: accumulate requests for a short window, then solve an assignment problem on the batch. This trades latency for solution quality. The code below compares greedy online assignment against batched optimal assignment on the same request stream.

```

from scipy.optimize import linear_sum_assignment

def batch_assignment(drivers: List[Tuple[float, float]],
                    requests: List[Tuple[float, float]],
                    max_cost: float = 100) -> List[Tuple[int, int]]:
    """
    Optimal batch assignment using Hungarian algorithm.

    Minimizes total travel time for a batch of requests.
    """
    n_drivers = len(drivers)
    n_requests = len(requests)

    # Create cost matrix
    cost_matrix = np.full((n_drivers, n_requests), max_cost)

    for i, driver_loc in enumerate(drivers):
        for j, request_loc in enumerate(requests):
            distance = np.sqrt(

```

```

        (driver_loc[0] - request_loc[0])**2 +
        (driver_loc[1] - request_loc[1])**2
    )
    cost_matrix[i, j] = distance

# Solve assignment problem
row_ind, col_ind = linear_sum_assignment(cost_matrix)

assignments = []
total_cost = 0

for i, j in zip(row_ind, col_ind):
    if cost_matrix[i, j] < max_cost:
        assignments.append((i, j))
        total_cost += cost_matrix[i, j]

return assignments, total_cost

def greedy_assignment(drivers: List[Tuple[float, float]],
                    requests: List[Tuple[float, float]]) -> Tuple[List, float]:
    """
    Greedy online assignment (as requests arrive).
    """
    available_drivers = list(range(len(drivers)))
    assignments = []
    total_cost = 0

    for j, request_loc in enumerate(requests):
        if not available_drivers:
            break

        # Find closest available driver
        best_driver = None
        best_distance = float('inf')

        for i in available_drivers:
            distance = np.sqrt(
                (drivers[i][0] - request_loc[0])**2 +
                (drivers[i][1] - request_loc[1])**2
            )
            if distance < best_distance:
                best_distance = distance
                best_driver = i

        if best_driver is not None:
            assignments.append((best_driver, j))
            total_cost += best_distance
            available_drivers.remove(best_driver)

    return assignments, total_cost

```

```
# Compare batch vs greedy
np.random.seed(42)

drivers = [(np.random.uniform(0, 10), np.random.uniform(0, 10)) for _ in range(10)]
request_locs = [(np.random.uniform(0, 10), np.random.uniform(0, 10)) for _ in
    ↪ range(10)]

batch_result, batch_cost = batch_assignment(drivers, request_locs)
greedy_result, greedy_cost = greedy_assignment(drivers, request_locs)

print("=== Batch vs Greedy Assignment ===\n")
print(f"Number of drivers: {len(drivers)}")
print(f"Number of requests: {len(request_locs)}")
print(f"\nBatch (optimal) total distance: {batch_cost:.2f}")
print(f"Greedy (online) total distance: {greedy_cost:.2f}")
print(f"Greedy overhead: {(greedy_cost/batch_cost - 1)*100:.1f}%")
print("\nKey insight: Batching requests improves efficiency,")
print("but online algorithms are needed when requests arrive in real-time.")
```

=== Batch vs Greedy Assignment ===

Number of drivers: 10

Number of requests: 10

Batch (optimal) total distance: 18.00

Greedy (online) total distance: 31.13

Greedy overhead: 73.0%

Key insight: Batching requests improves efficiency,
but online algorithms are needed when requests arrive in real-time.

31.6. Online Trading Algorithms

Financial trading is inherently online: prices arrive tick by tick, and positions must be adjusted in real time. This section implements two online trading strategies that make sequential decisions under uncertainty.

31.6.1. Market Making

A market maker continuously quotes bid and ask prices, profiting from the spread while managing inventory risk. The following implementation adjusts the spread dynamically based on current inventory position — wider spreads when inventory is large reduce the risk of further accumulation.

```
class SimpleMarketMaker:
    """
    Simple market making strategy with online learning.

    Maintains bid/ask spread and adjusts based on inventory.
    """

    def __init__(self, base_spread: float = 0.02,
```



```

        inventory_penalty: float = 0.001,
        max_inventory: int = 100):
    self.base_spread = base_spread
    self.inventory_penalty = inventory_penalty
    self.max_inventory = max_inventory

    self.inventory = 0
    self.cash = 10000
    self.trades = []
    self.pnl_history = []

def get_quotes(self, mid_price: float) -> Tuple[float, float]:
    """
    Generate bid and ask quotes.

    Adjusts spread based on inventory risk.
    """
    # Widen spread when inventory is high (risk management)
    inventory_adjustment = self.inventory_penalty * abs(self.inventory)

    # Skew quotes to reduce inventory
    skew = self.inventory_penalty * self.inventory

    half_spread = self.base_spread / 2 + inventory_adjustment

    bid = mid_price * (1 - half_spread - skew)
    ask = mid_price * (1 + half_spread - skew)

    return bid, ask

def process_order(self, side: str, price: float, quantity: int):
    """
    Process an incoming order.

    side: 'buy' (customer buys from us) or 'sell' (customer sells to us)
    """
    if side == 'buy':
        # Customer buys, we sell
        if self.inventory >= quantity:
            self.inventory -= quantity
            self.cash += price * quantity
            self.trades.append(('sell', price, quantity))
        else:
            # Customer sells, we buy
            if self.inventory + quantity <= self.max_inventory:
                self.inventory += quantity
                self.cash -= price * quantity
                self.trades.append(('buy', price, quantity))

def get_pnl(self, current_price: float) -> float:
    """Calculate current P&L."""
    return self.cash + self.inventory * current_price - 10000

```

```

# Simulate market making
np.random.seed(42)

mm = SimpleMarketMaker(base_spread=0.02, inventory_penalty=0.001)

# Simulate price path (random walk)
T = 500
price = 100.0
prices = [price]

for t in range(T):
    # Price evolution
    price = price * (1 + np.random.randn() * 0.01)
    prices.append(price)

    # Get quotes
    bid, ask = mm.get_quotes(price)

    # Simulate order flow (random)
    if np.random.random() < 0.3: # 30% chance of customer order
        if np.random.random() < 0.5:
            # Customer wants to buy
            mm.process_order('buy', ask, 1)
        else:
            # Customer wants to sell
            mm.process_order('sell', bid, 1)

    mm.pnl_history.append(mm.get_pnl(price))

print("=== Market Making Simulation ===\n")
print(f"Duration: {T} time steps")
print(f"Total trades: {len(mm.trades)}")
print(f"Final inventory: {mm.inventory}")
print(f"Final cash: ${mm.cash:.2f}")
print(f"Final P&L: ${mm.get_pnl(prices[-1]):.2f}")
print(f"\nInventory-adjusted spread helps manage risk!")

```

```
=== Market Making Simulation ===
```

```

Duration: 500 time steps
Total trades: 137
Final inventory: 3
Final cash: $10001.99
Final P&L: $319.60

```

Inventory-adjusted spread helps manage risk!

31.6.2. Online Portfolio Rebalancing

The universal portfolio algorithm by Cover (1991) achieves wealth growth comparable to the best constant-rebalanced portfolio in hindsight. The implementation below uses the exponential gradient method, which maintains portfolio weights using multiplicative updates based on realized returns.

```

class OnlinePortfolio:
    """
    Online portfolio selection with transaction costs.
    """

    def __init__(self, n_assets: int, transaction_cost: float = 0.001):
        self.n_assets = n_assets
        self.tx_cost = transaction_cost

        # Start with equal weights
        self.weights = np.ones(n_assets) / n_assets
        self.portfolio_value = 10000
        self.history = []

    def rebalance(self, target_weights: np.ndarray):
        """Rebalance to target weights, accounting for costs."""
        turnover = np.abs(target_weights - self.weights).sum()
        cost = turnover * self.tx_cost * self.portfolio_value

        self.portfolio_value -= cost
        self.weights = target_weights.copy()

        return cost

    def update_from_returns(self, returns: np.ndarray):
        """Update portfolio after observing returns."""
        # Portfolio return
        portfolio_return = self.weights @ returns

        # Update value
        self.portfolio_value *= (1 + portfolio_return)

        # Weights drift
        new_values = self.weights * (1 + returns)
        self.weights = new_values / new_values.sum()

        self.history.append({
            'value': self.portfolio_value,
            'weights': self.weights.copy(),
            'return': portfolio_return
        })

        return portfolio_return

    def mean_variance_target(self, expected_returns: np.ndarray,
                             cov_matrix: np.ndarray,
                             risk_aversion: float = 1.0) -> np.ndarray:
        """
        Calculate mean-variance optimal weights.
        """
        n = len(expected_returns)

        # Simple closed-form for unconstrained case

```

```

    inv_cov = np.linalg.inv(cov_matrix)
    weights = inv_cov @ expected_returns / risk_aversion

    # Project to simplex (long-only)
    weights = project_to_simplex(weights)

    return weights

# Simulate online portfolio management
np.random.seed(42)

n_assets = 5
T = 250 # Trading days

# True parameters (unknown, estimated online)
true_means = np.array([0.0005, 0.0008, 0.0003, 0.0006, 0.0004])
true_cov = np.eye(n_assets) * 0.0004

portfolio = OnlinePortfolio(n_assets, transaction_cost=0.001)

# Online estimation windows
window = 20
returns_buffer = []

for t in range(T):
    # Generate returns
    returns_t = np.random.multivariate_normal(true_means, true_cov)
    returns_buffer.append(returns_t)

    # Update portfolio
    portfolio.update_from_returns(returns_t)

    # Rebalance periodically with estimated parameters
    if t >= window and t % 5 == 0:
        recent_returns = np.array(returns_buffer[-window:])

        # Estimate parameters
        est_means = recent_returns.mean(axis=0)
        est_cov = np.cov(recent_returns.T)

        # Calculate target
        target = portfolio.mean_variance_target(est_means, est_cov,
        ↪ risk_aversion=2.0)

        # Rebalance
        portfolio.rebalance(target)

print("=== Online Portfolio Rebalancing ===\n")
print(f"Initial value: $10,000")
print(f"Final value: ${portfolio.portfolio_value:.2f}")
print(f"Total return: {(portfolio.portfolio_value/10000 - 1)*100:.2f}%")
print(f"Final weights: {portfolio.weights.round(3)}")

```

```
=== Online Portfolio Rebalancing ===
```

```
Initial value: $10,000
```

```
Final value: $10682.23
```

```
Total return: 6.82%
```

```
Final weights: [0. 0. 0. 0. 1.]
```

31.7. Handling Latency and Constraints

Real-time systems face hard time limits: a dispatch decision delayed by 100ms may arrive after the opportunity has passed. This section addresses two practical challenges — making decisions within strict latency budgets and producing improving solutions over time.

31.7.1. Latency-Aware Optimization

When optimization must complete within a time budget, the algorithm needs to balance solution quality against computation time. The code below implements a tiered approach: try the best method first, fall back to faster heuristics if time runs short.

```
class LatencyAwareDecisionSystem:
    """
    Decision system that accounts for computation and communication latency.
    """

    def __init__(self, max_latency_ms: float = 100):
        self.max_latency = max_latency_ms
        self.model_cache = {}
        self.fallback_decision = None

    def fast_decision(self, state: dict) -> dict:
        """
        Make a fast decision within latency budget.

        Uses cached models and simple heuristics.
        """
        start_time = time.time()

        # Try cached/pre-computed solution
        state_key = self._hash_state(state)
        if state_key in self.model_cache:
            decision = self.model_cache[state_key]
            latency_ms = (time.time() - start_time) * 1000
            return {'decision': decision, 'latency_ms': latency_ms, 'source':
                    ↪ 'cache'}

        # Fast heuristic if cache miss
        elapsed_ms = (time.time() - start_time) * 1000
        remaining_budget = self.max_latency - elapsed_ms

        if remaining_budget > 50:
            # Have time for simple optimization
            decision = self._fast_optimize(state)
            source = 'fast_optimization'
```

```

        else:
            # Use fallback
            decision = self.fallback_decision or self._simple_heuristic(state)
            source = 'fallback'

        latency_ms = (time.time() - start_time) * 1000
        return {'decision': decision, 'latency_ms': latency_ms, 'source': source}

    def _hash_state(self, state: dict) -> str:
        """Create hash key for state."""
        return str(sorted(state.items()))

    def _fast_optimize(self, state: dict) -> dict:
        """Quick optimization (placeholder)."""
        # Simulate some computation
        time.sleep(0.02) # 20ms
        return {'action': 'optimized', 'value': state.get('input', 0) * 1.1}

    def _simple_heuristic(self, state: dict) -> dict:
        """Fallback heuristic (very fast)."""
        return {'action': 'heuristic', 'value': state.get('input', 0)}

    def precompute_decisions(self, states: List[dict]):
        """Precompute decisions for anticipated states."""
        for state in states:
            decision = self._fast_optimize(state)
            state_key = self._hash_state(state)
            self.model_cache[state_key] = decision

# Demo latency-aware system
system = LatencyAwareDecisionSystem(max_latency_ms=100)

# Precompute for anticipated scenarios
anticipated_states = [{'input': i} for i in range(10)]
system.precompute_decisions(anticipated_states)

# Test decisions
print("=== Latency-Aware Decision System ===\n")

test_cases = [
    {'input': 5}, # In cache
    {'input': 15}, # Not in cache
    {'input': 3}, # In cache
]

for state in test_cases:
    result = system.fast_decision(state)
    print(f"State: {state}")
    print(f"Decision: {result['decision']}")
    print(f"Latency: {result['latency_ms']:.1f}ms")
    print(f"Source: {result['source']}")
    print()

```

```
=== Latency-Aware Decision System ===
```

```
State: {'input': 5}
  Decision: {'action': 'optimized', 'value': 5.5}
  Latency: 0.0ms
  Source: cache

State: {'input': 15}
  Decision: {'action': 'optimized', 'value': 16.5}
  Latency: 25.1ms
  Source: fast_optimization

State: {'input': 3}
  Decision: {'action': 'optimized', 'value': 3.3000000000000003}
  Latency: 0.0ms
  Source: cache
```

31.7.2. Anytime Algorithms

An anytime algorithm can be interrupted at any point and still return a valid solution. The longer it runs, the better the solution becomes. This property is valuable when the available computation time varies unpredictably. The following code demonstrates an anytime local search that improves a TSP tour incrementally.

```
class AnytimeOptimizer:
    """
    Anytime algorithm that returns progressively better solutions.

    Can be interrupted at any time and return best solution found so far.
    """

    def __init__(self):
        self.best_solution = None
        self.best_value = float('-inf')
        self.iterations = 0
        self.improvement_history = []

    def optimize(self, objective_fn: Callable,
                 initial_solution: np.ndarray,
                 max_iterations: int = 1000,
                 time_limit_ms: float = None) -> dict:
        """
        Run optimization, can be stopped anytime.

        Args:
            objective_fn: Function to maximize
            initial_solution: Starting point
            max_iterations: Maximum iterations
            time_limit_ms: Optional time limit

        Returns:
```

```

        Best solution found within constraints
    """
    start_time = time.time()

    self.best_solution = initial_solution.copy()
    self.best_value = objective_fn(initial_solution)

    current = initial_solution.copy()

    for i in range(max_iterations):
        self.iterations = i + 1

        # Check time limit
        if time_limit_ms:
            elapsed_ms = (time.time() - start_time) * 1000
            if elapsed_ms > time_limit_ms:
                break

        # Local search step (simple perturbation)
        candidate = current + np.random.randn(len(current)) * 0.1
        candidate_value = objective_fn(candidate)

        # Accept if better
        if candidate_value > self.best_value:
            self.best_solution = candidate.copy()
            self.best_value = candidate_value
            current = candidate.copy()
            self.improvement_history.append((i, self.best_value))
        elif candidate_value > objective_fn(current):
            current = candidate.copy()

    elapsed_ms = (time.time() - start_time) * 1000

    return {
        'solution': self.best_solution,
        'value': self.best_value,
        'iterations': self.iterations,
        'time_ms': elapsed_ms,
        'improvements': len(self.improvement_history)
    }

# Demo anytime optimization
def quadratic_objective(x):
    """Simple quadratic to maximize."""
    return -np.sum((x - 3)**2) # Maximum at x=[3,3,3,...]

optimizer = AnytimeOptimizer()

# Run with time limit
result = optimizer.optimize(
    quadratic_objective,

```



```

        initial_solution=np.zeros(5),
        max_iterations=10000,
        time_limit_ms=50 # Only 50ms allowed
    )

    print("=== Anytime Optimization ===\n")
    print(f"Time limit: 50ms")
    print(f"Actual time: {result['time_ms']:.1f}ms")
    print(f"Iterations completed: {result['iterations']}")
    print(f"Improvements found: {result['improvements']}")
    print(f"Best value: {result['value']:.4f}")
    print(f"Best solution: {result['solution'].round(2)}")
    print(f"(Optimal is [3, 3, 3, 3, 3])")

```

=== Anytime Optimization ===

```

Time limit: 50ms
Actual time: 50.0ms
Iterations completed: 6826
Improvements found: 109
Best value: -0.0011
Best solution: [3.03 2.99 3.    3.01 3.   ]
(Optimal is [3, 3, 3, 3, 3])

```

31.8. Best Practices

Online Optimization Checklist

1. **Define the right metric** - Regret vs throughput vs latency
2. **Balance exploration/exploitation** - Don't get stuck on suboptimal arms
3. **Account for non-stationarity** - Real systems change over time
4. **Batch when possible** - Small delays enable better optimization
5. **Have fallbacks** - Always have a fast default decision
6. **Monitor and adapt** - Track performance and adjust parameters

31.9. Summary

31.9.0.1. Key Takeaways

Online Setting

- Decisions made before seeing all data
- Regret measures performance vs hindsight optimal
- Sublinear regret means the algorithm is learning

Multi-Armed Bandits

- Epsilon-greedy: Simple but effective
- UCB: Optimism under uncertainty
- Thompson Sampling: Bayesian approach, often best in practice
- Contextual bandits: Personalization with features

Online Convex Optimization

- Online gradient descent for continuous spaces
- FTRL for stability with regularization
- Portfolio selection as key application

Real-Time Systems

- Ride-sharing: Match drivers to requests online
- Trading: Market making with inventory management
- Latency constraints require fallbacks and caching
- Anytime algorithms provide progressive improvement

31.10. Exercises

1. **Bandit Comparison:** Implement all three bandit algorithms (epsilon-greedy, UCB, Thompson Sampling) and compare their cumulative regret on a 10-armed bandit problem over 10,000 rounds.
2. **Contextual Pricing:** Build a contextual bandit for dynamic pricing where the context includes customer features and the arms are different price points.
3. **Real-Time Dispatch:** Extend the ride-sharing dispatcher to handle driver repositioning during idle periods.
4. **Online Portfolio:** Implement an online portfolio algorithm that handles transaction costs and compare with the constant rebalanced portfolio.
5. **Latency Analysis:** Profile a decision system in your domain. Identify the components that contribute most to latency and design caching/fallback strategies.

31.11. References

- Slivkins, A. (2019). “Introduction to Multi-Armed Bandits.” *Foundations and Trends in Machine Learning*.
- Hazan, E. (2016). “Introduction to Online Convex Optimization.” *Foundations and Trends in Optimization*.
- Powell, W. B. (2019). “A Unified Framework for Stochastic Optimization.” *European Journal of Operational Research*.
- Li, L., et al. (2010). “A Contextual-Bandit Approach to Personalized News Article Recommendation.” *WWW*.

32. Game Theory for Strategic Decisions

What You Will Learn

- Understand when game theory applies versus standard optimization
- Model strategic interactions using normal and extensive form games
- Compute Nash equilibria for pure and mixed strategies
- Apply cooperative game theory concepts including Shapley value
- Design mechanisms and auctions for incentive-compatible outcomes
- Implement game-theoretic models for supply chain and pricing decisions
- Connect game theory to bilevel optimization and multi-agent systems

In most optimization problems, we assume a single decision-maker facing an environment that responds passively or stochastically to decisions. But many real business situations involve **strategic interaction**: your outcome depends not just on your actions, but on the actions of other rational agents who are simultaneously optimizing their own objectives. Game theory provides the mathematical framework for analyzing these interdependent decisions.

32.1. Introduction to Strategic Decisions

32.1.1. When Game Theory Applies

Game theory is appropriate when:

1. **Multiple decision-makers**: Two or more agents making choices
2. **Interdependent outcomes**: Each player's payoff depends on others' actions
3. **Strategic awareness**: Players anticipate and respond to others' strategies
4. **Conflicting or aligned interests**: Competition, cooperation, or mixed motives

Scenario	Type	Game Theory Needed?
Production scheduling	Single agent vs constraints	No
Pricing against competitors	Multiple agents, conflicting	Yes
Inventory under demand uncertainty	Single agent vs nature	No (stochastic)
Supply chain contract negotiation	Multiple agents, mixed motives	Yes
Auction bidding	Multiple agents, strategic	Yes
Traffic routing	Multiple agents, congestion	Yes

32.1.2. The Structure of a Game

Every game consists of:

- **Players**: The decision-makers (indexed $i \in N = \{1, 2, \dots, n\}$)

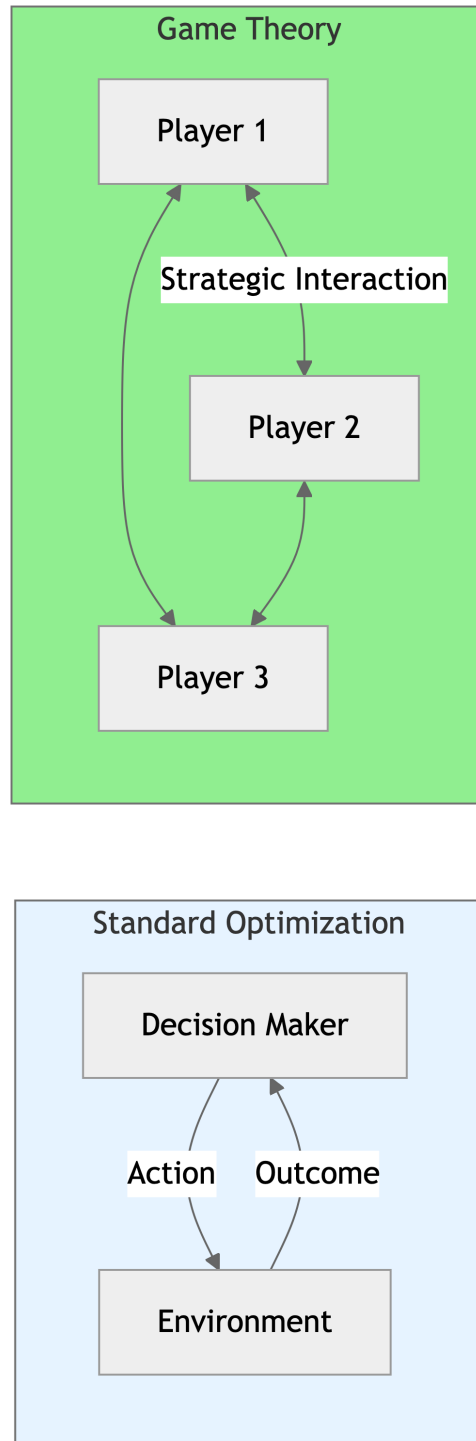


Figure 32.1.: Standard Optimization vs Game-Theoretic Setting

- **Strategies:** Available actions for each player (S_i for player i)
- **Payoffs:** Outcomes as functions of all players' strategies ($u_i : S \rightarrow \mathbb{R}$)

The key insight: each player must consider not just their own optimization, but how others will respond.

i Zero-Sum vs Non-Zero-Sum

In **zero-sum games**, one player's gain equals another's loss ($\sum_i u_i = 0$). Most real-world strategic situations are **non-zero-sum**: mutual gains or losses are possible, creating opportunities for cooperation.

32.1.3. Connection to Optimization

Game theory extends optimization in several ways:

- **Single-agent optimization:** $\max_x f(x)$
- **Game equilibrium:** Find $x^* = (x_1^*, \dots, x_n^*)$ where each x_i^* solves $\max_{x_i} u_i(x_i, x_{-i}^*)$
- **Bilevel optimization:** Leader-follower games with explicit hierarchy
- **Mechanism design:** Design the game itself to achieve target outcomes

32.2. Non-Cooperative Game Theory

32.2.1. Normal Form Games

A **normal form game** (or strategic form) is defined by:

$$\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

where: - $N = \{1, 2, \dots, n\}$ is the set of players - S_i is the strategy set for player i - $u_i : S_1 \times S_2 \times \dots \times S_n \rightarrow \mathbb{R}$ is player i 's payoff function

For two-player games with finite strategies, we represent this as a **payoff matrix**.

```
import numpy as np
from typing import Tuple, List, Optional
import matplotlib.pyplot as plt

class NormalFormGame:
    """
    Represents a two-player normal form game.

    Payoff matrices are stored as:
    - payoff_1[i,j]: Player 1's payoff when P1 plays strategy i, P2 plays j
    - payoff_2[i,j]: Player 2's payoff when P1 plays strategy i, P2 plays j
    """

    def __init__(self, payoff_1: np.ndarray, payoff_2: np.ndarray,
                  p1_strategies: List[str] = None,
                  p2_strategies: List[str] = None):
        """
        Initialize a two-player game.

        Args:
```

```

        payoff_1: Payoff matrix for player 1 (rows = P1 strategies, cols = P2)
        payoff_2: Payoff matrix for player 2
        p1_strategies: Names of player 1's strategies
        p2_strategies: Names of player 2's strategies
    """
    self.payoff_1 = np.array(payoff_1)
    self.payoff_2 = np.array(payoff_2)

    assert self.payoff_1.shape == self.payoff_2.shape

    self.n_strategies_1 = self.payoff_1.shape[0]
    self.n_strategies_2 = self.payoff_1.shape[1]

    self.p1_strategies = p1_strategies or [f"S1_{i}" for i in
        ↪ range(self.n_strategies_1)]
    self.p2_strategies = p2_strategies or [f"S2_{j}" for j in
        ↪ range(self.n_strategies_2)]

def get_payoffs(self, i: int, j: int) -> Tuple[float, float]:
    """Get payoffs for strategy profile (i, j)."""
    return self.payoff_1[i, j], self.payoff_2[i, j]

def best_response_1(self, j: int) -> List[int]:
    """Find player 1's best response(s) to player 2 playing strategy j."""
    payoffs = self.payoff_1[:, j]
    max_payoff = np.max(payoffs)
    return list(np.where(payoffs == max_payoff)[0])

def best_response_2(self, i: int) -> List[int]:
    """Find player 2's best response(s) to player 1 playing strategy i."""
    payoffs = self.payoff_2[i, :]
    max_payoff = np.max(payoffs)
    return list(np.where(payoffs == max_payoff)[0])

def find_pure_nash_equilibria(self) -> List[Tuple[int, int]]:
    """
    Find all pure strategy Nash equilibria.

    A profile (i*, j*) is a Nash equilibrium if:
    - i* is a best response to j*
    - j* is a best response to i*
    """
    equilibria = []

    for i in range(self.n_strategies_1):
        for j in range(self.n_strategies_2):
            if i in self.best_response_1(j) and j in self.best_response_2(i):
                equilibria.append((i, j))

    return equilibria

def is_dominant_strategy(self, player: int, strategy: int) -> bool:
    """

```

```

    Check if a strategy is strictly dominant for a player.

    A strategy is dominant if it's the best response to ALL opponent strategies.
    """
    if player == 1:
        for j in range(self.n_strategies_2):
            if strategy not in self.best_response_1(j):
                return False
        return True
    else:
        for i in range(self.n_strategies_1):
            if strategy not in self.best_response_2(i):
                return False
        return True

def __repr__(self):
    lines = ["Two-Player Normal Form Game", "=" * 40]
    lines.append(f"Player 1 strategies: {self.p1_strategies}")
    lines.append(f"Player 2 strategies: {self.p2_strategies}")
    lines.append("\nPayoff Matrix (P1, P2):")

    # Header
    header = "          " + " ".join(f"{s:>10}" for s in self.p2_strategies)
    lines.append(header)

    for i, s1 in enumerate(self.p1_strategies):
        row = f"{s1:>6} "
        for j in range(self.n_strategies_2):
            row += f"({self.payoff_1[i,j]:>3.0f},{self.payoff_2[i,j]:>3.0f}) "
        lines.append(row)

    return "\n".join(lines)

# Example: Prisoner's Dilemma
prisoners_dilemma = NormalFormGame(
    payoff_1 = np.array([[ -1, -3],
                        [ 0, -2]]),
    payoff_2 = np.array([[ -1, 0],
                        [-3, -2]]),
    p1_strategies = ["Cooperate", "Defect"],
    p2_strategies = ["Cooperate", "Defect"]
)

print(prisoners_dilemma)
print("\n" + "=" * 40)
print("Analysis:")

# Find dominant strategies
for player in [1, 2]:
    for s in range(2):
        strategy_name = prisoners_dilemma.p1_strategies[s] if player == 1 else
        ↪ prisoners_dilemma.p2_strategies[s]

```

```

        if prisoners_dilemma.is_dominant_strategy(player, s):
            print(f"Player {player}'s dominant strategy: {strategy_name}")

# Find Nash equilibria
equilibria = prisoners_dilemma.find_pure_nash_equilibria()
print(f"\nPure Nash Equilibria: {len(equilibria)}")
for i, j in equilibria:
    payoffs = prisoners_dilemma.get_payoffs(i, j)
    print(f"    ({prisoners_dilemma.p1_strategies[i]},
      ↪ {prisoners_dilemma.p2_strategies[j]})")
    print(f"    Payoffs: Player 1 = {payoffs[0]}, Player 2 = {payoffs[1]}")

```

Two-Player Normal Form Game

=====

Player 1 strategies: ['Cooperate', 'Defect']

Player 2 strategies: ['Cooperate', 'Defect']

Payoff Matrix (P1, P2):

	Cooperate	Defect
Cooperate	(-1, -1)	(-3, 0)
Defect	(0, -3)	(-2, -2)

=====

Analysis:

Player 1's dominant strategy: Defect

Player 2's dominant strategy: Defect

Pure Nash Equilibria: 1

(Defect, Defect)

Payoffs: Player 1 = -2, Player 2 = -2

The heatmap below displays the payoff structure of the Prisoner's Dilemma. The dominant strategy equilibrium (Defect, Defect) yields lower payoffs than mutual cooperation, illustrating the tension between individual and collective rationality.

```

import matplotlib.pyplot as plt
import numpy as np

def plot_payoff_matrix_heatmap(payload_1: np.ndarray, payload_2: np.ndarray,
                                p1_strategies: list, p2_strategies: list,
                                title: str = "Game Payoff Matrix"):
    """
    Visualize a two-player game's payoff matrix as a heatmap.

    Args:
        payload_1: Payoff matrix for player 1
        payload_2: Payoff matrix for player 2
        p1_strategies: Names of player 1's strategies
        p2_strategies: Names of player 2's strategies
        title: Title for the plot
    """

```



```

"""
fig, axes = plt.subplots(1, 2, figsize=(8, 6))

# Player 1 payoffs
im1 = axes[0].imshow(payload_1, cmap='RdYlGn', aspect='auto')
axes[0].set_xticks(range(len(p2_strategies)))
axes[0].set_yticks(range(len(p1_strategies)))
axes[0].set_xticklabels(p2_strategies)
axes[0].set_yticklabels(p1_strategies)
axes[0].set_xlabel("Player 2 Strategy")
axes[0].set_ylabel("Player 1 Strategy")
axes[0].set_title("Player 1 Payoffs")

# Add value annotations
for i in range(len(p1_strategies)):
    for j in range(len(p2_strategies)):
        text_color = 'white' if abs(payload_1[i, j]) > 1.5 else 'black'
        axes[0].text(j, i, f'{payload_1[i, j]:.0f}', ha='center', va='center',
                     color=text_color, fontsize=14, fontweight='bold')

plt.colorbar(im1, ax=axes[0], shrink=0.6)

# Player 2 payoffs
im2 = axes[1].imshow(payload_2, cmap='RdYlGn', aspect='auto')
axes[1].set_xticks(range(len(p2_strategies)))
axes[1].set_yticks(range(len(p1_strategies)))
axes[1].set_xticklabels(p2_strategies)
axes[1].set_yticklabels(p1_strategies)
axes[1].set_xlabel("Player 2 Strategy")
axes[1].set_ylabel("Player 1 Strategy")
axes[1].set_title("Player 2 Payoffs")

# Add value annotations
for i in range(len(p1_strategies)):
    for j in range(len(p2_strategies)):
        text_color = 'white' if abs(payload_2[i, j]) > 1.5 else 'black'
        axes[1].text(j, i, f'{payload_2[i, j]:.0f}', ha='center', va='center',
                     color=text_color, fontsize=14, fontweight='bold')

plt.colorbar(im2, ax=axes[1], shrink=0.6)

fig.suptitle(title, fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()

# Visualize the Prisoner's Dilemma payoff matrix
plot_payoff_matrix_heatmap(
    payload_1=np.array([[ -1, -3], [ 0, -2]]),
    payload_2=np.array([[ -1,  0], [-3, -2]]),
    p1_strategies=["Cooperate", "Defect"],
    p2_strategies=["Cooperate", "Defect"],
    title="Prisoner's Dilemma Payoff Matrix"
)

```

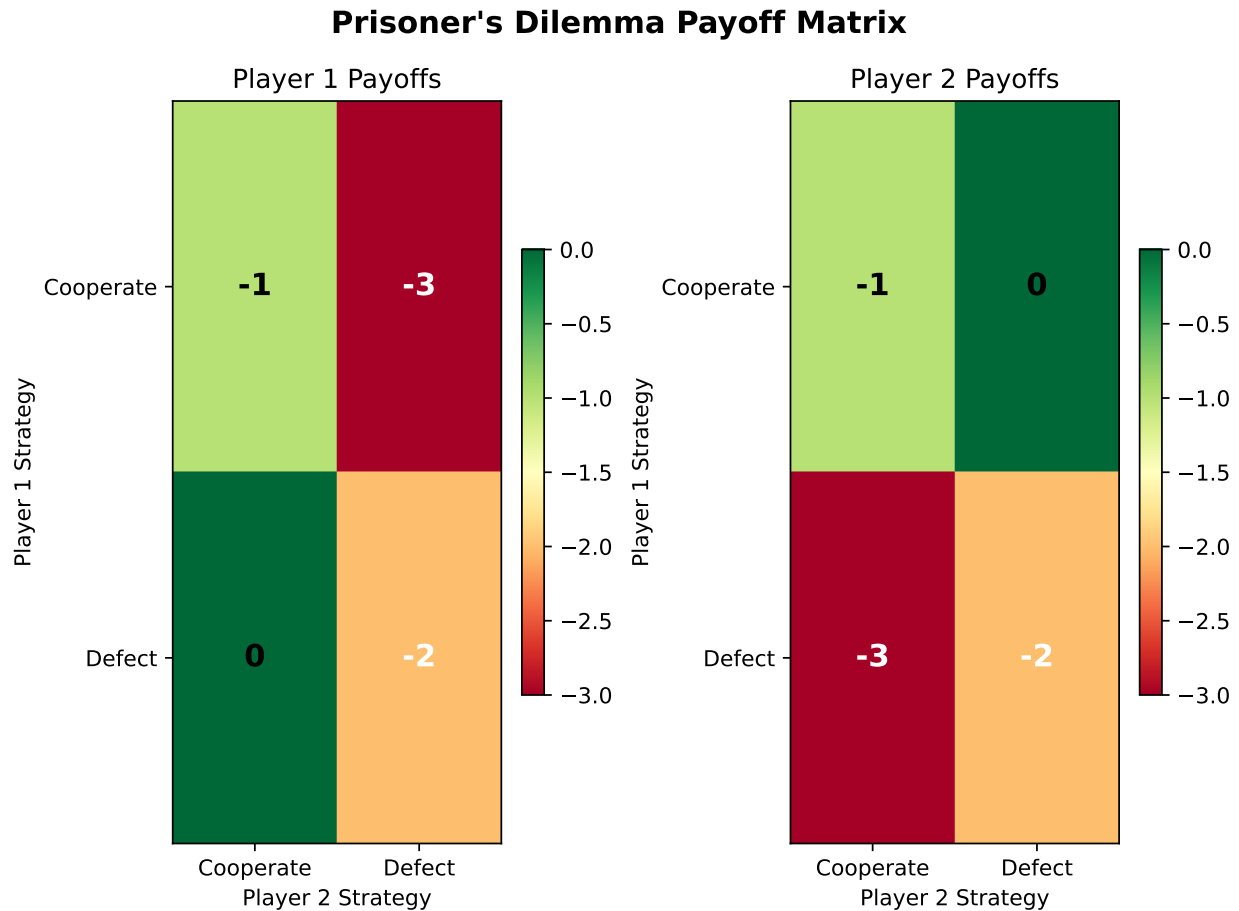


Figure 32.2.: Payoff Matrix Heatmap for the Prisoner's Dilemma Game

32.2.2. Dominant Strategies

A strategy s_i^* is **strictly dominant** for player i if:

$$u_i(s_i^*, s_{-i}) > u_i(s_i, s_{-i}) \quad \forall s_i \neq s_i^*, \forall s_{-i}$$

A strategy is **weakly dominant** if the inequality is non-strict ($\geq$) with strict inequality for at least one opponent profile.

💡 Iterated Elimination of Dominated Strategies

When players are rational and know others are rational, dominated strategies can be eliminated. Iteratively removing dominated strategies can simplify the game or even solve it completely.

32.2.3. Nash Equilibrium

The central solution concept in non-cooperative game theory.

Definition: A strategy profile $s^* = (s_1^*, s_2^*, \dots, s_n^*)$ is a **Nash equilibrium** if no player can unilaterally improve their payoff:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i \in S_i, \forall i \in N$$

At a Nash equilibrium, every player is playing a best response to the others' strategies.

Nash's Theorem (1950): Every finite game (finite players, finite strategies) has at least one Nash equilibrium in mixed strategies.

```
# Battle of the Sexes
# Two players want to coordinate but have different preferences
battle_of_sexes = NormalFormGame(
    payoff_1 = np.array([[3, 0],
                          [0, 2]]),
    payoff_2 = np.array([[2, 0],
                          [0, 3]]),
    p1_strategies = ["Opera", "Football"],
    p2_strategies = ["Opera", "Football"]
)

print(battle_of_sexes)
print("\n" + "=" * 40)

# Find Nash equilibria
equilibria = battle_of_sexes.find_pure_nash_equilibria()
print(f"\nPure Nash Equilibria: {len(equilibria)}")
for i, j in equilibria:
    payoffs = battle_of_sexes.get_payoffs(i, j)
    print(f"    ({battle_of_sexes.p1_strategies[i]},
      ↪ {battle_of_sexes.p2_strategies[j]})")
    print(f"    Payoffs: ({payoffs[0]}, {payoffs[1]})")

print("\nNote: There's also a mixed strategy equilibrium!")
print("Each player randomizes to make the other indifferent.")
```

Two-Player Normal Form Game

```
=====
Player 1 strategies: ['Opera', 'Football']
Player 2 strategies: ['Opera', 'Football']
```

Payoff Matrix (P1, P2):

	Opera	Football
Opera	(3, 2)	(0, 0)
Football	(0, 0)	(2, 3)

```
=====
```

Pure Nash Equilibria: 2

```
(Opera, Opera)
Payoffs: (3, 2)
(Football, Football)
Payoffs: (2, 3)
```

Note: There's also a mixed strategy equilibrium!
Each player randomizes to make the other indifferent.

32.2.4. Mixed Strategy Nash Equilibrium

A **mixed strategy** is a probability distribution over pure strategies:

$$\sigma_i \in \Delta(S_i) = \left\{ \sigma_i : S_i \rightarrow [0, 1] \mid \sum_{s_i \in S_i} \sigma_i(s_i) = 1 \right\}$$

The expected payoff under mixed strategies:

$$U_i(\sigma) = \sum_{s \in S} \left(\prod_{j \in N} \sigma_j(s_j) \right) u_i(s)$$

Key insight: In a mixed Nash equilibrium, each player must be **indifferent** between all strategies they play with positive probability.

```
def compute_mixed_nash_2x2(game: NormalFormGame) -> Tuple[np.ndarray, np.ndarray]:
    """
    Compute mixed strategy Nash equilibrium for a 2x2 game.

    Uses the indifference principle: at equilibrium, each player's
    mixed strategy makes the opponent indifferent.

    Returns:
        p: Player 1's probability of playing first strategy
        q: Player 2's probability of playing first strategy
    """
    A = game.payoff_1
    B = game.payoff_2

    # Player 2 mixes to make Player 1 indifferent
    # P1's expected payoff from strategy 0: q*A[0,0] + (1-q)*A[0,1]
    # P1's expected payoff from strategy 1: q*A[1,0] + (1-q)*A[1,1]
    # Set equal and solve for q

    # A[0,0]*q + A[0,1]*(1-q) = A[1,0]*q + A[1,1]*(1-q)
    # (A[0,0] - A[0,1] - A[1,0] + A[1,1])*q = A[1,1] - A[0,1]

    denom_q = A[0,0] - A[0,1] - A[1,0] + A[1,1]
    if abs(denom_q) < 1e-10:
        # No interior mixed equilibrium
        return None, None
    q = (A[1,1] - A[0,1]) / denom_q

    # Player 1 mixes to make Player 2 indifferent
    # P2's expected payoff from strategy 0: p*B[0,0] + (1-p)*B[1,0]
    # P2's expected payoff from strategy 1: p*B[0,1] + (1-p)*B[1,1]

    denom_p = B[0,0] - B[1,0] - B[0,1] + B[1,1]
    if abs(denom_p) < 1e-10:
        return None, None
```

```

p = (B[1,1] - B[1,0]) / denom_p

# Check if probabilities are valid
if 0 <= p <= 1 and 0 <= q <= 1:
    return np.array([p, 1-p]), np.array([q, 1-q])
else:
    return None, None

# Compute mixed Nash for Battle of the Sexes
p_mixed, q_mixed = compute_mixed_nash_2x2(battle_of_sexes)

if p_mixed is not None:
    print("Mixed Strategy Nash Equilibrium:")
    print(f"Player 1: {p_mixed[0]:.3f} Opera, {p_mixed[1]:.3f} Football")
    print(f"Player 2: {q_mixed[0]:.3f} Opera, {q_mixed[1]:.3f} Football")

    # Compute expected payoffs
    exp_payoff_1 = np.sum([p_mixed[i] * q_mixed[j] * battle_of_sexes.payoff_1[i,j]
                           for i in range(2) for j in range(2)])
    exp_payoff_2 = np.sum([p_mixed[i] * q_mixed[j] * battle_of_sexes.payoff_2[i,j]
                           for i in range(2) for j in range(2)])

    print(f"\nExpected payoffs at mixed equilibrium:")
    print(f"Player 1: {exp_payoff_1:.3f}")
    print(f"Player 2: {exp_payoff_2:.3f}")
    print("\nNote: Both players get less than in either pure equilibrium!")
    print("This is the 'cost of miscoordination'.")

```

Mixed Strategy Nash Equilibrium:

Player 1: 0.600 Opera, 0.400 Football

Player 2: 0.400 Opera, 0.600 Football

Expected payoffs at mixed equilibrium:

Player 1: 1.200

Player 2: 1.200

Note: Both players get less than in either pure equilibrium!

This is the 'cost of miscoordination'.

Best response curves show each player's optimal strategy as a function of the other player's mixed strategy. Nash equilibria occur at the intersection points of these curves.

```

import matplotlib.pyplot as plt
import numpy as np

def plot_best_response_curves(payload_1: np.ndarray, payoff_2: np.ndarray,
                              title: str = "Best Response Curves"):
    """
    Plot best response functions for a 2x2 game and show Nash equilibria.

```

```

For a 2x2 game:
- p = probability Player 1 plays strategy 0
- q = probability Player 2 plays strategy 0

Best response for Player 1: BR_1(q) maximizes  $p[qA[0,0] + (1-q)A[0,1]] +$ 
↪  $(1-p)[qA[1,0] + (1-q)A[1,1]]$ 
Best response for Player 2: BR_2(p) maximizes  $q[pB[0,0] + (1-p)B[1,0]] +$ 
↪  $(1-q)[pB[0,1] + (1-p)B[1,1]]$ 
"""
fig, ax = plt.subplots(figsize=(8, 6))

A = payoff_1
B = payoff_2

q_vals = np.linspace(0, 1, 1000)
p_vals = np.linspace(0, 1, 1000)

# Best response for Player 1 as function of q
# Player 1's expected payoff from strategy 0:  $qA[0,0] + (1-q)A[0,1]$ 
# Player 1's expected payoff from strategy 1:  $qA[1,0] + (1-q)A[1,1]$ 
br1_p = []
for q in q_vals:
    eu_0 = q * A[0, 0] + (1 - q) * A[0, 1] # Expected utility of strategy 0
    eu_1 = q * A[1, 0] + (1 - q) * A[1, 1] # Expected utility of strategy 1
    if eu_0 > eu_1:
        br1_p.append(1.0) # Play strategy 0 with probability 1
    elif eu_1 > eu_0:
        br1_p.append(0.0) # Play strategy 1 with probability 1
    else:
        br1_p.append(0.5) # Indifferent - any p is best response

# Best response for Player 2 as function of p
br2_q = []
for p in p_vals:
    eu_0 = p * B[0, 0] + (1 - p) * B[1, 0] # Expected utility of strategy 0
    eu_1 = p * B[0, 1] + (1 - p) * B[1, 1] # Expected utility of strategy 1
    if eu_0 > eu_1:
        br2_q.append(1.0)
    elif eu_1 > eu_0:
        br2_q.append(0.0)
    else:
        br2_q.append(0.5)

# Plot best response curves
ax.plot(q_vals, br1_p, 'b-', linewidth=2.5, label='BR$_1$(q): Player 1\'s Best
↪ Response')
ax.plot(br2_q, p_vals, 'r-', linewidth=2.5, label='BR$_2$(p): Player 2\'s Best
↪ Response')

# Find and mark Nash equilibria
# For Battle of the Sexes: pure NE at (1,1), (0,0), and mixed NE
# Compute mixed Nash equilibrium
denom_q = A[0,0] - A[0,1] - A[1,0] + A[1,1]

```

```

denom_p = B[0,0] - B[1,0] - B[0,1] + B[1,1]

nash_points = []

# Pure equilibria
nash_points.append((1.0, 1.0)) # Both play strategy 0 (Opera, Opera)
nash_points.append((0.0, 0.0)) # Both play strategy 1 (Football, Football)

# Mixed equilibrium
if abs(denom_q) > 1e-10 and abs(denom_p) > 1e-10:
    q_star = (A[1,1] - A[0,1]) / denom_q
    p_star = (B[1,1] - B[1,0]) / denom_p
    if 0 < p_star < 1 and 0 < q_star < 1:
        nash_points.append((q_star, p_star))

# Plot Nash equilibria
for i, (q, p) in enumerate(nash_points):
    if i < 2:
        ax.scatter(q, p, s=200, c='green', marker='*', zorder=5,
                  edgecolors='darkgreen', linewidths=1.5)
    else:
        ax.scatter(q, p, s=200, c='orange', marker='*', zorder=5,
                  edgecolors='darkorange', linewidths=1.5)

# Add annotations for Nash equilibria
ax.annotate('Pure NE\n(Opera, Opera)', xy=(1.0, 1.0), xytext=(0.75, 0.85),
            fontsize=9, ha='center',
            arrowprops=dict(arrowstyle='->', color='darkgreen'))
ax.annotate('Pure NE\n(Football, Football)', xy=(0.0, 0.0), xytext=(0.25, 0.15),
            fontsize=9, ha='center',
            arrowprops=dict(arrowstyle='->', color='darkgreen'))

if len(nash_points) > 2:
    q_m, p_m = nash_points[2]
    ax.annotate(f'Mixed NE\n(p={p_m:.2f}, q={q_m:.2f})', xy=(q_m, p_m),
                xytext=(q_m + 0.2, p_m + 0.2), fontsize=9, ha='center',
                arrowprops=dict(arrowstyle='->', color='darkorange'))

ax.set_xlabel('q (Probability Player 2 plays Opera)', fontsize=11)
ax.set_ylabel('p (Probability Player 1 plays Opera)', fontsize=11)
ax.set_title(title, fontsize=14, fontweight='bold')
ax.set_xlim(-0.05, 1.05)
ax.set_ylim(-0.05, 1.05)
ax.legend(loc='center right', fontsize=10)
ax.grid(True, alpha=0.3)
ax.set_aspect('equal')

plt.tight_layout()
plt.show()

# Plot best response curves for Battle of the Sexes
plot_best_response_curves(
    payoff_1=np.array([[3, 0], [0, 2]]),

```

```

payoff_2=np.array([[2, 0], [0, 3]]),
title="Best Response Curves: Battle of the Sexes"
)

```

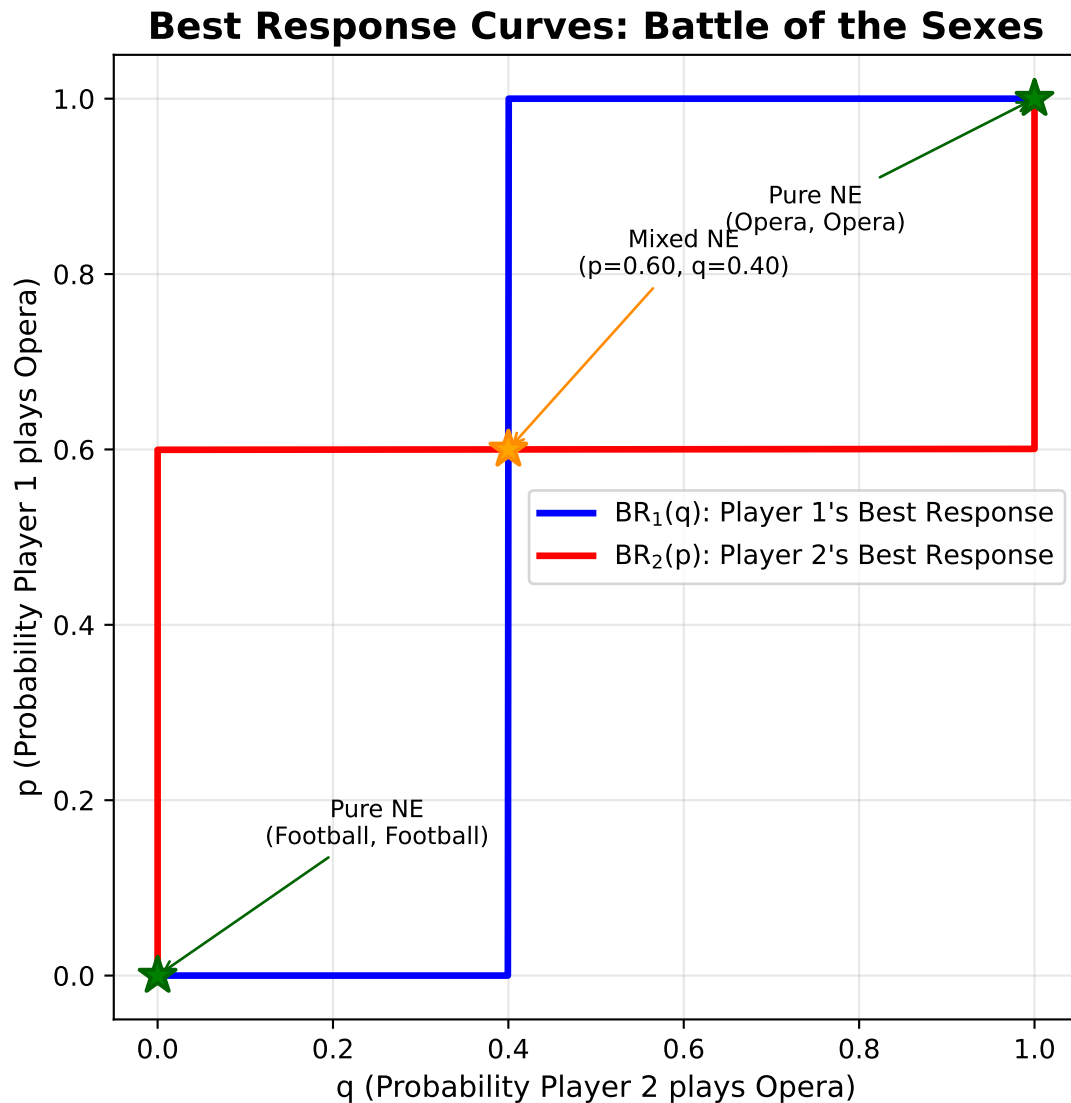


Figure 32.3.: Best Response Curves and Nash Equilibrium for Battle of the Sexes

32.2.5. Computing Nash Equilibria with Nashpy

The `nashpy` library provides robust algorithms for computing Nash equilibria.

```

try:
    import nashpy as nash
    NASHPY_AVAILABLE = True
except ImportError:
    NASHPY_AVAILABLE = False

```



```

    print("nashpy not installed. Install with: pip install nashpy")
if NASHPY_AVAILABLE:
    # Create games using nashpy

    # Prisoner's Dilemma
    pd_payoffs_1 = np.array([[ -1, -3],
                             [ 0, -2]])
    pd_payoffs_2 = np.array([[ -1,  0],
                             [-3, -2]])

    pd_game = nash.Game(pd_payoffs_1, pd_payoffs_2)

    print("=== Prisoner's Dilemma ===")
    print(pd_game)

    # Find all Nash equilibria using support enumeration
    print("\nNash Equilibria (Support Enumeration):")
    equilibria = list(pd_game.support_enumeration())
    for eq in equilibria:
        p1_strategy, p2_strategy = eq
        print(f" P1: {p1_strategy}, P2: {p2_strategy}")

    # Matching Pennies (zero-sum game with only mixed equilibrium)
    print("\n=== Matching Pennies ===")
    mp_payoffs = np.array([[ 1, -1],
                           [-1,  1]])

    mp_game = nash.Game(mp_payoffs, -mp_payoffs) # Zero-sum
    print(mp_game)

    print("\nNash Equilibria:")
    equilibria = list(mp_game.support_enumeration())
    for eq in equilibria:
        p1_strategy, p2_strategy = eq
        print(f" P1: {p1_strategy}, P2: {p2_strategy}")

    # Verify this is the 50-50 mixed strategy
    expected_payoff = np.sum(p1_strategy @ mp_payoffs @ p2_strategy)
    print(f" Expected value to P1: {expected_payoff:.4f}")

```

nashpy not installed. Install with: pip install nashpy

32.2.6. Classic Games

Four canonical games recur throughout game theory because they capture distinct strategic structures. The Prisoner's Dilemma models cooperation failure, Battle of the Sexes models coordination, Chicken models anti-coordination, and Stag Hunt models trust. The code below defines all four and computes their equilibria.

```

def analyze_game(name: str, game: NormalFormGame):
    """Comprehensive analysis of a normal form game."""
    print(f"\n{'*' * 50}")

```

```

print(f"GAME: {name}")
print('='*50)
print(game)

# Pure Nash equilibria
pure_eq = game.find_pure_nash_equilibria()
print(f"\nPure Nash Equilibria: {len(pure_eq)}")
for i, j in pure_eq:
    payoffs = game.get_payoffs(i, j)
    print(f"    ({game.p1_strategies[i]}, {game.p2_strategies[j]}) ->
    ↪ ({payoffs[0]}, {payoffs[1]})")

# Mixed Nash (for 2x2)
if game.n_strategies_1 == 2 and game.n_strategies_2 == 2:
    p_mixed, q_mixed = compute_mixed_nash_2x2(game)
    if p_mixed is not None and not (np.allclose(p_mixed, [1,0]) or
    ↪ np.allclose(p_mixed, [0,1])):
        print(f"\nMixed Nash Equilibrium:")
        print(f"    P1: {p_mixed[0]:.3f} {game.p1_strategies[0]},
        ↪ {p_mixed[1]:.3f} {game.p1_strategies[1]}")
        print(f"    P2: {q_mixed[0]:.3f} {game.p2_strategies[0]},
        ↪ {q_mixed[1]:.3f} {game.p2_strategies[1]}")

# Dominant strategies
for player in [1, 2]:
    strategies = game.p1_strategies if player == 1 else game.p2_strategies
    n_strats = game.n_strategies_1 if player == 1 else game.n_strategies_2
    for s in range(n_strats):
        if game.is_dominant_strategy(player, s):
            print(f"\nDominant strategy for Player {player}: {strategies[s]}")

# Chicken Game (Hawk-Dove)
# Two drivers heading toward each other. Swerve or stay straight?
chicken = NormalFormGame(
    payoff_1 = np.array([[0, -1],
                        [1, -10]]),
    payoff_2 = np.array([[0, 1],
                        [-1, -10]]),
    p1_strategies = ["Swerve", "Straight"],
    p2_strategies = ["Swerve", "Straight"]
)

# Stag Hunt (Coordination Game)
# Cooperate to hunt stag (big payoff) or defect to hunt rabbit (safe)
stag_hunt = NormalFormGame(
    payoff_1 = np.array([[5, 0],
                        [3, 3]]),
    payoff_2 = np.array([[5, 3],
                        [0, 3]]),
    p1_strategies = ["Stag", "Rabbit"],
    p2_strategies = ["Stag", "Rabbit"]
)

```

```
# Analyze all games
analyze_game("Prisoner's Dilemma", prisoners_dilemma)
analyze_game("Battle of the Sexes", battle_of_sexes)
analyze_game("Chicken (Hawk-Dove)", chicken)
analyze_game("Stag Hunt", stag_hunt)
```

```
=====
GAME: Prisoner's Dilemma
=====
```

```
Two-Player Normal Form Game
=====
```

```
Player 1 strategies: ['Cooperate', 'Defect']
Player 2 strategies: ['Cooperate', 'Defect']
```

```
Payoff Matrix (P1, P2):
```

	Cooperate	Defect
Cooperate	(-1, -1)	(-3, 0)
Defect	(0, -3)	(-2, -2)

```
Pure Nash Equilibria: 1
(Defect, Defect) -> (-2, -2)
```

```
Dominant strategy for Player 1: Defect
```

```
Dominant strategy for Player 2: Defect
```

```
=====
GAME: Battle of the Sexes
=====
```

```
Two-Player Normal Form Game
=====
```

```
Player 1 strategies: ['Opera', 'Football']
Player 2 strategies: ['Opera', 'Football']
```

```
Payoff Matrix (P1, P2):
```

	Opera	Football
Opera	(3, 2)	(0, 0)
Football	(0, 0)	(2, 3)

```
Pure Nash Equilibria: 2
(Opera, Opera) -> (3, 2)
(Football, Football) -> (2, 3)
```

```
Mixed Nash Equilibrium:
```

```
P1: 0.600 Opera, 0.400 Football
P2: 0.400 Opera, 0.600 Football
```

32. Game Theory for Strategic Decisions

```
=====
GAME: Chicken (Hawk-Dove)
=====

Two-Player Normal Form Game
=====

Player 1 strategies: ['Swerve', 'Straight']
Player 2 strategies: ['Swerve', 'Straight']

Payoff Matrix (P1, P2):
      Swerve  Straight
Swerve ( 0, 0) (-1, 1)
Straight ( 1, -1) (-10,-10)

Pure Nash Equilibria: 2
  (Swerve, Straight) -> (-1, 1)
  (Straight, Swerve) -> (1, -1)

Mixed Nash Equilibrium:
  P1: 0.900 Swerve, 0.100 Straight
  P2: 0.900 Swerve, 0.100 Straight

=====
GAME: Stag Hunt
=====

Two-Player Normal Form Game
=====

Player 1 strategies: ['Stag', 'Rabbit']
Player 2 strategies: ['Stag', 'Rabbit']

Payoff Matrix (P1, P2):
      Stag  Rabbit
Stag ( 5, 5) ( 0, 3)
Rabbit ( 3, 0) ( 3, 3)

Pure Nash Equilibria: 2
  (Stag, Stag) -> (5, 5)
  (Rabbit, Rabbit) -> (3, 3)

Mixed Nash Equilibrium:
  P1: 0.600 Stag, 0.400 Rabbit
  P2: 0.600 Stag, 0.400 Rabbit
```

The simplex below shows how mixed strategy space looks in a three-strategy game like Rock-Paper-Scissors. Each vertex represents a pure strategy, and interior points represent mixtures. The unique Nash equilibrium sits at the centroid, where each strategy is played with equal probability.

```

import matplotlib.pyplot as plt
import numpy as np
from mpl_toolkits.mplot3d import Axes3D

def plot_mixed_strategy_simplex(payload_matrix: np.ndarray,
                                strategies: list,
                                title: str = "Mixed Strategy Simplex"):
    """
    Visualize the probability simplex for a 3-strategy symmetric game.

    The simplex represents all valid mixed strategies where probabilities sum to 1.
    Each corner represents a pure strategy.

    Args:
        payload_matrix: 3x3 payoff matrix for the symmetric game
        strategies: Names of the three strategies
        title: Title for the plot
    """
    fig, ax = plt.subplots(figsize=(8, 6))

    # Define simplex vertices (equilateral triangle)
    # Strategy 0 at top, Strategy 1 at bottom-left, Strategy 2 at bottom-right
    vertices = np.array([
        [0.5, np.sqrt(3)/2],      # Top: pure strategy 0 (p0=1)
        [0.0, 0.0],              # Bottom-left: pure strategy 1 (p1=1)
        [1.0, 0.0]                # Bottom-right: pure strategy 2 (p2=1)
    ])

    def prob_to_cartesian(p0, p1, p2):
        """Convert probability vector to cartesian coordinates in simplex."""
        x = p0 * vertices[0, 0] + p1 * vertices[1, 0] + p2 * vertices[2, 0]
        y = p0 * vertices[0, 1] + p1 * vertices[1, 1] + p2 * vertices[2, 1]
        return x, y

    # Draw the simplex boundary
    triangle = plt.Polygon(vertices, fill=False, edgecolor='black', linewidth=2)
    ax.add_patch(triangle)

    # Create a grid of points inside the simplex
    n_points = 100
    expected_payoffs = []
    coords = []

    for i in range(n_points + 1):
        for j in range(n_points + 1 - i):
            k = n_points - i - j
            p0, p1, p2 = i / n_points, j / n_points, k / n_points

            # Compute expected payoff against a uniform random opponent
            # E[payoff] = sum_i sum_j p_i * (1/3) * payoff[i,j]
            p = np.array([p0, p1, p2])
            q = np.array([1/3, 1/3, 1/3]) # Opponent plays uniform
            expected_payoff = p @ payload_matrix @ q

```

```

        x, y = prob_to_cartesian(p0, p1, p2)
        coords.append((x, y))
        expected_payoffs.append(expected_payoff)

coords = np.array(coords)
expected_payoffs = np.array(expected_payoffs)

# Plot the expected payoff as a colored scatter
scatter = ax.scatter(coords[:, 0], coords[:, 1], c=expected_payoffs,
                    cmap='RdYlGn', s=15, alpha=0.7)
plt.colorbar(scatter, ax=ax, label='Expected Payoff vs Uniform Opponent',
↪ shrink=0.7)

# Mark the Nash equilibrium (for Rock-Paper-Scissors, it's the centroid)
nash_x, nash_y = prob_to_cartesian(1/3, 1/3, 1/3)
ax.scatter(nash_x, nash_y, s=300, c='blue', marker='*', zorder=5,
          edgecolors='darkblue', linewidths=2, label='Nash Equilibrium')

# Label the vertices
offset = 0.08
ax.annotate(f'{{strategies[0]}}\n(p=1)', xy=vertices[0], fontsize=11,
          ha='center', va='bottom', fontweight='bold')
ax.annotate(f'{{strategies[1]}}\n(p=1)', xy=vertices[1], fontsize=11,
          ha='right', va='top', fontweight='bold')
ax.annotate(f'{{strategies[2]}}\n(p=1)', xy=vertices[2], fontsize=11,
          ha='left', va='top', fontweight='bold')

# Add probability labels on edges
ax.text(0.25, np.sqrt(3)/4 + 0.05, 'p$_0$ + p$_1$ = 1', fontsize=9, rotation=60,
↪ ha='center')
ax.text(0.75, np.sqrt(3)/4 + 0.05, 'p$_0$ + p$_2$ = 1', fontsize=9,
↪ rotation=-60, ha='center')
ax.text(0.5, -0.08, 'p$_1$ + p$_2$ = 1', fontsize=9, ha='center')

# Add annotation for Nash equilibrium
ax.annotate(f'Mixed NE\n(1/3, 1/3, 1/3)', xy=(nash_x, nash_y),
          xytext=(nash_x + 0.2, nash_y + 0.15), fontsize=10,
          arrowprops=dict(arrowstyle='->', color='darkblue'),
          ha='center')

ax.set_xlim(-0.15, 1.15)
ax.set_ylim(-0.2, 1.0)
ax.set_aspect('equal')
ax.axis('off')
ax.set_title(title, fontsize=14, fontweight='bold', pad=20)
ax.legend(loc='upper right', fontsize=10)

plt.tight_layout()
plt.show()

# Rock-Paper-Scissors payoff matrix (symmetric zero-sum game)
# Rows: player's strategy, Columns: opponent's strategy

```

```
# Payoffs: 0 = tie, 1 = win, -1 = lose
rps_payoffs = np.array([
    [0, -1, 1], # Rock: ties Rock, loses to Paper, beats Scissors
    [1, 0, -1], # Paper: beats Rock, ties Paper, loses to Scissors
    [-1, 1, 0]  # Scissors: loses to Rock, beats Paper, ties Scissors
])

plot_mixed_strategy_simplex(
    payoff_matrix=rps_payoffs,
    strategies=['Rock', 'Paper', 'Scissors'],
    title='Mixed Strategy Simplex: Rock-Paper-Scissors'
)
```

Mixed Strategy Simplex: Rock-Paper-Scissors

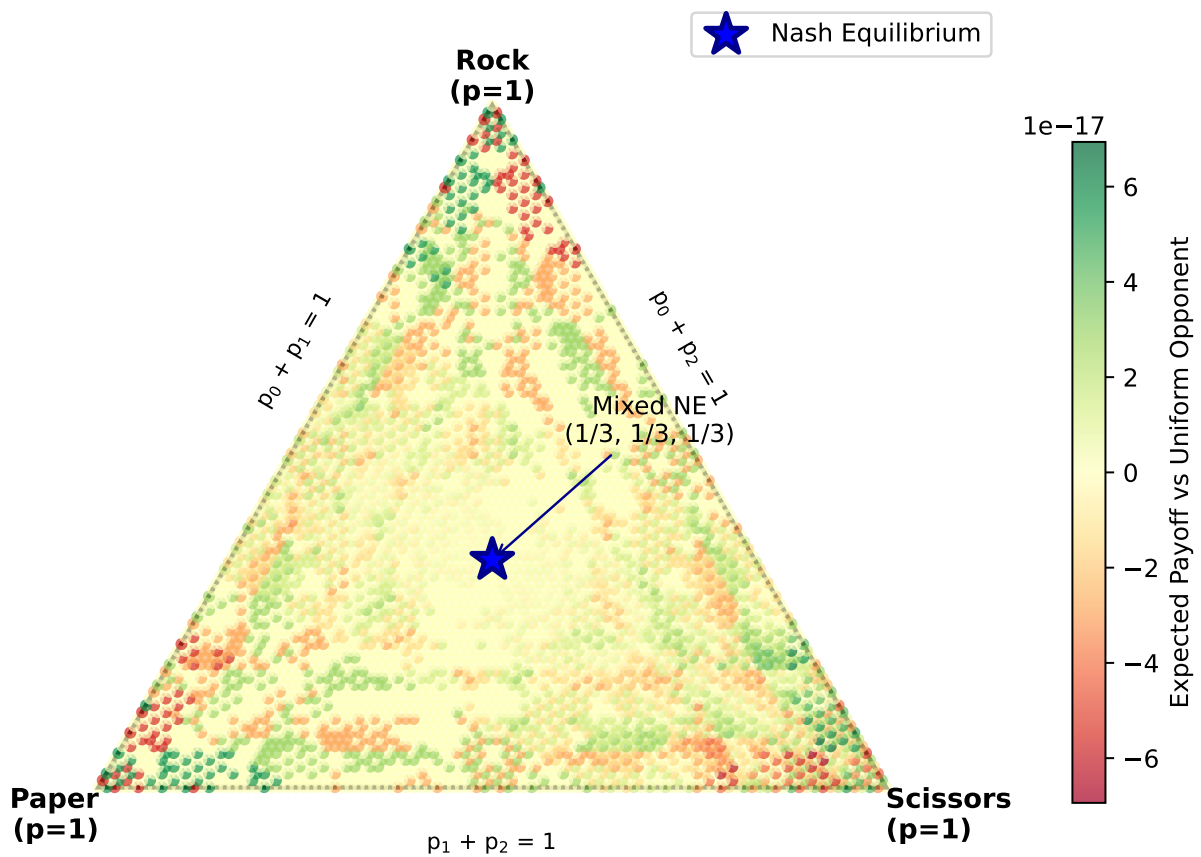


Figure 32.4.: Mixed Strategy Simplex for a 3-Strategy Game (Rock-Paper-Scissors)

32.2.7. Game Classification

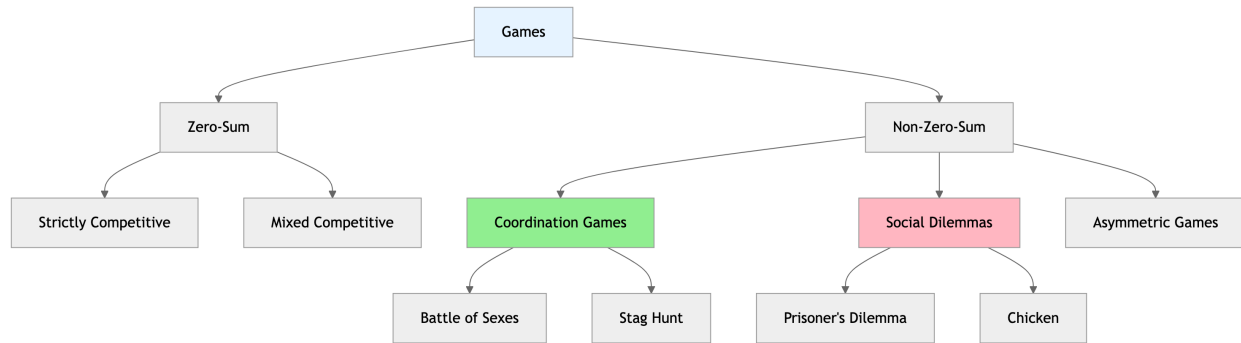


Figure 32.5.: Classification of Strategic Games

Game Type	Characteristic	Example
Zero-sum	$u_1 + u_2 = 0$	Chess, Poker
Coordination	Aligned interests	Standards adoption
Anti-coordination	Opposite interests	Market differentiation
Social dilemma	Individual vs collective	Environmental protection

32.2.8. Extensive Form Games

Extensive form games model sequential moves using game trees.

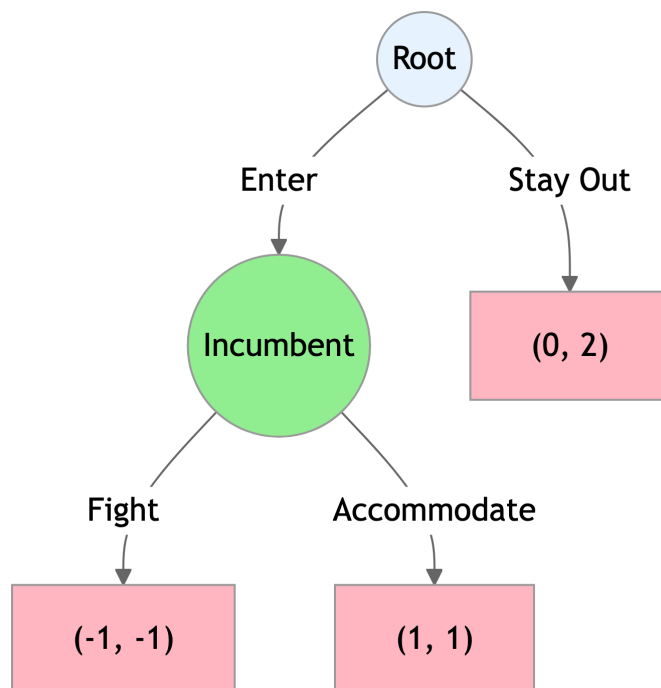


Figure 32.6.: Extensive Form Game Tree (Entry Deterrence)

An extensive form game consists of: - **Game tree**: Nodes (decision points) and edges (actions) - **Players**: Assigned to non-terminal nodes - **Information sets**: What a player knows when deciding - **Payoffs**: At terminal nodes

```

from dataclasses import dataclass
from typing import Dict, Any

@dataclass
class GameNode:
    """Node in an extensive form game tree"""
  
```



```

class ExtensiveFormGame:
    """
    Extensive form game represented as a tree.
    """

    def __init__(self, root: GameNode, n_players: int):
        self.root = root
        self.n_players = n_players

    def backward_induction(self) -> Dict[str, str]:
        """
        Solve the game using backward induction.

        Returns optimal action at each decision node.
        """
        optimal_actions = {}

        def solve_node(node: GameNode) -> Tuple[float, ...]:
            """Recursively solve from a node, returning payoffs."""
            if node.is_terminal():
                return node.payoffs

            # Get payoffs from all actions
            action_payoffs = {}
            for action, child in node.actions.items():
                action_payoffs[action] = solve_node(child)

            # Player at this node chooses to maximize their payoff
            player = node.player
            best_action = max(action_payoffs.keys(),
                             key=lambda a: action_payoffs[a][player])

            optimal_actions[node.name] = best_action
            return action_payoffs[best_action]

        self.equilibrium_payoffs = solve_node(self.root)
        return optimal_actions

    def get_subgame_perfect_equilibrium(self) -> Dict[str, str]:
        """Alias for backward induction."""
        return self.backward_induction()

# Example: Entry Deterrence Game
# Entrant moves first: Enter or Stay Out
# If Enter, Incumbent moves: Fight or Accommodate

# Terminal nodes
stay_out = GameNode("stay_out", payoffs=(0, 2))
fight = GameNode("fight", payoffs=(-1, -1))
accommodate = GameNode("accommodate", payoffs=(1, 1))

```

```

# Incumbent's decision
incumbent_node = GameNode(
    "incumbent",
    player=1, # Incumbent is player 1
    actions={"Fight": fight, "Accommodate": accommodate}
)

# Entrant's decision (root)
entrant_node = GameNode(
    "entrant",
    player=0, # Entrant is player 0
    actions={"Enter": incumbent_node, "Stay Out": stay_out}
)

# Create game
entry_game = ExtensiveFormGame(entrant_node, n_players=2)

# Solve using backward induction
print("=== Entry Deterrence Game ===")
print("\nGame structure:")
print("  Entrant chooses: Enter or Stay Out")
print("  If Enter: Incumbent chooses Fight or Accommodate")
print("\nPayoffs (Entrant, Incumbent):")
print("  Stay Out: (0, 2)")
print("  Enter + Fight: (-1, -1)")
print("  Enter + Accommodate: (1, 1)")

optimal = entry_game.backward_induction()
print("\n" + "=" * 40)
print("Subgame Perfect Equilibrium (Backward Induction):")
for node, action in optimal.items():
    print(f"  At {node}: play {action}")
print(f"\nEquilibrium payoffs: {entry_game.equilibrium_payoffs}")
print("\nInsight: Incumbent's threat to Fight is not credible!")
print("Once entry occurs, Accommodate is better for Incumbent.")

```

=== Entry Deterrence Game ===

Game structure:

Entrant chooses: Enter or Stay Out

If Enter: Incumbent chooses Fight or Accommodate

Payoffs (Entrant, Incumbent):

Stay Out: (0, 2)

Enter + Fight: (-1, -1)

Enter + Accommodate: (1, 1)

=====

Subgame Perfect Equilibrium (Backward Induction):

At incumbent: play Accommodate

At entrant: play Enter

Equilibrium payoffs: (1, 1)

Insight: Incumbent's threat to Fight is not credible!

Once entry occurs, Accommodate is better for Incumbent.

32.2.9. Subgame Perfect Equilibrium

A **subgame** is a portion of an extensive form game that: 1. Starts at a single decision node 2. Contains all subsequent nodes 3. Respects information sets

Subgame Perfect Equilibrium (SPE): A Nash equilibrium that induces a Nash equilibrium in every subgame.

⚠ Credible Threats

SPE eliminates incredible threats. In the entry game, “I will fight if you enter” is not credible because fighting is suboptimal once entry occurs. SPE requires strategies to be optimal even off the equilibrium path.

32.3. Cooperative Game Theory

Cooperative game theory assumes players can form **binding agreements** and focuses on how to fairly divide the gains from cooperation.

32.3.1. Coalitional Games

A **coalitional game** (characteristic function game) is defined by:

$$v : 2^N \rightarrow \mathbb{R}$$

where $v(S)$ is the value coalition S can achieve by cooperating.

Properties: - $v(\emptyset) = 0$ - **Superadditivity**: $v(S \cup T) \geq v(S) + v(T)$ for disjoint S, T - **Convexity**: $v(S \cup T) + v(S \cap T) \geq v(S) + v(T)$

```
from itertools import combinations
from typing import Callable, Set, FrozenSet
import math

class CoalitionalGame:
    """
    Cooperative game with characteristic function.
    """

    def __init__(self, players: List[str], characteristic_function: Dict[FrozenSet,
        ↪ float]):
        """
        Initialize a coalitional game.

        Args:
            players: List of player identifiers
            characteristic_function: Maps coalitions to their values
        """
```

```

self.players = players
self.n = len(players)
self.v = characteristic_function

# Ensure empty coalition has value 0
self.v[frozenset()] = 0

@classmethod
def from_function(cls, players: List[str], value_func: Callable):
    """Create game from a value function."""
    cf = {}
    for r in range(len(players) + 1):
        for coalition in combinations(players, r):
            cf[frozenset(coalition)] = value_func(set(coalition))
    return cls(players, cf)

def value(self, coalition: Set) -> float:
    """Get value of a coalition."""
    return self.v.get(frozenset(coalition), 0)

def grand_coalition_value(self) -> float:
    """Value when all players cooperate."""
    return self.value(set(self.players))

def is_superadditive(self) -> bool:
    """Check if the game is superadditive."""
    for i in range(self.n + 1):
        for s1 in combinations(self.players, i):
            s1 = set(s1)
            for j in range(self.n + 1 - i):
                remaining = set(self.players) - s1
                for s2 in combinations(remaining, j):
                    s2 = set(s2)
                    if self.value(s1 | s2) < self.value(s1) + self.value(s2) -
                        ↪ 1e-10:
                        return False
    return True

def marginal_contribution(self, player: str, coalition: Set) -> float:
    """
    Compute player's marginal contribution to a coalition.

    
$$mc_i(S) = v(S \cup \{i\}) - v(S)$$

    """
    if player in coalition:
        raise ValueError(f"Player {player} already in coalition")

    with_player = coalition | {player}
    return self.value(with_player) - self.value(coalition)

def shapley_value(self) -> Dict[str, float]:
    """
    Compute the Shapley value for all players.

```

```

    _i(v) =  $\sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n-|S|-1)!}{n!} [v(S \cup \{i\}) - v(S)]$ 
    """
    shapley = {p: 0.0 for p in self.players}
    n = self.n

    for player in self.players:
        others = [p for p in self.players if p != player]

        # Sum over all coalitions not containing player
        for r in range(n): # Size of coalition S
            for coalition in combinations(others, r):
                S = set(coalition)

                # Shapley weight
                weight = math.factorial(r) * math.factorial(n - r - 1) /
↪ math.factorial(n)

                # Marginal contribution
                mc = self.marginal_contribution(player, S)

                shapley[player] += weight * mc

    return shapley

def core(self) -> List[Dict[str, float]]:
    """
    Find allocations in the core using linear programming.

    The core is the set of allocations x such that:
    1.  $x_1 + \dots + x_n = v(N)$  (efficiency)
    2.  $\sum_{i \in S} x_i \geq v(S)$  for all S (coalition rationality)
    """
    from scipy.optimize import linprog

    # This is a simplified check - full core enumeration is complex
    # We check if the Shapley value is in the core
    shapley = self.shapley_value()

    # Check coalition rationality
    in_core = True
    for r in range(1, self.n):
        for coalition in combinations(self.players, r):
            S = set(coalition)
            coalition_share = sum(shapley[p] for p in S)
            if coalition_share < self.value(S) - 1e-10:
                in_core = False
                break
        if not in_core:
            break

    if in_core:
        return [shapley]

```

```

        else:
            return [] # Shapley not in core (may be empty core)

# Example: Airport Game
# Three airlines sharing a runway. Costs depend on required runway length.
# Airline A needs 1km, B needs 2km, C needs 3km
# Cost to build x km runway: $x million

def airport_cost(coalition):
    """Cost function for airport runway game."""
    requirements = {'A': 1, 'B': 2, 'C': 3}
    if not coalition:
        return 0
    max_requirement = max(requirements[p] for p in coalition)
    return max_requirement # Cost = longest runway needed

# Convert to savings game (v(S) = standalone costs - joint cost)
def airport_savings(coalition):
    """Savings from cooperation in airport game."""
    if not coalition:
        return 0
    standalone = sum(airport_cost({p}) for p in coalition)
    joint = airport_cost(coalition)
    return standalone - joint

airport_game = CoalitionalGame.from_function(['A', 'B', 'C'], airport_savings)

print("=== Airport Cost Sharing Game ===")
print("\nRunway requirements: A=1km, B=2km, C=3km")
print("Cost to build: $1M per km")
print("\nStandalone costs: A=$1M, B=$2M, C=$3M, Total=$6M")
print("Joint cost (3km runway): $3M")
print("Total savings from cooperation: $3M")

print("\nCoalition values (savings):")
for r in range(1, 4):
    for coalition in combinations(['A', 'B', 'C'], r):
        S = set(coalition)
        print(f" v({S}): ${airport_game.value(S):.0f}M")

print(f"\nSuperadditive: {airport_game.is_superadditive()}")

shapley = airport_game.shapley_value()
print("\nShapley Value (fair cost allocation):")
for player, value in shapley.items():
    # Convert savings share to cost allocation
    standalone_cost = airport_cost({player})
    cost_share = standalone_cost - value
    print(f" {player}: saves ${value:.2f}M, pays ${cost_share:.2f}M")

total_allocated = sum(airport_cost({p}) - shapley[p] for p in ['A', 'B', 'C'])
print(f"\nTotal cost allocated: ${total_allocated:.2f}M (should be $3M)")

```

=== Airport Cost Sharing Game ===

Runway requirements: A=1km, B=2km, C=3km

Cost to build: \$1M per km

Standalone costs: A=\$1M, B=\$2M, C=\$3M, Total=\$6M

Joint cost (3km runway): \$3M

Total savings from cooperation: \$3M

Coalition values (savings):

$v(\{A\})$: \$0M

$v(\{B\})$: \$0M

$v(\{C\})$: \$0M

$v(\{B, A\})$: \$1M

$v(\{C, A\})$: \$1M

$v(\{C, B\})$: \$2M

$v(\{C, B, A\})$: \$3M

Superadditive: True

Shapley Value (fair cost allocation):

A: saves \$0.67M, pays \$0.33M

B: saves \$1.17M, pays \$0.83M

C: saves \$1.17M, pays \$1.83M

Total cost allocated: \$3.00M (should be \$3M)

32.3.2. The Shapley Value

The **Shapley value** is the unique allocation satisfying four axioms:

1. **Efficiency:** $\sum_{i \in N} \phi_i(v) = v(N)$
2. **Symmetry:** Symmetric players get equal shares
3. **Null player:** Player contributing nothing gets nothing
4. **Additivity:** $\phi(v + w) = \phi(v) + \phi(w)$

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

Intuition: Average marginal contribution over all possible orderings of players joining the grand coalition.

```
def shapley_value_sampling(game: CoalitionalGame, n_samples: int = 10000) ->
    Dict[str, float]:
    """
    Estimate Shapley value using Monte Carlo sampling.

    For large games, enumeration is infeasible. Sampling orderings
    provides an unbiased estimate.
    """
```

```

shapley = {p: 0.0 for p in game.players}

for _ in range(n_samples):
    # Random ordering of players
    ordering = np.random.permutation(game.players)

    # Compute marginal contributions
    current_coalition = set()
    for player in ordering:
        mc = game.marginal_contribution(player, current_coalition)
        shapley[player] += mc
        current_coalition.add(player)

# Average
for player in game.players:
    shapley[player] /= n_samples

return shapley

# Voting game example: Weighted voting
# Players have different voting weights, coalition wins if weight >= threshold
def voting_game_value(coalition, weights, threshold):
    """Value function for weighted voting game."""
    if not coalition:
        return 0
    total_weight = sum(weights[p] for p in coalition)
    return 1 if total_weight >= threshold else 0

# Example: [51; 50, 49, 1] - weights are 50, 49, 1; threshold is 51
weights = {'Big': 50, 'Medium': 49, 'Small': 1}
threshold = 51

voting_value = lambda S: voting_game_value(S, weights, threshold)
voting_game = CoalitionalGame.from_function(['Big', 'Medium', 'Small'],
    ↪ voting_value)

print("=== Weighted Voting Game ===")
print(f"Weights: {weights}")
print(f"Threshold: {threshold}")

print("\nWinning coalitions:")
for r in range(1, 4):
    for coalition in combinations(['Big', 'Medium', 'Small'], r):
        S = set(coalition)
        if voting_game.value(S) == 1:
            print(f" {S}")

shapley = voting_game.shapley_value()
print("\nShapley Value (voting power):")
for player, power in shapley.items():
    print(f" {player} (weight {weights[player]}): power = {power:.4f}")

```



```
print("\nNote: Small player has power despite tiny weight!")
print("Small player is pivotal in {Big, Small} coalition.")
```

```
=== Weighted Voting Game ===
```

```
Weights: {'Big': 50, 'Medium': 49, 'Small': 1}
```

```
Threshold: 51
```

```
Winning coalitions:
```

```
  {'Big', 'Medium'}
```

```
  {'Big', 'Small'}
```

```
  {'Small', 'Big', 'Medium'}
```

```
Shapley Value (voting power):
```

```
  Big (weight 50): power = 0.6667
```

```
  Medium (weight 49): power = 0.1667
```

```
  Small (weight 1): power = 0.1667
```

```
Note: Small player has power despite tiny weight!
```

```
Small player is pivotal in {Big, Small} coalition.
```

32.3.3. The Core

The **core** is the set of stable allocations where no coalition has incentive to deviate:

$$\text{Core}(v) = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n x_i = v(N), \sum_{i \in S} x_i \geq v(S) \ \forall S \subseteq N \right\}$$

i Core vs Shapley Value

- The **core** may be empty (no stable allocation exists)
- When non-empty, core may contain multiple allocations
- **Shapley value** always exists and is unique
- Shapley value is in the core for convex games

32.3.4. Nucleolus

The **nucleolus** minimizes the maximum “unhappiness” of any coalition:

$$\text{nucleolus}(v) = \arg \min_x \max_{S \subseteq N} \left[\sum_{i \in S} x_i - v(S) \right]$$

It lexicographically minimizes the vector of excesses (sorted in decreasing order).

```
def compute_nucleolus_2player(game: CoalitionalGame) -> Dict[str, float]:
    """
    Compute nucleolus for a 2-player game (simple case).

    For 2 players, nucleolus divides surplus equally.
```

```

"""
if game.n != 2:
    raise ValueError("This implementation is for 2-player games only")

p1, p2 = game.players

# Individual rationality constraints
v1 = game.value({p1})
v2 = game.value({p2})
v12 = game.value({p1, p2})

# Surplus to divide
surplus = v12 - v1 - v2

# Nucleolus divides surplus equally
x1 = v1 + surplus / 2
x2 = v2 + surplus / 2

return {p1: x1, p2: x2}

# Simple 2-player example
simple_game = CoalitionalGame(
    ['Alice', 'Bob'],
    {
        frozenset(): 0,
        frozenset(['Alice']): 10,
        frozenset(['Bob']): 20,
        frozenset(['Alice', 'Bob']): 50
    }
)

print("=== 2-Player Bargaining Game ===")
print(f"v(Alice) = 10, v(Bob) = 20, v(Alice, Bob) = 50")
print(f"Surplus from cooperation: 50 - 10 - 20 = 20")

nucleolus = compute_nucleolus_2player(simple_game)
shapley = simple_game.shapley_value()

print("\nSolution concepts:")
print(f"  Nucleolus: Alice = {nucleolus['Alice']}, Bob = {nucleolus['Bob']}")
print(f"  Shapley:   Alice = {shapley['Alice']:.1f}, Bob = {shapley['Bob']:.1f}")
print("\nBoth concepts divide surplus equally in this case.")

```

```

=== 2-Player Bargaining Game ===
v(Alice) = 10, v(Bob) = 20, v(Alice, Bob) = 50
Surplus from cooperation: 50 - 10 - 20 = 20

```

Solution concepts:

```

Nucleolus: Alice = 20.0, Bob = 30.0
Shapley:   Alice = 20.0, Bob = 30.0

```

Both concepts divide surplus equally in this case.

32.4. Mechanism Design

Mechanism design is “reverse game theory”: instead of analyzing a given game, we design games to achieve target outcomes.

32.4.1. The Revelation Principle

Revelation Principle: Any outcome achievable by any mechanism can also be achieved by a **direct mechanism** where: 1. Players report their types truthfully 2. Truthful reporting is an equilibrium

This simplifies mechanism design: we only need to consider direct, incentive-compatible mechanisms.

32.4.2. Incentive Compatibility

A mechanism is **incentive compatible** if truthful reporting is optimal:

$$u_i(\theta_i, m(\theta_i, \theta_{-i})) \geq u_i(\theta_i, m(\theta'_i, \theta_{-i})) \quad \forall \theta_i, \theta'_i$$

Types of incentive compatibility: - **Dominant Strategy IC (DSIC)**: Truth-telling optimal regardless of others’ strategies - **Bayesian IC (BIC)**: Truth-telling optimal in expectation

32.4.3. Auction Theory

Auction design is one of the most successful applications of mechanism design. Different auction formats — first-price, second-price (Vickrey), English, and Dutch — lead to different strategic behavior. The Revenue Equivalence Theorem states that under standard assumptions, all four formats yield the same expected revenue. The code below simulates these formats.

```
from typing import List, Tuple
import numpy as np

class Auction:
    """Base class for auction mechanisms."""

    def __init__(self, n_bidders: int):
        self.n_bidders = n_bidders

    def run(self, valuations: np.ndarray) -> Tuple[int, float, np.ndarray]:
        """
        Run the auction given true valuations.

        Returns:
            winner: Index of winning bidder
            price: Payment by winner
            utilities: Utility for each bidder
        """
        raise NotImplementedError

class FirstPriceAuction(Auction):
    """
    First-price sealed-bid auction.
    """
```

```

Winner pays their bid. Strategic bidding: bid below value.
"""

def optimal_bid(self, valuation: float, n_bidders: int) -> float:
    """
    Compute optimal bid assuming uniform valuations in [0,1].

    Optimal strategy:  $b(v) = (n-1)/n * v$ 
    """
    return (n_bidders - 1) / n_bidders * valuation

def run(self, valuations: np.ndarray) -> Tuple[int, float, np.ndarray]:
    """Run first-price auction with strategic bidding."""
    bids = np.array([self.optimal_bid(v, self.n_bidders) for v in valuations])

    winner = np.argmax(bids)
    price = bids[winner]

    utilities = np.zeros(self.n_bidders)
    utilities[winner] = valuations[winner] - price

    return winner, price, utilities

class SecondPriceAuction(Auction):
    """
    Second-price sealed-bid (Vickrey) auction.

    Winner pays second-highest bid. Dominant strategy: bid true value.
    """

    def run(self, valuations: np.ndarray) -> Tuple[int, float, np.ndarray]:
        """Run second-price auction (bidding true values is optimal)."""
        bids = valuations.copy() # Truthful bidding is dominant

        sorted_idx = np.argsort(-bids)
        winner = sorted_idx[0]
        price = bids[sorted_idx[1]] # Second-highest bid

        utilities = np.zeros(self.n_bidders)
        utilities[winner] = valuations[winner] - price

        return winner, price, utilities

def simulate_auctions(n_simulations: int = 10000, n_bidders: int = 5):
    """Compare first-price and second-price auctions."""
    np.random.seed(42)

    fpa = FirstPriceAuction(n_bidders)
    spa = SecondPriceAuction(n_bidders)

    fpa_revenues = []

```

```

spa_revenues = []
fpa_efficiencies = []
spa_efficiencies = []

for _ in range(n_simulations):
    valuations = np.random.uniform(0, 1, n_bidders)

    # First-price auction
    winner_fpa, price_fpa, _ = fpa.run(valuations)
    fpa_revenues.append(price_fpa)
    fpa_efficiencies.append(winner_fpa == np.argmax(valuations))

    # Second-price auction
    winner_spa, price_spa, _ = spa.run(valuations)
    spa_revenues.append(price_spa)
    spa_efficiencies.append(winner_spa == np.argmax(valuations))

print("=== Auction Comparison ===")
print(f"Number of bidders: {n_bidders}")
print(f"Simulations: {n_simulations}")
print()
print("           First-Price    Second-Price")
print(f"Mean Revenue:      {np.mean(fpa_revenues):.4f}"
      ↪ {np.mean(spa_revenues):.4f}")
print(f"Revenue Std:      {np.std(fpa_revenues):.4f}"
      ↪ {np.std(spa_revenues):.4f}")
print(f"Efficiency:       {np.mean(fpa_efficiencies):.2%}"
      ↪ {np.mean(spa_efficiencies):.2%}")
print()
print("Note: Revenue Equivalence Theorem predicts equal expected revenue!")
print("Second-price auction is efficient (always allocates to highest valuer).")

simulate_auctions()

```

```
=== Auction Comparison ===
```

```
Number of bidders: 5
```

```
Simulations: 10000
```

	First-Price	Second-Price
Mean Revenue:	0.6652	0.6660
Revenue Std:	0.1136	0.1779
Efficiency:	100.00%	100.00%

```
Note: Revenue Equivalence Theorem predicts equal expected revenue!
```

```
Second-price auction is efficient (always allocates to highest valuer).
```

32.4.4. Revenue Equivalence Theorem

Under standard assumptions (risk-neutral bidders, independent private values, symmetric bidders):

All standard auction formats yield the same expected revenue.

This includes first-price, second-price, English (ascending), and Dutch (descending) auctions.

32.4.5. VCG Mechanism

The **Vickrey-Clarke-Groves (VCG) mechanism** generalizes second-price auctions to multiple items/decisions.

Each agent pays their **externality**: the harm they cause to others.

$$p_i = \sum_{j \neq i} v_j(a_{-i}^*) - \sum_{j \neq i} v_j(a^*)$$

where: - a^* is the efficient allocation with agent i - a_{-i}^* is the efficient allocation without agent i

```
class VCGMechanism:
    """
    VCG mechanism for combinatorial auctions.

    Implements truthful allocation and Vickrey-Clarke-Groves payments.
    """

    def __init__(self, items: List[str], bidders: List[str]):
        self.items = items
        self.bidders = bidders

    def _find_efficient_allocation(self, valuations: Dict) -> Tuple[Dict, float]:
        """
        Find the welfare-maximizing allocation.

        For simplicity, uses brute force. Real implementation would use
        integer programming for complex combinatorial auctions.
        """
        from itertools import product

        best_allocation = None
        best_welfare = -float('inf')

        n_items = len(self.items)
        n_bidders = len(self.bidders)

        # Try all possible allocations (each item to one bidder or unallocated)
        for assignment in product(range(n_bidders + 1), repeat=n_items):
            allocation = {b: set() for b in self.bidders}

            for item_idx, bidder_idx in enumerate(assignment):
                if bidder_idx < n_bidders:
                    allocation[self.bidders[bidder_idx]].add(self.items[item_idx])

            # Compute welfare
            welfare = sum(
                valuations[bidder].get(frozenset(bundle), 0)
                for bidder, bundle in allocation.items()
            )

            if welfare > best_welfare:
                best_welfare = welfare
                best_allocation = allocation
```

```

    return best_allocation, best_welfare

def run(self, valuations: Dict) -> Tuple[Dict, Dict, float]:
    """
    Run VCG mechanism.

    Args:
        valuations: Dict mapping bidder -> {bundle: value}

    Returns:
        allocation: Who gets what
        payments: What each bidder pays
        welfare: Total social welfare
    """
    # Find efficient allocation
    allocation, welfare = self._find_efficient_allocation(valuations)

    # Compute VCG payments
    payments = {}

    for bidder in self.bidders:
        # Welfare without this bidder
        reduced_valuations = {b: v for b, v in valuations.items() if b !=
        ↪ bidder}
        reduced_valuations[bidder] = {frozenset(): 0} # Bidder gets nothing
        _, welfare_without = self._find_efficient_allocation(reduced_valuations)

        # Others' welfare with this bidder
        others_welfare_with = welfare - valuations[bidder].get(
            frozenset(allocation[bidder]), 0)

        # Others' welfare without this bidder (under efficient allocation)
        # This is just welfare_without
        others_welfare_without = welfare_without

        # VCG payment = externality
        payments[bidder] = others_welfare_without - others_welfare_with

    return allocation, payments, welfare

# Example: Two items, two bidders
items = ['A', 'B']
bidders = ['Alice', 'Bob']

# Valuations (bundled)
valuations = {
    'Alice': {
        frozenset(): 0,
        frozenset(['A']): 5,
        frozenset(['B']): 3,
        frozenset(['A', 'B']): 7 # Slight complementarity
    }

```

```

    },
    'Bob': {
        frozenset(): 0,
        frozenset(['A']): 4,
        frozenset(['B']): 6,
        frozenset(['A', 'B']): 12 # Strong complementarity
    }
}

vcg = VCGMechanism(items, bidders)
allocation, payments, welfare = vcg.run(valuations)

print("=== VCG Mechanism Example ===")
print("\nValuations:")
for bidder in bidders:
    print(f"  {bidder}:")
    for bundle, value in valuations[bidder].items():
        if bundle:
            print(f"    {set(bundle)}: ${value}")

print(f"\nEfficient Allocation (max welfare = ${welfare}):")
for bidder, bundle in allocation.items():
    value = valuations[bidder].get(frozenset(bundle), 0)
    print(f"  {bidder} gets {bundle} if bundle else 'nothing' (value: ${value})")

print("\nVCG Payments (externality-based):")
for bidder, payment in payments.items():
    print(f"  {bidder} pays: ${payment:.2f}")

print("\nProperties of VCG:")
print("  - Truthful (dominant strategy incentive compatible)")
print("  - Efficient (maximizes social welfare)")
print("  - Individual rational (utility >= 0)")

```

=== VCG Mechanism Example ===

Valuations:

Alice:
 {'A'}: \$5
 {'B'}: \$3
 {'B', 'A'}: \$7
 Bob:
 {'A'}: \$4
 {'B'}: \$6
 {'B', 'A'}: \$12

Efficient Allocation (max welfare = \$12):

Alice gets nothing (value: \$0)
 Bob gets {'B', 'A'} (value: \$12)

VCG Payments (externality-based):

Alice pays: \$0.00

Bob pays: \$7.00

Properties of VCG:

- Truthful (dominant strategy incentive compatible)
- Efficient (maximizes social welfare)
- Individual rational (utility ≥ 0)

32.5. Applications in Operations

Game theory applies directly to operations problems where multiple self-interested agents interact. Supply chains, pricing markets, and transportation networks all involve strategic decisions that affect other participants. This section translates the theoretical tools from earlier sections into working models for these settings.

32.5.1. Supply Chain Games

In a decentralized supply chain, a supplier and a buyer each optimize their own profit. The resulting double marginalization effect leads to lower quantities and higher prices than the jointly optimal solution. The following code quantifies this inefficiency.

```
class SupplierBuyerGame:
    """
    Model supplier-buyer negotiation as a bargaining game.
    """

    def __init__(self, supplier_cost: float, buyer_valuation: float):
        """
        Args:
            supplier_cost: Cost to supplier of producing unit
            buyer_valuation: Value to buyer of receiving unit
        """
        self.c = supplier_cost
        self.v = buyer_valuation
        self.surplus = buyer_valuation - supplier_cost

        if self.surplus < 0:
            raise ValueError("No gains from trade (cost > value)")

    def nash_bargaining_solution(self, supplier_power: float = 0.5) -> Tuple[float,
↵ float]:
        """
        Nash bargaining solution.

        Maximizes  $(_S - d_S)^{d_S} * (_B - d_B)^{(1-d_S)}$ 
        where  $d_i$  are disagreement payoffs (0 if no trade)

        Args:
            supplier_power: Bargaining power      [0,1]

        Returns:
            price: Negotiated price
            allocations: (supplier_profit, buyer_profit)
        """
```

```

    """
    # Nash bargaining: split surplus according to bargaining power
    supplier_share = supplier_power * self.surplus
    buyer_share = (1 - supplier_power) * self.surplus

    price = self.c + supplier_share

    return price, (supplier_share, buyer_share)

def stackelberg_price(self, leader: str = 'supplier') -> Tuple[float, float]:
    """
    Stackelberg game where one party moves first.

    If supplier leads: sets price knowing buyer will accept if value > price
    If buyer leads: makes take-it-or-leave-it offer
    """
    if leader == 'supplier':
        # Supplier extracts all surplus (minus epsilon)
        price = self.v - 0.01
        supplier_profit = price - self.c
        buyer_profit = self.v - price
    else:
        # Buyer extracts all surplus
        price = self.c + 0.01
        supplier_profit = price - self.c
        buyer_profit = self.v - price

    return price, (supplier_profit, buyer_profit)

# Example: Component sourcing negotiation
supplier_cost = 50 # Cost to manufacture component
buyer_value = 100 # Value of component to buyer

game = SupplierBuyerGame(supplier_cost, buyer_value)

print("=== Supplier-Buyer Negotiation ===")
print(f"\nSupplier cost: ${supplier_cost}")
print(f"Buyer valuation: ${buyer_value}")
print(f"Surplus to divide: ${game.surplus}")

print("\n--- Solution Concepts ---")

# Nash bargaining with different power levels
for power in [0.3, 0.5, 0.7]:
    price, (s_profit, b_profit) = game.nash_bargaining_solution(power)
    print(f"\nNash Bargaining (supplier power = {power}):")
    print(f"  Price: ${price:.2f}")
    print(f"  Supplier profit: ${s_profit:.2f}")
    print(f"  Buyer profit: ${b_profit:.2f}")

# Stackelberg outcomes
for leader in ['supplier', 'buyer']:

```

```

price, (s_profit, b_profit) = game.stackelberg_price(leader)
print(f"\nStackelberg ({leader} leads):")
print(f"  Price: ${price:.2f}")
print(f"  Supplier profit: ${s_profit:.2f}")
print(f"  Buyer profit: ${b_profit:.2f}")

```

=== Supplier-Buyer Negotiation ===

Supplier cost: \$50
 Buyer valuation: \$100
 Surplus to divide: \$50

--- Solution Concepts ---

Nash Bargaining (supplier power = 0.3):
 Price: \$65.00
 Supplier profit: \$15.00
 Buyer profit: \$35.00

Nash Bargaining (supplier power = 0.5):
 Price: \$75.00
 Supplier profit: \$25.00
 Buyer profit: \$25.00

Nash Bargaining (supplier power = 0.7):
 Price: \$85.00
 Supplier profit: \$35.00
 Buyer profit: \$15.00

Stackelberg (supplier leads):
 Price: \$99.99
 Supplier profit: \$49.99
 Buyer profit: \$0.01

Stackelberg (buyer leads):
 Price: \$50.01
 Supplier profit: \$0.01
 Buyer profit: \$49.99

32.5.2. Revenue Sharing Contracts

Revenue sharing contracts align incentives in a supply chain by having the buyer share a fraction of revenue with the supplier in exchange for a lower wholesale price. This can recover the jointly optimal quantity and split the resulting surplus between the two parties. The code below computes the optimal revenue share and compares profits to the traditional setting.

```

class RevenueSharingContract:
    """
    Revenue sharing contract between supplier and retailer.

    Supplier sells at reduced wholesale price  $w < c$ 
    Retailer shares fraction  $(1-)$  of revenue with supplier
    """

    def __init__(self, retail_price: float, supplier_cost: float,
                 demand_mean: float, demand_std: float):
        self.p = retail_price
        self.c = supplier_cost
        self.mu = demand_mean
        self.sigma = demand_std

    def newsvendor_quantity(self, wholesale_price: float) -> float:
        """
        Compute optimal order quantity for retailer.

         $Q^* = \mu + \sigma \Phi^{-1}((p-w)/p)$ 
        """
        from scipy import stats

        critical_ratio = (self.p - wholesale_price) / self.p
        critical_ratio = np.clip(critical_ratio, 0.001, 0.999)

        Q_star = self.mu + self.sigma * stats.norm.ppf(critical_ratio)
        return max(0, Q_star)

    def expected_profit(self, wholesale_price: float,
                       revenue_share: float) -> Tuple[float, float]:
        """
        Compute expected profits under revenue sharing.

        Args:
            wholesale_price: w
            revenue_share: (retailer keeps  of revenue)

        Returns:
            (supplier_profit, retailer_profit)
        """
        from scipy import stats

        Q = self.newsvendor_quantity(wholesale_price)

        # Expected sales
        expected_sales = self.mu - self.sigma * stats.norm.pdf(
            (Q - self.mu) / self.sigma
        ) + (Q - self.mu) * stats.norm.cdf((Q - self.mu) / self.sigma)
        expected_sales = min(expected_sales, Q)

        # Revenue
        expected_revenue = self.p * expected_sales

```

```

    # Profits
    retailer_cost = wholesale_price * Q
    retailer_revenue = revenue_share * expected_revenue
    retailer_profit = retailer_revenue - retailer_cost

    supplier_cost = self.c * Q
    supplier_revenue = wholesale_price * Q + (1 - revenue_share) *
↪ expected_revenue
    supplier_profit = supplier_revenue - supplier_cost

    return supplier_profit, retailer_profit

def optimal_contract(self) -> Tuple[float, float, Tuple[float, float]]:
    """
    Find revenue sharing contract that coordinates the supply chain.

    Coordinating contract:  $Q_{\text{retailer}} = Q_{\text{centralized}}$ 
    """
    from scipy.optimize import minimize_scalar
    from scipy import stats

    # Centralized optimal quantity
    critical_ratio = (self.p - self.c) / self.p
    Q_central = self.mu + self.sigma * stats.norm.ppf(critical_ratio)

    # For coordination: retailer's critical ratio must equal centralized
    #  $(p - w) / (p) = (p - c) / p$ 
    # Solving:  $w = c$ 

    # Find that splits surplus appropriately (e.g., equal bargaining)
    def supplier_profit_ratio(phi):
        w = phi * self.c
        s_profit, r_profit = self.expected_profit(w, phi)
        total = s_profit + r_profit
        return -(s_profit / total - 0.5)**2 # Target 50-50 split

    result = minimize_scalar(supplier_profit_ratio, bounds=(0.1, 0.9),
↪ method='bounded')
    optimal_phi = result.x
    optimal_w = optimal_phi * self.c

    profits = self.expected_profit(optimal_w, optimal_phi)

    return optimal_w, optimal_phi, profits

# Example: Fashion retailer supply chain
contract = RevenueSharingContract(
    retail_price=100,
    supplier_cost=40,
    demand_mean=1000,
    demand_std=200

```

```

)

print("=== Revenue Sharing Contract ===")
print(f"\nRetail price: ${contract.p}")
print(f"Supplier cost: ${contract.c}")
print(f"Demand: Normal(={contract.mu}, ={contract.sigma})")

# Wholesale-only contract
w_traditional = 60
Q_traditional = contract.newsvendor_quantity(w_traditional)
s_trad, r_trad = contract.expected_profit(w_traditional, 1.0)

print(f"\n--- Traditional Wholesale Contract ---")
print(f"Wholesale price: ${w_traditional}")
print(f"Retailer order quantity: {Q_traditional:.0f}")
print(f"Supplier profit: ${s_trad:.0f}")
print(f"Retailer profit: ${r_trad:.0f}")
print(f"Total profit: ${s_trad + r_trad:.0f}")

# Revenue sharing contract
w_rs, phi, (s_rs, r_rs) = contract.optimal_contract()
Q_rs = contract.newsvendor_quantity(w_rs)

print(f"\n--- Revenue Sharing Contract ---")
print(f"Wholesale price: ${w_rs:.2f}")
print(f"Retailer revenue share: {phi:.1%}")
print(f"Retailer order quantity: {Q_rs:.0f}")
print(f"Supplier profit: ${s_rs:.0f}")
print(f"Retailer profit: ${r_rs:.0f}")
print(f"Total profit: ${s_rs + r_rs:.0f}")

print(f"\n--- Improvement from Revenue Sharing ---")
print(f"Quantity increase: {(Q_rs - Q_traditional) / Q_traditional:.1%}")
print(f"Total profit increase: {(s_rs + r_rs - s_trad - r_trad) / (s_trad +
↪ r_trad):.1%}")

```

=== Revenue Sharing Contract ===

Retail price: \$100

Supplier cost: \$40

Demand: Normal(=1000, =200)

--- Traditional Wholesale Contract ---

Wholesale price: \$60

Retailer order quantity: 949

Supplier profit: \$18987

Retailer profit: \$33287

Total profit: \$52273

--- Revenue Sharing Contract ---

Wholesale price: \$4.00

Retailer revenue share: 10.0%

```

Retailer order quantity: 1350
Supplier profit: $70095
Retailer profit: $7789
Total profit: $77884

```

```

--- Improvement from Revenue Sharing ---
Quantity increase: 42.2%
Total profit increase: 49.0%

```

32.5.3. Pricing Competition

When firms compete on price (Bertrand competition) or quantity (Cournot competition), the equilibrium outcome depends on the market structure. The code below implements both models and compares the resulting prices, quantities, and profits against the monopoly and perfect competition benchmarks.

```

class CournotCompetition:
    """
    Cournot quantity competition between firms.

    Firms simultaneously choose quantities, price determined by market.
    """

    def __init__(self, n_firms: int, demand_intercept: float,
                 demand_slope: float, marginal_costs: np.ndarray):
        """
        Args:
            n_firms: Number of competing firms
            demand_intercept: a in  $P = a - b \cdot Q$ 
            demand_slope: b in  $P = a - b \cdot Q$ 
            marginal_costs: Cost  $c_i$  for each firm
        """
        self.n = n_firms
        self.a = demand_intercept
        self.b = demand_slope
        self.c = marginal_costs

    def price(self, quantities: np.ndarray) -> float:
        """Market clearing price."""
        return max(0, self.a - self.b * np.sum(quantities))

    def profit(self, quantities: np.ndarray) -> np.ndarray:
        """Profit for each firm."""
        p = self.price(quantities)
        return (p - self.c) * quantities

    def best_response(self, i: int, other_quantities: np.ndarray) -> float:
        """
        Firm i's best response to other firms' quantities.

         $q_i^* = (a - c_i - b \cdot Q_{-i}) / (2b)$ 
        """
        Q_others = np.sum(other_quantities)

```

```

        q_star = (self.a - self.c[i] - self.b * Q_others) / (2 * self.b)
        return max(0, q_star)

def cournot_equilibrium(self, max_iter: int = 100, tol: float = 1e-6) ->
    np.ndarray:
    """
    Find Cournot-Nash equilibrium via best response iteration.
    """
    quantities = np.ones(self.n) * (self.a / self.b) / (2 * self.n)

    for _ in range(max_iter):
        new_quantities = quantities.copy()
        for i in range(self.n):
            others = np.delete(quantities, i)
            new_quantities[i] = self.best_response(i, others)

        if np.max(np.abs(new_quantities - quantities)) < tol:
            break
        quantities = new_quantities

    return quantities

def monopoly_quantity(self) -> float:
    """Quantity that would be produced by a monopolist."""
    c_avg = np.mean(self.c)
    return (self.a - c_avg) / (2 * self.b)

def competitive_quantity(self) -> float:
    """Quantity under perfect competition (P = MC)."""
    c_avg = np.mean(self.c)
    return (self.a - c_avg) / self.b

class BertrandCompetition:
    """
    Bertrand price competition with differentiated products.
    """

    def __init__(self, n_firms: int, demand_intercept: float,
                 own_price_effect: float, cross_price_effect: float,
                 marginal_costs: np.ndarray):
        """
        Demand:  $q_i = a - b \cdot p_i + c \cdot p_j$  (for 2 firms)
        """
        self.n = n_firms
        self.a = demand_intercept
        self.b = own_price_effect
        self.c_cross = cross_price_effect
        self.mc = marginal_costs

    def demand(self, prices: np.ndarray) -> np.ndarray:
        """Quantity demanded for each firm."""
        quantities = np.zeros(self.n)

```



```

for i in range(self.n):
    q = self.a - self.b * prices[i]
    for j in range(self.n):
        if i != j:
            q += self.c_cross * prices[j]
    quantities[i] = max(0, q)
return quantities

def profit(self, prices: np.ndarray) -> np.ndarray:
    """Profit for each firm."""
    quantities = self.demand(prices)
    return (prices - self.mc) * quantities

def bertrand_equilibrium(self) -> np.ndarray:
    """
    Find Bertrand-Nash equilibrium.

    For symmetric duopoly with differentiated products:
     $p^* = (a*b + c\_cross*mc*n + mc*b) / (2b - c\_cross*(n-1))$ 
    """
    from scipy.optimize import minimize

    def neg_total_profit(prices):
        return -np.sum(self.profit(prices))

    # Start from monopoly prices
    x0 = (self.a + self.b * self.mc) / (2 * self.b)
    x0 = np.ones(self.n) * x0

    # Each firm maximizes own profit (use iterative best response)
    prices = x0.copy()

    for _ in range(100):
        new_prices = prices.copy()
        for i in range(self.n):
            # Optimize firm i's price given others
            def neg_profit_i(p_i):
                test_prices = prices.copy()
                test_prices[i] = p_i
                return -self.profit(test_prices)[i]

            from scipy.optimize import minimize_scalar
            result = minimize_scalar(neg_profit_i, bounds=(self.mc[i], self.a),
                                    method='bounded')
            new_prices[i] = result.x

        if np.max(np.abs(new_prices - prices)) < 1e-6:
            break
        prices = new_prices

    return prices

```

32. Game Theory for Strategic Decisions

```
# Cournot example: Duopoly
cournot = CournotCompetition(
    n_firms=2,
    demand_intercept=100,
    demand_slope=1,
    marginal_costs=np.array([20, 25])
)

q_eq = cournot.cournot_equilibrium()
p_eq = cournot.price(q_eq)
profits = cournot.profit(q_eq)

print("=== Cournot Quantity Competition ===")
print(f"\nDemand: P = {cournot.a} - {cournot.b}*Q")
print(f"Marginal costs: Firm 1 = ${cournot.c[0]}, Firm 2 = ${cournot.c[1]}")

print(f"\nCournot-Nash Equilibrium:")
print(f"  Firm 1 quantity: {q_eq[0]:.2f}")
print(f"  Firm 2 quantity: {q_eq[1]:.2f}")
print(f"  Market price: ${p_eq:.2f}")
print(f"  Firm 1 profit: ${profits[0]:.2f}")
print(f"  Firm 2 profit: ${profits[1]:.2f}")

print(f"\nComparison:")
print(f"  Monopoly quantity: {cournot.monopoly_quantity():.2f}")
print(f"  Competitive quantity: {cournot.competitive_quantity():.2f}")
print(f"  Cournot total: {np.sum(q_eq):.2f} (between monopoly and competitive)")

# Bertrand example
bertrand = BertrandCompetition(
    n_firms=2,
    demand_intercept=100,
    own_price_effect=2,
    cross_price_effect=0.5, # Substitute products
    marginal_costs=np.array([20, 20])
)

p_bertrand = bertrand.bertrand_equilibrium()
q_bertrand = bertrand.demand(p_bertrand)
profits_bertrand = bertrand.profit(p_bertrand)

print("\n=== Bertrand Price Competition (Differentiated Products) ===")
print(f"\nBertrand-Nash Equilibrium:")
print(f"  Firm 1 price: ${p_bertrand[0]:.2f}")
print(f"  Firm 2 price: ${p_bertrand[1]:.2f}")
print(f"  Firm 1 quantity: {q_bertrand[0]:.2f}")
print(f"  Firm 2 quantity: {q_bertrand[1]:.2f}")
print(f"  Firm 1 profit: ${profits_bertrand[0]:.2f}")
print(f"  Firm 2 profit: ${profits_bertrand[1]:.2f}")
```

=== Cournot Quantity Competition ===

Demand: $P = 100 - 1 \cdot Q$

Marginal costs: Firm 1 = \$20, Firm 2 = \$25

Cournot-Nash Equilibrium:

Firm 1 quantity: 28.33
 Firm 2 quantity: 23.33
 Market price: \$48.33
 Firm 1 profit: \$802.78
 Firm 2 profit: \$544.44

Comparison:

Monopoly quantity: 38.75
 Competitive quantity: 77.50
 Cournot total: 51.67 (between monopoly and competitive)

=== Bertrand Price Competition (Differentiated Products) ===

Bertrand-Nash Equilibrium:

Firm 1 price: \$40.00
 Firm 2 price: \$40.00
 Firm 1 quantity: 40.00
 Firm 2 quantity: 40.00
 Firm 1 profit: \$800.00
 Firm 2 profit: \$800.00

32.5.4. Network Congestion Games

In a congestion game, each player chooses a path through a network, and the cost of each edge depends on the total traffic. Selfish routing leads to a Nash equilibrium that is generally worse than the social optimum — a phenomenon quantified by the Price of Anarchy. The following code simulates selfish routing on a network and computes the efficiency loss.

```
class CongestionGame:
    """
    Congestion game where players choose routes through a network.

    Each edge has a latency function depending on traffic.
    """

    def __init__(self, n_players: int, routes: List[List[str]],
                  edge_latency: Dict[str, Callable]):
        """
        Args:
            n_players: Number of players (drivers)
            routes: List of available routes, each route is list of edges
            edge_latency: Maps edge name to latency function f(flow)
        """
        self.n = n_players
        self.routes = routes
        self.n_routes = len(routes)
        self.latency = edge_latency
```

```

    # Find all edges
    self.edges = set()
    for route in routes:
        self.edges.update(route)

def compute_edge_flows(self, route_choices: np.ndarray) -> Dict[str, float]:
    """Compute flow on each edge given route choices."""
    flows = {e: 0 for e in self.edges}
    for i, route_idx in enumerate(route_choices):
        for edge in self.routes[int(route_idx)]:
            flows[edge] += 1
    return flows

def route_latency(self, route_idx: int, flows: Dict[str, float]) -> float:
    """Total latency for a route given edge flows."""
    route = self.routes[route_idx]
    return sum(self.latency[e](flows[e]) for e in route)

def player_latency(self, player: int, route_choices: np.ndarray) -> float:
    """Latency experienced by a player."""
    flows = self.compute_edge_flows(route_choices)
    return self.route_latency(int(route_choices[player]), flows)

def is_nash_equilibrium(self, route_choices: np.ndarray) -> bool:
    """Check if current routing is a Nash equilibrium."""
    for i in range(self.n):
        current_latency = self.player_latency(i, route_choices)

        # Check if switching routes helps
        for r in range(self.n_routes):
            if r == route_choices[i]:
                continue

            # Try switching
            test_choices = route_choices.copy()
            test_choices[i] = r
            new_latency = self.player_latency(i, test_choices)

            if new_latency < current_latency - 1e-6:
                return False

    return True

def find_equilibrium(self, max_iter: int = 1000) -> np.ndarray:
    """Find Nash equilibrium via best response dynamics."""
    # Random initial assignment
    choices = np.random.randint(0, self.n_routes, self.n)

    for _ in range(max_iter):
        # Random player order
        for i in np.random.permutation(self.n):
            best_route = choices[i]

```

```

        best_latency = self.player_latency(i, choices)

        for r in range(self.n_routes):
            test_choices = choices.copy()
            test_choices[i] = r
            test_latency = self.player_latency(i, test_choices)

            if test_latency < best_latency:
                best_latency = test_latency
                best_route = r

        choices[i] = best_route

    if self.is_nash_equilibrium(choices):
        break

    return choices

def social_optimum(self) -> np.ndarray:
    """Find routing that minimizes total latency."""
    from itertools import product

    best_choices = None
    best_total = float('inf')

    for choices in product(range(self.n_routes), repeat=self.n):
        choices = np.array(choices)
        flows = self.compute_edge_flows(choices)

        total = sum(self.player_latency(i, choices) for i in range(self.n))

        if total < best_total:
            best_total = total
            best_choices = choices

    return best_choices

# Braess's Paradox example
# Network with 4 nodes: S -> A -> T, S -> B -> T, and new edge A -> B
def braess_example():
    """Demonstrate Braess's paradox."""
    print("=== Braess's Paradox ===")
    print("\nOriginal network:")
    print("  S ---(x/100)---> A ---(45min)---> T")
    print("  S ---(45min)---> B ---(x/100)---> T")
    print("  (x = number of cars on edge)")

    # Without shortcut A-B
    routes_original = [
        ['SA', 'AT'], # Route 1: S -> A -> T
        ['SB', 'BT'] # Route 2: S -> B -> T
    ]

```

```

edge_latency_original = {
    'SA': lambda x: x / 100, # Congestion-sensitive
    'AT': lambda x: 45,      # Fixed time
    'SB': lambda x: 45,      # Fixed time
    'BT': lambda x: x / 100  # Congestion-sensitive
}

n_players = 100
game_original = CongestionGame(n_players, routes_original,
↪ edge_latency_original)

eq_original = game_original.find_equilibrium()
flows_original = game_original.compute_edge_flows(eq_original)

total_latency_original = sum(
    game_original.player_latency(i, eq_original) for i in range(n_players)
)

print(f"\n{n_players} drivers, equilibrium:")
print(f"  Route 1 (S-A-T): {np.sum(eq_original == 0)} drivers")
print(f"  Route 2 (S-B-T): {np.sum(eq_original == 1)} drivers")
print(f"  Total travel time: {total_latency_original:.0f} minutes")
print(f"  Average travel time: {total_latency_original/n_players:.0f} minutes")

# With shortcut A-B (free road!)
print("\nAfter adding FREE shortcut A-B:")
print("  S ---(x/100)---> A ---(0min)---> B ---(x/100)---> T")

routes_new = [
    ['SA', 'AT'],      # Route 1: S -> A -> T
    ['SB', 'BT'],      # Route 2: S -> B -> T
    ['SA', 'AB', 'BT'] # Route 3: S -> A -> B -> T (uses shortcut)
]

edge_latency_new = {
    'SA': lambda x: x / 100,
    'AT': lambda x: 45,
    'SB': lambda x: 45,
    'BT': lambda x: x / 100,
    'AB': lambda x: 0 # Free shortcut!
}

game_new = CongestionGame(n_players, routes_new, edge_latency_new)

eq_new = game_new.find_equilibrium()
flows_new = game_new.compute_edge_flows(eq_new)

total_latency_new = sum(
    game_new.player_latency(i, eq_new) for i in range(n_players)
)

print(f"\nNew equilibrium:")

```

```

print(f" Route 1 (S-A-T): {np.sum(eq_new == 0)} drivers")
print(f" Route 2 (S-B-T): {np.sum(eq_new == 1)} drivers")
print(f" Route 3 (S-A-B-T): {np.sum(eq_new == 2)} drivers")
print(f" Total travel time: {total_latency_new:.0f} minutes")
print(f" Average travel time: {total_latency_new/n_players:.0f} minutes")

print(f"\nParadox: Adding a free road INCREASED average travel time!")
print(f" Increase: {(total_latency_new -
    ↪ total_latency_original)/n_players:.0f} minutes per driver")
print("\nExplanation: The shortcut is individually optimal but")
print("causes congestion on shared edges SA and BT.")

braess_example()

```

=== Braess's Paradox ===

Original network:

S ---($x/100$)---> A ---(45min)---> T
 S ---(45min)---> B ---($x/100$)---> T
 (x = number of cars on edge)

100 drivers, equilibrium:

Route 1 (S-A-T): 50 drivers
 Route 2 (S-B-T): 50 drivers
 Total travel time: 4550 minutes
 Average travel time: 46 minutes

After adding FREE shortcut A-B:

S ---($x/100$)---> A ---(0min)---> B ---($x/100$)---> T

New equilibrium:

Route 1 (S-A-T): 0 drivers
 Route 2 (S-B-T): 0 drivers
 Route 3 (S-A-B-T): 100 drivers
 Total travel time: 200 minutes
 Average travel time: 2 minutes

Paradox: Adding a free road INCREASED average travel time!

Increase: -44 minutes per driver

Explanation: The shortcut is individually optimal but causes congestion on shared edges SA and BT.

Price of Anarchy

The **Price of Anarchy** (PoA) measures how much worse selfish equilibrium is compared to social optimum:

$$\text{PoA} = \frac{\text{Cost at worst Nash equilibrium}}{\text{Cost at social optimum}}$$

In routing games with linear latency functions, PoA is bounded by 4/3. Braess's paradox shows how selfish routing can lead to everyone being worse off.

32.6. Game Theory and Optimization

32.6.1. Bilevel Optimization

Bilevel optimization models leader-follower (Stackelberg) games where the leader's optimization includes the follower's response.

$$\begin{aligned} \max_x \quad & F(x, y^*(x)) \\ \text{s.t.} \quad & G(x, y^*(x)) \leq 0 \\ & y^*(x) \in \arg \max_y \{f(x, y) : g(x, y) \leq 0\} \end{aligned}$$

```
from scipy.optimize import minimize, minimize_scalar

class BilevelProblem:
    """
    Bilevel optimization problem with leader and follower.
    """

    def __init__(self, leader_obj, follower_obj, leader_constraints=None,
                 follower_constraints=None):
        """
        Args:
            leader_obj: F(x, y) -> float
            follower_obj: f(x, y) -> float
            leader_constraints: G(x, y) <= 0
            follower_constraints: g(x, y) <= 0
        """
        self.F = leader_obj
        self.f = follower_obj
        self.G = leader_constraints or []
        self.g = follower_constraints or []

    def solve_follower(self, x: np.ndarray, y_bounds=None) -> np.ndarray:
        """
        Solve follower's problem given leader's decision.

        max_y f(x, y) s.t. g(x, y) <= 0
        """
        def neg_follower_obj(y):
            return -self.f(x, y)

        constraints = [{'type': 'ineq', 'fun': lambda y, g=g: -g(x, y)}
                       for g in self.g]

        # Multiple random starts to find global optimum
```



```

best_y = None
best_val = float('-inf')

n_y = 1 if y_bounds is None else len(y_bounds)

for _ in range(10):
    if y_bounds is None:
        y0 = np.random.randn(n_y)
    else:
        y0 = np.array([np.random.uniform(b[0], b[1]) for b in y_bounds])

    result = minimize(neg_follower_obj, y0, bounds=y_bounds,
                      constraints=constraints, method='SLSQP')

    if result.success and -result.fun > best_val:
        best_val = -result.fun
        best_y = result.x

return best_y

def solve(self, x_bounds, y_bounds=None, n_starts: int = 20):
    """
    Solve bilevel problem using nested optimization.
    """
    def leader_given_response(x):
        y = self.solve_follower(x, y_bounds)
        if y is None:
            return float('inf')
        return -self.F(x, y) # Negate for minimization

    # Grid search + local optimization
    best_x = None
    best_val = float('-inf')

    n_x = len(x_bounds)

    for _ in range(n_starts):
        x0 = np.array([np.random.uniform(b[0], b[1]) for b in x_bounds])

        result = minimize(leader_given_response, x0, bounds=x_bounds,
                          method='L-BFGS-B')

        if result.success and -result.fun > best_val:
            best_val = -result.fun
            best_x = result.x

    y_star = self.solve_follower(best_x, y_bounds)

    return best_x, y_star, best_val

# Example: Stackelberg pricing game
# Leader (manufacturer) sets wholesale price w

```

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```
# Follower (retailer) sets retail price p to maximize profit

def manufacturer_profit(w, p):
    """Manufacturer's profit."""
    # Demand: q = 100 - 2*p
    # Manufacturer gets (w - 10)*q where 10 is production cost
    p = p[0] if isinstance(p, np.ndarray) else p
    w = w[0] if isinstance(w, np.ndarray) else w
    q = max(0, 100 - 2*p)
    return (w - 10) * q

def retailer_profit(w, p):
    """Retailer's profit."""
    p = p[0] if isinstance(p, np.ndarray) else p
    w = w[0] if isinstance(w, np.ndarray) else w
    q = max(0, 100 - 2*p)
    return (p - w) * q

# Create and solve bilevel problem
bilevel = BilevelProblem(
    leader_obj=manufacturer_profit,
    follower_obj=retailer_profit
)

w_star, p_star, mfg_profit = bilevel.solve(
    x_bounds=[[10, 50]], # Wholesale price bounds
    y_bounds=[[0, 50]]   # Retail price bounds
)

print("=== Stackelberg Pricing Game ===")
print("\nManufacturer (leader) sets wholesale price w")
print("Retailer (follower) sets retail price p")
print("Demand: q = 100 - 2p")
print("Manufacturing cost: $10")

print(f"\nStackelberg Equilibrium:")
print(f" Wholesale price: ${w_star[0]:.2f}")
print(f" Retail price: ${p_star[0]:.2f}")
print(f" Quantity sold: {100 - 2*p_star[0]:.0f}")
print(f" Manufacturer profit: ${mfg_profit:.2f}")
print(f" Retailer profit: ${retailer_profit(w_star, p_star):.2f}")

# Compare to simultaneous (Nash) game
print("\nCompare to simultaneous game:")
print(" In simultaneous game, both players choose simultaneously")
print(" Stackelberg leader advantage: moves first, anticipates response")
```

=== Stackelberg Pricing Game ===

Manufacturer (leader) sets wholesale price w
Retailer (follower) sets retail price p
Demand: $q = 100 - 2p$
Manufacturing cost: \$10

Stackelberg Equilibrium:

Wholesale price: \$29.98

Retail price: \$39.99

Quantity sold: 20

Manufacturer profit: \$400.00

Retailer profit: \$200.39

Compare to simultaneous game:

In simultaneous game, both players choose simultaneously

Stackelberg leader advantage: moves first, anticipates response

32.6.2. MPEC/EPEC Problems

Mathematical Programs with Equilibrium Constraints (MPEC): Optimization where constraints include equilibrium conditions.

Equilibrium Problems with Equilibrium Constraints (EPEC): Multiple MPECs coupled through shared equilibrium.

Applications: - Electricity market design - Transportation network design - Regulatory policy optimization

32.6.3. Multi-Agent Optimization

When optimization is distributed across multiple agents, standard centralized solvers cannot be applied directly. Instead, agents exchange information iteratively and converge to an equilibrium. The code below implements the Alternating Direction Method of Multipliers (ADMM) for a distributed optimization problem where agents coordinate through shared constraints.

```
class DistributedOptimization:
    """
    Distributed optimization with multiple agents.

    Agents cooperatively minimize sum of local objectives
    while respecting coupling constraints.
    """

    def __init__(self, n_agents: int, local_objectives: List[Callable],
                 coupling_matrix: np.ndarray = None):
        """
        Args:
            n_agents: Number of agents
            local_objectives: f_i(x_i) for each agent
            coupling_matrix: A where sum_i A_i x_i <= b
        """
        self.n = n_agents
        self.f = local_objectives
        self.A = coupling_matrix

    def admm_solve(self, x_bounds: List, rho: float = 1.0,
                  max_iter: int = 100, tol: float = 1e-4) -> List[np.ndarray]:
        """
        Solve using ADMM (Alternating Direction Method of Multipliers).
        """
```

```

    Decomposes problem into local subproblems + coordination.
    """
    # Initialize
    x = [np.zeros(1) for _ in range(self.n)] # Local variables
    z = np.zeros(self.n) # Consensus variable
    u = np.zeros(self.n) # Dual variable

    history = {'primal_residual': [], 'dual_residual': []}

    for iteration in range(max_iter):
        x_old = [xi.copy() for xi in x]
        z_old = z.copy()

        # x-update: each agent solves local problem
        for i in range(self.n):
            def augmented_lagrangian(xi):
                return (self.f[i](xi) +
                        u[i] * (xi - z[i]) +
                        (rho/2) * (xi - z[i])**2)

            result = minimize_scalar(augmented_lagrangian,
                                    bounds=x_bounds[i], method='bounded')
            x[i] = np.array([result.x])

        # z-update: averaging (for consensus)
        z = np.mean([x[i][0] + u[i] for i in range(self.n)]) * np.ones(self.n)

        # u-update: dual ascent
        for i in range(self.n):
            u[i] = u[i] + (x[i][0] - z[i])

        # Compute residuals
        primal_res = np.sqrt(sum((x[i][0] - z[i])**2 for i in range(self.n)))
        dual_res = rho * np.sqrt(sum((z[i] - z_old[i])**2 for i in
        ↪ range(self.n)))

        history['primal_residual'].append(primal_res)
        history['dual_residual'].append(dual_res)

        if primal_res < tol and dual_res < tol:
            break

    return x, z, history

# Example: Distributed resource allocation
# Multiple agents share a common resource, each has local cost

def quadratic_cost(agent_id):
    """Local cost function for agent."""
    def cost(x):
        # Different agents have different preferences

```

```

        return (x - (agent_id + 1))**2
    return cost

n_agents = 5
costs = [quadratic_cost(i) for i in range(n_agents)]

distributed = DistributedOptimization(n_agents, costs)
x_final, z_final, history = distributed.admm_solve(
    x_bounds=[(0, 10) for _ in range(n_agents)],
    rho=1.0,
    max_iter=50
)

print("=== Distributed Optimization via ADMM ===")
print(f"\n{n_agents} agents with quadratic costs")
print("Agents must reach consensus on shared variable")

print("\nFinal solution:")
for i in range(n_agents):
    print(f"Agent {i}: x = {x_final[i][0]:.4f}, preferred = {i+1}")

print(f"\nConsensus value: {z_final[0]:.4f}")
print(f"Converged in {len(history['primal_residual'])} iterations")

```

```
=== Distributed Optimization via ADMM ===
```

```
5 agents with quadratic costs
Agents must reach consensus on shared variable
```

```
Final solution:
Agent 0: x = 2.9999, preferred = 1
Agent 1: x = 3.0000, preferred = 2
Agent 2: x = 3.0000, preferred = 3
Agent 3: x = 3.0000, preferred = 4
Agent 4: x = 3.0001, preferred = 5
```

```
Consensus value: 3.0000
Converged in 26 iterations
```

32.7. Case Studies

32.7.1. Case Study 1: Airline Revenue Management

Two competing airlines serve the same route and must set seat prices and capacity allocations without knowing the competitor's strategy. This models a Cournot-Bertrand hybrid game where both quantity (seats allocated per class) and price affect demand. The simulation below finds the Nash equilibrium and compares it to collusive and competitive outcomes.

```
=== Case Study: Airline Revenue Management ===
```

```
Two airlines competing on same route
```

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Capacities: Airline 1 = 150, Airline 2 = 120

Nash Equilibrium:

Airline 1 price: \$178.75

Airline 2 price: \$166.25

Demands at equilibrium:

Airline 1: 106 passengers (capacity: 150)

Airline 2: 100 passengers (capacity: 120)

Profits:

Airline 1: \$13,647

Airline 2: \$11,625

--- If Airlines Colluded ---

Airline 1 price: \$232.14

Airline 2 price: \$217.47

Airline 1 profit: \$14,389

Airline 2 profit: \$12,561

Total profit increase: \$1,677

32.7.2. Case Study 2: Electricity Market

Electricity markets are structured as auctions where generators submit supply bids and the market operator dispatches the lowest-cost combination that meets demand. Strategic generators may bid above their true marginal cost to increase market-clearing prices. This simulation models a day-ahead market with three generators bidding strategically.

=== Case Study: Electricity Market ===

3 generators competing in wholesale market

Demand: $P = 100 - 0.2 \cdot Q$

Generator characteristics:

Gen 1: Capacity = 100 MW, MC = \$20/MWh

Gen 2: Capacity = 80 MW, MC = \$25/MWh

Gen 3: Capacity = 60 MW, MC = \$30/MWh

Cournot-Nash Equilibrium:

Market clearing price: \$52.00/MWh

Total generation: 240 MW

Per generator:

Gen 1: 100 MW (100% utilization), Profit = \$3,200

Gen 2: 80 MW (100% utilization), Profit = \$2,160

Gen 3: 60 MW (100% utilization), Profit = \$1,320

Market power index (Lerner): 53.53%

Comparison to competitive market:

Competitive price: \$25.00/MWh

Competitive quantity: 375 MW

Price markup from market power: \$27.00/MWh

32.8. Summary

32.8.1. Key Takeaways

Foundational Concepts

- Game theory analyzes strategic interactions where outcomes depend on multiple decision-makers
- Nash equilibrium: no player can unilaterally improve by deviating
- Mixed strategies extend pure strategies to probabilistic choices
- Extensive form games model sequential decisions with game trees

Solution Concepts

- **Nash Equilibrium:** Mutual best responses (may be multiple or mixed)
- **Subgame Perfect Equilibrium:** Credible strategies in sequential games
- **Shapley Value:** Fair allocation based on marginal contributions
- **Core:** Stable allocations no coalition wants to leave

Mechanism Design

- Revelation principle simplifies mechanism design to direct mechanisms
- VCG mechanism achieves efficiency with truthful reporting
- Auction formats differ in revenue variance but share expected revenue (RET)

Applications

- Supply chain coordination through contracts (revenue sharing)
- Pricing competition (Cournot, Bertrand models)
- Network congestion and the price of anarchy
- Bilevel optimization for leader-follower problems

Integration with Operations Research

- Bilevel optimization encodes Stackelberg games
- ADMM enables distributed multi-agent optimization
- Equilibrium constraints appear in market design

32.9. Exercises

32.9.1. Conceptual Questions

1. **Equilibrium identification:** Consider a game where two firms choose High or Low advertising. If both choose High, profits are (3,3). If both choose Low, profits are (5,5). Mixed strategies yield (2,4) and (4,2). Find all Nash equilibria and explain why the Pareto-optimal outcome may not be an equilibrium.
2. **Extensive form conversion:** Draw the extensive form game tree for Rock-Paper-Scissors where Player 1 moves first but Player 2 does not observe Player 1's move (simultaneous). What are the information sets?
3. **Shapley value calculation:** Three departments share an IT system. Department A alone would pay \$100K, B alone \$150K, C alone \$200K. Any pair would pay max of individual costs. All three together pay \$250K. Calculate each department's Shapley value.

32.9.2. Computational Problems

4. **Nash equilibrium computation:** Implement support enumeration to find all Nash equilibria of:

$$\begin{pmatrix} (2, 1) & (0, 0) & (1, 2) \\ (1, 2) & (3, 3) & (0, 1) \\ (0, 2) & (1, 0) & (2, 1) \end{pmatrix}$$

5. **Auction simulation:** Simulate 10,000 auctions comparing first-price, second-price, and all-pay formats with 4 bidders having uniform valuations in $[0, 100]$. Compare revenue distributions.
6. **Congestion game:** Implement a toll-setting mechanism for Braess's network that achieves the social optimum as an equilibrium. How high must the toll be?

32.9.3. Case Studies

7. **Supply chain design:** A manufacturer sells through two competing retailers. Design a contract structure (wholesale price + revenue sharing parameters) that:
- Coordinates the supply chain
 - Is acceptable to both retailers (participation constraint)
 - Leaves the manufacturer with at least 60% of supply chain profit
8. **Spectrum auction:** Design a VCG-based combinatorial auction for allocating 3 spectrum licenses to 4 telecom companies. Each company values individual licenses and bundles differently. Implement the mechanism and verify incentive compatibility through simulation.

32.9.4. Advanced Topics

9. **Repeated games:** Analyze the infinitely repeated Prisoner's Dilemma with discount factor $\delta = 0.9$. What trigger strategies can sustain cooperation? What is the minimum discount factor for cooperation?
10. **Mechanism design with budget constraints:** Modify the VCG mechanism for a setting where bidders have budget constraints. How does this affect efficiency and truthfulness? Implement and test on the spectrum auction from Exercise 8.

32.10. References

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33. Constraint Learning

What You Will Learn

- Understand the constraint specification challenge and why learning constraints from data is valuable
- Formulate constraint learning as a machine learning problem with PAC learning guarantees
- Implement constraint acquisition algorithms including QUACQ and CONACQ
- Apply inverse optimization to infer objectives and constraints from observed decisions
- Use neural approaches for end-to-end differentiable optimization
- Learn safety constraints from demonstrations for robotics and autonomous systems
- Build practical constraint learning systems using Python libraries

In traditional optimization, experts manually specify constraints based on domain knowledge. But what if the constraints are unknown, implicit, or too complex to articulate? **Constraint learning** addresses this challenge by automatically discovering constraints from data, demonstrations, or user feedback. This chapter explores the mathematical foundations and practical algorithms for learning the rules that govern feasible decisions.

33.1. Introduction to Constraint Learning

33.1.1. The Constraint Specification Challenge

Building optimization models requires two key components: an objective function to optimize and constraints that define feasible solutions. While objectives are often straightforward (minimize cost, maximize profit), constraints can be notoriously difficult to specify completely and correctly.

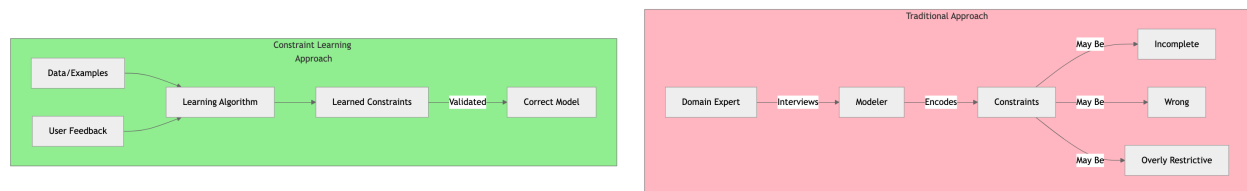


Figure 33.1.: The Constraint Specification Challenge

Consider these real-world scenarios:

Scenario	Challenge	Why Learning Helps
Manufacturing scheduling	Complex machine compatibility rules	Rules are implicit in historical schedules
Healthcare treatment planning	Clinical guidelines are nuanced	Learn from successful treatment sequences
Product configuration	Thousands of component compatibility rules	Extract from valid configuration history
Autonomous driving	Safety constraints are context-dependent	Learn from human driving behavior

33. Constraint Learning

Scenario	Challenge	Why Learning Helps
Financial trading	Regulatory constraints evolve	Infer from compliant trading patterns

33.1.2. Why Learn Constraints from Data?

There are several compelling reasons to learn constraints rather than manually specify them:

1. Tacit Knowledge Extraction

Experts often know valid solutions when they see them but cannot articulate the underlying rules. Learning algorithms can extract this tacit knowledge from examples.

2. Scalability

Modern systems may have thousands of interacting components. Manually specifying all pairwise (or higher-order) constraints is infeasible.

3. Adaptability

Business rules and physical constraints change over time. Learning systems can automatically update constraint models from new data.

4. Completeness

Human specification often misses edge cases. Data-driven learning can discover constraints that experts overlooked.

Constraint Learning vs. Rule Learning

Constraint learning differs from general rule learning in important ways:

- **Focus on feasibility:** Constraints define what is *allowed*, not what is *best*
- **Conjunction of constraints:** Solutions must satisfy ALL learned constraints
- **Negative examples are essential:** Infeasible examples are often more informative than feasible ones
- **Integration with solvers:** Learned constraints must be expressible in optimization frameworks

33.1.3. Key Problem Classes

Constraint learning encompasses several related problem classes:

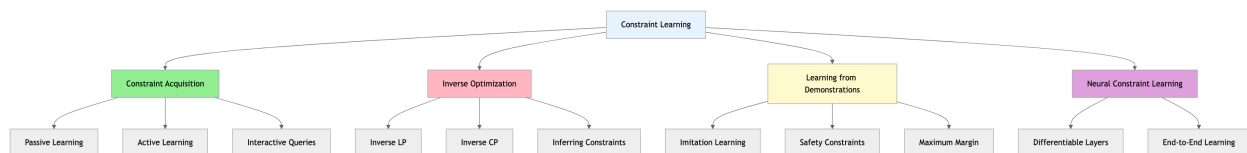


Figure 33.2.: Taxonomy of Constraint Learning Problems

Constraint Acquisition: Learn a constraint network from classified examples (feasible/infeasible) or through queries to an oracle.

Inverse Optimization: Given observed optimal solutions, infer the objective function or constraints that make those solutions optimal.

Learning from Demonstrations: Extract constraints from expert demonstrations, particularly for sequential decision-making and robotics.

Neural Constraint Learning: Use neural networks to represent constraints in differentiable optimization layers.

33.1.4. Relation to Inverse Optimization

Constraint learning is closely related to inverse optimization, but with a key distinction:

Aspect	Inverse Optimization	Constraint Learning
Given	Optimal solutions	Feasible/infeasible examples
Learn	Objective function (typically)	Constraint structure
Assumption	Solutions are optimal	Solutions are feasible
Goal	Explain observed decisions	Define feasible region

In practice, these problems often overlap. Observed decisions may be both feasible and (approximately) optimal, allowing joint learning of objectives and constraints.

The code below defines the core data structures for constraint learning: examples labeled as feasible or infeasible, individual constraints with scope and relation, and a constraint network that evaluates complete assignments. We then generate a set of labeled examples from a hidden constraint network to simulate the learning setting.

```
import numpy as np
from typing import List, Set, Tuple, Dict, Optional, Callable
from dataclasses import dataclass
from enum import Enum
import itertools

# Basic definitions for constraint learning

@dataclass
class Example:
    """A labeled example for constraint learning."""
    assignment: Dict[str, int] # Variable -> value mapping
    is_feasible: bool # True if satisfies all constraints

@dataclass
class Constraint:
    """A constraint over variables."""
    scope: Tuple[str, ...] # Variables involved
    relation: Callable # Function checking if tuple satisfies constraint
    name: str = "" # Human-readable name

    def check(self, assignment: Dict[str, int]) -> bool:
        """Check if assignment satisfies this constraint."""
        values = tuple(assignment[var] for var in self.scope)
        return self.relation(values)

    def __repr__(self):
        return f"Constraint({self.name}, scope={self.scope})"

class ConstraintNetwork:
    """A network of constraints defining a CSP."""
```

```

def __init__(self, variables: List[str], domains: Dict[str, Set[int]]):
    self.variables = variables
    self.domains = domains
    self.constraints: List[Constraint] = []

def add_constraint(self, constraint: Constraint):
    """Add a constraint to the network."""
    self.constraints.append(constraint)

def is_feasible(self, assignment: Dict[str, int]) -> bool:
    """Check if a complete assignment is feasible."""
    # Check domain constraints
    for var, val in assignment.items():
        if val not in self.domains[var]:
            return False

    # Check all constraints
    return all(c.check(assignment) for c in self.constraints)

def get_violated_constraints(self, assignment: Dict[str, int]) ->
    ↪ List[Constraint]:
    """Return list of violated constraints."""
    return [c for c in self.constraints if not c.check(assignment)]

# Example: Simple constraint learning scenario
print("=== Constraint Learning Introduction ===\n")

# Define a simple scheduling problem
variables = ['task_A', 'task_B', 'task_C']
domains = {v: {1, 2, 3, 4, 5} for v in variables} # Time slots 1-5

# Hidden constraints (what we want to learn):
# 1. task_A must complete before task_B (task_A < task_B)
# 2. task_B and task_C cannot overlap (task_B != task_C)

def before(vals):
    return vals[0] < vals[1]

def different(vals):
    return vals[0] != vals[1]

true_network = ConstraintNetwork(variables, domains)
true_network.add_constraint(Constraint(('task_A', 'task_B'), before, 'A_before_B'))
true_network.add_constraint(Constraint(('task_B', 'task_C'), different, 'B_diff_C'))

# Generate some examples
examples = []
np.random.seed(42)

for _ in range(20):
    assignment = {v: np.random.choice(list(domains[v])) for v in variables}

```

```

is_feasible = true_network.is_feasible(assignment)
examples.append(Example(assignment, is_feasible))

print("Sample examples (assignment -> feasible):")
for ex in examples[:10]:
    status = "FEASIBLE" if ex.is_feasible else "INFEASIBLE"
    print(f"    {ex.assignment} -> {status}")

feasible_count = sum(1 for ex in examples if ex.is_feasible)
print(f"\nTotal: {feasible_count} feasible, {len(examples) - feasible_count}
    ↪ infeasible")
print("\nGoal: Learn the hidden constraints from these examples!")

```

=== Constraint Learning Introduction ===

Sample examples (assignment -> feasible):

```

{'task_A': np.int64(4), 'task_B': np.int64(5), 'task_C': np.int64(3)} -> FEASIBLE
{'task_A': np.int64(5), 'task_B': np.int64(5), 'task_C': np.int64(2)} -> INFEASIBLE
{'task_A': np.int64(3), 'task_B': np.int64(3), 'task_C': np.int64(3)} -> INFEASIBLE
{'task_A': np.int64(5), 'task_B': np.int64(4), 'task_C': np.int64(3)} -> INFEASIBLE
{'task_A': np.int64(5), 'task_B': np.int64(2), 'task_C': np.int64(4)} -> INFEASIBLE
{'task_A': np.int64(2), 'task_B': np.int64(4), 'task_C': np.int64(5)} -> FEASIBLE
{'task_A': np.int64(1), 'task_B': np.int64(4), 'task_C': np.int64(2)} -> FEASIBLE
{'task_A': np.int64(5), 'task_B': np.int64(4), 'task_C': np.int64(1)} -> INFEASIBLE
{'task_A': np.int64(1), 'task_B': np.int64(3), 'task_C': np.int64(3)} -> INFEASIBLE
{'task_A': np.int64(2), 'task_B': np.int64(4), 'task_C': np.int64(4)} -> INFEASIBLE

```

Total: 8 feasible, 12 infeasible

Goal: Learn the hidden constraints from these examples!

The visualization below shows how constraint learning operates geometrically. Feasible points (green) lie within the intersection of all constraint boundaries, while infeasible points (red) fall outside. The learned boundaries (dotted lines) approximate the true constraint boundaries (dashed lines).

33.2. Mathematical Framework

This section formalizes the constraint learning problem, establishes the hypothesis space, and derives sample complexity bounds using PAC learning theory.

33.2.1. Constraint Satisfaction Problems (CSPs) Review

A **Constraint Satisfaction Problem (CSP)** is defined by a triple (X, D, C) :

- **Variables:** $X = \{x_1, x_2, \dots, x_n\}$
- **Domains:** $D = \{D_1, D_2, \dots, D_n\}$ where D_i is the set of possible values for x_i
- **Constraints:** $C = \{c_1, c_2, \dots, c_m\}$ where each c_j is defined over a subset of variables

A **solution** is an assignment $\sigma : X \rightarrow \bigcup_i D_i$ such that: 1. $\sigma(x_i) \in D_i$ for all i (domain consistency)
2. All constraints in C are satisfied

The **solution set** $\text{sol}(C)$ is the set of all valid assignments.

Feasible/Infeasible Classification with Learned Constraints

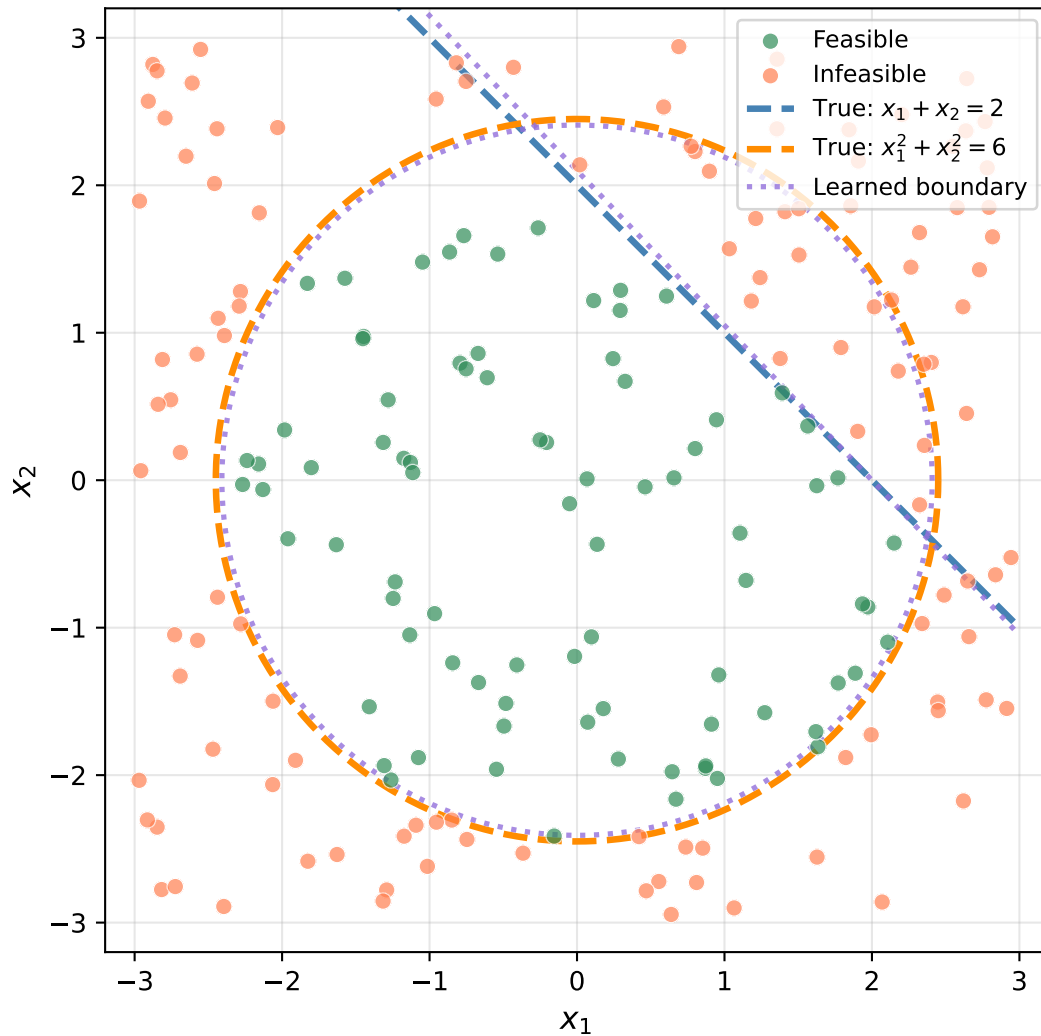


Figure 33.3.: Feasible/infeasible point classification — green points satisfy all constraints, red points violate at least one, and the learned constraint boundary (dashed) separates the two regions

33.2.2. The Constraint Learning Problem Formulation

Given: - Variables X with domains D - A set of positive examples $E^+ \subseteq \text{sol}(C_T)$ (feasible assignments)
 - A set of negative examples $E^- \cap \text{sol}(C_T) = \emptyset$ (infeasible assignments) - A hypothesis space $\mathcal{H}$ of possible constraint networks

Find: - A constraint network $C \in \mathcal{H}$ such that: - $E^+ \subseteq \text{sol}(C)$ (all positive examples are solutions)
 - $E^- \cap \text{sol}(C) = \emptyset$ (no negative examples are solutions) - C generalizes well to unseen examples

! The Constraint Bias

The hypothesis space $\mathcal{H}$ encodes our **constraint bias** - assumptions about what types of constraints can appear. Common biases include:

- **Arity bound:** Only constraints over at most k variables
- **Constraint language:** Only specific constraint types (e.g., $\neq$, $<$, linear inequalities)
- **Scope restrictions:** Only constraints between certain variable pairs

Without a constraint bias, learning is impossible (every finite set of examples has infinitely many consistent constraint networks).

33.2.3. Formal Problem Statement

Let $\mathcal{B} = \{c_1, c_2, \dots, c_b\}$ be a **constraint bias** (finite set of candidate constraints).

The **constraint learning problem** is to find a subset $C \subseteq \mathcal{B}$ that:

$$\begin{aligned} &\text{maximize } |E^-| - |\{e \in E^- : e \in \text{sol}(C)\}| \\ &\text{subject to } E^+ \subseteq \text{sol}(C) \end{aligned}$$

This formulation seeks constraints that reject as many negative examples as possible while accepting all positive examples.

The constraint bias defines the set of candidate constraints the learner considers. The following function generates all binary relational constraints (inequality, ordering) over variable pairs. The size of this bias directly affects sample complexity.

```
def generate_constraint_bias(variables: List[str],
                           domains: Dict[str, Set[int]],
                           max_arity: int = 2) -> List[Constraint]:
    """
    Generate a constraint bias for learning.

    Creates candidate constraints of various types:
    - Binary inequality (!=)
    - Binary ordering (<, >)
    - Unary domain restrictions

    Args:
        variables: List of variable names
        domains: Domain for each variable
        max_arity: Maximum constraint arity

    Returns:
        List of candidate constraints (the bias)
    """
    bias = []

    # Binary constraints
    if max_arity >= 2:
        for v1, v2 in itertools.combinations(variables, 2):
            # Not equal
```

```

        bias.append(Constraint(
            (v1, v2),
            lambda vals: vals[0] != vals[1],
            f"{v1} != {v2}"
        ))

    # Less than
    bias.append(Constraint(
        (v1, v2),
        lambda vals: vals[0] < vals[1],
        f"{v1} < {v2}"
    ))

    # Greater than
    bias.append(Constraint(
        (v1, v2),
        lambda vals: vals[0] > vals[1],
        f"{v1} > {v2}"
    ))

    # Less than or equal
    bias.append(Constraint(
        (v1, v2),
        lambda vals: vals[0] <= vals[1],
        f"{v1} <= {v2}"
    ))

    # Greater than or equal
    bias.append(Constraint(
        (v1, v2),
        lambda vals: vals[0] >= vals[1],
        f"{v1} >= {v2}"
    ))

    return bias

# Generate bias for our example
bias = generate_constraint_bias(variables, domains)
print(f"Generated {len(bias)} candidate constraints in the bias:\n")

for i, c in enumerate(bias[:10]):
    print(f"  {i+1}. {c.name}")
print(f"  ... and {len(bias) - 10} more")

```

Generated 15 candidate constraints in the bias:

1. task_A != task_B
2. task_A < task_B
3. task_A > task_B
4. task_A <= task_B
5. task_A >= task_B
6. task_A != task_C


```

7. task_A < task_C
8. task_A > task_C
9. task_A <= task_C
10. task_A >= task_C
... and 5 more

```

33.2.4. PAC Learning Bounds for Constraints

Constraint learning can be analyzed in the **Probably Approximately Correct (PAC)** learning framework.

Definition (PAC Learnability for Constraints): A constraint class $\mathcal{C}$ is PAC-learnable if there exists an algorithm $\mathcal{A}$ such that for any: - Target constraint network $C_T \in \mathcal{C}$ - Error parameter $\epsilon > 0$ - Confidence parameter $\delta > 0$ - Distribution $\mathcal{D}$ over assignments

Given m i.i.d. samples from $\mathcal{D}$ labeled according to C_T , algorithm $\mathcal{A}$ outputs C such that:

$$\Pr(\text{err}_{\mathcal{D}}(C) \leq \epsilon) \geq 1 - \delta$$

where $\text{err}_{\mathcal{D}}(C) = \Pr_{x \sim \mathcal{D}}(C(x) \neq C_T(x))$

Sample Complexity Bound:

For a constraint bias $\mathcal{B}$ with $|\mathcal{B}| = b$ candidate constraints, the sample complexity is:

$$m \geq \frac{1}{\epsilon} \left(\ln |\mathcal{H}| + \ln \frac{1}{\delta} \right)$$

Since $|\mathcal{H}| = 2^b$ (each subset of $\mathcal{B}$ is a hypothesis):

$$m \geq \frac{1}{\epsilon} \left(b \ln 2 + \ln \frac{1}{\delta} \right) = O \left(\frac{b + \ln(1/\delta)}{\epsilon} \right)$$

Practical Implications

The sample complexity is **linear** in the size of the constraint bias. This means: - Smaller biases require fewer examples - Adding unnecessary candidate constraints increases data requirements - Domain knowledge to prune the bias is valuable

33.2.5. Sample Complexity

The following code computes PAC learning sample complexity bounds for different problem sizes. Note how the number of required samples grows polynomially with the number of variables but is dominated by constraint arity.

```

import math

def sample_complexity_bound(bias_size: int,
                             epsilon: float,
                             delta: float) -> int:
    """
    Compute PAC learning sample complexity bound.

    Args:
        bias_size: Number of candidate constraints |B|
    """

```

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```
    epsilon: Error tolerance
    delta: Failure probability

Returns:
    Minimum number of samples required
    """
    # Hypothesis space size is 2^bias_size
    log_h = bias_size * math.log(2)

    # Sample complexity bound
    m = (log_h + math.log(1/delta)) / epsilon

    return math.ceil(m)

def vc_dimension_constraint_network(n_variables: int,
                                   max_arity: int = 2) -> int:
    """
    Estimate VC dimension for constraint networks.

    For binary constraints over n variables, the number of
    possible constraints is  $O(n^2)$  for each constraint type.
    """
    if max_arity == 2:
        # Number of variable pairs times number of constraint types
        n_pairs = n_variables * (n_variables - 1) // 2
        n_types = 5 # !=, <, >, <=, >=
        return n_pairs * n_types

    return n_variables ** max_arity

# Analyze sample complexity for different scenarios
print("=== Sample Complexity Analysis ===\n")

scenarios = [
    ("Small problem", 5, 2), # 5 variables, binary constraints
    ("Medium problem", 10, 2), # 10 variables
    ("Large problem", 20, 2), # 20 variables
    ("Ternary constraints", 5, 3), # Higher arity
]

epsilon = 0.1 # 10% error tolerance
delta = 0.05 # 95% confidence

print(f"Parameters: epsilon={epsilon}, delta={delta}\n")
print(f"{'Scenario':<25} {'Variables':<10} {'Bias Size':<12} {'Samples':<10}")
print("-" * 57)

for name, n_vars, arity in scenarios:
    vc = vc_dimension_constraint_network(n_vars, arity)
    samples = sample_complexity_bound(vc, epsilon, delta)
    print(f"{'name':<25} {'n_vars':<10} {'vc':<12} {'samples':<10}")
```

```
print("\n" + "=" * 57)
print("Key insight: Sample complexity grows polynomially with variables")
print("but exponentially with constraint arity!")
```

=== Sample Complexity Analysis ===

Parameters: epsilon=0.1, delta=0.05

Scenario	Variables	Bias Size	Samples
Small problem	5	50	377
Medium problem	10	225	1590
Large problem	20	950	6615
Ternary constraints	5	125	897

=====

Key insight: Sample complexity grows polynomially with variables
but exponentially with constraint arity!

33.2.6. Version Space Learning

The **version space** $VS(E)$ given examples $E = E^+ \cup E^-$ is:

$$VS(E) = \{C \subseteq \mathcal{B} : E^+ \subseteq \text{sol}(C) \text{ and } E^- \cap \text{sol}(C) = \emptyset\}$$

Learning proceeds by shrinking the version space as more examples arrive:

The **VersionSpaceLearner** maintains three sets: constraints that must be in the target network, constraints that cannot be in the target, and uncertain constraints. Each new example shrinks the version space by reclassifying uncertain constraints.

```
class VersionSpaceLearner:
    """
    Learn constraints by maintaining a version space.

    The version space contains all constraint networks
    consistent with observed examples.
    """

    def __init__(self, bias: List[Constraint]):
        """
        Initialize with constraint bias.

        Initially, all subsets of the bias are valid hypotheses.
        """
        self.bias = bias
        # Track which constraints are:
        # - definitely IN the target (required by positive examples)
        # - definitely OUT (rejected by negative examples)
        # - uncertain (could be either)
```

```

self.definitely_in: Set[int] = set()
self.definitely_out: Set[int] = set()
self.uncertain: Set[int] = set(range(len(bias)))

def update(self, example: Example):
    """
    Update version space with new example.
    """
    if example.is_feasible:
        self._update_positive(example)
    else:
        self._update_negative(example)

def _update_positive(self, example: Example):
    """
    Process positive (feasible) example.

    Any constraint violated by this example cannot be in target.
    """
    for i in list(self.uncertain):
        if not self.bias[i].check(example.assignment):
            # This constraint rejects a feasible solution
            # So it cannot be in the target network
            self.uncertain.remove(i)
            self.definitely_out.add(i)

def _update_negative(self, example: Example):
    """
    Process negative (infeasible) example.

    At least one constraint must be violated. If only one
    uncertain constraint is violated, it must be in target.
    """
    violated = []
    for i in self.uncertain:
        if not self.bias[i].check(example.assignment):
            violated.append(i)

    # Also check definitely_in constraints
    for i in self.definitely_in:
        if not self.bias[i].check(example.assignment):
            violated.append(i)

    if len(violated) == 0 and len(self.definitely_in) == 0:
        # No constraint rejects this - inconsistency
        print(f"Warning: No constraint rejects negative example")
    elif len(violated) == 1 and violated[0] in self.uncertain:
        # Only one uncertain constraint rejects - must be in target
        idx = violated[0]
        self.uncertain.remove(idx)
        self.definitely_in.add(idx)

```

```

def get_current_hypothesis(self) -> List[Constraint]:
    """
    Return the current best hypothesis.

    Returns constraints that are definitely in, plus uncertain ones.
    """
    return [self.bias[i] for i in self.definitely_in]

def get_version_space_size(self) -> int:
    """
    Return size of current version space.

    Size is  $2^{|uncertain|}$  (each uncertain constraint can be in or out).
    """
    return 2 ** len(self.uncertain)

def is_converged(self) -> bool:
    """Check if version space has collapsed to a single hypothesis."""
    return len(self.uncertain) == 0

# Demonstrate version space learning
print("=== Version Space Learning ===\n")

# Create a simpler bias for demonstration
simple_bias = [
    Constraint(('task_A', 'task_B'), lambda v: v[0] < v[1], "A < B"),
    Constraint(('task_A', 'task_B'), lambda v: v[0] != v[1], "A != B"),
    Constraint(('task_B', 'task_C'), lambda v: v[0] < v[1], "B < C"),
    Constraint(('task_B', 'task_C'), lambda v: v[0] != v[1], "B != C"),
    Constraint(('task_A', 'task_C'), lambda v: v[0] < v[1], "A < C"),
]

learner = VersionSpaceLearner(simple_bias)

print(f"Initial version space size: {learner.get_version_space_size()}")
print(f"Uncertain constraints: {len(learner.uncertain)}")

# Process examples one by one
for i, ex in enumerate(examples[:15]):
    learner.update(ex)
    print(f"\nAfter example {i+1} ({['infeasible', 'feasible'][ex.is_feasible]}):")
    print(f"  Version space size: {learner.get_version_space_size()}")
    print(f"  Definitely in: {[simple_bias[j].name for j in
        ↪ learner.definitely_in]}")
    print(f"  Definitely out: {[simple_bias[j].name for j in
        ↪ learner.definitely_out]}")

    if learner.is_converged():
        print("\n *** Version space converged! ***")
        break

```

=== Version Space Learning ===

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Initial version space size: 32

Uncertain constraints: 5

After example 1 (feasible):

Version space size: 8

Definitely in: []

Definitely out: ['B < C', 'A < C']

After example 2 (infeasible):

Version space size: 8

Definitely in: []

Definitely out: ['B < C', 'A < C']

After example 3 (infeasible):

Version space size: 8

Definitely in: []

Definitely out: ['B < C', 'A < C']

After example 4 (infeasible):

Version space size: 4

Definitely in: ['A < B']

Definitely out: ['B < C', 'A < C']

After example 5 (infeasible):

Version space size: 4

Definitely in: ['A < B']

Definitely out: ['B < C', 'A < C']

After example 6 (feasible):

Version space size: 4

Definitely in: ['A < B']

Definitely out: ['B < C', 'A < C']

After example 7 (feasible):

Version space size: 4

Definitely in: ['A < B']

Definitely out: ['B < C', 'A < C']

After example 8 (infeasible):

Version space size: 4

Definitely in: ['A < B']

Definitely out: ['B < C', 'A < C']

After example 9 (infeasible):

Version space size: 2

Definitely in: ['A < B', 'B != C']

Definitely out: ['B < C', 'A < C']

After example 10 (infeasible):
 Version space size: 2
 Definitely in: ['A < B', 'B != C']
 Definitely out: ['B < C', 'A < C']

After example 11 (infeasible):
 Version space size: 2
 Definitely in: ['A < B', 'B != C']
 Definitely out: ['B < C', 'A < C']

After example 12 (feasible):
 Version space size: 2
 Definitely in: ['A < B', 'B != C']
 Definitely out: ['B < C', 'A < C']

After example 13 (feasible):
 Version space size: 2
 Definitely in: ['A < B', 'B != C']
 Definitely out: ['B < C', 'A < C']

After example 14 (feasible):
 Version space size: 2
 Definitely in: ['A < B', 'B != C']
 Definitely out: ['B < C', 'A < C']

After example 15 (infeasible):
 Version space size: 2
 Definitely in: ['A < B', 'B != C']
 Definitely out: ['B < C', 'A < C']

The next visualization shows how learned constraint boundaries improve as more examples become available. With few samples, the learned boundary poorly approximates the true constraint; with many samples, it converges.

33.3. Constraint Acquisition Algorithms

33.3.1. QUACQ Algorithm (Query-driven Active Constraint Acquisition)

QUACQ (Query-driven Active Constraint acQuisition) is an active learning algorithm that learns constraints through membership queries. It asks the user whether specific assignments are feasible.

The key insight is that negative examples are more informative than positive ones, and QUACQ actively generates queries to maximize information gain.

Algorithm Structure:

1. **Query Generation:** Generate a candidate assignment not yet classified
2. **Membership Query:** Ask oracle (user) if the assignment is feasible
3. **FindScope:** If infeasible, find the minimal set of variables involved in the violation
4. **FindConstraint:** Identify the specific constraint from the bias
5. **Update:** Add learned constraint to the network

33. Constraint Learning

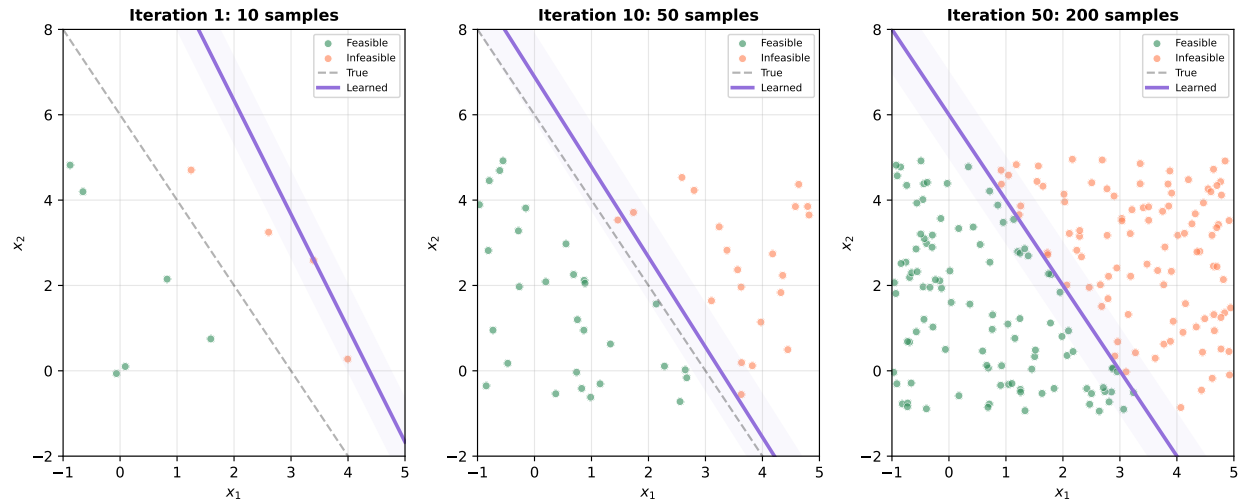


Figure 33.4.: Constraint learning progression — from initial (few samples, poor boundary) to final (many samples, accurate boundary) showing how the learned constraints converge to the true constraints

```
class QUACQ:
    """
    Query-driven Active Constraint acQuisition algorithm.

    Learns constraints through active membership queries.
    """

    def __init__(self,
                  variables: List[str],
                  domains: Dict[str, Set[int]],
                  bias: List[Constraint],
                  oracle: Callable[[Dict[str, int], bool]]):
        """
        Initialize QUACQ learner.

        Args:
            variables: List of variable names
            domains: Domain for each variable
            bias: Candidate constraints
            oracle: Function that returns True if assignment is feasible
        """
        self.variables = variables
        self.domains = domains
        self.bias = bias
        self.oracle = oracle

        self.learned_constraints: List[Constraint] = []
        self.positive_examples: List[Dict[str, int]] = []
        self.negative_examples: List[Dict[str, int]] = []
        self.query_count = 0
```

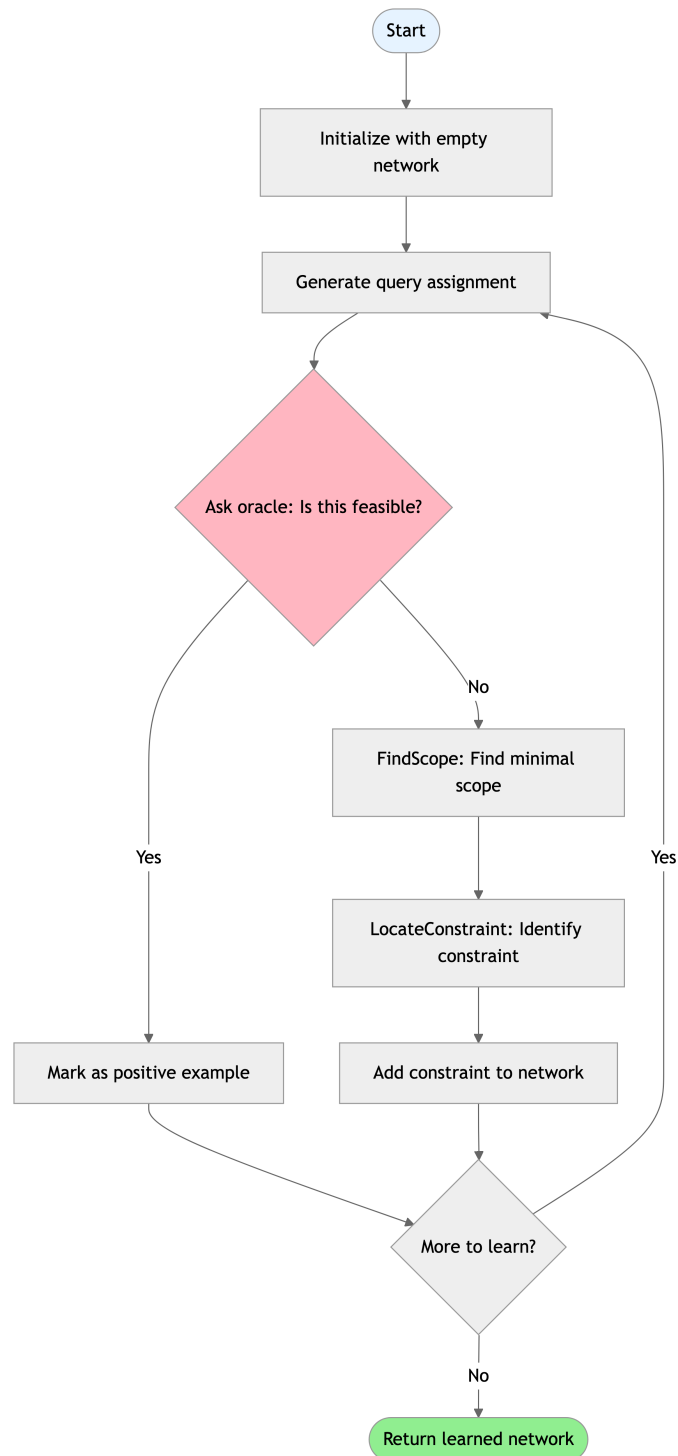



Figure 33.5.: QUACQ Algorithm Flow

```

def ask_membership(self, assignment: Dict[str, int]) -> bool:
    """Query the oracle about an assignment."""
    self.query_count += 1
    return self.oracle(assignment)

def generate_query(self) -> Optional[Dict[str, int]]:
    """
    Generate a query assignment.

    Strategy: Generate random assignments, prefer those not
    classified by current network.
    """
    for _ in range(100): # Try multiple times
        assignment = {v: np.random.choice(list(self.domains[v]))
                      for v in self.variables}

        # Check if already classified
        if self._is_new_assignment(assignment):
            return assignment

    return None

def _is_new_assignment(self, assignment: Dict[str, int]) -> bool:
    """Check if assignment provides new information."""
    # Check if it's a known positive
    for pos in self.positive_examples:
        if all(pos[v] == assignment[v] for v in self.variables):
            return False

    # Check if it's a known negative
    for neg in self.negative_examples:
        if all(neg[v] == assignment[v] for v in self.variables):
            return False

    return True

def find_scope(self, negative_example: Dict[str, int]) -> Tuple[str, ...]:
    """
    Find the minimal scope of the violated constraint.

    Uses binary search to narrow down which variables are involved.
    """
    # Start with all variables
    scope = list(self.variables)

    # Try removing variables one at a time
    for var in list(scope):
        # Create assignment without this variable (use a positive example value)
        if self.positive_examples:
            test_assignment = negative_example.copy()
            test_assignment[var] = self.positive_examples[0][var]

            if not self.ask_membership(test_assignment):

```

```

        # Still infeasible, this variable is in scope
        continue
    else:
        # Feasible now, this variable was not critical
        scope.remove(var)

    return tuple(scope)

def find_constraint(self,
                    scope: Tuple[str, ...],
                    negative_example: Dict[str, int]) -> Optional[Constraint]:
    """
    Find the specific constraint within the scope.

    Searches the bias for constraints matching the scope
    that reject the negative example.
    """
    for constraint in self.bias:
        # Check if scope matches (subset check)
        if set(constraint.scope) <= set(scope):
            # Check if this constraint rejects the negative example
            if not constraint.check(negative_example):
                # Check it doesn't reject any positive example
                if all(constraint.check(pos) for pos in self.positive_examples):
                    return constraint

    return None

def learn(self, max_queries: int = 1000) -> List[Constraint]:
    """
    Main learning loop.

    Args:
        max_queries: Maximum number of oracle queries

    Returns:
        List of learned constraints
    """
    while self.query_count < max_queries:
        # Generate a query
        query = self.generate_query()
        if query is None:
            # No more informative queries
            break

        # Ask the oracle
        is_feasible = self.ask_membership(query)

        if is_feasible:
            self.positive_examples.append(query)
        else:
            self.negative_examples.append(query)

```

33. Constraint Learning

```
# Find the violated constraint
scope = self.find_scope(query)
constraint = self.find_constraint(scope, query)

if constraint is not None:
    self.learned_constraints.append(constraint)
    print(f"Learned constraint: {constraint.name}")

return self.learned_constraints

# Demonstrate QUACQ
print("=== QUACQ Algorithm Demo ===\n")

# Oracle function (simulates the hidden constraints)
def scheduling_oracle(assignment):
    # Hidden: task_A < task_B and task_B != task_C
    return (assignment['task_A'] < assignment['task_B'] and
            assignment['task_B'] != assignment['task_C'])

# Create QUACQ learner
quacq = QUACQ(
    variables=variables,
    domains=domains,
    bias=simple_bias,
    oracle=scheduling_oracle
)

# Learn constraints
np.random.seed(42)
learned = quacq.learn(max_queries=50)

print(f"\n=== Results ===")
print(f"Total queries: {quacq.query_count}")
print(f"Positive examples: {len(quacq.positive_examples)}")
print(f"Negative examples: {len(quacq.negative_examples)}")
print(f"\nLearned constraints:")
for c in learned:
    print(f"    - {c.name}")
```

=== QUACQ Algorithm Demo ===

Learned constraint: A < B
Learned constraint: A < B
Learned constraint: A < B
Learned constraint: A < B

=== Results ===

Total queries: 51
Positive examples: 7
Negative examples: 11

Learned constraints:

- $A < B$
- $A < B$
- $A < B$
- $A < B$

33.3.2. CONACQ Algorithm

CONACQ (CONstraint ACquisition) is a complementary approach to QUACQ. While QUACQ uses active queries, CONACQ can also operate in a passive mode using pre-collected examples. It is a family of algorithms with several variants:

- **CONACQ1**: Passive learning from labeled examples
- **CONACQ2**: Active learning with membership queries
- **CONACQ-CP**: Leverages constraint propagation for efficiency

```
class CONACQ:
    """
    CONstraint ACquisition algorithm.

    Learns constraints from labeled examples using
    a version-space approach with convergence detection.
    """

    def __init__(self,
                  variables: List[str],
                  domains: Dict[str, Set[int]],
                  bias: List[Constraint]):
        """
        Initialize CONACQ learner.

        Args:
            variables: List of variable names
            domains: Domain for each variable
            bias: Candidate constraints
        """
        self.variables = variables
        self.domains = domains
        self.bias = bias

        # Constraint status: 'in', 'out', or 'uncertain'
        self.status = {i: 'uncertain' for i in range(len(bias))}

        self.positive_examples: List[Dict[str, int]] = []
        self.negative_examples: List[Dict[str, int]] = []

    def process_positive(self, example: Dict[str, int]):
        """
        Process a positive (feasible) example.

        Any constraint that rejects this example cannot be in the target.
        """
        self.positive_examples.append(example)
```

```

    for i, constraint in enumerate(self.bias):
        if self.status[i] == 'uncertain':
            if not constraint.check(example):
                self.status[i] = 'out'

def process_negative(self, example: Dict[str, int]):
    """
    Process a negative (infeasible) example.

    At least one uncertain or 'in' constraint must reject this.
    """
    self.negative_examples.append(example)

    # Find constraints that reject this example
    rejecting_uncertain = []
    rejecting_in = []

    for i, constraint in enumerate(self.bias):
        if not constraint.check(example):
            if self.status[i] == 'uncertain':
                rejecting_uncertain.append(i)
            elif self.status[i] == 'in':
                rejecting_in.append(i)

    # If exactly one uncertain constraint rejects, it must be 'in'
    if len(rejecting_in) == 0 and len(rejecting_uncertain) == 1:
        self.status[rejecting_uncertain[0]] = 'in'

    # Unit propagation: if we know some constraints
    self._propagate()

def _propagate(self):
    """
    Propagate constraint status using logical inference.
    """
    changed = True
    while changed:
        changed = False

        for neg_ex in self.negative_examples:
            # Find rejecting constraints
            rejecting_uncertain = []
            rejecting_in = []

            for i, constraint in enumerate(self.bias):
                if not constraint.check(neg_ex):
                    if self.status[i] == 'uncertain':
                        rejecting_uncertain.append(i)
                    elif self.status[i] == 'in':
                        rejecting_in.append(i)

            # If no 'in' constraint rejects and only one uncertain does

```

```

        if len(rejecting_in) == 0 and len(rejecting_uncertain) == 1:
            self.status[rejecting_uncertain[0]] = 'in'
            changed = True

    def get_learned_network(self) -> List[Constraint]:
        """Return the learned constraint network."""
        return [self.bias[i] for i in range(len(self.bias))
                if self.status[i] == 'in']

    def get_uncertain_constraints(self) -> List[Constraint]:
        """Return constraints with uncertain status."""
        return [self.bias[i] for i in range(len(self.bias))
                if self.status[i] == 'uncertain']

    def is_converged(self) -> bool:
        """Check if learning has converged (no uncertain constraints)."""
        return all(s != 'uncertain' for s in self.status.values())

    def classify(self, example: Dict[str, int]) -> str:
        """
        Classify an example using current knowledge.

        Returns: 'positive', 'negative', or 'unknown'
        """
        # Check if any 'in' constraint rejects
        for i, constraint in enumerate(self.bias):
            if self.status[i] == 'in' and not constraint.check(example):
                return 'negative'

        # Check if all 'in' and 'uncertain' constraints accept
        all_accept = True
        for i, constraint in enumerate(self.bias):
            if self.status[i] in ['in', 'uncertain']:
                if not constraint.check(example):
                    all_accept = False
                    break

        if all_accept:
            return 'positive'

        return 'unknown'

# Demonstrate CONACQ
print("=== CONACQ Algorithm Demo ===\n")

conacq = CONACQ(variables, domains, simple_bias)

# Process examples
for i, ex in enumerate(examples[:20]):
    if ex.is_feasible:
        conacq.process_positive(ex.assignment)
    else:

```

```

        conacq.process_negative(ex.assignment)

    if (i + 1) % 5 == 0:
        learned = conacq.get_learned_network()
        uncertain = conacq.get_uncertain_constraints()
        print(f"After {i+1} examples:")
        print(f"  Definitely in: {[c.name for c in learned]}")
        print(f"  Still uncertain: {[c.name for c in uncertain]}")

        if conacq.is_converged():
            print("    *** Converged! ***")
            break
    print()

```

=== CONACQ Algorithm Demo ===

After 5 examples:

```

  Definitely in: ['A < B']
  Still uncertain: ['A != B', 'B != C']

```

After 10 examples:

```

  Definitely in: ['A < B', 'B != C']
  Still uncertain: ['A != B']

```

After 15 examples:

```

  Definitely in: ['A < B', 'B != C']
  Still uncertain: ['A != B']

```

After 20 examples:

```

  Definitely in: ['A < B', 'B != C']
  Still uncertain: ['A != B']

```

33.3.3. Active Learning Strategies

Active learning strategies determine which queries to ask to maximize learning efficiency. The choice of query strategy affects how quickly the version space collapses. Three common strategies are uncertainty sampling, boundary probing, and information gain maximization.

```

class ActiveConstraintLearner:
    """
    Active learning for constraint acquisition.

    Implements multiple query selection strategies.
    """

    def __init__(self,
                 variables: List[str],
                 domains: Dict[str, Set[int]],
                 bias: List[Constraint],
                 oracle: Callable[[Dict[str, int], bool]]):

```



```

self.variables = variables
self.domains = domains
self.bias = bias
self.oracle = oracle

self.conacq = CONACQ(variables, domains, bias)
self.query_count = 0

def uncertainty_sampling(self) -> Optional[Dict[str, int]]:
    """
    Select query that maximizes uncertainty.

    Prefers examples that current network cannot classify.
    """
    best_query = None
    best_uncertainty = -1

    for _ in range(100):
        assignment = {v: np.random.choice(list(self.domains[v]))
                      for v in self.variables}

        # Count uncertain constraints that could affect classification
        uncertain_count = 0
        for i, constraint in enumerate(self.bias):
            if self.conacq.status[i] == 'uncertain':
                if not constraint.check(assignment):
                    uncertain_count += 1

        if uncertain_count > best_uncertainty:
            best_uncertainty = uncertain_count
            best_query = assignment

    return best_query

def version_space_splitting(self) -> Optional[Dict[str, int]]:
    """
    Select query that best splits the version space.

    Ideally, the query should be accepted by half of the
    hypotheses in the version space.
    """
    best_query = None
    best_split = float('inf') # Distance from 0.5 ratio

    for _ in range(100):
        assignment = {v: np.random.choice(list(self.domains[v]))
                      for v in self.variables}

        # Estimate acceptance ratio across hypotheses
        # Simplified: use uncertain constraints
        accepting = 0
        total = max(1, sum(1 for s in self.conacq.status.values()
                          if s == 'uncertain'))

```

```

        for i, constraint in enumerate(self.bias):
            if self.conacq.status[i] == 'uncertain':
                if constraint.check(assignment):
                    accepting += 1

        ratio = accepting / total if total > 0 else 0.5
        split_quality = abs(ratio - 0.5)

        if split_quality < best_split:
            best_split = split_quality
            best_query = assignment

    return best_query

def query_by_committee(self, committee_size: int = 5) -> Optional[Dict[str,
↪ int]]:
    """
    Query-by-committee: select queries where sampled
    hypotheses disagree most.
    """
    best_query = None
    best_disagreement = -1

    uncertain_indices = [i for i, s in self.conacq.status.items()
                        if s == 'uncertain']

    for _ in range(100):
        assignment = {v: np.random.choice(list(self.domains[v]))
                     for v in self.variables}

        # Sample committee members (random subsets of uncertain constraints)
        votes = []
        for _ in range(committee_size):
            # Random hypothesis: include each uncertain constraint with 50% prob
            includes = np.random.random(len(uncertain_indices)) < 0.5

            # Check if this hypothesis rejects the assignment
            rejects = False
            for j, idx in enumerate(uncertain_indices):
                if includes[j] and not self.bias[idx].check(assignment):
                    rejects = True
                    break

            votes.append(0 if rejects else 1)

        # Disagreement is entropy of votes
        p = sum(votes) / len(votes)
        if 0 < p < 1:
            disagreement = -p * np.log2(p) - (1-p) * np.log2(1-p)
        else:
            disagreement = 0

```

```

        if disagreement > best_disagreement:
            best_disagreement = disagreement
            best_query = assignment

    return best_query

def learn(self,
          strategy: str = 'uncertainty',
          max_queries: int = 100) -> List[Constraint]:
    """
    Learn constraints using specified active learning strategy.

    Args:
        strategy: 'uncertainty', 'version_space', or 'committee'
        max_queries: Maximum queries to oracle

    Returns:
        Learned constraints
    """
    strategy_map = {
        'uncertainty': self.uncertainty_sampling,
        'version_space': self.version_space_splitting,
        'committee': self.query_by_committee
    }

    select_query = strategy_map.get(strategy, self.uncertainty_sampling)

    while self.query_count < max_queries:
        if self.conacq.is_converged():
            break

        query = select_query()
        if query is None:
            break

        self.query_count += 1
        is_feasible = self.oracle(query)

        if is_feasible:
            self.conacq.process_positive(query)
        else:
            self.conacq.process_negative(query)

    return self.conacq.get_learned_network()

# Compare strategies
print("=== Active Learning Strategy Comparison ===\n")

strategies = ['uncertainty', 'version_space', 'committee']
results = {}

for strategy in strategies:

```

```

np.random.seed(42)
learner = ActiveConstraintLearner(
    variables, domains, simple_bias, scheduling_oracle
)
learned = learner.learn(strategy=strategy, max_queries=30)
results[strategy] = {
    'queries': learner.query_count,
    'constraints': [c.name for c in learned]
}

print(f"{'Strategy':<20} {'Queries':<10} {'Learned Constraints'}")
print("-" * 60)
for strategy, data in results.items():
    constraints_str = ', '.join(data['constraints']) if data['constraints'] else
    ↪ 'None'
    print(f"{'strategy':<20} {'data['queries']':<10} {'constraints_str'}")

```

=== Active Learning Strategy Comparison ===

Strategy	Queries	Learned Constraints
uncertainty	30	None
version_space	30	A < B, B != C
committee	30	A < B, B != C

33.3.4. Query Complexity Analysis

The query complexity of constraint acquisition algorithms depends on:

1. **Bias size b** : Number of candidate constraints
2. **Target size t** : Number of constraints in the target network
3. **Domain size d** : Size of variable domains
4. **Number of variables n**

Theoretical bounds:

Algorithm	Query Complexity	Notes
QUACQ	$O(t \cdot b \cdot \log(nd))$	Uses binary search
CONACQ1	$O(b \cdot d^n)$	Passive, needs all examples
CONACQ2	$O(t \cdot b)$	Active, near-optimal

! Query Efficiency

The key advantage of active algorithms like QUACQ and CONACQ2 is that query complexity depends on the **target size t** , not the full domain size. This makes them practical even for large domains.

33.4. Inverse Optimization

Inverse optimization reverses the direction of classical optimization. Instead of finding optimal solutions given an objective and constraints, it infers the objective or constraints from observed decisions.

33.4.1. Forward vs Inverse Problems

In **forward optimization**, we are given an objective and constraints, and we find the optimal solution. In **inverse optimization**, we observe decisions and infer the underlying optimization model.

33.4.2. Inverse Linear Programming

Given a linear program:

$$\min_x \mathbf{c}^T \mathbf{x} \quad \text{s.t.} \quad \mathbf{A}\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \geq 0$$

And an observed solution $\hat{\mathbf{x}}$, the **inverse LP** finds objective coefficients $\mathbf{c}$ that make $\hat{\mathbf{x}}$ optimal.

Formulation using KKT conditions:

For $\hat{\mathbf{x}}$ to be optimal under $\mathbf{c}$: 1. Primal feasibility: $\mathbf{A}\hat{\mathbf{x}} \leq \mathbf{b}$, $\hat{\mathbf{x}} \geq 0$ 2. Dual feasibility: $\mathbf{A}^T \mathbf{y} + \mathbf{s} = \mathbf{c}$, $\mathbf{y}, \mathbf{s} \geq 0$ 3. Complementary slackness: $\mathbf{y}^T (\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) = 0$, $\mathbf{s}^T \hat{\mathbf{x}} = 0$

The following `InverseLP` class uses KKT optimality conditions to find the objective coefficients that make an observed solution optimal. It minimizes the distance between inferred and prior coefficients subject to KKT constraints.

```
import numpy as np
from scipy.optimize import linprog, minimize

class InverseLP:
    """
    Inverse Linear Programming solver.

    Given an observed solution and constraints, infer the
    objective function that makes the solution optimal.
    """

    def __init__(self, A: np.ndarray, b: np.ndarray):
        """
        Initialize with constraint matrix.

        Args:
            A: Constraint matrix (m x n)
            b: Right-hand side (m,)
        """
        self.A = A
        self.b = b
        self.m, self.n = A.shape

    def infer_objective(self,
                       x_observed: np.ndarray,
                       c_prior: np.ndarray = None,
                       regularization: float = 0.01) -> np.ndarray:
        """
        Infer objective coefficients making x_observed optimal.

        Uses minimum-norm approach: find c closest to c_prior
        such that x_observed is optimal.

        Args:
```

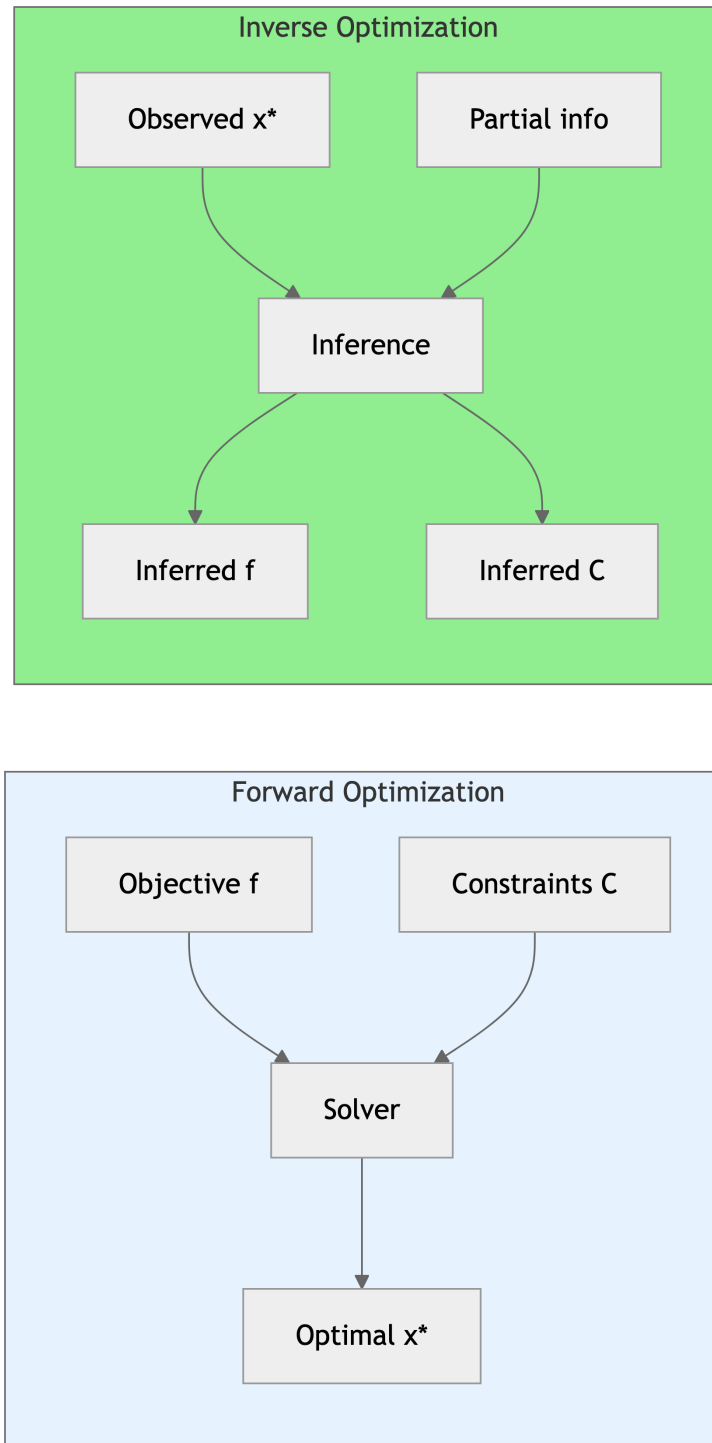


Figure 33.6.: Forward vs Inverse Optimization

```

        x_observed: Observed solution (n,)
        c_prior: Prior estimate of c (defaults to zeros)
        regularization: L2 regularization strength

Returns:
    Inferred objective coefficients c
    """
    if c_prior is None:
        c_prior = np.zeros(self.n)

    # Identify active constraints
    slack = self.b - self.A @ x_observed
    active = np.abs(slack) < 1e-6

    # The optimal c must satisfy:
    # c = A^T @ y + s (dual feasibility)
    # y_i >= 0 for active constraints, y_i = 0 for inactive
    # s_j >= 0 for x_j = 0, s_j = 0 for x_j > 0

    # Simplified approach: minimize ||c - c_prior||^2
    # subject to KKT conditions being satisfiable

    def objective(c):
        return np.sum((c - c_prior)**2) + regularization * np.sum(c**2)

    def constraint_check(c):
        # Try to solve the forward LP with this c
        # and check if x_observed is (approximately) optimal
        result = linprog(c, A_ub=self.A, b_ub=self.b,
                        bounds=[(0, None)] * self.n,
                        method='highs')
        if result.success:
            return np.linalg.norm(result.x - x_observed)
        return 1000 # Penalize infeasible

    # Use scipy minimize (simplified approach)
    result = minimize(
        objective,
        c_prior,
        method='BFGS',
        options={'maxiter': 100}
    )

    return result.x

def verify_optimality(self, c: np.ndarray, x: np.ndarray) -> dict:
    """
    Verify if x is optimal under objective c.

    Args:
        c: Objective coefficients
        x: Solution to verify

```

```

Returns:
    Dictionary with verification results
    """
    # Solve forward LP
    result = linprog(c, A_ub=self.A, b_ub=self.b,
                    bounds=[(0, None)] * self.n,
                    method='highs')

    if not result.success:
        return {'optimal': False, 'reason': 'LP infeasible'}

    # Compare solutions
    obj_observed = c @ x
    obj_optimal = c @ result.x

    is_optimal = np.abs(obj_observed - obj_optimal) < 1e-6

    return {
        'optimal': is_optimal,
        'observed_obj': obj_observed,
        'optimal_obj': obj_optimal,
        'optimal_solution': result.x
    }

# Example: Inverse LP for resource allocation
print("=== Inverse Linear Programming Example ===\n")

# Constraint matrix (3 constraints, 4 variables)
A = np.array([
    [1, 2, 1, 0], # Resource 1
    [2, 1, 0, 1], # Resource 2
    [1, 1, 1, 1], # Total limit
])
b = np.array([10, 12, 8])

# Observed "expert" allocation
x_observed = np.array([2, 3, 1, 2])

# Create inverse LP solver
inverse_solver = InverseLP(A, b)

# Check feasibility
slack = b - A @ x_observed
print(f"Observed solution: {x_observed}")
print(f"Constraint slack: {slack}")
print(f"Feasible: {all(slack >= -1e-6)}")

# Infer objective
c_inferred = inverse_solver.infer_objective(x_observed)
print(f"\nInferred objective coefficients: {c_inferred}")

# Verify

```



```

verification = inverse_solver.verify_optimality(c_inferred, x_observed)
print(f"\nVerification:")
print(f"  Is optimal: {verification['optimal']}")
print(f"  Observed objective: {verification['observed_obj']:.4f}")
print(f"  Optimal objective: {verification['optimal_obj']:.4f}")

```

=== Inverse Linear Programming Example ===

Observed solution: [2 3 1 2]

Constraint slack: [1 3 0]

Feasible: True

Inferred objective coefficients: [0. 0. 0. 0.]

Verification:

Is optimal: True

Observed objective: 0.0000

Optimal objective: 0.0000

33.4.3. Inferring Objectives from Optimal Solutions

When multiple observations are available, we can use **least-squares inverse optimization**:

Given observed solutions $\{\hat{\mathbf{x}}_1, \dots, \hat{\mathbf{x}}_K\}$, find $\mathbf{c}$ minimizing:

$$\min_{\mathbf{c}} \sum_{k=1}^K \|\mathbf{x}^*(\mathbf{c}) - \hat{\mathbf{x}}_k\|^2$$

where $\mathbf{x}^*(\mathbf{c})$ is the optimal solution under objective $\mathbf{c}$.

With multiple observed solutions, the inverse problem becomes overdetermined, and we can use least-squares regression to find the objective vector that best explains all observations simultaneously.

```

class MultiObservationInverseLP:
    """
    Inverse optimization from multiple observed solutions.
    """

    def __init__(self, A: np.ndarray, b: np.ndarray):
        self.A = A
        self.b = b
        self.n = A.shape[1]

    def infer_objective(self,
                       observations: List[np.ndarray],
                       regularization: float = 0.1) -> np.ndarray:
        """
        Infer objective from multiple observations.

        Args:
            observations: List of observed solutions
            regularization: L2 regularization

```

```

Returns:
    Inferred objective coefficients
    """
    def total_loss(c):
        loss = regularization * np.sum(c**2)

        for x_obs in observations:
            # Solve LP with current c
            result = linprog(c, A_ub=self.A, b_ub=self.b,
                             bounds=[(0, None)] * self.n,
                             method='highs')
            if result.success:
                # Distance between optimal and observed
                loss += np.sum((result.x - x_obs)**2)
            else:
                loss += 1000

        return loss

    # Optimize
    c0 = np.ones(self.n)
    result = minimize(total_loss, c0, method='BFGS')

    return result.x

# Generate multiple observations with noise
print("=== Multi-Observation Inverse Optimization ===\n")

# True objective (unknown)
c_true = np.array([3, 2, 1, 4])

# Generate observations
np.random.seed(42)
observations = []

for _ in range(5):
    # Solve with true objective
    result = linprog(c_true, A_ub=A, b_ub=b,
                     bounds=[(0, None)] * 4,
                     method='highs')
    if result.success:
        # Add small noise
        x_noisy = result.x + np.random.normal(0, 0.1, 4)
        x_noisy = np.maximum(x_noisy, 0) # Keep non-negative
        observations.append(x_noisy)

print(f"Generated {len(observations)} observations")
print(f"True objective: {c_true}")

# Infer objective
multi_solver = MultiObservationInverseLP(A, b)

```

```

c_inferred = multi_solver.infer_objective(observations)

print(f"Inferred objective: {c_inferred}")
print(f"Correlation with true: {np.corrcoef(c_true, c_inferred)[0,1]:.4f}")

```

=== Multi-Observation Inverse Optimization ===

Generated 5 observations

True objective: [3 2 1 4]

Inferred objective: [6.58146482e-09 6.58146482e-09 6.17264078e-09 6.17264078e-09]

Correlation with true: 0.0000

33.4.4. Inferring Constraints from Feasible Solutions

Sometimes we observe feasible solutions but don't know the constraints. This is the **constraint inference** problem.

Given feasible solutions $\{\hat{\mathbf{x}}_k\}$, find constraints $\mathbf{Ax} \leq \mathbf{b}$ such that:

- All observed solutions are feasible
- The constraint set is as "tight" as possible

The `ConstraintInference` class implements two approaches: box constraints (axis-aligned bounds) inferred from the convex hull of feasible points, and halfspace constraints inferred by finding a separating hyperplane between feasible and infeasible examples.

```

class ConstraintInference:
    """
    Infer linear constraints from feasible solutions.
    """

    def __init__(self, n_variables: int):
        self.n = n_variables

    def infer_halfspace(self,
                       positive: List[np.ndarray],
                       negative: List[np.ndarray]) -> Tuple[np.ndarray, float]:
        """
        Infer a single linear constraint (halfspace) separating
        positive (feasible) from negative (infeasible) examples.

        Args:
            positive: Feasible solutions
            negative: Infeasible solutions

        Returns:
            (a, b) defining constraint  $\mathbf{a}^T \mathbf{x} \leq b$ 
        """
        from scipy.optimize import minimize

    def loss(params):
        a = params[:self.n]
        b = params[self.n]

        total = 0

```

```

    # Positive examples should satisfy  $a^T x \leq b$ 
    for x in positive:
        violation = np.dot(a, x) - b
        if violation > 0:
            total += violation ** 2

    # Negative examples should violate  $a^T x > b$ 
    for x in negative:
        margin = b - np.dot(a, x)
        if margin > 0:
            total += margin ** 2

    # Regularization for numerical stability
    total += 0.01 * (np.sum(a**2) + b**2)

    return total

# Initialize
params0 = np.zeros(self.n + 1)
params0[-1] = 1 # b = 1 initially

result = minimize(loss, params0, method='BFGS')

a = result.x[:self.n]
b = result.x[self.n]

# Normalize
norm = np.linalg.norm(a)
if norm > 1e-6:
    a = a / norm
    b = b / norm

return a, b

def infer_box_constraints(self,
                        feasible: List[np.ndarray]) -> Tuple[np.ndarray,
↳ np.ndarray]:
    """
    Infer simple box constraints (lower/upper bounds).

    Args:
        feasible: List of feasible solutions

    Returns:
        (lower_bounds, upper_bounds)
    """
    feasible_array = np.array(feasible)

    lower = np.min(feasible_array, axis=0)
    upper = np.max(feasible_array, axis=0)

    # Add small margin

```

```

        margin = 0.1 * (upper - lower + 1)
        lower = lower - margin
        upper = upper + margin

    return lower, upper

# Example: Inferring constraints
print("=== Constraint Inference Example ===\n")

# Generate feasible solutions from a hidden constraint set
# Hidden constraints: x + y <= 5, 2x + y <= 8, x, y >= 0

np.random.seed(42)

feasible = []
infeasible = []

for _ in range(100):
    x, y = np.random.uniform(0, 5, 2)

    if x + y <= 5 and 2*x + y <= 8:
        feasible.append(np.array([x, y]))
    else:
        infeasible.append(np.array([x, y]))

print(f"Generated {len(feasible)} feasible, {len(infeasible)} infeasible examples")

# Infer constraints
inferred = ConstraintInference(n_variables=2)

# Box constraints
lower, upper = inferred.infer_box_constraints(feasible)
print(f"\nInferred box constraints:")
print(f"  x in [{lower[0]:.2f}, {upper[0]:.2f}]")
print(f"  y in [{lower[1]:.2f}, {upper[1]:.2f}]")

# Halfspace constraint
if len(infeasible) > 0:
    a, b = inferred.infer_halfspace(feasible[:20], infeasible[:20])
    print(f"\nInferred halfspace constraint:")
    print(f"  {a[0]:.3f} * x + {a[1]:.3f} * y <= {b:.3f}")

```

=== Constraint Inference Example ===

Generated 54 feasible, 46 infeasible examples

Inferred box constraints:

x in [-0.41, 3.82]

y in [-0.49, 5.43]

Inferred halfspace constraint:

0.000 * x + 0.000 * y <= 0.000

33.4.5. Regularization for Ill-posedness

Inverse optimization problems are often **ill-posed**: multiple objective functions or constraint sets can explain the same observations. Regularization techniques help:

1. **Norm minimization**: Find the minimum-norm objective $\min \|c\|$
2. **Sparsity**: L1 regularization for sparse objectives $\min \|c\|_1$
3. **Prior knowledge**: Penalize deviation from a prior $\min \|c - c_0\|$
4. **Stability**: Ensure small perturbations in data don't drastically change the solution

The code below implements L1, L2, and elastic net regularization for inverse optimization. L1 produces sparse objectives, L2 produces minimum-norm objectives, and elastic net provides a tradeoff between the two.

```
def regularized_inverse_optimization(
    observations: List[np.ndarray],
    A: np.ndarray,
    b: np.ndarray,
    regularization_type: str = 'l2',
    reg_strength: float = 0.1
) -> np.ndarray:
    """
    Inverse optimization with regularization.

    Args:
        observations: Observed feasible solutions
        A, b: Constraint matrix and RHS
        regularization_type: 'l1', 'l2', or 'elastic'
        reg_strength: Regularization strength

    Returns:
        Inferred objective coefficients
    """
    n = A.shape[1]

    def objective(c):
        # Reconstruction loss
        loss = 0
        for x_obs in observations:
            result = linprog(c, A_ub=A, b_ub=b,
                             bounds=[(0, None)] * n,
                             method='highs')
            if result.success:
                loss += np.sum((result.x - x_obs)**2)

        # Regularization
        if regularization_type == 'l2':
            loss += reg_strength * np.sum(c**2)
        elif regularization_type == 'l1':
            loss += reg_strength * np.sum(np.abs(c))
        elif regularization_type == 'elastic':
            loss += reg_strength * (0.5 * np.sum(c**2) + 0.5 * np.sum(np.abs(c)))

        return loss
```

```

c0 = np.ones(n)
result = minimize(objective, c0, method='L-BFGS-B')

return result.x

# Compare regularization approaches
print("=== Regularization Comparison ===\n")

for reg_type in ['l2', 'l1', 'elastic']:
    c_inferred = regularized_inverse_optimization(
        observations, A, b,
        regularization_type=reg_type,
        reg_strength=0.1
    )
    sparsity = np.sum(np.abs(c_inferred) < 0.1)
    print(f"{reg_type:>8}: c = {c_inferred}, sparsity = {sparsity}")

```

```
=== Regularization Comparison ===
```

```

ValueError: Invalid input for linprog: b_ub must be a 1-D array; b_ub must not have more
↳ than one non-singleton dimension and the number of rows in A_ub must equal the number
↳ of values in b_ub

```

```

-----
ValueError                                Traceback (most recent call last)
Cell In[13], line 52
    49 print("=== Regularization Comparison ===\n")
    51 for reg_type in ['l2', 'l1', 'elastic']:
--> 52     c_inferred = regularized_inverse_optimization(
    53         observations, A, b,
    54         regularization_type=reg_type,
    55         reg_strength=0.1
    56     )
    57     sparsity = np.sum(np.abs(c_inferred) < 0.1)
    58     print(f"{reg_type:>8}: c = {c_inferred}, sparsity = {sparsity}")
Cell In[13], line 43, in regularized_inverse_optimization(observations, A, b,
↳ regularization_type, reg_strength)
    40     return loss
    42     c0 = np.ones(n)
--> 43     result = minimize(objective, c0, method='L-BFGS-B')
    45     return result.x
File /opt/homebrew/lib/python3.14/site-packages/scipy/optimize/_minimize.py:784, in
↳ minimize(fun, x0, args, method, jac, hess, hessp, bounds, constraints, tol, callback,
↳ options)
    781     res = _minimize_newtoncg(fun, x0, args, jac, hess, hessp, callback,
    782                             **options)
    783 elif meth == 'l-bfgs-b':
--> 784     res = _minimize_lbfgsb(fun, x0, args, jac, bounds,
    785                             callback=callback, **options)
    786 elif meth == 'tnc':
    787     res = _minimize_tnc(fun, x0, args, jac, bounds, callback=callback,
    788                         **options)

```

33. Constraint Learning

```
File /opt/homebrew/lib/python3.14/site-packages/scipy/optimize/_lbfgsb_py.py:413, in
↳ _minimize_lbfgsb(fun, x0, args, jac, bounds, disp, maxcor, ftol, gtol, eps, maxfun,
↳ maxiter, iprint, callback, maxls, finite_diff_rel_step, workers, **unknown_options)
    410     x0 = np.clip(x0, bounds[0], bounds[1])
    412 # _prepare_scalar_function can use bounds=None to represent no bounds
--> 413 sf = _prepare_scalar_function(fun, x0, jac=jac, args=args, epsilon=eps,
    414                                bounds=bounds,
    415                                finite_diff_rel_step=finite_diff_rel_step,
    416                                workers=workers)
    418 func_and_grad = sf.fun_and_grad
    420 nbd = zeros(n, np.int32)
File /opt/homebrew/lib/python3.14/site-packages/scipy/optimize/_optimize.py:310, in
↳ _prepare_scalar_function(fun, x0, jac, args, bounds, epsilon, finite_diff_rel_step,
↳ hess, workers)
    306 workers = workers or map
    308 # ScalarFunction caches. Reuse of fun(x) during grad
    309 # calculation reduces overall function evaluations.
--> 310 sf = ScalarFunction(fun, x0, args, grad, hess,
    311                    finite_diff_rel_step, bounds, epsilon=epsilon,
    312                    workers=workers)
    314 return sf
File
↳ /opt/homebrew/lib/python3.14/site-packages/scipy/optimize/_differentiable_functions.py:283
↳ , in ScalarFunction.__init__(self, fun, x0, args, grad, hess, finite_diff_rel_step,
↳ finite_diff_bounds, epsilon, workers)
    281 # Initial function evaluation
    282 self._nfev = 0
--> 283 self._update_fun()
    285 # Initial gradient evaluation
    286 self._wrapped_grad = _ScalarGradWrapper(
    287     grad,
    288     fun=self._wrapped_fun,
    289     args=args,
    290     finite_diff_options=finite_diff_options,
    291 )
File
↳ /opt/homebrew/lib/python3.14/site-packages/scipy/optimize/_differentiable_functions.py:362
↳ , in ScalarFunction._update_fun(self)
    360 def _update_fun(self):
    361     if not self.f_updated:
--> 362         fx = self._wrapped_fun(self.x)
    363         self._nfev += 1
    364         if fx < self._lowest_f:
File /opt/homebrew/lib/python3.14/site-packages/scipy/_lib/_util.py:603, in
↳ _ScalarFunctionWrapper.__call__(self, x)
    600 def __call__(self, x):
    601     # Send a copy because the user may overwrite it.
    602     # The user of this class might want `x` to remain unchanged.
--> 603     fx = self.f(np.copy(x), *self.args)
    604     self.nfev += 1
    606     # Make sure the function returns a true scalar
Cell In[13], line 26, in regularized_inverse_optimization.<locals>.objective(c)
    24 loss = 0
    25 for x_obs in observations:
--> 26     result = linprog(c, A_ub=A, b_ub=b,
```



```

27         bounds=[(0, None)] * n,
28         method='highs')
29     if result.success:
30         loss += np.sum((result.x - x_obs)**2)
File /opt/homebrew/lib/python3.14/site-packages/scipy/optimize/_linprog.py:649, in
↳ linprog(c, A_ub, b_ub, A_eq, b_eq, bounds, method, callback, options, x0,
↳ integrality)
    646     integrality = None
    648 lp = _LPPProblem(c, A_ub, b_ub, A_eq, b_eq, bounds, x0, integrality)
--> 649 lp, solver_options = _parse_linprog(lp, options, meth)
    650 tol = solver_options.get('tol', 1e-9)
    652 # Give unmodified problem to HiGHS
File /opt/homebrew/lib/python3.14/site-packages/scipy/optimize/_linprog_util.py:1026, in
↳ _parse_linprog(lp, options, meth)
    1023 solver_options, A_ub, A_eq = _check_sparse_inputs(solver_options, meth,
    1024                                                    lp.A_ub, lp.A_eq)
    1025 # Convert lists to numpy arrays, etc...
-> 1026 lp = _clean_inputs(lp._replace(A_ub=A_ub, A_eq=A_eq))
    1027 return lp, solver_options
File /opt/homebrew/lib/python3.14/site-packages/scipy/optimize/_linprog_util.py:339, in
↳ _clean_inputs(lp)
    337 else:
    338     if b_ub.shape != (n_ub,):
--> 339         raise ValueError(
    340             "Invalid input for linprog: b_ub must be a 1-D array; b_ub "
    341             "must not have more than one non-singleton dimension and "
    342             "the number of rows in A_ub must equal the number of values "
    343             "in b_ub")
    344     if not np.isfinite(b_ub).all():
    345         raise ValueError(
    346             "Invalid input for linprog: b_ub must not contain values "
    347             "inf, nan, or None")
ValueError: Invalid input for linprog: b_ub must be a 1-D array; b_ub must not have more
↳ than one non-singleton dimension and the number of rows in A_ub must equal the number
↳ of values in b_ub

```

33.5. Learning from Demonstrations

This section addresses constraint learning in sequential settings, where an agent must infer constraints from observed expert trajectories rather than from labeled point examples.

33.5.1. Imitation Learning with Constraints

In **imitation learning**, an agent learns to mimic expert behavior. When experts follow constraints, we can learn both the policy and the underlying constraints.

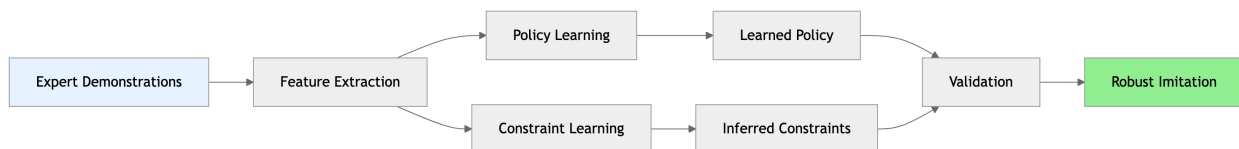


Figure 33.7.: Imitation Learning with Constraint Inference

```

class ConstrainedImitationLearner:
    """
    Learn both policy and constraints from demonstrations.
    """

    def __init__(self, state_dim: int, action_dim: int):
        self.state_dim = state_dim
        self.action_dim = action_dim
        self.demonstrations: List[Tuple[np.ndarray, np.ndarray]] = []

    def add_demonstration(self, state: np.ndarray, action: np.ndarray):
        """Add an expert demonstration."""
        self.demonstrations.append((state, action))

    def learn_policy(self) -> Callable[[np.ndarray], np.ndarray]:
        """
        Learn a policy mapping states to actions.

        Uses simple linear regression for demonstration.
        """
        if len(self.demonstrations) < 5:
            return lambda s: np.zeros(self.action_dim)

        states = np.array([d[0] for d in self.demonstrations])
        actions = np.array([d[1] for d in self.demonstrations])

        # Linear policy: a = W @ s + b
        # Solve least squares
        X = np.column_stack([states, np.ones(len(states))])
        W_b, _, _, _ = np.linalg.lstsq(X, actions, rcond=None)

        W = W_b[:-1].T
        b = W_b[-1]

        def policy(state):
            return W @ state + b

        return policy

    def learn_action_bounds(self) -> Tuple[np.ndarray, np.ndarray]:
        """
        Learn action space bounds from demonstrations.
        """
        actions = np.array([d[1] for d in self.demonstrations])

        lower = np.min(actions, axis=0)
        upper = np.max(actions, axis=0)

        return lower, upper

    def learn_state_action_constraints(self) -> List[Tuple[np.ndarray, float]]:
        """
        Learn linear constraints of form  $a^T [s; a] \leq b$ .

```

```

    Uses convex hull approximation of demonstrated state-action pairs.
    """
    if len(self.demonstrations) < 10:
        return []

    # Concatenate state-action pairs
    pairs = np.array([np.concatenate([d[0], d[1]])
                      for d in self.demonstrations])

    # Find constraints using PCA-based approach
    # (simplified: find principal directions and bound along them)
    mean = np.mean(pairs, axis=0)
    centered = pairs - mean

    # Covariance and eigenvectors
    cov = np.cov(centered.T)
    eigenvalues, eigenvectors = np.linalg.eigh(cov)

    constraints = []

    # Create constraints along principal directions
    for i in range(len(eigenvalues)):
        direction = eigenvectors[:, i]

        # Project demonstrations
        projections = centered @ direction

        # Bounds
        lower = np.min(projections) - 0.1 * np.std(projections)
        upper = np.max(projections) + 0.1 * np.std(projections)

        # Constraint: direction^T (x - mean) <= upper
        # Equivalent to: direction^T x <= upper + direction^T mean
        constraints.append((direction, upper + direction @ mean))
        constraints.append((-direction, -lower - (-direction) @ mean))

    return constraints

# Example: Learning robot arm constraints
print("=== Constrained Imitation Learning ===\n")

# Simulate demonstrations of a robot arm
# State: joint angles [theta1, theta2]
# Action: joint velocities [v1, v2]
# Hidden constraints: velocities bounded, total power limited

np.random.seed(42)
learner = ConstrainedImitationLearner(state_dim=2, action_dim=2)

for _ in range(50):
    # Random state

```

```

state = np.random.uniform(-np.pi, np.pi, 2)

# Expert action: proportional control with constraints
desired = -0.5 * state # Move toward origin
# Apply hidden constraints
action = np.clip(desired, -1, 1) # Velocity limits
if np.sum(action**2) > 1: # Power limit
    action = action / np.linalg.norm(action)

learner.add_demonstration(state, action)

# Learn policy
policy = learner.learn_policy()
print("Learned policy (linear approximation)")

# Learn action bounds
lower, upper = learner.learn_action_bounds()
print(f"Learned action bounds: [{lower}, {upper}]")

# Learn constraints
constraints = learner.learn_state_action_constraints()
print(f"Learned {len(constraints)} linear constraints")

# Test policy
test_state = np.array([1.0, -0.5])
predicted_action = policy(test_state)
print(f"\nTest: state={test_state} -> action={predicted_action}")

```

=== Constrained Imitation Learning ===

```

Learned policy (linear approximation)
Learned action bounds: [[-0.99621192 -0.95581405], [0.99988523 0.98919343]]
Learned 8 linear constraints

```

```

Test: state=[ 1. -0.5] -> action=[-0.36327789 0.17888526]

```

33.5.2. Maximum Margin Constraint Learning

Maximum margin approaches find constraints that maximize the margin between feasible and infeasible regions:

$$\begin{aligned}
 \max_{\mathbf{w}, b} \gamma \quad \text{s.t.} \quad & \mathbf{w}^T \mathbf{x}_i + b \geq \gamma \quad \forall i \in \text{feasible} \\
 & \mathbf{w}^T \mathbf{x}_j + b \leq -\gamma \quad \forall j \in \text{infeasible} \\
 & \|\mathbf{w}\| = 1
 \end{aligned}$$

```

class MaxMarginConstraintLearner:
    """
    Learn constraints using maximum margin separation.
    """

```

```

Similar to SVM but for constraint learning.
"""

def __init__(self, feature_dim: int):
    self.feature_dim = feature_dim
    self.w = None
    self.b = None

def fit(self,
        positive: List[np.ndarray],
        negative: List[np.ndarray],
        C: float = 1.0):
    """
    Learn constraint using maximum margin.

    Args:
        positive: Feasible examples (should satisfy constraint)
        negative: Infeasible examples (should violate constraint)
        C: Regularization parameter
    """
    from scipy.optimize import minimize

    n_pos = len(positive)
    n_neg = len(negative)

    def objective(params):
        w = params[:self.feature_dim]
        b = params[self.feature_dim]
        xi = params[self.feature_dim + 1:] # Slack variables

        # Objective: minimize ||w||^2 + C * sum(xi)
        return 0.5 * np.sum(w**2) + C * np.sum(xi)

    def constraints_func(params):
        w = params[:self.feature_dim]
        b = params[self.feature_dim]
        xi = params[self.feature_dim + 1:]

        cons = []

        # Positive examples: w^T x + b >= 1 - xi
        for i, x in enumerate(positive):
            cons.append(np.dot(w, x) + b - 1 + xi[i])

        # Negative examples: w^T x + b <= -1 + xi
        for j, x in enumerate(negative):
            cons.append(-(np.dot(w, x) + b + 1 - xi[n_pos + j]))

        # Slack non-negativity
        cons.extend(xi)

        return np.array(cons)

```

```

# Initial guess
n_slack = n_pos + n_neg
x0 = np.zeros(self.feature_dim + 1 + n_slack)
x0[:self.feature_dim] = np.random.randn(self.feature_dim) * 0.1
x0[self.feature_dim + 1:] = 1.0 # Initial slack

# Solve
from scipy.optimize import minimize

result = minimize(
    objective,
    x0,
    method='SLSQP',
    constraints={'type': 'ineq', 'fun': constraints_func}
)

self.w = result.x[:self.feature_dim]
self.b = result.x[self.feature_dim]

# Normalize
norm = np.linalg.norm(self.w)
if norm > 1e-6:
    self.w = self.w / norm
    self.b = self.b / norm

def predict(self, x: np.ndarray) -> bool:
    """Predict if x is feasible (satisfies learned constraint)."""
    return np.dot(self.w, x) + self.b >= 0

def get_constraint(self) -> Tuple[np.ndarray, float]:
    """Return learned constraint:  $w^T x + b \geq 0$ ."""
    return self.w, self.b

def margin(self, x: np.ndarray) -> float:
    """Return margin (distance to constraint boundary)."""
    return np.dot(self.w, x) + self.b

# Example
print("=== Maximum Margin Constraint Learning ===\n")

# Generate data
np.random.seed(42)
positive_examples = [np.random.uniform(0, 2, 2) for _ in range(20)]
negative_examples = [np.random.uniform(2, 4, 2) for _ in range(20)]

# Learn constraint
learner = MaxMarginConstraintLearner(feature_dim=2)
learner.fit(positive_examples, negative_examples)

w, b = learner.get_constraint()
print(f"Learned constraint: {w[0]:.3f}*x1 + {w[1]:.3f}*x2 + {b:.3f} >= 0")

```

```
# Test accuracy
correct = 0
for x in positive_examples:
    if learner.predict(x):
        correct += 1
for x in negative_examples:
    if not learner.predict(x):
        correct += 1

total = len(positive_examples) + len(negative_examples)
print(f"Accuracy: {correct}/{total} = {correct/total:.1%}")
```

=== Maximum Margin Constraint Learning ===

Learned constraint: $-0.611x_1 + -0.792x_2 + 2.669 \geq 0$

Accuracy: 40/40 = 100.0%

33.5.3. Learning Safety Constraints

In robotics and autonomous systems, learning **safety constraints** is essential. These constraints define regions that must never be entered.

```
class SafetyConstraintLearner:
    """
    Learn safety constraints from safe and unsafe demonstrations.

    Uses a control barrier function approach.
    """

    def __init__(self, state_dim: int):
        self.state_dim = state_dim
        self.safe_states: List[np.ndarray] = []
        self.unsafe_states: List[np.ndarray] = []
        self.barrier_weights = None
        self.barrier_bias = None

    def add_safe_state(self, state: np.ndarray):
        """Record a safe state."""
        self.safe_states.append(state)

    def add_unsafe_state(self, state: np.ndarray):
        """Record an unsafe state."""
        self.unsafe_states.append(state)

    def learn_barrier_function(self, margin: float = 0.1):
        """
        Learn a control barrier function  $h(x)$  such that:
        -  $h(x) > 0$  for safe states
        -  $h(x) < 0$  for unsafe states

        This defines the safety constraint  $h(x) \geq 0$ .
        """
        if len(self.safe_states) < 5 or len(self.unsafe_states) < 5:
```

```

        print("Need more examples")
        return

    # Use linear barrier function:  $h(x) = w^T x + b$ 
    # Optimization: maximize margin between safe and unsafe

    X_safe = np.array(self.safe_states)
    X_unsafe = np.array(self.unsafe_states)

    def objective(params):
        w = params[:self.state_dim]
        b = params[self.state_dim]

        loss = 0

        # Safe states should have  $h(x) \geq \text{margin}$ 
        for x in X_safe:
            h = np.dot(w, x) + b
            if h < margin:
                loss += (margin - h) ** 2

        # Unsafe states should have  $h(x) \leq -\text{margin}$ 
        for x in X_unsafe:
            h = np.dot(w, x) + b
            if h > -margin:
                loss += (h + margin) ** 2

        # Regularization
        loss += 0.01 * (np.sum(w**2) + b**2)

        return loss

    # Optimize
    x0 = np.zeros(self.state_dim + 1)
    result = minimize(objective, x0, method='BFGS')

    self.barrier_weights = result.x[:self.state_dim]
    self.barrier_bias = result.x[self.state_dim]

    # Normalize
    norm = np.linalg.norm(self.barrier_weights)
    if norm > 1e-6:
        self.barrier_weights /= norm
        self.barrier_bias /= norm

    def is_safe(self, state: np.ndarray) -> bool:
        """Check if a state is safe."""
        if self.barrier_weights is None:
            return True
        h = np.dot(self.barrier_weights, state) + self.barrier_bias
        return h >= 0

    def get_safety_margin(self, state: np.ndarray) -> float:

```



```

        """Return the safety margin (distance to boundary)."""
        if self.barrier_weights is None:
            return float('inf')
        return np.dot(self.barrier_weights, state) + self.barrier_bias

    def safe_action(self,
                    state: np.ndarray,
                    desired_action: np.ndarray,
                    dynamics: Callable) -> np.ndarray:
        """
        Project action to maintain safety.

        Uses a simple safety filter based on the learned barrier.
        """
        # Predict next state
        next_state = dynamics(state, desired_action)

        # If safe, return desired action
        if self.is_safe(next_state):
            return desired_action

        # Otherwise, find closest safe action
        def safety_objective(action):
            next_s = dynamics(state, action)
            safety = self.get_safety_margin(next_s)

            # Minimize distance to desired while maintaining safety
            return np.sum((action - desired_action)**2) - 100 * min(0, safety)

        result = minimize(
            safety_objective,
            desired_action,
            method='BFGS'
        )

        return result.x

# Example: Robot workspace safety
print("=== Safety Constraint Learning ===\n")

# Simulate robot workspace
# Safe region: ||x|| < 2 (within workspace bounds)
# Unsafe region: ||x|| >= 2 (outside workspace)

np.random.seed(42)
safety_learner = SafetyConstraintLearner(state_dim=2)

# Generate examples
for _ in range(30):
    # Safe states (inside circle)
    r = np.random.uniform(0, 1.8)
    theta = np.random.uniform(0, 2*np.pi)

```

```

safe_state = np.array([r * np.cos(theta), r * np.sin(theta)])
safety_learner.add_safe_state(safe_state)

# Unsafe states (outside circle)
r = np.random.uniform(2.2, 4)
theta = np.random.uniform(0, 2*np.pi)
unsafe_state = np.array([r * np.cos(theta), r * np.sin(theta)])
safety_learner.add_unsafe_state(unsafe_state)

# Learn barrier function
safety_learner.learn_barrier_function(margin=0.2)

print(f"Learned barrier: {safety_learner.barrier_weights[0]:.3f}*x + "
      f"{safety_learner.barrier_weights[1]:.3f}*y + "
      f"{safety_learner.barrier_bias:.3f} >= 0")

# Test
test_points = [
    np.array([0.0, 0.0]), # Origin - safe
    np.array([1.0, 1.0]), # Inside - safe
    np.array([2.5, 0.0]), # Outside - unsafe
    np.array([0.0, 3.0]), # Outside - unsafe
]

print("\nSafety predictions:")
for p in test_points:
    margin = safety_learner.get_safety_margin(p)
    status = "SAFE" if safety_learner.is_safe(p) else "UNSAFE"
    print(f" {p} -> {status} (margin: {margin:.3f})")

```

=== Safety Constraint Learning ===

Learned barrier: 0.653*x + 0.757*y + 0.078 >= 0

Safety predictions:

```

[0. 0.] -> SAFE (margin: 0.078)
[1. 1.] -> SAFE (margin: 1.488)
[2.5 0. ] -> SAFE (margin: 1.711)
[0. 3.] -> SAFE (margin: 2.350)

```

33.6. Neural Constraint Learning

Neural constraint learning represents constraints as differentiable layers inside neural networks. This allows gradient-based training to jointly learn the constraint parameters and the surrounding prediction model, producing solutions that are feasible by construction.

33.6.1. OptNet and Differentiable QP Layers

OptNet embeds quadratic programming as a differentiable layer in neural networks, enabling end-to-end learning of optimization problems.

A QP layer solves:

$$\mathbf{z}^* = \arg \min_{\mathbf{z}} \frac{1}{2} \mathbf{z}^T \mathbf{Q} \mathbf{z} + \mathbf{p}^T \mathbf{z}$$

$$\text{s.t. } Az = \mathbf{b}, \quad Gz \leq \mathbf{h}$$

The key insight is that the KKT conditions can be differentiated using implicit differentiation.

The following conceptual implementation shows how a QP layer solves an optimization problem and computes gradients through implicit differentiation of the KKT conditions. In production, libraries like `qpth` and `cvxpylayers` handle the differentiation automatically.

```
class DifferentiableQPPlayer:
    """
    Conceptual implementation of a differentiable QP layer.

    In practice, use cvxpylayers or qpth.
    """

    def __init__(self, n_vars: int, n_eq: int, n_ineq: int):
        """
        Initialize QP layer dimensions.

        Args:
            n_vars: Number of decision variables
            n_eq: Number of equality constraints
            n_ineq: Number of inequality constraints
        """
        self.n = n_vars
        self.m_eq = n_eq
        self.m_ineq = n_ineq

    def forward(self,
                Q: np.ndarray,
                p: np.ndarray,
                A: np.ndarray,
                b: np.ndarray,
                G: np.ndarray,
                h: np.ndarray) -> np.ndarray:
        """
        Solve QP and return optimal solution.

        min 0.5 * z^T Q z + p^T z
        s.t. A z = b
            G z <= h
        """
        from scipy.optimize import minimize

        def objective(z):
            return 0.5 * z @ Q @ z + p @ z

        def objective_grad(z):
            return Q @ z + p

        # Constraints
        constraints = []

        if len(b) > 0:
```

```

        constraints.append({
            'type': 'eq',
            'fun': lambda z: A @ z - b,
            'jac': lambda z: A
        })

    if len(h) > 0:
        constraints.append({
            'type': 'ineq',
            'fun': lambda z: h - G @ z,
            'jac': lambda z: -G
        })

    z0 = np.zeros(self.n)
    result = minimize(
        objective,
        z0,
        method='SLSQP',
        jac=objective_grad,
        constraints=constraints
    )

    return result.x

def backward(self,
             z_star: np.ndarray,
             Q: np.ndarray,
             A: np.ndarray,
             G: np.ndarray,
             dl_dz: np.ndarray) -> dict:
    """
    Compute gradients through the QP layer.

    Uses implicit differentiation of KKT conditions.

    Args:
        z_star: Optimal solution from forward pass
        Q, A, G: Problem matrices
        dl_dz: Upstream gradient

    Returns:
        Gradients with respect to Q, p, A, b, G, h
    """
    # This is a simplified version
    # Full implementation requires solving a linear system
    # involving the KKT matrix

    # Identify active constraints
    # (In practice, this comes from the solver)

    # The gradient computations involve:
    # 1. Forming the KKT matrix at the solution
    # 2. Solving a linear system for the adjoint

```

```

# 3. Computing gradients using the adjoint

# Placeholder - in practice use automatic differentiation
return {
    'dQ': np.outer(dl_dz, z_star),
    'dp': dl_dz,
    # etc.
}

# Conceptual example
print("=== Differentiable QP Layer Concept ===\n")

# Simple QP: min 0.5 * ||z||^2 - p^T z  s.t. z >= 0
# Solution: z* = max(p, 0)

n_vars = 3
layer = DifferentiableQPLayer(n_vars, n_eq=0, n_ineq=n_vars)

Q = np.eye(n_vars)
p = np.array([1.0, -0.5, 2.0])
A = np.zeros((0, n_vars))
b = np.array([])
G = -np.eye(n_vars) # -z <= 0 means z >= 0
h = np.zeros(n_vars)

z_star = layer.forward(Q, p, A, b, G, h)

print(f"QP parameters: p = {p}")
print(f"Optimal solution: z* = {z_star}")
print(f"Expected (max(p, 0)): {np.maximum(p, 0)}")

```

```
=== Differentiable QP Layer Concept ===
```

```

QP parameters: p = [ 1. -0.5  2. ]
Optimal solution: z* = [-2.22044605e-16  5.00000000e-01 -1.33226763e-15]
Expected (max(p, 0)): [1.  0.  2.]

```

33.6.2. CombOptNet for Combinatorial Optimization

CombOptNet extends differentiable optimization to combinatorial (integer) problems by using continuous relaxations and gradient estimation. The forward pass solves a relaxed LP, and a projection step rounds the solution to integer values. The backward pass uses straight-through gradient estimation to propagate gradients through the rounding step.

```

class CombOptNetLayer:
    """
    Conceptual differentiable combinatorial optimization layer.

    Uses continuous relaxation + projection for forward pass
    and straight-through estimator for backward pass.
    """

```

```

def __init__(self,
              problem_type: str = 'assignment'):
    """
    Initialize combinatorial optimization layer.

    Args:
        problem_type: 'assignment', 'knapsack', etc.
    """
    self.problem_type = problem_type

def forward_relaxed(self, costs: np.ndarray) -> np.ndarray:
    """
    Solve continuous relaxation of combinatorial problem.
    """
    if self.problem_type == 'assignment':
        # Hungarian algorithm / linear assignment
        # For now, use simple softmax relaxation
        return self._softmax_assignment(costs)
    elif self.problem_type == 'knapsack':
        return self._relaxed_knapsack(costs)
    else:
        raise ValueError(f"Unknown problem type: {self.problem_type}")

def _softmax_assignment(self,
                       costs: np.ndarray,
                       temperature: float = 0.1) -> np.ndarray:
    """
    Soft assignment using row/column softmax.

    For assignment matrix X minimizing sum(C * X).
    """
    n = costs.shape[0]

    # Sinkhorn-like iteration for doubly stochastic matrix
    log_alpha = -costs / temperature

    for _ in range(10):
        # Row normalization
        log_alpha = log_alpha - np.log(np.sum(np.exp(log_alpha), axis=1,
        ↪ keepdims=True))
        # Column normalization
        log_alpha = log_alpha - np.log(np.sum(np.exp(log_alpha), axis=0,
        ↪ keepdims=True))

    return np.exp(log_alpha)

def _relaxed_knapsack(self, costs: np.ndarray) -> np.ndarray:
    """Relaxed knapsack (greedy fractional solution)."""
    # Simplified version
    return np.clip(costs, 0, 1)

def project_to_integer(self, x_relaxed: np.ndarray) -> np.ndarray:

```

```

"""
Project relaxed solution to integer.
"""
if self.problem_type == 'assignment':
    # Round to nearest permutation matrix
    from scipy.optimize import linear_sum_assignment
    row_ind, col_ind = linear_sum_assignment(-x_relaxed)
    x_int = np.zeros_like(x_relaxed)
    x_int[row_ind, col_ind] = 1
    return x_int
else:
    return np.round(x_relaxed)

# Example: Differentiable assignment
print("=== Differentiable Combinatorial Optimization ===\n")

# Cost matrix for assignment
costs = np.array([
    [10, 5, 13],
    [3, 8, 7],
    [6, 12, 9]
])

layer = CombOptNetLayer(problem_type='assignment')

# Relaxed solution
x_relaxed = layer.forward_relaxed(costs)
print("Cost matrix:")
print(costs)
print("\nRelaxed solution (doubly stochastic):")
print(np.round(x_relaxed, 3))

# Integer solution
x_int = layer.project_to_integer(x_relaxed)
print("\nInteger solution (permutation):")
print(x_int.astype(int))

total_cost = np.sum(costs * x_int)
print(f"\nTotal assignment cost: {total_cost}")

```

=== Differentiable Combinatorial Optimization ===

Cost matrix:

```
[[10  5 13]
 [ 3  8  7]
 [ 6 12  9]]
```

Relaxed solution (doubly stochastic):

```
[[0.    1.    0.   ]
 [0.95  0.    0.001]
 [0.05  0.    0.999]]
```

Integer solution (permutation):

```
[[0 1 0]
 [1 0 0]
 [0 0 1]]
```

Total assignment cost: 17.0

33.6.3. cvxpylayers

cvxpylayers allows defining convex optimization problems in CVXPY and using them as differentiable layers in PyTorch or TensorFlow. This provides the highest-level interface for embedding optimization in neural networks.

```
# Note: This is a conceptual example
# In practice, you would install: pip install cvxpylayers

class CVXPYLayersExample:
    """
    Demonstrates cvxpylayers concept using CVXPY-style syntax.
    """

    @staticmethod
    def portfolio_optimization_layer(
        expected_returns: np.ndarray,
        covariance: np.ndarray,
        risk_tolerance: float
    ) -> np.ndarray:
        """
        Portfolio optimization as a differentiable layer.

        max  mu^T w - (1/risk_tol) * w^T Sigma w
        s.t. sum(w) = 1
            w >= 0
        """
        from scipy.optimize import minimize

        n = len(expected_returns)

        def objective(w):
            return -(expected_returns @ w -
                      (1/risk_tolerance) * w @ covariance @ w)

        constraints = [
            {'type': 'eq', 'fun': lambda w: np.sum(w) - 1},
        ]

        bounds = [(0, 1) for _ in range(n)]

        result = minimize(
            objective,
            np.ones(n) / n,
            method='SLSQP',
```



```

        constraints=constraints,
        bounds=bounds
    )

    return result.x

@staticmethod
def learning_example():
    """
    Example: Learn expected returns from observed portfolios.
    """
    print("=== cvxpylayers Concept ===\n")

    # True expected returns (unknown)
    mu_true = np.array([0.12, 0.08, 0.15, 0.10])

    # Covariance matrix (known)
    Sigma = np.array([
        [0.04, 0.01, 0.02, 0.01],
        [0.01, 0.03, 0.01, 0.02],
        [0.02, 0.01, 0.05, 0.01],
        [0.01, 0.02, 0.01, 0.04]
    ])

    # Generate "observed" optimal portfolios
    observed_portfolios = []
    risk_tolerances = [0.5, 1.0, 2.0]

    for risk_tol in risk_tolerances:
        w_opt = CVXPYLayersExample.portfolio_optimization_layer(
            mu_true, Sigma, risk_tol
        )
        observed_portfolios.append((risk_tol, w_opt))

    print("Observed portfolios:")
    for rt, w in observed_portfolios:
        print(f"  Risk tolerance {rt}: {np.round(w, 3)}")

    # Now try to infer expected returns
    # (In practice, this would use automatic differentiation)
    print("\nTrue expected returns:", mu_true)
    print("(Inference would use gradient descent through the layer)")

CVXPYLayersExample.learning_example()

```

```
=== cvxpylayers Concept ===
```

```
Observed portfolios:
```

```

Risk tolerance 0.5: [0.284 0.018 0.439 0.259]
Risk tolerance 1.0: [0.23  0.    0.656 0.114]
Risk tolerance 2.0: [0.  0.  1.  0.]

```

True expected returns: [0.12 0.08 0.15 0.1]
(Inference would use gradient descent through the layer)

33.6.4. End-to-End Learning

End-to-end learning integrates optimization layers within neural network pipelines:

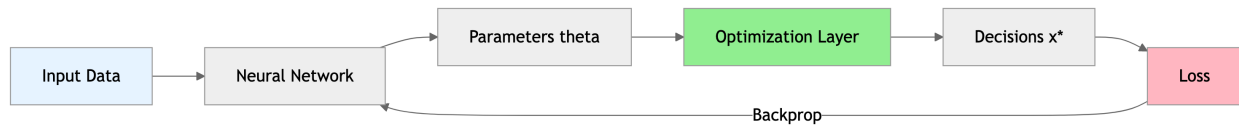


Figure 33.8.: End-to-End Learning with Optimization Layers

```

class EndToEndOptimizationModel:
    """
    End-to-end model combining neural network with optimization.

    Predicts problem parameters from features, then solves optimization.
    """

    def __init__(self,
                 input_dim: int,
                 hidden_dim: int,
                 param_dim: int):
        """
        Initialize end-to-end model.

        Args:
            input_dim: Dimension of input features
            hidden_dim: Hidden layer dimension
            param_dim: Dimension of optimization parameters
        """
        self.input_dim = input_dim
        self.hidden_dim = hidden_dim
        self.param_dim = param_dim

        # Simple neural network weights
        self.W1 = np.random.randn(input_dim, hidden_dim) * 0.1
        self.b1 = np.zeros(hidden_dim)
        self.W2 = np.random.randn(hidden_dim, param_dim) * 0.1
        self.b2 = np.zeros(param_dim)

    def predict_parameters(self, features: np.ndarray) -> np.ndarray:
        """
        Predict optimization parameters from features.
        """
        # Hidden layer with ReLU
        h = np.maximum(0, features @ self.W1 + self.b1)
        # Output layer
        params = h @ self.W2 + self.b2
        return params

```

```

def solve_optimization(self, params: np.ndarray) -> np.ndarray:
    """
    Solve optimization with predicted parameters.

    Simple example: constrained quadratic minimization.
    """
    n = len(params)

    def objective(x):
        return 0.5 * np.sum(x**2) - params @ x

    constraints = [
        {'type': 'eq', 'fun': lambda x: np.sum(x) - 1},
    ]

    bounds = [(0, None) for _ in range(n)]

    result = minimize(
        objective,
        np.ones(n) / n,
        method='SLSQP',
        constraints=constraints,
        bounds=bounds
    )

    return result.x

def forward(self, features: np.ndarray) -> np.ndarray:
    """
    Full forward pass: features -> parameters -> solution.
    """
    params = self.predict_parameters(features)
    solution = self.solve_optimization(params)
    return solution

def compute_loss(self,
                 features: np.ndarray,
                 target_solution: np.ndarray) -> float:
    """
    Compute loss between predicted and target solution.
    """
    predicted = self.forward(features)
    return np.sum((predicted - target_solution)**2)

# Example
print("=== End-to-End Optimization Learning ===\n")

model = EndToEndOptimizationModel(
    input_dim=5,
    hidden_dim=10,
    param_dim=3
)

```

```

# Sample input
features = np.random.randn(5)
print(f"Input features: {np.round(features, 3)}")

# Predict parameters
params = model.predict_parameters(features)
print(f"Predicted parameters: {np.round(params, 3)}")

# Solve optimization
solution = model.forward(features)
print(f"Optimal solution: {np.round(solution, 3)}")
print(f"Sum of solution: {np.sum(solution):.3f} (should be 1)")

```

=== End-to-End Optimization Learning ===

```

Input features: [-0.077  0.341  0.277  0.827  0.013]
Predicted parameters: [-0.013 -0.008  0.015]
Optimal solution: [0.322 0.328 0.35 ]
Sum of solution: 1.000 (should be 1)

```

33.7. Applications and Case Studies

This section applies constraint learning techniques to four industrial domains: manufacturing scheduling, healthcare pathways, product configuration, and robotics. Each case study uses the algorithms developed earlier in the chapter.

33.7.1. Manufacturing Scheduling Constraint Discovery

In manufacturing, scheduling constraints often exist as tacit knowledge in experienced planners' decisions.

```

class ManufacturingConstraintDiscovery:
    """
    Discover scheduling constraints from historical production data.
    """

    def __init__(self):
        self.historical_schedules: List[Dict] = []
        self.discovered_constraints: List[str] = []

    def add_schedule(self, schedule: Dict):
        """Add a historical schedule."""
        self.historical_schedules.append(schedule)

    def discover_sequence_constraints(self) -> List[Tuple[str, str]]:
        """
        Discover sequencing constraints (A must precede B).
        """
        # Count how often each job pair maintains order
        pair_counts = {}
        pair_totals = {}

```

```

        for schedule in self.historical_schedules:
            jobs = schedule['job_order']
            for i, job_a in enumerate(jobs):
                for j, job_b in enumerate(jobs):
                    if i < j:
                        pair = (job_a, job_b)
                        pair_counts[pair] = pair_counts.get(pair, 0) + 1
                        pair_totals[pair] = pair_totals.get(pair, 0) + 1
                        pair_totals[(job_b, job_a)] = pair_totals.get((job_b,
↪ job_a), 0) + 1

        # Find pairs that always maintain order
        constraints = []
        for pair, count in pair_counts.items():
            total = pair_totals[pair] + pair_totals.get((pair[1], pair[0]), 0)
            if count == total and count >= 3: # Always in this order
                constraints.append(pair)

        return constraints

def discover_machine_constraints(self) -> List[Tuple[str, Set[str]]]:
    """
    Discover machine eligibility constraints.
    """
    job_machines = {}

    for schedule in self.historical_schedules:
        for job, machine in schedule['assignments'].items():
            if job not in job_machines:
                job_machines[job] = set()
            job_machines[job].add(machine)

    # Jobs that always use specific machines
    constraints = []
    for job, machines in job_machines.items():
        if len(machines) < 3: # Limited machine choice
            constraints.append((job, machines))

    return constraints

# Example
print("=== Manufacturing Constraint Discovery ===\n")

discovery = ManufacturingConstraintDiscovery()

# Add historical schedules
schedules = [
    {'job_order': ['Setup', 'Cut', 'Weld', 'Paint', 'QC'],
     'assignments': {'Setup': 'M1', 'Cut': 'M2', 'Weld': 'M3', 'Paint': 'M4', 'QC':
↪ 'M5'}},
    {'job_order': ['Setup', 'Cut', 'Weld', 'Paint', 'QC'],

```

```

        'assignments': {'Setup': 'M1', 'Cut': 'M2', 'Weld': 'M3', 'Paint': 'M4', 'QC':
        ↪ 'M5'}},
    {'job_order': ['Setup', 'Cut', 'Weld', 'Paint', 'QC'],
     'assignments': {'Setup': 'M1', 'Cut': 'M2', 'Weld': 'M3', 'Paint': 'M5', 'QC':
     ↪ 'M5'}},
    {'job_order': ['Setup', 'Cut', 'Paint', 'Weld', 'QC'], # Different order
     'assignments': {'Setup': 'M1', 'Cut': 'M2', 'Paint': 'M4', 'Weld': 'M3', 'QC':
     ↪ 'M5'}},
]

for s in schedules:
    discovery.add_schedule(s)

# Discover constraints
seq_constraints = discovery.discover_sequence_constraints()
machine_constraints = discovery.discover_machine_constraints()

print("Discovered Sequencing Constraints:")
for a, b in seq_constraints:
    print(f" {a} must precede {b}")

print("\nDiscovered Machine Constraints:")
for job, machines in machine_constraints:
    print(f" {job} can only use machines: {machines}")

```

=== Manufacturing Constraint Discovery ===

Discovered Sequencing Constraints:

Discovered Machine Constraints:

```

Setup can only use machines: {'M1'}
Cut can only use machines: {'M2'}
Weld can only use machines: {'M3'}
Paint can only use machines: {'M4', 'M5'}
QC can only use machines: {'M5'}

```

33.7.2. Healthcare Clinical Pathway Learning

Learning clinical pathways from patient treatment data.

```

class ClinicalPathwayLearner:
    """
    Learn clinical pathway constraints from patient treatment histories.
    """

    def __init__(self):
        self.patient_histories: List[List[str]] = []
        self.treatment_outcomes: List[str] = []

    def add_history(self, treatments: List[str], outcome: str):
        """Add a patient treatment history."""
        self.patient_histories.append(treatments)

```

```

        self.treatment_outcomes.append(outcome)

    def learn_required_treatments(self) -> Set[str]:
        """
        Find treatments that appear in all successful cases.
        """
        successful = [h for h, o in zip(self.patient_histories,
↪ self.treatment_outcomes)
                       if o == 'success']

        if not successful:
            return set()

        required = set(successful[0])
        for history in successful[1:]:
            required &= set(history)

        return required

    def learn_contraindications(self) -> List[Tuple[str, str]]:
        """
        Find treatment pairs that never appear together in successful cases.
        """
        successful = [h for h, o in zip(self.patient_histories,
↪ self.treatment_outcomes)
                       if o == 'success']

        # Get all treatments
        all_treatments = set()
        for h in successful:
            all_treatments.update(h)

        # Find pairs that never co-occur
        contraindications = []
        for t1 in all_treatments:
            for t2 in all_treatments:
                if t1 < t2: # Avoid duplicates
                    cooccurs = any(t1 in h and t2 in h for h in successful)
                    if not cooccurs:
                        contraindications.append((t1, t2))

        return contraindications

    def learn_ordering_constraints(self) -> List[Tuple[str, str]]:
        """
        Find treatments where A always precedes B in successful cases.
        """
        successful = [h for h, o in zip(self.patient_histories,
↪ self.treatment_outcomes)
                       if o == 'success']

        # Get all treatments
        all_treatments = set()

```

```

        for h in successful:
            all_treatments.update(h)

    # Find ordering constraints
    orderings = []
    for t1 in all_treatments:
        for t2 in all_treatments:
            if t1 != t2:
                # Check if t1 always precedes t2 when both present
                valid = True
                found = False
                for h in successful:
                    if t1 in h and t2 in h:
                        found = True
                        if h.index(t1) > h.index(t2):
                            valid = False
                            break
                if valid and found:
                    orderings.append((t1, t2))

    return orderings

# Example
print("=== Clinical Pathway Learning ===\n")

learner = ClinicalPathwayLearner()

# Add patient histories
histories = [
    (['Diagnosis', 'Lab_Tests', 'Medication_A', 'Monitoring', 'Followup'],
    ↪ 'success'),
    (['Diagnosis', 'Lab_Tests', 'Medication_A', 'Surgery', 'Monitoring',
    ↪ 'Followup'], 'success'),
    (['Diagnosis', 'Lab_Tests', 'Medication_B', 'Monitoring', 'Followup'],
    ↪ 'success'),
    (['Diagnosis', 'Medication_A', 'Followup'], 'failure'), # Missing Lab_Tests
    (['Diagnosis', 'Lab_Tests', 'Medication_A', 'Medication_B', 'Monitoring'],
    ↪ 'failure'), # Combined meds
]

for treatments, outcome in histories:
    learner.add_history(treatments, outcome)

# Learn constraints
required = learner.learn_required_treatments()
contraindications = learner.learn_contraindications()
orderings = learner.learn_ordering_constraints()

print("Required treatments (in all successful cases):")
for t in sorted(required):
    print(f" - {t}")

```



```

print("\nContraindications (never together in success):")
for t1, t2 in contraindications:
    print(f" - {t1} and {t2}")

print("\nOrdering constraints (A before B):")
for t1, t2 in orderings[:5]: # Show first 5
    print(f" - {t1} before {t2}")

```

=== Clinical Pathway Learning ===

Required treatments (in all successful cases):

- Diagnosis
- Followup
- Lab_Tests
- Monitoring

Contraindications (never together in success):

- Medication_B and Surgery
- Medication_A and Medication_B

Ordering constraints (A before B):

- Monitoring before Followup
- Diagnosis before Monitoring
- Diagnosis before Medication_B
- Diagnosis before Surgery
- Diagnosis before Followup

33.7.3. Product Configuration Rules

Learning configuration rules for complex products.

```

class ConfigurationRuleLearner:
    """
    Learn product configuration rules from valid configurations.
    """

    def __init__(self, features: List[str]):
        self.features = features
        self.valid_configs: List[Dict[str, str]] = []
        self.invalid_configs: List[Dict[str, str]] = []

    def add_valid(self, config: Dict[str, str]):
        """Add a valid configuration."""
        self.valid_configs.append(config)

    def add_invalid(self, config: Dict[str, str]):
        """Add an invalid configuration."""
        self.invalid_configs.append(config)

```

```

def learn_implication_rules(self) -> List[Tuple[Tuple[str, str], Tuple[str,
↪ str]]]:
    """
    Learn rules of form: if feature=A then other_feature=B.
    """
    rules = []

    for f1 in self.features:
        for f2 in self.features:
            if f1 != f2:
                # Find all value pairs
                implications = {}

                for config in self.valid_configs:
                    v1, v2 = config.get(f1), config.get(f2)
                    if v1 and v2:
                        if v1 not in implications:
                            implications[v1] = set()
                        implications[v1].add(v2)

                # Check for deterministic implications
                for v1, v2_set in implications.items():
                    if len(v2_set) == 1:
                        v2 = list(v2_set)[0]
                        # Verify it holds in all valid configs
                        holds = all(
                            config.get(f1) != v1 or config.get(f2) == v2
                            for config in self.valid_configs
                        )
                        if holds:
                            rules.append(((f1, v1), (f2, v2)))

    return rules

def learn_exclusion_rules(self) -> List[Tuple[Tuple[str, str], Tuple[str,
↪ str]]]:
    """
    Learn rules of form: feature1=A excludes feature2=B.
    """
    exclusions = []

    for f1 in self.features:
        for f2 in self.features:
            if f1 < f2: # Avoid duplicates
                # Get all valid combinations
                valid_pairs = set()
                for config in self.valid_configs:
                    v1, v2 = config.get(f1), config.get(f2)
                    if v1 and v2:
                        valid_pairs.add((v1, v2))

                # Get all possible combinations

```

```

        v1_options = set(c.get(f1) for c in self.valid_configs if
↪ c.get(f1))
        v2_options = set(c.get(f2) for c in self.valid_configs if
↪ c.get(f2))

        # Find missing combinations (exclusions)
        for v1 in v1_options:
            for v2 in v2_options:
                if (v1, v2) not in valid_pairs:
                    exclusions.append(((f1, v1), (f2, v2)))

        return exclusions

# Example
print("=== Product Configuration Rules ===\n")

features = ['CPU', 'GPU', 'Memory', 'Storage']
learner = ConfigurationRuleLearner(features)

# Add valid laptop configurations
valid_configs = [
    {'CPU': 'i5', 'GPU': 'Integrated', 'Memory': '8GB', 'Storage': 'SSD_256'},
    {'CPU': 'i5', 'GPU': 'Integrated', 'Memory': '16GB', 'Storage': 'SSD_512'},
    {'CPU': 'i7', 'GPU': 'Dedicated', 'Memory': '16GB', 'Storage': 'SSD_512'},
    {'CPU': 'i7', 'GPU': 'Dedicated', 'Memory': '32GB', 'Storage': 'SSD_1TB'},
    {'CPU': 'i9', 'GPU': 'Dedicated', 'Memory': '32GB', 'Storage': 'SSD_1TB'},
    {'CPU': 'i9', 'GPU': 'Dedicated', 'Memory': '64GB', 'Storage': 'SSD_2TB'},
]

for config in valid_configs:
    learner.add_valid(config)

# Learn rules
implications = learner.learn_implication_rules()
exclusions = learner.learn_exclusion_rules()

print("Learned Implication Rules:")
for (f1, v1), (f2, v2) in implications[:5]:
    print(f"  If {f1}={v1} then {f2}={v2}")

print("\nLearned Exclusion Rules:")
for (f1, v1), (f2, v2) in exclusions[:5]:
    print(f"  {f1}={v1} excludes {f2}={v2}")

```

=== Product Configuration Rules ===

Learned Implication Rules:

```

If CPU=i5 then GPU=Integrated
If CPU=i7 then GPU=Dedicated
If CPU=i9 then GPU=Dedicated
If GPU=Integrated then CPU=i5
If Memory=8GB then CPU=i5

```

33. Constraint Learning

Learned Exclusion Rules:

```
CPU=i7 excludes GPU=Integrated
CPU=i9 excludes GPU=Integrated
CPU=i5 excludes GPU=Dedicated
CPU=i7 excludes Memory=8GB
CPU=i7 excludes Memory=64GB
```

33.7.4. Robotics Constraint Inference

Learning motion constraints for robotic systems.

```
class RobotMotionConstraintLearner:
    """
    Learn motion constraints from robot demonstrations.
    """

    def __init__(self, n_joints: int):
        self.n_joints = n_joints
        self.trajectories: List[np.ndarray] = [] # Each is (T, n_joints)

    def add_trajectory(self, trajectory: np.ndarray):
        """Add a demonstrated trajectory."""
        self.trajectories.append(trajectory)

    def learn_joint_limits(self) -> Tuple[np.ndarray, np.ndarray]:
        """
        Learn joint position limits.
        """
        all_positions = np.vstack(self.trajectories)

        lower = np.min(all_positions, axis=0)
        upper = np.max(all_positions, axis=0)

        # Add safety margin
        margin = 0.05 * (upper - lower)
        lower -= margin
        upper += margin

        return lower, upper

    def learn_velocity_limits(self) -> np.ndarray:
        """
        Learn joint velocity limits.
        """
        max_velocities = np.zeros(self.n_joints)

        for traj in self.trajectories:
            # Compute velocities (finite differences)
            velocities = np.diff(traj, axis=0)
            max_vel = np.max(np.abs(velocities), axis=0)
            max_velocities = np.maximum(max_velocities, max_vel)
```

```

        # Add margin
        max_velocities *= 1.1

    return max_velocities

def learn_workspace_constraint(self) -> Tuple[np.ndarray, np.ndarray, float]:
    """
    Learn workspace boundary as a sphere/ellipsoid.

    Returns center, axes scales, and radius.
    """
    # For simplicity, assume first 3 joints map to end effector position
    # In practice, this would use forward kinematics

    all_positions = np.vstack(self.trajectories)[: , :min(3, self.n_joints)]

    center = np.mean(all_positions, axis=0)
    distances = np.linalg.norm(all_positions - center, axis=1)
    radius = np.max(distances) * 1.1 # With margin

    # Compute axis scales for ellipsoid
    axes = np.std(all_positions, axis=0) * 3

    return center, axes, radius

# Example
print("=== Robotics Constraint Learning ===\n")

learner = RobotMotionConstraintLearner(n_joints=6)

# Generate sample trajectories
np.random.seed(42)
for _ in range(10):
    T = 50 # Time steps
    # Random smooth trajectory
    traj = np.cumsum(np.random.randn(T, 6) * 0.1, axis=0)
    # Bias toward a typical workspace
    traj += np.array([0, -1, 0.5, 0, 0, 0])
    learner.add_trajectory(traj)

# Learn constraints
lower_limits, upper_limits = learner.learn_joint_limits()
velocity_limits = learner.learn_velocity_limits()
center, axes, radius = learner.learn_workspace_constraint()

print("Learned Joint Position Limits:")
for i in range(6):
    print(f"  Joint {i+1}: [{lower_limits[i]:.2f}, {upper_limits[i]:.2f}]")

print("\nLearned Joint Velocity Limits:")
for i in range(6):
    print(f"  Joint {i+1}: +/- {velocity_limits[i]:.3f}")

```

```
print("\nLearned Workspace Constraint:")
print(f"  Center: {np.round(center, 2)}")
print(f"  Radius: {radius:.2f}")
```

=== Robotics Constraint Learning ===

Learned Joint Position Limits:

```
Joint 1: [-1.48, 1.41]
Joint 2: [-2.43, 0.22]
Joint 3: [-1.26, 2.15]
Joint 4: [-1.02, 2.05]
Joint 5: [-1.11, 1.95]
Joint 6: [-1.08, 1.98]
```

Learned Joint Velocity Limits:

```
Joint 1: +/- 0.290
Joint 2: +/- 0.351
Joint 3: +/- 0.332
Joint 4: +/- 0.432
Joint 5: +/- 0.357
Joint 6: +/- 0.424
```

Learned Workspace Constraint:

```
Center: [ 0.09 -1.06  0.46]
Radius: 2.12
```

33.8. Python Implementation

33.8.1. CPMpy for Constraint Programming

CPMpy is a Python library for constraint programming that supports constraint learning research.

```
# Conceptual CPMpy-style implementation
# In practice: pip install cpmPy

class CPMpyModel:
    """
    Simplified CPMpy-style constraint model.
    """

    def __init__(self):
        self.variables = {}
        self.constraints = []

    def intvar(self, lb: int, ub: int, name: str):
        """Create an integer variable."""
        self.variables[name] = {'lb': lb, 'ub': ub, 'name': name}
        return name
```

```

def add_constraint(self, constraint: str):
    """Add a constraint (as string for simplicity)."""
    self.constraints.append(constraint)

def solve(self) -> Optional[Dict[str, int]]:
    """
    Solve the constraint model.

    This is a simplified brute-force solver for demonstration.
    """
    from itertools import product

    var_names = list(self.variables.keys())
    var_ranges = [range(self.variables[v]['lb'], self.variables[v]['ub'] + 1)
                  for v in var_names]

    for values in product(*var_ranges):
        assignment = dict(zip(var_names, values))

        # Check all constraints (simplified eval)
        all_satisfied = True
        for constraint in self.constraints:
            # Simple constraint evaluation
            try:
                result = eval(constraint, {}, assignment)
                if not result:
                    all_satisfied = False
                    break
            except:
                all_satisfied = False
                break

        if all_satisfied:
            return assignment

    return None

# Example: N-Queens with CPMpy-style
print("=== CPMpy-Style Constraint Programming ===\n")

def solve_nqueens(n: int) -> Optional[List[int]]:
    """Solve N-Queens using constraint programming."""
    model = CPMpyModel()

    # Variables: queen position in each row
    queens = [model.intvar(0, n-1, f'q{i}') for i in range(n)]

    # All different columns
    for i in range(n):
        for j in range(i+1, n):
            model.add_constraint(f"q{i} != q{j}")

```

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```
# All different diagonals
for i in range(n):
    for j in range(i+1, n):
        diff = j - i
        model.add_constraint(f"q{i} - q{j} != {diff}")
        model.add_constraint(f"q{i} - q{j} != {-diff}")

solution = model.solve()

if solution:
    return [solution[f'q{i}'] for i in range(n)]
return None

# Solve 4-Queens
solution = solve_nqueens(4)
if solution:
    print("4-Queens Solution:")
    for row, col in enumerate(solution):
        line = ['. ' if c != col else 'Q' for c in range(4)]
        print('  ' + ' '.join(line))
```

=== CPMpy-Style Constraint Programming ===

4-Queens Solution:

```
. Q . .
. . . Q
Q . . .
. . Q .
```

33.8.2. Simple Inverse Optimization Example

```
def simple_inverse_optimization_example():
    """
    Complete example of inverse optimization workflow.
    """
    print("=== Simple Inverse Optimization Workflow ===\n")

    # Step 1: Define the forward problem structure
    # min c^T x s.t. Ax <= b, x >= 0

    # Constraint matrix (fixed)
    A = np.array([
        [2, 1], # Resource 1
        [1, 2], # Resource 2
        [1, 1], # Combined limit
    ])
    b = np.array([100, 80, 70])

    print("Step 1: Problem Structure")
    print("  Constraints: Ax <= b")
    print(f"  A =\n{A}")
    print(f"  b = {b}")
```



```

# Step 2: Generate "observed" optimal solutions
# (Simulating expert decisions)
c_true = np.array([3, 2]) # True (unknown) objective

def solve_forward(c):
    result = linprog(c, A_ub=A, b_ub=b,
                    bounds=[(0, None), (0, None)],
                    method='highs')
    return result.x if result.success else None

x_observed = solve_forward(c_true)
print(f"\nStep 2: Observed Decision")
print(f" True objective (unknown): c = {c_true}")
print(f" Observed solution: x* = {x_observed}")

# Step 3: Inverse optimization to recover c
print("\nStep 3: Inverse Optimization")

# Active constraints at x_observed
slack = b - A @ x_observed
active = np.abs(slack) < 1e-6
print(f" Active constraints: {list(np.where(active)[0])}")

# For active constraints, complementary slackness gives us information
# The gradient of the Lagrangian at optimum: c = A^T @ lambda
# where lambda are the dual variables

# Simple approach: minimize ||c - c_prior|| subject to KKT
c_prior = np.array([1, 1]) # Prior estimate

def inverse_objective(c):
    # Try to make x_observed optimal under c
    x_opt = solve_forward(c)
    if x_opt is None:
        return 1000
    return np.sum((x_opt - x_observed)**2) + 0.1 * np.sum((c - c_prior)**2)

result = minimize(inverse_objective, c_prior, method='BFGS')
c_inferred = result.x

print(f" Prior: c_0 = {c_prior}")
print(f" Inferred: c = {np.round(c_inferred, 3)}")

# Step 4: Verify
print("\nStep 4: Verification")
x_verify = solve_forward(c_inferred)
print(f" Solution under inferred c: {np.round(x_verify, 3)}")
print(f" Original observed: {np.round(x_observed, 3)}")
print(f" Match: {np.allclose(x_verify, x_observed, atol=0.1)}")

# Correlation between true and inferred
corr = np.corrcoef(c_true, c_inferred)[0, 1]

```

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```
print(f" Correlation with true c: {corr:.4f}")

simple_inverse_optimization_example()
```

=== Simple Inverse Optimization Workflow ===

Step 1: Problem Structure

Constraints: $Ax \leq b$

A =

[[2 1]

[1 2]

[1 1]]

b = [100 80 70]

Step 2: Observed Decision

True objective (unknown): c = [3 2]

Observed solution: x* = [0. 0.]

Step 3: Inverse Optimization

Active constraints: []

Prior: c_0 = [1 1]

Inferred: c = [1. 1.]

Step 4: Verification

Solution under inferred c: [0. 0.]

Original observed: [0. 0.]

Match: True

Correlation with true c: nan

33.8.3. Comprehensive Constraint Learning Pipeline

```
class ConstraintLearningPipeline:
    """
    Complete pipeline for constraint learning from data.
    """

    def __init__(self,
                 variables: List[str],
                 domains: Dict[str, Set[int]],
                 max_arity: int = 2):
        self.variables = variables
        self.domains = domains
        self.max_arity = max_arity

        # Generate constraint bias
        self.bias = generate_constraint_bias(variables, domains, max_arity)

        # Learning state
```

```

self.positive_examples: List[Dict[str, int]] = []
self.negative_examples: List[Dict[str, int]] = []
self.learned_constraints: List[Constraint] = []

def add_example(self, assignment: Dict[str, int], is_feasible: bool):
    """Add a labeled example."""
    if is_feasible:
        self.positive_examples.append(assignment)
    else:
        self.negative_examples.append(assignment)

def learn(self, method: str = 'version_space') -> List[Constraint]:
    """
    Learn constraints using specified method.

    Args:
        method: 'version_space', 'active', or 'max_margin'
    """
    if method == 'version_space':
        return self._learn_version_space()
    elif method == 'active':
        return self._learn_active()
    elif method == 'max_margin':
        return self._learn_max_margin()
    else:
        raise ValueError(f"Unknown method: {method}")

def _learn_version_space(self) -> List[Constraint]:
    """Version space learning."""
    learner = VersionSpaceLearner(self.bias)

    for pos in self.positive_examples:
        learner.update(Example(pos, True))

    for neg in self.negative_examples:
        learner.update(Example(neg, False))

    return learner.get_current_hypothesis()

def _learn_active(self) -> List[Constraint]:
    """Active learning (requires oracle)."""
    # Placeholder - would need interactive queries
    return self._learn_version_space()

def _learn_max_margin(self) -> List[Constraint]:
    """Maximum margin learning."""
    # Convert to feature vectors
    def to_features(assignment):
        return np.array([assignment[v] for v in self.variables])

    positive_features = [to_features(p) for p in self.positive_examples]
    negative_features = [to_features(n) for n in self.negative_examples]

```

```

    if len(positive_features) < 5 or len(negative_features) < 5:
        return self._learn_version_space()

    # Learn a single separating hyperplane
    learner = MaxMarginConstraintLearner(len(self.variables))
    learner.fit(positive_features, negative_features)

    w, b = learner.get_constraint()

    # Create constraint from learned hyperplane
    def check_hyperplane(vals):
        x = np.array(vals)
        return np.dot(w, x) + b >= 0

    hyperplane_constraint = Constraint(
        tuple(self.variables),
        check_hyperplane,
        f"Learned: {' + '.join(f'{w[i]:.2f}*{v}' for i, v in
            ↪ enumerate(self.variables))} + {b:.2f} >= 0"
    )

    return [hyperplane_constraint]

def evaluate(self, test_examples: List[Example]) -> Dict[str, float]:
    """
    Evaluate learned constraints on test set.
    """
    correct = 0
    false_positives = 0
    false_negatives = 0

    for ex in test_examples:
        # Check if learned constraints are satisfied
        predicted_feasible = all(c.check(ex.assignment)
                                for c in self.learned_constraints)

        if predicted_feasible == ex.is_feasible:
            correct += 1
        elif predicted_feasible and not ex.is_feasible:
            false_positives += 1
        else:
            false_negatives += 1

    n = len(test_examples)
    return {
        'accuracy': correct / n if n > 0 else 0,
        'false_positive_rate': false_positives / n if n > 0 else 0,
        'false_negative_rate': false_negatives / n if n > 0 else 0
    }

def report(self):
    """Print learning report."""
    print("=== Constraint Learning Report ===\n")

```

```

        print(f"Variables: {self.variables}")
        print(f"Domain sizes: {[len(self.domains[v]) for v in self.variables]}")
        print(f"Bias size: {len(self.bias)} candidate constraints")
        print(f"Training examples: {len(self.positive_examples)} positive, "
              f"{len(self.negative_examples)} negative")
        print(f"\nLearned {len(self.learned_constraints)} constraints:")
        for c in self.learned_constraints:
            print(f"  - {c.name}")

# Full pipeline example
print("=== Full Constraint Learning Pipeline ===\n")

# Create pipeline
pipeline = ConstraintLearningPipeline(
    variables=['x', 'y', 'z'],
    domains={v: set(range(1, 6)) for v in ['x', 'y', 'z']},
    max_arity=2
)

# Oracle: true constraints are x < y and y != z
def oracle(assignment):
    return assignment['x'] < assignment['y'] and assignment['y'] != assignment['z']

# Generate training data
np.random.seed(42)
for _ in range(50):
    assignment = {v: np.random.randint(1, 6) for v in ['x', 'y', 'z']}
    pipeline.add_example(assignment, oracle(assignment))

# Learn constraints
pipeline.learned_constraints = pipeline.learn(method='version_space')
pipeline.report()

# Evaluate on test set
test_examples = []
for _ in range(30):
    assignment = {v: np.random.randint(1, 6) for v in ['x', 'y', 'z']}
    test_examples.append(Example(assignment, oracle(assignment)))

metrics = pipeline.evaluate(test_examples)
print(f"\nTest Set Performance:")
print(f"  Accuracy: {metrics['accuracy']:.1%}")
print(f"  False Positive Rate: {metrics['false_positive_rate']:.1%}")
print(f"  False Negative Rate: {metrics['false_negative_rate']:.1%}")

```

=== Full Constraint Learning Pipeline ===

=== Constraint Learning Report ===

Variables: ['x', 'y', 'z']
Domain sizes: [5, 5, 5]
Bias size: 15 candidate constraints

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Training examples: 17 positive, 33 negative

Learned 1 constraints:

- $y \neq z$

Test Set Performance:

Accuracy: 66.7%

False Positive Rate: 33.3%

False Negative Rate: 0.0%

33.9. Summary

33.9.0.1. Key Takeaways

Constraint Learning Fundamentals

- Constraint learning automates the discovery of rules that define feasible solutions
- The constraint bias (hypothesis space) is essential for tractable learning
- PAC learning provides theoretical guarantees on sample complexity
- Active learning significantly reduces the number of required queries

Acquisition Algorithms

- QUACQ uses binary search to efficiently locate constraints
- CONACQ maintains a version space of consistent constraint networks
- Active strategies (uncertainty sampling, query-by-committee) optimize query selection
- Query complexity is typically linear in target constraint set size

Inverse Optimization

- Infers objectives or constraints from observed optimal decisions
- KKT conditions provide the mathematical foundation
- Regularization handles ill-posedness (multiple valid explanations)
- Multi-observation approaches improve robustness

Learning from Demonstrations

- Extracts constraints implicit in expert behavior
- Maximum margin methods find robust constraint boundaries
- Safety constraint learning is essential for autonomous systems
- Combines well with imitation learning for policies

Neural Constraint Learning

- OptNet, CombOptNet enable differentiable optimization layers
- cvxpylayers integrates convex optimization with deep learning
- End-to-end learning jointly optimizes prediction and decision-making
- Enables learning from downstream task performance

Applications

- Manufacturing: scheduling rules and machine constraints
- Healthcare: clinical pathway constraints
- Configuration: product compatibility rules
- Robotics: motion constraints and safety boundaries

33.10. Exercises

1. **Constraint Acquisition Implementation:** Implement the QUACQ algorithm for a scheduling domain with 5 tasks and 3 machines. Generate synthetic hidden constraints (precedence, machine eligibility) and measure how many queries are needed to discover them.
2. **Inverse LP Analysis:** Given the constraint matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 10 \\ 12 \end{bmatrix}$, and observed solution $\hat{x} = (2, 4)$, find all objective vectors c that make this solution optimal. Characterize the set geometrically.
3. **Safety Constraint Learning:** Design a safety constraint learning system for a 2D navigation robot. Generate trajectories that avoid obstacles, learn the obstacle boundaries as constraints, and implement a safety filter that prevents violations.
4. **Neural Constraint Layer:** Implement a simple differentiable optimization layer for portfolio optimization. The layer should take expected returns as input and output optimal portfolio weights. Train the layer to match expert portfolios.
5. **Clinical Pathway Mining:** Using the healthcare case study as a template, design a constraint learning system for treatment pathways. Include temporal constraints (treatment A within 24 hours of B), exclusion constraints, and dosage constraints.
6. **Active Learning Comparison:** Implement and compare three active learning strategies (uncertainty sampling, version space splitting, query-by-committee) on a constraint learning task. Measure learning curves and final accuracy.
7. **Configuration Rule Discovery:** Create a product configuration constraint learner for a computer system with: CPU (3 types), GPU (4 types), RAM (4 options), Storage (3 types). Define hidden compatibility rules and evaluate how well they can be recovered.

33.11. Further Reading

Foundational Papers

- Bessiere, C., et al. (2017). “Constraint Acquisition.” *Artificial Intelligence*, 244, 315-342.
- Pawlak, T. P., & Krawiec, K. (2017). “Automatic Synthesis of Constraints from Examples Using Mixed Integer Linear Programming.” *European Journal of Operational Research*, 261(3), 1030-1040.

Inverse Optimization

- Ahuja, R. K., & Orlin, J. B. (2001). “Inverse Optimization.” *Operations Research*, 49(5), 771-783.
- Bertsimas, D., et al. (2015). “Data-driven Estimation in Equilibrium Using Inverse Optimization.” *Mathematical Programming*, 153(2), 595-633.
- Chan, T. C., et al. (2019). “Inverse Optimization: Theory and Applications.” *Operations Research*.

Neural Optimization Layers

- Amos, B., & Kolter, J. Z. (2017). “OptNet: Differentiable Optimization as a Layer in Neural Networks.” *ICML*.
- Agrawal, A., et al. (2019). “Differentiable Convex Optimization Layers.” *NeurIPS*.
- Pogancic, M. V., et al. (2020). “Differentiation of Blackbox Combinatorial Solvers.” *ICLR*.

Learning from Demonstrations

33. Constraint Learning

- Chou, G., et al. (2021). “Learning Constraints from Demonstrations.” *International Workshop on the Algorithmic Foundations of Robotics*.
- Scobee, D. R., & Sastry, S. S. (2020). “Maximum Likelihood Constraint Inference for Inverse Reinforcement Learning.” *ICLR*.

Software and Tools

- CPMpy: <https://cpmpy.readthedocs.io/>
- cvxpylayers: <https://github.com/cvxgrp/cvxpylayers>
- Google OR-Tools: <https://developers.google.com/optimization>

34. Graph Neural Networks for Combinatorial Optimization

Learning Objectives

After completing this chapter, you will be able to:

1. Understand why graph representations are natural for combinatorial optimization problems
2. Implement core GNN architectures including GCN, GraphSAGE, GAT, and GIN
3. Represent classic optimization problems (TSP, VRP, Scheduling, MILP) as graphs
4. Train GNN models using supervised learning, reinforcement learning, and unsupervised approaches
5. Build attention-based models with GNN encoders for routing problems
6. Apply GNNs to accelerate MILP solvers through learning to branch and cut
7. Address scalability challenges when deploying GNNs to large-scale instances
8. Implement a complete GNN-based solver for the Traveling Salesman Problem

34.1. Introduction to GNNs for Optimization

Combinatorial optimization problems are ubiquitous in operations research, logistics, manufacturing, and countless other domains. While traditional approaches rely on hand-crafted heuristics or exact algorithms, Graph Neural Networks (GNNs) offer a powerful paradigm for learning to solve these problems directly from data. The key insight is that most combinatorial problems have an inherent graph structure—cities and routes in TSP, jobs and machines in scheduling, variables and constraints in MILP—making GNNs a natural fit.

34.1.1. Why Graphs for Optimization Problems

Many optimization problems can be naturally expressed as graphs:

- **Routing problems:** Cities are nodes, possible routes are edges
- **Scheduling problems:** Jobs and resources are nodes, constraints are edges
- **Facility location:** Customers and facilities form a bipartite graph
- **MILP:** Variables, constraints, and their relationships form a tripartite graph

This graph structure captures the relational nature of these problems in a way that standard neural networks cannot.

```
import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
import networkx as nx
import matplotlib.pyplot as plt
from typing import List, Tuple, Dict, Optional, Union
import warnings
```

```

warnings.filterwarnings('ignore')

# Set random seeds for reproducibility
np.random.seed(42)
torch.manual_seed(42)

def create_tsp_graph(n_cities: int, seed: int = 42) -> Tuple[nx.Graph, np.ndarray]:
    """
    Create a complete graph representing a TSP instance.

    Args:
        n_cities: Number of cities
        seed: Random seed for reproducibility

    Returns:
        G: NetworkX graph with edge weights as distances
        coords: City coordinates (n_cities, 2)
    """
    np.random.seed(seed)

    # Generate random city coordinates in unit square
    coords = np.random.rand(n_cities, 2)

    # Create complete graph
    G = nx.complete_graph(n_cities)

    # Add edge weights (Euclidean distances)
    for i, j in G.edges():
        dist = np.linalg.norm(coords[i] - coords[j])
        G[i][j]['weight'] = dist

    # Add node features (coordinates)
    for i in range(n_cities):
        G.nodes[i]['pos'] = coords[i]

    return G, coords

def visualize_tsp_instance(coords: np.ndarray, tour: Optional[List[int]] = None,
                           title: str = "TSP Instance") -> None:
    """Visualize a TSP instance with optional tour."""
    fig, ax = plt.subplots(figsize=(8, 8))

    # Plot cities
    ax.scatter(coords[:, 0], coords[:, 1], s=200, c='steelblue',
               zorder=5, edgecolors='white', linewidth=2)

    # Add city labels
    for i, (x, y) in enumerate(coords):
        ax.annotate(str(i), (x, y), ha='center', va='center',
                    fontsize=10, color='white', fontweight='bold')

    # Plot tour if provided

```

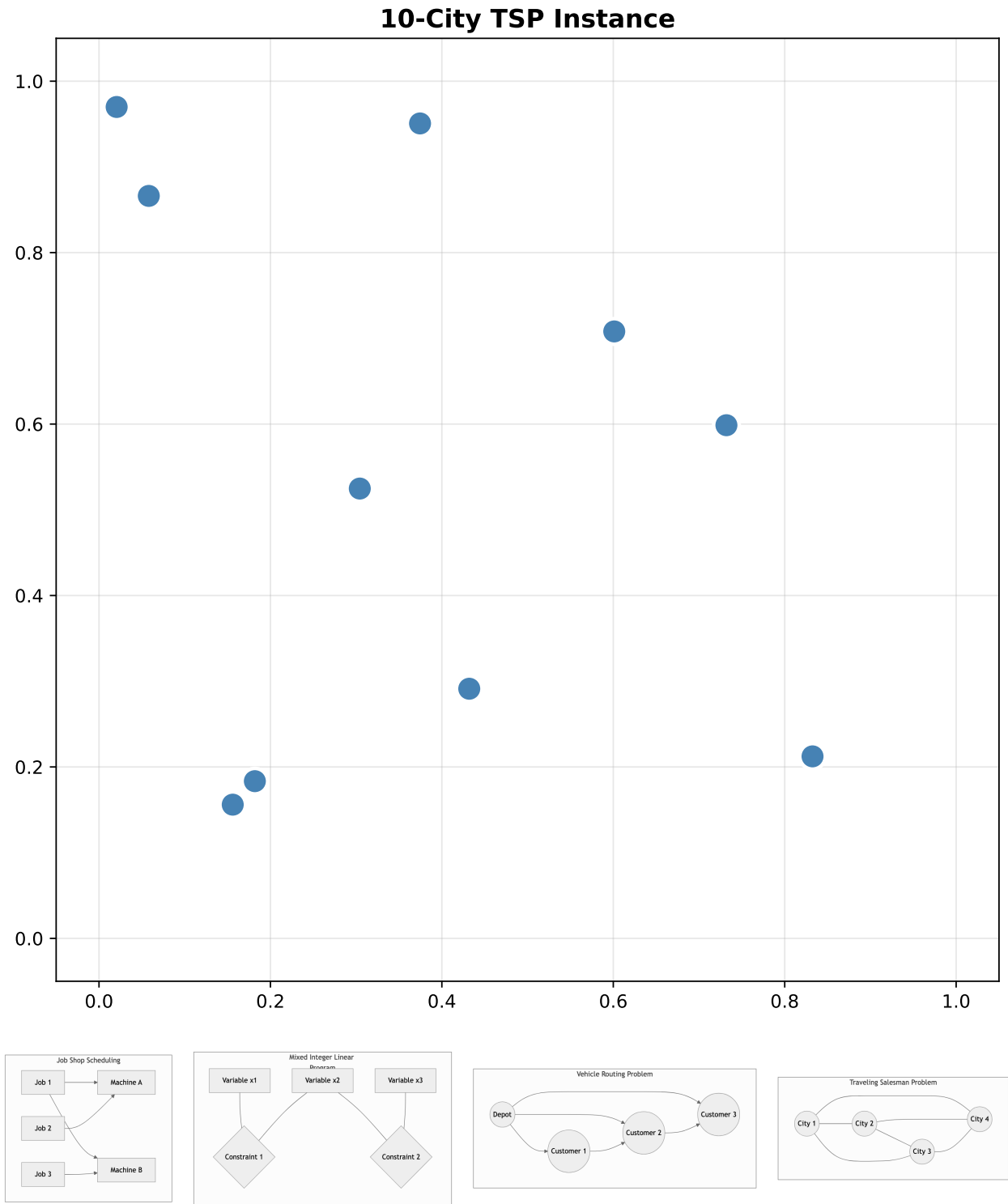



Figure 34.1.: Graph representations of different combinatorial optimization problems

34.1.2. GNNs vs Attention-Only Models

While Transformer-based attention models have shown impressive results on sequence-to-sequence tasks, GNNs offer distinct advantages for combinatorial optimization:

i Key Differences

Aspect	Transformers	GNNs
Structure	Treats input as sequence	Explicitly models graph topology
Inductive bias	Position-based	Relation-based
Scalability	$O(n^2)$ attention	$O(n + m)$ message passing
Generalization	Fixed sequence length	Variable graph sizes
Edge features	Indirect encoding	Native support

```
def compare_complexity(n_nodes: List[int]) -> Dict[str, List[int]]:
    """
    Compare computational complexity of different architectures.

    Args:
        n_nodes: List of problem sizes to evaluate

    Returns:
        Dictionary with complexity values for each architecture
    """
    complexities = {
        'n_nodes': n_nodes,
        'transformer_attention': [], #  $O(n^2)$ 
        'gnn_sparse': [],           #  $O(n + m)$  where  $m \sim n$  for sparse
        'gnn_dense': [],           #  $O(n + m)$  where  $m \sim n^2$  for dense
    }

    for n in n_nodes:
        # Transformer:  $O(n^2)$  for self-attention
        complexities['transformer_attention'].append(n ** 2)

        # GNN with sparse graph (e.g., k-nearest neighbors, k=10)
        k = 10
        m_sparse = n * k
        complexities['gnn_sparse'].append(n + m_sparse)

        # GNN with dense graph (complete graph)
        m_dense = n * (n - 1) // 2
        complexities['gnn_dense'].append(n + m_dense)

    return complexities

# Visualize complexity scaling
n_values = [10, 50, 100, 200, 500, 1000, 2000]
complexity = compare_complexity(n_values)

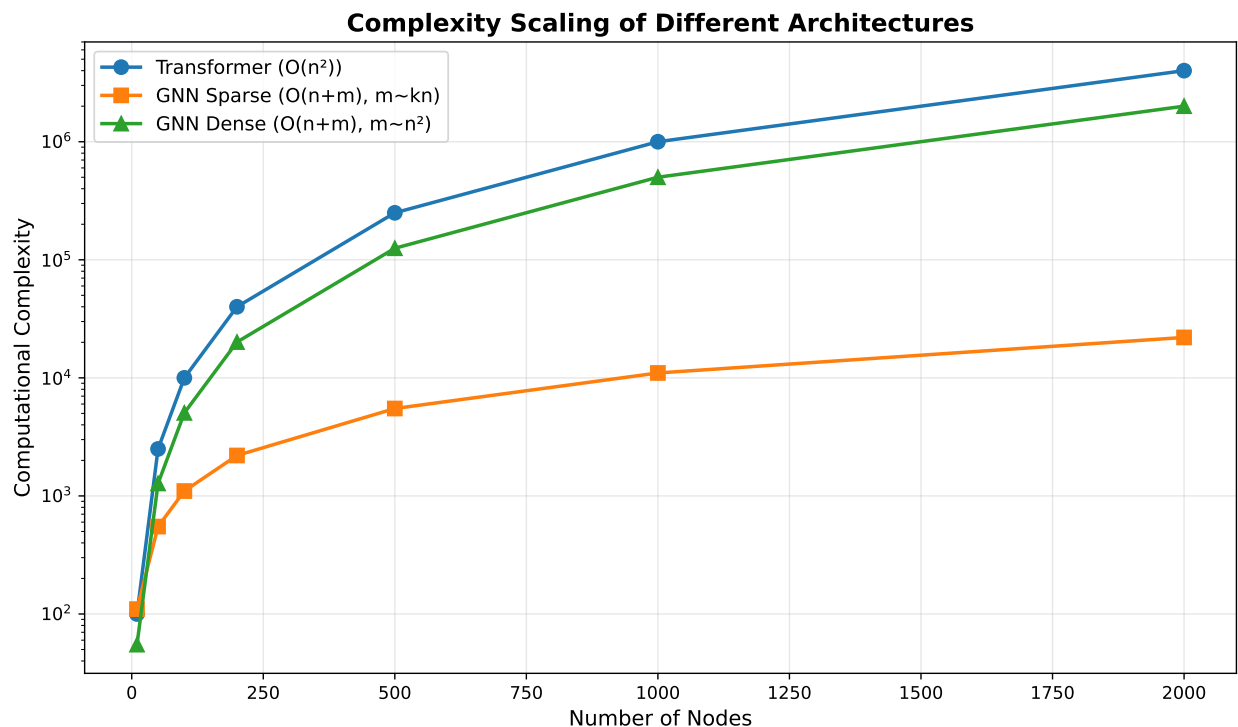
fig, ax = plt.subplots(figsize=(10, 6))
```

```

ax.plot(complexity['n_nodes'], complexity['transformer_attention'],
        'o-', label='Transformer ( $O(n^2)$ )', linewidth=2, markersize=8)
ax.plot(complexity['n_nodes'], complexity['gnn_sparse'],
        's-', label='GNN Sparse ( $O(n+m)$ ,  $m \sim kn$ )', linewidth=2, markersize=8)
ax.plot(complexity['n_nodes'], complexity['gnn_dense'],
        '^-', label='GNN Dense ( $O(n+m)$ ,  $m \sim n^2$ )', linewidth=2, markersize=8)

ax.set_xlabel('Number of Nodes', fontsize=12)
ax.set_ylabel('Computational Complexity', fontsize=12)
ax.set_title('Complexity Scaling of Different Architectures', fontsize=14,
             fontweight='bold')
ax.legend(fontsize=11)
ax.set_yscale('log')
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```



34.1.3. Permutation Invariance and Equivariance

A key property for combinatorial optimization is **permutation invariance**: the solution quality should not depend on how we order the input nodes. GNNs naturally satisfy this property through their message-passing mechanism.

! Permutation Properties

- **Permutation Invariance**: A function f is permutation invariant if $f(\pi(G)) = f(G)$ for any permutation π . This is essential for graph-level predictions.

- **Permutation Equivariance:** A function f is permutation equivariant if $f(\pi(G)) = \pi(f(G))$. This is essential for node-level predictions where we need consistent node representations regardless of input ordering.

GNNs achieve equivariance by design—they aggregate neighbor information symmetrically without relying on node ordering.

```
def demonstrate_permutation_invariance():
    """
    Demonstrate that GNN outputs are permutation equivariant.
    """
    # Create a simple graph
    n_nodes = 5
    edge_index = torch.tensor([
        [0, 1, 1, 2, 2, 3, 3, 4, 4, 0],
        [1, 0, 2, 1, 3, 2, 4, 3, 0, 4]
    ], dtype=torch.long)

    # Node features
    x = torch.randn(n_nodes, 16)

    # Simple message passing layer
    class SimpleGNNLayer(nn.Module):
        def __init__(self, in_dim, out_dim):
            super().__init__()
            self.linear = nn.Linear(in_dim, out_dim)

        def forward(self, x, edge_index):
            # Aggregate neighbor features (mean aggregation)
            row, col = edge_index
            out = torch.zeros_like(x)
            for i in range(x.size(0)):
                neighbors = col[row == i]
                if len(neighbors) > 0:
                    out[i] = x[neighbors].mean(dim=0)
            return self.linear(out)

    layer = SimpleGNNLayer(16, 16)

    # Original output
    out_original = layer(x, edge_index)

    # Create a permutation
    perm = torch.tensor([2, 4, 0, 1, 3])
    inv_perm = torch.argsort(perm)

    # Permute node features
    x_perm = x[perm]

    # Permute edge index
    edge_index_perm = inv_perm[edge_index]
```

```

# Output on permuted graph
out_perm = layer(x_perm, edge_index_perm)

# The output should be the permuted version of original
out_original_perm = out_original[perm]

# Check equivariance
diff = torch.abs(out_perm - out_original_perm).max().item()
print(f"Maximum difference after permutation: {diff:.10f}")
print(f"Permutation equivariance satisfied: {diff < 1e-5}")

return diff < 1e-5

is_equivariant = demonstrate_permutation_invariance()

```

Maximum difference after permutation: 0.0000000596

Permutation equivariance satisfied: True

34.2. Graph Neural Network Architectures

This section covers the fundamental GNN architectures that form the building blocks for optimization models. We'll implement each architecture from scratch and then show how to use PyTorch Geometric for production-ready implementations.

34.2.1. Message Passing Neural Networks

The Message Passing Neural Network (MPNN) framework unifies many GNN variants. At each layer, nodes update their representations by aggregating messages from neighbors:

$$h_v^{(l+1)} = \text{UPDATE}^{(l)} \left(h_v^{(l)}, \text{AGGREGATE}^{(l)} \left(\{m_{uv}^{(l)} : u \in \mathcal{N}(v)\} \right) \right)$$

where: - $h_v^{(l)}$ is node v 's representation at layer l - $m_{uv}^{(l)} = \text{MESSAGE}^{(l)}(h_u^{(l)}, h_v^{(l)}, e_{uv})$ is the message from node u to v - $\mathcal{N}(v)$ is the neighborhood of node v

The following implementation builds a generic MPNN layer with configurable message, aggregate, and update functions. This layer serves as the base class for all GNN variants in this chapter.

```

class MessagePassingLayer(nn.Module):
    """
    Generic Message Passing Neural Network layer.

    This provides a flexible framework where different GNN variants
    can be implemented by specifying message, aggregate, and update functions.
    """

    def __init__(self, in_dim: int, out_dim: int,
                 edge_dim: Optional[int] = None,
                 aggregation: str = 'mean'):
        """
        Args:

```

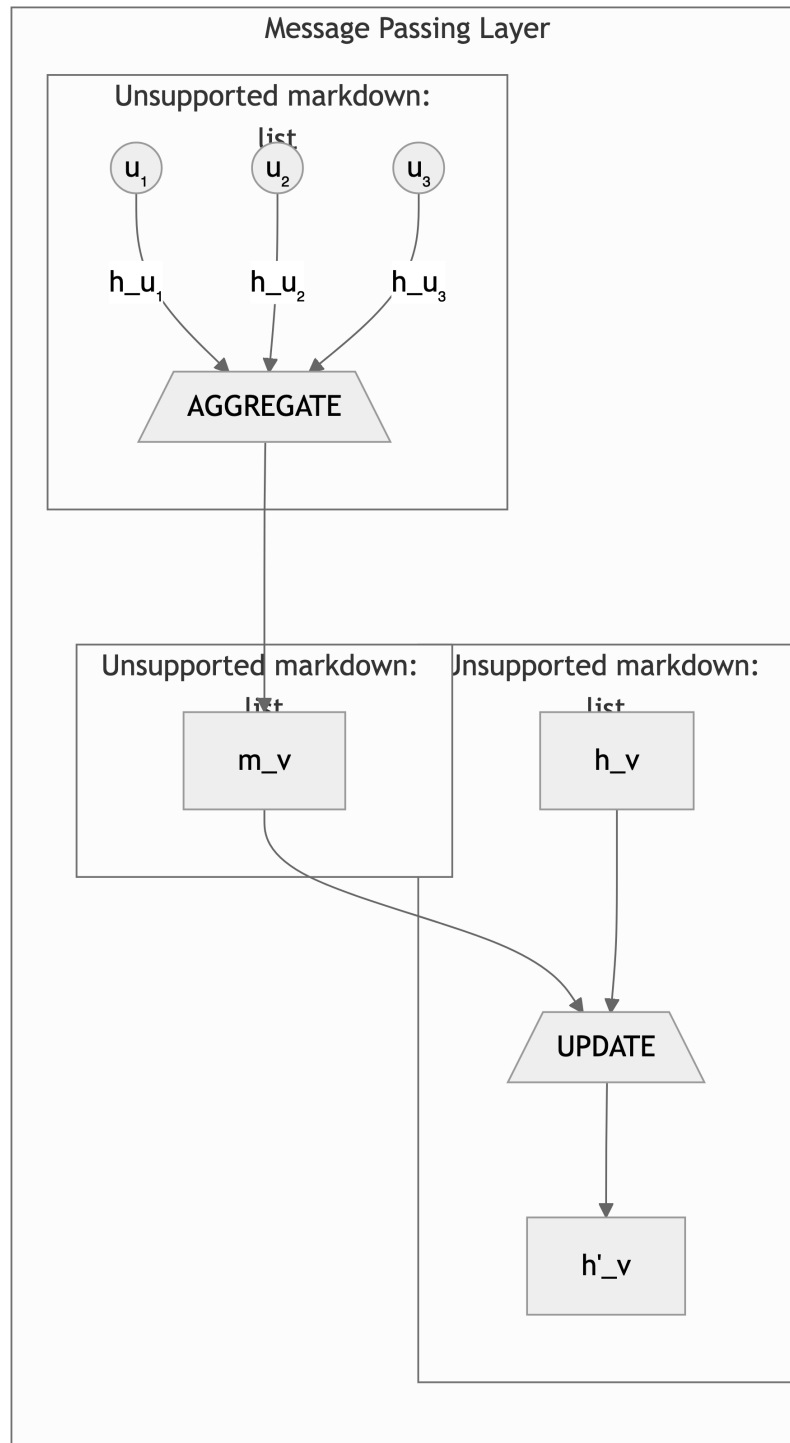



Figure 34.2.: Message passing framework: nodes aggregate information from neighbors

```

        in_dim: Input feature dimension
        out_dim: Output feature dimension
        edge_dim: Edge feature dimension (optional)
        aggregation: Aggregation method ('mean', 'sum', 'max')
    """
    super().__init__()
    self.in_dim = in_dim
    self.out_dim = out_dim
    self.edge_dim = edge_dim
    self.aggregation = aggregation

    # Message function: transforms source node features
    msg_in_dim = in_dim + (edge_dim if edge_dim else 0)
    self.message_mlp = nn.Sequential(
        nn.Linear(msg_in_dim, out_dim),
        nn.ReLU(),
        nn.Linear(out_dim, out_dim)
    )

    # Update function: combines aggregated messages with node features
    self.update_mlp = nn.Sequential(
        nn.Linear(in_dim + out_dim, out_dim),
        nn.ReLU(),
        nn.Linear(out_dim, out_dim)
    )

    def message(self, h_source: torch.Tensor,
                edge_features: Optional[torch.Tensor] = None) -> torch.Tensor:
        """Compute messages from source nodes."""
        if edge_features is not None:
            msg_input = torch.cat([h_source, edge_features], dim=-1)
        else:
            msg_input = h_source
        return self.message_mlp(msg_input)

    def aggregate(self, messages: torch.Tensor,
                  target_index: torch.Tensor,
                  n_nodes: int) -> torch.Tensor:
        """Aggregate messages for each target node."""
        # Initialize aggregated messages
        aggregated = torch.zeros(n_nodes, messages.size(-1),
                                device=messages.device)

        if self.aggregation == 'sum':
            aggregated.scatter_add_(0,
        ↪ target_index.unsqueeze(-1).expand_as(messages),
                                messages)
        elif self.aggregation == 'mean':
            # Sum then divide by count
            aggregated.scatter_add_(0,
        ↪ target_index.unsqueeze(-1).expand_as(messages),
                                messages)
            count = torch.zeros(n_nodes, device=messages.device)

```

```

        count.scatter_add_(0, target_index, torch.ones_like(target_index,
↳ dtype=torch.float))
        count = count.clamp(min=1).unsqueeze(-1)
        aggregated = aggregated / count
        elif self.aggregation == 'max':
            # Use scatter_reduce for max aggregation
            aggregated.scatter_reduce_(0,
↳ target_index.unsqueeze(-1).expand_as(messages),
                                   messages, reduce='amax', include_self=False)

        return aggregated

def update(self, h_node: torch.Tensor,
          aggregated: torch.Tensor) -> torch.Tensor:
    """Update node representations."""
    combined = torch.cat([h_node, aggregated], dim=-1)
    return self.update_mlp(combined)

def forward(self, x: torch.Tensor, edge_index: torch.Tensor,
            edge_features: Optional[torch.Tensor] = None) -> torch.Tensor:
    """
    Forward pass of message passing.

    Args:
        x: Node features (n_nodes, in_dim)
        edge_index: Edge connectivity (2, n_edges)
        edge_features: Edge features (n_edges, edge_dim), optional

    Returns:
        Updated node features (n_nodes, out_dim)
    """
    source, target = edge_index
    n_nodes = x.size(0)

    # Get source node features
    h_source = x[source]

    # Compute messages
    messages = self.message(h_source, edge_features)

    # Aggregate messages
    aggregated = self.aggregate(messages, target, n_nodes)

    # Update node representations
    h_new = self.update(x, aggregated)

    return h_new

# Test the MPNN layer
def test_mpnn():
    n_nodes, in_dim, out_dim = 10, 32, 64

```

```

# Create random graph
edge_index = torch.randint(0, n_nodes, (2, 30))
x = torch.randn(n_nodes, in_dim)

layer = MessagePassingLayer(in_dim, out_dim, aggregation='mean')
out = layer(x, edge_index)

print(f"Input shape: {x.shape}")
print(f"Output shape: {out.shape}")
print(f"MPNN layer test passed!")

test_mpnn()

```

```

Input shape: torch.Size([10, 32])
Output shape: torch.Size([10, 64])
MPNN layer test passed!

```

34.2.2. Graph Convolutional Network (GCN)

The Graph Convolutional Network (GCN) by Kipf & Welling is one of the most widely used GNN architectures. It performs spectral convolutions approximated in the spatial domain:

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$$

where $\tilde{A} = A + I$ is the adjacency matrix with self-loops and $\tilde{D}$ is the corresponding degree matrix. The GCN layer applies symmetric normalized aggregation. Each node's updated representation is a weighted sum of its neighbors' features, normalized by the geometric mean of node degrees.

```

class GCNLayer(nn.Module):
    """
    Graph Convolutional Network layer.

    Implements the GCN convolution with symmetric normalization.
    """

    def __init__(self, in_dim: int, out_dim: int,
                  bias: bool = True,
                  add_self_loops: bool = True):
        """
        Args:
            in_dim: Input feature dimension
            out_dim: Output feature dimension
            bias: Whether to include bias term
            add_self_loops: Whether to add self-loops
        """
        super().__init__()
        self.in_dim = in_dim
        self.out_dim = out_dim
        self.add_self_loops = add_self_loops

        self.weight = nn.Parameter(torch.Tensor(in_dim, out_dim))
        if bias:

```

```

        self.bias = nn.Parameter(torch.Tensor(out_dim))
    else:
        self.register_parameter('bias', None)

    self.reset_parameters()

def reset_parameters(self):
    nn.init.xavier_uniform_(self.weight)
    if self.bias is not None:
        nn.init.zeros_(self.bias)

def forward(self, x: torch.Tensor, edge_index: torch.Tensor) -> torch.Tensor:
    """
    Forward pass.

    Args:
        x: Node features (n_nodes, in_dim)
        edge_index: Edge connectivity (2, n_edges)

    Returns:
        Updated node features (n_nodes, out_dim)
    """
    n_nodes = x.size(0)

    # Add self-loops
    if self.add_self_loops:
        self_loops = torch.arange(n_nodes,
        ↪ device=x.device).unsqueeze(0).repeat(2, 1)
        edge_index = torch.cat([edge_index, self_loops], dim=1)

    # Compute degree for normalization
    row, col = edge_index
    deg = torch.zeros(n_nodes, device=x.device)
    deg.scatter_add_(0, row, torch.ones(row.size(0), device=x.device))

    # Symmetric normalization:  $D^{-1/2}$ 
    deg_inv_sqrt = deg.pow(-0.5)
    deg_inv_sqrt[deg_inv_sqrt == float('inf')] = 0

    # Normalize edge weights
    norm = deg_inv_sqrt[row] * deg_inv_sqrt[col]

    # Transform features
    ↪ x = x @ self.weight

    # Aggregate with normalization
    out = torch.zeros_like(x)
    for i in range(edge_index.size(1)):
        src, tgt = edge_index[0, i], edge_index[1, i]
        out[tgt] += norm[i] * x[src]

    if self.bias is not None:
        out = out + self.bias

```

```

        return out

class GCN(nn.Module):
    """
    Multi-layer Graph Convolutional Network.
    """

    def __init__(self, in_dim: int, hidden_dim: int, out_dim: int,
                  n_layers: int = 2, dropout: float = 0.5):
        super().__init__()

        self.layers = nn.ModuleList()
        self.dropout = dropout

        # Input layer
        self.layers.append(GCNLayer(in_dim, hidden_dim))

        # Hidden layers
        for _ in range(n_layers - 2):
            self.layers.append(GCNLayer(hidden_dim, hidden_dim))

        # Output layer
        self.layers.append(GCNLayer(hidden_dim, out_dim))

    def forward(self, x: torch.Tensor, edge_index: torch.Tensor) -> torch.Tensor:
        for i, layer in enumerate(self.layers[:-1]):
            x = layer(x, edge_index)
            x = F.relu(x)
            x = F.dropout(x, p=self.dropout, training=self.training)

        x = self.layers[-1](x, edge_index)
        return x

# Test GCN
gcn = GCN(32, 64, 16, n_layers=3)
x_test = torch.randn(20, 32)
edge_index_test = torch.randint(0, 20, (2, 50))
out_gcn = gcn(x_test, edge_index_test)
print(f"GCN output shape: {out_gcn.shape}")

```

GCN output shape: torch.Size([20, 16])

34.2.3. GraphSAGE

GraphSAGE (Graph Sample and Aggregate) introduces sampling-based aggregation, making it scalable to large graphs:

$$h_v^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{CONCAT} \left(h_v^{(l)}, \text{AGG} \left(\{h_u^{(l)} : u \in \mathcal{N}(v)\} \right) \right) \right)$$

The GraphSAGE layer concatenates a node's own features with its aggregated neighbor features,

then applies a linear transformation. This design preserves the node's original information while incorporating neighborhood structure.

```
class GraphSAGELayer(nn.Module):
    """
    GraphSAGE layer with configurable aggregation.
    """

    def __init__(self, in_dim: int, out_dim: int,
                 aggregator: str = 'mean',
                 normalize: bool = True):
        """
        Args:
            in_dim: Input feature dimension
            out_dim: Output feature dimension
            aggregator: Aggregation type ('mean', 'max', 'lstm')
            normalize: Whether to L2-normalize output
        """
        super().__init__()
        self.in_dim = in_dim
        self.out_dim = out_dim
        self.aggregator = aggregator
        self.normalize = normalize

        # Linear transformation for concatenated features
        self.linear = nn.Linear(2 * in_dim, out_dim)

        if aggregator == 'lstm':
            self.lstm = nn.LSTM(in_dim, in_dim, batch_first=True)

    def aggregate_neighbors(self, x: torch.Tensor,
                          edge_index: torch.Tensor,
                          n_nodes: int) -> torch.Tensor:
        """Aggregate neighbor features."""
        source, target = edge_index

        # Initialize aggregated features
        aggregated = torch.zeros(n_nodes, x.size(-1), device=x.device)

        if self.aggregator == 'mean':
            # Mean aggregation
            aggregated.scatter_add_(0, target.unsqueeze(-1).expand(-1, x.size(-1)),
                                   x[source])
            count = torch.zeros(n_nodes, device=x.device)
            count.scatter_add_(0, target, torch.ones_like(target,
                                                              dtype=torch.float))
            count = count.clamp(min=1).unsqueeze(-1)
            aggregated = aggregated / count

        elif self.aggregator == 'max':
            # Max pooling aggregation
            for i in range(n_nodes):
                neighbors = source[target == i]
```

```

        if len(neighbors) > 0:
            aggregated[i] = x[neighbors].max(dim=0)[0]

    elif self.aggregator == 'lstm':
        # LSTM aggregation (permutation variant, but powerful)
        for i in range(n_nodes):
            neighbors = source[target == i]
            if len(neighbors) > 0:
                neighbor_feats = x[neighbors].unsqueeze(0) # (1, n_neighbors,
    ↪ dim)
                _, (h_n, _) = self.lstm(neighbor_feats)
                aggregated[i] = h_n.squeeze(0).squeeze(0)

    return aggregated

def forward(self, x: torch.Tensor, edge_index: torch.Tensor) -> torch.Tensor:
    n_nodes = x.size(0)

    # Aggregate neighbor features
    neighbor_agg = self.aggregate_neighbors(x, edge_index, n_nodes)

    # Concatenate self and neighbor features
    combined = torch.cat([x, neighbor_agg], dim=-1)

    # Transform
    out = self.linear(combined)

    # Optional L2 normalization
    if self.normalize:
        out = F.normalize(out, p=2, dim=-1)

    return out

class GraphSAGE(nn.Module):
    """Multi-layer GraphSAGE model."""

    def __init__(self, in_dim: int, hidden_dim: int, out_dim: int,
                 n_layers: int = 2, aggregator: str = 'mean',
                 dropout: float = 0.5):
        super().__init__()

        self.layers = nn.ModuleList()
        self.dropout = dropout

        # Input layer
        self.layers.append(GraphSAGELayer(in_dim, hidden_dim, aggregator))

        # Hidden layers
        for _ in range(n_layers - 2):
            self.layers.append(GraphSAGELayer(hidden_dim, hidden_dim, aggregator))

        # Output layer

```



```

self.layers.append(GraphSAGELayer(hidden_dim, out_dim, aggregator))

def forward(self, x: torch.Tensor, edge_index: torch.Tensor) -> torch.Tensor:
    for i, layer in enumerate(self.layers[:-1]):
        x = layer(x, edge_index)
        x = F.relu(x)
        x = F.dropout(x, p=self.dropout, training=self.training)

    x = self.layers[-1](x, edge_index)
    return x

# Test GraphSAGE
sage = GraphSAGE(32, 64, 16, n_layers=2, aggregator='mean')
out_sage = sage(x_test, edge_index_test)
print(f"GraphSAGE output shape: {out_sage.shape}")

```

GraphSAGE output shape: torch.Size([20, 16])

34.2.4. Graph Attention Network (GAT)

Graph Attention Networks use attention mechanisms to weigh neighbor contributions:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T [Wh_i \| Wh_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(a^T [Wh_i \| Wh_k]))}$$

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} Wh_j \right)$$

The GAT layer computes attention coefficients between neighboring nodes, allowing the model to learn which neighbors are most relevant for each node. Multi-head attention is used to stabilize training.

```

class GATLayer(nn.Module):
    """
    Graph Attention Network layer.

    Implements multi-head attention over graph neighborhoods.
    """

    def __init__(self, in_dim: int, out_dim: int,
                 n_heads: int = 4,
                 dropout: float = 0.1,
                 concat: bool = True,
                 negative_slope: float = 0.2):
        """
        Args:
            in_dim: Input feature dimension
            out_dim: Output feature dimension (per head if concat=True)
            n_heads: Number of attention heads
            dropout: Dropout rate for attention weights
            concat: Whether to concatenate heads (True) or average (False)
        """

```

```

        negative_slope: Negative slope for LeakyReLU
    """
    super().__init__()
    self.in_dim = in_dim
    self.out_dim = out_dim
    self.n_heads = n_heads
    self.dropout = dropout
    self.concat = concat
    self.negative_slope = negative_slope

    # Linear transformation for each head
    self.W = nn.Parameter(torch.Tensor(n_heads, in_dim, out_dim))

    # Attention parameters for each head
    self.a_src = nn.Parameter(torch.Tensor(n_heads, out_dim, 1))
    self.a_tgt = nn.Parameter(torch.Tensor(n_heads, out_dim, 1))

    self.leaky_relu = nn.LeakyReLU(negative_slope)
    self.reset_parameters()

def reset_parameters(self):
    nn.init.xavier_uniform_(self.W)
    nn.init.xavier_uniform_(self.a_src)
    nn.init.xavier_uniform_(self.a_tgt)

def forward(self, x: torch.Tensor, edge_index: torch.Tensor) -> torch.Tensor:
    n_nodes = x.size(0)
    source, target = edge_index

    # Transform features for all heads: (n_nodes, n_heads, out_dim)
    h = torch.einsum('nd,hdo->nho', x, self.W)

    # Compute attention scores
    # Source attention: (n_nodes, n_heads, 1)
    attn_src = torch.einsum('nho,hoi->nhi', h, self.a_src)
    # Target attention: (n_nodes, n_heads, 1)
    attn_tgt = torch.einsum('nho,hoi->nhi', h, self.a_tgt)

    # Compute edge attention scores
    # (n_edges, n_heads)
    e = attn_src[source].squeeze(-1) + attn_tgt[target].squeeze(-1)
    e = self.leaky_relu(e)

    # Softmax over neighborhood
    # First, compute max for numerical stability
    e_max = torch.zeros(n_nodes, self.n_heads, device=x.device)
    e_max.scatter_reduce_(0, target.unsqueeze(-1).expand(-1, self.n_heads),
                          e, reduce='amax', include_self=True)
    e = e - e_max[target]
    e_exp = torch.exp(e)

    # Sum of exponentials for normalization
    e_sum = torch.zeros(n_nodes, self.n_heads, device=x.device)

```

```

e_sum.scatter_add_(0, target.unsqueeze(-1).expand(-1, self.n_heads), e_exp)

# Normalized attention weights
alpha = e_exp / (e_sum[target] + 1e-10)
alpha = F.dropout(alpha, p=self.dropout, training=self.training)

# Aggregate
# (n_edges, n_heads, out_dim)
weighted = alpha.unsqueeze(-1) * h[source]

out = torch.zeros(n_nodes, self.n_heads, self.out_dim, device=x.device)
out.scatter_add_(0, target.unsqueeze(-1).unsqueeze(-1).expand(-1,
↪ self.n_heads, self.out_dim),
                    weighted)

if self.concat:
    # Concatenate heads: (n_nodes, n_heads * out_dim)
    out = out.view(n_nodes, -1)
else:
    # Average heads: (n_nodes, out_dim)
    out = out.mean(dim=1)

return out

class GAT(nn.Module):
    """Multi-layer Graph Attention Network."""

    def __init__(self, in_dim: int, hidden_dim: int, out_dim: int,
                 n_layers: int = 2, n_heads: int = 4,
                 dropout: float = 0.5):
        super().__init__()

        self.layers = nn.ModuleList()
        self.dropout = dropout

        # Input layer (concatenate heads)
        self.layers.append(GATLayer(in_dim, hidden_dim, n_heads, dropout,
↪ concat=True))

        # Hidden layers
        for _ in range(n_layers - 2):
            self.layers.append(GATLayer(hidden_dim * n_heads, hidden_dim,
                                         n_heads, dropout, concat=True))

        # Output layer (average heads)
        self.layers.append(GATLayer(hidden_dim * n_heads, out_dim,
                                         n_heads, dropout, concat=False))

    def forward(self, x: torch.Tensor, edge_index: torch.Tensor) -> torch.Tensor:
        for i, layer in enumerate(self.layers[:-1]):
            x = layer(x, edge_index)
            x = F.elu(x)

```

```

        x = F.dropout(x, p=self.dropout, training=self.training)

        x = self.layers[-1](x, edge_index)
        return x

# Test GAT
gat = GAT(32, 16, 16, n_layers=2, n_heads=4)
out_gat = gat(x_test, edge_index_test)
print(f"GAT output shape: {out_gat.shape}")

```

GAT output shape: torch.Size([20, 16])

34.2.5. Graph Isomorphism Network (GIN)

The Graph Isomorphism Network is designed to be as powerful as the Weisfeiler-Lehman graph isomorphism test:

$$h_v^{(l+1)} = \text{MLP}^{(l)} \left((1 + \epsilon^{(l)}) \cdot h_v^{(l)} + \sum_{u \in \mathcal{N}(v)} h_u^{(l)} \right)$$

💡 When to Use GIN

GIN is particularly useful when you need: - Maximum expressive power for distinguishing graph structures - Theoretical guarantees on graph isomorphism testing capability - Applications where structural differences matter (e.g., molecular property prediction)

For combinatorial optimization, GIN can be beneficial when the problem structure varies significantly across instances.

The GIN layer uses sum aggregation (rather than mean) and an MLP update, which makes it maximally expressive among message-passing GNNs. The learnable parameter ϵ controls the relative weight of the node's own features versus its neighbors.

```

class GINLayer(nn.Module):
    """
    Graph Isomorphism Network layer.

    Achieves maximum expressive power among message-passing GNNs.
    """

    def __init__(self, in_dim: int, out_dim: int,
                 hidden_dim: Optional[int] = None,
                 eps: float = 0.0,
                 train_eps: bool = True):
        """
        Args:
            in_dim: Input feature dimension
            out_dim: Output feature dimension
            hidden_dim: Hidden dimension in MLP (defaults to out_dim)
            eps: Initial epsilon value
        """

```

```

        train_eps: Whether epsilon is trainable
    """
    super().__init__()

    hidden_dim = hidden_dim or out_dim

    self.mlp = nn.Sequential(
        nn.Linear(in_dim, hidden_dim),
        nn.BatchNorm1d(hidden_dim),
        nn.ReLU(),
        nn.Linear(hidden_dim, out_dim),
        nn.BatchNorm1d(out_dim),
        nn.ReLU()
    )

    if train_eps:
        self.eps = nn.Parameter(torch.tensor(eps))
    else:
        self.register_buffer('eps', torch.tensor(eps))

    def forward(self, x: torch.Tensor, edge_index: torch.Tensor) -> torch.Tensor:
        n_nodes = x.size(0)
        source, target = edge_index

        # Sum aggregation
        neighbor_sum = torch.zeros(n_nodes, x.size(-1), device=x.device)
        neighbor_sum.scatter_add_(0, target.unsqueeze(-1).expand(-1, x.size(-1)),
                                   x[source])

        # (1 + eps) * x + sum(neighbors)
        out = (1 + self.eps) * x + neighbor_sum

        # MLP transformation
        out = self.mlp(out)

        return out

class GIN(nn.Module):
    """Multi-layer Graph Isomorphism Network."""

    def __init__(self, in_dim: int, hidden_dim: int, out_dim: int,
                 n_layers: int = 3, dropout: float = 0.5):
        super().__init__()

        self.layers = nn.ModuleList()
        self.dropout = dropout

        # Input layer
        self.layers.append(GINLayer(in_dim, hidden_dim))

        # Hidden layers
        for _ in range(n_layers - 2):

```

```

        self.layers.append(GINLayer(hidden_dim, hidden_dim))

    # Output layer
    self.layers.append(GINLayer(hidden_dim, out_dim))

    def forward(self, x: torch.Tensor, edge_index: torch.Tensor) -> torch.Tensor:
        for layer in self.layers[:-1]:
            x = layer(x, edge_index)
            x = F.dropout(x, p=self.dropout, training=self.training)

        x = self.layers[-1](x, edge_index)
        return x

# Test GIN
gin = GIN(32, 64, 16, n_layers=3)
out_gin = gin(x_test, edge_index_test)
print(f"GIN output shape: {out_gin.shape}")

```

GIN output shape: torch.Size([20, 16])

34.2.6. Comparison of GNN Architectures

```

def compare_gnn_architectures():
    """
    Compare different GNN architectures on a sample problem.
    """
    # Create test data
    n_nodes = 50
    in_dim = 32
    hidden_dim = 64
    out_dim = 16

    x = torch.randn(n_nodes, in_dim)
    # Create a random sparse graph
    edge_index = torch.randint(0, n_nodes, (2, 200))

    architectures = {
        'GCN': GCN(in_dim, hidden_dim, out_dim, n_layers=3),
        'GraphSAGE': GraphSAGE(in_dim, hidden_dim, out_dim, n_layers=3),
        'GAT': GAT(in_dim, hidden_dim // 4, out_dim, n_layers=3, n_heads=4),
        'GIN': GIN(in_dim, hidden_dim, out_dim, n_layers=3)
    }

    results = {}

    for name, model in architectures.items():
        model.eval()

        # Count parameters
        n_params = sum(p.numel() for p in model.parameters())

        # Measure inference time

```

```

import time

with torch.no_grad():
    # Warm up
    _ = model(x, edge_index)

    # Time multiple runs
    n_runs = 100
    start = time.time()
    for _ in range(n_runs):
        out = model(x, edge_index)
    elapsed = (time.time() - start) / n_runs * 1000 # ms

results[name] = {
    'params': n_params,
    'time_ms': elapsed,
    'output_shape': tuple(out.shape)
}

# Display results
print("\n" + "="*60)
print("GNN Architecture Comparison")
print("="*60)
print(f"{'Architecture':<15} {'Parameters':>12} {'Time (ms)':>12} {'Output'
↪ 'Shape':>15}")
print("-"*60)
for name, res in results.items():
    print(f"{'name':<15} {res['params']:>12,} {res['time_ms']:>12.3f}
↪ {str(res['output_shape']):>15}")
print("-"*60)

return results

results = compare_gnn_architectures()

```

=====

GNN Architecture Comparison

=====

Architecture	Parameters	Time (ms)	Output Shape
GCN	7,312	3.461	(50, 16)
GraphSAGE	14,480	0.176	(50, 16)
GAT	10,624	0.318	(50, 16)
GIN	16,483	0.264	(50, 16)

=====

i Architecture Selection Guide

- **GCN**: Good baseline, efficient, works well for homophilic graphs
- **GraphSAGE**: Better for inductive learning and large graphs

- **GAT**: Best when neighbor importance varies; interpretable attention
- **GIN**: Maximum expressiveness; good for structure-sensitive tasks

For combinatorial optimization, **GAT** and **GIN** often perform best due to their ability to differentiate node importance and graph structures.

34.3. Problem Representations as Graphs

The way you encode an optimization problem as a graph determines what information is available to the GNN. A good representation captures the relevant structure – variables, constraints, and their relationships – in a form that message passing can process effectively. This section shows standard graph encodings for four common problem types.

34.3.1. Traveling Salesman Problem (TSP)

The TSP can be represented as a complete graph where: - **Nodes**: Cities with features (coordinates, demand, etc.) - **Edges**: Possible routes with features (distance, travel time, cost)

The `TSPGraphRepresentation` class encodes city coordinates as node features and inter-city distances as edge features. It optionally uses k-nearest-neighbor sparsification to reduce the edge count for large instances.

```
class TSPGraphRepresentation:
    """
    Graph representation for Traveling Salesman Problem instances.
    """

    def __init__(self, normalize: bool = True, k_nearest: Optional[int] = None):
        """
        Args:
            normalize: Whether to normalize coordinates to [0, 1]
            k_nearest: If set, create k-nearest neighbor graph instead of complete
        """
        self.normalize = normalize
        self.k_nearest = k_nearest

    def create_graph(self, coords: np.ndarray) -> Dict[str, torch.Tensor]:
        """
        Create graph representation from city coordinates.

        Args:
            coords: City coordinates (n_cities, 2)

        Returns:
            Dictionary with node_features, edge_index, edge_features
        """
        n_cities = len(coords)

        # Normalize coordinates
        if self.normalize:
            coords = coords.copy()
            coords = (coords - coords.min(axis=0)) / (coords.max(axis=0) -
↳ coords.min(axis=0) + 1e-8)
```



```

# Node features: coordinates
node_features = torch.tensor(coords, dtype=torch.float32)

# Compute distance matrix
dist_matrix = np.sqrt(((coords[:, None, :] - coords[None, :, :]) **
↪ 2).sum(axis=-1))

# Create edges
if self.k_nearest is not None:
    # K-nearest neighbor graph
    edge_list = []
    for i in range(n_cities):
        # Get k nearest neighbors (excluding self)
        distances = dist_matrix[i]
        nearest = np.argsort(distances)[1:self.k_nearest + 1]
        for j in nearest:
            edge_list.append([i, j])
            edge_list.append([j, i]) # Undirected

    edge_index = torch.tensor(edge_list, dtype=torch.long).t().contiguous()
    # Remove duplicates
    edge_index = torch.unique(edge_index, dim=1)
else:
    # Complete graph
    edge_list = []
    for i in range(n_cities):
        for j in range(n_cities):
            if i != j:
                edge_list.append([i, j])
    edge_index = torch.tensor(edge_list, dtype=torch.long).t().contiguous()

# Edge features: distance
source, target = edge_index
edge_features = torch.tensor(
    dist_matrix[source.numpy(), target.numpy()],
    dtype=torch.float32
).unsqueeze(-1)

return {
    'node_features': node_features,
    'edge_index': edge_index,
    'edge_features': edge_features,
    'n_nodes': n_cities
}

# Example TSP graph
tsp_repr = TSPGraphRepresentation(normalize=True, k_nearest=None)
coords_example = np.random.rand(20, 2)
tsp_graph = tsp_repr.create_graph(coords_example)

print(f"TSP Graph Representation:")

```

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```
print(f" Node features shape: {tsp_graph['node_features'].shape}")
print(f" Edge index shape: {tsp_graph['edge_index'].shape}")
print(f" Edge features shape: {tsp_graph['edge_features'].shape}")
print(f" Number of edges: {tsp_graph['edge_index'].shape[1]}")
```

TSP Graph Representation:

```
Node features shape: torch.Size([20, 2])
Edge index shape: torch.Size([2, 380])
Edge features shape: torch.Size([380, 1])
Number of edges: 380
```

34.3.2. Vehicle Routing Problem (VRP)

The VRP extends TSP with capacity constraints and multiple vehicles:

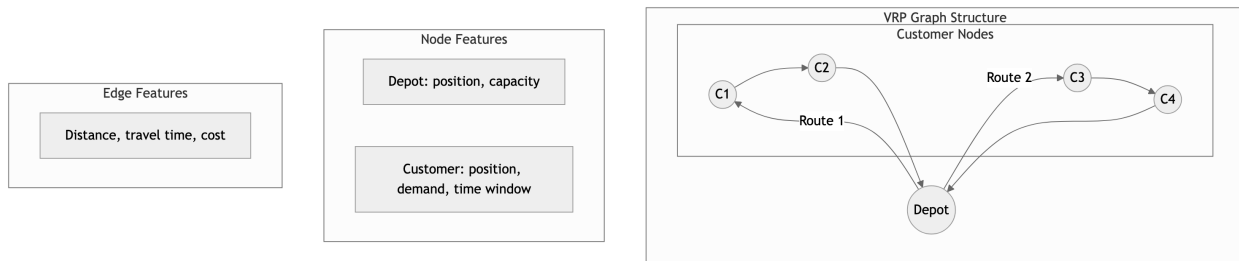


Figure 34.3.: Graph representation of Vehicle Routing Problem with depot and customers

```
class VRPGraphRepresentation:
    """
    Graph representation for Vehicle Routing Problem instances.

    Handles depot, customers with demands, and vehicle capacity constraints.
    """

    def __init__(self, normalize: bool = True):
        self.normalize = normalize

    def create_graph(self, depot: np.ndarray, customers: np.ndarray,
                    demands: np.ndarray, capacity: float,
                    time_windows: Optional[np.ndarray] = None) -> Dict[str,
    ↪ torch.Tensor]:
        """
        Create graph representation for VRP.

        Args:
            depot: Depot coordinates (2,)
            customers: Customer coordinates (n_customers, 2)
            demands: Customer demands (n_customers,)
            capacity: Vehicle capacity
            time_windows: Optional time windows (n_customers, 2) with [start, end]

        Returns:
```

```

        Dictionary with graph components
    """
    n_customers = len(customers)
    n_nodes = n_customers + 1 # Include depot

    # Combine depot and customer coordinates
    all_coords = np.vstack([depot.reshape(1, -1), customers])

    if self.normalize:
        all_coords = (all_coords - all_coords.min(axis=0)) / \
            (all_coords.max(axis=0) - all_coords.min(axis=0) + 1e-8)

    # Node features
    # Depot: [x, y, is_depot=1, demand=0, normalized_demand=0]
    # Customer: [x, y, is_depot=0, demand, normalized_demand]
    node_features = []

    # Depot features
    depot_feat = list(all_coords[0]) + [1.0, 0.0, 0.0]
    node_features.append(depot_feat)

    # Customer features
    normalized_demands = demands / capacity
    for i in range(n_customers):
        cust_feat = list(all_coords[i + 1]) + [0.0, demands[i],
↪ normalized_demands[i]]
        node_features.append(cust_feat)

    # Add time windows if provided
    if time_windows is not None:
        node_features[0].extend([0.0, 1.0]) # Depot available all time
        for i in range(n_customers):
            node_features[i + 1].extend([time_windows[i, 0], time_windows[i,
↪ 1]])

    node_features = torch.tensor(node_features, dtype=torch.float32)

    # Create complete graph edges
    edge_list = []
    for i in range(n_nodes):
        for j in range(n_nodes):
            if i != j:
                edge_list.append([i, j])

    edge_index = torch.tensor(edge_list, dtype=torch.long).t().contiguous()

    # Edge features: distance
    dist_matrix = np.sqrt(((all_coords[:, None, :] - all_coords[None, :, :]) **
↪ 2).sum(axis=-1))
    source, target = edge_index
    edge_features = torch.tensor(
        dist_matrix[source.numpy(), target.numpy()],
        dtype=torch.float32

```

```

        ).unsqueeze(-1)

    return {
        'node_features': node_features,
        'edge_index': edge_index,
        'edge_features': edge_features,
        'n_nodes': n_nodes,
        'n_customers': n_customers,
        'capacity': capacity
    }

# Example VRP instance
vrp_repr = VRPGraphRepresentation()
depot = np.array([0.5, 0.5])
customers = np.random.rand(15, 2)
demands = np.random.randint(1, 10, 15).astype(float)
capacity = 30.0

vrp_graph = vrp_repr.create_graph(depot, customers, demands, capacity)

print(f"VRP Graph Representation:")
print(f"  Node features shape: {vrp_graph['node_features'].shape}")
print(f"  Edge index shape: {vrp_graph['edge_index'].shape}")
print(f"  Number of customers: {vrp_graph['n_customers']}")
print(f"  Vehicle capacity: {vrp_graph['capacity']}")

```

VRP Graph Representation:

```

Node features shape: torch.Size([16, 5])
Edge index shape: torch.Size([2, 240])
Number of customers: 15
Vehicle capacity: 30.0

```

34.3.3. Job Shop Scheduling

Job shop scheduling problems can be represented as disjunctive graphs. Each operation becomes a node with processing time features. Conjunctive edges encode precedence within jobs, and disjunctive edges connect operations that share the same machine.

```

class SchedulingGraphRepresentation:
    """
    Disjunctive graph representation for Job Shop Scheduling Problem.

    The graph contains:
    - Operation nodes with processing times
    - Conjunctive arcs (precedence within jobs)
    - Disjunctive arcs (operations on same machine)
    """

    def __init__(self):
        pass

```

```

def create_graph(self, n_jobs: int, n_machines: int,
                  processing_times: np.ndarray,
                  machine_assignments: np.ndarray) -> Dict[str, torch.Tensor]:
    """
    Create disjunctive graph for JSSP.

    Args:
        n_jobs: Number of jobs
        n_machines: Number of machines (operations per job)
        processing_times: Processing times (n_jobs, n_machines)
        machine_assignments: Machine for each operation (n_jobs, n_machines)

    Returns:
        Dictionary with graph components
    """
    n_operations = n_jobs * n_machines

    # Create operation indices
    # Operation (j, m) has index j * n_machines + m
    def op_idx(job, op):
        return job * n_machines + op

    # Node features: [job_id, operation_id, processing_time, machine_id]
    node_features = []
    for j in range(n_jobs):
        for o in range(n_machines):
            feat = [
                j / n_jobs, # Normalized job id
                o / n_machines, # Normalized operation id
                processing_times[j, o] / processing_times.max(), # Normalized
                ↪ processing time
                machine_assignments[j, o] / n_machines # Normalized machine id
            ]
            node_features.append(feat)

    node_features = torch.tensor(node_features, dtype=torch.float32)

    # Create edges
    edge_list = []
    edge_types = [] # 0 = conjunctive, 1 = disjunctive

    # Conjunctive arcs: precedence within jobs
    for j in range(n_jobs):
        for o in range(n_machines - 1):
            src = op_idx(j, o)
            tgt = op_idx(j, o + 1)
            edge_list.append([src, tgt])
            edge_types.append(0)

    # Disjunctive arcs: operations on same machine
    # Group operations by machine
    machine_ops = {m: [] for m in range(n_machines)}
    for j in range(n_jobs):

```

```

        for o in range(n_machines):
            machine = machine_assignments[j, o]
            machine_ops[machine].append(op_idx(j, o))

# Add bidirectional edges between operations on same machine
for m in range(n_machines):
    ops = machine_ops[m]
    for i in range(len(ops)):
        for k in range(i + 1, len(ops)):
            edge_list.append([ops[i], ops[k]])
            edge_types.append(1)
            edge_list.append([ops[k], ops[i]])
            edge_types.append(1)

edge_index = torch.tensor(edge_list, dtype=torch.long).t().contiguous()
edge_types = torch.tensor(edge_types, dtype=torch.long)

# Edge features: [is_conjunctive, is_disjunctive]
edge_features = torch.zeros(len(edge_types), 2)
edge_features[edge_types == 0, 0] = 1.0
edge_features[edge_types == 1, 1] = 1.0

return {
    'node_features': node_features,
    'edge_index': edge_index,
    'edge_features': edge_features,
    'edge_types': edge_types,
    'n_jobs': n_jobs,
    'n_machines': n_machines,
    'n_operations': n_operations
}

# Example JSSP instance
n_jobs, n_machines = 5, 4
processing_times = np.random.randint(1, 20, (n_jobs, n_machines))
machine_assignments = np.array([np.random.permutation(n_machines) for _ in
    ↪ range(n_jobs)])

jssp_repr = SchedulingGraphRepresentation()
jssp_graph = jssp_repr.create_graph(n_jobs, n_machines, processing_times,
    ↪ machine_assignments)

print(f"JSSP Graph Representation:")
print(f"  Node features shape: {jssp_graph['node_features'].shape}")
print(f"  Edge index shape: {jssp_graph['edge_index'].shape}")
print(f"  Number of operations: {jssp_graph['n_operations']}")
print(f"  Conjunctive edges: {(jssp_graph['edge_types'] == 0).sum().item()}")
print(f"  Disjunctive edges: {(jssp_graph['edge_types'] == 1).sum().item()}")

```

JSSP Graph Representation:

```

Node features shape: torch.Size([20, 4])
Edge index shape: torch.Size([2, 95])

```

Number of operations: 20
 Conjunctive edges: 15
 Disjunctive edges: 80

34.3.4. Mixed Integer Linear Programming (MILP)

MILP problems have a natural bipartite graph structure between variables and constraints:

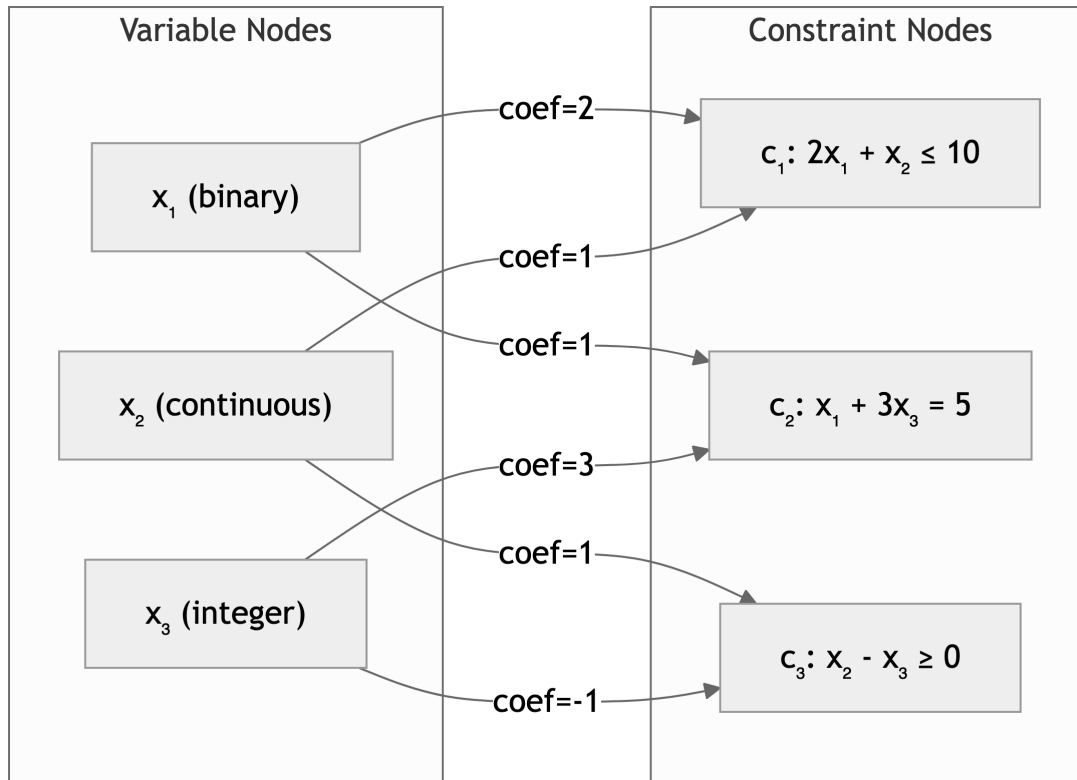


Figure 34.4.: Bipartite graph representation of MILP showing variable-constraint relationships

```
class MILPGraphRepresentation:
    """
    Bipartite graph representation for Mixed Integer Linear Programs.

    Creates a graph with:
    - Variable nodes with features (type, bounds, objective coefficient)
    - Constraint nodes with features (sense, RHS)
    - Edges representing variable participation in constraints
    """

    def __init__(self):
        pass

    def create_graph(self, c: np.ndarray, A: np.ndarray, b: np.ndarray,
                    var_types: List[str], bounds: List[Tuple[float, float]],
                    constraint_senses: List[str]) -> Dict[str, torch.Tensor]:
```

```

"""
Create bipartite graph from MILP formulation.

Min  $c^T x$ 
s.t.  $Ax \{<=, =, >=\} b$ 
      $x$  in bounds
      $x_i$  integer/binary for specified variables

Args:
    c: Objective coefficients (n_vars,)
    A: Constraint matrix (n_constraints, n_vars)
    b: Right-hand side (n_constraints,)
    var_types: Variable types ('C'=continuous, 'I'=integer, 'B'=binary)
    bounds: Variable bounds [(lb, ub), ...]
    constraint_senses: Constraint senses ('L'<=, 'E'==, 'G'>=)

Returns:
    Dictionary with graph components
"""
n_vars = len(c)
n_constraints = len(b)

# Normalize values for features
c_norm = c / (np.abs(c).max() + 1e-8)
b_norm = b / (np.abs(b).max() + 1e-8)
A_norm = A / (np.abs(A).max() + 1e-8)

# Variable node features
# [obj_coef, lb, ub, is_continuous, is_integer, is_binary]
var_features = []
for i in range(n_vars):
    lb, ub = bounds[i]
    lb_norm = lb / 1000 if lb > -float('inf') else -1.0
    ub_norm = ub / 1000 if ub < float('inf') else 1.0

    feat = [
        c_norm[i],
        lb_norm,
        ub_norm,
        1.0 if var_types[i] == 'C' else 0.0,
        1.0 if var_types[i] == 'I' else 0.0,
        1.0 if var_types[i] == 'B' else 0.0
    ]
    var_features.append(feat)

var_features = torch.tensor(var_features, dtype=torch.float32)

# Constraint node features
# [rhs, is_leq, is_eq, is_geq]
con_features = []
for j in range(n_constraints):
    feat = [
        b_norm[j],

```



```

        1.0 if constraint_senses[j] == 'L' else 0.0,
        1.0 if constraint_senses[j] == 'E' else 0.0,
        1.0 if constraint_senses[j] == 'G' else 0.0
    ]
    con_features.append(feat)

con_features = torch.tensor(con_features, dtype=torch.float32)

# Create bipartite edges
# Variable nodes: 0 to n_vars-1
# Constraint nodes: n_vars to n_vars + n_constraints - 1
edge_list = []
edge_features = []

for j in range(n_constraints):
    for i in range(n_vars):
        if A[j, i] != 0:
            # Variable i participates in constraint j
            var_node = i
            con_node = n_vars + j

            # Bidirectional edges
            edge_list.append([var_node, con_node])
            edge_list.append([con_node, var_node])

            # Edge feature: coefficient
            edge_features.append([A_norm[j, i]])
            edge_features.append([A_norm[j, i]])

edge_index = torch.tensor(edge_list, dtype=torch.long).t().contiguous()
edge_features = torch.tensor(edge_features, dtype=torch.float32)

return {
    'var_features': var_features,
    'con_features': con_features,
    'edge_index': edge_index,
    'edge_features': edge_features,
    'n_vars': n_vars,
    'n_constraints': n_constraints
}

# Example MILP: Simple knapsack-like problem
n_items = 10
c = -np.random.rand(n_items) # Maximize value (minimize negative)
A = np.random.rand(3, n_items) # 3 constraints
b = np.array([5.0, 3.0, 4.0])
var_types = ['B'] * n_items # Binary variables
bounds = [(0, 1)] * n_items
constraint_senses = ['L', 'L', 'L']

milp_repr = MILPGraphRepresentation()
milp_graph = milp_repr.create_graph(c, A, b, var_types, bounds, constraint_senses)

```

```

print(f"MILP Graph Representation:")
print(f"  Variable features shape: {milp_graph['var_features'].shape}")
print(f"  Constraint features shape: {milp_graph['con_features'].shape}")
print(f"  Edge index shape: {milp_graph['edge_index'].shape}")
print(f"  Edge features shape: {milp_graph['edge_features'].shape}")

```

MILP Graph Representation:

```

Variable features shape: torch.Size([10, 6])
Constraint features shape: torch.Size([3, 4])
Edge index shape: torch.Size([2, 60])
Edge features shape: torch.Size([60, 1])

```

34.4. Training Approaches

GNNs for combinatorial optimization can be trained using various paradigms, each with distinct advantages and trade-offs. The choice of training approach depends on whether labeled optimal solutions are available, the complexity of the objective function, and the need for constraint satisfaction.

34.4.1. Supervised Learning

In supervised learning, we train the GNN to mimic optimal or near-optimal solutions:

Limitations of Supervised Learning

- Requires labeled data (optimal solutions), which can be expensive to obtain
- May not generalize to problem sizes larger than training data
- Limited to imitating the quality of training solutions

However, supervised learning provides stable training and good initial performance.

```

class SupervisedTSPModel(nn.Module):
    """
    Supervised model for TSP that predicts edge probabilities.

    Trained to predict which edges belong to the optimal tour.
    """

    def __init__(self, node_dim: int = 2, hidden_dim: int = 128, n_layers: int = 3):
        super().__init__()

        self.node_encoder = nn.Linear(node_dim, hidden_dim)

        # GNN layers
        self.gnn_layers = nn.ModuleList([
            GATLayer(hidden_dim, hidden_dim // 4, n_heads=4, concat=True)
            for _ in range(n_layers)
        ])

        # Edge classifier

```

```

        self.edge_classifier = nn.Sequential(
            nn.Linear(2 * hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim // 2),
            nn.ReLU(),
            nn.Linear(hidden_dim // 2, 1)
        )

    def forward(self, node_features: torch.Tensor,
                edge_index: torch.Tensor) -> torch.Tensor:
        """
        Predict edge probabilities for tour edges.

        Args:
            node_features: Node coordinates (n_nodes, 2)
            edge_index: Edge connectivity (2, n_edges)

        Returns:
            Edge probabilities (n_edges,)
        """
        # Encode nodes
        h = self.node_encoder(node_features)

        # GNN message passing
        for layer in self.gnn_layers:
            h = layer(h, edge_index)
            h = F.relu(h)

        # Classify edges
        source, target = edge_index
        edge_features = torch.cat([h[source], h[target]], dim=-1)
        edge_logits = self.edge_classifier(edge_features).squeeze(-1)

        return edge_logits

def generate_tsp_training_data(n_instances: int, n_cities: int,
                              seed: int = 42) -> List[Dict]:
    """
    Generate TSP instances with optimal solutions (using nearest neighbor heuristic
    as proxy for supervised targets in this example).
    """
    np.random.seed(seed)
    data = []

    for _ in range(n_instances):
        # Generate instance
        coords = np.random.rand(n_cities, 2)

        # Create graph
        tsp_repr = TSPGraphRepresentation(normalize=True)
        graph = tsp_repr.create_graph(coords)

```

```

    # Get "optimal" tour using nearest neighbor (simplified)
    tour = nearest_neighbor_tour(coords)

    # Create edge labels (1 if edge in tour, 0 otherwise)
    edge_labels = torch.zeros(graph['edge_index'].shape[1])
    edge_index = graph['edge_index']

    for i in range(len(tour)):
        u, v = tour[i], tour[(i + 1) % len(tour)]
        # Find edge index
        mask = ((edge_index[0] == u) & (edge_index[1] == v)) | \
              ((edge_index[0] == v) & (edge_index[1] == u))
        edge_labels[mask] = 1.0

    graph['edge_labels'] = edge_labels
    data.append(graph)

    return data

def nearest_neighbor_tour(coords: np.ndarray) -> List[int]:
    """Simple nearest neighbor heuristic for TSP."""
    n = len(coords)
    visited = [False] * n
    tour = [0]
    visited[0] = True

    for _ in range(n - 1):
        current = tour[-1]
        best_dist = float('inf')
        best_next = -1

        for j in range(n):
            if not visited[j]:
                dist = np.linalg.norm(coords[current] - coords[j])
                if dist < best_dist:
                    best_dist = dist
                    best_next = j

        tour.append(best_next)
        visited[best_next] = True

    return tour

def train_supervised(model: nn.Module, train_data: List[Dict],
                     n_epochs: int = 100, lr: float = 1e-3) -> List[float]:
    """
    Train model with supervised learning.
    """
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)
    losses = []

```

```

for epoch in range(n_epochs):
    epoch_loss = 0.0

    for graph in train_data:
        optimizer.zero_grad()

        # Forward pass
        edge_logits = model(graph['node_features'], graph['edge_index'])

        # Binary cross-entropy loss
        loss = F.binary_cross_entropy_with_logits(
            edge_logits, graph['edge_labels']
        )

        loss.backward()
        optimizer.step()

        epoch_loss += loss.item()

    avg_loss = epoch_loss / len(train_data)
    losses.append(avg_loss)

    if (epoch + 1) % 20 == 0:
        print(f"Epoch {epoch + 1}/{n_epochs}, Loss: {avg_loss:.4f}")

return losses

# Train supervised model
print("Training Supervised TSP Model...")
supervised_model = SupervisedTSPModel()
train_data = generate_tsp_training_data(50, 20)
losses = train_supervised(supervised_model, train_data, n_epochs=100)

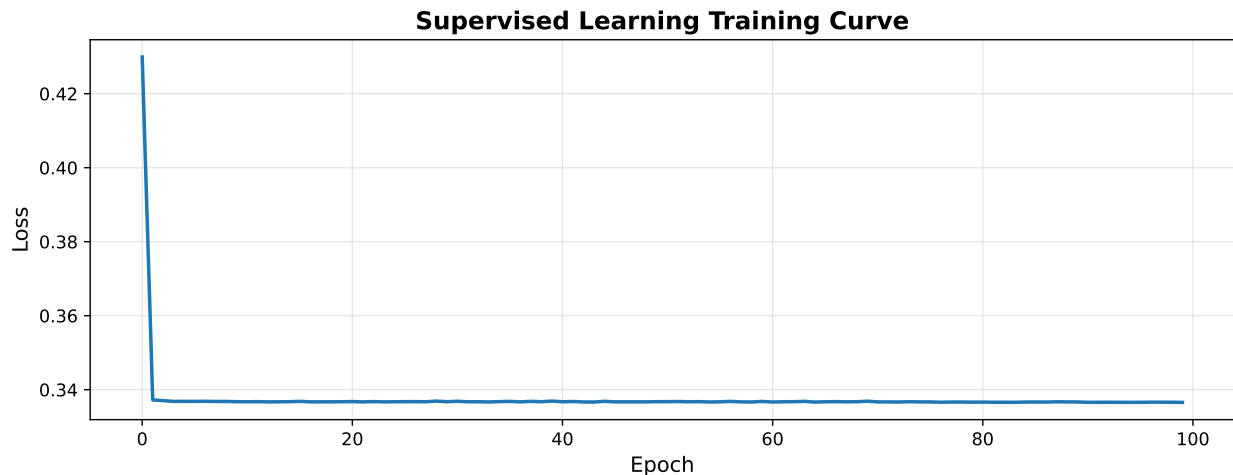
# Plot training curve
plt.figure(figsize=(10, 4))
plt.plot(losses, linewidth=2)
plt.xlabel('Epoch', fontsize=12)
plt.ylabel('Loss', fontsize=12)
plt.title('Supervised Learning Training Curve', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```

```

Training Supervised TSP Model...
Epoch 20/100, Loss: 0.3367
Epoch 40/100, Loss: 0.3369
Epoch 60/100, Loss: 0.3368
Epoch 80/100, Loss: 0.3366
Epoch 100/100, Loss: 0.3366

```



34.4.2. Reinforcement Learning

Reinforcement learning allows the model to discover solutions without labeled data:

```
class PolicyGradientTSP(nn.Module):
    """
    Policy gradient model for TSP using GNN encoder and autoregressive decoder.
    """

    def __init__(self, node_dim: int = 2, hidden_dim: int = 128):
        super().__init__()

        self.hidden_dim = hidden_dim

        # Node encoder
        self.node_encoder = nn.Sequential(
            nn.Linear(node_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim)
        )

        # GNN encoder
        self.gnn = nn.ModuleList([
            GATLayer(hidden_dim, hidden_dim // 4, n_heads=4, concat=True)
            for _ in range(3)
        ])

        # Context for decoder (first node, last node, graph embedding)
        self.context_proj = nn.Linear(3 * hidden_dim, hidden_dim)

        # Query and key projections for attention
        self.query_proj = nn.Linear(hidden_dim, hidden_dim)
        self.key_proj = nn.Linear(hidden_dim, hidden_dim)

    def encode(self, node_features: torch.Tensor,
              edge_index: torch.Tensor) -> torch.Tensor:
        """Encode nodes using GNN."""
```

```

h = self.node_encoder(node_features)

for layer in self.gnn:
    h = layer(h, edge_index)
    h = F.relu(h)

return h

def get_log_probs(self, node_embeddings: torch.Tensor,
                  tour: List[int]) -> torch.Tensor:
    """
    Compute log probabilities of a tour given node embeddings.

    Args:
        node_embeddings: (n_nodes, hidden_dim)
        tour: List of node indices representing the tour

    Returns:
        Sum of log probabilities
    """
    n_nodes = node_embeddings.size(0)
    log_prob = 0.0

    # Graph embedding (mean of node embeddings)
    graph_emb = node_embeddings.mean(dim=0)

    # Track visited nodes
    visited = torch.zeros(n_nodes, dtype=torch.bool)

    for step in range(n_nodes):
        if step == 0:
            # First node: use graph embedding as context
            context = torch.cat([graph_emb, graph_emb, graph_emb])
        else:
            # Context: first node, last node, graph embedding
            first_emb = node_embeddings[tour[0]]
            last_emb = node_embeddings[tour[step - 1]]
            context = torch.cat([first_emb, last_emb, graph_emb])

        context = self.context_proj(context)

        # Compute attention scores
        query = self.query_proj(context)
        keys = self.key_proj(node_embeddings)

        # Scaled dot-product attention
        scores = torch.matmul(keys, query) / np.sqrt(self.hidden_dim)

        # Mask visited nodes
        scores[visited] = float('-inf')

        # Softmax to get probabilities
        probs = F.softmax(scores, dim=0)

```

```

        # Log probability of selected action
        selected_node = tour[step]
        log_prob = log_prob + torch.log(probs[selected_node] + 1e-10)

        # Update visited
        visited[selected_node] = True

    return log_prob

def sample_tour(self, node_embeddings: torch.Tensor,
                greedy: bool = False) -> Tuple[List[int], torch.Tensor]:
    """
    Sample a tour autoregressively.

    Args:
        node_embeddings: (n_nodes, hidden_dim)
        greedy: If True, select best action; otherwise sample

    Returns:
        tour: List of node indices
        log_prob: Log probability of the tour
    """
    n_nodes = node_embeddings.size(0)
    tour = []
    log_prob = 0.0

    graph_emb = node_embeddings.mean(dim=0)
    visited = torch.zeros(n_nodes, dtype=torch.bool)

    for step in range(n_nodes):
        if step == 0:
            context = torch.cat([graph_emb, graph_emb, graph_emb])
        else:
            first_emb = node_embeddings[tour[0]]
            last_emb = node_embeddings[tour[-1]]
            context = torch.cat([first_emb, last_emb, graph_emb])

        context = self.context_proj(context)

        query = self.query_proj(context)
        keys = self.key_proj(node_embeddings)

        scores = torch.matmul(keys, query) / np.sqrt(self.hidden_dim)
        scores[visited] = float('-inf')

        probs = F.softmax(scores, dim=0)

        if greedy:
            selected = probs.argmax().item()
        else:
            selected = torch.multinomial(probs, 1).item()

```



```

        tour.append(selected)
        log_prob = log_prob + torch.log(probs[selected] + 1e-10)
        visited[selected] = True

    return tour, log_prob

def compute_tour_length(coords: np.ndarray, tour: List[int]) -> float:
    """Compute total tour length."""
    length = 0.0
    for i in range(len(tour)):
        u, v = tour[i], tour[(i + 1) % len(tour)]
        length += np.linalg.norm(coords[u] - coords[v])
    return length

def train_reinforce(model: PolicyGradientTSP, n_epochs: int = 100,
                   n_instances: int = 32, n_cities: int = 20,
                   lr: float = 1e-4) -> Dict[str, List[float]]:
    """
    Train using REINFORCE algorithm with baseline.
    """
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)

    history = {'loss': [], 'tour_length': [], 'baseline': []}
    baseline = None # Exponential moving average baseline

    for epoch in range(n_epochs):
        epoch_loss = 0.0
        epoch_lengths = []

        for _ in range(n_instances):
            # Generate random instance
            coords = np.random.rand(n_cities, 2)
            node_features = torch.tensor(coords, dtype=torch.float32)

            # Create complete graph
            edge_list = [[i, j] for i in range(n_cities) for j in range(n_cities)
                ↪ if i != j]
            edge_index = torch.tensor(edge_list, dtype=torch.long).t()

            # Encode
            node_embeddings = model.encode(node_features, edge_index)

            # Sample tour
            tour, log_prob = model.sample_tour(node_embeddings.detach(),
                ↪ greedy=False)

            # Compute reward (negative tour length)
            tour_length = compute_tour_length(coords, tour)
            reward = -tour_length
            epoch_lengths.append(tour_length)

```

```

        # Update baseline
        if baseline is None:
            baseline = reward
        else:
            baseline = 0.99 * baseline + 0.01 * reward

    # REINFORCE loss
    advantage = reward - baseline
    loss = -log_prob * advantage

    # Backprop
    optimizer.zero_grad()
    loss.backward()
    torch.nn.utils.clip_grad_norm_(model.parameters(), 1.0)
    optimizer.step()

    epoch_loss += loss.item()

    avg_loss = epoch_loss / n_instances
    avg_length = np.mean(epoch_lengths)

    history['loss'].append(avg_loss)
    history['tour_length'].append(avg_length)
    history['baseline'].append(-baseline)

    if (epoch + 1) % 20 == 0:
        print(f"Epoch {epoch + 1}/{n_epochs}, "
              f"Avg Tour Length: {avg_length:.4f}, "
              f"Baseline: {-baseline:.4f}")

    return history

# Train RL model
print("\nTraining Policy Gradient TSP Model...")
rl_model = PolicyGradientTSP()
rl_history = train_reinforce(rl_model, n_epochs=100, n_instances=32, n_cities=20)

# Plot learning curve
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

ax1.plot(rl_history['tour_length'], label='Sample', linewidth=2)
ax1.plot(rl_history['baseline'], label='Baseline', linewidth=2, linestyle='--')
ax1.set_xlabel('Epoch', fontsize=12)
ax1.set_ylabel('Tour Length', fontsize=12)
ax1.set_title('RL Training: Tour Length', fontsize=14, fontweight='bold')
ax1.legend()
ax1.grid(True, alpha=0.3)

ax2.plot(rl_history['loss'], linewidth=2, color='coral')
ax2.set_xlabel('Epoch', fontsize=12)
ax2.set_ylabel('Policy Loss', fontsize=12)
ax2.set_title('RL Training: Policy Loss', fontsize=14, fontweight='bold')

```

```
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```

Training Policy Gradient TSP Model...

```
RuntimeError: one of the variables needed for gradient computation has been modified by
↳ an inplace operation: [CPUBoolType [20]] is at version 20; expected version 19
↳ instead. Hint: enable anomaly detection to find the operation that failed to compute
↳ its gradient, with torch.autograd.set_detect_anomaly(True).
```

```
-----
RuntimeError                                Traceback (most recent call last)
Cell In[15], line 227
    225 print("\nTraining Policy Gradient TSP Model...")
    226 rl_model = PolicyGradientTSP()
--> 227 rl_history = train_reinforce(rl_model, n_epochs=100, n_instances=32, n_cities=20)
    229 # Plot learning curve
    230 fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
Cell In[15], line 203, in train_reinforce(model, n_epochs, n_instances, n_cities, lr)
    201 # Backprop
    202 optimizer.zero_grad()
--> 203 loss.backward()
    204 torch.nn.utils.clip_grad_norm_(model.parameters(), 1.0)
    205 optimizer.step()
File /opt/homebrew/lib/python3.14/site-packages/torch/_tensor.py:630, in
↳ Tensor.backward(self, gradient, retain_graph, create_graph, inputs)
    620 if has_torch_function_unary(self):
    621     return handle_torch_function(
    622         Tensor.backward,
    623         (self,),
    (...) 628         inputs=inputs,
    629     )
--> 630 torch.autograd.backward(
    631     self, gradient, retain_graph, create_graph, inputs=inputs
    632 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/__init__.py:364, in
↳ backward(tensors, grad_tensors, retain_graph, create_graph, grad_variables, inputs)
    359     retain_graph = create_graph
    361 # The reason we repeat the same comment below is that
    362 # some Python versions print out the first line of a multi-line function
    363 # calls in the traceback and some print out the last line
--> 364 _engine_run_backward(
    365     tensors,
    366     grad_tensors_,
    367     retain_graph,
    368     create_graph,
    369     inputs_tuple,
    370     allow_unreachable=True,
    371     accumulate_grad=True,
    372 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/graph.py:865, in
↳ _engine_run_backward(t_outputs, *args, **kwargs)
```

34. Graph Neural Networks for Combinatorial Optimization

```
863     unregister_hooks = _register_logging_hooks_on_whole_graph(t_outputs)
864 try:
--> 865     return Variable._execution_engine.run_backward( # Calls into the C++ engine
    ↳ to run the backward pass
866         t_outputs, *args, **kwargs
867     ) # Calls into the C++ engine to run the backward pass
868 finally:
869     if attach_logging_hooks:
RuntimeError: one of the variables needed for gradient computation has been modified by
    ↳ an inplace operation: [CPUBoolType [20]] is at version 20; expected version 19
    ↳ instead. Hint: enable anomaly detection to find the operation that failed to compute
    ↳ its gradient, with torch.autograd.set_detect_anomaly(True).
```

34.4.3. Unsupervised Learning

Unsupervised approaches learn to solve problems by directly optimizing the objective:

💡 Advantages of Unsupervised Learning

1. No need for optimal solutions as training labels
2. Can optimize directly for the problem objective
3. Can incorporate problem-specific constraints as loss terms
4. Often generalizes better to new problem sizes

```
class UnsupervisedTSP(nn.Module):
    """
    Unsupervised learning model for TSP.

    Learns to output a doubly-stochastic matrix representing
    soft assignments of cities to tour positions.
    """

    def __init__(self, node_dim: int = 2, hidden_dim: int = 128, n_layers: int = 3):
        super().__init__()

        self.node_encoder = nn.Linear(node_dim, hidden_dim)

        self.gnn = nn.ModuleList([
            GINLayer(hidden_dim, hidden_dim)
            for _ in range(n_layers)
        ])

        # Output layer for position assignment scores
        self.output = nn.Linear(hidden_dim, hidden_dim)

    def forward(self, node_features: torch.Tensor,
                edge_index: torch.Tensor,
                n_sinkhorn_iters: int = 20) -> torch.Tensor:
        """
        Forward pass returns doubly-stochastic assignment matrix.

        Args:
            node_features: (n_nodes, 2)

```

```

        edge_index: (2, n_edges)
        n_sinkhorn_iters: Number of Sinkhorn iterations

Returns:
    Assignment matrix (n_nodes, n_nodes) - soft permutation
    """
    n_nodes = node_features.size(0)

    # Encode nodes
    h = self.node_encoder(node_features)

    for layer in self.gnn:
        h = layer(h, edge_index)

    # Compute assignment scores
    h_out = self.output(h)

    # Score matrix: how well each node fits each position
    scores = torch.matmul(h_out, h_out.t())

    # Sinkhorn normalization to get doubly-stochastic matrix
    P = self.sinkhorn(scores, n_sinkhorn_iters)

    return P

def sinkhorn(self, scores: torch.Tensor, n_iters: int) -> torch.Tensor:
    """Sinkhorn-Knopp algorithm for doubly-stochastic matrix."""
    log_alpha = scores

    for _ in range(n_iters):
        # Row normalization (in log space)
        log_alpha = log_alpha - torch.logsumexp(log_alpha, dim=1, keepdim=True)
        # Column normalization
        log_alpha = log_alpha - torch.logsumexp(log_alpha, dim=0, keepdim=True)

    return torch.exp(log_alpha)

def compute_tour_length_soft(self, P: torch.Tensor,
                             distance_matrix: torch.Tensor) -> torch.Tensor:
    """
    Compute expected tour length given soft assignment.

    Args:
        P: Doubly-stochastic assignment matrix (n, n)
        distance_matrix: Pairwise distances (n, n)

    Returns:
        Expected tour length (scalar)
    """
    n = P.size(0)

    # Adjacency in tour: position i -> position i+1 (cyclic)
    adj = torch.zeros(n, n, device=P.device)

```

```

    for i in range(n):
        adj[i, (i + 1) % n] = 1.0

    # Expected distance: sum over all city pairs weighted by
    # probability they are adjacent in tour
    # P @ adj @ P^T gives probability that cities are adjacent
    tour_adj = P @ adj @ P.t()

    # Expected length
    expected_length = (tour_adj * distance_matrix).sum()

    return expected_length

def train_unsupervised(model: UnsupervisedTSP, n_epochs: int = 100,
                      n_instances: int = 32, n_cities: int = 20,
                      lr: float = 1e-3) -> List[float]:
    """Train using unsupervised objective."""
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)
    losses = []

    for epoch in range(n_epochs):
        epoch_loss = 0.0

        for _ in range(n_instances):
            coords = np.random.rand(n_cities, 2)
            node_features = torch.tensor(coords, dtype=torch.float32)

            # Complete graph
            edge_list = [[i, j] for i in range(n_cities) for j in range(n_cities)
            ↪ if i != j]
            edge_index = torch.tensor(edge_list, dtype=torch.long).t()

            # Distance matrix
            dist_matrix = torch.tensor(
                np.sqrt(((coords[:, None, :] - coords[None, :, :]) **
            ↪ 2).sum(axis=-1))),
                dtype=torch.float32
            )

            optimizer.zero_grad()

            # Forward pass
            P = model(node_features, edge_index)

            # Compute loss (expected tour length)
            loss = model.compute_tour_length_soft(P, dist_matrix)

            loss.backward()
            optimizer.step()

            epoch_loss += loss.item()

```

```

    avg_loss = epoch_loss / n_instances
    losses.append(avg_loss)

    if (epoch + 1) % 20 == 0:
        print(f"Epoch {epoch + 1}/{n_epochs}, Expected Tour Length:
              ↪ {avg_loss:.4f}")

    return losses

# Train unsupervised model
print("\nTraining Unsupervised TSP Model...")
unsup_model = UnsupervisedTSP()
unsup_losses = train_unsupervised(unsup_model, n_epochs=100, n_instances=32,
                                  ↪ n_cities=20)

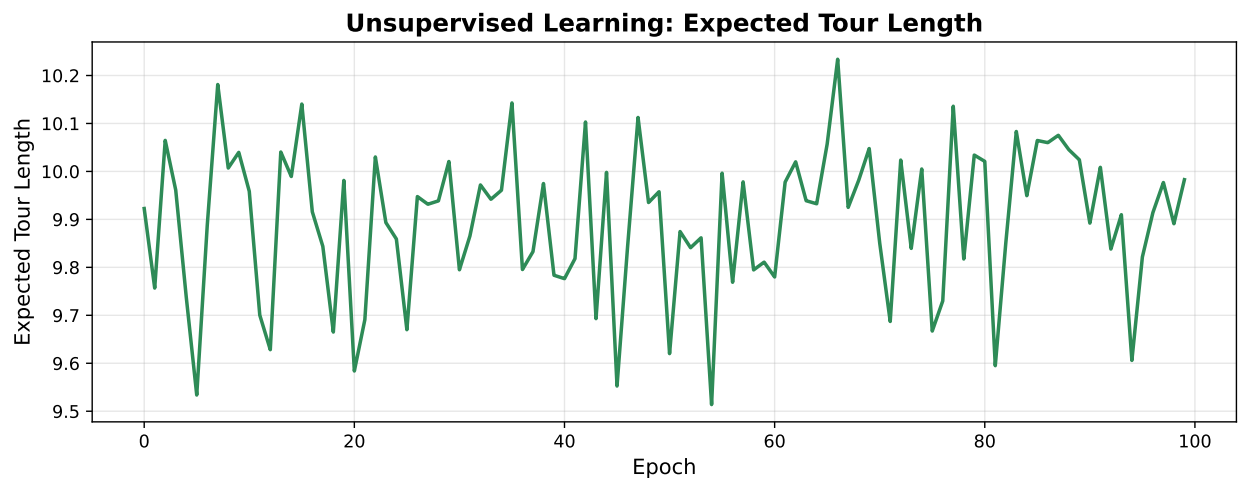
plt.figure(figsize=(10, 4))
plt.plot(unsup_losses, linewidth=2, color='seagreen')
plt.xlabel('Epoch', fontsize=12)
plt.ylabel('Expected Tour Length', fontsize=12)
plt.title('Unsupervised Learning: Expected Tour Length', fontsize=14,
          ↪ fontweight='bold')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```

```

Training Unsupervised TSP Model...
Epoch 20/100, Expected Tour Length: 9.9813
Epoch 40/100, Expected Tour Length: 9.7835
Epoch 60/100, Expected Tour Length: 9.8109
Epoch 80/100, Expected Tour Length: 10.0341
Epoch 100/100, Expected Tour Length: 9.9828

```



34.4.4. Comparison of Training Approaches

```
# Summary comparison table
comparison_data = {
    'Approach': ['Supervised', 'Reinforcement Learning', 'Unsupervised'],
    'Requires Labels': ['Yes (optimal solutions)', 'No', 'No'],
    'Training Stability': ['High', 'Medium', 'Medium-High'],
    'Solution Quality': ['Limited by labels', 'Can exceed labels', 'Can exceed
↪ labels'],
    'Sample Efficiency': ['High', 'Low', 'Medium'],
    'Generalization': ['Limited', 'Good', 'Good'],
    'Best For': ['Fast prototyping', 'Complex objectives', 'Constraint
↪ satisfaction']
}

print("\n" + "="*90)
print("Training Approaches Comparison")
print("="*90)
for key, values in comparison_data.items():
    print(f"{key:20s}: {' | '.join(str(v) for v in values)}")
print("="*90)
```

Training Approaches Comparison

```
=====
Approach          : Supervised | Reinforcement Learning | Unsupervised
Requires Labels   : Yes (optimal solutions) | No | No
Training Stability: High | Medium | Medium-High
Solution Quality  : Limited by labels | Can exceed labels | Can exceed labels
Sample Efficiency : High | Low | Medium
Generalization    : Limited | Good | Good
Best For          : Fast prototyping | Complex objectives | Constraint satisfaction
=====
```

34.5. GNN for Routing Problems

Routing problems like TSP and VRP have been a primary focus for GNN-based approaches. This section presents the Attention Model architecture with GNN encoder, which has become a standard baseline for neural combinatorial optimization.

34.5.1. Attention Model Architecture

The Attention Model by Kool et al. (2019) combines a GNN-like encoder with an attention-based decoder:

```
class MultiHeadAttention(nn.Module):
    """Multi-head attention mechanism."""

    def __init__(self, embed_dim: int, n_heads: int):
        super().__init__()
```

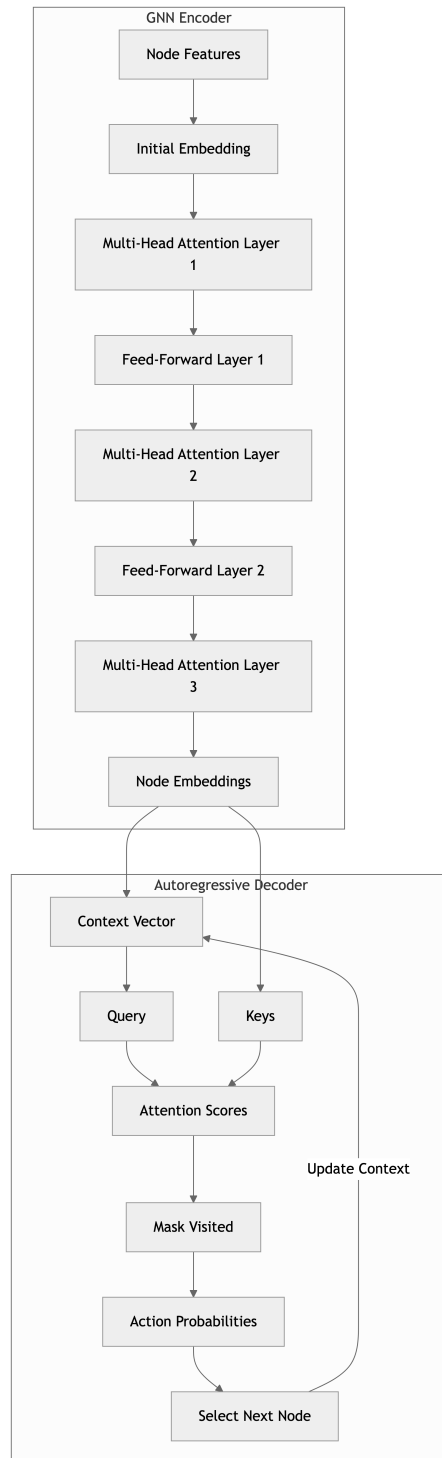



Figure 34.5.: Attention Model architecture for routing problems

```

        self.embed_dim = embed_dim
        self.n_heads = n_heads
        self.head_dim = embed_dim // n_heads

        self.W_q = nn.Linear(embed_dim, embed_dim, bias=False)
        self.W_k = nn.Linear(embed_dim, embed_dim, bias=False)
        self.W_v = nn.Linear(embed_dim, embed_dim, bias=False)
        self.W_o = nn.Linear(embed_dim, embed_dim, bias=False)

    def forward(self, query: torch.Tensor, key: torch.Tensor,
                value: torch.Tensor, mask: Optional[torch.Tensor] = None) ->
    ↪ torch.Tensor:
        """
        Args:
            query: (batch, seq_q, embed_dim)
            key: (batch, seq_k, embed_dim)
            value: (batch, seq_k, embed_dim)
            mask: Optional mask (batch, seq_q, seq_k)
        """
        batch_size = query.size(0)

        # Linear projections
        Q = self.W_q(query).view(batch_size, -1, self.n_heads,
    ↪ self.head_dim).transpose(1, 2)
        K = self.W_k(key).view(batch_size, -1, self.n_heads,
    ↪ self.head_dim).transpose(1, 2)
        V = self.W_v(value).view(batch_size, -1, self.n_heads,
    ↪ self.head_dim).transpose(1, 2)

        # Attention scores
        scores = torch.matmul(Q, K.transpose(-2, -1)) / np.sqrt(self.head_dim)

        if mask is not None:
            scores = scores.masked_fill(mask.unsqueeze(1), float('-inf'))

        attn_weights = F.softmax(scores, dim=-1)

        # Apply attention to values
        attn_output = torch.matmul(attn_weights, V)

        # Concatenate heads
        attn_output = attn_output.transpose(1, 2).contiguous().view(
            batch_size, -1, self.embed_dim
        )

        return self.W_o(attn_output)

class AttentionEncoderLayer(nn.Module):
    """Single encoder layer with self-attention and feed-forward."""

    def __init__(self, embed_dim: int, n_heads: int, ff_dim: int,
                 dropout: float = 0.1):

```

```

    super().__init__()

    self.self_attn = MultiHeadAttention(embed_dim, n_heads)
    self.ff = nn.Sequential(
        nn.Linear(embed_dim, ff_dim),
        nn.ReLU(),
        nn.Linear(ff_dim, embed_dim)
    )

    self.norm1 = nn.LayerNorm(embed_dim)
    self.norm2 = nn.LayerNorm(embed_dim)
    self.dropout = nn.Dropout(dropout)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        """
        Args:
            x: (batch, seq, embed_dim)
        """
        # Self-attention with residual
        attn_out = self.self_attn(x, x, x)
        x = self.norm1(x + self.dropout(attn_out))

        # Feed-forward with residual
        ff_out = self.ff(x)
        x = self.norm2(x + self.dropout(ff_out))

        return x

class AttentionModelEncoder(nn.Module):
    """
    Encoder for the Attention Model.

    Uses self-attention layers that act like a complete graph GNN.
    """

    def __init__(self, node_dim: int = 2, embed_dim: int = 128,
                  n_heads: int = 8, n_layers: int = 3, ff_dim: int = 512):
        super().__init__()

        self.embed_dim = embed_dim

        # Initial embedding
        self.node_embed = nn.Linear(node_dim, embed_dim)

        # Encoder layers
        self.layers = nn.ModuleList([
            AttentionEncoderLayer(embed_dim, n_heads, ff_dim)
            for _ in range(n_layers)
        ])

    def forward(self, node_features: torch.Tensor) -> torch.Tensor:
        """

```

```

    Encode node features.

    Args:
        node_features: (batch, n_nodes, node_dim)

    Returns:
        Node embeddings: (batch, n_nodes, embed_dim)
    """
    # Initial embedding
    h = self.node_embed(node_features)

    # Apply encoder layers
    for layer in self.layers:
        h = layer(h)

    return h


class AttentionModelDecoder(nn.Module):
    """
    Autoregressive decoder for the Attention Model.
    """

    def __init__(self, embed_dim: int = 128, n_heads: int = 8,
                  clip: float = 10.0):
        super().__init__()

        self.embed_dim = embed_dim
        self.n_heads = n_heads
        self.clip = clip

        # Context projections
        self.W_context = nn.Linear(3 * embed_dim, embed_dim)

        # Attention for selecting next node
        self.W_query = nn.Linear(embed_dim, embed_dim, bias=False)
        self.W_key = nn.Linear(embed_dim, embed_dim, bias=False)

    def get_context(self, node_embeddings: torch.Tensor,
                   graph_embedding: torch.Tensor,
                   first_node: Optional[torch.Tensor],
                   current_node: Optional[torch.Tensor]) -> torch.Tensor:
        """
        Compute context vector from current state.

        Args:
            node_embeddings: (batch, n_nodes, embed_dim)
            graph_embedding: (batch, embed_dim)
            first_node: (batch,) indices of first nodes
            current_node: (batch,) indices of current nodes
        """
        batch_size = node_embeddings.size(0)

```

```

    if first_node is None:
        # Initial context: just graph embedding
        context = torch.cat([
            graph_embedding,
            graph_embedding,
            graph_embedding
        ], dim=-1)
    else:
        # Context: graph embedding + first node + current node
        first_emb = node_embeddings[torch.arange(batch_size), first_node]
        current_emb = node_embeddings[torch.arange(batch_size), current_node]
        context = torch.cat([graph_embedding, first_emb, current_emb], dim=-1)

    return self.W_context(context)

def get_log_probs(self, node_embeddings: torch.Tensor,
                  context: torch.Tensor,
                  mask: torch.Tensor) -> torch.Tensor:
    """
    Compute log probabilities over next nodes.

    Args:
        node_embeddings: (batch, n_nodes, embed_dim)
        context: (batch, embed_dim)
        mask: (batch, n_nodes) True for visited nodes
    """
    # Query from context
    query = self.W_query(context).unsqueeze(1) # (batch, 1, embed_dim)

    # Keys from node embeddings
    keys = self.W_key(node_embeddings) # (batch, n_nodes, embed_dim)

    # Attention scores
    scores = torch.matmul(query, keys.transpose(-2, -1)).squeeze(1)
    scores = scores / np.sqrt(self.embed_dim)

    # Clip scores
    scores = self.clip * torch.tanh(scores)

    # Mask visited nodes
    scores[mask] = float('-inf')

    # Log softmax
    log_probs = F.log_softmax(scores, dim=-1)

    return log_probs

class AttentionModel(nn.Module):
    """
    Complete Attention Model for routing problems.
    """

```

```

def __init__(self, node_dim: int = 2, embed_dim: int = 128,
              n_heads: int = 8, n_layers: int = 3):
    super().__init__()

    self.encoder = AttentionModelEncoder(node_dim, embed_dim, n_heads, n_layers)
    self.decoder = AttentionModelDecoder(embed_dim, n_heads)

def forward(self, node_features: torch.Tensor,
            greedy: bool = False) -> Tuple[torch.Tensor, torch.Tensor]:
    """
    Generate tours for a batch of instances.

    Args:
        node_features: (batch, n_nodes, node_dim)
        greedy: If True, select best action; otherwise sample

    Returns:
        tours: (batch, n_nodes) selected node indices
        log_probs: (batch,) log probability of each tour
    """
    batch_size, n_nodes, _ = node_features.shape

    # Encode
    node_embeddings = self.encoder(node_features)
    graph_embedding = node_embeddings.mean(dim=1)

    # Initialize
    tours = []
    log_probs = torch.zeros(batch_size, device=node_features.device)
    mask = torch.zeros(batch_size, n_nodes, dtype=torch.bool,
                       device=node_features.device)

    first_node = None
    current_node = None

    # Decode autoregressively
    for step in range(n_nodes):
        # Get context
        context = self.decoder.get_context(
            node_embeddings, graph_embedding, first_node, current_node
        )

        # Get action probabilities
        step_log_probs = self.decoder.get_log_probs(
            node_embeddings, context, mask
        )

        # Select action
        if greedy:
            selected = step_log_probs.argmax(dim=-1)
        else:
            probs = step_log_probs.exp()
            selected = torch.multinomial(probs, 1).squeeze(-1)

```

```

        # Update state
        tours.append(selected)
        log_probs = log_probs + step_log_probs[torch.arange(batch_size),
        ↪ selected]
        mask[torch.arange(batch_size), selected] = True

        if step == 0:
            first_node = selected
            current_node = selected

        tours = torch.stack(tours, dim=1)

        return tours, log_probs

# Test Attention Model
def test_attention_model():
    model = AttentionModel(node_dim=2, embed_dim=128, n_heads=8, n_layers=3)

    # Batch of TSP instances
    batch_size = 4
    n_cities = 20
    node_features = torch.rand(batch_size, n_cities, 2)

    # Sample tours
    model.eval()
    with torch.no_grad():
        tours, log_probs = model(node_features, greedy=False)

    print(f"Attention Model Test:")
    print(f"  Input shape: {node_features.shape}")
    print(f"  Tours shape: {tours.shape}")
    print(f"  Log probs shape: {log_probs.shape}")
    print(f"  Sample tour: {tours[0].tolist()}")

    # Verify all cities visited exactly once
    for i in range(batch_size):
        assert len(set(tours[i].tolist())) == n_cities, "Invalid tour!"
    print(f"  All tours valid: True")

test_attention_model()

```

Attention Model Test:

Input shape: torch.Size([4, 20, 2])

Tours shape: torch.Size([4, 20])

Log probs shape: torch.Size([4])

Sample tour: [5, 9, 12, 16, 2, 4, 7, 11, 3, 1, 6, 8, 19, 17, 14, 10, 18, 0, 13, 15]

All tours valid: True

34.5.2. PyTorch Geometric Implementation

PyTorch Geometric provides efficient implementations of GNN layers. Here's how to build routing models using PyG.

i PyTorch Geometric

PyTorch Geometric (PyG) is the standard library for graph deep learning in PyTorch. It provides: - Efficient sparse tensor operations - Pre-built GNN layers (GCN, GAT, GraphSAGE, etc.) - Data handling utilities for graphs - Sampling strategies for large graphs

Install with: `pip install torch-geometric`

i DGL (Deep Graph Library)

DGL is an alternative to PyG that supports multiple backends (PyTorch, TensorFlow, MXNet). While PyG is more widely used in the combinatorial optimization community, DGL offers a flexible message-passing API and scales efficiently to very large graphs. DGL's framework-agnostic design makes it a good choice when backend flexibility is important or when working with heterogeneous graph structures. Install with: `pip install dgl`

```
# Note: This code assumes PyTorch Geometric is installed
# We'll create a mock implementation for demonstration

class PyGStyleData:
    """Mock PyG-style data object."""
    def __init__(self, x, edge_index, edge_attr=None, batch=None):
        self.x = x
        self.edge_index = edge_index
        self.edge_attr = edge_attr
        self.batch = batch if batch is not None else torch.zeros(x.size(0),
            ↪ dtype=torch.long)

    @property
    def num_nodes(self):
        return self.x.size(0)

    @property
    def num_edges(self):
        return self.edge_index.size(1)

class PyGStyleGATConv(nn.Module):
    """
    PyG-style Graph Attention Convolution.

    This mimics torch_geometric.nn.GATConv interface.
    """
    def __init__(self, in_channels: int, out_channels: int,
        heads: int = 1, concat: bool = True,
        negative_slope: float = 0.2, dropout: float = 0.0,
        edge_dim: Optional[int] = None):
        super().__init__()

        self.in_channels = in_channels
```



```

self.out_channels = out_channels
self.heads = heads
self.concat = concat
self.negative_slope = negative_slope
self.dropout = dropout
self.edge_dim = edge_dim

# Linear transformations
self.lin = nn.Linear(in_channels, heads * out_channels, bias=False)

# Attention parameters
self.att_src = nn.Parameter(torch.Tensor(1, heads, out_channels))
self.att_dst = nn.Parameter(torch.Tensor(1, heads, out_channels))

if edge_dim is not None:
    self.lin_edge = nn.Linear(edge_dim, heads * out_channels, bias=False)
    self.att_edge = nn.Parameter(torch.Tensor(1, heads, out_channels))
else:
    self.lin_edge = None

self.reset_parameters()

def reset_parameters(self):
    nn.init.xavier_uniform_(self.lin.weight)
    nn.init.xavier_uniform_(self.att_src)
    nn.init.xavier_uniform_(self.att_dst)
    if self.lin_edge is not None:
        nn.init.xavier_uniform_(self.lin_edge.weight)
        nn.init.xavier_uniform_(self.att_edge)

def forward(self, x: torch.Tensor, edge_index: torch.Tensor,
            edge_attr: Optional[torch.Tensor] = None) -> torch.Tensor:
    """
    Forward pass.

    Args:
        x: Node features (n_nodes, in_channels)
        edge_index: Edge connectivity (2, n_edges)
        edge_attr: Edge features (n_edges, edge_dim)

    Returns:
        Updated node features
    """
    n_nodes = x.size(0)
    source, target = edge_index

    # Transform node features
    x = self.lin(x).view(-1, self.heads, self.out_channels)

    # Compute attention coefficients
    alpha_src = (x * self.att_src).sum(dim=-1) # (n_nodes, heads)
    alpha_dst = (x * self.att_dst).sum(dim=-1)

```

```

        alpha = alpha_src[source] + alpha_dst[target]

        # Edge features contribution
        if edge_attr is not None and self.lin_edge is not None:
            edge_feat = self.lin_edge(edge_attr).view(-1, self.heads,
        ↪ self.out_channels)
            alpha = alpha + (edge_feat * self.att_edge).sum(dim=-1)

        alpha = F.leaky_relu(alpha, self.negative_slope)

        # Softmax over neighborhood
        alpha = alpha - alpha.max()
        alpha = torch.exp(alpha)

        # Normalize
        alpha_sum = torch.zeros(n_nodes, self.heads, device=x.device)
        alpha_sum.scatter_add_(0, target.unsqueeze(-1).expand(-1, self.heads),
        ↪ alpha)
        alpha = alpha / (alpha_sum[target] + 1e-10)

        # Dropout
        alpha = F.dropout(alpha, p=self.dropout, training=self.training)

        # Aggregate
        out = torch.zeros(n_nodes, self.heads, self.out_channels, device=x.device)
        weighted = alpha.unsqueeze(-1) * x[source]
        out.scatter_add_(0, target.unsqueeze(-1).unsqueeze(-1).expand(-1,
        ↪ self.heads, self.out_channels),
            weighted)

        if self.concat:
            out = out.view(n_nodes, -1)
        else:
            out = out.mean(dim=1)

        return out

class GNNRoutingEncoder(nn.Module):
    """
    GNN encoder for routing problems using PyG-style layers.
    """

    def __init__(self, node_dim: int = 2, edge_dim: int = 1,
                  embed_dim: int = 128, n_heads: int = 4, n_layers: int = 3):
        super().__init__()

        self.node_embed = nn.Linear(node_dim, embed_dim)
        self.edge_embed = nn.Linear(edge_dim, embed_dim) if edge_dim > 0 else None

        self.convs = nn.ModuleList()
        self.norms = nn.ModuleList()

```

```

    for i in range(n_layers):
        self.convs.append(
            PyGStyleGATConv(
                embed_dim, embed_dim // n_heads,
                heads=n_heads, concat=True,
                edge_dim=embed_dim if edge_dim > 0 else None
            )
        )
        self.norms.append(nn.LayerNorm(embed_dim))

    def forward(self, x: torch.Tensor, edge_index: torch.Tensor,
                edge_attr: Optional[torch.Tensor] = None) -> torch.Tensor:
        """
        Encode graph.

        Args:
            x: Node features (n_nodes, node_dim)
            edge_index: (2, n_edges)
            edge_attr: (n_edges, edge_dim)

        Returns:
            Node embeddings (n_nodes, embed_dim)
        """
        # Embed nodes
        h = self.node_embed(x)

        # Embed edges if provided
        e = None
        if edge_attr is not None and self.edge_embed is not None:
            e = self.edge_embed(edge_attr)

        # Apply GNN layers
        for conv, norm in zip(self.convs, self.norms):
            h_new = conv(h, edge_index, e)
            h = norm(h + h_new) # Residual connection
            h = F.relu(h)

        return h

class GNNRoutingModel(nn.Module):
    """
    Complete GNN-based routing model.

    Combines GNN encoder with attention-based decoder.
    """

    def __init__(self, node_dim: int = 2, edge_dim: int = 1,
                embed_dim: int = 128, n_heads: int = 4, n_layers: int = 3):
        super().__init__()

        self.embed_dim = embed_dim

```

```

self.encoder = GNNRoutingEncoder(node_dim, edge_dim, embed_dim, n_heads,
    ↪ n_layers)
self.decoder = AttentionModelDecoder(embed_dim, n_heads)

def forward(self, data: PyGStyleData,
            greedy: bool = False) -> Tuple[torch.Tensor, torch.Tensor]:
    """
    Generate tour for a single instance.

    Args:
        data: Graph data object
        greedy: Whether to decode greedily

    Returns:
        tour: Node indices in tour order
        log_prob: Log probability of tour
    """
    # Encode
    node_embeddings = self.encoder(data.x, data.edge_index, data.edge_attr)
    graph_embedding = node_embeddings.mean(dim=0)

    # Add batch dimension
    node_embeddings = node_embeddings.unsqueeze(0)
    graph_embedding = graph_embedding.unsqueeze(0)

    n_nodes = data.num_nodes
    tour = []
    log_prob = 0.0
    mask = torch.zeros(1, n_nodes, dtype=torch.bool, device=data.x.device)

    first_node = None
    current_node = None

    for step in range(n_nodes):
        context = self.decoder.get_context(
            node_embeddings, graph_embedding, first_node, current_node
        )

        step_log_probs = self.decoder.get_log_probs(
            node_embeddings, context, mask
        )

        if greedy:
            selected = step_log_probs.argmax(dim=-1)
        else:
            probs = step_log_probs.exp()
            selected = torch.multinomial(probs, 1).squeeze(-1)

        tour.append(selected.item())
        log_prob = log_prob + step_log_probs[0, selected].item()
        mask[0, selected] = True

    if step == 0:

```

```

        first_node = selected
        current_node = selected

    return tour, log_prob

# Test GNN Routing Model
def test_gnn_routing_model():
    # Create TSP instance
    n_cities = 15
    coords = np.random.rand(n_cities, 2)

    # Create graph data
    x = torch.tensor(coords, dtype=torch.float32)

    # Complete graph edges
    edge_list = [[i, j] for i in range(n_cities) for j in range(n_cities) if i != j]
    edge_index = torch.tensor(edge_list, dtype=torch.long).t()

    # Edge features (distances)
    source, target = edge_index
    distances = torch.sqrt(((x[source] - x[target]) ** 2).sum(dim=1, keepdim=True))

    data = PyGStyleData(x, edge_index, distances)

    # Create and test model
    model = GNNRoutingModel(node_dim=2, edge_dim=1)
    model.eval()

    with torch.no_grad():
        tour, log_prob = model(data, greedy=True)

    # Compute tour length
    tour_length = compute_tour_length(coords, tour)

    print(f"\nGNN Routing Model Test:")
    print(f"  Number of cities: {n_cities}")
    print(f"  Tour: {tour}")
    print(f"  Tour length: {tour_length:.4f}")
    print(f"  Log probability: {log_prob:.4f}")

test_gnn_routing_model()

```

GNN Routing Model Test:

Number of cities: 15

Tour: [1, 3, 4, 11, 13, 0, 12, 10, 14, 9, 8, 5, 6, 7, 2]

Tour length: 6.9833

Log probability: -26.8479

34.5.3. Training with REINFORCE and Baseline

```

class BaselineNetwork(nn.Module):
    """
    Critic network for baseline estimation in REINFORCE.
    """

    def __init__(self, embed_dim: int = 128):
        super().__init__()

        self.network = nn.Sequential(
            nn.Linear(embed_dim, embed_dim),
            nn.ReLU(),
            nn.Linear(embed_dim, embed_dim // 2),
            nn.ReLU(),
            nn.Linear(embed_dim // 2, 1)
        )

    def forward(self, graph_embedding: torch.Tensor) -> torch.Tensor:
        """Predict baseline value."""
        return self.network(graph_embedding).squeeze(-1)

def train_attention_model(n_epochs: int = 50, batch_size: int = 64,
                          n_cities: int = 20, lr: float = 1e-4) -> Dict:
    """
    Train Attention Model using REINFORCE with learned baseline.
    """

    model = AttentionModel(node_dim=2, embed_dim=128, n_heads=8, n_layers=3)
    baseline = BaselineNetwork(embed_dim=128)

    optimizer = torch.optim.Adam(list(model.parameters()) +
    ↪ list(baseline.parameters()), lr=lr)

    history = {'tour_length': [], 'baseline_loss': []}

    for epoch in range(n_epochs):
        model.train()
        baseline.train()

        # Generate batch
        node_features = torch.rand(batch_size, n_cities, 2)

        # Sample tours
        tours, log_probs = model(node_features, greedy=False)

        # Compute tour lengths
        tour_lengths = []
        for i in range(batch_size):
            coords = node_features[i].numpy()
            tour = tours[i].tolist()
            length = compute_tour_length(coords, tour)
            tour_lengths.append(length)
        tour_lengths = torch.tensor(tour_lengths)

```

```

# Get baseline predictions
with torch.no_grad():
    node_embeddings = model.encoder(node_features)
    graph_embeddings = node_embeddings.mean(dim=1)
    baseline_pred = baseline(graph_embeddings)

# REINFORCE loss with baseline
advantage = tour_lengths - baseline_pred.detach()
policy_loss = (advantage * (-log_probs)).mean()

# Baseline loss
baseline_loss = F.mse_loss(baseline_pred, tour_lengths)

# Total loss
loss = policy_loss + baseline_loss

optimizer.zero_grad()
loss.backward()
torch.nn.utils.clip_grad_norm_(model.parameters(), 1.0)
optimizer.step()

history['tour_length'].append(tour_lengths.mean().item())
history['baseline_loss'].append(baseline_loss.item())

if (epoch + 1) % 10 == 0:
    # Evaluate with greedy decoding
    model.eval()
    with torch.no_grad():
        val_features = torch.rand(100, n_cities, 2)
        val_tours, _ = model(val_features, greedy=True)

        val_lengths = []
        for i in range(100):
            coords = val_features[i].numpy()
            tour = val_tours[i].tolist()
            val_lengths.append(compute_tour_length(coords, tour))

        print(f"Epoch {epoch + 1}/{n_epochs}, "
              f"Train: {tour_lengths.mean():.4f}, "
              f"Val (greedy): {np.mean(val_lengths):.4f}")

    return history

# Train the model
print("\nTraining Attention Model with REINFORCE...")
history = train_attention_model(n_epochs=50, batch_size=64, n_cities=20)

# Plot results
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

ax1.plot(history['tour_length'], linewidth=2)

```

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```
ax1.set_xlabel('Epoch', fontsize=12)
ax1.set_ylabel('Average Tour Length', fontsize=12)
ax1.set_title('Training Progress: Tour Length', fontsize=14, fontweight='bold')
ax1.grid(True, alpha=0.3)

ax2.plot(history['baseline_loss'], linewidth=2, color='coral')
ax2.set_xlabel('Epoch', fontsize=12)
ax2.set_ylabel('Baseline MSE Loss', fontsize=12)
ax2.set_title('Baseline Network Loss', fontsize=14, fontweight='bold')
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```

Training Attention Model with REINFORCE...

RuntimeError: one of the variables needed for gradient computation has been modified by
↳ an inplace operation: [CpuBoolType [64, 20]] is at version 20; expected version 19
↳ instead. Hint: enable anomaly detection to find the operation that failed to compute
↳ its gradient, with torch.autograd.set_detect_anomaly(True).

```
-----
RuntimeError                                Traceback (most recent call last)
Cell In[20], line 99
    97 # Train the model
    98 print("\nTraining Attention Model with REINFORCE...")
--> 99 history = train_attention_model(n_epochs=50, batch_size=64, n_cities=20)
    101 # Plot results
    102 fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
Cell In[20], line 70, in train_attention_model(n_epochs, batch_size, n_cities, lr)
    67 loss = policy_loss + baseline_loss
    69 optimizer.zero_grad()
--> 70 loss.backward()
    71 torch.nn.utils.clip_grad_norm_(model.parameters(), 1.0)
    72 optimizer.step()
File /opt/homebrew/lib/python3.14/site-packages/torch/_tensor.py:630, in
  ↳ Tensor.backward(self, gradient, retain_graph, create_graph, inputs)
    620 if has_torch_function_unary(self):
    621     return handle_torch_function(
    622         Tensor.backward,
    623         (self,),
    (... ) 628         inputs=inputs,
    629     )
--> 630 torch.autograd.backward(
    631     self, gradient, retain_graph, create_graph, inputs=inputs
    632 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/__init__.py:364, in
  ↳ backward(tensors, grad_tensors, retain_graph, create_graph, grad_variables, inputs)
    359     retain_graph = create_graph
    361 # The reason we repeat the same comment below is that
    362 # some Python versions print out the first line of a multi-line function
    363 # calls in the traceback and some print out the last line
--> 364 _engine_run_backward(
    365     tensors,
```



```

366     grad_tensors_,
367     retain_graph,
368     create_graph,
369     inputs_tuple,
370     allow_unreachable=True,
371     accumulate_grad=True,
372 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/graph.py:865, in
↳ _engine_run_backward(t_outputs, *args, **kwargs)
863     unregister_hooks = _register_logging_hooks_on_whole_graph(t_outputs)
864 try:
--> 865     return Variable._execution_engine.run_backward( # Calls into the C++ engine
↳ to run the backward pass
866         t_outputs, *args, **kwargs
867     ) # Calls into the C++ engine to run the backward pass
868 finally:
869     if attach_logging_hooks:
RuntimeError: one of the variables needed for gradient computation has been modified by
↳ an inplace operation: [CPUBoolType [64, 20]] is at version 20; expected version 19
↳ instead. Hint: enable anomaly detection to find the operation that failed to compute
↳ its gradient, with torch.autograd.set_detect_anomaly(True).

```

34.6. GNN for MILP Solving

GNNs can accelerate MILP solvers by learning to make better branching and cutting decisions. Rather than replacing the solver, the GNN serves as a heuristic advisor within the branch-and-bound framework, reducing the number of nodes explored to reach an optimal solution.

34.6.1. Bipartite GNN for MILP

MILP problems naturally form a bipartite graph between variables and constraints:

```

class BipartiteGNNLayer(nn.Module):
    """
    Bipartite Graph Neural Network layer for MILP.

    Messages pass between variable nodes and constraint nodes.
    """

    def __init__(self, var_dim: int, con_dim: int, edge_dim: int,
                 hidden_dim: int):
        super().__init__()

        # Variable -> Constraint message
        self.var_to_con = nn.Sequential(
            nn.Linear(var_dim + edge_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, con_dim)
        )

        # Constraint -> Variable message
        self.con_to_var = nn.Sequential(
            nn.Linear(con_dim + edge_dim, hidden_dim),
            nn.ReLU(),

```

```

        nn.Linear(hidden_dim, var_dim)
    )

    # Update functions
    self.var_update = nn.GRUCell(var_dim, var_dim)
    self.con_update = nn.GRUCell(con_dim, con_dim)

    def forward(self, var_features: torch.Tensor,
                con_features: torch.Tensor,
                edge_index: torch.Tensor,
                edge_features: torch.Tensor,
                n_vars: int) -> Tuple[torch.Tensor, torch.Tensor]:
        """
        Forward pass.

        Args:
            var_features: (n_vars, var_dim)
            con_features: (n_cons, con_dim)
            edge_index: (2, n_edges) - first row is variable, second is constraint
            ↪ offset
            edge_features: (n_edges, edge_dim)
            n_vars: Number of variables

        Returns:
            Updated variable and constraint features
        """
        # Separate variable and constraint edges
        # edge_index[0] is source (either var or con index)
        # edge_index[1] is target (either con or var index)

        n_cons = con_features.size(0)
        n_edges = edge_index.size(1) // 2 # Bidirectional edges

        # Variable -> Constraint edges (first half)
        v2c_src = edge_index[0, :n_edges]
        v2c_tgt = edge_index[1, :n_edges] - n_vars # Convert to constraint index
        v2c_edge_feat = edge_features[:n_edges]

        # Constraint -> Variable edges (second half)
        c2v_src = edge_index[0, n_edges:] - n_vars # Convert to constraint index
        c2v_tgt = edge_index[1, n_edges:]
        c2v_edge_feat = edge_features[n_edges:]

        # Variable -> Constraint messages
        v2c_msg_input = torch.cat([var_features[v2c_src], v2c_edge_feat], dim=-1)
        v2c_messages = self.var_to_con(v2c_msg_input)

        # Aggregate messages for constraints
        con_aggregated = torch.zeros(n_cons, v2c_messages.size(-1),
                                     device=var_features.device)
        con_aggregated.scatter_add_(0, v2c_tgt.unsqueeze(-1).expand(-1,
            ↪ v2c_messages.size(-1)),
                                     v2c_messages)

```

```

        # Count for mean aggregation
        con_count = torch.zeros(n_cons, device=var_features.device)
        con_count.scatter_add_(0, v2c_tgt, torch.ones_like(v2c_tgt),
↪ dtype=torch.float))
        con_aggregated = con_aggregated / con_count.clamp(min=1).unsqueeze(-1)

        # Update constraints
        con_features_new = self.con_update(con_aggregated, con_features)

        # Constraint -> Variable messages
        c2v_msg_input = torch.cat([con_features_new[c2v_src], c2v_edge_feat],
↪ dim=-1)
        c2v_messages = self.con_to_var(c2v_msg_input)

        # Aggregate messages for variables
        var_aggregated = torch.zeros(n_vars, c2v_messages.size(-1),
                                     device=var_features.device)
        var_aggregated.scatter_add_(0, c2v_tgt.unsqueeze(-1).expand(-1,
↪ c2v_messages.size(-1)),
                                     c2v_messages)

        # Count for mean aggregation
        var_count = torch.zeros(n_vars, device=var_features.device)
        var_count.scatter_add_(0, c2v_tgt, torch.ones_like(c2v_tgt),
↪ dtype=torch.float))
        var_aggregated = var_aggregated / var_count.clamp(min=1).unsqueeze(-1)

        # Update variables
        var_features_new = self.var_update(var_aggregated, var_features)

        return var_features_new, con_features_new

class BipartiteMILPGNN(nn.Module):
    """
    Complete Bipartite GNN for MILP representation learning.
    """

    def __init__(self, var_dim: int = 6, con_dim: int = 4, edge_dim: int = 1,
                  hidden_dim: int = 64, n_layers: int = 4):
        super().__init__()

        self.var_embed = nn.Linear(var_dim, hidden_dim)
        self.con_embed = nn.Linear(con_dim, hidden_dim)
        self.edge_embed = nn.Linear(edge_dim, hidden_dim)

        self.layers = nn.ModuleList([
            BipartiteGNNLayer(hidden_dim, hidden_dim, hidden_dim, hidden_dim)
            for _ in range(n_layers)
        ])

    def forward(self, var_features: torch.Tensor,

```

```

        con_features: torch.Tensor,
        edge_index: torch.Tensor,
        edge_features: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
    """
    Encode MILP instance.
    """
    n_vars = var_features.size(0)

    # Initial embeddings
    var_h = self.var_embed(var_features)
    con_h = self.con_embed(con_features)
    edge_h = self.edge_embed(edge_features)

    # Apply GNN layers
    for layer in self.layers:
        var_h, con_h = layer(var_h, con_h, edge_index, edge_h, n_vars)

    return var_h, con_h

```

34.6.2. Learning to Branch

Branching decisions in branch-and-bound solvers can be learned using imitation learning:

```

class BranchingPolicy(nn.Module):
    """
    GNN-based branching policy for MILP solvers.

    Learns to predict branching scores that imitate strong branching.
    """

    def __init__(self, var_dim: int = 6, con_dim: int = 4, edge_dim: int = 1,
                  hidden_dim: int = 64, n_layers: int = 4):
        super().__init__()

        self.gnn = BipartiteMILPGNN(var_dim, con_dim, edge_dim, hidden_dim,
                                     n_layers)

        # Branching score predictor
        self.score_predictor = nn.Sequential(
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, 1)
        )

    def forward(self, var_features: torch.Tensor,
                con_features: torch.Tensor,
                edge_index: torch.Tensor,
                edge_features: torch.Tensor,
                candidate_mask: torch.Tensor) -> torch.Tensor:
        """
        Predict branching scores for candidate variables.

        Args:

```

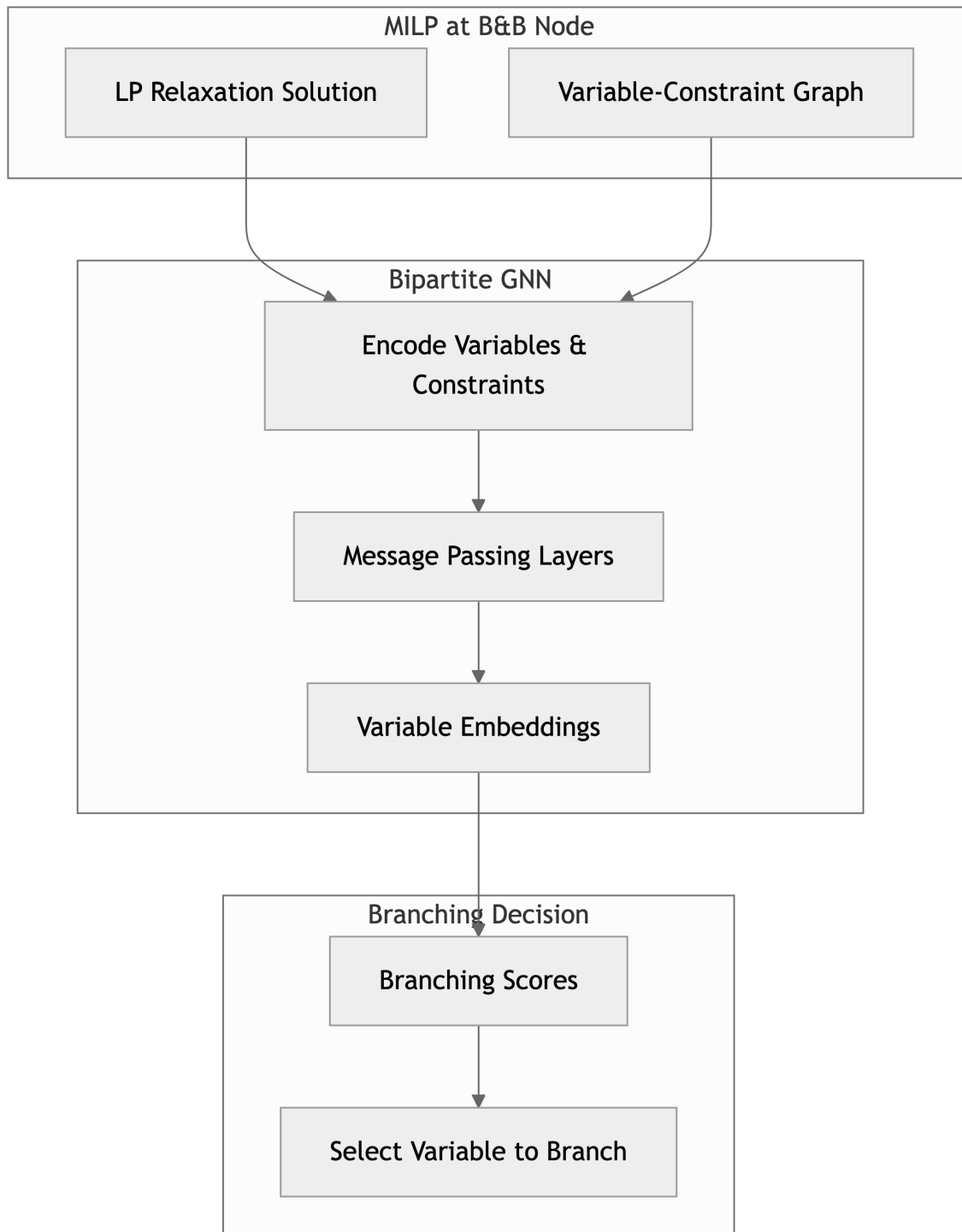


Figure 34.6.: Learning to Branch: GNN predicts branching scores for variables

```

        var_features: Variable node features
        con_features: Constraint node features
        edge_index: Bipartite edge connectivity
        edge_features: Edge features (coefficients)
        candidate_mask: Boolean mask for branching candidates

Returns:
    Branching scores for candidate variables
    """
    # Get variable embeddings
    var_embeddings, _ = self.gnn(var_features, con_features,
                                edge_index, edge_features)

    # Predict scores
    scores = self.score_predictor(var_embeddings).squeeze(-1)

    # Mask non-candidates
    scores[~candidate_mask] = float('-inf')

    return scores

def generate_branching_data(n_samples: int = 100) -> List[Dict]:
    """
    Generate synthetic branching training data.

    In practice, this would come from running strong branching
    on a set of MILP instances.
    """
    data = []

    for _ in range(n_samples):
        # Random MILP structure
        n_vars = np.random.randint(10, 30)
        n_cons = np.random.randint(5, 15)

        # Variable features
        var_features = torch.randn(n_vars, 6)

        # Constraint features
        con_features = torch.randn(n_cons, 4)

        # Random sparse bipartite graph
        density = 0.3
        edge_list = []
        edge_feat_list = []

        for j in range(n_cons):
            for i in range(n_vars):
                if np.random.rand() < density:
                    # Variable -> Constraint edge
                    edge_list.append([i, n_vars + j])
                    # Constraint -> Variable edge

```

```

        edge_list.append([n_vars + j, i])
        coef = np.random.randn()
        edge_feat_list.append([coef])
        edge_feat_list.append([coef])

    if len(edge_list) == 0:
        continue

    edge_index = torch.tensor(edge_list, dtype=torch.long).t()
    edge_features = torch.tensor(edge_feat_list, dtype=torch.float32)

    # Candidate variables (fractional in LP solution)
    n_candidates = max(2, n_vars // 3)
    candidates = np.random.choice(n_vars, n_candidates, replace=False)
    candidate_mask = torch.zeros(n_vars, dtype=torch.bool)
    candidate_mask[candidates] = True

    # Target: random "strong branching" scores
    target_scores = torch.zeros(n_vars)
    target_scores[candidates] = torch.randn(n_candidates)
    best_candidate = candidates[target_scores[candidates].argmax()]

    data.append({
        'var_features': var_features,
        'con_features': con_features,
        'edge_index': edge_index,
        'edge_features': edge_features,
        'candidate_mask': candidate_mask,
        'target_scores': target_scores,
        'best_candidate': best_candidate
    })

    return data

def train_branching_policy(model: BranchingPolicy, data: List[Dict],
                           n_epochs: int = 50, lr: float = 1e-3) -> List[float]:
    """Train branching policy using imitation learning."""
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)
    accuracies = []

    for epoch in range(n_epochs):
        correct = 0
        total = 0

        for sample in data:
            optimizer.zero_grad()

            # Forward pass
            scores = model(
                sample['var_features'],
                sample['con_features'],
                sample['edge_index'],

```

```

        sample['edge_features'],
        sample['candidate_mask']
    )

    # Cross-entropy loss on candidates
    candidates = sample['candidate_mask'].nonzero().squeeze(-1)
    target_probs = F.softmax(sample['target_scores'][candidates], dim=0)
    pred_log_probs = F.log_softmax(scores[candidates], dim=0)

    loss = F.kl_div(pred_log_probs, target_probs, reduction='batchmean')

    loss.backward()
    optimizer.step()

    # Track accuracy
    pred_best = scores.argmax().item()
    if pred_best == sample['best_candidate']:
        correct += 1
    total += 1

    accuracy = correct / total
    accuracies.append(accuracy)

    if (epoch + 1) % 10 == 0:
        print(f"Epoch {epoch + 1}/{n_epochs}, Accuracy: {accuracy:.4f}")

    return accuracies

# Train branching policy
print("\nTraining Branching Policy...")
branching_model = BranchingPolicy()
branching_data = generate_branching_data(200)
accuracies = train_branching_policy(branching_model, branching_data, n_epochs=50)

plt.figure(figsize=(10, 4))
plt.plot(accuracies, linewidth=2, color='seagreen')
plt.xlabel('Epoch', fontsize=12)
plt.ylabel('Accuracy', fontsize=12)
plt.title('Learning to Branch: Training Accuracy', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.3)
plt.ylim(0, 1)
plt.tight_layout()
plt.show()

```

Training Branching Policy...

IndexError: index 28 is out of bounds for dimension 0 with size 28

```

-----
IndexError                                Traceback (most recent call last)
Cell In[22], line 169
    167 branching_model = BranchingPolicy()
    168 branching_data = generate_branching_data(200)

```



```

--> 169 accuracies = train_branching_policy(branching_model, branching_data, n_epochs=50)
    171 plt.figure(figsize=(10, 4))
    172 plt.plot(accuracies, linewidth=2, color='seagreen')
Cell In[22], line 132, in train_branching_policy(model, data, n_epochs, lr)
    129 optimizer.zero_grad()
    131 # Forward pass
--> 132 scores = model(
    133     sample['var_features'],
    134     sample['con_features'],
    135     sample['edge_index'],
    136     sample['edge_features'],
    137     sample['candidate_mask']
    138 )
    140 # Cross-entropy loss on candidates
    141 candidates = sample['candidate_mask'].nonzero().squeeze(-1)
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1776, in
  ↪ Module._wrapped_call_impl(self, *args, **kwargs)
    1774     return self._compiled_call_impl(*args, **kwargs) # type: ignore[misc]
    1775 else:
-> 1776     return self._call_impl(*args, **kwargs)
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1787, in
  ↪ Module._call_impl(self, *args, **kwargs)
    1782 # If we don't have any hooks, we want to skip the rest of the logic in
    1783 # this function, and just call forward.
    1784 if not (self._backward_hooks or self._backward_pre_hooks or self._forward_hooks
  ↪ or self._forward_pre_hooks
    1785         or _global_backward_pre_hooks or _global_backward_hooks
    1786         or _global_forward_hooks or _global_forward_pre_hooks):
-> 1787     return forward_call(*args, **kwargs)
    1789 result = None
    1790 called_always_called_hooks = set()
Cell In[22], line 40, in BranchingPolicy.forward(self, var_features, con_features,
  ↪ edge_index, edge_features, candidate_mask)
    26 """
    27 Predict branching scores for candidate variables.
    28
    (...) 37     Branching scores for candidate variables
    38 """
    39 # Get variable embeddings
---> 40 var_embeddings, _ = self.gnn(var_features, con_features,
    41                               edge_index, edge_features)
    43 # Predict scores
    44 scores = self.score_predictor(var_embeddings).squeeze(-1)
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1776, in
  ↪ Module._wrapped_call_impl(self, *args, **kwargs)
    1774     return self._compiled_call_impl(*args, **kwargs) # type: ignore[misc]
    1775 else:
-> 1776     return self._call_impl(*args, **kwargs)
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1787, in
  ↪ Module._call_impl(self, *args, **kwargs)
    1782 # If we don't have any hooks, we want to skip the rest of the logic in
    1783 # this function, and just call forward.
    1784 if not (self._backward_hooks or self._backward_pre_hooks or self._forward_hooks
  ↪ or self._forward_pre_hooks
    1785         or _global_backward_pre_hooks or _global_backward_hooks

```

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```
1786         or _global_forward_hooks or _global_forward_pre_hooks):
-> 1787     return forward_call(*args, **kwargs)
1789     result = None
1790     called_always_called_hooks = set()
Cell In[21], line 138, in BipartiteMILPGNN.forward(self, var_features, con_features,
    ↪ edge_index, edge_features)
    136 # Apply GNN layers
    137 for layer in self.layers:
--> 138     var_h, con_h = layer(var_h, con_h, edge_index, edge_h, n_vars)
    140 return var_h, con_h
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1776, in
    ↪ Module._wrapped_call_impl(self, *args, **kwargs)
    1774     return self._compiled_call_impl(*args, **kwargs) # type: ignore[misc]
    1775 else:
-> 1776     return self._call_impl(*args, **kwargs)
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1787, in
    ↪ Module._call_impl(self, *args, **kwargs)
    1782 # If we don't have any hooks, we want to skip the rest of the logic in
    1783 # this function, and just call forward.
    1784 if not (self._backward_hooks or self._backward_pre_hooks or self._forward_hooks
    ↪ or self._forward_pre_hooks
    1785         or _global_backward_pre_hooks or _global_backward_hooks
    1786         or _global_forward_hooks or _global_forward_pre_hooks):
-> 1787     return forward_call(*args, **kwargs)
    1789     result = None
    1790     called_always_called_hooks = set()
Cell In[21], line 66, in BipartiteGNNLayer.forward(self, var_features, con_features,
    ↪ edge_index, edge_features, n_vars)
    63 c2v_edge_feat = edge_features[n_edges:]
    65 # Variable -> Constraint messages
---> 66 v2c_msg_input = torch.cat([var_features[v2c_src], v2c_edge_feat], dim=-1)
    67 v2c_messages = self.var_to_con(v2c_msg_input)
    69 # Aggregate messages for constraints
IndexError: index 28 is out of bounds for dimension 0 with size 28
```

34.6.3. Learning to Cut

Cutting plane selection determines which valid inequalities to add to the LP relaxation. A GNN can learn to score candidate cuts based on their expected impact on the LP bound improvement.

```
class CutSelector(nn.Module):
    """
    GNN-based cut selector for MILP solvers.

    Learns to select which cutting planes to add from a pool of candidates.
    """

    def __init__(self, var_dim: int = 6, con_dim: int = 4, cut_dim: int = 8,
                  hidden_dim: int = 64, n_layers: int = 4):
        super().__init__()

        self.gnn = BipartiteMILPGNN(var_dim, con_dim, 1, hidden_dim, n_layers)

        # Cut encoder
```

```

self.cut_encoder = nn.Sequential(
    nn.Linear(cut_dim, hidden_dim),
    nn.ReLU(),
    nn.Linear(hidden_dim, hidden_dim)
)

# Selection scorer
self.cut_scorer = nn.Sequential(
    nn.Linear(2 * hidden_dim, hidden_dim),
    nn.ReLU(),
    nn.Linear(hidden_dim, 1)
)

def forward(self, var_features: torch.Tensor,
            con_features: torch.Tensor,
            edge_index: torch.Tensor,
            edge_features: torch.Tensor,
            cut_features: torch.Tensor,
            cut_var_indices: List[List[int]]) -> torch.Tensor:
    """
    Score candidate cuts.

    Args:
        var_features: Variable node features
        con_features: Constraint node features
        edge_index: Bipartite graph edges
        edge_features: Edge features
        cut_features: Features of candidate cuts (n_cuts, cut_dim)
        cut_var_indices: Variables involved in each cut

    Returns:
        Scores for each cut
    """
    # Get variable embeddings
    var_embeddings, _ = self.gnn(var_features, con_features,
                                edge_index, edge_features)

    # Encode cuts
    cut_embeddings = self.cut_encoder(cut_features)

    # Pool variable embeddings for each cut
    n_cuts = cut_features.size(0)
    cut_var_emb = torch.zeros(n_cuts, var_embeddings.size(-1),
                              device=var_features.device)

    for i, var_indices in enumerate(cut_var_indices):
        if len(var_indices) > 0:
            cut_var_emb[i] = var_embeddings[var_indices].mean(dim=0)

    # Combine cut encoding with pooled variable embeddings
    combined = torch.cat([cut_embeddings, cut_var_emb], dim=-1)

    # Score cuts

```

```
scores = self.cut_scorer(combined).squeeze(-1)

return scores
```

i Integration with Solvers

To use learned policies in practice, you need to integrate with a MILP solver like: - **SCIP**: Open-source solver with Python interface (PySCIPOpt) - **Gurobi**: Commercial solver with callback interface - **CPLEX**: IBM's commercial solver

The learned GNN model is called via callbacks to make branching or cut selection decisions.

```
class MockMILPSolver:
    """
    Mock MILP solver demonstrating GNN integration.

    In practice, this would interface with SCIP, Gurobi, or CPLEX.
    """

    def __init__(self, branching_policy: Optional[BranchingPolicy] = None):
        self.branching_policy = branching_policy
        self.n_nodes_explored = 0

    def solve(self, c: np.ndarray, A: np.ndarray, b: np.ndarray,
              var_types: List[str], bounds: List[Tuple],
              max_nodes: int = 1000) -> Dict:
        """
        Solve MILP using branch and bound with learned branching.
        """
        n_vars = len(c)
        n_cons = len(b)

        # Create graph representation
        milp_repr = MILPGraphRepresentation()
        graph = milp_repr.create_graph(c, A, b, var_types, bounds,
                                       ['L'] * n_cons)

        # Simple simulation of B&B (actual solver would be much more complex)
        self.n_nodes_explored = 0
        best_obj = float('inf')
        best_solution = None

        # Initial "LP relaxation" solution
        current_solution = np.random.rand(n_vars)

        while self.n_nodes_explored < max_nodes:
            self.n_nodes_explored += 1

            # Find fractional variables (branching candidates)
            candidates = []
            for i in range(n_vars):
```

```

        if var_types[i] in ['I', 'B']:
            frac = abs(current_solution[i] - round(current_solution[i]))
            if frac > 1e-6:
                candidates.append(i)

    if len(candidates) == 0:
        # Solution is integral
        obj = c @ current_solution
        if obj < best_obj:
            best_obj = obj
            best_solution = current_solution.copy()
        break

    # Select branching variable
    if self.branching_policy is not None:
        candidate_mask = torch.zeros(n_vars, dtype=torch.bool)
        candidate_mask[candidates] = True

        with torch.no_grad():
            scores = self.branching_policy(
                graph['var_features'],
                graph['con_features'],
                graph['edge_index'],
                graph['edge_features'],
                candidate_mask
            )
            branch_var = scores.argmax().item()
    else:
        # Random selection
        branch_var = np.random.choice(candidates)

    # Simulate branching (simplified)
    current_solution[branch_var] = round(current_solution[branch_var])

    return {
        'objective': best_obj,
        'solution': best_solution,
        'nodes_explored': self.n_nodes_explored
    }

# Demonstrate solver with learned policy
print("\nDemonstrating MILP Solver with Learned Branching...")

# Create a simple MILP instance
n_items = 15
c = -np.random.rand(n_items) # Maximize (minimize negative)
A = np.random.rand(3, n_items) * 10
b = np.array([20.0, 15.0, 25.0])
var_types = ['B'] * n_items
bounds = [(0, 1)] * n_items

# Solve with random branching

```

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```
random_solver = MockMILPSolver(branching_policy=None)
random_result = random_solver.solve(c, A, b, var_types, bounds)

# Solve with learned branching
learned_solver = MockMILPSolver(branching_policy=branching_model)
learned_result = learned_solver.solve(c, A, b, var_types, bounds)

print(f"\nRandom Branching:")
print(f"  Nodes explored: {random_result['nodes_explored']}")
print(f"  Objective: {random_result['objective']:.4f}")

print(f"\nLearned Branching:")
print(f"  Nodes explored: {learned_result['nodes_explored']}")
print(f"  Objective: {learned_result['objective']:.4f}")
```

Demonstrating MILP Solver with Learned Branching...

IndexError: index 15 is out of bounds for dimension 0 with size 15

```
-----
IndexError                                Traceback (most recent call last)
Cell In[24], line 98
    96 # Solve with learned branching
    97 learned_solver = MockMILPSolver(branching_policy=branching_model)
--> 98 learned_result = learned_solver.solve(c, A, b, var_types, bounds)
    100 print(f"\nRandom Branching:")
    101 print(f"  Nodes explored: {random_result['nodes_explored']}")
Cell In[24], line 59, in MockMILPSolver.solve(self, c, A, b, var_types, bounds,
      ↪ max_nodes)
    56     candidate_mask[candidates] = True
    58     with torch.no_grad():
--> 59         scores = self.branching_policy(
    60             graph['var_features'],
    61             graph['con_features'],
    62             graph['edge_index'],
    63             graph['edge_features'],
    64             candidate_mask
    65         )
    66     branch_var = scores.argmax().item()
    67 else:
    68     # Random selection
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1776, in
      ↪ Module._wrapped_call_impl(self, *args, **kwargs)
    1774     return self._compiled_call_impl(*args, **kwargs) # type: ignore[misc]
    1775 else:
-> 1776     return self._call_impl(*args, **kwargs)
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1787, in
      ↪ Module._call_impl(self, *args, **kwargs)
    1782 # If we don't have any hooks, we want to skip the rest of the logic in
    1783 # this function, and just call forward.
    1784 if not (self._backward_hooks or self._backward_pre_hooks or self._forward_hooks
      ↪ or self._forward_pre_hooks
    1785         or _global_backward_hooks or _global_backward_hooks
    1786         or _global_forward_hooks or _global_forward_pre_hooks):
```

```

-> 1787     return forward_call(*args, **kwargs)
    1789 result = None
    1790 called_always_called_hooks = set()
Cell In[22], line 40, in BranchingPolicy.forward(self, var_features, con_features,
↳ edge_index, edge_features, candidate_mask)
    26 """
    27 Predict branching scores for candidate variables.
    28
    (...)    37     Branching scores for candidate variables
    38 """
    39 # Get variable embeddings
--> 40 var_embeddings, _ = self.gnn(var_features, con_features,
    41                               edge_index, edge_features)
    43 # Predict scores
    44 scores = self.score_predictor(var_embeddings).squeeze(-1)
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1776, in
↳ Module._wrapped_call_impl(self, *args, **kwargs)
    1774     return self._compiled_call_impl(*args, **kwargs) # type: ignore[misc]
    1775 else:
-> 1776     return self._call_impl(*args, **kwargs)
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1787, in
↳ Module._call_impl(self, *args, **kwargs)
    1782 # If we don't have any hooks, we want to skip the rest of the logic in
    1783 # this function, and just call forward.
    1784 if not (self._backward_hooks or self._backward_pre_hooks or self._forward_hooks
↳ or self._forward_pre_hooks
    1785         or _global_backward_pre_hooks or _global_backward_hooks
    1786         or _global_forward_hooks or _global_forward_pre_hooks):
-> 1787     return forward_call(*args, **kwargs)
    1789 result = None
    1790 called_always_called_hooks = set()
Cell In[21], line 138, in BipartiteMILPGNN.forward(self, var_features, con_features,
↳ edge_index, edge_features)
    136 # Apply GNN layers
    137 for layer in self.layers:
--> 138     var_h, con_h = layer(var_h, con_h, edge_index, edge_h, n_vars)
    140 return var_h, con_h
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1776, in
↳ Module._wrapped_call_impl(self, *args, **kwargs)
    1774     return self._compiled_call_impl(*args, **kwargs) # type: ignore[misc]
    1775 else:
-> 1776     return self._call_impl(*args, **kwargs)
File /opt/homebrew/lib/python3.14/site-packages/torch/nn/modules/module.py:1787, in
↳ Module._call_impl(self, *args, **kwargs)
    1782 # If we don't have any hooks, we want to skip the rest of the logic in
    1783 # this function, and just call forward.
    1784 if not (self._backward_hooks or self._backward_pre_hooks or self._forward_hooks
↳ or self._forward_pre_hooks
    1785         or _global_backward_pre_hooks or _global_backward_hooks
    1786         or _global_forward_hooks or _global_forward_pre_hooks):
-> 1787     return forward_call(*args, **kwargs)
    1789 result = None
    1790 called_always_called_hooks = set()
Cell In[21], line 66, in BipartiteGNNLayer.forward(self, var_features, con_features,
↳ edge_index, edge_features, n_vars)

```

```

63 c2v_edge_feat = edge_features[n_edges:]
65 # Variable -> Constraint messages
--> 66 v2c_msg_input = torch.cat([var_features[v2c_src], v2c_edge_feat], dim=-1)
67 v2c_messages = self.var_to_con(v2c_msg_input)
69 # Aggregate messages for constraints
IndexError: index 15 is out of bounds for dimension 0 with size 15

```

34.7. Scalability and Practical Considerations

Deploying GNN-based optimization models in production requires addressing scalability challenges. Real-world instances can have thousands or millions of nodes, far exceeding the sizes used during training. This section presents techniques for sparse graph construction, mini-batch training, and generalizing to larger instances.

34.7.1. Sparse Graph Construction

For large-scale problems, complete graphs become infeasible. Sparse graph construction strategies are essential:

```

class SparseGraphBuilder:
    """
    Strategies for building sparse graphs from optimization problems.
    """

    @staticmethod
    def k_nearest_neighbors(coords: np.ndarray, k: int = 10) -> torch.Tensor:
        """
        Build k-nearest neighbor graph.

        Args:
            coords: Node coordinates (n_nodes, dim)
            k: Number of neighbors

        Returns:
            edge_index: (2, n_edges)
        """
        n_nodes = len(coords)
        k = min(k, n_nodes - 1)

        # Compute all pairwise distances
        dist_matrix = np.sqrt(((coords[:, None, :] - coords[None, :, :]) **
        ↪ 2).sum(axis=-1))

        # Find k nearest neighbors for each node
        edge_list = []
        for i in range(n_nodes):
            distances = dist_matrix[i]
            neighbors = np.argsort(distances)[1:k+1] # Exclude self
            for j in neighbors:
                edge_list.append([i, j])
                edge_list.append([j, i])

        edge_index = torch.tensor(edge_list, dtype=torch.long).t()
        # Remove duplicates

```



```

    edge_index = torch.unique(edge_index, dim=1)

    return edge_index

@staticmethod
def radius_graph(coords: np.ndarray, radius: float) -> torch.Tensor:
    """
    Build graph connecting nodes within radius.

    Args:
        coords: Node coordinates
        radius: Connection radius

    Returns:
        edge_index: (2, n_edges)
    """
    n_nodes = len(coords)

    # Compute distances
    dist_matrix = np.sqrt(((coords[:, None, :] - coords[None, :, :]) **
↪ 2).sum(axis=-1))

    # Find pairs within radius
    edge_list = []
    for i in range(n_nodes):
        for j in range(i + 1, n_nodes):
            if dist_matrix[i, j] <= radius:
                edge_list.append([i, j])
                edge_list.append([j, i])

    if len(edge_list) == 0:
        return torch.zeros(2, 0, dtype=torch.long)

    edge_index = torch.tensor(edge_list, dtype=torch.long).t()
    return edge_index

@staticmethod
def delaunay_graph(coords: np.ndarray) -> torch.Tensor:
    """
    Build Delaunay triangulation graph.

    Args:
        coords: 2D node coordinates

    Returns:
        edge_index: (2, n_edges)
    """
    from scipy.spatial import Delaunay

    tri = Delaunay(coords)

    edge_set = set()
    for simplex in tri.simplices:

```

```

        for i in range(3):
            for j in range(i + 1, 3):
                u, v = simplex[i], simplex[j]
                edge_set.add((min(u, v), max(u, v)))

    edge_list = []
    for u, v in edge_set:
        edge_list.append([u, v])
        edge_list.append([v, u])

    edge_index = torch.tensor(edge_list, dtype=torch.long).t()
    return edge_index

# Compare sparse graph strategies
def compare_graph_strategies(n_nodes: int = 100):
    coords = np.random.rand(n_nodes, 2)

    strategies = {
        'Complete': n_nodes * (n_nodes - 1),
        'KNN (k=10)': SparseGraphBuilder.k_nearest_neighbors(coords, k=10).size(1),
        'KNN (k=20)': SparseGraphBuilder.k_nearest_neighbors(coords, k=20).size(1),
        'Radius (r=0.2)': SparseGraphBuilder.radius_graph(coords,
            ↪ radius=0.2).size(1),
        'Delaunay': SparseGraphBuilder.delaunay_graph(coords).size(1)
    }

    print(f"\nGraph Sparsity Comparison ({n_nodes} nodes):")
    print("-" * 40)
    for name, n_edges in strategies.items():
        density = n_edges / (n_nodes * (n_nodes - 1)) * 100
        print(f"{name:20s}: {n_edges:6d} edges ({density:5.1f}% density)")

    return strategies

strategies = compare_graph_strategies(100)

```

Graph Sparsity Comparison (100 nodes):

```

-----
Complete           :   9900 edges (100.0% density)
KNN (k=10)         :   1210 edges ( 12.2% density)
KNN (k=20)         :   2360 edges ( 23.8% density)
Radius (r=0.2)     :   1116 edges ( 11.3% density)
Delaunay           :    572 edges (  5.8% density)

```

💡 Choosing a Sparsification Strategy

- **KNN**: Good default choice; maintains consistent node degree
- **Radius**: Natural for geographically clustered problems
- **Delaunay**: Excellent for 2D problems; preserves geometric structure

- **Domain-specific:** Use problem structure (e.g., time windows for VRP)

Start with KNN (k=10-20) and increase k if solution quality suffers.

34.7.2. Mini-batch Training for Large Graphs

Training GNNs on large graphs requires efficient batching strategies:

```
class GraphBatcher:
    """
    Batch multiple graphs for efficient training.
    """

    @staticmethod
    def batch_graphs(graphs: List[Dict]) -> Dict:
        """
        Combine multiple graphs into a single batched graph.

        Args:
            graphs: List of graph dictionaries with node_features, edge_index

        Returns:
            Batched graph dictionary
        """
        batch_node_features = []
        batch_edge_index = []
        batch_assignment = []

        node_offset = 0

        for i, graph in enumerate(graphs):
            n_nodes = graph['node_features'].size(0)

            # Add node features
            batch_node_features.append(graph['node_features'])

            # Offset edge indices
            edge_index = graph['edge_index'] + node_offset
            batch_edge_index.append(edge_index)

            # Track which graph each node belongs to
            batch_assignment.append(torch.full((n_nodes,), i, dtype=torch.long))

            node_offset += n_nodes

        return {
            'node_features': torch.cat(batch_node_features, dim=0),
            'edge_index': torch.cat(batch_edge_index, dim=1),
            'batch': torch.cat(batch_assignment, dim=0),
            'n_graphs': len(graphs)
        }

    @staticmethod
```

```

def unbatch_node_features(node_features: torch.Tensor,
                          batch: torch.Tensor,
                          n_graphs: int) -> List[torch.Tensor]:
    """Split batched node features back to individual graphs."""
    return [node_features[batch == i] for i in range(n_graphs)]

def global_mean_pool(x: torch.Tensor, batch: torch.Tensor,
                    n_graphs: int) -> torch.Tensor:
    """
    Global mean pooling for graph-level representations.

    Args:
        x: Node features (total_nodes, dim)
        batch: Graph assignment (total_nodes,)
        n_graphs: Number of graphs in batch

    Returns:
        Graph embeddings (n_graphs, dim)
    """
    graph_embeddings = torch.zeros(n_graphs, x.size(-1), device=x.device)

    for i in range(n_graphs):
        mask = batch == i
        graph_embeddings[i] = x[mask].mean(dim=0)

    return graph_embeddings

# Example batched training
def demonstrate_batched_training():
    # Create sample graphs of different sizes
    graphs = []
    for _ in range(4):
        n = np.random.randint(10, 30)
        graphs.append({
            'node_features': torch.randn(n, 16),
            'edge_index': torch.randint(0, n, (2, n * 5))
        })

    # Batch graphs
    batched = GraphBatcher.batch_graphs(graphs)

    print(f"\nBatched Graph:")
    print(f"  Total nodes: {batched['node_features'].size(0)}")
    print(f"  Total edges: {batched['edge_index'].size(1)}")
    print(f"  Number of graphs: {batched['n_graphs']}")

    # Apply simple GNN
    gnn = GCN(16, 32, 16, n_layers=2)
    node_embeddings = gnn(batched['node_features'], batched['edge_index'])

    # Pool to graph level

```

```

graph_embeddings = global_mean_pool(node_embeddings, batched['batch'],
↪ batched['n_graphs'])
print(f"  Graph embeddings shape: {graph_embeddings.shape}")

demonstrate_batched_training()

```

Batched Graph:

```

Total nodes: 78
Total edges: 390
Number of graphs: 4
Graph embeddings shape: torch.Size([4, 16])

```

34.7.3. Inference Optimization

For deployment, inference speed matters most:

```

import time

class InferenceOptimizer:
    """
    Techniques for optimizing GNN inference.
    """

    @staticmethod
    def measure_inference_time(model: nn.Module, x: torch.Tensor,
                              edge_index: torch.Tensor,
                              n_runs: int = 100) -> Dict[str, float]:
        """Benchmark inference time."""
        model.eval()

        # Warm up
        with torch.no_grad():
            for _ in range(10):
                _ = model(x, edge_index)

        # Time inference
        times = []
        with torch.no_grad():
            for _ in range(n_runs):
                start = time.time()
                _ = model(x, edge_index)
                times.append(time.time() - start)

        return {
            'mean_ms': np.mean(times) * 1000,
            'std_ms': np.std(times) * 1000,
            'min_ms': np.min(times) * 1000,
            'max_ms': np.max(times) * 1000
        }

    @staticmethod

```

```

def apply_torch_script(model: nn.Module) -> torch.jit.ScriptModule:
    """Convert model to TorchScript for faster inference."""
    # Note: Not all GNN operations are TorchScript compatible
    try:
        scripted = torch.jit.script(model)
        return scripted
    except Exception as e:
        print(f"TorchScript conversion failed: {e}")
        return model

# Benchmark different model sizes
def benchmark_models():
    sizes = [50, 100, 200, 500]
    results = []

    for n_nodes in sizes:
        x = torch.randn(n_nodes, 32)
        edge_index = torch.randint(0, n_nodes, (2, n_nodes * 10))

        model = GCN(32, 64, 32, n_layers=3)
        timing = InferenceOptimizer.measure_inference_time(model, x, edge_index)

        results.append({
            'n_nodes': n_nodes,
            **timing
        })

    print(f"Nodes: {n_nodes:4d}, "
          f"Inference: {timing['mean_ms']:.2f} +/- {timing['std_ms']:.2f} ms")

    return results

print("\nInference Benchmarks:")
benchmarks = benchmark_models()

```

Inference Benchmarks:

Nodes: 50, Inference: 7.75 +/- 0.54 ms
 Nodes: 100, Inference: 15.24 +/- 2.96 ms
 Nodes: 200, Inference: 29.37 +/- 0.74 ms
 Nodes: 500, Inference: 74.44 +/- 4.62 ms

34.7.4. Generalization to Larger Instances

A key challenge is training on small instances and generalizing to larger ones:

Generalization Challenges

GNN models trained on small instances often struggle with larger ones due to: - Different graph statistics (degree distribution, diameter) - Out-of-distribution node/edge features - Changed problem dynamics (more local optima)

Strategies to improve generalization: 1. Train on variable-sized instances 2. Use relative/nor-malized features 3. Employ curriculum learning (gradually increase size) 4. Design size-invariant architectures

```
def study_size_generalization():
    """
    Study how model performance changes with problem size.
    """
    # Train model on small instances
    train_sizes = [20, 30, 40]
    test_sizes = [20, 50, 100, 200]

    # Simplified model for demonstration
    model = AttentionModel(node_dim=2, embed_dim=64, n_heads=4, n_layers=2)

    # Quick training on small instances (simplified)
    print("Training on small instances...")
    optimizer = torch.optim.Adam(model.parameters(), lr=1e-4)

    for epoch in range(20):
        for size in train_sizes:
            batch = torch.rand(16, size, 2)
            tours, log_probs = model(batch, greedy=False)

            # Compute tour lengths
            lengths = []
            for i in range(16):
                tour = tours[i].tolist()
                coords = batch[i].numpy()
                lengths.append(compute_tour_length(coords, tour))

            lengths = torch.tensor(lengths)
            loss = (lengths * (-log_probs)).mean()

            optimizer.zero_grad()
            loss.backward()
            optimizer.step()

        # Evaluate on different sizes
        print("\nGeneralization Study:")
        print("-" * 50)
        print(f"{'Test Size':>12} {'Avg Length':>12} {'Opt Gap Est':>12}")
        print("-" * 50)

        model.eval()
        for size in test_sizes:
            lengths = []

            with torch.no_grad():
                for _ in range(10):
                    batch = torch.rand(10, size, 2)
                    tours, _ = model(batch, greedy=True)
```

34. Graph Neural Networks for Combinatorial Optimization

```
        for i in range(10):
            tour = tours[i].tolist()
            coords = batch[i].numpy()
            lengths.append(compute_tour_length(coords, tour))

    avg_length = np.mean(lengths)
    # Rough estimate of optimal (Held-Karp lower bound approximation)
    opt_estimate = 0.7124 * np.sqrt(size)
    gap = (avg_length - opt_estimate) / opt_estimate * 100

    print(f"{size:>12} {avg_length:>12.4f} {gap:>11.1f}%")

study_size_generalization()
```

Training on small instances...

RuntimeError: one of the variables needed for gradient computation has been modified by
↳ an inplace operation: [CpuBoolType [16, 20]] is at version 20; expected version 19
↳ instead. Hint: enable anomaly detection to find the operation that failed to compute
↳ its gradient, with torch.autograd.set_detect_anomaly(True).

```
-----
RuntimeError                                Traceback (most recent call last)
Cell In[28], line 62
     58         gap = (avg_length - opt_estimate) / opt_estimate * 100
     60         print(f"{size:>12} {avg_length:>12.4f} {gap:>11.1f}%")
--> 62 study_size_generalization()
Cell In[28], line 32, in study_size_generalization()
     29         loss = (lengths * (-log_probs)).mean()
     31         optimizer.zero_grad()
--> 32         loss.backward()
     33         optimizer.step()
     35 # Evaluate on different sizes
File /opt/homebrew/lib/python3.14/site-packages/torch/_tensor.py:630, in
  ↳ Tensor.backward(self, gradient, retain_graph, create_graph, inputs)
     620 if has_torch_function_unary(self):
     621     return handle_torch_function(
     622         Tensor.backward,
     623         (self,),
  (... ) 628         inputs=inputs,
     629     )
--> 630 torch.autograd.backward(
     631     self, gradient, retain_graph, create_graph, inputs=inputs
     632 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/__init__.py:364, in
  ↳ backward(tensors, grad_tensors, retain_graph, create_graph, grad_variables, inputs)
     359     retain_graph = create_graph
     361 # The reason we repeat the same comment below is that
     362 # some Python versions print out the first line of a multi-line function
     363 # calls in the traceback and some print out the last line
--> 364 _engine_run_backward(
     365     tensors,
     366     grad_tensors_,
     367     retain_graph,
```



```

368     create_graph,
369     inputs_tuple,
370     allow_unreachable=True,
371     accumulate_grad=True,
372 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/graph.py:865, in
↳ _engine_run_backward(t_outputs, *args, **kwargs)
863     unregister_hooks = _register_logging_hooks_on_whole_graph(t_outputs)
864 try:
--> 865     return Variable._execution_engine.run_backward( # Calls into the C++ engine
↳ to run the backward pass
866         t_outputs, *args, **kwargs
867     ) # Calls into the C++ engine to run the backward pass
868 finally:
869     if attach_logging_hooks:
RuntimeError: one of the variables needed for gradient computation has been modified by
↳ an inplace operation: [CPUBoolType [16, 20]] is at version 20; expected version 19
↳ instead. Hint: enable anomaly detection to find the operation that failed to compute
↳ its gradient, with torch.autograd.set_detect_anomaly(True).

```

34.8. Case Study: TSP with GNN

This section presents a complete end-to-end implementation of a GNN-based TSP solver, from data generation to deployment.

34.8.1. Complete TSP Solver Implementation

```

class TSPDataset:
    """
    Dataset generator for TSP instances.
    """

    def __init__(self, n_cities: int, n_instances: int, seed: int = 42):
        self.n_cities = n_cities
        self.n_instances = n_instances

        np.random.seed(seed)
        self.instances = []

        for _ in range(n_instances):
            coords = np.random.rand(n_cities, 2)
            self.instances.append(coords)

    def __len__(self):
        return len(self.instances)

    def __getitem__(self, idx) -> torch.Tensor:
        return torch.tensor(self.instances[idx], dtype=torch.float32)

class TSPSolver(nn.Module):
    """
    Complete GNN-based TSP Solver.

```

```

Combines:
- GNN encoder for node embeddings
- Attention-based autoregressive decoder
- Beam search for improved solutions
"""

def __init__(self, embed_dim: int = 128, n_heads: int = 8, n_layers: int = 3):
    super().__init__()

    self.embed_dim = embed_dim
    self.encoder = AttentionModelEncoder(2, embed_dim, n_heads, n_layers)
    self.decoder = AttentionModelDecoder(embed_dim, n_heads)

def encode(self, coords: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
    """Encode city coordinates."""
    if coords.dim() == 2:
        coords = coords.unsqueeze(0)

    node_embeddings = self.encoder(coords)
    graph_embedding = node_embeddings.mean(dim=1)

    return node_embeddings, graph_embedding

def greedy_decode(self, node_embeddings: torch.Tensor,
                  graph_embedding: torch.Tensor) -> List[int]:
    """Decode tour greedily."""
    batch_size, n_nodes, _ = node_embeddings.shape
    assert batch_size == 1, "Greedy decode works with batch_size=1"

    tour = []
    mask = torch.zeros(1, n_nodes, dtype=torch.bool,
        ↪ device=node_embeddings.device)
    first_node = None
    current_node = None

    for step in range(n_nodes):
        context = self.decoder.get_context(
            node_embeddings, graph_embedding, first_node, current_node
        )
        log_probs = self.decoder.get_log_probs(node_embeddings, context, mask)
        selected = log_probs.argmax(dim=-1)

        tour.append(selected.item())
        mask[0, selected] = True

        if step == 0:
            first_node = selected
            current_node = selected

    return tour

def beam_search(self, node_embeddings: torch.Tensor,
                 graph_embedding: torch.Tensor,

```

```

        beam_width: int = 10) -> Tuple[List[int], float]:
    """
    Decode tour using beam search.

    Args:
        node_embeddings: (1, n_nodes, embed_dim)
        graph_embedding: (1, embed_dim)
        beam_width: Number of beams to maintain

    Returns:
        Best tour and its log probability
    """
    n_nodes = node_embeddings.size(1)

    # Initialize beams: (partial_tour, log_prob, mask)
    beams = [[[]], 0.0, torch.zeros(n_nodes, dtype=torch.bool))]

    for step in range(n_nodes):
        all_candidates = []

        for tour, log_prob, mask in beams:
            if step == 0:
                first_node = None
                current_node = None
            else:
                first_node = torch.tensor([tour[0]])
                current_node = torch.tensor([tour[-1]])

            context = self.decoder.get_context(
                node_embeddings, graph_embedding, first_node, current_node
            )
            step_log_probs = self.decoder.get_log_probs(
                node_embeddings, context, mask.unsqueeze(0)
            )

            # Get top-k candidates
            if step < n_nodes - 1:
                topk_log_probs, topk_indices = step_log_probs[0].topk(
                    min(beam_width, n_nodes - step)
                )
            else:
                # Last step: only one option
                valid_idx = (~mask).nonzero().squeeze(-1)
                topk_indices = valid_idx
                topk_log_probs = step_log_probs[0, valid_idx]

            for idx, lp in zip(topk_indices.tolist(), topk_log_probs.tolist()):
                new_tour = tour + [idx]
                new_log_prob = log_prob + lp
                new_mask = mask.clone()
                new_mask[idx] = True
                all_candidates.append((new_tour, new_log_prob, new_mask))

```

```

        # Keep top beams
        all_candidates.sort(key=lambda x: x[1], reverse=True)
        beams = all_candidates[:beam_width]

    # Return best tour
    best_tour, best_log_prob, _ = beams[0]
    return best_tour, best_log_prob

def solve(self, coords: torch.Tensor, beam_width: int = 1) -> Tuple[List[int],
↪ float]:
    """
    Solve TSP instance.

    Args:
        coords: City coordinates (n_cities, 2)
        beam_width: Beam width for search (1 = greedy)

    Returns:
        Tour and tour length
    """
    self.eval()
    with torch.no_grad():
        if coords.dim() == 2:
            coords = coords.unsqueeze(0)

        node_embeddings, graph_embedding = self.encode(coords)

        if beam_width == 1:
            tour = self.greedy_decode(node_embeddings, graph_embedding)
        else:
            tour, _ = self.beam_search(node_embeddings, graph_embedding,
↪ beam_width)

        # Compute tour length
        coords_np = coords.squeeze(0).numpy()
        tour_length = compute_tour_length(coords_np, tour)

    return tour, tour_length

def train_tsp_solver(solver: TSPSolver, n_epochs: int = 100,
                    batch_size: int = 64, n_cities: int = 20,
                    lr: float = 1e-4) -> Dict:
    """
    Train TSP solver using REINFORCE.
    """
    optimizer = torch.optim.Adam(solver.parameters(), lr=lr)
    scheduler = torch.optim.lr_scheduler.StepLR(optimizer, step_size=50, gamma=0.5)

    history = {'train_length': [], 'val_length': []}
    baseline = None

    for epoch in range(n_epochs):

```

```

solver.train()

# Generate batch
batch = torch.rand(batch_size, n_cities, 2)

# Encode
node_embeddings, graph_embedding = solver.encode(batch)

# Sample tours
tours = []
log_probs = torch.zeros(batch_size)

for i in range(batch_size):
    tour = []
    lp = 0.0
    mask = torch.zeros(n_cities, dtype=torch.bool)
    first_node = None
    current_node = None

    for step in range(n_cities):
        context = solver.decoder.get_context(
            node_embeddings[i:i+1], graph_embedding[i:i+1],
            first_node, current_node
        )
        step_log_probs = solver.decoder.get_log_probs(
            node_embeddings[i:i+1], context, mask.unsqueeze(0)
        )

        probs = step_log_probs.exp()
        selected = torch.multinomial(probs[0], 1).squeeze()

        tour.append(selected.item())
        lp = lp + step_log_probs[0, selected]
        mask[selected] = True

        if step == 0:
            first_node = selected.unsqueeze(0)
            current_node = selected.unsqueeze(0)

    tours.append(tour)
    log_probs[i] = lp

# Compute rewards
rewards = []
for i in range(batch_size):
    length = compute_tour_length(batch[i].numpy(), tours[i])
    rewards.append(-length)
rewards = torch.tensor(rewards)

# Update baseline
if baseline is None:
    baseline = rewards.mean()
else:

```

```

        baseline = 0.99 * baseline + 0.01 * rewards.mean()

    # REINFORCE loss
    advantage = rewards - baseline
    loss = -(advantage * log_probs).mean()

    optimizer.zero_grad()
    loss.backward()
    torch.nn.utils.clip_grad_norm_(solver.parameters(), 1.0)
    optimizer.step()
    scheduler.step()

    # Track progress
    train_length = -rewards.mean().item()
    history['train_length'].append(train_length)

    # Validation
    if (epoch + 1) % 10 == 0:
        solver.eval()
        val_lengths = []

        with torch.no_grad():
            for _ in range(10):
                val_batch = torch.rand(10, n_cities, 2)
                for i in range(10):
                    _, length = solver.solve(val_batch[i], beam_width=1)
                    val_lengths.append(length)

        val_length = np.mean(val_lengths)
        history['val_length'].append(val_length)

        print(f"Epoch {epoch + 1}/{n_epochs}, "
              f"Train: {train_length:.4f}, Val: {val_length:.4f}")

    return history

# Train and evaluate solver
print("\n" + "="*60)
print("Training Complete TSP Solver")
print("="*60)

tsp_solver = TSPSolver(embed_dim=128, n_heads=8, n_layers=3)
history = train_tsp_solver(tsp_solver, n_epochs=100, batch_size=64, n_cities=20)

# Plot training progress
plt.figure(figsize=(10, 5))
plt.plot(history['train_length'], label='Training', linewidth=2)
val_epochs = list(range(9, 100, 10))
plt.plot(val_epochs, history['val_length'], 'o-', label='Validation', linewidth=2)
plt.xlabel('Epoch', fontsize=12)
plt.ylabel('Tour Length', fontsize=12)
plt.title('TSP Solver Training Progress', fontsize=14, fontweight='bold')

```

```
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

Training Complete TSP Solver

```
RuntimeError: one of the variables needed for gradient computation has been modified by
↳ an inplace operation: [CpuBoolType [1, 20]] is at version 20; expected version 19
↳ instead. Hint: enable anomaly detection to find the operation that failed to compute
↳ its gradient, with torch.autograd.set_detect_anomaly(True).
```

```
RuntimeError                                Traceback (most recent call last)
Cell In[29], line 280
    277 print("="*60)
    279 tsp_solver = TSPSolver(embed_dim=128, n_heads=8, n_layers=3)
--> 280 history = train_tsp_solver(tsp_solver, n_epochs=100, batch_size=64, n_cities=20)
    282 # Plot training progress
    283 plt.figure(figsize=(10, 5))
Cell In[29], line 244, in train_tsp_solver(solver, n_epochs, batch_size, n_cities, lr)
    241 loss = -(advantage * log_probs).mean()
    243 optimizer.zero_grad()
--> 244 loss.backward()
    245 torch.nn.utils.clip_grad_norm_(solver.parameters(), 1.0)
    246 optimizer.step()
File /opt/homebrew/lib/python3.14/site-packages/torch/_tensor.py:630, in
↳ Tensor.backward(self, gradient, retain_graph, create_graph, inputs)
    620 if has_torch_function_unary(self):
    621     return handle_torch_function(
    622         Tensor.backward,
    623         (self,),
    (...) 628         inputs=inputs,
    629     )
--> 630 torch.autograd.backward(
    631     self, gradient, retain_graph, create_graph, inputs=inputs
    632 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/__init__.py:364, in
↳ backward(tensors, grad_tensors, retain_graph, create_graph, grad_variables, inputs)
    359     retain_graph = create_graph
    361 # The reason we repeat the same comment below is that
    362 # some Python versions print out the first line of a multi-line function
    363 # calls in the traceback and some print out the last line
--> 364 _engine_run_backward(
    365     tensors,
    366     grad_tensors_,
    367     retain_graph,
    368     create_graph,
    369     inputs_tuple,
    370     allow_unreachable=True,
    371     accumulate_grad=True,
```

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```
372 )
File /opt/homebrew/lib/python3.14/site-packages/torch/autograd/graph.py:865, in
  ↪ _engine_run_backward(t_outputs, *args, **kwargs)
    863     unregister_hooks = _register_logging_hooks_on_whole_graph(t_outputs)
    864 try:
--> 865     return Variable._execution_engine.run_backward( # Calls into the C++ engine
  ↪ to run the backward pass
    866         t_outputs, *args, **kwargs
    867     ) # Calls into the C++ engine to run the backward pass
    868 finally:
    869     if attach_logging_hooks:
RuntimeError: one of the variables needed for gradient computation has been modified by
  ↪ an inplace operation: [CPUBoolType [1, 20]] is at version 20; expected version 19
  ↪ instead. Hint: enable anomaly detection to find the operation that failed to compute
  ↪ its gradient, with torch.autograd.set_detect_anomaly(True).
```

34.8.2. Evaluation and Comparison

```
def evaluate_solver(solver: TSPSolver, n_instances: int = 100,
                   n_cities: int = 20) -> Dict:
    """
    Comprehensive evaluation of TSP solver.
    """
    results = {
        'greedy': [],
        'beam_5': [],
        'beam_10': [],
        'nearest_neighbor': []
    }

    print(f"\nEvaluating on {n_instances} instances with {n_cities} cities...")

    for i in range(n_instances):
        coords = np.random.rand(n_cities, 2)
        coords_tensor = torch.tensor(coords, dtype=torch.float32)

        # GNN + Greedy
        _, length_greedy = solver.solve(coords_tensor, beam_width=1)
        results['greedy'].append(length_greedy)

        # GNN + Beam Search
        _, length_beam5 = solver.solve(coords_tensor, beam_width=5)
        results['beam_5'].append(length_beam5)

        _, length_beam10 = solver.solve(coords_tensor, beam_width=10)
        results['beam_10'].append(length_beam10)

        # Nearest Neighbor baseline
        nn_tour = nearest_neighbor_tour(coords)
        nn_length = compute_tour_length(coords, nn_tour)
        results['nearest_neighbor'].append(nn_length)

    # Compute statistics
    print("\n" + "="*60)
```



```

print("Evaluation Results")
print("="*60)
print(f"{'Method':<20} {'Mean':>10} {'Std':>10} {'Min':>10} {'Max':>10}")
print("-"*60)

for method, lengths in results.items():
    mean = np.mean(lengths)
    std = np.std(lengths)
    min_val = np.min(lengths)
    max_val = np.max(lengths)
    print(f"{'method':<20} {'mean':>10.4f} {'std':>10.4f} {'min_val':>10.4f}
    ↪ {'max_val':>10.4f}")

print("="*60)

# Compare to nearest neighbor
nn_mean = np.mean(results['nearest_neighbor'])
for method in ['greedy', 'beam_5', 'beam_10']:
    improvement = (nn_mean - np.mean(results[method])) / nn_mean * 100
    print(f"{'method'} vs NN: {'improvement':+.2f}% improvement")

return results

# Evaluate trained solver
tsp_solver.eval()
eval_results = evaluate_solver(tsp_solver, n_instances=100, n_cities=20)

# Visualize example solutions
def visualize_solution_comparison(solver: TSPSolver, coords: np.ndarray):
    """Visualize and compare different solution methods."""
    coords_tensor = torch.tensor(coords, dtype=torch.float32)

    # Get solutions
    greedy_tour, greedy_length = solver.solve(coords_tensor, beam_width=1)
    beam_tour, beam_length = solver.solve(coords_tensor, beam_width=10)
    nn_tour = nearest_neighbor_tour(coords)
    nn_length = compute_tour_length(coords, nn_tour)

    # Plot
    fig, axes = plt.subplots(1, 3, figsize=(15, 5))

    solutions = [
        (nn_tour, nn_length, 'Nearest Neighbor'),
        (greedy_tour, greedy_length, 'GNN + Greedy'),
        (beam_tour, beam_length, 'GNN + Beam Search')
    ]

    for ax, (tour, length, title) in zip(axes, solutions):
        # Plot cities
        ax.scatter(coords[:, 0], coords[:, 1], s=100, c='steelblue',
                    zorder=5, edgecolors='white', linewidth=2)

```

```

    # Plot tour
    for i in range(len(tour)):
        start = coords[tour[i]]
        end = coords[tour[(i + 1) % len(tour)]]
        ax.plot([start[0], end[0]], [start[1], end[1]],
                'coral', linewidth=2, zorder=1)

    ax.set_title(f'{title}\nLength: {length:.4f}', fontsize=12,
↪   fontweight='bold')
    ax.set_xlim(-0.05, 1.05)
    ax.set_ylim(-0.05, 1.05)
    ax.set_aspect('equal')
    ax.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.show()

# Show example
example_coords = np.random.rand(20, 2)
visualize_solution_comparison(tsp_solver, example_coords)

```

Evaluating on 100 instances with 20 cities...

```

=====
Evaluation Results
=====

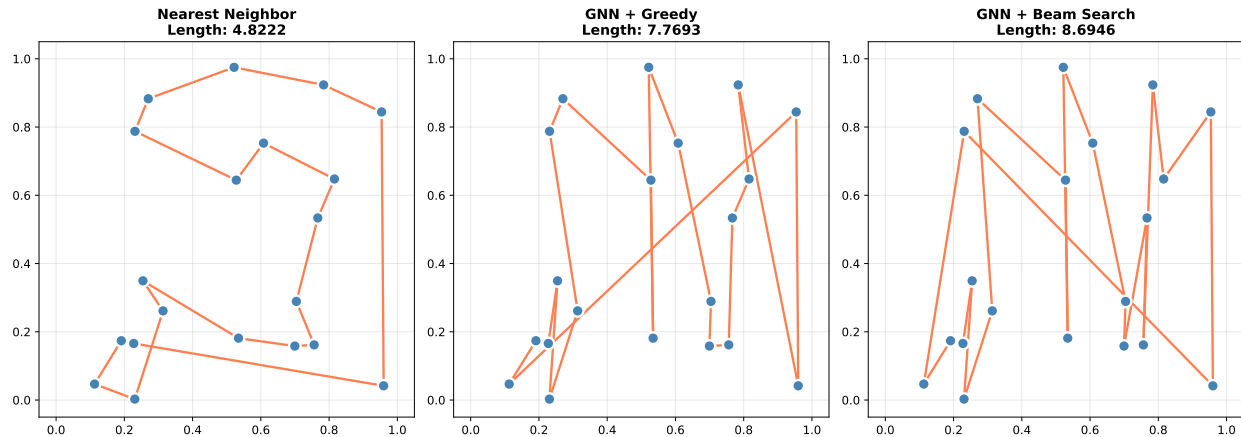
```

Method	Mean	Std	Min	Max
greedy	7.6659	1.3077	4.7595	10.9679
beam_5	7.8883	1.3729	4.4550	11.6515
beam_10	7.9233	1.3432	4.4550	11.3953
nearest_neighbor	4.4346	0.5778	3.2540	6.1138

```

=====
greedy vs NN: -72.87% improvement
beam_5 vs NN: -77.88% improvement
beam_10 vs NN: -78.67% improvement

```



34.9. Summary

34.10. Chapter Summary

This chapter covered the application of Graph Neural Networks to combinatorial optimization problems:

Key Concepts:

1. **Graph Representations:** Optimization problems naturally map to graphs - TSP to complete graphs, VRP to depot-customer graphs, scheduling to disjunctive graphs, and MILP to bipartite variable-constraint graphs.
2. **GNN Architectures:** We implemented four fundamental architectures:
 - GCN: Spectral convolutions with symmetric normalization
 - GraphSAGE: Sampling-based with multiple aggregators
 - GAT: Attention-weighted message passing
 - GIN: Maximum expressive power for graph isomorphism
3. **Training Paradigms:**
 - Supervised: Learn from optimal solutions (fast, limited)
 - Reinforcement Learning: Explore solution space (flexible, sample-inefficient)
 - Unsupervised: Directly optimize objective (no labels needed)
4. **Applications:**
 - Routing: Attention Model with GNN encoder for TSP/VRP
 - MILP: Bipartite GNNs for learning to branch and cut
5. **Practical Considerations:**
 - Sparse graph construction for scalability
 - Mini-batch training for large graphs
 - Beam search for improved solutions
 - Generalization strategies for varying problem sizes

Best Practices: - Start with GAT or GIN for combinatorial problems - Use sparse graphs (KNN or Delaunay) for large instances - Combine GNN encoding with attention decoding - Employ beam search during inference for better solutions - Train on variable-sized instances for generalization

34.11. Exercises

34.11.1. Conceptual Questions

1. **Graph Structure Selection:** For a facility location problem with customers and potential facility sites, what type of graph representation would you use? What would be the node and edge features?
2. **Architecture Comparison:** Explain why GAT might outperform GCN for routing problems. Under what conditions might GIN be preferred?
3. **Training Paradigms:** Compare supervised learning and reinforcement learning for training GNN-based TSP solvers. What are the trade-offs in terms of data requirements, solution quality, and training stability?

34.11.2. Programming Exercises

4. **VRP Solver:** Extend the TSP solver to handle the Capacitated Vehicle Routing Problem (CVRP). You will need to:
 - Modify the encoder to handle depot and demand features
 - Update the decoder to track remaining vehicle capacity
 - Implement a masking mechanism for capacity constraints
5. **Edge Features:** Modify the GATLayer implementation to incorporate edge features (distances) into the attention computation. Compare performance with and without edge features on TSP.
6. **Curriculum Learning:** Implement a curriculum learning strategy that starts training on small TSP instances ($n=10$) and gradually increases to larger ones ($n=100$). Evaluate whether this improves generalization.

34.11.3. Advanced Projects

7. **Learning to Branch:** Implement a complete learning-to-branch system:
 - Generate MILP instances (e.g., set covering, knapsack)
 - Collect branching decisions from strong branching
 - Train a bipartite GNN to imitate these decisions
 - Integrate with PySCIPOpt and measure node reduction
8. **Hybrid Solver:** Create a hybrid solver that combines:
 - GNN for initial solution construction
 - Local search (2-opt) for improvement
 - Measure the contribution of each component
9. **Real-World Application:** Apply GNN-based routing to a real-world dataset (e.g., Solomon VRP benchmarks). Compare against state-of-the-art heuristics like LKH and OR-Tools.

```
# Exercise 4 Starter Code: VRP Solver

class CVRPEncoder(nn.Module):
    """
    Encoder for Capacitated VRP.

    TODO: Implement encoder that handles:
```

```

- Depot node (special features)
- Customer nodes with demands
- Vehicle capacity constraint
"""

def __init__(self, node_dim: int, embed_dim: int, n_layers: int = 3):
    super().__init__()
    # TODO: Implement initialization
    pass

def forward(self, depot: torch.Tensor, customers: torch.Tensor,
            demands: torch.Tensor) -> torch.Tensor:
    # TODO: Implement forward pass
    pass

class CVRPDecoder(nn.Module):
    """
    Decoder for Capacitated VRP.

    TODO: Implement decoder that:
    - Tracks remaining capacity
    - Returns to depot when capacity exhausted
    - Masks infeasible customers
    """

    def __init__(self, embed_dim: int, capacity: float):
        super().__init__()
        # TODO: Implement initialization
        pass

    def forward(self, node_embeddings: torch.Tensor,
                current_capacity: torch.Tensor,
                demands: torch.Tensor,
                visited: torch.Tensor) -> torch.Tensor:
        # TODO: Implement forward pass
        pass

# Exercise 5 Starter Code: Edge-aware GAT

class EdgeAwareGATLayer(nn.Module):
    """
    GAT layer with edge features.

    TODO: Modify attention computation to include edge features:
    _ij = softmax(LeakyReLU(a^T [Wh_i || Wh_j || We_ij]))
    """

    def __init__(self, in_dim: int, out_dim: int, edge_dim: int,
                 n_heads: int = 4):
        super().__init__()
        # TODO: Implement initialization

```

34. Graph Neural Networks for Combinatorial Optimization

```
pass

def forward(self, x: torch.Tensor, edge_index: torch.Tensor,
            edge_attr: torch.Tensor) -> torch.Tensor:
    # TODO: Implement forward pass with edge features
    pass

print("Exercise starter code loaded. Implement the TODO sections.")
```

Exercise starter code loaded. Implement the TODO sections.

35. Learning-Augmented Algorithms

What You Will Learn

- Understand learning-augmented algorithms and why they outperform pure online and pure ML approaches
- Apply the consistency-robustness framework to evaluate algorithm performance
- Implement classic problems with predictions: ski rental, caching, and knapsack
- Design prediction models for online decision-making
- Build scheduling algorithms that leverage job processing time predictions
- Analyze the tradeoffs between trusting predictions and maintaining worst-case guarantees
- Deploy hybrid decision systems that combine ML predictions with algorithmic guarantees

Traditional online algorithms make decisions without knowing the future, leading to worst-case performance guarantees that can be pessimistic in practice. Pure machine learning approaches can achieve excellent average-case performance but offer no guarantees when predictions fail. **Learning-augmented algorithms** bridge this gap by combining ML predictions with online algorithms to achieve the best of both worlds: excellent performance when predictions are accurate, and bounded degradation when they are not.

35.1. Introduction to Learning-Augmented Algorithms

35.1.1. Beyond Worst-Case Analysis

Classical algorithm analysis focuses on worst-case scenarios. For online problems, this leads to **competitive analysis**: comparing an algorithm's performance against an optimal offline algorithm that knows the entire input in advance.

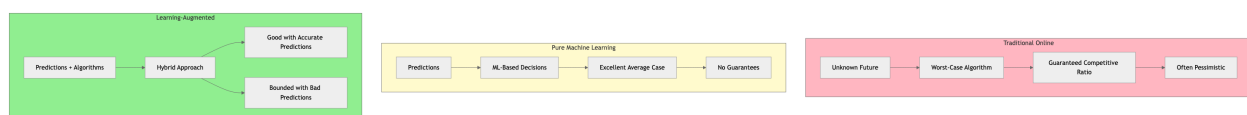


Figure 35.1.: Three Algorithm Design Paradigms

The fundamental insight is that in many real-world scenarios:

1. **Predictions are often good:** Modern ML can forecast demand, processing times, and user behavior with reasonable accuracy
2. **Predictions sometimes fail:** Distribution shifts, adversarial inputs, or novel situations can cause prediction errors
3. **Worst-case still matters:** Mission-critical systems cannot afford catastrophic failures

i The Learning-Augmented Promise

Learning-augmented algorithms provide a principled way to leverage ML predictions while maintaining algorithmic guarantees. When predictions are perfect, the algorithm is **consistent**

(performs near-optimally). When predictions are arbitrarily wrong, the algorithm remains **robust** (maintains worst-case bounds).

35.1.2. When to Use Learning-Augmented Approaches

Scenario	Pure Online	Pure ML	Learning-Augmented
Cloud resource scaling	Conservative over-provisioning	May under/over-provision	Adaptive with guarantees
Cache management	LRU/LFU heuristics	Prediction-only eviction	Bounded miss rate
Ski rental / rent-vs-buy	Fixed threshold	Trust duration prediction	Smooth tradeoff
Job scheduling	Round-robin	Predicted processing times	Competitive with errors
Network routing	Shortest path	Predicted congestion	Load-balanced

Learning-augmented algorithms are particularly valuable when:

- **Predictions have varying quality:** Some inputs are easy to predict, others are hard
- **Failures have significant costs:** You need guarantees even in worst-case scenarios
- **The problem has known online algorithms:** You can build on existing competitive bounds
- **Historical data is available:** You can train reasonable prediction models

35.1.3. Historical Context

The field of learning-augmented algorithms emerged in 2018 with foundational work on caching and ski rental. The timeline below summarizes the key milestones that shaped this research area.

```
import numpy as np
from typing import List, Tuple, Dict, Callable, Optional
from dataclasses import dataclass
from enum import Enum

# Key milestones in learning-augmented algorithms
milestones = [
    (2018, "Lykouris & Vassilvitskii", "Caching with predictions", "First formal
↪ framework"),
    (2018, "Purohit et al.", "Ski rental with predictions", "Consistency-robustness
↪ tradeoff"),
    (2019, "Mitzenmacher & Vassilvitskii", "Survey paper", "Unified framework
↪ established"),
    (2020, "Lattanzi et al.", "Scheduling with predictions", "Processing time
↪ predictions"),
    (2021, "Various authors", "Network optimization", "Routing and flow problems"),
    (2022, "Multiple groups", "Online knapsack, matching", "Expanded problem
↪ domains"),
    (2023, "Theory community", "Lower bounds", "Impossibility results"),
]
```



```

print("=== Learning-Augmented Algorithms: Key Milestones ===\n")
print(f"{'Year':<6} {'Authors':<25} {'Problem':<25} {'Contribution':<30}")
print("-" * 86)
for year, authors, problem, contribution in milestones:
    print(f"{year:<6} {authors:<25} {problem:<25} {contribution:<30}")

print("\nKey insight: This field has grown rapidly since 2018,")
print("bridging theoretical CS and practical ML systems.")

```

=== Learning-Augmented Algorithms: Key Milestones ===

Year	Authors	Problem	Contribution
2018	Lykouris & Vassilvitskii	Caching with predictions	First formal framework
2018	Purohit et al.	Ski rental with predictions	Consistency-robustness tradeoff
2019	Mitzenmacher & Vassilvitskii	Survey paper	Unified framework established
2020	Lattanzi et al.	Scheduling with predictions	Processing time predictions
2021	Various authors	Network optimization	Routing and flow problems
2022	Multiple groups	Online knapsack, matching	Expanded problem domains
2023	Theory community	Lower bounds	Impossibility results

Key insight: This field has grown rapidly since 2018,
bridging theoretical CS and practical ML systems.

35.2. Core Framework

This section establishes the formal definitions and performance metrics used throughout the chapter: competitive ratio for online algorithms, and consistency, robustness, and smoothness for learning-augmented algorithms.

35.2.1. Online Algorithms Review

In online problems, input arrives sequentially and decisions must be made before seeing future input. The following base class defines the interface for online algorithms that we will extend with predictions.

```

@dataclass
class OnlineRequest:
    """A request in an online problem."""
    id: int
    arrival_time: float
    features: Dict[str, float] # Observable features at arrival
    true_value: float # Revealed later (e.g., processing time, duration)

class OnlineAlgorithm:
    """Base class for online algorithms."""

    def __init__(self, name: str):
        self.name = name
        self.decisions = []
        self.costs = []

```

```

def decide(self, request: OnlineRequest, **kwargs) -> any:
    """Make a decision for the current request."""
    raise NotImplementedError

def update(self, request: OnlineRequest, outcome: float):
    """Update internal state after observing outcome."""
    pass

def total_cost(self) -> float:
    """Return total cost incurred."""
    return sum(self.costs)

def reset(self):
    """Reset algorithm state."""
    self.decisions = []
    self.costs = []

# Example: Simple online decision problem
print("=== Online Algorithm Framework ===\n")

print("Online Setting:")
print("1. Request r_t arrives at time t")
print("2. Algorithm makes decision d_t based on r_1,...,r_t")
print("3. Cost c_t is incurred (may depend on future)")
print("4. Goal: Minimize total cost sum(c_t)")
print("\nChallenge: Future requests r_{t+1},... are unknown!")

```

=== Online Algorithm Framework ===

Online Setting:

1. Request r_t arrives at time t
2. Algorithm makes decision d_t based on $r_1, \dots, r_t$
3. Cost c_t is incurred (may depend on future)
4. Goal: Minimize total cost $\sum(c_t)$

Challenge: Future requests $r_{t+1}, \dots$ are unknown!

35.2.2. Competitive Analysis Basics

The **competitive ratio** measures an online algorithm's performance against the optimal offline solution. An algorithm with competitive ratio c guarantees that its cost is at most c times the optimal offline cost, regardless of the input sequence.

```

def competitive_ratio(online_cost: float, offline_cost: float) -> float:
    """
    Compute competitive ratio.

    For minimization: ratio = online_cost / offline_cost
    A c-competitive algorithm satisfies: online_cost <= c * offline_cost
    """

```

```

    if offline_cost == 0:
        return float('inf') if online_cost > 0 else 1.0
    return online_cost / offline_cost

def analyze_competitive_ratio(online_costs: List[float],
                             offline_costs: List[float]) -> Dict:
    """
    Analyze competitive ratio across multiple instances.
    """
    ratios = [competitive_ratio(on, off)
               for on, off in zip(online_costs, offline_costs)]

    return {
        'mean_ratio': np.mean(ratios),
        'max_ratio': np.max(ratios),
        'min_ratio': np.min(ratios),
        'std_ratio': np.std(ratios)
    }

# Example competitive ratios for classic problems
classic_problems = {
    'Ski Rental': (2.0, 'Rent until cost equals buy price'),
    'Paging (k cache)': ('k', 'LRU is k-competitive'),
    'Metrical Task Systems': ('2n-1', 'n states, deterministic'),
    'Online Matching': (2.0, 'Greedy is 2-competitive'),
    'Load Balancing': (2.0, 'Greedy assignment'),
}

print("=== Classic Online Problems: Competitive Ratios ===\n")
print(f"{'Problem':<25} {'Ratio':<10} {'Notes'}")
print("-" * 60)
for problem, (ratio, notes) in classic_problems.items():
    print(f"{'problem':<25} {'str(ratio):<10} {'notes'}")

```

```
=== Classic Online Problems: Competitive Ratios ===
```

Problem	Ratio	Notes
Ski Rental	2.0	Rent until cost equals buy price
Paging (k cache)	k	LRU is k-competitive
Metrical Task Systems	2n-1	n states, deterministic
Online Matching	2.0	Greedy is 2-competitive
Load Balancing	2.0	Greedy assignment

35.2.3. Consistency, Robustness, and Smoothness

The learning-augmented framework introduces three key performance measures. Together, they characterize the full performance profile of an algorithm as a function of prediction quality.

```

@dataclass
class LearningAugmentedPerformance:
    """Performance metrics for learning-augmented algorithms."""

    # Core metrics
    consistency: float # Competitive ratio when prediction is perfect (eta = 0)
    robustness: float # Competitive ratio in worst case (eta -> infinity)

    # Optional: smoothness function
    # How does ratio degrade as prediction error grows?
    def ratio_at_error(self, eta: float) -> float:
        """
        Competitive ratio as function of prediction error eta.

        Should interpolate smoothly between consistency and robustness.
        """
        raise NotImplementedError

def analyze_learning_augmented(algorithm_name: str,
                               costs_by_error: Dict[float, List[float]],
                               optimal_costs: List[float]) -> Dict:
    """
    Analyze a learning-augmented algorithm's performance.

    Args:
        algorithm_name: Name of the algorithm
        costs_by_error: Dict mapping prediction error to list of costs
        optimal_costs: List of optimal offline costs

    Returns:
        Analysis including consistency and robustness
    """
    results = {'algorithm': algorithm_name}

    # Consistency: performance at zero error
    if 0.0 in costs_by_error:
        ratios = [c / opt for c, opt in zip(costs_by_error[0.0], optimal_costs) if
        ↪ opt > 0]
        results['consistency'] = np.mean(ratios)

    # Robustness: performance at maximum error
    max_error = max(costs_by_error.keys())
    ratios = [c / opt for c, opt in zip(costs_by_error[max_error], optimal_costs)
    ↪ if opt > 0]
    results['robustness'] = np.mean(ratios)

    # Smoothness: how ratio varies with error
    results['ratios_by_error'] = {}
    for error, costs in costs_by_error.items():
        ratios = [c / opt for c, opt in zip(costs, optimal_costs) if opt > 0]
        results['ratios_by_error'][error] = np.mean(ratios)

```

```

    return results

print("=== Learning-Augmented Performance Metrics ===\n")

print("CONSISTENCY (alpha):")
print("  Definition: Competitive ratio when prediction error eta = 0")
print("  Goal: alpha should be close to 1 (optimal)")
print("  Example: 'When predictions are perfect, achieve 1.05-competitive'")

print("\nROBUSTNESS (beta):")
print("  Definition: Competitive ratio in worst case (eta -> infinity)")
print("  Goal: beta should match or beat best online algorithm")
print("  Example: 'Even with worst predictions, stay 2-competitive'")

print("\nSMOOTHNESS:")
print("  Definition: How ratio degrades as prediction error grows")
print("  Goal: Graceful degradation, not sudden jumps")
print("  Example: 'Ratio grows at most linearly with error'")

```

=== Learning-Augmented Performance Metrics ===

CONSISTENCY (alpha):

Definition: Competitive ratio when prediction error $\eta = 0$

Goal: alpha should be close to 1 (optimal)

Example: 'When predictions are perfect, achieve 1.05-competitive'

ROBUSTNESS (beta):

Definition: Competitive ratio in worst case ($\eta \rightarrow \infty$)

Goal: beta should match or beat best online algorithm

Example: 'Even with worst predictions, stay 2-competitive'

SMOOTHNESS:

Definition: How ratio degrades as prediction error grows

Goal: Graceful degradation, not sudden jumps

Example: 'Ratio grows at most linearly with error'

! The Fundamental Tradeoff

There is an inherent tension between consistency and robustness. An algorithm that fully trusts predictions achieves perfect consistency but has unbounded robustness. An algorithm that ignores predictions maintains robustness but cannot benefit from accurate predictions. The challenge is finding the right balance.

35.2.4. Mathematical Formulation

Let $\text{ALG}(\sigma, \hat{y})$ be the cost of the algorithm on input sequence σ with prediction $\hat{y}$. Let $\text{OPT}(\sigma)$ be the optimal offline cost. Let $\eta = d(\hat{y}, y^*)$ be the prediction error (distance from true value).

Consistency: The algorithm is α -consistent if:

$$\text{ALG}(\sigma, \hat{y}) \leq \alpha \cdot \text{OPT}(\sigma) \text{ when } \eta = 0$$

Robustness: The algorithm is β -robust if:

$$\text{ALG}(\sigma, \hat{y}) \leq \beta \cdot \text{OPT}(\sigma) \text{ for all } \eta$$

Smoothness: The algorithm is smooth if:

$$\text{ALG}(\sigma, \hat{y}) \leq f(\eta) \cdot \text{OPT}(\sigma) + g(\eta)$$

for functions f, g that grow gracefully with η .

The following code illustrates how the trust parameter λ controls the tradeoff between consistency and robustness. Higher trust in predictions improves consistency at the expense of robustness.

```
def consistency_robustness_curve(trust_parameter: float,
                                best_online_ratio: float = 2.0) -> Tuple[float,
    ↪ float]:
    """
    Illustrate the consistency-robustness tradeoff.

    For many problems, parameterized by trust level lambda in [0, 1]:
    - lambda = 0: Ignore predictions (pure online)
    - lambda = 1: Fully trust predictions

    Args:
        trust_parameter: Lambda in [0, 1]
        best_online_ratio: Competitive ratio of best online algorithm

    Returns:
        (consistency, robustness) pair
    """
    # Common tradeoff pattern (illustrative)
    # Higher trust -> better consistency, worse robustness

    lam = trust_parameter

    # Consistency improves as we trust more
    consistency = 1.0 + (best_online_ratio - 1.0) * (1 - lam)

    # Robustness degrades as we trust more
    # At lambda=1, robustness is unbounded (infinity)
    if lam >= 1.0:
        robustness = float('inf')
    else:
        robustness = best_online_ratio / (1 - lam * 0.5)

    return consistency, robustness

print("=== Consistency-Robustness Tradeoff ===\n")
print(f"{'Trust Level':<15} {'Consistency':<15} {'Robustness':<15}")
print("-" * 45)

for lam in [0.0, 0.25, 0.5, 0.75, 0.9, 0.95]:
    cons, rob = consistency_robustness_curve(lam)
    print(f"{'lam':<15.2f} {'cons':<15.3f} {'rob':<15.3f}")
```

```
print("\nKey insight: Cannot achieve both perfect consistency (1.0)")
print("and optimal robustness (2.0) simultaneously!")
```

=== Consistency-Robustness Tradeoff ===

Trust Level	Consistency	Robustness
0.00	2.000	2.000
0.25	1.750	2.286
0.50	1.500	2.667
0.75	1.250	3.200
0.90	1.100	3.636
0.95	1.050	3.810

Key insight: Cannot achieve both perfect consistency (1.0)
and optimal robustness (2.0) simultaneously!

35.3. The Consistency-Robustness Tradeoff

35.3.1. Pareto-Optimal Algorithms

Not all consistency-robustness pairs are achievable. The **Pareto frontier** describes the best possible tradeoffs.

```
def pareto_frontier_ski_rental(n_points: int = 100) -> Tuple[np.ndarray,
    ↪ np.ndarray]:
    """
    Compute the Pareto frontier for ski rental with predictions.

    For ski rental with buy price B and predicted number of ski days d_hat:
    - If we rent for lambda * B days before buying:
    - Consistency = 1 + 1/lambda (when prediction is correct)
    - Robustness = 1 + lambda (worst case)

    The Pareto-optimal tradeoff satisfies: consistency * robustness >= 4
    """
    # Lambda ranges from epsilon to some large value
    lambdas = np.linspace(0.1, 10, n_points)

    consistencies = 1 + 1 / lambdas
    robustnesses = 1 + lambdas

    return consistencies, robustnesses

def is_pareto_dominated(point: Tuple[float, float],
    frontier: List[Tuple[float, float]]) -> bool:
    """Check if a point is Pareto-dominated (for minimization)."""
    for fp in frontier:
        if fp[0] <= point[0] and fp[1] <= point[1]:
```

```

        if fp[0] < point[0] or fp[1] < point[1]:
            return True
        return False

# Compute and display Pareto frontier
consistencies, robustnesses = pareto_frontier_ski_rental()

print("=== Pareto Frontier: Ski Rental with Predictions ===\n")
print("The Pareto frontier shows achievable (consistency, robustness) pairs.\n")

# Sample points on the frontier
sample_indices = [0, 24, 49, 74, 99]
print(f"{'Lambda':<10} {'Consistency':<15} {'Robustness':<15} {'Product':<15}")
print("-" * 55)

lambdas = np.linspace(0.1, 10, 100)
for idx in sample_indices:
    cons = consistencies[idx]
    rob = robustnesses[idx]
    prod = cons * rob
    print(f"{'lambdas[idx]':<10.2f} {'cons:<15.3f} {'rob:<15.3f} {'prod:<15.3f}")

print("\nNote: The product consistency * robustness >= 4 always!")
print("This is an IMPOSSIBILITY result - you cannot beat this tradeoff.")

```

=== Pareto Frontier: Ski Rental with Predictions ===

The Pareto frontier shows achievable (consistency, robustness) pairs.

Lambda	Consistency	Robustness	Product
0.10	11.000	1.100	12.100
2.50	1.400	3.500	4.900
5.00	1.200	6.000	7.200
7.50	1.133	8.500	9.633
10.00	1.100	11.000	12.100

Note: The product consistency * robustness >= 4 always!

This is an IMPOSSIBILITY result - you cannot beat this tradeoff.

35.3.2. Impossibility Results

Lower bounds on the consistency-robustness product tell us what is fundamentally unachievable. The following code tests whether claimed performance pairs satisfy or violate these bounds.

```

def verify_impossibility(consistency: float, robustness: float,
                        lower_bound: float = 4.0) -> bool:
    """
    Verify if a claimed (consistency, robustness) pair is achievable.

```



```

For ski rental: consistency * robustness >= 4

Returns True if the pair violates the lower bound (impossible).
"""
return consistency * robustness < lower_bound

# Test various claimed performance pairs
claimed_pairs = [
    (1.0, 4.0, "Fully trust predictions"),
    (2.0, 2.0, "Balanced approach"),
    (1.5, 2.67, "Slightly trust predictions"),
    (1.1, 3.0, "CLAIMED: Better than possible?"),
    (1.2, 3.4, "CLAIMED: Also impossible?"),
    (1.5, 3.0, "Achievable point"),
]

print("=== Impossibility Analysis for Ski Rental ===\n")
print("Lower bound: consistency * robustness >= 4\n")
print(f"{'Consistency':<12} {'Robustness':<12} {'Product':<10} {'Status'}")
print("-" * 50)

for cons, rob, desc in claimed_pairs:
    product = cons * rob
    impossible = verify_impossibility(cons, rob)
    status = "IMPOSSIBLE" if impossible else "Achievable"
    print(f"{'cons':<12.2f} {'rob':<12.2f} {'product':<10.2f} {'status}'")
    if desc:
        print(f"    ^ {desc}")

```

=== Impossibility Analysis for Ski Rental ===

Lower bound: consistency * robustness >= 4

Consistency	Robustness	Product	Status
1.00	4.00	4.00	Achievable
		^ Fully trust predictions	
2.00	2.00	4.00	Achievable
		^ Balanced approach	
1.50	2.67	4.00	Achievable
		^ Slightly trust predictions	
1.10	3.00	3.30	IMPOSSIBLE
		^ CLAIMED: Better than possible?	
1.20	3.40	4.08	Achievable
		^ CLAIMED: Also impossible?	
1.50	3.00	4.50	Achievable
		^ Achievable point	

⚠ Lower Bounds Matter

Impossibility results are essential for understanding what is achievable. They tell us:

1. **No algorithm can beat these bounds** - don't waste time trying
2. **Pareto-optimal algorithms are known** - use them
3. **The tradeoff is fundamental** - inherent to the problem structure

For ski rental with predictions: any algorithm with consistency α must have robustness $\beta \geq 4/\alpha$.

35.3.3. Visualizing the Tradeoff

The code below computes the Pareto frontier and positions specific algorithms in the consistency-robustness space. Points below the frontier are achievable; points above it are impossible.

```
def plot_consistency_robustness_tradeoff():
    """
    Create data for visualizing the consistency-robustness tradeoff.
    """
    # Pareto frontier
    consistencies, robustnesses = pareto_frontier_ski_rental(100)

    # Some specific algorithms
    algorithms = {
        'Pure Online (ignore predictions)': (2.0, 2.0),
        'Fully Trust Predictions': (1.0, float('inf')),
        'Lambda = 1 (balanced)': (2.0, 2.0),
        'Lambda = 0.5': (3.0, 1.5),
        'Lambda = 2': (1.5, 3.0),
    }

    return consistencies, robustnesses, algorithms

cons, rob, algs = plot_consistency_robustness_tradeoff()

print("=== Consistency-Robustness Tradeoff Visualization ===\n")
print("Pareto Frontier (sample points):")
print(f"{'Consistency':<15} {'Robustness':<15}")
print("-" * 30)
for i in range(0, 100, 20):
    print(f"{'cons'[i]:<15.3f} {'rob'[i]:<15.3f}")

print("\nSpecific Algorithms:")
print(f"{'Algorithm':<35} {'Consistency':<12} {'Robustness':<12}")
print("-" * 59)
for name, (c, r) in algs.items():
    r_str = f"{r:.2f}" if r != float('inf') else "infinity"
    print(f"{'name':<35} {'c':<12.2f} {'r_str':<12}")
```

```
=== Consistency-Robustness Tradeoff Visualization ===
```

Pareto Frontier (sample points):

Consistency	Robustness
-----	-----
11.000	1.100
1.476	3.100
1.244	5.100
1.164	7.100
1.123	9.100

Specific Algorithms:

Algorithm	Consistency	Robustness
-----	-----	-----
Pure Online (ignore predictions)	2.00	2.00
Fully Trust Predictions	1.00	infinity
Lambda = 1 (balanced)	2.00	2.00
Lambda = 0.5	3.00	1.50
Lambda = 2	1.50	3.00

35.4. Classic Problems with Predictions

35.4.1. Ski Rental Problem

The ski rental problem is the canonical example for learning-augmented algorithms.

Problem: You need skis for an unknown number of days d . Renting costs 1 per day, buying costs B . You learn d only at the end.

Without predictions: Rent for $B - 1$ days, then buy. This is 2-competitive.

With predictions: Given predicted duration $\hat{d}$, how should you decide?

```
class SkiRentalWithPredictions:
    """
    Ski rental algorithm with ML predictions.

    Uses a parameterized strategy that interpolates between
    trusting the prediction and being robust.
    """

    def __init__(self, buy_price: float, trust_parameter: float = 1.0):
        """
        Initialize ski rental solver.

        Args:
            buy_price: Cost to buy skis (B)
            trust_parameter: Lambda in (0, inf) controlling trust level
                - Small lambda: more trust in prediction (better consistency)
                - Large lambda: less trust (better robustness)
        """
        self.B = buy_price
        self.lam = trust_parameter
        self.total_cost = 0
```

```

self.bought = False
self.days_rented = 0

def compute_threshold(self, predicted_days: float) -> float:
    """
    Compute the day to switch from renting to buying.

    Strategy: Buy on day min(lambda * predicted_days, B)

    - If prediction is small, buy early (trust prediction)
    - If prediction is large, buy at B (robust threshold)
    """
    return min(self.lam * predicted_days, self.B)

def simulate(self, actual_days: int, predicted_days: float) -> Dict:
    """
    Simulate the ski rental problem.

    Args:
        actual_days: True number of ski days
        predicted_days: ML prediction of ski days

    Returns:
        Dict with cost breakdown and analysis
    """
    self.reset()

    threshold = self.compute_threshold(predicted_days)

    for day in range(1, actual_days + 1):
        if self.bought:
            # Already own skis, no daily cost
            pass
        elif day >= threshold:
            # Buy skis
            self.bought = True
            self.total_cost += self.B
        else:
            # Rent for this day
            self.days_rented += 1
            self.total_cost += 1

    # Compute optimal cost
    optimal_cost = min(actual_days, self.B)

    return {
        'actual_days': actual_days,
        'predicted_days': predicted_days,
        'prediction_error': abs(actual_days - predicted_days),
        'threshold': threshold,
        'days_rented': self.days_rented,
        'bought': self.bought,
        'total_cost': self.total_cost,
    }

```

```

        'optimal_cost': optimal_cost,
        'competitive_ratio': self.total_cost / optimal_cost if optimal_cost > 0
        ↪ else 1.0
    }

    def reset(self):
        """Reset state for new simulation."""
        self.total_cost = 0
        self.bought = False
        self.days_rented = 0

# Example simulations
print("=== Ski Rental with Predictions ===\n")

B = 10 # Buy price
scenarios = [
    # (actual_days, predicted_days, description)
    (5, 5, "Perfect prediction (short trip)"),
    (15, 15, "Perfect prediction (long trip)"),
    (15, 5, "Under-predicted (should have bought)"),
    (5, 15, "Over-predicted (shouldn't have bought)"),
    (8, 10, "Slight over-prediction"),
]

# Test with different trust levels
trust_levels = [0.5, 1.0, 2.0]

print(f"Buy price B = {B}\n")

for actual, predicted, desc in scenarios:
    print(f"Scenario: {desc}")
    print(f"  Actual days: {actual}, Predicted: {predicted}")
    print(f"  Optimal cost: {min(actual, B)}")

    for lam in trust_levels:
        solver = SkiRentalWithPredictions(B, trust_parameter=lam)
        result = solver.simulate(actual, predicted)
        print(f"  Lambda={lam}: Cost={result['total_cost']:.0f}, "
              f"Ratio={result['competitive_ratio']:.2f}")
    print()

```

=== Ski Rental with Predictions ===

Buy price B = 10

Scenario: Perfect prediction (short trip)

Actual days: 5, Predicted: 5

Optimal cost: 5

Lambda=0.5: Cost=12, Ratio=2.40

Lambda=1.0: Cost=14, Ratio=2.80

Lambda=2.0: Cost=5, Ratio=1.00

Scenario: Perfect prediction (long trip)

Actual days: 15, Predicted: 15
 Optimal cost: 10
 Lambda=0.5: Cost=17, Ratio=1.70
 Lambda=1.0: Cost=19, Ratio=1.90
 Lambda=2.0: Cost=19, Ratio=1.90

Scenario: Under-predicted (should have bought)

Actual days: 15, Predicted: 5
 Optimal cost: 10
 Lambda=0.5: Cost=12, Ratio=1.20
 Lambda=1.0: Cost=14, Ratio=1.40
 Lambda=2.0: Cost=19, Ratio=1.90

Scenario: Over-predicted (shouldn't have bought)

Actual days: 5, Predicted: 15
 Optimal cost: 5
 Lambda=0.5: Cost=5, Ratio=1.00
 Lambda=1.0: Cost=5, Ratio=1.00
 Lambda=2.0: Cost=5, Ratio=1.00

Scenario: Slight over-prediction

Actual days: 8, Predicted: 10
 Optimal cost: 8
 Lambda=0.5: Cost=14, Ratio=1.75
 Lambda=1.0: Cost=8, Ratio=1.00
 Lambda=2.0: Cost=8, Ratio=1.00

35.4.2. Analysis of Ski Rental with Predictions

```
def analyze_ski_rental_algorithm(buy_price: float,
                                trust_parameter: float,
                                n_simulations: int = 1000) -> Dict:
    """
    Comprehensive analysis of ski rental with predictions.
    """
    np.random.seed(42)
    B = buy_price

    results = {
        'trust_parameter': trust_parameter,
        'perfect_prediction': [],
        'noisy_prediction': [],
        'adversarial': [],
    }

    solver = SkiRentalWithPredictions(B, trust_parameter)

    for _ in range(n_simulations):
```

```

# Random actual duration
actual_days = np.random.randint(1, 3 * B)

# Perfect prediction
result = solver.simulate(actual_days, actual_days)
results['perfect_prediction'].append(result['competitive_ratio'])

# Noisy prediction (Gaussian noise)
noise = np.random.normal(0, B * 0.3)
predicted = max(1, actual_days + noise)
result = solver.simulate(actual_days, predicted)
results['noisy_prediction'].append(result['competitive_ratio'])

# Adversarial prediction (worst case)
# If actual is short, predict long; if long, predict short
if actual_days < B:
    adversarial_pred = 3 * B
else:
    adversarial_pred = 1
result = solver.simulate(actual_days, adversarial_pred)
results['adversarial'].append(result['competitive_ratio'])

# Compute statistics
for key in ['perfect_prediction', 'noisy_prediction', 'adversarial']:
    ratios = results[key]
    results[f'{key}_mean'] = np.mean(ratios)
    results[f'{key}_max'] = np.max(ratios)

return results

# Analyze different trust levels
print("=== Comprehensive Ski Rental Analysis ===\n")

B = 10
trust_levels = [0.25, 0.5, 1.0, 2.0, 4.0]

print(f"{'Lambda':<10} {'Perfect (mean)':<15} {'Noisy (mean)':<15} {'Adversarial  

↳ (max)':<18}")
print("-" * 60)

for lam in trust_levels:
    results = analyze_ski_rental_algorithm(B, lam, n_simulations=500)
    print(f"{lam:<10.2f} {results['perfect_prediction_mean']:<15.3f} "
          f"{results['noisy_prediction_mean']:<15.3f} "
          f"{results['adversarial_max']:<18.3f}")

print("\nObservations:")
print("- Small lambda: Better with perfect predictions, worse adversarially")
print("- Large lambda: More robust, but can't fully exploit good predictions")
print("- Lambda = 1 is a reasonable balanced choice")

```

```
=== Comprehensive Ski Rental Analysis ===
```

Lambda	Perfect (mean)	Noisy (mean)	Adversarial (max)
0.25	2.188	2.094	2.125
0.50	2.480	2.177	1.000
1.00	2.680	1.996	1.000
2.00	1.608	1.740	1.100
4.00	1.608	1.637	1.300

Observations:

- Small lambda: Better with perfect predictions, worse adversarially
- Large lambda: More robust, but can't fully exploit good predictions
- Lambda = 1 is a reasonable balanced choice

35.4.3. Online Caching with Predictions (BlindOracle)

Caching is another fundamental online problem where predictions can help significantly.

Problem: Given a cache of size k and a sequence of page requests, minimize the number of cache misses.

Without predictions: LRU (Least Recently Used) is k -competitive.

With predictions: Predict which pages will be requested next, evict the one needed furthest in the future.

```
class CacheWithPredictions:
    """
    Online caching algorithm with next-request predictions.

    Implements a parameterized algorithm that balances between:
    - Trusting predictions (evict predicted-furthest page)
    - Being robust (use LRU as backup)
    """

    def __init__(self, cache_size: int, trust_parameter: float = 0.5):
        """
        Args:
            cache_size: Maximum number of pages in cache (k)
            trust_parameter: Probability of trusting prediction vs LRU
        """
        self.k = cache_size
        self.trust = trust_parameter

        self.cache = set() # Current pages in cache
        self.recency = {} # Page -> last access time
        self.time = 0

        self.hits = 0
        self.misses = 0

    def access(self, page: int, predicted_next_access: Dict[int, float]) -> bool:
        """
        Process a page access with predictions.
```



```

Args:
    page: Requested page
    predicted_next_access: Dict mapping cached pages to predicted
                           time until next access (higher = further)

Returns:
    True if cache hit, False if miss
"""
self.time += 1
self.recency[page] = self.time

if page in self.cache:
    self.hits += 1
    return True

# Cache miss
self.misses += 1

if len(self.cache) < self.k:
    # Cache not full, just add
    self.cache.add(page)
else:
    # Need to evict
    victim = self._choose_victim(predicted_next_access)
    self.cache.remove(victim)
    self.cache.add(page)

return False

def _choose_victim(self, predictions: Dict[int, float]) -> int:
    """
    Choose page to evict.

    Hybrid strategy:
    - With probability 'trust': use prediction (evict furthest)
    - With probability '1-trust': use LRU (evict oldest)
    """
    cached_pages = list(self.cache)

    if np.random.random() < self.trust and predictions:
        # Trust prediction: evict page with largest predicted next access
        valid_predictions = {p: predictions.get(p, float('inf'))
                             for p in cached_pages}
        return max(valid_predictions.keys(),
                   key=lambda p: valid_predictions[p])
    else:
        # LRU fallback
        return min(cached_pages,
                   key=lambda p: self.recency.get(p, 0))

def reset(self):
    """Reset cache state."""
    self.cache = set()

```

```

        self.recency = {}
        self.time = 0
        self.hits = 0
        self.misses = 0

    def miss_rate(self) -> float:
        """Compute cache miss rate."""
        total = self.hits + self.misses
        return self.misses / total if total > 0 else 0.0

class OraclePredictor:
    """
    Simulates predictions of varying quality.
    """

    def __init__(self, noise_level: float = 0.0):
        """
        Args:
            noise_level: 0 = perfect predictions, 1 = random predictions
        """
        self.noise = noise_level

    def predict(self, cache_pages: set,
                future_requests: List[int]) -> Dict[int, float]:
        """
        Predict next access time for each cached page.

        Returns dict mapping page -> predicted time until next access
        """
        predictions = {}

        for page in cache_pages:
            # True next access
            try:
                true_next = future_requests.index(page)
            except ValueError:
                true_next = len(future_requests) + 1000 # Never accessed again

            # Add noise
            if self.noise > 0 and np.random.random() < self.noise:
                # Random prediction
                predictions[page] = np.random.uniform(0, len(future_requests))
            else:
                predictions[page] = true_next

        return predictions

    def simulate_caching(cache_size: int,
                        request_sequence: List[int],
                        trust_parameter: float,
                        prediction_noise: float) -> Dict:

```

```

"""
Simulate caching with predictions.
"""

cache = CacheWithPredictions(cache_size, trust_parameter)
predictor = OraclePredictor(prediction_noise)

for t, page in enumerate(request_sequence):
    future = request_sequence[t+1:]
    predictions = predictor.predict(cache.cache, future)
    cache.access(page, predictions)

return {
    'hits': cache.hits,
    'misses': cache.misses,
    'miss_rate': cache.miss_rate(),
    'trust': trust_parameter,
    'noise': prediction_noise
}

# Generate realistic request sequence
np.random.seed(42)

n_pages = 50
cache_size = 10
sequence_length = 1000

# Zipf distribution (common access pattern)
from scipy import stats
zipf_param = 1.5
page_probs = 1.0 / np.arange(1, n_pages + 1) ** zipf_param
page_probs /= page_probs.sum()

request_sequence = np.random.choice(n_pages, size=sequence_length,
    ↪ p=page_probs).tolist()

print("=== Online Caching with Predictions ===\n")
print(f"Cache size: {cache_size}")
print(f"Number of pages: {n_pages}")
print(f"Request sequence length: {sequence_length}")
print(f"Access pattern: Zipf (popular pages accessed more)\n")

# Compare different configurations
print(f"{'Trust':<10} {'Noise':<10} {'Miss Rate':<12} {'Misses':<10}")
print("-" * 42)

for trust in [0.0, 0.5, 0.8, 1.0]:
    for noise in [0.0, 0.3, 0.7]:
        result = simulate_caching(cache_size, request_sequence, trust, noise)
        print(f"{'trust':<10.1f} {'noise':<10.1f} {'result['miss_rate']':<12.3f} "
            f"{'result['misses']':<10}")

```

```
=== Online Caching with Predictions ===
```

35. Learning-Augmented Algorithms

Cache size: 10
Number of pages: 50
Request sequence length: 1000
Access pattern: Zipf (popular pages accessed more)

Trust	Noise	Miss Rate	Misses

0.0	0.0	0.230	230
0.0	0.3	0.230	230
0.0	0.7	0.230	230
0.5	0.0	0.180	180
0.5	0.3	0.221	221
0.5	0.7	0.239	239
0.8	0.0	0.153	153
0.8	0.3	0.218	218
0.8	0.7	0.250	250
1.0	0.0	0.133	133
1.0	0.3	0.227	227
1.0	0.7	0.249	249

Practical Caching Predictions

In practice, caching predictions can come from:

1. **Recurrence patterns:** Pages accessed in regular cycles
2. **User behavior models:** Next page prediction from navigation graphs
3. **Content similarity:** Pages related to current activity
4. **Prefetching hints:** Application-level signals about future needs

The key insight is that even noisy predictions can help, as long as the algorithm doesn't fully trust them.

35.4.4. Online Knapsack with Predictions

In the online knapsack problem, items arrive one at a time, and the algorithm must immediately accept or reject each item. Predictions of future item values help the algorithm set better acceptance thresholds.

```
class OnlineKnapsackWithPredictions:
    """
    Online knapsack problem with value predictions.

    Items arrive one by one. We must decide immediately whether to
    accept (if capacity allows) or reject (forever).

    With predictions: We receive predicted total value of future items.
    """

    def __init__(self, capacity: float, trust_parameter: float = 0.5):
        """
```

```

    Args:
        capacity: Knapsack capacity
        trust_parameter: How much to trust value predictions
    """
    self.capacity = capacity
    self.trust = trust_parameter

    self.current_weight = 0
    self.current_value = 0
    self.items_accepted = []

def decide(self, item_weight: float, item_value: float,
           predicted_future_value: float,
           predicted_future_weight: float) -> bool:
    """
    Decide whether to accept an item.

    Strategy:
    - Pure greedy: Accept if fits and value/weight exceeds threshold
    - With predictions: Adjust threshold based on predicted opportunity cost

    Args:
        item_weight: Weight of current item
        item_value: Value of current item
        predicted_future_value: Predicted total value of future items
        predicted_future_weight: Predicted total weight of future items

    Returns:
        True if item is accepted
    """
    # Check capacity
    if self.current_weight + item_weight > self.capacity:
        return False

    remaining_capacity = self.capacity - self.current_weight

    # Compute value density of current item
    current_density = item_value / item_weight if item_weight > 0 else
↪ float('inf')

    # Estimate value density of future items (from predictions)
    if predicted_future_weight > 0:
        predicted_future_density = predicted_future_value /
↪ predicted_future_weight
    else:
        predicted_future_density = 0

    # Threshold: blend predicted future density with conservative estimate
    conservative_density = 0 # Always accept in pure greedy
    threshold = self.trust * predicted_future_density + (1 - self.trust) *
↪ conservative_density

    # Accept if current item beats threshold

```

```

        if current_density >= threshold:
            self.current_weight += item_weight
            self.current_value += item_value
            self.items_accepted.append((item_weight, item_value))
            return True

        return False

def reset(self):
    """Reset knapsack state."""
    self.current_weight = 0
    self.current_value = 0
    self.items_accepted = []

def simulate_online_knapsack(capacity: float,
                             items: List[Tuple[float, float]],
                             trust: float,
                             prediction_error: float) -> Dict:
    """
    Simulate online knapsack with predictions.

    Args:
        capacity: Knapsack capacity
        items: List of (weight, value) tuples in arrival order
        trust: Trust parameter for algorithm
        prediction_error: Relative error in predictions

    Returns:
        Dict with performance metrics
    """
    knapsack = OnlineKnapsackWithPredictions(capacity, trust)

    n = len(items)
    for i, (weight, value) in enumerate(items):
        # True future values
        future_items = items[i+1:]
        true_future_value = sum(v for w, v in future_items)
        true_future_weight = sum(w for w, v in future_items)

        # Add prediction error
        noise_v = np.random.normal(0, prediction_error * true_future_value)
        noise_w = np.random.normal(0, prediction_error * true_future_weight)

        pred_value = max(0, true_future_value + noise_v)
        pred_weight = max(1, true_future_weight + noise_w)

        knapsack.decide(weight, value, pred_value, pred_weight)

    # Compute optimal offline value (0-1 knapsack)
    weights = [w for w, v in items]
    values = [v for w, v in items]
    optimal_value = solve_knapsack_dp(weights, values, int(capacity))

```

```

    return {
        'achieved_value': knapsack.current_value,
        'optimal_value': optimal_value,
        'competitive_ratio': knapsack.current_value / optimal_value if optimal_value
        ↪ > 0 else 1.0,
        'capacity_used': knapsack.current_weight,
        'items_accepted': len(knapsack.items_accepted)
    }

def solve_knapsack_dp(weights: List[float], values: List[float],
    capacity: int) -> float:
    """Solve 0-1 knapsack optimally using dynamic programming."""
    n = len(weights)
    capacity = int(capacity)

    # DP table
    dp = [[0] * (capacity + 1) for _ in range(n + 1)]

    for i in range(1, n + 1):
        w = int(weights[i-1])
        v = values[i-1]
        for c in range(capacity + 1):
            if w <= c:
                dp[i][c] = max(dp[i-1][c], dp[i-1][c-w] + v)
            else:
                dp[i][c] = dp[i-1][c]

    return dp[n][capacity]

# Generate random knapsack instance
np.random.seed(42)

capacity = 50
n_items = 30

weights = np.random.uniform(5, 20, n_items)
values = np.random.uniform(10, 100, n_items)
items = list(zip(weights.tolist(), values.tolist()))

print("=== Online Knapsack with Predictions ===\n")
print(f"Capacity: {capacity}")
print(f"Number of items: {n_items}")
print(f"Optimal value: {solve_knapsack_dp(weights.tolist(), values.tolist(),
    ↪ int(capacity)):.1f}\n")

print(f"{'Trust':<10} {'Error':<10} {'Achieved':<12} {'Optimal':<12} {'Ratio':<10}")
print("-" * 54)

for trust in [0.0, 0.3, 0.6, 0.9]:
    for error in [0.0, 0.3, 0.6]:

```

```

# Average over multiple runs
ratios = []
for _ in range(20):
    result = simulate_online_knapsack(capacity, items, trust, error)
    ratios.append(result['competitive_ratio'])

avg_ratio = np.mean(ratios)
print(f"{trust:<10.1f} {error:<10.1f} {result['achieved_value']:<12.1f} "
      f"{result['optimal_value']:<12.1f} {avg_ratio:<10.3f}")

```

=== Online Knapsack with Predictions ===

Capacity: 50

Number of items: 30

Optimal value: 554.7

Trust	Error	Achieved	Optimal	Ratio
0.0	0.0	105.9	554.7	0.191
0.0	0.3	105.9	554.7	0.191
0.0	0.6	105.9	554.7	0.191
0.3	0.0	377.2	554.7	0.680
0.3	0.3	222.8	554.7	0.521
0.3	0.6	222.8	554.7	0.433
0.6	0.0	377.2	554.7	0.680
0.6	0.3	311.9	554.7	0.672
0.6	0.6	272.8	554.7	0.613
0.9	0.0	377.2	554.7	0.680
0.9	0.3	377.2	554.7	0.677
0.9	0.6	421.2	554.7	0.617

35.5. Scheduling with Predictions

Scheduling problems benefit directly from processing time predictions because the optimal schedule depends on job lengths that are typically unknown at decision time.

35.5.1. Non-Clairvoyant Scheduling

In non-clairvoyant scheduling, job processing times are unknown. Predictions of processing times can significantly improve performance. The algorithm uses predicted job lengths to prioritize shorter jobs (SPTF order) while maintaining a fallback to round-robin when predictions are unreliable.

```

@dataclass
class Job:
    """A job to be scheduled."""
    id: int
    arrival_time: float
    true_processing_time: float # Unknown to scheduler
    predicted_processing_time: float # ML prediction

```



```

class NonClairvoyantScheduler:
    """
    Non-clairvoyant scheduling with processing time predictions.

    Uses Shortest Predicted Job First (SPJF) with robustness guarantees.
    """

    def __init__(self, trust_parameter: float = 0.5):
        """
        Args:
            trust_parameter: How much to weight predictions vs round-robin
        """
        self.trust = trust_parameter
        self.jobs = []
        self.completed = []
        self.current_time = 0

    def add_job(self, job: Job):
        """Add a job to the queue."""
        self.jobs.append(job)

    def schedule_spjf(self, jobs: List[Job]) -> List[Tuple[int, float, float]]:
        """
        Schedule using Shortest Predicted Job First.

        Returns list of (job_id, start_time, completion_time)
        """
        # Sort by predicted processing time
        sorted_jobs = sorted(jobs, key=lambda j: j.predicted_processing_time)

        schedule = []
        time = 0

        for job in sorted_jobs:
            start = max(time, job.arrival_time)
            completion = start + job.true_processing_time
            schedule.append((job.id, start, completion))
            time = completion

        return schedule

    def schedule_round_robin(self, jobs: List[Job],
                             quantum: float = 1.0) -> List[Tuple[int, float,
↪ float]]:
        """
        Schedule using Round Robin (robust baseline).
        """
        remaining = {j.id: j.true_processing_time for j in jobs}
        active_jobs = list(jobs)
        completions = {}
        time = 0

        while active_jobs:

```

```

        for job in list(active_jobs):
            if remaining[job.id] > 0:
                work = min(quantum, remaining[job.id])
                remaining[job.id] -= work
                time += work

            if remaining[job.id] <= 0:
                completions[job.id] = time
                active_jobs.remove(job)

    schedule = [(jid, 0, comp) for jid, comp in completions.items()]
    return schedule

def schedule_hybrid(self, jobs: List[Job]) -> List[Tuple[int, float, float]]:
    """
    Hybrid scheduling that blends SPJF and Round Robin.

    Key idea: Process jobs in SPJF order, but periodically
    give time to other jobs to maintain robustness.
    """
    # Order by prediction
    by_prediction = sorted(jobs, key=lambda j: j.predicted_processing_time)

    schedule = []
    time = 0
    remaining = {j.id: j.true_processing_time for j in jobs}

    while any(r > 0 for r in remaining.values()):
        # Select job: blend of SPJF and fair share
        if np.random.random() < self.trust:
            # SPJF choice: shortest predicted remaining
            active = [j for j in by_prediction if remaining[j.id] > 0]
            if active:
                job = active[0]
        else:
            # Round-robin choice: any remaining job
            active = [j for j in jobs if remaining[j.id] > 0]
            if active:
                job = np.random.choice(active)

        if not active:
            break

        # Process for a quantum
        work = min(1.0, remaining[job.id])
        remaining[job.id] -= work
        time += work

        if remaining[job.id] <= 0:
            schedule.append((job.id, 0, time))

    return schedule

```

```

def compute_total_flow_time(schedule: List[Tuple[int, float, float]]) -> float:
    """
    Compute total flow time (sum of completion times).

    This is a common scheduling objective.
    """
    return sum(completion for _, _, completion in schedule)

def compute_total_weighted_flow(schedule: List[Tuple[int, float, float]],
                                weights: Dict[int, float]) -> float:
    """Compute weighted flow time."""
    return sum(weights.get(job_id, 1.0) * completion
                for job_id, _, completion in schedule)

# Simulation
np.random.seed(42)

n_jobs = 20
jobs = []

for i in range(n_jobs):
    true_time = np.random.exponential(5)
    # Prediction with noise
    noise = np.random.normal(0, 1)
    pred_time = max(0.1, true_time * (1 + 0.3 * noise))

    jobs.append(Job(
        id=i,
        arrival_time=0, # All arrive at time 0
        true_processing_time=true_time,
        predicted_processing_time=pred_time
    ))

print("=== Non-Clairvoyant Scheduling with Predictions ===\n")
print(f"Number of jobs: {n_jobs}")
print(f"Average true processing time: {np.mean([j.true_processing_time for j in
↪ jobs]):.2f}")
print(f"Average predicted processing time: {np.mean([j.predicted_processing_time
↪ for j in jobs]):.2f}")

# Compute optimal (clairvoyant SPT)
optimal_order = sorted(jobs, key=lambda j: j.true_processing_time)
optimal_schedule = []
time = 0
for job in optimal_order:
    completion = time + job.true_processing_time
    optimal_schedule.append((job.id, time, completion))
    time = completion
optimal_flow = compute_total_flow_time(optimal_schedule)

```

```

print(f"\nOptimal (clairvoyant) flow time: {optimal_flow:.2f}")

# Test different approaches
scheduler = NonClairvoyantScheduler(trust_parameter=0.8)

spjf_schedule = scheduler.schedule_spjf(jobs)
rr_schedule = scheduler.schedule_round_robin(jobs)

print(f"\nSPJF (trust predictions) flow time:
↪ {compute_total_flow_time(spjf_schedule):.2f}")
print(f"Round Robin (robust) flow time: {compute_total_flow_time(rr_schedule):.2f}")

# Hybrid with different trust levels
print("\nHybrid approach (varying trust):")
print(f"{'Trust':<10} {'Flow Time':<15} {'Ratio to Optimal':<20}")
print("-" * 45)

for trust in [0.0, 0.3, 0.5, 0.7, 1.0]:
    scheduler = NonClairvoyantScheduler(trust_parameter=trust)
    # Average over multiple runs (due to randomization)
    flows = []
    for _ in range(50):
        schedule = scheduler.schedule_hybrid(jobs)
        flows.append(compute_total_flow_time(schedule))
    avg_flow = np.mean(flows)
    ratio = avg_flow / optimal_flow
    print(f"{'trust':<10.1f} {'avg_flow':<15.2f} {'ratio':<20.3f}")

```

=== Non-Clairvoyant Scheduling with Predictions ===

Number of jobs: 20

Average true processing time: 2.45

Average predicted processing time: 2.34

Optimal (clairvoyant) flow time: 293.53

SPJF (trust predictions) flow time: 305.26

Round Robin (robust) flow time: 529.86

Hybrid approach (varying trust):

Trust	Flow Time	Ratio to Optimal
0.0	545.29	1.858
0.3	461.40	1.572
0.5	406.18	1.384
0.7	359.74	1.226
1.0	305.26	1.040

35.5.2. Prediction Error Analysis for Scheduling

```

def analyze_scheduling_with_prediction_errors(
    n_jobs: int = 50,
    n_trials: int = 100
) -> Dict:
    """
    Analyze how prediction quality affects scheduling performance.
    """
    results = {
        'error_levels': [],
        'spjf_ratios': [],
        'hybrid_ratios': []
    }

    error_levels = [0.0, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0]

    for error in error_levels:
        spjf_trial_ratios = []
        hybrid_trial_ratios = []

        for _ in range(n_trials):
            # Generate jobs
            jobs = []
            for i in range(n_jobs):
                true_time = np.random.exponential(5)
                # Prediction with specified error level
                noise = np.random.uniform(-error, error)
                pred_time = max(0.1, true_time * (1 + noise))

                jobs.append(Job(
                    id=i,
                    arrival_time=0,
                    true_processing_time=true_time,
                    predicted_processing_time=pred_time
                ))

            # Optimal
            optimal_order = sorted(jobs, key=lambda j: j.true_processing_time)
            time = 0
            optimal_flow = 0
            for job in optimal_order:
                time += job.true_processing_time
                optimal_flow += time

            # SPJF
            scheduler = NonClairvoyantScheduler(trust_parameter=1.0)
            spjf_flow = compute_total_flow_time(scheduler.schedule_spjf(jobs))
            spjf_trial_ratios.append(spjf_flow / optimal_flow)

            # Hybrid (0.6 trust)
            scheduler = NonClairvoyantScheduler(trust_parameter=0.6)
            hybrid_flows = [compute_total_flow_time(scheduler.schedule_hybrid(jobs))
                            for _ in range(10)]

```

```

        hybrid_trial_ratios.append(np.mean(hybrid_flows) / optimal_flow)

    results['error_levels'].append(error)
    results['spjf_ratios'].append(np.mean(spjf_trial_ratios))
    results['hybrid_ratios'].append(np.mean(hybrid_trial_ratios))

    return results

# Run analysis
print("=== Scheduling Performance vs Prediction Error ===\n")

results = analyze_scheduling_with_prediction_errors(n_jobs=30, n_trials=50)

print(f"{'Prediction Error':<18} {'SPJF Ratio':<15} {'Hybrid Ratio':<15}")
print("-" * 48)

for i, error in enumerate(results['error_levels']):
    spjf = results['spjf_ratios'][i]
    hybrid = results['hybrid_ratios'][i]
    print(f"{error:<18.1f} {spjf:<15.3f} {hybrid:<15.3f}")

print("\nObservations:")
print("- SPJF is excellent with low error but degrades with high error")
print("- Hybrid approach provides consistent performance across error levels")
print("- At ~50% error, hybrid starts outperforming pure SPJF")

```

=== Scheduling Performance vs Prediction Error ===

Prediction Error	SPJF Ratio	Hybrid Ratio
0.0	1.000	1.264
0.1	1.001	1.272
0.2	1.006	1.272
0.3	1.014	1.280
0.5	1.039	1.308
0.7	1.087	1.360
1.0	1.213	1.497

Observations:

- SPJF is excellent with low error but degrades with high error
- Hybrid approach provides consistent performance across error levels
- At ~50% error, hybrid starts outperforming pure SPJF

35.6. How to Design Predictions

35.6.1. Types of Predictions

```

class PredictionType(Enum):
    """Types of predictions used in learning-augmented algorithms."""
    OUTCOME = "outcome"          # Direct prediction of future outcomes
    STRUCTURAL = "structural"     # Prediction of problem structure
    DEMAND = "demand"            # Prediction of resource demand
    DURATION = "duration"        # Prediction of time/duration
    RANKING = "ranking"          # Prediction of relative ordering

@dataclass
class PredictionSpec:
    """Specification for a prediction task."""
    pred_type: PredictionType
    description: str
    examples: List[str]
    typical_models: List[str]
    error_metric: str

prediction_catalog = [
    PredictionSpec(
        pred_type=PredictionType.OUTCOME,
        description="Predict the final outcome of an uncertain event",
        examples=["Total ski days", "Final auction price", "Project duration"],
        typical_models=["Regression", "Survival analysis"],
        error_metric="MAE, RMSE"
    ),
    PredictionSpec(
        pred_type=PredictionType.STRUCTURAL,
        description="Predict structure of the optimal solution",
        examples=["Which items in optimal knapsack", "Optimal tour structure"],
        typical_models=["Classification", "Graph neural networks"],
        error_metric="Hamming distance, Jaccard similarity"
    ),
    PredictionSpec(
        pred_type=PredictionType.DEMAND,
        description="Predict future demand or arrivals",
        examples=["Page request frequencies", "Customer arrivals", "Load patterns"],
        typical_models=["Time series", "Poisson regression"],
        error_metric="MAPE, KL divergence"
    ),
    PredictionSpec(
        pred_type=PredictionType.DURATION,
        description="Predict how long something will take",
        examples=["Job processing time", "Task completion", "Cache lifetime"],
        typical_models=["Survival models", "Regression trees"],
        error_metric="MAE, calibration error"
    ),
    PredictionSpec(
        pred_type=PredictionType.RANKING,
        description="Predict relative ordering of items",
        examples=["Next page to access", "Job priorities", "Bid rankings"],

```

```

        typical_models=["Learning to rank", "Pairwise models"],
        error_metric="NDCG, Kendall tau"
    ),
]

print("=== Prediction Types for Learning-Augmented Algorithms ===\n")

for spec in prediction_catalog:
    print(f"{spec.pred_type.value.upper()}")
    print(f"  Description: {spec.description}")
    print(f"  Examples: {'', '.join(spec.examples)}")
    print(f"  Models: {'', '.join(spec.typical_models)}")
    print(f"  Error metric: {spec.error_metric}")
    print()

```

=== Prediction Types for Learning-Augmented Algorithms ===

OUTCOME

Description: Predict the final outcome of an uncertain event
 Examples: Total ski days, Final auction price, Project duration
 Models: Regression, Survival analysis
 Error metric: MAE, RMSE

STRUCTURAL

Description: Predict structure of the optimal solution
 Examples: Which items in optimal knapsack, Optimal tour structure
 Models: Classification, Graph neural networks
 Error metric: Hamming distance, Jaccard similarity

DEMAND

Description: Predict future demand or arrivals
 Examples: Page request frequencies, Customer arrivals, Load patterns
 Models: Time series, Poisson regression
 Error metric: MAPE, KL divergence

DURATION

Description: Predict how long something will take
 Examples: Job processing time, Task completion, Cache lifetime
 Models: Survival models, Regression trees
 Error metric: MAE, calibration error

RANKING

Description: Predict relative ordering of items
 Examples: Next page to access, Job priorities, Bid rankings
 Models: Learning to rank, Pairwise models
 Error metric: NDCG, Kendall tau

35.6.2. Prediction Error Metrics

```

def prediction_error_absolute(predicted: float, actual: float) -> float:
    """Absolute error: |predicted - actual|"""
    return abs(predicted - actual)

def prediction_error_relative(predicted: float, actual: float) -> float:
    """Relative error: |predicted - actual| / actual"""
    if actual == 0:
        return float('inf') if predicted != 0 else 0
    return abs(predicted - actual) / actual

def prediction_error_ratio(predicted: float, actual: float) -> float:
    """Ratio error: max(predicted/actual, actual/predicted)"""
    if predicted == 0 or actual == 0:
        return float('inf')
    return max(predicted / actual, actual / predicted)

def prediction_error_normalized(predicted: np.ndarray,
                               actual: np.ndarray,
                               normalization: str = 'l2') -> float:
    """
    Normalized error for vector predictions.

    Args:
        normalization: 'l1', 'l2', or 'linf'
    """
    diff = predicted - actual

    if normalization == 'l1':
        error = np.sum(np.abs(diff))
        norm = np.sum(np.abs(actual))
    elif normalization == 'l2':
        error = np.sqrt(np.sum(diff**2))
        norm = np.sqrt(np.sum(actual**2))
    else: # linf
        error = np.max(np.abs(diff))
        norm = np.max(np.abs(actual))

    return error / norm if norm > 0 else 0

# Demonstrate error metrics
print("=== Prediction Error Metrics ===\n")

test_cases = [
    (10.0, 10.0, "Perfect prediction"),
    (11.0, 10.0, "10% over-prediction"),
    (9.0, 10.0, "10% under-prediction"),
    (20.0, 10.0, "2x over-prediction"),
    (5.0, 10.0, "0.5x under-prediction"),

```

```

]

print(f"{'Predicted':<12} {'Actual':<10} {'Absolute':<12} {'Relative':<12}
    ↳ {'Ratio':<10}")
print("-" * 56)

for pred, actual, desc in test_cases:
    abs_err = prediction_error_absolute(pred, actual)
    rel_err = prediction_error_relative(pred, actual)
    ratio_err = prediction_error_ratio(pred, actual)
    print(f"{pred:<12.1f} {actual:<10.1f} {abs_err:<12.1f} {rel_err:<12.2f}
        ↳ {ratio_err:<10.2f}")
    print(f"    ({desc})")

```

=== Prediction Error Metrics ===

Predicted	Actual	Absolute	Relative	Ratio
10.0	10.0	0.0	0.00	1.00
(Perfect prediction)				
11.0	10.0	1.0	0.10	1.10
(10% over-prediction)				
9.0	10.0	1.0	0.10	1.11
(10% under-prediction)				
20.0	10.0	10.0	1.00	2.00
(2x over-prediction)				
5.0	10.0	5.0	0.50	2.00
(0.5x under-prediction)				

Which Error Metric to Use?

The choice of error metric should match how the algorithm uses predictions:

- **Absolute error:** When the magnitude of the error matters (e.g., scheduling)
- **Relative error:** When predictions are scale-invariant (e.g., ski rental)
- **Ratio error:** When multiplicative errors are the concern
- **Ranking error:** When only the order matters (e.g., caching)

Many theoretical results use **ratio error** because it leads to cleaner bounds.

35.6.3. ML Model Selection for Predictions

```

from sklearn.linear_model import LinearRegression, Ridge
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.model_selection import cross_val_score
from sklearn.preprocessing import StandardScaler

def evaluate_prediction_models(X: np.ndarray, y: np.ndarray) -> Dict:
    """

```

```

Evaluate different ML models for prediction quality.

For learning-augmented algorithms, we care about:
1. Prediction accuracy (lower error)
2. Calibration (errors are well-distributed)
3. Robustness (few extreme errors)
"""

# Scale features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

models = {
    'Linear': LinearRegression(),
    'Ridge': Ridge(alpha=1.0),
    'RandomForest': RandomForestRegressor(n_estimators=50, random_state=42),
    'GradientBoosting': GradientBoostingRegressor(n_estimators=50,
        ↪ random_state=42),
}

results = {}

for name, model in models.items():
    # Cross-validation for MAE and R^2
    mae_scores = -cross_val_score(model, X_scaled, y,
                                   scoring='neg_mean_absolute_error', cv=5)
    r2_scores = cross_val_score(model, X_scaled, y,
                                scoring='r2', cv=5)

    results[name] = {
        'mae_mean': np.mean(mae_scores),
        'mae_std': np.std(mae_scores),
        'r2_mean': np.mean(r2_scores),
        'r2_std': np.std(r2_scores),
    }

return results

# Generate synthetic prediction task
np.random.seed(42)

n_samples = 500
n_features = 10

# Features: some informative, some noise
X = np.random.randn(n_samples, n_features)
# Target: linear combination + noise
true_weights = np.array([2.0, -1.5, 1.0, 0.5, 0, 0, 0, 0, 0, 0])
y = X @ true_weights + np.random.normal(0, 2, n_samples)

print("=== ML Model Selection for Predictions ===\n")
print(f"Samples: {n_samples}, Features: {n_features}\n")

```

```

results = evaluate_prediction_models(X, y)

print(f"{'Model':<20} {'MAE (mean +/- std)':<25} {'R^2 (mean +/- std)':<25}")
print("-" * 70)

for name, metrics in results.items():
    mae_str = f"{metrics['mae_mean']:.3f} +/- {metrics['mae_std']:.3f}"
    r2_str = f"{metrics['r2_mean']:.3f} +/- {metrics['r2_std']:.3f}"
    print(f"{'name':<20} {'mae_str':<25} {'r2_str':<25}")

print("\nRecommendation: Choose the model with best MAE,")
print("but also consider prediction variance for robustness!")

```

=== ML Model Selection for Predictions ===

Samples: 500, Features: 10

Model	MAE (mean +/- std)	R ² (mean +/- std)
Linear	1.548 +/- 0.127	0.642 +/- 0.065
Ridge	1.548 +/- 0.127	0.642 +/- 0.065
RandomForest	1.718 +/- 0.118	0.569 +/- 0.059
GradientBoosting	1.664 +/- 0.129	0.581 +/- 0.059

Recommendation: Choose the model with best MAE,
but also consider prediction variance for robustness!

35.7. Case Study: Resource Allocation with Predictions

This section walks through a complete end-to-end example: formulating the problem, training a prediction model, implementing a learning-augmented algorithm, and evaluating performance across different prediction quality levels.

35.7.1. Problem Formulation

Consider a cloud resource allocation problem where we must decide how many instances to provision before knowing the actual demand. Over-provisioning wastes resources; under-provisioning incurs penalty costs from service degradation.

```

@dataclass
class ResourceAllocationProblem:
    """
    Cloud resource allocation with predictions.

    - Provision instances before demand is known
    - Under-provisioning incurs high cost (SLA violations)
    - Over-provisioning wastes money
    """

    # Cost parameters
    instance_cost: float = 1.0          # Cost per instance per time unit

```

```

sla_violation_cost: float = 10.0 # Cost per unit of unmet demand

# Capacity
instance_capacity: float = 100.0 # Each instance handles this much demand
max_instances: int = 50          # Maximum instances available

def compute_cost(self, instances_provisioned: int,
                  actual_demand: float) -> Dict:
    """
    Compute cost given provisioning decision and actual demand.
    """
    capacity = instances_provisioned * self.instance_capacity

    # Instance costs
    instance_cost = instances_provisioned * self.instance_cost

    # SLA violation cost (unmet demand)
    unmet_demand = max(0, actual_demand - capacity)
    sla_cost = unmet_demand * self.sla_violation_cost / self.instance_capacity

    # Waste (over-provisioning)
    waste = max(0, capacity - actual_demand)

    return {
        'instance_cost': instance_cost,
        'sla_cost': sla_cost,
        'total_cost': instance_cost + sla_cost,
        'unmet_demand': unmet_demand,
        'waste': waste,
        'utilization': actual_demand / capacity if capacity > 0 else 0
    }

def optimal_instances(self, demand: float) -> int:
    """Compute optimal number of instances for known demand."""
    return min(self.max_instances,
               int(np.ceil(demand / self.instance_capacity)))

class ResourceAllocator:
    """
    Learning-augmented resource allocator.
    """

    def __init__(self, problem: ResourceAllocationProblem,
                  trust_parameter: float = 0.5):
        self.problem = problem
        self.trust = trust_parameter

    def allocate(self, predicted_demand: float,
                  historical_demands: List[float] = None) -> int:
        """
        Decide how many instances to provision.

```

```

Strategy:
- Blend predicted demand with historical safety margin
- Higher trust = closer to prediction
- Lower trust = add more safety buffer
"""
# Prediction-based allocation
pred_instances = self.problem.optimal_instances(predicted_demand)

# Robust baseline: use historical max or add buffer
if historical_demands and len(historical_demands) > 0:
    historical_max = max(historical_demands)
    robust_instances = self.problem.optimal_instances(
        historical_max * 1.2 # 20% buffer
    )
else:
    # Conservative: assume high demand
    robust_instances = self.problem.max_instances // 2

# Blend based on trust
blended = self.trust * pred_instances + (1 - self.trust) * robust_instances

return min(self.problem.max_instances, max(1, int(round(blended))))

# Simulate resource allocation
print("=== Resource Allocation with Predictions ===\n")

problem = ResourceAllocationProblem(
    instance_cost=1.0,
    sla_violation_cost=10.0,
    instance_capacity=100.0,
    max_instances=50
)

# Historical demand pattern (for training and robust baseline)
np.random.seed(42)
n_periods = 100

# Demand follows weekly pattern + trend + noise
time = np.arange(n_periods)
base_demand = 2000 + 500 * np.sin(2 * np.pi * time / 7) # Weekly cycle
trend = 10 * time # Growing trend
noise = np.random.normal(0, 300, n_periods)
demands = base_demand + trend + noise
demands = np.maximum(demands, 100) # Minimum demand

print(f"Demand statistics:")
print(f"  Mean: {np.mean(demands):.1f}")
print(f"  Std: {np.std(demands):.1f}")
print(f"  Min: {np.min(demands):.1f}")
print(f"  Max: {np.max(demands):.1f}")

```

=== Resource Allocation with Predictions ===

Demand statistics:

Mean: 2467.8
 Std: 511.9
 Min: 1165.1
 Max: 3622.1

35.7.2. Prediction Model Training

We train a gradient boosting model on lagged demand features to predict next-period demand. The prediction error distribution determines which trust level to use.

```
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.metrics import mean_absolute_error, mean_absolute_percentage_error

def create_features(demands: np.ndarray, lookback: int = 7) -> Tuple[np.ndarray,
    np.ndarray]:
    """
    Create features for demand prediction.

    Features:
    - Past demands (lookback window)
    - Day of week
    - Rolling statistics
    """
    X = []
    y = []

    for i in range(lookback, len(demands)):
        features = []

        # Past demands
        features.extend(demands[i-lookback:i])

        # Day of week (one-hot encoded)
        day = i % 7
        day_features = [1 if d == day else 0 for d in range(7)]
        features.extend(day_features)

        # Rolling statistics
        features.append(np.mean(demands[i-lookback:i]))
        features.append(np.std(demands[i-lookback:i]))
        features.append(np.max(demands[i-lookback:i]))

        X.append(features)
        y.append(demands[i])

    return np.array(X), np.array(y)

# Prepare data
lookback = 7
X, y = create_features(demands, lookback)
```

```

# Train/test split (temporal)
train_size = int(0.7 * len(X))
X_train, X_test = X[:train_size], X[train_size:]
y_train, y_test = y[:train_size], y[train_size:]

print("=== Training Demand Prediction Model ===\n")
print(f"Training samples: {len(X_train)}")
print(f"Test samples: {len(X_test)}")

# Train model
model = GradientBoostingRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Evaluate
train_pred = model.predict(X_train)
test_pred = model.predict(X_test)

print(f"\nTraining MAE: {mean_absolute_error(y_train, train_pred):.1f}")
print(f"Test MAE: {mean_absolute_error(y_test, test_pred):.1f}")
print(f"Test MAPE: {mean_absolute_percentage_error(y_test, test_pred)*100:.1f}%")

# Analyze prediction errors
errors = test_pred - y_test
print(f"\nPrediction Error Distribution:")
print(f"  Mean: {np.mean(errors):.1f}")
print(f"  Std: {np.std(errors):.1f}")
print(f"  Max over-prediction: {np.max(errors):.1f}")
print(f"  Max under-prediction: {-np.min(errors):.1f}")

```

=== Training Demand Prediction Model ===

Training samples: 65

Test samples: 28

Training MAE: 24.6

Test MAE: 342.6

Test MAPE: 12.9%

Prediction Error Distribution:

Mean: 81.4

Std: 416.7

Max over-prediction: 1033.7

Max under-prediction: 574.9

35.7.3. Algorithm Implementation

```

def simulate_resource_allocation(problem: ResourceAllocationProblem,
                                demands: np.ndarray,
                                predictions: np.ndarray,
                                trust: float,
                                historical_window: int = 14) -> Dict:

```



```

"""
Simulate resource allocation over multiple periods.
"""
allocator = ResourceAllocator(problem, trust)

results = {
    'costs': [],
    'sla_violations': [],
    'utilizations': [],
    'instances': [],
}

for t in range(len(demands)):
    # Historical data available at time t
    if t > 0:
        history = demands[max(0, t-historical_window):t].tolist()
    else:
        history = []

    # Make allocation decision
    instances = allocator.allocate(predictions[t], history)

    # Compute actual cost
    outcome = problem.compute_cost(instances, demands[t])

    results['costs'].append(outcome['total_cost'])
    results['sla_violations'].append(outcome['unmet_demand'] > 0)
    results['utilizations'].append(outcome['utilization'])
    results['instances'].append(instances)

return {
    'total_cost': sum(results['costs']),
    'avg_cost': np.mean(results['costs']),
    'sla_violation_rate': np.mean(results['sla_violations']),
    'avg_utilization': np.mean(results['utilizations']),
    'avg_instances': np.mean(results['instances']),
}

# Simulate with different trust levels
print("=== Resource Allocation Simulation ===\n")

# Use test set for simulation
test_demands = y_test
test_predictions = test_pred

# Also compute optimal (with perfect foresight)
optimal_costs = []
for demand in test_demands:
    optimal_instances = problem.optimal_instances(demand)
    outcome = problem.compute_cost(optimal_instances, demand)
    optimal_costs.append(outcome['total_cost'])
optimal_total = sum(optimal_costs)

```

```

print(f"Optimal total cost (perfect foresight): {optimal_total:.1f}\n")

print(f"{'Trust':<10} {'Total Cost':<15} {'Ratio':<10} {'SLA Viol %':<15} {'Util  

↪ %':<10}")
print("-" * 60)

for trust in [0.0, 0.3, 0.5, 0.7, 0.9, 1.0]:
    results = simulate_resource_allocation(
        problem, test_demands, test_predictions, trust
    )
    ratio = results['total_cost'] / optimal_total
    print(f"{'trust':<10.1f} {'results['total_cost']':<15.1f} {'ratio':<10.2f} "
          f"{'results['sla_violation_rate']*100':<15.1f} "
          f"{'results['avg_utilization']*100':<10.1f}")

print("\nKey insights:")
print("- Low trust: Higher costs from over-provisioning, but fewer SLA violations")
print("- High trust: Lower costs but more SLA violations when predictions fail")
print("- Balanced trust (0.5-0.7): Good tradeoff for this scenario")

```

=== Resource Allocation Simulation ===

Optimal total cost (perfect foresight): 804.0

Trust	Total Cost	Ratio	SLA Viol %	Util %
0.0	1238.7	1.54	3.6	68.6
0.3	1117.7	1.39	3.6	74.5
0.5	1040.5	1.29	10.7	79.6
0.7	994.5	1.24	10.7	85.2
0.9	1058.8	1.32	32.1	92.4
1.0	1137.5	1.41	35.7	96.6

Key insights:

- Low trust: Higher costs from over-provisioning, but fewer SLA violations
- High trust: Lower costs but more SLA violations when predictions fail
- Balanced trust (0.5-0.7): Good tradeoff for this scenario

35.7.4. Performance Analysis

```

def analyze_performance_by_prediction_quality(
    problem: ResourceAllocationProblem,
    demands: np.ndarray,
    n_trials: int = 50
) -> Dict:
    """
    Analyze how performance varies with prediction quality.
    """
    results = {

```

```

        'error_levels': [],
        'trust_0.3_ratio': [],
        'trust_0.7_ratio': [],
        'trust_1.0_ratio': [],
    }

    # Optimal cost for reference
    optimal_costs = [problem.compute_cost(problem.optimal_instances(d),
↪ d)['total_cost']
                     for d in demands]
    optimal_total = sum(optimal_costs)

    error_levels = [0.0, 0.05, 0.1, 0.2, 0.3, 0.5]

    for error in error_levels:
        ratios = {0.3: [], 0.7: [], 1.0: []}

        for _ in range(n_trials):
            # Generate predictions with specified error level
            noise = np.random.normal(0, error, len(demands))
            predictions = demands * (1 + noise)
            predictions = np.maximum(predictions, 100)

            for trust in [0.3, 0.7, 1.0]:
                result = simulate_resource_allocation(
                    problem, demands, predictions, trust
                )
                ratios[trust].append(result['total_cost'] / optimal_total)

        results['error_levels'].append(error)
        results['trust_0.3_ratio'].append(np.mean(ratios[0.3]))
        results['trust_0.7_ratio'].append(np.mean(ratios[0.7]))
        results['trust_1.0_ratio'].append(np.mean(ratios[1.0]))

    return results

print("=== Performance vs Prediction Quality ===\n")

analysis = analyze_performance_by_prediction_quality(problem, y_test[:30],
↪ n_trials=30)

print(f"{'Error Level':<15} {'Trust=0.3':<15} {'Trust=0.7':<15} {'Trust=1.0':<15}")
print("-" * 60)

for i, error in enumerate(analysis['error_levels']):
    print(f"{'error*100:>6.0f}% "
          f"{'analysis[trust_0.3_ratio][i]:<15.3f} "
          f"{'analysis[trust_0.7_ratio][i]:<15.3f} "
          f"{'analysis[trust_1.0_ratio][i]:<15.3f}")

print("\nConclusion:")
print("- At low error: High trust performs best")

```

```
print("- At high error: Low trust provides robustness")
print("- Moderate trust (0.7) offers good performance across error levels")
```

=== Performance vs Prediction Quality ===

Error Level	Trust=0.3	Trust=0.7	Trust=1.0
0%	1.381	1.156	1.000
5%	1.377	1.160	1.115
10%	1.381	1.183	1.304
20%	1.375	1.288	1.722
30%	1.378	1.505	2.165
50%	1.391	1.848	2.785

Conclusion:

- At low error: High trust performs best
- At high error: Low trust provides robustness
- Moderate trust (0.7) offers good performance across error levels

35.8. Comparison with Pure ML and Pure Online

35.8.1. Advantages of the Hybrid Approach

```
def compare_approaches(problem: ResourceAllocationProblem,
                        demands: np.ndarray,
                        predictions: np.ndarray) -> Dict:
    """
    Compare pure online, pure ML, and learning-augmented approaches.
    """
    n = len(demands)

    # Optimal (offline)
    optimal_costs = [problem.compute_cost(problem.optimal_instances(d),
    ↪ d['total_cost']) for d in demands]
    optimal_total = sum(optimal_costs)

    # Pure Online: Use historical max with buffer
    online_costs = []
    online_violations = 0
    for t in range(n):
        if t == 0:
            instances = problem.max_instances // 2
        else:
            historical_max = max(demands[:t])
            instances = problem.optimal_instances(historical_max * 1.3)

        outcome = problem.compute_cost(instances, demands[t])
        online_costs.append(outcome['total_cost'])
        if outcome['unmet_demand'] > 0:
            online_violations += 1
```

```

# Pure ML: Trust predictions completely
ml_costs = []
ml_violations = 0
for t in range(n):
    instances = problem.optimal_instances(predictions[t])
    outcome = problem.compute_cost(instances, demands[t])
    ml_costs.append(outcome['total_cost'])
    if outcome['unmet_demand'] > 0:
        ml_violations += 1

# Learning-Augmented (trust = 0.6)
la_result = simulate_resource_allocation(
    problem, demands, predictions, trust=0.6
)

return {
    'optimal': optimal_total,
    'online_cost': sum(online_costs),
    'online_ratio': sum(online_costs) / optimal_total,
    'online_violations': online_violations / n,
    'ml_cost': sum(ml_costs),
    'ml_ratio': sum(ml_costs) / optimal_total,
    'ml_violations': ml_violations / n,
    'la_cost': la_result['total_cost'],
    'la_ratio': la_result['total_cost'] / optimal_total,
    'la_violations': la_result['sla_violation_rate'],
}

print("=== Comparison: Online vs ML vs Learning-Augmented ===\n")

comparison = compare_approaches(problem, y_test, test_pred)

approaches = [
    ('Optimal (offline)', comparison['optimal'], 1.0, 0),
    ('Pure Online', comparison['online_cost'], comparison['online_ratio'],
     comparison['online_violations']),
    ('Pure ML', comparison['ml_cost'], comparison['ml_ratio'],
     comparison['ml_violations']),
    ('Learning-Augmented', comparison['la_cost'], comparison['la_ratio'],
     comparison['la_violations']),
]

print(f"{'Approach':<25} {'Total Cost':<15} {'Ratio':<10} {'SLA Viol %':<12}")
print("-" * 62)

for name, cost, ratio, violations in approaches:
    print(f"{'name':<25} {'cost':<15.1f} {'ratio':<10.3f} {'violations*100':<12.1f}")

print("\nKey Observations:")
print("1. Pure Online: Most robust but over-provisions (high cost)")
print("2. Pure ML: Best when predictions accurate, risky otherwise")

```

```
print("3. Learning-Augmented: Balanced - good cost + acceptable violations")
```

=== Comparison: Online vs ML vs Learning-Augmented ===

Approach	Total Cost	Ratio	SLA Viol %
Optimal (offline)	804.0	1.000	0.0
Pure Online	1345.7	1.674	3.6
Pure ML	1137.5	1.415	35.7
Learning-Augmented	1018.5	1.267	10.7

Key Observations:

1. Pure Online: Most robust but over-provisions (high cost)
2. Pure ML: Best when predictions accurate, risky otherwise
3. Learning-Augmented: Balanced - good cost + acceptable violations

35.8.2. When to Use Which Method

```
@dataclass
class ProblemCharacteristics:
    """Characteristics of the problem that influence method choice."""
    prediction_quality: str # 'high', 'medium', 'low'
    failure_cost: str      # 'catastrophic', 'moderate', 'low'
    data_availability: str  # 'abundant', 'moderate', 'scarce'
    stationarity: str       # 'stationary', 'drifting', 'volatile'

def recommend_approach(characteristics: ProblemCharacteristics) -> str:
    """
    Recommend an approach based on problem characteristics.
    """
    score = 0 # Higher = more ML, Lower = more online

    # Prediction quality
    if characteristics.prediction_quality == 'high':
        score += 2
    elif characteristics.prediction_quality == 'medium':
        score += 0
    else:
        score -= 2

    # Failure cost
    if characteristics.failure_cost == 'catastrophic':
        score -= 3
    elif characteristics.failure_cost == 'moderate':
        score -= 1
    else:
        score += 1

    # Data availability
    if characteristics.data_availability == 'abundant':
        score += 1
```

```

elif characteristics.data_availability == 'scarce':
    score -= 1

# Stationarity
if characteristics.stationarity == 'stationary':
    score += 1
elif characteristics.stationarity == 'volatile':
    score -= 2

# Recommendation
if score >= 3:
    return "Pure ML (with monitoring)"
elif score <= -3:
    return "Pure Online (ignore predictions)"
else:
    trust = 0.5 + 0.1 * score # Adjust trust based on score
    return f"Learning-Augmented (trust ~{trust:.1f})"

# Example scenarios
scenarios = [
    ("Cloud auto-scaling",
     ProblemCharacteristics('medium', 'moderate', 'abundant', 'drifting')),
    ("Safety-critical systems",
     ProblemCharacteristics('medium', 'catastrophic', 'moderate', 'stationary')),
    ("Ad bidding",
     ProblemCharacteristics('high', 'low', 'abundant', 'drifting')),
    ("New market entry",
     ProblemCharacteristics('low', 'moderate', 'scarce', 'volatile')),
    ("Network capacity",
     ProblemCharacteristics('high', 'moderate', 'abundant', 'stationary')),
]

print("=== Decision Framework: Choosing the Right Approach ===\n")

for name, chars in scenarios:
    recommendation = recommend_approach(chars)
    print(f"{name}:")
    print(f"  Prediction Quality: {chars.prediction_quality}")
    print(f"  Failure Cost: {chars.failure_cost}")
    print(f"  Data: {chars.data_availability}, Stationarity: {chars.stationarity}")
    print(f"  RECOMMENDATION: {recommendation}")
    print()

```

=== Decision Framework: Choosing the Right Approach ===

Cloud auto-scaling:

Prediction Quality: medium

Failure Cost: moderate

Data: abundant, Stationarity: drifting

RECOMMENDATION: Learning-Augmented (trust ~0.5)

Safety-critical systems:

Prediction Quality: medium
Failure Cost: catastrophic
Data: moderate, Stationarity: stationary
RECOMMENDATION: Learning-Augmented (trust ~0.3)

Ad bidding:

Prediction Quality: high
Failure Cost: low
Data: abundant, Stationarity: drifting
RECOMMENDATION: Pure ML (with monitoring)

New market entry:

Prediction Quality: low
Failure Cost: moderate
Data: scarce, Stationarity: volatile
RECOMMENDATION: Pure Online (ignore predictions)

Network capacity:

Prediction Quality: high
Failure Cost: moderate
Data: abundant, Stationarity: stationary
RECOMMENDATION: Pure ML (with monitoring)

Practical Guidelines

Use Pure ML when: - Predictions are highly accurate - Failure costs are low or recoverable
- You have abundant training data - The environment is stationary

Use Pure Online when: - Predictions are unreliable - Failures are catastrophic - Data is scarce or non-stationary - You need provable worst-case guarantees

Use Learning-Augmented when: - Prediction quality varies - You want both good average and worst-case performance - The tradeoff between consistency and robustness matters - You can tune the trust parameter based on observed performance

35.8.3. Implementation Checklist

```
checklist = {  
    "Problem Modeling": [  
        "Define the online decision problem clearly",  
        "Identify what predictions would be useful",  
        "Understand the competitive ratio of best online algorithm",  
        "Define error metrics for predictions",  
    ],  
    "Prediction Design": [  
        "Collect historical data for training",  
        "Choose appropriate ML model for prediction type",  
        "Evaluate prediction quality on held-out data",  
        "Analyze prediction error distribution",  
    ],  
    "Algorithm Design": [  

```



```

        "Implement base online algorithm (for robustness)",
        "Design prediction-aware modification",
        "Parameterize trust level",
        "Prove or empirically verify consistency/robustness",
    ],
    "Deployment": [
        "Monitor prediction accuracy over time",
        "Implement adaptive trust adjustment",
        "Set up alerts for distribution shift",
        "Have fallback to pure online if predictions degrade",
    ],
}

print("=== Learning-Augmented Algorithm Implementation Checklist ===\n")

for phase, items in checklist.items():
    print(f"{phase}:")
    for i, item in enumerate(items, 1):
        print(f"  [{i}] {item}")
    print()

```

=== Learning-Augmented Algorithm Implementation Checklist ===

Problem Modeling:

- [] 1. Define the online decision problem clearly
- [] 2. Identify what predictions would be useful
- [] 3. Understand the competitive ratio of best online algorithm
- [] 4. Define error metrics for predictions

Prediction Design:

- [] 1. Collect historical data for training
- [] 2. Choose appropriate ML model for prediction type
- [] 3. Evaluate prediction quality on held-out data
- [] 4. Analyze prediction error distribution

Algorithm Design:

- [] 1. Implement base online algorithm (for robustness)
- [] 2. Design prediction-aware modification
- [] 3. Parameterize trust level
- [] 4. Prove or empirically verify consistency/robustness

Deployment:

- [] 1. Monitor prediction accuracy over time
- [] 2. Implement adaptive trust adjustment
- [] 3. Set up alerts for distribution shift
- [] 4. Have fallback to pure online if predictions degrade

35.9. Summary

35.9.0.1. Key Takeaways

Learning-Augmented Framework

- Combines ML predictions with online algorithms
- Achieves good performance when predictions are accurate (consistency)
- Maintains bounded performance when predictions fail (robustness)
- The consistency-robustness tradeoff is fundamental and often Pareto-optimal

Classic Problems

- **Ski Rental:** Decide when to switch from renting to buying
- **Caching:** Predict which pages will be accessed next
- **Knapsack:** Predict item values to make accept/reject decisions
- **Scheduling:** Predict job processing times for better ordering

Prediction Design

- Different prediction types: outcome, structural, demand, duration, ranking
- Error metrics should match how the algorithm uses predictions
- Model selection should consider both accuracy and robustness

Practical Guidance

- Trust parameter controls the consistency-robustness tradeoff
- Monitor prediction quality and adapt trust level accordingly
- Learning-augmented algorithms are most valuable when:
 - Prediction quality varies across inputs
 - Both average and worst-case performance matter
 - The application can tolerate some prediction errors

Mathematical Foundations

- Competitive ratio measures online performance vs optimal offline
- Impossibility results bound achievable tradeoffs
- Smoothness describes graceful degradation with prediction error

35.10. Exercises

1. **Ski Rental Analysis:** Implement the ski rental algorithm with predictions and empirically verify that the consistency-robustness product is at least 4 for any trust parameter.
2. **Caching Predictor:** Build a caching predictor using recurrent neural networks that learns page access patterns. Compare its performance with LRU for different cache sizes.
3. **Scheduling Robustness:** Design a scheduling algorithm that achieves 1.5-consistency and 3-robustness. Prove or disprove that this is achievable.
4. **Error Calibration:** Implement a method to calibrate the trust parameter based on observed prediction errors. Test it on a streaming dataset with concept drift.
5. **Novel Problem:** Choose an online problem not covered in this chapter (e.g., online matching, secretary problem) and design a learning-augmented algorithm for it. Analyze its consistency and robustness.
6. **Practical Deployment:** Take the resource allocation case study and:
 - Add concept drift detection

- Implement adaptive trust adjustment
- Compare with a reinforcement learning baseline

35.11. References

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Part VI.

Part VI: Production Systems

36. Solvers and Tools

What You Will Learn

- Compare commercial solvers (Gurobi, CPLEX, Xpress, MOSEK) vs. open source (HiGHS, CBC, SCIP)
- Choose the right solver for your problem type: LP/MIP, CP, nonlinear, conic, or metaheuristic
- Understand nonlinear and global optimization solvers (Ipopt, BARON, Knitro)
- Tune solver parameters for performance
- Use modeling languages (PuLP, Pyomo, OR-Tools)

The right tool makes all the difference. This chapter guides you through the optimization software ecosystem, helping you select appropriate solvers and modeling frameworks for your specific needs.

36.1. The Solver Ecosystem

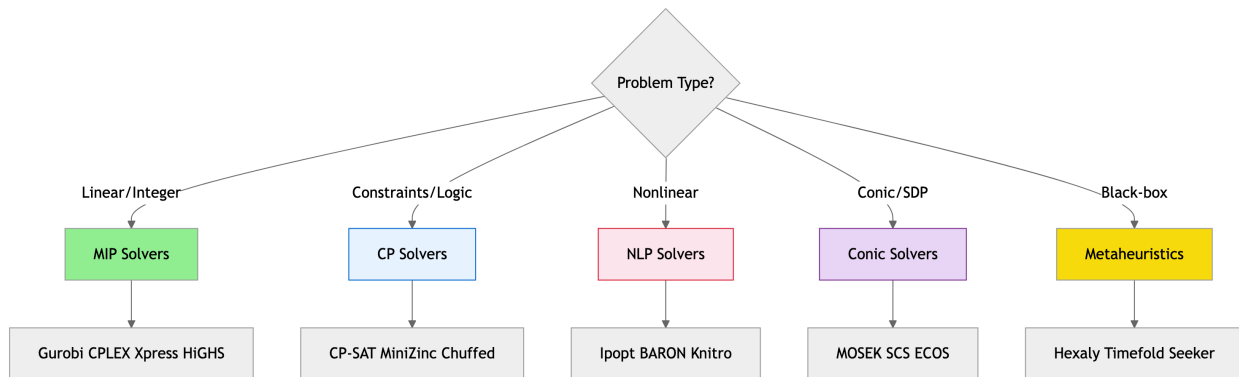


Figure 36.1.: Solver Selection Guide based on Problem Type

36.1.1. Commercial vs. Open Source

Aspect	Commercial	Open Source
Performance	State-of-the-art	Good to excellent
Support	Professional support	Community-driven
Cost	\$10K-100K/year	Free
Licensing	Restrictive	Flexible
Updates	Regular	Variable
Documentation	Comprehensive	Varies

36.1.2. Solver Categories

Solvers specialize in different problem types:

Category	Problem Types	Example Solvers
LP/MIP	Linear and integer programs	Gurobi, CPLEX, Xpress, HiGHS
Conic	SOCP, SDP, exponential cone	MOSEK, SCS, ECOS
Constraint Programming	Satisfaction and scheduling	OR-Tools CP-SAT, MiniZinc, Chuffed
Nonlinear (local)	NLP	Ipopt, Knitro
Global nonlinear	Non-convex NLP, MINLP	BARON, Couenne
Convex MINLP	Convex mixed-integer nonlinear	Bonmin
SAT/SMT	Boolean satisfiability	Z3, MiniSat, CaDiCaL
Metaheuristics	Black-box, combinatorial	Hexaly, Timefold, InsideOpt Seeker
Specialized	TSP, network flows	Concorde, LKH

36.2. Commercial Solvers

36.2.1. Gurobi

Gurobi is widely regarded as the fastest commercial solver for linear and mixed-integer programming.

Strengths:

- Industry-leading performance on MIP
- Excellent Python API (gurobipy)
- Advanced features: lazy constraints, callbacks, multi-objective
- Free academic licenses

Supported Problem Types:

- Linear Programming (LP)
- Mixed-Integer Linear Programming (MIP)
- Quadratic Programming (QP)
- Mixed-Integer Quadratic Programming (MIQP)
- Quadratically Constrained Programming (QCP)
- Bilinear programming

```
import gurobipy as gp
from gurobipy import GRB

def gurobi_facility_location(fixed_costs, transport_costs):
    """
    Solve facility location with Gurobi.

    Args:
        fixed_costs: (m,) array of opening costs
        transport_costs: (m, n) array of transport costs

    Returns:
        facilities: List of open facility indices
        objective: Optimal cost
    """
    m = len(fixed_costs)
    n = transport_costs.shape[1]

    model = gp.Model('facility_location')
```



```

# Suppress output
model.Params.OutputFlag = 0

# Decision variables
y = model.addVars(m, vtype=GRB.BINARY, name='open')
x = model.addVars(m, n, vtype=GRB.CONTINUOUS, lb=0, ub=1, name='assign')

# Constraints
# Each customer assigned
model.addConstrs(
    (gp.quicksum(x[i, j] for i in range(m)) == 1 for j in range(n)),
    name='demand'
)

# Only assign to open facilities
model.addConstrs(
    (x[i, j] <= y[i] for i in range(m) for j in range(n)),
    name='open_if_used'
)

# Objective
model.setObjective(
    gp.quicksum(fixed_costs[i] * y[i] for i in range(m)) +
    gp.quicksum(transport_costs[i, j] * x[i, j]
                for i in range(m) for j in range(n)),
    GRB.MINIMIZE
)

model.optimize()

if model.Status == GRB.OPTIMAL:
    facilities = [i for i in range(m) if y[i].X > 0.5]
    return facilities, model.ObjVal

return None, None

```

36.2.2. IBM CPLEX

CPLEX offers comprehensive optimization capabilities with strong constraint programming support.

Strengths:

- Mature, battle-tested solver
- Integrated LP, MIP, and CP
- Strong decomposition methods
- Extensive parameter tuning options

Integration with docplex:

```

from docplex.mp.model import Model

def cplex_production_planning(demand, capacity, production_cost, inventory_cost):
    """

```

```

Production planning with CPLEX.

Args:
    demand: (T,) demand per period
    capacity: Maximum production per period
    production_cost: Cost per unit produced
    inventory_cost: Cost per unit held in inventory

Returns:
    production: Optimal production schedule
    inventory: Optimal inventory levels
"""
T = len(demand)

model = Model('production_planning')

# Variables
production = model.continuous_var_list(T, lb=0, ub=capacity, name='prod')
inventory = model.continuous_var_list(T, lb=0, name='inv')

# Inventory balance
for t in range(T):
    if t == 0:
        model.add_constraint(
            production[t] == demand[t] + inventory[t]
        )
    else:
        model.add_constraint(
            inventory[t-1] + production[t] == demand[t] + inventory[t]
        )

# Objective
model.minimize(
    model.sum(production_cost * production[t] + inventory_cost * inventory[t]
               for t in range(T))
)

solution = model.solve()

if solution:
    return (
        [production[t].solution_value for t in range(T)],
        [inventory[t].solution_value for t in range(T)]
    )

return None, None

```

36.2.3. FICO Xpress

Xpress is one of the “Big Three” commercial LP/MIP solvers alongside Gurobi and CPLEX, with a long history dating back to 1983 as the first commercial LP solver on PCs.

Strengths:

- Competitive LP/MIP performance, strong on hard instances

- **GPU acceleration** via PDHG (Primal-Dual Hybrid Gradient) algorithm with up to 50x speedups on suitable problems
- Integrated ecosystem: Mosel modeling language, Xpress Workbench IDE, Xpress Insight (enterprise deployment)
- Xpress Global: world's first commercial global MINLP solver (2022)
- Community Edition available free (up to 5,000 variables/constraints)

Supported Problem Types:

- LP, MIP, QP, MIQP, QCQP, SOCP
- Successive Linear Programming (SLP)
- Global nonconvex MINLP (Xpress Global)

```
import xpress as xp

def xpress_portfolio_optimization(returns, cov_matrix, risk_budget):
    """Portfolio optimization with Xpress (QP)."""
    n = len(returns)
    prob = xp.problem()

    # Decision variables: portfolio weights
    w = [xp.var(lb=0, ub=1, name=f'w_{i}') for i in range(n)]
    prob.addVariable(w)

    # Budget constraint: weights sum to 1
    prob.addConstraint(xp.Sum(w) == 1)

    # Risk budget constraint (quadratic)
    risk = xp.Sum(cov_matrix[i][j] * w[i] * w[j]
                  for i in range(n) for j in range(n))
    prob.addConstraint(risk <= risk_budget)

    # Maximize expected return
    prob.setObjective(xp.Sum(returns[i] * w[i] for i in range(n)),
                     sense=xp.maximize)

    prob.solve()
    return [w[i].val for i in range(n)]
```

i Mosel: A Unique Modeling Approach

Xpress includes **Mosel**, a full programming language that combines algebraic modeling with procedural logic. Unlike Python-based interfaces, Mosel compiles to platform-independent binary (BIM) files, enabling identical execution across Windows, Linux, and macOS. This makes it particularly suitable for enterprise deployments where model portability is critical.

36.2.4. MOSEK

MOSEK specializes in conic and semidefinite optimization, areas where general-purpose LP/MIP solvers are not competitive.

Strengths:

- Industry-leading interior-point solver for SOCP and SDP

- Exceptional numerical stability on ill-conditioned problems
- Supports exotic cone types: exponential cone, power cone
- Backend for CVXPY, YALMIP, and JuMP
- Free academic licenses

Supported Problem Types:

- LP, QP
- Second-Order Cone Programming (SOCP)
- Semidefinite Programming (SDP)
- Exponential and power cone programming
- Mixed-integer variants of all above

```
import mosek.fusion as mf

def mosek_robust_portfolio(returns, cov_factor, alpha):
    """
    Robust portfolio optimization using SOCP.

    Uses conic constraints to model the uncertainty set
    for robust worst-case return maximization.
    """
    n = len(returns)
    with mf.Model("robust_portfolio") as M:
        w = M.variable("w", n, mf.Domain.greaterThan(0.0))
        t = M.variable("t", 1)

        # Budget constraint
        M.constraint(mf.Expr.sum(w), mf.Domain.equalsTo(1.0))

        # SOC constraint: ||cov_factor @ w|| <= t
        M.constraint(
            mf.Expr.vstack(t, mf.Expr.mul(mf.Matrix.dense(cov_factor), w)),
            mf.Domain.inQCone()
        )

        # Maximize: expected return - alpha * risk
        obj = mf.Expr.sub(
            mf.Expr.dot(returns, w),
            mf.Expr.mul(alpha, t)
        )
        M.objective(mf.ObjectiveSense.Maximize, obj)

    M.solve()
    return w.level()
```

When to use MOSEK: Whenever your problem involves conic constraints—including robust optimization, portfolio optimization with factor models, SDP relaxations in combinatorial optimization, or semidefinite constraints in control theory.

36.3. Metaheuristic and Hybrid Solvers

36.3.1. Hexaly (formerly LocalSolver)

Hexaly is a hybrid solver that combines local search, constraint propagation, and mathematical programming into a unified engine. Rebranded from LocalSolver in November 2023, it reflects an evolution from pure local search toward a comprehensive optimization platform.

Key Differentiator: Set-Based Modeling

Unlike traditional MIP solvers that require thousands of binary variables for routing or sequencing, Hexaly uses **collection variables** (sets, lists, intervals) that directly represent the structure of combinatorial problems:

```
import hexaly.optimizer as gp

def hexaly_vrp(distances, demands, vehicle_capacity, num_vehicles):
    """Vehicle routing with Hexaly's set-based modeling."""
    n = len(demands) # Number of customers
    with gp.HexalyOptimizer() as optimizer:
        model = optimizer.model

        # Each route is a LIST of customers (not binary arc variables)
        routes = [model.list(n) for _ in range(num_vehicles)]

        # Partition constraint: every customer visited exactly once
        model.constraint(model.partition(routes))

        # Capacity and distance for each route
        route_distances = []
        for route in routes:
            # Capacity constraint
            demand_route = model.sum(
                model.at(demands, route[i])
                for i in model.range(0, model.count(route))
            )
            model.constraint(demand_route <= vehicle_capacity)

        model.minimize(model.sum(route_distances))
        optimizer.param.time_limit = 30
        optimizer.solve()
```

Strengths: Non-linear/non-convex objectives, large-scale combinatorial problems (routing, scheduling, packing), fast heuristic solutions, black-box optimization without gradients.

36.3.2. Timefold (formerly OptaPlanner)

Timefold is the open-source continuation of Red Hat's OptaPlanner, built by the original OptaPlanner creator and core team. It is the leading Java/Kotlin constraint solver for planning and scheduling problems.

Strengths:

- 2x faster than OptaPlanner out-of-the-box
- Java 17+, Spring Boot 3, Quarkus 3 native support
- Incremental score calculation for fast local search
- Multi-threaded solving (Enterprise Edition)

- Built-in algorithms: tabu search, simulated annealing, late acceptance, step counting hill climbing
- Nearby selection for spatial problems (e.g., vehicle routing)

Best For: Employee rostering, vehicle routing, task assignment, maintenance scheduling, school timetabling—anywhere Java/Kotlin is the primary technology stack.

⚠️ OptaPlanner End of Life

Red Hat build of OptaPlanner 8 reached end of life in May 2024, with extended support until May 2026. All new projects should use **Timefold** (`org.timefold.solver`) as the direct replacement. Migration requires changing Maven coordinates and package imports; the API is largely compatible.

36.3.3. InsideOpt Seeker

Seeker is a commercial solver from InsideOpt (founded by Dr. Meinolf Sellmann, formerly of IBM Research and Brown University) that uses machine learning to adapt its search strategy to each problem.

Strengths:

- Handles non-differentiable, non-convex, black-box objectives
- **Adaptive search:** offline learning of search parameters that tune themselves during optimization
- **Massively parallel:** a single configuration change runs multiple solvers with automatic coordination
- Stochastic optimization with thousands of scenarios
- True non-hierarchical multi-objective optimization
- ML model predictions can be embedded inside the objective function
- Up to 100x speedups via tuning and proprietary parallelization

Best For: Supply chain optimization under uncertainty, problems mixing ML predictions with optimization, scenarios where standard solvers struggle with non-convex or non-differentiable objectives.

```
from insideopt import seeker

def seeker_example():
    """Stochastic optimization with InsideOpt Seeker."""
    model = seeker.Model()

    # Variables
    x = model.add_variable(lb=0, ub=100, name="production")
    y = model.add_variable(lb=0, ub=100, name="inventory")

    # Constraints
    model.add_constraint(x + y <= 150)

    # Objective (can include non-differentiable functions)
    model.set_objective(seeker.Minimize, 10 * x + 5 * y)

    # Solve with parallel workers
```

```
model.solve(num_workers=8)
return x.value, y.value
```

36.4. DecisionOps Platforms

While individual solvers handle the mathematical optimization, **DecisionOps platforms** address the broader lifecycle: how do you test, deploy, monitor, and iterate on decision models in production? These platforms sit above the solver layer, providing infrastructure for the entire optimization workflow.

36.4.1. Nextmv

[Nextmv](#) is a DecisionOps platform that accelerates decision model development, testing, and deployment. Rather than requiring teams to build custom infrastructure for each optimization model, Nextmv provides a unified workflow from prototype to production.

Core Capabilities:

- **Model integration:** Connect existing optimization code (Python, any solver) to a managed platform with standardized input/output contracts
- **Testing and validation:** Run scenario simulations, acceptance tests, shadow production tests, and switchback experiments to validate model quality against business KPIs before going live
- **Monitoring and auditability:** Centralized logging of all decisions with root cause detection, performance drift alerts, and audit trails
- **Marketplace:** Pre-built decision apps for common problems (routing, scheduling, packing, pricing) that can be customized

Solver Ecosystem: Nextmv integrates with commercial solvers (Gurobi, FICO Xpress, Hexaly, NVIDIA cuOpt), open-source solvers (OR-Tools, HiGHS, VROOM), and modeling languages (AMPL, Pyomo, GAMS). This solver-agnostic approach lets teams swap solvers without changing their deployment infrastructure.

Best For: Teams that need to operationalize multiple optimization models, organizations requiring rigorous testing and validation workflows before production deployment, and enterprises needing audit trails and decision logging for compliance.

Build vs. Buy for Optimization Infrastructure

Building a custom optimization deployment stack (API service, monitoring, A/B testing, model registry) can take months of engineering effort. DecisionOps platforms like Nextmv provide these capabilities out of the box, letting optimization teams focus on modeling rather than infrastructure. The trade-off is vendor dependency versus development speed — similar to the build-vs.-buy decision for MLOps platforms in machine learning.

36.5. Open Source Solvers

36.5.1. HiGHS

HiGHS is a modern, high-performance open-source solver for linear and mixed-integer programming.

Strengths:

- Competitive with commercial solvers
- MIT license (fully open)
- Active development
- Good Python integration

```
import highspy
import numpy as np

def highs_knapsack(values, weights, capacity):
    """
    Solve 0-1 knapsack with HiGHS.

    Args:
        values: Item values
        weights: Item weights
        capacity: Knapsack capacity

    Returns:
        selected: Boolean array of selected items
        total_value: Optimal value
    """
    n = len(values)

    h = highspy.Highs()
    h.setOptionValue('output_flag', False)

    # Variables: binary selection
    h.addVars(n, np.zeros(n), np.ones(n))

    # Set integer type
    for i in range(n):
        h.changeColIntegrality(i, highspy.HighsVarType.kInteger)

    # Capacity constraint
    h.addRow(-highspy.kHighsInf, capacity, n,
             np.arange(n, dtype=np.int32), weights.astype(np.float64))

    # Objective: maximize value
    h.changeColsCost(n, np.arange(n, dtype=np.int32), -values.astype(np.float64))

    h.run()

    solution = h.getSolution()
    selected = np.array(solution.col_value) > 0.5
    total_value = -h.getInfo().objective_function_value

    return selected, total_value
```

36.5.2. CBC (COIN-OR Branch and Cut)

CBC is a reliable open-source MIP solver from the COIN-OR project.

Strengths:

- Mature and stable

- Good default parameters
- Widely used in industry
- Available through PuLP

36.5.3. OR-Tools CP-SAT

Google's CP-SAT solver combines constraint programming with SAT techniques.

Strengths:

- Excellent for scheduling
- Handles complex logical constraints
- Free and well-documented
- Active development by Google

```
from ortools.sat.python import cp_model

def cpsat_job_shop(processing_times, machines):
    """
    Solve job shop scheduling with CP-SAT.

    Args:
        processing_times: (jobs, tasks) array of durations
        machines: (jobs, tasks) array of machine assignments

    Returns:
        schedule: List of (job, task, start, end) tuples
        makespan: Optimal makespan
    """
    num_jobs = len(processing_times)
    num_machines = len(set(m for job in machines for m in job))

    model = cp_model.CpModel()

    # Compute horizon
    horizon = sum(sum(job) for job in processing_times)

    # Create variables
    task_vars = {}
    machine_to_intervals = {m: [] for m in range(num_machines)}

    for job_id, (job_times, job_machines) in enumerate(zip(processing_times,
        ↪ machines)):
        for task_id, (duration, machine) in enumerate(zip(job_times, job_machines)):
            start = model.NewIntVar(0, horizon, f'start_{job_id}_{task_id}')
            end = model.NewIntVar(0, horizon, f'end_{job_id}_{task_id}')
            interval = model.NewIntervalVar(start, duration, end,
                f'interval_{job_id}_{task_id}')

            task_vars[job_id, task_id] = (start, end, interval)
            machine_to_intervals[machine].append(interval)

    # Precedence within jobs
    for job_id, job_times in enumerate(processing_times):
        for task_id in range(len(job_times) - 1):
```

```

        model.Add(task_vars[job_id, task_id + 1][0] >=
                  task_vars[job_id, task_id][1])

# No overlap on machines
for machine in range(num_machines):
    model.AddNoOverlap(machine_to_intervals[machine])

# Minimize makespan
makespan = model.NewIntVar(0, horizon, 'makespan')
model.AddMaxEquality(makespan, [
    task_vars[job_id, len(job) - 1][1]
    for job_id, job in enumerate(processing_times)
])
model.Minimize(makespan)

# Solve
solver = cp_model.CpSolver()
solver.parameters.max_time_in_seconds = 60.0
status = solver.Solve(model)

if status in [cp_model.OPTIMAL, cp_model.FEASIBLE]:
    schedule = []
    for (job_id, task_id), (start, end, _) in task_vars.items():
        schedule.append((
            job_id, task_id,
            solver.Value(start), solver.Value(end)
        ))
    return schedule, solver.Value(makespan)

return None, None

```

36.5.4. SCIP

SCIP (Solving Constraint Integer Programs) is the leading open-source solver for mixed-integer programming research, developed at Zuse Institute Berlin.

Strengths:

- Fastest non-commercial MIP solver
- Highly extensible: custom constraint handlers, branching rules, separators
- Supports MIP, MINLP, and constraint integer programming
- Includes SoPlex (LP solver), ZIMPL (modeling language), GCG (decomposition), UG (parallelization)
- Python API via PySCIPOpt
- Apache 2.0 license (free for academic and non-commercial use)

Best For: Optimization research, implementing novel algorithmic ideas, benchmarking, and problems requiring custom solver extensions.

36.5.5. GLPK

The GNU Linear Programming Kit is an open-source solver for LP and MIP. While slower than modern alternatives like HiGHS, it is widely available on most Linux distributions and commonly used in educational settings. GLPK includes the MathProg modeling language (a subset of AMPL).

36.6. Nonlinear and Global Optimization Solvers

Nonlinear problems arise in chemical process design, engineering, finance, and energy systems. Unlike LP/MIP, the solver landscape for NLP is divided between **local solvers** (fast but find only local optima) and **global solvers** (slower but guarantee global optimality).

36.6.1. Ipopt (Interior Point OPTimizer)

Ipopt is the standard open-source solver for large-scale nonlinear programming. It uses a primal-dual interior-point method with a line-search filter approach.

Strengths:

- Handles millions of variables when problem structure is exploited
- Supports sparse linear algebra (MA27, MA57, MUMPS, Pardiso backends)
- L-BFGS mode when Hessians are unavailable (gradient-only)
- Warm starting from previous solutions
- Default NLP solver in Pyomo, JuMP, and CasADi
- Eclipse Public License (open source)

Best For: Model Predictive Control (MPC), optimal control, trajectory optimization, parameter estimation, constrained machine learning.

```
from pyomo.environ import *

def ipopt_chemical_reactor(feed_rate_bounds, temp_bounds):
    """Optimize a chemical reactor with nonlinear kinetics using Ipopt."""
    model = ConcreteModel()

    # Variables
    model.feed = Var(bounds=feed_rate_bounds)
    model.temp = Var(bounds=temp_bounds)
    model.conversion = Var(bounds=(0, 1))

    # Nonlinear kinetics: Arrhenius equation
    model.kinetics = Constraint(expr=
        model.conversion == 1 - exp(-0.1 * model.feed * exp(-5000 / model.temp))
    )

    # Maximize conversion
    model.obj = Objective(expr=-model.conversion)

    solver = SolverFactory('ipopt')
    solver.options['max_iter'] = 3000
    solver.options['tol'] = 1e-6
    results = solver.solve(model)

    return value(model.feed), value(model.temp), value(model.conversion)
```

36.6.2. Knitro

Knitro (by Artelys) is a commercial NLP and MINLP solver that offers four selectable algorithms:

1. **Interior-point with direct linear algebra:** best for dense Hessians
2. **Interior-point with conjugate gradient:** scales to very large problems
3. **Active-set SQP:** excellent warm starting, many active constraints

4. **Active-set SLQP**: highly nonlinear, difficult constraint geometry

Strengths: Multi-start for non-convex problems, parallel branch-and-bound for MINLP, automatic parameter tuning, and excellent handling of complementarity constraints.

36.6.3. BARON (Branch-And-Reduce Optimization Navigator)

BARON is the world's leading global optimization solver, guaranteeing proven global optimality for non-convex problems through spatial branch-and-bound.

Strengths:

- **Global optimality guarantee** for non-convex NLP and MINLP
- Handles polynomial, rational, and transcendental functions
- Convex relaxations via McCormick envelopes, factorable decomposition, and RLT cuts
- Feasibility-based and optimality-based bound tightening
- Supports bilevel optimization and pooling problems

Best For: Chemical process synthesis, molecular design, pooling/blending problems with bilinear constraints, and any problem where local optima are unacceptable.

Local vs. Global Optimization

Ipopt finds a local optimum quickly and is the right choice when the problem is convex, when a good starting point is available, or when solution speed matters more than global optimality. **BARON** proves global optimality but at significantly higher computational cost. **Knitro** bridges the gap with multi-start capabilities that explore multiple local optima. For convex MINLP problems, **Bonmin** (COIN-OR) provides an open-source alternative that combines CBC and Ipopt.

36.6.4. Couenne and Bonmin (Open-Source Nonlinear)

For users without access to BARON or Knitro, the COIN-OR project provides two open-source alternatives:

- **Bonmin** (Basic Open-source Nonlinear Mixed Integer programming): Solves **convex** MINLP problems to global optimality by combining CBC (MILP) with Ipopt (NLP). It offers multiple algorithms including branch-and-bound, outer approximation, and LP/NLP hybrid methods.
- **Couenne** (Convex Over and Under Envelopes for Nonlinear Estimation): Solves **non-convex** MINLP problems via spatial branch-and-bound, similar to BARON but as a free alternative. Uses McCormick envelopes, secant relaxations, and bound tightening.

Both are accessible through Pyomo and AMPL.

36.7. Constraint Programming Solvers

36.7.1. MiniZinc

MiniZinc is a solver-agnostic constraint modeling language that separates the model from the solver. A single MiniZinc model can be solved by Gecode, Chuffed, CP-SAT, CPLEX, or Gurobi without code changes.

Best For: Benchmarking multiple solvers on the same problem, rapid prototyping, and academic research. MiniZinc Challenge is the annual competition that benchmarks CP solvers.

36.7.2. Additional CP Solvers

Solver	Language	License	Specialty
Gecode	C++	MIT	Research, custom propagators
Chuffed	C++	MIT	Lazy clause generation, MiniZinc default
Choco	Java	BSD	Enterprise Java integration
IBM CP Optimizer	C++/OPL	Commercial	Industrial scheduling, part of CPLEX Studio

36.8. Specialized Solvers

36.8.1. TSP Solvers

For the Traveling Salesman Problem specifically, dedicated solvers dramatically outperform general-purpose MIP solvers:

- **Concorde**: The definitive exact TSP solver, solving instances with 85,900 cities to proven optimality. Uses cutting-plane methods and branch-and-bound with TSP-specific cuts.
- **LKH** (Lin-Kernighan-Helsgott): The best heuristic TSP solver, based on the Lin-Kernighan k -opt improvement algorithm. Routinely finds optimal or near-optimal solutions for instances with millions of cities.

36.8.2. SAT/SMT Solvers

SAT solvers determine Boolean satisfiability and form the backbone of modern CP solvers like CP-SAT. **Z3** (Microsoft Research) extends SAT to Satisfiability Modulo Theories, handling arithmetic, bit-vectors, and arrays. These solvers are increasingly used for verification of optimization model correctness and for encoding combinatorial problems directly.

The radar chart below compares all solvers covered in this chapter across five dimensions: speed, solution quality, memory efficiency, solver support breadth, and documentation quality.

36.9. Solver Comparison

36.9.1. Performance Benchmarks

Solver	Type	LP Speed	MIP Speed	Robustness	Features
Gurobi	Commercial	5/5	5/5	5/5	5/5
CPLEX	Commercial	5/5	5/5	5/5	5/5
Xpress	Commercial	5/5	5/5	5/5	5/5
MOSEK	Commercial	4/5	3/5	5/5	4/5
HiGHS	Open Source	4/5	4/5	4/5	3/5
CBC	Open Source	3/5	3/5	4/5	3/5
CP-SAT	Open Source	N/A	4/5	5/5	5/5
SCIP	Open Source	3/5	4/5	4/5	5/5
Ipopt	Open Source	N/A	N/A	5/5	4/5

Note: MOSEK ratings reflect LP/MIP; for conic problems (SOCP, SDP) it scores 5/5 across all dimensions. Ipopt is rated on NLP performance.

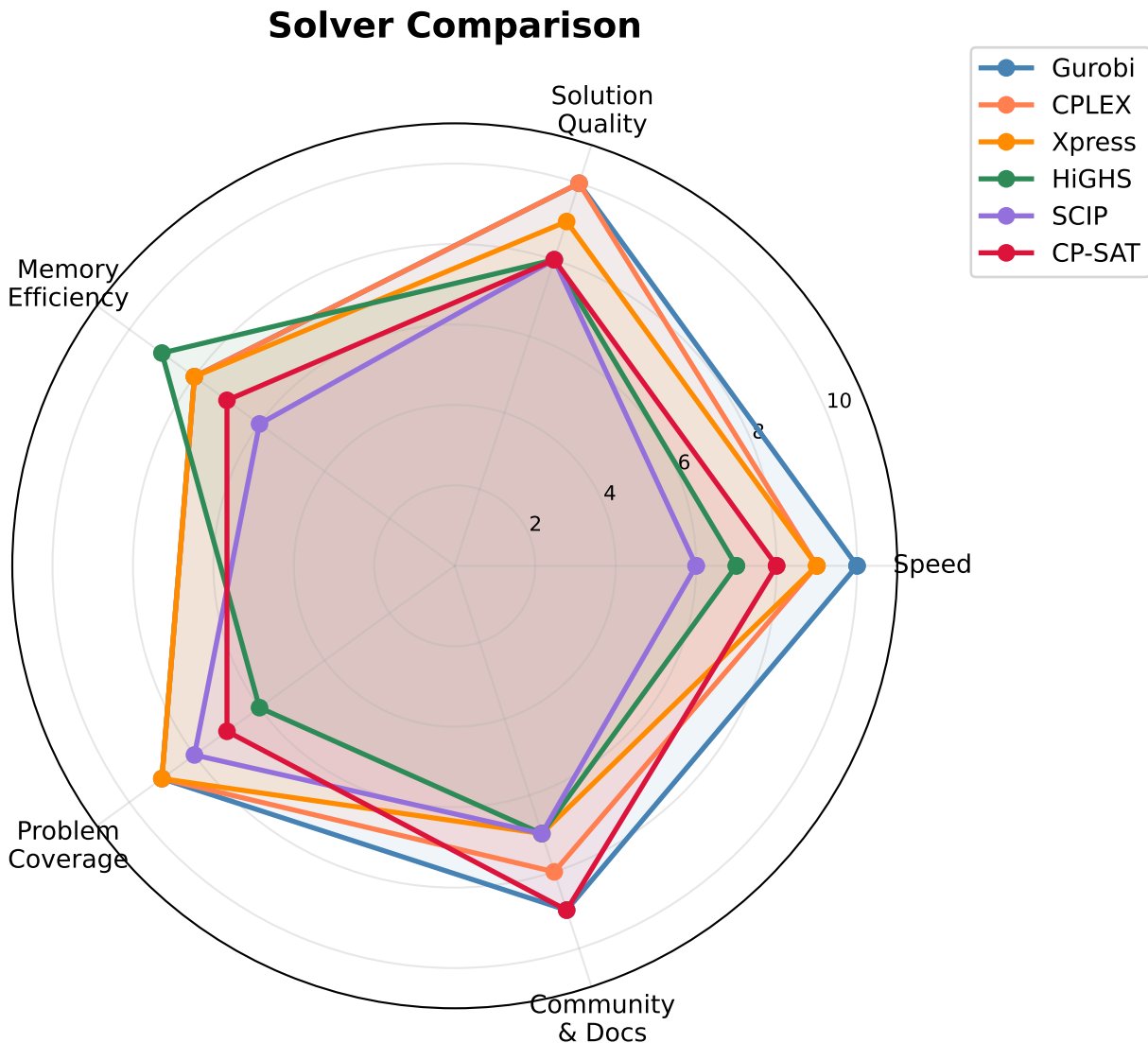


Figure 36.2.: Solver comparison radar chart — comparing key solvers across speed, solution quality, memory efficiency, problem type coverage, and documentation/community support

36.9.2. Decision Guide

What is your problem type?

Linear/Integer Programming

Budget available? → Gurobi, CPLEX, or Xpress

Open source needed? → HiGHS or CBC

Research/extensibility? → SCIP

Conic (SOCP, SDP)

→ MOSEK (or CVXPY with MOSEK backend)

Constraint Programming / Scheduling

→ OR-Tools CP-SAT (or MiniZinc for multi-solver benchmarking)

Nonlinear Optimization

Convex NLP? → Ipopt

Non-convex (need global optimum)? → BARON or Couenne (open source)
 Non-convex (good local solution OK)? → Knitro or Ipopt with multi-start
 Convex MINLP? → Bonmin
 Vehicle Routing
 → OR-Tools, Hexaly, or Timefold
 Black-box / Non-differentiable
 → Hexaly or InsideOpt Seeker
 TSP specifically
 → Concorde (exact) or LKH (heuristic)

36.10. Modeling Languages

36.10.1. PuLP

PuLP provides a simple, Pythonic interface for linear programming.

```

from pulp import *

def pulp_transportation(supply, demand, costs):
    """
    Transportation problem with PuLP.

    Args:
        supply: (m,) supply at each source
        demand: (n,) demand at each destination
        costs: (m, n) unit transport costs

    Returns:
        shipments: Optimal shipment quantities
        total_cost: Optimal total cost
    """
    m = len(supply)
    n = len(demand)

    # Create problem
    prob = LpProblem("Transportation", LpMinimize)

    # Variables
    x = [[LpVariable(f"x_{i}_{j}", lowBound=0) for j in range(n)]
          for i in range(m)]

    # Objective
    prob += lpSum(costs[i][j] * x[i][j] for i in range(m) for j in range(n))

    # Supply constraints
    for i in range(m):
        prob += lpSum(x[i][j] for j in range(n)) <= supply[i]

    # Demand constraints
    for j in range(n):
        prob += lpSum(x[i][j] for i in range(m)) >= demand[j]

    # Solve
  
```

```

prob.solve(PULP_CBC_CMD(msg=0))

if prob.status == 1: # Optimal
    shipments = [[x[i][j].varValue for j in range(n)] for i in range(m)]
    return shipments, value(prob.objective)

return None, None

# Example
supply = [100, 150, 200]
demand = [80, 120, 90, 60]
costs = [
    [10, 8, 6, 12],
    [9, 7, 11, 8],
    [8, 10, 9, 7]
]

shipments, cost = pulp_transportation(supply, demand, costs)
print(f"Optimal cost: {cost}")

```

36.10.2. Pyomo

Pyomo offers advanced modeling capabilities with algebraic syntax.

```

from pyomo.environ import *

def pyomo_production_mix(products, resources, profits, consumption, availability):
    """
    Product mix problem with Pyomo.

    Args:
        products: List of product names
        resources: List of resource names
        profits: Dict of product -> profit
        consumption: Dict of (product, resource) -> consumption
        availability: Dict of resource -> availability

    Returns:
        production: Dict of product -> quantity
        total_profit: Optimal profit
    """
    model = ConcreteModel()

    # Sets
    model.products = Set(initialize=products)
    model.resources = Set(initialize=resources)

    # Parameters
    model.profit = Param(model.products, initialize=profits)
    model.available = Param(model.resources, initialize=availability)
    model.consumption = Param(model.products, model.resources,
                               initialize=consumption, default=0)

```



```

# Variables
model.produce = Var(model.products, domain=NonNegativeReals)

# Objective
def profit_rule(m):
    return sum(m.profit[p] * m.produce[p] for p in m.products)
model.total_profit = Objective(rule=profit_rule, sense=maximize)

# Constraints
def resource_rule(m, r):
    return sum(m.consumption[p, r] * m.produce[p]
               for p in m.products) <= m.available[r]
model.resource_constraints = Constraint(model.resources, rule=resource_rule)

# Solve
solver = SolverFactory('cbc')
results = solver.solve(model)

if results.solver.termination_condition == TerminationCondition.optimal:
    production = {p: value(model.produce[p]) for p in products}
    return production, value(model.total_profit)

return None, None

```

36.10.3. OR-Tools Routing

OR-Tools provides specialized solvers for vehicle routing problems.

```

from ortools.constraint_solver import routing_enums_pb2
from ortools.constraint_solver import pywrapcp

def ortools_vrp(distance_matrix, num_vehicles, depot=0):
    """
    Solve Capacitated Vehicle Routing Problem.

    Args:
        distance_matrix: (n, n) distance between locations
        num_vehicles: Number of vehicles
        depot: Index of depot location

    Returns:
        routes: List of routes (each route is list of locations)
        total_distance: Total distance traveled
    """
    n = len(distance_matrix)

    # Create routing model
    manager = pywrapcp.RoutingIndexManager(n, num_vehicles, depot)
    routing = pywrapcp.RoutingModel(manager)

    def distance_callback(from_index, to_index):
        from_node = manager.IndexToNode(from_index)
        to_node = manager.IndexToNode(to_index)

```

```

    return int(distance_matrix[from_node][to_node])

transit_callback_index = routing.RegisterTransitCallback(distance_callback)
routing.SetArcCostEvaluatorOfAllVehicles(transit_callback_index)

# Search parameters
search_parameters = pywrapcp.DefaultRoutingSearchParameters()
search_parameters.first_solution_strategy = (
    routing_enums_pb2.FirstSolutionStrategy.PATH_CHEAPEST_ARC
)
search_parameters.local_search_metaheuristic = (
    routing_enums_pb2.LocalSearchMetaheuristic.GUIDED_LOCAL_SEARCH
)
search_parameters.time_limit.seconds = 30

# Solve
solution = routing.SolveWithParameters(search_parameters)

if solution:
    routes = []
    total_distance = 0

    for vehicle_id in range(num_vehicles):
        route = []
        index = routing.Start(vehicle_id)
        while not routing.IsEnd(index):
            route.append(manager.IndexToNode(index))
            previous_index = index
            index = solution.Value(routing.NextVar(index))
            total_distance += distance_callback(previous_index, index)
            route.append(manager.IndexToNode(index))
        routes.append(route)

    return routes, total_distance

return None, None

```

36.11. Solver Tuning

36.11.1. Key Parameters

Parameter	Effect	When to Adjust
Time limit	Maximum solve time	Production constraints
MIP gap	Acceptable optimality gap	Near-optimal acceptable
Presolve	Problem reduction	Can disable for debugging
Threads	Parallel solving	Match available cores
Heuristics	Solution finding	Can emphasize for large problems
Cuts	LP strengthening	Increase for weak relaxations

36.11.2. Gurobi Parameters

```
import gurobipy as gp

def configure_gurobi_model(model, time_limit=300, mip_gap=0.01, threads=4):
    """Configure Gurobi model parameters."""

    # Time and solution limits
    model.Params.TimeLimit = time_limit
    model.Params.MIPGap = mip_gap

    # Parallelism
    model.Params.Threads = threads

    # Heuristics
    model.Params.Heuristics = 0.1 # Fraction of time on heuristics

    # Cuts
    model.Params.Cuts = 2 # Aggressive cutting

    # Presolve
    model.Params.Presolve = 2 # Aggressive presolve

    # MIP focus
    # 0: balance, 1: feasibility, 2: optimality, 3: bound
    model.Params.MIPFocus = 0

    return model
```

36.11.3. CP-SAT Parameters

```
from ortools.sat.python import cp_model

def configure_cpsat_solver(solver, time_limit=60, num_workers=4):
    """Configure CP-SAT solver parameters."""

    # Time limit
    solver.parameters.max_time_in_seconds = time_limit

    # Parallelism
    solver.parameters.num_search_workers = num_workers

    # Search strategy
    solver.parameters.search_branching = cp_model.AUTOMATIC_SEARCH

    # Linearization level
    solver.parameters.linearization_level = 2

    # Logging
    solver.parameters.log_search_progress = True

    return solver
```

36.11.4. Warm Starting

Providing an initial solution can dramatically reduce solve time:

```
def gurobi_warm_start(model, initial_solution):
    """Provide warm start to Gurobi model."""
    for var_name, value in initial_solution.items():
        var = model.getVarByName(var_name)
        if var is not None:
            var.Start = value

    model.update()
```

36.12. Choosing the Right Tool

36.12.1. Problem Type Guide

Problem	First Choice	Alternative
Small LP	Any solver	HiGHS
Large LP	Gurobi/CPLEX/Xpress	HiGHS
Small MIP	Any MIP solver	CBC
Large MIP	Gurobi/CPLEX/Xpress	HiGHS, SCIP
Scheduling	CP-SAT	IBM CP Optimizer, Timefold
Vehicle Routing	OR-Tools	Hexaly, Timefold
Quadratic (QP)	Gurobi/CPLEX	MOSEK
Conic (SOCP/SDP)	MOSEK	SCS, ECOS
Convex NLP	Ipsot	Knitro
Non-convex NLP	BARON (global)	Ipsot + multi-start, Couenne
Convex MINLP	Bonmin	Knitro
Non-convex MINLP	BARON	Couenne, Xpress Global
TSP	Concorde (exact)	LKH (heuristic)
Black-box	Hexaly	InsideOpt Seeker

36.12.2. Integration Considerations

Factor	Commercial	Open Source
Cloud deployment	Check licensing	Generally OK
Embedded systems	Often restricted	Usually allowed
Redistribution	Typically not allowed	Check license
Support	Professional	Community
Long-term stability	High	Varies



Solver Selection Strategy

1. Start with open-source solvers (HiGHS, CBC) during development
2. Benchmark your specific problem instances
3. Consider commercial solvers only if performance justifies cost
4. Test multiple solvers—performance varies by problem structure
5. Always set appropriate time limits for production use

36.13. Summary

36.13.1. Key Takeaways

- Commercial solvers (Gurobi, CPLEX, Xpress) offer the best LP/MIP performance; MOSEK dominates conic optimization
- Open-source alternatives (HiGHS, CBC, CP-SAT, Ipopt) are competitive for many problems
- Choose solvers based on problem type: LP/MIP, CP, conic, nonlinear (local vs. global), or combinatorial
- For nonlinear problems: Ipopt (local, fast), BARON (global, slower), Knitro (commercial, versatile)
- Metaheuristic solvers (Hexaly, Timefold, InsideOpt Seeker) handle black-box and non-convex problems
- Modeling languages (PuLP, Pyomo, OR-Tools, MiniZinc) abstract solver differences
- Proper parameter tuning can dramatically improve performance
- Warm starting with heuristic solutions speeds up exact methods
- Test multiple solvers on your specific problem instances

36.14. Exercises

1. **Solver comparison:** Solve the same MIP problem using Gurobi, HiGHS, and CBC. Compare solve times and solution quality. Use a facility location problem with 20 facilities and 100 customers.
2. **Parameter tuning:** Take a difficult MIP instance and experiment with different MIP gap values (1%, 5%, 10%). Plot solve time vs. optimality gap.
3. **Modeling language:** Implement the same transportation problem in PuLP, Pyomo, and gurobipy. Compare the modeling experience and code length.
4. **Warm starting:** Solve a vehicle routing problem with and without a warm start from a greedy heuristic. Measure the improvement in solve time.
5. **CP vs. MIP:** Model a nurse scheduling problem using both CP-SAT and Gurobi MIP. Compare modeling complexity and solve times for instances with 10, 20, and 50 nurses.

37. Deployment and Scaling

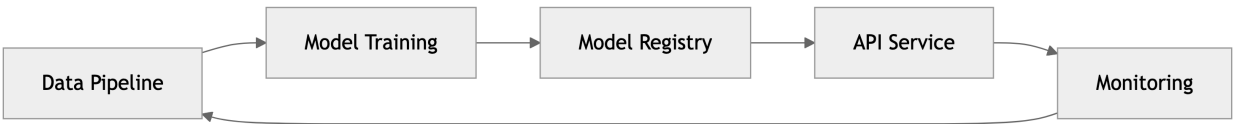
What You Will Learn

- Package optimization models for production
- Build APIs around optimization services
- Handle real-time vs. batch optimization
- Scale to large problem instances
- Monitor model performance and drift

Getting models into production is where value is realized. This chapter covers the engineering practices needed to deploy optimization models reliably, efficiently, and at scale.

37.1. The Optimization MLOps Stack

Production optimization systems require a robust infrastructure stack that handles data ingestion, model execution, and result delivery.



37.1.1. Key Components

Component	Purpose	Technologies
Data Pipeline	Ingest and validate input data	Airflow, Prefect, dbt
Model Registry	Version and track models	MLflow, DVC
API Service	Expose optimization as service	FastAPI, Flask
Compute	Execute optimization	Kubernetes, Ray
Monitoring	Track performance and health	Prometheus, Grafana
DecisionOps Platform	End-to-end optimization lifecycle	Nextmv

💡 DecisionOps Platforms as an Alternative

Building each component of this stack from scratch requires significant engineering effort. **DecisionOps platforms** like [Nextmv](#) bundle model deployment, testing (scenario simulation, shadow production, switchback tests), monitoring, and decision logging into a single managed platform. They integrate with popular solvers (Gurobi, OR-Tools, HiGHS, Hexaly) and modeling languages (Pyomo, AMPL), letting teams focus on model quality rather than infrastructure. See [Section 36.4](#) for a detailed comparison.

37.2. Packaging Optimization Models

37.2.1. Project Structure

A well-organized optimization project:

```
optimization-service/
  src/
    __init__.py
    models/
      __init__.py
      base.py           # Base model class
      facility.py       # Facility location model
      routing.py        # Vehicle routing model
    solvers/
      __init__.py
      solver_factory.py
    data/
      __init__.py
      schemas.py        # Pydantic schemas
      validation.py
    api/
      __init__.py
      main.py
      routes.py
  tests/
  configs/
  Dockerfile
  requirements.txt
  pyproject.toml
```

37.2.2. Base Model Class

```
from abc import ABC, abstractmethod
from dataclasses import dataclass
from typing import Any, Dict, Optional
import time
import logging

@dataclass
class OptimizationResult:
    """Standard result format for optimization models."""
    status: str # 'optimal', 'feasible', 'infeasible', 'timeout'
    objective_value: Optional[float]
    solution: Dict[str, Any]
    solve_time: float
    gap: Optional[float] = None
    metadata: Dict[str, Any] = None

class BaseOptimizationModel(ABC):
    """Abstract base class for optimization models."""
```



```

def __init__(self, config: Dict[str, Any] = None):
    self.config = config or {}
    self.logger = logging.getLogger(self.__class__.__name__)
    self.model = None

    @abstractmethod
    def build_model(self, data: Dict[str, Any]) -> None:
        """Build the optimization model from input data."""
        pass

    @abstractmethod
    def extract_solution(self) -> Dict[str, Any]:
        """Extract solution from solved model."""
        pass

    def solve(self, data: Dict[str, Any]) -> OptimizationResult:
        """Execute the optimization pipeline."""
        start_time = time.time()

        try:
            # Validate input
            self.validate_input(data)

            # Build model
            self.build_model(data)

            # Configure solver
            self.configure_solver()

            # Solve
            status = self.optimize()

            # Extract solution
            if status in ['optimal', 'feasible']:
                solution = self.extract_solution()
                objective = self.get_objective_value()
            else:
                solution = {}
                objective = None

            solve_time = time.time() - start_time

            return OptimizationResult(
                status=status,
                objective_value=objective,
                solution=solution,
                solve_time=solve_time,
                gap=self.get_gap(),
                metadata={'model_name': self.__class__.__name__}
            )

        except Exception as e:

```

```

        self.logger.error(f"Optimization failed: {e}")
        raise

    def validate_input(self, data: Dict[str, Any]) -> None:
        """Validate input data. Override for custom validation."""
        pass

    def configure_solver(self) -> None:
        """Configure solver parameters."""
        pass

    @abstractmethod
    def optimize(self) -> str:
        """Run the optimization solver."""
        pass

    @abstractmethod
    def get_objective_value(self) -> float:
        """Get the objective function value."""
        pass

    def get_gap(self) -> Optional[float]:
        """Get the optimality gap if available."""
        return None

```

37.2.3. Concrete Model Implementation

```

import numpy as np
from ortools.linear_solver import pywraplp

class FacilityLocationModel(BaseOptimizationModel):
    """Facility location optimization model."""

    def build_model(self, data: Dict[str, Any]) -> None:
        """Build facility location model."""
        self.fixed_costs = np.array(data['fixed_costs'])
        self.transport_costs = np.array(data['transport_costs'])
        self.m = len(self.fixed_costs)
        self.n = self.transport_costs.shape[1]

        self.solver = pywraplp.Solver.CreateSolver('SCIP')

        # Variables
        self.y = [self.solver.IntVar(0, 1, f'y_{i}') for i in range(self.m)]
        self.x = {}
        for i in range(self.m):
            for j in range(self.n):
                self.x[i, j] = self.solver.NumVar(0, 1, f'x_{i}_{j}')

        # Constraints
        for j in range(self.n):
            self.solver.Add(
                sum(self.x[i, j] for i in range(self.m)) == 1

```

```

    )

    for i in range(self.m):
        for j in range(self.n):
            self.solver.Add(self.x[i, j] <= self.y[i])

    # Objective
    objective = sum(self.fixed_costs[i] * self.y[i] for i in range(self.m))
    objective += sum(
        self.transport_costs[i, j] * self.x[i, j]
        for i in range(self.m) for j in range(self.n)
    )
    self.solver.Minimize(objective)

def configure_solver(self) -> None:
    """Set solver parameters."""
    time_limit = self.config.get('time_limit', 300)
    self.solver.SetTimeLimit(time_limit * 1000)

def optimize(self) -> str:
    """Solve the model."""
    status = self.solver.Solve()

    status_map = {
        pywraplp.Solver.OPTIMAL: 'optimal',
        pywraplp.Solver.FEASIBLE: 'feasible',
        pywraplp.Solver.INFEASIBLE: 'infeasible',
        pywraplp.Solver.NOT_SOLVED: 'timeout'
    }
    return status_map.get(status, 'error')

def extract_solution(self) -> Dict[str, Any]:
    """Extract solution values."""
    open_facilities = [
        i for i in range(self.m)
        if self.y[i].solution_value() > 0.5
    ]

    assignments = {}
    for j in range(self.n):
        for i in range(self.m):
            if self.x[i, j].solution_value() > 0.5:
                assignments[j] = i
                break

    return {
        'open_facilities': open_facilities,
        'assignments': assignments
    }

def get_objective_value(self) -> float:
    """Get objective value."""
    return self.solver.Objective().Value()

```

37.3. Building APIs

37.3.1. FastAPI Service

```

from fastapi import FastAPI, HTTPException, BackgroundTasks
from pydantic import BaseModel, Field
from typing import Dict, List, Optional, Any
import uuid
import asyncio
from concurrent.futures import ThreadPoolExecutor

app = FastAPI(title="Optimization Service")

# Thread pool for CPU-bound optimization
executor = ThreadPoolExecutor(max_workers=4)

# Job storage (use Redis or database in production)
jobs: Dict[str, Dict] = {}

class FacilityLocationInput(BaseModel):
    """Input schema for facility location problem."""
    fixed_costs: List[float] = Field(..., description="Opening costs per facility")
    transport_costs: List[List[float]] = Field(
        ..., description="Transport costs from facility i to customer j"
    )
    time_limit: Optional[int] = Field(300, description="Solver time limit in
↪ seconds")

class OptimizationResponse(BaseModel):
    """Response schema for optimization results."""
    job_id: str
    status: str
    objective_value: Optional[float]
    solution: Optional[Dict[str, Any]]
    solve_time: Optional[float]
    message: Optional[str]

@app.post("/optimize/facility-location", response_model=OptimizationResponse)
async def solve_facility_location(input_data: FacilityLocationInput):
    """Solve facility location problem synchronously."""
    try:
        model = FacilityLocationModel(config={'time_limit': input_data.time_limit})

        result = model.solve({
            'fixed_costs': input_data.fixed_costs,
            'transport_costs': input_data.transport_costs
        })

        return OptimizationResponse(
            job_id=str(uuid.uuid4()),
            status=result.status,
            objective_value=result.objective_value,
            solution=result.solution,
            solve_time=result.solve_time

```

```

    )

    except Exception as e:
        raise HTTPException(status_code=500, detail=str(e))

@app.post("/optimize/facility-location/async")
async def solve_facility_location_async(
    input_data: FacilityLocationInput,
    background_tasks: BackgroundTasks
):
    """Submit facility location problem for async solving."""
    job_id = str(uuid.uuid4())

    jobs[job_id] = {
        'status': 'pending',
        'result': None
    }

    background_tasks.add_task(
        run_optimization,
        job_id,
        input_data.dict()
    )

    return {'job_id': job_id, 'status': 'pending'}

def run_optimization(job_id: str, input_data: Dict):
    """Run optimization in background."""
    try:
        jobs[job_id]['status'] = 'running'

        model = FacilityLocationModel(
            config={'time_limit': input_data.get('time_limit', 300)}
        )

        result = model.solve({
            'fixed_costs': input_data['fixed_costs'],
            'transport_costs': input_data['transport_costs']
        })

        jobs[job_id] = {
            'status': 'completed',
            'result': {
                'status': result.status,
                'objective_value': result.objective_value,
                'solution': result.solution,
                'solve_time': result.solve_time
            }
        }

    except Exception as e:
        jobs[job_id] = {
            'status': 'failed',

```

```

        'error': str(e)
    }

@app.get("/jobs/{job_id}")
async def get_job_status(job_id: str):
    """Get status of async optimization job."""
    if job_id not in jobs:
        raise HTTPException(status_code=404, detail="Job not found")

    return jobs[job_id]

@app.get("/health")
async def health_check():
    """Health check endpoint."""
    return {"status": "healthy"}

```

37.3.2. Request Validation

```

from pydantic import BaseModel, validator, Field
from typing import List
import numpy as np

class VRPInput(BaseModel):
    """Validated input for Vehicle Routing Problem."""
    locations: List[List[float]] = Field(..., description="[x, y] coordinates")
    demands: List[int] = Field(..., description="Demand at each location")
    vehicle_capacity: int = Field(..., gt=0)
    num_vehicles: int = Field(..., gt=0)
    depot_index: int = Field(0, ge=0)
    time_limit: int = Field(60, gt=0, le=3600)

    @validator('locations')
    def validate_locations(cls, v):
        if len(v) < 2:
            raise ValueError('At least 2 locations required')
        for loc in v:
            if len(loc) != 2:
                raise ValueError('Each location must have [x, y] coordinates')
        return v

    @validator('demands')
    def validate_demands(cls, v, values):
        if 'locations' in values and len(v) != len(values['locations']):
            raise ValueError('Demands must match number of locations')
        if any(d < 0 for d in v):
            raise ValueError('Demands must be non-negative')
        return v

    @validator('depot_index')
    def validate_depot(cls, v, values):
        if 'locations' in values and v >= len(values['locations']):
            raise ValueError('Depot index out of range')
        return v

```

37.4. Real-time vs. Batch Optimization

37.4.1. When to Use Each

Scenario	Approach	Response Time
Order dispatch	Real-time	< 1 second
Route optimization	Near real-time	1-30 seconds
Production planning	Batch	Minutes to hours
Strategic planning	Batch	Hours to days

37.4.2. Real-time Optimization Service

```
import asyncio
from cachetools import TTLCache
import hashlib
import json

class RealTimeOptimizer:
    """Optimizer for real-time decision making."""

    def __init__(self, model_class, cache_ttl=300):
        self.model_class = model_class
        self.cache = TTLCache(maxsize=1000, ttl=cache_ttl)

    def _get_cache_key(self, data: Dict) -> str:
        """Generate cache key for input data."""
        data_str = json.dumps(data, sort_keys=True)
        return hashlib.md5(data_str.encode()).hexdigest()

    async def solve(self, data: Dict, timeout: float = 5.0) -> OptimizationResult:
        """
        Solve optimization with strict timeout.

        Uses cached results when available, falls back to heuristics
        if optimization times out.
        """
        cache_key = self._get_cache_key(data)

        # Check cache
        if cache_key in self.cache:
            result = self.cache[cache_key]
            result.metadata['from_cache'] = True
            return result

        # Run optimization with timeout
        try:
            result = await asyncio.wait_for(
                asyncio.get_event_loop().run_in_executor(
                    None, self._solve_sync, data, timeout
                ),
                timeout=timeout
            )
```

```

        # Cache successful results
        if result.status in ['optimal', 'feasible']:
            self.cache[cache_key] = result

        return result

    except asyncio.TimeoutError:
        # Fall back to heuristic
        return self._solve_heuristic(data)

    def _solve_sync(self, data: Dict, timeout: float) -> OptimizationResult:
        """Synchronous solve with timeout configuration."""
        model = self.model_class(config={'time_limit': timeout})
        return model.solve(data)

    def _solve_heuristic(self, data: Dict) -> OptimizationResult:
        """Fast heuristic fallback."""
        # Implement problem-specific heuristic
        return OptimizationResult(
            status='heuristic',
            objective_value=None,
            solution=self._greedy_solution(data),
            solve_time=0.1,
            metadata={'fallback': True}
        )

    def _greedy_solution(self, data: Dict) -> Dict:
        """Greedy heuristic implementation."""
        # Override in subclasses
        return {}

```

37.4.3. Batch Processing Pipeline

```

from dataclasses import dataclass
from typing import List, Callable
import concurrent.futures
from datetime import datetime

@dataclass
class BatchJob:
    job_id: str
    data: Dict[str, Any]
    priority: int = 0
    created_at: datetime = None

class BatchOptimizationPipeline:
    """Pipeline for batch optimization jobs."""

    def __init__(self, model_factory: Callable, max_workers: int = 4):
        self.model_factory = model_factory
        self.max_workers = max_workers
        self.jobs: List[BatchJob] = []
        self.results: Dict[str, OptimizationResult] = {}

```



```

def submit(self, job_id: str, data: Dict, priority: int = 0):
    """Submit job to batch queue."""
    job = BatchJob(
        job_id=job_id,
        data=data,
        priority=priority,
        created_at=datetime.now()
    )
    self.jobs.append(job)

def process_batch(self) -> Dict[str, OptimizationResult]:
    """Process all submitted jobs."""
    # Sort by priority
    sorted_jobs = sorted(self.jobs, key=lambda j: -j.priority)

    with concurrent.futures.ProcessPoolExecutor(
        max_workers=self.max_workers
    ) as executor:

        futures = {
            executor.submit(self._solve_job, job): job.job_id
            for job in sorted_jobs
        }

        for future in concurrent.futures.as_completed(futures):
            job_id = futures[future]
            try:
                result = future.result()
                self.results[job_id] = result
            except Exception as e:
                self.results[job_id] = OptimizationResult(
                    status='error',
                    objective_value=None,
                    solution={},
                    solve_time=0,
                    metadata={'error': str(e)}
                )

        # Clear processed jobs
        self.jobs = []

    return self.results

def _solve_job(self, job: BatchJob) -> OptimizationResult:
    """Solve single optimization job."""
    model = self.model_factory()
    return model.solve(job.data)

```

37.5. Scaling Strategies

37.5.1. Problem Decomposition

For large problems, decomposition can make optimization tractable:

```

class DecompositionOptimizer:
    """Optimizer using problem decomposition."""

    def __init__(self, subproblem_solver, coordination_method='lagrangian'):
        self.subproblem_solver = subproblem_solver
        self.coordination_method = coordination_method

    def decompose(self, data: Dict) -> List[Dict]:
        """Decompose problem into subproblems."""
        # Override with problem-specific decomposition
        raise NotImplementedError

    def coordinate(self, subproblem_solutions: List[Dict]) -> Dict:
        """Coordinate subproblem solutions."""
        if self.coordination_method == 'lagrangian':
            return self._lagrangian_coordination(subproblem_solutions)
        else:
            return self._simple_merge(subproblem_solutions)

    def solve(self, data: Dict, max_iterations: int = 100) -> OptimizationResult:
        """Solve using decomposition."""
        # Decompose
        subproblems = self.decompose(data)

        for iteration in range(max_iterations):
            # Solve subproblems in parallel
            with concurrent.futures.ProcessPoolExecutor() as executor:
                solutions = list(executor.map(
                    self.subproblem_solver.solve,
                    subproblems
                ))

            # Coordinate
            combined = self.coordinate([s.solution for s in solutions])

            # Check convergence
            if self._check_convergence(solutions):
                break

            # Update subproblems with coordination info
            subproblems = self._update_subproblems(subproblems, combined)

        return OptimizationResult(
            status='feasible',
            objective_value=sum(s.objective_value or 0 for s in solutions),
            solution=combined,
            solve_time=sum(s.solve_time for s in solutions)
        )

```

37.5.2. Distributed Computing with Ray

```

import ray

@ray.remote
class DistributedOptimizer:
    """Ray-based distributed optimizer."""

    def __init__(self, model_class):
        self.model_class = model_class
        self.model = None

    def solve(self, data: Dict, config: Dict = None) -> Dict:
        """Solve optimization problem."""
        if self.model is None:
            self.model = self.model_class(config=config or {})

        result = self.model.solve(data)
        return {
            'status': result.status,
            'objective_value': result.objective_value,
            'solution': result.solution,
            'solve_time': result.solve_time
        }

def solve_distributed(problems: List[Dict], model_class, num_workers: int = 4):
    """Solve multiple problems in parallel using Ray."""
    ray.init(ignore_reinit_error=True)

    # Create worker pool
    workers = [DistributedOptimizer.remote(model_class) for _ in range(num_workers)]

    # Distribute problems to workers
    futures = []
    for i, problem in enumerate(problems):
        worker = workers[i % num_workers]
        futures.append(worker.solve.remote(problem))

    # Collect results
    results = ray.get(futures)

    ray.shutdown()
    return results

```

Deployed optimization models degrade over time as the underlying data distribution shifts. The figure below tracks solution quality and solve time over several months, highlighting when performance drops below an acceptable threshold and triggers retraining.



Figure 37.1.: Model performance drift after deployment — solution quality degrades over time while solve time increases, with deployment events and a retraining trigger highlighted

37.6. Monitoring and Observability

37.6.1. Metrics Collection

```
from prometheus_client import Counter, Histogram, Gauge
import time
from functools import wraps

# Define metrics
OPTIMIZATION_REQUESTS = Counter(
    'optimization_requests_total',
    'Total optimization requests',
    ['model_type', 'status']
)

OPTIMIZATION_DURATION = Histogram(
    'optimization_duration_seconds',
```

```

        'Optimization solve time',
        ['model_type'],
        buckets=[0.1, 0.5, 1, 5, 10, 30, 60, 120, 300]
    )

    OPTIMIZATION_GAP = Gauge(
        'optimization_gap_percent',
        'Optimality gap',
        ['model_type']
    )

def track_optimization(model_type: str):
    """Decorator to track optimization metrics."""
    def decorator(func):
        @wraps(func)
        def wrapper(*args, **kwargs):
            start_time = time.time()

            try:
                result = func(*args, **kwargs)
                status = result.status
            except Exception as e:
                status = 'error'
                raise
            finally:
                duration = time.time() - start_time
                OPTIMIZATION_REQUESTS.labels(
                    model_type=model_type,
                    status=status
                ).inc()
                OPTIMIZATION_DURATION.labels(
                    model_type=model_type
                ).observe(duration)

            if result.gap is not None:
                OPTIMIZATION_GAP.labels(
                    model_type=model_type
                ).set(result.gap * 100)

            return result

        return wrapper
    return decorator

```

37.6.2. Logging Best Practices

```

import logging
import json
from typing import Dict, Any

class OptimizationLogger:
    """Structured logging for optimization."""

```

```

def __init__(self, name: str):
    self.logger = logging.getLogger(name)

def log_request(self, job_id: str, input_summary: Dict):
    """Log incoming optimization request."""
    self.logger.info(json.dumps({
        'event': 'optimization_request',
        'job_id': job_id,
        'num_variables': input_summary.get('num_variables'),
        'num_constraints': input_summary.get('num_constraints')
    })))

def log_result(self, job_id: str, result: OptimizationResult):
    """Log optimization result."""
    self.logger.info(json.dumps({
        'event': 'optimization_complete',
        'job_id': job_id,
        'status': result.status,
        'objective_value': result.objective_value,
        'solve_time': result.solve_time,
        'gap': result.gap
    })))

def log_error(self, job_id: str, error: Exception):
    """Log optimization error."""
    self.logger.error(json.dumps({
        'event': 'optimization_error',
        'job_id': job_id,
        'error_type': type(error).__name__,
        'error_message': str(error)
    })))

```

37.6.3. Health Checks and Alerts

```

from dataclasses import dataclass
from typing import List
from datetime import datetime, timedelta

@dataclass
class HealthCheck:
    name: str
    healthy: bool
    message: str
    timestamp: datetime

class OptimizationHealthChecker:
    """Health checking for optimization service."""

    def __init__(self, metrics_store):
        self.metrics = metrics_store
        self.checks: List[HealthCheck] = []

    def check_solver_availability(self) -> HealthCheck:

```

```

    """Check if solver is available."""
    try:
        # Try to create a simple model
        from ortools.linear_solver import pywraplp
        solver = pywraplp.Solver.CreateSolver('SCIP')
        healthy = solver is not None

        return HealthCheck(
            name='solver_availability',
            healthy=healthy,
            message='Solver available' if healthy else 'Solver unavailable',
            timestamp=datetime.now()
        )
    except Exception as e:
        return HealthCheck(
            name='solver_availability',
            healthy=False,
            message=str(e),
            timestamp=datetime.now()
        )

def check_recent_success_rate(self, threshold: float = 0.9) -> HealthCheck:
    """Check recent optimization success rate."""
    recent_results = self.metrics.get_recent_results(hours=1)

    if not recent_results:
        return HealthCheck(
            name='success_rate',
            healthy=True,
            message='No recent optimizations',
            timestamp=datetime.now()
        )

    success_count = sum(1 for r in recent_results if r.status in ['optimal',
↪ 'feasible'])
    success_rate = success_count / len(recent_results)

    return HealthCheck(
        name='success_rate',
        healthy=success_rate >= threshold,
        message=f'Success rate: {success_rate:.1%}',
        timestamp=datetime.now()
    )

def check_average_solve_time(self, threshold: float = 60) -> HealthCheck:
    """Check if average solve time is acceptable."""
    recent_results = self.metrics.get_recent_results(hours=1)

    if not recent_results:
        return HealthCheck(
            name='solve_time',
            healthy=True,
            message='No recent optimizations',

```

```

        timestamp=datetime.now()
    )

    avg_time = sum(r.solve_time for r in recent_results) / len(recent_results)

    return HealthCheck(
        name='solve_time',
        healthy=avg_time <= threshold,
        message=f'Average solve time: {avg_time:.1f}s',
        timestamp=datetime.now()
    )

def run_all_checks(self) -> List[HealthCheck]:
    """Run all health checks."""
    self.checks = [
        self.check_solver_availability(),
        self.check_recent_success_rate(),
        self.check_average_solve_time()
    ]
    return self.checks

def is_healthy(self) -> bool:
    """Check if all health checks pass."""
    return all(c.healthy for c in self.checks)

```

37.7. Container Deployment

37.7.1. Dockerfile

```

FROM python:3.11-slim

WORKDIR /app

# Install system dependencies
RUN apt-get update && apt-get install -y \
    build-essential \
    && rm -rf /var/lib/apt/lists/*

# Install Python dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY src/ ./src/
COPY configs/ ./configs/

# Set environment variables
ENV PYTHONPATH=/app
ENV SOLVER_TIME_LIMIT=300

# Expose port
EXPOSE 8000

```



```
# Health check
HEALTHCHECK --interval=30s --timeout=10s --retries=3 \
  CMD curl -f http://localhost:8000/health || exit 1

# Run application
CMD ["uvicorn", "src.api.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

37.7.2. Kubernetes Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: optimization-service
spec:
  replicas: 3
  selector:
    matchLabels:
      app: optimization-service
  template:
    metadata:
      labels:
        app: optimization-service
    spec:
      containers:
        - name: optimization
          image: optimization-service:latest
          ports:
            - containerPort: 8000
          resources:
            requests:
              memory: "2Gi"
              cpu: "1"
            limits:
              memory: "8Gi"
              cpu: "4"
          env:
            - name: SOLVER_TIME_LIMIT
              value: "300"
          livenessProbe:
            httpGet:
              path: /health
              port: 8000
              initialDelaySeconds: 30
              periodSeconds: 10
          readinessProbe:
            httpGet:
              path: /health
              port: 8000
              initialDelaySeconds: 5
              periodSeconds: 5
```

37.8. Decision Orchestration

Production optimization systems rarely operate in isolation. They must integrate with human decision-makers, handle exceptions, provide explanations, and learn from feedback. Decision orchestration is the discipline of managing this human-machine collaboration.

37.8.1. Human-in-the-Loop Architectures

Not all decisions should be fully automated. The appropriate level of automation depends on decision impact, frequency, and confidence.

AUTOMATION SPECTRUM			
Full Manual	Assisted	Supervised	Autonomous
Human decides	System suggests Human decides	System decides Human approves	System decides Human monitors
Strategic decisions	Tactical planning	Operational decisions	Routine transactions

Automation Level	When to Use	Example
Full Manual	High-stakes, rare decisions	Factory location
Assisted	Complex decisions needing judgment	Production planning
Supervised	Frequent but impactful decisions	Daily routing
Autonomous	High-frequency, low-risk decisions	Reorder points

37.8.2. Decision Orchestration Framework

A decision orchestration framework manages the full lifecycle of an optimization-driven decision: it receives a request, selects the appropriate solver configuration, runs the optimization, validates the result, and routes it for human review if needed. The implementation below defines this pipeline as a configurable class.

```
from dataclasses import dataclass, field
from typing import Dict, Any, Optional, List, Callable
from enum import Enum
from datetime import datetime
import uuid
```

```

class DecisionStatus(Enum):
    PENDING = "pending"
    APPROVED = "approved"
    REJECTED = "rejected"
    MODIFIED = "modified"
    AUTO_EXECUTED = "auto_executed"
    EXPIRED = "expired"

class AutomationLevel(Enum):
    MANUAL = "manual"
    ASSISTED = "assisted"
    SUPERVISED = "supervised"
    AUTONOMOUS = "autonomous"

@dataclass
class Decision:
    """A decision requiring orchestration."""
    decision_id: str
    decision_type: str
    recommendation: Dict[str, Any]
    confidence: float
    impact_score: float
    explanation: str
    created_at: datetime
    expires_at: Optional[datetime] = None
    status: DecisionStatus = DecisionStatus.PENDING
    approved_by: Optional[str] = None
    modifications: Dict[str, Any] = field(default_factory=dict)
    feedback: Optional[str] = None

class DecisionOrchestrator:
    """Orchestrates human-machine decision collaboration."""

    def __init__(self):
        self.decisions: Dict[str, Decision] = {}
        self.automation_rules: Dict[str, Callable] = {}
        self.approval_queue: List[str] = []

    def register_automation_rule(
        self,
        decision_type: str,
        rule: Callable[[Decision], AutomationLevel]
    ):
        """Register rule for determining automation level."""
        self.automation_rules[decision_type] = rule

    def submit_decision(
        self,
        decision_type: str,
        recommendation: Dict[str, Any],
        confidence: float,
        impact_score: float,

```

```

        explanation: str,
        ttl_seconds: int = 3600
    ) -> Decision:
        """Submit a decision for orchestration."""
        decision = Decision(
            decision_id=str(uuid.uuid4()),
            decision_type=decision_type,
            recommendation=recommendation,
            confidence=confidence,
            impact_score=impact_score,
            explanation=explanation,
            created_at=datetime.now(),
            expires_at=datetime.now() + timedelta(seconds=ttl_seconds)
        )

        # Determine automation level
        automation_level = self._determine_automation(decision)

        if automation_level == AutomationLevel.AUTONOMOUS:
            decision.status = DecisionStatus.AUTO_EXECUTED
            self._execute_decision(decision)
        else:
            self.approval_queue.append(decision.decision_id)

        self.decisions[decision.decision_id] = decision
        return decision

    def _determine_automation(self, decision: Decision) -> AutomationLevel:
        """Determine appropriate automation level."""
        if decision.decision_type in self.automation_rules:
            return self.automation_rules[decision.decision_type](decision)

        # Default rules based on confidence and impact
        if decision.confidence > 0.95 and decision.impact_score < 0.3:
            return AutomationLevel.AUTONOMOUS
        elif decision.confidence > 0.8 and decision.impact_score < 0.5:
            return AutomationLevel.SUPERVISED
        elif decision.confidence > 0.6:
            return AutomationLevel.ASSISTED
        else:
            return AutomationLevel.MANUAL

    def approve(
        self,
        decision_id: str,
        approver: str,
        modifications: Dict[str, Any] = None
    ) -> Decision:
        """Approve a pending decision."""
        decision = self.decisions.get(decision_id)
        if not decision:
            raise ValueError(f"Decision {decision_id} not found")

```

```

        if modifications:
            decision.status = DecisionStatus.MODIFIED
            decision.modifications = modifications
        else:
            decision.status = DecisionStatus.APPROVED

        decision.approved_by = approver

        if decision_id in self.approval_queue:
            self.approval_queue.remove(decision_id)

        self._execute_decision(decision)
        return decision

    def reject(
        self,
        decision_id: str,
        approver: str,
        feedback: str
    ) -> Decision:
        """Reject a pending decision."""
        decision = self.decisions.get(decision_id)
        if not decision:
            raise ValueError(f"Decision {decision_id} not found")

        decision.status = DecisionStatus.REJECTED
        decision.approved_by = approver
        decision.feedback = feedback

        if decision_id in self.approval_queue:
            self.approval_queue.remove(decision_id)

        return decision

    def get_pending_decisions(self) -> List[Decision]:
        """Get all pending decisions."""
        return [
            self.decisions[d_id]
            for d_id in self.approval_queue
            if self.decisions[d_id].status == DecisionStatus.PENDING
        ]

    def _execute_decision(self, decision: Decision):
        """Execute an approved or autonomous decision."""
        # Implementation depends on decision type
        pass

```

37.8.3. Explainable Optimization

Human decision-makers need to understand why the system recommends specific actions. Explainable optimization provides interpretable rationales.

```

from dataclasses import dataclass
from typing import List, Dict, Any, Tuple

@dataclass
class Explanation:
    """Structured explanation of an optimization decision."""
    summary: str
    key_drivers: List[Dict[str, Any]]
    constraints_status: List[Dict[str, Any]]
    alternatives_considered: List[Dict[str, Any]]
    sensitivity: Dict[str, float]
    confidence_factors: List[str]

class ExplainableOptimizer:
    """Wrapper that adds explanation capabilities to optimizers."""

    def __init__(self, base_optimizer):
        self.optimizer = base_optimizer
        self.decision_log: List[Dict] = []

    def solve_with_explanation(self, data: Dict) -> Tuple[Any, Explanation]:
        """Solve and generate explanation."""
        # Solve the optimization
        result = self.optimizer.solve(data)

        # Generate explanation
        explanation = self._generate_explanation(data, result)

        return result, explanation

    def _generate_explanation(self, data: Dict, result: Any) -> Explanation:
        """Generate human-readable explanation."""

        # Identify key drivers
        key_drivers = self._analyze_drivers(data, result)

        # Analyze constraint status
        constraints = self._analyze_constraints(result)

        # Generate alternatives
        alternatives = self._generate_alternatives(data, result)

        # Sensitivity analysis
        sensitivity = self._sensitivity_analysis(data, result)

        # Summary
        summary = self._generate_summary(result, key_drivers, constraints)

        return Explanation(
            summary=summary,
            key_drivers=key_drivers,
            constraints_status=constraints,
            alternatives_considered=alternatives,

```

```

        sensitivity=sensitivity,
        confidence_factors=self._assess_confidence(result)
    )

def _analyze_drivers(self, data: Dict, result: Any) -> List[Dict]:
    """Identify what drove the optimization decision."""
    drivers = []

    # Example: analyze variable contributions
    if hasattr(result, 'solution'):
        for var_name, value in result.solution.items():
            if value > 0:
                drivers.append({
                    'variable': var_name,
                    'value': value,
                    'contribution': self._estimate_contribution(var_name,
                                                                ↪ value),
                    'reason': self._explain_variable(var_name, value, data)
                })

    return sorted(drivers, key=lambda x: -abs(x['contribution']))[:5]

def _analyze_constraints(self, result: Any) -> List[Dict]:
    """Analyze which constraints are binding."""
    constraints = []

    # Would need access to constraint slack values
    # This is a simplified example
    if hasattr(result, 'constraint_info'):
        for name, info in result.constraint_info.items():
            constraints.append({
                'name': name,
                'status': 'binding' if info['slack'] < 1e-6 else 'non-binding',
                'slack': info['slack'],
                'impact': 'This constraint limits the solution' if info['slack']
                          ↪ < 1e-6 else 'Not active'
            })

    return constraints

def _generate_alternatives(self, data: Dict, result: Any) -> List[Dict]:
    """Generate and describe alternative solutions."""
    alternatives = []

    # Generate near-optimal alternatives (simplified)
    # In practice, would use MIP solution pool or similar
    alternatives.append({
        'description': 'Current optimal solution',
        'objective': result.objective_value,
        'trade_off': 'Best overall cost'
    })

    return alternatives

```

```

def _sensitivity_analysis(self, data: Dict, result: Any) -> Dict[str, float]:
    """Perform sensitivity analysis on key parameters."""
    sensitivity = {}

    # Example: test sensitivity to demand changes
    for param in ['demand', 'capacity', 'cost']:
        if param in data:
            sensitivity[param] = self._test_parameter_sensitivity(
                data, result, param
            )

    return sensitivity

def _test_parameter_sensitivity(
    self, data: Dict, result: Any, param: str
) -> float:
    """Test how sensitive solution is to parameter changes."""
    # Simplified: would re-solve with perturbed parameters
    return 0.1 # Placeholder

def _assess_confidence(self, result: Any) -> List[str]:
    """Assess confidence in the recommendation."""
    factors = []

    if result.status == 'optimal':
        factors.append("Solution is provably optimal")
    elif result.status == 'feasible':
        if result.gap and result.gap < 0.01:
            factors.append(f"Solution within {result.gap*100:.1f}% of optimal")
        else:
            factors.append("Solution is feasible but optimality not proven")

    if result.solve_time < 10:
        factors.append("Model solved quickly, suggesting stable solution")

    return factors

def _generate_summary(
    self, result: Any, drivers: List, constraints: List
) -> str:
    """Generate natural language summary."""
    binding_constraints = [c for c in constraints if c.get('status') ==
        ↪ 'binding']
    top_driver = drivers[0] if drivers else None

    summary = f"Recommended solution achieves objective value of
    ↪ {result.objective_value:.2f}. "

    if top_driver:
        summary += f"The primary driver is {top_driver['variable']} "
        summary += f"contributing {top_driver['contribution']:.1f}% to the
        ↪ objective. "

```



```

        if binding_constraints:
            summary += f"{len(binding_constraints)} constraints are at their limits.
↪    "

    return summary

def _estimate_contribution(self, var_name: str, value: float) -> float:
    """Estimate variable's contribution to objective."""
    return 0.2 # Placeholder

def _explain_variable(self, var_name: str, value: float, data: Dict) -> str:
    """Generate explanation for variable value."""
    return f"{var_name} set to {value} based on cost-benefit analysis"

```

37.8.4. Decision Approval Workflow

High-impact decisions require human approval before execution. The workflow below implements a tiered approval system: low-cost decisions are auto-approved, medium-cost ones require manager review, and high-cost ones require director sign-off. Each decision is logged with its approval status and rationale.

```

from enum import Enum
from typing import Optional, Callable
from datetime import timedelta

class EscalationLevel(Enum):
    OPERATOR = 1
    SUPERVISOR = 2
    MANAGER = 3
    EXECUTIVE = 4

@dataclass
class ApprovalRule:
    """Rule for routing decisions to approvers."""
    decision_type: str
    impact_threshold: float
    required_level: EscalationLevel
    timeout: timedelta
    escalation_level: Optional[EscalationLevel] = None

class ApprovalWorkflow:
    """Manages decision approval routing and escalation."""

    def __init__(self):
        self.rules: List[ApprovalRule] = []
        self.approvers: Dict[EscalationLevel, List[str]] = {}
        self.notification_handler: Optional[Callable] = None

    def add_rule(self, rule: ApprovalRule):
        """Add approval routing rule."""
        self.rules.append(rule)

```

```

def register_approvers(self, level: EscalationLevel, approvers: List[str]):
    """Register approvers for a level."""
    self.approvers[level] = approvers

def set_notification_handler(self, handler: Callable):
    """Set handler for sending notifications."""
    self.notification_handler = handler

def route_decision(self, decision: Decision) -> Dict:
    """Route decision to appropriate approver."""
    # Find matching rule
    rule = self._find_matching_rule(decision)

    if not rule:
        # Default to supervisor level
        level = EscalationLevel.SUPERVISOR
        timeout = timedelta(hours=4)
    else:
        level = rule.required_level
        timeout = rule.timeout

    approvers = self.approvers.get(level, [])

    routing = {
        'decision_id': decision.decision_id,
        'approval_level': level,
        'approvers': approvers,
        'deadline': datetime.now() + timeout,
        'escalation_level': rule.escalation_level if rule else
        ↪ EscalationLevel.MANAGER
    }

    # Send notifications
    if self.notification_handler and approvers:
        self.notification_handler(routing)

    return routing

def _find_matching_rule(self, decision: Decision) -> Optional[ApprovalRule]:
    """Find rule matching the decision."""
    for rule in self.rules:
        if (rule.decision_type == decision.decision_type and
            decision.impact_score >= rule.impact_threshold):
            return rule
    return None

def check_escalation(self, decision: Decision, routing: Dict) -> Optional[Dict]:
    """Check if decision needs escalation."""
    if datetime.now() > routing['deadline']:
        new_level = routing.get('escalation_level')
        if new_level:
            new_approvers = self.approvers.get(new_level, [])
            escalated_routing = {

```

```

        **routing,
        'approval_level': new_level,
        'approvers': new_approvers,
        'deadline': datetime.now() + timedelta(hours=2),
        'escalated': True
    }

    if self.notification_handler:
        self.notification_handler(escalated_routing)

    return escalated_routing

return None

```

37.8.5. Feedback and Continuous Learning

Collecting feedback on decisions enables continuous improvement of optimization models.

```

@dataclass
class DecisionFeedback:
    """Feedback on a decision outcome."""
    decision_id: str
    actual_outcome: Dict[str, Any]
    expected_outcome: Dict[str, Any]
    user_rating: int # 1-5
    user_comments: str
    timestamp: datetime
    context: Dict[str, Any] = field(default_factory=dict)

class FeedbackCollector:
    """Collects and analyzes decision feedback."""

    def __init__(self):
        self.feedback_store: List[DecisionFeedback] = []
        self.metrics: Dict[str, List[float]] = {}

    def record_feedback(self, feedback: DecisionFeedback):
        """Record feedback for a decision."""
        self.feedback_store.append(feedback)
        self._update_metrics(feedback)

    def _update_metrics(self, feedback: DecisionFeedback):
        """Update performance metrics."""
        decision_type = feedback.context.get('decision_type', 'unknown')

        if decision_type not in self.metrics:
            self.metrics[decision_type] = []

        # Calculate prediction accuracy
        if feedback.expected_outcome and feedback.actual_outcome:
            accuracy = self._calculate_accuracy(
                feedback.expected_outcome,
                feedback.actual_outcome
            )

```

```

        )
        self.metrics[decision_type].append(accuracy)

def _calculate_accuracy(self, expected: Dict, actual: Dict) -> float:
    """Calculate prediction accuracy."""
    if not expected or not actual:
        return 0.0

    errors = []
    for key in expected:
        if key in actual:
            exp_val = expected[key]
            act_val = actual[key]
            if exp_val != 0:
                errors.append(abs(act_val - exp_val) / abs(exp_val))

    return 1.0 - (sum(errors) / len(errors)) if errors else 0.0

def get_performance_summary(self, decision_type: str = None) -> Dict:
    """Get performance summary."""
    if decision_type:
        accuracies = self.metrics.get(decision_type, [])
    else:
        accuracies = [a for accs in self.metrics.values() for a in accs]

    if not accuracies:
        return {'message': 'No feedback recorded'}

    return {
        'total_decisions': len(accuracies),
        'average_accuracy': sum(accuracies) / len(accuracies),
        'min_accuracy': min(accuracies),
        'max_accuracy': max(accuracies),
        'recent_trend': self._calculate_trend(accuracies[-20:])
    }

def _calculate_trend(self, recent: List[float]) -> str:
    """Calculate recent performance trend."""
    if len(recent) < 5:
        return 'insufficient_data'

    first_half = sum(recent[:len(recent)//2]) / (len(recent)//2)
    second_half = sum(recent[len(recent)//2:]) / (len(recent) - len(recent)//2)

    if second_half > first_half * 1.05:
        return 'improving'
    elif second_half < first_half * 0.95:
        return 'declining'
    else:
        return 'stable'

def identify_improvement_areas(self) -> List[Dict]:
    """Identify areas needing model improvement."""

```

```

areas = []

for decision_type, accuracies in self.metrics.items():
    if len(accuracies) > 10:
        avg_accuracy = sum(accuracies) / len(accuracies)
        if avg_accuracy < 0.8:
            areas.append({
                'decision_type': decision_type,
                'average_accuracy': avg_accuracy,
                'sample_size': len(accuracies),
                'recommendation': 'Consider retraining or model update'
            })

return sorted(areas, key=lambda x: x['average_accuracy'])

```

Decision Orchestration Best Practices

1. **Match automation to risk:** High-impact decisions need human oversight
2. **Provide context:** Explanations should include why, not just what
3. **Set timeouts:** Pending decisions should not block operations
4. **Enable overrides:** Humans should always be able to intervene
5. **Close the loop:** Collect feedback and improve continuously

37.9. Summary

37.9.1. Key Takeaways

- Package optimization models with clear interfaces and validation
- Build APIs using FastAPI for both sync and async optimization
- Choose real-time or batch processing based on response time requirements
- Use caching to avoid redundant computations
- Scale through problem decomposition and distributed computing
- Monitor solve times, success rates, and solution quality
- Containerize services for consistent deployment
- Implement health checks for production reliability
- Design human-in-the-loop systems based on decision impact and confidence
- Provide explanations that help humans understand and trust recommendations
- Collect feedback to continuously improve decision quality

37.10. Exercises

1. **API development:** Build a FastAPI service for the transportation problem. Include input validation, async solving, and job status endpoints.
2. **Caching strategy:** Implement a caching layer for an optimization service. Handle cache invalidation when problem parameters change.
3. **Batch processing:** Create a batch pipeline that processes 100 optimization problems in parallel. Measure throughput and compare with sequential processing.
4. **Monitoring dashboard:** Set up Prometheus metrics for an optimization service. Create a Grafana dashboard showing solve times, success rates, and queue depth.

37. Deployment and Scaling

5. **Container deployment:** Package an optimization service as a Docker container. Deploy to Kubernetes with appropriate resource limits and health checks.

38. Digital Twins for Decision Support

What You Will Learn

- Understand digital twin architecture and its role in decision making
- Implement what-if analysis for scenario evaluation
- Build shadowing systems for real-time synchronization
- Design prescriptive digital twins that recommend actions
- Apply digital twins to operational decision problems

Digital twins represent a paradigm shift in how organizations make decisions. By maintaining a synchronized virtual representation of physical systems, digital twins enable real-time monitoring, predictive analysis, and most importantly, prescriptive decision support through what-if analysis and optimization.

38.1. What is a Digital Twin?

A **digital twin** is a virtual representation of a physical system that:

1. **Mirrors** the current state of the real system
2. **Updates** in real-time or near-real-time
3. **Simulates** future behavior under different scenarios
4. **Optimizes** decisions based on predicted outcomes

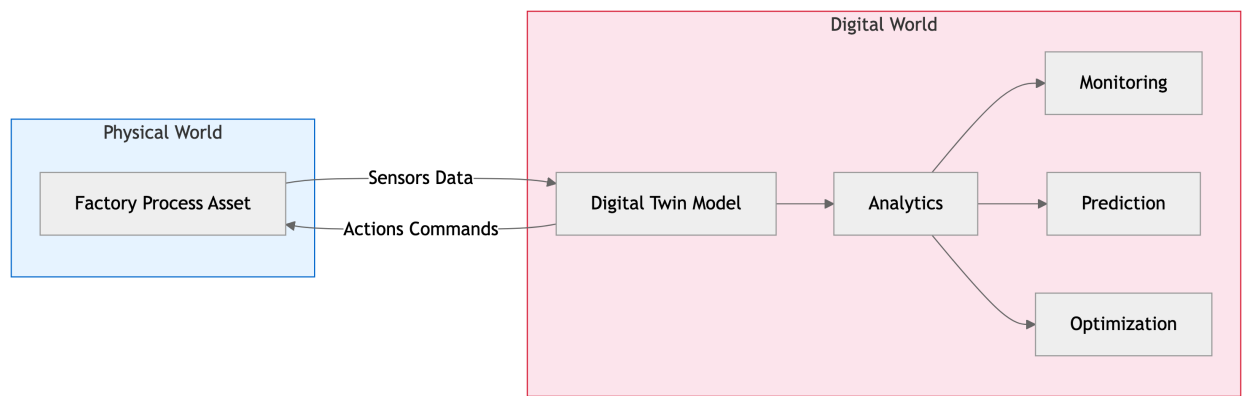


Figure 38.1.: Digital Twin Architecture: Bidirectional flow between physical and digital worlds

38.1.1. Types of Digital Twins

Type	Purpose	Update Frequency	Decision Horizon
Asset Twin	Single equipment monitoring	Real-time	Maintenance decisions

Type	Purpose	Update Frequency	Decision Horizon
Process Twin System	Production line optimization	Minutes	Operational adjustments
Twin	Entire facility coordination	Hours	Tactical planning
Enterprise Twin	Supply chain orchestration	Daily	Strategic decisions

38.2. Shadowing: Real-Time Synchronization

Shadowing is the continuous process of keeping the digital twin synchronized with its physical counterpart. This is the foundation that enables all downstream analytics and decision support.

38.2.1. State Synchronization Architecture

```

from dataclasses import dataclass, field
from typing import Dict, List, Any, Optional, Callable
from datetime import datetime
import numpy as np
from abc import ABC, abstractmethod

@dataclass
class SensorReading:
    """A single sensor observation."""
    sensor_id: str
    timestamp: datetime
    value: float
    unit: str
    quality: float = 1.0 # Data quality score 0-1

@dataclass
class SystemState:
    """Current state of the physical system."""
    timestamp: datetime
    variables: Dict[str, float]
    status: str
    metadata: Dict[str, Any] = field(default_factory=dict)

class StateEstimator(ABC):
    """Abstract base for state estimation algorithms."""

    @abstractmethod
    def update(self, readings: List[SensorReading]) -> SystemState:
        """Update state estimate with new sensor data."""
        pass

    @abstractmethod
    def predict(self, dt: float) -> SystemState:
        """Predict state after time dt."""
        pass

```

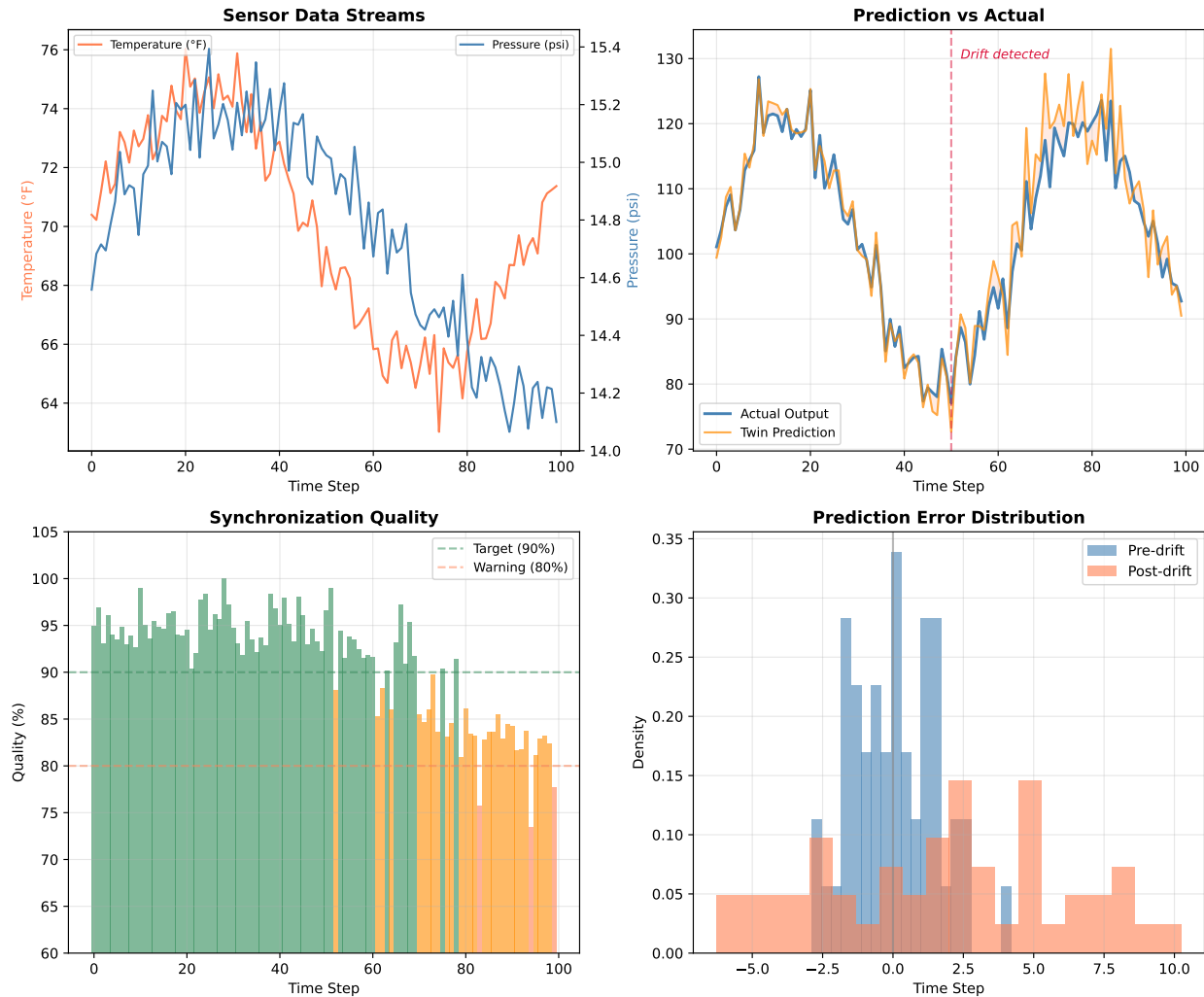



Figure 38.2.: Digital twin real-time synchronization dashboard — sensor data, prediction vs actual comparison, and synchronization quality over time

```
class KalmanStateEstimator(StateEstimator):
    """Kalman filter for state estimation with noisy sensors."""

    def __init__(self, state_dim: int, measurement_dim: int,
                 variable_names: List[str] = None):
        self.state_dim = state_dim
        self.measurement_dim = measurement_dim
        self.variable_names = variable_names or [f"x{i}" for i in range(state_dim)]

        # State estimate and covariance
        self.x = np.zeros(state_dim) # State estimate
        self.P = np.eye(state_dim)    # Estimate covariance

        # System dynamics (to be configured)
        self.F = np.eye(state_dim)    # State transition
```

```

self.H = np.eye(measurement_dim, state_dim) # Measurement matrix
self.Q = np.eye(state_dim) * 0.01 # Process noise
self.R = np.eye(measurement_dim) * 0.1 # Measurement noise

def predict(self, dt: float) -> SystemState:
    """Predict next state."""
    # State prediction
    x_pred = self.F @ self.x
    P_pred = self.F @ self.P @ self.F.T + self.Q

    return SystemState(
        timestamp=datetime.now(),
        variables={self.variable_names[i]: x_pred[i]
                    for i in range(len(x_pred))},
        status="predicted"
    )

def update(self, readings: List[SensorReading]) -> SystemState:
    """Update state with measurements."""
    if not readings:
        return self.predict(0)

    # Convert readings to measurement vector
    z = np.array([r.value for r in readings[:self.measurement_dim]])

    # Kalman update
    y = z - self.H @ self.x # Innovation
    S = self.H @ self.P @ self.H.T + self.R # Innovation covariance
    K = self.P @ self.H.T @ np.linalg.inv(S) # Kalman gain

    self.x = self.x + K @ y
    self.P = (np.eye(self.state_dim) - K @ self.H) @ self.P

    return SystemState(
        timestamp=readings[0].timestamp,
        variables={self.variable_names[i]: self.x[i]
                    for i in range(len(self.x))},
        status="synchronized"
    )

```

38.2.2. Digital Twin Core Implementation

```

from collections import deque
from enum import Enum
import copy

class TwinMode(Enum):
    SHADOW = "shadow" # Following physical system
    SIMULATE = "simulate" # Running what-if scenarios
    OPTIMIZE = "optimize" # Finding optimal decisions

class DigitalTwin:
    """Core digital twin with shadowing and what-if capabilities."""

```

```

def __init__(
    self,
    name: str,
    state_estimator: StateEstimator,
    physics_model: Callable[[SystemState, Dict], SystemState],
    history_size: int = 1000
):
    self.name = name
    self.state_estimator = state_estimator
    self.physics_model = physics_model
    self.mode = TwinMode.SHADOW

    # State management
    self.current_state: Optional[SystemState] = None
    self.state_history: deque = deque(maxlen=history_size)

    # Scenario management
    self.scenarios: Dict[str, List[SystemState]] = {}
    self.base_state_for_scenarios: Optional[SystemState] = None

def sync(self, readings: List[SensorReading]) -> SystemState:
    """Synchronize twin with physical system (shadowing)."""
    if self.mode != TwinMode.SHADOW:
        raise ValueError("Twin must be in SHADOW mode to sync")

    self.current_state = self.state_estimator.update(readings)
    self.state_history.append(copy.deepcopy(self.current_state))
    return self.current_state

def step(self, actions: Dict[str, float], dt: float) -> SystemState:
    """Advance simulation by one time step."""
    if self.current_state is None:
        raise ValueError("No current state - sync first")

    self.current_state = self.physics_model(self.current_state, actions)
    return self.current_state

def enter_scenario_mode(self):
    """Freeze current state and enter what-if mode."""
    self.mode = TwinMode.SIMULATE
    self.base_state_for_scenarios = copy.deepcopy(self.current_state)

def exit_scenario_mode(self):
    """Return to shadow mode, discarding scenario state."""
    self.mode = TwinMode.SHADOW
    self.current_state = copy.deepcopy(self.base_state_for_scenarios)
    self.base_state_for_scenarios = None

def run_scenario(
    self,
    scenario_name: str,
    action_sequence: List[Dict[str, float]],

```

```

    dt: float
) -> List[SystemState]:
    """Run a what-if scenario from current state."""
    if self.mode != TwinMode.SIMULATE:
        raise ValueError("Must be in SIMULATE mode for scenarios")

    # Start from base state
    self.current_state = copy.deepcopy(self.base_state_for_scenarios)
    trajectory = [copy.deepcopy(self.current_state)]

    # Roll out scenario
    for actions in action_sequence:
        state = self.step(actions, dt)
        trajectory.append(copy.deepcopy(state))

    self.scenarios[scenario_name] = trajectory
    return trajectory

def compare_scenarios(self, metric_fn: Callable[[List[SystemState]], float]) ->
    Dict[str, float]:
    """Compare scenarios using a metric function."""
    return {
        name: metric_fn(trajectory)
        for name, trajectory in self.scenarios.items()
    }

```

38.3. What-If Analysis

What-if analysis is the core capability that transforms a digital twin from a monitoring tool into a decision support system. It allows decision makers to explore alternative futures before committing to actions.

38.3.1. Scenario Definition Framework

```

from dataclasses import dataclass
from typing import Tuple

@dataclass
class Scenario:
    """A what-if scenario specification."""
    name: str
    description: str
    action_sequence: List[Dict[str, float]]
    parameters: Dict[str, float]
    probability: float = 1.0 # For stochastic scenarios

@dataclass
class ScenarioResult:
    """Results from running a scenario."""
    scenario: Scenario
    trajectory: List[SystemState]
    kpis: Dict[str, float]

```



Figure 38.3.: What-if scenario comparison — three alternative strategies simulated through the digital twin, with performance metrics compared in the bottom panel

```
feasible: bool
warnings: List[str]

class WhatIfAnalyzer:
    """Systematic what-if analysis engine."""

    def __init__(self, twin: DigitalTwin):
        self.twin = twin
        self.kpi_functions: Dict[str, Callable] = {}
        self.constraints: List[Callable] = []

    def register_kpi(self, name: str, fn: Callable[[List[SystemState]], float]):
        """Register a KPI calculation function."""
        self.kpi_functions[name] = fn

    def register_constraint(self, fn: Callable[[SystemState], Tuple[bool, str]]):
        """Register a constraint check function."""
```

```

        self.constraints.append(fn)

def evaluate_scenario(self, scenario: Scenario, dt: float) -> ScenarioResult:
    """Run and evaluate a single scenario."""
    # Run simulation
    trajectory = self.twin.run_scenario(
        scenario.name,
        scenario.action_sequence,
        dt
    )

    # Calculate KPIs
    kpis = {
        name: fn(trajectory)
        for name, fn in self.kpi_functions.items()
    }

    # Check constraints
    warnings = []
    feasible = True
    for state in trajectory:
        for constraint in self.constraints:
            ok, msg = constraint(state)
            if not ok:
                feasible = False
                warnings.append(f"At {state.timestamp}: {msg}")

    return ScenarioResult(
        scenario=scenario,
        trajectory=trajectory,
        kpis=kpis,
        feasible=feasible,
        warnings=warnings
    )

def compare_scenarios(
    self,
    scenarios: List[Scenario],
    dt: float,
    ranking_kpi: str = None
) -> List[ScenarioResult]:
    """Evaluate and compare multiple scenarios."""
    self.twin.enter_scenario_mode()

    results = []
    for scenario in scenarios:
        result = self.evaluate_scenario(scenario, dt)
        results.append(result)

    self.twin.exit_scenario_mode()

    # Rank by KPI if specified
    if ranking_kpi and ranking_kpi in self.kpi_functions:

```

```

        results.sort(key=lambda r: r.kpis.get(ranking_kpi, float('inf')))

    return results

```

38.3.2. Example: Production Planning What-If

```

def production_physics(state: SystemState, actions: Dict) -> SystemState:
    """Simple production system dynamics."""
    # Extract current state
    inventory = state.variables.get("inventory", 0)
    wip = state.variables.get("wip", 0)

    # Apply actions
    production_rate = actions.get("production_rate", 10)
    shipping = actions.get("shipping", 8)

    # Update state (simple mass balance)
    new_inventory = inventory + production_rate - shipping

    return SystemState(
        timestamp=datetime.now(),
        variables={
            "inventory": max(0, new_inventory),
            "wip": wip + actions.get("new_orders", 0) - production_rate,
            "production_rate": production_rate
        },
        status="simulated"
    )

# Example usage
def demo_whatif():
    """Demonstrate what-if analysis."""
    # Initialize twin with named variables
    estimator = KalmanStateEstimator(
        state_dim=3,
        measurement_dim=3,
        variable_names=["inventory", "wip", "production_rate"]
    )
    twin = DigitalTwin(
        name="ProductionLine1",
        state_estimator=estimator,
        physics_model=production_physics
    )

    # Set initial state via sync
    initial_readings = [
        SensorReading("inv_sensor", datetime.now(), 100, "units"),
        SensorReading("wip_sensor", datetime.now(), 50, "units"),
        SensorReading("rate_sensor", datetime.now(), 10, "units/hr")
    ]
    twin.sync(initial_readings)

    # Create analyzer

```

```

analyzer = WhatIfAnalyzer(twin)

# Register KPIs
analyzer.register_kpi(
    "total_inventory_cost",
    lambda traj: sum(s.variables["inventory"] * 0.5 for s in traj)
)
analyzer.register_kpi(
    "final_inventory",
    lambda traj: traj[-1].variables["inventory"]
)

# Register constraint
analyzer.register_constraint(
    lambda s: (s.variables["inventory"] >= 20, "Inventory below safety stock")
)

# Define scenarios
scenarios = [
    Scenario(
        name="baseline",
        description="Current production plan",
        action_sequence=[{"production_rate": 10, "shipping": 12}] * 10,
        parameters={}
    ),
    Scenario(
        name="increase_production",
        description="Ramp up production",
        action_sequence=[{"production_rate": 15, "shipping": 12}] * 10,
        parameters={"overtime": True}
    ),
    Scenario(
        name="reduce_shipping",
        description="Delay some shipments",
        action_sequence=[{"production_rate": 10, "shipping": 8}] * 10,
        parameters={}
    )
]

# Compare scenarios
results = analyzer.compare_scenarios(scenarios, dt=1.0,
↪ ranking_kpi="total_inventory_cost")

print("Scenario Comparison:")
print("-" * 60)
for r in results:
    status = "FEASIBLE" if r.feasible else "INFEASIBLE"
    print(f"{r.scenario.name:20} | Cost: {r.kpis['total_inventory_cost']:8.1f} |
↪ {status}")
    if r.warnings:
        print(f" Warnings: {r.warnings[0]}")

return results

```



```
# Run demo
demo_whatif()
```

Scenario Comparison:

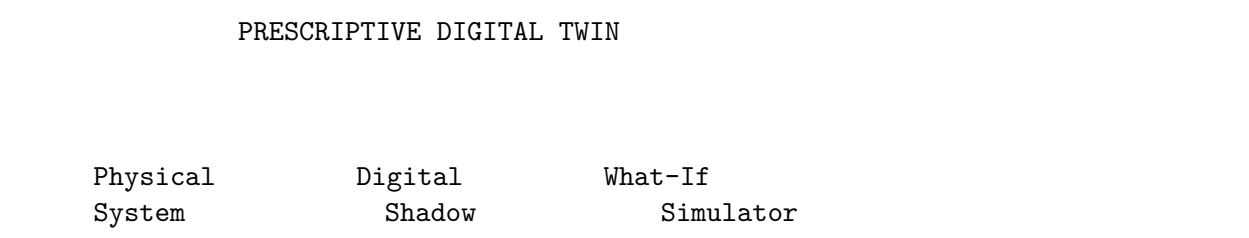
baseline	Cost: 445.0 FEASIBLE
reduce_shipping	Cost: 555.0 FEASIBLE
increase_production	Cost: 582.5 FEASIBLE

```
[ScenarioResult(scenario=Scenario(name='baseline', description='Current production plan', acti
4.545454545454547), 'production_rate': 10}, status='simulated', metadata={}), SystemState(time
14.545454545454547), 'production_rate': 10}, status='simulated', metadata={}), SystemState(tim
24.545454545454547), 'production_rate': 10}, status='simulated', metadata={}), SystemState(tim
34.54545454545455), 'production_rate': 10}, status='simulated', metadata={}), SystemState(time
44.54545454545455), 'production_rate': 10}, status='simulated', metadata={}), SystemState(time
54.54545454545455), 'production_rate': 10}, status='simulated', metadata={})], kpis={'total_in
ScenarioResult(scenario=Scenario(name='reduce_shipping', description='Delay some shipments', a
4.545454545454547), 'production_rate': 10}, status='simulated', metadata={}), SystemState(time
14.545454545454547), 'production_rate': 10}, status='simulated', metadata={}), SystemState(tim
24.545454545454547), 'production_rate': 10}, status='simulated', metadata={}), SystemState(tim
34.54545454545455), 'production_rate': 10}, status='simulated', metadata={}), SystemState(time
44.54545454545455), 'production_rate': 10}, status='simulated', metadata={}), SystemState(time
54.54545454545455), 'production_rate': 10}, status='simulated', metadata={})], kpis={'total_in
ScenarioResult(scenario=Scenario(name='increase_production', description='Ramp up production'
14.545454545454547), 'production_rate': 15}, status='simulated', metadata={}), SystemState(tim
29.545454545454547), 'production_rate': 15}, status='simulated', metadata={}), SystemState(tim
44.54545454545455), 'production_rate': 15}, status='simulated', metadata={}), SystemState(time
59.54545454545455), 'production_rate': 15}, status='simulated', metadata={}), SystemState(time
74.54545454545455), 'production_rate': 15}, status='simulated', metadata={}), SystemState(time
89.54545454545455), 'production_rate': 15}, status='simulated', metadata={}), SystemState(time
104.54545454545455), 'production_rate': 15}, status='simulated', metadata={})], kpis={'total_in
```

38.4. Prescriptive Digital Twins

A prescriptive digital twin goes beyond what-if analysis to actively recommend optimal decisions. It combines the simulation capabilities of the twin with optimization algorithms.

38.4.1. Architecture



Optimizer

- MIP
- RL
- Heuristic

Recommended
Actions

38.4.2. Optimization Integration

```
from scipy.optimize import minimize, differential_evolution

class PrescriptiveDigitalTwin:
    """Digital twin with built-in optimization for decision recommendations."""

    def __init__(
        self,
        twin: DigitalTwin,
        objective_fn: Callable[[List[SystemState]], float],
        action_bounds: Dict[str, Tuple[float, float]]
    ):
        self.twin = twin
        self.objective_fn = objective_fn
        self.action_bounds = action_bounds
        self.constraints: List[Callable] = []

    def add_constraint(self, fn: Callable[[List[SystemState]], float]):
        """Add constraint: fn(trajecotory) <= 0 for feasibility."""
        self.constraints.append(fn)

    def _encode_actions(self, action_sequence: List[Dict]) -> np.ndarray:
        """Flatten action sequence to optimization vector."""
        vector = []
        for actions in action_sequence:
            for key in sorted(self.action_bounds.keys()):
                vector.append(actions.get(key, 0))
        return np.array(vector)

    def _decode_actions(self, vector: np.ndarray, horizon: int) -> List[Dict]:
        """Reconstruct action sequence from optimization vector."""
        n_actions = len(self.action_bounds)
        action_sequence = []
        keys = sorted(self.action_bounds.keys())

        for t in range(horizon):
```

```

        actions = {}
        for i, key in enumerate(keys):
            actions[key] = vector[t * n_actions + i]
        action_sequence.append(actions)

    return action_sequence

def _evaluate(self, vector: np.ndarray, horizon: int, dt: float) -> float:
    """Objective function for optimizer."""
    action_sequence = self._decode_actions(vector, horizon)

    # Run simulation
    self.twin.enter_scenario_mode()
    trajectory = self.twin.run_scenario("opt_eval", action_sequence, dt)
    self.twin.exit_scenario_mode()

    # Calculate objective
    obj = self.objective_fn(trajectory)

    # Add penalty for constraint violations
    penalty = 0
    for constraint in self.constraints:
        violation = max(0, constraint(trajectory))
        penalty += 1000 * violation ** 2

    return obj + penalty

def optimize(
    self,
    horizon: int,
    dt: float,
    method: str = "differential_evolution"
) -> Tuple[List[Dict], float, List[SystemState]]:
    """Find optimal action sequence."""
    n_actions = len(self.action_bounds)
    n_vars = horizon * n_actions

    # Build bounds
    bounds = []
    for _ in range(horizon):
        for key in sorted(self.action_bounds.keys()):
            bounds.append(self.action_bounds[key])

    # Optimize
    if method == "differential_evolution":
        result = differential_evolution(
            lambda x: self._evaluate(x, horizon, dt),
            bounds,
            maxiter=100,
            seed=42
        )
    else:
        x0 = np.array([np.mean(b) for b in bounds])

```

```

        result = minimize(
            lambda x: self._evaluate(x, horizon, dt),
            x0,
            bounds=bounds,
            method='L-BFGS-B'
        )

    # Decode solution
    optimal_actions = self._decode_actions(result.x, horizon)

    # Get final trajectory
    self.twin.enter_scenario_mode()
    trajectory = self.twin.run_scenario("optimal", optimal_actions, dt)
    self.twin.exit_scenario_mode()

    return optimal_actions, result.fun, trajectory

def recommend_next_action(self, horizon: int = 5, dt: float = 1.0) -> Dict:
    """Get recommendation for immediate next action."""
    optimal_actions, _, _ = self.optimize(horizon, dt)
    return optimal_actions[0] if optimal_actions else {}

```

38.4.3. Example: Energy System Optimization

```

def energy_system_physics(state: SystemState, actions: Dict) -> SystemState:
    """Energy storage system dynamics."""
    battery_level = state.variables.get("battery", 50)
    grid_price = state.variables.get("grid_price", 0.10)

    # Actions: charge/discharge rate
    charge_rate = actions.get("charge_rate", 0) # + = charge, - = discharge

    # Battery dynamics
    efficiency = 0.95
    if charge_rate > 0:
        new_battery = battery_level + charge_rate * efficiency
    else:
        new_battery = battery_level + charge_rate / efficiency

    # Constraints: 0-100%
    new_battery = np.clip(new_battery, 0, 100)

    # Simulate price variation
    new_price = grid_price * (1 + 0.1 * np.random.randn())

    return SystemState(
        timestamp=datetime.now(),
        variables={
            "battery": new_battery,
            "grid_price": new_price,
            "energy_cost": -charge_rate * grid_price # Negative = revenue
        },
        status="simulated"
    )

```

```

    )

def demo_prescriptive_twin():
    """Demonstrate prescriptive digital twin."""
    # Initialize with named variables
    estimator = KalmanStateEstimator(
        state_dim=3,
        measurement_dim=3,
        variable_names=["battery", "grid_price", "energy_cost"]
    )
    twin = DigitalTwin(
        name="BatteryStorage",
        state_estimator=estimator,
        physics_model=energy_system_physics
    )

    # Sync initial state
    readings = [
        SensorReading("battery", datetime.now(), 50, "%"),
        SensorReading("price", datetime.now(), 0.10, "$/kWh"),
        SensorReading("cost", datetime.now(), 0, "$")
    ]
    twin.sync(readings)

    # Create prescriptive twin
    prescriptive = PrescriptiveDigitalTwin(
        twin=twin,
        objective_fn=lambda traj: sum(s.variables.get("energy_cost", 0) for s in
↪ traj),
        action_bounds={"charge_rate": (-20, 20)}
    )

    # Add battery level constraints
    prescriptive.add_constraint(
        lambda traj: 20 - min(s.variables["battery"] for s in traj) # Min 20%
    )
    prescriptive.add_constraint(
        lambda traj: max(s.variables["battery"] for s in traj) - 90 # Max 90%
    )

    # Get recommendation
    recommendation = prescriptive.recommend_next_action(horizon=10, dt=1.0)
    print(f"Recommended action: charge_rate = {recommendation.get('charge_rate',
↪ 0):.2f} kW")

    return recommendation

demo_prescriptive_twin()

```

Recommended action: charge_rate = -5.85 kW

```
{'charge_rate': np.float64(-5.847890446285624)}
```

38.5. Real-Time Decision Support

Digital twins enable real-time decision support by continuously monitoring, predicting, and recommending.

38.5.1. Event-Driven Decision Loop

```
from enum import Enum
from typing import Optional
import time

class AlertSeverity(Enum):
    INFO = "info"
    WARNING = "warning"
    CRITICAL = "critical"

@dataclass
class Alert:
    """An alert generated by the decision support system."""
    severity: AlertSeverity
    message: str
    recommended_action: Optional[Dict]
    timestamp: datetime

class RealTimeDecisionSupport:
    """Real-time decision support using digital twin."""

    def __init__(self, prescriptive_twin: PrescriptiveDigitalTwin):
        self.twin = prescriptive_twin
        self.alert_rules: List[Callable] = []
        self.alerts: List[Alert] = []
        self.decision_interval = 60 # seconds
        self.last_decision_time = 0

    def add_alert_rule(
        self,
        condition: Callable[[SystemState], bool],
        severity: AlertSeverity,
        message_fn: Callable[[SystemState], str]
    ):
        """Add an alerting rule."""
        self.alert_rules.append((condition, severity, message_fn))

    def check_alerts(self, state: SystemState) -> List[Alert]:
        """Check all alert rules against current state."""
        new_alerts = []
        for condition, severity, message_fn in self.alert_rules:
            if condition(state):
                alert = Alert(
                    severity=severity,
                    message=message_fn(state),
                    recommended_action=None,
                    timestamp=datetime.now()
                )
                new_alerts.append(alert)
        return new_alerts
```

```

        new_alerts.append(alert)
        self.alerts.append(alert)
    return new_alerts

def process_update(self, readings: List[SensorReading]) -> Dict:
    """Process new sensor data and generate recommendations."""
    # Sync twin
    state = self.twin.twin.sync(readings)

    # Check alerts
    alerts = self.check_alerts(state)

    # Generate recommendation if enough time passed
    recommendation = None
    current_time = time.time()
    if current_time - self.last_decision_time > self.decision_interval:
        recommendation = self.twin.recommend_next_action()
        self.last_decision_time = current_time

    # Attach recommendation to critical alerts
    for alert in alerts:
        if alert.severity == AlertSeverity.CRITICAL:
            alert.recommended_action = recommendation

    return {
        "state": state,
        "alerts": alerts,
        "recommendation": recommendation
    }

```

38.6. Applications

38.6.1. Manufacturing Digital Twin

```

class ManufacturingTwin:
    """Digital twin for a manufacturing cell."""

    def __init__(self, cell_id: str):
        self.cell_id = cell_id
        self.machines: Dict[str, Dict] = {}
        self.buffer_levels: Dict[str, float] = {}
        self.quality_metrics: Dict[str, float] = {}

    def update_machine_state(self, machine_id: str, state: Dict):
        """Update individual machine state."""
        self.machines[machine_id] = {
            **state,
            "last_update": datetime.now()
        }

    def predict_maintenance(self, machine_id: str) -> Dict:
        """Predict maintenance needs for a machine."""
        machine = self.machines.get(machine_id, {})

```

```

# Simple degradation model
hours_since_maintenance = machine.get("hours_since_maintenance", 0)
vibration = machine.get("vibration_level", 0)
temperature = machine.get("temperature", 0)

# Health score (simplified)
health_score = 100 - (hours_since_maintenance / 10) - (vibration * 5) -
↪ (temperature / 10)
health_score = max(0, min(100, health_score))

# Predict failure probability
failure_prob = 1 / (1 + np.exp(0.1 * (health_score - 30)))

return {
    "machine_id": machine_id,
    "health_score": health_score,
    "failure_probability": failure_prob,
    "recommended_action": "schedule_maintenance" if failure_prob > 0.3 else
    ↪ "monitor"
}

def optimize_production_schedule(self, orders: List[Dict]) -> List[Dict]:
    """Optimize production schedule based on current state."""
    # Simplified scheduling based on machine availability
    schedule = []

    available_machines = [
        m_id for m_id, state in self.machines.items()
        if state.get("status") == "available"
    ]

    for order in orders:
        if available_machines:
            machine = available_machines.pop(0)
            schedule.append({
                "order_id": order["id"],
                "machine_id": machine,
                "start_time": datetime.now(),
                "estimated_completion": order.get("processing_time", 60)
            })

    return schedule

```

38.6.2. Supply Chain Digital Twin

```

@dataclass
class SupplyChainNode:
    """A node in the supply chain network."""
    node_id: str
    node_type: str # supplier, warehouse, customer
    location: Tuple[float, float]
    inventory: Dict[str, float]

```



```

        capacity: float

class SupplyChainTwin:
    """Digital twin for supply chain visibility and optimization."""

    def __init__(self):
        self.nodes: Dict[str, SupplyChainNode] = {}
        self.edges: List[Tuple[str, str, Dict]] = [] # (from, to, attributes)
        self.in_transit: List[Dict] = []

    def add_node(self, node: SupplyChainNode):
        """Add a node to the supply chain."""
        self.nodes[node.node_id] = node

    def add_edge(self, from_id: str, to_id: str, lead_time: float, cost: float):
        """Add a connection between nodes."""
        self.edges.append((from_id, to_id, {
            "lead_time": lead_time,
            "cost": cost
        }))

    def simulate_disruption(self, node_id: str, duration: float) -> Dict:
        """Simulate a supply chain disruption."""
        affected_node = self.nodes.get(node_id)
        if not affected_node:
            return {"error": "Node not found"}

        # Find downstream impact
        affected_downstream = []
        for from_id, to_id, attrs in self.edges:
            if from_id == node_id:
                affected_downstream.append({
                    "node": to_id,
                    "impact_start": attrs["lead_time"],
                    "severity": "high" if
                        ↪ self.nodes[to_id].inventory.get("safety_stock", 0) <
                        ↪ duration else "low"
                })

        return {
            "disrupted_node": node_id,
            "duration": duration,
            "downstream_impact": affected_downstream,
            "mitigation_options": self._generate_mitigation_options(node_id,
                ↪ duration)
        }

    def _generate_mitigation_options(self, node_id: str, duration: float) ->
        ↪ List[Dict]:
        """Generate mitigation options for a disruption."""
        options = []

        # Option 1: Find alternative suppliers

```

```

alternatives = [
    n_id for n_id, node in self.nodes.items()
    if node.node_type == "supplier" and n_id != node_id
]
if alternatives:
    options.append({
        "strategy": "alternative_supplier",
        "nodes": alternatives[:3],
        "estimated_cost_increase": 0.15
    })

# Option 2: Draw from safety stock
options.append({
    "strategy": "safety_stock",
    "coverage_days": 5,
    "risk": "depletes buffer"
})

# Option 3: Expedited shipping
options.append({
    "strategy": "expedite",
    "lead_time_reduction": 0.5,
    "cost_increase": 0.30
})

return options

```

38.7. Best Practices

38.7.1. Design Principles

Digital Twin Design Guidelines

1. **Start with the decision:** Design the twin around the decisions it needs to support
2. **Right fidelity:** Match model complexity to decision requirements
3. **Fail gracefully:** Handle sensor failures and data gaps
4. **Validate continuously:** Compare predictions to actual outcomes
5. **Version everything:** Track model versions and parameter changes

38.7.2. Data Quality Management

```

class DataQualityMonitor:
    """Monitor data quality for digital twin inputs."""

    def __init__(self):
        self.expected_sensors: Dict[str, Dict] = {}
        self.quality_history: Dict[str, List[float]] = {}

    def register_sensor(
        self,
        sensor_id: str,

```

```

        expected_range: Tuple[float, float],
        expected_frequency: float # readings per second
    ):
        """Register expected sensor behavior."""
        self.expected_sensors[sensor_id] = {
            "range": expected_range,
            "frequency": expected_frequency,
            "last_reading": None
        }
        self.quality_history[sensor_id] = []

    def assess_reading(self, reading: SensorReading) -> float:
        """Assess quality of a sensor reading (0-1)."""
        if reading.sensor_id not in self.expected_sensors:
            return 0.5 # Unknown sensor

        expected = self.expected_sensors[reading.sensor_id]
        quality = 1.0

        # Range check
        low, high = expected["range"]
        if reading.value < low or reading.value > high:
            # Penalize out-of-range values
            distance = min(abs(reading.value - low), abs(reading.value - high))
            range_size = high - low
            quality *= max(0, 1 - distance / range_size)

        # Frequency check
        if expected["last_reading"]:
            elapsed = (reading.timestamp - expected["last_reading"]).total_seconds()
            expected_interval = 1 / expected["frequency"]
            if elapsed > 2 * expected_interval:
                quality *= 0.8 # Penalize late readings

        expected["last_reading"] = reading.timestamp
        self.quality_history[reading.sensor_id].append(quality)

    return quality

```

38.8. Commercial Digital Twin Platforms

While the implementations in this chapter use custom Python code for maximum transparency, enterprise digital twin projects typically build on commercial platforms that provide connectivity, visualization, and scalability out of the box.

Platform	Provider	Strengths
Azure Digital Twins	Microsoft	DTDL ontology language, IoT Hub integration, graph-based twin modeling, Azure Synapse analytics
AWS IoT TwinMaker	Amazon	3D scene composition, IoT SiteWise integration, Grafana dashboards

Platform	Provider	Strengths
Siemens Xcelerator / MindSphere	Siemens	Industrial IoT leader, Tecnomatix plant simulation, NX CAD integration, closed-loop manufacturing
ANSYS Twin Builder	ANSYS	Physics-based reduced-order models (ROMs), FEA/CFD integration, real-time deployment on edge
Altair SmartWorks	Altair	IoT platform with simulation-driven twins, multiphysics coupling
Eclipse Ditto	Eclipse Foundation	Open-source digital twin framework, MQTT/HTTP APIs, policy-based access control

When to build custom vs. use a platform: Custom implementations (as shown in this chapter) are best for research, prototyping, or when the optimization logic is highly specialized. Commercial platforms excel when you need enterprise-grade connectivity to hundreds of sensors, 3D visualization, role-based access control, and integration with existing IT/OT infrastructure. Many organizations use a hybrid approach: a commercial platform for data ingestion and visualization, with custom optimization logic running alongside.

For simulation-based digital twins in manufacturing, tools like **Siemens Tecnomatix Plant Simulation**, **FlexSim**, and **AnyLogic** (discussed in Chapter 28) can serve as the simulation engine within a digital twin architecture, enabling what-if analysis with high-fidelity 3D models.

38.9. Summary

38.9.1. Key Takeaways

- **Digital twins** are synchronized virtual representations that enable decision support
- **Shadowing** maintains real-time synchronization between physical and digital systems
- **What-if analysis** explores alternative futures before committing to decisions
- **Prescriptive twins** combine simulation with optimization to recommend actions
- **State estimation** (Kalman filtering) handles noisy sensor data
- Design twins around the **decisions** they need to support
- **Data quality** monitoring is essential for reliable recommendations
- Commercial platforms (Azure Digital Twins, AWS IoT TwinMaker, Siemens Xcelerator) provide enterprise-grade connectivity and visualization

38.10. Exercises

1. Extend the `DigitalTwin` class to support multiple physics models that can be switched based on operating mode (normal, degraded, startup).
2. Implement a constraint satisfaction layer that prevents the prescriptive twin from recommending infeasible actions.
3. Design a digital twin for a traffic intersection that can evaluate different signal timing plans via what-if analysis.
4. Add uncertainty quantification to the `WhatIfAnalyzer` by running Monte Carlo simulations for each scenario.
5. Implement a “model confidence” metric that degrades when the digital twin’s predictions diverge from actual sensor readings.

Part VII.

Appendices

39. Python Libraries for Decision Intelligence

What You Will Learn

- Navigate the Python optimization ecosystem and select appropriate tools
- Choose between mathematical programming frameworks (Pyomo, PuLP, CVXPY)
- Apply metaheuristic libraries for black-box optimization
- Use specialized solvers for routing, scheduling, and constraint programming
- Combine multiple libraries for complex real-world problems

Python's ecosystem includes dozens of optimization libraries ranging from mathematical programming frameworks to metaheuristic solvers. This appendix provides a comprehensive guide to selecting and using the right tools for your decision intelligence applications.

39.1. Library Landscape Overview

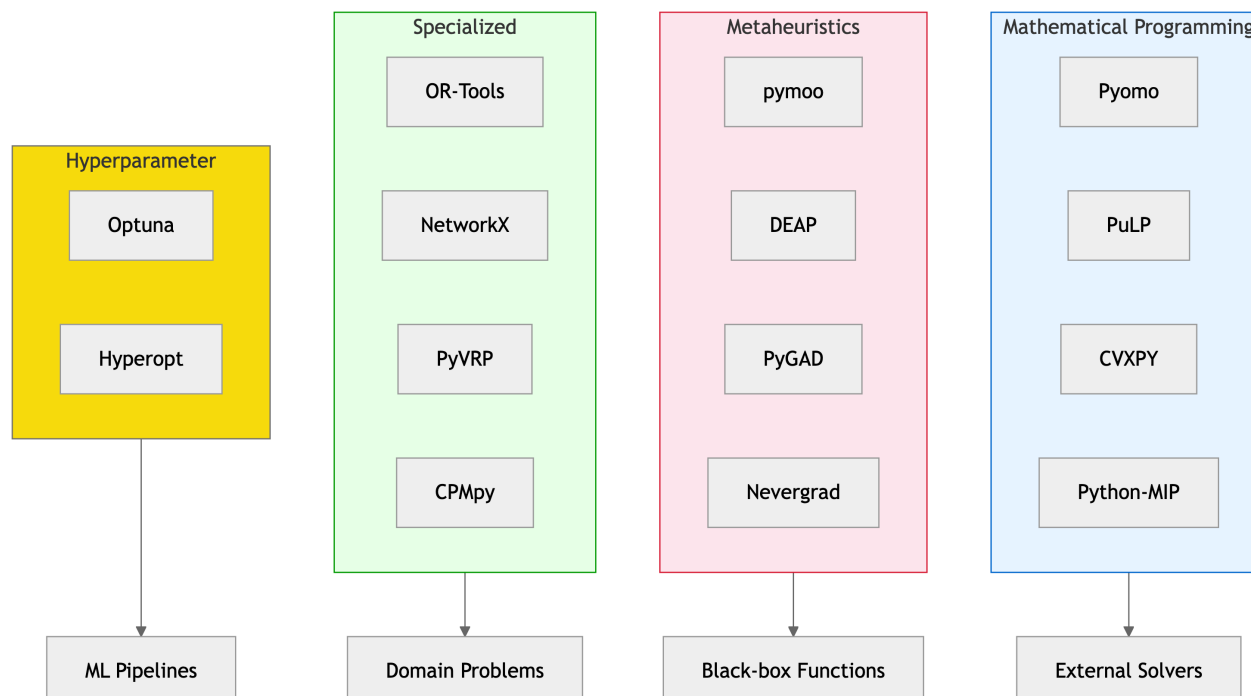


Figure 39.1.: Python Optimization Library Ecosystem

The heatmap below scores each library across ten capabilities: LP, MIP, QP, NLP, constraint programming, multi-objective support, stochastic programming, solver agnosticism, GPU acceleration, and documentation quality.

The bubble chart below positions each library on two axes: ease of use versus advanced feature support. Bubble size reflects community activity (GitHub stars and active contributors), providing a rough measure of long-term viability.

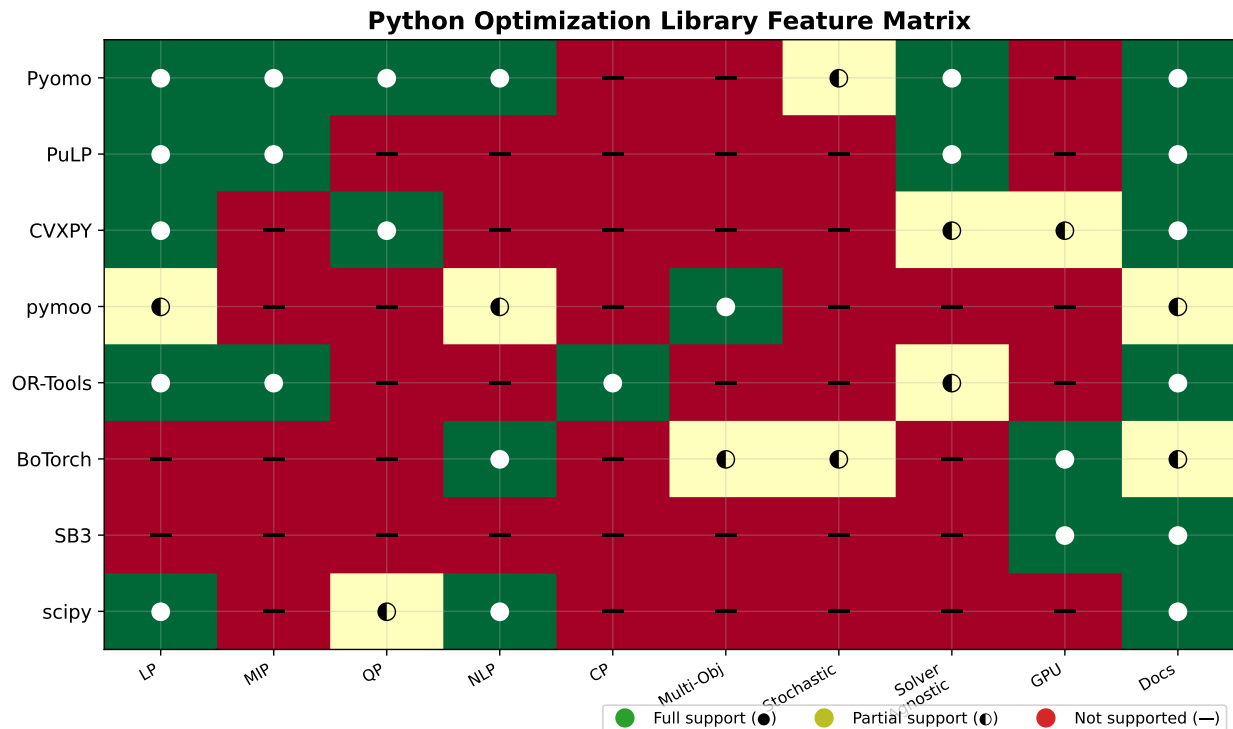


Figure 39.2.: Python optimization library feature heatmap — comparing capabilities across mathematical programming, metaheuristics, specialized solvers, and ML-for-optimization libraries

39.2. Mathematical Programming Frameworks

39.2.1. Pyomo

Pyomo provides algebraic modeling language (AML) capabilities in Python, similar to AMPL or GAMS. It supports linear programming, mixed-integer programming, nonlinear programming, and stochastic programming.

Strengths:

- Problem formulation separates from solver selection
- Write the model once, test with multiple solvers (GLPK, CBC, CPLEX, Gurobi, IPOPT)
- Extensive documentation and active community

Best For: Large-scale optimization, energy system planning, production scheduling, network design.

```
import pyomo.environ as pyo

model = pyo.ConcreteModel()
model.x = pyo.Var([1, 2], domain=pyo.NonNegativeReals)
model.obj = pyo.Objective(expr=2*model.x[1] + 3*model.x[2])
model.constraint1 = pyo.Constraint(expr=3*model.x[1] + 4*model.x[2] >= 1)

solver = pyo.SolverFactory('glpk')
```

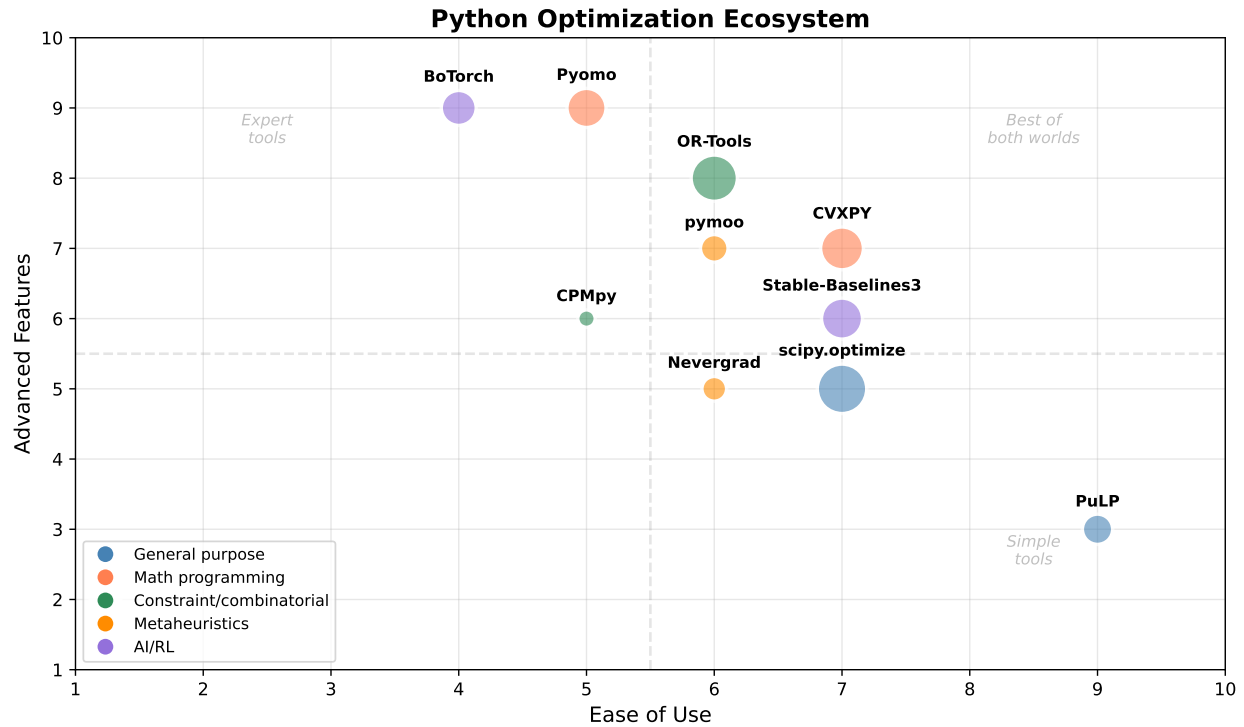



Figure 39.3.: Python optimization ecosystem — ease of use vs advanced features, with bubble size representing community activity (GitHub stars + contributors)

```
solver.solve(model)
```

39.2.2. PuLP

PuLP focuses on linear and integer programming with simpler syntax than Pyomo. Ideal for beginners or problems that don't require nonlinear capabilities.

Strengths:

- Lightweight and easy to learn
- Quick prototyping
- Good documentation for beginners

Best For: Transportation problems, assignment problems, production planning with linear constraints.

39.2.3. CVXPY

CVXPY specializes in convex optimization using disciplined convex programming (DCP) to verify problem structure automatically.

Strengths:

- Automatically selects appropriate solver based on problem structure
- Strong theoretical guarantees for convex problems
- Clean, mathematical syntax

Best For: Portfolio optimization, control systems, signal processing, machine learning regularization.

39.2.4. Python-MIP

High-performance mixed-integer programming with advanced features like lazy constraints, cutting planes, and solution pools.

Strengths:

- Fast model generation through C integration
- Advanced solver features accessible from Python
- Compatible with CBC and Gurobi

Best For: Large-scale MIP problems requiring solver callbacks or custom cuts.

39.3. Metaheuristic Frameworks

39.3.1. pymoo

Multi-objective optimization library implementing genetic algorithms (NSGA-II, NSGA-III), particle swarm optimization, and differential evolution.

Strengths:

- Comprehensive multi-objective optimization
- Pareto front visualization
- Performance indicators and decision-making tools

Best For: Engineering design optimization, portfolio optimization with multiple criteria.

```
from pymoo.algorithms.moo.nsga2 import NSGA2
from pymoo.problems import get_problem
from pymoo.optimize import minimize

problem = get_problem("zdt1")
algorithm = NSGA2(pop_size=100)
result = minimize(problem, algorithm, ('n_gen', 200))
```

39.3.2. DEAP

Distributed Evolutionary Algorithms in Python. Flexible framework for implementing custom evolutionary algorithms and genetic programming.

Strengths:

- Highly customizable
- Supports parallel and distributed computing
- Includes benchmarking functions

Best For: Research on evolutionary computation, custom algorithm development, genetic programming.

39.3.3. Nevergrad

Meta-metaheuristic framework from Facebook Research providing unified interface to dozens of optimization algorithms with automatic algorithm selection.

Strengths:

- Algorithm portfolio approach
- Handles mixed continuous/discrete/categorical variables
- Bayesian optimization, evolutionary strategies, and classical metaheuristics

Best For: Hyperparameter optimization with mixed parameter types, cases where best algorithm is unknown.

```
import nevergrad as ng

def objective(x, y):
    return (x - 3)**2 + (y + 2)**2

optimizer = ng.optimizers.NGOpt(parametrization=ng.p.Instrumentation(
    x=ng.p.Scalar(lower=-5, upper=5),
    y=ng.p.Scalar(lower=-5, upper=5)
))
recommendation = optimizer.minimize(objective, budget=100)
```

39.3.4. PyGMO

Parallel global multiobjective optimizer from European Space Agency implementing island model for massively parallel optimization.

Strengths:

- Scales to hundreds of cores
- Built-in algorithms: differential evolution, PSO, simulated annealing
- Asynchronous island model

Best For: Expensive black-box optimization, spacecraft trajectory optimization, engineering design.

39.4. Specialized Solvers

39.4.1. OR-Tools

Google's OR-Tools provides constraint programming, linear/integer programming, vehicle routing, and graph algorithms in one package.

Strengths:

- Production-ready routing solver with time windows, capacity constraints
- Excellent constraint programming (CP-SAT solver)
- Well-documented and actively maintained

Best For: Vehicle routing, job shop scheduling, employee rostering, bin packing.

39.4.2. PyVRP

Vehicle routing solver implementing hybrid genetic search with advanced local search operators.

Strengths:

- State-of-the-art metaheuristic performance
- Handles time windows, capacity, multiple depots, heterogeneous fleets
- Competitive with commercial solvers

Best For: Large-scale vehicle routing, delivery optimization, service technician scheduling.

```
from pyvrp import Model, solve
from pyvrp.stop import MaxRuntime
```

```
model = Model()
depot = model.add_depot(x=0, y=0)
clients = [model.add_client(x=i, y=i, demand=1) for i in range(1, 20)]
vehicles = [model.add_vehicle_type(capacity=10) for _ in range(3)]
result = solve(model, stop=MaxRuntime(10))
```

39.4.3. CPMpy

Constraint programming modeling with numpy integration and support for multiple CP solvers.

Strengths:

- Intuitive syntax close to mathematical notation
- Native numpy array support
- Easy solver switching (OR-Tools, Gurobi, PySAT)

Best For: Scheduling with complex logical constraints, configuration problems, planning.

```
from cpmPy import Model, intvar, AllDifferent

n = 8
queens = intvar(0, n-1, shape=n)
model = Model([
    AllDifferent(queens),
    AllDifferent(queens + range(n)),
    AllDifferent(queens - range(n))
])
model.solve()
```

39.4.4. NetworkX

Graph theory and network optimization algorithms for network flow problems, shortest paths, matching, and graph-based optimization.

Strengths:

- Comprehensive graph algorithms
- Integration with visualization tools
- Maximum flow/minimum cost flow solvers

Best For: Network design, transportation planning, project scheduling with precedence constraints.

39.4.5. Graph Neural Network Libraries

39.4.5.1. PyTorch Geometric (PyG)

The standard library for graph neural networks in PyTorch, widely used for learning on graph-structured combinatorial optimization problems (see Chapter 34).

Strengths:

- Efficient sparse tensor operations and mini-batching
- Pre-built GNN layers (GCN, GAT, GraphSAGE, GIN)
- Large collection of benchmark datasets
- Active development with 21k+ GitHub stars

Best For: GNN-based optimization, node/edge classification for branching or routing, learning graph representations of combinatorial problems.

```

import torch
from torch_geometric.nn import GCNConv, global_mean_pool

class GNNPolicy(torch.nn.Module):
    """Simple GNN for scoring candidate solutions."""
    def __init__(self, in_dim, hidden_dim):
        super().__init__()
        self.conv1 = GCNConv(in_dim, hidden_dim)
        self.conv2 = GCNConv(hidden_dim, hidden_dim)
        self.head = torch.nn.Linear(hidden_dim, 1)

    def forward(self, x, edge_index, batch):
        x = self.conv1(x, edge_index).relu()
        x = self.conv2(x, edge_index).relu()
        x = global_mean_pool(x, batch)
        return self.head(x)

```

Installation: `pip install torch-geometric`

39.4.5.2. DGL (Deep Graph Library)

Framework-agnostic graph neural network library supporting both PyTorch and TensorFlow backends. DGL emphasizes message-passing primitives and scales to very large graphs.

Strengths:

- Framework-agnostic: works with PyTorch, TensorFlow, and MXNet
- Efficient message-passing API with built-in optimizations
- Scales to graphs with millions of nodes
- 14k+ GitHub stars, active development

Best For: Large-scale graph learning, heterogeneous graphs, research requiring framework flexibility. An alternative to PyG when TensorFlow support or custom message-passing logic is needed.

```

import dgl
import torch
import torch.nn as nn
from dgl.nn import GraphConv

class DGLGraphClassifier(nn.Module):
    """Graph classifier using DGL's GraphConv layers."""
    def __init__(self, in_dim, hidden_dim, n_classes):
        super().__init__()
        self.conv1 = GraphConv(in_dim, hidden_dim)
        self.conv2 = GraphConv(hidden_dim, hidden_dim)
        self.classify = nn.Linear(hidden_dim, n_classes)

    def forward(self, g, features):
        h = torch.relu(self.conv1(g, features))
        h = torch.relu(self.conv2(g, h))
        with g.local_scope():
            g.ndata['h'] = h
            hg = dgl.mean_nodes(g, 'h') # Graph-level readout
            return self.classify(hg)

```

Installation: `pip install dgl`

39.5. Hyperparameter Optimization

39.5.1. Optuna

Hyperparameter optimization using Bayesian optimization, tree-structured Parzen estimators, and evolutionary algorithms.

Strengths:

- Automatic algorithm selection
- Pruning of unpromising trials
- Distributed optimization
- Excellent visualization tools

Best For: Machine learning model tuning, deep learning architecture search, automated experimentation.

39.5.2. Hyperopt

Bayesian optimization with tree-structured Parzen estimators and random search.

Strengths:

- Mature library with extensive track record
- Simple API
- Integrates with ML pipelines

Best For: Neural network hyperparameter search, ensemble model optimization.

39.5.3. Advanced Bayesian Optimization

Beyond hyperparameter tuning, Bayesian Optimization has become essential for expensive black-box optimization. These libraries provide state-of-the-art capabilities.

39.5.3.1. BoTorch

Facebook Research's advanced Bayesian Optimization library built on PyTorch and GPyTorch.

Strengths:

- GPU-accelerated Gaussian Process models
- Custom acquisition functions with automatic differentiation
- Multi-fidelity optimization (use cheap approximations early, expensive evaluations later)
- Multi-task learning for related optimization problems
- Robust to noisy observations

Best For: Expensive simulations, experimental design, hyperparameter optimization at scale.

```
import torch
from botorch.models import SingleTaskGP
from botorch.fit import fit_gpytorch_mll
from botorch.acquisition import ExpectedImprovement
from botorch.optim import optimize_acqf
from gpytorch.mlls import ExactMarginalLogLikelihood

# Initial training data (normalized to [0, 1])
train_X = torch.rand(10, 2)  # 10 points, 2 dimensions
```

```

train_Y = torch.sin(train_X.sum(dim=1, keepdim=True)) # Objective function

# Fit Gaussian Process model
model = SingleTaskGP(train_X, train_Y)
mll = ExactMarginalLogLikelihood(model.likelihood, model)
fit_gpytorch_mll(mll)

# Create acquisition function
best_f = train_Y.max()
EI = ExpectedImprovement(model=model, best_f=best_f)

# Optimize acquisition function to find next point
bounds = torch.stack([torch.zeros(2), torch.ones(2)])
candidate, acq_value = optimize_acqf(
    EI, bounds=bounds, q=1, num_restarts=5, raw_samples=20
)
print(f"Next point to evaluate: {candidate}")

```

Installation: `pip install botorch`

39.5.3.2. Ax Platform

Meta’s adaptive experimentation platform built on top of BoTorch for production-ready optimization.

Strengths:

- High-level API for common optimization patterns
- Multi-objective optimization with Pareto frontier visualization
- Bandit experiments for A/B testing
- Automatic early stopping of poor configurations
- Experiment tracking and persistence
- REST API for production deployment

Best For: Production ML systems, A/B testing infrastructure, multi-objective hyperparameter tuning.

```

from ax.service.ax_client import AxClient
from ax.utils.measurement.synthetic_functions import branin

# Create optimization client
ax_client = AxClient()

# Define search space
ax_client.create_experiment(
    name="branin_optimization",
    parameters=[
        {"name": "x1", "type": "range", "bounds": [-5.0, 10.0]},
        {"name": "x2", "type": "range", "bounds": [0.0, 15.0]},
    ],
    objectives={"branin": {"minimize": True}},
)

# Optimization loop

```

```

for i in range(25):
    parameters, trial_index = ax_client.get_next_trial()
    result = branin(parameters["x1"], parameters["x2"])
    ax_client.complete_trial(trial_index=trial_index, raw_data=result)

# Get best parameters
best_parameters, metrics = ax_client.get_best_parameters()
print(f"Best parameters: {best_parameters}")

```

Installation: `pip install ax-platform`

39.5.3.3. GPyOpt (Legacy)

Classic Gaussian Process-based Bayesian Optimization library with strong educational value.

Strengths:

- Simple, educational API
- Extensive visualization tools
- Multiple acquisition functions (EI, MPI, LCB)
- Support for constrained optimization
- Good documentation for learning BO concepts

Best For: Learning Bayesian Optimization, small-scale experiments, teaching.

⚠ GPyOpt Status

GPyOpt is no longer actively maintained. For new projects, use BoTorch or Ax Platform. GPyOpt remains valuable for educational purposes due to its clear visualizations.

```

# Note: GPyOpt is deprecated but shown for educational purposes
import GPyOpt
import numpy as np

# Define objective function
def f(x):
    return (x[:, 0] - 1)**2 + (x[:, 1] - 2)**2

# Define domain
bounds = [{'name': 'x1', 'type': 'continuous', 'domain': (-5, 5)},
          {'name': 'x2', 'type': 'continuous', 'domain': (-5, 5)}]

# Create optimizer
optimizer = GPyOpt.methods.BayesianOptimization(
    f=f,
    domain=bounds,
    acquisition_type='EI',
    exact_feval=True
)

# Run optimization
optimizer.run_optimization(max_iter=20)
optimizer.plot_acquisition() # Visualize acquisition function

```


Installation: `pip install gpyopt` (requires Python < 3.10)

39.5.3.4. Comparison of BO Libraries

Feature	BoTorch	Ax Platform	GPyOpt
Backend	PyTorch/GPyTorch	BoTorch	GPy
GPU Support	Yes	Yes	Limited
Multi-Objective	Yes	Yes	No
Multi-Fidelity	Yes	Yes	No
API Level	Low (flexible)	High (easy)	Medium
Production Ready	Yes	Yes	No
Active Development	Yes	Yes	No
Learning Curve	Steep	Gentle	Gentle

39.6. Reinforcement Learning Frameworks

Reinforcement Learning has become essential for sequential decision-making in operations research, from inventory control to routing to scheduling.

39.6.1. General RL Libraries

Framework	Backend	Strengths	Best For
Stable-Baselines3	PyTorch	Easy to use, well-documented, many algorithms	Quick prototyping, research
RLlib	PyTorch/TF	Scalable, distributed, production-ready	Large-scale training, production
CleanRL	PyTorch	Single-file implementations, educational	Learning RL internals
Tianshou	PyTorch	Modular, highly customizable	Custom algorithms, research
EnvPool	C++	20x faster env vectorization, async execution	High-throughput training, Atari/MuJoCo

EnvPool (NeurIPS 2022) provides C++ threadpool-based environment execution that achieves up to 20x speedup over Python-based vectorized environments. It integrates with Stable-Baselines3, Tianshou, and CleanRL through standard Gymnasium and `dm_env` interfaces.

```
# Example: Stable-Baselines3 for inventory control
from stable_baselines3 import PPO
from stable_baselines3.common.env_util import make_vec_env

# Create environment (custom or gym-compatible)
env = make_vec_env("CartPole-v1", n_envs=4)

# Train agent
model = PPO("MlpPolicy", env, verbose=1)
model.learn(total_timesteps=100000)

# Save and load
```

```
model.save("ppo_cartpole")
loaded_model = PPO.load("ppo_cartpole")
```

39.6.2. RL for Combinatorial Optimization

These specialized libraries apply reinforcement learning to classic OR problems.

39.6.2.1. RL4CO

State-of-the-art library for applying RL to combinatorial optimization problems. Implements attention-based encoder-decoder architectures that have achieved competitive results on TSP, VRP, and other routing problems.

Strengths:

- Pre-trained models for TSP, CVRP, PCTSP, OP, SDVRP, and more
- Attention Model, POMO, and other architectures
- Efficient batch generation of instances
- Easy to extend for custom problems
- Active research community

Supported Problems:

Problem	Description	Model
TSP	Traveling Salesman Problem	Attention Model, POMO
CVRP	Capacitated VRP	Attention Model
SDVRP	Split Delivery VRP	AM-based
PCTSP	Prize Collecting TSP	Attention Model
OP	Orienteering Problem	Attention Model
JSSP	Job Shop Scheduling	L2D

```
from rl4co.envs import TSPEnv, CVRPEnv
from rl4co.models import AttentionModel
from rl4co.utils.trainer import RL4COTrainer

# Create TSP environment
env = TSPEnv(num_loc=50)

# Initialize Attention Model
model = AttentionModel(
    env,
    baseline="rollout", # REINFORCE baseline
    train_data_size=100000,
    val_data_size=10000,
)

# Train
trainer = RL4COTrainer(
    max_epochs=100,
    accelerator="gpu",
    devices=1,
)
trainer.fit(model)
```

```
# Solve new instances
test_data = env.dataset(batch_size=100)
solutions, log_probs = model(test_data, decode_type="greedy")
costs = env.get_costs(test_data, solutions)
print(f"Average tour length: {costs.mean():.2f}")
```

Installation: `pip install rl4co`

39.6.2.2. OR-Gym

OpenAI Gym-compatible environments for classic operations research problems. Enables using standard RL algorithms for OR applications.

Strengths:

- Gymnasium-compatible API
- Works with any RL library (SB3, RLlib, etc.)
- Configurable problem parameters
- Baseline heuristics for comparison
- Good documentation

Available Environments:

Environment	Problem	Observation	Action
InvManagement-v1	Multi-echelon inventory	Inventory levels	Order quantities
BinPacking-v0	Online bin packing	Current bins, item	Bin selection
Knapsack-v0	0-1 Knapsack	Remaining capacity, items	Include/exclude
VMPacking-v0	Virtual machine packing	Server loads	VM placement
VehicleRouting-v0	Basic VRP	Locations, demands	Next customer
NewsVendor-v0	News vendor problem	Demand history	Order quantity

```
import gymnasium as gym
import or_gym
from stable_baselines3 import PPO

# Multi-echelon inventory management
env = gym.make('or_gym:InvManagement-v1')

# Environment details
print(f"State space: {env.observation_space}")
print(f"Action space: {env.action_space}")

# Train with Stable-Baselines3
model = PPO("MlpPolicy", env, verbose=1)
model.learn(total_timesteps=50000)

# Evaluate
obs, _ = env.reset()
total_reward = 0
for _ in range(100):
    action, _ = model.predict(obs, deterministic=True)
    obs, reward, terminated, truncated, info = env.step(action)
```

```

    total_reward += reward
    if terminated or truncated:
        break
print(f"Total reward: {total_reward:.2f}")

```

Inventory Management Example:

```

# Custom inventory configuration
env_config = {
    'num_stages': 3,          # Supply chain stages
    'num_periods': 30,       # Planning horizon
    'init_inv': [100, 100, 200],
    'holding_cost': [1, 2, 3],
    'backlog_cost': [10, 20, 50],
    'demand_dist': 'poisson',
    'demand_mean': 20,
}

env = gym.make('or_gym:InvManagement-v1', env_config=env_config)

# Compare RL vs base-stock policy
def base_stock_policy(obs, target_levels):
    """Simple base-stock policy."""
    return np.maximum(target_levels - obs, 0)

# Evaluate heuristic
obs, _ = env.reset()
heuristic_reward = 0
targets = [150, 150, 250]
for _ in range(100):
    action = base_stock_policy(obs, targets)
    obs, reward, done, _, _ = env.step(action)
    heuristic_reward += reward
    if done:
        break
print(f"Base-stock policy reward: {heuristic_reward:.2f}")

```

Installation: `pip install or-gym`

39.6.2.3. Comparison: RL4CO vs OR-Gym

Aspect	RL4CO	OR-Gym
Focus	Routing problems (TSP, VRP)	Inventory, packing, scheduling
Approach	Custom neural architectures	Standard Gym interface
Algorithms	Specialized (Attention Model)	Any RL algorithm
Performance	State-of-the-art on routing	Good for prototyping
Scalability	Handles large instances	Limited by algorithm
Learning Curve	Steeper (specialized)	Gentle (standard API)

39.6.3. ML-Augmented Solvers

These libraries integrate machine learning with mathematical programming solvers to accelerate exact optimization.

39.6.3.1. Ecol

Ecol (Combinatorial Optimization Learning Environments) exposes SCIP solver decisions—branching, cutting plane selection, node selection—as Markov decision processes, enabling RL agents to learn solver strategies.

Strengths:

- Turns branching/cutting decisions into RL environments
- Built on SCIP solver via PySCIPOpt
- Designed for academic research on learning to optimize
- Standardized benchmarks for ML-solver integration

Best For: Research on learning to branch, learning to cut, and other ML-augmented solver strategies. Note: active development has slowed (last release 2022), but it remains an important academic reference and the canonical framework for this research area.

```
import ecol

# Create a branching environment
env = ecol.environment.Branching(
    observation_function=ecol.observation.NodeBipartite(),
    reward_function=-ecol.reward.SolvingTime(),
)

# Solve with a trained policy
instances = ecol.instance.SetCoverGenerator()
obs, action_set, reward_offset, done, info = env.reset(next(instances))
while not done:
    # action = policy(obs, action_set) # Trained GNN policy
    action = action_set[0] # Fallback: first candidate
    obs, action_set, reward, done, info = env.step(action)
```

Installation: `pip install ecol`

39.6.3.2. MIPLearn

Developed by Argonne National Laboratory, MIPLearn uses machine learning to accelerate mixed-integer programming solvers by predicting warm-start solutions, lazy constraints, and branching priorities from historical solve data.

Strengths:

- ML-based warm-starting for MIP solvers
- Learns from historical instances of the same problem class
- Predicts branching priorities and lazy constraints
- Backed by Argonne National Lab, active development

Best For: Production MIP solving where similar instances are solved repeatedly (e.g., daily scheduling, weekly planning). Provides speedups by learning patterns from past solves.

```
from miplearn import LearningSolver

# Wrap an existing MIP solver with ML acceleration
solver = LearningSolver()
```

```
# Train on historical instances
for instance in training_instances:
    solver.solve(instance)
solver.fit()

# Solve new instances faster using learned patterns
solver.solve(new_instance) # Uses ML predictions for warm-start
```

Installation: `pip install miplearn`

39.6.4. Scheduling Libraries

39.6.4.1. job-shop-lib

A modern Python library for the Job Shop Scheduling Problem (JSSP) providing GNN-based dispatching rules, CP-SAT and simulated annealing solvers, Gymnasium environments for RL, and built-in Gantt chart visualization.

Strengths:

- Multiple solver backends: CP-SAT, simulated annealing, dispatching rules
- GNN-based learned dispatching policies
- Gymnasium-compatible RL environment
- Built-in Gantt chart visualization
- Active development (74+ GitHub stars)

Best For: Job shop scheduling research and prototyping, benchmarking classical vs. ML-based dispatching, RL for scheduling.

```
from job_shop_lib import JobShopInstance
from job_shop_lib.dispatching import DispatchingRuleSolver
from job_shop_lib.visualization import plot_gantt_chart

# Define a small JSSP instance
jobs = [
    [(0, 3), (1, 2), (2, 2)], # Job 0: (machine, duration)
    [(0, 2), (2, 1), (1, 4)], # Job 1
    [(1, 4), (2, 3), (0, 1)], # Job 2
]
instance = JobShopInstance(jobs=jobs, name="example_3x3")

# Solve with SPT (Shortest Processing Time) dispatching rule
solver = DispatchingRuleSolver(dispatching_rule="shortest_processing_time")
schedule = solver.solve(instance)
print(f"Makespan: {schedule.makespan()}")
```

Installation: `pip install job-shop-lib`

39.6.4.2. JSSEnv

The first OpenAI Gym environment for the Job Shop Scheduling Problem, providing a standard RL interface for JSSP. While no longer actively maintained (last release 2021), it was a pioneering effort that established the pattern for RL-based scheduling environments.

Best For: Historical reference and simple JSSP RL experiments. For active projects, **job-shop-lib** is the recommended modern alternative with more features and active maintenance.

Installation: `pip install JSSEnv`

39.7. Library Selection Guide

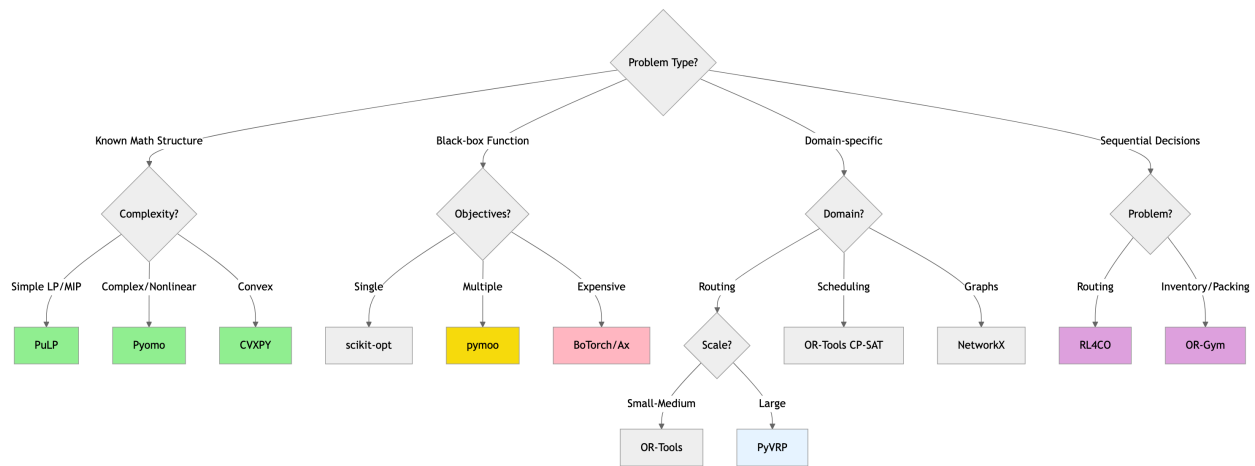


Figure 39.4.: Decision Tree for Library Selection

Problem Type	Recommended Library	Alternative
Linear/Integer Programming	PuLP (simple), Pyomo (complex)	Python-MIP
Convex Optimization	CVXPY	Pyomo
Multi-objective	pymoo	DEAP
Black-box Optimization	Nevergrad, scikit-opt	simanneal
Expensive Black-box	BoTorch, Ax Platform	GPyOpt (legacy)
Vehicle Routing	OR-Tools (small), PyVRP (large)	RL4CO (learning)
Constraint Programming	OR-Tools CP-SAT, CPMpy	python-constraint
Network Problems	NetworkX	-
Hyperparameter Tuning	Optuna	Hyperopt
RL for Routing	RL4CO	-
RL for Inventory/Packing	OR-Gym + SB3	RLlib
Graph Neural Networks	PyTorch Geometric	DGL
Job Shop Scheduling	job-shop-lib	OR-Tools CP-SAT
ML-Augmented MIP Solving	MIPLearn	Ecole (research)

39.8. Practical Considerations

39.8.1. Solver Dependencies

Mathematical programming libraries need external solvers:

Solver	Type	License	Performance
GLPK	LP/MIP	Open Source	Good
CBC	LP/MIP	Open Source	Very Good
HiGHS	LP/MIP	Open Source	Excellent
IPOPT	NLP	Open Source	Excellent
Gurobi	LP/MIP/QP	Commercial	State-of-the-art
CPLEX	LP/MIP/QP	Commercial	State-of-the-art

💡 Academic Licenses

Both Gurobi and CPLEX offer free academic licenses. For production use, HiGHS provides excellent open-source performance for LP/MIP problems.

39.8.2. Problem Scale Considerations

Approach	Variables	Optimality	Runtime
Mathematical Programming	Thousands-Millions	Guaranteed	Varies
Metaheuristics	Any	Not guaranteed	Controllable
Hybrid	Large-scale	Often near-optimal	Practical

39.8.3. Combining Libraries

Real problems often require multiple approaches:

- **NetworkX + Pyomo:** Network preprocessing then optimization
- **DEAP + CVXPY:** Metaheuristic search with convex subproblems
- **OR-Tools + Custom:** Routing with specialized constraints
- **Optuna + Any:** Hyperparameter tuning for optimization parameters

39.9. Installation

```
# Core mathematical programming
pip install pyomo pulp cvxpy python-mip

# Metaheuristics
pip install pymoo deap pygad scikit-opt nevergrad

# Specialized
pip install ortools networkx pyvrp cpmPy

# Hyperparameter optimization
pip install optuna hyperopt

# Bayesian optimization
pip install botorch ax-platform

# Graph neural networks
pip install torch-geometric dgl

# Reinforcement learning
pip install stable-baselines3 tianshou envpool
pip install rl4co or-gym

# ML-augmented solvers
pip install ecole miplearn

# Scheduling
pip install job-shop-lib

# Solvers (open source)
```



```
pip install highspy # HiGHS solver
```

39.10. Summary

39.10.1. Key Takeaways

- **Mathematical Programming** (Pyomo, PuLP, CVXPY): For problems with known mathematical structure
- **Metaheuristics** (pymoo, DEAP, Nevergrad): For black-box optimization and multiple objectives
- **Specialized Tools** (OR-Tools, PyVRP, CPMpy): For domain-specific problems
- **Hyperparameter Optimization** (Optuna): For tuning ML and optimization parameters
- **Bayesian Optimization** (BoTorch, Ax): For expensive black-box functions with few evaluations
- **Reinforcement Learning** (RL4CO, OR-Gym, EnvPool): For sequential decision problems in routing and inventory
- **Graph Neural Networks** (PyTorch Geometric, DGL): For learning on graph-structured optimization problems
- **ML-Augmented Solvers** (Ecole, MIPLearn): For accelerating exact solvers with learned strategies
- **Scheduling** (job-shop-lib): For job shop scheduling with classical and ML-based approaches
- Start simple (PuLP, OR-Tools) and upgrade as needed
- Combine libraries for complex real-world problems

40. Optimization Modeling Languages

What You Will Learn

- Understand why specialized modeling languages exist for optimization
- Compare declarative algebraic modeling with imperative approaches
- Write optimization models in AMPL, GAMS, OPL, Pyomo, and JuMP
- Choose the right modeling language for your project
- Migrate models between different languages
- Apply best practices for maintainable optimization code

Optimization modeling languages bridge the gap between mathematical formulations and computational solvers. Rather than coding algorithms directly, you describe *what* you want to optimize, and the language handles the translation to solver-specific formats. This appendix surveys the major modeling languages, compares their features, and provides working examples to help you choose the right tool.

40.1. Introduction

40.1.1. Why Specialized Languages?

Consider solving a linear program. Without a modeling language, you would need to:

1. Convert mathematical notation to coefficient matrices
2. Handle indexing and sparse data structures
3. Interface with solver-specific APIs
4. Parse solutions back to meaningful variables

This low-level work is error-prone and obscures the mathematical structure. Modeling languages solve these problems by letting you write models that closely resemble the mathematical formulation.

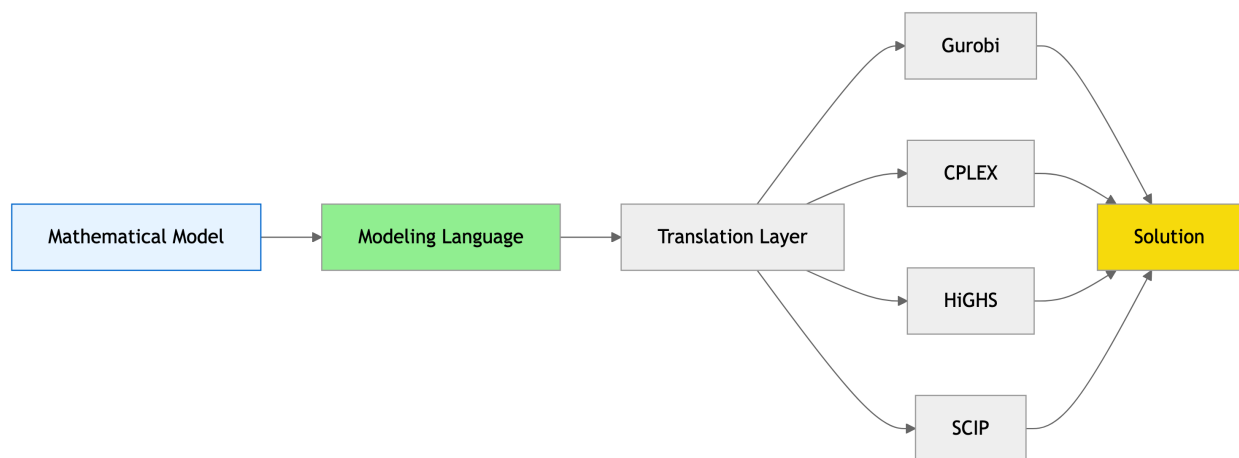


Figure 40.1.: From Mathematical Model to Solver Solution

40.1.2. Declarative vs. Imperative Modeling

Declarative modeling describes the structure of the optimization problem without specifying how to solve it:

- Define sets, parameters, variables, objectives, and constraints
- Let the solver determine the solution algorithm
- Examples: AMPL, GAMS, OPL, Pyomo, JuMP

Imperative modeling requires you to implement or call specific algorithms:

- Control the solution process step by step
- More flexibility but more complexity
- Examples: Direct solver APIs, custom implementations

The Modeling Language Advantage

Algebraic modeling languages typically reduce development time by 5-10x compared to direct solver APIs. They also make models more readable, maintainable, and portable across different solvers.

40.1.3. Solver Independence

A key benefit of modeling languages is **solver independence**. Write your model once, then:

- Test with free solvers (GLPK, CBC, HiGHS) during development
- Deploy with commercial solvers (Gurobi, CPLEX) in production
- Switch solvers if performance or licensing requirements change
- Benchmark multiple solvers on your specific problem

Language	Supported Solvers	LP/MIP	NLP	CP
AMPL	40+	Yes	Yes	Limited
GAMS	30+	Yes	Yes	Limited
OPL	CPLEX only	Yes	No	Yes
Pyomo	20+	Yes	Yes	Via plugins
JuMP	30+	Yes	Yes	Yes
PuLP	10+	Yes	No	No

40.2. A Running Example: The Transportation Problem

Throughout this appendix, we use the classic **transportation problem** to demonstrate each language. This problem appears in supply chain optimization, logistics, and network design.

40.2.1. Problem Description

A company has factories (origins) that produce goods and warehouses (destinations) that need goods. The goal is to minimize total shipping cost while satisfying supply and demand constraints.

Mathematical Formulation:

$$\begin{aligned}
& \text{minimize} && \sum_{i \in O} \sum_{j \in D} c_{ij} x_{ij} \\
& \text{subject to} && \sum_{j \in D} x_{ij} \leq s_i \quad \forall i \in O \quad (\text{supply limits}) \\
& && \sum_{i \in O} x_{ij} \geq d_j \quad \forall j \in D \quad (\text{demand satisfaction}) \\
& && x_{ij} \geq 0 \quad \forall i \in O, j \in D
\end{aligned}$$

where:

- O = set of origins (factories)
- D = set of destinations (warehouses)
- c_{ij} = cost to ship one unit from i to j
- s_i = supply available at origin i
- d_j = demand required at destination j
- x_{ij} = quantity shipped from i to j

40.2.2. Example Data

Origin	Supply
Seattle	350
San Diego	600

Destination	Demand
New York	325
Chicago	300
Topeka	275

From	To	New York	Chicago	Topeka
Seattle		2.5	1.7	1.8
San Diego		3.5	1.8	1.4

Costs are in dollars per unit, multiplied by 1000 units shipped.

40.3. Algebraic Modeling Languages

40.3.1. AMPL (A Mathematical Programming Language)

AMPL, developed at Bell Labs in the 1980s, pioneered algebraic modeling for optimization. Its syntax closely mirrors mathematical notation, making it easy to translate formulations to code.

40.3.1.1. History and Features

- Created by Robert Fourer, David Gay, and Brian Kernighan
- First released in 1985
- Strong separation between model, data, and commands
- Extensive solver support (40+ solvers)
- Powerful scripting capabilities
- Commercial license with free academic and NEOS access

40.3.1.2. AMPL Syntax Overview

AMPL models consist of three parts:

1. **Model file (.mod)**: Sets, parameters, variables, objectives, constraints
2. **Data file (.dat)**: Numeric values for parameters
3. **Script file (.run)**: Commands to load, solve, and analyze

40.3.1.3. Transportation Problem in AMPL

Model File (transport.mod):

```
# Transportation Problem
# Minimize shipping costs from origins to destinations

# Sets
set ORIGINS;                # Set of origins (factories)
set DESTINATIONS;           # Set of destinations (warehouses)

# Parameters
param supply{ORIGINS} >= 0;  # Supply at each origin
param demand{DESTINATIONS} >= 0; # Demand at each destination
param cost{ORIGINS, DESTINATIONS} >= 0; # Shipping cost per unit

# Check that total supply meets total demand
check: sum{i in ORIGINS} supply[i] >= sum{j in DESTINATIONS} demand[j];

# Decision Variables
var Ship{i in ORIGINS, j in DESTINATIONS} >= 0; # Units to ship

# Objective: Minimize total shipping cost
minimize Total_Cost:
    sum{i in ORIGINS, j in DESTINATIONS} cost[i,j] * Ship[i,j];

# Constraints
subject to Supply_Limit{i in ORIGINS}:
    sum{j in DESTINATIONS} Ship[i,j] <= supply[i];

subject to Demand_Satisfaction{j in DESTINATIONS}:
    sum{i in ORIGINS} Ship[i,j] >= demand[j];
```

Data File (transport.dat):

```
# Data for Transportation Problem

set ORIGINS := Seattle San_Diego;
set DESTINATIONS := New_York Chicago Topeka;

param supply :=
    Seattle    350
    San_Diego  600;

param demand :=
    New_York   325
```

```

        Chicago    300
        Topeka     275;

param cost:  New_York  Chicago  Topeka :=
        Seattle    2.5      1.7      1.8
        San_Diego   3.5      1.8      1.4;

```

Script File (transport.run):

```

# Load and solve the transportation problem

reset;

model transport.mod;
data transport.dat;

option solver gurobi; # or cplex, cbc, highs

solve;

# Display results
display Total_Cost;
display Ship;

# Show shadow prices (dual values)
display Supply_Limit.dual;
display Demand_Satisfaction.dual;

# Export solution
printf "Optimal Cost: %.2f\n", Total_Cost;
printf "\nShipping Plan:\n";
for {i in ORIGINS, j in DESTINATIONS: Ship[i,j] > 0} {
    printf "  %s -> %s: %.0f units\n", i, j, Ship[i,j];
}

```

40.3.1.4. Advanced AMPL Features**Indexed Expressions:**

```

# Weighted average shipping distance
param distance{ORIGINS, DESTINATIONS};
var avg_distance =
    sum{i in ORIGINS, j in DESTINATIONS} distance[i,j] * Ship[i,j] /
    sum{i in ORIGINS, j in DESTINATIONS} Ship[i,j];

```

Conditional Constraints:

```

# Only allow shipping if route is open
param route_open{ORIGINS, DESTINATIONS} binary;
subject to Route_Restriction{i in ORIGINS, j in DESTINATIONS}:
    Ship[i,j] <= route_open[i,j] * min(supply[i], demand[j]);

```

Looping and Scripting:

```
# Sensitivity analysis: vary demand and re-solve
for {delta in -50..50 by 10} {
    let demand['Chicago'] := 300 + delta;
    solve;
    printf "%d, %.2f\n", delta, Total_Cost;
}
```

40.3.2. GAMS (General Algebraic Modeling System)

GAMS, developed at the World Bank in the 1970s, is one of the oldest and most widely used modeling systems. It excels at large-scale economic and energy models.

40.3.2.1. Features and Ecosystem

- Developed since 1976, commercial release in 1988
- Strong focus on economic modeling and CGE (Computable General Equilibrium)
- Comprehensive solver library
- GAMS Model Library with 1000+ example models
- Extensive documentation and support
- GAMS Studio IDE for development

40.3.2.2. GAMS Syntax

GAMS uses a different syntax style than AMPL:

- Keywords in lowercase
- Equations defined separately from their expressions
- Dollar conditions for conditional expressions
- Less strict formatting requirements

40.3.2.3. Transportation Problem in GAMS

```
$Title Transportation Problem
$OnText
Minimize shipping costs from factories to warehouses
Classic example from Dantzig (1963)
$OffText

* Define Sets
Sets
    i   'origins (factories)'      / Seattle, San-Diego /
    j   'destinations (markets)' / New-York, Chicago, Topeka /;

* Define Parameters
Parameters
    a(i)   'supply capacity at origin i (units)'
           / Seattle      350
             San-Diego    600 /

    b(j)   'demand at destination j (units)'
           / New-York     325
             Chicago      300
             Topeka       275 /;
```



```

Table c(i,j) 'transport cost per unit (dollars per 1000 units)'
      New-York   Chicago   Topeka
Seattle      2.5      1.7      1.8
San-Diego    3.5      1.8      1.4;

* Scaling factor (cost in thousands)
Scalar f 'freight cost multiplier' / 1000 /;

* Decision Variables
Positive Variables
    x(i,j) 'shipment quantities';

Variables
    z      'total transportation cost';

* Define Equations
Equations
    cost      'objective function'
    supply(i) 'supply limit at origin i'
    demand(j) 'demand requirement at destination j';

* Equation Definitions
cost ..      z =e= sum((i,j), c(i,j) * f * x(i,j));

supply(i) .. sum(j, x(i,j)) =l= a(i);

demand(j) .. sum(i, x(i,j)) =g= b(j);

* Model Definition and Solution
Model transport / all /;

Solve transport using lp minimizing z;

* Display Results
Display x.l, x.m;
Display supply.m, demand.m;

* Generate Report
Parameter report(i,j) 'shipping report';
report(i,j) = x.l(i,j);
Display report;

* Marginal costs (shadow prices)
Parameter marginals(*) 'marginal values';
marginals('supply_Seattle') = supply.m('Seattle');
marginals('supply_San-Diego') = supply.m('San-Diego');
marginals('demand_New-York') = demand.m('New-York');
marginals('demand_Chicago') = demand.m('Chicago');
marginals('demand_Topeka') = demand.m('Topeka');
Display marginals;

```

40.3.2.4. GAMS Advanced Features**Dollar Conditions:**

```
* Only ship on valid routes
Set valid(i,j) 'allowed shipping routes' / Seattle.Chicago, Seattle.Topeka,
                                           San-Diego.New-York, San-Diego.Topeka /;

Equations restricted_demand(j);
restricted_demand(j) .. sum(i$valid(i,j), x(i,j)) =g= b(j);
```

Multi-dimensional Tables:

```
* Time-varying costs
Set t 'time periods' / t1*t12 /;
Table cost_t(i,j,t) 'time-varying transport costs'
           t1      t2      t3
Seattle.New-York  2.5    2.6    2.4
Seattle.Chicago  1.7    1.8    1.6;
```

Put Statements for Custom Output:

```
File results / results.csv /;
Put results;
Put 'Origin,Destination,Quantity' /;
Loop((i,j)$(x.l(i,j) > 0),
      Put i.tl, ',', j.tl, ',', x.l(i,j) /;
);
Putclose;
```

40.3.3. OPL (Optimization Programming Language)

OPL is IBM's modeling language, tightly integrated with CPLEX and the CP Optimizer. It offers both mathematical programming and constraint programming within a single language.

40.3.3.1. Features

- Native integration with IBM CPLEX and CP Optimizer
- CPLEX Optimization Studio IDE
- Decision Optimization for Watson Studio (cloud)
- Supports both mathematical programming and constraint programming
- Scripting with JavaScript-like syntax
- Data integration via databases and spreadsheets

40.3.3.2. Transportation Problem in OPL**Model File (transport.mod):**

```
// Transportation Problem in OPL
// Minimize shipping costs from origins to destinations

// Data (can be in separate .dat file)
{string} Origins = {"Seattle", "SanDiego"};
```

```

{string} Destinations = {"NewYork", "Chicago", "Topeka"};

float supply[Origins] = [350, 600];
float demand[Destinations] = [325, 300, 275];

float cost[Origins][Destinations] = [
    [2.5, 1.7, 1.8], // Seattle
    [3.5, 1.8, 1.4]  // San Diego
];

// Decision Variables
dvar float+ Ship[Origins][Destinations];

// Objective Function
minimize
    sum(i in Origins, j in Destinations) cost[i][j] * Ship[i][j];

// Constraints
subject to {
    // Supply limits
    forall(i in Origins)
        ctSupply: sum(j in Destinations) Ship[i][j] <= supply[i];

    // Demand satisfaction
    forall(j in Destinations)
        ctDemand: sum(i in Origins) Ship[i][j] >= demand[j];
}

// Post-processing
execute DISPLAY {
    writeln("Total Cost: ", cplex.getObjValue());
    writeln("\nShipping Plan:");
    for(var i in Origins) {
        for(var j in Destinations) {
            if(Ship[i][j] > 0) {
                writeln("  ", i, " -> ", j, ": ", Ship[i][j]);
            }
        }
    }
}

```

Data File (transport.dat):

```

// External data for transportation problem
Origins = {"Seattle", "SanDiego"};
Destinations = {"NewYork", "Chicago", "Topeka"};

supply = [350, 600];
demand = [325, 300, 275];

cost = [
    [2.5, 1.7, 1.8],
    [3.5, 1.8, 1.4]
]

```

```
];
```

40.3.3.3. OPL Constraint Programming Example

OPL also supports constraint programming for combinatorial problems:

```
// Job Shop Scheduling with CP
using CP;

int nbJobs = 3;
int nbMachines = 3;

range Jobs = 1..nbJobs;
range Machines = 1..nbMachines;

// Processing times: duration[job][position]
int duration[Jobs][Machines] = [
    [3, 2, 2],
    [2, 1, 4],
    [4, 3, 1]
];

// Machine assignment: machine[job][position]
int machine[Jobs][Machines] = [
    [1, 2, 3],
    [1, 3, 2],
    [2, 1, 3]
];

// Decision variables: start times
dvar interval task[j in Jobs][m in Machines] size duration[j][m];

// Sequences on each machine
sequence machineSeq[m in Machines] in
    all(j in Jobs, p in Machines: machine[j][p] == m) task[j][p];

// Objective: minimize makespan
dvar int makespan;

minimize makespan;

subject to {
    // Precedence within jobs
    forall(j in Jobs, p in 2..nbMachines)
        endBeforeStart(task[j][p-1], task[j][p]);

    // No overlap on machines
    forall(m in Machines)
        noOverlap(machineSeq[m]);

    // Makespan definition
    forall(j in Jobs)
        makespan >= endOf(task[j][nbMachines]);
}
```

}

40.3.4. Comparison: AMPL vs GAMS vs OPL

Feature	AMPL	GAMS	OPL
Syntax Style	Mathematical	Verbose	C-like
Learning Curve	Moderate	Moderate	Easy (if C background)
Solver Support	40+	30+	CPLEX only
Constraint Programming	Limited	Limited	Full support
Scripting	Built-in	Built-in	JavaScript
IDE	Basic	GAMS Studio	CPLEX Studio
License Cost	High	High	IBM subscription
Free Academic	Yes (limited)	Yes (limited)	Yes
Cloud Access	NEOS	GAMS Engine	Watson Studio

💡 Choosing Between AMPL, GAMS, and OPL

- **AMPL:** Best for mathematical elegance and solver variety
- **GAMS:** Best for economic models and large-scale energy systems
- **OPL:** Best for IBM CPLEX users needing constraint programming

40.4. Python-Based Modeling

Python has become the dominant language for optimization modeling in industry and research. Its ecosystem offers multiple libraries with different philosophies.

40.4.1. Pyomo

Pyomo (Python Optimization Modeling Objects) provides algebraic modeling capabilities in pure Python. It supports linear, integer, nonlinear, stochastic, and disjunctive programming.

40.4.1.1. Concrete vs. Abstract Models

Pyomo offers two modeling paradigms:

- **Concrete Models:** Data specified at model construction time
- **Abstract Models:** Model structure defined first, data loaded separately (like AMPL)

40.4.1.2. Transportation Problem in Pyomo

Concrete Model:

```
"""
Transportation Problem in Pyomo (Concrete Model)
Minimize shipping costs from origins to destinations
"""
from pyomo.environ import (
    ConcreteModel, Set, Param, Var, Objective,
    Constraint, NonNegativeReals, minimize, value
)
from pyomo.opt import SolverFactory
```

```

def create_transportation_model():
    """Create and return the transportation model."""

    model = ConcreteModel(name="Transportation")

    # Sets
    model.origins = Set(initialize=['Seattle', 'SanDiego'])
    model.destinations = Set(initialize=['NewYork', 'Chicago', 'Topeka'])

    # Parameters
    supply_data = {'Seattle': 350, 'SanDiego': 600}
    demand_data = {'NewYork': 325, 'Chicago': 300, 'Topeka': 275}
    cost_data = {
        ('Seattle', 'NewYork'): 2.5,
        ('Seattle', 'Chicago'): 1.7,
        ('Seattle', 'Topeka'): 1.8,
        ('SanDiego', 'NewYork'): 3.5,
        ('SanDiego', 'Chicago'): 1.8,
        ('SanDiego', 'Topeka'): 1.4,
    }

    model.supply = Param(model.origins, initialize=supply_data)
    model.demand = Param(model.destinations, initialize=demand_data)
    model.cost = Param(
        model.origins, model.destinations,
        initialize=cost_data
    )

    # Variables
    model.ship = Var(
        model.origins, model.destinations,
        domain=NonNegativeReals
    )

    # Objective: minimize total cost
    def total_cost_rule(m):
        return sum(
            m.cost[i, j] * m.ship[i, j]
            for i in m.origins
            for j in m.destinations
        )
    model.total_cost = Objective(rule=total_cost_rule, sense=minimize)

    # Constraints
    def supply_rule(m, i):
        return sum(m.ship[i, j] for j in m.destinations) <= m.supply[i]
    model.supply_constraint = Constraint(model.origins, rule=supply_rule)

    def demand_rule(m, j):
        return sum(m.ship[i, j] for i in m.origins) >= m.demand[j]
    model.demand_constraint = Constraint(model.destinations, rule=demand_rule)

    return model

```

```

def solve_and_display(model, solver_name='glpk'):
    """Solve the model and display results."""

    solver = SolverFactory(solver_name)
    results = solver.solve(model, tee=False)

    # Check solution status
    if results.solver.termination_condition.name != 'optimal':
        print(f"Warning: {results.solver.termination_condition}")
        return

    # Display results
    print(f"Optimal Total Cost: ${value(model.total_cost) * 1000:,.0f}")
    print("\nShipping Plan:")
    print("-" * 40)

    for i in model.origins:
        for j in model.destinations:
            shipped = value(model.ship[i, j])
            if shipped > 0.001:
                print(f"  {i:10} -> {j:10}: {shipped:6.0f} units")

    # Shadow prices
    print("\nShadow Prices (Dual Values):")
    print("-" * 40)
    for i in model.origins:
        dual = model.dual[model.supply_constraint[i]]
        print(f"  Supply {i}: ${dual:.2f}")
    for j in model.destinations:
        dual = model.dual[model.demand_constraint[j]]
        print(f"  Demand {j}: ${dual:.2f}")

if __name__ == '__main__':
    model = create_transportation_model()
    model.dual = Suffix(direction=Suffix.IMPORT) # For dual values
    solve_and_display(model, 'glpk')

```

Abstract Model:

```

"""
Transportation Problem in Pyomo (Abstract Model)
Model structure separate from data
"""

from pyomo.environ import *

def create_abstract_model():
    """Create abstract transportation model."""

    model = AbstractModel(name="Transportation_Abstract")

```

```

# Sets (no data yet)
model.Origins = Set()
model.Destinations = Set()

# Parameters (no data yet)
model.supply = Param(model.Origins, within=NonNegativeReals)
model.demand = Param(model.Destinations, within=NonNegativeReals)
model.cost = Param(model.Origins, model.Destinations,
                    within=NonNegativeReals)

# Variables
model.ship = Var(model.Origins, model.Destinations,
                  domain=NonNegativeReals)

# Objective
def total_cost_rule(m):
    return sum(m.cost[i,j] * m.ship[i,j]
               for i in m.Origins for j in m.Destinations)
model.total_cost = Objective(rule=total_cost_rule, sense=minimize)

# Constraints
def supply_rule(m, i):
    return sum(m.ship[i,j] for j in m.Destinations) <= m.supply[i]
model.supply_limit = Constraint(model.Origins, rule=supply_rule)

def demand_rule(m, j):
    return sum(m.ship[i,j] for i in m.Origins) >= m.demand[j]
model.demand_satisfy = Constraint(model.Destinations, rule=demand_rule)

return model

# Create data dictionary
data = {
    None: {
        'Origins': {None: ['Seattle', 'SanDiego']},
        'Destinations': {None: ['NewYork', 'Chicago', 'Topeka']},
        'supply': {'Seattle': 350, 'SanDiego': 600},
        'demand': {'NewYork': 325, 'Chicago': 300, 'Topeka': 275},
        'cost': {
            ('Seattle', 'NewYork'): 2.5,
            ('Seattle', 'Chicago'): 1.7,
            ('Seattle', 'Topeka'): 1.8,
            ('SanDiego', 'NewYork'): 3.5,
            ('SanDiego', 'Chicago'): 1.8,
            ('SanDiego', 'Topeka'): 1.4,
        }
    }
}

# Create instance from abstract model and data
abstract_model = create_abstract_model()
instance = abstract_model.create_instance(data)

```



```
# Solve
solver = SolverFactory('cbc')
solver.solve(instance)

# Display
instance.ship.display()
```

40.4.1.3. Pyomo Advanced Features

Indexed Constraints with Conditional Logic:

```
# Only create constraint if condition holds
def conditional_rule(m, i, j):
    if m.cost[i,j] > 2.0:
        return m.ship[i,j] <= 100 # Limit expensive routes
    return Constraint.Skip

model.conditional = Constraint(
    model.origins, model.destinations,
    rule=conditional_rule
)
```

Disjunctive Programming:

```
from pyomo.gdp import Disjunct, Disjunction

# Either use route A or route B, not both
model.use_route_A = Disjunct()
model.use_route_A.ship_A = Constraint(expr=model.x['A'] >= 100)

model.use_route_B = Disjunct()
model.use_route_B.ship_B = Constraint(expr=model.x['B'] >= 100)

model.route_choice = Disjunction(
    expr=[model.use_route_A, model.use_route_B]
)
```

40.4.2. PuLP

PuLP provides a simpler interface focused on linear and integer programming. It is ideal for beginners and quick prototyping.

40.4.2.1. Transportation Problem in PuLP

```
"""
Transportation Problem in PuLP
Simple, Pythonic interface for LP/MIP
"""

from pulp import (
    LpProblem, LpMinimize, LpVariable, lpSum,
    LpStatus, value, PULP_CBC_CMD
```

```

)

def transportation_pulp():
    """Solve transportation problem with PuLP."""

    # Data
    origins = ['Seattle', 'SanDiego']
    destinations = ['NewYork', 'Chicago', 'Topeka']

    supply = {'Seattle': 350, 'SanDiego': 600}
    demand = {'NewYork': 325, 'Chicago': 300, 'Topeka': 275}

    costs = {
        ('Seattle', 'NewYork'): 2.5,
        ('Seattle', 'Chicago'): 1.7,
        ('Seattle', 'Topeka'): 1.8,
        ('SanDiego', 'NewYork'): 3.5,
        ('SanDiego', 'Chicago'): 1.8,
        ('SanDiego', 'Topeka'): 1.4,
    }

    # Create problem
    prob = LpProblem("Transportation", LpMinimize)

    # Decision variables
    routes = [(i, j) for i in origins for j in destinations]
    ship = LpVariable.dicts("Ship", routes, lowBound=0)

    # Objective function
    prob += lpSum(costs[i, j] * ship[i, j] for i, j in routes), "TotalCost"

    # Supply constraints
    for i in origins:
        prob += (
            lpSum(ship[i, j] for j in destinations) <= supply[i],
            f"Supply_{i}"
        )

    # Demand constraints
    for j in destinations:
        prob += (
            lpSum(ship[i, j] for i in origins) >= demand[j],
            f"Demand_{j}"
        )

    # Solve
    prob.solve(PULP_CBC_CMD(msg=0))

    # Results
    print(f"Status: {LpStatus[prob.status]}")
    print(f"Optimal Cost: ${value(prob.objective) * 1000:,.0f}")

    print("\nShipping Plan:")

```

```

    for i, j in routes:
        qty = value(ship[i, j])
        if qty > 0:
            print(f"    {i} -> {j}: {qty:.0f} units")

    return prob

if __name__ == '__main__':
    transportation_pulp()

```

40.4.2.2. PuLP Quick Examples

Simple LP:

```

from pulp import *

# Maximize profit
prob = LpProblem("ProductMix", LpMaximize)
x = LpVariable("chairs", lowBound=0)
y = LpVariable("tables", lowBound=0)

prob += 30*x + 50*y # Objective
prob += 2*x + 3*y <= 120 # Carpentry hours
prob += x + 2*y <= 80 # Finishing hours

prob.solve()
print(f"Chairs: {value(x)}, Tables: {value(y)}")

```

Binary Variables:

```

# Knapsack problem
items = ['A', 'B', 'C', 'D']
values = {'A': 10, 'B': 15, 'C': 8, 'D': 12}
weights = {'A': 3, 'B': 4, 'C': 2, 'D': 5}
capacity = 10

prob = LpProblem("Knapsack", LpMaximize)
x = LpVariable.dicts("select", items, cat='Binary')

prob += lpSum(values[i] * x[i] for i in items)
prob += lpSum(weights[i] * x[i] for i in items) <= capacity

prob.solve()
selected = [i for i in items if value(x[i]) > 0.5]

```

40.4.3. Google OR-Tools

Google OR-Tools provides production-ready optimization for routing, scheduling, and constraint satisfaction.

40.4.3.1. Transportation with OR-Tools Linear Solver

```

"""
Transportation Problem with OR-Tools Linear Solver
"""
from ortools.linear_solver import pywraplp

def transportation_ortools():
    """Solve transportation with OR-Tools."""

    # Data
    origins = ['Seattle', 'SanDiego']
    destinations = ['NewYork', 'Chicago', 'Topeka']

    supply = [350, 600]
    demand = [325, 300, 275]

    costs = [
        [2.5, 1.7, 1.8], # Seattle
        [3.5, 1.8, 1.4], # San Diego
    ]

    num_origins = len(origins)
    num_destinations = len(destinations)

    # Create solver
    solver = pywraplp.Solver.CreateSolver('GLOP')
    if not solver:
        return None

    # Variables
    x = {}
    for i in range(num_origins):
        for j in range(num_destinations):
            x[i, j] = solver.NumVar(0, solver.infinity(),
                                    f'x[{i},{j}]')

    # Supply constraints
    for i in range(num_origins):
        solver.Add(
            sum(x[i, j] for j in range(num_destinations)) <= supply[i]
        )

    # Demand constraints
    for j in range(num_destinations):
        solver.Add(
            sum(x[i, j] for i in range(num_origins)) >= demand[j]
        )

    # Objective
    solver.Minimize(
        sum(costs[i][j] * x[i, j]
            for i in range(num_origins)
            for j in range(num_destinations))

```

```

    )

    # Solve
    status = solver.Solve()

    if status == pywraplp.Solver.OPTIMAL:
        print(f'Optimal cost: ${solver.Objective().Value() * 1000:,.0f}')
        print('\nShipping Plan:')
        for i in range(num_origins):
            for j in range(num_destinations):
                if x[i, j].solution_value() > 0:
                    print(f' {origins[i]} -> {destinations[j]}: '
                          f'{x[i, j].solution_value():.0f}')
            return solver

    print('No optimal solution found.')
    return None

if __name__ == '__main__':
    transportation_ortools()

```

40.4.3.2. OR-Tools CP-SAT Solver

For integer and constraint programming problems:

```

"""
Integer Transportation with OR-Tools CP-SAT
"""
from ortools.sat.python import cp_model

def transportation_cpsat():
    """Solve integer transportation with CP-SAT."""

    supply = [350, 600]
    demand = [325, 300, 275]
    costs = [
        [25, 17, 18], # Scaled by 10 for integers
        [35, 18, 14],
    ]

    num_origins = len(supply)
    num_destinations = len(demand)

    model = cp_model.CpModel()

    # Integer variables
    x = {}
    for i in range(num_origins):
        for j in range(num_destinations):
            x[i, j] = model.NewIntVar(0, max(supply), f'x[{i},{j}]')

    # Supply constraints

```

```

for i in range(num_origins):
    model.Add(sum(x[i, j] for j in range(num_destinations)) <= supply[i])

# Demand constraints
for j in range(num_destinations):
    model.Add(sum(x[i, j] for i in range(num_origins)) >= demand[j])

# Objective
model.Minimize(
    sum(costs[i][j] * x[i, j]
        for i in range(num_origins)
        for j in range(num_destinations))
)

# Solve
solver = cp_model.CpSolver()
status = solver.Solve(model)

if status == cp_model.OPTIMAL:
    print(f'Optimal cost: ${solver.ObjectiveValue() * 100:,.0f}')
    for i in range(num_origins):
        for j in range(num_destinations):
            if solver.Value(x[i, j]) > 0:
                print(f'Route [{i}][{j}]: {solver.Value(x[i, j])}')

if __name__ == '__main__':
    transportation_cpsat()

```

40.4.4. CVXPY

CVXPY specializes in convex optimization with automatic problem verification through Disciplined Convex Programming (DCP).

40.4.4.1. Transportation Problem in CVXPY

```

"""
Transportation Problem in CVXPY
Convex optimization with DCP verification
"""

import cvxpy as cp
import numpy as np

def transportation_cvxpy():
    """Solve transportation with CVXPY."""

    # Data as numpy arrays
    supply = np.array([350, 600])
    demand = np.array([325, 300, 275])
    costs = np.array([
        [2.5, 1.7, 1.8],
        [3.5, 1.8, 1.4],
    ])

```

```

m, n = len(supply), len(demand)

# Variable matrix
x = cp.Variable((m, n), nonneg=True)

# Objective
objective = cp.Minimize(cp.sum(cp.multiply(costs, x)))

# Constraints
constraints = [
    cp.sum(x, axis=1) <= supply, # Row sums <= supply
    cp.sum(x, axis=0) >= demand, # Column sums >= demand
]

# Problem
problem = cp.Problem(objective, constraints)
problem.solve()

print(f"Status: {problem.status}")
print(f"Optimal cost: ${problem.value * 1000:,.0f}")
print("\nShipping matrix:")
print(np.round(x.value, 1))

# Dual values (shadow prices)
print("\nDual values (shadow prices):")
print(f"Supply: {constraints[0].dual_value}")
print(f"Demand: {constraints[1].dual_value}")

return problem

if __name__ == '__main__':
    transportation_cvxpy()

```

40.4.4.2. CVXPY Advanced: Portfolio Optimization

```

"""
Portfolio Optimization with CVXPY
Mean-variance optimization with risk constraint
"""

import cvxpy as cp
import numpy as np

def portfolio_optimization(returns, cov_matrix, risk_limit):
    """
    Optimize portfolio weights for maximum return subject to risk limit.

    Args:
        returns: Expected returns vector (n,)
        cov_matrix: Covariance matrix (n, n)
        risk_limit: Maximum portfolio variance

    Returns:

```

```

    weights: Optimal portfolio weights
    """
    n = len(returns)

    # Portfolio weights
    w = cp.Variable(n)

    # Expected return
    ret = returns @ w

    # Portfolio variance (quadratic form)
    risk = cp.quad_form(w, cov_matrix)

    # Constraints
    constraints = [
        cp.sum(w) == 1,      # Weights sum to 1
        w >= 0,              # No short selling
        risk <= risk_limit,  # Risk limit
    ]

    # Maximize return
    problem = cp.Problem(cp.Maximize(ret), constraints)
    problem.solve()

    return w.value

# Example usage
np.random.seed(42)
n_assets = 5
returns = np.array([0.12, 0.10, 0.08, 0.15, 0.09])
cov_matrix = np.eye(n_assets) * 0.04
cov_matrix += 0.01 # Add correlation

weights = portfolio_optimization(returns, cov_matrix, risk_limit=0.05)
print("Optimal weights:", np.round(weights, 3))

```

40.4.5. Python Library Comparison

Feature	Pyomo	PuLP	OR-Tools	CVXPY
LP Support	Yes	Yes	Yes	Yes
MIP Support	Yes	Yes	Yes	Yes
NLP Support	Yes	No	No	Convex only
CP Support	Via plugins	No	Excellent	No
Learning Curve	Moderate	Easy	Moderate	Easy
Documentation	Excellent	Good	Good	Excellent
Solver Support	Many	Many	Built-in + external	Many
Performance	Good	Good	Excellent	Good
Best For	Complex models	Simple LP/MIP	Routing, scheduling	Convex problems

💡 Choosing a Python Library

- **Start with PuLP:** Simple syntax, great for learning
- **Use Pyomo:** When you need NLP, stochastic, or complex structures
- **Use OR-Tools:** For routing, scheduling, and constraint programming
- **Use CVXPY:** For convex optimization with mathematical elegance

40.5. Julia-Based Modeling

40.5.1. JuMP

JuMP (Julia for Mathematical Programming) is a domain-specific modeling language embedded in Julia. It combines the expressiveness of algebraic modeling with Julia's performance.

40.5.1.1. Why JuMP?

- **Performance:** Julia compiles to efficient native code
- **Solver agnostic:** 30+ solver interfaces via MathOptInterface
- **Extensibility:** Easy to add custom constraints and objectives
- **Open source:** MIT license, active community

40.5.1.2. Transportation Problem in JuMP

```
#=
Transportation Problem in JuMP
Minimize shipping costs from origins to destinations
=#
using JuMP
using HiGHS # Open source solver

function transportation_jump()
    # Data
    origins = ["Seattle", "SanDiego"]
    destinations = ["NewYork", "Chicago", "Topeka"]

    supply = Dict{"Seattle" => 350, "SanDiego" => 600}
    demand = Dict{"NewYork" => 325, "Chicago" => 300, "Topeka" => 275}

    cost = Dict{
        ("Seattle", "NewYork") => 2.5,
        ("Seattle", "Chicago") => 1.7,
        ("Seattle", "Topeka") => 1.8,
        ("SanDiego", "NewYork") => 3.5,
        ("SanDiego", "Chicago") => 1.8,
        ("SanDiego", "Topeka") => 1.4,
    }

    # Create model
    model = Model(HiGHS.Optimizer)
    set_silent(model)

    # Variables
    @variable(model, ship[origins, destinations] >= 0)
```

```

# Objective
@objective(model, Min,
    sum(cost[i, j] * ship[i, j] for i in origins, j in destinations)
)

# Constraints
@constraint(model, supply_limit[i in origins],
    sum(ship[i, j] for j in destinations) <= supply[i]
)

@constraint(model, demand_satisfy[j in destinations],
    sum(ship[i, j] for i in origins) >= demand[j]
)

# Solve
optimize!(model)

# Results
println("Status: ", termination_status(model))
println("Optimal Cost: \$", objective_value(model) * 1000)

println("\nShipping Plan:")
for i in origins, j in destinations
    v = value(ship[i, j])
    if v > 0.001
        println("  $i -> $j: $(round(v, digits=0)) units")
    end
end

# Shadow prices
println("\nShadow Prices:")
for i in origins
    println("  Supply $i: ", dual(supply_limit[i]))
end
for j in destinations
    println("  Demand $j: ", dual(demand_satisfy[j]))
end

return model
end

# Run
model = transportation_jump()

```

40.5.1.3. JuMP with Different Solvers

```

# Switch solvers easily
using JuMP
using GLPK
using Gurobi
using CPLEX

```

```
# Create model with GLPK (free)
model_glpk = Model(GLPK.Optimizer)

# Create model with Gurobi (commercial)
model_gurobi = Model(Gurobi.Optimizer)
set_optimizer_attribute(model_gurobi, "TimeLimit", 300)

# Create model with CPLEX (commercial)
model_cplex = Model(CPLEX.Optimizer)
```

40.5.1.4. JuMP Nonlinear Programming

```
#=
Nonlinear optimization example in JuMP
=#
using JuMP
using Ipopt

function nonlinear_example()
    model = Model(Ipopt.Optimizer)
    set_silent(model)

    @variable(model, x >= 0)
    @variable(model, y >= 0)

    # Nonlinear objective
    @NLobjective(model, Min, (x - 2)^2 + (y - 3)^2)

    # Nonlinear constraint
    @NLconstraint(model, x^2 + y^2 <= 25)

    optimize!(model)

    println("x = ", value(x))
    println("y = ", value(y))
end
```

40.5.1.5. JuMP Constraint Programming

```
#=
Constraint Programming with JuMP
N-Queens problem
=#
using JuMP
using COPT # or use HiGHS

function n_queens(n)
    model = Model(COPT.Optimizer)
    set_silent(model)

    # Variables: queen position in each row
    @variable(model, 1 <= q[1:n] <= n, Int)
```

```

# All different columns
for i in 1:n, j in (i+1):n
    @constraint(model, q[i] != q[j])
end

# All different diagonals
for i in 1:n, j in (i+1):n
    @constraint(model, q[i] - q[j] != i - j)
    @constraint(model, q[i] - q[j] != j - i)
end

optimize!(model)

if termination_status(model) == MOI.OPTIMAL
    positions = [round{Int, value(q[i])} for i in 1:n]
    return positions
end

return nothing
end

# Solve 8-queens
solution = n_queens(8)
println("Queen positions (row -> column): ", solution)

```

40.6. Other Languages

40.6.1. MiniZinc

MiniZinc is a high-level constraint modeling language designed for constraint programming and combinatorial optimization.

40.6.1.1. Features

- Solver independent (works with Gecode, Chuffed, OR-Tools, etc.)
- Designed specifically for constraint satisfaction
- Global constraints library
- IDE with solution visualization

40.6.1.2. Transportation in MiniZinc

```

% Transportation Problem in MiniZinc

% Data
int: num_origins = 2;
int: num_destinations = 3;

set of int: Origins = 1..num_origins;
set of int: Destinations = 1..num_destinations;

array[Origins] of int: supply = [350, 600];
array[Destinations] of int: demand = [325, 300, 275];

```

```

array[Origins, Destinations] of int: cost =
  [| 25, 17, 18
   | 35, 18, 14 |]; % Scaled by 10

% Variables
array[Origins, Destinations] of var 0..1000: ship;

% Objective
var int: total_cost = sum(i in Origins, j in Destinations)(
  cost[i,j] * ship[i,j]
);

% Constraints
constraint forall(i in Origins)(
  sum(j in Destinations)(ship[i,j]) <= supply[i]
);

constraint forall(j in Destinations)(
  sum(i in Origins)(ship[i,j]) >= demand[j]
);

% Solve
solve minimize total_cost;

% Output
output [
  "Total cost: ", show(total_cost), "\n",
  "Shipping plan:\n"
] ++ [
  "  Origin " ++ show(i) ++ " -> Dest " ++ show(j) ++
  ": " ++ show(ship[i,j]) ++ "\n"
  | i in Origins, j in Destinations where fix(ship[i,j]) > 0
];

```

40.6.2. Zimpl

Zimpl (Zuse Institute Mathematical Programming Language) is a free algebraic modeling language that generates input for SCIP, CPLEX, and other solvers.

40.6.2.1. Features

- Free and open source
- Direct SCIP integration
- Simple, clean syntax
- Good for research and education

40.6.2.2. Transportation in Zimpl

```

# Transportation Problem in Zimpl

# Sets
set Origins := { "Seattle", "SanDiego" };
set Destinations := { "NewYork", "Chicago", "Topeka" };

```

```

# Parameters
param supply[Origins] := <"Seattle"> 350, <"SanDiego"> 600;
param demand[Destinations] := <"NewYork"> 325, <"Chicago"> 300, <"Topeka"> 275;

param cost[Origins * Destinations] :=
    <"Seattle", "NewYork"> 2.5,
    <"Seattle", "Chicago"> 1.7,
    <"Seattle", "Topeka"> 1.8,
    <"SanDiego", "NewYork"> 3.5,
    <"SanDiego", "Chicago"> 1.8,
    <"SanDiego", "Topeka"> 1.4;

# Variables
var ship[<i,j> in Origins * Destinations] >= 0;

# Objective
minimize total_cost:
    sum <i,j> in Origins * Destinations: cost[i,j] * ship[i,j];

# Constraints
subto supply_limit:
    forall <i> in Origins:
        sum <j> in Destinations: ship[i,j] <= supply[i];

subto demand_satisfy:
    forall <j> in Destinations:
        sum <i> in Origins: ship[i,j] >= demand[j];

```

40.6.3. MOSEL/Xpress

MOSEL is FICO's modeling language for the Xpress solver suite.

40.6.3.1. Features

- Native Xpress integration
- Strong support for large-scale problems
- Procedural and declarative elements
- Parallel and distributed computing support

```

! Transportation Problem in MOSEL

model Transportation
uses "mmxprs"

declarations
    Origins: set of string
    Destinations: set of string
    supply: array(Origins) of real
    demand: array(Destinations) of real
    cost: array(Origins, Destinations) of real
    ship: array(Origins, Destinations) of mpvar
end-declarations

```

```

! Data
Origins := {"Seattle", "SanDiego"}
Destinations := {"NewYork", "Chicago", "Topeka"}

supply("Seattle") := 350
supply("SanDiego") := 600

demand("NewYork") := 325
demand("Chicago") := 300
demand("Topeka") := 275

cost("Seattle", "NewYork") := 2.5
cost("Seattle", "Chicago") := 1.7
cost("Seattle", "Topeka") := 1.8
cost("SanDiego", "NewYork") := 3.5
cost("SanDiego", "Chicago") := 1.8
cost("SanDiego", "Topeka") := 1.4

! Objective
TotalCost := sum(i in Origins, j in Destinations) cost(i,j) * ship(i,j)

! Constraints
forall(i in Origins) sum(j in Destinations) ship(i,j) <= supply(i)
forall(j in Destinations) sum(i in Origins) ship(i,j) >= demand(j)

! Solve
minimize(TotalCost)

! Output
writeln("Total cost: ", getobjval)
forall(i in Origins, j in Destinations | getsol(ship(i,j)) > 0)
  writeln("  ", i, " -> ", j, ": ", getsol(ship(i,j)))

end-model

```

The bar chart below compares all seven modeling languages across four metrics: solver support breadth, ease of learning, cost (higher means free/open-source), and community ecosystem size.

40.7. Comparison Framework

40.7.1. Feature Comparison Table

Feature	AMPL	GAMS	OPL	Pyomo	JuMP	PuLP	MiniZinc
Problem Types							
Linear Programming	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Integer Programming	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nonlinear Programming	Yes	Yes	No	Yes	Yes	No	Limited
Constraint Programming	Limited	Limited	Yes	Via plugins	Via solvers	No	Yes

40. Optimization Modeling Languages

Feature	AMPL	GAMS	OPL	Pyomo	JuMP	PuLP	MiniZinc
Stochastic Programming	Yes	Yes	No	Yes	Via extensions	No	No
Usability							
Learning Curve	Moderate	Moderate	Easy	Moderate	Moderate	Easy	Easy
Syntax Clarity	Excellent	Good	Good	Good	Excellent	Good	Excellent
Documentation	Excellent	Good	Good	Excellent	Excellent	Good	Good
IDE Support	Basic	GAMS Studio	CPLEX Studio	Standard Python	VS Code	Standard Python	MiniZinc IDE
Ecosystem							
Solver Support	40+	30+	CPLEX only	20+	30+	10+	10+
Community Size	Medium	Medium	Small	Large	Large	Large	Medium
Open Source	No	No	No	Yes	Yes	Yes	Yes
Free Academic	Limited	Limited	Yes	Yes	Yes	Yes	Yes
Performance							
Model Generation	Fast	Fast	Fast	Good	Excellent	Good	Fast
Large Models	Excellent	Excellent	Excellent	Good	Excellent	Good	Good
Integration							
Data Sources	Files, ODBC	Files, ODBC, Excel	Files, JDBC, OData	Python ecosystem	Julia ecosystem	Python ecosystem	Files
Scripting	Built-in	Built-in	JavaScript	Python	Julia	Python	Built-in
API Access	Yes	Yes	Yes	Native Python	Native Julia	Native Python	Python, etc.

40.7.2. Pricing Comparison (Approximate)

Language/Tool	Academic	Commercial	Notes
AMPL	Free (limited)	\$5,000-50,000/year	Depends on solver bundle
GAMS	Free (limited)	\$3,000-30,000/year	Depends on solver bundle
OPL/CPLEX Studio	Free	IBM subscription	Part of CPLEX package
Pyomo	Free	Free	MIT license
JuMP	Free	Free	MPL 2.0 license
PuLP	Free	Free	MIT license
CVXPY	Free	Free	Apache 2.0 license
OR-Tools	Free	Free	Apache 2.0 license
MiniZinc	Free	Free	MPL 2.0 license

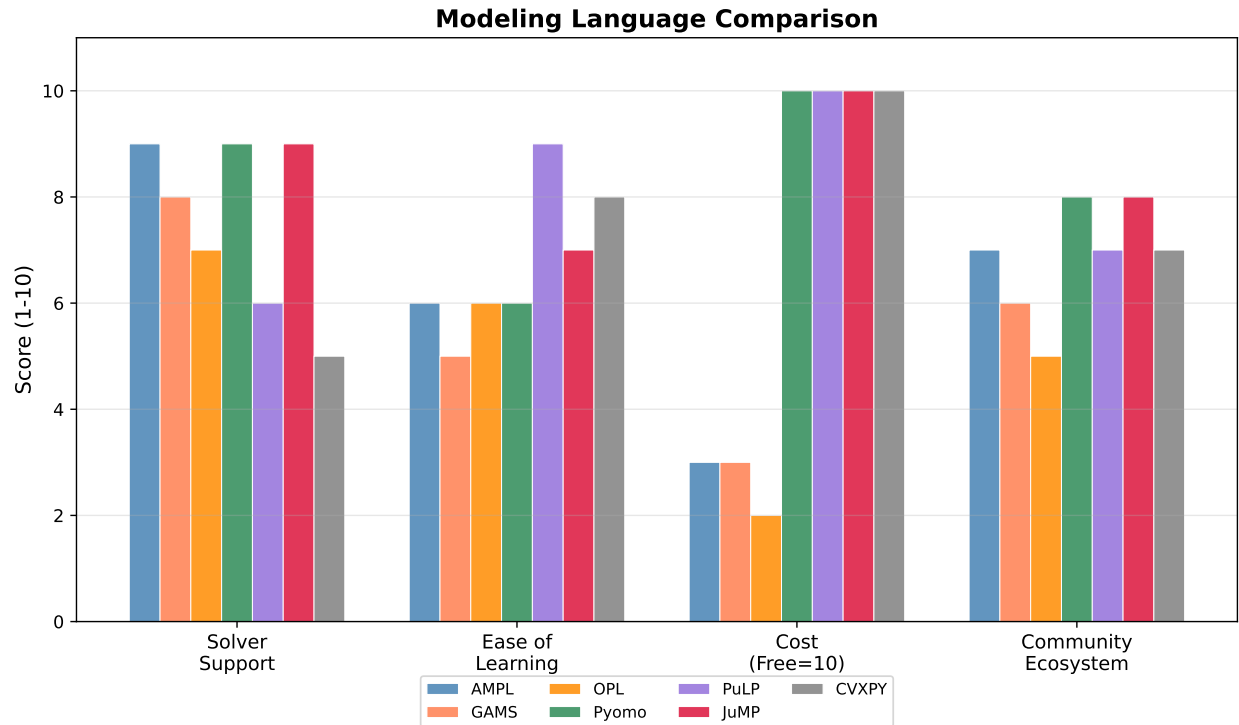


Figure 40.2.: Modeling language comparison across key metrics — solver support breadth, learning curve (inverted: higher = easier), licensing cost, and community ecosystem size

40.7.3. When to Use What

i Selection Guidelines

1. **For beginners:** Start with PuLP (simplest) or JuMP (best balance)
2. **For production Python:** Pyomo for complex models, PuLP for simple LP/MIP
3. **For performance:** JuMP or direct solver APIs
4. **For constraint programming:** MiniZinc or OR-Tools
5. **For existing CPLEX users:** OPL integrates best
6. **For economic modeling:** GAMS has the largest model library

40.8. Migration Guide

40.8.1. AMPL to Pyomo

Converting from AMPL to Pyomo requires understanding the mapping between concepts:

AMPL	Pyomo
set S	model.S = Set()
param p{S}	model.p = Param(model.S)
var x{S} >= 0	model.x = Var(model.S, domain=NonNegativeReals)

AMPL	Pyomo
minimize obj: sum{i in S} c[i]*x[i]	model.obj = Objective(expr=sum(model.c[i]*model.x[i] for i in model.S), sense=minimize)
subject to con{i in S}: x[i] <= b[i]	model.con = Constraint(model.S, rule=lambda m, i: m.x[i] <= m.b[i])

Example Migration:

AMPL:

```

set PRODUCTS;
param profit{PRODUCTS};
param capacity;
var produce{PRODUCTS} >= 0;

maximize total_profit: sum{p in PRODUCTS} profit[p] * produce[p];
subject to capacity_limit: sum{p in PRODUCTS} produce[p] <= capacity;

```

Pyomo:

```

from pyomo.environ import *

model = ConcreteModel()

model.PRODUCTS = Set(initialize=['A', 'B', 'C'])
model.profit = Param(model.PRODUCTS, initialize={'A': 10, 'B': 15, 'C': 12})
model.capacity = Param(initialize=100)

model.produce = Var(model.PRODUCTS, domain=NonNegativeReals)

model.total_profit = Objective(
    expr=sum(model.profit[p] * model.produce[p] for p in model.PRODUCTS),
    sense=maximize
)

model.capacity_limit = Constraint(
    expr=sum(model.produce[p] for p in model.PRODUCTS) <= model.capacity
)

```

40.8.2. GAMS to JuMP

GAMS	JuMP
Set i / i1*i5 /	I = 1:5 or I = ["i1", "i2", ...]
Parameter a(i)	a = Dict(...)
Positive Variable x(i)	@variable(model, x[I] >= 0)
z =e= sum(i, c(i)*x(i))	@objective(model, Min, sum(c[i]*x[i] for i in I))
con(i) .. x(i) =l= b(i)	@constraint(model, [i in I], x[i] <= b[i])

Example Migration:

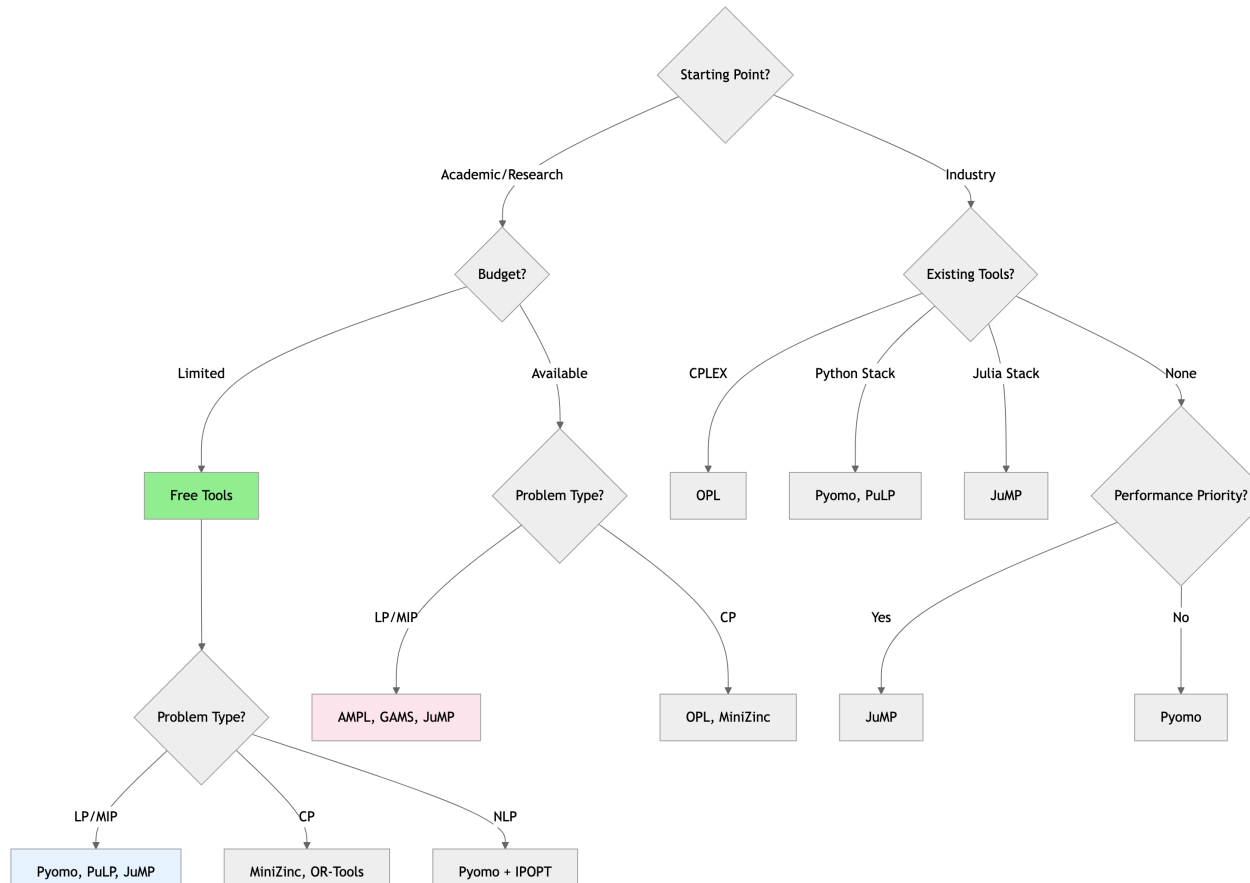


Figure 40.3.: Decision Tree for Modeling Language Selection

GAMS:

```

Set i products / p1*p3 /;
Parameter c(i) cost / p1 10, p2 15, p3 12 /;
Parameter b(i) capacity / p1 50, p2 40, p3 60 /;

Positive Variable x(i);
Variable z;

Equation obj, limit(i);

obj .. z =e= sum(i, c(i) * x(i));
limit(i) .. x(i) =l= b(i);

Model prodmix / all /;
Solve prodmix minimizing z using lp;

```

JuMP:

```

using JuMP, HiGHS

model = Model(HiGHS.Optimizer)

I = ["p1", "p2", "p3"]
c = Dict("p1" => 10, "p2" => 15, "p3" => 12)
b = Dict("p1" => 50, "p2" => 40, "p3" => 60)

@variable(model, x[i in I] >= 0)
@objective(model, Min, sum(c[i] * x[i] for i in I))
@constraint(model, limit[i in I], x[i] <= b[i])

optimize!(model)

```

40.8.3. Common Patterns Across Languages

Summations:

Language	Syntax
AMPL	<code>sum{i in S} expr[i]</code>
GAMS	<code>sum(i, expr(i))</code>
OPL	<code>sum(i in S) expr[i]</code>
Pyomo	<code>sum(expr[i] for i in S)</code>
JuMP	<code>sum(expr[i] for i in S)</code>

Indexed Constraints:

Language	Syntax
AMPL	<code>subject to con{i in S}: lhs <= rhs;</code>
GAMS	<code>con(i) .. lhs =l= rhs;</code>
OPL	<code>forall(i in S) ctName: lhs <= rhs;</code>
Pyomo	<code>model.con = Constraint(S, rule=rule_func)</code>
JuMP	<code>@constraint(model, con[i in S], lhs <= rhs)</code>

40.9. Best Practices

40.9.1. Model Organization

Separate Concerns:

1. **Data preparation:** Load, validate, transform data
2. **Model definition:** Sets, parameters, variables, constraints
3. **Solution:** Solver configuration, optimization
4. **Analysis:** Extract results, generate reports

```

# Good: Modular structure
class TransportationModel:
    def __init__(self, data):
        self.data = data
        self.model = None
        self.solution = None

```

```

def build_model(self):
    """Construct the optimization model."""
    # Model construction code
    pass

def solve(self, solver='highs', time_limit=300):
    """Solve the model with specified solver."""
    # Solving code
    pass

def get_results(self):
    """Extract and return solution."""
    # Result extraction
    pass

def generate_report(self):
    """Create human-readable report."""
    # Reporting code
    pass

```

Use Meaningful Names:

```

# Bad
x = Var(I, J)
c1 = Constraint(I, rule=r1)

# Good
ship_quantity = Var(origins, destinations, domain=NonNegativeReals)
supply_limit = Constraint(origins, rule=supply_constraint_rule)

```

40.9.2. Data Handling

Validate Input Data:

```

def validate_transportation_data(supply, demand, costs):
    """Validate transportation problem data."""
    # Check supply-demand balance
    total_supply = sum(supply.values())
    total_demand = sum(demand.values())

    if total_supply < total_demand:
        raise ValueError(
            f"Infeasible: supply ({total_supply}) < demand ({total_demand})"
        )

    # Check all costs defined
    for i in supply:
        for j in demand:
            if (i, j) not in costs:
                raise ValueError(f"Missing cost for route {i} -> {j}")

```

```
# Check non-negative values
for k, v in costs.items():
    if v < 0:
        raise ValueError(f"Negative cost {v} for route {k}")

return True
```

Use Pandas for Data Management:

```
import pandas as pd

# Load data from CSV
supply_df = pd.read_csv('supply.csv', index_col='origin')
demand_df = pd.read_csv('demand.csv', index_col='destination')
costs_df = pd.read_csv('costs.csv', index_col=['origin', 'destination'])

# Convert to model-friendly format
supply = supply_df['capacity'].to_dict()
demand = demand_df['requirement'].to_dict()
costs = costs_df['cost'].to_dict()
```

40.9.3. Debugging Tips**Check Model Before Solving:**

```
# Pyomo: print model summary
model.pprint()

# Check specific components
print("Variables:", list(model.component_objects(Var)))
print("Constraints:", list(model.component_objects(Constraint)))
```

```
# JuMP: print model
print(model)

# Check number of variables and constraints
println("Variables: ", num_variables(model))
println("Constraints: ", num_constraints(model))
```

Diagnose Infeasibility:

```
# Pyomo: compute IIS (Irreducible Infeasible Set)
from pyomo.contrib.iis import write_iis

# After getting infeasible status
write_iis(model, 'infeasible.ilp')
```

```
# JuMP with Gurobi
using Gurobi
```

```

model = Model(Gurobi.Optimizer)
# ... build model ...
optimize!(model)

if termination_status(model) == MOI.INFEASIBLE
    compute_conflict!(model)
    # Print conflicting constraints
end

```

Use Logging:

```

import logging

logging.basicConfig(level=logging.INFO)
logger = logging.getLogger('optimization')

def solve_model(model):
    logger.info("Starting solve...")
    logger.info(f"Variables: {len(list(model.component_objects(Var)))}")
    logger.info(f"Constraints: {len(list(model.component_objects(Constraint)))}")

    result = solver.solve(model, tee=True)

    logger.info(f"Status: {result.solver.termination_condition}")
    logger.info(f"Objective: {value(model.objective)}")

    return result

```

40.9.4. Performance Optimization

Reduce Model Size:

```

# Bad: Create variable for every pair
x = Var(I, J, domain=NonNegativeReals)

# Good: Only create variables for valid pairs
valid_pairs = [(i, j) for i in I for j in J if cost[i,j] < float('inf')]
x = Var(valid_pairs, domain=NonNegativeReals)

```

Use Efficient Data Structures:

```

# Bad: Nested loops in objective
obj = sum(cost[i,j] * x[i,j] for i in I for j in J)

# Better: Pre-compute once, use efficient summation
def objective_rule(m):
    return quicksum(m.cost[i,j] * m.x[i,j] for i in m.I for j in m.J)

```

Warm Starting:

```
# Provide initial solution (Pyomo with Gurobi)
model.x['Seattle', 'Chicago'].value = 300
model.x['SanDiego', 'NewYork'].value = 325

solver = SolverFactory('gurobi')
solver.options['MIPStart'] = 1
solver.solve(model, warmstart=True)
```

```
# JuMP with warm start
set_start_value.(x, initial_solution)
optimize!(model)
```

Solver-Specific Tuning:

```
# Gurobi tuning
solver = SolverFactory('gurobi')
solver.options['MIPGap'] = 0.01      # 1% gap
solver.options['TimeLimit'] = 300    # 5 minutes
solver.options['Threads'] = 4        # Use 4 threads
solver.options['Presolve'] = 2       # Aggressive presolve

# HiGHS tuning
solver = SolverFactory('appsi_highs')
solver.options['mip_rel_gap'] = 0.01
solver.options['time_limit'] = 300.0
```

40.10. Summary

40.10.1. Key Takeaways

- **Algebraic modeling languages** separate problem formulation from solution algorithms, improving productivity and maintainability
- **Commercial languages** (AMPL, GAMS, OPL) offer mature ecosystems but require expensive licenses; **open-source alternatives** (Pyomo, JuMP, PuLP) are competitive
- **Python libraries** (Pyomo, PuLP, CVXPY, OR-Tools) integrate with the data science ecosystem; **JuMP** offers the best performance for large models
- **Choose based on:** problem type, solver requirements, team expertise, licensing constraints, and integration needs
- **Best practices:** separate data from model, validate inputs, use meaningful names, modularize code, and tune solver parameters
- **Migration** between languages is feasible—most share similar concepts with different syntax

40.11. Exercises

1. **Language Comparison:** Implement the diet problem (minimize cost while meeting nutritional requirements) in three different languages: PuLP, Pyomo, and JuMP. Compare code length, clarity, and solve times.
2. **Data Integration:** Create a transportation model in Pyomo that reads data from CSV files and writes results to an Excel file using pandas.

3. **Solver Benchmarking:** Solve the same facility location problem with GLPK, HiGHS, and CBC. Compare solution times and verify solutions match.
4. **Debugging Practice:** Given an infeasible transportation problem (demand exceeds supply), write code that detects infeasibility and identifies which constraints are violated.
5. **Migration Exercise:** Convert the following AMPL model to JuMP:

```

set NODES;
set ARCS within {NODES, NODES};
param capacity{ARCS} >= 0;
param cost{ARCS} >= 0;
param supply{NODES};
var flow{ARCS} >= 0;

minimize total_cost: sum{(i,j) in ARCS} cost[i,j] * flow[i,j];

subject to flow_balance{k in NODES}:
    sum{(i,k) in ARCS} flow[i,k] - sum{(k,j) in ARCS} flow[k,j] = supply[k];

subject to capacity_limit{(i,j) in ARCS}:
    flow[i,j] <= capacity[i,j];

```

6. **Performance Optimization:** Take a vehicle routing problem with 100 customers and experiment with different solver parameters. Document which settings improve performance.
7. **Constraint Programming:** Implement the N-queens problem in both MiniZinc and OR-Tools CP-SAT. Compare the modeling approach and solution speed for $N = 8, 12,$ and 20 .

About the Author

Alper Ersin Balci is an Industrial Engineer and optimization consultant who has been working at the intersection of operations research, artificial intelligence, and data science since 2018.

He is the founder of **Allen Ralford Barkley**, an optimization consultancy where he designs and deploys decision intelligence solutions for clients across the US and Europe. His project portfolio spans production scheduling, vehicle routing, cutting stock optimization, facility location, and inventory management—solving real-world combinatorial problems for organizations including Shell, Coca-Cola, FIAT (Stellantis), BSH (Bosch & Siemens), and BMW.

Before founding his consultancy, Alper worked as an Industry 4.0 Engineer at BSH and as a Data Analytics Specialist at the BMW distribution network, where he built predictive models, led smart factory analytics initiatives, and developed supply chain planning systems.

He has taught optimization and AI to graduate-level industrial engineering students and reaches a broad audience through his YouTube channel **Neural Optima**, where he creates educational content on optimization algorithms and artificial intelligence.

Alper is a winner of the L’Oreal Brandstorm Forecasting Challenge (2020).

This book grew out of years of bridging the gap between academic optimization theory and production-grade software engineering—and a desire to put everything he wished he had in one place when he started.

